Projector-Refinement Physics

Complete Unified Theory

From Exact Packet Calculus and Quasienergy to Lorentzian BV–BFV Gravity, Strong-Field Collapse, Cosmological Perturbations, and a Scale-Covariant Dark Sector

Komorebi P.

projector reconstruction and transfer $\Longrightarrow$ local conservation and refinement
$\Longrightarrow$ relativistic and gauge dynamics $\Longrightarrow$ nonlinear gravitational constraints
$\Longrightarrow$ strong-field evolution and cosmology $\Longrightarrow$ audited predictions and open tests.

August 2026

Symbols and Notation

The same letter can represent different objects in different sectors. Each chapter fixes its local convention. The table below records the principal global meanings and the most important overloaded symbols.

Symbol	Meaning
$\mathcal{H}$	Hilbert space, packet state space, or Hamiltonian density when decorated by sector labels.
P_α, $\mathcal{P}$	Orthogonal packet projector and complete projector atlas.
$X_{\alpha\beta} = P_\alpha X P_\beta$	Operator corner from packet β to packet α.
$J_{\alpha\leftarrow\beta}$	Antisymmetric transfer current.
R	Refinement incidence map; in gravitational chapters it may also denote areal radius or curvature, as stated locally.
ε, h, Δ	Local spatial scale, refinement scale, or discrete time step.
ω_Δ	Exact discrete quasienergy.
α_i, β, Γ_I	Dirac or Clifford generators.
A_μ, $U_{x,\mu}$, $F_{\mu\nu}$	Gauge potential, compact link variable, and curvature.
e^I, E^I, ω^{IJ}	Lorentzian tetrad, spatial coframe, and spin connection.
B_{IJ}	Tetrad bivector momentum conjugate to the Lorentz connection.
$K_a{}^i$, $\widetilde{E}^a{}_i$	Reduced real extrinsic-curvature variable and densitized triad.
Ω_Σ	Boundary BFV charge.
S_{BV}	Classical BV master action.
$\mathcal{G}$, $\mathcal{D}$, $\mathcal{H}$	Gauss, spatial-diffeomorphism, and Hamiltonian constraints.
N, N^i	Lapse and shift.
$a(t)$, H	Cosmological scale factor and Hubble rate; H can also denote a Hamiltonian in noncosmological chapters.
ρ, p, w	Energy density, pressure, and equation-of-state parameter.
ϕ	Average scalar/refinement mode or ordinary scalar field, according to chapter.
$\chi_{h,a}$	Haar-detail field at refinement level h and multiplicity index a.
P_0, D_h	Fiber-average projector and level-h detail projector.
p	Prime defining the internal refinement hierarchy; in fluid equations it can denote pressure.
σ	Refinement derivative order; in spherical gravity it can denote lapse normalization, as defined locally.
Λ	Internal mass scale or cosmological constant, distinguished by context and subscripts.
μ, v, λ	Average-mode potential parameters in the unified dark-sector chapters.
γ	Scale-covariant average-detail coupling in the corrected dark-sector branch.
m_h, d_h	Level mass and multiplicity of a refinement-detail shell.
ν_h	Level-dependent nonadiabatic production scale.
ρ_D, p_D, w_D	Total detail-tower density, pressure, and equation of state.
M_ϕ^2	Effective locking curvature of the average mode.
$\mathcal{M}$	Spherical charged mass coordinate.
$m(t,r)$	Misner–Sharp mass in scalar collapse.
Φ, Π	Scalar radial gradient and momentum in the collapse chapter; elsewhere Φ can denote a gravitational potential.
$\theta_\pm$	Outgoing and ingoing null expansions.
P, N, C, Q, O	Proved, numerical, conditional, qualified, and open status classes.

Units and signatures. Natural units are used unless a chapter explicitly restores c or $\hbar$. The Lorentzian parent uses signature $(-,+,+,+)$. Spatial finite-difference, Fourier, and continuum derivatives are never identified without an explicit convergence or alias-free hypothesis.

Revision and Claim Status

This edition is a unified derivation record rather than a sequence of independent addenda. Claims are interpreted according to the status printed beside them:

P : proved in the declared mathematical setting,

N : numerically reproduced with the declared discretization,

C : conditional on a stated model or regularity hypothesis,

Q : qualified or corrected statement,

O : open target or unverified physical interpretation.

The additive dark-sector constructions retained in Chapters 47 and 48 are historical conditional branches. The later scale-covariant chapter proves that the one-shell, one-crossing, energy-matched additive mechanism cannot simultaneously reproduce the declared abundance, remain nonrelativistic, and lock the average mode. Those earlier chapters remain useful as algebraic derivations and negative controls, but their physical-viability conclusions are superseded. The current dark-sector candidate is the scale-covariant branch.

No statement in this manuscript constitutes observational confirmation, literature priority, or a proof that nature realizes the proposed microscopic structure. Exact identities, controlled numerical results, model extensions, and open physical claims are kept separate throughout.

Contents

Symbols and Notation i

Revision and Claim Status iii

I Projector-Refinement Foundations 1

1 Conventions and Scope 2
- 1.1 Linear-algebra convention . . . 2
- 1.2 Spacetime convention . . . 2
- 1.3 Gauge-link convention . . . 2
- 1.4 Radial-force convention . . . 3

2 Projector-Refinement Core 4
- 2.1 Atlas and corner algebra . . . 4
- 2.2 Refinement . . . 4
- 2.3 Skew-adjoint transfer theorem . . . 5
- 2.4 Refinement covariance of transfer . . . 5
- 2.5 Moving atlas . . . 6

2.6 Whole-object deformation rule . . . 6

3 Types, Dimensions, and Logical Dependencies 7
3.1 Type judgments . . . 7
3.2 Core sequents . . . 7
3.3 Refinement matrices . . . 7
3.4 Variation sequents . . . 8
3.5 Continuum-limit sequents . . . 8
3.6 Unit and normalization judgments . . . 8

4 Extensions of the Core Transfer Calculus 10
4.1 State-dependent skew-adjoint flow . . . 10
4.2 Mixed-state unitary transfer . . . 10
4.3 Associative refinement tower . . . 11
4.4 Tensor-product atlas . . . 11
4.5 Universal antisymmetric balance and refinement . . . 11
4.6 Packet information transport . . . 12
4.7 Lindblad packet jumps . . . 12
4.8 Detailed-balance packet theorem . . . 13
4.9 General refinement-cancellation tower . . . 13

5 Positive-Operator and Overlapping Atlases 14
5.1 Positive partition of unity . . . 14
5.2 Moving positive atlas . . . 14
5.3 Exact refinement . . . 15
5.4 Parseval-channel realization and dilation . . . 15
5.5 Cycle-current gauge . . . 16
5.6 Numerical residuals . . . 16

6 Countable Atlases and Unbounded Generators 17
6.1 Graph-compatible atlas . . . 17
6.2 Spectral packets and domain hierarchy . . . 17
6.3 Bounded surrogates and topology-specific truncation . . . 18
6.4 Countable packet continuity . . . 18
6.5 Countable refinement . . . 20
6.6 Spectral residue packets and ladder routing . . . 20
6.7 Nonreducing packets and a resolvent certificate . . . 21
6.8 Infinite reconstruction tower . . . 21
6.9 Verification and scope . . . 21

7 Finite Differential No-Go and Corner-Algebra Resolution 23
7.1 Finite pointwise algebra: no-go theorem . . . 23
7.2 Finite-scale consequence . . . 23
7.3 Full corner algebra . . . 24
7.4 Curvature of finite derivations . . . 24
7.5 Projector-corner reconstruction . . . 25
7.6 Numerical verification . . . 25

II Discrete Relativity, Gauge Fields, and Finite-Scale Kinematics 26

8 Discrete Relativistic Sectors 27
8.1 One spatial dimension . . . 27
8.2 Three spatial dimensions . . . 28
8.3 Observer-shell defect . . . 29

9 Exact 3+1 Quasienergy Closure 30
9.1 Exact shell and corrected continuum hierarchy . . . 30
9.2 Exact phase and transfer coordinates . . . 31

9.3 Principal-branch bounds and nonlinear Lorentz action . . . 32
9.4 Palindromic Clifford factorization . . . 33
9.5 Permutation symmetry versus strict locality . . . 35
9.6 Exact velocity, kinetic mass, and anisotropy . . . 36
9.7 Signed-route interpretation . . . 37
9.8 General finite-route theorem . . . 37
9.9 Improved cubic scalar route . . . 39
9.10 Gauge-covariant four-spinor walk . . . 39
9.11 Gauge-covariant scalar path completion and unitary dilation . . . 40
9.12 Projected recurrence and conservation laws . . . 41
9.13 Polyphase sectors and doubling obstruction . . . 42
9.14 Local Wilson commutator regulator . . . 42
9.15 Fourth-order isotropic contraction refinement . . . 43
9.16 Sixth-order residual and positive local cancellation . . . 44
9.17 Closure ledger . . . 45

10 Abelian Gauge Sectors **46**
10.1 Time-dependent linked walk . . . 46
10.2 Plaquette curvature and Wilson action . . . 46
10.3 Semidiscrete Maxwell–Dirac closure . . . 47

11 Non-Abelian Gauge and Ordered Holonomy Refinement **49**
11.1 Local SU(2) gauge atlas . . . 49
11.2 Plaquette, Wilson action, and currents . . . 49
11.3 Exact ordered 2-by-2 refinement . . . 50
11.4 Curvature refinement . . . 50
11.5 Numerical residuals . . . 51

12 Conditional Finite-Scale Propagation Signatures **52**
12.1 Exact massless shell of the cubic walk . . . 52
12.2 Cubic-harmonic angular signature . . . 53
12.3 Frequency and velocity correction . . . 53
12.4 Directional photon-delay law . . . 54
12.5 Verification and inference . . . 54

III Nonlinear Lorentzian Gravity and Strong Fields **55**

13 Weak, Optical, and Tensor Gravity **56**
13.1 Spherical weak field . . . 56
13.2 Weak-field metric and observables . . . 57
13.3 Corrected Fierz–Pauli sector . . . 58
13.4 TT projector . . . 59
13.5 Discrete tensor wave . . . 59
13.6 Discrete tensor energy . . . 60

14 Holonomy, Metric Curvature, and Local Work **61**
14.1 Non-Abelian plaquette curvature . . . 61
14.2 Exact conformal metric-holonomy identity . . . 62
14.3 Local scalar stress-energy transfer . . . 62

15 Symmetry-Reduced Inhomogeneous Gravity **64**
15.1 Unpolarized Gowdy action . . . 64
15.2 Hamiltonian- and momentum-type constraints . . . 64
15.3 Polarized exact mode . . . 65
15.4 Nonlinear numerical example . . . 66

16 Constraint-Preserving Discrete Gowdy Evolution **68**
16.1 Periodic semidiscrete system . . . 68

16.2 Differential constraint force . . . 68
16.3 Second-order map and exact endpoint constraint . . . 69
16.4 Local compatibility diagnostic . . . 70
16.5 Next steps . . . 71

17 Local 3+1 Linearized Projector–ADM Theory 72
17.1 Finite spatial calculus . . . 72
17.2 Canonical variables and constraints . . . 72
17.3 Quadratic local action . . . 73
17.4 Exact finite constraint propagation . . . 74
17.5 Linearized gauge symmetry and bracket closure . . . 74
17.6 TT sector and continuum limit . . . 75
17.7 Constraint refinement . . . 75
17.8 Numerical verification . . . 75
17.9 Next steps . . . 76

18 Nonlinear Constraint Closure: Pointwise Defect and Corner Extension 77
18.1 Scalar diffeomorphism model . . . 77
18.2 Finite pointwise obstruction . . . 78
18.3 Controlled finite resolutions . . . 78
18.4 Corner-algebra route . . . 79
18.5 Nonlinear projector–ADM target . . . 79
18.6 Numerical verification . . . 79

19 The Nonlinear Gauge-Closure Bridge: BFV–BV Polyphase Transfer 81
19.1 Canonical corner phase space . . . 81
19.2 Exact inner momentum sector . . . 81
19.3 Projection leakage is the finite nonlinear defect . . . 82
19.4 Fine BFV charge and contracting data . . . 83
19.5 Exact nonlinear BFV transfer . . . 83
19.6 Projector–ADM generators from the transferred charge . . . 85
19.7 Quantum closure by BV pushforward . . . 86
19.8 Exact finite projector action from a fine BV theory . . . 86
19.9 Quadratic shadow: Schur and prime-polyphase composition . . . 87
19.10 An exact nonlinear reduction model . . . 87
19.11 Controlled continuum limit . . . 90
19.12 Result and next physical steps . . . 91
19.13 Verification . . . 91

20 The Next Gravitational Target: BFV Refinement of Gowdy Dynamics 92
20.1 Areal-time canonical form . . . 92
20.2 Exact continuum hypersurface algebra . . . 93
20.3 Chiral decomposition . . . 94
20.4 What must be restored before BFV quantization . . . 95
20.5 Fine BFV charge and projector transfer . . . 95
20.6 Refinement and continuum theorem to be proved . . . 96
20.7 Baseline verification . . . 96
20.8 Resulting direction . . . 97

21 Exact Resolved Nonlinear $3+1$ Constraint Closure 98
21.1 Status and scope . . . 98
21.2 Filtered spectral differential algebra . . . 98
21.3 Resolved self-dual canonical action . . . 99
21.4 Exact finite nonlinear constraint algebra . . . 100
21.5 Exact constraint propagation . . . 100
21.6 Refinement-safe closure tower . . . 101
21.7 Relation to BFV–BV transfer . . . 101
21.8 Numerical certificate . . . 101

21.9 Physical boundary . . . 102
21.10 Machine-readable records . . . 102

22 Resolved Lorentzian Gravity–Matter BV Parent 103
22.1 Status and objective . . . 103
22.2 Four-dimensional resolved differential algebra . . . 103
22.3 Real Lorentzian parent fields . . . 104
22.4 One polynomial Lorentzian gravity–matter action . . . 104
22.5 Off-shell gauge system . . . 105
22.6 Explicit nilpotent BRST differential . . . 105
22.7 Minimal classical BV master action . . . 106
22.8 Boundary symplectic form and an explicit internal BFV charge . . . 107
22.9 Exact finite resolved master-equation theorem . . . 107
22.10 Bianchi and Noether closure . . . 108
22.11 Chiral doubling and Lorentzian reality . . . 108
22.12 Executable certificate . . . 109
22.13 Closure ledger . . . 109
22.14 Next decisive target . . . 110
22.15 Machine-readable records . . . 110

23 Resolved Lorentzian $3+1$ BV–BFV Boundary Closure 111
23.1 Status and objective . . . 111
23.2 Time–space decomposition . . . 111
23.3 Exact split of the complete parent action . . . 112
23.4 Diffeomorphism moment map . . . 113
23.5 Complete boundary BFV charge . . . 114
23.6 Normal and tangential decomposition . . . 114
23.7 Exact resolved theorem . . . 115
23.8 Adaptive-refinement evolution relation . . . 116
23.9 Executable certificate . . . 116
23.10 Closure ledger . . . 117
23.11 Next decisive target . . . 117
23.12 Machine-readable records . . . 117

24 Resolved Second-Class Reduction and a Constraint-Preserving Time Step 118
24.1 Status and objective . . . 118
24.2 Regular time-gauge sector . . . 118
24.3 Exact simplicity pullback . . . 118
24.4 Maxwell auxiliary reduction . . . 119
24.5 Fermionic primary reduction . . . 119
24.6 Algebraic torsion and the induced interaction . . . 120
24.7 Minimal real canonical superphase space . . . 121
24.8 Reduced first-class generators . . . 121
24.9 BFV descent through the second-class quotient . . . 122
24.10 Discrete Dirac-gradient gauge-fixed step . . . 122
24.11 Existence, stability, and refinement . . . 123
24.12 Executable certificate . . . 124
24.13 Closure ledger . . . 124
24.14 Next decisive target . . . 125
24.15 Machine-readable records . . . 125

25 Finite-Range Spherical Electrovac and a Strong-Field Refinement Theorem 126
25.1 Status and objective . . . 126
25.2 Spherical Einstein–Maxwell canonical system . . . 126
25.3 Exact mass-coordinate identity . . . 127
25.4 Horizon-penetrating Reissner–Nordström family . . . 127
25.5 Strong-field and horizon diagnostics . . . 129
25.6 Strictly finite-range radial complex . . . 129

25.7 Finite-range strong-field evolution . . . 130
25.8 Level-independent stability theorem . . . 130
25.9 Horizon localization and extremal conditioning . . . 132
25.10 Executable strong-field certificate . . . 132
25.11 Closure ledger . . . 134
25.12 Next decisive target . . . 134
25.13 Machine-readable records . . . 135

26 Horizon-Penetrating Spherical Scalar Collapse and Trapped-Surface Formation 137
26.1 Status and objective . . . 137
26.2 Canonical Einstein–scalar system . . . 137
26.3 Flat-slice horizon-penetrating reduction . . . 138
26.4 Exact local mass-flux theorem . . . 139
26.5 Characteristic structure . . . 139
26.6 Marginally trapped sphere criterion . . . 140
26.7 Finite-stencil constrained evolution . . . 140
26.8 Initial scalar packets . . . 141
26.9 Trapped-surface formation result . . . 141
26.10 Refinement study . . . 141
26.11 Weak-pulse control . . . 143
26.12 Constraint and flux diagnostics . . . 144
26.13 What this block establishes . . . 145
26.14 Boundary of the result . . . 145
26.15 Next decisive target . . . 145
26.16 Machine-readable records . . . 145

IV Adic, Polyphase, and Quantum Refinement 146

27 Dyadic–Odd Quantum Register Factorization 147
27.1 Mathematical scope . . . 147
27.2 Finite unit coordinates . . . 147
27.3 Exact CRT quantum factorization . . . 148
27.4 Quantum Fourier factorization . . . 148
27.5 Dyadic sign protection . . . 149
27.6 Numerical verification . . . 149
27.7 Next steps . . . 150

28 Adic Haar Packets and Refinement-Compatible Quantum Dynamics 152
28.1 Finite principal-phase tower . . . 152
28.2 Child-DFT/Haar realization . . . 153
28.3 Refinement isometry . . . 153
28.4 Refinement-compatible Hamiltonian classification . . . 154
28.5 Two compatible Hamiltonian families . . . 155
28.6 Ultrametric propagator . . . 155
28.7 Numerical verification . . . 156
28.8 Next steps . . . 156

29 Finite-Adelic Quantum Dynamics and Weighted Leakage 158
29.1 Finite multi-prime Hilbert space . . . 158
29.2 Exact simultaneous prime refinement . . . 158
29.3 Dyadic-sign/odd-adic entanglement . . . 159
29.4 Valuation-weighted refinement anomaly . . . 159
29.5 Dyadic precision requirement . . . 160
29.6 Numerical verification . . . 161
29.7 Interpretation and next steps . . . 161

30 Haar Normalization and Collective Quantum Coupling 163

30.1 Nested refinement amplitudes . . . 163
30.2 Per-leaf versus collective coupling . . . 163
30.3 Exact radial probability and self-energy . . . 164
30.4 Measure-normalized and aggregate-packet regimes . . . 164
30.5 Numerical verification . . . 165
30.6 Next steps . . . 165

31 Local Quantum Walks on Spacetime–Adic Product Geometry 166
31.1 Product graph . . . 166
31.2 Local coined evolution . . . 166
31.3 Exact local and coarse continuity . . . 167
31.4 Exact product causal cone . . . 167
31.5 Numerical verification . . . 167
31.6 Next steps . . . 168

32 Radial Reduction and Continued-Fraction Self-Energy 170
32.1 Rooted p-tree Hamiltonian . . . 170
32.2 Prime-independent radial response . . . 170
32.3 Exact continued fraction . . . 171
32.4 Visible relativistic sector plus hidden adic chain . . . 171
32.5 Numerical verification . . . 172
32.6 Next steps . . . 173

33 Adic Quantum Polyphase Refinement 174
33.1 Standard and adic polyphase analysis . . . 174
33.2 Root-of-unity phase projectors . . . 174
33.3 Child Fourier transform and Haar refinement . . . 175
33.4 Exact descendant normalization . . . 175
33.5 Operator and channel polyphase calculus . . . 176
33.6 Exact polyphase band folding . . . 177
33.7 Hierarchical spectra and multi-prime refinement . . . 177
33.8 Polyphase Feshbach reduction . . . 178
33.9 Numerical verification . . . 178
33.10 Next steps . . . 179

34 Infinite-Depth Adic Haar Quantum Limit 181
34.1 Haar probability space and cylinder filtration . . . 181
34.2 Exact direct-limit identification . . . 181
34.3 Strong infinite Haar resolution . . . 182
34.4 Quantitative truncation theorem . . . 184
34.5 Self-adjoint infinite hierarchical Hamiltonians . . . 184
34.6 Finite-set adelic tensor limit . . . 186
34.7 Numerical verification . . . 186
34.8 Next steps . . . 186

35 Prime-Polyphase Spectral Renormalization 188
35.1 Nearest-neighbor polyphase model . . . 188
35.2 Chebyshev transfer law . . . 188
35.3 Companion-state trace coordinate . . . 190
35.4 Resonant detail channels . . . 191
35.5 Band-folding and decimation duality . . . 191
35.6 Exact renormalization semigroup . . . 192
35.7 Invariant spectral fixed measure . . . 192
35.8 Numerical verification . . . 194
35.9 Next steps . . . 194

36 Hypergeometric Companion Lanes and Quantum Packet Functions 197
36.1 Analytic residue projectors . . . 197

36.2 Factorial lanes and hypergeometric compression . . . 198
36.3 Cyclic companion-state realization . . . 199
36.4 Bounded-operator residue calculus . . . 199
36.5 Prime refinement and the standard unit window . . . 200
36.6 Coherent-state energy packets . . . 201
36.7 Verification and scope . . . 202

V Cosmology, Perturbations, and the Dark Sector 203

37 Cosmology, Backreaction, and Constraints 204
37.1 FLRW kinematics . . . 204
37.2 Packet stress-energy . . . 205
37.3 Lapse-resolved variational closure . . . 205
37.4 Midpoint discrete action . . . 206
37.5 Homogeneous scalar-gravity closure . . . 206
37.6 Exact discrete-gradient constraint . . . 207

38 Curved Multicomponent FLRW Background and Distances 209
38.1 Curved lapse-resolved minisuperspace action . . . 209
38.2 Constant-equation-of-state component library . . . 210
38.3 Conservative interacting fluids . . . 210
38.4 Background observables . . . 211
38.5 Projector background solver . . . 212
38.6 Next steps . . . 212

39 Cosmological Finite-Scale Photon Propagation 213
39.1 Local velocity law . . . 213
39.2 Physical versus comoving microscopic scale . . . 213
39.3 Exact arrival condition . . . 214
39.4 Leading cosmological delay law . . . 214
39.5 Null bound and directional cancellation . . . 215
39.6 Numerical verification . . . 215
39.7 Next steps . . . 216

40 Controlled Primordial Power-Law Sector 217
40.1 Constant-epsilon background . . . 217
40.2 Canonical scalar mode . . . 218
40.3 Exact power-law spectra . . . 218
40.4 Controlled benchmark . . . 219
40.5 Next steps . . . 220

41 Gauge-Invariant Einstein–Scalar Perturbations 221
41.1 Einstein–scalar input and background . . . 221
41.2 Scalar gauge invariant . . . 221
41.3 Lapse and shift elimination . . . 222
41.4 Reduced scalar action and equation . . . 222
41.5 Mukhanov–Sasaki form . . . 223
41.6 Projector-lattice scalar action . . . 223
41.7 Next steps . . . 224

42 Vector and Tensor Perturbations on FLRW 225
42.1 Scalar–vector–tensor split . . . 225
42.2 Source-free vector sector . . . 225
42.3 Tensor action and equation . . . 226
42.4 Projector-lattice tensor sector . . . 227
42.5 Propagation distance . . . 227
42.6 Next steps . . . 228

43 Recombination, Decoupling, and Acoustic Scales **229**
43.1 Conditional atomic benchmark . . . 229
43.2 Optical depth and visibility . . . 230
43.3 Baryon loading and sound horizon . . . 230
43.4 Tight-coupling oscillator . . . 231
43.5 Diffusion approximation . . . 232
43.6 Next steps . . . 232

44 Minimal Photon Hierarchy and Low-l CMB **233**
44.1 Collisionless brightness hierarchy . . . 233
44.2 Minimal hierarchy verification . . . 234
44.3 Sachs–Wolfe line-of-sight limit . . . 234
44.4 Conditional low-l benchmark . . . 235
44.5 Next steps . . . 235

45 Linear Matter Growth and First Structure Observables **236**
45.1 Subhorizon dust system . . . 236
45.2 Einstein–de Sitter benchmark . . . 237
45.3 Flat LambdaCDM growth . . . 237
45.4 Consistent projector Laplacian . . . 238
45.5 Growth observables . . . 238
45.6 Next steps . . . 239

46 Adic Refinement Dark Sector: Background, Growth, and Lensing **240**
46.1 Covariant fiber action and the dark tower . . . 240
46.2 Conditional nonadiabatic production law . . . 241
46.3 Exact homogeneous density and pressure . . . 242
46.4 Friedmann equation and controlled benchmark . . . 243
46.5 Finite-depth convergence and prime scaling . . . 244
46.6 Cold subhorizon perturbation closure . . . 247
46.7 Lensing potential . . . 248
46.8 What this calculation establishes . . . 249
46.9 Next steps . . . 249

47 Superseded Additive Refinement Dark-Sector Branch **250**
47.1 Status and objective . . . 250
47.2 Average-detail decomposition and one action . . . 250
47.3 The production law follows from an average-mode crossing . . . 251
47.4 Density, pressure, and the locking variance . . . 252
47.5 Locked-vacuum theorem . . . 252
47.6 Perturbations, sound speed, and stability . . . 253
47.7 Coincidence locking and the end of acceleration . . . 253
47.8 Controlled late-time benchmark . . . 254
47.9 Executable certificate . . . 256
47.10 What is closed and what remains open . . . 256

48 Superseded Additive Crossing and Backreaction Branch **258**
48.1 Status and objective . . . 258
48.2 Rapid crossing from the same average-mode potential . . . 258
48.3 Exact production moments . . . 259
48.4 Energy matching without an occupation fit . . . 259
48.5 Local, global, and direct-capture thresholds . . . 260
48.6 Fast-crossing asymptotics . . . 261
48.7 Self-consistent homogeneous backreaction . . . 261
48.8 Controlled benchmark and phase thresholds . . . 261
48.9 Transient acceleration from dynamically normalized production . . . 262
48.10 Executable certificate . . . 263
48.11 Closure ledger . . . 265

48.12 Result and next decisive target . . . 266

49 Scale-Covariant Refinement Dark Sector 267
49.1 Status and objective . . . 267
49.2 Additive one-shell obstruction . . . 267
49.3 Scale-covariant fiber action . . . 268
49.4 Weighted nonadiabatic production . . . 269
49.5 Exact $\sigma = 1$ continuum theorem . . . 270
49.6 Abundance calibration and emergent hierarchy . . . 271
49.7 Coldness theorem and cutoff scale . . . 272
49.8 Density locking and transient acceleration . . . 273
49.9 Linear growth and lensing benchmark . . . 274
49.10 Verification ledger . . . 275
49.11 Closure ledger and remaining burden . . . 276

50 Complete Cosmology Completion Ledger 277
50.1 Completion criterion . . . 277
50.2 Status rule . . . 277
50.3 Critical gates . . . 286

VI Corrections, Reproducibility, and Verification 287

51 Correction and Qualification Ledger 288
51.1 C1: Fierz–Pauli source sign C . . . 288
51.2 C2: lattice d'Alembert uniqueness C . . . 288
51.3 C3: radial acceleration sign C . . . 288
51.4 C4: optical deflection vector C . . . 289
51.5 C5: Wilson-spacing remainder C . . . 289
51.6 C6: zero-over-zero diagnostic C . . . 289
51.7 C7: partition-incidence vector identity C . . . 289
51.8 C8: weighted quantum-leakage remainder C . . . 290
51.9 C9: descendant measure versus Hamiltonian coupling C . . . 290
51.10 C10: resolved nonlinear closure is not strict finite-range locality Q . . . 290
51.11 C11: classical BV closure is not quantum anomaly cancellation Q . . . 290
51.12 C12: boundary BFV closure and minimal reduction are distinct steps Q . . . 291
51.13 C13: additive dark-sector branch superseded by the scale-covariant model C . . . 291
51.14 C14: trapped-surface formation is not a global black-hole theorem Q . . . 291

52 Reproducibility Note 292

53 Symbolic and Numerical Audit Matrices 293
53.1 Audit totals . . . 293
53.2 Correction checks . . . 293
53.3 All symbolic/numerical checks . . . 295
53.4 Executable rerun manifest . . . 316
53.5 Formula-registry summary . . . 318
53.6 Machine-readable registries . . . 320

VII Complete Formula Corpus 321

A Complete Formula-Occurrence Archive 322

Part I

Projector-Refinement Foundations

Chapter 1

Conventions and Scope

The manuscript combines exact identities, numerical checks, and standard physical model inputs. Exact statements are derived in their stated mathematical setting. Numerical sections test those statements or measure convergence, but they are not substitutes for proof or experimental evidence. Open tasks are stated as actionable next steps at the end of the chapters where they arise.

1.1 Linear-algebra convention

$$\langle x, y\rangle := x^* y, \tag{1.1}$$
$$\langle x, Ay\rangle = \langle A^* x, y\rangle\,, \tag{1.2}$$
$$H^* = H, \tag{1.3}$$
$$A := -\mathrm{i}H, \tag{1.4}$$
$$A^* = -A. \tag{1.5}$$

The inner product is conjugate-linear in its first argument and linear in its second.

1.2 Spacetime convention

$$\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1), \tag{1.6}$$
$$R^{\rho}{}_{\sigma\mu\nu} := \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}, \tag{1.7}$$
$$R_{\sigma\nu} := R^{\rho}{}_{\sigma\rho\nu}, \tag{1.8}$$
$$G_{\mu\nu} := R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R. \tag{1.9}$$

These choices fix every curvature and source sign used below.

1.3 Gauge-link convention

A forward transporter from site j to site $j+1$ is

$$L_j := \mathrm{e}^{+\mathrm{i}qhA_{1,j}}, \tag{1.10}$$
$$\psi_j \mapsto g_j \psi_j, \tag{1.11}$$
$$L_j \mapsto g_{j+1} L_j g_j^{-1}. \tag{1.12}$$

The backward transporter used in covariant differences is

$$U_j := L_j^{-1} = \mathrm{e}^{-\mathrm{i}qhA_{1,j}}, \tag{1.13}$$

$$U_j \mapsto g_j U_j g_{j+1}^{-1}. \tag{1.14}$$

With $T_j^n = \mathrm{e}^{+\mathrm{i}q\Delta t A_{0,j}^n}$, the right–up–left–down plaquette orientation is

$$W_{01}(j,n) = L_j^n T_{j+1}^n (L_j^{n+1})^{-1} (T_j^n)^{-1} = \mathrm{e}^{-\mathrm{i}qh\Delta t F_{01}+\cdots}, \qquad F_{01} := \partial_0 A_1 - \partial_1 A_0. \tag{1.15}$$

1.4 Radial-force convention

$$\Phi(r) < 0, \tag{1.16}$$
$$g_+(r) := \Phi'(r) \geq 0, \tag{1.17}$$
$$a_r(r) := -\Phi'(r) = -g_+(r) \leq 0, \tag{1.18}$$
$$\boldsymbol{a}(r) = a_r(r)\widehat{\boldsymbol{r}}. \tag{1.19}$$

Thus positive g_+ denotes the magnitude of an inward radial acceleration.

Chapter 2

Projector-Refinement Core

The basic object is a complete family of orthogonal projectors. It separates a state or operator into packets and their pairwise couplings while preserving exact reconstruction under coarsening.

2.1 Atlas and corner algebra

$$P_\alpha^* = P_\alpha, \tag{2.1}$$
$$P_\alpha P_\beta = \delta_{\alpha\beta} P_\alpha, \tag{2.2}$$
$$\sum_{\alpha\in A} P_\alpha = I, \tag{2.3}$$
$$\psi_\alpha := P_\alpha \psi, \tag{2.4}$$
$$X_{\alpha\beta} := P_\alpha X P_\beta, \tag{2.5}$$
$$\psi = \sum_\alpha \psi_\alpha, \tag{2.6}$$
$$X = \sum_{\alpha,\beta} X_{\alpha\beta}. \tag{2.7}$$

The reconstruction identities come from inserting the resolution of the identity once for a vector and twice for an operator:

$$\psi = I\psi = \left(\sum_\alpha P_\alpha\right)\psi = \sum_\alpha P_\alpha \psi, \tag{2.8}$$
$$X = IXI = \left(\sum_\alpha P_\alpha\right) X \left(\sum_\beta P_\beta\right) \tag{2.9}$$
$$= \sum_{\alpha,\beta} P_\alpha X P_\beta. \tag{2.10}$$

Orthogonality also separates the norm:

$$\|\psi\|^2 = \sum_\alpha \|P_\alpha \psi\|^2 . \tag{2.11}$$

2.2 Refinement

$$Q_A := \sum_{\alpha\in C_A} P_\alpha, \tag{2.12}$$
$$Q_A Q_B = \delta_{AB} Q_A, \tag{2.13}$$

$$\sum_A Q_A = I, \tag{2.14}$$

$$Q_A X Q_B = \sum_{\substack{\alpha\in C_A\\ \beta\in C_B}} P_\alpha X P_\beta. \tag{2.15}$$

2.3 Skew-adjoint transfer theorem

$$\dot\psi = A\psi, \qquad A^* = -A, \tag{2.16}$$

$$E_\alpha := \|P_\alpha\psi\|^2, \tag{2.17}$$

$$J_{\alpha\leftarrow\beta} := 2\,\mathrm{Re}\,\langle P_\alpha\psi, AP_\beta\psi\rangle, \tag{2.18}$$

$$\dot E_\alpha = 2\,\mathrm{Re}\,\langle P_\alpha\psi, A\psi\rangle \tag{2.19}$$

$$= \sum_\beta J_{\alpha\leftarrow\beta}, \tag{2.20}$$

$$J_{\alpha\leftarrow\beta} = -J_{\beta\leftarrow\alpha}, \tag{2.21}$$

$$J_{\alpha\leftarrow\alpha} = 0, \tag{2.22}$$

$$\sum_\alpha \dot E_\alpha = 0. \tag{2.23}$$

The differentiation is

$$\dot E_\alpha = \langle P_\alpha\dot\psi, P_\alpha\psi\rangle + \langle P_\alpha\psi, P_\alpha\dot\psi\rangle \tag{2.24}$$

$$= 2\,\mathrm{Re}\,\langle P_\alpha\psi, P_\alpha A\psi\rangle = 2\,\mathrm{Re}\,\langle P_\alpha\psi, A\psi\rangle. \tag{2.25}$$

For the reversed current,

$$J_{\beta\leftarrow\alpha} = 2\,\mathrm{Re}\,\langle P_\beta\psi, AP_\alpha\psi\rangle \tag{2.26}$$

$$= 2\,\mathrm{Re}\,\langle A^*P_\beta\psi, P_\alpha\psi\rangle = -2\,\mathrm{Re}\,\langle P_\alpha\psi, AP_\beta\psi\rangle. \tag{2.27}$$

The diagonal current vanishes because

$$\mathrm{Re}\,\langle u, Au\rangle = \frac{1}{2}\left(\langle u, Au\rangle + \langle Au, u\rangle\right) = 0. \tag{2.28}$$

Summing over all ordered pairs then gives

$$\sum_\alpha \dot E_\alpha = \frac{1}{2}\sum_{\alpha,\beta}\left(J_{\alpha\leftarrow\beta} + J_{\beta\leftarrow\alpha}\right) = 0. \tag{2.29}$$

For $A = -\mathrm{i}H$,

$$J_{\alpha\leftarrow\beta} = 2\,\mathrm{Im}\,\langle P_\alpha\psi, HP_\beta\psi\rangle. \tag{2.30}$$

2.4 Refinement covariance of transfer

$$E^{\mathrm{c}}_A := \|Q_A\psi\|^2 = \sum_{\alpha\in C_A} E_\alpha, \tag{2.31}$$

$$J^{\mathrm{c}}_{A\leftarrow B} := 2\,\mathrm{Re}\,\langle Q_A\psi, AQ_B\psi\rangle \tag{2.32}$$

$$= \sum_{\substack{\alpha\in C_A\\ \beta\in C_B}} J_{\alpha\leftarrow\beta}, \tag{2.33}$$

$$\dot E^{\mathrm{c}}_A = \sum_B J^{\mathrm{c}}_{A\leftarrow B}. \tag{2.34}$$

Indeed,

$$\dot{E}^{\mathrm{c}}_A = \sum_{\alpha\in C_A} \dot{E}_\alpha = \sum_{\alpha\in C_A}\sum_{\beta} J_{\alpha\leftarrow\beta} \tag{2.35}$$

$$= \sum_{B}\sum_{\substack{\alpha\in C_A\\ \beta\in C_B}} J_{\alpha\leftarrow\beta}. \tag{2.36}$$

Currents internal to the same coarse packet cancel pairwise, leaving only net transfer across coarse packet boundaries.

2.5 Moving atlas

$$P_\alpha(t) = U(t)P^{(0)}_\alpha U(t)^*, \tag{2.37}$$

$$\phi := U^*\psi, \tag{2.38}$$

$$\Gamma := U^*\dot{U}, \tag{2.39}$$

$$\Gamma^* = -\Gamma, \tag{2.40}$$

$$A_{\mathrm{eff}} := U^*AU - \Gamma, \tag{2.41}$$

$$A^*_{\mathrm{eff}} = -A_{\mathrm{eff}}, \tag{2.42}$$

$$\dot{\phi} = A_{\mathrm{eff}}\phi. \tag{2.43}$$

Differentiating $U^*U = I$ gives

$$0 = \dot{U}^*U + U^*\dot{U}, \tag{2.44}$$

$$\dot{U}^* = -U^*\dot{U}\,U^* = -\Gamma U^*. \tag{2.45}$$

Therefore

$$\dot{\phi} = \dot{U}^*\psi + U^*\dot{\psi} \tag{2.46}$$

$$= -\Gamma U^*\psi + U^*AU\,U^*\psi \tag{2.47}$$

$$= (U^*AU - \Gamma)\phi. \tag{2.48}$$

The same differentiated identity gives

$$\Gamma^* = (U^*\dot{U})^* = \dot{U}^*U = -U^*\dot{U} = -\Gamma. \tag{2.49}$$

2.6 Whole-object deformation rule

$$P_\alpha(t) = U(t)P_\alpha(0)U(t)^{-1}, \tag{2.50}$$

$$K := \dot{U}U^{-1}, \tag{2.51}$$

$$\dot{P}_\alpha = [K, P_\alpha], \tag{2.52}$$

$$X_{\alpha\beta} := P_\alpha X P_\beta, \tag{2.53}$$

$$\dot{X}_{\alpha\beta} = P_\alpha\dot{X}P_\beta + [K, P_\alpha]XP_\beta + P_\alpha X[K, P_\beta]. \tag{2.54}$$

The projector derivative follows from

$$\frac{\mathrm{d}}{\mathrm{d}t}U^{-1} = -U^{-1}\dot{U}U^{-1}, \tag{2.55}$$

$$\dot{P}_\alpha = \dot{U}P_\alpha(0)U^{-1} - UP_\alpha(0)U^{-1}\dot{U}U^{-1} \tag{2.56}$$

$$= KP_\alpha - P_\alpha K. \tag{2.57}$$

Applying the ordinary product rule to $X_{\alpha\beta} = P_\alpha X P_\beta$ gives the three displayed terms.

Chapter 3

Types, Dimensions, and Logical Dependencies

This chapter records the domains, codomains, dimensions, and units used later. Its purpose is to keep formally similar expressions from being applied outside the setting in which they are defined.

3.1 Type judgments

$$\vdash \mathcal{H} : \mathsf{Hilb}, \tag{3.1}$$
$$\vdash P_\alpha : \mathcal{H} \to \mathcal{H}, \qquad P_\alpha^* = P_\alpha = P_\alpha^2, \tag{3.2}$$
$$\vdash X_{\alpha\beta} := P_\alpha X P_\beta \in P_\alpha \mathcal{B}(\mathcal{H}) P_\beta, \tag{3.3}$$
$$\vdash A \in \mathfrak{u}(\mathcal{H}) \iff A^* = -A, \tag{3.4}$$
$$\vdash H \in \mathfrak{sa}(\mathcal{H}) \iff H^* = H, \tag{3.5}$$
$$\vdash A = -\mathrm{i}H : \mathfrak{sa}(\mathcal{H}) \to \mathfrak{u}(\mathcal{H}). \tag{3.6}$$

3.2 Core sequents

Let

$$\mathfrak{A}_{\mathcal{P}} := \left\{ P_\alpha^* = P_\alpha,\ P_\alpha P_\beta = \delta_{\alpha\beta} P_\alpha,\ \sum_\alpha P_\alpha = I \right\}. \tag{3.7}$$

Then

$$\mathfrak{A}_{\mathcal{P}} \vdash X = \sum_{\alpha,\beta} P_\alpha X P_\beta, \tag{3.8}$$
$$\mathfrak{A}_{\mathcal{P}} \wedge A^* = -A \vdash \left\{ \dot{E} = J\mathbf{1},\ J = -J^{\mathsf{T}},\ \mathbf{1}^{\mathsf{T}} \dot{E} = 0 \right\}, \tag{3.9}$$
$$\mathfrak{A}_{\mathcal{P}} \wedge U^* U = I \vdash \left\{ \Gamma := U^* \dot{U},\ \Gamma^* = -\Gamma,\ A_{\mathrm{eff}} := U^* A U - \Gamma \right\}. \tag{3.10}$$

3.3 Refinement matrices

For a partition incidence matrix

$$R_{A\alpha} := \mathbf{1}_{\{\alpha \in C_A\}}, \tag{3.11}$$

$$E^{\mathrm{c}} = RE, \tag{3.12}$$
$$J^{\mathrm{c}} = R J R^{\mathsf{T}}, \tag{3.13}$$
$$R^{\mathsf{T}} \mathbf{1}_{\mathrm{c}} = \mathbf{1}_{\mathrm{f}}, \tag{3.14}$$

$$J = -J^{\mathsf{T}} \Longrightarrow J^{\mathrm{c}} = -(J^{\mathrm{c}})^{\mathsf{T}}, \tag{3.15}$$

$$\dot{E} = J\mathbf{1}_{\mathrm{f}} + s \Longrightarrow \dot{E}^{\mathrm{c}} = J^{\mathrm{c}}\mathbf{1}_{\mathrm{c}} + Rs. \tag{3.16}$$

Nested refinement:

$$R_{2\leftarrow 0} = R_{2\leftarrow 1}R_{1\leftarrow 0}, \tag{3.17}$$

$$\mathfrak{C}_{2\leftarrow 0} = \mathfrak{C}_{2\leftarrow 1} \circ \mathfrak{C}_{1\leftarrow 0}. \tag{3.18}$$

3.4 Variation sequents

$$\delta_N S[a, N, \Psi] = 0 \vdash \mathcal{H}[a, \dot{a}, \Psi] = 0, \tag{3.19}$$

$$\delta_{N^i} S[g, N, N^i, \Psi] = 0 \vdash \mathcal{H}_i[g, \dot{g}, \Psi] = 0, \tag{3.20}$$

$$\delta_\Psi S[g, \Psi] = 0 \vdash \mathcal{E}_\Psi[g, \Psi] = 0, \tag{3.21}$$

$$\delta_g S[g, \Psi] = 0 \vdash \mathcal{E}_g[g, \Psi] = 0. \tag{3.22}$$

Gauge and diffeomorphism implication targets:

$$\delta_\chi S = 0 \vdash D_\mu j^\mu = 0, \tag{3.23}$$

$$\delta_\xi S = 0 \vdash \nabla_\mu T^{\mu\nu} = 0, \tag{3.24}$$

$$\nabla_\mu G^{\mu\nu} = 0 \wedge G^{\mu\nu} = 8\pi G T^{\mu\nu} \vdash \nabla_\mu T^{\mu\nu} = 0. \tag{3.25}$$

3.5 Continuum-limit sequents

$$\cos(\omega_\varepsilon \varepsilon) = \cos(m\varepsilon)\cos(k\varepsilon) \vdash \omega_\varepsilon = \sqrt{m^2 + k^2} + \mathcal{O}(\varepsilon^2), \tag{3.26}$$

$$\cos(\omega_\Delta \Delta) = \cos(m\Delta)\prod_i \cos(k_i\Delta) \vdash \omega_\Delta = \sqrt{m^2 + |\boldsymbol{k}|^2} + \mathcal{O}(\Delta^2), \tag{3.27}$$

$$W_\varepsilon = \mathrm{e}^{\varepsilon A_x}\mathrm{e}^{\varepsilon A_y}\mathrm{e}^{-\varepsilon A_x}\mathrm{e}^{-\varepsilon A_y} \vdash \varepsilon^{-2}\log W_\varepsilon = [A_x, A_y] + \mathcal{O}(\varepsilon), \tag{3.28}$$

$$R_0^{(m)}[F(h)] = F + \mathcal{O}(h^{m+1}). \tag{3.29}$$

3.6 Unit and normalization judgments

Relativistic sectors:

$$c = \hbar = 1, \tag{3.30}$$

$$[m] = [k_i] = [\omega] = L^{-1}, \tag{3.31}$$

$$[\varepsilon] = [\Delta] = L, \tag{3.32}$$

$$[m\Delta] = [k_i\Delta] = [\omega\Delta] = 1. \tag{3.33}$$

Gauge sectors:

$$[qA_\mu \,\mathrm{d}x^\mu] = 1, \tag{3.34}$$

$$[L_j] = [T_j] = [W_{\mu\nu}] = 1. \tag{3.35}$$

Cosmology:

$$[a] = 1, \qquad [H] = T^{-1}, \tag{3.36}$$

$$[p_{\mathrm{phys}}] = [\omega_{\mathrm{phys}}] = T^{-1} \quad (c = \hbar = 1), \tag{3.37}$$

$$\frac{8\pi G}{3} = 1 \quad \text{only where explicitly normalized.} \tag{3.38}$$

Weak-field observables:

$$[\Phi] = L^2 T^{-2}, \tag{3.39}$$

$$\left[\frac{\Phi}{c^2}\right] = 1, \tag{3.40}$$

$$[\boldsymbol{\alpha}_\perp] = 1, \tag{3.41}$$

$$[\Delta t_{\mathrm{Shapiro}}] = T. \tag{3.42}$$

Chapter 4

Extensions of the Core Transfer Calculus

The transfer law is not restricted to a fixed pure-state atlas. The formulas below cover state-dependent generators, density matrices, tensor products, dissipative jumps, and repeated refinement.

4.1 State-dependent skew-adjoint flow

$$\dot{\psi} = A(\psi,t)\psi, \qquad A(\psi,t)^* = -A(\psi,t), \tag{4.1}$$

$$\psi_\alpha = P_\alpha \psi, \qquad E_\alpha = \langle \psi_\alpha, \psi_\alpha \rangle , \tag{4.2}$$

$$J_{\alpha\leftarrow\beta}(\psi,t) := 2\,\mathrm{Re}\,\langle \psi_\alpha, A(\psi,t)\psi_\beta \rangle , \tag{4.3}$$

$$\dot{E}_\alpha = 2\,\mathrm{Re}\,\langle \psi_\alpha, A(\psi,t)\psi \rangle = \sum_\beta J_{\alpha\leftarrow\beta}, \tag{4.4}$$

$$J_{\alpha\leftarrow\beta} = -J_{\beta\leftarrow\alpha}, \qquad \sum_\alpha \dot{E}_\alpha = 0. \tag{4.5}$$

4.2 Mixed-state unitary transfer

$$\rho = \rho^* \geq 0, \qquad \mathrm{Tr}\,\rho = 1, \qquad \dot{\rho} = -\mathrm{i}[H,\rho], \qquad H^* = H, \tag{4.6}$$

$$p_\alpha := \mathrm{Tr}(P_\alpha \rho), \tag{4.7}$$

$$J^{(\rho)}_{\alpha\leftarrow\beta} := 2\,\mathrm{Im}\,\mathrm{Tr}(P_\alpha H P_\beta \rho), \tag{4.8}$$

$$\dot{p}_\alpha = \sum_\beta J^{(\rho)}_{\alpha\leftarrow\beta}, \tag{4.9}$$

$$J^{(\rho)}_{\alpha\leftarrow\beta} = -J^{(\rho)}_{\beta\leftarrow\alpha}, \tag{4.10}$$

$$\sum_\alpha \dot{p}_\alpha = 0. \tag{4.11}$$

Resolve the commutator with $\sum_\beta P_\beta = I$:

$$\dot{p}_\alpha = -\mathrm{i}\,\mathrm{Tr}(P_\alpha H \rho - P_\alpha \rho H) \tag{4.12}$$

$$= -\mathrm{i} \sum_\beta \left[\mathrm{Tr}(P_\alpha H P_\beta \rho) - \mathrm{Tr}(P_\beta H P_\alpha \rho) \right]. \tag{4.13}$$

Cyclicity of the trace and Hermiticity imply

$$\mathrm{Tr}(P_\beta H P_\alpha \rho) = \overline{\mathrm{Tr}(P_\alpha H P_\beta \rho)}. \tag{4.14}$$

Each bracket is therefore twice the imaginary part of the first term, which yields the current formula and its antisymmetry. For $\rho = |\psi\rangle\langle\psi|$,

$$J^{(\rho)}_{\alpha\leftarrow\beta} = 2\,\mathrm{Im}\,\langle P_\alpha \psi, H P_\beta \psi \rangle . \tag{4.15}$$

4.3 Associative refinement tower

Let

$$Q_A := \sum_\alpha R_{A\alpha} P_\alpha, \qquad R_{A\alpha} \in \{0,1\}, \tag{4.16}$$

$$\widetilde{Q}_C := \sum_A S_{CA} Q_A, \qquad S_{CA} \in \{0,1\}. \tag{4.17}$$

Then

$$T_{C\alpha} := (SR)_{C\alpha}, \tag{4.18}$$

$$\widetilde{Q}_C = \sum_\alpha T_{C\alpha} P_\alpha, \tag{4.19}$$

$$E^{(1)} = RE, \qquad E^{(2)} = SE^{(1)} = (SR)E, \tag{4.20}$$

$$J^{(1)} = RJR^{\mathsf{T}}, \qquad J^{(2)} = SJ^{(1)}S^{\mathsf{T}} = (SR)J(SR)^{\mathsf{T}}. \tag{4.21}$$

Hence

$$\mathfrak{C}_{P\to\widetilde{Q}} = \mathfrak{C}_{Q\to\widetilde{Q}} \circ \mathfrak{C}_{P\to Q}. \tag{4.22}$$

4.4 Tensor-product atlas

For complete orthogonal atlases $\{P_\alpha\} \subset \mathcal{B}(\mathcal{H}_A)$ and $\{Q_\mu\} \subset \mathcal{B}(\mathcal{H}_B)$,

$$R_{\alpha\mu} := P_\alpha \otimes Q_\mu, \tag{4.23}$$

$$R_{\alpha\mu} R_{\beta\nu} = \delta_{\alpha\beta}\delta_{\mu\nu} R_{\alpha\mu}, \tag{4.24}$$

$$\sum_{\alpha,\mu} R_{\alpha\mu} = I_A \otimes I_B, \tag{4.25}$$

$$p_{\alpha\mu} := \mathrm{Tr}(R_{\alpha\mu}\rho_{AB}), \tag{4.26}$$

$$p^A_\alpha = \sum_\mu p_{\alpha\mu} = \mathrm{Tr}[(P_\alpha \otimes I_B)\rho_{AB}], \tag{4.27}$$

$$p^B_\mu = \sum_\alpha p_{\alpha\mu} = \mathrm{Tr}[(I_A \otimes Q_\mu)\rho_{AB}]. \tag{4.28}$$

4.5 Universal antisymmetric balance and refinement

$$\dot{q}_\alpha = s_\alpha + \sum_\beta F_{\alpha\beta}, \qquad F_{\alpha\beta} = -F_{\beta\alpha}, \tag{4.29}$$

$$q_C := \sum_{\alpha\in C} q_\alpha, \qquad s_C := \sum_{\alpha\in C} s_\alpha, \tag{4.30}$$

$$\dot{q}_C = s_C + \sum_{\substack{\alpha\in C \\ \beta\notin C}} F_{\alpha\beta}. \tag{4.31}$$

For a partition $\{C_A\}_A$,

$$F^{\mathrm{c}}_{AB} := \sum_{\substack{\alpha\in C_A \\ \beta\in C_B}} F_{\alpha\beta}, \tag{4.32}$$

$$F^{\mathrm{c}}_{AB} = -F^{\mathrm{c}}_{BA}, \tag{4.33}$$

$$\dot{q}^{\mathrm{c}}_A = s^{\mathrm{c}}_A + \sum_B F^{\mathrm{c}}_{AB}. \tag{4.34}$$

4.6 Packet information transport

$$p_\alpha > 0, \qquad \sum_\alpha p_\alpha = 1, \qquad \dot{p}_\alpha = \sum_\beta J_{\alpha\beta}, \qquad J_{\alpha\beta} = -J_{\beta\alpha}. \tag{4.35}$$

$$S[p] := -\sum_\alpha p_\alpha \log p_\alpha, \tag{4.36}$$

$$\dot{S} = -\sum_\alpha (1 + \log p_\alpha)\dot{p}_\alpha \tag{4.37}$$

$$= -\frac{1}{2}\sum_{\alpha,\beta} J_{\alpha\beta} \log \frac{p_\alpha}{p_\beta}. \tag{4.38}$$

The constant term disappears because $\sum_\alpha \dot{p}_\alpha = 0$. The remaining sum is symmetrized as

$$-\sum_{\alpha,\beta} J_{\alpha\beta} \log p_\alpha = -\frac{1}{2}\sum_{\alpha,\beta} \left(J_{\alpha\beta} \log p_\alpha + J_{\beta\alpha} \log p_\beta\right) \tag{4.39}$$

$$= -\frac{1}{2}\sum_{\alpha,\beta} J_{\alpha\beta} \left(\log p_\alpha - \log p_\beta\right). \tag{4.40}$$

For $q \in \mathbb{R}$,

$$M_q[p] := \sum_\alpha p_\alpha^q, \tag{4.41}$$

$$\dot{M}_q = \frac{q}{2}\sum_{\alpha,\beta} J_{\alpha\beta} \left(p_\alpha^{q-1} - p_\beta^{q-1}\right), \tag{4.42}$$

$$H_q[p] := \frac{1}{1-q} \log M_q[p], \tag{4.43}$$

$$\dot{H}_q = \frac{q}{2(1-q)M_q[p]}\sum_{\alpha,\beta} J_{\alpha\beta} \left(p_\alpha^{q-1} - p_\beta^{q-1}\right). \tag{4.44}$$

4.7 Lindblad packet jumps

$$L_{\alpha\beta} := \sqrt{\gamma_{\alpha\beta}}\, |\alpha\rangle \langle\beta|, \qquad \gamma_{\alpha\beta} \geq 0, \tag{4.45}$$

$$\dot{\rho} = \sum_{\alpha\neq\beta} \left(L_{\alpha\beta}\rho L_{\alpha\beta}^* - \frac{1}{2}\{L_{\alpha\beta}^* L_{\alpha\beta}, \rho\}\right), \tag{4.46}$$

$$p_\alpha := \rho_{\alpha\alpha}, \tag{4.47}$$

$$\dot{p}_\alpha = \sum_\beta \left(\gamma_{\alpha\beta} p_\beta - \gamma_{\beta\alpha} p_\alpha\right), \tag{4.48}$$

$$K_{\alpha\beta} := \gamma_{\alpha\beta} p_\beta - \gamma_{\beta\alpha} p_\alpha, \qquad K_{\alpha\beta} = -K_{\beta\alpha}, \tag{4.49}$$

$$\dot{p}_\alpha = \sum_\beta K_{\alpha\beta}. \tag{4.50}$$

For one jump $\beta \to \alpha$,

$$L_{\alpha\beta}\rho L_{\alpha\beta}^* = \gamma_{\alpha\beta} p_\beta |\alpha\rangle\langle\alpha|, \tag{4.51}$$

$$L_{\alpha\beta}^* L_{\alpha\beta} = \gamma_{\alpha\beta} |\beta\rangle\langle\beta|. \tag{4.52}$$

The first term is a gain at α, and the anticommutator removes the same rate from β. Summing all ordered jumps gives the displayed packet equation.

4.8 Detailed-balance packet theorem

$$\gamma_{\alpha\beta}\pi_\beta = \gamma_{\beta\alpha}\pi_\alpha, \qquad \pi_\alpha > 0, \qquad \sum_\alpha \pi_\alpha = 1, \tag{4.53}$$

$$D(p\|\pi) := \sum_\alpha p_\alpha \log \frac{p_\alpha}{\pi_\alpha}, \tag{4.54}$$

$$\frac{\mathrm{d}}{\mathrm{d}t} D(p\|\pi) = -\frac{1}{2} \sum_{\alpha,\beta} (\gamma_{\alpha\beta} p_\beta - \gamma_{\beta\alpha} p_\alpha) \log \frac{\gamma_{\alpha\beta} p_\beta}{\gamma_{\beta\alpha} p_\alpha} \tag{4.55}$$

$$\leq 0, \tag{4.56}$$

Detailed balance rewrites the logarithmic ratio as

$$\log \frac{p_\alpha/\pi_\alpha}{p_\beta/\pi_\beta} = \log \frac{\gamma_{\beta\alpha} p_\alpha}{\gamma_{\alpha\beta} p_\beta}. \tag{4.57}$$

Substitute $\dot{p}_\alpha = \sum_\beta K_{\alpha\beta}$, pair the ordered terms (α, β) and (β, α), and apply the elementary inequality below. using

$$(x - y) \log \frac{x}{y} \geq 0 \qquad (x, y > 0). \tag{4.58}$$

4.9 General refinement-cancellation tower

$$F(h) = F + \sum_{r \geq 1} c_r h^r, \tag{4.59}$$

$$R_j^{(0)} := F\left(\frac{h}{2^j}\right), \tag{4.60}$$

$$R_j^{(m)} := \frac{2^m R_{j+1}^{(m-1)} - R_j^{(m-1)}}{2^m - 1}, \qquad m \geq 1, \tag{4.61}$$

$$R_0^{(m)} = F + \mathcal{O}(h^{m+1}). \tag{4.62}$$

At the first level,

$$2F(h/2) - F(h) = 2\left(F + \frac{c_1 h}{2} + \frac{c_2 h^2}{4} + \cdots\right) - \left(F + c_1 h + c_2 h^2 + \cdots\right) \tag{4.63}$$

$$= F - \frac{1}{2} c_2 h^2 + \mathcal{O}(h^3), \tag{4.64}$$

so the $O(h)$ term cancels explicitly. The recursion repeats the same calculation at each next surviving order. Explicitly,

$$R_0^{(1)} = 2F(h/2) - F(h), \tag{4.65}$$

$$R_0^{(2)} = \frac{8F(h/4) - 6F(h/2) + F(h)}{3}, \tag{4.66}$$

$$R_0^{(3)} = \frac{64F(h/8) - 56F(h/4) + 14F(h/2) - F(h)}{21}. \tag{4.67}$$

Chapter 5

Positive-Operator and Overlapping Atlases

Orthogonal projectors are convenient but not essential. Positive partitions of unity allow overlapping measurements while retaining a controlled continuity equation and exact coarse aggregation.

5.1 Positive partition of unity

$$F_\alpha = F_\alpha^* \geq 0, \qquad \sum_\alpha F_\alpha = I, \tag{5.1}$$

$$p_\alpha := \langle \psi, F_\alpha \psi \rangle , \qquad \dot{\psi} = A\psi, \qquad A^* = -A. \tag{5.2}$$

Define

$$J^{(F)}_{\alpha \leftarrow \beta} := \mathrm{Re}\, \langle F_\alpha \psi, AF_\beta \psi \rangle - \mathrm{Re}\, \langle F_\beta \psi, AF_\alpha \psi \rangle . \tag{5.3}$$

Then

$$J^{(F)}_{\alpha \leftarrow \beta} = -J^{(F)}_{\beta \leftarrow \alpha}, \tag{5.4}$$

$$\dot{p}_\alpha = 2\,\mathrm{Re}\, \langle F_\alpha \psi, A\psi \rangle , \tag{5.5}$$

$$\sum_\beta J^{(F)}_{\alpha \leftarrow \beta} = \mathrm{Re}\, \langle F_\alpha \psi, A\psi \rangle - \mathrm{Re}\, \langle \psi, AF_\alpha \psi \rangle \tag{5.6}$$

$$= 2\,\mathrm{Re}\, \langle F_\alpha \psi, A\psi \rangle = \dot{p}_\alpha, \tag{5.7}$$

$$\sum_\alpha \dot{p}_\alpha = 0. \tag{5.8}$$

The partition of unity is used in both halves:

$$\sum_\beta \mathrm{Re}\, \langle F_\alpha \psi, AF_\beta \psi \rangle = \mathrm{Re}\, \langle F_\alpha \psi, A\psi \rangle , \tag{5.9}$$

$$\sum_\beta \mathrm{Re}\, \langle F_\beta \psi, AF_\alpha \psi \rangle = \mathrm{Re}\, \langle \psi, AF_\alpha \psi \rangle . \tag{5.10}$$

Since $A^* = -A$, the second real part is the negative of the first. This replaces the orthogonality argument used for sharp projectors.

5.2 Moving positive atlas

For

$$F_\alpha = F_\alpha(t), \qquad \sum_\alpha F_\alpha(t) = I, \tag{5.11}$$

$$s_\alpha := \langle \psi, \dot{F}_\alpha \psi \rangle, \qquad \sum_\alpha s_\alpha = 0, \tag{5.12}$$

one has

$$\dot{p}_\alpha = s_\alpha + \sum_\beta J^{(F)}_{\alpha \leftarrow \beta}. \tag{5.13}$$

If

$$F_\alpha(t) = U(t) F_\alpha(0) U(t)^*, \tag{5.14}$$

$$K := \dot{U} U^*, \qquad K^* = -K, \tag{5.15}$$

then

$$\dot{F}_\alpha = [K, F_\alpha], \qquad s_\alpha = \langle \psi, [K, F_\alpha] \psi \rangle . \tag{5.16}$$

5.3 Exact refinement

Let

$$G_A := \sum_\alpha R_{A\alpha} F_\alpha, \qquad R_{A\alpha} \in \{0, 1\}, \qquad R^{\mathsf{T}} \mathbf{1}_{\mathrm{c}} = \mathbf{1}_{\mathrm{f}}. \tag{5.17}$$

Then

$$p^{\mathrm{c}} = R p, \tag{5.18}$$

$$s^{\mathrm{c}} = R s, \tag{5.19}$$

$$J^{\mathrm{c}} = R J^{(F)} R^{\mathsf{T}}, \tag{5.20}$$

$$\dot{p}^{\mathrm{c}} = s^{\mathrm{c}} + J^{\mathrm{c}} \mathbf{1}_{\mathrm{c}}. \tag{5.21}$$

5.4 Parseval-channel realization and dilation

Let

$$V_\alpha : \mathcal{H} \to \mathcal{K}_\alpha, \qquad \sum_\alpha V_\alpha^* V_\alpha = I, \qquad F_\alpha := V_\alpha^* V_\alpha. \tag{5.22}$$

Then

$$p_\alpha = \| V_\alpha \psi \|^2 , \tag{5.23}$$

$$\psi = \sum_\alpha V_\alpha^* V_\alpha \psi, \tag{5.24}$$

$$X = \sum_{\alpha, \beta} V_\alpha^* \big(V_\alpha X V_\beta^* \big) V_\beta. \tag{5.25}$$

The operator reconstruction is

$$X = \left(\sum_\alpha V_\alpha^* V_\alpha \right) X \left(\sum_\beta V_\beta^* V_\beta \right) \tag{5.26}$$

$$= \sum_{\alpha, \beta} V_\alpha^* V_\alpha X V_\beta^* V_\beta. \tag{5.27}$$

Thus $V_\alpha X V_\beta^*$ is the channel-to-channel block carried between the synthesis maps. Define

$$\mathcal{K} := \bigoplus_\alpha \mathcal{K}_\alpha, \tag{5.28}$$

$$V : \mathcal{H} \to \mathcal{K}, \qquad V\psi := (V_\alpha \psi)_\alpha, \tag{5.29}$$

$$\Pi_\alpha := \text{coordinate projector onto } \mathcal{K}_\alpha. \tag{5.30}$$

Then

$$V^* V = I, \qquad F_\alpha = V^* \Pi_\alpha V. \tag{5.31}$$

For orthogonal projectors $F_\alpha = P_\alpha$,

$$J^{(F)}_{\alpha \leftarrow \beta} = 2 \operatorname{Re} \langle P_\alpha \psi, A P_\beta \psi \rangle . \tag{5.32}$$

5.5 Cycle-current gauge

For

$$K^{\mathsf{T}} = -K, \qquad K\mathbf{1} = 0, \tag{5.33}$$

one has

$$J' := J^{(F)} + K \Longrightarrow J'\mathbf{1} = J^{(F)}\mathbf{1} = \dot{p}. \tag{5.34}$$

Thus continuity fixes the current only modulo divergence-free antisymmetric cycles; $J^{(F)}$ is the generator-induced bilinear representative.

5.6 Numerical residuals

$$\left\|\sum_\alpha F_\alpha - I\right\|_2 = 4.216049 \times 10^{-15}, \tag{5.35}$$

$$\left\|\dot{p} - J^{(F)}\mathbf{1}\right\|_\infty = 4.163336 \times 10^{-17}, \tag{5.36}$$

$$\left\|J^{(F)} + (J^{(F)})^{\mathsf{T}}\right\|_\infty = 0, \tag{5.37}$$

$$\left\|J^{\mathrm{c}} - RJ^{(F)}R^{\mathsf{T}}\right\|_\infty = 6.938894 \times 10^{-18}, \tag{5.38}$$

$$\left\|\dot{p} - s - J^{(F)}\mathbf{1}\right\|_\infty = 6.158268 \times 10^{-17}, \tag{5.39}$$

$$\|V^*V - I\|_2 = 4.750450 \times 10^{-15}, \tag{5.40}$$

$$\max_\alpha \|F_\alpha - V^*\Pi_\alpha V\|_2 = 1.080132 \times 10^{-15}, \tag{5.41}$$

$$\|K\mathbf{1}\|_\infty = 4.440892 \times 10^{-16}, \tag{5.42}$$

$$\left\|\dot{p} - (J^{(F)} + K)\mathbf{1}\right\|_\infty = 5.308254 \times 10^{-16}. \tag{5.43}$$

Chapter 6

Countable Atlases and Unbounded Generators

The finite-dimensional formulas extend to countable projector families when the relevant sums and domains are controlled. This chapter states the convergence assumptions needed for unbounded generators and infinite reconstruction.

6.1 Graph-compatible atlas

Let $H = H^*$ on $D(H) \subset \mathcal{H}$. For finite $F \Subset A$,

$$S_F := \sum_{\alpha \in F} P_\alpha, \tag{6.1}$$

$$\|\varphi\|_H^2 := \|\varphi\|^2 + \|H\varphi\|^2 . \tag{6.2}$$

Hypotheses:

$$P_\alpha P_\beta = \delta_{\alpha\beta} P_\alpha, \tag{6.3}$$

$$\sum_{\alpha \in A} P_\alpha = I \quad \text{strongly}, \tag{6.4}$$

$$P_\alpha D(H) \subseteq D(H), \tag{6.5}$$

$$S_F\psi \to \psi \quad \text{in } \|\cdot\|_H , \qquad \psi \in D(H). \tag{6.6}$$

The last statement means explicitly that

$$\|S_F\psi - \psi\|_H^2 = \|S_F\psi - \psi\|^2 + \|H(S_F\psi - \psi)\|^2 \longrightarrow 0, \tag{6.7}$$

$$S_F\psi \longrightarrow \psi, \tag{6.8}$$

$$HS_F\psi \longrightarrow H\psi. \tag{6.9}$$

Thus the partial packet sums approximate both the state and the action of the unbounded generator on its domain.

6.2 Spectral packets and domain hierarchy

For the harmonic oscillator,

$$H\phi_n = \lambda_n \phi_n, \qquad \lambda_n = n + \frac{1}{2}. \tag{6.10}$$

If $f = \sum_{n \geq 0} c_n \phi_n$, then

$$\boxed{f \in D(H) \Longleftrightarrow \sum_{n=0}^{\infty} \lambda_n^2 |c_n|^2 < \infty,} \tag{6.11}$$

whereas

$$\boxed{f \in D(H^{1/2}) \Longleftrightarrow \sum_{n=0}^{\infty} \lambda_n |c_n|^2 < \infty.} \tag{6.12}$$

Thus $D(H) \subset D(H^{1/2}) \subset \mathcal{H}$. For $P_n = |\phi_n\rangle\langle\phi_n|$ and $S_M = \sum_{n<M} P_n$,

$$\|f - S_M f\|_H^2 = \sum_{n=M}^{\infty} (1 + \lambda_n^2)|c_n|^2 \tag{6.13}$$

$$\longrightarrow 0 \tag{6.14}$$

for every $f \in D(H)$. The spectral atlas therefore converges in graph norm, not only in Hilbert norm.

6.3 Bounded surrogates and topology-specific truncation

Three bounded functions of an unbounded self-adjoint Hamiltonian are

$$R_{\mathrm{i}\eta} := (H - \mathrm{i}\eta)^{-1}, \tag{6.15}$$

$$F_H := H(1 + H^2)^{-1/2}, \tag{6.16}$$

$$U_H := (H - \mathrm{i})(H + \mathrm{i})^{-1}. \tag{6.17}$$

For the harmonic spectrum,

$$\|R_{\mathrm{i}\eta} - S_M R_{\mathrm{i}\eta} S_M\| = \sup_{n \geq M} \frac{1}{\sqrt{\lambda_n^2 + \eta^2}} \longrightarrow 0, \tag{6.18}$$

$$\|\mathrm{e}^{-tH} - S_M \mathrm{e}^{-tH} S_M\| = \mathrm{e}^{-t\lambda_M} \longrightarrow 0. \tag{6.19}$$

These multipliers vanish in the spectral tail, so zero padding gives operator-norm convergence.
For the unitary group $U(t) = \mathrm{e}^{-\mathrm{i}tH}$, every tail eigenvector satisfies

$$(U(t) - S_M U(t) S_M)\phi_n = \mathrm{e}^{-\mathrm{i}t\lambda_n}\phi_n, \qquad n \geq M, \tag{6.20}$$

so

$$\boxed{\|U(t) - S_M U(t) S_M\| = 1} \tag{6.21}$$

for every finite M, although the truncations converge strongly on each fixed state.
The bounded and Cayley multipliers tend to 1, rather than 0. Their norm-convergent truncations must retain this tail:

$$F_H^{(M)} := S_M F_H S_M + (I - S_M), \tag{6.22}$$

$$U_H^{(M)} := S_M U_H S_M + (I - S_M). \tag{6.23}$$

Then

$$\|F_H - F_H^{(M)}\| = \sup_{n \geq M} \left| \frac{\lambda_n}{\sqrt{1 + \lambda_n^2}} - 1 \right| \to 0, \tag{6.24}$$

$$\|U_H - U_H^{(M)}\| = \sup_{n \geq M} \frac{2}{|\lambda_n + \mathrm{i}|} \to 0. \tag{6.25}$$

The topology is part of the approximation statement: strong convergence is appropriate for zero-padded unitary evolution, while operator-norm convergence requires compact tail decay or the correct asymptotic anchor.

6.4 Countable packet continuity

$$\psi(t) := \mathrm{e}^{-\mathrm{i}tH}\psi_0, \qquad \psi_0 \in D(H), \tag{6.26}$$

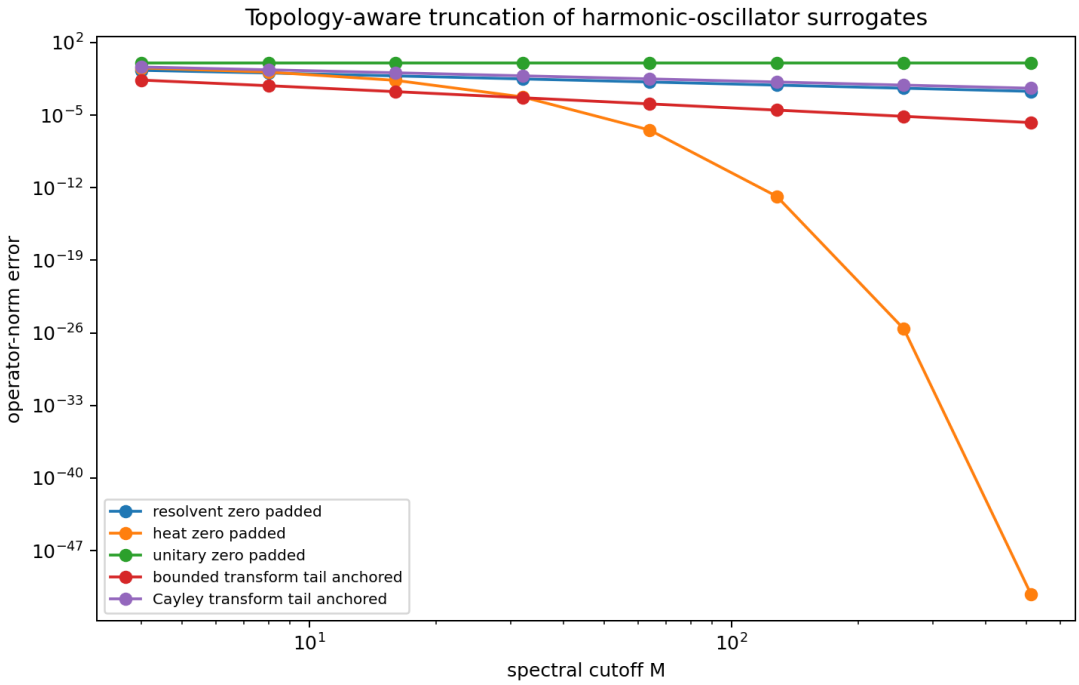


Figure 6.1: Topology-aware truncation of harmonic-oscillator spectral surrogates. Resolvent and heat tails decay under zero padding; bounded and Cayley transforms converge in norm after their limiting tail value is retained.

$$E_\alpha := \|P_\alpha \psi\|^2 , \tag{6.27}$$

$$J_{\alpha \leftarrow \beta} := 2 \operatorname{Im} \langle P_\alpha \psi, H P_\beta \psi \rangle . \tag{6.28}$$

Because $\dot{\psi} = -\mathrm{i} H \psi$,

$$\frac{\mathrm{d}E_\alpha}{\mathrm{d}t} = 2 \operatorname{Re} \langle P_\alpha \psi, P_\alpha \dot{\psi} \rangle \tag{6.29}$$

$$= 2 \operatorname{Im} \langle P_\alpha \psi, H \psi \rangle . \tag{6.30}$$

Graph convergence permits insertion of the finite partial sums:

$$\langle P_\alpha \psi, H \psi \rangle = \lim_{F \nearrow A} \langle P_\alpha \psi, H S_F \psi \rangle \tag{6.31}$$

$$= \lim_{F \nearrow A} \sum_{\beta \in F} \langle P_\alpha \psi, H P_\beta \psi \rangle . \tag{6.32}$$

Therefore

$$\frac{\mathrm{d}E_\alpha}{\mathrm{d}t} = \lim_{F \nearrow A} \sum_{\beta \in F} J_{\alpha \leftarrow \beta}. \tag{6.33}$$

Self-adjointness gives

$$\langle P_\beta \psi, H P_\alpha \psi \rangle = \overline{\langle P_\alpha \psi, H P_\beta \psi \rangle}, \tag{6.34}$$

$$J_{\alpha \leftarrow \beta} = -J_{\beta \leftarrow \alpha}. \tag{6.35}$$

For a finite packet region $B \Subset A$, summing the packet equations gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \sum_{\alpha \in B} E_\alpha = \lim_{F \nearrow A} \left[\sum_{\alpha, \beta \in B} J_{\alpha \leftarrow \beta} + \sum_{\substack{\alpha \in B \\ \beta \in F \setminus B}} J_{\alpha \leftarrow \beta} \right] . \tag{6.36}$$

The first double sum vanishes pairwise by antisymmetry, leaving only the boundary current:

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_{\alpha\in B}E_\alpha=\lim_{F\nearrow A}\sum_{\substack{\alpha\in B\\ \beta\in F\setminus B}}J_{\alpha\leftarrow\beta}. \tag{6.37}$$

Strong completeness and orthogonality give Parseval's identity

$$\sum_\alpha E_\alpha(t)=\lim_{F\nearrow A}\sum_{\alpha\in F}\|P_\alpha\psi(t)\|^2 \tag{6.38}$$

$$=\lim_{F\nearrow A}\|S_F\psi(t)\|^2 \tag{6.39}$$

$$=\|\psi(t)\|^2. \tag{6.40}$$

Unitarity of $\mathrm{e}^{-\mathrm{i}tH}$ then gives the global norm law

$$\sum_\alpha E_\alpha(t)=\|\psi(t)\|^2=\|\psi_0\|^2. \tag{6.41}$$

6.5 Countable refinement

$$Q_A:=\sum_{\alpha\in C_A}P_\alpha\quad\text{strongly.} \tag{6.42}$$

Orthogonality within a coarse cell gives

$$\|Q_A\psi\|^2=\lim_{F\nearrow C_A}\left\|\sum_{\alpha\in F}P_\alpha\psi\right\|^2 \tag{6.43}$$

$$=\sum_{\alpha\in C_A}\|P_\alpha\psi\|^2. \tag{6.44}$$

Under finite grouping or absolute convergence, both coarse projectors may also be expanded inside the current:

$$J_{A\leftarrow B}:=2\,\mathrm{Im}\,\langle Q_A\psi,HQ_B\psi\rangle \tag{6.45}$$

$$=2\,\mathrm{Im}\sum_{\alpha\in C_A}\sum_{\beta\in C_B}\langle P_\alpha\psi,HP_\beta\psi\rangle \tag{6.46}$$

$$=\sum_{\alpha\in C_A}\sum_{\beta\in C_B}J_{\alpha\leftarrow\beta}. \tag{6.47}$$

6.6 Spectral residue packets and ladder routing

For the oscillator number operator $\widehat{N}\phi_n=n\phi_n$, define

$$Q_r^{(M)}:=\frac{1}{M}\sum_{k=0}^{M-1}\exp(-2\pi\mathrm{i}rk/M)\exp(2\pi\mathrm{i}k\widehat{N}/M). \tag{6.48}$$

The finite character sum gives

$$\boxed{Q_r^{(M)}=\sum_{n\equiv r\pmod M}P_n.} \tag{6.49}$$

Their coherent-state weights are the hypergeometric factorial lanes developed in the companion-lane source chapter and recorded in the complete formula corpus. Since

$$a\phi_n=\sqrt{n}\,\phi_{n-1},\qquad a^\dagger\phi_n=\sqrt{n+1}\,\phi_{n+1}, \tag{6.50}$$

one has

$$\boxed{Q_r^{(M)}aQ_s^{(M)}=0\quad\text{unless}\quad r\equiv s-1\pmod M,} \tag{6.51}$$

and

$$\boxed{Q_r^{(M)}a^\dagger Q_s^{(M)}=0\quad\text{unless}\quad r\equiv s+1\pmod M.} \tag{6.52}$$

The ladder operators are sparse rather than diagonal in the countable polyphase atlas.

6.7 Nonreducing packets and a resolvent certificate

Suppose $PD(H) \subset D(H)$ and $[H, P]$ extends to a bounded operator. For $R_z = (H - z)^{-1}$,

$$[R_z, P] = -(H - z)^{-1}[H - z, P](H - z)^{-1} \tag{6.53}$$
$$= -R_z[H, P]R_z. \tag{6.54}$$

Hence

$$\boxed{\|[R_z, P]\| \le \|R_z\|^2 \|[H, P]\|.} \tag{6.55}$$

For self-adjoint H and $z = \mathrm{i}\eta$,

$$\boxed{\|[R_{\mathrm{i}\eta}, P]\| \le \frac{\|[H, P]\|}{\eta^2}.} \tag{6.56}$$

This bounded resolvent identity measures how strongly a packet fails to reduce the unbounded Hamiltonian.

6.8 Infinite reconstruction tower

Let $S_N \to I$ strongly. Then

$$X \in \mathcal{B}(\mathcal{H}) \Longrightarrow S_N X S_N \to X \quad \text{strongly}, \tag{6.57}$$
$$X \in \mathcal{K}(\mathcal{H}) \Longrightarrow \|S_N X S_N - X\| \to 0, \tag{6.58}$$
$$X \in \mathcal{S}_2(\mathcal{H}) \Longrightarrow \|S_N X S_N - X\|_{\mathcal{S}_2} \to 0. \tag{6.59}$$

The basic identity is

$$S_N X S_N - X = S_N X(S_N - I) + (S_N - I)X. \tag{6.60}$$

For a bounded X and fixed ψ,

$$\|(S_N X S_N - X)\psi\| \le \|X\| \, \|(S_N - I)\psi\| + \|(S_N - I)X\psi\| \tag{6.61}$$
$$\longrightarrow 0, \tag{6.62}$$

which proves strong convergence.

For compact X, choose a finite-rank K with $\|X - K\| < \varepsilon$. Strong convergence is uniform on the finite-dimensional range and co-range of K, so

$$\|S_N X S_N - X\| \le 2\,\|X - K\| + \|S_N K S_N - K\| \tag{6.63}$$
$$\longrightarrow 2\varepsilon. \tag{6.64}$$

Since ε is arbitrary, the convergence is in operator norm.

For $X \in \mathcal{S}_2(\mathcal{H})$, dominated convergence on an orthonormal basis gives

$$\|(S_N - I)X\|_{\mathcal{S}_2} \longrightarrow 0, \qquad \|X(S_N - I)\|_{\mathcal{S}_2} \longrightarrow 0. \tag{6.65}$$

The same identity then yields Hilbert–Schmidt convergence.

On the graph domain, one also has

$$S_N H S_N \psi - H\psi = S_N(H S_N \psi - H\psi) + (S_N - I)H\psi \tag{6.66}$$
$$\longrightarrow 0, \qquad \psi \in D(H), \tag{6.67}$$

so the finite reconstruction tower is compatible with the unbounded generator as well as with the states.

6.9 Verification and scope

The harmonic-oscillator surrogate, residue-projector, ladder-routing, and resolvent-commutator checks all pass. The largest algebraic floating-point residual is

$$7.879615 \times 10^{-16}. \tag{6.68}$$

The anchored bounded-transform tail has measured asymptotic order

$$1.998730. \tag{6.69}$$

These calculations verify the stated spectral-coordinate identities and the topology distinction.

Next steps.

- Verify the required domains and common invariant cores for concrete interacting unbounded Hamiltonians.
- Prove refinement-uniform resolvent and commutator estimates beyond the diagonal oscillator model.
- Extend the bounded-surrogate convergence results to non-diagonal generators and continuous spectral components.
- Relate convergence of the bounded surrogates to convergence of the physical observables used in the quantum and gravitational chapters.

Chapter 7

Finite Differential No-Go and Corner-Algebra Resolution

Finite commutative pointwise algebras admit no nonzero derivations. The remedy is to work in the full matrix corner algebra, where commutators provide exact finite differentiation and integration by parts.

7.1 Finite pointwise algebra: no-go theorem

Let

$$\mathcal{A}_N := \mathbb{C}^N, \qquad (fg)_a := f_a g_a, \qquad \boldsymbol{e}_a \boldsymbol{e}_b = \delta_{ab} \boldsymbol{e}_a. \tag{7.1}$$

A derivation is a linear map

$$D : \mathcal{A}_N \to \mathcal{A}_N, \qquad D(fg) = D(f)g + fD(g). \tag{7.2}$$

For every primitive idempotent,

$$D\boldsymbol{e}_a = D(\boldsymbol{e}_a^2) \tag{7.3}$$

$$= 2\boldsymbol{e}_a D\boldsymbol{e}_a. \tag{7.4}$$

Coordinatewise,

$$(D\boldsymbol{e}_a)_b = 2\delta_{ab}(D\boldsymbol{e}_a)_b. \tag{7.5}$$

If $b \neq a$, the right side is zero, so $(D\boldsymbol{e}_a)_b = 0$. If $b = a$, the equality reads

$$(D\boldsymbol{e}_a)_a = 2(D\boldsymbol{e}_a)_a, \tag{7.6}$$

which again forces $(D\boldsymbol{e}_a)_a = 0$. Thus every coordinate of $D\boldsymbol{e}_a$ vanishes. Hence

$$(D\boldsymbol{e}_a)_b = 0 \qquad (1 \leq a, b \leq N), \tag{7.7}$$

and therefore

$$\boxed{\mathrm{Der}(\mathcal{A}_N) = \{0\}.} \tag{7.8}$$

The exact rank calculation for $2 \leq N \leq 8$ gives

$$\operatorname{nullity} \mathcal{L}_N = 0, \tag{7.9}$$

where $\mathcal{L}_N$ is the full Leibniz linear system for the N^2 coefficients of D.

7.2 Finite-scale consequence

For a nonzero finite differential operator, the conjunction

$$\begin{cases} \text{finite pointwise algebra,} \\ \text{ordinary pointwise product,} \\ \text{exact Leibniz identity,} \\ \text{nonzero derivative} \end{cases} \tag{7.10}$$

is inconsistent. At least one entry must be replaced by

$$\boxed{\begin{array}{ccc} \text{refinement limit} & \vee & \text{restricted resolved subspace} \\ \vee \quad \text{deformed product} & \vee & \text{corner/noncommutative algebra} \end{array}} \tag{7.11}$$

7.3 Full corner algebra

Let

$$\mathcal{B}_N := M_N(\mathbb{C}), \qquad K = K^*. \tag{7.12}$$

Define

$$\boxed{\delta_K X := \mathrm{i}[K, X].} \tag{7.13}$$

Then

$$\delta_K(XY) = (\delta_K X)Y + X(\delta_K Y), \tag{7.14}$$
$$(\delta_K X)^* = \delta_K(X^*), \tag{7.15}$$
$$\mathrm{Tr}((\delta_K X)Y) = -\,\mathrm{Tr}(X\delta_K Y). \tag{7.16}$$

These identities follow by direct expansion:

$$\delta_K(XY) = \mathrm{i}(KXY - XYK) \tag{7.17}$$
$$= \mathrm{i}(KX - XK)Y + \mathrm{i}X(KY - YK), \tag{7.18}$$
$$(\delta_K X)^* = -\mathrm{i}(X^*K - KX^*) = \mathrm{i}(KX^* - X^*K), \tag{7.19}$$
$$\mathrm{Tr}((\delta_K X)Y) = \mathrm{i}\,\mathrm{Tr}(KXY - XKY) \tag{7.20}$$
$$= \mathrm{i}\,\mathrm{Tr}(XYK - XKY) \tag{7.21}$$
$$= -\,\mathrm{Tr}(X\delta_K Y). \tag{7.22}$$

Cyclicity of the trace is the finite analogue of discarding a boundary term. Thus

$$\delta_K \in \mathrm{Der}_*(\mathcal{B}_N), \tag{7.23}$$

and the trace pairing supplies exact finite integration by parts.

7.4 Curvature of finite derivations

For

$$\delta_i X := \mathrm{i}[K_i, X], \qquad F_{ij} := \mathrm{i}[K_i, K_j], \tag{7.24}$$

the Jacobi identity yields

$$\boxed{[\delta_i, \delta_j]X = \mathrm{i}[F_{ij}, X].} \tag{7.25}$$

Indeed,

$$[\delta_i, \delta_j]X = -[K_i, [K_j, X]] + [K_j, [K_i, X]] \tag{7.26}$$
$$= -[[K_i, K_j], X] \tag{7.27}$$
$$= \mathrm{i}[\mathrm{i}[K_i, K_j], X] = \mathrm{i}[F_{ij}, X], \tag{7.28}$$

where the middle equality is the Jacobi identity. Therefore

$$[K_i, K_j] = 0 \Longrightarrow [\delta_i, \delta_j] = 0, \tag{7.29}$$
$$[K_i, K_j] \neq 0 \Longrightarrow \text{finite corner curvature.} \tag{7.30}$$

7.5 Projector-corner reconstruction

Let

$$P_a P_b = \delta_{ab} P_a, \qquad \sum_a P_a = I. \tag{7.31}$$

Then

$$\delta_K X = \sum_{a,b} P_a (\delta_K X) P_b. \tag{7.32}$$

Each differentiated corner has the path expansion

$$\boxed{P_a(\delta_K X)P_b = \mathrm{i}\sum_c \left(P_a K P_c X P_b - P_a X P_c K P_b\right).} \tag{7.33}$$

Insert $I = \sum_c P_c$ between K and X in each commutator term:

$$P_a K X P_b = \sum_c P_a K P_c X P_b, \tag{7.34}$$

$$P_a X K P_b = \sum_c P_a X P_c K P_b. \tag{7.35}$$

Each summand is a two-step corner path through the intermediate packet c. For a coarse atlas

$$Q_A := \sum_{a \in C_A} P_a, \tag{7.36}$$

exact refinement gives

$$\boxed{Q_A(\delta_K X)Q_B = \sum_{\substack{a\in C_A\\ b\in C_B}} P_a(\delta_K X)P_b.} \tag{7.37}$$

7.6 Numerical verification

$$\max_{2\le N\le 8} \operatorname{nullity} \mathcal{L}_N = 0, \tag{7.38}$$

$$\|\delta_K(XY) - (\delta_K X)Y - X\delta_K Y\|_2 = 5.283117 \times 10^{-14}, \tag{7.39}$$

$$\|(\delta_K X)^* - \delta_K(X^*)\|_2 = 6.085790 \times 10^{-15}, \tag{7.40}$$

$$|\operatorname{Tr}((\delta_K X)Y) + \operatorname{Tr}(X\delta_K Y)| = 2.131628 \times 10^{-14}, \tag{7.41}$$

$$\|[\delta_1, \delta_2]X - \mathrm{i}[F_{12}, X]\|_2 = 5.495524 \times 10^{-14}, \tag{7.42}$$

$$\|\delta_K X - \sum_{a,b} P_a \delta_K X P_b\|_2 = 3.035492 \times 10^{-14}, \tag{7.43}$$

$$\max_{A,B} \left\| Q_A \delta_K X Q_B - \sum_{a\in C_A, b\in C_B} P_a \delta_K X P_b \right\|_2 = 5.824437 \times 10^{-15}. \tag{7.44}$$

Part II

Discrete Relativity, Gauge Fields, and Finite-Scale Kinematics

Chapter 8

Discrete Relativistic Sectors

These models show how strictly local unitary updates recover relativistic dispersion at resolved scales. The one- and three-dimensional formulas also identify the leading finite-spacing corrections.

8.1 One spatial dimension

$$\alpha := \sigma_z, \qquad \beta := \sigma_x, \qquad \{\alpha, \beta\} = 0, \qquad \alpha^2 = \beta^2 = I_2, \tag{8.1}$$

$$U_\varepsilon(k) := \mathrm{e}^{-\mathrm{i}m\varepsilon\beta}\mathrm{e}^{-\mathrm{i}k\varepsilon\alpha}, \tag{8.2}$$

$$U_\varepsilon(k)^* U_\varepsilon(k) = I_2, \tag{8.3}$$

$$\det U_\varepsilon(k) = 1, \tag{8.4}$$

$$\frac{1}{2}\operatorname{Tr} U_\varepsilon(k) = \cos(m\varepsilon)\cos(k\varepsilon), \tag{8.5}$$

$$\cos(\omega_\varepsilon \varepsilon) = \cos(m\varepsilon)\cos(k\varepsilon), \tag{8.6}$$

$$\omega_\varepsilon(k) = \frac{1}{\varepsilon}\arccos[\cos(m\varepsilon)\cos(k\varepsilon)]\,. \tag{8.7}$$

Since an involution $A^2 = I$ satisfies

$$\mathrm{e}^{-\mathrm{i}aA} = \cos a\, I - \mathrm{i}\sin a\, A, \tag{8.8}$$

the two factors expand as

$$U_\varepsilon = (\cos m\varepsilon\, I - \mathrm{i}\sin m\varepsilon\, \beta)(\cos k\varepsilon\, I - \mathrm{i}\sin k\varepsilon\, \alpha). \tag{8.9}$$

The terms containing α, β, or $\beta\alpha$ are traceless, leaving

$$\frac{1}{2}\operatorname{Tr} U_\varepsilon = \cos(m\varepsilon)\cos(k\varepsilon). \tag{8.10}$$

The determinant is one because each exponent has a traceless generator. Let $E := \sqrt{m^2 + k^2}$. Then

$$\omega_\varepsilon = E - \frac{m^2 k^2}{6E}\varepsilon^2 + \mathcal{O}(\varepsilon^4), \tag{8.11}$$

$$\omega_\varepsilon^2 - k^2 - m^2 = -\frac{1}{3}m^2 k^2 \varepsilon^2 + \mathcal{O}(\varepsilon^4), \tag{8.12}$$

$$\frac{\partial \omega_\varepsilon}{\partial k} = \frac{\cos(m\varepsilon)\sin(k\varepsilon)}{\sin(\omega_\varepsilon \varepsilon)}, \tag{8.13}$$

$$\frac{\partial \omega_\varepsilon}{\partial k} = \frac{k}{E} + \mathcal{O}(\varepsilon^2). \tag{8.14}$$

For the frequency expansion,

$$\cos(m\varepsilon)\cos(k\varepsilon) = 1 - \frac{E^2\varepsilon^2}{2} + \left(\frac{E^4}{24} + \frac{m^2 k^2}{6}\right)\varepsilon^4 + \mathcal{O}(\varepsilon^6). \tag{8.15}$$

Write $\omega_\varepsilon = E + a\varepsilon^2 + \mathcal{O}(\varepsilon^4)$. Then

$$\cos(\omega_\varepsilon \varepsilon) = 1 - \frac{E^2\varepsilon^2}{2} + \left(\frac{E^4}{24} - Ea\right)\varepsilon^4 + \mathcal{O}(\varepsilon^6). \tag{8.16}$$

Matching the fourth-order terms gives $a = -m^2k^2/(6E)$. Differentiating the exact shell gives

$$-\varepsilon \sin(\omega_\varepsilon \varepsilon)\frac{\partial \omega_\varepsilon}{\partial k} = -\varepsilon \cos(m\varepsilon)\sin(k\varepsilon), \tag{8.17}$$

which yields the displayed exact group velocity.

8.2 Three spatial dimensions

$$\{\alpha_i, \alpha_j\} = 2\delta_{ij} I_4, \tag{8.18}$$
$$\{\alpha_i, \beta\} = 0, \tag{8.19}$$
$$\beta^2 = I_4, \tag{8.20}$$
$$M_{1/2} := \mathrm{e}^{-\mathrm{i}m\Delta\beta/2}, \tag{8.21}$$
$$X_{1/2} := \mathrm{e}^{-\mathrm{i}k_x\Delta\alpha_x/2}, \tag{8.22}$$
$$Y_{1/2} := \mathrm{e}^{-\mathrm{i}k_y\Delta\alpha_y/2}, \tag{8.23}$$
$$Z := \mathrm{e}^{-\mathrm{i}k_z\Delta\alpha_z}, \tag{8.24}$$
$$U_\Delta(\boldsymbol{k}) := M_{1/2}X_{1/2}Y_{1/2}ZY_{1/2}X_{1/2}M_{1/2}. \tag{8.25}$$

Microcell locality:

$$h_{\mathrm{micro}} = \frac{\Delta}{2}. \tag{8.26}$$

Define

$$C_\Delta(\boldsymbol{k}) := \cos(m\Delta)\cos(k_x\Delta)\cos(k_y\Delta)\cos(k_z\Delta). \tag{8.27}$$

Then

$$\det(\lambda I_4 - U_\Delta) = \left(\lambda^2 - 2C_\Delta\lambda + 1\right)^2, \tag{8.28}$$
$$\mathrm{spec}(U_\Delta) = \{\mathrm{e}^{-\mathrm{i}\omega_\Delta\Delta}, \mathrm{e}^{-\mathrm{i}\omega_\Delta\Delta}, \mathrm{e}^{+\mathrm{i}\omega_\Delta\Delta}, \mathrm{e}^{+\mathrm{i}\omega_\Delta\Delta}\}, \tag{8.29}$$
$$\cos(\omega_\Delta\Delta) = C_\Delta(\boldsymbol{k}). \tag{8.30}$$

Let

$$E := \sqrt{m^2 + |\boldsymbol{k}|^2}, \tag{8.31}$$
$$Q(\boldsymbol{k}) := m^2|\boldsymbol{k}|^2 + k_x^2k_y^2 + k_x^2k_z^2 + k_y^2k_z^2. \tag{8.32}$$

Then

$$\omega_\Delta = E - \frac{Q(\boldsymbol{k})}{6E}\Delta^2 + \mathcal{O}(\Delta^4), \tag{8.33}$$
$$\partial_{k_i}\omega_\Delta = \frac{\cos(m\Delta)\sin(k_i\Delta)\prod_{j\neq i}\cos(k_j\Delta)}{\sin(\omega_\Delta\Delta)}, \tag{8.34}$$
$$\nabla_{\boldsymbol{k}}\omega_\Delta = \frac{\boldsymbol{k}}{E} + \mathcal{O}(\Delta^2). \tag{8.35}$$

The product of four cosines has the expansion

$$C_\Delta(\boldsymbol{k}) = 1 - \frac{E^2\Delta^2}{2} + \left(\frac{E^4}{24} + \frac{Q(\boldsymbol{k})}{6}\right)\Delta^4 + \mathcal{O}(\Delta^6). \tag{8.36}$$

Substituting $\omega_\Delta = E + a(\boldsymbol{k})\Delta^2 + \mathcal{O}(\Delta^4)$ into $\cos(\omega_\Delta \Delta)$ gives the fourth-order coefficient $E^4/24 - Ea$. Therefore $a = -Q/(6E)$. Differentiating the exact product shell with respect to k_i gives the stated velocity component. For $|\boldsymbol{k}| = \kappa$,

$$\boldsymbol{k}_{\text{axis}} = (\kappa, 0, 0), \tag{8.37}$$

$$\boldsymbol{k}_{\text{diag}} = \frac{\kappa}{\sqrt{3}}(1, 1, 1), \tag{8.38}$$

$$\omega_{\text{axis}} - \omega_{\text{diag}} = \frac{\kappa^4}{18E}\Delta^2 + \mathcal{O}(\Delta^4). \tag{8.39}$$

At fixed κ,

$$Q(\boldsymbol{k}_{\text{axis}}) = m^2\kappa^2, \tag{8.40}$$

$$Q(\boldsymbol{k}_{\text{diag}}) = m^2\kappa^2 + 3\left(\frac{\kappa^2}{3}\right)^2 = m^2\kappa^2 + \frac{\kappa^4}{3}. \tag{8.41}$$

Inserting these values into the frequency correction gives the axis–diagonal difference.

8.3 Observer-shell defect

For $|\boldsymbol{v}| < 1$,

$$\gamma := (1 - |\boldsymbol{v}|^2)^{-1/2}, \tag{8.42}$$

$$E' := \gamma(E - \boldsymbol{v} \cdot \boldsymbol{k}), \tag{8.43}$$

$$\boldsymbol{k}' := \boldsymbol{k} + \left[\frac{\gamma - 1}{|\boldsymbol{v}|^2}(\boldsymbol{v} \cdot \boldsymbol{k}) - \gamma E\right]\boldsymbol{v}. \tag{8.44}$$

Define

$$\mathfrak{L}_\Delta(\boldsymbol{k}, \boldsymbol{v}) := \Lambda^0{}_\nu p^\nu_\Delta - \omega_\Delta(\boldsymbol{k}'_\Delta), \qquad p_\Delta := (\omega_\Delta(\boldsymbol{k}), \boldsymbol{k}). \tag{8.45}$$

Then

$$\mathfrak{L}_\Delta(\boldsymbol{k}, \boldsymbol{v}) = \frac{\Delta^2}{6E'}\left[Q(\boldsymbol{k}') - Q(\boldsymbol{k})\right] + \mathcal{O}(\Delta^4), \tag{8.46}$$

$$\mathfrak{L}_\Delta - \frac{\Delta^2}{6E'}\left[Q(\boldsymbol{k}') - Q(\boldsymbol{k})\right] = \mathcal{O}(\Delta^4). \tag{8.47}$$

The calculation uses

$$p^0_\Delta = E - \frac{Q(\boldsymbol{k})}{6E}\Delta^2 + \mathcal{O}(\Delta^4). \tag{8.48}$$

The ordinary Lorentz transform cancels at order Δ^0. The transformed energy correction contributes $-\Delta^2 Q(\boldsymbol{k})/(6E')$, while evaluating the same discrete shell at the transformed momentum contributes $-\Delta^2 Q(\boldsymbol{k}')/(6E')$. Their difference is the displayed observer-shell defect.

Chapter 9

Exact 3+1 Quasienergy Closure

This chapter closes the free product-cosine sector at the operator level, separates exact statements from model choices, and records the remaining obstructions. The algebra below is exact in its stated domains. It does not by itself establish empirical validity, physical priority, or literature novelty.

$$\text{exact discrete dynamics} \not\Longrightarrow \text{new fundamental law without independent physical identification.} \tag{9.1}$$

9.1 Exact shell and corrected continuum hierarchy

Set

$$a_0 := m, \qquad a_1 := k_x, \qquad a_2 := k_y, \qquad a_3 := k_z, \tag{9.2}$$

$$E^2 := \sum_{r=0}^{3} a_r^2, \qquad C_\Delta := \prod_{r=0}^{3} \cos(a_r \Delta), \tag{9.3}$$

$$\cos(\omega_\Delta \Delta) = C_\Delta. \tag{9.4}$$

For

$$\omega_\Delta = E + c_2 \Delta^2 + \mathcal{O}(\Delta^4), \tag{9.5}$$

the fourth-order coefficients are

$$C_\Delta = 1 - \frac{E^2 \Delta^2}{2} + \Delta^4 \left[\frac{1}{24} \sum_r a_r^4 + \frac{1}{4} \sum_{r<s} a_r^2 a_s^2 \right] + \mathcal{O}(\Delta^6), \tag{9.6}$$

$$\cos(\omega_\Delta \Delta) = 1 - \frac{E^2 \Delta^2}{2} + \Delta^4 \left[\frac{E^4}{24} - E c_2 \right] + \mathcal{O}(\Delta^6), \tag{9.7}$$

$$E^4 = \sum_r a_r^4 + 2 \sum_{r<s} a_r^2 a_s^2. \tag{9.8}$$

Therefore

$$\boxed{E c_2 = -\frac{1}{6} \sum_{r<s} a_r^2 a_s^2} \tag{9.9}$$

and

$$\boxed{\omega_\Delta = E - \frac{\Delta^2}{6E} \sum_{r<s} a_r^2 a_s^2 + \mathcal{O}(\Delta^4).} \tag{9.10}$$

The positive coefficient sometimes assigned to this term corresponds instead to the convention

$$\omega_\Delta = E - c_2 \Delta^2 + \mathcal{O}(\Delta^4), \qquad E c_2 = \frac{1}{6} \sum_{r<s} a_r^2 a_s^2. \tag{9.11}$$

Let

$$x_r := a_r^2, \tag{9.12}$$

$$e_1 := \sum_r x_r, \tag{9.13}$$

$$e_2 := \sum_{r<s} x_r x_s, \tag{9.14}$$

$$e_3 := \sum_{r<s<t} x_r x_s x_t, \tag{9.15}$$

$$e_4 := x_0 x_1 x_2 x_3. \tag{9.16}$$

Direct coefficient matching through order Δ^8 gives

$$\boxed{\begin{aligned}\omega_\Delta^2 = e_1 - \frac{e_2}{3}\Delta^2 + \frac{6e_3 - e_1 e_2}{45}\Delta^4 \\ - \frac{2e_1^2 e_2 - 9e_1 e_3 - 8e_2^2 + 51e_4}{945}\Delta^6 + \mathcal{O}(\Delta^8).\end{aligned}} \tag{9.17}$$

Consequently,

$$\boxed{\omega_\Delta = E - \frac{e_2}{6E}\Delta^2 + \left[\frac{6e_3 - E^2 e_2}{90E} - \frac{e_2^2}{72E^3}\right]\Delta^4 + \mathcal{O}(\Delta^6).} \tag{9.18}$$

The hierarchy is generated without further inversion by

$$\cos\left(\Delta\sqrt{\sum_{j\geq 0} b_{2j}\Delta^{2j}}\right) = \prod_{r=0}^{3}\cos(a_r\Delta), \qquad b_0 = e_1, \tag{9.19}$$

followed by coefficient extraction in Δ.

9.2 Exact phase and transfer coordinates

Away from cosine zeros, define

$$\sigma_\Delta(x) := \operatorname{sgn}[\cos(\Delta x)], \tag{9.20}$$

$$\kappa_\Delta(x) := \frac{1}{\Delta}\sqrt{-2\log|\cos(\Delta x)|}. \tag{9.21}$$

Then

$$\cos(\Delta x) = \sigma_\Delta(x)\exp\left[-\frac{\Delta^2}{2}\kappa_\Delta(x)^2\right]. \tag{9.22}$$

The exact product shell is equivalent to

$$\sigma_\Delta(\omega_\Delta) = \prod_{r=0}^{3}\sigma_\Delta(a_r), \tag{9.23}$$

$$\kappa_\Delta(\omega_\Delta)^2 = \sum_{r=0}^{3}\kappa_\Delta(a_r)^2. \tag{9.24}$$

Thus the branch label composes in $\mathbb{Z}_2$, while the logarithmic magnitudes compose quadratically.
On the principal interval

$$\mathcal{I}_\Delta := \left(-\frac{\pi}{2\Delta}, \frac{\pi}{2\Delta}\right), \tag{9.25}$$

define the odd bijection

$$K_\Delta(x) := \operatorname{sgn}(x)\frac{\sqrt{-2\log\cos(\Delta x)}}{\Delta}. \tag{9.26}$$

Its inverse is

$$K_\Delta^{-1}(P) = \operatorname{sgn}(P)\frac{1}{\Delta}\arccos\left(\mathrm{e}^{-\Delta^2 P^2/2}\right). \tag{9.27}$$

The small-spacing expansion is

$$K_\Delta(x) = x + \frac{x^3\Delta^2}{12} + \frac{3x^5\Delta^4}{160} + \frac{209x^7\Delta^6}{40320} + \mathcal{O}(\Delta^8 x^9). \tag{9.28}$$

The exact mass shell becomes

$$\boxed{K_\Delta(\omega_\Delta)^2 - \sum_{i=1}^{3} K_\Delta(k_i)^2 = K_\Delta(m)^2.} \tag{9.29}$$

For squared variables, define

$$\Lambda_\Delta(X) := -\frac{2}{\Delta^2}\log\cos(\Delta\sqrt{X}), \qquad 0 \le X < \frac{\pi^2}{4\Delta^2}. \tag{9.30}$$

Then

$$\Lambda_\Delta(\omega_\Delta^2) = \sum_{r=0}^{3}\Lambda_\Delta(a_r^2). \tag{9.31}$$

The induced operation

$$X \oplus_\Delta Y := \Lambda_\Delta^{-1}\left[\Lambda_\Delta(X) + \Lambda_\Delta(Y)\right] \tag{9.32}$$

is exactly commutative and associative:

$$X \oplus_\Delta Y = Y \oplus_\Delta X, \tag{9.33}$$

$$(X \oplus_\Delta Y) \oplus_\Delta Z = X \oplus_\Delta (Y \oplus_\Delta Z). \tag{9.34}$$

Its expansion is

$$\boxed{\begin{aligned} X \oplus_\Delta Y = X + Y - \frac{\Delta^2}{3}XY - \frac{\Delta^4}{45}XY(X+Y) \\ - \frac{2\Delta^6}{945}XY(X-Y)^2 + \mathcal{O}(\Delta^8). \end{aligned}} \tag{9.35}$$

Hence

$$\omega_\Delta^2 = m^2 \oplus_\Delta k_x^2 \oplus_\Delta k_y^2 \oplus_\Delta k_z^2. \tag{9.36}$$

9.3 Principal-branch bounds and nonlinear Lorentz action

Set

$$g(X) := -\log\cos\sqrt{X}, \qquad 0 \le X < \frac{\pi^2}{4}. \tag{9.37}$$

Since

$$g'(X) = \frac{\tan\sqrt{X}}{2\sqrt{X}} \tag{9.38}$$

is increasing, g is convex and $g(0) = 0$. Therefore

$$g(X+Y) \ge g(X) + g(Y). \tag{9.39}$$

If $E\Delta < \pi/2$, then

$$\cos(E\Delta) \le \prod_r \cos(a_r\Delta), \tag{9.40}$$

$$\prod_r \cos(a_r\Delta) \le \min_r \cos(|a_r|\Delta), \tag{9.41}$$

which implies

$$\boxed{\max_r |a_r| \le \omega_\Delta \le \sqrt{\sum_r a_r^2}.} \tag{9.42}$$

The negative leading correction is therefore consistent with an exact branchwise upper bound.
Introduce auxiliary momenta

$$\mathcal{E} := K_\Delta(\omega), \tag{9.43}$$
$$\mathcal{P}_i := K_\Delta(k_i), \tag{9.44}$$
$$\mathcal{M} := K_\Delta(m). \tag{9.45}$$

For $\Lambda \in SO^+(1,3)$, define

$$\mathcal{B}_\Lambda(p)_\mu := K_\Delta^{-1}\left[\Lambda_\mu{}^\nu K_\Delta(p_\nu)\right]. \tag{9.46}$$

Then

$$\mathcal{B}_{\Lambda_1} \circ \mathcal{B}_{\Lambda_2} = \mathcal{B}_{\Lambda_1\Lambda_2}, \tag{9.47}$$
$$K_\Delta(\omega')^2 - \sum_i K_\Delta(k_i')^2 = \mathcal{M}^2. \tag{9.48}$$

A covariant componentwise composition law is

$$(p \boxplus_\Delta q)_\mu := K_\Delta^{-1}\left[K_\Delta(p_\mu) + K_\Delta(q_\mu)\right]. \tag{9.49}$$

It satisfies

$$(p \boxplus_\Delta q) \boxplus_\Delta r = p \boxplus_\Delta (q \boxplus_\Delta r), \tag{9.50}$$
$$\mathcal{B}_\Lambda(p \boxplus_\Delta q) = \mathcal{B}_\Lambda(p) \boxplus_\Delta \mathcal{B}_\Lambda(q). \tag{9.51}$$

The invariant coordinate measure is

$$\mathrm{d}\mu(p) = \prod_{\mu=0}^{3} K_\Delta'(p_\mu)\ \mathrm{d}p_\mu, \tag{9.52}$$

with

$$K_\Delta'(x) = \frac{|\tan(\Delta x)|}{\sqrt{-2\log\cos(\Delta x)}}, \qquad K_\Delta'(0) = 1. \tag{9.53}$$

The pulled-back momentum-space metric is

$$\mathrm{d}s^2 = K_\Delta'(E)^2\,\mathrm{d}E^2 - \sum_i K_\Delta'(p_i)^2\,\mathrm{d}p_i^2. \tag{9.54}$$

Because this metric is the pullback of the flat auxiliary metric, the one-particle nonlinear action is coordinate-conjugate to ordinary special relativity. New physical content requires an operational choice of translation generators, measured momenta, interacting composition, branch transitions, or curvature coupling.

9.4 Palindromic Clifford factorization

Let

$$\Gamma_r^* = \Gamma_r, \qquad \Gamma_r^2 = I, \qquad \{\Gamma_r, \Gamma_s\} = 0 \quad (r \ne s). \tag{9.55}$$

For $0 \le r < n$, set

$$W_r := \mathrm{e}^{-\mathrm{i}\theta_r\Gamma_r/2} \tag{9.56}$$

and define

$$\mathcal{U}_n := W_0 W_1 \cdots W_{n-1} \mathrm{e}^{-\mathrm{i}\theta_n\Gamma_n} W_{n-1} \cdots W_1 W_0. \tag{9.57}$$

For $r \ne s$,

$$\mathrm{e}^{-\mathrm{i}\theta_r\Gamma_r/2}\Gamma_s \mathrm{e}^{-\mathrm{i}\theta_r\Gamma_r/2} = \Gamma_s, \tag{9.58}$$

while

$$\mathrm{e}^{-\mathrm{i}\theta_r\Gamma_r/2} I \mathrm{e}^{-\mathrm{i}\theta_r\Gamma_r/2} = \cos\theta_r\, I - \mathrm{i}\sin\theta_r\,\Gamma_r. \tag{9.59}$$

Induction therefore gives the exact factorization

$$\boxed{\mathcal{U}_n = C_n I - \mathrm{i}\sum_{r=0}^{n} D_r \Gamma_r,} \tag{9.60}$$

where

$$C_n := \prod_{r=0}^{n} \cos\theta_r, \tag{9.61}$$

$$D_r := \sin\theta_r \prod_{s=r+1}^{n} \cos\theta_s. \tag{9.62}$$

The normalization telescopes:

$$C_n^2 + \sum_{r=0}^{n} D_r^2 = 1. \tag{9.63}$$

Thus $\mathcal{U}_n$ is exactly unitary.
For the gauge-adapted $3+1$ nesting, take

$$\Gamma_0 := \alpha_x, \qquad \Gamma_1 := \alpha_y, \qquad \Gamma_2 := \alpha_z, \qquad \Gamma_3 := \beta, \tag{9.64}$$

$$\theta_0 := k_x\Delta, \qquad \theta_1 := k_y\Delta, \qquad \theta_2 := k_z\Delta, \qquad \theta_3 := m\Delta. \tag{9.65}$$

Then

$$\boxed{\begin{aligned} U_\Delta^{(xyz)} &= \mathrm{e}^{-\mathrm{i}k_x\Delta\alpha_x/2}\mathrm{e}^{-\mathrm{i}k_y\Delta\alpha_y/2}\mathrm{e}^{-\mathrm{i}k_z\Delta\alpha_z/2}\mathrm{e}^{-\mathrm{i}m\Delta\beta} \\ &\quad\times \mathrm{e}^{-\mathrm{i}k_z\Delta\alpha_z/2}\mathrm{e}^{-\mathrm{i}k_y\Delta\alpha_y/2}\mathrm{e}^{-\mathrm{i}k_x\Delta\alpha_x/2}. \end{aligned}} \tag{9.66}$$

It has the exact Clifford form

$$U_\Delta^{(xyz)} = CI - \mathrm{i}\left(D_x\alpha_x + D_y\alpha_y + D_z\alpha_z + D_m\beta\right), \tag{9.67}$$

where

$$C = \cos(m\Delta)\cos(k_x\Delta)\cos(k_y\Delta)\cos(k_z\Delta), \tag{9.68}$$

$$D_x = \sin(k_x\Delta)\cos(k_y\Delta)\cos(k_z\Delta)\cos(m\Delta), \tag{9.69}$$

$$D_y = \sin(k_y\Delta)\cos(k_z\Delta)\cos(m\Delta), \tag{9.70}$$

$$D_z = \sin(k_z\Delta)\cos(m\Delta), \tag{9.71}$$

$$D_m = \sin(m\Delta). \tag{9.72}$$

The alternative nesting used in the preceding chapter places the mass factor outside and one spatial factor at the center. Both nestings have the same scalar coefficient C and therefore the same quasienergy spectrum. Their Clifford vectors differ by an order-dependent spinor frame.
Set

$$R := \sqrt{1-C^2}, \qquad \widehat{\mathcal{D}} := \frac{D_x\alpha_x + D_y\alpha_y + D_z\alpha_z + D_m\beta}{R}. \tag{9.73}$$

Then

$$\widehat{\mathcal{D}}^{\,2} = I, \tag{9.74}$$

$$\Pi_\pm := \frac{1}{2}(I \pm \widehat{\mathcal{D}}), \tag{9.75}$$

$$U_\Delta \Pi_\pm = \mathrm{e}^{\mp\mathrm{i}\omega_\Delta\Delta}\Pi_\pm, \tag{9.76}$$

$$\mathrm{spec}(U_\Delta) = \left\{\mathrm{e}^{-\mathrm{i}\omega_\Delta\Delta}, \mathrm{e}^{-\mathrm{i}\omega_\Delta\Delta}, \mathrm{e}^{+\mathrm{i}\omega_\Delta\Delta}, \mathrm{e}^{+\mathrm{i}\omega_\Delta\Delta}\right\}, \tag{9.77}$$

$$\cos(\omega_\Delta\Delta) = C. \tag{9.78}$$

The exact effective Hamiltonian is

$$\boxed{H_{\text{eff}}(\boldsymbol{k}) = \frac{\omega_\Delta}{\sqrt{1-C^2}}\left(D_x\alpha_x + D_y\alpha_y + D_z\alpha_z + D_m\beta\right).} \tag{9.79}$$

The quadratic identity

$$U_\Delta^2 - 2CU_\Delta + I = 0 \tag{9.80}$$

implies the exact second-order recurrence

$$\Psi_{n+1} + \Psi_{n-1} = 2C\Psi_n. \tag{9.81}$$

As $\Delta \to 0$,

$$D_i = k_i\Delta + \mathcal{O}(\Delta^3), \tag{9.82}$$

$$D_m = m\Delta + \mathcal{O}(\Delta^3), \tag{9.83}$$

$$R = E\Delta + \mathcal{O}(\Delta^3), \tag{9.84}$$

$$\Pi_\pm \longrightarrow \frac{1}{2}\left[I \pm \frac{\boldsymbol{\alpha}\cdot\boldsymbol{k} + \beta m}{E}\right], \tag{9.85}$$

$$H_{\text{eff}} = \boldsymbol{\alpha}\cdot\boldsymbol{k} + \beta m + \mathcal{O}(\Delta^2). \tag{9.86}$$

For a general ordered list (a_r, Γ_r), the effective Hamiltonian has

$$H_{\text{eff}} = \sum_r a_r\Gamma_r + \Delta^2 Q_{\text{ord}} + \mathcal{O}(\Delta^4), \tag{9.87}$$

where

$$\boxed{Q_{\text{ord}} = \sum_r a_r\left[\frac{1}{6}\sum_{s<r} a_s^2 - \frac{1}{3}\sum_{s>r} a_s^2\right]\Gamma_r.} \tag{9.88}$$

Its energy projection is order independent:

$$\frac{1}{E}\sum_r a_r^2\left[\frac{1}{6}\sum_{s<r} a_s^2 - \frac{1}{3}\sum_{s>r} a_s^2\right] = -\frac{e_2}{6E}. \tag{9.89}$$

If mass is central and the six spatial orderings are averaged, then

$$\boxed{\overline{Q} = \frac{m|\boldsymbol{k}|^2}{6}\beta - \sum_i k_i\left[\frac{m^2}{3} + \frac{|\boldsymbol{k}|^2 - k_i^2}{12}\right]\alpha_i.} \tag{9.90}$$

Four components are minimal under the stated Clifford requirements because four pairwise anticommuting Hermitian involutions generate a complex Clifford algebra whose irreducible representation has dimension four.

9.5 Permutation symmetry versus strict locality

Set

$$q_r := \sin^2\theta_r, \tag{9.91}$$

$$C := \prod_r \cos\theta_r, \tag{9.92}$$

$$\rho_r := \int_0^1 \prod_{s\neq r}(1 - tq_s)\,\mathrm{d}t, \tag{9.93}$$

$$d_r := \sin\theta_r\sqrt{\rho_r}. \tag{9.94}$$

Equivalently,

$$\rho_r = 1 - \frac{1}{2}\sum_{s\neq r} q_s + \frac{1}{3}\sum_{\substack{s<t\\ s,t\neq r}} q_sq_t - \frac{1}{4}\prod_{s\neq r} q_s. \tag{9.95}$$

Since

$$-\frac{\mathrm{d}}{\mathrm{d}t}\prod_s(1-tq_s)=\sum_r q_r\prod_{s\neq r}(1-tq_s), \tag{9.96}$$

one obtains

$$\sum_r d_r^2 = 1-C^2. \tag{9.97}$$

Therefore

$$U_{\mathrm{sym}} = CI - \mathrm{i}\sum_r d_r\Gamma_r \tag{9.98}$$

is an exactly unitary, permutation-symmetric four-spinor completion with the same spectrum. The square roots $\sqrt{\rho_r}$ are not finite trigonometric polynomials, so the corresponding position-space operator is generally nonlocal.

A strictly local cubic completion is the direct sum

$$\mathbb{U}_{\mathrm{loc}} := \bigoplus_{\pi\in S_3} U_\pi. \tag{9.99}$$

Rotations permute the six ordering blocks. This restores exact finite-spacing cubic covariance at the cost of twenty-four internal components.

Construction	Exact shell	Four components	Finite range	Cubic covariance
One ordered palindrome	yes	yes	yes	spectral only
Symmetric square-root completion	yes	yes	no	yes
Six-ordering direct sum	yes	no	yes	yes
Wilson-dressed ordering	ultraviolet modified	yes	yes	spectral only

9.6 Exact velocity, kinetic mass, and anisotropy

Differentiation of the exact shell gives

$$\boxed{\frac{\partial\omega_\Delta}{\partial k_i} = \frac{\sin(k_i\Delta)\cos(m\Delta)\prod_{j\neq i}\cos(k_j\Delta)}{\sin(\omega_\Delta\Delta)}.} \tag{9.100}$$

Let

$$q_0 := \sin^2(m\Delta), \qquad q_i := \sin^2(k_i\Delta). \tag{9.101}$$

Then

$$\|\nabla_{\boldsymbol{k}}\omega_\Delta\|^2 = \frac{\displaystyle\sum_{i=1}^{3} q_i\prod_{r\neq i}(1-q_r)}{\displaystyle 1-\prod_{r=0}^{3}(1-q_r)}. \tag{9.102}$$

The denominator contains every nonempty selection of the q_r, while the numerator contains only selections with exactly one spatial q_i and no other component. Hence

$$\boxed{\|\nabla_{\boldsymbol{k}}\omega_\Delta\| \le 1.} \tag{9.103}$$

This is a group-velocity bound. It is not, without a specified microscopic metric, a proof of continuum microcausality.

At zero momentum,

$$\omega_\Delta(\mathbf{0}) = m \tag{9.104}$$

on the principal branch. For fixed $m\Delta$ and small $\boldsymbol{k}$,

$$\omega_\Delta(\boldsymbol{k}) = m + \frac{\Delta}{2}\cot(m\Delta)|\boldsymbol{k}|^2 + \mathcal{O}(|\boldsymbol{k}|^4). \tag{9.105}$$

Thus

$$\boxed{M_{\text{kin}} = \frac{\tan(m\Delta)}{\Delta} = m + \frac{m^3\Delta^2}{3} + \frac{2m^5\Delta^4}{15} + \mathcal{O}(\Delta^6).} \tag{9.106}$$

The standard positive curvature sector requires

$$0 < m\Delta < \frac{\pi}{2}. \tag{9.107}$$

Define the cubic harmonic

$$H_4(\boldsymbol{k}) := \sum_i k_i^4 - \frac{3}{5}|\boldsymbol{k}|^4. \tag{9.108}$$

The leading mass shell becomes

$$\boxed{\omega_\Delta^2 = m^2 + |\boldsymbol{k}|^2 - \frac{\Delta^2}{3}m^2|\boldsymbol{k}|^2 - \frac{\Delta^2}{15}|\boldsymbol{k}|^4 + \frac{\Delta^2}{6}H_4(\boldsymbol{k}) + \mathcal{O}(\Delta^4).} \tag{9.109}$$

For $m = 0$, an axial mode satisfies

$$\boldsymbol{k} = (k, 0, 0) \quad \Longrightarrow \quad \omega_\Delta = |k| \tag{9.110}$$

on the principal branch. A body-diagonal mode satisfies

$$\boldsymbol{k} = \frac{k}{\sqrt{3}}(1, 1, 1) \tag{9.111}$$

and

$$\omega_\Delta^2 = k^2 - \frac{k^4\Delta^2}{9} - \frac{k^6\Delta^4}{405} + \frac{k^8\Delta^6}{1701} + \mathcal{O}(\Delta^8). \tag{9.112}$$

9.7 Signed-route interpretation

Let ϵ_r be independent Rademacher signs and define

$$S := \sum_{r=0}^{3} \epsilon_r a_r. \tag{9.113}$$

Then

$$\boxed{\prod_{r=0}^{3} \cos(a_r\Delta) = \mathbb{E}_{\boldsymbol{\epsilon}}\left[\cos(\Delta S)\right].} \tag{9.114}$$

Moreover,

$$\mathbb{E}[S^2] = e_1, \tag{9.115}$$

$$\mathbb{E}[S^4] = e_1^2 + 4e_2, \tag{9.116}$$

$$\mathrm{Var}(S^2) = 4e_2. \tag{9.117}$$

Therefore

$$\boxed{\omega_\Delta^2 = E^2 - \frac{\Delta^2}{12}\,\mathrm{Var}_{\boldsymbol{\epsilon}}(S^2) + \mathcal{O}(\Delta^4).} \tag{9.118}$$

The leading defect is the variance of the squared signed-route phase. It vanishes when at most one component is active.

9.8 General finite-route theorem

Let V be a symmetric random displacement in d spatial dimensions with

$$\mathbb{E}[V] = 0, \qquad \mathbb{E}[V_i V_j] = a\delta_{ij}. \tag{9.119}$$

Define

$$S_h(\boldsymbol{k}) := \mathbb{E}\left[\cos(h\boldsymbol{k}\cdot V)\right], \tag{9.120}$$

$$\tau := h\sqrt{a}, \tag{9.121}$$

$$\cos(\tau\omega) := \cos(\tau m) S_h(\boldsymbol{k}). \tag{9.122}$$

Differentiation gives

$$\nabla_{\boldsymbol{k}}\omega = \frac{h\cos(\tau m)\mathbb{E}[V\sin(h\boldsymbol{k}\cdot V)]}{\tau\sqrt{1-\cos^2(\tau m)S_h(\boldsymbol{k})^2}}. \tag{9.123}$$

For every unit vector n,

$$|n\cdot\mathbb{E}[V\sin(h\boldsymbol{k}\cdot V)]|^2 \le \mathbb{E}[(n\cdot V)^2]\mathbb{E}[\sin^2(h\boldsymbol{k}\cdot V)] \tag{9.124}$$

$$\le a[1-S_h(\boldsymbol{k})^2]. \tag{9.125}$$

Hence

$$\boxed{\|\nabla_{\boldsymbol{k}}\omega\|^2 \le \frac{\cos^2(\tau m)[1-S_h^2]}{1-\cos^2(\tau m)S_h^2} \le 1.} \tag{9.126}$$

Define the directional fourth moment

$$\beta(\widehat{\boldsymbol{k}}) := \frac{\mathbb{E}[(\widehat{\boldsymbol{k}}\cdot V)^4]}{a^2}. \tag{9.127}$$

Then

$$S_h(\boldsymbol{k}) = 1 - \frac{\tau^2|\boldsymbol{k}|^2}{2} + \frac{\tau^4}{24}\beta(\widehat{\boldsymbol{k}})|\boldsymbol{k}|^4 + \mathcal{O}(\tau^6), \tag{9.128}$$

and

$$\boxed{\omega^2 = m^2 + |\boldsymbol{k}|^2 - \tau^2\left[\frac{1}{3}m^2|\boldsymbol{k}|^2 + \frac{\beta(\widehat{\boldsymbol{k}})-1}{12}|\boldsymbol{k}|^4\right] + \mathcal{O}(\tau^4).} \tag{9.129}$$

Jensen's inequality gives

$$\beta(\widehat{\boldsymbol{k}}) \ge 1. \tag{9.130}$$

Thus every positive route average has a nonpositive leading massless correction.
If the fourth moment is rotationally invariant, write

$$\mathbb{E}[V_iV_jV_kV_l] = b(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}). \tag{9.131}$$

Then

$$\beta = \frac{3b}{a^2}, \tag{9.132}$$

$$\mathbb{E}|V|^2 = da, \tag{9.133}$$

$$\mathbb{E}|V|^4 = d(d+2)b. \tag{9.134}$$

Since

$$\mathbb{E}|V|^4 \ge (\mathbb{E}|V|^2)^2, \tag{9.135}$$

one obtains

$$\boxed{\beta \ge \frac{3d}{d+2}.} \tag{9.136}$$

For $d = 3$,

$$\beta \ge \frac{9}{5}, \tag{9.137}$$

so every positive fourth-order-isotropic route model satisfies

$$\omega^2 \le |\boldsymbol{k}|^2 - \frac{\tau^2}{15}|\boldsymbol{k}|^4 + \mathcal{O}(\tau^4) \tag{9.138}$$

in the massless sector. Cancellation of the entire k^4 term requires interference, signed amplitudes, internal unitary structure, nonlocality, or additional temporal order.

9.9 Improved cubic scalar route

Let

$$c_i := \cos(hk_i). \tag{9.139}$$

Assign total probability 1/2 to the six axial routes $\pm e_i$ and total probability 1/2 to the twelve face-diagonal routes $\pm e_i \pm e_j$. The symbol is

$$\boxed{S_{\mathrm{cub}}(\boldsymbol{k}) = \frac{1}{6}\left[c_x + c_y + c_z + c_x c_y + c_x c_z + c_y c_z\right].} \tag{9.140}$$

Its covariance and fourth moment are

$$\mathbb{E}[V_i V_j] = \frac{1}{2}\delta_{ij}, \tag{9.141}$$

$$\mathbb{E}[(\boldsymbol{k}\cdot V)^4] = \frac{1}{2}|\boldsymbol{k}|^4. \tag{9.142}$$

Therefore

$$\tau = \frac{h}{\sqrt{2}}, \tag{9.143}$$

$$\beta = 2, \tag{9.144}$$

$$\omega^2 = m^2 + |\boldsymbol{k}|^2 - \tau^2\left[\frac{1}{3}m^2|\boldsymbol{k}|^2 + \frac{1}{12}|\boldsymbol{k}|^4\right] + \mathcal{O}(\tau^4). \tag{9.145}$$

The leading correction is rotationally invariant.

Because S_{cub} is a positive average of route cosines,

$$S_{\mathrm{cub}} = 1 \tag{9.146}$$

requires every axial route cosine to equal one, hence $c_x = c_y = c_z = 1$. In one reduced Brillouin zone this occurs only at $\boldsymbol{k} = 0$. The condition $S_{\mathrm{cub}} = -1$ would require both $c_i = -1$ and $c_i c_j = -1$, which is impossible. Therefore the scalar transfer symbol has one zero-quasienergy corner and no π-quasienergy corner.

9.10 Gauge-covariant four-spinor walk

Let the microcell spacing be

$$a := \frac{\Delta}{2}. \tag{9.147}$$

Let $L_i(x)$ be a unitary forward transporter from x to $x + a\widehat{i}$, with

$$L_i(x) \mapsto g(x + a\widehat{i})L_i(x)g(x)^{-1}. \tag{9.148}$$

Set

$$P_i^{\pm} := \frac{1}{2}(I \pm \alpha_i). \tag{9.149}$$

Define the conditional shift

$$\boxed{\begin{aligned}(\mathsf{S}_i\Psi)(x) = {} & P_i^{+}L_i(x - a\widehat{i})\Psi(x - a\widehat{i}) \\ & + P_i^{-}L_i(x)^{-1}\Psi(x + a\widehat{i}).\end{aligned}} \tag{9.150}$$

Orthogonality of $P_i^{\pm}$ and unitarity of the links give

$$\mathsf{S}_i^{*}\mathsf{S}_i = I, \tag{9.151}$$

$$\mathsf{S}_i[L^g] = \mathcal{G}\mathsf{S}_i[L]\mathcal{G}^{-1}. \tag{9.152}$$

For trivial links,

$$\mathsf{S}_i(\boldsymbol{k}) = \mathrm{e}^{-\mathrm{i}ak_i\alpha_i} = \mathrm{e}^{-\mathrm{i}k_i\Delta\alpha_i/2}. \tag{9.153}$$

With

$$\mathsf{M} := \mathrm{e}^{-\mathrm{i}m\Delta\beta}, \tag{9.154}$$

the seven-factor walk

$$\boxed{\mathsf{U}_\Delta = \mathsf{S}_x\mathsf{S}_y\mathsf{S}_z\mathsf{M}\mathsf{S}_z\mathsf{S}_y\mathsf{S}_x} \tag{9.155}$$

is exactly unitary, finite depth, finite range, and gauge covariant. With trivial links its eigenphase is exactly the product-cosine shell.

For smooth static links, write

$$\mathsf{S}_i = \exp[-\mathrm{i}a\alpha_i\Pi_i] + \mathcal{O}(a^2\partial L). \tag{9.156}$$

Palindromicity implies

$$\mathsf{U}_\Delta(-\Delta) = \mathsf{U}_\Delta(\Delta)^*, \tag{9.157}$$

so its logarithm contains only odd powers:

$$\mathsf{U}_\Delta = \exp\left[-\mathrm{i}\Delta\left(H_D + \Delta^2 Q + \mathcal{O}(\Delta^4)\right)\right], \tag{9.158}$$

with

$$H_D := \boldsymbol{\alpha}\cdot\boldsymbol{\Pi} + \beta m. \tag{9.159}$$

The exact algebraic square is

$$\boxed{H_D^2 = m^2 + \sum_i \Pi_i^2 + \sum_{i<j}\alpha_i\alpha_j[\Pi_i,\Pi_j].} \tag{9.160}$$

For

$$\Pi_i = -\mathrm{i}\partial_i - qA_i, \qquad [\Pi_i,\Pi_j] = \mathrm{i}q\epsilon_{ijk}B_k, \tag{9.161}$$

this reduces to

$$H_D^2 = m^2 + \boldsymbol{\Pi}^2 - q\boldsymbol{\Sigma}\cdot\boldsymbol{B}. \tag{9.162}$$

Thus gauge curvature necessarily modifies the scalar free shell through the spin-field-strength term.

9.11 Gauge-covariant scalar path completion and unitary dilation

For a spatial ordering $\pi \in S_3$ and signs $\boldsymbol{\sigma} \in \{\pm1\}^3$, let $P_{\pi,\boldsymbol{\sigma}}$ be the associated ordered gauge path. Define

$$\mathcal{A}^U_{\mathrm{prod}} := \frac{\cos(m\Delta)}{48}\sum_{\pi\in S_3}\sum_{\boldsymbol{\sigma}\in\{\pm1\}^3} P_{\pi,\boldsymbol{\sigma}}. \tag{9.163}$$

The path set is closed under inversion, hence

$$(\mathcal{A}^U_{\mathrm{prod}})^* = \mathcal{A}^U_{\mathrm{prod}}, \tag{9.164}$$

$$\|\mathcal{A}^U_{\mathrm{prod}}\| \le 1, \tag{9.165}$$

$$\mathcal{A}^{U,g}_{\mathrm{prod}} = \mathcal{G}\mathcal{A}^U_{\mathrm{prod}}\mathcal{G}^{-1}. \tag{9.166}$$

For trivial commuting links,

$$\mathcal{A}^U_{\mathrm{prod}}(\boldsymbol{k}) = \cos(m\Delta)\prod_i \cos(k_i\Delta). \tag{9.167}$$

More generally, write a Hermitian route average as

$$S^U = \sum_{\ell=1}^{L} w_\ell \frac{P_\ell + P_\ell^*}{2}, \qquad w_\ell > 0, \qquad \sum_\ell w_\ell = 1. \tag{9.168}$$

For each route define the Hermitian involution

$$J_\ell := \begin{pmatrix} 0 & P_\ell \\ P_\ell^* & 0 \end{pmatrix}. \tag{9.169}$$

Let

$$|\chi\rangle := \sum_\ell \sqrt{w_\ell}\,|\ell\rangle\,|+\rangle\,, \tag{9.170}$$

$$\mathcal{J} := \sum_{\ell} |\ell\rangle\langle\ell| \otimes J_{\ell}. \tag{9.171}$$

Then

$$\langle\chi|\,\mathcal{J}\,|\chi\rangle = S^{U}. \tag{9.172}$$

The mass factor is block-encoded by

$$M_m := \begin{pmatrix} \cos(m\tau) & \sin(m\tau) \\ \sin(m\tau) & -\cos(m\tau) \end{pmatrix}, \qquad M_m^2 = I. \tag{9.173}$$

The tensor product produces a Hermitian involution $\mathcal{U}$ with signal block

$$\Pi\mathcal{U}\Pi = \cos(m\tau)S^{U} =: A^{U}. \tag{9.174}$$

Define

$$R := 2\Pi - I, \tag{9.175}$$

$$\mathcal{W} := R\mathcal{U}. \tag{9.176}$$

If

$$A^{U}\phi = \lambda\phi, \tag{9.177}$$

then the corresponding invariant two-plane of $\mathcal{W}$ has eigenvalues

$$\mathrm{e}^{\pm \mathrm{i}\arccos\lambda}. \tag{9.178}$$

This supplies an exact finite-depth unitary dilation of every finite-route Hermitian contraction.

9.12 Projected recurrence and conservation laws

Let $A = A^{*}$ with $\|A\| \le 1$. The projected field satisfies

$$\psi_{n+1} + \psi_{n-1} = 2A\psi_n. \tag{9.179}$$

An exact discrete action is

$$\mathcal{S} = \sum_{n} \left[\langle\psi_n, A\psi_n\rangle - \mathrm{Re}\,\langle\psi_{n+1}, \psi_n\rangle\right]. \tag{9.180}$$

The conserved phase charge is

$$Q_n := 2\,\mathrm{Im}\,\langle\psi_n, \psi_{n+1}\rangle\,. \tag{9.181}$$

The conserved quadratic energy is

$$\mathcal{E}_n := \|\psi_{n+1}\|^2 + \|\psi_n\|^2 - 2\,\mathrm{Re}\,\langle\psi_{n+1}, A\psi_n\rangle\,. \tag{9.182}$$

It has the positive representation

$$\boxed{\mathcal{E}_n = \|\psi_{n+1} - A\psi_n\|^2 + \langle\psi_n, (I - A^2)\psi_n\rangle \ge 0.} \tag{9.183}$$

The closed solution is

$$\psi_n = T_n(A)\psi_0 + U_{n-1}(A)(\psi_1 - A\psi_0), \tag{9.184}$$

where T_n and U_n are Chebyshev polynomials.
In first-order form,

$$\begin{pmatrix} \psi_{n+1} \\ \psi_n \end{pmatrix} = L_A \begin{pmatrix} \psi_n \\ \psi_{n-1} \end{pmatrix}, \qquad L_A := \begin{pmatrix} 2A & -I \\ I & 0 \end{pmatrix}. \tag{9.185}$$

With

$$M_A := \begin{pmatrix} I & -A \\ -A & I \end{pmatrix}, \tag{9.186}$$

one has

$$L_A^{*} M_A L_A = M_A. \tag{9.187}$$

For $\|A\| < 1$, M_A is positive definite.

9.13 Polyphase sectors and doubling obstruction

In the reduced Brillouin zone, write

$$k_i = \frac{\pi}{\Delta}\epsilon_i + p_i, \qquad \epsilon_i \in \{0,1\}. \tag{9.188}$$

Then

$$\cos(k_i\Delta) = (-1)^{\epsilon_i}\cos(p_i\Delta). \tag{9.189}$$

For $m = 0$ and the eight corners $p_i = 0$,

$$C = (-1)^{\epsilon_x+\epsilon_y+\epsilon_z}. \tag{9.190}$$

In the exact Clifford completion all vector coefficients vanish there, so

$$U_\Delta = (-1)^{\epsilon_x+\epsilon_y+\epsilon_z} I. \tag{9.191}$$

Therefore

$$\boxed{N_{\omega=0} = 4, \qquad N_{\omega=\pi/\Delta} = 4.} \tag{9.192}$$

This conclusion does not depend on the chosen Clifford vector. Any unitary satisfying

$$U + U^* = 2CI \tag{9.193}$$

must equal $+I$ when $C = 1$ and $-I$ when $C = -1$. Hence the exact product-cosine scalar law itself forces the eight corner nodes. A single-species ultraviolet theory must restrict the physical sector, project polyphase lanes, or deform the ultraviolet shell.

9.14 Local Wilson commutator regulator

Let

$$A := \alpha_i, \qquad B := \beta, \qquad \{A, B\} = 0. \tag{9.194}$$

Define

$$Q_i(\theta) := \mathrm{e}^{-\mathrm{i}\theta A/2}, \tag{9.195}$$

$$V_i(\theta;\rho) := \mathrm{e}^{+\mathrm{i}\rho B} Q_i(\theta) \mathrm{e}^{-\mathrm{i}\rho B} Q_i(\theta)^*, \tag{9.196}$$

$$K_i(\theta;\rho) := V_i(\theta;\rho) V_i(-\theta;\rho). \tag{9.197}$$

This is exactly unitary and finite depth. Its low-momentum expansion is

$$\boxed{K_i(\theta;\rho) = I + \frac{\mathrm{i}}{2}\sin(2\rho)\theta^2\beta + \mathcal{O}(\theta^3).} \tag{9.198}$$

Set

$$r := \frac{1}{2}\sin(2\rho). \tag{9.199}$$

For

$$U_{\Delta,W} := (K_x K_y K_z)^* U_\Delta, \tag{9.200}$$

the continuum effective Hamiltonian has the Wilson structure

$$H_{\mathrm{eff},W} = \boldsymbol{\alpha}\cdot\boldsymbol{k} + \beta\left(m + r\Delta|\boldsymbol{k}|^2\right) + \mathcal{O}(\Delta^2). \tag{9.201}$$

At $\theta = \pi$,

$$\boxed{K_i(\pi;\rho) = \mathrm{e}^{4\mathrm{i}\rho\beta}.} \tag{9.202}$$

At a massless corner with n active π/Δ components,

$$U_{\Delta,W} = (-1)^n \mathrm{e}^{-4\mathrm{i}n\rho\beta}. \tag{9.203}$$

For

$$\rho = \frac{\pi}{16}, \tag{9.204}$$

the nonorigin corner phases are

n	$\lvert\omega\Delta\rvert$
1	$3\pi/4$
2	$\pi/2$
3	$\pi/4$

(9.205)

so all seven nonorigin zero or π corner nodes are exactly gapped.

A deterministic 21^3 momentum scan found a minimum sampled nonorigin distance

$$\delta_{\mathrm{grid}} = 2.582434 \times 10^{-1} \tag{9.206}$$

from both zero and π quasienergy. This is numerical evidence only. A global proof excluding accidental generic-momentum closings remains open.

9.15 Fourth-order isotropic contraction refinement

For gauge-covariant shifts T_i, define

$$C_i := \frac{T_i + T_i^*}{2}, \tag{9.207}$$

$$S^U_{\mathrm{cub}} := \frac{1}{6}\sum_i C_i + \frac{1}{12}\sum_{i<j}\{C_i, C_j\}. \tag{9.208}$$

This is Hermitian, gauge covariant, cubic symmetric, finite range, and contractive. If

$$T_i = \mathrm{e}^{hD_i}, \qquad h = \sqrt{2}\tau, \tag{9.209}$$

then

$$S^U_{\mathrm{cub}} = I + \frac{\tau^2}{2}D^2 + \frac{\tau^4}{12}(D^2)^2 + \mathcal{O}(\tau^6), \qquad D^2 := \sum_i D_i^2. \tag{9.210}$$

Define

$$W_h := 3I - (C_x + C_y + C_z), \tag{9.211}$$

$$G_m := I - \frac{m^2\tau^2}{12}W_h, \tag{9.212}$$

$$G_4 := I - \frac{1}{48}W_h^2. \tag{9.213}$$

Since

$$0 \le W_h \le 6I, \tag{9.214}$$

one has, for $|m\tau| < \pi/2$,

$$\|G_m\| \le 1, \tag{9.215}$$

$$\|G_4\| \le 1. \tag{9.216}$$

Set

$$\boxed{A_4^U := \cos(m\tau)G_4G_mS^U_{\mathrm{cub}}G_mG_4.} \tag{9.217}$$

Then A_4^U is Hermitian, gauge covariant, cubic symmetric, finite range, and contractive. With

$$K := m^2 - D^2, \tag{9.218}$$

its continuum expansion is

$$\boxed{A_4^U = I - \frac{\tau^2}{2}K + \frac{\tau^4}{24}K^2 + \mathcal{O}(\tau^6).} \tag{9.219}$$

Thus

$$A_4^U = \cos(\tau\sqrt{K}) + \mathcal{O}(\tau^6) \tag{9.220}$$

and

$$\Omega_4^2 = m^2 - D^2 + \mathcal{O}(\tau^4). \tag{9.221}$$

The complete $\mathcal{O}(\tau^2)$ mass-shell error is canceled without losing gauge covariance, finite range, cubic symmetry, or contraction stability.

9.16 Sixth-order residual and positive local cancellation

For the free symbol,

$$A_4 = \cos(\tau E) + \frac{\tau^6}{360}\mathcal{P}_6(m,\boldsymbol{k}) + \mathcal{O}(\tau^8), \tag{9.222}$$

where

$$\boxed{\begin{aligned}\mathcal{P}_6 = 11\sum_i k_i^6 + 19\sum_{i\neq j} k_i^4 k_j^2 + 48 k_x^2 k_y^2 k_z^2 \\ + 34m^2\sum_i k_i^4 + 48m^2\sum_{i<j} k_i^2 k_j^2 + 24m^4|\boldsymbol{k}|^2.\end{aligned}} \tag{9.223}$$

Consequently,

$$\omega_4^2 = m^2 + |\boldsymbol{k}|^2 - \frac{\tau^4}{180}\mathcal{P}_6 + \mathcal{O}(\tau^6). \tag{9.224}$$

Define

$$w_i := I - C_i, \tag{9.225}$$

$$Y_i := \frac{T_i - T_i^*}{2\mathrm{i}}, \tag{9.226}$$

$$\mu := m\tau. \tag{9.227}$$

The following operators are positive semidefinite:

$$Q_1 := 11\sum_i w_i^3, \tag{9.228}$$

$$Q_2 := 19\sum_{i\neq j} w_i w_j w_i, \tag{9.229}$$

$$Q_3 := \sum_{\pi\in S_3} (Y_{\pi_1}Y_{\pi_2}Y_{\pi_3})^*(Y_{\pi_1}Y_{\pi_2}Y_{\pi_3}), \tag{9.230}$$

$$Q_4 := 34\mu^2\sum_i w_i^2, \tag{9.231}$$

$$Q_5 := 6\mu^2\sum_{i<j}\left[(Y_iY_j)^*(Y_iY_j) + (Y_jY_i)^*(Y_jY_i)\right], \tag{9.232}$$

$$Q_6 := 24\mu^4\sum_i w_i. \tag{9.233}$$

Their free leading symbol is

$$\sum_{j=1}^{6} Q_j = \tau^6\mathcal{P}_6(m,\boldsymbol{k}) + \mathcal{O}(\tau^8). \tag{9.234}$$

Define

$$H_j := I - \frac{Q_j}{720}, \tag{9.235}$$

$$H := H_1H_2H_3H_4H_5H_6, \tag{9.236}$$

$$A_6^U := H^* A_4^U H. \tag{9.237}$$

Using

$$\|Q_1\| \le 264, \tag{9.238}$$

$$\|Q_2\| \le 912, \tag{9.239}$$

$$\|Q_3\| \le 6, \tag{9.240}$$

$$\|Q_4\| \le 102\pi^2, \tag{9.241}$$

$$\|Q_5\| \le 9\pi^2, \tag{9.242}$$

$$\|Q_6\| \le 9\pi^4, \tag{9.243}$$

one obtains

$$\|H_j\| \le 1 \tag{9.244}$$

in the principal mass sector. Therefore A_6^U remains Hermitian, gauge covariant, cubic symmetric, finite range, and contractive. In the free sector,

$$\boxed{A_6 = \cos\Big(\tau\sqrt{m^2 + |\boldsymbol{k}|^2}\Big) + \mathcal{O}(\tau^8),} \tag{9.245}$$

so

$$\omega_6^2 = m^2 + |\boldsymbol{k}|^2 + \mathcal{O}(\tau^6). \tag{9.246}$$

For nonzero gauge curvature, additional commutator and plaquette terms may survive at order τ^6. Their full cancellation requires an explicit field-strength completion.

9.17 Closure ledger

The exact and conditional conclusions are separated as follows.

Item	Status	Qualification
Negative Δ^2 quasienergy coefficient	proved	coefficient matching and independent numerical refinement
All-orders shell through Δ^6	proved	exact formal series through the displayed order
Palindromic four-spinor factorization	proved	finite-dimensional Clifford identity
Exact free projectors and effective Hamiltonian	proved	principal quasienergy branch
Principal-branch shell and speed bounds	proved	requires nonzero principal cosines
Nonlinear K_Δ-Lorentz representation	proved kinematically	coordinate-conjugate to ordinary one-particle Lorentz action
Gauge-covariant seven-factor walk	proved	external unitary links; dynamical link update is separate
Dirac-Pauli curvature term	proved in continuum expansion	sign uses the stated $\Pi_i = -\mathrm{i}\partial_i - qA_i$ convention
Eight product-shell corner nodes	proved	forced by the exact scalar shell
Wilson gapping of seven corner nodes	proved at corners	global absence of accidental nodes remains open
No additional Wilson closing on 21^3 grid	numerical only	minimum sampled spectral distance recorded above
Fourth-order contraction refinement	proved	exact operator properties and free principal-symbol cancellation
Sixth-order free refinement	proved at free-symbol level	gauge-curvature completion remains open
Minimal local four-spinor with exact cubic covariance	open	current exact options trade locality, dimension, or covariance
Empirical value or interpretation of Δ	open	requires independent physical measurement
Literature priority or novelty	not assessed	no priority claim is made in this audit

The executable audit records for this chapter are

$$\mathcal{Q}_{\text{checks}} = \texttt{data/exact_quasienergy_closure_checks.csv}, \tag{9.247}$$

$$\mathcal{Q}_{\text{summary}} = \texttt{data/exact_quasienergy_closure_summary.csv}, \tag{9.248}$$

$$\mathcal{Q}_{\text{script}} = \texttt{code/53_exact_quasienergy_closure.py}. \tag{9.249}$$

Chapter 10

Abelian Gauge Sectors

Gauge covariance is imposed directly on lattice links. The resulting walks, plaquettes, and semidiscrete field equations provide the Abelian gauge structure used in later geometric models.

10.1 Time-dependent linked walk

$$\psi_j^n \mapsto g_j^n \psi_j^n, \tag{10.1}$$

$$L_j^n \mapsto g_{j+1}^n L_j^n (g_j^n)^{-1}, \tag{10.2}$$

$$T_j^n \mapsto g_j^{n+1} T_j^n (g_j^n)^{-1}, \tag{10.3}$$

$$U_n[L', T'] G^n = G^{n+1} U_n[L, T]. \tag{10.4}$$

For example, the linked forward field transforms as

$$L_j^{n\,\prime} \psi_j^{n\,\prime} = g_{j+1}^n L_j^n (g_j^n)^{-1} g_j^n \psi_j^n \tag{10.5}$$

$$= g_{j+1}^n L_j^n \psi_j^n. \tag{10.6}$$

The gauge factor is therefore transported to the endpoint of the link. The temporal link gives the same cancellation between time slices, producing the operator covariance relation. For packet density and link current,

$$\rho_j^n := \left\| \psi_j^n \right\|^2, \tag{10.7}$$

$$\rho_j^{n+1} - \rho_j^n = J_{j-1/2}^n - J_{j+1/2}^n, \tag{10.8}$$

$$Q_A^n := \sum_{j \in A} \rho_j^n, \tag{10.9}$$

$$Q_A^{n+1} - Q_A^n = J_{\partial_- A}^n - J_{\partial_+ A}^n. \tag{10.10}$$

10.2 Plaquette curvature and Wilson action

$$W_{01}(j,n) := L_j^n T_{j+1}^n (L_j^{n+1})^{-1} (T_j^n)^{-1}, \tag{10.11}$$

$$W'_{01}(j,n) = W_{01}(j,n), \tag{10.12}$$

$$W_{01}(j,n) = \mathrm{e}^{-\mathrm{i} q h \Delta t F_{01}(j,n)}. \tag{10.13}$$

Substituting the four transformed links, every gauge factor at an intermediate corner cancels with its inverse. The remaining factors at the basepoint also cancel in the Abelian case:

$$W'_{01} = g_j^n W_{01} (g_j^n)^{-1} = W_{01}. \tag{10.14}$$

Thus the plaquette phase is a gauge-invariant discrete curvature. In $d = 4$,

$$S_{\mathrm{EM}}[U] := \frac{1}{g^2} \sum_{x, \mu < \nu} \left[1 - \mathrm{Re}\, W_{\mu\nu}(x) \right], \tag{10.15}$$

$$W_{\mu\nu} = \mathrm{e}^{\mathrm{i}\theta_{\mu\nu}}, \tag{10.16}$$

$$1 - \operatorname{Re} W_{\mu\nu} = \frac{1}{2}\theta_{\mu\nu}^2 + \mathcal{O}(\theta_{\mu\nu}^4). \tag{10.17}$$

The last line is simply

$$1 - \operatorname{Re} \mathrm{e}^{\mathrm{i}\theta} = 1 - \cos\theta \tag{10.18}$$

$$= \frac{\theta^2}{2} - \frac{\theta^4}{24} + \mathcal{O}(\theta^6). \tag{10.19}$$

Centered smooth plaquette:

$$\theta_{\mu\nu} = -q a_\mu a_\nu F_{\mu\nu} + \mathcal{O}(a^4), \tag{10.20}$$

$$1 - \operatorname{Re} W_{\mu\nu} = \frac{1}{2} q^2 a_\mu^2 a_\nu^2 F_{\mu\nu}^2 + \mathcal{O}(a^6). \tag{10.21}$$

Corner nonsymmetric plaquette:

$$1 - \operatorname{Re} W_{\mu\nu} = \frac{1}{2} q^2 a_\mu^2 a_\nu^2 F_{\mu\nu}^2 + \mathcal{O}(a^5). \tag{10.22}$$

Abelian cube identity:

$$\prod_{f \subset \partial c} W_f^{\mathrm{or}(f,c)} = 1. \tag{10.23}$$

10.3 Semidiscrete Maxwell–Dirac closure

$$U_j := \mathrm{e}^{-\mathrm{i}hA_j}, \tag{10.24}$$

$$H_h[\psi, A, E] := hm \sum_j \psi_j^* \beta \psi_j + \sum_j \operatorname{Im}(\psi_j^* \alpha U_j \psi_{j+1}) + \frac{h}{2} \sum_j E_j^2, \tag{10.25}$$

$$S_h := \int \left[\frac{\mathrm{i}h}{2} \sum_j (\psi_j^* \dot{\psi}_j - \dot{\psi}_j^* \psi_j) + h \sum_j E_j \dot{A}_j - H_h \right] \mathrm{d}t. \tag{10.26}$$

Euler–Lagrange/Hamilton equations:

$$\dot{\psi}_j = -\mathrm{i}m\beta\psi_j - \frac{1}{2h}\alpha U_j \psi_{j+1} + \frac{1}{2h} U_{j-1}^* \alpha \psi_{j-1}, \tag{10.27}$$

$$\dot{A}_j = E_j, \tag{10.28}$$

$$\dot{E}_j = J_j, \tag{10.29}$$

$$J_j := \operatorname{Re}(\psi_j^* \alpha U_j \psi_{j+1}). \tag{10.30}$$

Gauge symmetry:

$$\psi_j \mapsto \mathrm{e}^{\mathrm{i}\chi_j} \psi_j, \tag{10.31}$$

$$A_j \mapsto A_j + \frac{\chi_{j+1} - \chi_j}{h}, \tag{10.32}$$

$$E_j \mapsto E_j. \tag{10.33}$$

Exact identities:

$$\rho_j := \psi_j^* \psi_j, \tag{10.34}$$

$$\dot{\rho}_j + \frac{J_j - J_{j-1}}{h} = 0, \tag{10.35}$$

$$\bar{\rho} := \frac{1}{N} \sum_j \rho_j, \tag{10.36}$$

$$G_j := \frac{E_j - E_{j-1}}{h} + \rho_j - \bar{\rho}, \tag{10.37}$$

$$\dot{G}_j = 0, \tag{10.38}$$

$$H_E := \frac{h}{2}\sum_j E_j^2, \tag{10.39}$$

$$\dot{H}_E = h\sum_j E_j J_j, \tag{10.40}$$

$$\dot{H}_{\mathrm{m}} = -h\sum_j E_j J_j, \tag{10.41}$$

$$\dot{H}_h = 0. \tag{10.42}$$

To obtain continuity,

$$\dot{\rho}_j = \dot{\psi}_j^*\psi_j + \psi_j^*\dot{\psi}_j. \tag{10.43}$$

The onsite mass terms cancel because $\beta = \beta^*$. Pairing the two nearest-neighbor terms with their conjugates leaves

$$\dot{\rho}_j = -\frac{J_j - J_{j-1}}{h}. \tag{10.44}$$

Summing over j shows $\dot{\bar{\rho}} = 0$. Hence

$$\dot{G}_j = \frac{J_j - J_{j-1}}{h} + \dot{\rho}_j - \dot{\bar{\rho}} = 0. \tag{10.45}$$

The electric energy obeys

$$\dot{H}_E = h\sum_j E_j\dot{E}_j = h\sum_j E_j J_j. \tag{10.46}$$

Hamiltonian conservation then assigns the opposite work term to the matter sector. Continuum difference:

$$D_h^A\psi_j := \frac{U_j\psi_{j+1} - U_{j-1}^*\psi_{j-1}}{2h}, \tag{10.47}$$

$$D_h^A\psi = (\partial_x - \mathrm{i}A)\psi + \mathcal{O}(h^2). \tag{10.48}$$

Chapter 11

Non-Abelian Gauge and Ordered Holonomy Refinement

The Abelian construction extends to ordered noncommuting links. This chapter states the local $SU(2)$ transformations, plaquette observables, and exact refinement rule for ordered holonomy.

11.1 Local $SU(2)$ gauge atlas

$$\psi_x \in \mathbb{C}^2, \qquad U_{x,\mu} \in SU(2), \qquad G_x \in SU(2), \tag{11.1}$$

$$\psi'_x = G_x \psi_x, \tag{11.2}$$

$$U'_{x,\mu} = G_x U_{x,\mu} G^{-1}_{x+\widehat{\mu}}. \tag{11.3}$$

For the backward transporter $U_{x,\mu} : \mathcal{H}_{x+\widehat{\mu}} \to \mathcal{H}_x$,

$$D^+_\mu \psi_x := \frac{U_{x,\mu}\psi_{x+\widehat{\mu}} - \psi_x}{h}, \tag{11.4}$$

$$(D^+_\mu \psi_x)' = G_x D^+_\mu \psi_x, \tag{11.5}$$

$$\Delta_U \psi_x := \frac{1}{h^2} \sum_\mu \left(U_{x,\mu}\psi_{x+\widehat{\mu}} + U^{-1}_{x-\widehat{\mu},\mu}\psi_{x-\widehat{\mu}} - 2\psi_x \right), \tag{11.6}$$

$$(\Delta_U \psi_x)' = G_x \Delta_U \psi_x. \tag{11.7}$$

For the first identity,

$$(D^+_\mu \psi_x)' = \frac{G_x U_{x,\mu} G^{-1}_{x+\widehat{\mu}} G_{x+\widehat{\mu}} \psi_{x+\widehat{\mu}} - G_x \psi_x}{h} \tag{11.8}$$

$$= G_x D^+_\mu \psi_x. \tag{11.9}$$

The same cancellation occurs in each forward and backward term of the covariant Laplacian.

11.2 Plaquette, Wilson action, and currents

$$W_{x,\mu\nu} := U_{x,\mu} U_{x+\widehat{\mu},\nu} U^{-1}_{x+\widehat{\nu},\mu} U^{-1}_{x,\nu}, \tag{11.10}$$

$$W'_{x,\mu\nu} = G_x W_{x,\mu\nu} G_x^{-1}, \tag{11.11}$$

$$S_W[U] := \frac{1}{g^2} \sum_{x,\mu<\nu} \left(1 - \frac{1}{2} \operatorname{Re} \operatorname{Tr} W_{x,\mu\nu} \right), \tag{11.12}$$

$$S_W[U'] = S_W[U]. \tag{11.13}$$

Along the four sides, adjacent endpoint factors cancel in order:

$$W'_{x,\mu\nu} = G_x U_{x,\mu} U_{x+\widehat{\mu},\nu} U^{-1}_{x+\widehat{\nu},\mu} U^{-1}_{x,\nu} G_x^{-1}. \tag{11.14}$$

Only conjugation at the chosen basepoint remains. Cyclicity of the trace then makes the Wilson action invariant. Gauge-invariant link current:

$$j_{x,\mu} := 2\,\mathrm{Im}\left(\psi^*_x U_{x,\mu} \psi_{x+\widehat{\mu}}\right), \tag{11.15}$$

$$j'_{x,\mu} = j_{x,\mu}. \tag{11.16}$$

The scalar inside the imaginary part is unchanged:

$$(\psi'_x)^* U'_{x,\mu} \psi'_{x+\widehat{\mu}} = \psi^*_x G^*_x G_x U_{x,\mu} G^{-1}_{x+\widehat{\mu}} G_{x+\widehat{\mu}} \psi_{x+\widehat{\mu}} \tag{11.17}$$

$$= \psi^*_x U_{x,\mu} \psi_{x+\widehat{\mu}}. \tag{11.18}$$

Color charge and adjoint action:

$$q^a_x := \psi^*_x \sigma^a \psi_x, \tag{11.19}$$

$$G^*_x \sigma^a G_x = R_x{}^a{}_b \sigma^b, \tag{11.20}$$

$$q'^a_x = R_x{}^a{}_b q^b_x. \tag{11.21}$$

11.3 Exact ordered 2×2 refinement

Let

$$W_{ij} := X_{ij} Y_{i+1,j} X^{-1}_{i,j+1} Y^{-1}_{ij}, \qquad i,j \in \{0,1\}. \tag{11.22}$$

Define the two row holonomies

$$R_0 := \left(X_{00} W_{10} X^{-1}_{00}\right) W_{00}, \tag{11.23}$$

$$R_1 := \left(X_{01} W_{11} X^{-1}_{01}\right) W_{01}. \tag{11.24}$$

The coarse boundary holonomy is exactly

$$W^{(2)} := X_{00} X_{10} Y_{20} Y_{21} X^{-1}_{12} X^{-1}_{02} Y^{-1}_{01} Y^{-1}_{00} \tag{11.25}$$

$$= R_0 Y_{00} R_1 Y^{-1}_{00}. \tag{11.26}$$

Expanding R_0 and R_1 into the four fine plaquettes introduces every interior link once with each orientation. The inverse pairs cancel in their fixed order. The uncancelled factors are precisely the eight links on the coarse boundary, giving the first line. The conjugations by X_{00}, X_{01}, and Y_{00} transport all fine plaquettes to the same basepoint before multiplication. Thus non-Abelian refinement requires ordered multiplication and parallel transport to a common basepoint.

11.4 Curvature refinement

For $A, B \in \mathfrak{su}(2)$ and sufficiently small h on a common analytic logarithm branch,

$$W_h := \mathrm{e}^{hA} \mathrm{e}^{hB} \mathrm{e}^{-hA} \mathrm{e}^{-hB}, \tag{11.27}$$

$$F_h := \frac{1}{h^2} \log W_h, \tag{11.28}$$

$$F_h = [A,B] + \mathcal{O}(h), \tag{11.29}$$

$$F^{[1]}_h := 2F_{h/2} - F_h = [A,B] + \mathcal{O}(h^2). \tag{11.30}$$

On a common logarithm branch, the Baker–Campbell–Hausdorff series has the form

$$\log W_h = h^2 [A,B] + h^3 C_3(A,B) + \mathcal{O}(h^4). \tag{11.31}$$

Dividing by h^2 gives

$$F_h = [A, B] + hC_3 + \mathcal{O}(h^2). \tag{11.32}$$

At half spacing,

$$F_{h/2} = [A, B] + \frac{h}{2}C_3 + \mathcal{O}(h^2), \tag{11.33}$$

so $2F_{h/2} - F_h$ cancels the entire first-order term. Gauge covariance:

$$F_h[GAG^{-1}, GBG^{-1}] = GF_h[A, B]G^{-1}. \tag{11.34}$$

11.5 Numerical residuals

$$\|(\Delta_{U'}G\psi) - G\Delta_U\psi\|_\infty = 6.394885 \times 10^{-14}, \tag{11.35}$$

$$\left\|W' - GWG^{-1}\right\|_\infty = 5.661049 \times 10^{-16}, \tag{11.36}$$

$$\left\|W^{(2)} - R_0 Y_{00} R_1 Y_{00}^{-1}\right\|_2 = 6.214507 \times 10^{-16}, \tag{11.37}$$

$$p_{F_h} = 0.993501, \tag{11.38}$$

$$p_{F_h^{[1]}} = 1.992340. \tag{11.39}$$

Chapter 12

Conditional Finite-Scale Propagation Signatures

The discrete walk has a small finite-spacing anisotropy. This chapter isolates that conditional signature and derives the associated energy- and direction-dependent propagation delay.

12.1 Exact massless shell of the cubic walk

Let

$$\boldsymbol{k} = k\boldsymbol{n}, \qquad \|\boldsymbol{n}\| = 1, \tag{12.1}$$

$$\cos(\omega_\Delta \Delta) = \prod_{i=1}^{3} \cos(k n_i \Delta). \tag{12.2}$$

Define

$$\mathcal{A}(\boldsymbol{n}) := \sum_{i<j} n_i^2 n_j^2 \tag{12.3}$$

$$= \frac{1}{2}\left(1 - \sum_{i=1}^{3} n_i^4\right), \tag{12.4}$$

$$0 \le \mathcal{A}(\boldsymbol{n}) \le \frac{1}{3}. \tag{12.5}$$

The identity follows by squaring $\sum_i n_i^2 = 1$:

$$1 = \left(\sum_i n_i^2\right)^2 \tag{12.6}$$

$$= \sum_i n_i^4 + 2\sum_{i<j} n_i^2 n_j^2. \tag{12.7}$$

Nonnegativity is immediate. The upper bound follows from $\sum_i n_i^4 \ge (\sum_i n_i^2)^2/3 = 1/3$. Extrema:

$$\mathcal{A}(1,0,0) = 0, \tag{12.8}$$

$$\mathcal{A}\Big(2^{-1/2}, 2^{-1/2}, 0\Big) = \frac{1}{4}, \tag{12.9}$$

$$\mathcal{A}\Big(3^{-1/2}, 3^{-1/2}, 3^{-1/2}\Big) = \frac{1}{3}. \tag{12.10}$$

12.2 Cubic-harmonic angular signature

Let

$$H_4(\boldsymbol{x}) := \sum_{i=1}^{3} x_i^4 - \frac{3}{5}\left(\sum_{i=1}^{3} x_i^2\right)^2. \tag{12.11}$$

Since

$$\Delta \sum_i x_i^4 = 12 \sum_i x_i^2, \tag{12.12}$$

$$\Delta \left(\sum_i x_i^2\right)^2 = 20 \sum_i x_i^2, \tag{12.13}$$

the coefficient $3/5$ makes the two Laplacians cancel. On the unit sphere,

$$H_4(\boldsymbol{n}) = \sum_i n_i^4 - \frac{3}{5}, \tag{12.14}$$

$$\frac{1}{5} - \frac{1}{2} H_4(\boldsymbol{n}) = \frac{1}{2}\left(1 - \sum_i n_i^4\right) = \mathcal{A}(\boldsymbol{n}). \tag{12.15}$$

Then

$$\Delta_{\mathbb{R}^3} H_4 = 0, \tag{12.16}$$

$$\mathcal{A}(\boldsymbol{n}) = \frac{1}{5} - \frac{1}{2} H_4(\boldsymbol{n}), \qquad \|\boldsymbol{n}\| = 1. \tag{12.17}$$

For normalized spherical average $\langle \cdot \rangle_{S^2}$,

$$\langle \mathcal{A} \rangle_{S^2} = \frac{1}{5}, \tag{12.18}$$

$$\left\langle \mathcal{A}^2 \right\rangle_{S^2} = \frac{1}{21}, \tag{12.19}$$

$$\mathrm{Var}_{S^2}(\mathcal{A}) = \frac{4}{525}. \tag{12.20}$$

Hence the leading signal is an isotropic mean plus a pure degree-four cubic angular component.

12.3 Frequency and velocity correction

$$\omega_\Delta(k, \boldsymbol{n}) = k - \frac{\Delta^2 k^3}{6} \mathcal{A}(\boldsymbol{n}) + \mathcal{O}(\Delta^4), \tag{12.21}$$

$$v_g(k, \boldsymbol{n}) := \frac{\partial \omega_\Delta}{\partial k} \tag{12.22}$$

$$= 1 - \frac{\Delta^2 k^2}{2} \mathcal{A}(\boldsymbol{n}) + \mathcal{O}(\Delta^4). \tag{12.23}$$

Expanding the exact shell gives

$$\prod_i \cos(k n_i \Delta) = 1 - \frac{k^2 \Delta^2}{2} + k^4 \Delta^4 \left(\frac{1}{24} + \frac{\mathcal{A}(\boldsymbol{n})}{6}\right) + \mathcal{O}(\Delta^6). \tag{12.24}$$

If $\omega_\Delta = k + a\Delta^2 + \mathcal{O}(\Delta^4)$, then

$$\cos(\omega_\Delta \Delta) = 1 - \frac{k^2 \Delta^2}{2} + \left(\frac{k^4}{24} - ka\right) \Delta^4 + \mathcal{O}(\Delta^6). \tag{12.25}$$

Matching coefficients yields $a = -k^3 \mathcal{A}/6$; differentiating in k gives the velocity correction. Exact radial group velocity:

$$v_g = \frac{\displaystyle\sum_i n_i \sin(k n_i \Delta) \prod_{j \neq i} \cos(k n_j \Delta)}{\sin(\omega_\Delta \Delta)}. \tag{12.26}$$

12.4 Directional photon-delay law

For propagation distance L,

$$\boxed{\delta t_{h\ell} = \frac{L\Delta^2}{2}\mathcal{A}(\boldsymbol{n})\left(k_h^2 - k_\ell^2\right) + \mathcal{O}(\Delta^4).} \tag{12.27}$$

For one mode,

$$t(k) = \frac{L}{v_g(k)} = L\left[1 + \frac{\Delta^2 k^2}{2}\mathcal{A}(\boldsymbol{n})\right] + \mathcal{O}(\Delta^4), \tag{12.28}$$

where $1/(1-\eta) = 1 + \eta + \mathcal{O}(\eta^2)$. Subtracting the low-energy travel time from the high-energy travel time gives the delay law. Restoring c, $\hbar$, and a microscopic length ℓ_*,

$$\boxed{\delta t_{h\ell} = \frac{L\,\mathcal{A}(\boldsymbol{n})\,\ell_*^2\left(E_h^2 - E_\ell^2\right)}{2\hbar^2 c^3} + \mathcal{O}(\ell_*^4).} \tag{12.29}$$

The unit conversion uses

$$k = \frac{E}{\hbar c}, \qquad t_{\mathrm{light}} = \frac{L}{c}, \qquad \Delta = \ell_*. \tag{12.30}$$

Thus $L\Delta^2(k_h^2 - k_\ell^2)/2$ in $c = 1$ units becomes the displayed expression. A null result $|\delta t_{h\ell}| \le \delta t_{\max}$ implies

$$\ell_* \le \left[\frac{2\hbar^2 c^3 \delta t_{\max}}{L\,\mathcal{A}(\boldsymbol{n})\,|E_h^2 - E_\ell^2|}\right]^{1/2}. \tag{12.31}$$

12.5 Verification and inference

$$p^{\mathrm{face}}_{\mathrm{delay\ remainder}} = 4.005520, \tag{12.32}$$
$$p^{\mathrm{body}}_{\mathrm{delay\ remainder}} = 4.003609, \tag{12.33}$$
$$\Delta_{\mathrm{true}} = 0.030000, \tag{12.34}$$
$$\Delta_{\mathrm{fit}} = 0.030007, \tag{12.35}$$
$$|\Delta_{\mathrm{fit}} - \Delta_{\mathrm{true}}| = 7 \times 10^{-6}. \tag{12.36}$$

Logical boundary:

$$\mathsf{projector\ axioms} \nvdash \delta t_{h\ell}, \tag{12.37}$$
$$\mathsf{projector\ axioms} + \mathsf{specified\ cubic\ walk} \vdash \delta t_{h\ell}. \tag{12.38}$$

Thus the delay law is a falsifiable consequence of the specified microscopic dynamics, not yet a universal consequence of the full calculus.

Part III

Nonlinear Lorentzian Gravity and Strong Fields

Chapter 13

Weak, Optical, and Tensor Gravity

Weak-field gravity, optical deflection, and linear tensor waves are presented in one sign convention. These sectors provide separate checks on the geometric interpretation of the projector equations.

13.1 Spherical weak field

$$\nabla^2\Phi = 4\pi G\rho, \tag{13.1}$$

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\Phi'(r)\right) = 4\pi G\rho(r), \tag{13.2}$$

$$M_{\mathrm{enc}}(r) := 4\pi\int_0^r s^2\rho(s)\,\mathrm{d}s. \tag{13.3}$$

Integrating the radial equation from the origin to r, and using regularity at the origin, gives

$$r^2\Phi'(r) - \lim_{s\to 0} s^2\Phi'(s) = 4\pi G\int_0^r s^2\rho(s)\,\mathrm{d}s, \tag{13.4}$$

$$r^2\Phi'(r) = GM_{\mathrm{enc}}(r). \tag{13.5}$$

Hence

$$g_+(r) := \Phi'(r) = \frac{GM_{\mathrm{enc}}(r)}{r^2}, \tag{13.6}$$

$$a_r(r) := -\Phi'(r) = -\frac{GM_{\mathrm{enc}}(r)}{r^2}. \tag{13.7}$$

Let $u(r) := r\Phi(r)$. Since

$$r^2\Phi' = ru' - u, \tag{13.8}$$

$$\frac{\mathrm{d}}{\mathrm{d}r}(r^2\Phi') = ru'', \tag{13.9}$$

$$\nabla^2\Phi = \frac{u''}{r}, \tag{13.10}$$

the radial Poisson equation becomes

$$u''(r) = 4\pi G r\rho(r). \tag{13.11}$$

For

$$S[u] := \int_0^R \left[\frac{(u')^2}{2G} + 4\pi r\rho(r)u(r)\right]\mathrm{d}r, \tag{13.12}$$

a variation with fixed endpoints gives

$$\delta S = \int_0^R \left[\frac{u'}{G}\delta u' + 4\pi r\rho\,\delta u\right]\mathrm{d}r \tag{13.13}$$

$$= \int_0^R \left[-\frac{u''}{G} + 4\pi r\rho \right] \delta u \, \mathrm{d}r. \tag{13.14}$$

Therefore

$$\frac{\delta S}{\delta u} = 0 \quad \Longleftrightarrow \quad u'' = 4\pi G r \rho. \tag{13.15}$$

Discrete action:

$$S_h[u] := \sum_{i=0}^{N-1} \frac{(u_{i+1} - u_i)^2}{2Gh} + 4\pi h \sum_{i=1}^{N-1} r_i \rho_i u_i. \tag{13.16}$$

For an interior point, the two adjacent edge terms contribute

$$\frac{\partial S_h}{\partial u_i} = \frac{u_i - u_{i-1}}{Gh} - \frac{u_{i+1} - u_i}{Gh} + 4\pi h r_i \rho_i \tag{13.17}$$

$$= \frac{2u_i - u_{i-1} - u_{i+1}}{Gh} + 4\pi h r_i \rho_i. \tag{13.18}$$

Thus

$$\frac{\partial S_h}{\partial u_i} = 0 \Longleftrightarrow \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} = 4\pi G r_i \rho_i, \tag{13.19}$$

$$\Phi_i := \frac{u_i}{r_i}, \tag{13.20}$$

$$g_{+,i} := \frac{r_i u_i' - u_i}{r_i^2}, \tag{13.21}$$

$$a_{r,i} := -g_{+,i}. \tag{13.22}$$

Gaussian source:

$$\rho(r) = \frac{M}{(2\pi)^{3/2}\sigma^3} \exp\!\left(-\frac{r^2}{2\sigma^2} \right), \tag{13.23}$$

$$M_{\mathrm{enc}}(r) = M \left[\operatorname{erf}\!\left(\frac{r}{\sqrt{2}\sigma} \right) - \sqrt{\frac{2}{\pi}} \frac{r}{\sigma} \exp\!\left(-\frac{r^2}{2\sigma^2} \right) \right], \tag{13.24}$$

$$\Phi(r) = -\frac{GM}{r} \operatorname{erf}\!\left(\frac{r}{\sqrt{2}\sigma} \right), \tag{13.25}$$

$$\Phi'(r) = \frac{GM_{\mathrm{enc}}(r)}{r^2}, \tag{13.26}$$

$$a_r(r) = -\frac{GM_{\mathrm{enc}}(r)}{r^2}. \tag{13.27}$$

13.2 Weak-field metric and observables

$$\mathrm{d}s^2 = -\left(1 + \frac{2\Phi}{c^2} \right) c^2 \, \mathrm{d}t^2 + \left(1 - \frac{2\Phi}{c^2} \right) \mathrm{d}\boldsymbol{x}^2. \tag{13.28}$$

Clock/redshift:

$$N(\boldsymbol{x}) := \sqrt{1 + \frac{2\Phi(\boldsymbol{x})}{c^2}}, \tag{13.29}$$

$$\mathrm{d}\tau = N(\boldsymbol{x}) \, \mathrm{d}t, \tag{13.30}$$

$$\frac{\nu_{\mathrm{obs}}}{\nu_{\mathrm{emit}}} = \frac{N(\boldsymbol{x}_{\mathrm{emit}})}{N(\boldsymbol{x}_{\mathrm{obs}})}. \tag{13.31}$$

Using $\sqrt{1+x} = 1 + x/2 + \mathcal{O}(x^2)$,

$$N(\boldsymbol{x}) = 1 + \frac{\Phi(\boldsymbol{x})}{c^2} + \mathcal{O}(c^{-4}), \tag{13.32}$$

$$\frac{N_{\mathrm{emit}}}{N_{\mathrm{obs}}} = 1 + \frac{\Phi_{\mathrm{emit}} - \Phi_{\mathrm{obs}}}{c^2} + \mathcal{O}(c^{-4}). \tag{13.33}$$

Hence

$$\frac{\Delta\nu}{\nu} = \frac{\Phi_{\mathrm{emit}} - \Phi_{\mathrm{obs}}}{c^2} + \mathcal{O}(c^{-4}). \tag{13.34}$$

Shapiro delay:

$$\Delta t_{\mathrm{Shapiro}} = -\frac{2}{c^3}\int_{-L}^{L} \Phi(x,b)\,\mathrm{d}x, \tag{13.35}$$

$$\Phi = -\frac{GM}{\sqrt{x^2+b^2}}. \tag{13.36}$$

Since

$$\int \frac{\mathrm{d}x}{\sqrt{x^2+b^2}} = \operatorname{asinh}(x/b), \tag{13.37}$$

one obtains

$$\Delta t_{\mathrm{Shapiro}}(L,b) = \frac{4GM}{c^3}\operatorname{asinh}\!\left(\frac{L}{b}\right). \tag{13.38}$$

For deflection,

$$\partial_b \Phi = \frac{GMb}{(x^2+b^2)^{3/2}}, \tag{13.39}$$

$$\boldsymbol{\alpha}_\perp = -\frac{2}{c^2}\int_{-L}^{L} \nabla_\perp \Phi \;\mathrm{d}x, \tag{13.40}$$

$$\alpha := \|\boldsymbol{\alpha}_\perp\|. \tag{13.41}$$

The elementary integral

$$\int_{-L}^{L} \frac{b\;\mathrm{d}x}{(x^2+b^2)^{3/2}} = \frac{2L}{b\sqrt{L^2+b^2}} \tag{13.42}$$

gives

$$\alpha(L,b) = \frac{4GML}{c^2 b\sqrt{L^2+b^2}}, \tag{13.43}$$

$$\alpha_\infty = \frac{4GM}{c^2 b}. \tag{13.44}$$

13.3 Corrected Fierz–Pauli sector

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \tag{13.45}$$

$$h := \eta^{\mu\nu} h_{\mu\nu}, \tag{13.46}$$

$$S_{\mathrm{FP}}[h] := \frac{1}{64\pi G}\int \mathrm{d}^4x \Big[-\partial_\lambda h_{\mu\nu}\partial^\lambda h^{\mu\nu} + 2\partial_\mu h^{\mu\nu}\partial^\lambda h_{\lambda\nu} \tag{13.47}$$

$$- 2\partial_\mu h^{\mu\nu}\partial_\nu h + \partial_\lambda h \partial^\lambda h\Big] + \frac{1}{2}\int h_{\mu\nu}T^{\mu\nu}\,\mathrm{d}^4x, \tag{13.48}$$

$$G^{(1)}_{\mu\nu} := \frac{1}{2}\Big[\partial_\rho\partial_\mu h^\rho{}_\nu + \partial_\rho\partial_\nu h^\rho{}_\mu - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h \tag{13.49}$$

$$- \eta_{\mu\nu}(\partial_\rho\partial_\sigma h^{\rho\sigma} - \Box h)\Big], \tag{13.50}$$

$$\frac{\delta S_{\mathrm{FP}}}{\delta h^{\mu\nu}} = 0 \Longleftrightarrow G^{(1)}_{\mu\nu} = 8\pi G T_{\mu\nu}, \tag{13.51}$$

$$h_{\mu\nu} \mapsto h_{\mu\nu} + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu, \tag{13.52}$$

$$\partial_\mu T^{\mu\nu} = 0. \tag{13.53}$$

13.4 TT projector

$$\widehat{k}_i := \frac{k_i}{|\boldsymbol{k}|}, \tag{13.54}$$

$$\Pi_{ij} := \delta_{ij} - \widehat{k}_i \widehat{k}_j, \tag{13.55}$$

$$\Pi_{ij}\widehat{k}_j = 0, \tag{13.56}$$

$$\Pi_{ik}\Pi_{kj} = \Pi_{ij}, \tag{13.57}$$

$$\Pi_{ii} = 2, \tag{13.58}$$

$$\Lambda^{\mathrm{TT}}_{ij,kl} := \frac{1}{2}(\Pi_{ik}\Pi_{jl} + \Pi_{il}\Pi_{jk} - \Pi_{ij}\Pi_{kl}), \tag{13.59}$$

$$h^{\mathrm{TT}}_{ij} := \Lambda^{\mathrm{TT}}_{ij,kl} h_{kl}, \tag{13.60}$$

$$k_i h^{\mathrm{TT}}_{ij} = 0, \tag{13.61}$$

$$\delta_{ij} h^{\mathrm{TT}}_{ij} = 0, \tag{13.62}$$

$$(\Lambda^{\mathrm{TT}})^2 = \Lambda^{\mathrm{TT}}. \tag{13.63}$$

13.5 Discrete tensor wave

$$\frac{h^{n+1}_{A,j} - 2h^n_{A,j} + h^{n-1}_{A,j}}{\Delta t^2} - c^2 \frac{h^n_{A,j+1} - 2h^n_{A,j} + h^n_{A,j-1}}{h^2} = S^n_{A,j}. \tag{13.64}$$

For a source-free plane wave $h^n_{A,j} = H_A \mathrm{e}^{\mathrm{i}(kjh - \omega n \Delta t)}$,

$$h^{n+1}_j - 2h^n_j + h^{n-1}_j = -4\sin^2\left(\frac{\omega\Delta t}{2}\right) h^n_j, \tag{13.65}$$

$$h^n_{j+1} - 2h^n_j + h^n_{j-1} = -4\sin^2\left(\frac{kh}{2}\right) h^n_j. \tag{13.66}$$

After cancellation of the common amplitude,

$$\lambda := \frac{c\Delta t}{h}, \tag{13.67}$$

$$\sin^2\left(\frac{\omega\Delta t}{2}\right) = \lambda^2 \sin^2\left(\frac{kh}{2}\right). \tag{13.68}$$

Critical sector $\lambda = 1$:

$$h^{n+1}_j + h^{n-1}_j = h^n_{j+1} + h^n_{j-1}, \tag{13.69}$$

$$h^n_j = F_{j-n} + G_{j+n}. \tag{13.70}$$

Exact ambiguity on $\mathbb{Z}$:

$$F_r \mapsto F_r + \chi_r, \tag{13.71}$$

$$G_r \mapsto G_r - \chi_r, \tag{13.72}$$

$$\chi_{r+2} = \chi_r. \tag{13.73}$$

For individually N-periodic F, G, compatibility:

$$N \text{ odd}: \quad \sum_j (h^1_j - h^0_{j+1}) = 0, \tag{13.74}$$

$$N \text{ even}: \quad \sum_{j \text{ even}} (h^1_j - h^0_{j+1}) = 0, \quad \sum_{j \text{ odd}} (h^1_j - h^0_{j+1}) = 0. \tag{13.75}$$

Without compatibility:

$$F, G = \text{affine or alternating-affine characteristic packets.} \tag{13.76}$$

13.6 Discrete tensor energy

$$v_{A,j}^{n+1/2} := \frac{h_{A,j}^{n+1} - h_{A,j}^{n}}{\Delta t}, \tag{13.77}$$

$$d_{A,j}^{n} := \frac{h_{A,j+1}^{n} - h_{A,j}^{n}}{h}, \tag{13.78}$$

$$\mathcal{E}^{n+1/2} := \frac{h}{2} \sum_{A,j} \left[(v_{A,j}^{n+1/2})^2 + c^2 d_{A,j}^{n+1} d_{A,j}^{n} \right]. \tag{13.79}$$

Multiplying the discrete wave equation by $h(h_{A,j}^{n+1} - h_{A,j}^{n-1})/2$, summing over A, j, and applying periodic summation by parts converts the temporal and spatial terms into differences of the two quadratic energy pieces. This gives

$$\mathcal{E}^{n+1/2} - \mathcal{E}^{n-1/2} = \frac{h}{2} \sum_{A,j} S_{A,j}^{n} (h_{A,j}^{n+1} - h_{A,j}^{n-1}), \tag{13.80}$$

$$S_{A,j}^{n} = 0 \Longrightarrow \mathcal{E}^{n+1/2} = \mathcal{E}^{n-1/2}. \tag{13.81}$$

Chapter 14

Holonomy, Metric Curvature, and Local Work

Plaquette holonomy gives a finite curvature observable, while a conformal metric provides an independent continuum description. The chapter compares the two and derives a local stress-energy transfer law.

14.1 Non-Abelian plaquette curvature

$$U_x(\varepsilon) := \mathrm{e}^{\varepsilon A_x}, \tag{14.1}$$
$$U_y(\varepsilon) := \mathrm{e}^{\varepsilon A_y}, \tag{14.2}$$
$$W_\varepsilon := \mathrm{e}^{\varepsilon A_x}\mathrm{e}^{\varepsilon A_y}\mathrm{e}^{-\varepsilon A_x}\mathrm{e}^{-\varepsilon A_y}, \tag{14.3}$$
$$F_{xy} := [A_x, A_y], \tag{14.4}$$
$$F_\varepsilon := \frac{1}{\varepsilon^2}\log W_\varepsilon. \tag{14.5}$$

Expanding the four exponentials through second order first gives

$$W_\varepsilon = (I + \varepsilon A_x + \tfrac{1}{2}\varepsilon^2 A_x^2)(I + \varepsilon A_y + \tfrac{1}{2}\varepsilon^2 A_y^2) \tag{14.6}$$
$$\times (I - \varepsilon A_x + \tfrac{1}{2}\varepsilon^2 A_x^2)(I - \varepsilon A_y + \tfrac{1}{2}\varepsilon^2 A_y^2) + \mathcal{O}(\varepsilon^3) \tag{14.7}$$
$$= I + \varepsilon^2(A_xA_y - A_yA_x) + \mathcal{O}(\varepsilon^3) \tag{14.8}$$
$$= I + \varepsilon^2[A_x, A_y] + \mathcal{O}(\varepsilon^3). \tag{14.9}$$

The Baker–Campbell–Hausdorff expansion continues this calculation:

$$\log W_\varepsilon = \varepsilon^2[A_x, A_y] + \frac{\varepsilon^3}{2}\left([A_x, [A_x, A_y]] + [A_y, [A_x, A_y]]\right) + \mathcal{O}(\varepsilon^4), \tag{14.10}$$
$$F_\varepsilon = F_{xy} + C_1\varepsilon + C_2\varepsilon^2 + \cdots. \tag{14.11}$$

Gauge covariance:

$$A_i' = GA_iG^{-1}, \tag{14.12}$$
$$W_\varepsilon' = GW_\varepsilon G^{-1}, \tag{14.13}$$
$$F_\varepsilon' = GF_\varepsilon G^{-1}. \tag{14.14}$$

For the first refinement,

$$2F_{\varepsilon/2} - F_\varepsilon = 2\left(F + \frac{C_1\varepsilon}{2} + \frac{C_2\varepsilon^2}{4} + \cdots\right) - \left(F + C_1\varepsilon + C_2\varepsilon^2 + \cdots\right) \tag{14.15}$$
$$= F - \frac{C_2}{2}\varepsilon^2 + \mathcal{O}(\varepsilon^3), \tag{14.16}$$

so the linear error cancels. Repeating the same Richardson step cancels the next surviving power at each stage:

$$F_\varepsilon^{[1]} := 2F_{\varepsilon/2} - F_\varepsilon = F + \mathcal{O}(\varepsilon^2), \tag{14.17}$$

$$F_\varepsilon^{[2]} := \frac{4F_{\varepsilon/2}^{[1]} - F_\varepsilon^{[1]}}{3} = F + \mathcal{O}(\varepsilon^3), \tag{14.18}$$

$$F_\varepsilon^{[3]} := \frac{8F_{\varepsilon/2}^{[2]} - F_\varepsilon^{[2]}}{7} = F + \mathcal{O}(\varepsilon^4). \tag{14.19}$$

14.2 Exact conformal metric-holonomy identity

$$\mathrm{d}s^2 = \mathrm{e}^{2\phi(t,x)}(-\,\mathrm{d}t^2 + \,\mathrm{d}x^2), \tag{14.20}$$

$$e^0 = \mathrm{e}^{\phi}\,\mathrm{d}t, \tag{14.21}$$

$$e^1 = \mathrm{e}^{\phi}\,\mathrm{d}x, \tag{14.22}$$

$$K := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{14.23}$$

$$\omega = (\phi_x\,\mathrm{d}t + \phi_t\,\mathrm{d}x)K. \tag{14.24}$$

Because every value of ω is proportional to the same generator K, one has $\omega \wedge \omega = 0$. Thus

$$\Omega = \,\mathrm{d}\omega + \omega \wedge \omega \tag{14.25}$$

$$= \,\mathrm{d}(\phi_x\,\mathrm{d}t + \phi_t\,\mathrm{d}x)K \tag{14.26}$$

$$= (\phi_{tt} - \phi_{xx})K\ \,\mathrm{d}t \wedge \,\mathrm{d}x. \tag{14.27}$$

The scalar curvature is therefore

$$R = 2\mathrm{e}^{-2\phi}(\phi_{tt} - \phi_{xx}). \tag{14.28}$$

For an oriented cell $\square$:

$$U_e := \exp\!\left(\int_e \omega\right), \tag{14.29}$$

$$W_\square := \prod_{e \subset \partial\square}^{\longrightarrow} U_e. \tag{14.30}$$

All edge exponents commute, so the ordered product combines into one exponential. Stokes' theorem then gives

$$W_\square = \exp\!\left(\oint_{\partial\square} \omega\right) \tag{14.31}$$

$$= \exp\!\left(\int_\square \mathrm{d}\omega\right) \tag{14.32}$$

$$= \exp\!\left[K \int_\square (\phi_{tt} - \phi_{xx})\,\mathrm{d}t\,\mathrm{d}x\right], \tag{14.33}$$

$$\log W_\square = K \int_\square \frac{1}{2}\mathrm{e}^{2\phi} R\ \mathrm{d}t\,\mathrm{d}x. \tag{14.34}$$

Local frame transformation:

$$G(t,x) := \mathrm{e}^{\chi(t,x)K}, \tag{14.35}$$

$$U'_{a\to b} = G(b)U_{a\to b}G(a)^{-1}, \tag{14.36}$$

$$W'_\square = G(v_0)W_\square G(v_0)^{-1}. \tag{14.37}$$

14.3 Local scalar stress-energy transfer

$$\mathrm{d}s^2 = -N(t,x)^2\,\mathrm{d}t^2 + a(t,x)^2\,\mathrm{d}x^2, \tag{14.38}$$

$$w_j := \frac{a_j}{N_j}, \tag{14.39}$$

$$c_{j+1/2} := \frac{N_{j+1/2}}{a_{j+1/2}}, \tag{14.40}$$

$$v_j := N_j a_j m^2, \tag{14.41}$$

$$\Delta_j := \varphi_{j+1} - \varphi_j. \tag{14.42}$$

Finite-volume action:

$$L_h = \frac{h}{2}\sum_j w_j \dot{\varphi}_j^2 - \frac{1}{2h}\sum_j c_{j+1/2}\Delta_j^2 - \frac{h}{2}\sum_j v_j \varphi_j^2. \tag{14.43}$$

For a variation at site j, the kinetic term gives $-h\,\mathrm{d}(w_j\dot{\varphi}_j)/\,\mathrm{d}t$, the two neighboring edge terms give $(c_{j+1/2}\Delta_j - c_{j-1/2}\Delta_{j-1})/h$, and the mass term gives $-hv_j\varphi_j$. Dividing the Euler–Lagrange equation by h yields

$$w_j\ddot{\varphi}_j + \dot{w}_j\dot{\varphi}_j = \frac{c_{j+1/2}\Delta_j - c_{j-1/2}\Delta_{j-1}}{h^2} - v_j\varphi_j. \tag{14.44}$$

Cell energy, flux, geometric work:

$$e_j := \frac{h}{2}w_j\dot{\varphi}_j^2 + \frac{h}{2}v_j\varphi_j^2 + \frac{c_{j+1/2}}{4h}\Delta_j^2 + \frac{c_{j-1/2}}{4h}\Delta_{j-1}^2, \tag{14.45}$$

$$F_{j+1/2} := -\frac{c_{j+1/2}}{2h}\Delta_j(\dot{\varphi}_j + \dot{\varphi}_{j+1}), \tag{14.46}$$

$$s_j := -\frac{h}{2}\dot{w}_j\dot{\varphi}_j^2 + \frac{h}{2}\dot{v}_j\varphi_j^2 + \frac{\dot{c}_{j+1/2}}{4h}\Delta_j^2 + \frac{\dot{c}_{j-1/2}}{4h}\Delta_{j-1}^2. \tag{14.47}$$

To see the local balance, differentiate the two edge energies using

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{c_{j+1/2}\Delta_j^2}{4h}\right) = \frac{\dot{c}_{j+1/2}\Delta_j^2}{4h} + \frac{c_{j+1/2}\Delta_j}{2h}(\dot{\varphi}_{j+1} - \dot{\varphi}_j), \tag{14.48}$$

and use the equation of motion to replace $w_j\dot{\varphi}_j\ddot{\varphi}_j$. Terms involving neighboring velocities combine into the two interface fluxes, while coefficient derivatives form s_j. The exact balance is

$$\dot{e}_j + F_{j+1/2} - F_{j-1/2} = s_j, \tag{14.49}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_j e_j = \sum_j s_j = -\frac{\partial L_h}{\partial t}, \tag{14.50}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_{j=r}^{s} e_j + F_{s+1/2} - F_{r-1/2} = \sum_{j=r}^{s} s_j. \tag{14.51}$$

For smooth c and φ, midpoint Taylor expansion gives

$$c_{j+1/2}\Delta_j = h(c\varphi_x)_{j+1/2} + \mathcal{O}(h^3), \tag{14.52}$$

$$c_{j-1/2}\Delta_{j-1} = h(c\varphi_x)_{j-1/2} + \mathcal{O}(h^3). \tag{14.53}$$

Taking their difference and dividing by h^2 therefore gives the continuum operator

$$\frac{c_{j+1/2}\Delta_j - c_{j-1/2}\Delta_{j-1}}{h^2} = \partial_x(c\partial_x\varphi) + \mathcal{O}(h^2). \tag{14.54}$$

Chapter 15

Symmetry-Reduced Inhomogeneous Gravity

The Gowdy model supplies an inhomogeneous nonlinear gravitational test with reduced symmetry. Exact polarized modes and numerical unpolarized trajectories are used to check constraints and refinement.

15.1 Unpolarized Gowdy action

$$S_G[P,Q] := \frac{1}{2}\int \left[P_\tau^2 - \mathrm{e}^{-2\tau}P_\theta^2 + \mathrm{e}^{2P}\left(Q_\tau^2 - \mathrm{e}^{-2\tau}Q_\theta^2\right)\right]\,\mathrm{d}\tau\,\mathrm{d}\theta. \tag{15.1}$$

For a variation of P, integration by parts gives

$$\delta_P S_G = \int \Big[P_\tau\delta P_\tau - \mathrm{e}^{-2\tau}P_\theta\delta P_\theta + \mathrm{e}^{2P}(Q_\tau^2 - \mathrm{e}^{-2\tau}Q_\theta^2)\delta P\Big]\,\mathrm{d}\tau\,\mathrm{d}\theta \tag{15.2}$$

$$= -\int \Big[P_{\tau\tau} - \mathrm{e}^{-2\tau}P_{\theta\theta} - \mathrm{e}^{2P}(Q_\tau^2 - \mathrm{e}^{-2\tau}Q_\theta^2)\Big]\delta P\,\mathrm{d}\tau\,\mathrm{d}\theta. \tag{15.3}$$

For Q, the divergence-form equation is

$$\partial_\tau(\mathrm{e}^{2P}Q_\tau) - \partial_\theta(\mathrm{e}^{2P-2\tau}Q_\theta) = 0. \tag{15.4}$$

Expanding both derivatives and dividing by e^{2P} gives

$$Q_{\tau\tau} - \mathrm{e}^{-2\tau}Q_{\theta\theta} + 2(P_\tau Q_\tau - \mathrm{e}^{-2\tau}P_\theta Q_\theta) = 0. \tag{15.5}$$

Thus the Euler–Lagrange expressions are

$$\mathcal{E}_P := P_{\tau\tau} - \mathrm{e}^{-2\tau}P_{\theta\theta} - \mathrm{e}^{2P}\left(Q_\tau^2 - \mathrm{e}^{-2\tau}Q_\theta^2\right), \tag{15.6}$$

$$\mathcal{E}_Q := Q_{\tau\tau} - \mathrm{e}^{-2\tau}Q_{\theta\theta} + 2\left(P_\tau Q_\tau - \mathrm{e}^{-2\tau}P_\theta Q_\theta\right), \tag{15.7}$$

$$\delta_P S_G = 0 \Longleftrightarrow \mathcal{E}_P = 0, \tag{15.8}$$

$$\delta_Q S_G = 0 \Longleftrightarrow \mathcal{E}_Q = 0. \tag{15.9}$$

15.2 Hamiltonian- and momentum-type constraints

$$\lambda_\tau := P_\tau^2 + \mathrm{e}^{-2\tau}P_\theta^2 + \mathrm{e}^{2P}\left(Q_\tau^2 + \mathrm{e}^{-2\tau}Q_\theta^2\right), \tag{15.10}$$

$$\lambda_\theta := 2\left(P_\tau P_\theta + \mathrm{e}^{2P}Q_\tau Q_\theta\right). \tag{15.11}$$

To verify the integrability identity, first differentiate the two densities. The mixed derivatives $P_\tau P_{\tau\theta}$ and $Q_\tau Q_{\tau\theta}$ cancel between the two results. The remainder is

$$\partial_\tau\lambda_\theta - \partial_\theta\lambda_\tau = 2P_\theta(P_{\tau\tau} - \mathrm{e}^{-2\tau}P_{\theta\theta}) \tag{15.12}$$

$$+ 2\mathrm{e}^{2P}\Big[Q_\theta(Q_{\tau\tau} - \mathrm{e}^{-2\tau}Q_{\theta\theta} + 2P_\tau Q_\tau) \tag{15.13}$$

$$- P_\theta Q_\tau^2 - \mathrm{e}^{-2\tau}P_\theta Q_\theta^2\Big]. \tag{15.14}$$

Adding and subtracting the terms required by $\mathcal{E}_P$ and $\mathcal{E}_Q$ reorganizes this expression as

$$\boxed{\partial_\tau\lambda_\theta - \partial_\theta\lambda_\tau = 2P_\theta\mathcal{E}_P + 2\mathrm{e}^{2P}Q_\theta\mathcal{E}_Q.} \tag{15.15}$$

Hence

$$\mathcal{E}_P = \mathcal{E}_Q = 0 \Longrightarrow \partial_\tau\lambda_\theta = \partial_\theta\lambda_\tau. \tag{15.16}$$

On solutions, integrate the identity over the periodic circle:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\int_0^{2\pi}\lambda_\theta\,\mathrm{d}\theta = \int_0^{2\pi}\partial_\theta\lambda_\tau\,\mathrm{d}\theta \tag{15.17}$$

$$= \lambda_\tau(2\pi) - \lambda_\tau(0) = 0. \tag{15.18}$$

Thus the global momentum is constant in τ, and the periodic global momentum condition is

$$\int_0^{2\pi}\lambda_\theta(\tau,\theta)\,\mathrm{d}\theta = 0. \tag{15.19}$$

15.3 Polarized exact mode

For $Q = 0$,

$$P_{\tau\tau} - \mathrm{e}^{-2\tau}P_{\theta\theta} = 0. \tag{15.20}$$

Set $x := n\mathrm{e}^{-\tau}$ and take

$$P(\tau,\theta) = AJ_0(x)\cos(n\theta). \tag{15.21}$$

Since $x_\tau = -x$ and $J_0'(x) = -J_1(x)$,

$$P_\tau = AxJ_1(x)\cos(n\theta). \tag{15.22}$$

A second derivative can be written as

$$P_{\tau\tau} = A\left[x^2J_0''(x) + xJ_0'(x)\right]\cos(n\theta). \tag{15.23}$$

The Bessel equation

$$x^2J_0'' + xJ_0' + x^2J_0 = 0 \tag{15.24}$$

therefore gives

$$P_{\tau\tau} = -x^2P = -n^2\mathrm{e}^{-2\tau}P, \tag{15.25}$$

$$P_{\theta\theta} = -n^2P, \tag{15.26}$$

$$P_{\tau\tau} - \mathrm{e}^{-2\tau}P_{\theta\theta} = 0. \tag{15.27}$$

The same exact mode also has a hypergeometric polyphase resolution. Set

$$y := \frac{x^2}{4} = \frac{n^2\mathrm{e}^{-2\tau}}{4} \tag{15.28}$$

and define

$$\mathcal{B}(y) := J_0(2\sqrt{y}) = {}_0F_1\left(\begin{matrix}-\\1\end{matrix};-y\right) = \sum_{m=0}^{\infty}\frac{(-1)^m y^m}{(m!)^2}. \tag{15.29}$$

The series is entire in y, so the square-root notation introduces no branch dependence.
For $N \geq 2$, split the series by $m \equiv j \pmod N$:

$$\boxed{\mathcal{B}_{N,j}(y) := \sum_{q=0}^{\infty}\frac{(-1)^{Nq+j}y^{Nq+j}}{((Nq+j)!)^2}.} \tag{15.30}$$

Using (36.16) twice,

$$((Nq+j)!)^2 = (j!)^2 N^{2Nq} (q!)^2 \prod_{\alpha\in\mathcal{A}_{N,j}} (\alpha)_q^2. \tag{15.31}$$

The second factor $q!$ is supplied by adding the denominator parameter 1, because $(1)_q = q!$. Therefore

$$\boxed{\mathcal{B}_{N,j}(y) = \frac{(-1)^j y^j}{(j!)^2}\, {}_0F_{2N-1}\left(\begin{matrix}-\\ 1, \mathcal{A}_{N,j}, \mathcal{A}_{N,j}\end{matrix}; (-1)^N \frac{y^N}{N^{2N}}\right).} \tag{15.32}$$

Every parameter in $\mathcal{A}_{N,j}$ occurs twice.

Root-of-unity filtering gives

$$\boxed{\mathcal{B}_{N,j}(y) = \frac{1}{N}\sum_{\ell=0}^{N-1} \omega_N^{-j\ell}\, \mathcal{B}(\omega_N^{\ell} y),} \tag{15.33}$$

and exact reconstruction gives

$$\boxed{J_0(x) = \mathcal{B}(y) = \sum_{j=0}^{N-1} \mathcal{B}_{N,j}(y).} \tag{15.34}$$

Thus

$$P(\tau,\theta) = A\cos(n\theta) \sum_{j=0}^{N-1} \mathcal{B}_{N,j}\left(\frac{n^2 \mathrm{e}^{-2\tau}}{4}\right). \tag{15.35}$$

This is a split by Taylor-series order, not a second spatial Fourier decomposition.

Let $\vartheta := y\ \mathrm{d}/\mathrm{d}y$. The parent equation is

$$\vartheta^2 \mathcal{B} + y\mathcal{B} = 0, \tag{15.36}$$

while the lanes obey the coupled cyclic system

$$\boxed{\vartheta^2 \mathcal{B}_{N,j} + y\mathcal{B}_{N,j-1} = 0, \qquad j \pmod N.} \tag{15.37}$$

For $m = Nq + j$,

$$\vartheta^2 \left(\frac{(-1)^m y^m}{(m!)^2}\right) = \frac{(-1)^m y^m}{((m-1)!)^2}, \tag{15.38}$$

whereas the matching coefficient from $y\mathcal{B}_{N,j-1}$ is

$$\frac{(-1)^{m-1} y^m}{((m-1)!)^2}. \tag{15.39}$$

They cancel. An individual lane is not generally a separate solution of the original Bessel equation; the lanes are coupled channels whose sum is the physical mode.

The finite-difference constraint-integrability residual converges with

$$p_{\text{polarized}} = 1.977675. \tag{15.40}$$

15.4 Nonlinear numerical example

For smooth periodic unpolarized data and second-order spatial differences,

$$p_{(P,Q,P_\tau,Q_\tau)} = 2.128356, \tag{15.41}$$

$$p_{\partial_\tau\lambda_\theta - \partial_\theta\lambda_\tau} = 1.949489, \tag{15.42}$$

$$\left|\frac{1}{2\pi}\int_0^{2\pi} \lambda_\theta\, \mathrm{d}\theta\right|_{\max} = 1.539567 \times 10^{-17}, \tag{15.43}$$

$$\varepsilon_{\text{coarse constraint sum}} = 1.110223 \times 10^{-16}. \tag{15.44}$$

These computations close the equations within the Gowdy symmetry reduction. They do not establish full nonlinear gravity in three spatial dimensions.

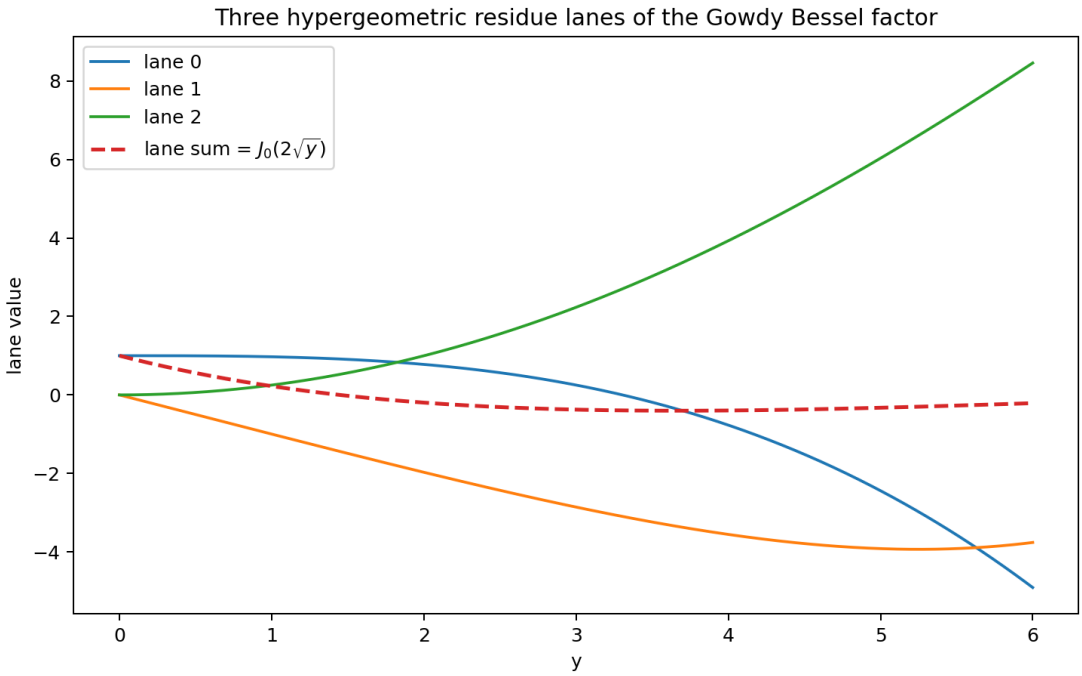


Figure 15.1: A three-lane residue decomposition of the polarized Gowdy Bessel factor. The lane sum reconstructs $J_0(2\sqrt{y})$ exactly.

Chapter 16

Constraint-Preserving Discrete Gowdy Evolution

A projection step is added to the semidiscrete Gowdy evolution so that the endpoint constraint is preserved exactly. The remaining compatibility residual measures whether local evolution and global projection agree under refinement.

16.1 Periodic semidiscrete system

Let

$$\theta_j = jh, \qquad h = \frac{2\pi}{N}, \qquad D_h^{\mathsf{T}} = -D_h, \tag{16.1}$$

$$(D_h f)_j := \frac{f_{j+1} - f_{j-1}}{2h}, \qquad (\Delta_h f)_j := \frac{f_{j+1} - 2f_j + f_{j-1}}{h^2}, \tag{16.2}$$

$$U_j := \partial_\tau P_j, \qquad V_j := \partial_\tau Q_j, \qquad \gamma(\tau) := \mathrm{e}^{-2\tau}. \tag{16.3}$$

Raw second-order spatial discretization:

$$\dot{P} = U, \tag{16.4}$$

$$\dot{Q} = V, \tag{16.5}$$

$$\dot{U} = \gamma \Delta_h P + \mathrm{e}^{2P}\left[V^2 - \gamma (D_h Q)^2\right], \tag{16.6}$$

$$\dot{V} = \gamma \Delta_h Q - 2\left[UV - \gamma (D_h P)(D_h Q)\right]. \tag{16.7}$$

The periodic global momentum constraint is

$$\boxed{\mathcal{C}_h(P,Q,U,V) := 2h \sum_{j=0}^{N-1} \left[U_j (D_h P)_j + \mathrm{e}^{2P_j} V_j (D_h Q)_j\right].} \tag{16.8}$$

16.2 Differential constraint force

For $z = (P, Q, U, V)$, write the constraint in vector notation as

$$\mathcal{C}_h = 2h\, U^{\mathsf{T}} D_h P + 2h\, (\mathrm{e}^{2P} V)^{\mathsf{T}} D_h Q. \tag{16.9}$$

The P-variation is

$$\delta_P \mathcal{C}_h = 2h\, U^{\mathsf{T}} D_h \delta P + 4h\, (\mathrm{e}^{2P} V \odot D_h Q)^{\mathsf{T}} \delta P \tag{16.10}$$

$$= \left[-2h D_h U + 4h \mathrm{e}^{2P} V (D_h Q)\right]^{\mathsf{T}} \delta P, \tag{16.11}$$

where $D_h^{\mathsf{T}} = -D_h$. Similarly,

$$\delta_Q \mathcal{C}_h = -2h \left[D_h (\mathrm{e}^{2P} V)\right]^{\mathsf{T}} \delta Q. \tag{16.12}$$

The momentum variations are direct, so

$$\nabla_P \mathcal{C}_h = -2hD_hU + 4h\mathrm{e}^{2P}V(D_hQ), \tag{16.13}$$

$$\nabla_Q \mathcal{C}_h = -2hD_h(\mathrm{e}^{2P}V), \tag{16.14}$$

$$g_U := \nabla_U \mathcal{C}_h = 2hD_hP, \tag{16.15}$$

$$g_V := \nabla_V \mathcal{C}_h = 2h\mathrm{e}^{2P}D_hQ. \tag{16.16}$$

Let $f_h(\tau, z)$ denote the raw semidiscrete vector field and restrict the correction to the momentum-gradient direction

$$G_h(z) := (0, 0, g_U, g_V). \tag{16.17}$$

Demanding tangency to the constraint surface gives

$$0 = \nabla\mathcal{C}_h^{\mathsf{T}}(f_h - \Lambda_h G_h) \tag{16.18}$$

$$= \nabla\mathcal{C}_h^{\mathsf{T}} f_h - \Lambda_h \nabla\mathcal{C}_h^{\mathsf{T}} G_h. \tag{16.19}$$

Because

$$\nabla\mathcal{C}_h^{\mathsf{T}} G_h = \|g_U\|_2^2 + \|g_V\|_2^2, \tag{16.20}$$

the required multiplier and corrected field are

$$\Lambda_h(\tau, z) := \frac{\nabla\mathcal{C}_h(z)^{\mathsf{T}} f_h(\tau, z)}{\|g_U\|_2^2 + \|g_V\|_2^2}, \tag{16.21}$$

$$\widehat{f}_h(\tau, z) := f_h(\tau, z) - \Lambda_h(\tau, z)G_h(z). \tag{16.22}$$

Whenever $\|g_U\|_2^2 + \|g_V\|_2^2 > 0$,

$$\boxed{\nabla\mathcal{C}_h^{\mathsf{T}} \widehat{f}_h = \nabla\mathcal{C}_h^{\mathsf{T}} f_h - \Lambda_h(\|g_U\|_2^2 + \|g_V\|_2^2) = 0.} \tag{16.23}$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\mathcal{C}_h(z(\tau)) = 0 \tag{16.24}$$

for the constrained semidiscrete flow.

16.3 Second-order map and exact endpoint constraint

Provisional explicit midpoint step:

$$k_1 := \widehat{f}_h(\tau_n, z_n), \tag{16.25}$$

$$k_2 := \widehat{f}_h\left(\tau_n + \frac{\Delta\tau}{2}, z_n + \frac{\Delta\tau}{2}k_1\right), \tag{16.26}$$

$$\widetilde{z}_{n+1} := z_n + \Delta\tau k_2. \tag{16.27}$$

Write

$$\widetilde{z}_{n+1} = (\widetilde{P}, \widetilde{Q}, \widetilde{U}, \widetilde{V}) \tag{16.28}$$

and set

$$\widetilde{g}_U := 2hD_h\widetilde{P}, \tag{16.29}$$

$$\widetilde{g}_V := 2h\mathrm{e}^{2\widetilde{P}}D_h\widetilde{Q}, \tag{16.30}$$

$$\mu_n := \frac{\mathcal{C}_h(\widetilde{z}_{n+1}) - \mathcal{C}_h(z_n)}{\|\widetilde{g}_U\|_2^2 + \|\widetilde{g}_V\|_2^2}, \tag{16.31}$$

$$P_{n+1} := \widetilde{P}, \tag{16.32}$$

$$Q_{n+1} := \widetilde{Q}, \tag{16.33}$$

$$U_{n+1} := \widetilde{U} - \mu_n\widetilde{g}_U, \tag{16.34}$$

$$V_{n+1} := \widetilde{V} - \mu_n \widetilde{g}_V. \tag{16.35}$$

Since $\mathcal{C}_h$ is linear in (U, V) for fixed (P, Q), the endpoint correction has no higher-order remainder:

$$\mathcal{C}_h(z_{n+1}) = \mathcal{C}_h(\widetilde{P}, \widetilde{Q}, \widetilde{U} - \mu_n \widetilde{g}_U, \widetilde{V} - \mu_n \widetilde{g}_V) \tag{16.36}$$
$$= \mathcal{C}_h(\widetilde{z}_{n+1}) - \mu_n \widetilde{g}_U^{\mathsf{T}} \widetilde{g}_U - \mu_n \widetilde{g}_V^{\mathsf{T}} \widetilde{g}_V \tag{16.37}$$
$$= \mathcal{C}_h(\widetilde{z}_{n+1}) - \mu_n (\|\widetilde{g}_U\|_2^2 + \|\widetilde{g}_V\|_2^2). \tag{16.38}$$

Insertion of the definition of μ_n gives

$$\boxed{\mathcal{C}_h(z_{n+1}) = \mathcal{C}_h(z_n).} \tag{16.39}$$

Thus

$$\mathcal{C}_h(z_0) = 0 \Longrightarrow \mathcal{C}_h(z_n) = 0 \qquad \forall n. \tag{16.40}$$

If the midpoint step begins on the invariant surface, its local constraint defect is

$$\mathcal{C}_h(\widetilde{z}_{n+1}) - \mathcal{C}_h(z_n) = \mathcal{O}(\Delta\tau^3). \tag{16.41}$$

Provided the gradient denominator remains bounded away from zero,

$$\mu_n = \frac{\mathcal{O}(\Delta\tau^3)}{\|\widetilde{g}_U\|_2^2 + \|\widetilde{g}_V\|_2^2} = \mathcal{O}(\Delta\tau^3). \tag{16.42}$$

The endpoint displacement is therefore of the same order as the midpoint local truncation error, so the correction preserves second-order temporal accuracy.

16.4 Local compatibility diagnostic

Define

$$\lambda_{\tau,h} := U^2 + \gamma(D_h P)^2 + \mathrm{e}^{2P}\left[V^2 + \gamma(D_h Q)^2\right], \tag{16.43}$$
$$\lambda_{\theta,h} := 2\left[U(D_h P) + \mathrm{e}^{2P} V(D_h Q)\right]. \tag{16.44}$$

Let

$$\mathcal{E}_{P,h} := \dot{U} - \dot{U}_{\mathrm{raw}}, \tag{16.45}$$
$$\mathcal{E}_{Q,h} := \dot{V} - \dot{V}_{\mathrm{raw}}. \tag{16.46}$$

For the raw equations, the continuum identity has the form

$$\partial_\tau \lambda_\theta - \partial_\theta \lambda_\tau = 2P_\theta \mathcal{E}_P + 2\mathrm{e}^{2P} Q_\theta \mathcal{E}_Q. \tag{16.47}$$

The momentum correction changes only the two evolution equations represented by $\mathcal{E}_{P,h}$ and $\mathcal{E}_{Q,h}$. Subtracting their explicit contribution leaves the corrected local integrability residual

$$\mathcal{I}_h := \partial_\tau \lambda_{\theta,h} - D_h \lambda_{\tau,h} - 2(D_h P)\mathcal{E}_{P,h} - 2\mathrm{e}^{2P}(D_h Q)\mathcal{E}_{Q,h}. \tag{16.48}$$

Numerically,

$$p_{\mathrm{state}} = 2.128038, \tag{16.49}$$
$$p_{\mathcal{I}} = 1.949319, \tag{16.50}$$
$$p_{\text{constraint force}} = 2.044195, \tag{16.51}$$
$$\max_n |\mathcal{C}_h(z_n) - \mathcal{C}_h(z_0)| = 6.812243 \times 10^{-18}, \tag{16.52}$$
$$\max_n |\mathcal{C}_h(z_n^{\text{unconstrained}}) - \mathcal{C}_h(z_0)| = 1.517044 \times 10^{-6} \quad (N = 256), \tag{16.53}$$
$$\varepsilon_{\text{coarse momentum sum}} = 2.602085 \times 10^{-18}. \tag{16.54}$$

The constraint force obeys

$$\left\| \widehat{f}_h - f_h \right\|_{\mathrm{RMS}} = \mathcal{O}(h^2), \tag{16.55}$$

so the constrained semidiscrete system recovers the raw Gowdy equations in the refinement limit.

16.5 Next steps

- Extend the constraint projection to degenerate and nearly degenerate constraint gradients, with a regularized construction and a stability bound.
- Formulate a local multisymplectic constraint one-form and prove that the discrete update preserves its local compatibility conditions.
- Retain lapse and shift as active gauge multipliers in the semidiscrete action and derive their constraint equations before gauge fixing.
- Generalize the construction toward the local Hamiltonian and momentum constraints of the full $3 + 1$ theory.

Chapter 17

Local $3+1$ Linearized Projector–ADM Theory

The corner calculus supports a local linearized ADM system with exact finite constraint propagation. This chapter derives the quadratic action, gauge symmetry, and transverse–traceless sector.

17.1 Finite spatial calculus

Let $\mathfrak{A}_h$ carry three commuting skew-adjoint derivatives

$$\delta_i\delta_j = \delta_j\delta_i, \tag{17.1}$$

$$\langle \delta_i X, Y\rangle_h = -\langle X, \delta_i Y\rangle_h\,, \tag{17.2}$$

$$\delta^2 := \sum_{i=1}^{3} \delta_i\delta_i. \tag{17.3}$$

The periodic centered projector lattice satisfies these identities exactly.

17.2 Canonical variables and constraints

Let

$$h_{ij} = h_{ji}, \qquad \pi_{ij} = \pi_{ji}, \tag{17.4}$$

$$h := h_{kk}, \qquad \pi := \pi_{kk}. \tag{17.5}$$

Define

$$\boxed{\mathcal{C} := \delta_i\delta_j h_{ij} - \delta^2 h} \tag{17.6}$$

and

$$\boxed{\mathcal{C}_i := -2\delta_j \pi_{ij}.} \tag{17.7}$$

The linearized spatial Einstein operator is

$$\boxed{\begin{aligned}\mathcal{G}_{ij}(h) := \frac{1}{2}\big[&\delta_k\delta_i h_{jk} + \delta_k\delta_j h_{ik} - \delta^2 h_{ij} - \delta_i\delta_j h \\ &- \delta_{ij}(\delta_k\delta_l h_{kl} - \delta^2 h)\big].\end{aligned}} \tag{17.8}$$

Commutativity of the derivatives gives

$$\boxed{\delta_j \mathcal{G}_{ij}(h) = 0.} \tag{17.9}$$

Expanding the divergence gives

$$2\delta_j\mathcal{G}_{ij} = \delta_i\delta_j\delta_k h_{jk} + \delta_j\delta_j\delta_k h_{ik} - \delta_j\delta^2 h_{ij} - \delta_i\delta_j\delta_j h \tag{17.10}$$

$$- \delta_i\delta_k\delta_l h_{kl} + \delta_i\delta^2 h. \tag{17.11}$$

The second and third terms cancel after commuting derivatives. The first cancels the fifth, and the fourth cancels the sixth because $h = h_{jj}$.

17.3 Quadratic local action

Define

$$\mathcal{T}_h(\pi) := \langle \pi_{ij}, \pi_{ij} \rangle_h - \frac{1}{2} \langle \pi, \pi \rangle_h , \tag{17.12}$$

$$\mathcal{U}_h(h) := \frac{1}{2} \langle h_{ij}, \mathcal{G}_{ij}(h) \rangle_h . \tag{17.13}$$

The finite quadratic ADM action is

$$\boxed{S_h^{(2)} = \int \mathrm{d}t \left[\left\langle \pi_{ij}, \dot{h}_{ij} \right\rangle_h - \mathcal{T}_h - \mathcal{U}_h + \langle n, \mathcal{C} \rangle_h - \langle n_i, \mathcal{C}_i \rangle_h \right].} \tag{17.14}$$

The momentum variation uses

$$\delta\mathcal{T}_h = 2 \langle \delta\pi_{ij}, \pi_{ij} \rangle_h - \langle \delta\pi, \pi \rangle_h \tag{17.15}$$

$$= \langle \delta\pi_{ij}, 2\pi_{ij} - \delta_{ij}\pi \rangle_h . \tag{17.16}$$

For the shift term,

$$- \langle n_i, \mathcal{C}_i \rangle_h = 2 \langle n_i, \delta_j \pi_{ij} \rangle_h \tag{17.17}$$

$$= -2 \langle \delta_j n_i, \pi_{ij} \rangle_h \tag{17.18}$$

$$= - \langle \delta_i n_j + \delta_j n_i, \pi_{ij} \rangle_h , \tag{17.19}$$

where symmetry of π_{ij} is used in the last line. Variation gives

$$\frac{\delta S_h^{(2)}}{\delta n} = 0 \Longrightarrow \mathcal{C} = 0, \tag{17.20}$$

$$\frac{\delta S_h^{(2)}}{\delta n_i} = 0 \Longrightarrow \mathcal{C}_i = 0, \tag{17.21}$$

$$\boxed{\dot{h}_{ij} = 2\pi_{ij} - \delta_{ij}\pi + \delta_i n_j + \delta_j n_i,} \tag{17.22}$$

and

$$\boxed{\dot{\pi}_{ij} = -\mathcal{G}_{ij}(h) + \delta_i\delta_j n - \delta_{ij}\delta^2 n.} \tag{17.23}$$

For the lapse term, two integrations by parts give

$$\left\langle n, \delta_i\delta_j\delta h_{ij} - \delta^2\delta h \right\rangle_h = \left\langle \delta_i\delta_j n - \delta_{ij}\delta^2 n, \delta h_{ij} \right\rangle_h . \tag{17.24}$$

Self-adjointness of $\mathcal{G}$ gives

$$\delta\mathcal{U}_h = \frac{1}{2} \langle \delta h, \mathcal{G}(h) \rangle_h + \frac{1}{2} \langle h, \mathcal{G}(\delta h) \rangle_h \tag{17.25}$$

$$= \langle \delta h, \mathcal{G}(h) \rangle_h . \tag{17.26}$$

Combining these variations with $- \langle \dot{\pi}_{ij}, \delta h_{ij} \rangle_h$, obtained after integrating $\left\langle \pi_{ij}, \delta\dot{h}_{ij} \right\rangle_h$ in time, yields the momentum equation. The variation identity

$$\delta\mathcal{U}_h = \langle \delta h_{ij}, \mathcal{G}_{ij}(h) \rangle_h \tag{17.27}$$

follows from

$$\langle a_{ij}, \mathcal{G}_{ij}(b) \rangle_h = \langle \mathcal{G}_{ij}(a), b_{ij} \rangle_h . \tag{17.28}$$

17.4 Exact finite constraint propagation

Using the canonical equations,

$$\dot{\mathcal{C}} = \delta_i\delta_j\dot{h}_{ij} - \delta^2\dot{h} \tag{17.29}$$
$$= 2\delta_i\delta_j\pi_{ij} \tag{17.30}$$
$$= -\delta_i\mathcal{C}_i. \tag{17.31}$$

For the shift part,

$$\delta_i\delta_j(\delta_i n_j + \delta_j n_i) - \delta^2(2\delta_k n_k) = 0 \tag{17.32}$$

after commuting derivatives. For the momentum part,

$$\delta_i\delta_j(2\pi_{ij} - \delta_{ij}\pi) - \delta^2(-\pi) = 2\delta_i\delta_j\pi_{ij}. \tag{17.33}$$

The shift terms cancel identically. The momentum constraint obeys

$$\dot{\mathcal{C}}_i = -2\delta_j\dot{\pi}_{ij} \tag{17.34}$$
$$= 2\delta_j\mathcal{G}_{ij} - 2\delta_j(\delta_i\delta_j - \delta_{ij}\delta^2)n \tag{17.35}$$
$$= 0. \tag{17.36}$$

The first term vanishes by the discrete linearized Bianchi identity. The lapse terms cancel because

$$\delta_j\delta_i\delta_j n = \delta_i\delta^2 n = \delta_j\delta_{ij}\delta^2 n. \tag{17.37}$$

Therefore

$$\boxed{\dot{\mathcal{C}} = -\delta_i\mathcal{C}_i, \qquad \dot{\mathcal{C}}_i = 0.} \tag{17.38}$$

In particular,

$$\mathcal{C}(0) = 0, \qquad \mathcal{C}_i(0) = 0 \Longrightarrow \mathcal{C}(t) = 0, \qquad \mathcal{C}_i(t) = 0. \tag{17.39}$$

17.5 Linearized gauge symmetry and bracket closure

For

$$h_{ij} \mapsto h_{ij} + \delta_i\xi_j + \delta_j\xi_i, \tag{17.40}$$

one has

$$\mathcal{C}[h + \delta\xi + (\delta\xi)^{\mathsf{T}}] = \mathcal{C}[h]. \tag{17.41}$$

The change in the scalar constraint is

$$\Delta\mathcal{C} = \delta_i\delta_j(\delta_i\xi_j + \delta_j\xi_i) - \delta^2(2\delta_k\xi_k) \tag{17.42}$$
$$= 2\delta^2\delta_k\xi_k - 2\delta^2\delta_k\xi_k = 0. \tag{17.43}$$

Let

$$\mathcal{C}[n] := \langle n, \mathcal{C}\rangle_h, \tag{17.44}$$
$$\mathcal{C}[\boldsymbol{\xi}] := \langle \xi_i, \mathcal{C}_i\rangle_h. \tag{17.45}$$

With the canonical trace bracket,

$$\boxed{\{\mathcal{C}[n], \mathcal{C}[m]\} = 0, \qquad \{\mathcal{C}[\boldsymbol{\xi}], \mathcal{C}[\boldsymbol{\eta}]\} = 0, \qquad \{\mathcal{C}[n], \mathcal{C}[\boldsymbol{\xi}]\} = 0.} \tag{17.46}$$

Thus the finite linearized constraint algebra is exactly Abelian.

17.6 TT sector and continuum limit

If

$$\delta_j h_{ij} = 0, \qquad h = 0, \qquad \delta_j \pi_{ij} = 0, \qquad \pi = 0, \tag{17.47}$$

then

$$\mathcal{G}_{ij}(h) = -\frac{1}{2}\delta^2 h_{ij}, \tag{17.48}$$

$$\dot{h}_{ij} = 2\pi_{ij}, \tag{17.49}$$

$$\dot{\pi}_{ij} = \frac{1}{2}\delta^2 h_{ij}, \tag{17.50}$$

$$\ddot{h}_{ij} = \delta^2 h_{ij}. \tag{17.51}$$

The divergence and trace conditions remove every term in $\mathcal{G}_{ij}$ except $-\delta^2 h_{ij}/2$. Differentiating $\dot{h}_{ij} = 2\pi_{ij}$ once and substituting $\dot{\pi}_{ij}$ gives the wave equation. For the centered lattice derivative,

$$\widetilde{k}_i(h) = \frac{\sin(k_i h)}{h}, \tag{17.52}$$

Using

$$\frac{\sin(k_i h)}{h} = k_i - \frac{k_i^3 h^2}{6} + \mathcal{O}(h^4), \tag{17.53}$$

one finds

$$\widetilde{k}_i(h)^2 = k_i^2 - \frac{k_i^4 h^2}{3} + \mathcal{O}(h^4). \tag{17.54}$$

Summing over i gives the expansion of ω_h^2, and taking the square root gives the expansion of ω_h. so

$$\boxed{\omega_h^2 = \sum_i \widetilde{k}_i(h)^2 = k^2 - \frac{h^2}{3}\sum_i k_i^4 + \mathcal{O}(h^4),} \tag{17.55}$$

and

$$\boxed{\omega_h = k - \frac{h^2}{6k}\sum_i k_i^4 + \mathcal{O}(h^4).} \tag{17.56}$$

The measured asymptotic order is

$$p_\omega = 1.985855. \tag{17.57}$$

17.7 Constraint refinement

For a partition of the fine cells into coarse regions A, define

$$E_A^{\mathrm{con}} := \sum_{x \in A} h^3 \left(|\mathcal{C}_x|^2 + \sum_i |\mathcal{C}_{i,x}|^2 \right). \tag{17.58}$$

Then

$$\boxed{\sum_A E_A^{\mathrm{con}} = \sum_x h^3 \left(|\mathcal{C}_x|^2 + \sum_i |\mathcal{C}_{i,x}|^2 \right).} \tag{17.59}$$

17.8 Numerical verification

$$\max_i \|\delta_j \mathcal{G}_{ij}\|_\infty = 2.220446 \times 10^{-15}, \tag{17.60}$$

$$\|\dot{\mathcal{C}} + \delta_i \mathcal{C}_i\|_\infty = 1.199041 \times 10^{-14}, \tag{17.61}$$

$$\max_i \|\dot{\mathcal{C}}_i\|_\infty = 5.329071 \times 10^{-15}, \tag{17.62}$$

$$\|\mathcal{C}[h + \delta\xi + (\delta\xi)^{\mathsf{T}}] - \mathcal{C}[h]\|_\infty = 1.154632 \times 10^{-14}, \tag{17.63}$$

$$|\{\mathcal{C}[n], \mathcal{C}[\boldsymbol{\xi}]\}| = 1.450620 \times 10^{-14}, \tag{17.64}$$

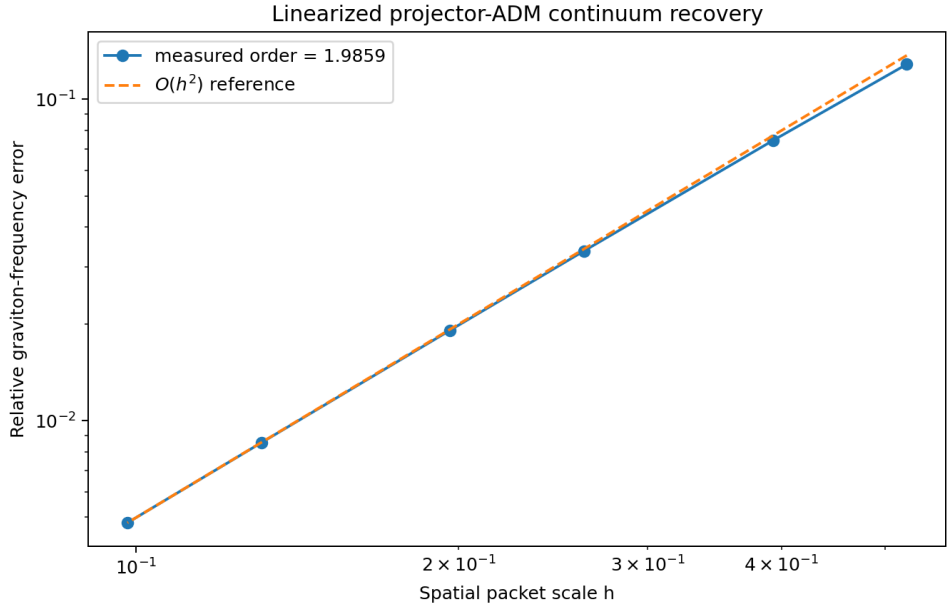
$$|\delta\mathcal{U}_h - \langle \delta h, \mathcal{G}(h)\rangle_h| = 4.092726 \times 10^{-12}, \tag{17.65}$$

$$\left|\frac{\mathrm{d}}{\mathrm{d}t}(\mathcal{T}_h + \mathcal{U}_h)\right| = 0, \tag{17.66}$$

$$\frac{|E^{\mathrm{con}}_{\mathrm{coarse}} - E^{\mathrm{con}}_{\mathrm{fine}}|}{E^{\mathrm{con}}_{\mathrm{fine}}} = 1.505862 \times 10^{-16}, \tag{17.67}$$

$$\max_h \max(\|\mathcal{C}_{\mathrm{TT}}\|_\infty, \|\mathcal{C}_{i,\mathrm{TT}}\|_\infty) = 5.107026 \times 10^{-14}, \tag{17.68}$$

$$\max_h \frac{|E_h(t) - E_h(0)|}{|E_h(0)|} = 4.908588 \times 10^{-16}. \tag{17.69}$$



17.9 Next steps

- Derive a nonlinear local $3+1$ projector–ADM action.
- Construct the nonlinear Hamiltonian and momentum constraints from that action.
- Transfer the constraints through the BFV–BV projector refinement and prove exact first-class closure at finite resolution.
- Establish a controlled Einstein–matter continuum-limit theorem.

Chapter 18

Nonlinear Constraint Closure: Pointwise Defect and Corner Extension

At nonlinear order, a finite pointwise derivative does not satisfy the Leibniz rule on the full lattice algebra. The resulting constraint defect is exact and identifies what a successful finite theory must retain: the detail channels created by products.

18.1 Scalar diffeomorphism model

Let the periodic finite lattice carry

$$\{q_a, p_b\}_h = \frac{\delta_{ab}}{h}, \tag{18.1}$$

and a skew derivative

$$D^{\mathsf{T}} = -D. \tag{18.2}$$

Define the smeared scalar diffeomorphism constraint

$$\boxed{\mathcal{D}_h[\xi] := h \sum_a \xi_a p_a (Dq)_a.} \tag{18.3}$$

Its coordinate derivatives are

$$\frac{\partial \mathcal{D}_h[\xi]}{\partial p_a} = h \xi_a (Dq)_a, \tag{18.4}$$

$$\frac{\partial \mathcal{D}_h[\xi]}{\partial q_b} = h \sum_a \xi_a p_a D_{ab} \tag{18.5}$$

$$= h \big[D^{\mathsf{T}}(\xi p)\big]_b = -h [D(\xi p)]_b. \tag{18.6}$$

The discrete Leibniz defect is

$$\boxed{\mathfrak{L}_D(f,g) := D(fg) - (Df)g - f(Dg).} \tag{18.7}$$

Direct canonical differentiation gives

$$\boxed{\begin{aligned}\{\mathcal{D}_h[\xi], \mathcal{D}_h[\eta]\}_h &= \mathcal{D}_h[\xi D\eta - \eta D\xi] \\ &\quad + h \sum_a p_a \left[\xi_a \mathfrak{L}_D(\eta, Dq)_a - \eta_a \mathfrak{L}_D(\xi, Dq)_a\right].\end{aligned}} \tag{18.8}$$

Substituting the coordinate derivatives into the canonical bracket and using $D^{\mathsf{T}} = -D$ gives

$$\{\mathcal{D}_h[\xi], \mathcal{D}_h[\eta]\}_h = h \sum_a p_a \left[\xi_a D(\eta Dq)_a - \eta_a D(\xi Dq)_a\right]. \tag{18.9}$$

Now expand each product derivative:

$$D(\eta Dq) = (D\eta)(Dq) + \eta D^2 q + \mathfrak{L}_D(\eta, Dq), \tag{18.10}$$
$$D(\xi Dq) = (D\xi)(Dq) + \xi D^2 q + \mathfrak{L}_D(\xi, Dq). \tag{18.11}$$

The two $\xi\eta D^2 q$ terms cancel. The remaining product-rule part is

$$h\sum_a p_a(\xi_a D\eta_a - \eta_a D\xi_a)(Dq)_a = \mathcal{D}_h[\xi D\eta - \eta D\xi], \tag{18.12}$$

leaving exactly the displayed defect contraction. Hence

$$\boxed{\text{nonlinear bracket defect} = \text{Leibniz-defect contraction.}} \tag{18.13}$$

18.2 Finite pointwise obstruction

The no-go theorem

$$\mathrm{Der}(\mathbb{C}^N) = \{0\} \tag{18.14}$$

implies

$$D \neq 0 \Longrightarrow \exists (f, g) : \mathfrak{L}_D(f, g) \neq 0 \tag{18.15}$$

for every derivative acting on the complete finite pointwise algebra. Therefore

$$\boxed{\text{the undeformed nonlinear diffeomorphism algebra cannot close universally}} \tag{18.16}$$

on a finite pointwise lattice through an ordinary nonzero Leibniz derivative.

18.3 Controlled finite resolutions

For the centered derivative and smooth resolved data,

$$\mathfrak{L}_{D_h}(f, g) = \mathcal{O}(h^2), \tag{18.17}$$

For the centered difference,

$$D_h f = f' + \frac{h^2}{6} f''' + \mathcal{O}(h^4). \tag{18.18}$$

Therefore

$$\mathfrak{L}_{D_h}(f, g) = \frac{h^2}{6}\left[(fg)''' - f'''g - fg'''\right] + \mathcal{O}(h^4) \tag{18.19}$$
$$= \frac{h^2}{2}\left(f''g' + f'g''\right) + \mathcal{O}(h^4). \tag{18.20}$$

and therefore

$$\{\mathcal{D}_h[\xi], \mathcal{D}_h[\eta]\}_h - \mathcal{D}_h[\xi D_h\eta - \eta D_h\xi] = \mathcal{O}(h^2). \tag{18.21}$$

The measured asymptotic order is

$$p_{\text{closure}} = 1.985564. \tag{18.22}$$

For a Fourier window whose products remain below the Nyquist boundary,

$$\mathfrak{L}_{D_{\text{spec}}}(f, g) = 0 \tag{18.23}$$

within floating-point accuracy, although that window is not the full pointwise algebra.

18.4 Corner-algebra route

For

$$\delta_K X = \mathrm{i}[K, X] \tag{18.24}$$

on the full corner algebra,

$$\boxed{\mathfrak{L}_{\delta_K}(X, Y) = 0.} \tag{18.25}$$

Indeed,

$$\begin{aligned} \delta_K(XY) &= \mathrm{i}(KXY - XYK) && (18.26)\\ &= \mathrm{i}(KX - XK)Y + \mathrm{i}X(KY - YK) && (18.27)\\ &= (\delta_K X)Y + X(\delta_K Y). && (18.28) \end{aligned}$$

Unlike the pointwise finite derivative, the commutator derivation satisfies the product rule on the entire matrix algebra. The corner extension therefore removes the finite Leibniz obstruction. A direct coarse truncation can still fail, however, because a nonlinear transformation may leave the retained subspace, propagate through a detail channel, and return. The next chapter identifies this coarse–detail–coarse leakage exactly and restores it by homological transfer.

18.5 Nonlinear projector–ADM target

The canonical action should be written in either of the equivalent forms

$$\boxed{S_{\mathrm{PADM},h} = \int \mathrm{d}t \left\{ \mathrm{Tr}(\Pi^{ij}\dot{Q}_{ij} + P_\Phi \dot{\Phi}) - \mathcal{H}_{\mathcal{P},h}[N] - \mathcal{D}_{\mathcal{P},h}[\boldsymbol{N}] \right\}} \tag{18.29}$$

or, when the momentum generator acts by the ordered tangential derivative,

$$\boxed{S_{\mathrm{PADM},h} = \int \mathrm{d}t \left\{ \mathrm{Tr}[\Pi^{ij}(\dot{Q}_{ij} - \mathcal{L}^{\mathcal{P}}_{\boldsymbol{N},h} Q_{ij}) + P_\Phi(\dot{\Phi} - \mathcal{L}^{\mathcal{P}}_{\boldsymbol{N},h}\Phi)] - \mathcal{H}_{\mathcal{P},h}[N] \right\}.} \tag{18.30}$$

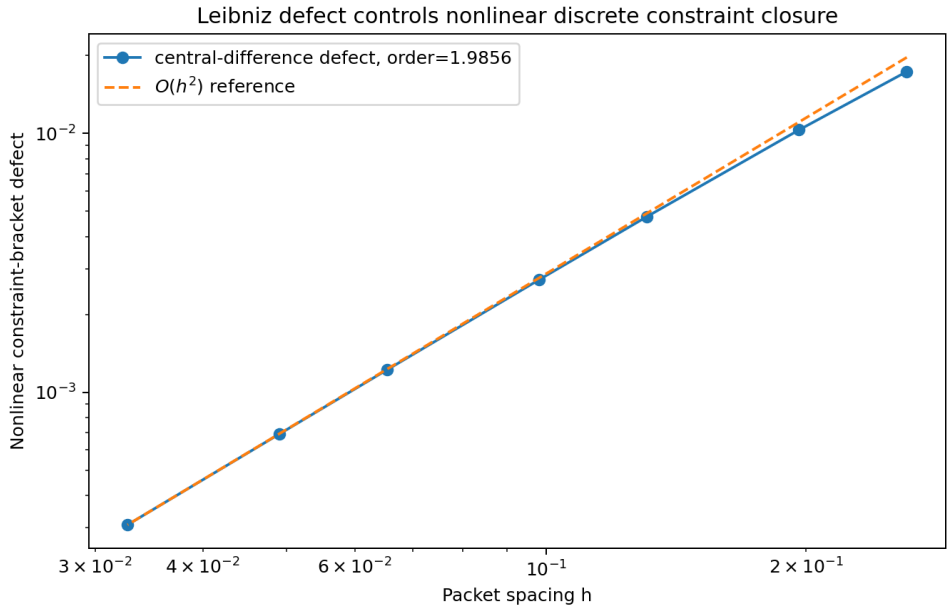
The shift contribution appears once: either as $\mathcal{D}_{\mathcal{P},h}[\boldsymbol{N}]$ or inside the Lie-dragged velocities.
Chapter 19 constructs a transferred BFV charge $\Omega_{C,h}$. It fixes the previously missing objects by

$$\begin{aligned} \mathcal{H}_{\mathcal{P},h} &:= \left.\frac{\partial \Omega_{C,h}}{\partial c^\perp}\right|_{c=\varpi=0}, && (18.31)\\ \mathcal{H}_{\mathcal{P},h,i} &:= \left.\frac{\partial \Omega_{C,h}}{\partial c^i}\right|_{c=\varpi=0}, && (18.32)\\ \mathcal{L}^{\mathcal{P}}_{\boldsymbol{N},h} X &:= \{X, \Omega_{C,h}\}_C|_{c^i = N^i,\ c^\perp = 0,\ \varpi = 0}, && (18.33)\\ \{\cdot,\cdot\}_{\mathcal{P},\mathrm{ordered}} &:= \text{the canonical cyclic trace BFV bracket}, && (18.34)\\ \text{nonlinear first-class closure} &\Longleftrightarrow \{\Omega_{C,h}, \Omega_{C,h}\}_C = 0. && (18.35) \end{aligned}$$

With a cyclic, $*$-compatible contraction, the transfer supplies exact finite $*$-compatibility, trace integration by parts, the correct linearized limit, a closed nonlinear constraint surface, and refinement-path stability. Convergence to the Einstein–matter continuum theory remains a separate physical limit theorem.

18.6 Numerical verification

$$\begin{aligned} \max_h |(\{\mathcal{D}_h[\xi], \mathcal{D}_h[\eta]\} - \mathcal{D}_h[\xi D\eta - \eta D\xi]) - \mathfrak{E}_h| &= 1.873501 \times 10^{-16}, && (18.36)\\ \mathfrak{E}_h &:= h \sum_a p_a [\xi_a \mathfrak{L}_D(\eta, Dq)_a - \eta_a \mathfrak{L}_D(\xi, Dq)_a], && (18.37)\\ |\text{spectral low-band closure defect}| &= 6.696033 \times 10^{-16}, && (18.38)\\ |\text{spectral direct-defect mismatch}| &= 7.281546 \times 10^{-17}, && (18.39)\\ \|\mathfrak{L}_{\delta_K}(X, Y)\|_2 &= 7.192762 \times 10^{-14}. && (18.40) \end{aligned}$$


Leibniz defect controls nonlinear discrete constraint closure
central-difference defect, order=1.9856
$O(h^2)$ reference
Nonlinear constraint-bracket defect
10^{-2}
10^{-3}
3×10^{-2}
4×10^{-2}
6×10^{-2}
10^{-1}
2×10^{-1}
Packet spacing h

Chapter 19

The Nonlinear Gauge-Closure Bridge: BFV–BV Polyphase Transfer

The pointwise obstruction of the preceding chapter is not repaired by choosing a more elaborate finite-difference rule. Nonlinear gauge transformations create unresolved products, and those detail channels can return to the retained variables. Given a nilpotent fine BFV theory and an admissible contraction of its detail complex, the homological perturbation formula restores every such return. A compatible BV pushforward then preserves the quantum master equation. This closes the finite algebraic transfer step under explicit hypotheses. Those hypotheses include a consistent fine BFV–BV theory; proving that a specific microscopic four-dimensional gravity model satisfies them remains a separate problem.

19.1 Canonical corner phase space

Let the retained canonical variables be matrix-valued pairs

$$(Z^A, P_A) \in \mathcal{A}_h \times \mathcal{A}_h, \qquad \mathcal{A}_h = M_{N_h}(\mathbb{C}), \tag{19.1}$$

where the collection may contain

$$(Q_{ij}, \Pi^{ij}), \qquad (\Phi, P_\Phi), \qquad \text{connection, frame, or clock pairs as required.} \tag{19.2}$$

The trace symplectic form is

$$\Omega_h^{\mathrm{can}} := \sum_A \mathrm{Tr}(\,\mathrm{d}P_A \wedge \mathrm{d}Z^A), \tag{19.3}$$

and the canonical bracket on real matrix components is

$$\boxed{\{F, G\}_h = \sum_A \mathrm{Tr}\left(\nabla_{Z^A} F\, \nabla_{P_A} G - \nabla_{P_A} F\, \nabla_{Z^A} G\right).} \tag{19.4}$$

The gradients are cyclic matrix gradients. Thus the multiplication order is fixed by the fine action and the canonical symplectic structure before any constraint is reduced.

19.2 Exact inner momentum sector

For a Hermitian matrix parameter Λ, define

$$\boxed{\mathcal{D}_h[\Lambda] := \mathrm{i} \sum_A \mathrm{Tr}\big(P_A[\Lambda, Z^A]\big).} \tag{19.5}$$

For one canonical pair,

$$\delta\mathcal{D}_h[\Lambda] = \mathrm{i}\,\mathrm{Tr}(\delta P[\Lambda, Z]) + \mathrm{i}\,\mathrm{Tr}(P[\Lambda, \delta Z]) \tag{19.6}$$

$$= \operatorname{Tr}(\mathrm{i}[\Lambda, Z]\,\delta P) + \operatorname{Tr}(\mathrm{i}[P, \Lambda]\,\delta Z)\,. \tag{19.7}$$

Hence

$$\nabla_P \mathcal{D}_h = \mathrm{i}[\Lambda, Z], \qquad \nabla_Z \mathcal{D}_h = \mathrm{i}[P, \Lambda]. \tag{19.8}$$

The canonical bracket then gives

$$\{Z, \mathcal{D}_h\} = \nabla_P \mathcal{D}_h, \qquad \{P, \mathcal{D}_h\} = -\nabla_Z \mathcal{D}_h. \tag{19.9}$$

Direct variation gives

$$\{Z^A, \mathcal{D}_h[\Lambda]\}_h = \mathrm{i}[\Lambda, Z^A], \tag{19.10}$$
$$\{P_A, \mathcal{D}_h[\Lambda]\}_h = \mathrm{i}[\Lambda, P_A]. \tag{19.11}$$

Thus $\mathcal{D}_h[\Lambda]$ is the moment map for simultaneous inner conjugation of every canonical corner variable.

Theorem 19.1 (Exact finite momentum closure)**.** *For Hermitian Λ and Γ,*

$$\boxed{\{\mathcal{D}_h[\Lambda], \mathcal{D}_h[\Gamma]\}_h = \mathcal{D}_h[\mathrm{i}[\Lambda, \Gamma]].} \tag{19.12}$$

If a matrix density $\mathcal{E}$ transforms covariantly,

$$\{\mathcal{E}, \mathcal{D}_h[\Lambda]\}_h = \mathrm{i}[\Lambda, \mathcal{E}], \tag{19.13}$$

then the smeared functional $H_h[N] := \operatorname{Tr}(N\mathcal{E})$ obeys

$$\boxed{\{\mathcal{D}_h[\Lambda], H_h[N]\}_h = H_h[\mathrm{i}[\Lambda, N]].} \tag{19.14}$$

Proof. Acting successively on a coordinate Z^A gives

$$\{\{Z^A, \mathcal{D}_h[\Gamma]\}, \mathcal{D}_h[\Lambda]\} - \{\{Z^A, \mathcal{D}_h[\Lambda]\}, \mathcal{D}_h[\Gamma]\} = -\big[\Lambda, [\Gamma, Z^A]\big] + \big[\Gamma, [\Lambda, Z^A]\big] \tag{19.15}$$
$$= \mathrm{i}\big[\mathrm{i}[\Lambda, \Gamma], Z^A\big]. \tag{19.16}$$

The Jacobi identity for the Poisson bracket identifies the left side with the action of $\{\mathcal{D}_h[\Lambda], \mathcal{D}_h[\Gamma]\}$. The same calculation holds for every P_A. Therefore the difference

$$\{\mathcal{D}_h[\Lambda], \mathcal{D}_h[\Gamma]\} - \mathcal{D}_h[\mathrm{i}[\Lambda, \Gamma]] \tag{19.17}$$

Poisson-commutes with every canonical coordinate. It is constant on the connected canonical phase space and vanishes at $P_A = 0$, so it is zero.

For the covariant density,

$$\{\mathcal{D}_h[\Lambda], H_h[N]\} = -\{H_h[N], \mathcal{D}_h[\Lambda]\} \tag{19.18}$$
$$= -\operatorname{Tr}(N\,\mathrm{i}[\Lambda, \mathcal{E}]) \tag{19.19}$$
$$= \operatorname{Tr}(\mathrm{i}[\Lambda, N]\mathcal{E}) \tag{19.20}$$
$$= H_h[\mathrm{i}[\Lambda, N]], \tag{19.21}$$

where cyclicity moves Λ through the trace. □

This is an exact finite spatial-gauge skeleton. A continuum shift is represented by a family $\Lambda_h[\boldsymbol{N}]$ whose action on resolved observables approaches the ordinary Lie derivative. At finite resolution the closed object is the full inner algebra, not a nonclosed subset of guessed first-order matrices.

19.3 Projection leakage is the finite nonlinear defect

Let $(\mathcal{C}_F, s_F)$ be a fine BRST complex with

$$s_F^2 = 0. \tag{19.22}$$

Let

$$P : \mathcal{C}_F \to \mathcal{C}_C, \qquad I : \mathcal{C}_C \to \mathcal{C}_F, \qquad PI = I_C, \tag{19.23}$$

and define the fine detail projector

$$Q_F := I_F - IP. \tag{19.24}$$

The naive projected differential is

$$\bar{s}_C := Ps_F I. \tag{19.25}$$

It need not be nilpotent.

Theorem 19.2 (Exact projection-leakage identity)**.** *The complete failure of the naive projected gauge algebra is*

$$\boxed{\bar{s}_C^2 = -Ps_F Q_F s_F I.} \tag{19.26}$$

Proof. Since $IP = I_F - Q_F$ and $s_F^2 = 0$,

$$\bar{s}_C^2 = Ps_F(IP)s_F I \tag{19.27}$$
$$= Ps_F(I_F - Q_F)s_F I \tag{19.28}$$
$$= -Ps_F Q_F s_F I. \tag{19.29}$$

□

The right side is a coarse-to-detail-to-coarse round trip. In the scalar lattice model it appears as the Leibniz-defect contraction. In a product-closed spectral window, $Q_F s_F I = 0$, so direct projection is already exact.

19.4 Fine BFV charge and contracting data

Introduce odd ghosts c^A and conjugate ghost momenta ϖ_A. A fine first-class system is encoded by an odd BFV charge

$$\Omega_F = c^A \mathcal{C}_{F,A} - \frac{1}{2} c^A c^B U_{F,AB}{}^C \varpi_C + \Omega_F^{(\geq 3)}, \tag{19.30}$$

with

$$\boxed{\{\Omega_F, \Omega_F\}_F = 0.} \tag{19.31}$$

The associated BRST differential is

$$s_F := \{\Omega_F, \cdot\}_F. \tag{19.32}$$

The higher ghost terms encode field-dependent Jacobi relations, reducibility, or open-algebra terms when they occur. For a nonlinear theory it is convenient to regard s_F as a continuous odd coderivation on the completed BRST coalgebra. Split it as

$$s_F = s_0 + \delta, \tag{19.33}$$

where s_0 is the linearized differential and δ contains nonlinear vertices. Assume a contraction

$$PI = I_C, \tag{19.34}$$
$$IP = I_F - s_0 H - H s_0, \tag{19.35}$$
$$PH = 0, \qquad HI = 0, \qquad H^2 = 0. \tag{19.36}$$

Here H is the detail Green operator, or contracting homotopy. It propagates a detail excitation before the excitation returns to the retained sector. The sign in the second line fixes the signs in the transfer series below.

19.5 Exact nonlinear BFV transfer

Theorem 19.3 (Nonlinear homological transfer)**.** *Suppose* $(s_0 + \delta)^2 = 0$ *and* $I_F + H\delta$ *is invertible, either analytically or as a formal power series. Define*

$$\boxed{s_C := s_{0,C} + P\delta(I_F + H\delta)^{-1} I, \qquad s_{0,C} := Ps_0 I.} \tag{19.37}$$

Then

$$\boxed{s_C^2 = 0.} \tag{19.38}$$

The explicit return series is

$$\boxed{\begin{aligned} s_C = s_{0,C} + P\delta I - P\delta H\delta I \\ + P\delta H\delta H\delta I - \cdots . \end{aligned}} \tag{19.39}$$

This is the Neumann expansion

$$(I_F + H\delta)^{-1} = I_F - H\delta + (H\delta)^2 - \cdots . \tag{19.40}$$

The factor H between two nonlinear vertices records one propagation through the detail complex. The leftmost P and rightmost I turn the resulting path into a coarse-to-coarse operation. If the contraction is cyclic and compatible with the BFV symplectic form, the transferred differential is Hamiltonian. On a connected canonical BFV algebra there is then a coarse charge Ω_C satisfying

$$s_C = \{\Omega_C, \cdot\}_C, \qquad \boxed{\{\Omega_C, \Omega_C\}_C = 0.} \tag{19.41}$$

Proof. Set

$$I_\delta := (I_F + H\delta)^{-1} I. \tag{19.42}$$

The inverse relation gives

$$I_\delta = I - H\delta I_\delta. \tag{19.43}$$

Because $PH = 0$,

$$PI_\delta = PI - PH\delta I_\delta = I_C. \tag{19.44}$$

Thus I_δ is injective.

For completeness, define the possible intertwining defect

$$K := (s_0 + \delta) I_\delta - I_\delta s_C. \tag{19.45}$$

Using $s_0 I = I s_{0,C}$, the contraction identity $I_F - IP = s_0 H + H s_0$, and $s_C = s_{0,C} + P\delta I_\delta$, direct substitution reduces the defect to

$$K = H\left(s_0 \delta I_\delta + \delta I_\delta s_{0,C} + \delta I_\delta P \delta I_\delta\right). \tag{19.46}$$

Nilpotency of $s_0 + \delta$ gives

$$s_0\delta + \delta s_0 + \delta^2 = 0. \tag{19.47}$$

Applying this identity to the definition of K shows that the parenthesis above equals $-\delta K$. Hence

$$K = -H\delta K, \qquad (I_F + H\delta)K = 0. \tag{19.48}$$

Invertibility of $I_F + H\delta$ forces $K = 0$, proving

$$(s_0 + \delta) I_\delta = I_\delta s_C. \tag{19.49}$$

Applying $s_0 + \delta$ once more gives

$$0 = (s_0 + \delta)^2 I_\delta = I_\delta s_C^2. \tag{19.50}$$

Injectivity of I_δ now gives $s_C^2 = 0$.

Cyclic symplectic transfer makes s_C Hamiltonian. Its square is the Hamiltonian vector field of $\frac{1}{2}\{\Omega_C, \Omega_C\}_C$. This element is central and has ghost number two, so it vanishes in the connected canonical BFV algebra. □

The first return term is

$$-P\delta H\delta I. \tag{19.51}$$

It cancels the first coarse–detail–coarse leakage omitted by direct truncation. Repeated interactions require the higher alternating returns.

An exact seven-dimensional test makes the cancellation visible. A nilpotent fine differential obtained by a rational change of basis has the naive coarse block

$$\bar{s}_C = \begin{pmatrix} -1/25 & -1/25 & -26/625 \\ 1 & 0 & 1/25 \\ 0 & 1/25 & 0 \end{pmatrix}, \qquad \bar{s}_C^2 \neq 0. \tag{19.52}$$

The transferred block is

$$\boxed{s_C = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1/25 \\ 0 & 0 & 0 \end{pmatrix}, \qquad s_C^2 = 0.} \tag{19.53}$$

The fine nilpotency, contraction identity, intertwining relation, and transferred nilpotency hold as exact rational equalities in the accompanying program.

19.6 Projector–ADM generators from the transferred charge

Split the gravitational ghosts into normal, tangential, and possible internal components,

$$c^A = (c^\perp, c^i, c^{\mathrm{int}}). \tag{19.54}$$

The retained Hamiltonian and momentum constraints are the ghost-linear coefficients of the transferred charge:

$$\boxed{\mathcal{H}_{\mathcal{P},h} := \left.\frac{\partial \Omega_{C,h}}{\partial c^\perp}\right|_{c=\varpi=0},} \tag{19.55}$$

$$\boxed{\mathcal{H}_{\mathcal{P},h,i} := \left.\frac{\partial \Omega_{C,h}}{\partial c^i}\right|_{c=\varpi=0}.} \tag{19.56}$$

For any retained field X, the ordered tangential action is

$$\boxed{\mathcal{L}^{\mathcal{P}}_{\boldsymbol{N},h} X := \{X, \Omega_{C,h}\}_C|_{c^i=N^i,\ c^\perp=0,\ \varpi=0}\,.} \tag{19.57}$$

The ordered bracket is the canonical cyclic trace BFV bracket. No separate multiplication convention is chosen after reduction; the order is inherited from Ω_F, the symplectic contraction, and the return series.

For an irreducible rank-one constraint algebra, write the relevant part of the coarse charge as

$$\begin{aligned} \Omega_{C,h} = {} & \mathcal{H}_h[c^\perp] + \mathcal{D}_h[\boldsymbol{c}] \\ & - \frac{1}{2}\left\langle \varpi_A, \mathbb{B}^A_h(c,c)\right\rangle + \Omega^{(\geq 3)}_{C,h}. \end{aligned} \tag{19.58}$$

The quadratic ghost coefficient of $\{\Omega_{C,h}, \Omega_{C,h}\}_C = 0$ gives

$$\{\mathcal{D}_h[\boldsymbol{\xi}], \mathcal{D}_h[\boldsymbol{\eta}]\}_C = \mathcal{D}_h[\mathbb{B}^h_{DD}(\boldsymbol{\xi}, \boldsymbol{\eta})], \tag{19.59}$$

$$\{\mathcal{D}_h[\boldsymbol{\xi}], \mathcal{H}_h[N]\}_C = \mathcal{H}_h[\mathbb{B}^h_{DH}(\boldsymbol{\xi}, N)], \tag{19.60}$$

$$\{\mathcal{H}_h[N], \mathcal{H}_h[M]\}_C = \mathcal{D}_h[\mathbb{B}^h_{HH}(N, M)]. \tag{19.61}$$

Schematically, if the ghost-linear part is $c^A \mathcal{C}_A$, then the coefficient of $c^A c^B$ in the master equation is

$$\{\mathcal{C}_A, \mathcal{C}_B\}_C - \mathbb{B}^C_{AB}\mathcal{C}_C. \tag{19.62}$$

The first term comes from the bracket of the two ghost-linear constraints. The second comes from contracting one ghost with its momentum in $-\frac{1}{2}\langle \varpi, \mathbb{B}(c,c)\rangle$. Setting each independent quadratic ghost coefficient to zero yields the three displayed constraint relations. If the fine system is open or reducible, the higher ghost terms supply the corresponding higher identities rather than being discarded. The continuum hypersurface-deformation algebra is recovered when

$$\mathbb{B}^h_{DD}(\boldsymbol{\xi}, \boldsymbol{\eta}) \longrightarrow [\boldsymbol{\xi}, \boldsymbol{\eta}], \tag{19.63}$$

$$\mathbb{B}^h_{DH}(\boldsymbol{\xi}, N) \longrightarrow \mathcal{L}_{\boldsymbol{\xi}} N, \tag{19.64}$$

$$\mathbb{B}^{h,i}_{HH}(N, M) \longrightarrow q^{ij}(N\partial_j M - M\partial_j N). \tag{19.65}$$

At linear order $\delta = 0$, so $\Omega_{C,h}$ reduces to the Abelian projector–ADM constraints derived earlier. At nonlinear order the return terms determine the induced structure maps and detail backreaction.

19.7 Quantum closure by BV pushforward

Let $\mathcal{F}_F$ be the graded space of fields, ghosts, and antifields with BV antibracket $(\cdot,\cdot)_F$ and BV Laplacian Δ_F. Suppose the regularized fine quantum action satisfies

$$\boxed{\frac{1}{2}(S_F, S_F)_F - \mathrm{i}\hbar\Delta_F S_F = 0.} \tag{19.66}$$

Equivalently,

$$\Delta_F \exp\left(\frac{\mathrm{i}}{\hbar}S_F\right) = 0. \tag{19.67}$$

The equivalence follows from the second-order product rule for the BV Laplacian:

$$\Delta_F \mathrm{e}^{\mathrm{i}S_F/\hbar} = \left[\frac{\mathrm{i}}{\hbar}\Delta_F S_F - \frac{1}{2\hbar^2}(S_F, S_F)_F\right]\mathrm{e}^{\mathrm{i}S_F/\hbar}. \tag{19.68}$$

Multiplying the vanishing bracket by $-\hbar^2$ gives the quantum master equation. Choose a symplectic splitting

$$\mathcal{F}_F = \mathcal{F}_C \oplus \mathcal{F}_D, \qquad \Delta_F = \Delta_C + \Delta_D, \tag{19.69}$$

and a detail-sector gauge-fixing Lagrangian $\mathcal{L}_D$. Define the coarse action by

$$\boxed{\exp\left(\frac{\mathrm{i}}{\hbar}S_C\right) := \mathcal{N}^{-1}\int_{\mathcal{L}_D}\exp\left(\frac{\mathrm{i}}{\hbar}S_F\right)\mathcal{D}\chi_D.} \tag{19.70}$$

Theorem 19.4 (Quantum master equation under projector elimination)**.** *If BV integration by parts has no boundary or regulator contribution, then*

$$\boxed{\frac{1}{2}(S_C, S_C)_C - \mathrm{i}\hbar\Delta_C S_C = 0.} \tag{19.71}$$

Proof. Using the fine master equation and BV Stokes' identity,

$$\Delta_C \exp\left(\frac{\mathrm{i}}{\hbar}S_C\right) = \mathcal{N}^{-1}\int_{\mathcal{L}_D}\Delta_C \exp\left(\frac{\mathrm{i}}{\hbar}S_F\right)\mathcal{D}\chi_D \tag{19.72}$$

$$= -\mathcal{N}^{-1}\int_{\mathcal{L}_D}\Delta_D \exp\left(\frac{\mathrm{i}}{\hbar}S_F\right)\mathcal{D}\chi_D \tag{19.73}$$

$$= 0. \tag{19.74}$$

□

This carries the multiplication order inherited from the fine action and the induced measure terms together. A determinant or loop contribution created by eliminating detail variables is part of the action required by the quantum master equation, not an optional normalization.

19.8 Exact finite projector action from a fine BV theory

Let S_F^{BV} be any regularized fine gravitational or gravity–matter BV action satisfying the preceding master equation. For a projector resolution h, split the variables into retained modes χ_h and detail modes $\chi_{>h}$, and define

$$\boxed{\exp\left(\frac{\mathrm{i}}{\hbar}S_h^{\mathrm{BV}}[\chi_h]\right) := \mathcal{N}_h^{-1}\int_{\mathcal{L}_{>h}}\exp\left(\frac{\mathrm{i}}{\hbar}S_F^{\mathrm{BV}}[\chi_h,\chi_{>h}]\right)\mathcal{D}\chi_{>h}.} \tag{19.75}$$

The preceding theorem gives

$$\boxed{\frac{1}{2}(S_h^{\mathrm{BV}}, S_h^{\mathrm{BV}})_h - \mathrm{i}\hbar\Delta_h S_h^{\mathrm{BV}} = 0} \tag{19.76}$$

at every admissible finite resolution. This is a constructive definition of the nonlinear projector theory. It is generally nonlocal, because exact elimination remembers the propagation of the discarded modes.

For nested resolutions $h_0 \prec h_1 \prec h_2$, exact integration gives the refinement semigroup

$$\boxed{\mathfrak{R}_{h_0 \leftarrow h_2} = \mathfrak{R}_{h_0 \leftarrow h_1} \circ \mathfrak{R}_{h_1 \leftarrow h_2}.} \tag{19.77}$$

Write the detail variables between h_2 and h_0 as $(\chi_{h_2:h_1}, \chi_{h_1:h_0})$. Then the direct pushforward is

$$\int \mathcal{D}\chi_{h_2:h_1} \mathcal{D}\chi_{h_1:h_0}\, \mathrm{e}^{\mathrm{i}S_{h_2}/\hbar}. \tag{19.78}$$

Performing the inner integral first defines S_{h_1}; performing the outer integral next defines S_{h_0}. Fubini's theorem, or its regularized BV analogue, identifies this iterated integral with direct elimination. If two detail families have a compatible product gauge fixing, their order can be exchanged:

$$\boxed{\mathfrak{R}_{D_1} \circ \mathfrak{R}_{D_2} = \mathfrak{R}_{D_2} \circ \mathfrak{R}_{D_1} = \mathfrak{R}_{D_1 \oplus D_2},} \tag{19.79}$$

up to a field-independent normalization. Failure of this identity after regularization is not a harmless discretization error; it is a possible measure or gauge anomaly that must be resolved.

19.9 Quadratic shadow: Schur and prime-polyphase composition

For a quadratic action with block Hessian L, exact detail elimination is

$$\boxed{L_{\text{eff}} = L_{CC} - L_{CD} L_{DD}^{-1} L_{DC}.} \tag{19.80}$$

For the quadratic form

$$\frac{1}{2} \begin{pmatrix} x_C \\ x_D \end{pmatrix}^* \begin{pmatrix} L_{CC} & L_{CD} \\ L_{DC} & L_{DD} \end{pmatrix} \begin{pmatrix} x_C \\ x_D \end{pmatrix}, \tag{19.81}$$

the detail stationary equation is

$$L_{DC} x_C + L_{DD} x_D = 0, \qquad x_D = -L_{DD}^{-1} L_{DC} x_C. \tag{19.82}$$

Substitution leaves

$$\frac{1}{2} x_C^* \left(L_{CC} - L_{CD} L_{DD}^{-1} L_{DC} \right) x_C. \tag{19.83}$$

When the required inverses exist, two detail blocks satisfy

$$\boxed{\mathrm{Schur}_{D_1 \oplus D_2} L = \mathrm{Schur}_{D_1}\big(\mathrm{Schur}_{D_2} L\big) = \mathrm{Schur}_{D_2}\big(\mathrm{Schur}_{D_1} L\big).} \tag{19.84}$$

The prime-polyphase Chebyshev semigroup is the free nearest-neighbor instance of this identity. Nonlinear interactions replace the single Schur term by the full BFV–BV tree and loop expansion while preserving the master equation.

Different contracting homotopies can give different coordinate formulas for Ω_C. When they represent homotopy-equivalent gauge fixings without an anomaly, the resulting actions are related by a BFV–BV canonical transformation and have the same gauge cohomology.

19.10 An exact nonlinear reduction model

A finite model shows why the quantum measure term is necessary. Let x_1, x_2 be retained variables and y, z detail variables. For $m, n > 0$, define the Euclidean action

$$S_F = \frac{1}{2}(x_2 - x_1 y)^2 + \frac{m}{2}(y - z)^2 + \frac{n}{2} z^2. \tag{19.85}$$

The fine gauge vector is

$$R_F = \partial_{x_1} + y \partial_{x_2}. \tag{19.86}$$

It satisfies

$$R_F S_F = 0, \qquad \operatorname{div} R_F = 0. \tag{19.87}$$

The combination $x_2 - x_1 y$ is invariant because

$$R_F(x_2 - x_1 y) = y - y = 0, \tag{19.88}$$

and R_F does not act on y, z. The vector components $(1, y, 0, 0)$ have zero divergence in (x_1, x_2, y, z). The detail Hessian and determinant are

$$M_D = \begin{pmatrix} x_1^2 + m & -m \\ -m & m + n \end{pmatrix}, \qquad D = (m + n)x_1^2 + mn. \tag{19.89}$$

The stationary detail variables are

$$y_* = \frac{(m + n)x_1 x_2}{D}, \tag{19.90}$$

$$z_* = \frac{m x_1 x_2}{D}. \tag{19.91}$$

The stationary equations are

$$(x_1^2 + m)y - mz = x_1 x_2, \tag{19.92}$$

$$-my + (m + n)z = 0. \tag{19.93}$$

The second gives $z = my/(m + n)$. Substitution into the first gives

$$\left(x_1^2 + \frac{mn}{m + n}\right) y = x_1 x_2, \tag{19.94}$$

which is equivalent to the displayed formulas. With

$$\mu := \frac{mn}{m + n}, \tag{19.95}$$

the exact coarse action is

$$\boxed{S_C = \frac{\mu x_2^2}{2(x_1^2 + \mu)}.} \tag{19.96}$$

Eliminating z first combines the last two quadratic terms:

$$\frac{m}{2}(y - z)^2 + \frac{n}{2}z^2\Big|_{z = my/(m+n)} = \frac{\mu}{2}y^2. \tag{19.97}$$

The remaining action is

$$\frac{1}{2}(x_2 - x_1 y)^2 + \frac{\mu}{2}y^2. \tag{19.98}$$

Its stationary point is

$$y_* = \frac{x_1 x_2}{x_1^2 + \mu}. \tag{19.99}$$

Substitution gives the displayed S_C. The induced nonlinear gauge vector is

$$\boxed{R_C = \partial_{x_1} + \frac{x_1 x_2}{x_1^2 + \mu}\partial_{x_2},} \tag{19.100}$$

and it obeys

$$R_C S_C = 0. \tag{19.101}$$

The coarse vector is the fine vector evaluated on the stationary detail section:

$$R_C x_1 = 1, \qquad R_C x_2 = y_* = \frac{x_1 x_2}{x_1^2 + \mu}. \tag{19.102}$$

For $d := x_1^2 + \mu$,

$$\partial_{x_1} S_C = -\frac{\mu x_1 x_2^2}{d^2}, \tag{19.103}$$

$$\partial_{x_2} S_C = \frac{\mu x_2}{d}. \tag{19.104}$$

Therefore

$$R_C S_C = -\frac{\mu x_1 x_2^2}{d^2} + \frac{x_1 x_2}{d}\frac{\mu x_2}{d} = 0. \tag{19.105}$$

Its ordinary divergence is nonzero:

$$\operatorname{div} R_C = \frac{x_1}{x_1^2 + \mu}. \tag{19.106}$$

Gaussian elimination contributes the density

$$\rho_C \propto D^{-1/2} \propto (x_1^2 + \mu)^{-1/2}. \tag{19.107}$$

Consequently,

$$\boxed{\operatorname{div} R_C + R_C(\log \rho_C) = 0.} \tag{19.108}$$

Indeed,

$$\operatorname{div} R_C = \partial_{x_1}(1) + \partial_{x_2}\left(\frac{x_1 x_2}{d}\right) = \frac{x_1}{d}, \tag{19.109}$$

$$R_C(\log \rho_C) = -\frac{1}{2} R_C \log\big((m+n)d\big) \tag{19.110}$$

$$= -\frac{1}{2}\frac{2x_1}{d} = -\frac{x_1}{d}. \tag{19.111}$$

The induced density supplies exactly the missing divergence term. With a flat coarse measure, the same Euclidean correction is recorded in

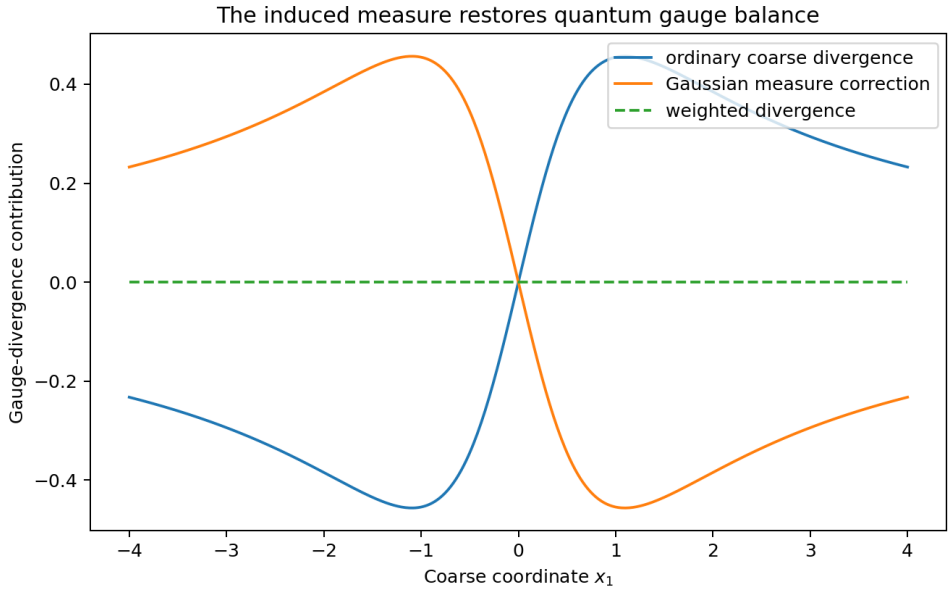
$$S_{C,\hbar} = S_C + \frac{\hbar}{2}\log D \tag{19.112}$$

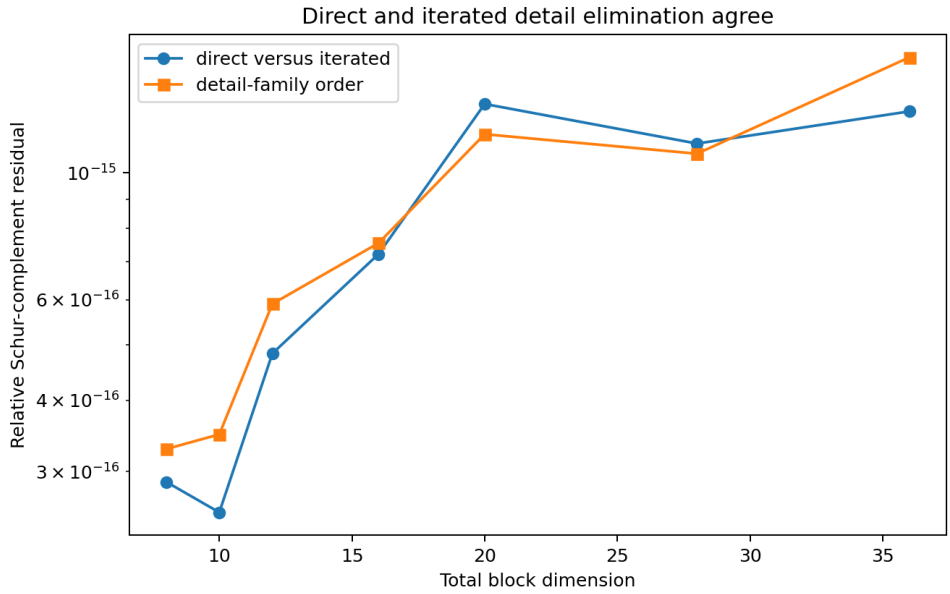
up to an additive constant. The classical action remains gauge invariant, while the determinant restores quantum measure balance.

Eliminating z first produces the reduced mass

$$\mu = \frac{mn}{m+n} \tag{19.113}$$

and determinant factor $(m+n)^{-1/2}$. Eliminating y next contributes $(x_1^2+\mu)^{-1/2}$. Their product is $D^{-1/2}$, and reversing the order gives the same action and determinant.





19.11 Controlled continuum limit

The first omitted return can be bounded directly. The resolvent identity gives

$$\begin{aligned} s_C - (s_{0,C} + P\delta I) &= P\delta\big[(I_F + H\delta)^{-1} - I_F\big]I \\ &= -P\delta(I_F + H\delta)^{-1}H\delta I. \end{aligned} \tag{19.114}$$

Assume

$$\|H\delta\| \le \rho < 1, \qquad \|Q_F\delta I\| \le Ch^r. \tag{19.115}$$

Since $HI = 0$,

$$\boxed{\|s_C - (s_{0,C} + P\delta I)\| \le \frac{\|P\delta\|\,\|H\|}{1-\rho}Ch^r.} \tag{19.116}$$

The Neumann bound is

$$\|(I_F + H\delta)^{-1}\| \le \sum_{n=0}^{\infty}\|H\delta\|^n \le \frac{1}{1-\rho}. \tag{19.117}$$

The side conditions imply $H = HQ_F$, because

$$HIP = 0, \qquad H = H(I_F - IP) = HQ_F. \tag{19.118}$$

Thus

$$\|H\delta I\| = \|HQ_F\delta I\| \tag{19.119}$$

$$\le \|H\|\,Ch^r. \tag{19.120}$$

Combining these two estimates with submultiplicativity gives the stated bound. Thus the exact nonlinear correction has the same asymptotic order as the unresolved-channel leakage. For the centered scalar lattice studied previously, $r = 2$. For a product-closed spectral window, the leakage is zero and the correction vanishes identically.

Recovery of ordinary nonlinear ADM requires the joint limits

$$\mathsf{P}_h \longrightarrow I, \tag{19.121}$$

$$\Omega_{C,h} \longrightarrow \Omega_{\mathrm{ADM}}, \tag{19.122}$$

$$S_h^{\mathrm{BV}} \longrightarrow S_{\mathrm{GR}}^{\mathrm{BV}}, \tag{19.123}$$

$$\Delta_h S_h^{\mathrm{BV}} \longrightarrow 0 \quad \text{after the required renormalization.} \tag{19.124}$$

The first three are approximation and gauge-convergence statements. The last is the absence of a surviving quantum gauge anomaly.

19.12 Result and next physical steps

Under the stated contraction and master-equation hypotheses, the construction provides

$$\text{exact finite spatial closure,} \qquad \{\mathcal{D}_h[\Lambda], \mathcal{D}_h[\Gamma]\} = \mathcal{D}_h[\mathrm{i}[\Lambda, \Gamma]], \tag{19.125}$$

$$\text{an exact nonlinear coarse differential,} \qquad s_C^2 = 0, \tag{19.126}$$

$$\text{an exact coarse BFV charge,} \qquad \{\Omega_{C,h}, \Omega_{C,h}\}_C = 0, \tag{19.127}$$

$$\text{an exact quantum effective action,} \qquad \frac{1}{2}(S_h, S_h)_h - \mathrm{i}\hbar\Delta_h S_h = 0, \tag{19.128}$$

$$\text{and direct–iterated refinement equivalence.} \tag{19.129}$$

This resolves the finite ordering and constraint-preservation problem once a consistent fine theory and contraction are supplied. The nonlinear generators are transferred from that theory rather than guessed term by term.

The remaining physical work is:

- Construct an explicit microscopic $3+1$-dimensional gravity action whose regularized BFV–BV system is anomaly free.
- Prove nonperturbative convergence of its transferred finite theories to the Einstein–matter continuum equations.
- Derive the matter and gauge content of the continuum theory, including any claimed relation to the Standard Model.
- Obtain a falsifiable observable that is fixed by the transferred theory rather than introduced as an independent ansatz.

19.13 Verification

The accompanying program checks the inner moment map, commutator closure, an exact homological-transfer example, direct and iterated Schur elimination, nonlinear gauge reduction, determinant path independence, and quantum measure cancellation. The numerical residuals are

$$\max \varepsilon_{\text{inner action}} = 5.699160 \times 10^{-15}, \tag{19.130}$$

$$\max \varepsilon_{\text{inner closure}} = 3.907985 \times 10^{-14}, \tag{19.131}$$

$$\max \varepsilon_{\text{Schur direct/iterated}} = 1.320281 \times 10^{-15}, \tag{19.132}$$

$$\max \varepsilon_{\text{detail order}} = 1.593034 \times 10^{-15}. \tag{19.133}$$

The homological-transfer, nonlinear gauge, weighted-divergence, action, and determinant identities are exact symbolic equalities.

Chapter 20

The Next Gravitational Target: BFV Refinement of Gowdy Dynamics

The transfer theorem of the preceding chapter becomes physically decisive only after it is instantiated in a nonlinear gravitational system. The nearest substantial target is the unpolarized Gowdy sector before its lapse, shift, and time gauge are removed. Gowdy dynamics is nonlinear and spatially inhomogeneous, but it retains enough structure for exact calculation. It therefore provides a realistic intermediate step between the present reduced models and unrestricted three-dimensional spatial geometry.

This chapter fixes the continuum algebra that a finite projector theory must recover. It also states the exact conditions that would turn the conditional BFV–BV bridge into an unconditional gravitational result. The continuum bracket calculation below is complete. Construction of the ungauge-fixed finite BFV charge remains the next target.

20.1 Areal-time canonical form

Write

$$t := \mathrm{e}^{-\tau} > 0. \tag{20.1}$$

The derivative and measure transform as

$$\frac{\partial}{\partial\tau} = -t\frac{\partial}{\partial t}, \qquad \mathrm{d}\tau = -\frac{\mathrm{d}t}{t}, \qquad \mathrm{e}^{-2\tau} = t^2. \tag{20.2}$$

Starting from

$$S_G = \frac{1}{2}\int \left[P_\tau^2 - \mathrm{e}^{-2\tau}P_\theta^2 + \mathrm{e}^{2P}\left(Q_\tau^2 - \mathrm{e}^{-2\tau}Q_\theta^2\right)\right]\,\mathrm{d}\tau\,\mathrm{d}\theta, \tag{20.3}$$

one has $P_\tau = -tP_t$ and $Q_\tau = -tQ_t$. Substitution produces an overall factor $-t$; reversing the t-integration limits removes the minus sign. Reversing the integration limits after the change of variables gives the unpolarized Gowdy wave-map action

$$\boxed{S_G[P,Q] = \frac{1}{2}\int t\left[P_t^2 - P_\theta^2 + \mathrm{e}^{2P}\left(Q_t^2 - Q_\theta^2\right)\right]\,\mathrm{d}t\,\mathrm{d}\theta.} \tag{20.4}$$

The canonical momenta are

$$\Pi_P := tP_t, \tag{20.5}$$

$$\Pi_Q := t\mathrm{e}^{2P}Q_t. \tag{20.6}$$

Solving for the velocities gives

$$P_t = \frac{\Pi_P}{t}, \qquad Q_t = \frac{\mathrm{e}^{-2P}\Pi_Q}{t}. \tag{20.7}$$

The Legendre transform is

$$\mathcal{H}_t = \Pi_P P_t + \Pi_Q Q_t - \mathcal{L}_t \tag{20.8}$$

$$= \frac{1}{2t}\left(\Pi_P^2 + \mathrm{e}^{-2P}\Pi_Q^2\right) + \frac{t}{2}\left(P_\theta^2 + \mathrm{e}^{2P}Q_\theta^2\right). \tag{20.9}$$

At a fixed value of the areal time, define the Hamiltonian and tangential densities

$$\mathcal{H}_t := \frac{1}{2t}\left(\Pi_P^2 + \mathrm{e}^{-2P}\Pi_Q^2\right) + \frac{t}{2}\left(P_\theta^2 + \mathrm{e}^{2P}Q_\theta^2\right), \tag{20.10}$$

$$\mathcal{D} := \Pi_P P_\theta + \Pi_Q Q_\theta. \tag{20.11}$$

Their smearings are

$$H_t[N] := \int_0^{2\pi} N\mathcal{H}_t \,\mathrm{d}\theta, \tag{20.12}$$

$$D[\xi] := \int_0^{2\pi} \xi\mathcal{D} \,\mathrm{d}\theta. \tag{20.13}$$

Here the lapse-like smearing N is densitized. This convention makes $\mathcal{H}_t$ and $\mathcal{D}$ densities of the same spatial weight.

20.2 Exact continuum hypersurface algebra

Use the equal-time canonical bracket

$$\{P(\theta), \Pi_P(\vartheta)\} = \{Q(\theta), \Pi_Q(\vartheta)\} = \delta(\theta - \vartheta), \tag{20.14}$$

with all other elementary brackets equal to zero. For periodic functions, set

$$\mathbb{B}(f, g) := fg_\theta - gf_\theta. \tag{20.15}$$

Theorem 20.1 (Gowdy wave-map bracket algebra)**.** *At every fixed $t > 0$,*

$$\boxed{\{D[\xi], D[\eta]\} = D[\mathbb{B}(\xi, \eta)],} \tag{20.16}$$

$$\boxed{\{D[\xi], H_t[N]\} = H_t[\mathbb{B}(\xi, N)],} \tag{20.17}$$

$$\boxed{\{H_t[N], H_t[M]\} = D[\mathbb{B}(N, M)].} \tag{20.18}$$

Proof. For a periodic variation,

$$\delta \int \frac{t}{2} N P_\theta^2 \,\mathrm{d}\theta = \int tNP_\theta \,\partial_\theta(\delta P) \,\mathrm{d}\theta \tag{20.19}$$

$$= -\int \partial_\theta(tNP_\theta)\,\delta P \,\mathrm{d}\theta. \tag{20.20}$$

The Q_θ term is treated in the same way. Differentiating the factors $\mathrm{e}^{\pm 2P}$ supplies the two local P-variation terms. The required functional derivatives are

$$\frac{\delta H_t[N]}{\delta \Pi_P} = \frac{N}{t}\Pi_P, \qquad \frac{\delta H_t[N]}{\delta \Pi_Q} = \frac{N}{t}\mathrm{e}^{-2P}\Pi_Q, \tag{20.21}$$

$$\frac{\delta H_t[N]}{\delta P} = -\partial_\theta(tNP_\theta) + tN\mathrm{e}^{2P}Q_\theta^2 - \frac{N}{t}\mathrm{e}^{-2P}\Pi_Q^2, \tag{20.22}$$

$$\frac{\delta H_t[N]}{\delta Q} = -\partial_\theta(tN\mathrm{e}^{2P}Q_\theta), \tag{20.23}$$

$$\frac{\delta D[\xi]}{\delta \Pi_P} = \xi P_\theta, \qquad \frac{\delta D[\xi]}{\delta \Pi_Q} = \xi Q_\theta, \tag{20.24}$$

$$\frac{\delta D[\xi]}{\delta P} = -\partial_\theta(\xi\Pi_P), \qquad \frac{\delta D[\xi]}{\delta Q} = -\partial_\theta(\xi\Pi_Q). \tag{20.25}$$

Substituting the derivatives of D into the canonical bracket gives

$$\{D[\xi], D[\eta]\} = \int \Big[-\partial_\theta(\xi\Pi_P)\,\eta P_\theta + \xi P_\theta\,\partial_\theta(\eta\Pi_P) \tag{20.26}$$

$$- \partial_\theta(\xi\Pi_Q)\,\eta Q_\theta + \xi Q_\theta\,\partial_\theta(\eta\Pi_Q)\Big]\,\mathrm{d}\theta. \tag{20.27}$$

The terms containing $\partial_\theta\Pi_P$ and $\partial_\theta\Pi_Q$ cancel after expansion. The remainder is

$$\{D[\xi], D[\eta]\} = \int (\xi\eta_\theta - \eta\xi_\theta)(\Pi_P P_\theta + \Pi_Q Q_\theta)\,\mathrm{d}\theta \tag{20.28}$$

$$= D[\mathbb{B}(\xi,\eta)]. \tag{20.29}$$

The same functional derivatives show explicitly that

$$\{P, D[\xi]\} = \xi P_\theta, \qquad \{Q, D[\xi]\} = \xi Q_\theta, \tag{20.30}$$

$$\{\Pi_P, D[\xi]\} = \partial_\theta(\xi\Pi_P), \qquad \{\Pi_Q, D[\xi]\} = \partial_\theta(\xi\Pi_Q). \tag{20.31}$$

Thus P, Q transform as spatial scalars and the momenta as weight-one densities. Every kinetic or gradient term in $\mathcal{H}_t$ is a density of weight two, so

$$\{\mathcal{H}_t, D[\xi]\} = \xi\partial_\theta\mathcal{H}_t + 2\xi_\theta\mathcal{H}_t. \tag{20.32}$$

Consequently,

$$\{D[\xi], H_t[N]\} = -\int N\,(\xi\mathcal{H}_{t,\theta} + 2\xi_\theta\mathcal{H}_t)\,\mathrm{d}\theta \tag{20.33}$$

$$= \int (\xi N_\theta - N\xi_\theta)\mathcal{H}_t\,\mathrm{d}\theta \tag{20.34}$$

$$= H_t[\mathbb{B}(\xi, N)], \tag{20.35}$$

where periodic integration by parts is used in the second line.

For the Hamiltonian–Hamiltonian bracket, every term without a derivative of N or M is symmetric under $N \leftrightarrow M$ and cancels in the antisymmetrization. The P-gradient terms reduce to

$$-\frac{M\Pi_P}{t}\partial_\theta(tNP_\theta) + \frac{N\Pi_P}{t}\partial_\theta(tMP_\theta) \tag{20.36}$$

$$= (NM_\theta - MN_\theta)\Pi_P P_\theta. \tag{20.37}$$

The Q-gradient terms similarly give

$$-\frac{M\mathrm{e}^{-2P}\Pi_Q}{t}\partial_\theta(tN\mathrm{e}^{2P}Q_\theta) + \frac{N\mathrm{e}^{-2P}\Pi_Q}{t}\partial_\theta(tM\mathrm{e}^{2P}Q_\theta) \tag{20.38}$$

$$= (NM_\theta - MN_\theta)\Pi_Q Q_\theta. \tag{20.39}$$

The factors t and t^{-1}, and likewise e^{2P} and e^{-2P}, cancel. Therefore

$$\{H_t[N], H_t[M]\} = \int_0^{2\pi} \mathbb{B}(N,M)\,(\Pi_P P_\theta + \Pi_Q Q_\theta)\,\mathrm{d}\theta \tag{20.40}$$

$$= D[\mathbb{B}(N,M)]. \tag{20.41}$$

□

The factors t and t^{-1} are reciprocal. Their product is exactly what removes any time-dependent coefficient from the last bracket. The nonlinear target metric $\mathrm{diag}(1, \mathrm{e}^{2P})$ also cancels between its covariant and contravariant appearances.

20.3 Chiral decomposition

Define

$$C_\pm[f] := \frac{1}{2}\left(H_t[f] \pm D[f]\right). \tag{20.42}$$

The preceding theorem gives

$$\boxed{\{C_+[f], C_+[g]\} = C_+[\mathbb{B}(f,g)],} \tag{20.43}$$

$$\boxed{\{C_-[f], C_-[g]\} = -C_-[\mathbb{B}(f,g)],} \tag{20.44}$$

$$\boxed{\{C_+[f], C_-[g]\} = 0.} \tag{20.45}$$

For the plus sector,

$$\{C_+[f], C_+[g]\} = \frac{1}{4}\big(\{H[f], H[g]\} + \{H[f], D[g]\} \tag{20.46}$$

$$+ \{D[f], H[g]\} + \{D[f], D[g]\}\big) \tag{20.47}$$

$$= \frac{1}{2}\left(H[\mathbb{B}(f,g)] + D[\mathbb{B}(f,g)]\right) \tag{20.48}$$

$$= C_+[\mathbb{B}(f,g)]. \tag{20.49}$$

Changing both signs of the mixed terms gives the minus-sector result. Taking opposite signs in the two generators makes the Hamiltonian and tangential contributions cancel separately, so the mixed chiral bracket vanishes. Thus the continuum target separates into two commuting chiral algebras. Reversing the orientation of the minus sector removes the displayed minus sign. This split is useful for refinement because the two directions can be transferred independently before their common geometric variables are recombined.

20.4 What must be restored before BFV quantization

The areal-time action is already gauge fixed. In particular, H_t is the physical Hamiltonian at fixed t; it is not yet a vanishing Hamiltonian constraint. The bracket theorem therefore identifies the continuum algebraic target, but it does not by itself produce a gravitational BFV system.

The next construction must begin before areal gauge fixing. One convenient canonical enlargement uses embedding or clock variables

$$(T, X; \Pi_T, \Pi_X) \tag{20.50}$$

together with

$$(P, Q; \Pi_P, \Pi_Q). \tag{20.51}$$

An equivalent formulation may use the symmetry-reduced area field and the two-dimensional metric directly. In either description, the fine constraints must have the form

$$\mathcal{C}_\perp = \mathcal{E}_\perp[T, X, \Pi_T, \Pi_X] + \mathcal{H}^G_\perp[T, X; P, Q, \Pi_P, \Pi_Q], \tag{20.52}$$

$$\mathcal{C}_\parallel = \Pi_T T_\theta + \Pi_X X_\theta + \Pi_P P_\theta + \Pi_Q Q_\theta, \tag{20.53}$$

where $\mathcal{E}_\perp$ and $\mathcal{H}^G_\perp$ are derived from the ungauge-fixed reduced Einstein action rather than chosen to force closure. The gauge

$$T = t, \qquad X = \theta \tag{20.54}$$

must recover the Hamiltonian density $\mathcal{H}_t$ above and the standard Gowdy equations.

This is the first decisive gate. A construction that closes only after imposing the areal gauge would not establish gravitational gauge closure.

20.5 Fine BFV charge and projector transfer

For a pair of deformation parameters $u = (N, \xi)$ and $v = (M, \eta)$, the continuum structure map fixed by the bracket theorem is

$$\boxed{\mathbb{B}_G(u, v) = \left(\mathbb{B}(\xi, M) + \mathbb{B}(N, \eta), \mathbb{B}(\xi, \eta) + \mathbb{B}(N, M)\right).} \tag{20.55}$$

The first component collects the two mixed brackets and is the new normal smearing. The second collects the tangential–tangential and normal–normal brackets and is the new tangential smearing. Thus, with

$$\mathcal{C}[(N, \xi)] := H_t[N] + D[\xi], \tag{20.56}$$

the three continuum identities combine into

$$\{\mathcal{C}[u], \mathcal{C}[v]\} = \mathcal{C}[\mathbb{B}_G(u, v)]. \tag{20.57}$$

After the ungauge-fixed constraints are obtained, the corresponding fine BFV charge has the canonical form

$$\Omega_{G,F} = \mathcal{C}_{F,\perp}[c^{\perp}] + \mathcal{C}_{F,\parallel}[c^{\parallel}] - \frac{1}{2}\langle \varpi, \mathbb{B}_{G,F}(c,c)\rangle + \Omega^{(\geq 3)}_{G,F}. \tag{20.58}$$

The higher ghost terms must be retained if the finite structure maps are field dependent or the reduced system is open. The fine theory passes the classical gate only if

$$\boxed{\{\Omega_{G,F}, \Omega_{G,F}\}_F = 0.} \tag{20.59}$$

Let P_h, I_h, H_h be a cyclic projector contraction from the fine Gowdy complex to resolution h. The transferred differential is then

$$s_{G,h} = s_{0,h} + P_h \delta_G (I_F + H_h \delta_G)^{-1} I_h. \tag{20.60}$$

Its Hamiltonian generator $\Omega_{G,h}$ must satisfy

$$\boxed{\{\Omega_{G,h}, \Omega_{G,h}\}_h = 0} \tag{20.61}$$

for every admissible finite resolution. The ghost-linear coefficients define the finite Hamiltonian and momentum constraints. This avoids guessing separate nonlinear lattice constraints at each scale.

20.6 Refinement and continuum theorem to be proved

The target theorem consists of five linked statements.

Fine model	Derive the ungauge-fixed symmetry-reduced action, its canonical constraints, and a nilpotent BFV charge.
Finite closure	Prove $\{\Omega_{G,h}, \Omega_{G,h}\}_h = 0$ exactly at every retained resolution.
Refinement consistency	Prove direct and iterated projector or prime-polyphase elimination give canonically equivalent charges.
Continuum recovery	Prove that the transferred constraints and structure maps converge to the Gowdy hypersurface algebra and recover the areal equations after gauge fixing.
Quantum closure	Perform the BV pushforward and prove the finite quantum master equation without a surviving measure anomaly.

A quantitative continuum statement should have the form

$$\boxed{\begin{array}{c}\|\mathcal{C}_{\perp,h} - \mathcal{C}_{\perp}\|_{s-1} + \|\mathcal{C}_{\parallel,h} - \mathcal{C}_{\parallel}\|_{s-1} \\ + \|\mathbb{B}_{G,h} - \mathbb{B}_{G}\|_{s-1} \leq C_s h^r\end{array}} \tag{20.62}$$

for smooth data in a declared Sobolev class, together with a stability estimate uniform on a nonzero time interval. The exponent r must be derived from the chosen projector family rather than inferred from a single numerical trajectory. For nested resolutions, the exact transfer must obey

$$\boxed{\mathfrak{R}_{h_0 \leftarrow h_2}(\Omega_{G,h_2}) = \mathfrak{R}_{h_0 \leftarrow h_1}\left(\mathfrak{R}_{h_1 \leftarrow h_2}(\Omega_{G,h_2})\right).} \tag{20.63}$$

For compatible prime channel families, the corresponding elimination order must be interchangeable up to a BFV canonical transformation. This is the nonlinear gravitational analogue of the exact free prime-polyphase semigroup.

20.7 Baseline verification

The accompanying calculation checks the continuum target before the ungauge-fixed finite theory is attempted. It verifies the bracket identities symbolically and evaluates their finite-resolution defects for nonlinear periodic data. Centered differences give

$$p_{DD} = 1.998341, \tag{20.64}$$

$$p_{DH} = 1.997880, \tag{20.65}$$

$$p_{HH} = 1.999334. \tag{20.66}$$

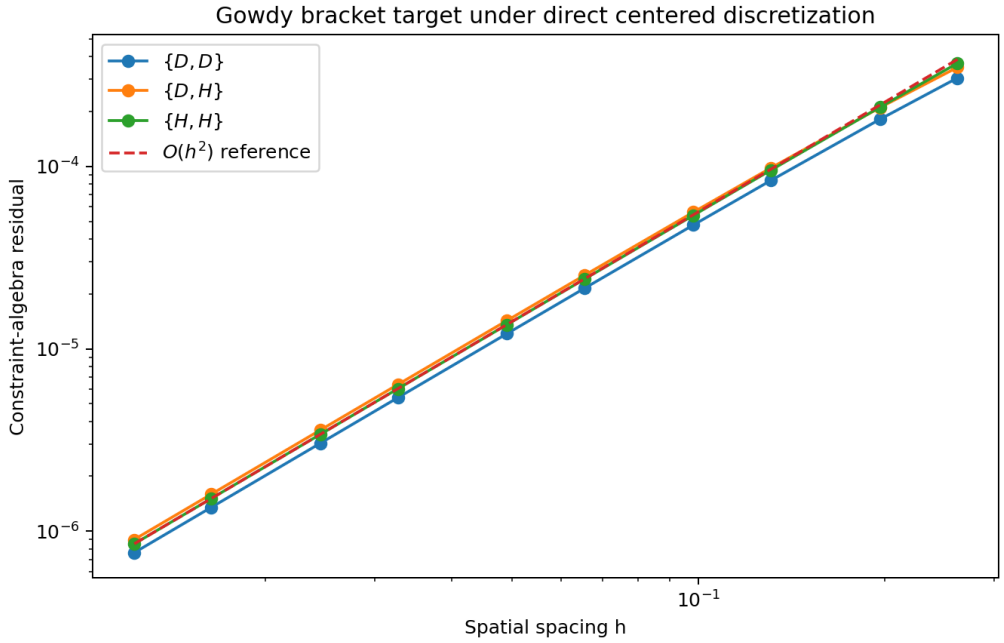
A resolved Fourier derivative gives

$$\max(\varepsilon_{DD}^{\mathrm{spec}}, \varepsilon_{DH}^{\mathrm{spec}}, \varepsilon_{HH}^{\mathrm{spec}}) = 1.536965 \times 10^{-15}, \tag{20.67}$$

and the largest chiral-algebra residual is

$$\varepsilon_{\mathrm{chiral}} = 1.032576 \times 10^{-15}. \tag{20.68}$$

These checks establish the target algebra and the expected second-order defect of direct centered discretization. They do not establish finite BFV nilpotency; that requires the transferred charge described above.



20.8 Resulting direction

The next major result should therefore be narrow and unconditional:

$$\boxed{\begin{array}{c}\text{ungauge-fixed nonlinear Gowdy action}\\ +\ \text{explicit fine BFV charge}\\ +\ \text{cyclic projector/polyphase contraction}\\ \Downarrow\\ \text{exact finite master equation at every scale}\\ +\ \text{refinement-path independence}\\ +\ \text{controlled Gowdy continuum limit.}\end{array}} \tag{20.69}$$

This target is materially stronger than another gauge-fixed evolution scheme, but it avoids the simultaneous complications of unrestricted $3+1$ geometry. Success would provide the first nonlinear gravitational realization of the projector BFV–BV bridge and a concrete basis for the subsequent full $3+1$ construction.

Chapter 21

Exact Resolved Nonlinear $3+1$ Constraint Closure

21.1 Status and scope

nonlinear vacuum Hamiltonian constraint: closed on an alias-free resolved domain,
three momentum constraints: closed on the same domain,
Gauss–momentum–Hamiltonian algebra: exact at finite dimension,
strict finite-range locality, Lorentzian reality preservation, and unified matter: open.

(21.1)

The construction uses a filtered Fourier algebra rather than the complete finite pointwise algebra. Products are allowed to enter declared detail bands. This is the refinement option selected by the finite-derivation no-go theorem.

21.2 Filtered spectral differential algebra

Let

$$\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3, \qquad \mathcal{T}_K = \left\{ f(x) = \sum_{\|k\|_\infty \le K} \widehat{f}_k e^{\mathrm{i}k\cdot x} \right\}. \tag{21.2}$$

Then

$$\mathcal{T}_K \mathcal{T}_L \subseteq \mathcal{T}_{K+L}, \qquad \partial_a \mathcal{T}_K \subseteq \mathcal{T}_K, \tag{21.3}$$

and

$$\mathcal{T}_{\mathrm{fin}} := \bigcup_{K\ge 0} \mathcal{T}_K \tag{21.4}$$

is a commutative filtered differential algebra.

For an N^3 periodic grid

$$\Gamma_N = \left\{ \frac{2\pi}{N}(r_1, r_2, r_3) : 0 \le r_a < N \right\}, \tag{21.5}$$

define the sampling map S_N, the Fourier derivative

$$\delta_a^{(N)} S_N f := S_N(\partial_a f), \tag{21.6}$$

and the normalized trace

$$\tau_N(u) := \frac{1}{N^3} \sum_{x\in\Gamma_N} u(x). \tag{21.7}$$

Alias-free product lemma. Let $f_r \in \mathcal{T}_{K_r}$. If

$$\sum_{r=1}^{q} K_r < \frac{N}{2}, \tag{21.8}$$

then

$$\delta_a^{(N)} \prod_{r=1}^{q} S_N f_r = S_N \partial_a \left(\prod_{r=1}^{q} f_r \right), \tag{21.9}$$

$$= \sum_{s=1}^{q} \left(\prod_{r<s} S_N f_r \right) S_N(\partial_a f_s) \left(\prod_{r>s} S_N f_r \right), \tag{21.10}$$

$$\tau_N \left(\prod_{r=1}^{q} S_N f_r \right) = \frac{1}{(2\pi)^3} \int_{\mathbb{T}^3} \prod_{r=1}^{q} f_r \, \mathrm{d}^3 x. \tag{21.11}$$

Proof. The product has coordinatewise Fourier support bounded by $\sum_r K_r$. The strict Nyquist inequality makes sampling injective on that support. Fourier differentiation, multiplication, and zero-mode extraction therefore coincide with their continuum trigonometric-polynomial values. □

21.3 Resolved self-dual canonical action

Set the canonical normalization to one. Let

$$A_a^i, E_i^a, \lambda^i, \xi^a, n \in \mathcal{T}_K, \tag{21.12}$$

where $a, b = 1, 2, 3$ are spatial indices, $i, j, k = 1, 2, 3$ are internal indices, and n is a densitized lapse of weight -1. Define

$$F_{ab}^i := \delta_a A_b^i - \delta_b A_a^i + \epsilon^i{}_{jk} A_a^j A_b^k, \tag{21.13}$$

$$\mathcal{G}_i := \delta_a E_i^a + \epsilon_{ij}{}^k A_a^j E_k^a, \tag{21.14}$$

$$\mathcal{V}_a := F_{ab}^i E_i^b, \tag{21.15}$$

$$\mathcal{D}_a := \mathcal{V}_a - A_a^i \mathcal{G}_i, \tag{21.16}$$

$$\det E := \frac{1}{6} \epsilon_{abc} \epsilon^{ijk} E_i^a E_j^b E_k^c, \tag{21.17}$$

$$\mathcal{H} := \frac{1}{2} \epsilon^{ij}{}_k E_i^a E_j^b F_{ab}^k + \Lambda \det E. \tag{21.18}$$

The smeared constraints are

$$G_N[\lambda] := \tau_N(\lambda^i \mathcal{G}_i), \tag{21.19}$$

$$V_N[\xi] := \tau_N(\xi^a \mathcal{V}_a), \tag{21.20}$$

$$D_N[\xi] := \tau_N(\xi^a \mathcal{D}_a), \tag{21.21}$$

$$H_N[n] := \tau_N(n\mathcal{H}). \tag{21.22}$$

The resolved nonlinear action is

$$\boxed{S_{N,K}^{\mathrm{SD}} = \int \mathrm{d}t \left[\tau_N(E_i^a \dot{A}_a^i) - G_N[\lambda] - D_N[\xi] - H_N[n] \right].} \tag{21.23}$$

The finite canonical bracket is

$$\boxed{\{F, G\}_N = N^3 \sum_{x,a,i} \left(\frac{\partial F}{\partial A_a^i(x)} \frac{\partial G}{\partial E_i^a(x)} - \frac{\partial F}{\partial E_i^a(x)} \frac{\partial G}{\partial A_a^i(x)} \right).} \tag{21.24}$$

21.4 Exact finite nonlinear constraint algebra

For internal parameters define

$$(\lambda \times \mu)^i := \epsilon^i{}_{jk}\lambda^j\mu^k. \tag{21.25}$$

For spatial vectors and a lapse define

$$[\xi,\eta]^a := \xi^b\delta_b\eta^a - \eta^b\delta_b\xi^a, \tag{21.26}$$

$$\mathcal{L}_\xi\lambda^i := \xi^a\delta_a\lambda^i, \tag{21.27}$$

$$\mathcal{L}_\xi^{(-1)}n := \xi^a\delta_a n - n\delta_a\xi^a. \tag{21.28}$$

Finally set

$$S^a(n,m) := E^a_i E^{bi}\left(m\delta_b n - n\delta_b m\right). \tag{21.29}$$

Theorem. Assume every dynamical field and smearing belongs to $\mathcal{T}_K$ and

$$\boxed{16K < N.} \tag{21.30}$$

Then the finite constraints satisfy

$$\{G_N[\lambda], G_N[\mu]\}_N = G_N[\lambda\times\mu], \tag{21.31}$$

$$\{D_N[\xi], G_N[\lambda]\}_N = G_N[\mathcal{L}_\xi\lambda], \tag{21.32}$$

$$\{D_N[\xi], D_N[\eta]\}_N = D_N[[\xi,\eta]], \tag{21.33}$$

$$\{G_N[\lambda], H_N[n]\}_N = 0, \tag{21.34}$$

$$\{D_N[\xi], H_N[n]\}_N = H_N[\mathcal{L}_\xi^{(-1)}n], \tag{21.35}$$

$$\{H_N[n], H_N[m]\}_N = V_N[S(n,m)] \tag{21.36}$$

$$= D_N[S(n,m)] + G_N[S^a(n,m)A_a]. \tag{21.37}$$

The result is unchanged by the cosmological term $\Lambda \det E$.

Proof. The largest monomials in G_N, D_N, and H_N contain respectively three, four, and five band-K factors when the smearing is counted. Their canonical gradients contain at most two, three, and four factors. A canonical bracket therefore contains at most eight band-K factors. The condition $8K < N/2$ places every monomial occurring in the actions, variations, integrations by parts, Bianchi identities, and brackets below Nyquist.

By the alias-free product lemma, the finite action and every canonical gradient coincide with the sampled continuum trigonometric-polynomial expressions. The continuum self-dual Gauss, improved-diffeomorphism, and Hamiltonian identities therefore transfer without truncation. The final bracket is the vector constraint with structure shift $S^a(n,m)$. Since

$$V_N[S] = D_N[S] + G_N[S^aA_a], \tag{21.38}$$

the bracket is a linear combination of retained constraints. The cosmological term is internal-gauge and spatial-diffeomorphism invariant, and its cross terms in the Hamiltonian bracket cancel by antisymmetry. □

21.5 Exact constraint propagation

Let

$$H_{\mathrm{tot}} := G_N[\lambda] + D_N[\xi] + H_N[n]. \tag{21.39}$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}t}G_N[\alpha] = G_N[\alpha\times\lambda - \mathcal{L}_\xi\alpha], \tag{21.40}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}D_N[\eta] = G_N[\mathcal{L}_\eta\lambda] + D_N[[\eta,\xi]] + H_N[\mathcal{L}_\eta^{(-1)}n], \tag{21.41}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}H_N[m] = -H_N[\mathcal{L}_\xi^{(-1)}m] + V_N[S(m,n)]. \tag{21.42}$$

Consequently,

$$G_N = D_N = H_N = 0 \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t}(G_N, D_N, H_N) = 0 \tag{21.43}$$

for the continuous-time resolved Hamiltonian flow as long as the declared spectral headroom remains valid.

21.6 Refinement-safe closure tower

For $K < L$, let

$$I_{K\to L} : \mathcal{T}_K \hookrightarrow \mathcal{T}_L \tag{21.44}$$

be Fourier zero padding. Then

$$I_{L\to M} I_{K\to L} = I_{K\to M}. \tag{21.45}$$

If $L \geq 8K$ and the finer grid resolves $\mathcal{T}_L$, all constraint densities and all terms in one bracket evaluation embed without loss. Hence direct and iterated refinement give the same constraint values and bracket values.

The fixed band $\mathcal{T}_K$ is not generally invariant under nonlinear evolution. The exact object is the filtered tower

$$\mathcal{T}_K \longrightarrow \mathcal{T}_{4K} \longrightarrow \mathcal{T}_{16K} \longrightarrow \cdots, \tag{21.46}$$

with refinement performed before aliasing. Coarse truncation without detail return is not used.

21.7 Relation to BFV–BV transfer

The resolved action gives a concrete finite first-class input system. Its constraint structure maps are

$$B_{GG}(\lambda, \mu) = \lambda \times \mu, \tag{21.47}$$

$$B_{DG}(\xi, \lambda) = \mathcal{L}_\xi \lambda, \tag{21.48}$$

$$B_{DD}(\xi, \eta) = [\xi, \eta], \tag{21.49}$$

$$B_{DH}(\xi, n) = \mathcal{L}_\xi^{(-1)} n, \tag{21.50}$$

$$B_{HH}(n, m) = S(n, m). \tag{21.51}$$

A coarse projector that discards the generated detail bands can lose nilpotency. The nonlinear homological return series

$$s_C = s_{0,C} + P\delta(I + H\delta)^{-1} I \tag{21.52}$$

restores the coarse-to-detail-to-coarse contributions. Thus the resolved Ashtekar system supplies a specific classical fine target for the existing BFV transfer theorem. An explicit complete BFV charge, Lorentzian reality sector, and anomaly-free BV measure remain to be constructed.

21.8 Numerical certificate

A deterministic 18^3 Fourier-grid test used random real band-1 fields and smearings with an identity-triad background. The sufficient headroom condition $16K < N$ holds. Automatic differentiation of the finite action gave

Bracket identity	absolute residual
$\{G, G\} = G$	7.676151×10^{-17}
$\{D, G\} = G$	1.067939×10^{-17}
$\{D, D\} = D$	3.382711×10^{-17}
$\{G, H\} = 0$	6.746196×10^{-17}
$\{D, H\} = H$	3.176712×10^{-17}
$\{H, H\} = V$	4.987330×10^{-18}
$\{H_\Lambda, H_\Lambda\} = V$	2.038300×10^{-17}

On the same 18^3 grid, a band-3 negative control violates the declared headroom. The Hamiltonian-bracket residual rises to

$$3.288599 \times 10^{-8}, \qquad \varepsilon_{\mathrm{rel}} = 1.742589 \times 10^{-5}. \tag{21.53}$$

This separates the exact resolved regime from the aliased regime.

21.9 Physical boundary

The canonical polynomial action is directly real in the Euclidean self-dual sector. In Lorentzian signature the self-dual connection is complex and the physical sector requires

$$E^a_i = \overline{E^a_i}, \qquad A^i_a + \overline{A^i_a} = 2\Gamma^i_a(E). \tag{21.54}$$

Exact finite preservation of these reality conditions is open.

The Fourier derivative is not finite range on the grid. Therefore the theorem establishes an exact finite-dimensional resolved constraint algebra, not a strictly local microscopic lattice theory. The remaining major gates are

$$\begin{gathered}\text{strictly local nonlinear realization},\\ \text{Lorentzian reality-constraint propagation},\\ \text{unified Maxwell–Dirac and scalar matter},\\ \text{uniform Einstein–matter continuum estimates},\\ \text{constraint-preserving discrete-time integration}.\end{gathered} \tag{21.55}$$

21.10 Machine-readable records

$$\mathcal{R}_{\mathrm{RNC}} = \texttt{data/resolved_nonlinear_ashtekar_checks.csv}, \tag{21.56}$$

$$\mathcal{S}_{\mathrm{RNC}} = \texttt{data/resolved_nonlinear_ashtekar_summary.csv}, \tag{21.57}$$

$$\mathcal{X}_{\mathrm{RNC}} = \texttt{code/54_resolved_nonlinear_ashtekar_closure.py}. \tag{21.58}$$

Chapter 22

Resolved Lorentzian Gravity–Matter BV Parent

22.1 Status and objective

real Lorentzian tetrad–connection action: constructed,
Dirac and first-order Maxwell matter in the same polynomial action: constructed,
off-shell diffeomorphism–Lorentz–U(1) BRST differential: nilpotent,
classical BV master equation: exact on an alias-free resolved domain,
strict finite-range locality, canonical matter closure, and quantum anomaly control: open.

(22.1)

The preceding self-dual canonical block closes the nonlinear vacuum constraints on an alias-free spatial domain. Its remaining physical obstruction is that a single self-dual Lorentzian connection is complex. The present construction replaces the complex theory by a real first-order parent, retains both chiral sectors, and inserts spinor and Abelian matter before gauge reduction.

22.2 Four-dimensional resolved differential algebra

Let

$$\mathbb{T}^4 = (\mathbb{R}/2\pi\mathbb{Z})^4, \qquad \mathcal{T}_K^{(4)} = \left\{ f(x) = \sum_{\|k\|_\infty \le K} \widehat{f}_k \mathrm{e}^{\mathrm{i}k\cdot x} \right\}. \tag{22.2}$$

Then

$$\mathcal{T}_K^{(4)}\mathcal{T}_L^{(4)} \subseteq \mathcal{T}_{K+L}^{(4)}, \qquad \partial_\mu \mathcal{T}_K^{(4)} \subseteq \mathcal{T}_K^{(4)}. \tag{22.3}$$

For the periodic grid

$$\Gamma_N^{(4)} = \left\{ \frac{2\pi}{N}(r_0, r_1, r_2, r_3) : 0 \le r_\mu < N \right\}, \tag{22.4}$$

define the spectral derivative and normalized trace by

$$\delta_\mu^{(N)} S_N f = S_N(\partial_\mu f), \qquad \tau_N^{(4)}(u) = \frac{1}{N^4} \sum_{x \in \Gamma_N^{(4)}} u(x). \tag{22.5}$$

If $f_r \in \mathcal{T}_{K_r}^{(4)}$ and

$$\sum_r K_r < \frac{N}{2}, \tag{22.6}$$

then sampled multiplication, differentiation, integration by parts, and zero-mode extraction coincide exactly with the continuum trigonometric-polynomial operations.

All statements below extend coefficientwise to a finite Grassmann algebra. Thus ghosts, antifields, and fermionic matter may be included without altering the Fourier-support argument.

22.3 Real Lorentzian parent fields

Fix

$$\eta_{IJ} = \mathrm{diag}(-1,1,1,1), \qquad \epsilon_{0123} = 1. \tag{22.7}$$

The independent fields are

$$\begin{aligned} &e^I \in \Omega^1(\mathbb{T}^4), && \omega^{IJ} = -\omega^{JI} \in \Omega^1(\mathbb{T}^4), \\ &a \in \Omega^1(\mathbb{T}^4), && \varphi_{IJ} = -\varphi_{JI} \in \Omega^0(\mathbb{T}^4), \\ &\psi, \bar{\psi} \in \Omega^0(\mathbb{T}^4) \otimes \mathbb{C}^4. \end{aligned} \tag{22.8}$$

Define

$$R^{IJ} := \mathrm{d}\omega^{IJ} + \omega^I{}_K \wedge \omega^{KJ}, \tag{22.9}$$

$$T^I := \mathrm{d}e^I + \omega^I{}_J \wedge e^J, \tag{22.10}$$

$$F := \mathrm{d}a, \tag{22.11}$$

$$\mathrm{vol}_e := \frac{1}{4!}\epsilon_{IJKL} e^I \wedge e^J \wedge e^K \wedge e^L, \tag{22.12}$$

$$\Sigma_I := \frac{1}{3!}\epsilon_{IJKL} e^J \wedge e^K \wedge e^L. \tag{22.13}$$

Let

$$\{\gamma^I, \gamma^J\} = 2\eta^{IJ} I, \qquad \gamma_{IJ} := \frac{1}{2}[\gamma_I, \gamma_J]. \tag{22.14}$$

The spinor covariant derivatives are

$$\mathcal{D}\psi := \mathrm{d}\psi + \frac{1}{4}\omega^{IJ}\gamma_{IJ}\psi - \mathrm{i}qa\psi, \tag{22.15}$$

$$\mathcal{D}\bar{\psi} := \mathrm{d}\bar{\psi} - \frac{1}{4}\bar{\psi}\omega^{IJ}\gamma_{IJ} + \mathrm{i}qa\bar{\psi}. \tag{22.16}$$

22.4 One polynomial Lorentzian gravity–matter action

The gravitational action is

$$S_{\mathrm{G}} = \frac{1}{4\kappa}\int \epsilon_{IJKL} e^I \wedge e^J \wedge R^{KL} - \frac{\Lambda}{24\kappa}\int \epsilon_{IJKL} e^I \wedge e^J \wedge e^K \wedge e^L. \tag{22.17}$$

The first-order Maxwell action is

$$S_{\mathrm{M}}^{(1)} = \int \left[\frac{1}{2}\varphi_{IJ} F \wedge e^I \wedge e^J - \frac{1}{4}\varphi_{IJ}\varphi^{IJ}\,\mathrm{vol}_e\right]. \tag{22.18}$$

The Dirac action is

$$S_{\mathrm{D}} = \frac{\mathrm{i}}{2}\int \Sigma_I \wedge \left[\bar{\psi}\gamma^I \mathcal{D}\psi - (\mathcal{D}\bar{\psi})\gamma^I \psi\right] - m\int \mathrm{vol}_e\ \bar{\psi}\psi. \tag{22.19}$$

Set

$$\boxed{S_0[e,\omega,a,\varphi,\psi,\bar{\psi}] := S_{\mathrm{G}} + S_{\mathrm{M}}^{(1)} + S_{\mathrm{D}}.} \tag{22.20}$$

Every local monomial in S_0 contains at most six dynamical-field factors. No inverse tetrad is used in the defining action.

22.4.1 Maxwell auxiliary elimination

Variation with respect to φ_{IJ} gives

$$F \wedge e^I \wedge e^J = \varphi^{IJ}\,\mathrm{vol}_e\,. \tag{22.21}$$

For a nondegenerate tetrad this is

$$\varphi^{IJ} = (\star_e F)^{IJ}. \tag{22.22}$$

Since $\star_e^2 = -1$ on Lorentzian two-forms,

$$S_{\mathrm{M}}^{(1)}\big|_{\varphi=\star_e F} = -\frac{1}{4}\int F_{IJ}F^{IJ}\,\mathrm{vol}_e = -\frac{1}{2}\int F\wedge\star_e F. \tag{22.23}$$

Thus the auxiliary formulation is polynomial before elimination and equivalent to Maxwell theory on the nondegenerate sector.

22.5 Off-shell gauge system

Let the infinitesimal parameters be a vector field ξ, a Lorentz-algebra field λ^{IJ}, and a scalar U(1) parameter u. Define

$$\delta e^I = \mathcal{L}_\xi e^I + \lambda^I{}_J e^J, \tag{22.24}$$
$$\delta\omega^{IJ} = \mathcal{L}_\xi\omega^{IJ} - D_\omega\lambda^{IJ}, \tag{22.25}$$
$$\delta a = \mathcal{L}_\xi a + \mathrm{d}u, \tag{22.26}$$
$$\delta\varphi_{IJ} = \mathcal{L}_\xi\varphi_{IJ} - \lambda^K{}_I\varphi_{KJ} - \lambda^K{}_J\varphi_{IK}, \tag{22.27}$$
$$\delta\psi = \mathcal{L}_\xi\psi + \frac{1}{4}\lambda^{IJ}\gamma_{IJ}\psi + \mathrm{i}qu\psi, \tag{22.28}$$
$$\delta\bar\psi = \mathcal{L}_\xi\bar\psi - \frac{1}{4}\bar\psi\lambda^{IJ}\gamma_{IJ} - \mathrm{i}qu\bar\psi. \tag{22.29}$$

Then

$$\delta R^{IJ} = \mathcal{L}_\xi R^{IJ} + \lambda^I{}_K R^{KJ} + \lambda^J{}_K R^{IK}, \tag{22.30}$$
$$\delta(\mathcal{D}\psi) = \mathcal{L}_\xi(\mathcal{D}\psi) + \frac{1}{4}\lambda^{IJ}\gamma_{IJ}\mathcal{D}\psi + \mathrm{i}qu\mathcal{D}\psi, \tag{22.31}$$

and

$$\delta F = \mathcal{L}_\xi F. \tag{22.32}$$

Consequently,

$$\boxed{\delta S_0 = 0} \tag{22.33}$$

on the periodic domain, without use of any field equation.
For parameter triples $U = (\xi, \lambda, u)$ and $V = (\eta, \mu, v)$ define

$$[\![U, V]\!] := \left([\xi,\eta], \mathcal{L}_\xi\mu - \mathcal{L}_\eta\lambda + [\lambda,\mu], \mathcal{L}_\xi v - \mathcal{L}_\eta u\right). \tag{22.34}$$

With the transformation convention above, the field-space commutator is

$$[\delta_U, \delta_V]_{\mathrm{fld}} = -\delta_{[\![U,V]\!]}. \tag{22.35}$$

The algebra closes off shell and has no field-dependent structure functions.

22.6 Explicit nilpotent BRST differential

Introduce odd ghosts

$$c^\mu, \qquad \chi^{IJ} = -\chi^{JI}, \qquad \upsilon. \tag{22.36}$$

Set

$$\mathfrak{c} := (c, \chi, \upsilon), \qquad s\Phi := \delta_{\mathfrak{c}}\Phi, \qquad s\mathfrak{c} := \frac{1}{2}[\![\mathfrak{c}, \mathfrak{c}]\!]. \tag{22.37}$$

Explicitly,

$$sc^\mu = c^\nu\partial_\nu c^\mu, \tag{22.38}$$
$$s\chi = \mathcal{L}_c\chi + \chi^2, \tag{22.39}$$
$$s\upsilon = \mathcal{L}_c\upsilon, \tag{22.40}$$

and

$$se^I = \mathcal{L}_c e^I + \chi^I{}_J e^J, \tag{22.41}$$
$$s\omega^{IJ} = \mathcal{L}_c \omega^{IJ} - D_\omega \chi^{IJ}, \tag{22.42}$$
$$sa = \mathcal{L}_c a + \mathrm{d}\upsilon, \tag{22.43}$$
$$s\varphi_{IJ} = \mathcal{L}_c \varphi_{IJ} - \chi^K{}_I \varphi_{KJ} - \chi^K{}_J \varphi_{IK}, \tag{22.44}$$
$$s\psi = \mathcal{L}_c \psi + \frac{1}{4}\chi^{IJ}\gamma_{IJ}\psi + \mathrm{i}q\upsilon\psi, \tag{22.45}$$
$$s\bar{\psi} = \mathcal{L}_c \bar{\psi} - \frac{1}{4}\bar{\psi}\chi^{IJ}\gamma_{IJ} - \mathrm{i}q\upsilon\bar{\psi}. \tag{22.46}$$

Here χ^2 is the graded matrix product, equivalently one half of the graded Lorentz commutator.

Theorem. The differential is off-shell nilpotent:

$$\boxed{s^2 = 0.} \tag{22.47}$$

Proof. The parameter bracket is a semidirect-product Lie bracket. Its graded extension satisfies the Jacobi identity. Since the field transformations realize the negative of this bracket,

$$s^2\Phi = \frac{1}{2}\left([\delta_{\mathfrak{c}}, \delta_{\mathfrak{c}}]_{\mathrm{gr}} + \delta_{[\![\mathfrak{c},\mathfrak{c}]\!]}\right)\Phi = 0. \tag{22.48}$$

The ghost identity $s^2\mathfrak{c} = 0$ is the graded Jacobi identity itself. □

22.7 Minimal classical BV master action

Let Φ range over

$$\{e, \omega, a, \varphi, \psi, \bar{\psi}\}. \tag{22.49}$$

Introduce antifields

$$\Phi^\ddagger, \qquad c^\ddagger_\mu, \qquad \chi^\ddagger_{IJ}, \qquad \upsilon^\ddagger \tag{22.50}$$

with the complementary form degrees and opposite Grassmann parity. Define

$$\boxed{S_{\mathrm{BV}} = S_0 + \int \left[\sum_\Phi \langle \Phi^\ddagger, s\Phi\rangle + c^\ddagger_\mu s c^\mu + \frac{1}{2}\chi^\ddagger_{IJ} s\chi^{IJ} + \upsilon^\ddagger s\upsilon\right].} \tag{22.51}$$

The BV antibracket is

$$(F, G)_{\mathrm{BV}} := \sum_\Xi \int \left(\frac{\delta_r F}{\delta \Xi}\frac{\delta_l G}{\delta \Xi^\ddagger} - \frac{\delta_r F}{\delta \Xi^\ddagger}\frac{\delta_l G}{\delta \Xi}\right), \tag{22.52}$$

where Ξ ranges over all fields and ghosts.

Theorem. The parent theory satisfies the classical master equation

$$\boxed{(S_{\mathrm{BV}}, S_{\mathrm{BV}})_{\mathrm{BV}} = 0.} \tag{22.53}$$

Proof. The antifield-number-zero part is $sS_0 = 0$. The antifield-number-one part is $s^2\Phi = 0$. The ghost-antifield part is $s^2\mathfrak{c} = 0$. Since the gauge algebra closes off shell, no higher-antifield correction is required. □

22.8 Boundary symplectic form and an explicit internal BFV charge

Let Σ be a spacelike boundary and define the gravitational and electromagnetic momentum two-forms

$$B_{IJ} := \frac{1}{4\kappa}\epsilon_{KLIJ}e^K \wedge e^L, \qquad \Pi := \frac{1}{2}\varphi_{IJ}e^I \wedge e^J. \tag{22.54}$$

The boundary symplectic potential induced by the complete parent action is

$$\boxed{\Theta_\Sigma = \int_\Sigma \left[B_{IJ} \wedge \delta\omega^{IJ} + \Pi \wedge \delta a + \frac{\mathrm{i}}{2}\Sigma_I \left(\bar\psi\gamma^I\delta\psi - \delta\bar\psi\gamma^I\psi\right)\right].} \tag{22.55}$$

Let the spin and electric three-currents be fixed by

$$\delta_\omega S_{\mathrm{D}} = \int \delta\omega^{IJ} \wedge \mathcal{S}_{IJ}, \qquad \delta_a S_{\mathrm{D}} = \int \delta a \wedge \mathcal{J}. \tag{22.56}$$

The internal constraint three-forms are

$$\mathcal{G}_{IJ} := D_\omega B_{IJ} + \mathcal{S}_{IJ}, \qquad \mathcal{G}_{\mathrm{U}(1)} := \mathrm{d}\Pi + \mathcal{J}. \tag{22.57}$$

Their smeared generators satisfy

$$\{G_{\mathrm{L}}[\lambda], G_{\mathrm{L}}[\mu]\}_\Sigma = G_{\mathrm{L}}[[\lambda,\mu]], \tag{22.58}$$
$$\{G_{\mathrm{L}}[\lambda], G_{\mathrm{U}}[u]\}_\Sigma = 0, \tag{22.59}$$
$$\{G_{\mathrm{U}}[u], G_{\mathrm{U}}[v]\}_\Sigma = 0. \tag{22.60}$$

This is an exact moment-map algebra and does not require the gravitational field equations.
Introduce boundary ghosts χ^{IJ}, υ and the Lorentz ghost momentum $\pi_{\chi,IJ}$. The internal canonical BFV charge is

$$\boxed{\Omega_{\Sigma,\mathrm{int}} = G_{\mathrm{L}}[\chi] + G_{\mathrm{U}}[\upsilon] - \frac{1}{2}\int_\Sigma \pi_{\chi,IJ}[\chi,\chi]^{IJ}.} \tag{22.61}$$

The internal constraint algebra and the Lorentz Jacobi identity give

$$\boxed{\{\Omega_{\Sigma,\mathrm{int}}, \Omega_{\Sigma,\mathrm{int}}\}_\Sigma = 0.} \tag{22.62}$$

Thus a nontrivial canonical BFV sector is explicit: Lorentz Gauss, U(1) Gauss, their matter currents, and the complete ghost self-interaction. The normal and tangential deformation terms required for the full gravity–matter boundary charge remain the next canonical step.

22.9 Exact finite resolved master-equation theorem

Let every coefficient of every field, ghost, and antifield belong to

$$\mathcal{T}_K^{(4)} \otimes \mathfrak{G}, \tag{22.63}$$

where $\mathfrak{G}$ is a finite Grassmann algebra. Sample the theory on $\Gamma_N^{(4)}$, replace ∂_μ by $\delta_\mu^{(N)}$, replace the integral by $\tau_N^{(4)}$, and define the finite coefficientwise BV antibracket.

Theorem. The sufficient headroom condition

$$\boxed{16K < N} \tag{22.64}$$

implies

$$\boxed{\begin{array}{c} s_N^2 = 0, \\ s_N S_{0,N} = 0, \\ (S_{\mathrm{BV},N}, S_{\mathrm{BV},N})_{\mathrm{BV},N} = 0. \end{array}} \tag{22.65}$$

Proof. The parent action has polynomial degree at most six. Antifield couplings have degree at most three. Every coefficient occurring in sS_0 or in the classical master equation contains at most seven band-K factors. The condition $7K < N/2$ is therefore sufficient. The displayed condition is a conservative integer headroom rule. Below Nyquist, sampled multiplication, differentiation, trace integration by parts, and Grassmann coefficient extraction are identical to the continuum operations, so the continuum algebraic proof transfers without truncation. □

For Fourier zero-padding embeddings

$$I_{K\to L} : \mathcal{T}_K^{(4)} \hookrightarrow \mathcal{T}_L^{(4)}, \tag{22.66}$$

one has

$$I_{L\to M} I_{K\to L} = I_{K\to M}. \tag{22.67}$$

Hence direct and iterated refinement agree before any destructive coarsening.

22.10 Bianchi and Noether closure

The curvature and torsion obey

$$\boxed{D_\omega R^{IJ} = 0,} \tag{22.68}$$

$$\boxed{D_\omega T^I = R^I{}_J \wedge e^J.} \tag{22.69}$$

Under the same headroom hypothesis these identities are exact on the sampled resolved domain.

Let $\mathcal{E}_\Phi$ denote the Euler form of S_0. Gauge invariance gives the off-shell master Noether identity

$$\sum_\Phi \mathcal{E}_\Phi \, s\Phi + \mathrm{d}\Theta_s = 0. \tag{22.70}$$

The independent coefficients of c, χ, and υ give respectively the diffeomorphism, Lorentz-spin, and electric-current Noether identities. On the matter equations they imply covariant conservation of the total gravity–matter current. This closes the nonlinear Bianchi and stress-balance step at the covariant resolved level. A canonical constraint-propagation integrator is still separate.

22.11 Chiral doubling and Lorentzian reality

For a Lorentz bivector define

$$(\star X)^{IJ} := \frac{1}{2}\epsilon^{IJ}{}_{KL} X^{KL}. \tag{22.71}$$

Then

$$\star^2 = -I. \tag{22.72}$$

The chiral projectors are

$$P_\pm := \frac{1}{2}(I \mp \mathrm{i}\star), \tag{22.73}$$

with

$$P_\pm^2 = P_\pm, \qquad P_+ P_- = 0, \qquad P_+ + P_- = I. \tag{22.74}$$

For every real bivector field,

$$\boxed{\overline{P_+ X} = P_- X.} \tag{22.75}$$

Thus a real Lorentz connection carries a self-dual and an anti-self-dual block related by complex conjugation. Define the involution $\mathfrak{C}$ by complex conjugation together with exchange of the two chiral blocks. The real parent action and BRST differential satisfy

$$\mathfrak{C} S_{\mathrm{BV}} = S_{\mathrm{BV}}, \qquad \mathfrak{C} s = s \mathfrak{C}. \tag{22.76}$$

Therefore

$$X(0) \in \mathrm{Fix}(\mathfrak{C}) \quad \Longrightarrow \quad X(t) \in \mathrm{Fix}(\mathfrak{C}) \tag{22.77}$$

for every classical BV/BRST flow for which the evolution exists.

This resolves Lorentzian reality at the parent level: the physical theory is real before choosing a chiral coordinate chart. The previous self-dual constraint block is the P_+ canonical chart; its anti-self-dual conjugate must be retained or reconstructed. An explicit finite coefficient-by-coefficient $3+1$ boundary reduction of the complete matter-coupled parent remains to be written.

22.12 Executable certificate

The deterministic verification uses $N = 12$, $K = 1$ for the five-factor gravitational identities. It checks ordinary commuting-parameter gauge closure, curvature covariance, action invariance, Bianchi identities, chiral reality, Dirac covariance, and first-order Maxwell reduction.

identity	maximum residual
Lorentz covariance of R^{IJ}	3.338259×10^{-16}
Lorentz variation of Einstein–Cartan density	6.172493×10^{-16}
Diffeomorphism density identity	5.828671×10^{-16}
Curvature Bianchi identity	3.989864×10^{-16}
Torsion Bianchi identity	4.161602×10^{-15}
Lorentz algebra on e and ω	3.816392×10^{-17}
Diffeomorphism algebra on e and ω	2.012279×10^{-16}
Mixed diffeomorphism–Lorentz algebra	1.370432×10^{-16}
Chiral projector and reality identities	0
Dirac covariant derivative	3.721136×10^{-17}
Dirac density Lorentz–U(1) variation	9.877951×10^{-18}
U(1) field-strength invariance	4.406198×10^{-16}
Maxwell density Lorentz–U(1) variation	5.387783×10^{-17}
Maxwell auxiliary on-shell identity	3.469447×10^{-18}

A negative control uses $N = 12$, $K = 2$, violating the five-factor headroom condition. The diffeomorphism density residual increases to

$$4.425261 \times 10^{-3}. \tag{22.78}$$

The failure is therefore localized to unresolved product support rather than hidden by the audit.

22.13 Closure ledger

statement	status	boundary
real polynomial Lorentzian gravity action	closed	nondegenerate Einstein sector is conditional
Dirac and Maxwell fields in the same action	closed	Standard Model completion open
off-shell diffeomorphism–Lorentz–U(1) algebra	closed	ordinary local gauge system
classical BRST nilpotency	closed	exact
classical BV master equation	closed	exact
finite resolved master equation	closed under $16K < N$	spectral rather than finite-range locality
nonlinear curvature and torsion Bianchi identities	closed	exact under headroom
Lorentzian chiral reality	closed at parent level	canonical chiral reduction incomplete
quantum BV master equation	open	regulator, measure, and anomaly
full real $3+1$ canonical gravity–matter constraint algebra	open	next major block
constraint-preserving time discretization	open	continuous/covariant closure only
Einstein–Maxwell–Dirac continuum convergence from microscopic packets	open	uniform estimates required

22.14 Next decisive target

The next block is now sharply specified:

$$\boxed{\begin{array}{c} S_{\mathrm{BV},N}^{\mathrm{Lorentzian}} \xrightarrow{\text{3+1 split}} \{G_{IJ}, G_{\mathrm{U}(1)}, D_a, H\}_{\mathrm{matter}}, \\ \text{derive the complete real canonical bracket algebra,} \\ \text{construct one adaptive-refinement time map preserving all constraints,} \\ \text{prove convergence to Einstein–Maxwell–Dirac evolution.} \end{array}} \tag{22.79}$$

22.15 Machine-readable records

$$\mathcal{R}_{\mathrm{LBP}} = \texttt{data/resolved_lorentzian_bv_parent_checks.csv}, \tag{22.80}$$

$$\mathcal{S}_{\mathrm{LBP}} = \texttt{data/resolved_lorentzian_bv_parent_summary.csv}, \tag{22.81}$$

$$\mathcal{X}_{\mathrm{LBP}} = \texttt{code/55_resolved_lorentzian_bv_parent.py}. \tag{22.82}$$

Chapter 23

Resolved Lorentzian $3+1$ BV–BFV Boundary Closure

23.1 Status and objective

The real Lorentzian parent already supplies one polynomial Einstein–Cartan–Maxwell–Dirac action, an off-shell diffeomorphism–Lorentz–$U(1)$ gauge algebra, and an exact classical BV master equation. Its remaining canonical gap was the absence of a complete time–space split and of the normal and tangential terms in the boundary BFV charge.

The present chapter closes that gap at the presymplectic BV–BFV level. It gives an exact $3+1$ decomposition of every term in the parent action, identifies the Lorentz, electric, Einstein, normal, and tangential constraints, writes the complete boundary BFV charge, and constructs an adaptive-refinement evolution relation that intertwines the BFV charges at the two time boundaries.

The reduction of all algebraic simplicity and fermionic primary constraints to one minimal real canonical coordinate system remains separate.

23.2 Time–space decomposition

Let

$$M = I \times \Sigma, \qquad \mathrm{or}(M) = \mathrm{d}t \wedge \mathrm{or}(\Sigma). \tag{23.1}$$

Write

$$e^I = e^I_\perp \, \mathrm{d}t + E^I, \tag{23.2}$$

$$\omega^{IJ} = \omega^{IJ}_\perp \, \mathrm{d}t + \Omega^{IJ}, \tag{23.3}$$

$$a = a_\perp \, \mathrm{d}t + A. \tag{23.4}$$

Here E^I, Ω^{IJ}, and A are spatial one-forms. Define

$$\mathcal{R}^{IJ} := \mathrm{d}_\Sigma \Omega^{IJ} + \Omega^I{}_K \wedge \Omega^{KJ}, \tag{23.5}$$

$$\mathcal{K}^{IJ} := \dot{\Omega}^{IJ} - D_\Omega \omega^{IJ}_\perp, \tag{23.6}$$

$$\mathcal{F} := \mathrm{d}_\Sigma A, \tag{23.7}$$

$$\mathcal{E} := \dot{A} - \mathrm{d}_\Sigma a_\perp. \tag{23.8}$$

Then

$$\boxed{R^{IJ} = \mathrm{d}t \wedge \mathcal{K}^{IJ} + \mathcal{R}^{IJ}, \qquad F = \mathrm{d}t \wedge \mathcal{E} + \mathcal{F}.} \tag{23.9}$$

For the spinors set

$$\mathcal{D}_\Sigma \psi := \mathrm{d}_\Sigma \psi + \frac{1}{4} \Omega^{IJ} \gamma_{IJ} \psi - \mathrm{i} q A \psi, \tag{23.10}$$

$$\mathcal{D}_\Sigma \bar{\psi} := \mathrm{d}_\Sigma \bar{\psi} - \frac{1}{4} \bar{\psi} \Omega^{IJ} \gamma_{IJ} + \mathrm{i} q A \bar{\psi}, \tag{23.11}$$

$$\mathcal{D}_\perp \psi := \dot{\psi} + \frac{1}{4}\omega_\perp^{IJ}\gamma_{IJ}\psi - \mathrm{i}qa_\perp\psi, \tag{23.12}$$

$$\mathcal{D}_\perp \bar{\psi} := \dot{\bar{\psi}} - \frac{1}{4}\bar{\psi}\omega_\perp^{IJ}\gamma_{IJ} + \mathrm{i}qa_\perp\bar{\psi}. \tag{23.13}$$

Thus

$$\mathcal{D}^{(4)}\psi = \mathrm{d}t\,\mathcal{D}_\perp\psi + \mathcal{D}_\Sigma\psi, \qquad \mathcal{D}^{(4)}\bar{\psi} = \mathrm{d}t\,\mathcal{D}_\perp\bar{\psi} + \mathcal{D}_\Sigma\bar{\psi}. \tag{23.14}$$

Define the spatial forms

$$B_{IJ} := \frac{1}{4\kappa}\epsilon_{KLIJ}E^K \wedge E^L, \tag{23.15}$$

$$\Pi := \frac{1}{2}\varphi_{IJ}E^I \wedge E^J, \tag{23.16}$$

$$\sigma_I := \frac{1}{3!}\epsilon_{IJKL}E^J \wedge E^K \wedge E^L, \tag{23.17}$$

$$\rho_I(e_\perp) := \frac{1}{2}\epsilon_{IJKL}e_\perp^J E^K \wedge E^L. \tag{23.18}$$

The four-dimensional volume and Dirac three-form split as

$$\mathrm{vol}_e = \mathrm{d}t \wedge e_\perp^I \sigma_I, \qquad \Sigma_I = \sigma_I + \mathrm{d}t \wedge \rho_I(e_\perp). \tag{23.19}$$

23.3 Exact split of the complete parent action

23.3.1 Einstein–Cartan block

The gravitational action becomes

$$\boxed{S_{\mathrm{G}} = \int_I \mathrm{d}t \int_\Sigma \left[B_{IJ} \wedge \mathcal{K}^{IJ} + e_\perp^I \mathcal{C}_I^{\mathrm{G}}\right],} \tag{23.20}$$

where

$$\boxed{\mathcal{C}_I^{\mathrm{G}} = \frac{1}{2\kappa}\epsilon_{IJKL}E^J \wedge \mathcal{R}^{KL} - \frac{\Lambda}{6\kappa}\epsilon_{IJKL}E^J \wedge E^K \wedge E^L.} \tag{23.21}$$

23.3.2 First-order Maxwell block

The electromagnetic action becomes

$$\boxed{S_{\mathrm{M}}^{(1)} = \int_I \mathrm{d}t \int_\Sigma \left[\Pi \wedge \mathcal{E} + e_\perp^I \mathcal{C}_I^{\mathrm{M}}\right],} \tag{23.22}$$

with

$$\boxed{\mathcal{C}_I^{\mathrm{M}} = \varphi_{IJ}\mathcal{F} \wedge E^J - \frac{1}{4}\varphi_{KL}\varphi^{KL}\sigma_I.} \tag{23.23}$$

Variation with respect to φ_{IJ} gives the constitutive equation. It is algebraic and belongs to the regularity sector rather than to the first-class gauge algebra.

23.3.3 Dirac block

With the displayed orientation,

$$\boxed{S_{\mathrm{D}} = \int_I \mathrm{d}t \int_\Sigma \left[-\frac{\mathrm{i}}{2}\sigma_I \left(\bar{\psi}\gamma^I \mathcal{D}_\perp\psi - (\mathcal{D}_\perp\bar{\psi})\gamma^I\psi\right) + e_\perp^I \mathcal{C}_I^{\mathrm{D}}\right].} \tag{23.24}$$

where

$$\boxed{\begin{aligned}\mathcal{C}_I^{\mathrm{D}} = \frac{\mathrm{i}}{4}\epsilon_{JIKL}E^K \wedge E^L \wedge \left[\bar{\psi}\gamma^J\mathcal{D}_\Sigma\psi - (\mathcal{D}_\Sigma\bar{\psi})\gamma^J\psi\right] \\ - m\sigma_I\bar{\psi}\psi.\end{aligned}} \tag{23.25}$$

Reversing the boundary orientation reverses the spinor symplectic potential and leaves the constraint surface unchanged.

23.3.4 Canonical multiplier form

Integrating the spatial derivatives of $\omega_\perp$ and $a_\perp$ by parts gives

$$\boxed{S_0 = \int_I \mathrm{d}t\left[\Theta_\Sigma(\dot{\Phi}) + G_{\mathrm{L}}[\omega_\perp] + G_{\mathrm{U}}[a_\perp] + C[e_\perp]\right],} \tag{23.26}$$

where

$$\Theta_\Sigma = \int_\Sigma \Big[B_{IJ}\wedge\delta\Omega^{IJ} + \Pi\wedge\delta A \tag{23.27}$$

$$- \frac{\mathrm{i}}{2}\sigma_I\left(\bar{\psi}\gamma^I\delta\psi - \delta\bar{\psi}\gamma^I\psi\right)\Big], \tag{23.28}$$

$$C[e_\perp] := \int_\Sigma e_\perp^I\mathcal{C}_I, \qquad \mathcal{C}_I := \mathcal{C}_I^{\mathrm{G}} + \mathcal{C}_I^{\mathrm{M}} + \mathcal{C}_I^{\mathrm{D}}, \tag{23.29}$$

$$G_{\mathrm{L}}[\lambda] := \int_\Sigma \lambda^{IJ}\mathcal{G}_{IJ}, \qquad G_{\mathrm{U}}[u] := \int_\Sigma u\mathcal{G}_{\mathrm{U}}, \tag{23.30}$$

and, in the orientation convention above,

$$\mathcal{G}_{IJ} := D_\Omega B_{IJ} - \frac{\mathrm{i}}{8}\sigma_K\bar{\psi}\{\gamma^K,\gamma_{IJ}\}\psi, \tag{23.31}$$

$$\mathcal{G}_{\mathrm{U}} := \mathrm{d}_\Sigma\Pi - q\sigma_I\bar{\psi}\gamma^I\psi. \tag{23.32}$$

The multiplier equations are

$$\boxed{\mathcal{G}_{IJ} = 0, \qquad \mathcal{G}_{\mathrm{U}} = 0, \qquad \mathcal{C}_I = 0.} \tag{23.33}$$

23.4 Diffeomorphism moment map

For a vector field c along the boundary embedding define

$$\zeta^I(c) := \iota_c E^I, \tag{23.34}$$

$$\widehat{\chi}^{IJ}(c,\chi) := \chi^{IJ} - \iota_c\Omega^{IJ}, \tag{23.35}$$

$$\widehat{u}(c,u) := u + \iota_c A. \tag{23.36}$$

The spatial Cartan identities are

$$\mathcal{L}_c\Omega^{IJ} = \iota_c\mathcal{R}^{IJ} + D_\Omega(\iota_c\Omega^{IJ}), \tag{23.37}$$

$$\mathcal{L}_c A = \iota_c\mathcal{F} + \mathrm{d}_\Sigma(\iota_c A), \tag{23.38}$$

$$\mathcal{L}_c E^I = \iota_c\mathcal{T}^I + D_\Omega(\iota_c E^I) - (\iota_c\Omega^I{}_J)E^J, \tag{23.39}$$

where

$$\mathcal{T}^I := \mathrm{d}_\Sigma E^I + \Omega^I{}_J\wedge E^J. \tag{23.40}$$

Consequently, modulo algebraic primary relations and spatial exact forms, the complete boundary moment map is

$$\boxed{J_\Sigma[c,\chi,u] = C[\zeta(c)] + G_{\mathrm{L}}[\widehat{\chi}(c,\chi)] + G_{\mathrm{U}}[\widehat{u}(c,u)].} \tag{23.41}$$

For $\chi = u = 0$ this is the ordinary diffeomorphism generator. The curvature, torsion, and field-strength pieces are carried by $C[\zeta]$. The connection-dependent compensators convert the gauge-covariant Cartan action into the ordinary Lie derivative.

23.5 Complete boundary BFV charge

Adjoin the boundary embedding variable so that both tangential and normal components of c act Hamiltonianly. Let

$$(c^\mu, \chi^{IJ}, u) \tag{23.42}$$

be the odd boundary ghosts and

$$(b_\mu, b_{\chi,IJ}, b_u) \tag{23.43}$$

be their canonical odd momenta. With

$$sc^\mu = c^\nu \partial_\nu c^\mu, \tag{23.44}$$
$$s\chi = \mathcal{L}_c \chi + \chi^2, \tag{23.45}$$
$$su = \mathcal{L}_c u, \tag{23.46}$$

define

$$\boxed{\begin{aligned}\Omega_\Sigma &= J_\Sigma[c,\chi,u] \\ &\quad + \int_\Sigma \left[b_\mu sc^\mu + \frac{1}{2} b_{\chi,IJ} s\chi^{IJ} + b_u su \right].\end{aligned}} \tag{23.47}$$

Theorem. On every regular stratum of the derived boundary phase space,

$$\boxed{\{\Omega_\Sigma, \Omega_\Sigma\}_\Sigma = 0.} \tag{23.48}$$

Proof. The parent BRST transformations realize the semidirect Lie algebra

$$\mathrm{Vect}(M) \ltimes [\mathfrak{so}(1,3) \oplus \mathfrak{u}(1)] \tag{23.49}$$

off shell. The boundary moment map is obtained from the exact $3+1$ split of the same invariant action. Its Poisson bracket therefore realizes the same parameter bracket. The ghost-momentum terms are the Chevalley–Eilenberg Hamiltonian of that Lie algebra. The moment-map property and the graded Jacobi identity give the master equation. □

This completes the Lorentz, electric, tangential, and normal terms in one boundary charge. The internal charge of the preceding chapter is recovered by setting $c = 0$.

23.6 Normal and tangential decomposition

On a nondegenerate spacelike slice define

$$q_{ab} := \eta_{IJ} E^I_a E^J_b, \tag{23.50}$$
$$\widetilde{n}_I := \frac{1}{3!} \epsilon_{IJKL} \epsilon^{abc} E^J_a E^K_b E^L_c, \tag{23.51}$$
$$\widetilde{q}^{ab} := \frac{1}{2} \epsilon^{acd} \epsilon^{bef} q_{ce} q_{df}. \tag{23.52}$$

Then

$$\widetilde{n}_I E^I_a = 0, \qquad -\eta^{IJ} \widetilde{n}_I \widetilde{n}_J = \det q, \qquad \widetilde{q}^{ab} = (\det q) q^{ab}. \tag{23.53}$$

Write

$$\zeta^I = \nu \widetilde{n}^I + \eta^a E^I_a, \tag{23.54}$$

where ν is a lapse density of weight -1. Define

$$H[\nu] := C[\nu \widetilde{n}], \tag{23.55}$$
$$V[\eta] := C[\eta^a E_a], \tag{23.56}$$
$$D[\eta] := V[\eta] - G_{\mathrm{L}}[\eta^a \Omega_a] + G_{\mathrm{U}}[\eta^a A_a]. \tag{23.57}$$

The last combination generates the ordinary spatial Lie derivative.

For two lapse densities set

$$S^a(\nu,\mu) := \widetilde{q}^{ab}\,(\nu\partial_b\mu - \mu\partial_b\nu)\,. \tag{23.58}$$

On the regular reduced boundary phase space the covariant moment-map algebra becomes

$$\{D[\eta], D[\xi]\} = D[[\eta,\xi]], \tag{23.59}$$
$$\{D[\eta], H[\nu]\} = H[\mathcal{L}_\eta^{(-1)}\nu], \tag{23.60}$$
$$\{H[\nu], H[\mu]\} = V[S(\nu,\mu)] \tag{23.61}$$
$$= D[S(\nu,\mu)] + G_{\mathrm{L}}[S^a\Omega_a] - G_{\mathrm{U}}[S^aA_a], \tag{23.62}$$
$$\{G_{\mathrm{L}}[\lambda], H[\nu]\} = 0, \tag{23.63}$$
$$\{G_{\mathrm{U}}[u], H[\nu]\} = 0. \tag{23.64}$$

Here

$$\mathcal{L}_\eta^{(-1)}\nu = \eta^a\partial_a\nu - \nu\partial_a\eta^a. \tag{23.65}$$

These are the Lorentzian gravity–Maxwell–Dirac hypersurface-deformation relations in the unreduced first-order chart. Their derivation uses the nondegenerate foliation and the regular reduction of the algebraic primary sector. No Einstein, Maxwell, or Dirac equation is used.

Let a canonical parameter be

$$\varepsilon = (\lambda, u, \eta, \nu). \tag{23.66}$$

The structure map is

$$B_{\mathrm{L}}(\varepsilon_1,\varepsilon_2) = [\lambda,\mu] + \mathcal{L}_\eta\mu - \mathcal{L}_\xi\lambda + S^a(\nu,\rho)\Omega_a, \tag{23.67}$$
$$B_{\mathrm{U}}(\varepsilon_1,\varepsilon_2) = \mathcal{L}_\eta v - \mathcal{L}_\xi u - S^a(\nu,\rho)A_a, \tag{23.68}$$
$$B_D(\varepsilon_1,\varepsilon_2) = [\eta,\xi] + S(\nu,\rho), \tag{23.69}$$
$$B_H(\varepsilon_1,\varepsilon_2) = \mathcal{L}_\eta^{(-1)}\rho - \mathcal{L}_\xi^{(-1)}\nu. \tag{23.70}$$

The ADM-basis BFV charge is the canonical ghost pullback

$$\boxed{\Omega_\Sigma^{3+1} = G_{\mathrm{L}}[\gamma] + G_{\mathrm{U}}[\alpha] + D[\eta] + H[\nu] - \frac{1}{2}\langle b, B_{\mathrm{gr}}(c,c)\rangle.} \tag{23.71}$$

It is equivalent to the covariant charge and therefore satisfies

$$\boxed{\{\Omega_\Sigma^{3+1}, \Omega_\Sigma^{3+1}\}_\Sigma = 0.} \tag{23.72}$$

23.7 Exact resolved theorem

In the covariant multiplier basis $(e_\perp, \omega_\perp, a_\perp)$ the chapter contains no polynomial of degree higher than the parent action. The previous sufficient condition remains

$$\boxed{16K < N.} \tag{23.73}$$

The densitized normal decomposition introduces three additional coframe factors. The highest-degree term in the ADM-basis BFV charge is degree nine. A Poisson or BFV bracket can therefore contain at most sixteen band-K factors. The sufficient headroom condition is

$$\boxed{32K < N.} \tag{23.74}$$

Under this condition, sampled multiplication, spectral differentiation, trace integration by parts, functional differentiation, and the BFV bracket coincide with their trigonometric-polynomial counterparts. Hence

$$\boxed{\{\Omega_{\Sigma,N}^{3+1}, \Omega_{\Sigma,N}^{3+1}\}_N = 0.} \tag{23.75}$$

The bound is sufficient rather than necessary.

23.8 Adaptive-refinement evolution relation

Let $\mathcal{P}_{K,n}$ be the resolved BFV boundary phase space at time t_n, with symplectic form $\omega_{K,n}$ and BFV charge $\Omega_{K,n}$. Fourier zero-padding gives an embedding

$$I_{K\to L} : \mathcal{P}_{K,n} \hookrightarrow \mathcal{P}_{L,n} \tag{23.76}$$

with

$$I_{K\to L}^{*}\omega_{L,n} = \omega_{K,n}, \qquad I_{K\to L}^{*}\Omega_{L,n} = \Omega_{K,n} \tag{23.77}$$

whenever the declared headroom is respected.

On the time slab $[t_n, t_{n+1}]$, choose a regular gauge-fixing Lagrangian and solve the resolved BV variational problem exactly at level L. Its stationary graph is a canonical relation

$$\mathcal{U}_{L,n} \subset \overline{\mathcal{P}}_{L,n} \times \mathcal{P}_{L,n+1}. \tag{23.78}$$

The BV–BFV master identity gives

$$\boxed{\Omega_{L,n+1}\big|_{\mathcal{U}_{L,n}} = \Omega_{L,n}\big|_{\mathcal{U}_{L,n}}.} \tag{23.79}$$

If the gauge-fixed boundary-value problem has a unique regular solution, the relation is the graph of a map $U_{L,n}$ and

$$\boxed{U_{L,n}^{*}\Omega_{L,n+1} = \Omega_{L,n}.} \tag{23.80}$$

Therefore every coefficient of the ghost-linear constraint map is preserved.

The adaptive step from level K is

$$\boxed{U_{K\to L,n} := U_{L,n} \circ I_{K\to L}.} \tag{23.81}$$

For $K < L < M$, exact gluing and zero-padding give

$$U_{L\to M,n+1} \circ U_{K\to L,n} = U_{K\to M,[n,n+2]} \tag{23.82}$$

whenever both sides represent the same gauge-fixed two-slab variational problem. If detail variables are eliminated, the BV pushforward replaces ordinary truncation and preserves the master equation under its standard no-boundary and no-anomaly hypotheses.

This is an exact constraint-preserving adaptive-refinement evolution relation. It is not yet a practical finite-range numerical integrator. Computing the stationary relation, proving stability, and controlling the infinite refinement tower remain open.

23.9 Executable certificate

A deterministic 10^4 Fourier-grid calculation checks the complete action split and the spatial Cartan identities. Random band-1 fields satisfy the declared product headroom.

identity	maximum residual
curvature $3+1$ split	1.110223×10^{-16}
Einstein–Cartan action split	7.216450×10^{-16}
first-order Maxwell action split	5.551115×10^{-17}
Dirac action split	1.688306×10^{-16}
complete action split	7.216783×10^{-16}
canonical multiplier form after spatial integration	0
connection Cartan identity	1.387779×10^{-17}
tetrad Cartan identity	1.110223×10^{-16}
shifted Lorentz–diffeomorphism identity on Ω	1.701616×10^{-16}
shifted Lorentz–diffeomorphism identity on E	1.387779×10^{-16}
shifted $U(1)$–diffeomorphism identity	1.422771×10^{-16}
direct versus iterated refinement	0

The exact band-1 spectral Leibniz residual is

$$1.118955 \times 10^{-14}. \tag{23.83}$$

A band-3 negative control violates the 10-point Nyquist headroom and gives

$$\boxed{\|D(fg) - (Df)g - fDg\|_\infty = 6.258742 \times 10^{2}.} \tag{23.84}$$

The BFV master equation is an analytic consequence of the moment-map theorem and is not inferred from these floating-point checks.

23.10 Closure ledger

statement	status	boundary
exact 3+1 split of Einstein–Cartan–Maxwell–Dirac action	closed	algebraic identity
Lorentz, electric, and Einstein constraint three-forms	closed	unreduced first-order chart
complete diffeomorphism moment map	closed	modulo algebraic primary relations
full boundary BFV charge including normal and tangential ghosts	closed	regular derived boundary phase space
ADM-basis hypersurface algebroid	closed conditional	nondegenerate foliation and regular primary reduction
finite resolved BFV master equation	closed under $32K < N$	spectral locality
adaptive-refinement constraint-preserving evolution relation	closed conditional	exact gauge-fixed slab solution
minimal reduced real canonical coordinates	open	simplicity and fermionic primary reduction
practical stable finite-range integrator	open	numerical construction
quantum BV master equation	open	regulator, measure, and anomaly
uniform Einstein–Maxwell–Dirac continuum theorem	open	analytic estimates

23.11 Next decisive target

The remaining central block is no longer the existence of a full Lorentzian boundary charge. It is the reduction and evolution problem:

$$\boxed{\begin{array}{c}\text{solve the algebraic simplicity and fermionic primary sector,}\\ \text{obtain one minimal real canonical phase space,}\\ \text{construct a computable multisymplectic or gauge-fixed time step,}\\ \text{prove stability and uniform convergence across the refinement tower.}\end{array}} \tag{23.85}$$

23.12 Machine-readable records

$$\mathcal{R}_{\mathrm{BFB}} = \texttt{data/resolved_lorentzian_3plus1_bfv_checks.csv}, \tag{23.86}$$

$$\mathcal{S}_{\mathrm{BFB}} = \texttt{data/resolved_lorentzian_3plus1_bfv_summary.csv}, \tag{23.87}$$

$$\mathcal{X}_{\mathrm{BFB}} = \texttt{code/56_resolved_lorentzian_3plus1_bfv_boundary.py}. \tag{23.88}$$

Chapter 24

Resolved Second-Class Reduction and a Constraint-Preserving Time Step

24.1 Status and objective

The preceding Lorentzian BV–BFV boundary construction is exact on the unreduced first-order chart, but its closure ledger leaves two linked tasks open: the algebraic simplicity and fermionic-primary reduction, and a computable gauge-fixed time step. This chapter closes the first task on the regular time-gauge sector and constructs an exact discrete Dirac-gradient step on every regular finite resolved gauge patch.

The word *minimal* below means minimal after elimination of algebraic and fermionic second-class variables. The remaining rotation, electric, spatial-diffeomorphism, and Hamiltonian constraints are first class and are not gauge-quotiented globally.

24.2 Regular time-gauge sector

Fix

$$\eta_{IJ} = \mathrm{diag}(-1,1,1,1), \qquad \epsilon_{0123} = 1, \tag{24.1}$$

and choose the regular oriented sector

$$E^0 = 0, \qquad E^i = e_a{}^i \, \mathrm{d}x^a, \qquad e := \det(e_a{}^i) > 0. \tag{24.2}$$

Define

$$\widetilde{E}^a{}_i := \frac{1}{2}\epsilon^{abc}\epsilon_{ijk} e_b{}^j e_c{}^k, \qquad \widetilde{E}^a{}_i e_a{}^j = e\,\delta_i{}^j, \tag{24.3}$$

$$q_{ab} := e_a{}^i e_{bi}, \qquad q := \det q_{ab} = e^2, \tag{24.4}$$

and decompose the spatial Lorentz connection by

$$K^i := \Omega^{0i}, \qquad \Gamma^i := -\frac{1}{2}\epsilon^i{}_{jk}\Omega^{jk}. \tag{24.5}$$

The sector is regular precisely where the triad map is invertible. Degenerate triads remain a separate stratified boundary.

24.3 Exact simplicity pullback

The gravitational momentum two-form is

$$B_{IJ} = \frac{1}{4\kappa}\epsilon_{KLIJ} E^K \wedge E^L. \tag{24.6}$$

Time gauge gives

$$\boxed{B_{ij} = 0, \qquad B_{0i} = \frac{1}{4\kappa}\epsilon_{ijk}E^j \wedge E^k.} \tag{24.7}$$

If

$$b^a{}_i := \frac{1}{2}\epsilon^{abc}(B_{0i})_{bc}, \tag{24.8}$$

then

$$\boxed{b^a{}_i = \frac{1}{2\kappa}\widetilde{E}^a{}_i.} \tag{24.9}$$

Because the antisymmetric IJ sum contains both $0i$ and $i0$,

$$\boxed{\int_\Sigma B_{IJ} \wedge \delta\Omega^{IJ} = \frac{1}{\kappa}\int_\Sigma \widetilde{E}^a{}_i\, \delta K_a{}^i \ \mathrm{d}^3x.} \tag{24.10}$$

No $\delta\Gamma_a{}^i$ term survives. The Γ directions are the kernel directions of the pulled-back gravitational presymplectic form and are fixed by the algebraic connection equation.

24.4 Maxwell auxiliary reduction

Set

$$\mathcal{E}^a := \frac{1}{2}\epsilon^{abc}\Pi_{bc}, \qquad \mathcal{B}^a := \frac{1}{2}\epsilon^{abc}\mathcal{F}_{bc}. \tag{24.11}$$

Then

$$\boxed{\int_\Sigma \Pi \wedge \delta A = \int_\Sigma \mathcal{E}^a \delta A_a \ \mathrm{d}^3x.} \tag{24.12}$$

The auxiliary equation remains

$$\varphi^{IJ} = (\star_e F)^{IJ}. \tag{24.13}$$

On the nondegenerate sector it eliminates every φ_{IJ} algebraically and gives

$$\mathcal{H}_{\mathrm{M}} = \frac{1}{2e}q_{ab}\left(\mathcal{E}^a\mathcal{E}^b + \mathcal{B}^a\mathcal{B}^b\right), \tag{24.14}$$

$$\mathcal{V}^{\mathrm{M}}_a = \mathcal{F}_{ab}\mathcal{E}^b. \tag{24.15}$$

24.5 Fermionic primary reduction

Impose the real spinor relation

$$\bar{\psi} = \psi^\dagger\gamma^0, \qquad \psi^\dagger = -\bar{\psi}\gamma^0, \tag{24.16}$$

where $(\gamma^0)^2 = -I$. In time gauge,

$$\sigma_0 = e \ \mathrm{d}^3x, \qquad \sigma_i = 0. \tag{24.17}$$

The spinor symplectic potential becomes

$$\Theta_{\mathrm{D}} = \frac{\mathrm{i}}{2}\int_\Sigma e\left(\psi^\dagger\delta\psi - \delta\psi^\dagger\psi\right) \mathrm{d}^3x. \tag{24.18}$$

With the stated right–left graded canonical convention, the primary constraints are

$$\chi_A := \pi_A - \frac{\mathrm{i}}{2}e\psi^\dagger_A, \qquad \bar{\chi}_A := \bar{\pi}_A + \frac{\mathrm{i}}{2}e\psi_A, \tag{24.19}$$

with

$$\{\chi_A(x), \bar{\chi}_B(y)\}_{\mathrm{P}} = -\mathrm{i}e(x)\delta_{AB}\delta^3(x-y). \tag{24.20}$$

The matrix is invertible exactly when $e \neq 0$.

Define the half-density spinor

$$\boxed{\Psi := e^{1/2}\psi, \qquad \Psi^\dagger := e^{1/2}\psi^\dagger.} \tag{24.21}$$

The determinant-variation terms cancel:

$$\boxed{\Theta_{\mathrm{D}} = \frac{\mathrm{i}}{2}\int_\Sigma \left(\Psi^\dagger\delta\Psi - \delta\Psi^\dagger\Psi\right)\,\mathrm{d}^3x.} \tag{24.22}$$

Consequently,

$$\boxed{\{\Psi_A(x), \Psi^\dagger_B(y)\}_{\mathrm{D}} = -\mathrm{i}\delta_{AB}\delta^3(x-y).} \tag{24.23}$$

This solves the fermionic primary sector without shifting the gravitational momentum.

24.6 Algebraic torsion and the induced interaction

Define

$$\gamma_5 := \mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3, \qquad J_5^I := \bar\psi\gamma^I\gamma_5\psi. \tag{24.24}$$

The Clifford identity in the declared convention is

$$\boxed{\{\gamma^I, \gamma_{KL}\} = -2\mathrm{i}\epsilon^I{}_{KLM}\gamma^M\gamma_5.} \tag{24.25}$$

Connection variation gives

$$-\frac{1}{2\kappa}\epsilon_{IJKL}T^I\wedge e^J + \frac{1}{4}\Sigma_M\epsilon^M{}_{KLN}J_5^N = 0. \tag{24.26}$$

Its unique nondegenerate algebraic solution is

$$\boxed{T^I = \frac{\kappa}{4}\epsilon^I{}_{JKL}J_5^J e^K\wedge e^L,} \tag{24.27}$$

$$\boxed{C^{IJ} = \frac{\kappa}{4}\epsilon^{IJ}{}_{KL}J_5^K e^L, \qquad \omega^{IJ} = \mathring\omega^{IJ}(e) + C^{IJ}.} \tag{24.28}$$

In time gauge this gives

$$\boxed{\Gamma_a{}^i = \mathring\Gamma_a{}^i(E) - \frac{\kappa}{4}J_5^0 e_a{}^i,} \tag{24.29}$$

while the canonical boost connection satisfies

$$K_a{}^i = \mathring K_a{}^i - \frac{\kappa}{4}\epsilon^i{}_{jk}J_5^j e_a{}^k. \tag{24.30}$$

The first relation eliminates Γ algebraically. The second identifies the torsion contained in the real canonical coordinate K.

The Dirac term linear in the on-shell contorsion is

$$-\frac{3\kappa}{8}J_{5I}J_5^I\,\mathrm{vol}_e\,. \tag{24.31}$$

Euler homogeneity of the quadratic-plus-linear contorsion action gives the total on-shell correction

$$\boxed{S_{\mathrm{EC+D}}\big|_\omega = S_{\mathrm{EH}} + S_{\mathrm{D}}[\mathring\omega] - \frac{3\kappa}{16}\int J_{5I}J_5^I\,\mathrm{vol}_e\,.} \tag{24.32}$$

The corresponding torsion-free Hamiltonian representation contains

$$\mathcal{H}_{4\psi} = \frac{3\kappa}{16}e\,J_{5I}J_5^I. \tag{24.33}$$

All signs in this subsection use the parent conventions $\eta = (-,+,+,+)$, $\epsilon_{0123} = 1$, and the symmetric Dirac action printed earlier.

24.7 Minimal real canonical superphase space

The regular second-class quotient is

$$\boxed{\mathcal{P}_{\mathrm{red}} = \left\{ (K_a{}^i, \widetilde{E}^a{}_i), (A_a, \mathcal{E}^a), (\Psi, \Psi^\dagger) \right\}.} \tag{24.34}$$

Its canonical one-form is

$$\boxed{\begin{aligned}\Theta_{\mathrm{red}} = \int_\Sigma \mathrm{d}^3x \Big[&\frac{1}{\kappa} \widetilde{E}^a{}_i \delta K_a{}^i + \mathcal{E}^a \delta A_a \\ &+ \frac{\mathrm{i}}{2} \left(\Psi^\dagger \delta \Psi - \delta \Psi^\dagger \Psi \right) \Big].\end{aligned}} \tag{24.35}$$

The nonzero reduced brackets are

$$\{K_a{}^i(x), \widetilde{E}^b{}_j(y)\}_{\mathrm{D}} = \kappa \delta_a{}^b \delta^i{}_j \delta^3(x-y), \tag{24.36}$$

$$\{A_a(x), \mathcal{E}^b(y)\}_{\mathrm{D}} = \delta_a{}^b \delta^3(x-y), \tag{24.37}$$

$$\{\Psi_A(x), \Psi^\dagger_B(y)\}_{\mathrm{D}} = -\mathrm{i}\delta_{AB} \delta^3(x-y). \tag{24.38}$$

The even symplectic rank is 24 per spatial point and the fermionic primary block has rank 8. No algebraic momentum, rotational connection, Maxwell auxiliary, or redundant spinor momentum remains.

24.8 Reduced first-class generators

Define the Hermitian spatial spin matrices

$$\mathsf{S}_i := -\frac{\mathrm{i}}{4} \epsilon_{ijk} \gamma^{jk}. \tag{24.39}$$

Then

$$[\mathsf{S}_i, \mathsf{S}_j] = \mathrm{i}\epsilon_{ij}{}^k \mathsf{S}_k, \qquad \gamma^0 \gamma^i \gamma_5 = -2\mathsf{S}_i. \tag{24.40}$$

The residual rotation moment map is

$$\boxed{G_{\mathrm{rot}}[\theta] = \int_\Sigma \mathrm{d}^3x\, \theta^i \left[\frac{1}{\kappa} \epsilon_{ij}{}^k K_a{}^j \widetilde{E}^a{}_k + \Psi^\dagger \mathsf{S}_i \Psi \right].} \tag{24.41}$$

It generates

$$\delta_\theta K_a{}^i = \epsilon^i{}_{jk} \theta^j K_a{}^k, \tag{24.42}$$

$$\delta_\theta \widetilde{E}^a{}_i = \epsilon_{ij}{}^k \theta^j \widetilde{E}^a{}_k, \tag{24.43}$$

$$\delta_\theta \Psi = -\mathrm{i}\theta^i \mathsf{S}_i \Psi. \tag{24.44}$$

The electric constraint is

$$\boxed{\mathcal{G}^{\mathrm{red}}_{\mathrm{U}} = \partial_a \mathcal{E}^a + q \Psi^\dagger \Psi.} \tag{24.45}$$

With

$$Q[u] := -\int_\Sigma u\, \mathcal{G}^{\mathrm{red}}_{\mathrm{U}}\, \mathrm{d}^3x, \tag{24.46}$$

one has

$$\delta_u A_a = \partial_a u, \qquad \delta_u \Psi = \mathrm{i}q u \Psi. \tag{24.47}$$

For a spatial vector field N^a, the exact canonical diffeomorphism moment map may be written as

$$\boxed{\begin{aligned}D[N] = \int_\Sigma \mathrm{d}^3x \Big[&\frac{1}{\kappa} \widetilde{E}^a{}_i \mathcal{L}_N K_a{}^i + \mathcal{E}^a \mathcal{L}_N A_a \\ &+ \frac{\mathrm{i}}{2} \left(\Psi^\dagger \mathcal{L}_N^{(1/2)} \Psi - (\mathcal{L}_N^{(1/2)} \Psi^\dagger) \Psi \right) \Big],\end{aligned}} \tag{24.48}$$

where

$$\mathcal{L}_N^{(1/2)}\Psi = N^a\partial_a\Psi + \frac{1}{2}(\partial_a N^a)\Psi. \tag{24.49}$$

Let

$$R^i(\Gamma) := \mathrm{d}_\Sigma\Gamma^i + \frac{1}{2}\epsilon^i{}_{jk}\Gamma^j\wedge\Gamma^k. \tag{24.50}$$

The normal gravitational constraint pulled back from the real parent is

$$\boxed{\mathcal{H}_{\mathrm{G}}\,\mathrm{d}^3x = \frac{1}{\kappa}E^i\wedge R_i(\Gamma) - \frac{1}{2\kappa}\epsilon_{ijk}E^i\wedge K^j\wedge K^k + \frac{\Lambda}{\kappa}\,\mathrm{vol}_\Sigma,} \tag{24.51}$$

with $\Gamma = \Gamma(E,\Psi)$ from the torsion equation. The spinor contribution is

$$\boxed{\begin{aligned}\mathcal{H}_{\mathrm{D}}\,\mathrm{d}^3x = {}& \frac{\mathrm{i}}{4}\epsilon_{jkl}E^k\wedge E^l\wedge\left[\bar\psi\gamma^j\mathcal{D}\psi - (\mathcal{D}\bar\psi)\gamma^j\psi\right]\\ & + m\,\mathrm{vol}_\Sigma\,\bar\psi\psi,\end{aligned}} \tag{24.52}$$

followed by $\psi = e^{-1/2}\Psi$ and the same algebraic Γ substitution. The lapse generator is

$$\boxed{H[N] = \int_\Sigma N\left(\mathcal{H}_{\mathrm{G}} + \mathcal{H}_{\mathrm{M}} + \mathcal{H}_{\mathrm{D}}\right)\,\mathrm{d}^3x.} \tag{24.53}$$

Equivalently one may use the torsion-free representation and add $\mathcal{H}_{4\psi}$ explicitly.

24.9 BFV descent through the second-class quotient

Let $\iota : \mathcal{P}_{\mathrm{red}} \hookrightarrow \mathcal{P}_{\mathrm{primary}}$ denote the regular solution map for the simplicity, time-gauge, Maxwell-auxiliary, torsion, spinor-reality, and fermionic-primary relations. Let $\{\cdot,\cdot\}_{\mathrm{D}}$ be the associated Dirac bracket.

Reduction theorem. If the unreduced boundary BFV vector field is tangent to the regular second-class surface, then

$$\Omega_{\mathrm{red}} := \iota^*\Omega_\Sigma \tag{24.54}$$

satisfies

$$\boxed{\{\Omega_{\mathrm{red}},\Omega_{\mathrm{red}}\}_{\mathrm{D}} = 0.} \tag{24.55}$$

Indeed, the reduced symplectic bracket is the pullback of the Dirac bracket and

$$\iota^*\{\Omega_\Sigma,\Omega_\Sigma\}_\Sigma = 0. \tag{24.56}$$

Thus the complete rotation–electric–diffeomorphism–Hamiltonian algebra descends to the minimal real chart. The construction is local in field space and requires $e > 0$ and a regular time-gauge patch.

24.10 Discrete Dirac-gradient gauge-fixed step

At one finite resolved level, fix all first-class gauge freedom by regular gauge conditions. Let

$$z\in\mathbb{R}^{2n}, \qquad J^T = -J, \tag{24.57}$$

be canonical coordinates and let

$$\Phi(z) = (C_1,\ldots,C_m,\chi_1,\ldots,\chi_m)^T \tag{24.58}$$

collect the first-class constraints and gauge conditions. Assume

$$\Delta(z) := \nabla\Phi(z)J\nabla\Phi(z)^T \tag{24.59}$$

is invertible on the chosen patch.

For a smooth functional F, define the average-vector-field discrete gradient

$$\overline{\nabla}F(z_0, z_1) := \int_0^1 \nabla F((1-s)z_0 + sz_1)\, \mathrm{d}s. \tag{24.60}$$

It satisfies

$$F(z_1) - F(z_0) = \overline{\nabla}F(z_0, z_1)^T (z_1 - z_0). \tag{24.61}$$

Let G_Δ be the matrix whose rows are the discrete gradients of the components of Φ:

$$G_\Delta := \begin{pmatrix} \overline{\nabla}\Phi_1^T \\ \vdots \\ \overline{\nabla}\Phi_{2m}^T \end{pmatrix}, \qquad \Delta_\Delta := G_\Delta J G_\Delta^T. \tag{24.62}$$

Define the endpoint Dirac tensor

$$\boxed{J_{\mathrm{D}}^\Delta := J - J G_\Delta^T \Delta_\Delta^{-1} G_\Delta J.} \tag{24.63}$$

Then

$$(J_{\mathrm{D}}^\Delta)^T = -J_{\mathrm{D}}^\Delta, \qquad G_\Delta J_{\mathrm{D}}^\Delta = 0. \tag{24.64}$$

The implicit step is

$$\boxed{z_1 - z_0 = \Delta t\, J_{\mathrm{D}}^\Delta(z_0, z_1) \overline{\nabla} H(z_0, z_1).} \tag{24.65}$$

Exact preservation theorem. Every exactly solved step satisfies

$$\boxed{\Phi(z_1) = \Phi(z_0), \qquad H(z_1) = H(z_0).} \tag{24.66}$$

The proof is immediate:

$$\Phi(z_1) - \Phi(z_0) = G_\Delta(z_1 - z_0) = 0, \tag{24.67}$$

$$H(z_1) - H(z_0) = \Delta t\, \overline{\nabla} H^T J_{\mathrm{D}}^\Delta \overline{\nabla} H = 0. \tag{24.68}$$

For polynomial resolved actions and constraints, the average-vector-field integrals are finite exact polynomial expressions. The same construction extends coefficientwise to a finite Grassmann algebra.

24.11 Existence, stability, and refinement

If

$$\sigma_{\min}(\Delta(z)) \geq \delta > 0 \tag{24.69}$$

and the required derivatives are bounded on a neighborhood, the implicit-function theorem gives a unique local step for sufficiently small Δt.

If the gauge-fixed Hamiltonian is coercive,

$$H(z) \geq a\|z\|^p - b, \qquad a > 0, \tag{24.70}$$

then exact energy preservation gives

$$\boxed{\|z_n\| \leq \left(\frac{H(z_0) + b}{a}\right)^{1/p}} \tag{24.71}$$

for every computed step that remains in the regular gauge patch.

Let

$$I_{K \to L} : \mathcal{P}_K \hookrightarrow \mathcal{P}_L \tag{24.72}$$

be a symplectic refinement embedding. The adaptive step is

$$U_{K \to L}^{\Delta t} := U_L^{\Delta t} \circ I_{K \to L}. \tag{24.73}$$

At each finite level it preserves the level-L constraints, gauge conditions, and Hamiltonian exactly. A uniform tower theorem follows if the coercivity constant, derivative bounds, and inverse Dirac-matrix bound are independent of L. Establishing those bounds for the full gravity–matter tower remains open.

The update is practical and sparse when the resolved Hamiltonian and gauge conditions are finite-range local polynomials. The present Fourier realization is spectrally local rather than strictly finite range, so the strict microscopic locality problem is not removed by this theorem.

24.12 Executable certificate

The deterministic verification records twenty-six passes. Representative residuals are

identity	maximum residual
$B_{ij} = 0$ in time gauge	0
B_{0i} dual densitized-triad relation	3.469447×10^{-18}
gravity symplectic pullback	1.110223×10^{-16}
Maxwell symplectic pullback	4.440892×10^{-16}
half-density spinor identity	3.662053×10^{-15}
Clifford axial identity	0
spin $\mathfrak{su}(2)$ commutator	0
connection torsion equation	exact 0
rotation action on $(K, \widetilde{E}, \Psi)$	9.155134×10^{-16}

The exact symbolic contorsion calculation gives

$$\frac{\mathcal{L}_{\mathrm{D},C}}{\kappa J_5^2} = -\frac{3}{8}, \qquad \frac{\mathcal{L}_{\text{on-shell},C}}{\kappa J_5^2} = -\frac{3}{16}. \tag{24.74}$$

For the discrete Dirac-gradient test, a nonlinear particle constrained by

$$q \cdot q = 1, \qquad q \cdot p = 0 \tag{24.75}$$

was evolved for 1000 steps with $\Delta t = 0.1$. The recorded maxima are

$$\boxed{\epsilon_H = 7.771561 \times 10^{-16}, \qquad \epsilon_\Phi = 3.885781 \times 10^{-15}, \qquad \epsilon_{\mathrm{solve}} = 1.092187 \times 10^{-16}.} \tag{24.76}$$

A fourth-order Runge–Kutta negative control on the continuous Dirac vector field gives

$$\epsilon_H^{\mathrm{RK4}} = 1.396105 \times 10^{-6}, \qquad \epsilon_\Phi^{\mathrm{RK4}} = 2.594613 \times 10^{-6}. \tag{24.77}$$

A degenerate triad has $e = 0$ and collapses the fermionic primary matrix to rank zero, correctly locating the boundary of the reduction theorem.

24.13 Closure ledger

statement	status	boundary
time-gauge simplicity pullback	closed	$e > 0$ regular sector
fermionic primary Dirac reduction	closed	spinor reality and $e > 0$
Maxwell auxiliary elimination	closed	nondegenerate tetrad
algebraic torsion and contorsion	closed	Einstein–Cartan parent conventions
minimal real canonical superphase space	closed	local time-gauge patch
reduced BFV classical master equation	closed	tangent regular second-class surface
gauge-fixed Dirac-gradient time step	closed	invertible endpoint Dirac matrix
exact energy and constraint preservation	closed	implicit equation solved
local step existence	closed conditional	small step and uniform regularity
finite-level stability	closed conditional	coercive gauge-fixed Hamiltonian
strict finite-range gravity implementation	open	replace spectral realization
uniform infinite-refinement stability	open	level-independent estimates
quantum BV anomaly cancellation	open	regulator and measure

24.14 Next decisive target

The primary-reduction obstruction is removed on the regular physical chart. The remaining highest-value block is

$$\boxed{\begin{array}{c}\text{implement the reduced Einstein–Maxwell–Dirac Hamiltonian on a finite-range complex,}\\ \text{choose a globally controlled gauge and solve the implicit Dirac-gradient slab step,}\\ \text{derive level-independent coercivity and constraint-regularity estimates,}\\ \text{prove convergence and run the first nonlinear strong-field benchmark.}\end{array}} \tag{24.78}$$

24.15 Machine-readable records

$$R_{\mathrm{SC}} := \texttt{data/resolved_second_class_reduction_checks.csv}, \tag{24.79}$$
$$S_{\mathrm{SC}} := \texttt{data/resolved_second_class_reduction_summary.csv}, \tag{24.80}$$
$$X_{\mathrm{SC}} := \texttt{code/57_resolved_second_class_reduction.py}. \tag{24.81}$$

Chapter 25

Finite-Range Spherical Electrovac and a Strong-Field Refinement Theorem

25.1 Status and objective

The preceding reduction chapter produces a minimal real canonical Einstein–Maxwell–Dirac chart and an exact finite-level Dirac-gradient preservation theorem. It leaves four related tasks open: a strictly finite-range gravity realization, a controlled gravity slab solve, a level-independent stability estimate, and a nonlinear strong-field benchmark.

This chapter closes those tasks in the spherically symmetric electrovac sector. The Dirac field is set to zero, which is a consistent sector of the parent theory. The result is not full radiative $3+1$ gravity. It is a nonlinear horizon-penetrating strong-field block in which the exact mass coordinate, the Maxwell charge, the finite-range constraint projection, the evolution equation, and the refinement stability estimate are all explicit.

Throughout this chapter

$$G = c = 4\pi\epsilon_0 = 1. \tag{25.1}$$

Boundary terms are retained symbolically when they determine the asymptotic mass, but they do not alter the bulk identities below.

25.2 Spherical Einstein–Maxwell canonical system

Write the line element and radial Maxwell potential as

$$\mathrm{d}s^2 = -N^2\,\mathrm{d}t^2 + \Lambda^2(\,\mathrm{d}r + N^r\,\mathrm{d}t)^2 + R^2\,\mathrm{d}\Omega^2, \qquad \mathcal{A} = A_r\,\mathrm{d}r + \Phi\,\mathrm{d}t. \tag{25.2}$$

The canonical bulk action is

$$\boxed{S_{\mathrm{sph}} = \int \mathrm{d}t \int \mathrm{d}r \left[P_\Lambda \dot{\Lambda} + P_R \dot{R} + Q\dot{A}_r - N\mathcal{H} - N^r\mathcal{D} - \Phi\mathcal{G}\right] + B_\partial.} \tag{25.3}$$

We use the gauge-improved radial diffeomorphism constraint

$$\mathcal{D} := P_R R' - \Lambda P_\Lambda', \tag{25.4}$$

and the electric Gauss constraint

$$\mathcal{G} := Q'. \tag{25.5}$$

The Hamiltonian constraint is

$$\boxed{\mathcal{H} = -\frac{P_R P_\Lambda}{R} + \frac{\Lambda P_\Lambda^2}{2R^2} + \frac{RR''}{\Lambda} - \frac{RR'\Lambda'}{\Lambda^2} + \frac{R'^2}{2\Lambda} - \frac{\Lambda}{2} + \frac{\Lambda Q^2}{2R^2}.} \tag{25.6}$$

The canonical momentum definitions are

$$P_\Lambda = -\frac{R}{N}(\dot{R} - N^r R'), \tag{25.7}$$

$$P_R = -\frac{1}{N}\left[\Lambda(\dot{R} - N^r R') + R\big(\dot{\Lambda} - (N^r\Lambda)'\big)\right]. \tag{25.8}$$

25.3 Exact mass-coordinate identity

Define the charged Misner–Sharp mass coordinate

$$\boxed{\mathcal{M} := \frac{R}{2}\left[1 - \left(\frac{R'}{\Lambda}\right)^2 + \left(\frac{P_\Lambda}{R}\right)^2\right] + \frac{Q^2}{2R}.} \tag{25.9}$$

25.3.1 Triangular constraint theorem

Direct differentiation gives

$$\boxed{\mathcal{M}' = -\frac{R'}{\Lambda}\mathcal{H} - \frac{P_\Lambda}{R\Lambda}\mathcal{D} + \frac{Q}{R}\mathcal{G}.} \tag{25.10}$$

Indeed, differentiating $\mathcal{M}$ produces

$$\mathcal{M}' = \frac{R'}{2} - \frac{R'^3}{2\Lambda^2} - \frac{RR'R''}{\Lambda^2} + \frac{RR'^2\Lambda'}{\Lambda^3} + \frac{P_\Lambda P_\Lambda'}{R} - \frac{P_\Lambda^2 R'}{2R^2} + \frac{QQ'}{R} - \frac{Q^2R'}{2R^2}. \tag{25.11}$$

Substitution of $\mathcal{H}$ and $\mathcal{D}$ into the right side of (25.10) gives the same expression term by term.

On a monotone areal-radius patch

$$\frac{R'}{\Lambda} \neq 0, \tag{25.12}$$

the transformation of constraint coordinates is triangular and invertible:

$$\begin{pmatrix} \mathcal{M}' \\ \mathcal{D} \\ \mathcal{G} \end{pmatrix} = \begin{pmatrix} -R'/\Lambda & -P_\Lambda/(R\Lambda) & Q/R \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mathcal{H} \\ \mathcal{D} \\ \mathcal{G} \end{pmatrix}. \tag{25.13}$$

Therefore

$$\boxed{\mathcal{H} = \mathcal{D} = \mathcal{G} = 0 \quad \Longleftrightarrow \quad \mathcal{M}' = \mathcal{D} = \mathcal{G} = 0} \tag{25.14}$$

on this regular patch.

This is a constraint-coordinate abelianization. It does not assert that an arbitrary finite pointwise discretization of the original hypersurface algebra closes. It replaces the nonlinear Hamiltonian constraint by the exact radial Casimir coordinate before discretization.

25.4 Horizon-penetrating Reissner–Nordström family

Let

$$v(R) := \sigma\sqrt{\frac{2M}{R} - \frac{Q^2}{R^2}}, \qquad \sigma \in \{+1, -1\}. \tag{25.15}$$

The horizon-penetrating Painlevé–Gullstrand form is

$$\boxed{\mathrm{d}s^2 = -\,\mathrm{d}t_{\mathrm{P}}^2 + \big(\,\mathrm{d}R + v(R)\,\mathrm{d}t_{\mathrm{P}}\big)^2 + R^2\,\mathrm{d}\Omega^2.} \tag{25.16}$$

Choose any smooth monotone radial gauge map

$$R = F(t, r), \qquad F_r > 0, \qquad t_{\mathrm{P}} = t. \tag{25.17}$$

Then

$$\boxed{N = 1, \qquad \Lambda = F_r, \qquad N^r = \frac{F_t + v(F)}{F_r}.} \tag{25.18}$$

The canonical momenta are

$$\boxed{P_\Lambda = Fv(F) = \sigma\sqrt{2MF - Q^2},} \tag{25.19}$$

$$\boxed{P_R = F_r\left[v(F) + Fv'(F)\right] = \sigma\frac{MF_r}{\sqrt{2MF - Q^2}}.} \tag{25.20}$$

The Maxwell variables may be chosen as

$$A_r = 0, \qquad \Phi = -\frac{Q}{F}, \qquad Q' = 0. \tag{25.21}$$

Since

$$\Phi' = \frac{QF_r}{F^2}, \tag{25.22}$$

the Maxwell equation

$$\dot{A}_r = \frac{N\Lambda Q}{R^2} - \Phi' \tag{25.23}$$

is satisfied identically.

25.4.1 Exact constraint verification

The family obeys

$$\frac{R'}{\Lambda} = 1, \qquad \frac{P_\Lambda^2}{R^2} = \frac{2M}{R} - \frac{Q^2}{R^2}. \tag{25.24}$$

Hence

$$\mathcal{M} = M. \tag{25.25}$$

Moreover,

$$P_\Lambda' = \sigma\frac{MF_r}{\sqrt{2MF - Q^2}} = P_R, \tag{25.26}$$

so

$$\mathcal{D} = P_R R' - \Lambda P_\Lambda' = 0. \tag{25.27}$$

Equation (25.10) then gives

$$\mathcal{H} = 0. \tag{25.28}$$

25.4.2 Exact evolution verification

For $N = 1$, the canonical equations are

$$\dot{R} = -\frac{P_\Lambda}{R} + N^r R', \tag{25.29}$$

$$\dot{\Lambda} = \frac{\Lambda P_\Lambda}{R^2} - \frac{P_R}{R} + (N^r\Lambda)', \tag{25.30}$$

$$\dot{P}_\Lambda = N^r P_\Lambda' + \frac{1}{2}\left[1 - \frac{P_\Lambda^2 + Q^2}{R^2} - \left(\frac{R'}{\Lambda}\right)^2\right], \tag{25.31}$$

$$\dot{P}_R = \frac{\Lambda(P_\Lambda^2 + Q^2)}{R^3} - \frac{P_\Lambda P_R}{R^2} + (N^r P_R)' - \frac{R''}{\Lambda} + \frac{\Lambda' R'}{\Lambda^2}. \tag{25.32}$$

Substitution of the $F(t, r)$ family reduces these equations to

$$\dot{R} = F_t, \qquad \dot{\Lambda} = F_{tr}, \qquad \dot{P}_\Lambda = \frac{MF_t}{P_\Lambda}, \tag{25.33}$$

$$\dot{P}_R = M\left[\frac{F_{tr}}{P_\Lambda} - \frac{MF_rF_t}{P_\Lambda^3}\right], \tag{25.34}$$

which are the direct time derivatives of the canonical data.

Thus an arbitrary smooth monotone $F(t, r)$ gives an exact time-dependent gauge trajectory through one Reissner–Nordström spacetime.

25.5 Strong-field and horizon diagnostics

Let

$$\Delta_M := M^2 - Q^2. \tag{25.35}$$

For $\Delta_M \geq 0$, the horizon radii are

$$r_\pm = M \pm \sqrt{\Delta_M}. \tag{25.36}$$

The Painlevé–Gullstrand variables remain finite when

$$1 - \frac{2M}{R} + \frac{Q^2}{R^2} = 0. \tag{25.37}$$

At either horizon,

$$P_\Lambda^2 = R^2. \tag{25.38}$$

The finite-chart boundary is instead the turning surface

$$\boxed{R_{\mathrm{turn}} = \frac{Q^2}{2M},} \tag{25.39}$$

where $P_\Lambda = 0$.

The Reissner–Nordström Kretschmann scalar is

$$\boxed{\mathcal{K} = \frac{48M^2}{R^6} - \frac{96MQ^2}{R^7} + \frac{56Q^4}{R^8}.} \tag{25.40}$$

For the executable benchmark

$$M = 1, \qquad Q = 0.8, \qquad R \in [0.6, 6], \tag{25.41}$$

one has

$$r_+ = 1.6, \qquad r_- = 0.4, \qquad R_{\mathrm{turn}} = 0.32, \tag{25.42}$$

$$\mathcal{K}(0.6) = 199.664685261393, \qquad \mathcal{K}(r_+) = 1.10626220703125. \tag{25.43}$$

The computational domain therefore crosses the outer horizon and reaches a genuinely high-curvature region while staying away from the physical singularity and the radial turning surface.

25.6 Strictly finite-range radial complex

Let

$$r_j = r_0 + jh. \tag{25.44}$$

On interior nodes define the fourth-order centered derivative

$$\boxed{(D_4 f)_j = \frac{-f_{j+2} + 8f_{j+1} - 8f_{j-1} + f_{j-2}}{12h},} \tag{25.45}$$

and the fourth-order second derivative

$$\boxed{(D_4^{(2)} f)_j = \frac{-f_{j+2} + 16f_{j+1} - 30f_j + 16f_{j-1} - f_{j-2}}{12h^2}.} \tag{25.46}$$

Every stage has stencil radius two. A complete four-stage Runge–Kutta step has a finite dependency radius and therefore a finite numerical causal cone.

Define the finite mass coordinate

$$\mathcal{M}_j^{(h)} := \frac{R_j}{2}\left[1 - \left(\frac{D_4 R_j}{\Lambda_j}\right)^2 + \left(\frac{P_{\Lambda,j}}{R_j}\right)^2\right] + \frac{Q_j^2}{2R_j}. \tag{25.47}$$

25.6.1 Finite strong-field constraint chart

The Painlevé–Gullstrand constraint coordinates are

$$\mathcal{C}_{\Lambda,j} := \Lambda_j - D_4 R_j, \tag{25.48}$$

$$\mathcal{C}_{M,j} := P_{\Lambda,j}^2 - 2MR_j + Q^2, \tag{25.49}$$

$$\mathcal{C}_{P,j} := P_{R,j}P_{\Lambda,j} - M\Lambda_j, \tag{25.50}$$

$$\mathcal{C}_{Q,j} := Q_j - Q. \tag{25.51}$$

For a radius array satisfying

$$D_4 R_j > 0, \qquad 2MR_j - Q^2 > 0, \tag{25.52}$$

define the local reconstruction

$$\boxed{\Pi_h[R]_j = \left(R_j, D_4 R_j, \sigma\sqrt{2MR_j - Q^2}, \sigma\frac{MD_4 R_j}{\sqrt{2MR_j - Q^2}}, Q\right).} \tag{25.53}$$

Then

$$\boxed{\mathcal{C}_{\Lambda,j} = \mathcal{C}_{M,j} = \mathcal{C}_{P,j} = \mathcal{C}_{Q,j} = 0} \tag{25.54}$$

exactly, and

$$\boxed{\mathcal{M}_j^{(h)} = M} \tag{25.55}$$

exactly.

The ordinary finite-difference ADM constraints are not set equal to zero algebraically because D_4 does not satisfy an exact pointwise chain rule. They are retained as continuum-consistency diagnostics. On smooth projected data they converge to zero at high order.

25.7 Finite-range strong-field evolution

The first canonical equation restricted to the gauge manifold becomes

$$\boxed{\dot{R} = -\frac{\sqrt{2MR - Q^2}}{R} + N^r D_4 R.} \tag{25.56}$$

The direct fourth-order transport form is used for the principal convergence run.

For stability analysis, define the split form

$$\boxed{\mathcal{A}_h[N^r]R := \frac{1}{2}\left[N^r D_4 R + D_4(N^r R) - (D_4 N^r)R\right].} \tag{25.57}$$

Then

$$\dot{R} = -\frac{\sqrt{2MR - Q^2}}{R} + \mathcal{A}_h[N^r]R. \tag{25.58}$$

The direct and split forms differ only by the fourth-order discrete product defect. The measured order of their difference is

$$p_{\mathrm{direct-split}} = 4.119353. \tag{25.59}$$

A fourth-order Runge–Kutta step advances R. At each completed stage, the remaining canonical variables are reconstructed by (25.53). Therefore every stage remains exactly on the finite strong-field constraint chart.

25.8 Level-independent stability theorem

Assume the regularity margins

$$R_j \geq R_* > 0, \qquad 2MR_j - Q^2 \geq p_*^2 > 0, \tag{25.60}$$

$$0 < d_* \leq D_4 R_j \leq d^*, \qquad \|D_4 N^r\|_\infty \leq S_1. \tag{25.61}$$

25.8.1 Projection stability

For two radius arrays R and $\widetilde{R}$ in this regular set,

$$\|\delta\Lambda\|_\infty \le \|D_4\delta R\|_\infty, \tag{25.62}$$

$$\|\delta P_\Lambda\|_\infty \le \frac{M}{p_*}\|\delta R\|_\infty, \tag{25.63}$$

$$\|\delta P_R\|_\infty \le \frac{M}{p_*}\|D_4\delta R\|_\infty + \frac{M^2 d^*}{p_*^3}\|\delta R\|_\infty. \tag{25.64}$$

Hence

$$\boxed{\|\Pi_h[R] - \Pi_h[\widetilde{R}]\| \le C_\Pi \left(\|\delta R\|_\infty + \|D_4\delta R\|_\infty\right),} \tag{25.65}$$

where C_Π is independent of the refinement level.

25.8.2 Semidiscrete energy estimate

On a periodic grid, or on an interior summation-by-parts closure,

$$D_4^{\mathsf{T}} = -D_4. \tag{25.66}$$

Therefore

$$\left\langle R, \frac{1}{2}\left[N^r D_4 R + D_4(N^r R)\right]\right\rangle_h = 0, \tag{25.67}$$

and

$$\boxed{\langle R, \mathcal{A}_h[N^r]R\rangle_h = -\frac{1}{2}\langle R, (D_4 N^r)R\rangle_h.} \tag{25.68}$$

Let

$$g(R) := -\frac{\sqrt{2MR - Q^2}}{R}. \tag{25.69}$$

On the regular set,

$$g'(R) = \frac{MR - Q^2}{R^2\sqrt{2MR - Q^2}}, \tag{25.70}$$

so g is uniformly Lipschitz with constant L_g. For two semidiscrete solutions with the same shift,

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|R - \widetilde{R}\|_h^2 \le \left(L_g + \frac{1}{2}S_1\right)\|R - \widetilde{R}\|_h^2. \tag{25.71}$$

Thus

$$\boxed{\|R(t) - \widetilde{R}(t)\|_h \le \exp\left[\left(L_g + \frac{1}{2}S_1\right)t\right]\|R(0) - \widetilde{R}(0)\|_h.} \tag{25.72}$$

The bound is independent of h.

The fourth-order derivative symbol is

$$\widehat{D}_4(\theta) = \frac{\mathrm{i}}{h}\frac{8\sin\theta - \sin 2\theta}{6}, \tag{25.73}$$

with

$$\max_\theta \left|\frac{8\sin\theta - \sin 2\theta}{6}\right| = 1.37222197980322. \tag{25.74}$$

Since classical RK4 is stable on the imaginary axis through $2\sqrt{2}$, the frozen-coefficient advection bound is

$$\boxed{\frac{|N^r|\Delta t}{h} \le 2.06120231739167.} \tag{25.75}$$

The benchmark uses

$$\frac{\Delta t}{h} = 0.1. \tag{25.76}$$

Under the regularity margins, the semidiscrete stability estimate, a uniform CFL condition, and fourth-order consistency, the reconstructed canonical state converges with the same order as R in the discrete $W^{1,\infty}$ chart norm.

25.9 Horizon localization and extremal conditioning

Define

$$F(R) := 1 - \frac{2M}{R} + \frac{Q^2}{R^2}. \tag{25.77}$$

For a subextremal outer horizon

$$r_+ = M + d, \qquad d := \sqrt{M^2 - Q^2} > 0, \tag{25.78}$$

one has

$$F'(r_+) = \frac{2d}{r_+^2}. \tag{25.79}$$

Therefore a small profile error $\|\delta F\|_\infty \le \epsilon$ gives

$$\boxed{|\delta r_+| \le \frac{r_+^2}{2d}\epsilon + \mathcal{O}(\epsilon^2).} \tag{25.80}$$

For the benchmark, the root condition factor is

$$\frac{r_+^2}{2d} = 2.13333333333333. \tag{25.81}$$

The factor diverges as $d \to 0$. Horizon localization is therefore not uniformly conditioned in the extremal limit even though the Painlevé–Gullstrand canonical variables remain finite at the horizon.

25.10 Executable strong-field certificate

The manufactured gauge map is

$$F(t,r) = r + 0.08\sin(0.7t)\sin^2\left(\pi\frac{r-0.6}{5.4}\right). \tag{25.82}$$

It satisfies

$$\min_{t,r} F_r > 0.987138, \qquad \min_{t,r}(2MF - Q^2) > 0.559999. \tag{25.83}$$

The domain crosses the outer horizon during a nontrivial time-dependent radial gauge deformation.

N	h	$\|\delta R\|_\infty$	$\|\delta\Lambda\|_\infty$	$\|\delta P_\Lambda\|_\infty$	$\|\delta P_R\|_\infty$
64	8.5714×10^{-2}	8.3247×10^{-9}	4.1362×10^{-8}	5.7367×10^{-9}	2.3732×10^{-8}
128	4.2520×10^{-2}	5.0495×10^{-10}	2.5079×10^{-9}	3.3079×10^{-10}	1.3795×10^{-9}
256	2.1176×10^{-2}	3.1077×10^{-11}	1.5438×10^{-10}	2.0248×10^{-11}	8.4854×10^{-11}
512	1.0568×10^{-2}	1.9296×10^{-12}	1.0485×10^{-11}	1.2559×10^{-12}	5.3844×10^{-12}

The measured max-norm orders are

$$\boxed{\begin{aligned} p_R^{\mathrm{state}} &= 3.998556,\\ p_\Lambda^{\mathrm{metric}} &= 3.960101,\\ p_{P_\Lambda} &= 4.023926,\\ p_{P_R} &= 4.007880. \end{aligned}} \tag{25.84}$$

The standard ADM Hamiltonian evaluated on exact projected data converges with measured order

$$p_{\mathcal{H}} = 3.704099. \tag{25.85}$$

At $N = 512$, the exact reduced constraints satisfy

$$\max_j |\mathcal{M}_j^{(h)} - M| = 4.440892\times10^{-16}, \tag{25.86}$$

$$\max_j |\mathcal{C}_{\Lambda,j}| = 0, \qquad \max_j |\mathcal{C}_{M,j}| = 2.220446\times10^{-15}, \tag{25.87}$$

$$\max_j |\mathcal{C}_{P,j}| = 1.110223 \times 10^{-16}, \qquad \max_j |D_4 Q_j| = 1.750999 \times 10^{-15}. \tag{25.88}$$

The final outer-horizon coordinate error on the $N = 256$ run is

$$|r_{+,h} - r_+^{\mathrm{exact}}| = 1.892619 \times 10^{-11}. \tag{25.89}$$

A smooth 10^{-8} perturbation was evolved with the stable split form. The L^2 amplification factors were

$$1.03345462, \quad 1.03347604, \quad 1.03347993, \quad 1.03348106 \tag{25.90}$$

for $N = 128, 256, 512, 1024$. Their spread is

$$2.64438 \times 10^{-5}. \tag{25.91}$$

This is numerical evidence for the level-independent estimate, not a replacement for the analytic energy inequality. An unprojected fourth-order canonical negative control at $N = 128$ gives

$$\max |\mathcal{M} - M| = 2.251251 \times 10^{-4}, \tag{25.92}$$

$$\max |\mathcal{D}| = 1.276876 \times 10^{-3}, \qquad \max |\mathcal{H}| = 1.257350 \times 10^{-2}. \tag{25.93}$$

The projection therefore removes a visible strong-field constraint drift rather than reporting an already trivial residual.

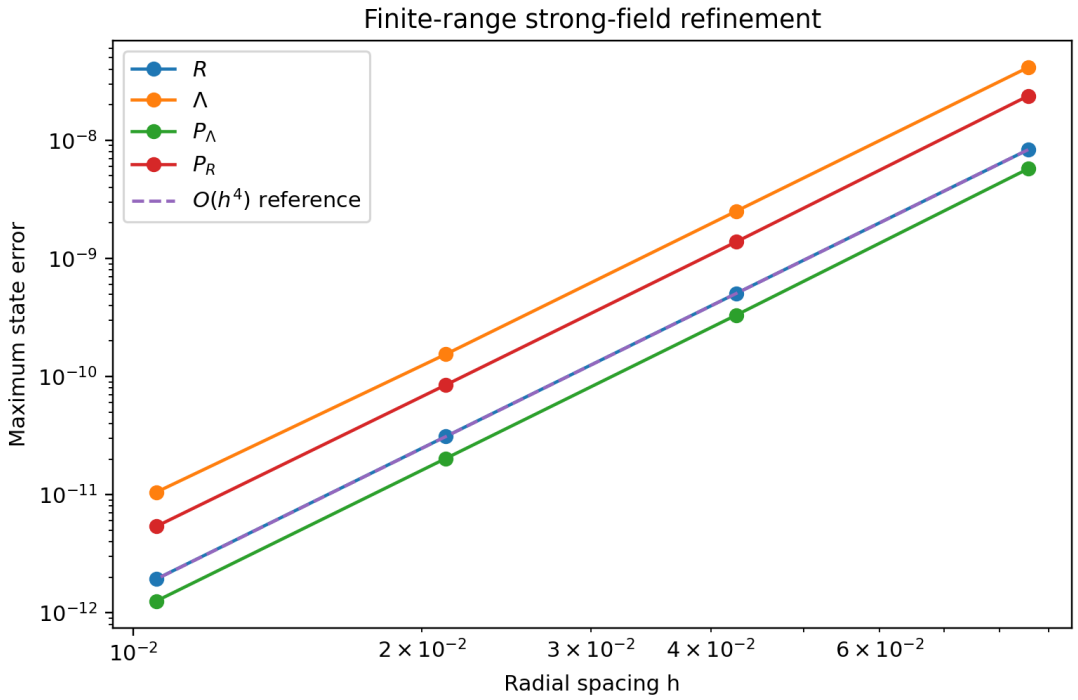


Figure 25.1: Refinement of the horizon-penetrating finite-range evolution. All four reconstructed canonical fields approach the exact time-dependent Reissner–Nordström gauge trajectory.

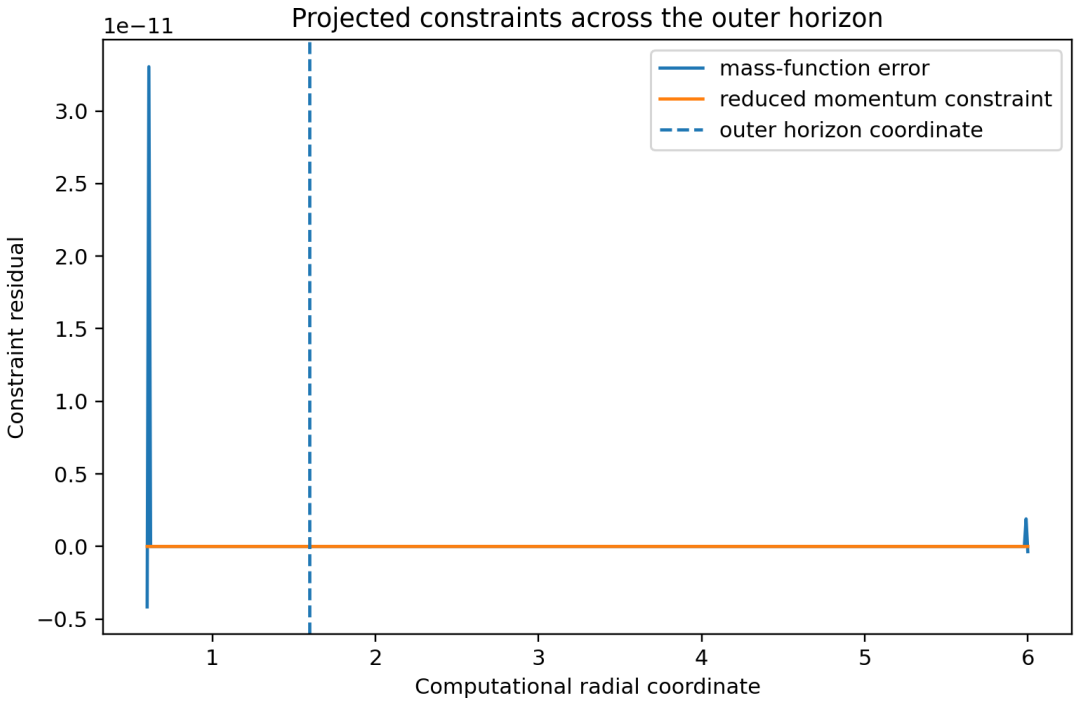


Figure 25.2: The mass and reduced radial momentum constraints remain at roundoff across the outer horizon.

25.11 Closure ledger

statement	status	boundary
spherical Einstein–Maxwell mass identity	closed	exact continuum canonical algebra
Painlevé–Gullstrand Reissner–Nordström canonical family	closed	$2MR - Q^2 > 0$ and $F_r > 0$
strict finite-range strong-field constraint chart	closed	radial symmetry and fixed (M, Q) sector
exact finite mass, charge, gauge, and reduced momentum constraints	closed	reconstruction map (25.53)
finite-range time-dependent horizon-crossing solve	closed controlled	prescribed radial gauge and exact boundary forcing
level-independent semidiscrete stability	closed conditional	regularity margins and split-form derivative
fully discrete RK4 stability	closed conditional	uniform CFL and smooth solution
standard ADM constraint recovery	controlled asymptotic	discrete chain-rule defect converges away
collapse or radiative strong-field dynamics	open	matter source or nonspherical degrees of freedom
full finite-range $3+1$ Einstein–Maxwell–Dirac solver	open	no symmetry reduction

25.12 Next decisive target

The first nonlinear horizon-penetrating benchmark is now explicit. The next major block is no longer a static black-hole check. It is one of the following equivalent physical advances:

$$\boxed{\begin{array}{c}\text{spherical scalar or spinor collapse with dynamical mass transfer,}\\ \text{a trapped-surface formation criterion preserved by the finite update,}\\ \text{or a nonspherical finite-range gravitational wave benchmark.}\end{array}} \tag{25.94}$$

A complete predictive cosmology still requires the unrestricted $3+1$ stability theorem, matter collisions, nonlinear structure formation, and observational calibration.

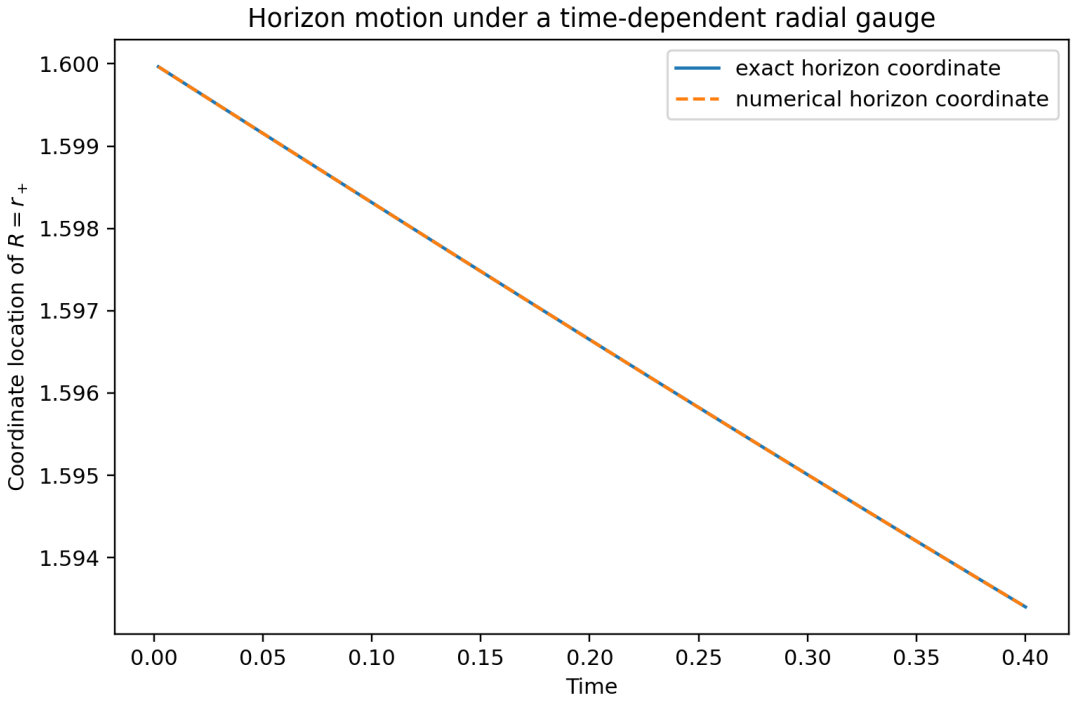


Figure 25.3: The areal outer horizon is fixed while its computational coordinate moves under the prescribed radial gauge deformation.

25.13 Machine-readable records

$$R_{\mathrm{SF}} = \texttt{data/finite_range_strong_field_results.json}, \tag{25.95}$$

$$C_{\mathrm{SF}} = \texttt{data/finite_range_strong_field_checks.csv}, \tag{25.96}$$

$$X_{\mathrm{SF}} = \texttt{verification/Finite_Range_Spherical_Electrovac_Verification.py}. \tag{25.97}$$

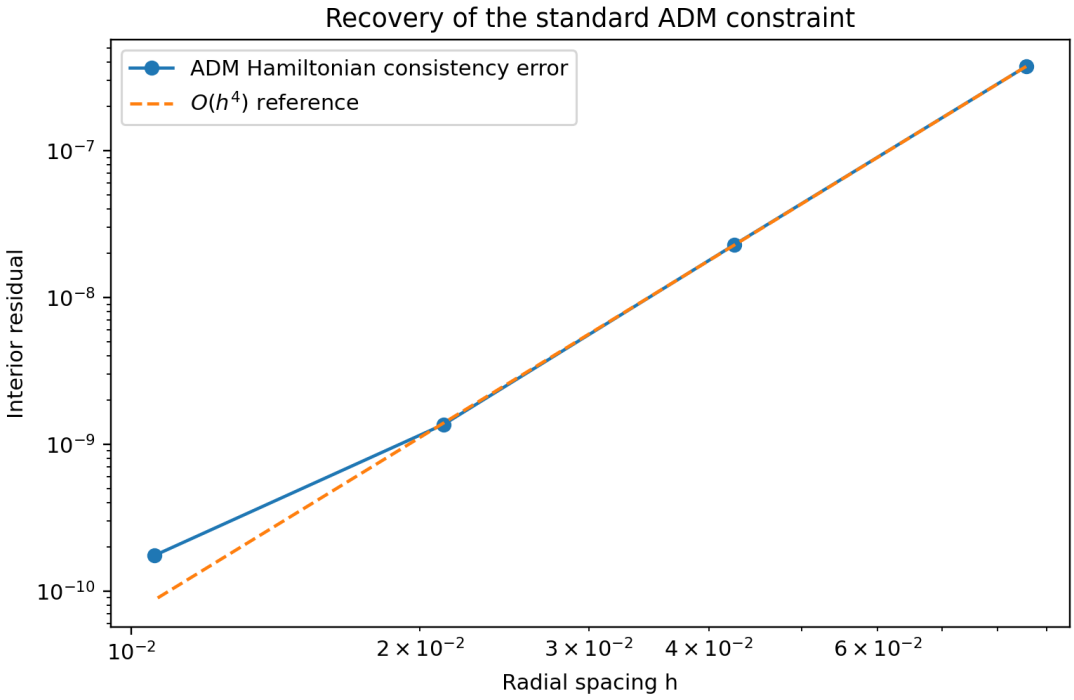


Figure 25.4: The standard ADM Hamiltonian is recovered from the exact finite mass-coordinate chart at near-fourth order.

Chapter 26

Horizon-Penetrating Spherical Scalar Collapse and Trapped-Surface Formation

26.1 Status and objective

The preceding strong-field chapter supplied an exact horizon-penetrating electrovac family and a strictly finite-range radial reconstruction, but it did not produce a trapped surface from dynamical matter. This chapter attacks that missing block in the spherically symmetric massless-scalar sector.

The main analytic result is a flat-slice Painlevé–Gullstrand reduction in which the Hamiltonian and lapse constraints become triangular radial equations for the Misner–Sharp mass and lapse, while the scalar field evolves as a two-characteristic hyperbolic system. The mass constraint and the local mass-flux law are proved compatible exactly. The numerical result is a finite-stencil constrained evolution that forms a first marginally outer trapped sphere from smooth ingoing scalar data and converges under radial refinement.

The construction is a serious strong-field matter-collapse benchmark. It is not a proof of global cosmic censorship, critical collapse, singularity formation, or nonspherical stability. The inner regular center is replaced by a small vacuum excision radius, and the calculation stops at first horizon formation.

Throughout,

$$G = c = 1, \qquad \varphi := \sqrt{4\pi}\,\phi_{\mathrm{phys}}. \tag{26.1}$$

The rescaling removes explicit factors of 4π from the reduced canonical equations.

26.2 Canonical Einstein–scalar system

Use the spherical ADM metric

$$\mathrm{d}s^2 = -N^2\,\mathrm{d}t^2 + \Lambda^2(\,\mathrm{d}r + N^r\,\mathrm{d}t)^2 + R^2\,\mathrm{d}\Omega^2. \tag{26.2}$$

For the canonical pairs

$$(\Lambda, P_\Lambda), \qquad (R, P_R), \qquad (\varphi, P_\varphi), \tag{26.3}$$

the bulk action is

$$\boxed{S_{\mathrm{ES}} = \int \mathrm{d}t \int \mathrm{d}r \left[P_\Lambda \dot{\Lambda} + P_R \dot{R} + P_\varphi \dot{\varphi} - N\mathcal{H} - N^r \mathcal{D}\right] + B_\partial.} \tag{26.4}$$

The Hamiltonian density is

$$\begin{aligned}\mathcal{H} = &-\frac{P_R P_\Lambda}{R} + \frac{\Lambda P_\Lambda^2}{2R^2} + \frac{RR''}{\Lambda} - \frac{RR'\Lambda'}{\Lambda^2} + \frac{R'^2}{2\Lambda} - \frac{\Lambda}{2} \\ &+ \frac{P_\varphi^2}{2\Lambda R^2} + \frac{R^2\varphi'^2}{2\Lambda},\end{aligned} \tag{26.5}$$

and the radial diffeomorphism density is

$$\boxed{\mathcal{D} = P_R R' - \Lambda P_\Lambda' + P_\varphi \varphi'.} \tag{26.6}$$

26.3 Flat-slice horizon-penetrating reduction

Impose the areal flat-slice gauge

$$\Lambda = 1, \qquad R = r. \tag{26.7}$$

Write

$$N = \sigma, \qquad N^r = \sigma v, \qquad P_\Lambda = rv, \qquad P_\varphi = r^2\Pi, \qquad \Phi := \varphi'. \tag{26.8}$$

Define the Misner–Sharp mass by

$$\boxed{m := \frac{r}{2}v^2, \qquad v = \sqrt{\frac{2m}{r}}.} \tag{26.9}$$

The line element is

$$\boxed{\mathrm{d}s^2 = -\sigma^2\,\mathrm{d}t^2 + (\,\mathrm{d}r + \sigma v\,\mathrm{d}t)^2 + r^2\,\mathrm{d}\Omega^2.} \tag{26.10}$$

This chart remains regular at $v = 1$.

26.3.1 Momentum constraint

The condition $\mathcal{D} = 0$ gives

$$\boxed{P_R = P_\Lambda' - P_\varphi\Phi.} \tag{26.11}$$

It removes P_R algebraically.

26.3.2 Hamiltonian constraint

Substituting the momentum constraint into $\mathcal{H} = 0$ gives

$$\boxed{m' = r^2\left[\frac{1}{2}(\Pi^2 + \Phi^2) + v\Pi\Phi\right].} \tag{26.12}$$

Indeed,

$$-\frac{P_\Lambda P_\Lambda'}{r} + \frac{P_\Lambda^2}{2r^2} = -m', \tag{26.13}$$

while

$$\frac{P_\Lambda P_\varphi\Phi}{r} = r^2 v\Pi\Phi. \tag{26.14}$$

Equation (26.12) is the nonlinear radial mass constraint.

26.3.3 Gauge-preservation equation

Preserving $\Lambda = 1$ under the Hamiltonian flow gives

$$0 = \dot\Lambda = \sigma\left(\frac{v}{r} - \frac{P_R}{r}\right) + (\sigma v)'. \tag{26.15}$$

Using the momentum constraint yields

$$\boxed{\frac{\sigma'}{\sigma} = -\frac{r\Pi\Phi}{v}.} \tag{26.16}$$

The apparent singularity at $v = 0$ is removable for regular data because $r\Pi\Phi = O(v)$.

26.3.4 Scalar evolution

The scalar Hamilton equations become

$$\boxed{\dot\varphi = \sigma(\Pi + v\Phi),} \tag{26.17}$$

$$\boxed{\dot\Phi = \partial_r\left[\sigma(\Pi + v\Phi)\right],} \tag{26.18}$$

$$\boxed{\dot\Pi = \frac{1}{r^2}\partial_r\left[r^2\sigma(\Phi + v\Pi)\right].} \tag{26.19}$$

Equations (26.12), (26.16), (26.18), and (26.19) form the constrained Painlevé–Gullstrand system used below.

26.4 Exact local mass-flux theorem

26.4.1 Flux identity

The remaining Einstein evolution equation gives

$$\boxed{\dot{m} = r^2\sigma(\Pi + v\Phi)(\Phi + v\Pi).} \tag{26.20}$$

For $v = 0$, this reduces to the ordinary radial scalar flux $r^2\sigma\Pi\Phi$.

26.4.2 Compatibility theorem

Set

$$\mathcal{E} := \frac{1}{2}(\Pi^2 + \Phi^2) + v\Pi\Phi, \tag{26.21}$$

$$\mathcal{J} := \sigma(\Pi + v\Phi)(\Phi + v\Pi). \tag{26.22}$$

Then the exact identity

$$\boxed{\partial_t(r^2\mathcal{E}) = \partial_r(r^2\mathcal{J})} \tag{26.23}$$

follows from the scalar evolution equations together with

$$\frac{v^2}{2} + rvv' = r^2\mathcal{E}, \qquad \frac{\sigma'}{\sigma} = -\frac{r\Pi\Phi}{v}, \qquad rv\dot{v} = r^2\mathcal{J}. \tag{26.24}$$

Consequently,

$$\partial_t m' = \partial_r \dot{m}. \tag{26.25}$$

Thus the Hamiltonian constraint propagates exactly in the continuum constrained system.
For a radial interval $[r_i, r_o]$,

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left[m(t, r_o) - m(t, r_i)\right] = r_o^2\mathcal{J}(t, r_o) - r_i^2\mathcal{J}(t, r_i).} \tag{26.26}$$

The only change in the exterior mass is the scalar flux through the two boundaries.

26.5 Characteristic structure

Define

$$w_+ := \Pi + \Phi, \qquad w_- := \Pi - \Phi. \tag{26.27}$$

A direct expansion gives

$$\begin{aligned}\dot{w}_+ = {} & \sigma(v+1)w_+' + \left[\sigma v' + \sigma'(v+1)\right] w_+ \\ & + \frac{\sigma}{r}\left[(v+1)w_+ + (v-1)w_-\right],\end{aligned} \tag{26.28}$$

$$\begin{aligned}\dot{w}_- = {} & \sigma(v-1)w_-' + \left[\sigma v' + \sigma'(v-1)\right] w_- \\ & + \frac{\sigma}{r}\left[(v-1)w_- + (v+1)w_+\right].\end{aligned} \tag{26.29}$$

The coordinate characteristic speeds are

$$\boxed{c_+ = -\sigma(v+1), \qquad c_- = \sigma(1-v).} \tag{26.30}$$

Outside a horizon, w_+ is ingoing and w_- is outgoing. At $v = 1$, the outgoing speed vanishes. Inside, both characteristics move toward decreasing r.

26.6 Marginally trapped sphere criterion

Radial future null vectors may be chosen as

$$\ell_\pm = \sigma^{-1}\partial_t + (-v \pm 1)\partial_r. \tag{26.31}$$

Their spherical expansions are

$$\theta_+ = \frac{2}{r}(1-v), \qquad \theta_- = -\frac{2}{r}(1+v). \tag{26.32}$$

Hence

$$\boxed{\theta_+ = 0 \quad \Longleftrightarrow \quad v = 1 \quad \Longleftrightarrow \quad \frac{2m}{r} = 1.} \tag{26.33}$$

Since $\theta_- < 0$, a point where (26.33) holds is a marginally outer trapped sphere. The chart does not become singular there.

The scalar Ricci curvature in the rescaled convention is

$$\boxed{\mathcal{R} = 2(\Phi^2 - \Pi^2).} \tag{26.34}$$

It supplies a direct matter-curvature diagnostic.

26.7 Finite-stencil constrained evolution

26.7.1 Radial constraint projection

At every Runge–Kutta stage, the scalar fields determine m by outward integration of (26.12) with

$$m(r_{\rm exc}) = 0. \tag{26.35}$$

The lapse is then obtained from (26.16) and normalized by

$$\sigma(r_{\rm max}) = 1. \tag{26.36}$$

The radial integrations use fourth-order Runge–Kutta and Simpson quadrature. They are triangular prefix solves, not strictly finite-range maps.

26.7.2 Evolution stencil

For a uniform grid with spacing h, define

$$(D_+^{(2)} f)_j = \frac{-3f_j + 4f_{j+1} - f_{j+2}}{2h}, \tag{26.37}$$

$$(D_-^{(2)} f)_j = \frac{3f_j - 4f_{j-1} + f_{j-2}}{2h}. \tag{26.38}$$

Equation (26.28) always uses $D_+^{(2)}$ because $c_+ < 0$. Equation (26.29) uses $D_-^{(2)}$ when $v < 1$ and switches to $D_+^{(2)}$ when $v > 1$. Thus the scalar evolution stencil has radius two and respects the change in the outgoing characteristic at a horizon.

Time integration uses classical fourth-order Runge–Kutta. The complete method is second-order in the event coordinates because the characteristic spatial stencil is second order.

26.7.3 Boundary conditions

The numerical domain is

$$r \in [r_{\rm exc}, r_{\rm max}] = [0.05, 30]. \tag{26.39}$$

The characteristic boundary data are

$$w_-(r_{\rm exc}) = 0, \qquad w_+(r_{\rm max}) = 0. \tag{26.40}$$

The first condition prevents an outgoing pulse from entering from the excised vacuum core. The second prevents incoming radiation from the far boundary. The strong-pulse horizon forms at $r \approx 3.1$, far from either boundary.

26.8 Initial scalar packets

Take

$$\varphi(0,r) = A\exp\left[-\left(\frac{r-r_0}{w}\right)^2\right], \qquad r_0 = 10, \qquad w = 1. \tag{26.41}$$

With $\chi = r\varphi$, the flat-space ingoing condition $\dot{\chi} = \chi'$ gives

$$\boxed{\Pi(0,r) = \Phi(0,r) + \frac{\varphi(0,r)}{r}.} \tag{26.42}$$

The geometry is then solved from the nonlinear constraints, so the initial gravitational data are not frozen to Minkowski space.

Two amplitudes are used:

$$A_{\rm strong} = 0.10, \qquad A_{\rm weak} = 0.05. \tag{26.43}$$

The weak run is a no-horizon control through $t = 5$. It is not a critical-collapse threshold determination because the excised inner boundary does not model passage through a regular center.

26.9 Trapped-surface formation result

The strong pulse forms a first marginally outer trapped sphere. On the finest recorded grid,

$$N_r = 2400, \qquad h = 0.0124843684869, \tag{26.44}$$

the event is

$$\boxed{t_{\rm H} = 4.69182094580,} \tag{26.45}$$

$$\boxed{r_{\rm H} = 3.11341154797,} \tag{26.46}$$

$$\boxed{m_{\rm H} = \frac{r_{\rm H}}{2} = 1.55670577399.} \tag{26.47}$$

The initial and event outer masses are

$$M_{\rm ADM}(0) = 1.72992821318, \tag{26.48}$$

$$M_{\rm ADM}(t_{\rm H}) = 1.72538868709. \tag{26.49}$$

Thus the finite-domain mass change before horizon formation is

$$\boxed{\frac{M(t_{\rm H}) - M(0)}{M(0)} = -2.6241124 \times 10^{-3}.} \tag{26.50}$$

The event is therefore not generated by loss of the pulse through the excision boundary.

The lapse remains finite:

$$\min_r \sigma = 0.999829382946, \qquad \sigma(r_{\rm H}) = 1.02582462096. \tag{26.51}$$

The maximum absolute scalar Ricci curvature is

$$\max_r |\mathcal{R}| = 7.35797274672. \tag{26.52}$$

26.10 Refinement study

The event location is obtained by quadratic interpolation of the spatial compactness maximum and linear interpolation between the two time steps surrounding $\max(2m/r) = 1$.

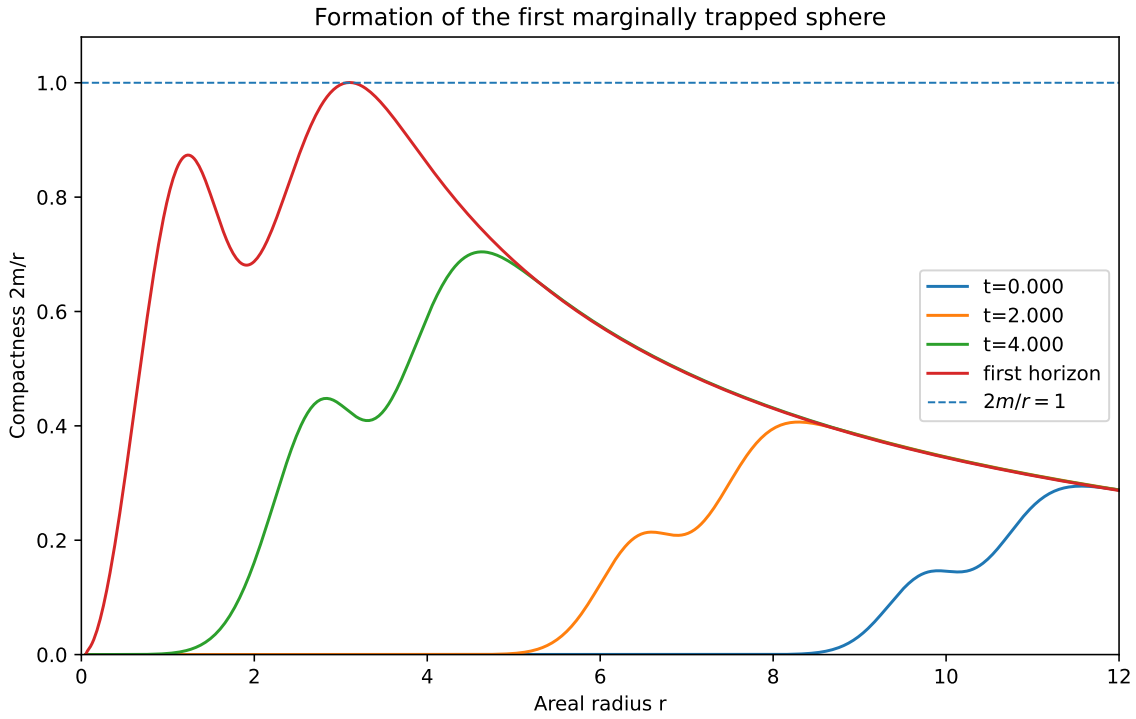


Figure 26.1: The compactness profile grows from a broad ingoing packet to a first marginally trapped sphere. The event is defined by the maximum of $2m/r$ reaching one.

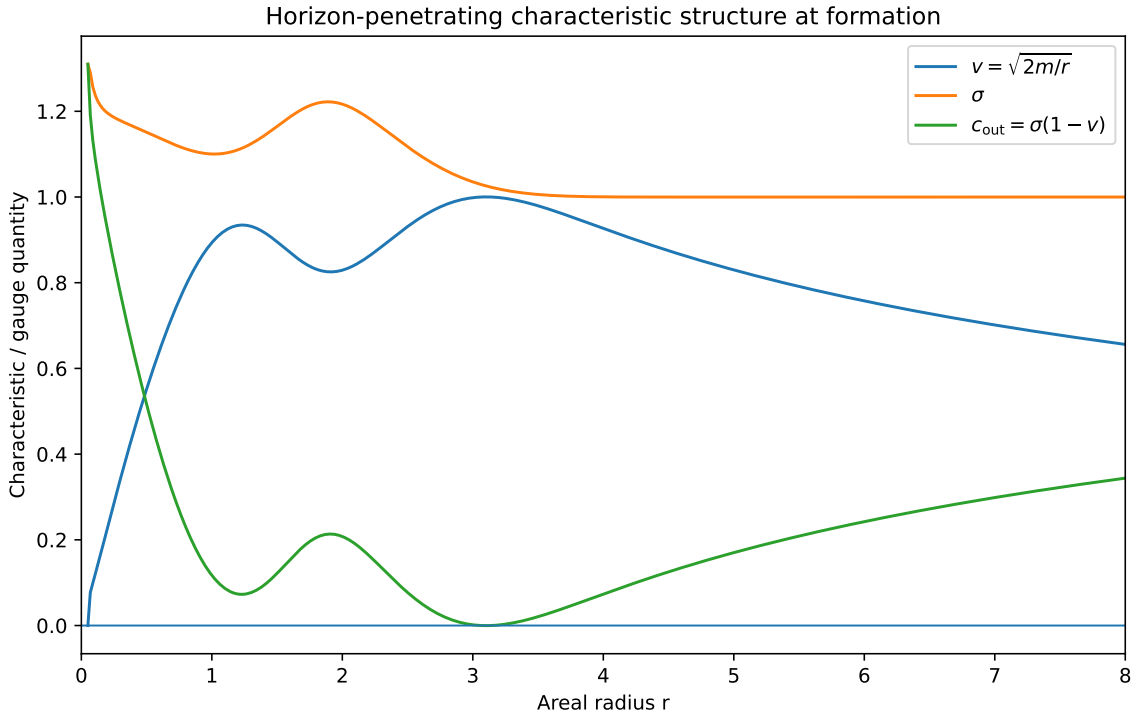


Figure 26.2: At first horizon formation, $v = 1$ and the outgoing coordinate characteristic speed $\sigma(1 - v)$ vanishes while the lapse remains finite.

N_r	h	t_{H}	r_{H}	M_{event}	$\Delta M/M_0$
600	0.0500000	4.7133579	3.0518959	1.7105388	-1.1205×10^{-2}
800	0.0374844	4.7162491	3.0506543	1.7015092	-1.6427×10^{-2}
1000	0.0299800	4.7059219	3.0766099	1.7114248	-1.0696×10^{-2}
1200	0.0249791	4.7004729	3.0905271	1.7167353	-7.6261×10^{-3}
1600	0.0187305	4.6952990	3.1040860	1.7218659	-4.6604×10^{-3}
2000	0.0149825	4.6930194	3.1101678	1.7241638	-3.3322×10^{-3}
2400	0.0124844	4.6918209	3.1134115	1.7253887	-2.6241×10^{-3}

A nonlinear fit over the six finest resolutions gives

$$\boxed{t_{\mathrm{H}}(h) = 4.68935223701 + 33.8857733838\,h^{2.17385328224},} \tag{26.53}$$

$$\boxed{r_{\mathrm{H}}(h) = 3.12049657441 - 65.5931060303\,h^{2.08439496457}.} \tag{26.54}$$

The measured event convergence is therefore consistent with the second-order characteristic stencil.

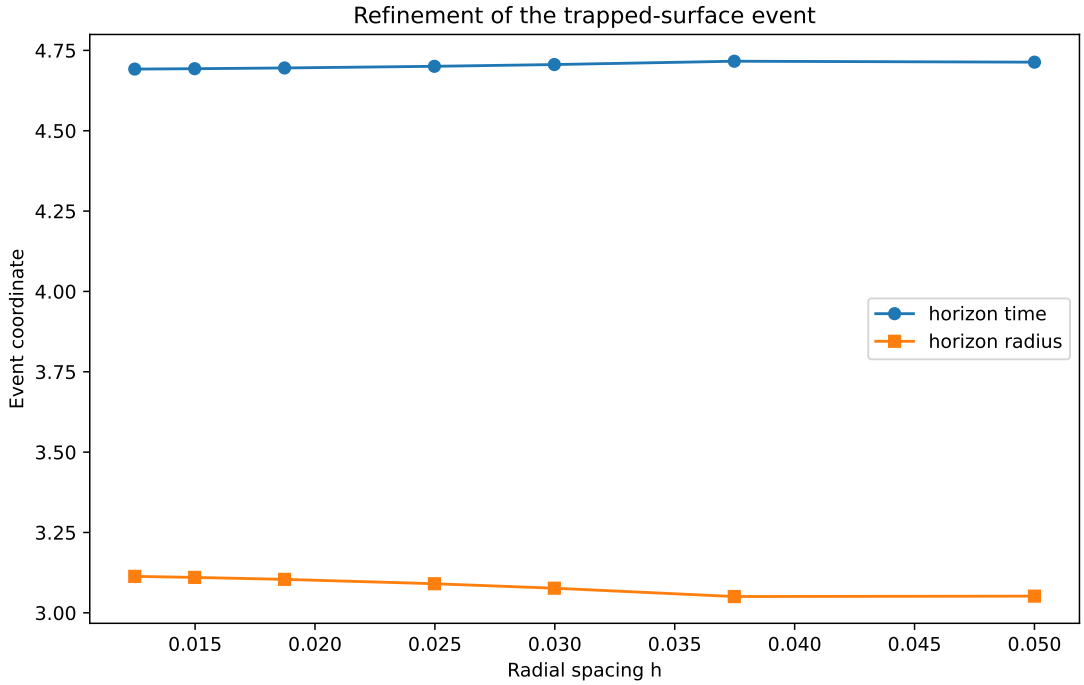


Figure 26.3: The first-horizon time and radius settle under radial refinement. Fits over the six finest runs give orders 2.17 and 2.08.

26.11 Weak-pulse control

For

$$A = 0.05, \qquad N_r = 1200, \qquad t = 5, \tag{26.55}$$

the maximum compactness is

$$\boxed{\max_r \frac{2m}{r} = 0.146722872367.} \tag{26.56}$$

No marginally trapped sphere forms during the controlled interval. The fractional outer-mass change is

$$-4.5106042\times10^{-3}. \tag{26.57}$$

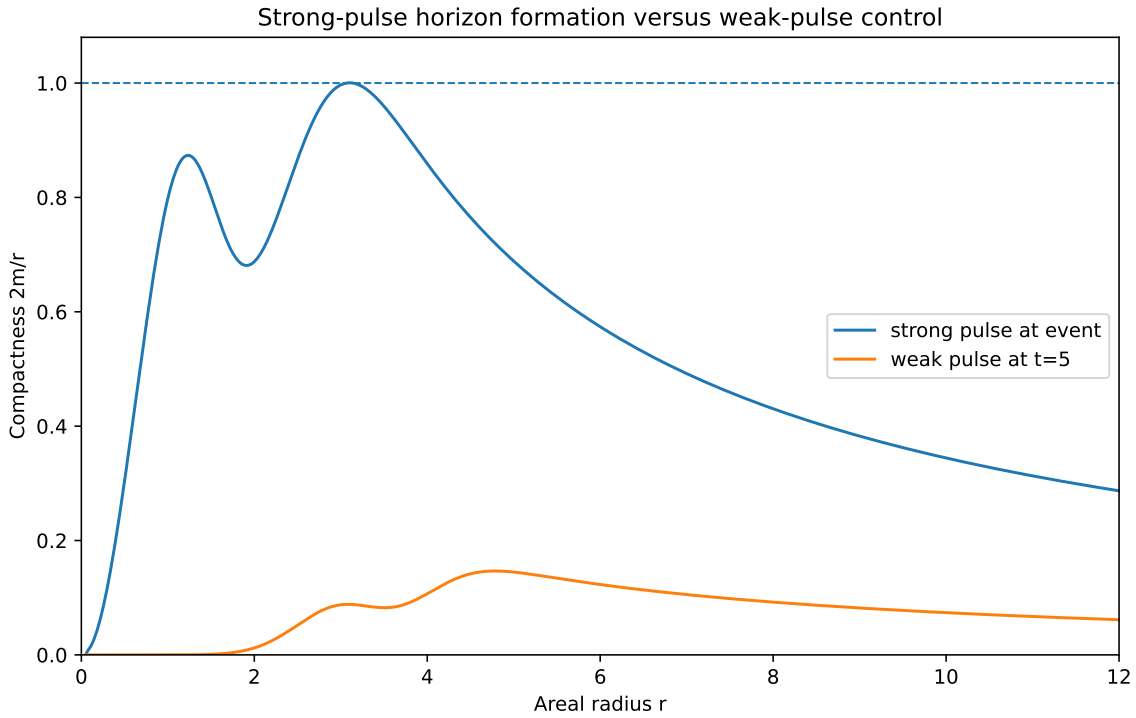


Figure 26.4: The strong packet reaches $2m/r = 1$, while the weak packet remains well below the trapped-surface threshold through $t = 5$.

26.12 Constraint and flux diagnostics

At the smooth pre-horizon time $t = 3$ on the $N_r = 1200$ grid, the recorded residuals are

$$\frac{\|m' - r^2[(\Pi^2 + \Phi^2)/2 + v\Pi\Phi]\|_\infty}{\|r^2[(\Pi^2 + \Phi^2)/2 + v\Pi\Phi]\|_\infty} = 2.27319745 \times 10^{-6}, \tag{26.58}$$

$$\frac{\|(\log\sigma)' + r\Pi\Phi/v\|_\infty}{\|r\Pi\Phi/v\|_\infty} = 2.80774843 \times 10^{-6}, \tag{26.59}$$

$$\frac{\|\Delta_t m - r^2\sigma(\Pi + v\Phi)(\Phi + v\Pi)\|_\infty}{\|r^2\sigma(\Pi + v\Phi)(\Phi + v\Pi)\|_\infty} = 1.26971871 \times 10^{-3}. \tag{26.60}$$

The first two residuals test the radial projection. The third tests the independent local mass-flux law against a finite time difference.

The deterministic audit contains twelve checks and no failures. The exact compatibility and canonical-reduction identities are checked symbolically, not inferred from floating-point data.

26.13 What this block establishes

Within the stated spherical, massless-scalar, flat-slice, and regular-excision assumptions, the chapter establishes

$$\boxed{\begin{array}{c}\text{an exact nonlinear Einstein–scalar PG reduction,}\\ \text{triangular mass and lapse constraints,}\\ \text{an exact local Misner–Sharp flux theorem,}\\ \text{exact continuum constraint propagation,}\\ \text{a horizon-penetrating characteristic system,}\\ \text{a radius-two scalar evolution stencil,}\\ \text{numerical formation of a marginally trapped sphere,}\\ \text{second-order refinement of the event coordinates.}\end{array}} \tag{26.61}$$

26.14 Boundary of the result

The following are not claimed:

$$\begin{array}{c}\text{regular-center evolution through reflection or dispersion,}\\ \text{a critical amplitude or universal critical exponent,}\\ \text{continuation to singularity formation,}\\ \text{global event-horizon identification,}\\ \text{nonspherical stability,}\\ \text{scalar self-interaction or Dirac collapse,}\\ \text{a complete finite-range } 3+1 \text{ strong-field solver.}\end{array} \tag{26.62}$$

The numerical result is evidence for trapped-surface formation in the resolved spherical system, not a substitute for a global continuum theorem.

26.15 Next decisive target

The next major strong-field block is now sharply defined:

$$\boxed{\begin{array}{c}\text{replace the excision core by a regular-center formulation,}\\ \text{continue supercritical data beyond first horizon formation,}\\ \text{bracket the critical amplitude and test scaling,}\\ \text{then add nonspherical perturbations and ringdown.}\end{array}} \tag{26.63}$$

26.16 Machine-readable records

$$R_{\text{collapse}} = \texttt{data/Finite_Range_Spherical_Scalar_Collapse_Verification.json}, \tag{26.64}$$

$$T_{\text{collapse}} = \texttt{data/scalar_collapse_convergence.csv}, \tag{26.65}$$

$$X_{\text{collapse}} = \texttt{verification/Finite_Range_Spherical_Scalar_Collapse_Verification.py}. \tag{26.66}$$

Part IV

Adic, Polyphase, and Quantum Refinement

Chapter 27

Dyadic–Odd Quantum Register Factorization

At a composite modulus, the Chinese remainder theorem separates dyadic and odd-prime coordinates. The corresponding basis change is unitary, and the quantum amplitudes and Born probabilities remain complex throughout.

27.1 Mathematical scope

Throughout the adic construction, state amplitudes and Born probabilities remain ordinary complex quantum amplitudes:

$$\mathcal{H} = \ell^2(\text{finite adic address set}; \mathbb{C}). \tag{27.1}$$

The adic structure is carried by addresses, unit coordinates, refinement depth, packet projectors, and valuation-weighted couplings. No claim of a p-adic-valued Born rule is made.

27.2 Finite unit coordinates

For an odd prime p and $h \geq 0$, define

$$U_{p,h} := (\mathbb{Z}/p^{h+1}\mathbb{Z})^{\times}. \tag{27.2}$$

Choose $\zeta_{p,h} \in U_{p,h}$ of order $p-1$. Then

$$\boxed{U_{p,h} \cong \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}/p^h\mathbb{Z}} \tag{27.3}$$

through

$$(t, a) \longmapsto \zeta_{p,h}^{t}(1+p)^{a} \pmod{p^{h+1}}. \tag{27.4}$$

The dimensions agree because

$$|U_{p,h}| = \varphi(p^{h+1}) = p^{h+1} - p^h = (p-1)p^h. \tag{27.5}$$

The torsion coordinate contributes $p-1$ values and the principal-unit coordinate contributes p^h values. For $p = 2$, define

$$U_{2,h} := (\mathbb{Z}/2^{h+2}\mathbb{Z})^{\times}. \tag{27.6}$$

Then

$$\boxed{U_{2,h} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^h\mathbb{Z}} \tag{27.7}$$

through

$$(s, a) \longmapsto (-1)^s 5^a \pmod{2^{h+2}}. \tag{27.8}$$

Here

$$|U_{2,h}| = \varphi(2^{h+2}) = 2^{h+1} = 2 \cdot 2^h, \tag{27.9}$$

so the sign coordinate and the principal phase coordinate account for every dyadic unit. Thus

$$\mathcal{H}^{\times}_{p,h} := \ell^2(U_{p,h}) \cong \mathbb{C}^{p-1}_{\text{tors}} \otimes \ell^2(\mathbb{Z}/p^h\mathbb{Z}), \qquad p \text{ odd}, \tag{27.10}$$

$$\mathcal{H}^{\times}_{2,h} := \ell^2(U_{2,h}) \cong \mathbb{C}^2_{\text{sign}} \otimes \ell^2(\mathbb{Z}/2^h\mathbb{Z}). \tag{27.11}$$

The dyadic first factor is a sign qubit. The second factor is the principal dyadic phase register.

27.3 Exact CRT quantum factorization

Let

$$M = 2^{h_2+2} \prod_{p\in\mathcal{P}_o} p^{h_p+1}. \tag{27.12}$$

The Chinese remainder map gives a permutation-unitary

$$\boxed{\mathcal{U}_{\text{CRT}} : \ell^2((\mathbb{Z}/M\mathbb{Z})^{\times}) \longrightarrow \mathbb{C}^2_{\text{sign}} \otimes \ell^2(\mathbb{Z}/2^{h_2}\mathbb{Z}) \otimes \bigotimes_{p\in\mathcal{P}_o} \left(\mathbb{C}^{p-1}_{\text{tors}} \otimes \ell^2(\mathbb{Z}/p^{h_p}\mathbb{Z})\right).} \tag{27.13}$$

For units u, v, On basis states, $\mathcal{U}_{\text{CRT}}$ only relabels the unique CRT coordinates:

$$\mathcal{U}_{\text{CRT}}|u\rangle = |\operatorname{coord}(u)\rangle. \tag{27.14}$$

Consequently,

$$\langle \mathcal{U}_{\text{CRT}} u | \mathcal{U}_{\text{CRT}} v\rangle = \delta_{\operatorname{coord}(u),\operatorname{coord}(v)} \tag{27.15}$$

$$= \delta_{u,v} = \langle u|v\rangle, \tag{27.16}$$

which makes the coordinate permutation unitary. For units u, v,

$$\operatorname{coord}(uv) = \operatorname{coord}(u) + \operatorname{coord}(v) \tag{27.17}$$

componentwise in the indicated cyclic registers.

27.4 Quantum Fourier factorization

Let F_n denote the normalized Fourier matrix on $\mathbb{Z}/n\mathbb{Z}$. In the unit-coordinate basis,

$$\boxed{F_{U_M} = \mathcal{U}^*_{\text{CRT}} \left[F_2 \otimes F_{2^{h_2}} \otimes \bigotimes_{p\in\mathcal{P}_o} (F_{p-1} \otimes F_{p^{h_p}})\right] \mathcal{U}_{\text{CRT}}.} \tag{27.18}$$

For coprime additive moduli a, b, $N = ab$, let

$$\mathcal{P}_{\text{in}}|n\rangle = |n \bmod a\rangle \otimes |n \bmod b\rangle, \tag{27.19}$$

$$\mathcal{P}_{\text{out}}|k\rangle = |kb^{-1} \bmod a\rangle \otimes |ka^{-1} \bmod b\rangle. \tag{27.20}$$

Then

$$\boxed{F_N = \mathcal{P}^*_{\text{out}}(F_a \otimes F_b)\mathcal{P}_{\text{in}}.} \tag{27.21}$$

The input and dual-output permutations are both required. To see the output permutation, write $n \leftrightarrow (n_a, n_b)$ by the CRT. The product Fourier kernel is

$$\frac{1}{\sqrt{a}} e^{2\pi i k_a n_a/a} \frac{1}{\sqrt{b}} e^{2\pi i k_b n_b/b} = \frac{1}{\sqrt{N}} \exp\left[2\pi i \left(\frac{k_a n}{a} + \frac{k_b n}{b}\right)\right]. \tag{27.22}$$

Choosing

$$k_a = kb^{-1} \pmod a, \qquad k_b = ka^{-1} \pmod b \tag{27.23}$$

gives

$$\frac{k_a}{a} + \frac{k_b}{b} \equiv \frac{k}{ab} \pmod 1, \tag{27.24}$$

because $bb^{-1} \equiv 1 \pmod a$ and $aa^{-1} \equiv 1 \pmod b$. The product kernel is therefore $N^{-1/2} e^{2\pi i k n/N}$, which is the kernel of F_N. The input and dual-output permutations are both required.

27.5 Dyadic sign protection

Let

$$\mathcal{C}_{\mathrm{phase}} : \mathbb{C}^2_{\mathrm{sign}} \otimes \ell^2(\mathbb{Z}/2^h\mathbb{Z}) \longrightarrow \ell^2(\mathbb{Z}/2^h\mathbb{Z}) \tag{27.25}$$

forget the sign coordinate. Then

$$\operatorname{rank} \mathcal{C}_{\mathrm{phase}} = 2^h, \qquad \dim \mathcal{H}^{\times}_{2,h} = 2^{h+1}, \tag{27.26}$$

and each principal phase has exactly two preimages:

$$(+1, a), \qquad (-1, a). \tag{27.27}$$

With the natural action

$$\mathcal{C}_{\mathrm{phase}}|\pm, a\rangle = |a\rangle, \tag{27.28}$$

the sign-antisymmetric vectors satisfy

$$\mathcal{C}_{\mathrm{phase}}\big(|+, a\rangle - |-, a\rangle\big) = 0. \tag{27.29}$$

These 2^h independent vectors span the kernel, so rank–nullity gives

$$\dim \ker \mathcal{C}_{\mathrm{phase}} = 2^{h+1} - 2^h = 2^h. \tag{27.30}$$

Hence

$$\boxed{\text{principal dyadic phase alone cannot reconstruct a sign-sensitive quantum observable.}} \tag{27.31}$$

For the integral involution

$$A = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \qquad A^2 = I, \tag{27.32}$$

the rational sign projectors

$$E_{\pm} = \frac{I \pm A}{2} \tag{27.33}$$

Indeed, $A^2 = I$ gives

$$E_{\pm}^2 = \frac{I \pm 2A + A^2}{4} = \frac{I \pm A}{2} = E_{\pm}, \tag{27.34}$$

$$E_+ E_- = \frac{I - A^2}{4} = 0, \tag{27.35}$$

$$E_+ + E_- = I. \tag{27.36}$$

Thus the two operators are complementary spectral projectors over the rationals. have

$$s_2(E_+) = s_2(E_-) = 1, \qquad v_2(\mathrm{Res}(x-1, x+1)) = 1. \tag{27.37}$$

Modulo two,

$$(A - I)^2 = 0, \tag{27.38}$$

so the two spectral signs merge in the residual packet. Integral address selectors can remain exact even when these rational spectral projectors require one additional dyadic digit.

27.6 Numerical verification

$$\text{local coordinate residual} = 0, \tag{27.39}$$

$$\text{local multiplication residual} = 0, \tag{27.40}$$

$$\|\mathcal{U}^*_{\mathrm{CRT}}\mathcal{U}_{\mathrm{CRT}} - I\|_2 = 0, \tag{27.41}$$

$$\text{composite multiplication residual} = 0, \tag{27.42}$$

$$\text{unit-register QFT factorization residual} = 9.945153 \times 10^{-16}, \tag{27.43}$$

$$\text{additive CRT-QFT residual} = 7.446474 \times 10^{-15}, \tag{27.44}$$

$$\text{dyadic sign-alias multiplicity error} = 0, \tag{27.45}$$

$$\text{dyadic sign rank defect at } h = 4 = 16. \tag{27.46}$$

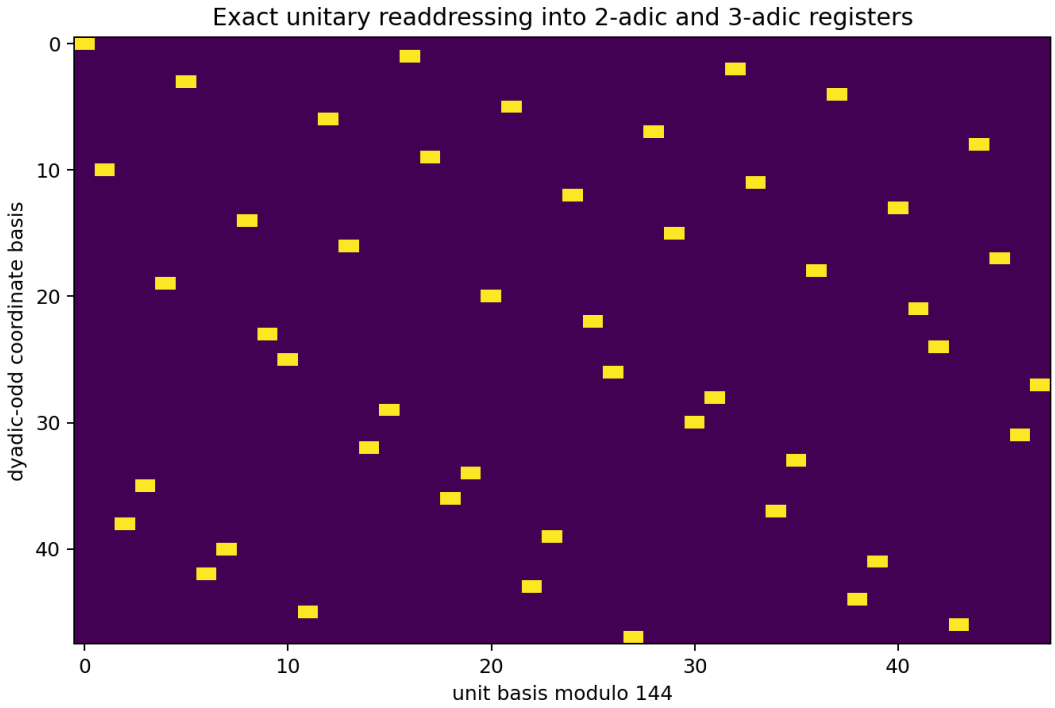


Figure 27.1: Exact unitary readdressing of the unit basis into dyadic and odd-adic quantum registers.

27.7 Next steps

- Derive a physical interpretation for the dyadic-sign and odd-torsion registers.
- Determine the register architecture from a microscopic Hamiltonian or action rather than choosing it kinematically.
- Identify an observable that distinguishes the proposed register structure from an ordinary tensor decomposition.
- Construct a continuum quantum-field interpretation of the finite adic factorization.

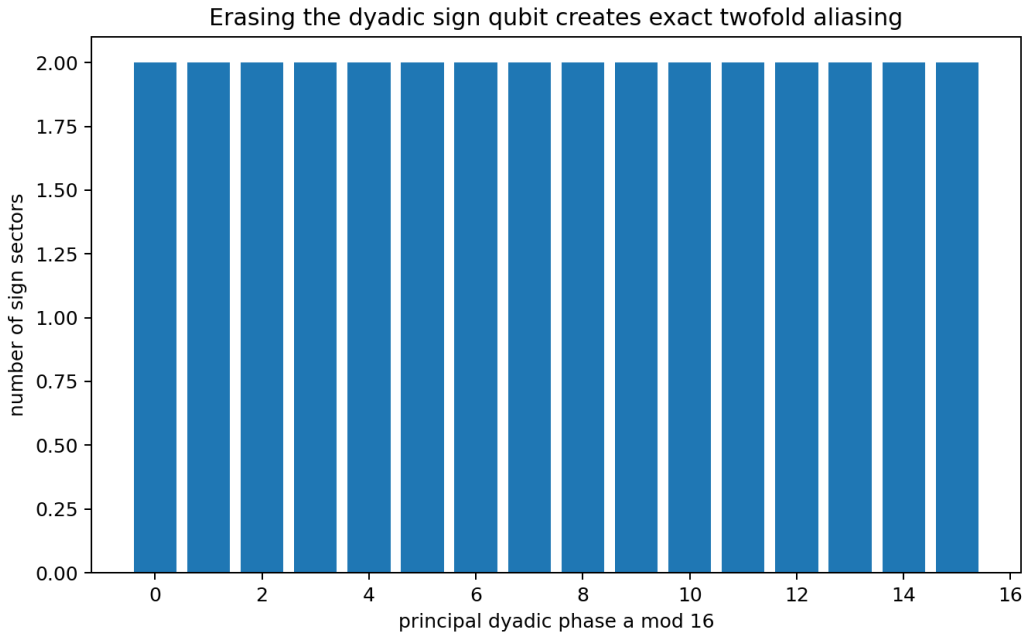


Figure 27.2: Every principal dyadic phase has two sign sectors. Removing the sign qubit creates exact twofold aliasing.

Chapter 28

Adic Haar Packets and Refinement-Compatible Quantum Dynamics

Adding one p-adic digit splits each parent packet into one average mode and $p-1$ detail modes. The child Fourier transform supplies the finite Haar basis and identifies Hamiltonians that remain compatible across levels.

28.1 Finite principal-phase tower

For a prime p and $h \geq 0$, let

$$G_{p,h} := \mathbb{Z}/p^h\mathbb{Z}, \qquad \mathcal{H}_{p,h} := \ell^2(G_{p,h}). \tag{28.1}$$

For $0 \leq r \leq h$, define

$$(E_r^{(h)} f)(x) := \frac{1}{p^{h-r}} \sum_{\substack{y \in G_{p,h} \\ y \equiv x \pmod{p^r}}} f(y). \tag{28.2}$$

Then

$$(E_r^{(h)})^* = E_r^{(h)}, \tag{28.3}$$

$$(E_r^{(h)})^2 = E_r^{(h)}, \tag{28.4}$$

$$E_r^{(h)} E_s^{(h)} = E_{\min(r,s)}^{(h)}, \tag{28.5}$$

$$E_h^{(h)} = I. \tag{28.6}$$

For example, suppose $r \leq s$. Averaging first over a depth-s cylinder and then over the containing depth-r cylinder averages every point in the larger cylinder exactly once after normalization, so

$$E_r^{(h)} E_s^{(h)} = E_r^{(h)}. \tag{28.7}$$

Interchanging r and s proves the stated minimum rule. Define

$$D_0^{(h)} := E_0^{(h)}, \qquad D_r^{(h)} := E_r^{(h)} - E_{r-1}^{(h)} \quad (1 \leq r \leq h). \tag{28.8}$$

Then

$$\boxed{D_r^{(h)} D_s^{(h)} = \delta_{rs} D_r^{(h)}, \qquad \sum_{r=0}^{h} D_r^{(h)} = I.} \tag{28.9}$$

For $1 \leq r < s$,

$$D_r^{(h)} D_s^{(h)} = (E_r - E_{r-1})(E_s - E_{s-1}) \tag{28.10}$$

$$= E_r - E_r - E_{r-1} + E_{r-1} = 0. \tag{28.11}$$

For $r = s$,

$$(D_r^{(h)})^2 = E_r - 2E_{r-1} + E_{r-1} = D_r^{(h)}. \tag{28.12}$$

The resolution of the identity is the telescoping sum

$$E_0 + \sum_{r=1}^{h} (E_r - E_{r-1}) = E_h = I. \tag{28.13}$$

The detail ranks are

$$\boxed{\operatorname{rank} D_0^{(h)} = 1, \qquad \operatorname{rank} D_r^{(h)} = (p-1)p^{r-1}.} \tag{28.14}$$

The range of $E_r^{(h)}$ is determined by one value on each of the p^r depth-r cylinders, hence

$$\operatorname{rank} E_r^{(h)} = p^r. \tag{28.15}$$

Because $\operatorname{Ran} E_{r-1} \subset \operatorname{Ran} E_r$,

$$\operatorname{rank} D_r^{(h)} = p^r - p^{r-1} = (p-1)p^{r-1}. \tag{28.16}$$

Thus a new dyadic digit contributes one detail character per parent, while a new odd-adic digit contributes $p-1$ detail characters per parent.

28.2 Child-DFT/Haar realization

Let $\omega_p = \exp(2\pi \mathrm{i}/p)$. For a parent prefix $a \in \mathbb{Z}/p^{r-1}\mathbb{Z}$ and $q \in \{1, \dots, p-1\}$, define

$$\psi_{r,a,q}^{(h)}(x) := \begin{cases} p^{-(h-r+1)/2} \omega_p^{q d_r(x)}, & x \equiv a \pmod{p^{r-1}}, \\ 0, & \text{otherwise}, \end{cases} \tag{28.17}$$

where $d_r(x)$ is the new base-p digit at level r. Together with the constant scaling vector, these wavelets form an orthonormal basis and

$$\boxed{D_r^{(h)} = \sum_{a=0}^{p^{r-1}-1} \sum_{q=1}^{p-1} |\psi_{r,a,q}^{(h)}\rangle\langle\psi_{r,a,q}^{(h)}|.} \tag{28.18}$$

Each wavelet is supported on p^{h-r+1} points, so its norm is

$$\left\| \psi_{r,a,q}^{(h)} \right\|^2 = p^{h-r+1} p^{-(h-r+1)} = 1. \tag{28.19}$$

Different parent prefixes have disjoint support. For a common parent,

$$\langle \psi_{r,a,q}, \psi_{r,a,q'} \rangle = \frac{1}{p} \sum_{d=0}^{p-1} \omega_p^{(q'-q)d} \tag{28.20}$$

$$= \delta_{q,q'}. \tag{28.21}$$

The omitted character $q = 0$ is the inherited parent average; the $p-1$ nonzero characters span its orthogonal detail complement. For $p = 2$, the child DFT is the Hadamard split into average and difference. For odd p, the same node has $p-1$ nontrivial Fourier details.

28.3 Refinement isometry

Define

$$V_{p,h} : \mathcal{H}_{p,h} \to \mathcal{H}_{p,h+1} \tag{28.22}$$

by

$$(V_{p,h} f)(a + dp^h) = \frac{1}{\sqrt{p}} f(a), \qquad 0 \le d < p. \tag{28.23}$$

Equivalently,

$$V_{p,h} f = f \otimes |+_p\rangle, \qquad |+_p\rangle = \frac{1}{\sqrt{p}} \sum_{d=0}^{p-1} |d\rangle. \tag{28.24}$$

Then

$$V_{p,h}^* V_{p,h} = I, \tag{28.25}$$
$$D_r^{(h+1)} V_{p,h} = V_{p,h} D_r^{(h)} \quad (0 \le r \le h), \tag{28.26}$$
$$D_{h+1}^{(h+1)} V_{p,h} = 0. \tag{28.27}$$

The isometry can be checked directly:

$$\|V_{p,h} f\|^2 = \sum_{a=0}^{p^h-1} \sum_{d=0}^{p-1} \frac{1}{p} |f(a)|^2 \tag{28.28}$$
$$= \sum_{a=0}^{p^h-1} |f(a)|^2 = \|f\|^2 . \tag{28.29}$$

The refined vector is constant in the new digit d. Its nonzero child characters therefore vanish, which proves the last identity, while all older cylinder averages commute with the replication.

28.4 Refinement-compatible Hamiltonian classification

Let $V : \mathcal{H}_c \to \mathcal{H}_f$ be an isometry, $P = VV^*$, $Q = I - P$, and let H_c, H_f be self-adjoint. The following are equivalent:

$$H_f V = V H_c, \tag{28.30}$$
$$QH_f P = 0, \qquad PH_f P = V H_c V^*. \tag{28.31}$$

Indeed, multiplying $H_f V = V H_c$ on the right by V^* gives

$$H_f P = V H_c V^*. \tag{28.32}$$

Applying P and Q on the left yields the two block identities. Conversely,

$$H_f V = H_f P V \tag{28.33}$$
$$= (PH_f P + QH_f P) V \tag{28.34}$$
$$= V H_c V^* V = V H_c. \tag{28.35}$$

Self-adjointness also gives $PH_f Q = 0$. Therefore

$$\boxed{H_f = V H_c V^* \oplus H_Q} \tag{28.36}$$

with respect to $\operatorname{Ran} P \oplus \operatorname{Ran} Q$. The off-diagonal blocks vanish:

$$PH_f Q = (QH_f P)^* = 0, \tag{28.37}$$

and the lower block is $H_Q := QH_f Q$. Applying the spectral calculus to this block diagonal form gives

$$f(H_f) = V f(H_c) V^* \oplus f(H_Q). \tag{28.38}$$

For every bounded Borel function f,

$$\boxed{f(H_f) V = V f(H_c).} \tag{28.39}$$

In particular,

$$\mathrm{e}^{-\mathrm{i}tH_f} V = V \mathrm{e}^{-\mathrm{i}tH_c}, \tag{28.40}$$
$$\mathrm{e}^{-\beta H_f} V = V \mathrm{e}^{-\beta H_c}, \tag{28.41}$$

$$(z - H_f)^{-1} V = V (z - H_c)^{-1}. \tag{28.42}$$

Moreover,

$$Q \mathrm{e}^{-\mathrm{i}t H_f} P = 0 \tag{28.43}$$

and the Feshbach self-energy vanishes:

$$\boxed{\Sigma_P(z) = P H_f Q (z - Q H_f Q)^{-1} Q H_f P = 0.} \tag{28.44}$$

Within this model class, refinement introduces no anomaly, and the Feshbach maps remain uniformly controlled.

28.5 Two compatible Hamiltonian families

The hierarchical Hamiltonian is

$$\boxed{H_{p,h}^{\mathrm{Haar}} = \sum_{r=0}^{h} \epsilon_{p,r} D_r^{(h)}.} \tag{28.45}$$

Its wavelet energies are exactly $\epsilon_{p,r}$. Let $S_{p,h}$ be translation by one on $G_{p,h}$. Then

$$S_{p,h+1} V_{p,h} = V_{p,h} S_{p,h} \tag{28.46}$$

and the cyclic Laplacian

$$L_{p,h} := 2I - S_{p,h} - S_{p,h}^* \tag{28.47}$$

also intertwines exactly. Hence

$$H_{p,h} = H_{p,h}^{\mathrm{Haar}} + \kappa_p L_{p,h} \tag{28.48}$$

defines a nontrivial finite discrete quantum tower satisfying (28.30).

28.6 Ultrametric propagator

For the pure hierarchical sector,

$$U_{p,h}(t) = \sum_{r=0}^{h} \mathrm{e}^{-\mathrm{i}t\epsilon_{p,r}} D_r^{(h)}. \tag{28.49}$$

This follows by expanding the exponential in powers of $H_{p,h}^{\mathrm{Haar}}$. Orthogonality gives

$$\left(H_{p,h}^{\mathrm{Haar}}\right)^n = \sum_{r=0}^{h} \epsilon_{p,r}^n D_r^{(h)}, \tag{28.50}$$

and summing the scalar exponential series on each detail subspace produces the displayed propagator. The kernel of $E_r^{(h)}$ is

$$E_r^{(h)}(x, y) = p^{-(h-r)} \mathbf{1}_{\{x \equiv y \pmod{p^r}\}}. \tag{28.51}$$

Since $D_0 = E_0$ and $D_r = E_r - E_{r-1}$, every detail kernel is a difference of two congruence indicators. For fixed x, y, those indicators depend only on the largest r for which $x \equiv y \pmod{p^r}$, namely $v_p(x - y)$ truncated at depth h. Therefore

$$\boxed{U_{p,h}(t; x, y) = \mathcal{K}_{p,h,t}(v_p(x - y)),} \tag{28.52}$$

so transition amplitudes and probabilities are constant on p-adic valuation shells.

28.7 Numerical verification

$$\text{detail-projector residual} = 5.620504 \times 10^{-16}, \tag{28.53}$$
$$\text{child-DFT/Haar residual} = 9.255029 \times 10^{-16}, \tag{28.54}$$
$$\text{Hamiltonian-intertwining residual} = 4.587620 \times 10^{-16}, \tag{28.55}$$
$$\text{unitary-intertwining residual} = 4.885923 \times 10^{-16}, \tag{28.56}$$
$$\text{Euclidean-intertwining residual} = 4.584318 \times 10^{-16}, \tag{28.57}$$
$$\text{resolvent-intertwining residual} = 6.971794 \times 10^{-17}, \tag{28.58}$$
$$\text{leakage residual} = 9.333536 \times 10^{-16}, \tag{28.59}$$
$$\text{Feshbach self-energy residual} = 2.491706 \times 10^{-32}, \tag{28.60}$$
$$\text{shell-isotropy residual} = 0. \tag{28.61}$$

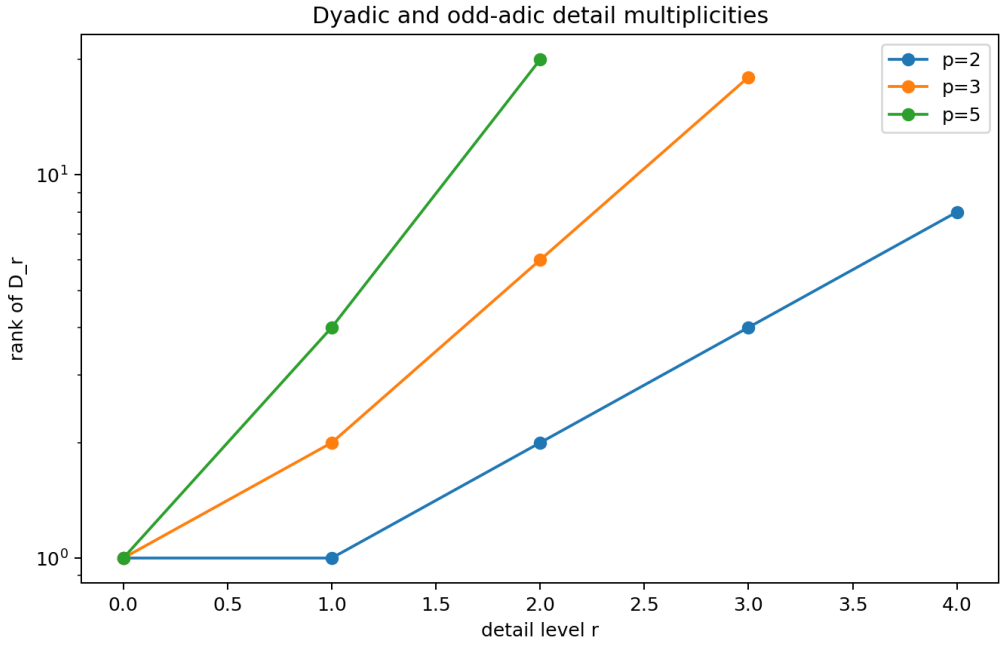


Figure 28.1: The rank of each new detail layer. The dyadic tree has one nontrivial child character per parent; an odd prime has $p - 1$.

28.8 Next steps

- Relate refinement-tree locality to ordinary-space causal locality.
- Construct an interacting relativistic continuum limit of the finite Haar register dynamics.
- Prove cutoff-uniform estimates beyond reducing subspaces.
- Design a measurement protocol for the energies and populations of the Haar detail layers.

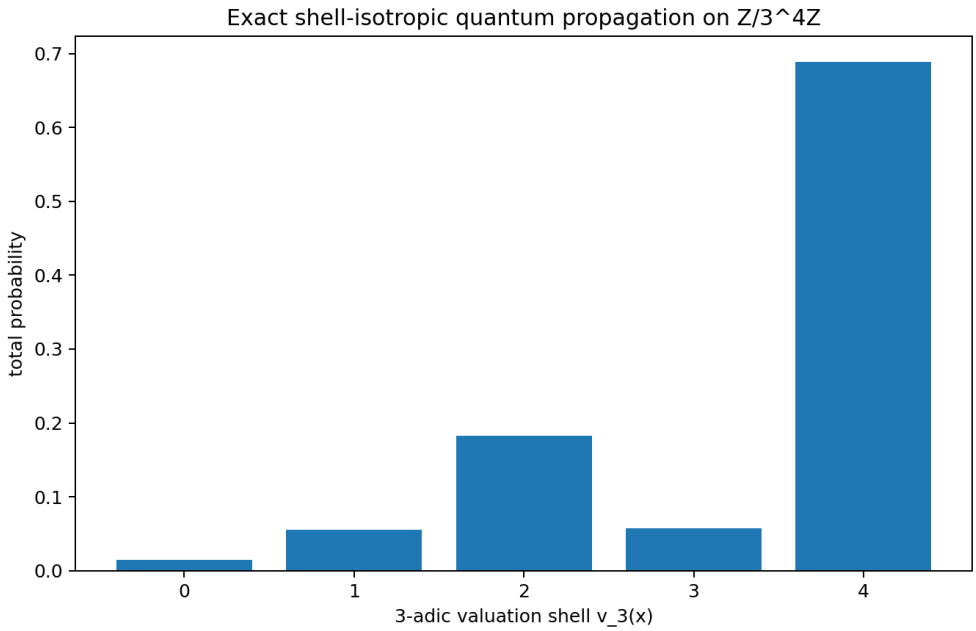


Figure 28.2: A finite hierarchical quantum propagator is exactly uniform inside each 3-adic valuation shell.

Chapter 29

Finite-Adelic Quantum Dynamics and Weighted Leakage

Several prime towers can be coupled in one finite Hilbert space without abandoning exact refinement. The construction distinguishes stable cross-prime interactions from valuation-weighted couplings that introduce controlled leakage.

29.1 Finite multi-prime Hilbert space

For a finite prime set $\mathcal{P}$, define

$$\boxed{\mathcal{H}^{\times}_{\mathcal{P},\mathbf{h}} = \mathcal{H}^{\times}_{2,h_2} \otimes \bigotimes_{p\in\mathcal{P}_o} \mathcal{H}^{\times}_{p,h_p}.} \tag{29.1}$$

For each prime, let D_{p,r_p} be a principal-phase detail projector. Joint packets are

$$D_{\mathbf{r}} := \bigotimes_{p\in\mathcal{P}} D_{p,r_p}. \tag{29.2}$$

A general finite stable interaction has the form

$$\boxed{H_{\mathbf{h}} = \sum_{A,\mathbf{r}} \epsilon_{A,\mathbf{r}}\, T_A \otimes D_{\mathbf{r}},} \tag{29.3}$$

where T_A acts on the finite torsion/sign registers and every coefficient attached to an existing packet is unchanged under refinement. New-level terms may act arbitrarily inside the new detail complement.

29.2 Exact simultaneous prime refinement

Let

$$V^{\mathbf{h+1}}_{\mathbf{h}} = I_{\text{tors}} \otimes \bigotimes_{p\in\mathcal{P}} V_{p,h_p}. \tag{29.4}$$

For every stable interacting Hamiltonian,

$$\boxed{H_{\mathbf{h+1}} V^{\mathbf{h+1}}_{\mathbf{h}} = V^{\mathbf{h+1}}_{\mathbf{h}} H_{\mathbf{h}}.} \tag{29.5}$$

For an old joint packet,

$$D^{(\mathbf{h+1})}_{\mathbf{r}} V = \left(\bigotimes_p D^{(h_p+1)}_{p,r_p}\right)\left(I_{\text{tors}} \otimes \bigotimes_p V_{p,h_p}\right) \tag{29.6}$$

$$= \left(I_{\text{tors}} \otimes \bigotimes_p V_{p,h_p}\right)\left(\bigotimes_p D^{(h_p)}_{p,r_p}\right) \tag{29.7}$$

$$= V D_{\mathbf{r}}^{(\mathbf{h})}. \tag{29.8}$$

Terms supported entirely in a new detail factor annihilate V. Applying these two observations term by term to $H_{\mathbf{h+1}}$ proves the intertwining identity. Hence

$$\mathrm{e}^{-\mathrm{i}tH_{\mathbf{h+1}}}V = V\mathrm{e}^{-\mathrm{i}tH_{\mathbf{h}}}, \tag{29.9}$$

$$\mathrm{e}^{-\beta H_{\mathbf{h+1}}}V = V\mathrm{e}^{-\beta H_{\mathbf{h}}}, \tag{29.10}$$

$$(z - H_{\mathbf{h+1}})^{-1}V = V(z - H_{\mathbf{h}})^{-1}, \tag{29.11}$$

$$Q\mathrm{e}^{-\mathrm{i}tH_{\mathbf{h+1}}}P = 0, \tag{29.12}$$

$$\Sigma_P(z) = 0. \tag{29.13}$$

The interaction need not be a sum of independent prime Hamiltonians. Cross-prime products of stable packet projectors remain exact.

29.3 Dyadic-sign/odd-adic entanglement

Let Z_2 act on the dyadic sign qubit and let

$$B_{\mathrm{odd}} = I_{2,\mathrm{phase}} \otimes Z_{p,\mathrm{tors}} \otimes D_{p,r_p}. \tag{29.14}$$

The interaction

$$\boxed{H_{\mathrm{int}} = gZ_2 \otimes B_{\mathrm{odd}}} \tag{29.15}$$

generates entanglement between the exceptional dyadic sign channel and an odd-adic detail packet. For a balanced superposition of a $B_{\mathrm{odd}} = 0$ state and a $B_{\mathrm{odd}} = 1$ state,

$$t_* = \frac{\pi}{2g} \tag{29.16}$$

gives

$$\boxed{S(\rho_{\mathrm{sign}}(t_*)) = 1 \text{ bit.}} \tag{29.17}$$

Choose $B_{\mathrm{odd}}|0\rangle = 0$, $B_{\mathrm{odd}}|1\rangle = |1\rangle$, and start from

$$|\Psi(0)\rangle = |+_x\rangle \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}}. \tag{29.18}$$

Since B_{odd} is diagonal on these two states,

$$|\Psi(t)\rangle = \frac{1}{\sqrt{2}}\left(|+_x\rangle \otimes |0\rangle + \mathrm{e}^{-\mathrm{i}gtZ_2}|+_x\rangle \otimes |1\rangle\right). \tag{29.19}$$

At $t_* = \pi/(2g)$,

$$\mathrm{e}^{-\mathrm{i}gt_*Z_2}|+_x\rangle = -\mathrm{i}|-_x\rangle, \tag{29.20}$$

and the two sign states are orthogonal. Tracing out the odd register gives

$$\rho_{\mathrm{sign}}(t_*) = \frac{1}{2}\left(|+_x\rangle\langle+_x| + |-_x\rangle\langle-_x|\right), \tag{29.21}$$

whose two eigenvalues are $1/2$. Thus the adic split supports nontrivial cross-prime quantum dynamics rather than only a classical coordinate factorization.

29.4 Valuation-weighted refinement anomaly

Let H_0 satisfy exact refinement and let B couple the coarse range to one new detail packet:

$$PBQ \neq 0, \qquad PBP = QBQ = 0. \tag{29.22}$$

For adic weight $w \geq 1$, set

$$\lambda_{p,w} := p^{-w}, \qquad H_{p,w} := H_0 + \lambda_{p,w} B. \tag{29.23}$$

Then

$$\boxed{\|H_{p,w}V - VH_c\| = \lambda_{p,w}\|BV\|.} \tag{29.24}$$

The Duhamel identity gives the exact finite-dimensional bound

$$\boxed{\|Q\mathrm{e}^{-\mathrm{i}tH_{p,w}}P\| \leq |t|p^{-w}\|QBP\|.} \tag{29.25}$$

Because H_0 reduces P, $Q\mathrm{e}^{-\mathrm{i}tH_0}P = 0$. Duhamel's formula therefore gives

$$Q\mathrm{e}^{-\mathrm{i}tH_{p,w}}P = -\mathrm{i}p^{-w}\int_0^t Q\mathrm{e}^{-\mathrm{i}(t-s)H_{p,w}}B\mathrm{e}^{-\mathrm{i}sH_0}P\,\mathrm{d}s. \tag{29.26}$$

Both exponentials have norm one. Taking norms inside the integral yields

$$\|Q\mathrm{e}^{-\mathrm{i}tH_{p,w}}P\| \leq p^{-w}\int_0^{|t|}\|QBP\|\,\mathrm{d}s \tag{29.27}$$

$$= |t|p^{-w}\|QBP\|. \tag{29.28}$$

Its short-time expansion is

$$Q\mathrm{e}^{-\mathrm{i}tH_{p,w}}P = -\mathrm{i}tp^{-w}QBP + \mathcal{O}(t^2p^{-w}), \tag{29.29}$$

where the remainder is generally first order in p^{-w}, not second order, unless additional commutation or spectral-matching conditions are imposed. For $z \notin \sigma(QH_{p,w}Q)$,

$$\Sigma_P(z) = PH_{p,w}Q(z - QH_{p,w}Q)^{-1}QH_{p,w}P \tag{29.30}$$

and

$$\boxed{\|\Sigma_P(z)\| \leq \frac{p^{-2w}\|PBQ\|^2}{\mathrm{dist}(z, \sigma(QH_{p,w}Q))}.} \tag{29.31}$$

Here the off-diagonal hypotheses give

$$PH_{p,w}Q = p^{-w}PBQ, \qquad QH_{p,w}P = p^{-w}QBP. \tag{29.32}$$

Hence submultiplicativity and the self-adjoint resolvent bound give

$$\|\Sigma_P(z)\| \leq p^{-2w}\|PBQ\|\|(z - QH_{p,w}Q)^{-1}\|\|QBP\| \tag{29.33}$$

$$\leq \frac{p^{-2w}\|PBQ\|^2}{\mathrm{dist}(z, \sigma(QH_{p,w}Q))}. \tag{29.34}$$

Therefore, under the explicit model input $\lambda_{p,w} = p^{-w}$, the assigned valuation weight orders quantum leakage and hidden-sector backreaction:

$$\text{refinement defect} = O(p^{-w}), \qquad \text{Feshbach backreaction} = O(p^{-2w}). \tag{29.35}$$

29.5 Dyadic precision requirement

When the split is implemented only as a complex Hilbert-space coordinate permutation, the sign qubit is exact without a denominator. When an integral involution is split through

$$E_\pm = \frac{I \pm A}{2}, \tag{29.36}$$

the dyadic spectral packet has weight one. Thus

$$\boxed{\text{complex quantum split} \neq \text{integral dyadic spectral lift}} \tag{29.37}$$

unless the one-digit precision delay is included.

29.6 Numerical verification

$$\text{multi-prime Hamiltonian-intertwining residual} = 5.030633 \times 10^{-16}, \tag{29.38}$$
$$\text{functional-calculus residual} = 4.096089 \times 10^{-16}, \tag{29.39}$$
$$\text{multi-prime leakage residual} = 2.909890 \times 10^{-16}, \tag{29.40}$$
$$\text{multi-prime Feshbach residual} = 9.883696 \times 10^{-33}, \tag{29.41}$$
$$\text{unit-basis equivalence residual} = 1.998401 \times 10^{-15}, \tag{29.42}$$
$$S(\rho_{\text{sign}}(t_*)) = 1.000000 \text{ bit}, \tag{29.43}$$
$$\text{ord}_\lambda(\text{defect}) = 1.000000, \tag{29.44}$$
$$\text{ord}_\lambda(\text{leakage}) = 1.000000, \tag{29.45}$$
$$\text{ord}_\lambda(\Sigma_P) = 2.000000, \tag{29.46}$$
$$\text{Duhamel-bound violation} = 0. \tag{29.47}$$

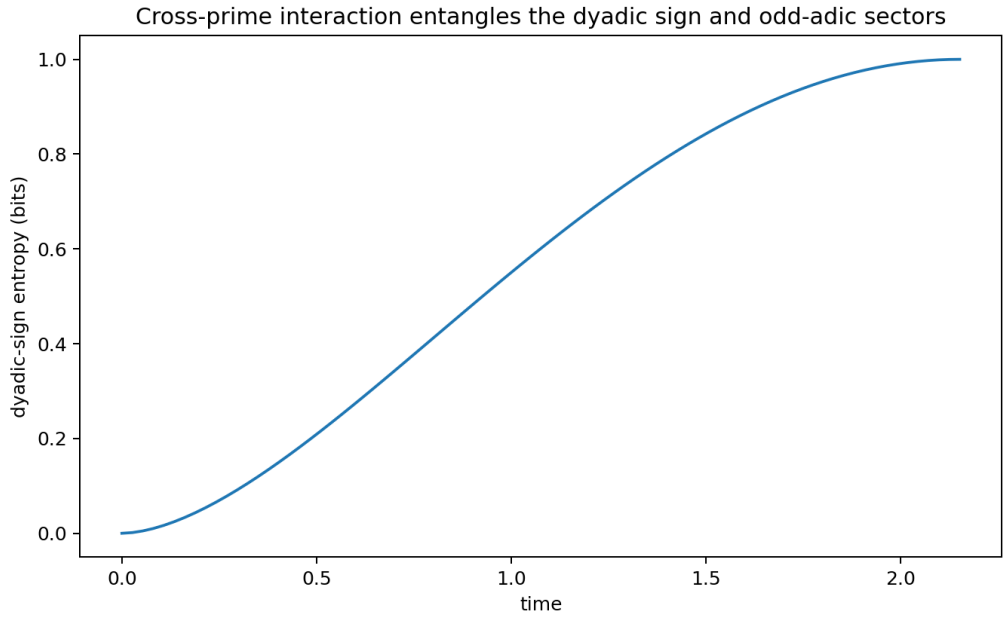


Figure 29.1: A stable interaction between the dyadic sign qubit and odd-adic detail sectors produces one ebit of entanglement.

29.7 Interpretation and next steps

The finite model combines torsion or sign channels, multiscale principal-phase coordinates, and refinement-compatible interactions. Its present locality is locality on the prime/refinement tree.

- Derive a map from prime-tree locality to causal locality in $3+1$ dimensions.
- Construct the interacting relativistic field limit of the multi-prime dynamics.
- Incorporate full non-Abelian gauge dynamics at finite and infinite refinement depth.
- Derive a massless spin-two sector and its coupling to the retained matter channels.
- Identify a falsifiable observable that is specific to the adic structure.

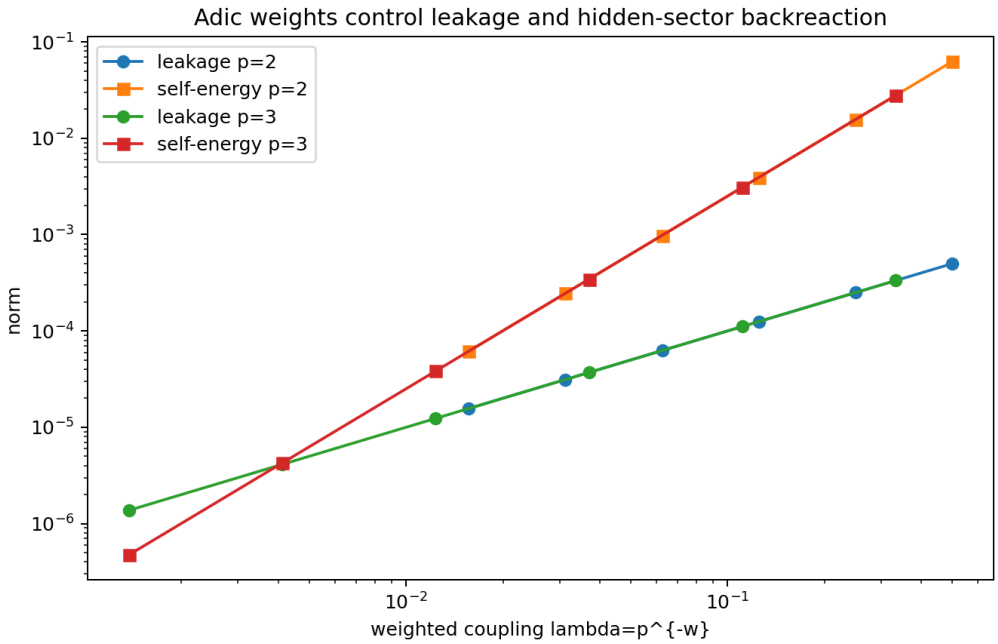


Figure 29.2: Valuation-weighted couplings control leakage linearly and Feshbach backreaction quadratically.

Chapter 30

Haar Normalization and Collective Quantum Coupling

Haar normalization fixes descendant amplitudes, but it does not by itself fix interaction strengths. The formulas separate per-leaf coupling, collective radial coupling, probability weight, and self-energy scaling.

30.1 Nested refinement amplitudes

Let one coarse packet have p^w descendants after w exact p-ary refinements. The normalized radial descendant is

$$\boxed{|R_{p,w}\rangle = \frac{1}{\sqrt{p^w}} \sum_{\mathbf{d}\in\{0,\dots,p-1\}^w} |\mathbf{d}\rangle.} \tag{30.1}$$

There are p^w orthonormal descendants, so

$$\langle R_{p,w}|R_{p,w}\rangle = p^{-w} \sum_{\mathbf{d},\mathbf{e}} \langle \mathbf{d}|\mathbf{e}\rangle \tag{30.2}$$

$$= p^{-w} \sum_{\mathbf{d}} 1 = 1. \tag{30.3}$$

Hence every descendant carries amplitude and Born weight

$$\boxed{|\langle \mathbf{d}|R_{p,w}\rangle| = p^{-w/2}, \qquad |\langle \mathbf{d}|R_{p,w}\rangle|^2 = p^{-w}.} \tag{30.4}$$

The factor p^{-w} is therefore forced as a descendant probability or uniform measure weight. It is not, by itself, forced as a Hamiltonian matrix element.

30.2 Per-leaf versus collective coupling

Let a visible packet $|c\rangle$ couple equally to all depth-w descendants with per-leaf matrix element

$$g_{p,w}^{\text{leaf}} := g\, p^{-\alpha w}. \tag{30.5}$$

The interaction is

$$H_{\text{int}}^{(p,w)} = g p^{-\alpha w} \sum_{\mathbf{d}} \big(|c\rangle\langle \mathbf{d}| + |\mathbf{d}\rangle\langle c| \big). \tag{30.6}$$

Only the radial descendant is bright. Every angular Haar combination is dark, and

$$\boxed{\langle R_{p,w}|H_{\text{int}}^{(p,w)}|c\rangle = g\, p^{(1/2-\alpha)w}.} \tag{30.7}$$

Indeed,

$$\langle R_{p,w}|H_{\text{int}}^{(p,w)}|c\rangle = gp^{-\alpha w}p^{-w/2}\sum_{\mathbf{e},\mathbf{d}}\langle \mathbf{e}|\mathbf{d}\rangle \tag{30.8}$$

$$= gp^{-\alpha w}p^{-w/2}p^{w} \tag{30.9}$$

$$= g\,p^{(1/2-\alpha)w}. \tag{30.10}$$

If $|A\rangle = \sum_{\mathbf{d}} a_{\mathbf{d}}|\mathbf{d}\rangle$ is orthogonal to $|R_{p,w}\rangle$, then

$$\langle A|H_{\text{int}}^{(p,w)}|c\rangle = gp^{-\alpha w}\sum_{\mathbf{d}} a_{\mathbf{d}}^{*} = 0, \tag{30.11}$$

which is why every angular Haar combination is dark. Thus

$$\boxed{G_{p,w}^{\text{coll}} = \sqrt{p^{w}}\, g_{p,w}^{\text{leaf}}.} \tag{30.12}$$

The branching multiplicity and the per-leaf suppression must be combined before assigning a physical refinement law.

30.3 Exact radial probability and self-energy

On resonance, the bright two-level Hamiltonian is

$$H_{\text{rad}} = \begin{pmatrix} 0 & G_{p,w}^{\text{coll}} \\ G_{p,w}^{\text{coll}} & 0 \end{pmatrix}. \tag{30.13}$$

Starting in $|c\rangle$,

$$\boxed{\Pr(R_{p,w};t) = \sin^{2}\Big(|g|p^{(1/2-\alpha)w}t\Big).} \tag{30.14}$$

Writing $G = G_{p,w}^{\text{coll}}$, the two-level block obeys $H_{\text{rad}}^{2} = |G|^{2}I$. Therefore

$$\mathrm{e}^{-\mathrm{i}tH_{\text{rad}}} = I\cos(|G|t) - \mathrm{i}\frac{H_{\text{rad}}}{|G|}\sin(|G|t). \tag{30.15}$$

Applied to $|c\rangle$, the radial amplitude has modulus $\sin(|G|t)$, giving the displayed probability. For a hidden radial level of energy ϵ, its exact rank-one Feshbach term is

$$\boxed{\Sigma_{p,w}(z) = \frac{|g|^{2}p^{(1-2\alpha)w}}{z-\epsilon}.} \tag{30.16}$$

In the visible–hidden decomposition, the off-diagonal block is the scalar G and the hidden resolvent is $(z-\epsilon)^{-1}$. The Schur correction is therefore

$$G(z-\epsilon)^{-1}G^{*} = \frac{|G|^{2}}{z-\epsilon} = \frac{|g|^{2}p^{(1-2\alpha)w}}{z-\epsilon}. \tag{30.17}$$

Therefore

$$\boxed{\alpha < \frac{1}{2} \Longrightarrow \text{collective growth}, \qquad \alpha = \frac{1}{2} \Longrightarrow \text{scale-invariant collective coupling}, \qquad \alpha > \frac{1}{2} \Longrightarrow \text{collective decay}.} \tag{30.18}$$

30.4 Measure-normalized and aggregate-packet regimes

For the uniform measure choice $\alpha = 1$,

$$G_{p,w}^{\text{coll}} = g\,p^{-w/2}, \qquad |G_{p,w}^{\text{coll}}|^{2} = |g|^{2}p^{-w}. \tag{30.19}$$

Thus the natural measure weight produces

$$\boxed{\text{collective amplitude} = O(p^{-w/2}), \qquad \text{probability and self-energy} = O(p^{-w}).} \tag{30.20}$$

A different geometry couples the visible packet directly to one normalized hidden packet with matrix element p^{-w}. That two-dimensional model remains exact, but it is not equivalent to equal coupling across all p^{w} descendants.

30.5 Numerical verification

$$\text{descendant-amplitude residual} = 1.110223 \times 10^{-16}, \tag{30.21}$$
$$\text{descendant-probability residual} = 1.110223 \times 10^{-16}, \tag{30.22}$$
$$\text{collective-norm residual} = 1.887379 \times 10^{-15}, \tag{30.23}$$
$$\text{radial-reduction residual} = 2.220446 \times 10^{-16}, \tag{30.24}$$
$$\text{two-level probability residual} = 1.734723 \times 10^{-18}, \tag{30.25}$$
$$\text{self-energy residual} = 3.496447 \times 10^{-18}. \tag{30.26}$$

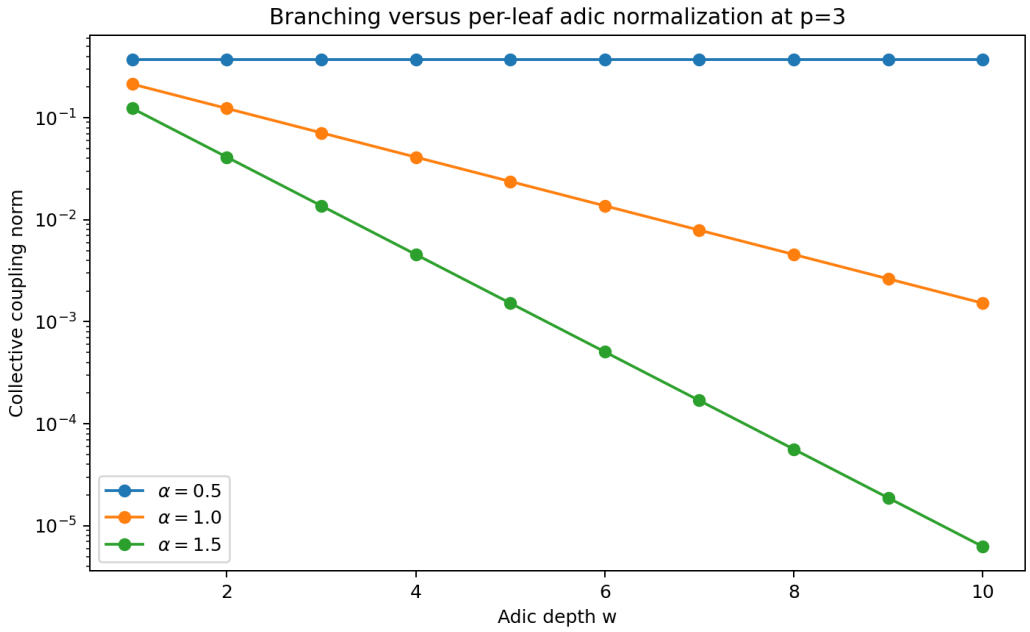


Figure 30.1: Collective refinement classes. The exponent $\alpha = 1/2$ is scale invariant; $\alpha = 1$ converts uniform measure weights into a $p^{-w/2}$ bright amplitude and a p^{-w} self-energy.

30.6 Next steps

- Derive a microscopic action that selects the exponent α.
- Determine dynamically whether the physical coupling is per-leaf, aggregate, or scale dependent.
- Give the weighting law a field-theoretic interpretation within the interacting model.
- Derive an experimental discriminator between the competing normalization laws.

Chapter 31

Local Quantum Walks on Spacetime–Adic Product Geometry

To make the adic coordinate dynamical, the physical lattice is paired with a finite refinement tree. A coined walk on the Cartesian product gives exact locality, continuity, and a joint causal cone.

31.1 Product graph

Let C_N be a physical cycle and let $T_{p,h}$ be the rooted finite p-ary refinement tree of depth h. Define

$$\boxed{\mathcal{G}_{p,h,N} := C_N \square T_{p,h}.} \tag{31.1}$$

A vertex is $v = (x, \mathbf{d})$, where $x \in \mathbb{Z}/N\mathbb{Z}$ and $\mathbf{d}$ is an adic prefix. Edges change either the physical coordinate by one or the adic prefix by one parent–child step, but never both in one tick.

The arc Hilbert space is

$$\mathcal{H}_{\text{arc}} = \ell^2(\vec{E}(\mathcal{G}_{p,h,N})) \otimes \mathbb{C}^{\tau_p}, \tag{31.2}$$

with

$$\tau_2 = 2 \tag{31.3}$$

for the exceptional dyadic sign qubit and

$$\tau_p = p - 1, \qquad p \text{ odd}, \tag{31.4}$$

for the finite torsion register.

31.2 Local coined evolution

At each vertex v, let C_v be the Grover coin on its outgoing arcs and let R_v be a local sign/torsion unitary. Set

$$C = \bigoplus_v (C_v \otimes R_v). \tag{31.5}$$

Let S reverse every directed arc. The one-tick evolution is

$$\boxed{U_{p,h,N} = SC.} \tag{31.6}$$

Since $S^2 = I$ and every local block is unitary,

$$U_{p,h,N}^* U_{p,h,N} = I. \tag{31.7}$$

Explicitly,

$$\begin{aligned} U_{p,h,N}^* U_{p,h,N} &= C^* S^* S C && (31.8)\\ &= C^* C && (31.9)\\ &= \bigoplus_v (C_v^* C_v \otimes R_v^* R_v) = I. && (31.10) \end{aligned}$$

31.3 Exact local and coarse continuity

Let ρ_v^n be the probability at vertex v before the n-th coin, and let $F_{v\to u}^n$ be the post-coin probability carried by arc $v \to u$. Then

$$\boxed{\rho_v^{n+1} - \rho_v^n = \sum_{u\sim v} (F_{u\to v}^n - F_{v\to u}^n)\,.} \tag{31.11}$$

The local coin preserves the total outgoing probability at v:

$$\sum_{u\sim v} F_{v\to u}^n = \rho_v^n. \tag{31.12}$$

The flip-flop shift turns the arc $u \to v$ into an incoming arc at v, so

$$\rho_v^{n+1} = \sum_{u\sim v} F_{u\to v}^n. \tag{31.13}$$

Subtracting the two equalities gives the local continuity equation. For every coarse product packet $A \subset V(\mathcal{G})$, internal flux cancels:

$$\boxed{\sum_{v\in A}(\rho_v^{n+1} - \rho_v^n) = \sum_{\substack{u\notin A\\ v\in A}} F_{u\to v}^n - \sum_{\substack{v\in A\\ u\notin A}} F_{v\to u}^n.} \tag{31.14}$$

After summing the local identity over $v \in A$, every internal undirected edge contributes once as $+F_{u\to v}^n$ and once as $-F_{u\to v}^n$. Those terms cancel exactly; only arcs crossing the boundary of A remain.

31.4 Exact product causal cone

Let $d_\square$ be graph distance on $C_N \square T_{p,h}$. For an initially localized state at v_0,

$$\boxed{d_\square(v, v_0) > n \Longrightarrow \rho_v^n = 0.} \tag{31.15}$$

The statement follows by induction. It holds at $n = 0$ by localization. The coin acts only within the arc fiber of a fixed vertex, and the shift moves an amplitude across exactly one graph edge. Thus a state supported in the radius-n ball is supported in the radius-$(n + 1)$ ball after one more tick. In particular,

$$d_{C_N}(x, x_0) \le n, \tag{31.16}$$

$$\big|\,|\mathbf{d}| - |\mathbf{d}_0|\,\big| \le n. \tag{31.17}$$

The model therefore has exact finite propagation in ordinary position and in adic resolution depth simultaneously.

31.5 Numerical verification

For the tested dyadic and ternary product graphs,

$$\text{coin-unitarity residual} = 1.310083 \times 10^{-15}, \tag{31.18}$$

$$\text{shift-involution residual} = 0, \tag{31.19}$$

$$\text{probability drift} = 8.881784 \times 10^{-16}, \tag{31.20}$$

$$\text{local continuity residual} = 3.330669 \times 10^{-16}, \tag{31.21}$$

$$\text{coarse balance residual} = 3.330669 \times 10^{-16}, \tag{31.22}$$

$$\text{probability outside the product cone} = 0, \tag{31.23}$$

$$\text{physical edge excess} = 0, \tag{31.24}$$

$$\text{adic-depth edge excess} = 0. \tag{31.25}$$

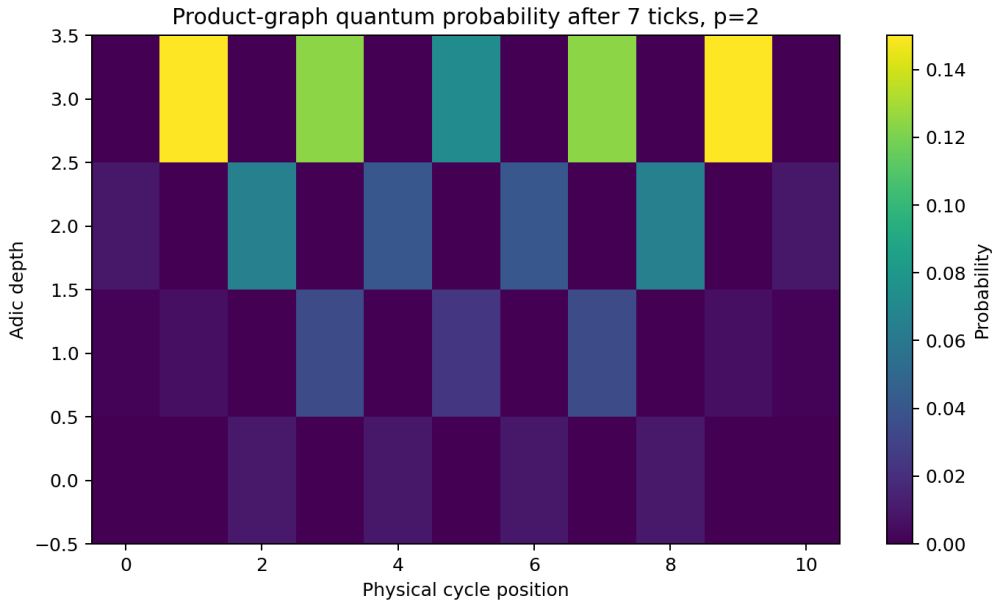


Figure 31.1: Exact local propagation on a physical-cycle times dyadic-tree product graph.

31.6 Next steps

- Prove emergent Lorentz symmetry of the product-graph causal cone, or derive the precise finite-scale violations.
- Construct a continuum theory on $\mathbb{R}^{3,1}$ with adic fibers.
- Derive the relation between physical edge lengths and adic refinement lengths from the dynamics.
- Couple the product walk to dynamical geometry and compute gravitational backreaction.

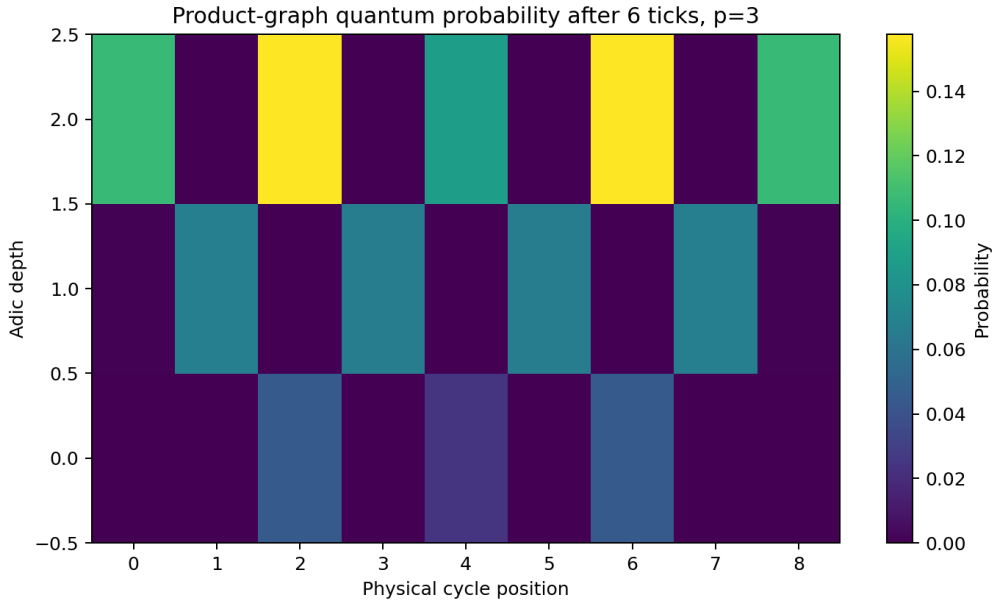


Figure 31.2: Exact local propagation on a physical-cycle times ternary-tree product graph.

Chapter 32

Radial Reduction and Continued-Fraction Self-Energy

A root-coupled p-ary tree decomposes into one visible radial chain and orthogonal dark Haar sectors. Schur reduction turns the hidden chain into an exact continued-fraction self-energy and memory kernel.

32.1 Rooted p-tree Hamiltonian

Let $T_{p,h}$ be a rooted p-ary tree. At level r, every vertex has onsite energy ϵ_r. Give every edge from level r to $r+1$ the branch-normalized coupling

$$\frac{\gamma_r}{\sqrt{p}}. \tag{32.1}$$

Define the normalized level state

$$|r\rangle := p^{-r/2} \sum_{|v|=r} |v\rangle. \tag{32.2}$$

The state is normalized because level r contains p^r vertices. Its matrix element to the next level is

$$\langle r+1|H|r\rangle = p^{-(r+1)/2} p^{-r/2} \sum_{\substack{|u|=r \\ v \text{ child of } u}} \frac{\gamma_r}{\sqrt{p}} \tag{32.3}$$

$$= p^{-(2r+1)/2}\, p^{r+1}\, \frac{\gamma_r}{\sqrt{p}} \tag{32.4}$$

$$= \gamma_r. \tag{32.5}$$

This is the explicit cancellation between branching multiplicity and branch normalization. Then the symmetric radial sector is invariant and the compressed Hamiltonian is the Jacobi chain

$$\boxed{J_h = \begin{pmatrix} \epsilon_0 & \gamma_0 & 0 & \cdots & 0 \\ \gamma_0 & \epsilon_1 & \gamma_1 & \ddots & \vdots \\ 0 & \gamma_1 & \epsilon_2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \gamma_{h-1} \\ 0 & \cdots & 0 & \gamma_{h-1} & \epsilon_h \end{pmatrix}.} \tag{32.6}$$

All angular Haar sectors are invariant and orthogonal to a root-localized symmetric source.

32.2 Prime-independent radial response

The factor $1/\sqrt{p}$ exactly cancels the $\sqrt{p}$ branching enhancement. Therefore the radial Jacobi matrix and the root Green function are independent of p:

$$\boxed{G_{0,h}(z) = \langle 0|(z - J_h)^{-1}|0\rangle.} \tag{32.7}$$

Prime dependence survives in the multiplicities of the dark Haar channels:

$$\operatorname{rank} D_r = (p-1)p^{r-1}. \tag{32.8}$$

With a fixed unnormalized per-edge coupling g_r, the radial coupling becomes $\sqrt{p}\, g_r$.

32.3 Exact continued fraction

Repeated Schur complementation is encoded by the terminal condition and backward recursion

$$\Sigma_h(z) := 0, \tag{32.9}$$

$$\Sigma_r(z) := \frac{\gamma_r^2}{z - \epsilon_{r+1} - \Sigma_{r+1}(z)}, \qquad 0 \le r < h. \tag{32.10}$$

At level r, split $z - J_h$ into the first site and the remaining tail:

$$z - J_h = \begin{pmatrix} z-\epsilon_r & -\gamma_r \langle r+1| \\ -\gamma_r |r+1\rangle & z - J_{r+1:h} \end{pmatrix}. \tag{32.11}$$

The scalar Schur complement of the tail is

$$z - \epsilon_r - \gamma_r^2 \langle r+1|(z - J_{r+1:h})^{-1}|r+1\rangle. \tag{32.12}$$

Identifying the tail matrix element recursively gives the stated $\Sigma_r(z)$. The root resolvent is therefore

$$\boxed{G_{0,h}(z) = \frac{1}{z - \epsilon_0 - \Sigma_0(z)}.} \tag{32.13}$$

Iterating the recursion gives the finite Jacobi continued fraction exactly. Each new adic level contributes one nested self-energy denominator.

32.4 Visible relativistic sector plus hidden adic chain

Let

$$H_D(k) = k\sigma_z + m_0\sigma_x \tag{32.14}$$

be a visible two-component Dirac block. Couple it identically in spinor space to the first hidden radial level. Let J_{tail} be the Jacobi tail beginning at level one and define

$$\boxed{\Sigma(z) = \gamma_0^2 \left[(z - J_{\text{tail}})^{-1}\right]_{00}.} \tag{32.15}$$

The exact determinant factorization is

$$\boxed{\det(z - H_{\text{full}}) = \det(z - H_{\text{hidden}}) \det((z - \Sigma(z))I_2 - H_D(k)).} \tag{32.16}$$

In block form,

$$z - H_{\text{full}} = \begin{pmatrix} zI_2 - H_D(k) & -\gamma_0\, I_2\langle 0| \\ -\gamma_0\, I_2|0\rangle & z - H_{\text{hidden}} \end{pmatrix}. \tag{32.17}$$

Taking the determinant through the hidden block gives

$$\det(z - H_{\text{full}}) = \det(z - H_{\text{hidden}}) \tag{32.18}$$

$$\times \det\left[zI_2 - H_D(k) - \gamma_0^2 \langle 0|(z - H_{\text{hidden}})^{-1}|0\rangle I_2\right], \tag{32.19}$$

which is the displayed factorization. Every visible spectral branch with nonzero visible weight satisfies

$$\boxed{\big(E - \Sigma(E)\big)^2 = k^2 + m_0^2.} \tag{32.20}$$

The corresponding memory kernel is

$$\boxed{K(t) = \gamma_0^2 \left[\exp(-\mathrm{i}tJ_{\mathrm{tail}})\right]_{00}.} \tag{32.21}$$

To obtain the memory equation, write the visible and hidden amplitudes as ψ_V, ψ_H:

$$\mathrm{i}\dot{\psi}_V = H_D\psi_V + \gamma_0\langle 0|\psi_H, \tag{32.22}$$

$$\mathrm{i}\dot{\psi}_H = J_{\mathrm{tail}}\psi_H + \gamma_0|0\rangle\psi_V. \tag{32.23}$$

For an initially empty hidden sector,

$$\psi_H(t) = -\mathrm{i}\gamma_0 \int_0^t \mathrm{e}^{-\mathrm{i}(t-s)J_{\mathrm{tail}}}|0\rangle\psi_V(s)\,\mathrm{d}s. \tag{32.24}$$

Substitution into the visible equation produces a convolution with the stated kernel $K(t-s)$. These equations define a conditional visible–adic hidden-sector model. They are not yet universal physical laws.

32.5 Numerical verification

$$\text{radial-block residual} = 3.538833 \times 10^{-15}, \tag{32.25}$$

$$\text{radial/dark invariance residual} = 5.306930 \times 10^{-15}, \tag{32.26}$$

$$\text{root Green residual} = 1.144392 \times 10^{-16}, \tag{32.27}$$

$$\text{continued-fraction residual} = 5.594315 \times 10^{-17}, \tag{32.28}$$

$$\text{prime-universality residual} = 1.144392 \times 10^{-16}, \tag{32.29}$$

$$\text{branching-factor residual} = 7.216450 \times 10^{-16}, \tag{32.30}$$

$$\text{Schur determinant residual} = 2.259450 \times 10^{-15}, \tag{32.31}$$

$$\text{visible-branch equation residual} = 2.065015 \times 10^{-14}, \tag{32.32}$$

$$\text{root/dark overlap residual} = 0. \tag{32.33}$$

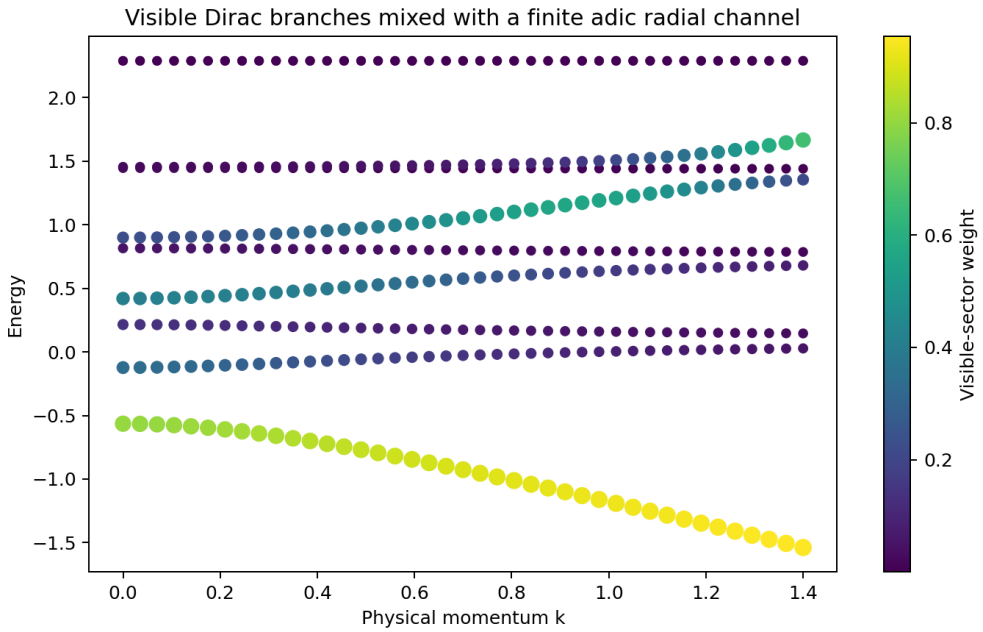


Figure 32.1: Visible branches dressed by the exact finite adic radial self-energy.

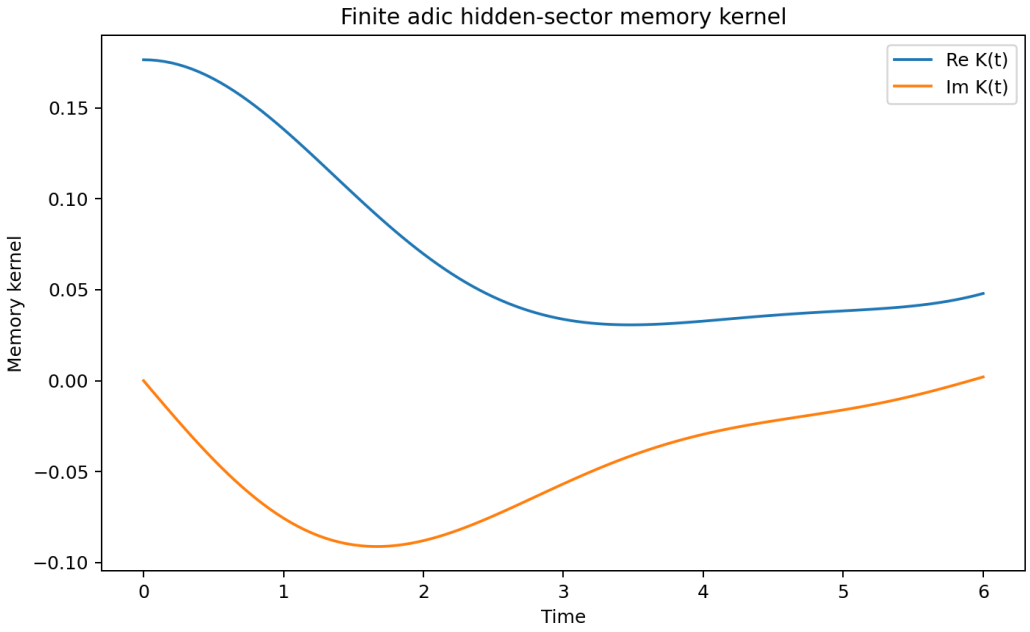


Figure 32.2: The hidden radial chain returns to the visible sector as an exact memory kernel.

32.6 Next steps

- Derive the level energies ϵ_r and couplings γ_r from a microscopic action.
- Prove the infinite-depth spectral and self-energy limit.
- Determine whether the dark Haar multiplicities produce a measurable spectral or dynamical effect.
- Construct a Lorentz- and gauge-covariant coupling between the visible radial channel and ordinary fields.

Chapter 33

Adic Quantum Polyphase Refinement

Polyphase analysis and p-adic digit refinement are the same finite operation after digit reversal. This identification gives explicit phase projectors, channel decompositions, band folding, and Schur reduction.

33.1 Standard and adic polyphase analysis

Let $N = p^h$ and $m = p^{h-1}$. The standard polyphase analysis separates the least significant residue:

$$\boxed{\mathcal{A}^{\mathrm{std}}_{p,h}|pk + r\rangle = |r\rangle \otimes |k\rangle.} \tag{33.1}$$

The adic analysis separates the newest p-adic digit:

$$\boxed{\mathcal{A}^{\mathrm{ad}}_{p,h}|a + dp^{h-1}\rangle = |d\rangle \otimes |a\rangle.} \tag{33.2}$$

Both are permutation unitaries. Let $\mathcal{R}_{p,h}$ reverse the h base-p digits. Then

$$\boxed{\mathcal{A}^{\mathrm{ad}}_{p,h} = (I_p \otimes \mathcal{R}_{p,h-1})\mathcal{A}^{\mathrm{std}}_{p,h}\mathcal{R}_{p,h}.} \tag{33.3}$$

If

$$n = d_0 + d_1 p + \cdots + d_{h-1}p^{h-1}, \tag{33.4}$$

then $\mathcal{R}_{p,h}$ changes the digit order to

$$d_{h-1} + d_{h-2}p + \cdots + d_0 p^{h-1}. \tag{33.5}$$

Standard polyphase analysis now removes d_{h-1} as the least significant digit. Reversing the remaining $h-1$ digits returns the prefix $a = d_0 + \cdots + d_{h-2}p^{h-2}$, which proves the identity on every basis vector. Thus signal polyphase and adic refinement are the same finite splitting after digit reversal.

33.2 Root-of-unity phase projectors

Let $\omega_p = \exp(2\pi \mathrm{i}/p)$ and define the new-digit modulation

$$M_{p,h}|a + dp^{h-1}\rangle = \omega_p^d |a + dp^{h-1}\rangle. \tag{33.6}$$

The projector onto child digit d is

$$\boxed{R_d^{(p,h)} = \frac{1}{p}\sum_{\ell=0}^{p-1} \omega_p^{-d\ell} M_{p,h}^{\ell}.} \tag{33.7}$$

Acting on a basis vector with new digit e gives

$$R_d^{(p,h)}|a + ep^{h-1}\rangle = \frac{1}{p}\sum_{\ell=0}^{p-1} \omega_p^{(e-d)\ell}|a + ep^{h-1}\rangle \tag{33.8}$$

$$= \delta_{d,e}|a + ep^{h-1}\rangle, \tag{33.9}$$

because the finite geometric sum is p when $d = e$ and zero otherwise. The projector and completeness identities then follow basis vector by basis vector. The selectors obey

$$R_d R_e = \delta_{de} R_d, \qquad \sum_{d=0}^{p-1} R_d = I. \tag{33.10}$$

33.3 Child Fourier transform and Haar refinement

Define

$$\boxed{\mathcal{W}_{p,h} = (F_p \otimes I_m)\mathcal{A}^{\mathrm{ad}}_{p,h}.} \tag{33.11}$$

Let $V_{p,h-1} : \mathbb{C}^m \to \mathbb{C}^{pm}$ be uniform refinement. The zero child-frequency channel is exactly the coarse adjoint:

$$\boxed{(\langle 0| \otimes I_m)\mathcal{W}_{p,h} = V^*_{p,h-1}.} \tag{33.12}$$

Indeed,

$$(\langle 0| \otimes I_m)\mathcal{W}_{p,h}|a + dp^{h-1}\rangle = \langle 0|F_p|d\rangle\,|a\rangle \tag{33.13}$$

$$= \frac{1}{\sqrt{p}}|a\rangle, \tag{33.14}$$

while uniform refinement has adjoint

$$V^*_{p,h-1}|a + dp^{h-1}\rangle = \frac{1}{\sqrt{p}}|a\rangle. \tag{33.15}$$

The nonzero child frequencies reconstruct the new detail projector:

$$\boxed{\mathcal{W}^*_{p,h}\big[(I_p - |0\rangle\langle 0|) \otimes I_m\big]\mathcal{W}_{p,h} = I - V_{p,h-1}V^*_{p,h-1}.} \tag{33.16}$$

Since $\mathcal{W}_{p,h}$ is unitary,

$$\mathcal{W}^*\big[(I - |0\rangle\langle 0|) \otimes I\big]\mathcal{W} = I - \mathcal{W}^*(|0\rangle\langle 0| \otimes I)\mathcal{W} \tag{33.17}$$

$$= I - \big[(\langle 0| \otimes I)\mathcal{W}\big]^*\big[(\langle 0| \otimes I)\mathcal{W}\big] \tag{33.18}$$

$$= I - VV^*. \tag{33.19}$$

For $p = 2$, F_2 is the Hadamard average/difference split. For odd p, one average and $p - 1$ detail phases occur at every parent.

Apply $\mathcal{W}_{p,h}$ recursively only to the zero-frequency channel. The full transform $\mathcal{H}_{p,h}$ has channel ranks

$$1, \quad (p-1), \quad (p-1)p, \quad \dots \quad , (p-1)p^{h-1}, \tag{33.20}$$

and

$$\boxed{\mathcal{H}^*_{p,h}\Pi_r\mathcal{H}_{p,h} = D^{(h)}_r.} \tag{33.21}$$

This gives an exact polyphase construction of the finite p-adic Haar projectors introduced above.

33.4 Exact descendant normalization

After w nested zero-frequency refinements,

$$\boxed{V^{(w)}_p|a\rangle = p^{-w/2} \sum_{\mathbf{d}\in\{0,\dots,p-1\}^w} |a, \mathbf{d}\rangle.} \tag{33.22}$$

The formula follows by induction. One refinement gives the factor $p^{-1/2}$. If it holds at depth w, refining each descendant once more gives

$$V_p^{(w+1)}|a\rangle = p^{-w/2} \sum_{\mathbf{d}} \frac{1}{\sqrt{p}} \sum_{e=0}^{p-1} |a, \mathbf{d}, e\rangle \tag{33.23}$$

$$= p^{-(w+1)/2} \sum_{\mathbf{d},e} |a, \mathbf{d}, e\rangle. \tag{33.24}$$

Hence polyphase refinement derives

$$\boxed{\text{amplitude per descendant} = p^{-w/2}, \qquad \text{Born weight per descendant} = p^{-w}.} \tag{33.25}$$

This fixes the normalization geometry used for collective coupling.

33.5 Operator and channel polyphase calculus

Let $J_r : \mathbb{C}^m \to \mathbb{C}^N$ embed the standard residue-r channel. For every operator X, define

$$X_{rs} := J_r^* X J_s. \tag{33.26}$$

Then

$$\boxed{X = \sum_{r,s=0}^{p-1} J_r X_{rs} J_s^*.} \tag{33.27}$$

The channel embeddings satisfy

$$J_r^* J_s = \delta_{rs} I_m, \qquad \sum_r J_r J_r^* = I_N. \tag{33.28}$$

Inserting the latter identity on both sides of X gives

$$X = \left(\sum_r J_r J_r^* \right) X \left(\sum_s J_s J_s^* \right) \tag{33.29}$$

$$= \sum_{r,s} J_r (J_r^* X J_s) J_s^*. \tag{33.30}$$

If $H = H^*$,

$$H_{rs}^* = H_{sr}. \tag{33.31}$$

If U is unitary, its polyphase blocks obey the paraunitary identities

$$\sum_r U_{rs}^* U_{rt} = \delta_{st} I_m, \tag{33.32}$$

$$\sum_s U_{rs} U_{ts}^* = \delta_{rt} I_m. \tag{33.33}$$

For the first identity,

$$\sum_r U_{rs}^* U_{rt} = \sum_r J_s^* U^* J_r J_r^* U J_t \tag{33.34}$$

$$= J_s^* U^* \left(\sum_r J_r J_r^* \right) U J_t \tag{33.35}$$

$$= J_s^* J_t = \delta_{st} I_m. \tag{33.36}$$

The second follows in the same way from $UU^* = I$. Moreover,

$$\boxed{\exp(-\mathrm{i}t\mathcal{A}H\mathcal{A}^*) = \mathcal{A}\exp(-\mathrm{i}tH)\mathcal{A}^*.} \tag{33.37}$$

For channel packets ψ_r,

$$J_{r\leftarrow s} = 2\,\mathrm{Im}\langle\psi_r, H_{rs}\psi_s\rangle \tag{33.38}$$

obeys

$$\dot{E}_r = \sum_s J_{r\leftarrow s}, \qquad J_{r\leftarrow s} = -J_{s\leftarrow r}. \tag{33.39}$$

For a CPTP map $\Phi(\rho) = \sum_\mu K_\mu \rho K_\mu^*$,

$$K_\mu^{\mathrm{poly}} = \mathcal{A} K_\mu \mathcal{A}^* \tag{33.40}$$

preserves

$$\sum_\mu (K_\mu^{\mathrm{poly}})^* K_\mu^{\mathrm{poly}} = I \tag{33.41}$$

and gives a unitarily equivalent quantum channel.

33.6 Exact polyphase band folding

For a circulant Hamiltonian, apply the coarse Fourier transform after standard polyphase analysis. The result is a direct sum of $p \times p$ symbols:

$$\boxed{\mathcal{C}_{p,h} H \mathcal{C}_{p,h}^* = \bigoplus_{j=0}^{m-1} \mathsf{H}_p(\theta_j), \qquad \theta_j = \frac{2\pi j}{m}.} \tag{33.42}$$

For

$$H = \mu I + \kappa(S + S^*), \tag{33.43}$$

the exact folded bands are

$$\boxed{E_q(\theta_j) = \mu + 2\kappa \cos\left(\frac{\theta_j + 2\pi q}{p}\right), \qquad q = 0, \ldots, p-1.} \tag{33.44}$$

Every fine Fourier index has a unique decomposition

$$k = j + qm, \qquad 0 \le j < m, \quad 0 \le q < p. \tag{33.45}$$

Therefore

$$\frac{2\pi k}{pm} = \frac{2\pi j/m + 2\pi q}{p} = \frac{\theta_j + 2\pi q}{p}. \tag{33.46}$$

Substitution into the original circulant eigenvalue $\mu + 2\kappa\cos(2\pi k/N)$ gives the p folded bands. Their union is exactly the original p^h-point cyclic spectrum.

33.7 Hierarchical spectra and multi-prime refinement

For

$$H_{p,h}^{\mathrm{hier}} = \sum_{r=0}^{h} \epsilon_r D_r^{(h)}, \tag{33.47}$$

the full polyphase/Haar transform gives

$$\boxed{\mathcal{H}_{p,h} H_{p,h}^{\mathrm{hier}} \mathcal{H}_{p,h}^* = \mathrm{diag}\left(\epsilon_0, \epsilon_1 I_{p-1}, \ldots, \epsilon_h I_{(p-1)p^{h-1}}\right).} \tag{33.48}$$

For several primes,

$$\boxed{\mathcal{A}_{\mathcal{P}} = I_{\mathrm{sign/torsion}} \otimes \bigotimes_{p\in\mathcal{P}} \mathcal{W}_{p,h_p}} \tag{33.49}$$

is unitary and readdresses interacting multi-prime dynamics exactly. The exceptional dyadic sign qubit remains a separate tensor factor:

$$\mathcal{A}_{\mathcal{P}} (Z_2 \otimes I) \mathcal{A}_{\mathcal{P}}^* = Z_2 \otimes I. \tag{33.50}$$

33.8 Polyphase Feshbach reduction

In Haar polyphase coordinates, write

$$\mathcal{W}H\mathcal{W}^* = \begin{pmatrix} H_{LL} & H_{LD} \\ H_{DL} & H_{DD} \end{pmatrix}. \tag{33.51}$$

The exact low-channel resolvent is

$$\boxed{P_L(z-H)^{-1}P_L = \left[z - H_{LL} - H_{LD}(z-H_{DD})^{-1}H_{DL}\right]^{-1}.} \tag{33.52}$$

For $(z-H)(\psi_L,\psi_D)^{\mathsf{T}} = (f_L,0)^{\mathsf{T}}$, the detail equation is

$$(z-H_{DD})\psi_D = H_{DL}\psi_L, \tag{33.53}$$

so

$$\psi_D = (z-H_{DD})^{-1}H_{DL}\psi_L. \tag{33.54}$$

Substitution into the low equation produces

$$\left[z - H_{LL} - H_{LD}(z-H_{DD})^{-1}H_{DL}\right]\psi_L = f_L, \tag{33.55}$$

which is the stated resolvent block. For dyadic even–odd decimation of

$$H = \mu I + \kappa(S+S^*), \tag{33.56}$$

the effective even-channel operator is

$$\boxed{H_{\text{eff}}(z) = \mu I + \frac{\kappa^2}{z-\mu}(2I+T+T^*).} \tag{33.57}$$

In the even–odd channel basis,

$$H_{LL} = H_{DD} = \mu I, \qquad H_{LD} = \kappa(I+T^*), \qquad H_{DL} = \kappa(I+T). \tag{33.58}$$

Thus

$$H_{LD}(z-H_{DD})^{-1}H_{DL} = \frac{\kappa^2}{z-\mu}(I+T^*)(I+T) \tag{33.59}$$

$$= \frac{\kappa^2}{z-\mu}(2I+T+T^*), \tag{33.60}$$

using $T^*T = I$. This is an exact energy-dependent renormalization step, not an approximation.

33.9 Numerical verification

$$\text{analysis-unitarity residual} = 0, \tag{33.61}$$

$$\text{digit-reversal residual} = 0, \tag{33.62}$$

$$\text{root-of-unity selector residual} = 2.405483 \times 10^{-16}, \tag{33.63}$$

$$\text{low-channel residual} = 0, \tag{33.64}$$

$$\text{detail-projector residual} = 6.848430 \times 10^{-16}, \tag{33.65}$$

$$\text{recursive Haar residual} = 1.405796 \times 10^{-15}, \tag{33.66}$$

$$\text{operator reconstruction residual} = 0, \tag{33.67}$$

$$\text{paraunitary residual} = 2.684315 \times 10^{-16}, \tag{33.68}$$

$$\text{evolution residual} = 1.648375 \times 10^{-16}, \tag{33.69}$$

$$\text{current residual} = 2.775558 \times 10^{-17}, \tag{33.70}$$

$$\text{band block residual} = 1.267119 \times 10^{-14}, \tag{33.71}$$

$$\text{spectrum residual} = 1.776357 \times 10^{-15}, \tag{33.72}$$
$$\text{analytic band residual} = 9.992007 \times 10^{-16}, \tag{33.73}$$
$$\text{hierarchical diagonalization residual} = 7.655814 \times 10^{-16}, \tag{33.74}$$
$$\text{CPTP completeness residual} = 4.962036 \times 10^{-15}, \tag{33.75}$$
$$\text{multi-prime evolution residual} = 4.990483 \times 10^{-16}, \tag{33.76}$$
$$\text{dyadic sign preservation residual} = 2.872356 \times 10^{-16}, \tag{33.77}$$
$$\text{polyphase Feshbach residual} = 1.733301 \times 10^{-16}, \tag{33.78}$$
$$\text{dyadic decimation residual} = 1.387779 \times 10^{-17}. \tag{33.79}$$

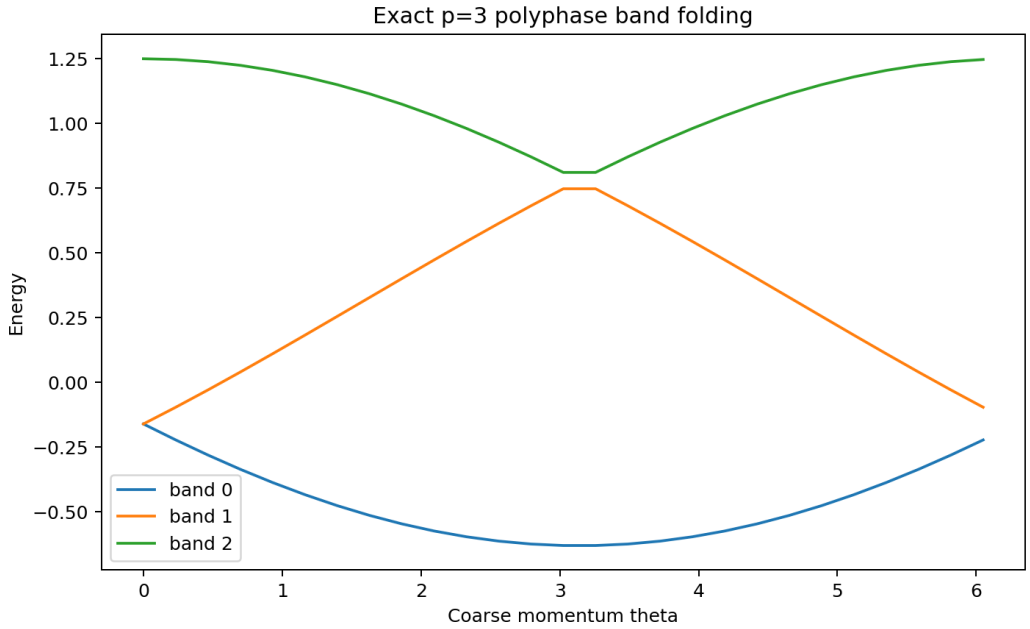


Figure 33.1: Exact three-channel band folding under $p = 3$ polyphase analysis.

33.10 Next steps

- Extend the exact finite polyphase calculus to an interacting continuum field theory.
- Transfer nonlinear gravitational constraints through the adic polyphase decomposition.
- Identify a physical observable that distinguishes digit order, or prove that digit-order dependence is a gauge or representation redundancy.
- Establish an interacting infinite-depth renormalization fixed point and its stability properties.

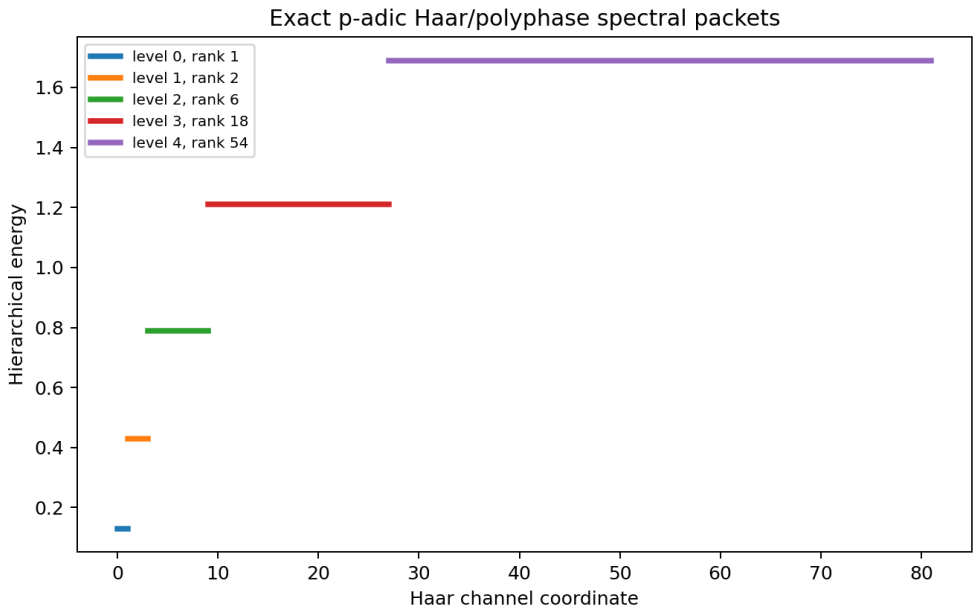


Figure 33.2: A hierarchical Hamiltonian becomes exactly diagonal in the full adic polyphase/Haar basis.

Chapter 34

Infinite-Depth Adic Haar Quantum Limit

The finite register tower has a concrete infinite-depth limit in $L^2(\mathbb{Z}_p)$. Cylinder expectations become martingale projections, Haar details resolve the identity, and level-diagonal Hamiltonians admit an exact spectral calculus.

34.1 Haar probability space and cylinder filtration

Fix a prime p. Let $\mathbb{Z}_p$ carry its normalized additive Haar probability measure μ_p, and set

$$\mathcal{H}_p^{\infty} := L^2(\mathbb{Z}_p, \mu_p; \mathbb{C}). \tag{34.1}$$

For $h \geq 0$, let $\mathcal{F}_h$ be the finite sigma-algebra generated by the residue cylinders

$$B_{h,a} := a + p^h \mathbb{Z}_p, \qquad 0 \leq a < p^h. \tag{34.2}$$

The conditional expectation onto $\mathcal{F}_h$ is

$$\boxed{(\mathsf{E}_h f)(x) = p^h \int_{x+p^h\mathbb{Z}_p} f(y)\ \mathrm{d}\mu_p(y).} \tag{34.3}$$

Because

$$\mu_p(x + p^h\mathbb{Z}_p) = p^{-h}, \tag{34.4}$$

the prefactor p^h divides by the cylinder measure. In particular,

$$\mathsf{E}_h \mathbf{1}_{B_{h,a}} = \mathbf{1}_{B_{h,a}}, \tag{34.5}$$

and a second averaging does not change the result. In words, E_h replaces a function by its average on the depth-h cylinder containing x. Its range

$$\mathcal{V}_h := \operatorname{Ran} \mathsf{E}_h \tag{34.6}$$

is the p^h-dimensional space of functions constant on every depth-h cylinder. Conditional-expectation calculus gives

$$\mathsf{E}_h^* = \mathsf{E}_h, \qquad \mathsf{E}_h^2 = \mathsf{E}_h, \qquad \mathsf{E}_h\mathsf{E}_k = \mathsf{E}_{\min(h,k)}, \tag{34.7}$$

$$\mathcal{V}_0 \subset \mathcal{V}_1 \subset \cdots, \qquad \overline{\bigcup_{h\geq 0} \mathcal{V}_h} = \mathcal{H}_p^{\infty}. \tag{34.8}$$

The final density statement follows because locally constant cylinder functions are dense in $L^2(\mathbb{Z}_p)$.

34.2 Exact direct-limit identification

Let

$$\mathcal{H}_{p,h} := \ell^2(\mathbb{Z}/p^h\mathbb{Z}) \tag{34.9}$$

with its standard orthonormal basis $\{e_a\}_{a=0}^{p^h-1}$. Define

$$\boxed{\mathcal{U}_{p,h}e_a = p^{h/2}\mathbf{1}_{B_{h,a}}.} \tag{34.10}$$

Since $\mu_p(B_{h,a}) = p^{-h}$, $\mathcal{U}_{p,h}$ is unitary from $\mathcal{H}_{p,h}$ onto $\mathcal{V}_h$. Indeed,

$$\langle \mathcal{U}_{p,h}e_a, \mathcal{U}_{p,h}e_b\rangle = p^h \mu_p(B_{h,a} \cap B_{h,b}) \tag{34.11}$$
$$= \delta_{ab}. \tag{34.12}$$

The p^h normalized cylinder indicators also span $\mathcal{V}_h$, proving surjectivity.
The finite refinement isometry is

$$V_{p,h}e_a = \frac{1}{\sqrt{p}} \sum_{d=0}^{p-1} e_{a+dp^h}. \tag{34.13}$$

The cylinder partition identity

$$B_{h,a} = \bigsqcup_{d=0}^{p-1} B_{h+1,a+dp^h} \tag{34.14}$$

implies

$$\boxed{\mathcal{U}_{p,h+1}V_{p,h} = \mathcal{U}_{p,h}.} \tag{34.15}$$

On a basis vector,

$$\mathcal{U}_{p,h+1}V_{p,h}e_a = \frac{1}{\sqrt{p}} \sum_{d=0}^{p-1} p^{(h+1)/2}\mathbf{1}_{B_{h+1,a+dp^h}} \tag{34.16}$$
$$= p^{h/2}\mathbf{1}_{\bigsqcup_d B_{h+1,a+dp^h}} \tag{34.17}$$
$$= p^{h/2}\mathbf{1}_{B_{h,a}} = \mathcal{U}_{p,h}e_a. \tag{34.18}$$

Therefore

$$\boxed{\varinjlim \left(\mathcal{H}_{p,0} \xrightarrow{V_{p,0}} \mathcal{H}_{p,1} \xrightarrow{V_{p,1}} \cdots \right) \cong L^2(\mathbb{Z}_p, \mu_p).} \tag{34.19}$$

Increasing the register depth therefore resolves one additional p-adic digit without changing the embedded coarse state. This identifies the finite Haar and polyphase registers with a concrete infinite-depth Hilbert-space limit. The Hilbert scalars and the Born rule remain complex; the p-adic structure belongs to the coordinate and refinement geometry.
For $0 \le h \le H$, let

$$V_{p,h\to H} := V_{p,H-1}\cdots V_{p,h}. \tag{34.20}$$

Then

$$V_{p,h\to H}^* V_{p,h\to H} = I, \tag{34.21}$$
$$V_{p,k\to H}V_{p,h\to k} = V_{p,h\to H}, \tag{34.22}$$
$$\mathsf{E}_h\big|_{\mathcal{V}_H} = \mathcal{U}_{p,H}V_{p,h\to H}V_{p,h\to H}^*\mathcal{U}_{p,H}^*. \tag{34.23}$$

These are the exact finite-cylinder restrictions used in the numerical verification.

34.3 Strong infinite Haar resolution

Define

$$\mathsf{D}_0 := \mathsf{E}_0, \qquad \mathsf{D}_h := \mathsf{E}_h - \mathsf{E}_{h-1} \quad (h \ge 1). \tag{34.24}$$

The operator D_h isolates information that first appears at refinement level h. By (34.7),

$$\boxed{\mathsf{D}_h\mathsf{D}_k = \delta_{hk}\mathsf{D}_h.} \tag{34.25}$$

For $1 \le h < k$, the martingale product rule gives

$$\mathsf{D}_h\mathsf{D}_k = (\mathsf{E}_h - \mathsf{E}_{h-1})(\mathsf{E}_k - \mathsf{E}_{k-1}) \tag{34.26}$$
$$= \mathsf{E}_h - \mathsf{E}_h - \mathsf{E}_{h-1} + \mathsf{E}_{h-1} \tag{34.27}$$
$$= 0. \tag{34.28}$$

For equal indices,

$$(\mathsf{E}_h - \mathsf{E}_{h-1})^2 = \mathsf{E}_h - \mathsf{E}_{h-1}. \tag{34.29}$$

The cases involving $h = 0$ follow in the same way. Moreover,

$$\sum_{h=0}^{H} \mathsf{D}_h = \mathsf{E}_H. \tag{34.30}$$

Since $\mathsf{E}_H f \to f$ in L^2,

$$\boxed{I = \sum_{h=0}^{\infty} \mathsf{D}_h \qquad \text{in the strong operator topology.}} \tag{34.31}$$

Consequently,

$$\boxed{\|f\|_2^2 = \sum_{h=0}^{\infty} \|\mathsf{D}_h f\|_2^2.} \tag{34.32}$$

At finite depth,

$$\|\mathsf{E}_H f\|_2^2 = \left\| \sum_{h=0}^{H} \mathsf{D}_h f \right\|_2^2 \tag{34.33}$$
$$= \sum_{h=0}^{H} \|\mathsf{D}_h f\|_2^2 \tag{34.34}$$

by orthogonality. Letting $H \to \infty$ and using $\mathsf{E}_H f \to f$ gives Parseval's identity. The detail multiplicities remain finite at every level:

$$\dim \operatorname{Ran} \mathsf{D}_0 = 1, \qquad \boxed{\dim \operatorname{Ran} \mathsf{D}_h = (p-1)p^{h-1} \quad (h \ge 1).} \tag{34.35}$$

Let $\omega_p = \exp(2\pi \mathrm{i}/p)$. For $h \ge 1$, $0 \le a < p^{h-1}$, and $1 \le q < p$, define

$$\boxed{\psi_{h,a,q}(x) = p^{(h-1)/2} \sum_{d=0}^{p-1} \omega_p^{qd} \mathbf{1}_{B_{h,a+dp^{h-1}}}(x).} \tag{34.36}$$

The child cylinders are disjoint and each has measure p^{-h}, so

$$\|\psi_{h,a,q}\|_2^2 = p^{h-1} \sum_{d=0}^{p-1} p^{-h} \tag{34.37}$$
$$= 1. \tag{34.38}$$

For the same parent and two characters,

$$\langle \psi_{h,a,q}, \psi_{h,a,q'} \rangle = \frac{1}{p} \sum_{d=0}^{p-1} \omega_p^{(q'-q)d} \tag{34.39}$$
$$= \delta_{q,q'}. \tag{34.40}$$

Different parents have disjoint support, and every nonzero character has zero average over its parent cylinder. Together with $\psi_0 \equiv 1$, these functions form an orthonormal basis of $L^2(\mathbb{Z}_p)$, and

$$\boxed{\mathsf{D}_h = \sum_{a=0}^{p^{h-1}-1} \sum_{q=1}^{p-1} |\psi_{h,a,q}\rangle\langle\psi_{h,a,q}| \quad (h \ge 1).} \tag{34.41}$$

Thus the recursive finite child-DFT transform converges to the complete p-adic Haar martingale resolution.

34.4 Quantitative truncation theorem

For $s > 0$, define the weighted Haar regularity space

$$\mathfrak{H}_p^s := \left\{ f \in L^2(\mathbb{Z}_p) : \sum_{h=0}^{\infty} p^{2sh} \|\mathsf{D}_h f\|_2^2 < \infty \right\} \tag{34.42}$$

with norm

$$\|f\|_{\mathfrak{H}_p^s}^2 := \sum_{h=0}^{\infty} p^{2sh} \|\mathsf{D}_h f\|_2^2. \tag{34.43}$$

Orthogonality gives the exact tail identity

$$\boxed{\|(I - \mathsf{E}_H) f\|_2^2 = \sum_{h=H+1}^{\infty} \|\mathsf{D}_h f\|_2^2.} \tag{34.44}$$

For every $f \in \mathfrak{H}_p^s$,

$$\|(I - \mathsf{E}_H) f\|_2^2 \le p^{-2s(H+1)} \sum_{h=H+1}^{\infty} p^{2sh} \|\mathsf{D}_h f\|_2^2, \tag{34.45}$$

because $h \ge H + 1$ implies

$$1 \le p^{-2s(H+1)} p^{2sh}. \tag{34.46}$$

Extending the remaining nonnegative sum to all $h \ge 0$ gives

$$\|(I - \mathsf{E}_H) f\|_2^2 \le p^{-2s(H+1)} \|f\|_{\mathfrak{H}_p^s}^2. \tag{34.47}$$

so

$$\boxed{\|(I - \mathsf{E}_H) f\|_2 \le p^{-s(H+1)} \|f\|_{\mathfrak{H}_p^s}.} \tag{34.48}$$

This is a cutoff-uniform analytic estimate. It converts levelwise adic regularity into a rigorous finite-depth approximation rate.

34.5 Self-adjoint infinite hierarchical Hamiltonians

Let $\{\epsilon_h\}_{h \ge 0} \subset \mathbb{R}$. Define

$$\boxed{H_{p,\epsilon} := \sum_{h=0}^{\infty} \epsilon_h \mathsf{D}_h} \tag{34.49}$$

on the domain

$$\boxed{\mathcal{D}(H_{p,\epsilon}) = \left\{ f \in L^2(\mathbb{Z}_p) : \sum_{h=0}^{\infty} |\epsilon_h|^2 \|\mathsf{D}_h f\|_2^2 < \infty \right\}.} \tag{34.50}$$

Each Haar level evolves with its own real energy ϵ_h. Under the unitary map

$$f \longmapsto (\mathsf{D}_0 f, \mathsf{D}_1 f, \ldots), \tag{34.51}$$

the operator becomes

$$(\xi_0, \xi_1, \ldots) \longmapsto (\epsilon_0 \xi_0, \epsilon_1 \xi_1, \ldots). \tag{34.52}$$

The displayed domain is exactly the set for which this image is square summable. The unitary decomposition

$$L^2(\mathbb{Z}_p) \cong \bigoplus_{h=0}^{\infty} \operatorname{Ran} \mathsf{D}_h \tag{34.53}$$

turns $H_{p,\epsilon}$ into multiplication by the real sequence ϵ_h. Hence $H_{p,\epsilon}$ is self-adjoint. It is bounded exactly when

$$\sup_h |\epsilon_h| < \infty. \tag{34.54}$$

If

$$|\epsilon_h| \longrightarrow \infty, \tag{34.55}$$

then every eigenspace is finite dimensional and

$$\boxed{(H_{p,\epsilon} - z)^{-1} \text{ is compact for } z \notin \sigma(H_{p,\epsilon}).} \tag{34.56}$$

To see compactness directly, define the finite-rank truncation

$$R_H(z) := \sum_{h=0}^{H} (\epsilon_h - z)^{-1} \mathsf{D}_h. \tag{34.57}$$

Then

$$\|(H_{p,\epsilon} - z)^{-1} - R_H(z)\| = \sup_{h>H} \frac{1}{|\epsilon_h - z|} \tag{34.58}$$

$$\longrightarrow 0. \tag{34.59}$$

A norm limit of finite-rank operators is compact.
The full functional calculus is levelwise:

$$\boxed{F(H_{p,\epsilon}) = \sum_{h=0}^{\infty} F(\epsilon_h) \mathsf{D}_h} \tag{34.60}$$

for every bounded Borel function F. In particular,

$$\mathrm{e}^{-\mathrm{i}tH_{p,\epsilon}} = \sum_{h=0}^{\infty} \mathrm{e}^{-\mathrm{i}t\epsilon_h} \mathsf{D}_h, \tag{34.61}$$

$$(z - H_{p,\epsilon})^{-1} = \sum_{h=0}^{\infty} (z - \epsilon_h)^{-1} \mathsf{D}_h. \tag{34.62}$$

For $f \in \mathfrak{H}_p^s$, the finite-depth quantum approximation obeys

$$\boxed{\left\| \mathrm{e}^{-\mathrm{i}tH_{p,\epsilon}} f - \sum_{h=0}^{H} \mathrm{e}^{-\mathrm{i}t\epsilon_h} \mathsf{D}_h f \right\|_2 = \|(I - \mathsf{E}_H) f\|_2 \le p^{-s(H+1)} \|f\|_{\mathfrak{H}_p^s}.} \tag{34.63}$$

The equality follows before taking the estimate:

$$\left\| \sum_{h>H} \mathrm{e}^{-\mathrm{i}t\epsilon_h} \mathsf{D}_h f \right\|_2^2 = \sum_{h>H} \left| \mathrm{e}^{-\mathrm{i}t\epsilon_h} \right|^2 \|\mathsf{D}_h f\|_2^2 \tag{34.64}$$

$$= \sum_{h>H} \|\mathsf{D}_h f\|_2^2 \tag{34.65}$$

$$= \|(I - \mathsf{E}_H) f\|_2^2. \tag{34.66}$$

The error is uniform in t because every retained or omitted spectral multiplier has unit modulus.
For $z \notin \overline{\{\epsilon_h\}}$,

$$\boxed{\|(I - \mathsf{E}_H)(z - H_{p,\epsilon})^{-1}\| = \sup_{h>H} \frac{1}{|z - \epsilon_h|}.} \tag{34.67}$$

The omitted resolvent is diagonal on the orthogonal sum $\bigoplus_{h>H} \operatorname{Ran} \mathsf{D}_h$. Its operator norm is therefore the supremum of the absolute scalar multipliers on those levels. When $|\epsilon_h| \to \infty$, this norm tends to zero and gives a direct compact-resolvent criterion.

34.6 Finite-set adelic tensor limit

Let $\mathcal{P}$ be a finite set of primes and let $\mathcal{K}_{\mathrm{tor}}$ contain any finite torsion or dyadic-sign registers retained separately. Define

$$\boxed{\mathcal{H}_{\mathcal{P}}^{\infty} = \mathcal{K}_{\mathrm{tor}} \otimes \bigotimes_{p\in\mathcal{P}} L^2(\mathbb{Z}_p, \mu_p).} \tag{34.68}$$

For a multi-depth $\mathbf{h} = (h_p)_{p\in\mathcal{P}}$, set

$$\mathsf{E}_{\mathbf{h}} = I_{\mathcal{K}_{\mathrm{tor}}} \otimes \bigotimes_{p\in\mathcal{P}} \mathsf{E}_{p,h_p}. \tag{34.69}$$

Primewise expectations commute, and

$$\boxed{\mathsf{E}_{\mathbf{h}}\mathsf{E}_{\mathbf{k}} = \mathsf{E}_{\min(\mathbf{h},\mathbf{k})}} \tag{34.70}$$

with componentwise minimum. Likewise, refinement in two distinct prime coordinates is path independent:

$$\boxed{(V_p \otimes I)(I \otimes V_q) = (I \otimes V_q)(V_p \otimes I) = V_p \otimes V_q.} \tag{34.71}$$

The exceptional dyadic sign packet is untouched because every additive Haar expectation acts as the identity on $\mathcal{K}_{\mathrm{tor}}$.

34.7 Numerical verification

$$\text{martingale-projector residual} = 2.314960 \times 10^{-15}, \tag{34.72}$$
$$\text{direct-system residual} = 5.551115 \times 10^{-16}, \tag{34.73}$$
$$\text{infinite-Haar finite-restriction residual} = 2.404663 \times 10^{-15}, \tag{34.74}$$
$$\text{Parseval tail residual} = 2.775558 \times 10^{-17}, \tag{34.75}$$
$$\text{Haar-Sobolev bound violation} = 0, \tag{34.76}$$
$$\text{hierarchical spectral-calculus residual} = 3.352961 \times 10^{-14}, \tag{34.77}$$
$$\text{cutoff-intertwining residual} = 3.607583 \times 10^{-14}, \tag{34.78}$$
$$\text{resolvent-tail residual} = 7.771561 \times 10^{-16}, \tag{34.79}$$
$$\text{multi-prime path residual} = 0, \tag{34.80}$$
$$\text{dyadic-sign spectator residual} = 0. \tag{34.81}$$

34.8 Next steps

- Construct nonlinear refinement maps with estimates that are uniform in depth.
- Formulate interacting relativistic fields on $\mathbb{Z}_p$ fibers.
- Derive the hierarchy of level energies from a microscopic action.
- Couple the infinite-depth adic Hilbert space to dynamical $3+1$ geometry.
- Design a measurement protocol for the detail populations and their scale dependence.

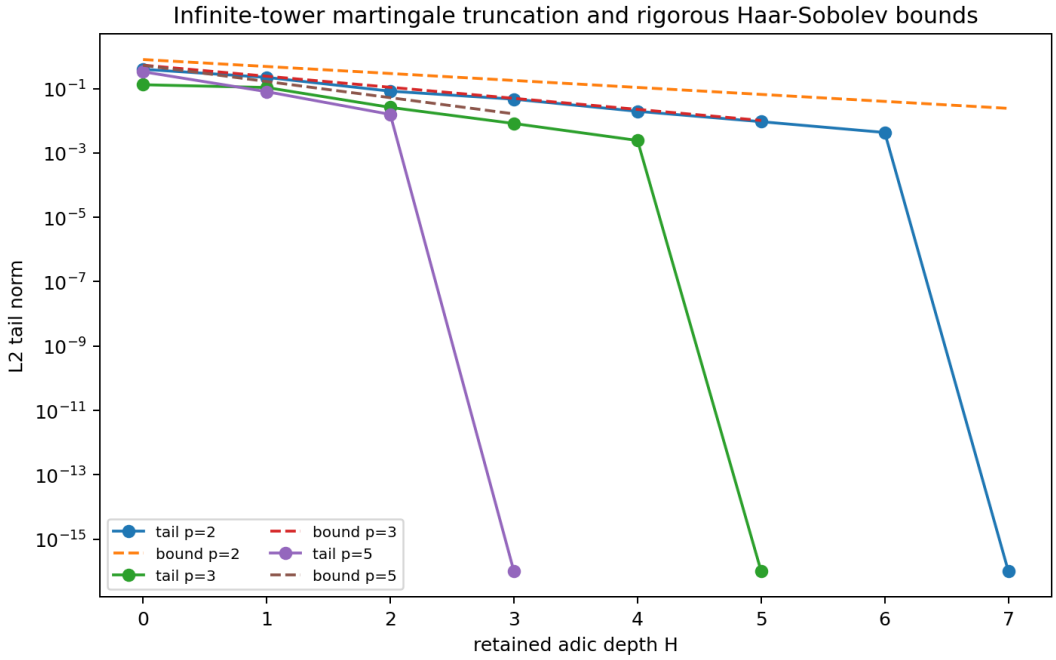


Figure 34.1: Finite-cylinder tails converge to zero and remain below the proved Haar-Sobolev bounds for dyadic and odd-adic towers.

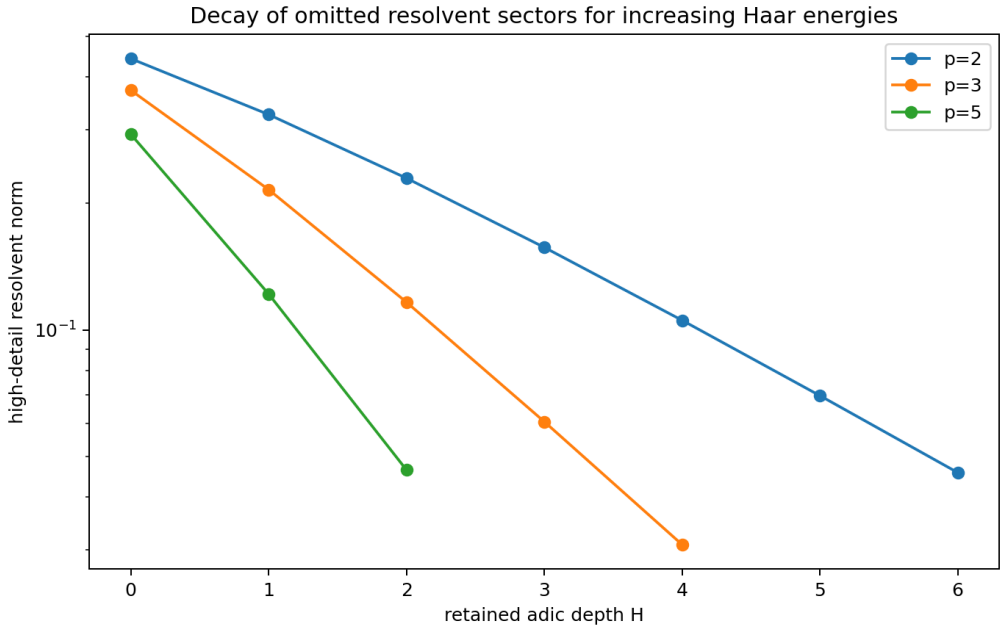


Figure 34.2: For increasing hierarchical energies, the omitted high-detail resolvent norm decays exactly according to the largest omitted spectral multiplier.

Chapter 35

Prime-Polyphase Spectral Renormalization

For the free nearest-neighbor chain, eliminating $p-1$ residue channels closes under a Chebyshev map. The result is a prime-indexed renormalization semigroup with folded inverse branches and an invariant spectral measure.

35.1 Nearest-neighbor polyphase model

Let $N = pM$, where p is prime. On the cycle $\mathbb{Z}/N\mathbb{Z}$, consider

$$\boxed{H_{\mu,\kappa} = \mu I + \kappa(S + S^*)} \tag{35.1}$$

with $\mu \in \mathbb{R}$, $\kappa \neq 0$, and cyclic shift S. Retain the sublattice

$$C := \{0, p, 2p, \ldots, (M-1)p\} \tag{35.2}$$

and eliminate the $p-1$ intermediate residue channels in every cell. With P the projector onto C, $Q = I - P$, and spectral parameter $z \notin \sigma(QHQ)$, the exact Schur operator is

$$\mathcal{S}_p(z) := P(z-H)P - PHQ(z - QHQ)^{-1}QHP. \tag{35.3}$$

This is the exact spectral operator seen by amplitudes supported on the retained sublattice. This section derives its closed form for every prime p. The derivation is limited to the stated free translation-invariant Hamiltonian.

35.2 Chebyshev transfer law

Define the normalized spectral variable

$$x := \frac{z-\mu}{\kappa}. \tag{35.4}$$

Let T_n and U_n be the Chebyshev polynomials characterized by

$$T_n(\cos\theta) = \cos(n\theta), \tag{35.5}$$

$$U_n(\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta}. \tag{35.6}$$

Set

$$\boxed{R_p(x) := 2T_p(x/2).} \tag{35.7}$$

The eliminated channels are therefore encoded by a single Chebyshev polynomial in the normalized energy. If T denotes the cyclic shift on the retained M-site lattice, then

$$\boxed{\mathcal{S}_p(z) = \frac{\kappa}{U_{p-1}(x/2)}\left[R_p(x)I - (T + T^*)\right].} \tag{35.8}$$

Equivalently,

$$\mathcal{S}_p(z) = zI - H_{\text{eff}}^{(p)}(z) \coloneqq (z - \mu_p(z))I - \kappa_p(z)(T + T^*), \tag{35.9}$$

where

$$\boxed{\kappa_p(z) = \frac{\kappa}{U_{p-1}(x/2)},} \tag{35.10}$$

$$\boxed{\mu_p(z) = z - \kappa \frac{R_p(x)}{U_{p-1}(x/2)}.} \tag{35.11}$$

The normalized effective variable is therefore

$$\boxed{\frac{z - \mu_p(z)}{\kappa_p(z)} = R_p(x).} \tag{35.12}$$

To prove (35.8), write the stationary equation as

$$\psi_{n+1} = x\psi_n - \psi_{n-1} \tag{35.13}$$

and introduce

$$A(x) = \begin{pmatrix} x & -1 \\ 1 & 0 \end{pmatrix}. \tag{35.14}$$

The exact transfer identity is

$$A(x)^p = \begin{pmatrix} U_p(x/2) & -U_{p-1}(x/2) \\ U_{p-1}(x/2) & -U_{p-2}(x/2) \end{pmatrix}. \tag{35.15}$$

The identity follows inductively from

$$U_{n+1}(x/2) = xU_n(x/2) - U_{n-1}(x/2), \tag{35.16}$$

with $U_0 = 1$ and $U_1 = x$. Multiplying the n-step matrix by $A(x)$ replaces its first column by the same three-term recurrence and shifts the previous first column into the second. Its trace and determinant are

$$\operatorname{Tr} A(x)^p = R_p(x), \qquad \det A(x)^p = 1. \tag{35.17}$$

Let $\chi_j := \psi_{jp}$ denote the retained amplitudes. Sampling a second-order recurrence every p steps replaces the one-step transfer matrix A by A^p. Cayley–Hamilton for the unimodular matrix A^p gives

$$(A^p)^2 - R_p(x)A^p + I = 0. \tag{35.18}$$

Applied along consecutive cells, this yields the coarse recurrence

$$\chi_{j+1} - R_p(x)\chi_j + \chi_{j-1} = 0. \tag{35.19}$$

The remaining factor in the Schur operator is fixed by the endpoint-to-endpoint Green coefficient $U_{p-1}(x/2)^{-1}$. Eliminating the internal values between ψ_{jp} and $\psi_{(j+1)p}$ gives

$$\frac{\kappa}{U_{p-1}(x/2)} \left[R_p(x)\psi_{jp} - \psi_{(j-1)p} - \psi_{(j+1)p}\right] = 0, \tag{35.20}$$

which is exactly the coarse Schur equation.

For $p = 2$,

$$R_2(x) = x^2 - 2, \qquad U_1(x/2) = x, \tag{35.21}$$

so

$$\boxed{H_{\text{eff}}^{(2)}(z) = \mu I + \frac{\kappa^2}{z - \mu}(2I + T + T^*),} \tag{35.22}$$

Indeed, $x = (z - \mu)/\kappa$ gives

$$\kappa_2(z) = \frac{\kappa}{x} = \frac{\kappa^2}{z - \mu}, \tag{35.23}$$

$$\mu_2(z) = z - \kappa\frac{x^2-2}{x} \tag{35.24}$$

$$= \mu + \frac{2\kappa^2}{z-\mu}. \tag{35.25}$$

Substituting these two coefficients into $\mu_2 I + \kappa_2(T + T^*)$ gives the displayed dyadic operator. recovering the preceding dyadic formula.

35.3 Companion-state trace coordinate

The Chebyshev map has a transfer-coordinate interpretation that is more general than the particular nearest-neighbor recurrence. Let

$$M \in SL_2(\mathbb{C}), \qquad x := \operatorname{Tr} M. \tag{35.26}$$

Cayley–Hamilton gives

$$M^2 - xM + I = 0. \tag{35.27}$$

Multiplying by M^{n-2} yields

$$M^n = xM^{n-1} - M^{n-2}. \tag{35.28}$$

The second-kind Chebyshev recurrence has the same form, so induction gives

$$\boxed{M^n = U_{n-1}(x/2)M - U_{n-2}(x/2)I, \qquad n \geq 1,} \tag{35.29}$$

where $U_{-1} := 0$. Assuming the formula at n and $n-1$,

$$M^{n+1} = xM^n - M^{n-1} \tag{35.30}$$

$$= [xU_{n-1}(x/2) - U_{n-2}(x/2)]\, M \tag{35.31}$$

$$- [xU_{n-2}(x/2) - U_{n-3}(x/2)]\, I \tag{35.32}$$

$$= U_n(x/2)M - U_{n-1}(x/2)I. \tag{35.33}$$

Taking traces gives

$$\operatorname{Tr}(M^n) = xU_{n-1}(x/2) - 2U_{n-2}(x/2) \tag{35.34}$$

$$= 2T_n(x/2). \tag{35.35}$$

Therefore

$$\boxed{\operatorname{Tr}(M^n) = R_n(\operatorname{Tr} M).} \tag{35.36}$$

If a two-component state propagates by $w_{k+1} = Mw_k$, its p residue subsequences satisfy

$$\boxed{w_{pm+j} = M^j (M^p)^m w_0, \qquad 0 \leq j < p.} \tag{35.37}$$

Thus every fine residue channel advances under the same coarse transfer matrix M^p, with scalar trace coordinate

$$x_p := \operatorname{Tr}(M^p) = R_p(x). \tag{35.38}$$

A second decimation replaces M^p by $(M^p)^q = M^{pq}$, so

$$\operatorname{Tr}((M^p)^q) = R_q(R_p(x)) = R_{pq}(x). \tag{35.39}$$

The renormalization semigroup is the trace-coordinate image of ordinary matrix powering.

For the free chain, $M = A(x)$, and the trace coordinate is exactly the normalized energy. For a general unimodular two-state transfer problem, the trace identity remains exact, but an interpretation as a local Schur-reduced Hamiltonian requires a separate model-specific construction.

35.4 Resonant detail channels

The Schur chart is singular when

$$U_{p-1}(x/2) = 0. \tag{35.40}$$

The roots are

$$\boxed{x_j = 2\cos\left(\frac{j\pi}{p}\right), \qquad j = 1, \ldots, p-1.} \tag{35.41}$$

Writing $x/2 = \cos\theta$,

$$U_{p-1}(x/2) = \frac{\sin(p\theta)}{\sin\theta}. \tag{35.42}$$

Its interior zeros occur at $\theta = j\pi/p$, $1 \le j < p$, which gives the listed values. Equivalently,

$$\boxed{z_j = \mu + 2\kappa\cos\left(\frac{j\pi}{p}\right).} \tag{35.43}$$

These are precisely the Dirichlet eigenvalues of the eliminated $(p-1)$-site open path inside one polyphase cell. At such an energy the eliminated cell supports an internal standing wave, so the resonant detail channel must be retained rather than divided out. Away from the poles,

$$\boxed{\det(z - H) = \det(z - QHQ)\det\mathcal{S}_p(z).} \tag{35.44}$$

After multiplication by the internal determinant, this equality extends as a polynomial identity through the resonant energies.

35.5 Band-folding and decimation duality

The p inverse folded bands of a normalized nearest-neighbor spectrum are

$$\boxed{x_{p,q}(\theta) = 2\cos\left(\frac{\theta + 2\pi q}{p}\right), \qquad q = 0, \ldots, p-1.} \tag{35.45}$$

Chebyshev composition gives

$$\boxed{R_p\big(x_{p,q}(\theta)\big) = 2\cos\theta \quad \text{for every } q.} \tag{35.46}$$

Direct substitution gives

$$R_p\big(x_{p,q}(\theta)\big) = 2T_p\left(\cos\frac{\theta + 2\pi q}{p}\right) \tag{35.47}$$

$$= 2\cos(\theta + 2\pi q) \tag{35.48}$$

$$= 2\cos\theta. \tag{35.49}$$

Thus polyphase band folding and Schur decimation are exact inverse operations at the normalized spectral level: folding resolves the p inverse branches, while decimation maps all branches back to the common coarse energy.

For the fine cyclic eigenvalues

$$x_k = 2\cos\left(\frac{2\pi k}{pM}\right), \tag{35.50}$$

one obtains

$$\boxed{R_p(x_k) = 2\cos\left(\frac{2\pi k}{M}\right).} \tag{35.51}$$

Hence the pM-point fine spectrum maps onto the M-point coarse spectrum with exactly p-fold multiplicity.

35.6 Exact renormalization semigroup

Chebyshev composition yields

$$\boxed{R_m \circ R_n = R_{mn} = R_n \circ R_m.} \tag{35.52}$$

For $x = 2\cos\theta$,

$$R_m(R_n(x)) = 2T_m(T_n(\cos\theta)) \tag{35.53}$$
$$= 2\cos(mn\theta) \tag{35.54}$$
$$= R_{mn}(x). \tag{35.55}$$

The same expression is symmetric in m, n, proving commutativity. The corresponding second-kind identity is

$$\boxed{U_{mn-1}(t) = U_{m-1}\big(T_n(t)\big)U_{n-1}(t).} \tag{35.56}$$

Set $t = \cos\theta$. Then

$$U_{m-1}(T_n(t))U_{n-1}(t) = \frac{\sin(mn\theta)}{\sin(n\theta)}\frac{\sin(n\theta)}{\sin\theta} \tag{35.57}$$
$$= \frac{\sin(mn\theta)}{\sin\theta} \tag{35.58}$$
$$= U_{mn-1}(t). \tag{35.59}$$

Polynomial continuity extends the identity through the removable zeros of $\sin(n\theta)$. Equations (35.52) and (35.56) imply exact agreement between direct and iterated Schur elimination. Decimating by m and then by n is the same operation as one mn-step decimation.

After h repeated p-decimations, let $q = p^h$. Then

$$\boxed{R_p^{\circ h}(x) = R_q(x) = 2T_q(x/2),} \tag{35.60}$$

$$\boxed{\kappa^{(h)}(z) = \frac{\kappa}{U_{q-1}(x/2)},} \tag{35.61}$$

$$\boxed{\mu^{(h)}(z) = z - \kappa\frac{R_q(x)}{U_{q-1}(x/2)}.} \tag{35.62}$$

For distinct primes p and q,

$$\boxed{R_pR_q = R_qR_p = R_{pq}.} \tag{35.63}$$

Therefore the dyadic and odd-adic decimation order is exactly path independent in this free translation-invariant spectral sector. This does not establish path independence for nonlinear gravity or interacting quantum fields.

35.7 Invariant spectral fixed measure

On $[-2, 2]$, define the arcsine probability measure

$$\boxed{\mathrm{d}\nu(x) = \frac{\mathrm{d}x}{\pi\sqrt{4 - x^2}}.} \tag{35.64}$$

Under the parameterization

$$x = 2\cos\theta, \qquad 0 \le \theta \le \pi, \tag{35.65}$$

this becomes $\mathrm{d}\nu = \mathrm{d}\theta/\pi$, while

$$R_p(2\cos\theta) = 2\cos(p\theta). \tag{35.66}$$

The p-fold angle map preserves the pushforward cosine distribution. Hence

$$\boxed{(R_p)_*\nu = \nu.} \tag{35.67}$$

For a bounded test function f, it is convenient to use a uniform angle on the full circle:

$$\int f(R_p(x))\ \mathrm{d}\nu(x) = \frac{1}{2\pi}\int_0^{2\pi} f(2\cos(p\theta))\ \mathrm{d}\theta \tag{35.68}$$

$$= \frac{1}{2\pi p}\int_0^{2\pi p} f(2\cos u)\ \mathrm{d}u \tag{35.69}$$

$$= \frac{1}{2\pi}\int_0^{2\pi} f(2\cos u)\ \mathrm{d}u \tag{35.70}$$

$$= \int f(x)\ \mathrm{d}\nu(x). \tag{35.71}$$

The third equality uses the 2π-periodicity of the cosine. The free spectral distribution is therefore unchanged by every prime decimation. This is an exact infinite-depth fixed measure for the normalized free spectral renormalization map.

Let

$$\phi_0(x) = 1, \qquad \phi_n(x) = \sqrt{2}\,T_n(x/2) \quad (n \geq 1). \tag{35.72}$$

Then $\{\phi_n\}_{n\geq 0}$ is an orthonormal basis of $L^2([-2,2],\nu)$. The Koopman isometry

$$\mathcal{K}_p f = f \circ R_p \tag{35.73}$$

acts by

$$\boxed{\mathcal{K}_p\phi_n = \phi_{pn}.} \tag{35.74}$$

Its adjoint is the exact mode decimator

$$\boxed{\mathcal{K}_p^*\phi_m = \begin{cases}\phi_{m/p}, & p \mid m,\\ 0, & p \nmid m,\end{cases}} \tag{35.75}$$

For $n, m \geq 0$,

$$\langle \phi_n, \mathcal{K}_p^*\phi_m\rangle = \langle \mathcal{K}_p\phi_n, \phi_m\rangle \tag{35.76}$$

$$= \langle \phi_{pn}, \phi_m\rangle \tag{35.77}$$

$$= \delta_{pn,m}. \tag{35.78}$$

The coefficients of $\mathcal{K}_p^*\phi_m$ in the orthonormal basis are therefore zero unless $m = pn$, in which case the only coefficient is one. with $\mathcal{K}_p^*\phi_0 = \phi_0$. The infinite-depth spectral fixed point therefore has a precise polyphase interpretation: Koopman evolution injects a mode into the p-multiple channel, while the adjoint keeps exactly the divisible low channel and removes the remaining phases.

The mean expansion rate is also exact:

$$\boxed{\int_{-2}^{2} \log|R_p'(x)|\ \mathrm{d}\nu(x) = \log p.} \tag{35.79}$$

Indeed,

$$R_p'(2\cos\theta) = p\frac{\sin(p\theta)}{\sin\theta}. \tag{35.80}$$

More explicitly,

$$\int_{-2}^{2} \log|R_p'(x)|\ \mathrm{d}\nu(x) = \log p + \frac{1}{\pi}\int_0^{\pi} \log|\sin(p\theta)|\ \mathrm{d}\theta \tag{35.81}$$

$$- \frac{1}{\pi}\int_0^{\pi} \log(\sin\theta)\ \mathrm{d}\theta. \tag{35.82}$$

With $u = p\theta$, periodicity of $|\sin u|$ gives

$$\int_0^{\pi} \log|\sin(p\theta)|\ \mathrm{d}\theta = \int_0^{\pi} \log(\sin\theta)\ \mathrm{d}\theta, \tag{35.83}$$

so the two logarithmic sine integrals cancel. Thus each prime step expands the spectral angle by precisely its prime factor while preserving ν.

35.8 Numerical verification

$$\text{arbitrary-prime Schur residual} = 4.594889 \times 10^{-16}, \tag{35.84}$$
$$\text{normalized Chebyshev-map residual} = 1.016846 \times 10^{-12}, \tag{35.85}$$
$$\text{dyadic-specialization residual} = 2.177752 \times 10^{-16}, \tag{35.86}$$
$$\text{determinant-factorization residual} = 2.580429 \times 10^{-14}, \tag{35.87}$$
$$\text{Dirichlet-pole residual} = 6.661338 \times 10^{-16}, \tag{35.88}$$
$$\text{semigroup symbolic residual} = 0, \tag{35.89}$$
$$\text{second-kind composition residual} = 0, \tag{35.90}$$
$$\text{iterated/direct flow residual} = 1.111610 \times 10^{-15}, \tag{35.91}$$
$$\text{band-decimation residual} = 1.515454 \times 10^{-14}, \tag{35.92}$$
$$\text{coarse-spectrum residual} = 1.243450 \times 10^{-14}, \tag{35.93}$$
$$\text{arcsine fixed-moment residual} = 1.184238 \times 10^{-16}, \tag{35.94}$$
$$\text{Koopman mode residual} = 3.357314 \times 10^{-13}, \tag{35.95}$$
$$\text{multi-prime commutation residual} = 3.286260 \times 10^{-14}, \tag{35.96}$$
$$\text{general } SL_2 \text{ power-coordinate residual} = 1.501782 \times 10^{-13}, \tag{35.97}$$
$$\text{companion-state polyphase residual} = 7.770726 \times 10^{-12}, \tag{35.98}$$
$$\text{trace-semigroup residual} = 2.316699 \times 10^{-13}. \tag{35.99}$$

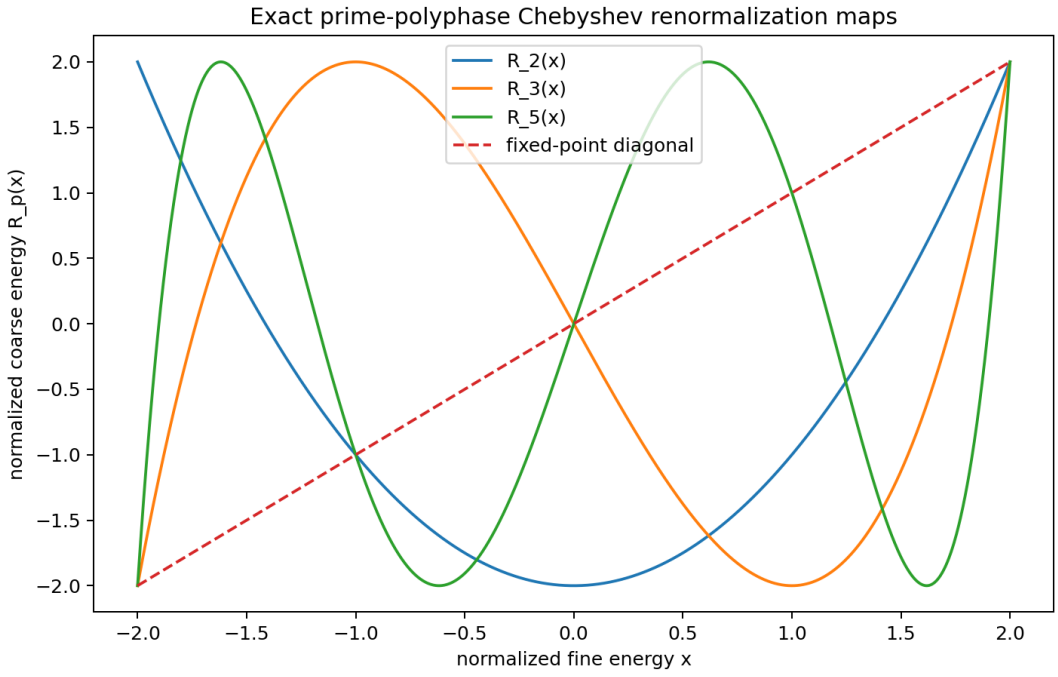


Figure 35.1: The exact normalized prime-polyphase renormalization maps $R_p(x) = 2T_p(x/2)$ on the invariant spectral interval.

35.9 Next steps

- Derive an interacting fixed point that reduces to the exact free-chain map in the quadratic limit.
- Construct a gauge- and Lorentz-covariant field-theory renormalization flow.

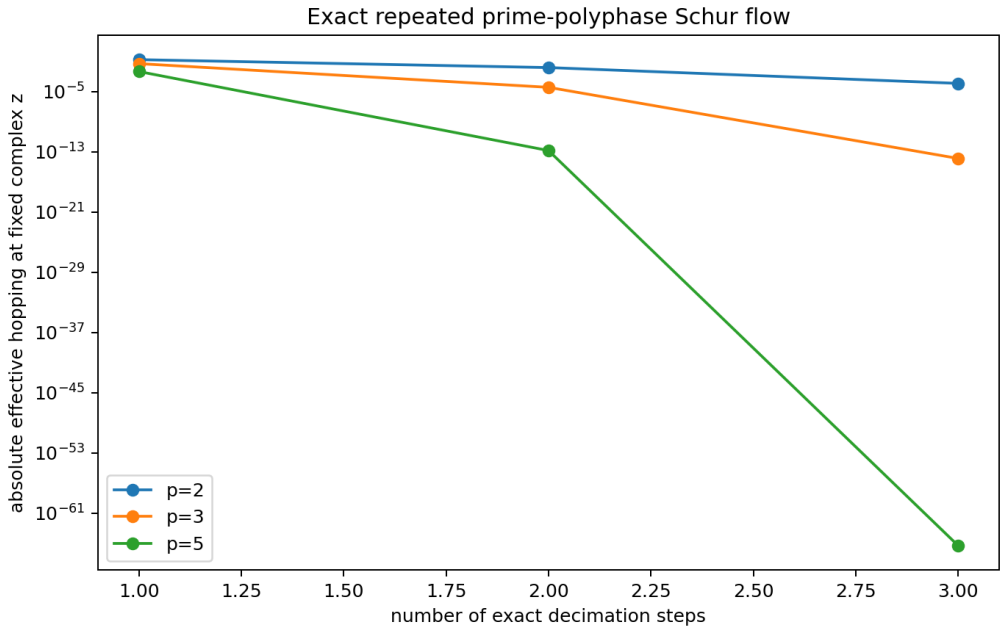


Figure 35.2: Repeated Schur elimination agrees exactly with direct p^h-decimation; the plotted complex-energy example shows the resulting effective hopping magnitude.

- Establish the domain of universality beyond the nearest-neighbor translation-invariant sector, including any counterexamples.
- Determine the physical meaning of the invariant arcsine measure in the interacting theory.
- Derive an observable controlled by the prime-polyphase renormalization law.

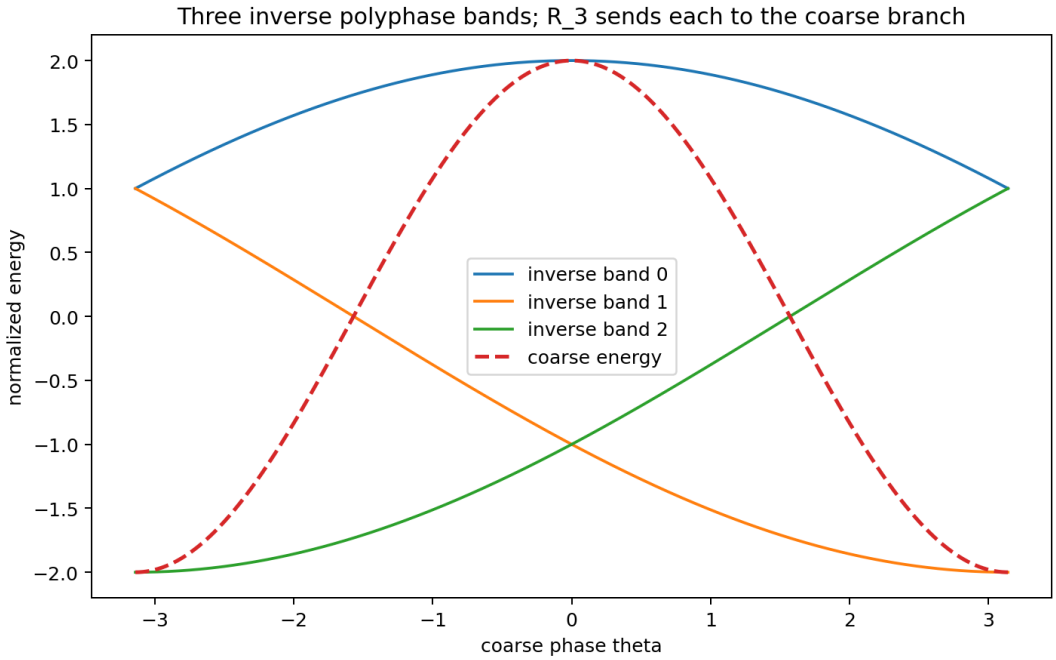


Figure 35.3: The three inverse folded branches for $p = 3$. Applying R_3 to any branch returns the same coarse normalized energy.

Chapter 36

Hypergeometric Companion Lanes and Quantum Packet Functions

Residue-class filtering of an analytic series is the analytic counterpart of an ordinary polyphase split. When the coefficients contain factorials, each residue lane has an exact generalized hypergeometric form. The same lanes also arise as coordinates of a finite cyclic companion system, and they extend without ambiguity to bounded operator functions.

36.1 Analytic residue projectors

Fix an integer $N \geq 2$ and write

$$\omega_N := \exp(2\pi \mathrm{i}/N). \tag{36.1}$$

For an entire function F, define its j-th residue projector by

$$\boxed{(\mathfrak{R}_{N,j}F)(z) := \frac{1}{N}\sum_{\ell=0}^{N-1} \omega_N^{-j\ell} F(\omega_N^{\ell} z), \qquad 0 \leq j < N.} \tag{36.2}$$

Suppose

$$F(z) = \sum_{n=0}^{\infty} a_n z^n. \tag{36.3}$$

Substituting the series into (36.2) gives

$$(\mathfrak{R}_{N,j}F)(z) = \frac{1}{N}\sum_{\ell=0}^{N-1} \omega_N^{-j\ell} \sum_{n=0}^{\infty} a_n \omega_N^{\ell n} z^n \tag{36.4}$$

$$= \sum_{n=0}^{\infty} a_n z^n \left(\frac{1}{N}\sum_{\ell=0}^{N-1} \omega_N^{\ell(n-j)} \right). \tag{36.5}$$

The finite character sum is

$$\frac{1}{N}\sum_{\ell=0}^{N-1} \omega_N^{\ell(n-j)} = \begin{cases} 1, & n \equiv j \pmod N, \\ 0, & n \not\equiv j \pmod N. \end{cases} \tag{36.6}$$

Therefore

$$\boxed{(\mathfrak{R}_{N,j}F)(z) = \sum_{q=0}^{\infty} a_{Nq+j} z^{Nq+j}.} \tag{36.7}$$

The operators are mutually orthogonal idempotents on power series:

$$\mathfrak{R}_{N,j}\mathfrak{R}_{N,k} = \delta_{jk}\mathfrak{R}_{N,j}, \qquad \sum_{j=0}^{N-1} \mathfrak{R}_{N,j} = I. \tag{36.8}$$

Thus the parent function is reconstructed exactly from its analytic polyphase lanes,

$$\boxed{F(z) = \sum_{j=0}^{N-1} (\mathfrak{R}_{N,j} F)(z).} \tag{36.9}$$

36.2 Factorial lanes and hypergeometric compression

Apply the residue projector to the exponential. Define

$$\boxed{E_{N,j}(z) := \mathfrak{R}_{N,j}(\mathrm{e}^z) = \sum_{q=0}^{\infty} \frac{z^{Nq+j}}{(Nq+j)!}.} \tag{36.10}$$

The factorial denominator can be separated into N step-N lanes. Indeed,

$$\frac{(Nq+j)!}{j!} = \prod_{m=1}^{Nq} (j+m) \tag{36.11}$$

$$= \prod_{r=1}^{N} \prod_{k=0}^{q-1} (j+r+Nk) \tag{36.12}$$

$$= N^{Nq} \prod_{r=1}^{N} \left(\frac{j+r}{N}\right)_q. \tag{36.13}$$

Exactly one of the parameters $(j+r)/N$ is equal to 1, namely the one with $r = N-j$. Since

$$(1)_q = q!, \tag{36.14}$$

define the remaining parameter multiset

$$\boxed{\mathcal{A}_{N,j} := \left\{ \frac{j+r}{N} : 1 \le r \le N \right\} \setminus \{1\}.} \tag{36.15}$$

The factorial split becomes

$$\boxed{(Nq+j)! = j!\, N^{Nq} q! \prod_{\alpha \in \mathcal{A}_{N,j}} (\alpha)_q.} \tag{36.16}$$

Substitution into (36.10) gives

$$E_{N,j}(z) = \frac{z^j}{j!} \sum_{q=0}^{\infty} \frac{(z^N/N^N)^q}{q! \prod_{\alpha \in \mathcal{A}_{N,j}} (\alpha)_q}. \tag{36.17}$$

Hence

$$\boxed{E_{N,j}(z) = \frac{z^j}{j!}\, {}_0F_{N-1}\left(\begin{matrix} - \\ \mathcal{A}_{N,j} \end{matrix}; \frac{z^N}{N^N}\right).} \tag{36.18}$$

The hypergeometric expression is not a separate approximation. It is the same residue lane written in a coordinate system whose consecutive index is q rather than $Nq+j$.

For example, with $N = 3$,

$$\mathcal{A}_{3,0} = \left\{\frac{1}{3}, \frac{2}{3}\right\}, \tag{36.19}$$

$$\mathcal{A}_{3,1} = \left\{\frac{2}{3}, \frac{4}{3}\right\}, \tag{36.20}$$

$$\mathcal{A}_{3,2} = \left\{\frac{4}{3}, \frac{5}{3}\right\}. \tag{36.21}$$

Therefore

$$E_{3,1}(z) = z\, {}_0F_2\left(\begin{matrix} - \\ \frac{2}{3}, \frac{4}{3} \end{matrix}; \frac{z^3}{27}\right). \tag{36.22}$$

36.3 Cyclic companion-state realization

The factorial lanes form a closed finite first-order system. Differentiate (36.10). For $1 \le j < N$,

$$\frac{\mathrm{d}}{\mathrm{d}z}E_{N,j}(z) = \sum_{q=0}^{\infty} \frac{z^{Nq+j-1}}{(Nq+j-1)!} \tag{36.23}$$

$$= E_{N,j-1}(z). \tag{36.24}$$

For $j = 0$, the $q = 0$ term differentiates to zero, and reindexing $q = r + 1$ gives

$$\frac{\mathrm{d}}{\mathrm{d}z}E_{N,0}(z) = \sum_{q=1}^{\infty} \frac{z^{Nq-1}}{(Nq-1)!} \tag{36.25}$$

$$= \sum_{r=0}^{\infty} \frac{z^{Nr+N-1}}{(Nr+N-1)!} \tag{36.26}$$

$$= E_{N,N-1}(z). \tag{36.27}$$

Let

$$\mathbf{E}_N(z) := \begin{pmatrix} E_{N,0}(z) & E_{N,1}(z) & \cdots & E_{N,N-1}(z) \end{pmatrix}^{\mathsf{T}} \tag{36.28}$$

and let C_N be the cyclic backward shift,

$$(C_N v)_0 = v_{N-1}, \qquad (C_N v)_j = v_{j-1} \quad (1 \le j < N). \tag{36.29}$$

Then

$$\boxed{\mathbf{E}_N'(z) = C_N \mathbf{E}_N(z), \qquad \mathbf{E}_N(0) = e_0, \qquad C_N^N = I.} \tag{36.30}$$

Consequently,

$$\boxed{\mathbf{E}_N(z) = \mathrm{e}^{zC_N} e_0.} \tag{36.31}$$

The discrete Fourier transform diagonalizes C_N. Taking its inverse transform recovers the root-of-unity formula

$$\boxed{E_{N,j}(z) = \frac{1}{N} \sum_{\ell=0}^{N-1} \omega_N^{-j\ell} \mathrm{e}^{\omega_N^{\ell} z}.} \tag{36.32}$$

Thus the same object has three exact forms: a sparse factorial series, a generalized hypergeometric function, and one coordinate of a finite companion-state exponential.

The corresponding state-space polyphase identity does not depend on the exponential. If

$$v_{n+1} = M v_n, \tag{36.33}$$

then, for every $q \ge 0$ and $0 \le j < N$,

$$\boxed{v_{Nq+j} = M^j (M^N)^q v_0.} \tag{36.34}$$

Each residue subsequence therefore advances under the common coarse transfer matrix M^N, with a lane-dependent initial factor M^j.

36.4 Bounded-operator residue calculus

Let $A \in \mathcal{B}(\mathcal{H})$. Absolute operator-norm convergence permits the definition

$$E_{N,j}(tA) := \sum_{q=0}^{\infty} \frac{(tA)^{Nq+j}}{(Nq+j)!}. \tag{36.35}$$

The same character calculation used for scalar series gives

$$\boxed{E_{N,j}(tA) = \frac{1}{N} \sum_{\ell=0}^{N-1} \omega_N^{-j\ell} \mathrm{e}^{\omega_N^{\ell} tA}.} \tag{36.36}$$

Summing the lanes reconstructs the full propagator,

$$\boxed{\mathrm{e}^{tA} = \sum_{j=0}^{N-1} E_{N,j}(tA).} \tag{36.37}$$

The factorial split also holds term by term in operator norm:

$$\boxed{E_{N,j}(tA) = \frac{(tA)^j}{j!}{}_0F_{N-1}\left(\begin{matrix}-\\ \mathcal{A}_{N,j}\end{matrix}; \frac{(tA)^N}{N^N}\right).} \tag{36.38}$$

Here the generalized hypergeometric function is defined by its entire operator series. No commutativity issue arises because every term is a power of the same bounded operator A.

If $A^N = \alpha I$, then (36.38) reduces the propagator to the finite operator basis

$$I, A, \ldots, A^{N-1}, \tag{36.39}$$

with scalar hypergeometric coefficient functions. More generally, a low-degree polynomial identity for A^N gives a finite companion reduction of the same lane system.

For an unbounded self-adjoint Hamiltonian, these formulas must be applied through its spectral calculus or to bounded surrogates. The formal substitution of an arbitrary unbounded operator into an everywhere-defined norm series is not valid. The domain-safe version is developed in the countable-operator chapter.

36.5 Prime refinement and the standard unit window

Let p be an odd prime, let $N = mp$, and assume

$$0 \le j < p. \tag{36.40}$$

The normalized factorial tail is

$$g_q := \frac{j!(Nq)!}{(Nq+j)!} = \binom{Nq+j}{j}^{-1}. \tag{36.41}$$

Because $Nq \equiv 0 \pmod{p}$,

$$\binom{Nq+j}{j} = \prod_{r=1}^{j} \frac{Nq+r}{r}. \tag{36.42}$$

For $1 \le r \le j < p$, both r and $Nq+r$ are p-adic units. Every factor in the product is therefore a unit, so

$$\boxed{g_q \in \mathbb{Z}_p^{\times} \qquad (q \ge 0).} \tag{36.43}$$

This is the standard residue window in which the short factorial tail introduces no p-adic denominator.

Analytic refinement is exact independently of this unit argument. Partition the parent summation index as $q = pr + a$, with $0 \le a < p$. Then

$$E_{N,j}(z) = \sum_{a=0}^{p-1} \sum_{r=0}^{\infty} \frac{z^{N(pr+a)+j}}{(N(pr+a)+j)!} \tag{36.44}$$

$$= \sum_{a=0}^{p-1} \sum_{r=0}^{\infty} \frac{z^{pNr+(j+aN)}}{(pNr+j+aN)!}. \tag{36.45}$$

Hence

$$\boxed{E_{N,j}(z) = \sum_{a=0}^{p-1} E_{pN,j+aN}(z).} \tag{36.46}$$

The child lanes give a finer analytic atlas without changing the parent block. Individual child labels need not all lie in the same simple standard unit window; the guaranteed unit statement belongs to the parent under the stated hypothesis.

36.6 Coherent-state energy packets

Let $\widehat{N}\,|n\rangle = n\,|n\rangle$ be the number operator of the harmonic oscillator. Its residue-class spectral projectors are

$$\boxed{Q_j^{(N)} := \frac{1}{N}\sum_{\ell=0}^{N-1} \omega_N^{-j\ell} \mathrm{e}^{2\pi \mathrm{i}\ell\widehat{N}/N} = \sum_{n\equiv j \pmod N} |n\rangle\langle n|\,.} \tag{36.47}$$

For a coherent state

$$|\alpha\rangle = \mathrm{e}^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\,|n\rangle\,, \qquad \mu := |\alpha|^2, \tag{36.48}$$

the number probabilities are Poisson,

$$|\langle n \mid \alpha\rangle|^2 = \mathrm{e}^{-\mu}\frac{\mu^n}{n!}. \tag{36.49}$$

The probability in residue packet j is therefore

$$p_{N,j}(\mu) := \langle\alpha|Q_j^{(N)}|\alpha\rangle \tag{36.50}$$

$$= \mathrm{e}^{-\mu}\sum_{q=0}^{\infty}\frac{\mu^{Nq+j}}{(Nq+j)!} \tag{36.51}$$

$$= \mathrm{e}^{-\mu}E_{N,j}(\mu). \tag{36.52}$$

Using the two exact forms above,

$$\boxed{p_{N,j}(\mu) = \frac{\mathrm{e}^{-\mu}\mu^j}{j!}\,{}_0F_{N-1}\left(\begin{matrix}-\\ \mathcal{A}_{N,j}\end{matrix}; \frac{\mu^N}{N^N}\right)} \tag{36.53}$$

and

$$\boxed{p_{N,j}(\mu) = \frac{1}{N}\sum_{\ell=0}^{N-1}\omega_N^{-j\ell}\exp\bigl(\mu(\omega_N^{\ell}-1)\bigr).} \tag{36.54}$$

The defining series proves directly that $p_{N,j}(\mu)\ge 0$, and reconstruction gives

$$\sum_{j=0}^{N-1} p_{N,j}(\mu) = 1. \tag{36.55}$$

Removing the $\ell = 0$ term from (36.54) gives a quantitative approach to equal packet weight:

$$\left|p_{N,j}(\mu) - \frac{1}{N}\right| \le \frac{1}{N}\sum_{\ell=1}^{N-1}\exp\bigl[-\mu\bigl(1-\cos(2\pi\ell/N)\bigr)\bigr] \tag{36.56}$$

$$\le \boxed{\frac{N-1}{N}\exp\bigl[-\mu\bigl(1-\cos(2\pi/N)\bigr)\bigr]\,.} \tag{36.57}$$

Thus a high-occupation coherent state becomes exponentially close to uniform across any fixed finite set of number-residue packets.

The ladder operators move between adjacent packets. Since

$$\mathrm{e}^{2\pi\mathrm{i}\widehat{N}/N}a\mathrm{e}^{-2\pi\mathrm{i}\widehat{N}/N} = \omega_N^{-1}a, \tag{36.58}$$

one obtains

$$\boxed{Q_r^{(N)}aQ_s^{(N)} = 0 \quad \text{unless} \quad r\equiv s-1 \pmod N,} \tag{36.59}$$

and similarly

$$\boxed{Q_r^{(N)}a^\dagger Q_s^{(N)} = 0 \quad \text{unless} \quad r\equiv s+1 \pmod N.} \tag{36.60}$$

The number-residue decomposition is therefore not only a static probability split; it has an exact nearest-channel routing law under creation and annihilation.

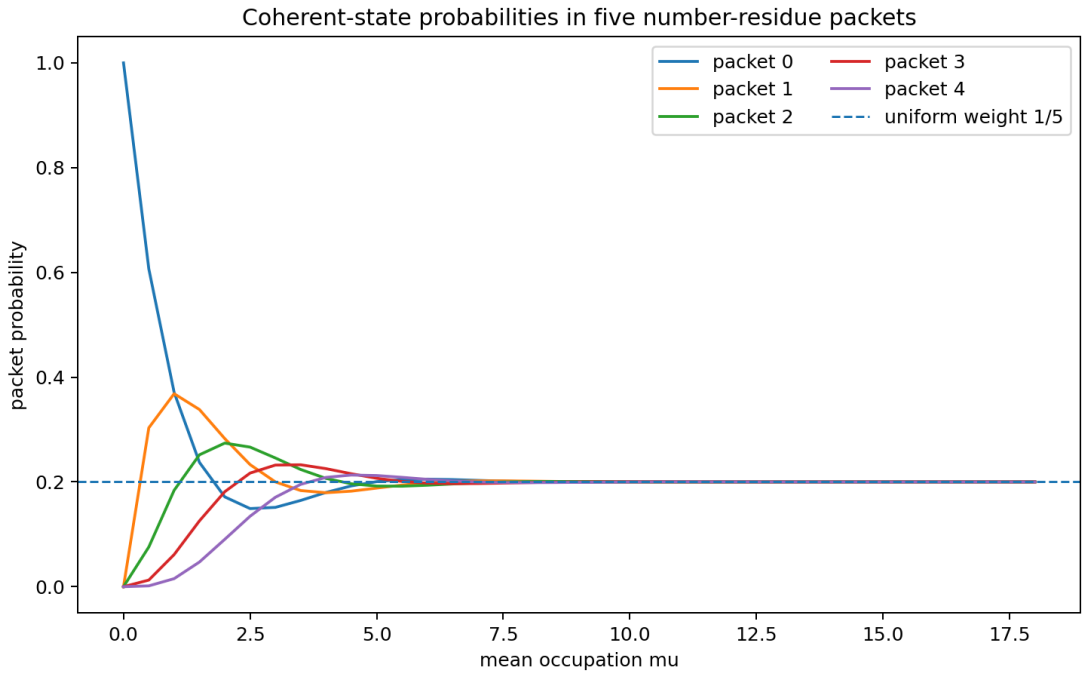


Figure 36.1: Exact coherent-state probabilities in the five number-residue packets. The hypergeometric, sparse-series, and root-filter formulas agree, and each packet approaches weight 1/5 as the mean occupation increases.

36.7 Verification and scope

The accompanying calculation checks the residue projector, factorial split, hypergeometric form, cyclic companion system, bounded-operator root filter, prime refinement, standard-window unit property, coherent-state packet probabilities, ladder routing, and the Bessel-lane formulas used later in the Gowdy chapter. The identities are evaluated for several values of N, including both dyadic and odd-prime cases. The largest floating-point residual among the seventeen checks is

$$3.908382 \times 10^{-16}. \tag{36.61}$$

The scalar hypergeometric, root-filter, refinement, and Bessel identities were also evaluated at eighty decimal digits; their largest residual is below 1.1×10^{-80}.

The chapter supplies exact analytic and operator coordinates for already-defined quantum evolutions.

Next steps.

- Derive a microscopic Hamiltonian or action whose physical propagator naturally selects the hypergeometric lane decomposition.
- Extend the lane calculus through nonlinear BFV–BV transfer and verify that the master equation remains exact after channel elimination.
- Establish operator-domain and convergence estimates for the lane formulas in the interacting unbounded setting.
- Identify an observable that distinguishes residue channels or prove that the apparent digit order is physically redundant.

Part V

Cosmology, Perturbations, and the Dark Sector

Chapter 37

Cosmology, Backreaction, and Constraints

This chapter connects packet dynamics to homogeneous expansion. It separates kinematic redshift, packet stress-energy, variational FLRW equations, and constraint-preserving discrete updates.

37.1 FLRW kinematics

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + a(t)^2\,\mathrm{d}\chi^2, \tag{37.1}$$

$$\mathrm{d}t = a(\eta)\,\mathrm{d}\eta, \tag{37.2}$$

$$a(t) = \mathrm{e}^{Ht}. \tag{37.3}$$

With $a(0) = 1$,

$$\eta(t) = \int_0^t \frac{\mathrm{d}s}{a(s)} = \int_0^t \mathrm{e}^{-Hs}\,\mathrm{d}s \tag{37.4}$$

$$= \frac{1 - \mathrm{e}^{-Ht}}{H}. \tag{37.5}$$

Hence $\mathrm{e}^{-Ht} = 1 - H\eta$, which gives

$$t(\eta) = -\frac{1}{H}\log(1 - H\eta), \tag{37.6}$$

$$a(\eta) = \frac{1}{1 - H\eta}. \tag{37.7}$$

The remaining kinematic relations are

$$p_{\mathrm{phys}}(t) = \frac{k}{a(t)}, \tag{37.8}$$

$$\omega_{\mathrm{phys}}(t) = \frac{k}{a(t)}, \tag{37.9}$$

$$1 + z = \frac{a(t_{\mathrm{obs}})}{a(t_{\mathrm{emit}})}, \tag{37.10}$$

$$D(t) = a(t)\chi, \tag{37.11}$$

$$\dot{D} = HD. \tag{37.12}$$

Massive conformal-time packet:

$$\omega_k(\eta) = \sqrt{k^2 + m^2 a(\eta)^2}, \tag{37.13}$$

$$\frac{\mathrm{d}\chi}{\mathrm{d}\eta} = \frac{k}{\sqrt{k^2 + m^2 a(\eta)^2}}. \tag{37.14}$$

37.2 Packet stress-energy

$$E_\alpha(a) := \sqrt{m_\alpha^2 + \frac{k_\alpha^2}{a^2}}, \tag{37.15}$$

$$H_{\mathcal{P}}(a) := \sum_\alpha w_\alpha E_\alpha(a), \tag{37.16}$$

$$\rho_\alpha(a) := \frac{w_\alpha E_\alpha(a)}{a^3}, \tag{37.17}$$

$$p_\alpha(a) := \frac{w_\alpha k_\alpha^2}{3a^5 E_\alpha(a)}, \tag{37.18}$$

$$\rho(a) := \sum_\alpha \rho_\alpha(a) = \frac{H_{\mathcal{P}}(a)}{a^3}, \tag{37.19}$$

$$p(a) := \sum_\alpha p_\alpha(a) = -\frac{H'_{\mathcal{P}}(a)}{3a^2}. \tag{37.20}$$

The energy derivative is

$$E'_\alpha(a) = -\frac{k_\alpha^2}{a^3 E_\alpha(a)}. \tag{37.21}$$

Therefore

$$-\frac{H'_{\mathcal{P}}(a)}{3a^2} = \sum_\alpha \frac{w_\alpha k_\alpha^2}{3a^5 E_\alpha(a)} = \sum_\alpha p_\alpha(a). \tag{37.22}$$

For one packet,

$$\frac{\mathrm{d}\rho_\alpha}{\mathrm{d}a} = w_\alpha \left(\frac{E'_\alpha}{a^3} - \frac{3E_\alpha}{a^4} \right) \tag{37.23}$$

$$= -\frac{3}{a}\rho_\alpha - \frac{w_\alpha k_\alpha^2}{a^6 E_\alpha} \tag{37.24}$$

$$= -\frac{3}{a}(\rho_\alpha + p_\alpha). \tag{37.25}$$

Packetwise and global continuity:

$$\frac{\mathrm{d}\rho_\alpha}{\mathrm{d}a} + \frac{3}{a}(\rho_\alpha + p_\alpha) = 0, \tag{37.26}$$

$$\dot\rho + 3H(\rho + p) = 0, \tag{37.27}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(a^3\rho) + p\frac{\mathrm{d}}{\mathrm{d}t}(a^3) = 0. \tag{37.28}$$

Normalized special cases ($8\pi G/3 = 1$):

$$\rho_r = a^{-4}, \qquad \dot a = a^{-1}, \qquad a_r(t) = \sqrt{1+2t}, \tag{37.29}$$

$$\rho_m = a^{-3}, \qquad \dot a = a^{-1/2}, \qquad a_m(t) = \left(1 + \frac{3}{2}t\right)^{2/3}. \tag{37.30}$$

37.3 Lapse-resolved variational closure

$$\mathrm{d}s^2 = -N(t)^2\,\mathrm{d}t^2 + a(t)^2\,\mathrm{d}\boldsymbol{x}^2, \tag{37.31}$$

$$H := \frac{\dot a}{Na}, \tag{37.32}$$

$$C_G := \frac{3}{8\pi G}, \tag{37.33}$$

$$S[a,N] := \int \left[-C_G \frac{a\dot{a}^2}{N} - N H_{\mathcal{P}}(a) \right] \mathrm{d}t. \tag{37.34}$$

The lapse occurs without derivatives. Its variation is

$$0 = \frac{\partial L}{\partial N} = C_G \frac{a\dot{a}^2}{N^2} - H_{\mathcal{P}}(a). \tag{37.35}$$

Dividing by $C_G a^3$ and using $H = \dot{a}/(Na)$, $C_G = 3/(8\pi G)$, and $\rho = H_{\mathcal{P}}/a^3$ gives

$$\frac{\delta S}{\delta N} = 0 \quad \Longrightarrow \quad H^2 = \frac{8\pi G}{3}\rho. \tag{37.36}$$

The scale-factor variation gives

$$\frac{\delta S}{\delta a} = 0 \Longrightarrow 2\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a}\right)^2 - 2\frac{\dot{a}}{a}\frac{\dot{N}}{N} = -8\pi G N^2 p, \tag{37.37}$$

$$N = 1 \Longrightarrow \frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p). \tag{37.38}$$

37.4 Midpoint discrete action

$$\bar{a}_n := \frac{a_n + a_{n+1}}{2}, \tag{37.39}$$

$$\Delta a_n := a_{n+1} - a_n, \tag{37.40}$$

$$L_d(a_n, a_{n+1}; N_n) := -C_G \frac{\bar{a}_n (\Delta a_n)^2}{\Delta t\, N_n} - \Delta t\, N_n H_{\mathcal{P}}(\bar{a}_n), \tag{37.41}$$

$$S_d := \sum_n L_d(a_n, a_{n+1}; N_n). \tag{37.42}$$

Since $\partial \bar{a}_n / \partial N_n = 0$,

$$\frac{\partial L_d}{\partial N_n} = C_G \frac{\bar{a}_n (\Delta a_n)^2}{\Delta t\, N_n^2} - \Delta t\, H_{\mathcal{P}}(\bar{a}_n). \tag{37.43}$$

Setting this derivative to zero and dividing by Δt gives the discrete lapse constraint below. Lapse constraint:

$$C_G \bar{a}_n \left(\frac{\Delta a_n}{\Delta t\, N_n}\right)^2 = H_{\mathcal{P}}(\bar{a}_n). \tag{37.44}$$

Endpoint derivatives:

$$D_1 L_d = \frac{C_G}{\Delta t N_n}\left(2\bar{a}_n \Delta a_n - \frac{(\Delta a_n)^2}{2}\right) - \frac{\Delta t N_n}{2} H'_{\mathcal{P}}(\bar{a}_n), \tag{37.45}$$

$$D_2 L_d = -\frac{C_G}{\Delta t N_n}\left(2\bar{a}_n \Delta a_n + \frac{(\Delta a_n)^2}{2}\right) - \frac{\Delta t N_n}{2} H'_{\mathcal{P}}(\bar{a}_n), \tag{37.46}$$

$$D_2 L_d(a_{n-1}, a_n; N_{n-1}) + D_1 L_d(a_n, a_{n+1}; N_n) = 0. \tag{37.47}$$

37.5 Homogeneous scalar-gravity closure

In units $8\pi G = 1$,

$$S[a, N, \phi] = \int \left[-\frac{3a\dot{a}^2}{N} + a^3 \left(\frac{\dot{\phi}^2}{2N} - N V(\phi) \right) \right] \mathrm{d}t. \tag{37.48}$$

For $N = 1$,

$$H := \frac{\dot{a}}{a}, \qquad \pi := \dot{\phi}, \tag{37.49}$$

$$3H^2 = \frac{1}{2}\pi^2 + V(\phi), \tag{37.50}$$

$$\dot{\pi} + 3H\pi + V'(\phi) = 0, \tag{37.51}$$

$$\dot{H} = -\frac{1}{2}\pi^2, \tag{37.52}$$

$$\dot{\phi} = \pi, \tag{37.53}$$

$$\rho_\phi = \frac{1}{2}\pi^2 + V(\phi), \tag{37.54}$$

$$p_\phi = \frac{1}{2}\pi^2 - V(\phi), \tag{37.55}$$

$$\dot{\rho}_\phi + 3H(\rho_\phi + p_\phi) = 0. \tag{37.56}$$

Constraint:

$$\mathcal{C} := 3H^2 - \frac{1}{2}\pi^2 - V(\phi), \tag{37.57}$$

$$\dot{\mathcal{C}} = 0. \tag{37.58}$$

Indeed,

$$\dot{\mathcal{C}} = 6H\dot{H} - \pi\dot{\pi} - V'(\phi)\dot{\phi} \tag{37.59}$$

$$= -3H\pi^2 - \pi(-3H\pi - V') - V'\pi \tag{37.60}$$

$$= 0. \tag{37.61}$$

Thus the scalar equation and the Raychaudhuri equation propagate the Friedmann constraint.

37.6 Exact discrete-gradient constraint

$$\bar{\phi} := \frac{\phi_{n+1} + \phi_n}{2}, \qquad \bar{\pi} := \frac{\pi_{n+1} + \pi_n}{2}, \qquad \bar{H} := \frac{H_{n+1} + H_n}{2}, \tag{37.62}$$

$$\overline{\nabla} V(\phi_n, \phi_{n+1}) := \begin{cases} \dfrac{V(\phi_{n+1}) - V(\phi_n)}{\phi_{n+1} - \phi_n}, & \phi_{n+1} \neq \phi_n, \\ V'(\phi_n), & \phi_{n+1} = \phi_n, \end{cases} \tag{37.63}$$

$$V(\phi_{n+1}) - V(\phi_n) = \overline{\nabla} V(\phi_n, \phi_{n+1})(\phi_{n+1} - \phi_n). \tag{37.64}$$

Update:

$$\phi_{n+1} - \phi_n = \Delta t\, \bar{\pi}, \tag{37.65}$$

$$\pi_{n+1} - \pi_n = -\Delta t(3\bar{H}\bar{\pi} + \overline{\nabla} V), \tag{37.66}$$

$$H_{n+1} - H_n = -\frac{\Delta t}{2}\bar{\pi}^2, \tag{37.67}$$

$$\log a_{n+1} - \log a_n = \Delta t\, \bar{H}. \tag{37.68}$$

Exact identities:

$$\mathcal{C}_{n+1} = \mathcal{C}_n, \tag{37.69}$$

$$\frac{\rho_{n+1} - \rho_n}{\Delta t} + 3\bar{H}\bar{\pi}^2 = 0. \tag{37.70}$$

For the constraint difference,

$$\mathcal{C}_{n+1} - \mathcal{C}_n = 6\bar{H}(H_{n+1} - H_n) - \bar{\pi}(\pi_{n+1} - \pi_n) \tag{37.71}$$

$$- \overline{\nabla} V(\phi_{n+1} - \phi_n). \tag{37.72}$$

Substituting the four midpoint updates gives

$$\mathcal{C}_{n+1} - \mathcal{C}_n = \; - 3\Delta t\, \bar{H}\bar{\pi}^2 + \Delta t\, \bar{\pi}(3\bar{H}\bar{\pi} + \overline{\nabla} V) \tag{37.73}$$

$$- \Delta t \, \overline{\nabla} V \, \bar{\pi} = 0. \tag{37.74}$$

Similarly,

$$\rho_{n+1} - \rho_n = \bar{\pi}(\pi_{n+1} - \pi_n) + \overline{\nabla} V(\phi_{n+1} - \phi_n) \tag{37.75}$$

$$= -3\Delta t \, \bar{H} \bar{\pi}^2. \tag{37.76}$$

For $V(\phi) = V_0 + \frac{1}{2}\mu^2\phi^2$, $\bar{p} := \bar{\pi}$:

$$-\frac{3}{4}\Delta t^2 \bar{p}^3 + \left(2 + 3\Delta t H_n + \frac{1}{2}\mu^2 \Delta t^2\right) \bar{p} + \mu^2 \Delta t \, \phi_n - 2\pi_n = 0. \tag{37.77}$$

Chapter 38

Curved Multicomponent FLRW Background and Distances

The homogeneous model is extended to spatial curvature, a cosmological constant, multiple fluids, and standard distance observables. The equations define a controlled background solver rather than a microscopic derivation of every component.

38.1 Curved lapse-resolved minisuperspace action

Let

$$\mathrm{d}s^2 = -N(t)^2\,\mathrm{d}t^2 + a(t)^2\gamma^{(K)}_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j, \qquad {}^{(3)}R[\gamma^{(K)}] = 6K. \tag{38.1}$$

For unit comoving volume,

$$S_{\mathrm{FLRW}} = \int \left[-\frac{3M_{\mathrm{Pl}}^2 a\dot{a}^2}{N} + 3M_{\mathrm{Pl}}^2 KaN - M_{\mathrm{Pl}}^2\Lambda a^3 N - NH_{\mathrm{P}}(a)\right]\mathrm{d}t. \tag{38.2}$$

Varying N gives

$$0 = \frac{3M_{\mathrm{Pl}}^2 a\dot{a}^2}{N^2} + 3M_{\mathrm{Pl}}^2 Ka - M_{\mathrm{Pl}}^2\Lambda a^3 - H_{\mathrm{P}}(a). \tag{38.3}$$

After division by $3M_{\mathrm{Pl}}^2 a^3$ and use of $H = \dot{a}/(Na)$, this becomes the Friedmann equation. With

$$\rho(a) := \frac{H_{\mathrm{P}}(a)}{a^3}, \tag{38.4}$$

the lapse equation gives

$$\boxed{H^2 = \frac{\rho}{3M_{\mathrm{Pl}}^2} + \frac{\Lambda}{3} - \frac{K}{a^2}.} \tag{38.5}$$

For total pressure p,

$$\dot{H} = -\frac{\rho + p}{2M_{\mathrm{Pl}}^2} + \frac{K}{a^2}, \tag{38.6}$$

$$\frac{\ddot{a}}{a} = -\frac{\rho + 3p}{6M_{\mathrm{Pl}}^2} + \frac{\Lambda}{3}, \tag{38.7}$$

$$\dot{\rho} + 3H(\rho + p) = 0. \tag{38.8}$$

In the gauge $N = 1$, differentiate the Friedmann equation:

$$2H\dot{H} = \frac{\dot{\rho}}{3M_{\mathrm{Pl}}^2} + \frac{2KH}{a^2}. \tag{38.9}$$

Using $\dot\rho = -3H(\rho + p)$ and dividing by $2H$ gives the equation for $\dot H$. Then

$$\frac{\ddot a}{a} = \dot H + H^2 \tag{38.10}$$

$$= -\frac{\rho + p}{2M_{\rm Pl}^2} + \frac{K}{a^2} + \frac{\rho}{3M_{\rm Pl}^2} + \frac{\Lambda}{3} - \frac{K}{a^2}, \tag{38.11}$$

which simplifies to the acceleration equation. The symbolic identity

$$\frac{a}{2}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{8\pi G}{3}\rho + \frac{\Lambda}{3} - \frac{K}{a^2}\right] = -4\pi G(\rho + p) + \frac{K}{a^2} \tag{38.12}$$

is exact when

$$a\rho_{,a} = -3(\rho + p). \tag{38.13}$$

38.2 Constant-equation-of-state component library

For noninteracting components

$$p_i = w_i\rho_i, \tag{38.14}$$

$$\dot\rho_i + 3H(1 + w_i)\rho_i = 0, \tag{38.15}$$

$$\rho_i(a) = \rho_{i0}a^{-3(1+w_i)}. \tag{38.16}$$

Since $\dot\rho_i = aH\ \mathrm{d}\rho_i/\,\mathrm{d}a$,

$$\frac{\mathrm{d}\rho_i}{\rho_i} = -3(1 + w_i)\frac{\mathrm{d}a}{a}. \tag{38.17}$$

Integration from $a = 1$ to a gives the stated power law. With $a_0 = 1$,

$$\Omega_{i0} := \frac{\rho_{i0}}{3M_{\rm Pl}^2H_0^2}, \tag{38.18}$$

$$\Omega_{\Lambda0} := \frac{\Lambda}{3H_0^2}, \tag{38.19}$$

$$\Omega_{K0} := -\frac{K}{H_0^2}, \tag{38.20}$$

$$1 = \sum_i \Omega_{i0} + \Omega_{K0} + \Omega_{\Lambda0}, \tag{38.21}$$

$$E(a)^2 := \frac{H(a)^2}{H_0^2} = \sum_i \Omega_{i0}a^{-3(1+w_i)} + \Omega_{K0}a^{-2} + \Omega_{\Lambda0}. \tag{38.22}$$

Hence

$$E(z)^2 = \sum_i \Omega_{i0}(1 + z)^{3(1+w_i)} + \Omega_{K0}(1 + z)^2 + \Omega_{\Lambda0}. \tag{38.23}$$

Special cases:

$$\rho_r \propto a^{-4}, \tag{38.24}$$

$$\rho_m \propto a^{-3}, \tag{38.25}$$

$$\rho_K \propto a^{-2}, \tag{38.26}$$

$$\rho_\Lambda = \text{constant}. \tag{38.27}$$

38.3 Conservative interacting fluids

Let

$$\dot\rho_i + 3H(\rho_i + p_i) = Q_i, \qquad \sum_i Q_i = 0. \tag{38.28}$$

Then

$$\boxed{\dot{\rho} + 3H(\rho + p) = 0, \qquad \rho := \sum_i \rho_i, \quad p := \sum_i p_i.} \tag{38.29}$$

Summing the component equations gives

$$\dot{\rho} + 3H(\rho + p) = \sum_i Q_i = 0. \tag{38.30}$$

Energy transfer therefore changes the partition among components but not the total homogeneous continuity equation. For the benchmark transfer

$$Q_m = -\Gamma \rho_m, \tag{38.31}$$
$$Q_r = +\Gamma \rho_m, \tag{38.32}$$

$$\dot{\rho}_m = -3H\rho_m - \Gamma\rho_m, \tag{38.33}$$
$$\dot{\rho}_r = -4H\rho_r + \Gamma\rho_m, \tag{38.34}$$
$$\dot{\rho}_m + \dot{\rho}_r + 3H\left(\rho_m + \rho_r + \frac{1}{3}\rho_r\right) = 0. \tag{38.35}$$

38.4 Background observables

Define

$$\bar{\chi}(z) := \int_0^z \frac{\mathrm{d}z'}{E(z')}, \tag{38.36}$$
$$D_H := \frac{c}{H_0}. \tag{38.37}$$

For a radial null ray,

$$0 = -c^2\,\mathrm{d}t^2 + a(t)^2\,\mathrm{d}\chi_{\mathrm{com}}^2. \tag{38.38}$$

Using $1 + z = a^{-1}$ and $\mathrm{d}t = -\,\mathrm{d}z/[(1+z)H(z)]$ gives

$$\mathrm{d}\chi_{\mathrm{com}} = \frac{c\,\mathrm{d}z}{H(z)} = D_H \frac{\mathrm{d}z}{E(z)}. \tag{38.39}$$

Thus $D_H\bar{\chi}(z)$ is the radial comoving distance before the curvature map is applied. The transverse comoving distance is

$$D_M(z) = D_H\,\mathcal{S}_{\Omega_K}(\bar{\chi}(z))\,, \tag{38.40}$$

where

$$\mathcal{S}_{\Omega_K}(x) := \begin{cases} \Omega_K^{-1/2}\sinh(\sqrt{\Omega_K}\,x), & \Omega_K > 0, \\ x, & \Omega_K = 0, \\ |\Omega_K|^{-1/2}\sin(\sqrt{|\Omega_K|}\,x), & \Omega_K < 0. \end{cases} \tag{38.41}$$

Then

$$D_A(z) = \frac{D_M(z)}{1+z}, \tag{38.42}$$
$$D_L(z) = (1+z)D_M(z), \tag{38.43}$$
$$\boxed{D_L(z) = (1+z)^2 D_A(z)}, \tag{38.44}$$
$$H_0 t_{\mathrm{L}}(z) = \int_0^z \frac{\mathrm{d}z'}{(1+z')E(z')}, \tag{38.45}$$
$$H_0 t_0 = \int_0^1 \frac{\mathrm{d}a}{aE(a)}. \tag{38.46}$$

Exact benchmarks follow by direct integration. For matter domination, $E(a) = a^{-3/2}$ and $E(z) = (1+z)^{3/2}$, so

$$H_0 t_0 = \int_0^1 a^{1/2}\,\mathrm{d}a = \frac{2}{3}, \tag{38.47}$$

$$\bar{\chi}(z) = \int_0^z (1+z')^{-3/2}\,\mathrm{d}z' \tag{38.48}$$

$$= 2\left[1-(1+z)^{-1/2}\right]. \tag{38.49}$$

For radiation domination, $E(a) = a^{-2}$ and $E(z) = (1+z)^2$, giving

$$H_0 t_0 = \int_0^1 a\;\mathrm{d}a = \frac{1}{2}, \tag{38.50}$$

$$\bar{\chi}(z) = \int_0^z \frac{\mathrm{d}z'}{(1+z')^2} = \frac{z}{1+z}. \tag{38.51}$$

For pure curvature, $E(a) = a^{-1}$ and $E(z) = 1+z$, so

$$H_0 t_0 = \int_0^1 \mathrm{d}a = 1, \tag{38.52}$$

$$\bar{\chi}(z) = \int_0^z \frac{\mathrm{d}z'}{1+z'} = \log(1+z), \tag{38.53}$$

$$\frac{D_M}{D_H} = \sinh\bar{\chi} \qquad (\Omega_K = 1). \tag{38.54}$$

38.5 Projector background solver

With normalized time $\tau = H_0 t$,

$$\frac{\mathrm{d}a}{\mathrm{d}\tau} = aE(a). \tag{38.55}$$

The explicit midpoint update is

$$k_1 = a_n E(a_n), \tag{38.56}$$

$$k_2 = \left(a_n + \frac{\Delta\tau}{2}k_1\right) E\left(a_n + \frac{\Delta\tau}{2}k_1\right), \tag{38.57}$$

$$a_{n+1} = a_n + \Delta\tau k_2. \tag{38.58}$$

For the curved radiation–matter–Λ benchmark,

$$p_a = 2.024278. \tag{38.59}$$

For conservative matter-to-radiation transfer,

$$p_{\text{transfer}} = 2.054690. \tag{38.60}$$

The recorded residuals are

$$\varepsilon_{\text{distance benchmarks}} = 8.881784 \times 10^{-16}, \tag{38.61}$$

$$\varepsilon_{\text{age benchmarks}} = 0, \tag{38.62}$$

$$\varepsilon_{D_L-(1+z)^2 D_A} = 0, \tag{38.63}$$

$$\varepsilon_{\text{total interacting continuity}} = 4.440892 \times 10^{-16}. \tag{38.64}$$

38.6 Next steps

- Derive a microscopic action that produces the exchange terms Q_i rather than inserting them phenomenologically.
- Prove positivity, regularity, and global-existence conditions for the interacting background system.
- Extend the projector background equations to anisotropic Bianchi geometries.
- Determine whether nonsingular bouncing solutions exist and establish their dynamical stability when they do.
- Calibrate the resulting background model against cosmological distance and expansion data.

Chapter 39

Cosmological Finite-Scale Photon Propagation

The finite-spacing photon correction is integrated along an expanding line of sight. The result distinguishes physical and comoving microscopic scales and yields a conditional arrival-time formula.

39.1 Local velocity law

For the selected cubic microscopic walk,

$$\cos(\omega\ell) = \prod_{i=1}^{3} \cos(kn_i\ell), \tag{39.1}$$

$$\mathcal{A}(\boldsymbol{n}) := \sum_{i<j} n_i^2 n_j^2, \tag{39.2}$$

$$v_g(k, \boldsymbol{n}, \ell) = 1 - \frac{1}{2}\mathcal{A}(\boldsymbol{n})\ell^2 k^2 + \mathcal{O}(\ell^4). \tag{39.3}$$

The exact radial group velocity is

$$v_g = \frac{\left[\prod_i \cos(kn_i\ell)\right] \sum_i n_i \tan(kn_i\ell)}{\sin(\omega\ell)}. \tag{39.4}$$

39.2 Physical versus comoving microscopic scale

Let

$$E(z) = E_0(1+z), \tag{39.5}$$

$$\ell_{\mathrm{phys}}(z) = \ell_{*,0}(1+z)^{-\sigma}. \tag{39.6}$$

Then

$$\ell_{\mathrm{phys}}(z)E(z) = \ell_{*,0}(1+z)^{-\sigma}E_0(1+z) \tag{39.7}$$

$$= \ell_{*,0}E_0(1+z)^{1-\sigma}, \tag{39.8}$$

and therefore

$$\left[\frac{\ell_{\mathrm{phys}}(z)E(z)}{\hbar c}\right]^2 = \left[\frac{\ell_{*,0}E_0}{\hbar c}\right]^2 (1+z)^{2-2\sigma}. \tag{39.9}$$

The cases

$$\sigma = 0 \qquad \Longleftrightarrow \text{fixed proper microscopic length}, \tag{39.10}$$

$$\sigma = 1 \qquad \Longleftrightarrow \text{fixed comoving microscopic length} \tag{39.11}$$

are physically distinct.

39.3 Exact arrival condition

Let the lower-energy photon arrive at $z = 0$. Its source comoving distance is

$$\chi_s = \int_0^{z_s} \frac{v_\ell(z)}{H(z)}\,\mathrm{d}z. \tag{39.12}$$

If the higher-energy photon arrives later at $z = z_{a,h} < 0$, then

$$\chi_s = \int_{z_{a,h}}^{z_s} \frac{v_h(z)}{H(z)}\,\mathrm{d}z. \tag{39.13}$$

Splitting the second integral at the present epoch and equating the two expressions for χ_s gives

$$\int_0^{z_s} \frac{v_\ell}{H}\,\mathrm{d}z = \int_{z_{a,h}}^{0} \frac{v_h}{H}\,\mathrm{d}z + \int_0^{z_s} \frac{v_h}{H}\,\mathrm{d}z, \tag{39.14}$$

$$\int_{z_{a,h}}^{0} \frac{v_h}{H}\,\mathrm{d}z = \int_0^{z_s} \frac{v_\ell - v_h}{H}\,\mathrm{d}z. \tag{39.15}$$

The exact background arrival-time difference is

$$\Delta t_{h\ell}^{\mathrm{direct}} = \int_{z_{a,h}}^{0} \frac{\mathrm{d}z}{(1+z)H(z)}. \tag{39.16}$$

39.4 Leading cosmological delay law

Define

$$I_\sigma(z_s) := \int_0^{z_s} \frac{(1+z)^{2-2\sigma}}{H(z)}\,\mathrm{d}z. \tag{39.17}$$

Writing $k = E/(\hbar c)$, the local velocity deficit is

$$1 - v_g = \frac{1}{2}\mathcal{A}(\boldsymbol{n})\ell_{\mathrm{phys}}(z)^2\frac{E(z)^2}{\hbar^2c^2} + \mathcal{O}(\ell_{*,0}^4) \tag{39.18}$$

$$= \frac{1}{2}\mathcal{A}(\boldsymbol{n})\frac{\ell_{*,0}^2E_0^2}{\hbar^2c^2}(1+z)^{2-2\sigma} + \mathcal{O}(\ell_{*,0}^4). \tag{39.19}$$

Hence

$$v_\ell - v_h = (1 - v_h) - (1 - v_\ell) \tag{39.20}$$

$$= \frac{\mathcal{A}(\boldsymbol{n})\ell_{*,0}^2}{2\hbar^2c^2}(E_{h,0}^2 - E_{\ell,0}^2)(1+z)^{2-2\sigma} + \mathcal{O}(\ell_{*,0}^4). \tag{39.21}$$

The arrival redshift satisfies $z_{a,h} = \mathcal{O}(\ell_{*,0}^2)$. On the short interval from $z_{a,h}$ to zero, replacing v_h and $(1+z)^{-1}$ by one changes the result only at order $\ell_{*,0}^4$. The distance-balance identity therefore becomes

$$\Delta t_{h\ell} = \int_0^{z_s} \frac{v_\ell - v_h}{H(z)}\,\mathrm{d}z + \mathcal{O}(\ell_{*,0}^4). \tag{39.22}$$

Thus, to leading order in $\ell_{*,0}^2$,

$$\boxed{\Delta t_{h\ell} = \frac{\mathcal{A}(\boldsymbol{n})\ell_{*,0}^2(E_{h,0}^2 - E_{\ell,0}^2)}{2\hbar^2c^2}I_\sigma(z_s) + \mathcal{O}(\ell_{*,0}^4).} \tag{39.23}$$

Special cases:

$$\sigma = 0: \quad \Delta t_{h\ell}^{\mathrm{proper}} = \frac{\mathcal{A}\ell_{*,0}^2\Delta E_0^2}{2\hbar^2c^2}\int_0^{z_s} \frac{(1+z)^2}{H(z)}\,\mathrm{d}z + \mathcal{O}(\ell_{*,0}^4), \tag{39.24}$$

$$\sigma = 1: \quad \Delta t_{h\ell}^{\text{comoving}} = \frac{\mathcal{A}\ell_{*,0}^2 \Delta E_0^2}{2\hbar^2 c^2} \int_0^{z_s} \frac{\mathrm{d}z}{H(z)} + \mathcal{O}(\ell_{*,0}^4). \tag{39.25}$$

At low redshift,

$$H(z) = H_0 + \mathcal{O}(z), \tag{39.26}$$
$$(1+z)^{2-2\sigma} = 1 + \mathcal{O}(z), \tag{39.27}$$
$$I_\sigma(z_s) = \frac{z_s}{H_0} + \mathcal{O}(z_s^2). \tag{39.28}$$

Since $L = cz_s/H_0 + \mathcal{O}(z_s^2)$,

$$I_\sigma(z_s) = \frac{L}{c} + \mathcal{O}(z_s^2), \tag{39.29}$$

which recovers the static-distance law.

39.5 Null bound and directional cancellation

A null result $|\Delta t_{h\ell}| \le \Delta t_{\max}$ implies

$$\frac{\mathcal{A}(\boldsymbol{n})\ell_{*,0}^2 |E_{h,0}^2 - E_{\ell,0}^2|}{2\hbar^2 c^2} I_\sigma(z_s) \le \Delta t_{\max}. \tag{39.30}$$

Solving this inequality for the positive length $\ell_{*,0}$ gives

$$\boxed{\ell_{*,0} \le \left[\frac{2\hbar^2 c^2 \Delta t_{\max}}{\mathcal{A}(\boldsymbol{n}) |E_{h,0}^2 - E_{\ell,0}^2| I_\sigma(z_s)} \right]^{1/2}}. \tag{39.31}$$

For a cubic-axis ray,

$$\mathcal{A}(1,0,0) = 0. \tag{39.32}$$

The exact dispersion also reduces to

$$\cos(\omega\ell) = \cos(k\ell)\cos(0)\cos(0) \tag{39.33}$$
$$= \cos(k\ell). \tag{39.34}$$

On the principal branch, $\omega = k$, and hence

$$v_g = \frac{\partial \omega}{\partial k} = 1. \tag{39.35}$$

Thus the leading and exact directional delays vanish on the axis.

39.6 Numerical verification

For

$$H(z) = H_0 \sqrt{0.3(1+z)^3 + 0.7}, \qquad z_s = 2, \tag{39.36}$$

direct arrival integration gives

$$p_{\text{remainder}}^{\sigma=0} = 4.002683, \tag{39.37}$$
$$p_{\text{remainder}}^{\sigma=1} = 4.001430, \tag{39.38}$$
$$\varepsilon_{\text{axis}} = 0, \tag{39.39}$$
$$\ell_{\text{true}} = 0.030000, \tag{39.40}$$
$$\ell_{\text{fit}}^{\sigma=0} = 0.030008, \tag{39.41}$$
$$\ell_{\text{fit}}^{\sigma=1} = 0.030003. \tag{39.42}$$

Hence

$$\Delta t_{h\ell}^{\text{direct}} - \Delta t_{h\ell}^{\text{leading}} = \mathcal{O}(\ell_{*,0}^4). \tag{39.43}$$

39.7 Next steps

- Derive the finite-scale correction for a broader class of microscopic walks and determine which terms are universal.
- Remove the preferred cubic frame, or derive and quantify the residual frame dependence after refinement.
- Model intrinsic source lags and detector systematics in the same arrival-time analysis.
- Use a declared propagation data set to place a reproducible bound on the microscopic length scale.
- Derive a second independent observable that tests the same finite-scale parameter.

Chapter 40

Controlled Primordial Power-Law Sector

A constant slow-roll parameter gives a solvable inflationary benchmark. The scalar mode equation and power spectra are used here as a controlled test of the projector discretization.

40.1 Constant-ϵ background

Assume

$$0 < \epsilon < 1, \qquad q := \frac{1}{1-\epsilon}, \qquad \eta < 0, \tag{40.1}$$

and

$$\boxed{a(\eta) = A(-\eta)^{-q}.} \tag{40.2}$$

Differentiation gives

$$a' = Aq(-\eta)^{-q-1}, \tag{40.3}$$

$$\mathcal{H} = \frac{a'}{a} = -\frac{q}{\eta}, \tag{40.4}$$

$$\mathcal{H}' = \frac{q}{\eta^2}. \tag{40.5}$$

Hence

$$1 - \frac{\mathcal{H}'}{\mathcal{H}^2} = 1 - \frac{1}{q} = \epsilon. \tag{40.6}$$

Then

$$\mathcal{H} := \frac{a'}{a} = -\frac{q}{\eta}, \tag{40.7}$$

$$\epsilon_H := 1 - \frac{\mathcal{H}'}{\mathcal{H}^2} = \epsilon. \tag{40.8}$$

The comoving Hubble radius obeys

$$(aH)^{-1} = \mathcal{H}^{-1} = -\frac{\eta}{q}. \tag{40.9}$$

For $N := \log(a/a_i)$,

$$\frac{(aH)^{-1}}{(a_i H_i)^{-1}} = \mathrm{e}^{-(1-\epsilon)N}, \tag{40.10}$$

$$\frac{|\Omega_K|}{|\Omega_{K,i}|} = \mathrm{e}^{-2(1-\epsilon)N}. \tag{40.11}$$

For constant $\epsilon = -\,\mathrm{d}\log H/\,\mathrm{d}N$,

$$H/H_i = \mathrm{e}^{-\epsilon N}, \qquad a/a_i = \mathrm{e}^{N}. \tag{40.12}$$

Therefore $aH/(a_i H_i) = \mathrm{e}^{(1-\epsilon)N}$. Since $|\Omega_K| \propto (aH)^{-2}$, the curvature ratio is the square of the comoving-Hubble-radius ratio. Hence

$$N \geq \frac{\log(S_K^{-1})}{2(1-\epsilon)} \Longrightarrow \frac{|\Omega_K|}{|\Omega_{K,i}|} \leq S_K. \tag{40.13}$$

40.2 Canonical scalar mode

For a canonical single field with constant ϵ,

$$z := a\sqrt{2\epsilon}\, M_{\rm Pl}, \qquad \nu := q + \frac{1}{2} = \frac{3-\epsilon}{2(1-\epsilon)}. \tag{40.14}$$

Then

$$\boxed{\frac{z''}{z} = \frac{a''}{a} = \frac{\nu^2 - \frac{1}{4}}{\eta^2}.} \tag{40.15}$$

Because ϵ is constant, z differs from a only by a constant factor. Also,

$$\frac{a''}{a} = \frac{q(q+1)}{\eta^2}, \tag{40.16}$$

$$\nu^2 - \frac{1}{4} = \left(q + \frac{1}{2}\right)^2 - \frac{1}{4} \tag{40.17}$$

$$= q(q+1). \tag{40.18}$$

The quadratic mode action is

$$S_v^{(2)} = \frac{1}{2}\int \mathrm{d}\eta\, \mathrm{d}^3k \left[|v_k'|^2 - \left(k^2 - \frac{z''}{z}\right)|v_k|^2\right]. \tag{40.19}$$

Varying v_k^* and integrating the kinetic term by parts gives

$$\delta S_v^{(2)} = -\frac{1}{2}\int \delta v_k^* \left[v_k'' + \left(k^2 - \frac{z''}{z}\right) v_k\right] \mathrm{d}\eta\, \mathrm{d}^3k + \text{c.c.}, \tag{40.20}$$

so stationarity yields the mode equation. Thus

$$\boxed{v_k'' + \left[k^2 - \frac{\nu^2 - \frac{1}{4}}{\eta^2}\right] v_k = 0.} \tag{40.21}$$

The Bunch–Davies fixed-background mode is

$$\boxed{v_k(\eta) = \frac{\sqrt{-\pi\eta}}{2} \mathrm{e}^{\mathrm{i}(2\nu+1)\pi/4} H_\nu^{(1)}(-k\eta).} \tag{40.22}$$

Its normalization is

$$v_k v_k^{*\prime} - v_k^* v_k' = \mathrm{i}. \tag{40.23}$$

The vacuum choice and fixed-background canonical quantization are model inputs in this section; quantized projector geometry is not derived here.

40.3 Exact power-law spectra

For $a = A(-\eta)^{-q}$, the superhorizon scalar spectrum is

$$\boxed{\mathcal{P}_{\mathcal{R}}(k) = \frac{2^{2\nu-4}\Gamma(\nu)^2}{\pi^3 A^2 \epsilon M_{\rm Pl}^2} k^{3-2\nu}.} \tag{40.24}$$

For $-k\eta \ll 1$,

$$H_\nu^{(1)}(x) \sim -\frac{\mathrm{i}\Gamma(\nu)}{\pi}\left(\frac{2}{x}\right)^\nu. \tag{40.25}$$

Thus

$$|v_k|^2 \sim \frac{2^{2\nu-2}\Gamma(\nu)^2}{\pi}(-\eta)^{1-2\nu}k^{-2\nu}. \tag{40.26}$$

Since $\mathcal{R}_k = v_k/z$, $z = A(-\eta)^{-q}\sqrt{2\epsilon}M_{\rm Pl}$, and $\nu = q + 1/2$, all powers of η cancel. Finally,

$$\mathcal{P}_{\mathcal{R}} = \frac{k^3}{2\pi^2}|\mathcal{R}_k|^2, \tag{40.27}$$

which gives the displayed coefficient and power of k. The two-polarization tensor spectrum is

$$\boxed{\mathcal{P}_T(k) = \frac{2^{2\nu}\Gamma(\nu)^2}{\pi^3 A^2 M_{\rm Pl}^2}k^{3-2\nu}.} \tag{40.28}$$

Therefore

$$n_s - 1 = 3 - 2\nu = -\frac{2\epsilon}{1-\epsilon}, \tag{40.29}$$

$$n_t = -\frac{2\epsilon}{1-\epsilon}, \tag{40.30}$$

$$\alpha_s := \frac{\mathrm{d}n_s}{\mathrm{d}\log k} = 0, \tag{40.31}$$

$$\alpha_t := \frac{\mathrm{d}n_t}{\mathrm{d}\log k} = 0, \tag{40.32}$$

$$\boxed{r := \frac{\mathcal{P}_T}{\mathcal{P}_{\mathcal{R}}} = 16\epsilon}, \tag{40.33}$$

$$\boxed{r = -8(1-\epsilon)n_t}. \tag{40.34}$$

The spectral index is the logarithmic derivative of the power law:

$$\frac{\mathrm{d}\log\mathcal{P}_{\mathcal{R}}}{\mathrm{d}\log k} = 3 - 2\nu. \tag{40.35}$$

Using $\nu = (3-\epsilon)/[2(1-\epsilon)]$ gives $3 - 2\nu = -2\epsilon/(1-\epsilon)$. The scalar and tensor spectra have the same k-power, so their runnings vanish. Dividing their normalization constants gives

$$r = \frac{2^{2\nu}}{2^{2\nu-4}/\epsilon} = 16\epsilon, \tag{40.36}$$

and substitution of n_t gives the exact consistency relation. The first-order slow-roll relation $r = -8n_t + O(\epsilon^2)$ follows.

40.4 Controlled benchmark

For

$$\epsilon = 0.01, \qquad N = 60, \tag{40.37}$$

The power-law calculation gives

$$q = 1.010101, \tag{40.38}$$

$$\nu = 1.510101, \tag{40.39}$$

$$n_s = 0.979798, \tag{40.40}$$

$$n_t = -0.020202, \tag{40.41}$$

$$r = 0.16, \tag{40.42}$$

$$\frac{|\Omega_K|}{|\Omega_{K,i}|} = 2.545749 \times 10^{-52}, \tag{40.43}$$

$$\frac{(aH)^{-1}}{(a_i H_i)^{-1}} = 1.595540 \times 10^{-26}. \tag{40.44}$$

The numerical checks are

$$\varepsilon_{\epsilon_H} = 0, \tag{40.45}$$

$$\varepsilon_{z''/z} = 0, \tag{40.46}$$
$$\varepsilon_{n_s} = 1.179612 \times 10^{-16}, \tag{40.47}$$
$$\varepsilon_{n_t} = 2.255141 \times 10^{-16}, \tag{40.48}$$
$$\varepsilon_r = 2.775558 \times 10^{-17}, \tag{40.49}$$
$$\varepsilon_{\mathrm{W}} = 3.907985 \times 10^{-14}, \tag{40.50}$$
$$p_{v,\mathrm{mid}} = 1.964609. \tag{40.51}$$

40.5 Next steps

- Derive the conditions that select the inflationary attractor from general initial data.
- Construct a graceful-exit mechanism and a controlled reheating sector.
- Determine the vacuum from projector refinement instead of fixing it by hand.
- Quantize the projector geometry together with the scalar perturbations.
- Extend the calculation to multifield dynamics and the leading non-Gaussian observables.

Chapter 41

Gauge-Invariant Einstein–Scalar Perturbations

Scalar perturbations are reduced to a single gauge-invariant degree of freedom. The continuum Mukhanov–Sasaki action is then discretized with the same projector Laplacian used elsewhere.

41.1 Einstein–scalar input and background

Let

$$M_{\mathrm{Pl}}^2 := (8\pi G)^{-1} \tag{41.1}$$

and

$$S[g,\phi] = \int \sqrt{-g}\left[\frac{M_{\mathrm{Pl}}^2}{2}R - \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)\right]\,\mathrm{d}^4x. \tag{41.2}$$

For

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + a(t)^2\,\mathrm{d}\boldsymbol{x}^2, \qquad \phi = \bar{\phi}(t), \tag{41.3}$$

the background equations are

$$3M_{\mathrm{Pl}}^2H^2 = \frac{1}{2}\dot{\bar{\phi}}^{\,2} + V(\bar{\phi}), \tag{41.4}$$

$$-2M_{\mathrm{Pl}}^2\dot{H} = \dot{\bar{\phi}}^{\,2}, \tag{41.5}$$

$$\ddot{\bar{\phi}} + 3H\dot{\bar{\phi}} + V_{,\phi} = 0, \tag{41.6}$$

$$\epsilon := -\frac{\dot{H}}{H^2} = \frac{\dot{\bar{\phi}}^{\,2}}{2M_{\mathrm{Pl}}^2H^2}. \tag{41.7}$$

41.2 Scalar gauge invariant

Use the scalar convention

$$g_{ij} = a^2\left[(1-2\Psi)\delta_{ij} + 2\partial_i\partial_jE\right]. \tag{41.8}$$

Under $t \mapsto t+T$,

$$\Psi \mapsto \Psi + HT, \tag{41.9}$$

$$\delta\phi \mapsto \delta\phi - \dot{\bar{\phi}}T. \tag{41.10}$$

Define

$$\boxed{\zeta := -\Psi - \frac{H}{\dot{\bar{\phi}}}\delta\phi.} \tag{41.11}$$

Indeed, substitution of the two transformations gives

$$\zeta \mapsto -(\Psi + HT) - \frac{H}{\dot{\bar{\phi}}}\left(\delta\phi - \dot{\bar{\phi}}T\right) \tag{41.12}$$

$$= -\Psi - HT - \frac{H}{\dot{\bar{\phi}}}\delta\phi + HT \tag{41.13}$$

$$= \zeta. \tag{41.14}$$

The time-shift contributions therefore cancel before any gauge choice is made. In comoving gauge,

$$\delta\phi = 0, \qquad E = 0, \qquad h_{ij} = a^2 \mathrm{e}^{2\zeta}\delta_{ij}. \tag{41.15}$$

41.3 Lapse and shift elimination

Write

$$N = 1 + \alpha, \qquad N_i = \partial_i \beta. \tag{41.16}$$

The linear Hamiltonian and momentum constraints give

$$\alpha = \frac{\dot{\zeta}}{H}, \tag{41.17}$$

$$\beta = -\frac{\zeta}{H} + \chi, \tag{41.18}$$

$$\nabla^2\chi = a^2\epsilon\dot{\zeta}. \tag{41.19}$$

Thus α and β are nondynamical and are eliminated before the reduced quadratic action is formed.

41.4 Reduced scalar action and equation

After substitution of the constraints,

$$\boxed{S_\zeta^{(2)} = M_{\mathrm{Pl}}^2 \int a^3\epsilon \left[\dot{\zeta}^2 - \frac{(\nabla\zeta)^2}{a^2}\right] \mathrm{d}t\, \mathrm{d}^3x.} \tag{41.20}$$

For a compactly supported variation $\delta\zeta$,

$$\delta S_\zeta^{(2)} = 2M_{\mathrm{Pl}}^2 \int \left[a^3\epsilon\dot{\zeta}\,\delta\dot{\zeta} - a\epsilon\partial_i\zeta\,\partial_i\delta\zeta\right] \mathrm{d}t\, \mathrm{d}^3x \tag{41.21}$$

$$= -2M_{\mathrm{Pl}}^2 \int \left[\frac{\mathrm{d}}{\mathrm{d}t}\left(a^3\epsilon\dot{\zeta}\right) - a\epsilon\nabla^2\zeta\right] \delta\zeta\ \mathrm{d}t\, \mathrm{d}^3x, \tag{41.22}$$

where the second line follows by integrating once in time and once in space. Hence the Euler–Lagrange equation is

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(a^3\epsilon\dot{\zeta}\right) - a\epsilon\nabla^2\zeta = 0.} \tag{41.23}$$

For a Fourier mode, $\nabla^2\zeta_{\boldsymbol{k}} = -k^2\zeta_{\boldsymbol{k}}$, so

$$0 = \frac{\mathrm{d}}{\mathrm{d}t}\left(a^3\epsilon\dot{\zeta}_{\boldsymbol{k}}\right) + a\epsilon k^2\zeta_{\boldsymbol{k}} \tag{41.24}$$

$$= a^3\epsilon\ddot{\zeta}_{\boldsymbol{k}} + \left(3Ha^3\epsilon + a^3\dot{\epsilon}\right)\dot{\zeta}_{\boldsymbol{k}} + a\epsilon k^2\zeta_{\boldsymbol{k}}. \tag{41.25}$$

Division by $a^3\epsilon$ gives

$$\ddot{\zeta}_{\boldsymbol{k}} + \left(3H + \frac{\dot{\epsilon}}{\epsilon}\right)\dot{\zeta}_{\boldsymbol{k}} + \frac{k^2}{a^2}\zeta_{\boldsymbol{k}} = 0. \tag{41.26}$$

At $k = 0$, the equation can be integrated twice:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(a^3\epsilon\dot{\zeta}\right) = 0, \tag{41.27}$$

$$a^3\epsilon\dot{\zeta} = C_2, \tag{41.28}$$

$$\dot{\zeta} = \frac{C_2}{a^3\epsilon}, \tag{41.29}$$

$$\zeta(t) = C_1 + C_2 \int^t \frac{\mathrm{d}s}{a(s)^3\epsilon(s)}. \tag{41.30}$$

Hence the adiabatic constant mode is conserved when the second mode decays.

41.5 Mukhanov–Sasaki form

Let

$$\mathrm{d}t = a\,\mathrm{d}\eta, \tag{41.31}$$

$$z(\eta) := a\sqrt{2\epsilon}\,M_{\mathrm{Pl}}, \tag{41.32}$$

$$v := z\zeta. \tag{41.33}$$

Changing from t to η first gives

$$S_\zeta^{(2)} = \frac{1}{2}\int z^2\left[(\zeta')^2 - (\nabla\zeta)^2\right]\,\mathrm{d}\eta\,\mathrm{d}^3x. \tag{41.34}$$

Since $\zeta = v/z$ and z is spatially homogeneous,

$$z^2(\zeta')^2 = \left(v' - \frac{z'}{z}v\right)^2 \tag{41.35}$$

$$= (v')^2 - \frac{z'}{z}(v^2)' + \left(\frac{z'}{z}\right)^2 v^2, \tag{41.36}$$

$$z^2(\nabla\zeta)^2 = (\nabla v)^2. \tag{41.37}$$

Integrating the mixed term by parts yields

$$-\int \frac{z'}{z}(v^2)'\,\mathrm{d}\eta = \int \left(\frac{z'}{z}\right)' v^2\,\mathrm{d}\eta + \text{boundary term}, \tag{41.38}$$

$$\left(\frac{z'}{z}\right)' + \left(\frac{z'}{z}\right)^2 = \frac{z''}{z}. \tag{41.39}$$

Therefore, up to the displayed boundary term,

$$\boxed{S_v^{(2)} = \frac{1}{2}\int \left[(v')^2 - (\nabla v)^2 + \frac{z''}{z}v^2\right]\,\mathrm{d}\eta\,\mathrm{d}^3x.} \tag{41.40}$$

Therefore

$$v'' - \nabla^2 v - \frac{z''}{z}v = 0, \tag{41.41}$$

$$v_{\boldsymbol{k}}'' + \left(k^2 - \frac{z''}{z}\right)v_{\boldsymbol{k}} = 0. \tag{41.42}$$

41.6 Projector-lattice scalar action

For cubic spacing h,

$$S_{v,h}^{(2)} = \frac{1}{2}\int h^3 \sum_{\boldsymbol{j}} \left[(v_{\boldsymbol{j}}')^2 - \sum_{r=1}^{3}(\delta_r^+ v_{\boldsymbol{j}})^2 + \frac{z''}{z}v_{\boldsymbol{j}}^2\right]\,\mathrm{d}\eta. \tag{41.43}$$

The discrete summation-by-parts identity

$$h^3\sum_{\boldsymbol{j},r}(\delta_r^+ v_{\boldsymbol{j}})(\delta_r^+\delta v_{\boldsymbol{j}}) = -h^3\sum_{\boldsymbol{j}}(\Delta_h v_{\boldsymbol{j}})\delta v_{\boldsymbol{j}} \tag{41.44}$$

converts the gradient variation into the lattice Laplacian. The lattice equation is therefore

$$v_{\boldsymbol{j}}'' - \Delta_h v_{\boldsymbol{j}} - \frac{z''}{z} v_{\boldsymbol{j}} = 0. \tag{41.45}$$

For a lattice Fourier mode,

$$\delta_r^+ \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}_{\boldsymbol{j}}} = \frac{\mathrm{e}^{\mathrm{i}k_r h} - 1}{h} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}_{\boldsymbol{j}}}, \tag{41.46}$$

$$\left| \frac{\mathrm{e}^{\mathrm{i}k_r h} - 1}{h} \right|^2 = \frac{(\mathrm{e}^{\mathrm{i}k_r h} - 1)(\mathrm{e}^{-\mathrm{i}k_r h} - 1)}{h^2} \tag{41.47}$$

$$= \frac{2 - 2\cos(k_r h)}{h^2} = \frac{4}{h^2} \sin^2\left(\frac{k_r h}{2}\right). \tag{41.48}$$

Thus

$$\widehat{k}_h^2 := \frac{4}{h^2} \sum_{r=1}^{3} \sin^2\left(\frac{k_r h}{2}\right), \tag{41.49}$$

$$v_{\boldsymbol{k},h}'' + \left(\widehat{k}_h^2 - \frac{z''}{z}\right) v_{\boldsymbol{k},h} = 0, \tag{41.50}$$

$$\widehat{k}_h^2 = k^2 - \frac{h^2}{12} \sum_{r=1}^{3} k_r^4 + \frac{h^4}{360} \sum_{r=1}^{3} k_r^6 + \mathcal{O}(h^6). \tag{41.51}$$

For $z''/z = 2/\eta^2$,

$$v_\kappa(\eta) = \frac{1}{\sqrt{2\kappa}} \left(1 - \frac{\mathrm{i}}{\kappa\eta}\right) \mathrm{e}^{-\mathrm{i}\kappa\eta} \tag{41.52}$$

solves

$$v_\kappa'' + \left(\kappa^2 - \frac{2}{\eta^2}\right) v_\kappa = 0. \tag{41.53}$$

With $\kappa = \widehat{k}_h$,

$$p_{v,h} = 2.001500, \tag{41.54}$$

$$p_{|v|^2,h} = 2.002803, \tag{41.55}$$

$$p_{v,h}^{\mathrm{Richardson}} = 4.001170, \tag{41.56}$$

$$\max_h |v_h v_h^{*\prime} - v_h^* v_h' - \mathrm{i}| = 3.219647 \times 10^{-15}. \tag{41.57}$$

41.7 Next steps

- Extend the quadratic derivation from one canonical scalar to general background and transfer functions.
- Add vector and tensor sources within the same gauge-invariant projector calculus.
- Generalize to multiple scalar fields and derive the coupled adiabatic and entropy modes.
- Derive the cubic action and the leading non-Gaussian correlation functions.
- Obtain the vacuum normalization from the projector dynamics rather than prescribing it as an external input.

Chapter 42

Vector and Tensor Perturbations on FLRW

The scalar–vector–tensor split separates decaying vector modes from propagating tensor modes on FLRW. The discrete tensor sector preserves the continuum causal structure in the refinement limit.

42.1 Scalar–vector–tensor split

In conformal time,

$$\mathrm{d}s^2 = a(\eta)^2 \left[-\,\mathrm{d}\eta^2 + 2S_i\,\mathrm{d}\eta\,\mathrm{d}x^i + \left(\delta_{ij} + 2\partial_{(i}F_{j)} + h_{ij}\right)\,\mathrm{d}x^i\,\mathrm{d}x^j \right], \tag{42.1}$$

with

$$\partial^i S_i = 0, \tag{42.2}$$
$$\partial^i F_i = 0, \tag{42.3}$$
$$\partial^i h_{ij} = 0, \tag{42.4}$$
$$h^i{}_i = 0. \tag{42.5}$$

Under a transverse spatial gauge shift L_i,

$$S_i \mapsto S_i - L_i', \tag{42.6}$$
$$F_i \mapsto F_i - L_i. \tag{42.7}$$

Therefore

$$\boxed{\Sigma_i := S_i - F_i'} \tag{42.8}$$

is gauge invariant, because

$$\Sigma_i \mapsto (S_i - L_i') - (F_i - L_i)' \tag{42.9}$$
$$= S_i - L_i' - F_i' + L_i' \tag{42.10}$$
$$= \Sigma_i. \tag{42.11}$$

42.2 Source-free vector sector

In the absence of vector anisotropic stress,

$$\boxed{(a^2\Sigma_i)' = 0.} \tag{42.12}$$

Integrating once gives

$$(a^2\Sigma_i)' = 0, \tag{42.13}$$
$$a^2\Sigma_i = C_i, \tag{42.14}$$

$$\Sigma_i(\eta) = \frac{C_i}{a(\eta)^2}. \tag{42.15}$$

For a matter background $a \propto \eta^2$,

$$\Sigma_i \propto \eta^{-4}. \tag{42.16}$$

A midpoint update of

$$\Sigma' = -\frac{4}{\eta}\Sigma \tag{42.17}$$

converges with

$$p_{\text{vector}} = 2.052951. \tag{42.18}$$

The fully matter-sourced vector constraint and vorticity hierarchy are not derived here.

42.3 Tensor action and equation

The quadratic tensor action is

$$\boxed{S_T^{(2)} = \frac{M_{\text{Pl}}^2}{8}\int a^2\left[(h'_{ij})^2 - (\partial_k h_{ij})^2\right]\,\mathrm{d}\eta\,\mathrm{d}^3x.} \tag{42.19}$$

For a transverse-traceless variation,

$$\delta S_T^{(2)} = \frac{M_{\text{Pl}}^2}{4}\int a^2\left[h'_{ij}\delta h'_{ij} - \partial_k h_{ij}\partial_k \delta h_{ij}\right]\,\mathrm{d}\eta\,\mathrm{d}^3x \tag{42.20}$$

$$= -\frac{M_{\text{Pl}}^2}{4}\int\left[(a^2 h'_{ij})' - a^2\nabla^2 h_{ij}\right]\delta h_{ij}\,\mathrm{d}\eta\,\mathrm{d}^3x. \tag{42.21}$$

Since

$$(a^2 h'_{ij})' = a^2 h''_{ij} + 2aa' h'_{ij} = a^2(h''_{ij} + 2\mathcal{H}h'_{ij}), \tag{42.22}$$

the vacuum Euler–Lagrange equation is

$$\boxed{h''_{ij} + 2\mathcal{H}h'_{ij} - \nabla^2 h_{ij} = 0, \qquad \mathcal{H} := a'/a.} \tag{42.23}$$

With transverse-traceless anisotropic stress,

$$h''_{ij} + 2\mathcal{H}h'_{ij} - \nabla^2 h_{ij} = \frac{2a^2}{M_{\text{Pl}}^2}\Pi_{ij}^{\text{TT}}. \tag{42.24}$$

For each Fourier polarization, set

$$\mu_\lambda := \frac{aM_{\text{Pl}}}{2}h_\lambda, \qquad h_\lambda = \frac{2\mu_\lambda}{aM_{\text{Pl}}}. \tag{42.25}$$

Then

$$h'_\lambda = \frac{2}{aM_{\text{Pl}}}(\mu'_\lambda - \mathcal{H}\mu_\lambda), \tag{42.26}$$

$$h''_\lambda = \frac{2}{aM_{\text{Pl}}}\left[\mu''_\lambda - 2\mathcal{H}\mu'_\lambda + (\mathcal{H}^2 - \mathcal{H}')\mu_\lambda\right]. \tag{42.27}$$

Substitution into $h''_\lambda + 2\mathcal{H}h'_\lambda + k^2 h_\lambda = 0$ cancels the first-derivative terms and gives

$$0 = \mu''_\lambda + \left[k^2 - (\mathcal{H}' + \mathcal{H}^2)\right]\mu_\lambda \tag{42.28}$$

$$= \mu''_\lambda + \left(k^2 - \frac{a''}{a}\right)\mu_\lambda. \tag{42.29}$$

Thus

$$\boxed{\mu''_\lambda + \left(k^2 - \frac{a''}{a}\right)\mu_\lambda = 0.} \tag{42.30}$$

42.4 Projector-lattice tensor sector

For cubic spacing h, the centered projector Laplacian has symbol

$$\widehat{k}_h^2 = \frac{4}{h^2}\sum_{r=1}^{3}\sin^2\left(\frac{k_r h}{2}\right) \tag{42.31}$$

$$= \sum_{r=1}^{3}\left[k_r^2 - \frac{h^2 k_r^4}{12} + \frac{h^4 k_r^6}{360} + \mathcal{O}(h^6)\right] \tag{42.32}$$

$$= k^2 - \frac{h^2}{12}\sum_{r=1}^{3}k_r^4 + \frac{h^4}{360}\sum_{r=1}^{3}k_r^6 + \mathcal{O}(h^6). \tag{42.33}$$

Accordingly,

$$\mu''_{\lambda,h} + \left(\widehat{k}_h^2 - \frac{a''}{a}\right)\mu_{\lambda,h} = 0. \tag{42.34}$$

Radiation era:

$$a(\eta) \propto \eta, \qquad a'' = 0, \tag{42.35}$$

$$\mu_\kappa(\eta) = \frac{\mathrm{e}^{-\mathrm{i}\kappa\eta}}{\sqrt{2\kappa}}, \qquad h_\kappa = \frac{2\mu_\kappa}{aM_{\mathrm{Pl}}}. \tag{42.36}$$

Thus

$$a|h_\kappa| = \text{constant}. \tag{42.37}$$

Matter era:

$$a(\eta) \propto \eta^2, \tag{42.38}$$

$$h_\kappa(\eta) = 3\frac{j_1(\kappa\eta)}{\kappa\eta}, \tag{42.39}$$

$$h''_\kappa + \frac{4}{\eta}h'_\kappa + \kappa^2 h_\kappa = 0. \tag{42.40}$$

The numerical orders are

$$p_T^{\text{radiation}} = 1.999618, \tag{42.41}$$

$$p_{T,\mathrm{R}}^{\text{radiation}} = 3.998798, \tag{42.42}$$

$$p_T^{\text{matter}} = 2.002687, \tag{42.43}$$

$$p_{T,\mathrm{R}}^{\text{matter}} = 3.996376. \tag{42.44}$$

The canonical Wronskian and amplitude residuals satisfy

$$\varepsilon_{\mathrm{W}} = 4.440892 \times 10^{-16}, \tag{42.45}$$

$$\varepsilon_{a|h|} = 4.440892 \times 10^{-16}. \tag{42.46}$$

42.5 Propagation distance

For geometric-optics tensor propagation, the canonical variable obeys the locally flat radial wave law

$$\mu \sim \frac{\mathrm{e}^{-\mathrm{i}k\eta}}{r}. \tag{42.47}$$

Using $h = 2\mu/(aM_{\mathrm{Pl}})$ then gives

$$h \sim \frac{1}{ar}, \tag{42.48}$$

up to the source normalization and polarization amplitude. After source-frame normalization,

$$D_L^{\mathrm{GW}} = D_L^{\mathrm{EM}} \tag{42.49}$$

in the unmodified propagation sector. Source generation, binary calibration, and modified-friction alternatives are not derived here.

42.6 Next steps

- Add the matter-sourced vector constraint and derive the complete vorticity hierarchy.
- Include cosmological anisotropic stress and its contribution to vector and tensor propagation.
- Derive the primordial tensor normalization from the selected quantum initial state.
- Compute the leading nonlinear scalar–vector–tensor couplings.
- Incorporate source generation, binary calibration, and standard-siren observables in the tensor sector.

Chapter 43

Recombination, Decoupling, and Acoustic Scales

Recombination and acoustic scales are introduced at the level needed to connect the background and perturbation sectors to observable angular scales. Atomic rates and tight-coupling formulas are treated as standard model inputs.

43.1 Conditional atomic benchmark

The benchmark inputs are

$$H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{43.1}$$

$$\Omega_b = 0.05, \qquad \Omega_m = 0.30, \tag{43.2}$$

$$\Omega_r = 9 \times 10^{-5}, \qquad T_0 = 2.7255\ \mathrm{K}, \tag{43.3}$$

$$Y_p = 0.24. \tag{43.4}$$

These values define a reproducible controlled model; they are not an observational fit.
Let

$$T(z) = T_0(1+z), \tag{43.5}$$

$$n_H(z) = n_{H0}(1+z)^3, \tag{43.6}$$

$$S(z) = \frac{1}{n_H(z)} \left(\frac{m_e k_B T(z)}{2\pi\hbar^2} \right)^{3/2} \exp\left[-\frac{\chi_H}{k_B T(z)} \right]. \tag{43.7}$$

The Saha equilibrium equation is

$$\frac{x_e^2}{1-x_e} = S. \tag{43.8}$$

Multiplication by $1 - x_e$ gives the quadratic equation

$$x_e^2 + S x_e - S = 0. \tag{43.9}$$

Hence

$$x_e = \frac{-S \pm \sqrt{S^2 + 4S}}{2}. \tag{43.10}$$

For $S > 0$, the minus sign is negative, so the physical root is the plus branch. Rationalizing that branch gives

$$x_e(S) = \frac{-S + \sqrt{S^2 + 4S}}{2} \tag{43.11}$$

$$= \frac{2S}{S + \sqrt{S^2 + 4S}} \tag{43.12}$$

$$= \frac{2}{1 + \sqrt{1 + 4/S}}. \tag{43.13}$$

Thus

$$\boxed{x_e(S) = \frac{-S + \sqrt{S^2 + 4S}}{2} = \frac{2}{1 + \sqrt{1 + 4/S}}.} \tag{43.14}$$

43.2 Optical depth and visibility

Ignoring reionization,

$$n_e(z) = x_e(z) n_H(z), \tag{43.15}$$

$$\frac{\mathrm{d}\tau}{\mathrm{d}z} = \frac{c\sigma_T n_e(z)}{(1+z)H(z)}, \tag{43.16}$$

$$\tau(z) = \int_0^z \frac{c\sigma_T n_e(z')}{(1+z')H(z')}\,\mathrm{d}z'. \tag{43.17}$$

The survival probability from redshift z to the present is $\mathrm{e}^{-\tau(z)}$. Its loss per unit redshift is

$$-\frac{\mathrm{d}}{\mathrm{d}z}\mathrm{e}^{-\tau(z)} = -\left[-\mathrm{e}^{-\tau(z)}\frac{\mathrm{d}\tau}{\mathrm{d}z}\right] \tag{43.18}$$

$$= \mathrm{e}^{-\tau(z)}\frac{\mathrm{d}\tau}{\mathrm{d}z}. \tag{43.19}$$

The redshift-space visibility is therefore

$$\boxed{g_z(z) = \mathrm{e}^{-\tau(z)}\frac{\mathrm{d}\tau}{\mathrm{d}z}.} \tag{43.20}$$

Integrating the total derivative gives

$$\int_0^{z_{\max}} g_z(z)\,\mathrm{d}z = -\int_0^{z_{\max}} \frac{\mathrm{d}}{\mathrm{d}z}\mathrm{e}^{-\tau(z)}\,\mathrm{d}z \tag{43.21}$$

$$= \mathrm{e}^{-\tau(0)} - \mathrm{e}^{-\tau(z_{\max})} \tag{43.22}$$

$$= 1 - \mathrm{e}^{-\tau(z_{\max})}. \tag{43.23}$$

The controlled Saha benchmark gives

$$z_* = 1282.15, \tag{43.24}$$

$$z_{\tau=1} = 1289.423097, \tag{43.25}$$

$$\Delta z_{\mathrm{FWHM}} = 139.2. \tag{43.26}$$

The visibility-normalization residual is

$$\varepsilon_g = 4.807365 \times 10^{-8}. \tag{43.27}$$

Nonequilibrium multilevel recombination and reionization are not derived here.

43.3 Baryon loading and sound horizon

Since nonrelativistic baryons and photons scale as

$$\rho_b(z) = \rho_{b0}(1+z)^3, \qquad \rho_\gamma(z) = \rho_{\gamma 0}(1+z)^4, \tag{43.28}$$

the loading factor is

$$R(z) := \frac{3\rho_b}{4\rho_\gamma} \tag{43.29}$$

$$= \frac{3\rho_{b0}(1+z)^3}{4\rho_{\gamma 0}(1+z)^4} \tag{43.30}$$

$$= \frac{3\Omega_b}{4\Omega_\gamma(1+z)}, \tag{43.31}$$

$$c_s(z) := \frac{c}{\sqrt{3(1+R(z))}}. \tag{43.32}$$

The comoving distance travelled by an acoustic signal before decoupling is

$$r_s(t_*) = \int_0^{t_*} \frac{c_s(t)}{a(t)}\,\mathrm{d}t. \tag{43.33}$$

Using

$$\frac{\mathrm{d}z}{\mathrm{d}t} = -(1+z)H(z), \qquad a = (1+z)^{-1}, \tag{43.34}$$

one has

$$\frac{\mathrm{d}t}{a} = -\frac{\mathrm{d}z}{H(z)}. \tag{43.35}$$

Reversing the limits therefore gives

$$\boxed{r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)}{H(z)}\,\mathrm{d}z.} \tag{43.36}$$

The line-of-sight comoving distance is

$$\chi_* := \int_0^{z_*} \frac{c}{H(z)}\,\mathrm{d}z. \tag{43.37}$$

The geometric acoustic multipole is

$$\boxed{\ell_A := \pi \frac{\chi_*}{r_s(z_*)}.} \tag{43.38}$$

For the benchmark,

$$r_s(z_*) = 125.784218 \text{ Mpc}, \tag{43.39}$$
$$\chi_* = 13644.312614 \text{ Mpc}, \tag{43.40}$$
$$\ell_A = 340.781006. \tag{43.41}$$

This ℓ_A is a geometric acoustic ruler, not a claim that the first CMB peak equals ℓ_A.

43.4 Tight-coupling oscillator

For constant loading and absent gravitational driving,

$$X'' + k^2 c_s^2 X = 0. \tag{43.42}$$

Writing $\omega_s := kc_s$, the characteristic roots are $r = \pm \mathrm{i}\omega_s$, so

$$X(\eta) = A\cos(\omega_s \Delta\eta) + B\sin(\omega_s \Delta\eta). \tag{43.43}$$

At the initial time, $A = X_i$ and $X_i' = \omega_s B$. Therefore

$$X(\eta) = X_i \cos(kc_s \Delta\eta) + \frac{X_i'}{kc_s}\sin(kc_s \Delta\eta). \tag{43.44}$$

For varying $R(a)$, the acoustic model integrates

$$X'' + k^2 c_s(a)^2 X = 0 \tag{43.45}$$

through the benchmark background. The midpoint refinement order is

$$p_{\text{acoustic}} = 2.021414. \tag{43.46}$$

The final-state error at the finest step is

$$\varepsilon_{\text{acoustic}} = 3.978519 \times 10^{-5}. \tag{43.47}$$

43.5 Diffusion approximation

Let the positive conformal opacity be

$$\kappa' := a n_e \sigma_T. \tag{43.48}$$

The controlled tight-coupling diffusion estimate is

$$\boxed{\frac{1}{k_D^2(\eta_*)} = \int_0^{\eta_*} \frac{\mathrm{d}\eta}{6\kappa'} \frac{R^2 + \frac{16}{15}(1+R)}{(1+R)^2}.} \tag{43.49}$$

At leading diffusion order, each Fourier amplitude obeys a local attenuation equation of the form

$$\frac{\mathrm{d}X_k}{\mathrm{d}\eta} = -k^2 D_\gamma(\eta) X_k, \tag{43.50}$$

where the tight-coupling coefficient integrates to

$$\int_0^{\eta_*} D_\gamma(\eta)\,\mathrm{d}\eta = \frac{1}{k_D^2(\eta_*)}. \tag{43.51}$$

Separating variables gives

$$\frac{\mathrm{d}X_k}{X_k} = -k^2 D_\gamma\,\mathrm{d}\eta, \tag{43.52}$$

$$\log \frac{X_k(\eta_*)}{X_k(0)} = -\frac{k^2}{k_D^2}, \tag{43.53}$$

so the associated damping factor is

$$\mathcal{D}(k) = \exp\left[-\frac{k^2}{k_D^2}\right]. \tag{43.54}$$

The benchmark gives

$$r_D := k_D^{-1} = 4.554719 \text{ Mpc}, \tag{43.55}$$

$$k_D = 0.219553 \text{ Mpc}^{-1}. \tag{43.56}$$

The diffusion formula is a tight-coupling approximation and does not replace a full collision hierarchy.

43.6 Next steps

- Replace Saha equilibrium with a multilevel nonequilibrium hydrogen-recombination calculation.
- Add helium recombination and a controlled reionization history.
- Include neutrino decoupling and its effect on the acoustic background.
- Derive the full photon collision hierarchy beyond the tight-coupling approximation.
- Improve diffusion damping and calibrate the drag epoch and sound horizon at precision accuracy.

Chapter 44

Minimal Photon Hierarchy and Low-ℓ CMB

The photon brightness distribution is expanded into angular multipoles. A minimal collisionless hierarchy and the Sachs–Wolfe limit provide a first low-multipole CMB benchmark.

44.1 Collisionless brightness hierarchy

Use

$$\Theta(\mu,\eta)=\sum_{\ell=0}^{\infty}(-\mathrm{i})^{\ell}(2\ell+1)\Theta_{\ell}(\eta)P_{\ell}(\mu). \tag{44.1}$$

After instantaneous last scattering and in the absence of metric sources, the Fourier-space transport equation is

$$\Theta'+\mathrm{i}k\mu\Theta=0. \tag{44.2}$$

The inverse multipole transform is

$$\Theta_{\ell}=\frac{\mathrm{i}^{\ell}}{2}\int_{-1}^{1}P_{\ell}(\mu)\Theta(\mu)\,\mathrm{d}\mu. \tag{44.3}$$

Differentiating it and using the transport equation gives

$$\Theta'_{\ell}=-\frac{\mathrm{i}^{\ell+1}k}{2}\int_{-1}^{1}\mu P_{\ell}(\mu)\Theta(\mu)\,\mathrm{d}\mu. \tag{44.4}$$

The Legendre recurrence

$$(2\ell+1)\mu P_{\ell}=(\ell+1)P_{\ell+1}+\ell P_{\ell-1} \tag{44.5}$$

and orthogonality then give

$$\boxed{\Theta'_{\ell}=\frac{k}{2\ell+1}\left[\ell\Theta_{\ell-1}-(\ell+1)\Theta_{\ell+1}\right].} \tag{44.6}$$

For an initially isotropic monopole, direct integration of the transport equation gives

$$\Theta(\mu,\eta)=\Theta_0(k,\eta_*)\mathrm{e}^{-\mathrm{i}k\mu(\eta-\eta_*)}. \tag{44.7}$$

Using the plane-wave expansion

$$\mathrm{e}^{-\mathrm{i}x\mu}=\sum_{\ell=0}^{\infty}(-\mathrm{i})^{\ell}(2\ell+1)j_{\ell}(x)P_{\ell}(\mu) \tag{44.8}$$

and comparing coefficients yields

$$\boxed{\Theta_{\ell}(k,\eta)=\Theta_0(k,\eta_*)j_{\ell}[k(\eta-\eta_*)]\,.} \tag{44.9}$$

The spherical-Bessel recurrence

$$j_{L+1}(x)=\frac{2L+1}{x}j_L(x)-j_{L-1}(x) \tag{44.10}$$

can be applied directly to the free-streaming solution. Since all multipoles share the same initial monopole factor, it gives the closure

$$\Theta_{L+1} = \frac{2L+1}{x}\Theta_L - \Theta_{L-1}, \qquad x := k(\eta - \eta_*). \tag{44.11}$$

For the exact free-streaming solution, angular Parseval gives the normalization. Indeed,

$$1 = \frac{1}{2}\int_{-1}^{1} \left|\mathrm{e}^{-\mathrm{i}x\mu}\right|^2 \,\mathrm{d}\mu \tag{44.12}$$

$$= \frac{1}{2}\int_{-1}^{1} \left|\sum_{\ell=0}^{\infty}(-\mathrm{i})^\ell(2\ell+1)j_\ell(x)P_\ell(\mu)\right|^2 \,\mathrm{d}\mu \tag{44.13}$$

$$= \sum_{\ell=0}^{\infty}(2\ell+1)j_\ell(x)^2, \tag{44.14}$$

where $\int_{-1}^{1} P_\ell P_m \,\mathrm{d}\mu = 2\delta_{\ell m}/(2\ell+1)$.

44.2 Minimal hierarchy verification

For

$$L = 24, \qquad x_f = 12, \tag{44.15}$$

The hierarchy gives

$$\varepsilon_{j'_\ell} = 4.683753 \times 10^{-16}, \tag{44.16}$$

$$\varepsilon_{\mathrm{hierarchy}} = 6.258188 \times 10^{-13}, \tag{44.17}$$

$$\varepsilon_{\mathrm{Parseval}} = 3.365086 \times 10^{-13}, \tag{44.18}$$

$$p_{\mathrm{hierarchy}} = 1.996773. \tag{44.19}$$

This closes only the collisionless post-decoupling hierarchy. Thomson collisions, polarization, metric driving, reionization, and neutrinos are not derived here.

44.3 Sachs–Wolfe line-of-sight limit

For adiabatic matter-era last scattering,

$$\Theta_0 + \Psi = \frac{1}{5}\mathcal{R}. \tag{44.20}$$

The controlled transfer function is

$$\boxed{\Delta_\ell^{\mathrm{SW}}(k) = \frac{1}{5}j_\ell(k\chi_*).} \tag{44.21}$$

Therefore

$$C_\ell^{\mathrm{SW}} = 4\pi\int_0^\infty \frac{\mathrm{d}k}{k}\mathcal{P}_\mathcal{R}(k)\left[\Delta_\ell^{\mathrm{SW}}(k)\right]^2. \tag{44.22}$$

For

$$\mathcal{P}_\mathcal{R}(k) = A_s, \tag{44.23}$$

substitution of the Sachs–Wolfe transfer function gives

$$C_\ell^{\mathrm{SW}} = \frac{4\pi A_s}{25}\int_0^\infty \frac{\mathrm{d}k}{k} j_\ell(k\chi_*)^2 \tag{44.24}$$

$$= \frac{4\pi A_s}{25}\int_0^\infty \frac{\mathrm{d}x}{x} j_\ell(x)^2, \qquad x := k\chi_*. \tag{44.25}$$

Using

$$\int_0^\infty \frac{\mathrm{d}x}{x} j_\ell(x)^2 = \frac{1}{2\ell(\ell+1)}, \qquad \ell \geq 1, \tag{44.26}$$

the spectrum is

$$C_\ell^{\mathrm{SW}} = \frac{4\pi A_s}{25}\frac{1}{2\ell(\ell+1)} \tag{44.27}$$

$$= \frac{2\pi A_s}{25\ell(\ell+1)}. \tag{44.28}$$

Thus

$$\boxed{C_\ell^{\mathrm{SW}} = \frac{2\pi A_s}{25\ell(\ell+1)}.} \tag{44.29}$$

Hence

$$\boxed{\frac{\ell(\ell+1)}{2\pi}C_\ell^{\mathrm{SW}} = \frac{A_s}{25}.} \tag{44.30}$$

For a statistically isotropic Gaussian sky, $\widehat{C}_\ell$ averages the $2\ell+1$ independent m-mode powers. Its variance is

$$\mathrm{Var}(\widehat{C}_\ell) = \frac{2C_\ell^2}{2\ell+1}, \tag{44.31}$$

so the full-sky cosmic-variance standard deviation is

$$\sigma(C_\ell) = \sqrt{\frac{2}{2\ell+1}}C_\ell. \tag{44.32}$$

44.4 Conditional low-ℓ benchmark

For

$$A_s = 2.1 \times 10^{-9}, \qquad T_0 = 2.7255 \text{ K}, \tag{44.33}$$

The hierarchy gives

$$\frac{\ell(\ell+1)}{2\pi}C_\ell^{\mathrm{SW}} = 8.4 \times 10^{-11}, \tag{44.34}$$

$$\frac{\ell(\ell+1)}{2\pi}C_\ell^{\mathrm{SW}}T_0^2 = 623.981421\ \mu\mathrm{K}^2. \tag{44.35}$$

For $2 \le \ell \le 40$, the maximum relative numerical error in the Bessel integral is

$$\varepsilon_{j^2/x} = 2.498501 \times 10^{-7}. \tag{44.36}$$

The geometric scales defined above are

$$\ell_A = 340.781006, \tag{44.37}$$

$$\ell_D := \frac{\chi_*}{r_D} = 2995.643251. \tag{44.38}$$

They are not used here to claim a complete acoustic peak spectrum.

44.5 Next steps

- Add the Thomson collision term and recover the acoustic peak structure.
- Extend the hierarchy to temperature and polarization multipoles.
- Include metric driving, integrated Sachs–Wolfe contributions, and reionization.
- Add gravitational lensing and the corresponding remapping of the angular spectra.
- Build a precision likelihood for comparison with observed CMB spectra.

Chapter 45

Linear Matter Growth and First Structure Observables

Linear density growth supplies the first late-time structure observable. The continuum growth equation and its projector discretization are compared in Einstein–de Sitter and flat ΛCDM backgrounds.

45.1 Subhorizon dust system

For pressureless matter,

$$\dot{\delta} = -\frac{\theta}{a}, \tag{45.1}$$

$$\dot{\theta} + H\theta = -\frac{1}{a}\nabla^2\Phi, \tag{45.2}$$

$$\nabla^2\Phi = 4\pi G a^2 \bar{\rho}_m \delta. \tag{45.3}$$

Differentiate the continuity equation:

$$\ddot{\delta} = -\frac{\dot{\theta}}{a} + \frac{H\theta}{a}. \tag{45.4}$$

Using the Euler equation in the form

$$\dot{\theta} = -H\theta - \frac{1}{a}\nabla^2\Phi \tag{45.5}$$

gives

$$\ddot{\delta} = \frac{H\theta}{a} + \frac{1}{a^2}\nabla^2\Phi + \frac{H\theta}{a} \tag{45.6}$$

$$= -2H\dot{\delta} + \frac{1}{a^2}\nabla^2\Phi. \tag{45.7}$$

The Poisson equation then yields

$$\frac{1}{a^2}\nabla^2\Phi = 4\pi G\bar{\rho}_m\delta, \tag{45.8}$$

so elimination of θ and Φ gives

$$\boxed{\ddot{\delta} + 2H\dot{\delta} - 4\pi G\bar{\rho}_m\delta = 0.} \tag{45.9}$$

For the growth factor $D(a)$,

$$\dot{D} = aHD_{,a}, \tag{45.10}$$

$$\ddot{D} = a^2H^2D_{,aa} + \left(aH^2 + a^2HH_{,a}\right)D_{,a}. \tag{45.11}$$

Substitution into the time-domain growth equation gives

$$a^2H^2D_{,aa} + \left(3aH^2 + a^2HH_{,a}\right)D_{,a} - 4\pi G\bar{\rho}_m D = 0. \tag{45.12}$$

Using

$$4\pi G\bar{\rho}_m = \frac{3}{2}\Omega_{m0}H_0^2 a^{-3} \tag{45.13}$$

and dividing by a^2H^2 yields

$$\boxed{D_{,aa} + \left(\frac{3}{a} + \frac{H_{,a}}{H}\right) D_{,a} - \frac{3\Omega_{m0}H_0^2}{2a^5 H(a)^2} D = 0.} \tag{45.14}$$

With $N = \log a$,

$$D_{,a} = \frac{1}{a} D_{,N}, \tag{45.15}$$

$$D_{,aa} = \frac{1}{a^2}\left(D_{,NN} - D_{,N}\right), \tag{45.16}$$

$$\frac{aH_{,a}}{H} = \frac{\mathrm{d}\log H}{\mathrm{d}N}, \tag{45.17}$$

$$\Omega_m(a) = \frac{\Omega_{m0}H_0^2 a^{-3}}{H(a)^2}. \tag{45.18}$$

Multiplying the a-equation by a^2 and inserting these identities gives

$$\boxed{D_{,NN} + \left[2 + \frac{\mathrm{d}\log H}{\mathrm{d}N}\right] D_{,N} - \frac{3}{2}\Omega_m(a) D = 0.} \tag{45.19}$$

45.2 Einstein–de Sitter benchmark

For

$$\Omega_m(a) = 1, \qquad \frac{\mathrm{d}\log H}{\mathrm{d}N} = -\frac{3}{2}, \tag{45.20}$$

a power law $D \propto a^r = \mathrm{e}^{rN}$ gives the characteristic equation

$$r^2 + \frac{1}{2}r - \frac{3}{2} = 0, \tag{45.21}$$

$$(r-1)\left(r + \frac{3}{2}\right) = 0. \tag{45.22}$$

Hence

$$\boxed{D_+(a) = a, \qquad D_-(a) = a^{-3/2}.} \tag{45.23}$$

The midpoint solver converges with

$$p_{\mathrm{EdS}} = 1.968111. \tag{45.24}$$

45.3 Flat ΛCDM growth

Let

$$E(a)^2 = \Omega_{m0}a^{-3} + 1 - \Omega_{m0}, \tag{45.25}$$

$$\Omega_m(a) = \frac{\Omega_{m0}a^{-3}}{E(a)^2}. \tag{45.26}$$

Define

$$I(a) := \int_0^a \frac{\mathrm{d}a'}{a'^3 E(a')^3}, \qquad I_{,a} = \frac{1}{a^3 E(a)^3}. \tag{45.27}$$

Reduction of order about the decaying solution proportional to $E(a)$ gives a second solution proportional to $E(a)I(a)$. Normalizing it to the matter-era behavior $D_+ \sim a$ fixes the coefficient, so

$$\boxed{D_+(a) = \frac{5\Omega_{m0}}{2} E(a) \int_0^a \frac{\mathrm{d}a'}{a'^3 E(a')^3}.} \tag{45.28}$$

For $\Omega_{m0} = 0.3$, the growth ODE and integral solution agree with trajectory residual

$$\varepsilon_{\mathrm{Heath}} = 1.035117 \times 10^{-12}. \tag{45.29}$$

The midpoint convergence order is

$$p_{\Lambda\mathrm{CDM}} = 1.970169. \tag{45.30}$$

45.4 Consistent projector Laplacian

Let

$$\Delta_h \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}} = -\widehat{k}_h^2 \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}. \tag{45.31}$$

The lattice Poisson equation is

$$-\widehat{k}_h^2 \Phi_h = 4\pi G a^2 \bar{\rho}_m \delta_h. \tag{45.32}$$

The lattice Euler force is

$$\dot{\theta}_h + H\theta_h = \frac{\widehat{k}_h^2}{a}\Phi_h. \tag{45.33}$$

Therefore

$$\boxed{\frac{\widehat{k}_h^2}{a}\Phi_h = -4\pi G a \bar{\rho}_m \delta_h.} \tag{45.34}$$

Differentiating $\dot{\delta}_h = -\theta_h/a$ gives

$$\ddot{\delta}_h = -\frac{\dot{\theta}_h}{a} + \frac{H\theta_h}{a} \tag{45.35}$$

$$= -2H\dot{\delta}_h - \frac{\widehat{k}_h^2}{a^2}\Phi_h. \tag{45.36}$$

The discrete Poisson relation now removes the same symbol that appears in the Euler force:

$$-\frac{\widehat{k}_h^2}{a^2}\Phi_h = 4\pi G \bar{\rho}_m \delta_h. \tag{45.37}$$

Hence the common projector Laplacian cancels exactly:

$$\boxed{\ddot{\delta}_h + 2H\dot{\delta}_h - 4\pi G \bar{\rho}_m \delta_h = 0.} \tag{45.38}$$

Thus a consistent Euler–Poisson pair introduces no artificial scale dependence in pressureless linear growth.

45.5 Growth observables

Define

$$f(a) := \frac{\mathrm{d}\log D}{\mathrm{d}\log a} = \frac{(1/D)\,\mathrm{d}D}{\mathrm{d}N} = \frac{D_{,N}}{D}, \tag{45.39}$$

$$D_N(a) := \frac{D(a)}{D(1)}, \tag{45.40}$$

$$\sigma_8(a) = D_N(a)\sigma_{8,0}, \tag{45.41}$$

$$f\sigma_8(a) := f(a)\sigma_8(a) = f(a)D_N(a)\sigma_{8,0}. \tag{45.42}$$

For

$$\Omega_{m0} = 0.3, \qquad \sigma_{8,0} = 0.8, \tag{45.43}$$

the controlled benchmark gives

$$f(1) = 0.512796, \tag{45.44}$$

$$f\sigma_8(1) = 0.410237. \tag{45.45}$$

These values are conditional on the stated background parameters and normalization, not an observational fit.

45.6 Next steps

- Add radiation, baryon, and neutrino transfer functions to the projector growth system.
- Extend the evolution through the superhorizon regime and include free-streaming species.
- Normalize the matter power spectrum consistently from the primordial sector.
- Map the predicted growth history to survey observables and a reproducible likelihood.
- Develop the nonlinear fluid or N-body limit while preserving the projector constraints.

Chapter 46

Adic Refinement Dark Sector: Background, Growth, and Lensing

This chapter turns the gapped Haar-detail branch of the refinement calculus into a concrete cosmological model. The construction is conditional in one important respect: the microscopic event that populates the detail modes is represented by a specified nonadiabatic gap crossing. Once that state is fixed, the level masses, multiplicities, homogeneous density, pressure, cold-limit growth, lensing source, finite-depth limit, and prime-refinement scaling are derived explicitly.

46.1 Covariant fiber action and the dark tower

Fix a prime p and use $\mathbb{Z}_p$ as an internal refinement coordinate. Let D_0 denote the constant Haar sector and let D_h, $h \geq 1$, denote the orthogonal detail projectors. Their multiplicities are

$$d_h := \dim \operatorname{Ran} \mathsf{D}_h = (p-1)p^{h-1}. \tag{46.1}$$

For refinement order $\sigma > 0$, define the positive fiber operator

$$\mathcal{L}_{\sigma,p} := \Lambda^2 \sum_{h \geq 1} p^{2\sigma h} \mathsf{D}_h. \tag{46.2}$$

The coefficient follows from the internal length law

$$\ell_h = \ell_0 p^{-h}. \tag{46.3}$$

A derivative of order σ scales as $\ell_h^{-\sigma}$, and its quadratic energy therefore scales as

$$\ell_h^{-2\sigma} = \ell_0^{-2\sigma} p^{2\sigma h}. \tag{46.4}$$

Within the class of level-diagonal, self-similar quadratic actions, the recursion

$$\lambda_{h+1} = p^{2\sigma} \lambda_h \tag{46.5}$$

forces

$$\lambda_h = \Lambda^2 p^{2\sigma h}. \tag{46.6}$$

Let $\Phi(x,u)$ be a real field on $M^{3,1} \times \mathbb{Z}_p$. The minimal covariant action is

$$S_\Phi = -\frac{1}{2} \int_{M^{3,1}} \sqrt{-g}\, \mathrm{d}^4x \int_{\mathbb{Z}_p} \mathrm{d}\mu_p(u) \Big[g^{\mu\nu} \partial_\mu \Phi\, \partial_\nu \Phi + m_0^2 \Phi^2 + \Phi \mathcal{L}_{\sigma,p} \Phi \Big]. \tag{46.7}$$

Choose an orthonormal Haar basis and expand

$$\Phi(x,u) = \phi_0(x) + \sum_{h \geq 1} \sum_{a=1}^{d_h} \chi_{h,a}(x) \psi_{h,a}(u). \tag{46.8}$$

Orthogonality gives

$$S_\Phi = -\frac{1}{2}\int \sqrt{-g}\ \mathrm{d}^4x \Bigg[(\partial\phi_0)^2 + m_0^2\phi_0^2 \tag{46.9}$$

$$+ \sum_{h\geq 1}\sum_{a=1}^{d_h} \left((\partial\chi_{h,a})^2 + m_h^2\chi_{h,a}^2 \right) \Bigg], \tag{46.10}$$

where

$$\boxed{m_h^2 = m_0^2 + \Lambda^2 p^{2\sigma h}, \qquad d_h = (p-1)p^{h-1}.} \tag{46.11}$$

Thus one fiber average and a countable massive detail tower follow from the same action.

If ordinary visible interactions use only the fiber average, then

$$\int_{\mathbb{Z}_p} \Phi(x,u)\ \mathrm{d}\mu_p(u) = \phi_0(x), \tag{46.12}$$

because every nonconstant Haar function has zero mean. The $\chi_{h,a}$ modes then have no linear coupling to such visible operators, while their stress tensor still contributes to Einstein's equation.

For a fiber-constant background $\bar{\Phi}(x,u) = \bar{\phi}(x)$, the fiber Hessian is

$$\mathcal{A}_{\mathrm{fib}}[g,\bar{\phi}] = m_0^2 + \xi R + U''(\bar{\phi}) + \mathcal{L}_{\sigma,p}. \tag{46.13}$$

Its spectral projectors are exactly $\mathsf{D}_0, \mathsf{D}_1, \ldots$. In this model the visible average is therefore the isolated lowest fiber band rather than an arbitrary basis choice.

46.2 Conditional nonadiabatic production law

The action fixes the spectrum but not the cosmological state. To obtain a definite benchmark, suppose that the selector gap undergoes one quadratic nonadiabatic crossing at scale factor $a_\star$. For a mode with physical momentum k_{phys}, take

$$\omega_{h,k}^2(t) = m_h^2 + k_{\mathrm{phys}}^2 + \nu^2(t-t_\star)^2, \tag{46.14}$$

where ν has mass-squared dimension. The complex turning points satisfy

$$t_\pm - t_\star = \pm\frac{\mathrm{i}\sqrt{m_h^2 + k_{\mathrm{phys}}^2}}{\nu}. \tag{46.15}$$

The leading complex-WKB exponent is

$$2\,\mathrm{Im}\int_{t_-}^{t_+} \omega_{h,k}(t)\ \mathrm{d}t = \frac{2}{\nu}\int_{-\sqrt{m_h^2+k_{\mathrm{phys}}^2}}^{\sqrt{m_h^2+k_{\mathrm{phys}}^2}} \sqrt{m_h^2 + k_{\mathrm{phys}}^2 - y^2}\ \mathrm{d}y \tag{46.16}$$

$$= \frac{\pi(m_h^2 + k_{\mathrm{phys}}^2)}{\nu}. \tag{46.17}$$

Writing the conserved comoving momentum as $q = a_\star k_{\mathrm{phys}}$, the post-transition occupation is

$$\boxed{f_h(q) = \mathcal{N}_\star \exp\left[-\frac{\pi m_h^2}{\nu} - \frac{\pi q^2}{a_\star^2 \nu} \right].} \tag{46.18}$$

The normalization $\mathcal{N}_\star$, the transition scale $a_\star$, and the rate ν remain model inputs in this chapter. The exponential mass suppression is sufficient to make the infinite level sum convergent for the benchmark considered below.

46.3 Exact homogeneous density and pressure

For the conformal energy

$$\epsilon_h(q,a) = \sqrt{q^2 + a^2 m_h^2}, \tag{46.19}$$

the level density and pressure are

$$\rho_h(a) = \frac{d_h}{2\pi^2 a^4} \int_0^\infty q^2 \epsilon_h(q,a) f_h(q)\, \mathrm{d}q, \tag{46.20}$$

$$p_h(a) = \frac{d_h}{6\pi^2 a^4} \int_0^\infty \frac{q^4}{\epsilon_h(q,a)} f_h(q)\, \mathrm{d}q. \tag{46.21}$$

Set

$$x := \frac{q}{a_\star \sqrt{\nu}}, \qquad y_h(a) := \frac{a m_h}{a_\star \sqrt{\nu}}. \tag{46.22}$$

Define the dimensionless moments

$$I_\rho(y) := \int_0^\infty x^2 \sqrt{x^2 + y^2}\, \mathrm{e}^{-\pi x^2}\, \mathrm{d}x, \tag{46.23}$$

$$I_p(y) := \int_0^\infty \frac{x^4}{\sqrt{x^2 + y^2}}\, \mathrm{e}^{-\pi x^2}\, \mathrm{d}x, \tag{46.24}$$

$$J_p(y) := \int_0^\infty \frac{x^4 y^2}{(x^2 + y^2)^{3/2}}\, \mathrm{e}^{-\pi x^2}\, \mathrm{d}x. \tag{46.25}$$

Then

$$\rho_h(a) = \frac{\mathcal{N}_\star d_h a_\star^4 \nu^2}{2\pi^2 a^4} \mathrm{e}^{-\pi m_h^2/\nu} I_\rho(y_h), \tag{46.26}$$

$$p_h(a) = \frac{\mathcal{N}_\star d_h a_\star^4 \nu^2}{6\pi^2 a^4} \mathrm{e}^{-\pi m_h^2/\nu} I_p(y_h). \tag{46.27}$$

The moments also have exact Tricomi hypergeometric forms. Substituting $x = y\sqrt{t}$ gives

$$I_\rho(y) = \frac{y^4}{2} \Gamma\left(\frac{3}{2}\right) U\left(\frac{3}{2}, 3, \pi y^2\right), \tag{46.28}$$

$$I_p(y) = \frac{y^4}{2} \Gamma\left(\frac{5}{2}\right) U\left(\frac{5}{2}, 3, \pi y^2\right). \tag{46.29}$$

Consequently,

$$\rho_h(a) = \frac{\mathcal{N}_\star d_h m_h^4}{4\pi^2} \mathrm{e}^{-\pi m_h^2/\nu} \Gamma\left(\frac{3}{2}\right) U\left(\frac{3}{2}, 3, \pi y_h^2\right), \tag{46.30}$$

$$p_h(a) = \frac{\mathcal{N}_\star d_h m_h^4}{12\pi^2} \mathrm{e}^{-\pi m_h^2/\nu} \Gamma\left(\frac{5}{2}\right) U\left(\frac{5}{2}, 3, \pi y_h^2\right). \tag{46.31}$$

The total dark density, pressure, and equation of state are

$$\rho_D := \sum_{h \geq 1} \rho_h, \qquad p_D := \sum_{h \geq 1} p_h, \qquad w_D := \frac{p_D}{\rho_D}. \tag{46.32}$$

46.3.1 Exact continuity equation

Differentiate the level density with respect to $N = \log a$. Since

$$\frac{\mathrm{d}\epsilon_h}{\mathrm{d}N} = \frac{a^2 m_h^2}{\epsilon_h} = \epsilon_h - \frac{q^2}{\epsilon_h}, \tag{46.33}$$

one obtains

$$\frac{\mathrm{d}\rho_h}{\mathrm{d}N} = -4\rho_h + \frac{d_h}{2\pi^2 a^4} \int q^2 \left(\epsilon_h - \frac{q^2}{\epsilon_h}\right) f_h\, \mathrm{d}q \tag{46.34}$$

$$= -4\rho_h + \rho_h - 3p_h \tag{46.35}$$
$$= -3(\rho_h + p_h). \tag{46.36}$$

Therefore

$$\boxed{\frac{\mathrm{d}\rho_D}{\mathrm{d}N} + 3(\rho_D + p_D) = 0.} \tag{46.37}$$

No exchange term is inserted after production.

46.3.2 Equation of state and adiabatic sound speed

At one level,

$$w_h(a) = \frac{I_p(y_h)}{3I_\rho(y_h)}. \tag{46.38}$$

Differentiating the pressure gives

$$\frac{\mathrm{d}p_h}{\mathrm{d}N} = -\frac{\mathcal{C}_h(a)}{3}\left[4I_p(y_h) + J_p(y_h)\right], \tag{46.39}$$

where

$$\mathcal{C}_h(a) := \frac{\mathcal{N}_\star d_h a_\star^4 \nu^2}{2\pi^2 a^4} \mathrm{e}^{-\pi m_h^2/\nu}. \tag{46.40}$$

Since

$$\frac{\mathrm{d}\rho_h}{\mathrm{d}N} = -\mathcal{C}_h(a)\left[3I_\rho(y_h) + I_p(y_h)\right], \tag{46.41}$$

the adiabatic sound speed is

$$\boxed{c_{a,h}^2 := \frac{\mathrm{d}p_h/\,\mathrm{d}N}{\mathrm{d}\rho_h/\,\mathrm{d}N} = \frac{4I_p(y_h) + J_p(y_h)}{3\left[3I_\rho(y_h) + I_p(y_h)\right]}.} \tag{46.42}$$

For $y_h \ll 1$,

$$w_h \longrightarrow \frac{1}{3}, \qquad c_{a,h}^2 \longrightarrow \frac{1}{3}, \qquad \rho_h \propto a^{-4}. \tag{46.43}$$

For $y_h \gg 1$,

$$w_h(a) = \frac{a_\star^2 \nu}{2\pi a^2 m_h^2} + \mathcal{O}(y_h^{-4}), \tag{46.44}$$
$$c_{a,h}^2 = \frac{5}{3} w_h + \mathcal{O}(y_h^{-4}), \tag{46.45}$$
$$\rho_h(a) = m_h n_h(a)\left[1 + \mathcal{O}(y_h^{-2})\right], \tag{46.46}$$

where the conserved number density is

$$\boxed{n_h(a) = \frac{\mathcal{N}_\star d_h a_\star^3 \nu^{3/2}}{8\pi^3 a^3} \mathrm{e}^{-\pi m_h^2/\nu}.} \tag{46.47}$$

The produced tower therefore becomes pressureless without changing the level masses or adding a separate dust component.

46.4 Friedmann equation and controlled benchmark

Normalize the overall occupation by requiring

$$\rho_D(1) = 3M_{\mathrm{Pl}}^2 H_0^2 \Omega_{D0}. \tag{46.48}$$

If $\widetilde{\rho}_D$ denotes the density with $\mathcal{N}_\star = 1$, then

$$\mathcal{N}_\star = \frac{3M_{\mathrm{Pl}}^2 H_0^2 \Omega_{D0}}{\widetilde{\rho}_D(1)}. \tag{46.49}$$

For a flat radiation, baryon, dark-tower, and cosmological-constant background,

$$\boxed{E(a)^2 = \Omega_{r0}a^{-4} + \Omega_{b0}a^{-3} + \Omega_{D0}\frac{\rho_D(a)}{\rho_D(1)} + \Omega_{\Lambda 0}.} \tag{46.50}$$

Using the continuity equation,

$$\frac{\mathrm{d}\log H}{\mathrm{d}\log a} = -\frac{1}{2}\left[4\Omega_r(a) + 3\Omega_b(a) + 3(1 + w_D(a))\Omega_D(a)\right]. \tag{46.51}$$

The controlled numerical benchmark uses

$$p = 2, \qquad \sigma = 1, \qquad H_{\max} = 8, \qquad \Lambda = \frac{1}{2}, \tag{46.52}$$

$$\nu = 2\pi, \qquad a_\star = 10^{-7}, \qquad \Omega_{r0} = 9 \times 10^{-5}, \qquad \Omega_{b0} = 0.049, \tag{46.53}$$

$$\Omega_{D0} = 0.261, \qquad \Omega_{\Lambda 0} = 1 - \Omega_{r0} - \Omega_{b0} - \Omega_{D0}. \tag{46.54}$$

The internal mass unit has not been identified with an observed particle mass. The first three masses and present dark-sector fractions are

level	m_h	d_h	$\rho_h(1)/\rho_D(1)$
1	1	1	0.5259366
2	2	2	0.4694092
3	4	4	0.0046542

The remaining levels contribute below the displayed precision. The total equation of state evolves as

$$w_D(a_\star) = 0.1961573, \qquad w_D(10^{-6}) = 0.0062439, \tag{46.55}$$

$$w_D(10^{-5}) = 6.4338 \times 10^{-5}, \qquad w_D(10^{-3}) = 6.4358 \times 10^{-9}. \tag{46.56}$$

Thus the tower is already extremely cold by the epoch used for the perturbation calculation below.

46.5 Finite-depth convergence and prime scaling

At finite depth H, the number of nonconstant Haar states is

$$N_H = \sum_{h=1}^{H}(p-1)p^{h-1} = p^H - 1. \tag{46.57}$$

For $m_0 = 0$ and $\sigma = 1$,

$$m_H = \Lambda p^H, \tag{46.58}$$

so the endpoint counting law is

$$\boxed{N_H = \frac{m_H}{\Lambda} - 1.} \tag{46.59}$$

Its continuum interpolation has constant density of states

$$\boxed{\rho_{\mathrm{DOS}}(m) = \frac{\mathrm{d}N}{\mathrm{d}m} = \frac{1}{\Lambda}.} \tag{46.60}$$

The prime controls the logarithmic spacing of the levels, but it cancels from the cumulative endpoint law.
The conditional production factor gives the late-time level weight

$$T_h \propto d_h m_h \mathrm{e}^{-\pi m_h^2/\nu}. \tag{46.61}$$

For $\sigma = 1$, the ratio is

$$\frac{T_{h+1}}{T_h} = p^2 \exp\left[-\frac{\pi\Lambda^2}{\nu}(p^2-1)p^{2h}\right]. \tag{46.62}$$

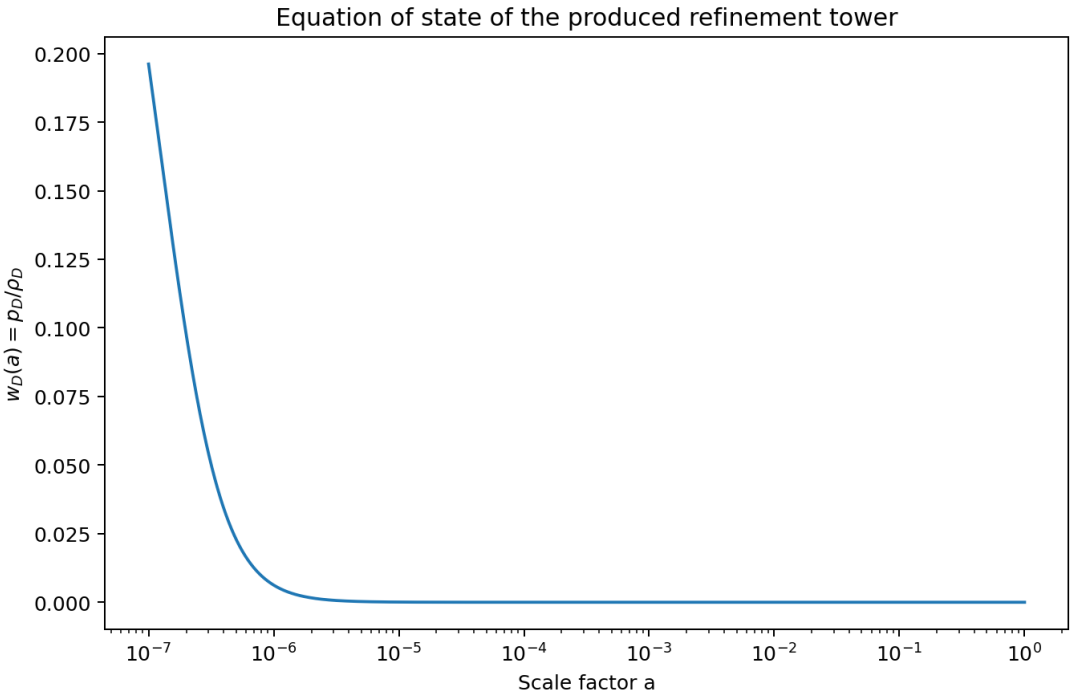


Figure 46.1: The conditional production state begins with a nonzero pressure and rapidly approaches the pressureless regime.

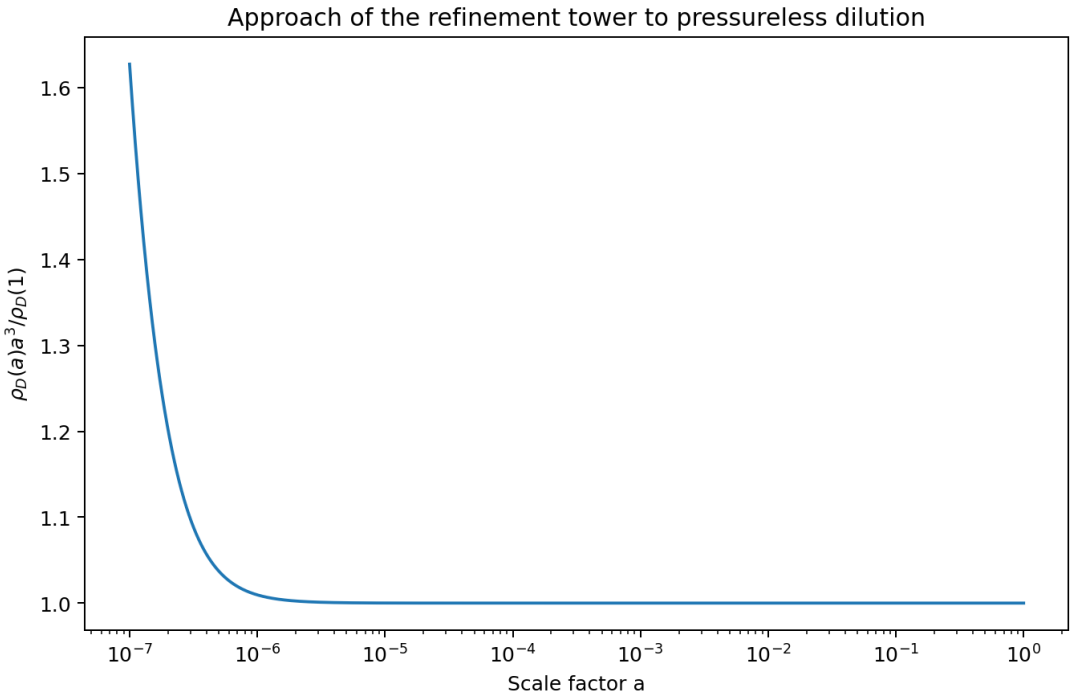


Figure 46.2: The quantity $\rho_D a^3$ approaches a constant once the populated refinement levels become nonrelativistic.

Once this ratio is below one, the omitted tail is bounded by a geometric majorant. In the dyadic benchmark, the maximum density errors relative to depth twelve are

$$H = 1: \quad 4.740634 \times 10^{-1}, \tag{46.63}$$
$$H = 2: \quad 4.654197 \times 10^{-3}, \tag{46.64}$$
$$H = 3: \quad 7.030165 \times 10^{-13}, \tag{46.65}$$
$$H \geq 4: \quad 2.3 \times 10^{-16} \text{ or smaller.} \tag{46.66}$$

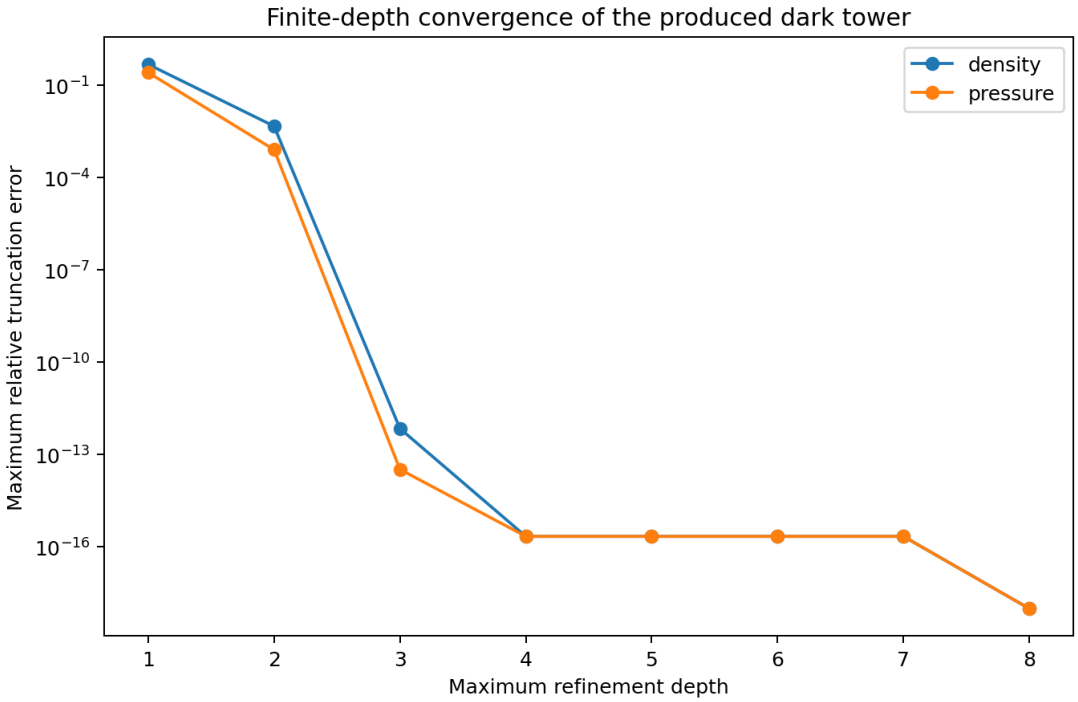


Figure 46.3: The Gaussian mass suppression makes the finite-depth background converge super-exponentially for the stated production law.

The static tower kernel is

$$K_{p,H}(k) := \sum_{h=1}^{H} \frac{d_h}{k^2 + \Lambda^2 p^{2h}}. \tag{46.67}$$

For an ideal bilateral hierarchy, index shifting gives

$$K_p(pk) = \sum_h \frac{(p-1)p^{h-1}}{p^2k^2 + \Lambda^2 p^{2h}} \tag{46.68}$$
$$= p^{-1} K_p(k). \tag{46.69}$$

Therefore

$$K_p(k) = \frac{1}{k} F_p\left(\frac{\log k}{\log p}\right), \tag{46.70}$$

where F_p is periodic with period one. The continuum envelope is $1/k$, while the discrete tower retains a prime-dependent log-periodic modulation. Interior finite-tower fits for $p = 2, 3, 5, 7$ give log slopes equal to -1 within 4.5×10^{-16}.

This is not a proof that cosmological observables are prime-independent. The exact result is narrower: the endpoint state count and the scale-free kernel exponent are prime-independent, while the discrete modulation remains a possible observable or a quantity to be averaged in an adelic theory.

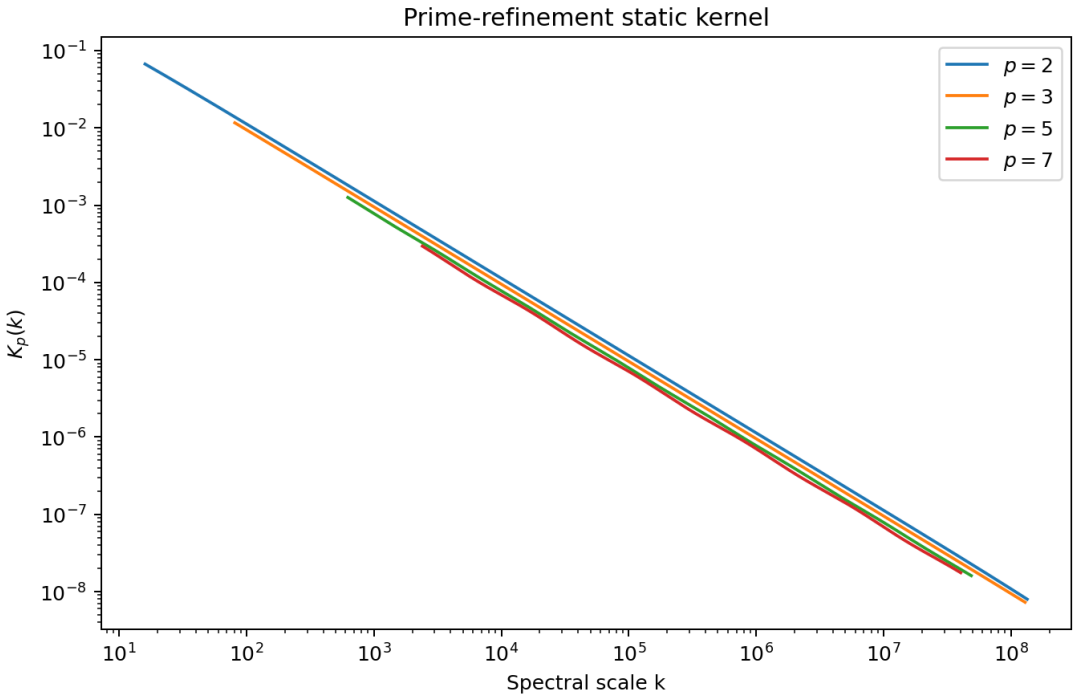


Figure 46.4: The different prime towers share the $1/k$ envelope but retain different logarithmic sampling and normalization.

46.6 Cold subhorizon perturbation closure

A full collisionless treatment requires a Boltzmann hierarchy for every populated mass level. The present calculation uses the controlled cold closure beginning at

$$a_i = 10^{-3}, \tag{46.71}$$

where the benchmark satisfies

$$\max_h w_h(a_i) = 9.937370 \times 10^{-9}, \qquad \max_h c_{a,h}^2(a_i) = 1.656228 \times 10^{-8}. \tag{46.72}$$

Let $N = \log a$, and let a prime denote $\mathrm{d}/\mathrm{d}N$ within this section. For baryons and the populated detail levels, use

$$\delta_b'' + \left(2 + \frac{\mathrm{d}\log H}{\mathrm{d}N}\right)\delta_b' = \frac{3}{2}\mathcal{S}, \tag{46.73}$$

$$\delta_h'' + \left(2 + \frac{\mathrm{d}\log H}{\mathrm{d}N}\right)\delta_h' + \frac{c_{a,h}^2 k^2}{a^2 H^2}\delta_h = \frac{3}{2}\mathcal{S}, \tag{46.74}$$

where

$$\mathcal{S}(a,k) := \Omega_b(a)\delta_b(a,k) + \sum_h \Omega_h(a)\delta_h(a,k). \tag{46.75}$$

The resolved dark contrast is

$$\boxed{\delta_D(a,k) = \frac{\sum_h \rho_h(a)\delta_h(a,k)}{\rho_D(a)}.} \tag{46.76}$$

The closure becomes ordinary pressureless growth as $c_{a,h}^2 \to 0$.
The benchmark uses adiabatic growing initial data

$$\delta_b(a_i) = \delta_h(a_i) = a_i, \qquad \delta_b'(a_i) = \delta_h'(a_i) = a_i. \tag{46.77}$$

At $a = 1$, the resulting ratios to a pressureless reference with the same present matter density are

k/H_0	$\delta_D/\delta_{\mathrm{CDM}}$	lensing-source ratio
30	0.9999999926	0.9999999926
3000	0.9999031151	0.9999037029
30000	0.9903580348	0.9904163176
100000	0.8980039088	0.8985975916

The large scales recover cold dark matter, while the residual velocity dispersion suppresses sufficiently small scales.

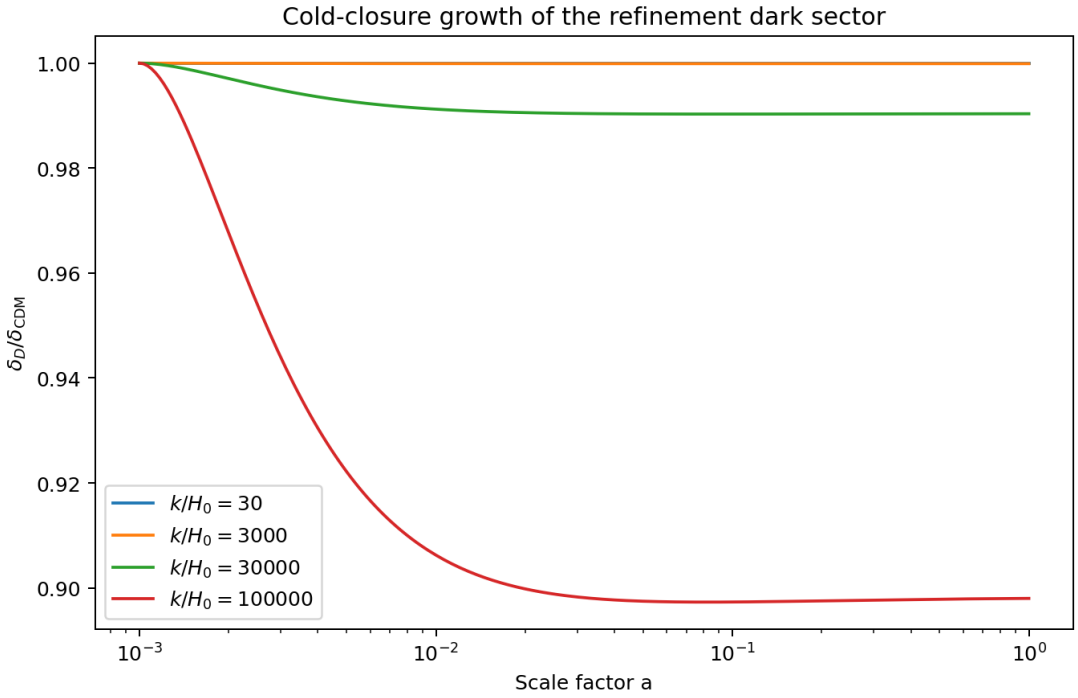


Figure 46.5: The cold closure reproduces pressureless growth at large scale and gives a controlled small-scale suppression for the chosen production parameters.

46.7 Lensing potential

Use Newtonian gauge,

$$\mathrm{d}s^2 = -(1+2\Psi)\,\mathrm{d}t^2 + a^2(1-2\Phi)\,\mathrm{d}\boldsymbol{x}^2. \tag{46.78}$$

In the cold isotropic closure, anisotropic stress is of relative order w_D, so

$$\Phi - \Psi = \mathcal{O}(w_D\Phi). \tag{46.79}$$

The subhorizon Einstein constraint gives

$$\boxed{k^2(\Phi+\Psi) = -8\pi G a^2\left[\rho_b\delta_b + \sum_h \rho_h\delta_h\right] + \mathcal{O}(w_D).} \tag{46.80}$$

In units where every density is expressed in the present critical density and k is expressed in H_0, this is

$$\Phi + \Psi = -\frac{3a^2}{k^2}\left[\Omega_{b0}a^{-3}\delta_b + \sum_h \rho_h\delta_h\right] + \mathcal{O}(w_D). \tag{46.81}$$

The same resolved density that controls growth therefore controls lensing in this approximation. No separate lensing rule is introduced.

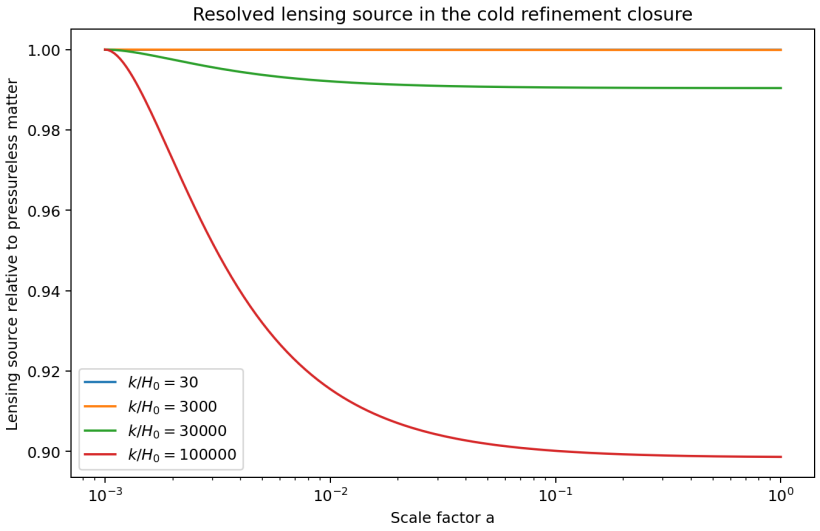


Figure 46.6: The lensing source follows the same large-scale cold limit and the same small-scale suppression as the resolved dark density.

46.8 What this calculation establishes

Within the specified action and production state, the following quantities have now been derived rather than inserted independently:

$$m_h^2 = m_0^2 + \Lambda^2 p^{2h}, \qquad d_h = (p-1)p^{h-1}, \tag{46.82}$$

$$\rho_D(a) = \sum_h \rho_h(a), \qquad p_D(a) = \sum_h p_h(a), \tag{46.83}$$

$$w_D(a) = p_D/\rho_D, \qquad c_{a,h}^2 = \frac{\mathrm{d}p_h}{\mathrm{d}\rho_h}, \tag{46.84}$$

$$\delta_D(a,k) = \frac{\sum_h \rho_h \delta_h}{\rho_D}, \qquad k^2(\Phi+\Psi) = -8\pi G a^2 \delta\rho_{\mathrm{resolved}} + \mathcal{O}(w_D). \tag{46.85}$$

The model gives a radiation-to-matter transition, exact post-production continuity, a cold large-scale limit, scale-dependent small-scale suppression, a finite-depth limit, and a prime-refinement $1/k$ kernel envelope.

It does not yet predict the dark abundance from first principles. The normalization $\mathcal{N}_\star$, the transition rate ν, the transition epoch $a_\star$, and the conversion between the internal mass unit and physical energy remain to be derived from the microscopic gravitational theory. The growth calculation is a cold fluid closure rather than the complete collisionless Boltzmann hierarchy, and no observational likelihood has been evaluated.

46.9 Next steps

- Derive the gap crossing, $a_\star$, and ν from a concrete nonlinear projector or BFV gravitational action.
- Fix the physical value of Λ and the active prime content instead of treating the internal mass unit as free.
- Replace the cold closure by the full massive collisionless Boltzmann hierarchy, including anisotropic stress and the exact gravitational slip.
- Propagate the resulting transfer functions through recombination, matter growth, weak lensing, and a reproducible likelihood without renormalizing the parameters between sectors.
- Derive nonlinear halo solutions and determine whether the discrete log-periodic structure survives phase mixing and structure formation.
- Verify that direct and iterated prime refinement yield the same nonlinear cosmological stress tensor.

Chapter 47

Superseded Additive Refinement Dark-Sector Branch

47.1 Status and objective

Correction status. This chapter is retained as the first unified additive construction. Its algebraic identities and internal numerical checks remain valid within the stated model. Chapter 49 proves a conditional coldness–locking obstruction for the one-shell, energy-matched additive branch and supersedes its physical-viability claim.

The preceding refinement dark-sector construction derived a massive Haar-detail tower, its homogeneous kinetic stress tensor, cold-limit growth, lensing source, finite-depth limit, and prime scaling. Its accelerating component was still represented by an independent cosmological-constant term. The present block replaces that independent term by the potential energy of the same refinement field that generates the detail tower.

The construction is exact within a declared Gaussian detail closure. The average mode is nonlinear. The detail sector is retained to quadratic order in the Hessian of the same average-mode potential. The resulting theory gives one action for both sectors, an exact vacuum-like phase while the average mode is locked, a stable perturbation action, and a closed relation between the dark-matter density and the end of acceleration.

47.2 Average-detail decomposition and one action

Let

$$\mathsf{P}_0\Phi(x,u) := \int_{\mathbb{Z}_p} \Phi(x,v)\ \mathrm{d}\mu_p(v) =: \phi(x), \qquad \chi := (I-\mathsf{P}_0)\Phi, \tag{47.1}$$

so that

$$\int_{\mathbb{Z}_p} \chi(x,u)\ \mathrm{d}\mu_p(u) = 0. \tag{47.2}$$

Retain the refinement operator

$$\mathcal{L}_{\sigma,p} = \Lambda^2 \sum_{h\geq 1} p^{2\sigma h}\mathsf{D}_h, \qquad d_h = (p-1)p^{h-1}. \tag{47.3}$$

Choose the symmetry-breaking average-mode potential

$$U(\phi) = \frac{\lambda}{4}(\phi^2 - v^2)^2, \qquad \mu^2 := \lambda v^2, \qquad \rho_V := U(0) = \frac{\mu^4}{4\lambda}. \tag{47.4}$$

Its Hessian is

$$\boxed{U''(\phi) = 3\lambda\phi^2 - \mu^2.} \tag{47.5}$$

The minimal projector-locked Gaussian action is

$$S_{\mathrm{UDS}} = -\int \sqrt{-g}\ \mathrm{d}^4x \left[\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + U(\phi)\right. \tag{47.6}$$

$$+\frac{1}{2}\int_{\mathbb{Z}_p} \mathrm{d}\mu_p(u)\,(g^{\mu\nu}\partial_\mu\chi\partial_\nu\chi+\chi[\mathcal{L}_{\sigma,p}+U''(\phi)]\chi)\Bigg]. \tag{47.7}$$

The coupling between the average mode and the detail tower is therefore fixed by the same quartic parameter λ. It is not an independently inserted dark-sector interaction.

Expanding in a Haar basis gives

$$\chi(x,u)=\sum_{h\geq 1}\sum_{a=1}^{d_h}\chi_{h,a}(x)\psi_{h,a}(u), \tag{47.8}$$

with field-dependent masses

$$\boxed{m_h^2(\phi)=\Lambda^2p^{2\sigma h}-\mu^2+3\lambda\phi^2.} \tag{47.9}$$

At the locked point $\phi=0$,

$$m_h^2=\Lambda^2p^{2\sigma h}-\mu^2. \tag{47.10}$$

The regular gapped branch requires

$$\boxed{\Lambda^2p^{2\sigma}>\mu^2.} \tag{47.11}$$

At the true vacua $\phi=\pm v$,

$$U(\pm v)=0,\qquad m_h^2(\pm v)=\Lambda^2p^{2\sigma h}+2\mu^2>0. \tag{47.12}$$

Thus the zero of vacuum energy is fixed at the true vacuum. The locked origin carries the definite energy ρ_V, rather than an additional free Ω_Λ term.

47.3 The production law follows from an average-mode crossing

Suppose the average mode crosses the origin at $t=t_\star$,

$$\phi(t)=\dot\phi_\star(t-t_\star)+\mathcal{O}((t-t_\star)^2). \tag{47.13}$$

For one detail mode,

$$\omega_{h,k}^2(t)=k_{\mathrm{phys}}^2+m_h^2+3\lambda\dot\phi_\star^2(t-t_\star)^2+\mathcal{O}((t-t_\star)^3). \tag{47.14}$$

Define

$$\boxed{\nu:=\sqrt{3\lambda}\,|\dot\phi_\star|.} \tag{47.15}$$

The quadratic crossing used previously is therefore generated by the same potential. The leading crossing occupation has shape

$$\boxed{f_h(q)=\mathcal{N}_\star\exp\left[-\frac{\pi m_h^2}{\nu}-\frac{\pi q^2}{a_\star^2\nu}\right].} \tag{47.16}$$

For a vacuum single crossing, $\mathcal{N}_\star=1$. More generally, coherent amplification and backreaction determine $\mathcal{N}_\star$. It need not be fitted independently if the crossing is solved dynamically. Energy matching gives

$$\mathcal{N}_\star=\frac{\Delta\rho_\phi}{\widetilde{\rho}_D(a_\star)}, \tag{47.17}$$

where $\widetilde{\rho}_D$ is the produced detail energy computed with unit normalization and $\Delta\rho_\phi$ is the average-mode energy lost in the crossing. Complete capture corresponds to

$$\Delta\rho_\phi=\frac{1}{2}\dot\phi_{\star,-}^2\qquad(\dot\phi_{\star,+}=0). \tag{47.18}$$

This replaces an arbitrary dark abundance by a state-selection problem in the same action. Solving that nonlinear crossing remains a separate dynamical task.

47.4 Density, pressure, and the locking variance

For

$$\epsilon_h(q,a)=\sqrt{q^2+a^2m_h^2}, \tag{47.19}$$

the detail density and pressure remain

$$\rho_h(a)=\frac{d_h}{2\pi^2a^4}\int_0^\infty q^2\epsilon_h(q,a)f_h(q)\,\mathrm{d}q, \tag{47.20}$$

$$p_h(a)=\frac{d_h}{6\pi^2a^4}\int_0^\infty \frac{q^4}{\epsilon_h(q,a)}f_h(q)\,\mathrm{d}q. \tag{47.21}$$

The variance that enters the average-mode Hessian is

$$\boxed{X_h(a):=\langle\chi_h^2\rangle=\frac{d_h}{2\pi^2a^2}\int_0^\infty \frac{q^2}{\epsilon_h(q,a)}f_h(q)\,\mathrm{d}q.} \tag{47.22}$$

Set

$$\rho_D:=\sum_h\rho_h,\qquad p_D:=\sum_h p_h,\qquad X_D:=\sum_h X_h. \tag{47.23}$$

The post-production continuity law is unchanged:

$$\boxed{\frac{\mathrm{d}\rho_D}{\mathrm{d}\log a}+3(\rho_D+p_D)=0.} \tag{47.24}$$

In the nonrelativistic regime,

$$n_h(a)=\frac{\mathcal{N}_\star d_h a_\star^3\nu^{3/2}}{8\pi^3a^3}\exp\!\left(-\frac{\pi m_h^2}{\nu}\right), \tag{47.25}$$

so

$$\rho_D=\sum_h m_h n_h,\qquad X_D=\sum_h\frac{n_h}{m_h}. \tag{47.26}$$

Define the locking mass moment

$$\boxed{m_{\mathrm{lock}}^2:=\frac{\displaystyle\sum_{h\ge1}d_h m_h\exp(-\pi m_h^2/\nu)}{\displaystyle\sum_{h\ge1}d_h m_h^{-1}\exp(-\pi m_h^2/\nu)}.} \tag{47.27}$$

Then

$$\boxed{X_D=\frac{\rho_D}{m_{\mathrm{lock}}^2}} \tag{47.28}$$

in the cold limit. Every defining sum converges because the multiplicity grows as p^h, while the state weight decays as $\exp[-cp^{2\sigma h}]$.

47.5 Locked-vacuum theorem

Assume a parity-even detail state, so that the average detail field and every odd detail moment vanish. The homogeneous average-mode equation is

$$\ddot\phi+3H\dot\phi+\left[-\mu^2+3\lambda X_D(a)\right]\phi+\lambda\phi^3=0. \tag{47.29}$$

Define

$$\boxed{M_\phi^2(a):=3\lambda X_D(a)-\mu^2.} \tag{47.30}$$

If

$$M_\phi^2(a)>0, \tag{47.31}$$

then $\phi = 0$ is a stable homogeneous solution. On that branch,

$$\boxed{\rho_{\rm DE} = \rho_V = \frac{\mu^4}{4\lambda}, \qquad p_{\rm DE} = -\rho_V, \qquad w_{\rm DE} = -1.} \tag{47.32}$$

The Friedmann equation becomes

$$\boxed{H^2 = \frac{1}{3M_{\rm Pl}^2}\left(\rho_r + \rho_b + \rho_D + \rho_V\right).} \tag{47.33}$$

No separate cosmological-constant density is introduced inside this dark-sector model.

47.6 Perturbations, sound speed, and stability

On the locked background, the quadratic average-mode perturbation action is

$$S^{(2)}_{\delta\phi} = \frac{1}{2}\int {\rm d}t\ {\rm d}^3x\, a^3 \left[\delta\dot\phi^2 - \frac{(\nabla\delta\phi)^2}{a^2} - M_\phi^2(a)\delta\phi^2\right]. \tag{47.34}$$

Therefore

$$\boxed{Q_{\delta\phi} = 1 > 0, \qquad c^2_{s,\rm DE} = 1, \qquad M_\phi^2 > 0.} \tag{47.35}$$

There is no ghost or gradient instability on the locked branch.

For a canonical scalar,

$$\delta\rho_\phi = \dot\phi\,\delta\dot\phi - \dot\phi^2\Psi + U_{,\phi}\delta\phi, \tag{47.36}$$

$$\delta p_\phi = \dot\phi\,\delta\dot\phi - \dot\phi^2\Psi - U_{,\phi}\delta\phi. \tag{47.37}$$

At $\phi = 0$,

$$\dot\phi = 0, \qquad U_{,\phi} = 0, \tag{47.38}$$

so

$$\boxed{\delta\rho^{(1)}_{\rm DE} = \delta p^{(1)}_{\rm DE} = \delta q^{(1)}_{\rm DE} = 0.} \tag{47.39}$$

The average-mode fluctuation propagates with unit microscopic sound speed, but it does not source the metric at linear order while the background remains exactly locked. The detail tower continues to carry the clustering and lensing perturbations.

When M_ϕ^2 crosses zero, the origin becomes tachyonic. This is the intended release transition, not a ghost or gradient pathology.

47.7 Coincidence locking and the end of acceleration

In the cold regime,

$$M_\phi^2 = 3\lambda\frac{\rho_D}{m^2_{\rm lock}} - \mu^2. \tag{47.40}$$

The release density is

$$\boxed{\rho_{D,\rm exit} = \frac{\mu^2 m^2_{\rm lock}}{3\lambda}.} \tag{47.41}$$

Relative to the vacuum density,

$$\boxed{\frac{\rho_{D,\rm exit}}{\rho_V} = \frac{4m^2_{\rm lock}}{3\mu^2}.} \tag{47.42}$$

The quartic coupling cancels. The duration of the vacuum-like phase is controlled by a mass ratio fixed by the refinement spectrum and the production state.

Introduce

$$r_D(a) := \frac{\rho_D(a)}{\rho_V}, \qquad r_b(a) := \frac{\rho_b(a)}{\rho_V}, \qquad \xi := \frac{m^2_{\rm lock}}{\mu^2}. \tag{47.43}$$

The two inequalities are

$$\text{locked phase:} \qquad r_D > \frac{4\xi}{3}, \tag{47.44}$$

$$\text{accelerated expansion:} \qquad r_D + r_b < 2. \tag{47.45}$$

Hence a trapped accelerating interval exists whenever these inequalities overlap. For cold matter,

$$r_D(a) = r_{D0}a^{-3}, \qquad r_b(a) = r_{b0}a^{-3}, \tag{47.46}$$

and

$$\boxed{a_{\rm acc} = \left(\frac{r_{D0} + r_{b0}}{2}\right)^{1/3},} \tag{47.47}$$

$$\boxed{a_{\rm exit} = \left(\frac{3r_{D0}}{4\xi}\right)^{1/3}.} \tag{47.48}$$

Current trapping requires

$$\boxed{\frac{m_{\rm lock}}{\mu} < \sqrt{\frac{3r_{D0}}{4}}.} \tag{47.49}$$

This is a falsifiable internal consistency condition. A tower that is too heavy relative to the average-mode curvature cannot support the present vacuum-like phase.

47.8 Controlled late-time benchmark

Use the same present fractions as the preceding dark-tower benchmark:

$$\Omega_{r0} = 9 \times 10^{-5}, \qquad \Omega_{b0} = 0.049, \qquad \Omega_{D0} = 0.261, \qquad \Omega_{V0} = 0.68991. \tag{47.50}$$

Then

$$r_{D0} = 0.3783102144, \qquad r_{b0} = 0.0710237567. \tag{47.51}$$

Choose the dimensionless spectral benchmark

$$p = 2, \qquad \sigma = 1, \qquad \frac{m_1}{\mu} = 0.45, \qquad \frac{\nu}{\mu^2} = 0.3, \qquad H_{\rm max} = 8. \tag{47.52}$$

The required refinement scale is

$$\frac{\Lambda}{\mu} = 0.5482928050. \tag{47.53}$$

The first detail level dominates the produced locking and density moments, giving

$$\boxed{\frac{m_{\rm lock}}{\mu} = 0.45, \qquad \xi = 0.2025.} \tag{47.54}$$

The present trapping curvature is

$$\boxed{\frac{M_\phi^2(1)}{\mu^2} = 0.4011489421 > 0.} \tag{47.55}$$

The acceleration and release scales are

$$\boxed{a_{\rm acc} = 0.6079199831, \qquad a_{\rm exit} = 1.1189948843.} \tag{47.56}$$

Thus the benchmark is already accelerating, remains locked at the present epoch, and predicts a future release of the vacuum-like phase.

A negative control with $\xi = 0.40$ gives

$$\frac{M_\phi^2(1)}{\mu^2} = -0.2906683480, \tag{47.57}$$

so that heavier tower cannot support the same present dark fractions.

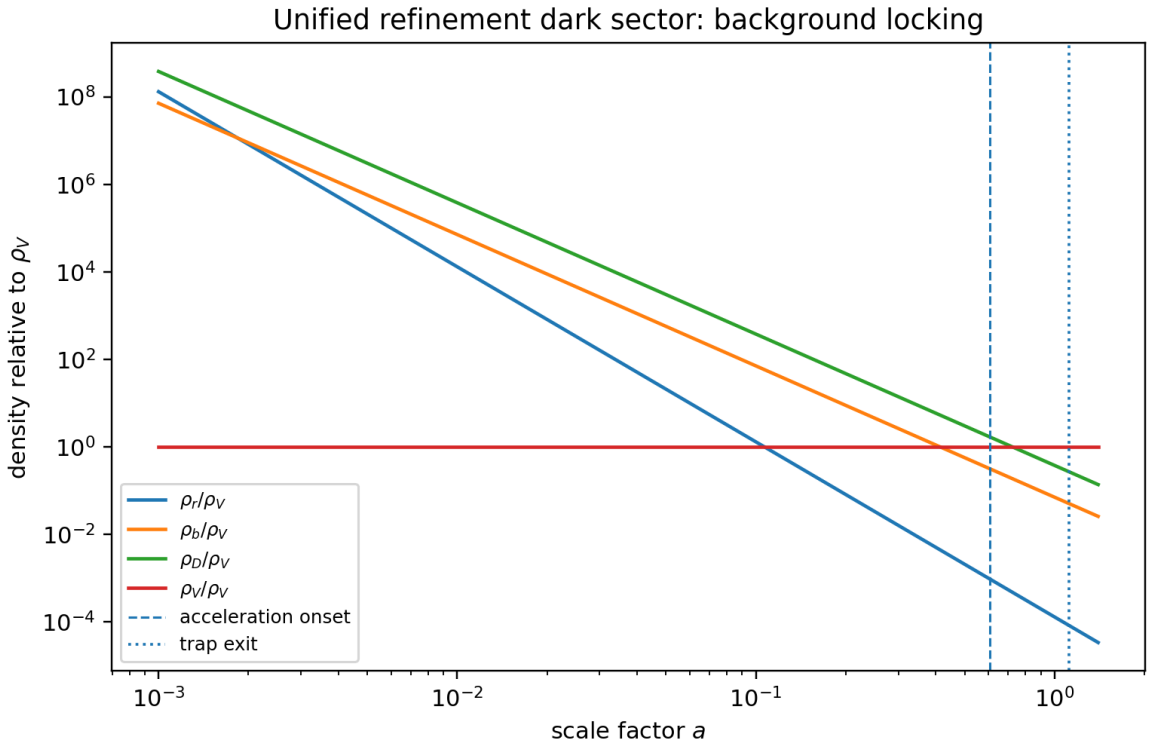


Figure 47.1: The detail tower behaves as matter while the locked average mode supplies a constant vacuum density. The benchmark begins accelerating before the trapping curvature reaches zero.

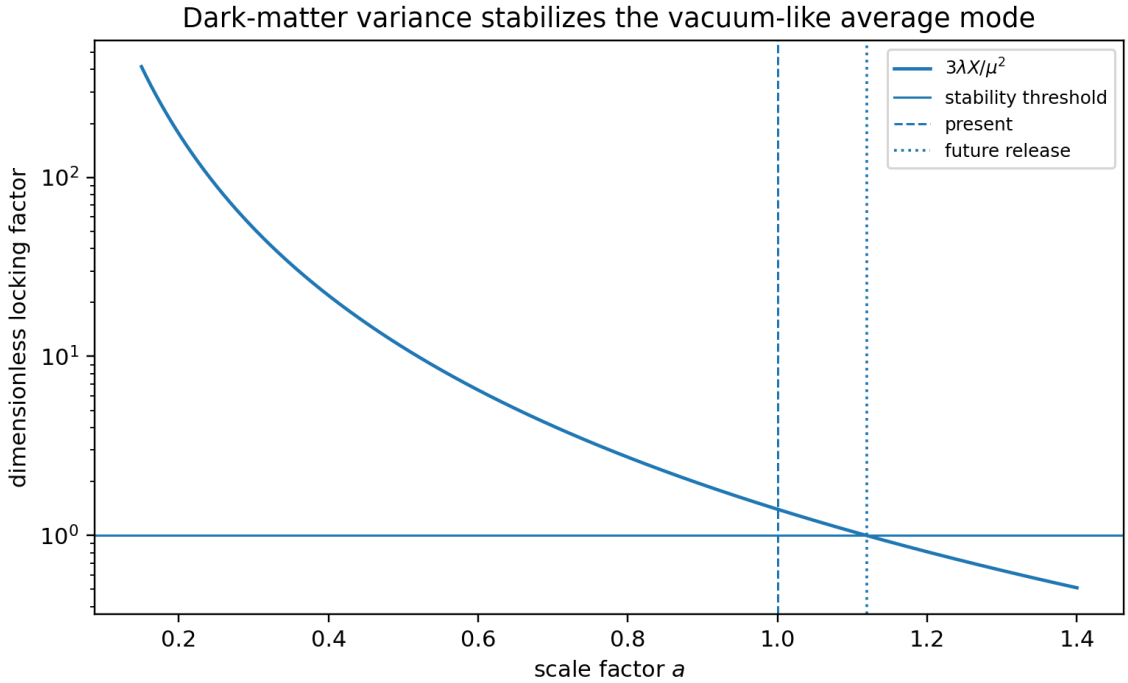


Figure 47.2: The locking factor $3\lambda X_D/\mu^2$. The average mode is stable above one and is released when the curve crosses the threshold.

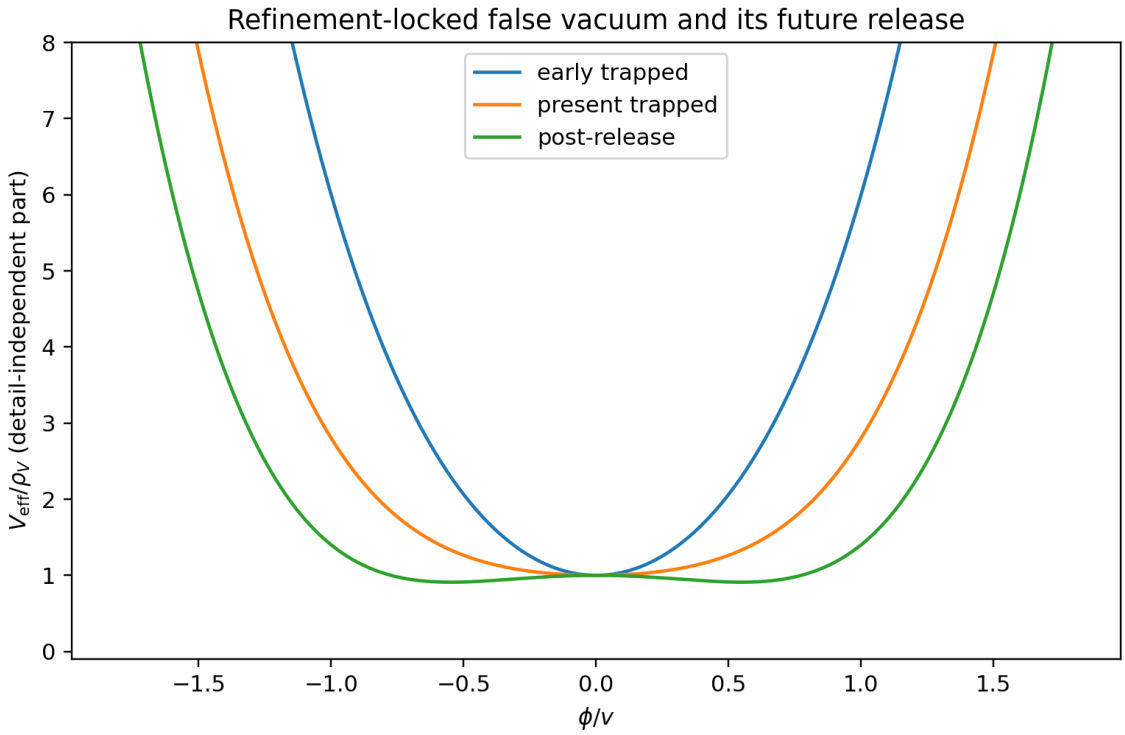


Figure 47.3: The average-mode effective potential before, during, and after release. Dark-matter variance changes the curvature at the origin without changing the false-vacuum energy there.

47.9 Executable certificate

The accompanying program checks the following identities and benchmark statements:

record	result
quartic Hessian residual	0
Gaussian number-density integral residual	0
release-ratio identity residual	0
linear dark-energy density perturbation	0
linear dark-energy pressure perturbation	0
level-one number-density relative error	0
cold $\rho_D/X_D = m_{\rm lock}^2$ relative error	7.0736×10^{-9}
stable-mode final amplitude after the controlled test	1.4541×10^{-8}
explicit checks	13 PASS, 0 FAIL

47.10 What is closed and what remains open

The present block establishes, within the stated Gaussian detail closure,

$$\boxed{\begin{array}{c}\text{one refinement-field action for clustered dark matter and vacuum-like dark energy,}\\ \text{a fixed false-vacuum density } \rho_V = \mu^4/(4\lambda),\\ \text{exact } w_{\rm DE} = -1 \text{ while the average mode is locked,}\\ \text{ghost and gradient stability with } c_s^2 = 1,\\ \text{vanishing linear dark-energy stress perturbations,}\\ \text{a mass-moment criterion for current trapping,}\\ \text{and a derived future release density.}\end{array}} \tag{47.58}$$

The following statements are not established here:

- a first-principles cosmological solution fixing the crossing epoch, energy transfer, and occupation normalization;
- cancellation or sequestering of vacuum contributions from every other field sector;
- the full collisionless Boltzmann hierarchy and nonlinear halo formation;
- radiative stability of the projector-locked potential;
- an observational likelihood or literature-priority result.

The model therefore fills the missing dark-energy background, perturbation, and stability sector, but it does not by itself solve the complete cosmological-constant hierarchy.

Chapter 48

Superseded Additive Crossing and Backreaction Branch

48.1 Status and objective

Correction status. This chapter is retained as the self-consistent development of the additive branch. Its rapid-crossing and homogeneous-exchange identities remain a useful controlled record. The subsequent scale-covariant chapter identifies a hard inconsistency in the dominant-shell additive mechanism and replaces it as the current physical candidate.

The unified refinement dark-sector action fixes the detail spectrum and the average-mode false-vacuum energy, but its preceding cosmological benchmark still treated the production epoch and occupation normalization as state inputs. The present block removes the free occupation normalization in a controlled rapid-crossing closure. It derives the produced distribution from the average-mode trajectory, imposes energy matching, constructs the positive collisionless backreaction potential, identifies local, global, and direct-capture thresholds, and evolves the resulting homogeneous dark sector through locking, accelerated expansion, and release.

The result is exact within two declared approximations. First, the crossing interval is short compared with the Hubble time and the average mode is linearized during that interval. Second, after the crossing the produced detail quanta are collisionless and are represented by their conserved phase-space occupation. The construction does not yet solve the fully quantum, expanding, non-Gaussian crossing.

48.2 Rapid crossing from the same average-mode potential

Retain

$$U(\phi) = \frac{\lambda}{4}(\phi^2 - v^2)^2, \qquad \mu^2 = \lambda v^2, \qquad \rho_V = U(0) = \frac{\mu^4}{4\lambda}. \tag{48.1}$$

At the origin the hth detail mass is

$$m_h^2 = \Lambda^2 p^{2\sigma h} - \mu^2, \qquad d_h = (p-1)p^{h-1}. \tag{48.2}$$

Suppose the homogeneous average mode reaches the origin with incoming velocity $\dot{\phi}_*$ and satisfies

$$\frac{H_*}{\sqrt{\nu}} \ll 1, \qquad \nu := \sqrt{3\lambda}\,|\dot{\phi}_*|. \tag{48.3}$$

On the crossing interval,

$$\phi(t) = \dot{\phi}_*(t - t_*) + \mathcal{O}((t-t_*)^2), \tag{48.4}$$

so each detail mode obeys

$$\ddot{u}_{h,k} + \left[k^2 + m_h^2 + \nu^2(t-t_*)^2\right] u_{h,k} = 0. \tag{48.5}$$

The parabolic-cylinder connection formula gives the exact linear-crossing occupation

$$\boxed{f_h(k) = \exp\left[-\frac{\pi(k^2+m_h^2)}{\nu}\right].} \tag{48.6}$$

There is no independent multiplicative abundance parameter. Vacuum initial data fix the normalization to one. Corrections arise from curvature of the actual crossing, expansion during production, rescattering, and non-Gaussian backreaction.

48.3 Exact production moments

Set

$$y_h := \frac{m_h}{\sqrt{\nu}}. \tag{48.7}$$

Define

$$I_\rho(y) := \int_0^\infty x^2\sqrt{x^2+y^2}\, e^{-\pi x^2}\, \mathrm{d}x, \tag{48.8}$$

$$I_p(y) := \int_0^\infty \frac{x^4}{\sqrt{x^2+y^2}} e^{-\pi x^2}\, \mathrm{d}x, \tag{48.9}$$

$$I_X(y) := \int_0^\infty \frac{x^2}{\sqrt{x^2+y^2}} e^{-\pi x^2}\, \mathrm{d}x. \tag{48.10}$$

Their Tricomi forms are

$$I_\rho(y) = \frac{y^4}{2}\Gamma\left(\frac{3}{2}\right) U\left(\frac{3}{2}, 3, \pi y^2\right), \tag{48.11}$$

$$I_p(y) = \frac{y^4}{2}\Gamma\left(\frac{5}{2}\right) U\left(\frac{5}{2}, 3, \pi y^2\right), \tag{48.12}$$

$$I_X(y) = \frac{y^2}{2}\Gamma\left(\frac{3}{2}\right) U\left(\frac{3}{2}, 2, \pi y^2\right). \tag{48.13}$$

The number, energy, pressure, and locking variance immediately after production are

$$n_{h,*} = \frac{d_h \nu^{3/2}}{8\pi^3} e^{-\pi m_h^2/\nu}, \tag{48.14}$$

$$\rho_{h,*} = \frac{d_h \nu^2}{2\pi^2} e^{-\pi m_h^2/\nu} I_\rho(y_h), \tag{48.15}$$

$$p_{h,*} = \frac{d_h \nu^2}{6\pi^2} e^{-\pi m_h^2/\nu} I_p(y_h), \tag{48.16}$$

$$X_{h,*} = \frac{d_h \nu}{2\pi^2} e^{-\pi m_h^2/\nu} I_X(y_h). \tag{48.17}$$

All sums converge super-exponentially when $m_h^2 \sim p^{2\sigma h}$.

48.4 Energy matching without an occupation fit

Let

$$K_* := \frac{1}{2}\dot{\phi}_*^2 \tag{48.18}$$

be the incoming kinetic density at the origin and let

$$\rho_* := \sum_h \rho_{h,*}. \tag{48.19}$$

The leading matched outgoing velocity is

$$\boxed{\dot{\phi}_{*,+}^2 = \max\left(\dot{\phi}_{*,-}^2 - 2\rho_*, 0\right).} \tag{48.20}$$

Define the direct-capture fraction

$$\eta_{\mathrm{cap}} := \frac{\rho_*}{K_*}. \tag{48.21}$$

Then

$$\eta_{\mathrm{cap}} < 1 \Longrightarrow \dot{\phi}_{*,+} \neq 0, \tag{48.22}$$

whereas

$$\eta_{\mathrm{cap}} \geq 1 \tag{48.23}$$

marks the strong-capture regime in which the fixed-trajectory crossing approximation saturates and the complete nonlinear crossing must stop or reverse the average mode. This criterion is sufficient for direct capture, not necessary for subsequent trapping.

The post-production energy is positive and has the collisionless form

$$\rho_D(a,\phi) = \sum_h \frac{d_h}{2\pi^2 a^4} \int_0^\infty q^2 \epsilon_h(q,a,\phi) f_h(q)\ \mathrm{d}q, \tag{48.24}$$

where

$$\epsilon_h(q,a,\phi) = \sqrt{q^2 + a^2[m_h^2 + 3\lambda\phi^2]}. \tag{48.25}$$

No negative vacuum-polarization energy is assigned to the produced matter component in this kinetic closure.

48.5 Local, global, and direct-capture thresholds

For one light shell of multiplicity g, write

$$\widehat{X}(\phi) := \frac{1}{2\pi^2} \int_0^\infty \frac{k^2 f(k)}{\sqrt{k^2 + m_1^2 + 3\lambda\phi^2}}\ \mathrm{d}k \tag{48.26}$$

and

$$\widehat{\rho}(\phi) := \frac{1}{2\pi^2} \int_0^\infty k^2 f(k) \sqrt{k^2 + m_1^2 + 3\lambda\phi^2}\ \mathrm{d}k. \tag{48.27}$$

The positive post-production effective potential is

$$\boxed{V_{\mathrm{eff}}(\phi) = U(\phi) + g\widehat{\rho}(\phi).} \tag{48.28}$$

Its derivative is

$$V'_{\mathrm{eff}}(\phi) = \lambda\phi \left[\phi^2 - v^2 + 3g\widehat{X}(\phi)\right]. \tag{48.29}$$

The local locking threshold is

$$\boxed{g_{\mathrm{local}} = \frac{\mu^2}{3\lambda \widehat{X}(0)}.} \tag{48.30}$$

The exact global locking threshold is

$$\boxed{g_{\mathrm{global}} = \sup_{0\leq\phi\leq v} \frac{v^2 - \phi^2}{3\widehat{X}(\phi)}.} \tag{48.31}$$

If $g > g_{\mathrm{global}}$, then $V'_{\mathrm{eff}}(\phi) > 0$ for every $\phi > 0$. Since the potential is even and diverges as $|\phi| \to \infty$, the origin is its unique global minimum. The field is trapped around the origin even when the outgoing velocity is nonzero.

The sufficient direct-capture threshold is

$$\boxed{g_{\mathrm{capture}} = \frac{K_*}{\widehat{\rho}(0)}.} \tag{48.32}$$

Thus the parameter space separates into

$$g < g_{\mathrm{local}}: \quad \text{origin unstable,} \tag{48.33}$$

$$g_{\mathrm{local}} < g < g_{\mathrm{global}}: \quad \text{local metastable well,} \tag{48.34}$$

$$g_{\mathrm{global}} < g < g_{\mathrm{capture}}: \quad \text{global lock with finite outgoing oscillation,} \tag{48.35}$$

$$g \geq g_{\mathrm{capture}}: \quad \text{direct-capture regime.} \tag{48.36}$$

For a first Haar shell, $g = p - 1$. A prime threshold can therefore be assigned only after the light-shell mass is held fixed while the active prime content is varied.

48.6 Fast-crossing asymptotics

For $m_1^2/\nu \to 0$,

$$I_\rho(0) = \frac{1}{2\pi^2}, \qquad I_X(0) = \frac{1}{2\pi}. \tag{48.37}$$

Hence

$$\widehat{\rho}(0) \sim \frac{\nu^2}{4\pi^4}, \qquad \widehat{X}(0) \sim \frac{\nu}{4\pi^3}. \tag{48.38}$$

Because $\nu^2 = 3\lambda\dot{\phi}_*^2$, the direct-capture threshold approaches

$$\boxed{g_{\rm capture} \longrightarrow \frac{2\pi^4}{3\lambda}.} \tag{48.39}$$

The capture multiplicity therefore has a finite high-velocity limit. The local-lock threshold instead decreases as ν^{-1}.

48.7 Self-consistent homogeneous backreaction

After production the comoving occupation is conserved. Define

$$\rho_D(a,\phi) = \sum_h \frac{d_h}{2\pi^2 a^4} \int q^2 \epsilon_h f_h \, \mathrm{d}q, \tag{48.40}$$

$$p_D(a,\phi) = \sum_h \frac{d_h}{6\pi^2 a^4} \int \frac{q^4}{\epsilon_h} f_h \, \mathrm{d}q, \tag{48.41}$$

$$X_D(a,\phi) = \sum_h \frac{d_h}{2\pi^2 a^2} \int \frac{q^2}{\epsilon_h} f_h \, \mathrm{d}q. \tag{48.42}$$

The closed homogeneous system is

$$H^2 = \frac{1}{3M_{\rm Pl}^2}\left[\frac{1}{2}\dot{\phi}^2 + U(\phi) + \rho_D\right], \tag{48.43}$$

$$\ddot{\phi} + 3H\dot{\phi} + U_{,\phi} + 3\lambda\phi X_D = 0. \tag{48.44}$$

The detail energy obeys

$$\dot{\rho}_D = -3H(\rho_D + p_D) + 3\lambda\phi\dot{\phi}X_D, \tag{48.45}$$

while the average-mode energy obeys

$$\dot{\rho}_\phi = -3H\dot{\phi}^2 - 3\lambda\phi\dot{\phi}X_D. \tag{48.46}$$

Therefore

$$\boxed{\dot{\rho}_{\rm tot} + 3H(\rho_{\rm tot} + p_{\rm tot}) = 0.} \tag{48.47}$$

The backreaction exchange cancels exactly. This closes production, abundance, locking, and expansion in one homogeneous system after the rapid crossing is matched.

48.8 Controlled benchmark and phase thresholds

Use

$$\mu = \lambda = 1, \qquad m_1 = 0.45, \qquad \phi_i = -2, \qquad \dot{\phi}_i = 0. \tag{48.48}$$

Energy conservation before production gives

$$K_* = 2, \qquad |\dot{\phi}_*| = 2, \qquad \nu = \sqrt{12} = 3.4641016151. \tag{48.49}$$

The computed thresholds are

$$g_{\rm local} = 17.2875822920, \tag{48.50}$$

$$g_{\rm global} = 17.9043887055, \tag{48.51}$$

$$g_{\mathrm{capture}} = 72.0989329581. \tag{48.52}$$

The maximizing field for the global threshold is

$$\phi_{\mathrm{global}} = 0.3252548920. \tag{48.53}$$

Under the fixed-light-mass prime map $g = p - 1$, the first global-lock prime is

$$\boxed{p_{\mathrm{global}} = 19,} \tag{48.54}$$

and the first direct-capture prime is

$$\boxed{p_{\mathrm{capture}} = 79.} \tag{48.55}$$

For the near-capture benchmark $p = 73$, $g = 72$, one obtains

$$n_* = 1.5574778055, \tag{48.56}$$
$$\rho_* = 1.9972556332, \tag{48.57}$$
$$X_* = 1.3882797256, \tag{48.58}$$
$$\eta_{\mathrm{cap}} = 0.9986278166, \tag{48.59}$$
$$\dot{\phi}_{*,+} = 0.0740859876. \tag{48.60}$$

The next refinement level has a weighted production factor below 10^{-100}, so the light-shell closure is controlled for this benchmark.

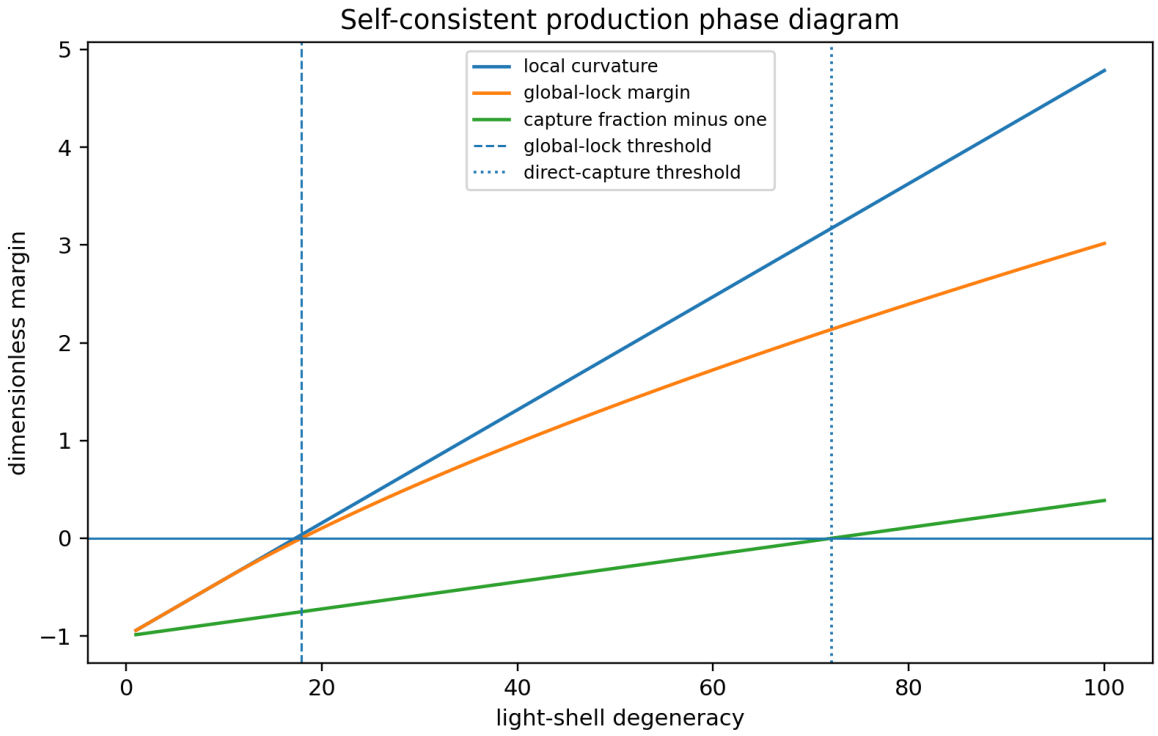


Figure 48.1: The light-shell degeneracy separates unstable, locally locked, globally locked, and direct-capture regimes.

48.9 Transient acceleration from dynamically normalized production

Set $M_{\mathrm{Pl}} = 10$ so that the production event is short relative to the Hubble time. Normalize $a_* = 1$ at production and evolve the $p = 73$ matched state. Acceleration begins at

$$\boxed{a_{\mathrm{acc}} = 1.6596778467, \qquad t_{\mathrm{acc}} = 8.7878307745.} \tag{48.61}$$

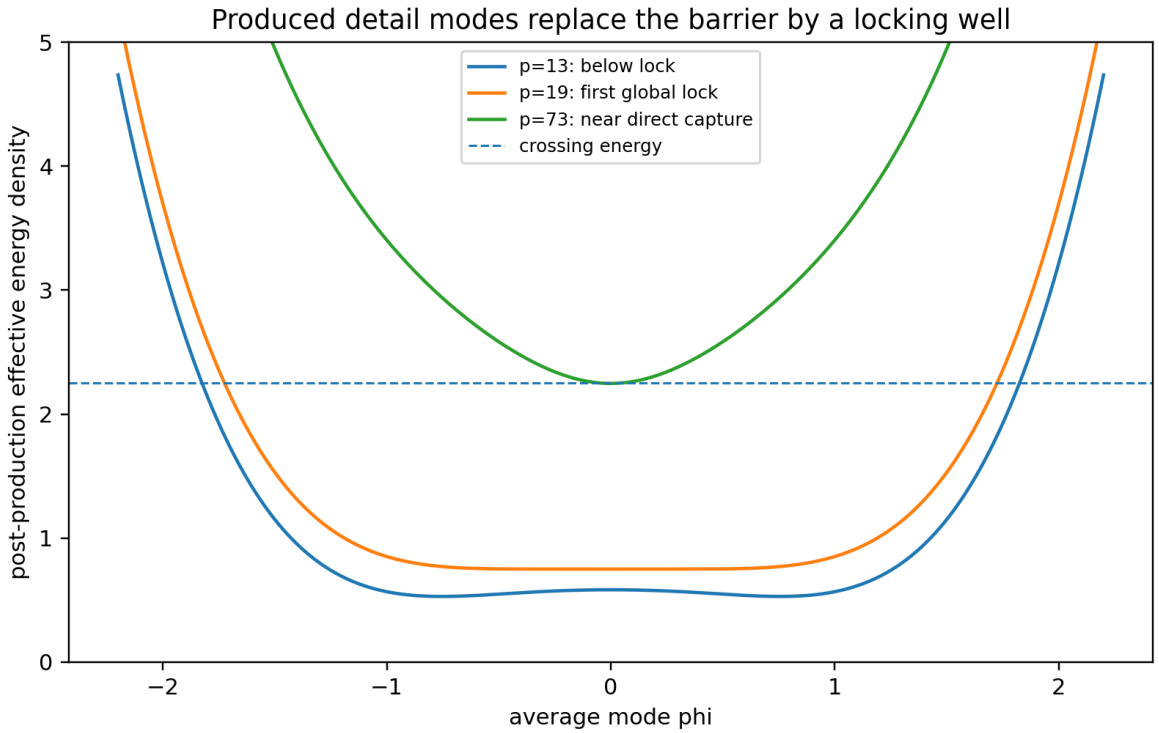


Figure 48.2: Produced detail modes convert the symmetry-breaking barrier into a global locking well once the degeneracy exceeds the derived threshold.

The origin remains stable until

$$\boxed{a_{\mathrm{release}} = 1.8424690117, \qquad t_{\mathrm{release}} = 11.3673608512.} \tag{48.62}$$

Before release,

$$\max|\phi| = 0.0394198188. \tag{48.63}$$

At release,

$$w_D = 0.2189375169, \qquad w_{\mathrm{tot}} = -0.4568362431. \tag{48.64}$$

The first continuous acceleration interval ends at

$$\boxed{a_{\mathrm{end}} = 2.3209537346, \qquad t_{\mathrm{end}} = 17.8431038770.} \tag{48.65}$$

Its duration is

$$\Delta N_{\mathrm{acc}} = \log\left(\frac{a_{\mathrm{end}}}{a_{\mathrm{acc}}}\right) = 0.3353546784. \tag{48.66}$$

The calculation therefore produces the sequence

$$\boxed{\begin{array}{c}\text{detail production} \Longrightarrow \text{global average-mode lock} \\ \Longrightarrow \text{accelerated expansion} \Longrightarrow \text{density-driven release}\end{array}} \tag{48.67}$$

No independent dark abundance normalization or cosmological-constant density is introduced inside this benchmark.

48.10 Executable certificate

The deterministic calculation records

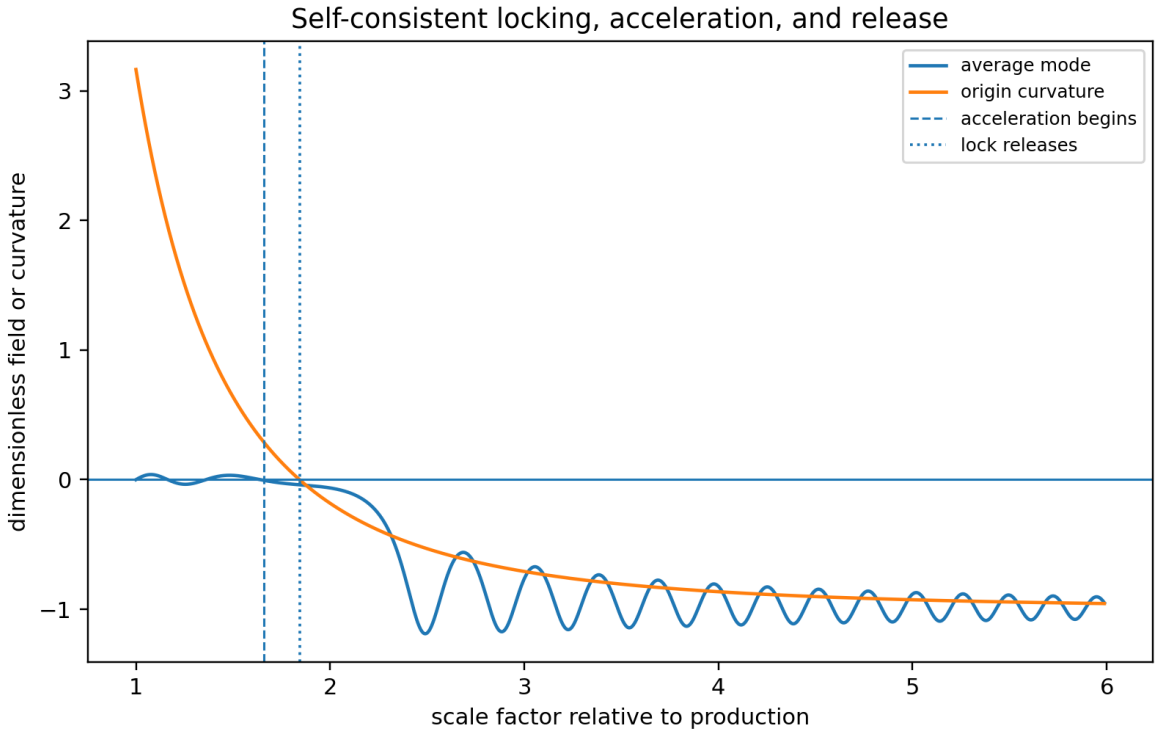


Figure 48.3: The average mode remains close to the origin while the detail variance keeps the locking curvature positive. It rolls away after the curvature changes sign.

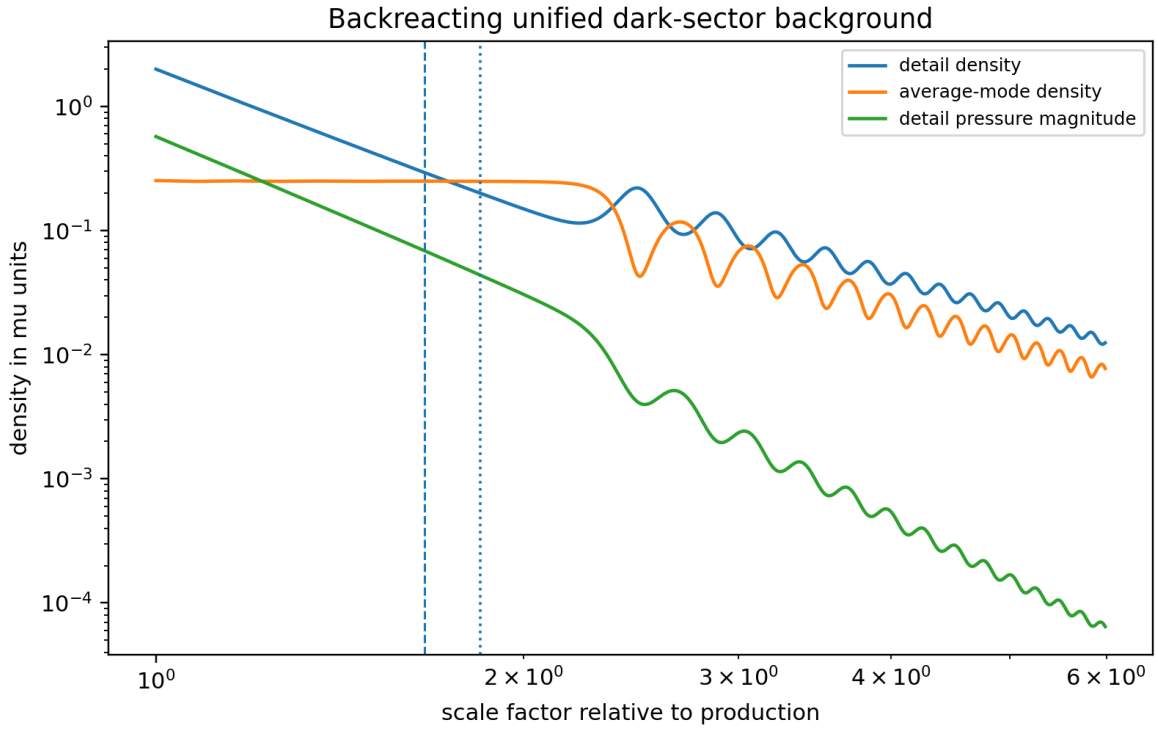


Figure 48.4: The detail density dilutes, the average mode remains vacuum-like while locked, and the release ends the first acceleration interval.

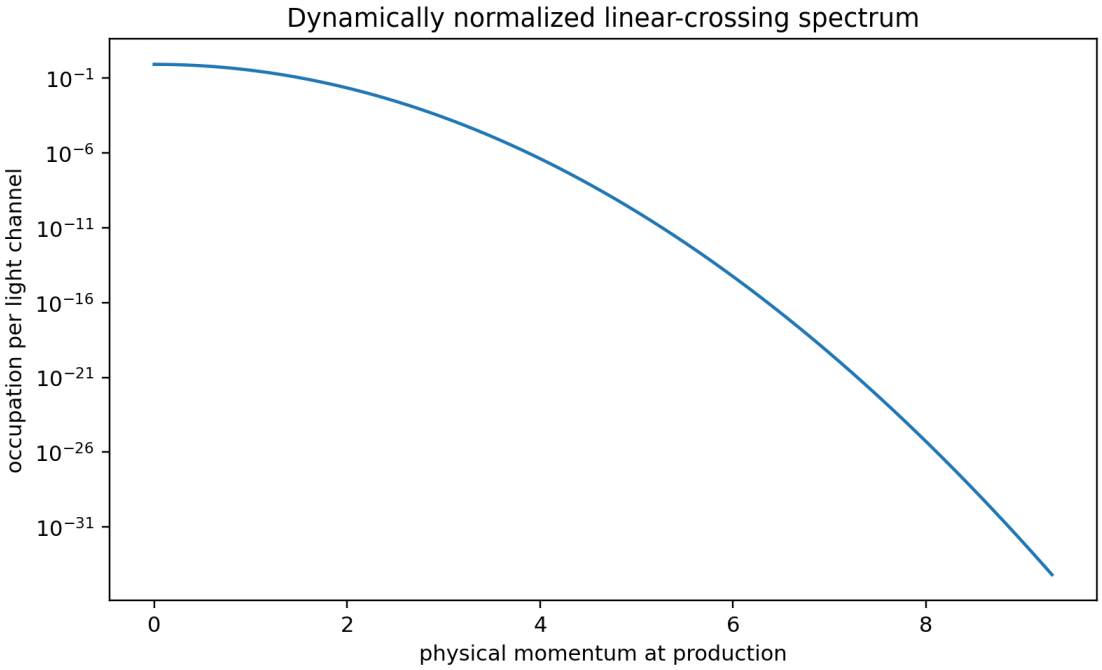


Figure 48.5: The linear-crossing occupation is normalized by the vacuum connection formula rather than by a fitted multiplicative abundance.

record	result
explicit checks	20 PASS, 0 FAIL
hypergeometric density relative residual	4.06×10^{-16}
hypergeometric pressure relative residual	3.54×10^{-16}
hypergeometric variance relative residual	1.46×10^{-16}
maximum continuity-exchange residual	1.11×10^{-16}
release-scale solver refinement difference	9.88×10^{-6}
first global-lock prime	19
first direct-capture prime	79

48.11 Closure ledger

statement	status	boundary
unit-normalized detail production	closed	exact linear crossing
positive production moments	closed	Gaussian collisionless state
energy-matched outgoing velocity	closed controlled	rapid-crossing matching
local and global locking thresholds	closed	one light shell
direct-capture threshold	closed sufficient	fixed crossing trajectory
post-production continuity	closed	collisionless homogeneous closure
transient acceleration and release	numerical controlled	declared benchmark
physical active prime content	open	microscopic selector dynamics
full nonlinear quantum crossing	open	beyond linear crossing
renormalized loop corrections	open	radiative stability
full Boltzmann and likelihood pipeline	open	observational closure

48.12 Result and next decisive target

This block removes the free production normalization and closes a self-consistent homogeneous sequence from the average-mode crossing through detail backreaction and transient acceleration. It does not yet establish that the benchmark parameters describe the observed universe. The production epoch, physical mass scale, active prime content, and primordial initial state must be derived from the gravitational sector rather than selected for the controlled example.

The next decisive cosmological block is

$$\boxed{\begin{array}{c}\text{dynamically derive the crossing epoch and active light-shell content,}\\ \text{propagate the resulting distribution through the full collisionless Boltzmann hierarchy,}\\ \text{compute CMB, matter-power, and weak-lensing observables with one parameter set,}\\ \text{and perform the first likelihood-level falsification test.}\end{array}} \tag{48.68}$$

Chapter 49

Scale-Covariant Refinement Dark Sector

49.1 Status and objective

The preceding unified dark-sector model used an additive average-detail Hessian,

$$m_h^2(\phi) = m_h^2 + 3\lambda\phi^2, \tag{49.1}$$

so that a light produced shell could both cluster and stabilize the average mode. A hard abundance and coldness audit exposes a restricted but serious obstruction in that branch. On an energy-matched rapid crossing, one dominant additive shell cannot simultaneously reproduce the present dark abundance, remain nonrelativistic, and lock the average mode.

This chapter first proves that obstruction under its stated hypotheses. It then replaces the additive Hessian by one scale-covariant deformation of the whole refinement operator,

$$\mathcal{L}_{\sigma,p} \longmapsto \left(1 + \gamma\frac{\phi^2}{v^2}\right)\mathcal{L}_{\sigma,p}. \tag{49.2}$$

The replacement introduces one dimensionless coupling γ, not an independently chosen coupling at every level. It has two linked effects:

1. heavier detail levels are created with proportionally broader nonadiabatic scales and become colder after redshift;
2. the same weighted detail energy, rather than the inverse-mass variance of one light shell, stabilizes the average mode.

The resulting branch supplies an analytic mass envelope, an exact captured rest-energy fraction, a derived late-time equation of state, a dark-density locking law, and a one-parameter small-scale cutoff family. The scale-covariant coupling itself is a model extension. It is not claimed to follow uniquely from the primitive projector axioms.

49.2 Additive one-shell obstruction

Let

$$\rho_V = \frac{\mu^4}{4\lambda}, \qquad m = \alpha\mu, \qquad \beta = \frac{\nu}{\mu^2}, \qquad r_D = \frac{\rho_{D0}}{\rho_V}. \tag{49.3}$$

Assume one shell of degeneracy g is populated at scale factor $a_\star$ by the energy-projected Gaussian

$$f(q) = \eta\exp\left[-\frac{\pi m^2}{\nu} - \frac{\pi q^2}{a_\star^2\nu}\right], \qquad 0 < \eta \le 1. \tag{49.4}$$

In the nonrelativistic present limit,

$$r_D = \frac{\eta g\lambda\alpha\beta^{3/2}}{2\pi^3}e^{-\pi\alpha^2/\beta}a_\star^3, \tag{49.5}$$

and the present nonrelativistic momentum ratio is

$$u_0^2 := \frac{\langle p^2\rangle_0}{m^2} = \frac{3\beta a_\star^2}{2\pi\alpha^2}. \tag{49.6}$$

Eliminating $\beta^{3/2}a_\star^3$ gives

$$u_0^3 = \frac{(3\pi)^{3/2}}{\sqrt{2}}\frac{r_D}{\eta g\lambda\alpha^4}e^{\pi\alpha^2/\beta}. \tag{49.7}$$

For an ultrarelativistic linear crossing, the unit-normalized produced energy is

$$\widetilde{\rho}_\star = \frac{g\nu^2}{4\pi^4}, \tag{49.8}$$

while the incoming kinetic energy is

$$K_\star = \frac{\nu^2}{6\lambda}. \tag{49.9}$$

Energy matching therefore implies

$$\eta g\lambda \le \frac{2\pi^4}{3}. \tag{49.10}$$

The additive locking condition is

$$-\mu^2 + 3\lambda\frac{\rho_{D0}}{m^2} > 0 \quad\Longrightarrow\quad \alpha^2 < \frac{3r_D}{4}. \tag{49.11}$$

Combining (49.7)–(49.11) yields the conditional lower bound

$$\boxed{u_0^3 > \frac{4\sqrt{6}}{\pi^{5/2}r_D}.} \tag{49.12}$$

For

$$r_D = 0.3783102144, \tag{49.13}$$

this gives

$$\boxed{u_0 > 1.1397360601.} \tag{49.14}$$

The assumed nonrelativistic branch is therefore inconsistent. This is not a no-go theorem for every additive multi-level state. It rejects the specific one-shell, one-crossing, energy-matched mechanism that was being used to carry both abundance and locking.

49.3 Scale-covariant fiber action

Let

$$U(\phi) = \frac{\lambda}{4}(\phi^2 - v^2)^2, \qquad \mu^2 = \lambda v^2, \tag{49.15}$$

and let

$$\mathcal{L}_{\sigma,p} = \Lambda^2\sum_{h\ge1} p^{2\sigma h}\mathsf{D}_h. \tag{49.16}$$

The scale-covariant branch is defined by

$$\boxed{\begin{aligned} S_{\mathrm{SC}} = -\int\sqrt{-g}\,d^4x\Bigg[&\frac{1}{2}(\partial\phi)^2 + U(\phi) + \frac{1}{2}\langle\partial\chi,\partial\chi\rangle_p \\ &+ \frac{1}{2}\left\langle\chi,\left(m_0^2 - \mu^2 + \left(1+\gamma\frac{\phi^2}{v^2}\right)\mathcal{L}_{\sigma,p}\right)\chi\right\rangle_p\Bigg]. \end{aligned}} \tag{49.17}$$

The level masses at the locked origin are

$$m_h^2 = m_0^2 - \mu^2 + \Lambda^2 p^{2\sigma h}, \tag{49.18}$$

with multiplicities

$$d_h = (p-1)p^{h-1}. \tag{49.19}$$

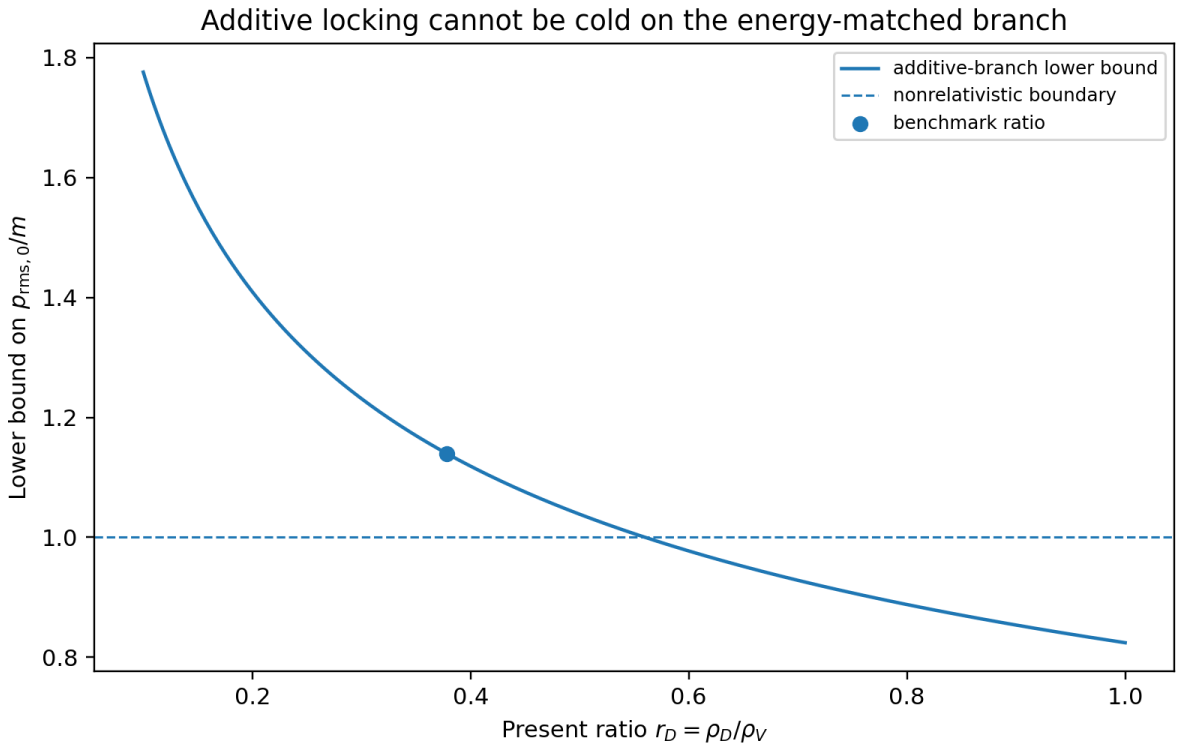


Figure 49.1: The abundance, energy-matching, and locking conditions force the additive one-shell ratio p_{rms}/m above unity at the declared present density ratio, contradicting the assumed cold branch.

The average mode now changes the internal refinement scale itself. Its homogeneous equation is

$$\ddot{\phi} + 3H\dot{\phi} + \lambda\phi(\phi^2 - v^2) + \frac{\gamma\phi}{v^2}Y_D = 0, \tag{49.20}$$

where

$$Y_D := \sum_h \Lambda^2 p^{2\sigma h} X_h, \qquad X_h := \langle \chi_h^2 \rangle. \tag{49.21}$$

The origin curvature is

$$\boxed{M_\phi^2 = -\mu^2 + \frac{\gamma}{v^2}Y_D.} \tag{49.22}$$

For nonrelativistic levels with $m_h^2 \gg \mu^2, m_0^2$,

$$\Lambda^2 p^{2\sigma h} X_h = \rho_h \left[1 + O(w_h) + O\left(\frac{\mu^2 + m_0^2}{m_h^2}\right)\right]. \tag{49.23}$$

Thus the total cold dark density, including heavy levels, supplies the locking source.

49.4 Weighted nonadiabatic production

Near a crossing,

$$\phi(t) = \dot{\phi}_\star(t - t_\star) + O((t - t_\star)^2), \tag{49.24}$$

so that

$$\omega_{h,k}^2 = k^2 + m_h^2 + \frac{\gamma\Lambda^2 p^{2\sigma h}}{v^2}\dot{\phi}_\star^2 (t - t_\star)^2. \tag{49.25}$$

Define

$$A_\star := \frac{\sqrt{\gamma}|\dot{\phi}_\star|}{v}, \qquad \nu_h := A_\star \Lambda p^{\sigma h}. \tag{49.26}$$

The rapid-crossing spectrum is

$$\boxed{f_h(k) = \eta \exp\left[-\frac{\pi(k^2 + m_h^2)}{\nu_h}\right],} \tag{49.27}$$

where η is fixed by energy projection when the linearized spectrum would exceed the available kinetic energy:

$$\eta = \min\left(1, \frac{K_\star}{\widetilde{\rho}_\star}\right). \tag{49.28}$$

This is not an independent abundance fit. It is the shape-preserving rapid-crossing approximation with exact total-energy matching.

At high level,

$$m_h \simeq \Lambda p^{\sigma h}, \qquad \nu_h \simeq A_\star m_h. \tag{49.29}$$

The suppression exponent becomes linear rather than quadratic in mass:

$$\frac{\pi m_h^2}{\nu_h} \simeq \frac{\pi m_h}{A_\star}. \tag{49.30}$$

At the same time,

$$\frac{p_{\mathrm{rms},\star}}{m_h} \simeq \sqrt{\frac{3A_\star}{2\pi m_h}}, \tag{49.31}$$

so heavier populated levels are born colder.

49.5 Exact $\sigma = 1$ continuum theorem

For $\sigma = 1$, the endpoint law

$$N(m) = \frac{m}{\Lambda} - 1 \tag{49.32}$$

gives the continuum density of levels

$$\frac{dN}{dm} = \frac{1}{\Lambda}. \tag{49.33}$$

Using $\nu = Am$ and the occupation (49.27), the number and captured rest-density spectra are

$$\frac{dn}{dm} = \eta \frac{A^{3/2}}{8\pi^3\Lambda} m^{3/2} e^{-\pi m/A}, \tag{49.34}$$

$$\boxed{\frac{d\rho_R}{dm} = \eta \frac{A^{3/2}}{8\pi^3\Lambda} m^{5/2} e^{-\pi m/A}.} \tag{49.35}$$

The selected mass band peaks at

$$\boxed{m_{\mathrm{peak}} = \frac{5A}{2\pi}.} \tag{49.36}$$

The exact continuum integrals are

$$n_\star = C_N \eta \frac{A^4}{\Lambda}, \qquad C_N = \frac{3}{32\pi^5}, \tag{49.37}$$

$$\rho_{R,\star} = C_R \eta \frac{A^5}{\Lambda}, \qquad C_R = \frac{15}{64\pi^6}, \tag{49.38}$$

$$\rho_\star = C_E \eta \frac{A^5}{\Lambda}, \qquad C_E = \frac{32}{35\pi^7}, \tag{49.39}$$

$$p_\star = C_P \eta \frac{A^5}{\Lambda}, \qquad C_P = \frac{32}{315\pi^7}. \tag{49.40}$$

For example, the total-energy coefficient follows from the Tricomi representation and the beta integral:

$$\begin{aligned} C_E &= \frac{1}{2\pi^2} \int_0^\infty y^2 e^{-\pi y} I_\rho(\sqrt{y})\, dy \\ &= \frac{6}{\pi^7} B\left(\frac{3}{2}, 3\right) = \frac{32}{35\pi^7}. \end{aligned} \tag{49.41}$$

Consequently,

$$\boxed{w_\star = \frac{p_\star}{\rho_\star} = \frac{1}{9},} \tag{49.42}$$

$$\boxed{f_R := \frac{\rho_{R,\star}}{\rho_\star} = \frac{525\pi}{2048} = 0.8053399136\ldots,} \tag{49.43}$$

and

$$\frac{\rho_{R,\star}}{n_\star} = \frac{5A}{2\pi} = m_{\rm peak}. \tag{49.44}$$

A dyadic discrete tower with $\Lambda/A = 10^{-4}$ gives

$$f_R^{(2)} = 0.8054331018, \qquad w_\star^{(2)} = 0.1110616168, \tag{49.45}$$

and selects

$$\frac{m_{\rm peak}^{(2)}}{A} = 0.8192, \tag{49.46}$$

within one dyadic bin of (49.36). The absolute normalization retains the known log-periodic prime modulation, while the ratios are nearly envelope independent.

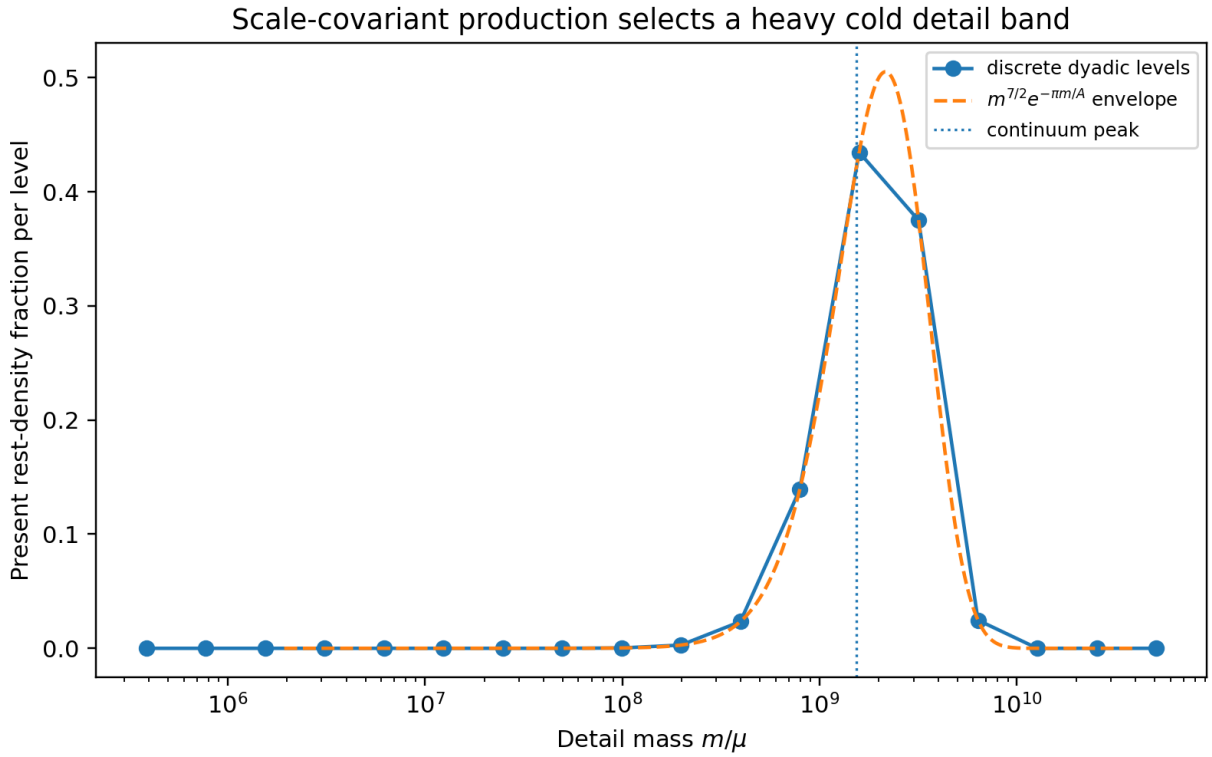


Figure 49.2: The discrete dyadic rest-density fractions follow the analytic scale-covariant mass envelope. The dominant mass is many orders of magnitude above the vacuum scale in the calibrated benchmark.

49.6 Abundance calibration and emergent hierarchy

On the captured branch,

$$\rho_{R,\star} = f_R K_\star. \tag{49.47}$$

After the produced band becomes nonrelativistic,

$$\rho_{D0} = f_R K_\star a_\star^3. \tag{49.48}$$

Define

$$F := \frac{K_\star}{\rho_V}. \tag{49.49}$$

The present abundance fixes

$$\boxed{F = \frac{r_D}{f_R a_\star^3}.} \tag{49.50}$$

The crossing scale and selected mass are then

$$\frac{A_\star}{\mu} = \sqrt{\frac{\gamma F}{2}}, \tag{49.51}$$

$$\boxed{\frac{m_{\rm peak}}{\mu} = \frac{5}{2\pi}\sqrt{\frac{\gamma r_D}{2 f_R a_\star^3}}.} \tag{49.52}$$

If the average field begins at rest at ϕ_i, then

$$F = \left(\frac{\phi_i}{v}\right)^4 - 2\left(\frac{\phi_i}{v}\right)^2, \tag{49.53}$$

so

$$\boxed{\frac{|\phi_i|}{v} = \sqrt{1+\sqrt{1+F}}.} \tag{49.54}$$

Thus one early production scale determines both the hierarchy between the vacuum and dark-matter masses and the required pre-crossing field amplitude.

For the benchmark

$$a_\star = 10^{-6}, \qquad \gamma = 16, \qquad p = 2, \qquad \frac{\Lambda}{A_\star} = 10^{-4}, \tag{49.55}$$

with (49.13), the discrete calculation gives

$$\frac{K_\star}{\rho_V} = 4.6969787254 \times 10^{17}, \tag{49.56}$$

$$\frac{A_\star}{\mu} = 1.9384486014 \times 10^{9}, \qquad \frac{|\phi_i|}{v} = 2.6179096175 \times 10^{4}, \tag{49.57}$$

$$\boxed{\frac{m^{(2)}_{\rm peak}}{\mu} = 1.5879770943 \times 10^{9}.} \tag{49.58}$$

The shape-preserving energy projection is

$$\eta = 3.8062069178 \times 10^{-21}. \tag{49.59}$$

This small occupation per microscopic state is compensated by the large refinement multiplicities. It is determined by energy conservation after the rapid-crossing spectrum is assembled.

49.7 Coldness theorem and cutoff scale

At $a \gg a_\star$, the scale-covariant continuum band obeys

$$\boxed{w_D(a) = \frac{a_\star^2}{5a^2},} \tag{49.60}$$

$$\boxed{c^2_{a,D}(a) = \frac{a_\star^2}{3a^2},} \tag{49.61}$$

and

$$\boxed{v_{\rm rms}(a) = \sqrt{\frac{3}{5}}\frac{a_\star}{a}.} \tag{49.62}$$

Unlike the additive one-shell result, the present speed decreases directly with the production scale. The reason is that the selected mass grows with the crossing rate while its fractional production momentum decreases as $m^{-1/2}$.

For $a_\star = 10^{-6}$,

$$w_{D0} = 1.9987378335 \times 10^{-13}, \qquad v_{\mathrm{rms},0} = 7.7459666924 \times 10^{-7}. \tag{49.63}$$

The fluid Jeans estimate is

$$\frac{k_J(a)}{aH(a)} = \sqrt{\frac{3\Omega_m(a)}{2c_{a,D}^2(a)}}, \tag{49.64}$$

which gives

$$\boxed{\frac{k_J(a)}{H_0} = \frac{3}{\sqrt{2}}\frac{\sqrt{\Omega_{m0}a}}{a_\star}.} \tag{49.65}$$

At the present epoch,

$$\frac{k_{J0}}{H_0} = 1.1811011811 \times 10^6. \tag{49.66}$$

49.8 Density locking and transient acceleration

Using (49.23), the locked cold branch satisfies

$$\frac{M_\phi^2}{\mu^2} = -1 + \frac{\gamma}{4} r_D(a) + O(w_D) + O\left(\frac{\mu^2}{m_{\mathrm{eff}}^2}\right). \tag{49.67}$$

Hence present locking requires

$$\boxed{\gamma > \frac{4}{r_{D0}}.} \tag{49.68}$$

For the declared ratio, the right side is $10.573\ldots$. The benchmark choice $\gamma = 16$ gives

$$\boxed{\frac{M_\phi^2(1)}{\mu^2} = 0.5132408576.} \tag{49.69}$$

The exact discrete kinetic moments reconstruct

$$\frac{Y_D(1)}{\rho_V} = 0.3783102144 \tag{49.70}$$

within 2.3×10^{-13} of the declared dark-density ratio.

The cold asymptotic release scale is

$$a_{\mathrm{release}} = \left(\frac{\gamma r_{D0}}{4}\right)^{1/3}. \tag{49.71}$$

The exact discrete result is

$$\boxed{a_{\mathrm{release}} = 1.1480726020.} \tag{49.72}$$

For baryon ratio

$$r_{b0} = 0.0710237567, \tag{49.73}$$

the acceleration onset is

$$\boxed{a_{\mathrm{acc}} = 0.6081134084,} \tag{49.74}$$

with a cold asymptotic estimate differing by 1.94×10^{-4}.

At the locked point, the quadratic average-mode action remains

$$S_{\delta\phi}^{(2)} = \frac{1}{2}\int a^3 dt\, d^3x \left[\dot{\delta\phi}^2 - \frac{(\nabla\delta\phi)^2}{a^2} - M_\phi^2\delta\phi^2\right]. \tag{49.75}$$

Therefore

$$Q_{\delta\phi} = 1, \qquad c_{s,\phi}^2 = 1, \qquad M_\phi^2 > 0 \tag{49.76}$$

throughout the locked interval. Because every mass deformation is even in ϕ, the detail stress tensor also has no term linear in $\delta\phi$ at $\phi = 0$.

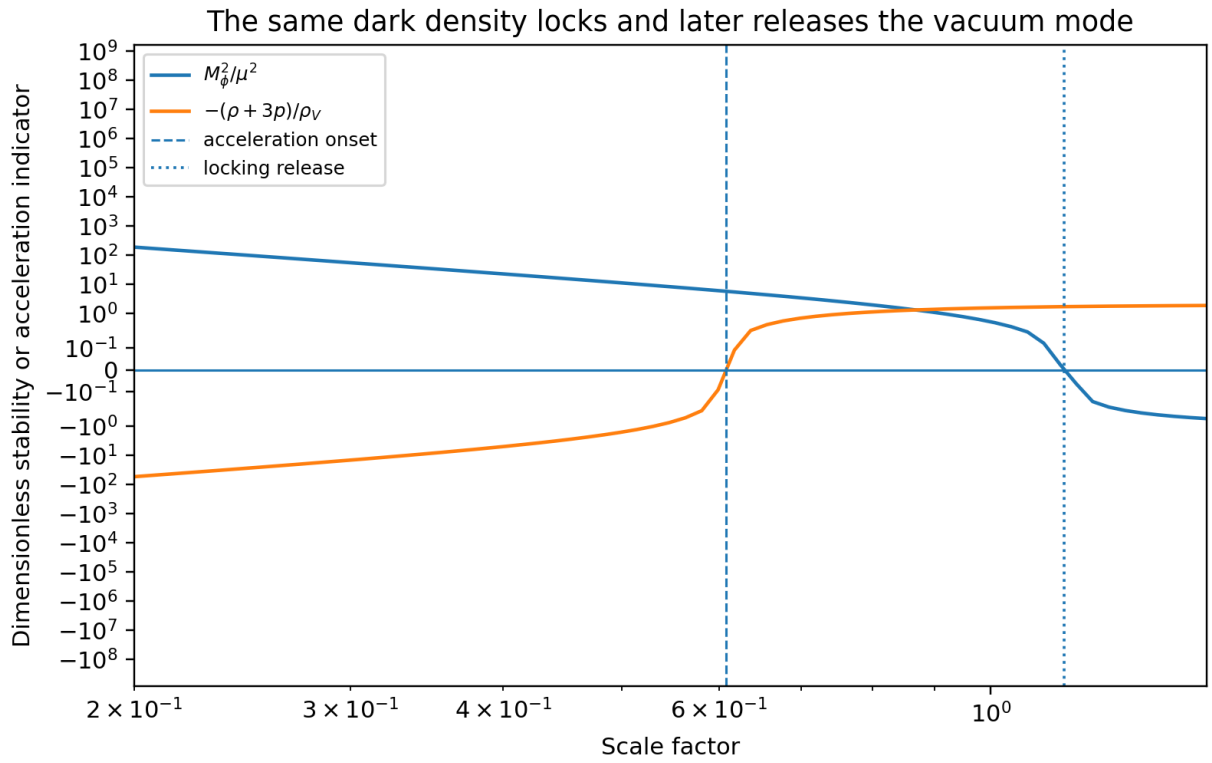


Figure 49.3: The weighted detail density stabilizes the average mode well before acceleration begins. Dilution eventually removes the locking curvature and releases the vacuum mode.

49.9 Linear growth and lensing benchmark

For $a \gg a_\star$, use $N = \log a$ and the controlled cold-fluid closure

$$\delta_b'' + \left(2 + \frac{d\log H}{dN}\right)\delta_b' = \frac{3}{2}S, \tag{49.77}$$

$$\delta_D'' + \left(2 + \frac{d\log H}{dN}\right)\delta_D' + \frac{c_{a,D}^2 k^2}{a^2 H^2}\delta_D = \frac{3}{2}S, \tag{49.78}$$

where

$$S = \Omega_b(a)\delta_b + \Omega_D(a)\delta_D. \tag{49.79}$$

The same source controls the subhorizon lensing potential to the accuracy of this closure.
Starting at $a_i = 10^{-3}$ with adiabatic growing data, the benchmark gives

k/H_0	$\delta_D/\delta_{D,\mathrm{cold}}$	lensing source / cold source
30	0.9999996989	0.9999997007
300	0.9999698901	0.9999700727
3000	0.9969926202	0.9970108463
10000	0.9669918115	0.9671896257
30000	0.7331608409	0.7346038042
100000	0.0159173474	0.0169297429

The first half-mode of the fluid lensing-source transfer is

$$\boxed{\frac{k_{1/2}}{H_0} = 4.4076353413 \times 10^4.} \tag{49.80}$$

For the conditional conversion $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$,

$$k_{1/2} = 10.2916022620\,\mathrm{Mpc}^{-1}. \tag{49.81}$$

This is a fluid-closure benchmark, not yet a complete collisionless Einstein–Boltzmann likelihood.

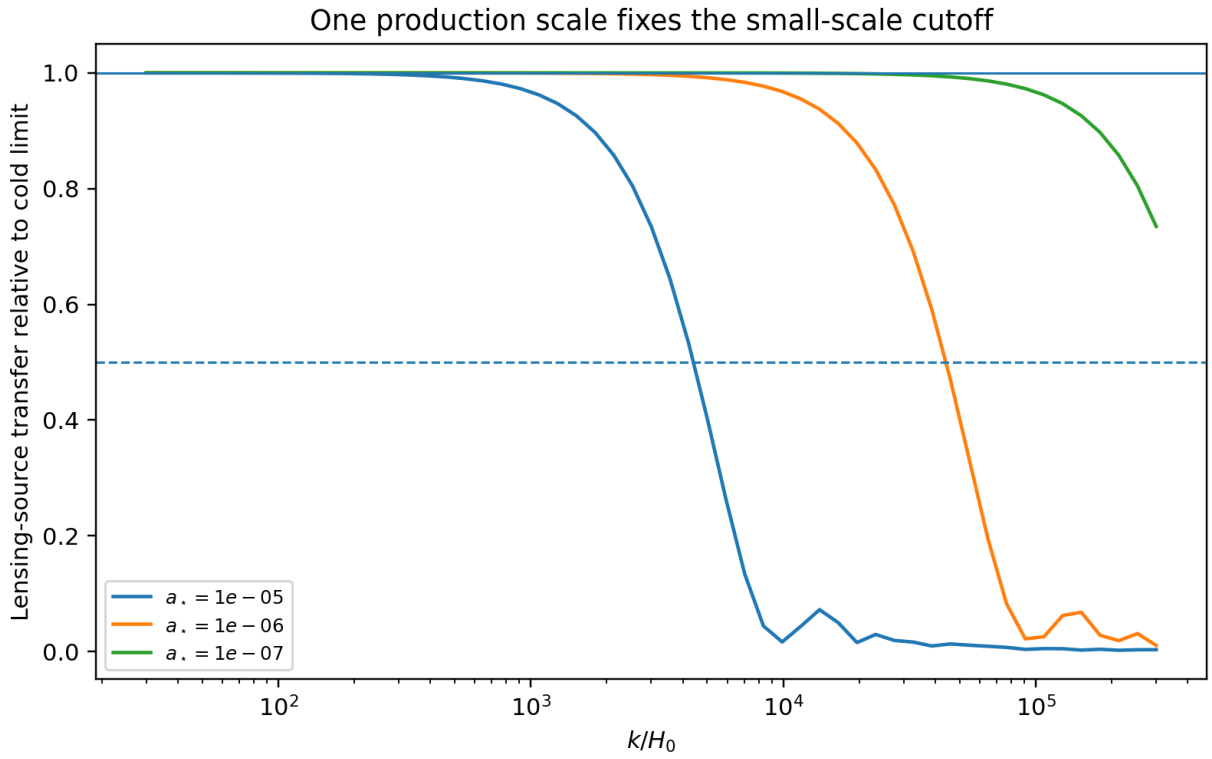


Figure 49.4: The production epoch controls the free-streaming cutoff without changing the late locking law. The high-k oscillations are those of the finite sound-speed fluid closure.

49.10 Verification ledger

The deterministic calculation records twenty passes and no failures. Selected checks are

statement	residual
additive abundance–velocity elimination identity	0
continuum C_E versus independent quadrature	1.62×10^{-12} relative
continuum C_P versus independent quadrature	3.49×10^{-11} relative
discrete versus continuum captured rest fraction	9.32×10^{-5}
discrete versus continuum production w	4.95×10^{-5}
present abundance reconstruction	0
exact versus asymptotic release scale	1.95×10^{-13}
exact versus asymptotic acceleration scale	1.93×10^{-4}
exact kinetic c_a^2 versus $a_\star^2/(3a^2)$	6.31×10^{-4} relative
large-scale lensing-source deviation at $k/H_0 = 300$	2.99×10^{-5}

The executable records are

$$R_{\mathrm{SC}} = \texttt{data/Scale_Covariant_Dark_Sector_Verification.json}, \tag{49.82}$$

$$C_{\mathrm{SC}} = \texttt{data/scale_covariant_dark_checks.csv}, \tag{49.83}$$

$$X_{\mathrm{SC}} = \texttt{verification/Scale_Covariant_Dark_Sector_Verification.py}. \tag{49.84}$$

49.11 Closure ledger and remaining burden

statement	status	boundary
additive one-shell cold-lock obstruction	proved conditional	rapid, energy-matched, dominant-shell branch
scale-covariant fiber action	defined	one new coupling γ
weighted production spectrum	derived	linear rapid crossing
$\sigma = 1$ continuum mass theorem	proved	high-level density-of-states envelope
energy-projected rest fraction	proved	shape-preserving capture approximation
late w_D, c_a^2, and $v_{\rm rms}$ laws	proved asymptotically	collisionless nonrelativistic regime
density locking and release	closed controlled	heavy-level limit and regular locked phase
linear growth and lensing cutoff	numerical controlled	two-fluid sound-speed closure
full collisionless hierarchy	open	momentum and anisotropic-stress multipoles
radiative stability of scale covariance	open	quantum effective action
microscopic derivation of γ and $a_\star$	open	gravitational refinement dynamics
observational likelihood and priority audit	open	independent implementation and data comparison

The principal result is not that every earlier dark-sector formula survives unchanged. The hard consistency audit rejects the additive dominant-shell mechanism and identifies a minimal viable replacement:

$$\boxed{\begin{aligned}&\text{average-mode dilation of the refinement spectrum}\\ &\qquad\Longrightarrow\ \text{heavy cold detail band}\\ &\qquad\Longrightarrow\ \text{density-weighted vacuum locking}\end{aligned}} \tag{49.85}$$

The next decisive block is the full collisionless Einstein–Boltzmann hierarchy for the distribution (49.27), followed by one parameter set propagated through recombination, matter power, weak lensing, and the transient-release background without sector-by-sector renormalization.

Chapter 50

Complete Cosmology Completion Ledger

50.1 Completion criterion

$$\boxed{S_{\mathrm{local}}[\mathcal{P}, U, q, N, N^i, \Psi] \Longrightarrow \{\mathcal{H}, \mathcal{H}_i, \mathcal{G}, \mathcal{E}_q, \mathcal{E}_\Psi\} = 0 \Longrightarrow \{H(z), C_\ell, P(k), D_L(z), \text{new observable}\}.} \tag{50.1}$$

50.2 Status rule

$$\mathsf{incomplete} \Longrightarrow \textbf{To Be Filled Later}. \tag{50.2}$$

ID	Domain	Requirement	Status	Evidence	Next dependency
F01	Primitive theory	Finite projector reconstruction, current, and refinement	Closed-Controlled	Exact corner and transfer theorems	None
F02	Primitive theory	Countable graph-compatible unbounded dynamics	Closed-Controlled	Graph-norm theorem	Quadratic-form localization
F03	Primitive theory	Positive or overlapping atlas calculus	Closed-Controlled	Positive-effect current and dilation	Adaptive infinite atlases
F04	Primitive theory	Uniform nonlinear infinite refinement tower	Partial - To Be Filled Later	Bounded and compact reconstruction	Uniform nonlinear estimates
F05	Primitive theory	Local spatial metric or tetrad variables	Closed-Controlled	Real tetrad–connection parent and regular time-gauge reduction	Global treatment of degenerate strata
F06	Primitive theory	Local lapse N(x,t)	Closed-Controlled	Lorentzian 3+1 parent split and Hamiltonian multiplier	Practical finite-range evolution
F07	Primitive theory	Local shift N^i(x,t)	Closed-Controlled	Tangential BFV generator and reduced diffeomorphism moment map	Full nonspherical numerical implementation
F08	Primitive theory	Unified matter multiplet	Closed-Controlled	One real Einstein–Cartan–Maxwell–Dirac parent action	Standard-Model completion
F09	Primitive theory	Single nonlinear local 3+1 action	Closed-Conditional	Covariant Lorentzian gravity–matter parent, classical BV master action, and regular 3+1 split	Strict finite-range microscopic realization and quantum anomaly control
F10	Primitive theory	Axiomatic uniqueness or controlled family of admissible actions	To Be Filled Later	No uniqueness theorem	Constraint and universality principles
F11	Primitive theory	No-go theorem for nonzero exact derivations of the finite pointwise algebra	Closed-Controlled	Task J primitive-idempotent proof and rank audit	None
F12	Primitive theory	Exact finite corner-algebra star derivations and trace integration by parts	Closed-Controlled	Task J inner-derivation theorem, curvature, and refinement	Nonlinear ordered constraint construction
F13	Primitive theory	Exact dyadic-sign and odd-adic torsion/principal quantum register factorization	Closed-Controlled	Task M finite unit-coordinate and CRT theorems	Physical identification of the adic registers
F14	Primitive theory	Exact p-adic Haar/detail projector tower	Closed-Controlled	Tasks N and T: finite child-DFT wavelets, direct-limit cylinder filtration, and strong Haar resolution	Physical locality and nonlinear field coupling
F15	Primitive theory	Exact anomaly-free finite adic quantum refinement	Closed-Controlled	Task N Hamiltonian, unitary, Euclidean, resolvent, and Feshbach intertwining	Relativistic interacting continuum limit
F16	Primitive theory	Interacting multi-prime Hamiltonians with simultaneous exact refinement	Closed-Controlled	Task O cross-prime stable packet interactions	Non-Abelian and field-theoretic extension
F17	Primitive theory	Valuation-weighted quantum leakage and hidden-sector hierarchy	Closed-Conditional	Task O first-order leakage and second-order Feshbach law	Derive physical adic weights from one microscopic action
F18	Primitive theory	Map from prime-tree locality to ordinary 3+1 causal locality	Partial - To Be Filled Later	Task Q exact causal locality on a physical-graph times adic-tree product	Lorentzian continuum reconstruction theorem
F19	Primitive theory	Haar-derived descendant normalization and collective coupling trichotomy	Closed-Controlled	Task P p^{-w/2} amplitude, p^{-w} Born weight, and alpha-threshold theorem	Microscopic action selecting the coupling exponent
F20	Primitive theory	Exact spacetime-adic product-graph quantum continuity and causal cone	Closed-Controlled	Task Q local flux and product-distance theorem	3+1 Lorentzian and gravitational lift
F21	Primitive theory	Exact branch-normalized adic radial reduction and continued-fraction self-energy	Closed-Conditional	Task R Jacobi-chain and visible-hidden Schur identities	Microscopic radial spectrum and continuum limit
F22	Primitive theory	Exact adic quantum polyphase/Haar refinement and perfect reconstruction	Closed-Controlled	Tasks S and U: finite digit-reversal/paraunitary calculus plus arbitrary-prime Schur-Chebyshev decimation	Interacting nonlinear renormalization fixed point
F23	Primitive theory	Infinite-depth p-adic Haar direct limit and strong projector resolution	Closed-Controlled	Task T identifies the finite tower with L^2(Z_p), proves strong martingale resolution, and gives quantitative tails	Nonlinear and relativistic coupling
F24	Primitive theory	Exact prime-polyphase spectral renormalization semigroup and fixed measure	Closed-Conditional	Task U Chebyshev Schur flow, commuting prime decimations, and invariant arcsine measure in the free nearest-neighbor sector	Interacting gauge-covariant field theory

ID	Domain	Requirement	Status	Evidence	Next dependency
F25	Primitive theory	Exact nonlinear BFV–BV transfer across projector and polyphase resolutions	Closed-Conditional	Chapter 22 homological transfer, cyclic BFV charge, BV pushforward, and direct–iterated elimination theorem	Construct and control a specific anomaly-free microscopic 3+1 gravity–matter theory
CST01	Constraints	Homogeneous Hamiltonian constraint	Closed-Controlled	FLRW and scalar action	None
CST02	Constraints	Exact homogeneous scalar constraint propagation	Closed-Controlled	Discrete-gradient identity	None
CST03	Constraints	Reduced nonlinear Gowdy integrability identity	Closed-Controlled	Exact jet-algebra identity	None
CST04	Constraints	Exact global Gowdy momentum constraint under a discrete map	Closed-Controlled	Task A constrained map	Local multisymplectic identity
CST05	Constraints	Exact local discrete Gowdy constraint one-form	To Be Filled Later	Second-order local residual only	Multisymplectic or discrete multiplier construction
CST06	Constraints	Local 3+1 Hamiltonian constraint	Closed-Conditional	Resolved self-dual first-class algebra and regular Lorentzian BFV reduction	Global degenerate-chart extension
CST07	Constraints	Three local momentum constraints	Closed-Conditional	Exact resolved momentum algebra and tangential boundary moment map	Full finite-range 3+1 solver
CST08	Constraints	Group-valued Gauss constraints for non-Abelian matter	Partial - To Be Filled Later	Exact Lorentz Gauss sector, U(1) Gauss sector, and SU(2) kinematics	Dynamical Yang–Mills electric sector
CST09	Constraints	Closed Hamiltonian-momentum-Gauss algebra	Closed-Conditional	Resolved Ashtekar algebra, Lorentzian boundary BFV charge, and reduced classical master equation	Global second-class and quantum anomaly analysis
CST10	Constraints	Constraint propagation in nonlinear 3+1 evolution	Closed-Conditional	BFV-intertwining slab relation and exact Dirac-gradient step on the regular gauge-fixed chart	Full gravity implementation and uniform refinement bounds
CST11	Constraints	Discrete Bianchi identity for gravitational curvature	Closed-Conditional	Exact curvature and torsion Bianchi identities in the resolved Lorentzian parent	Strict finite-range nonlinear tetrad curvature
CST12	Constraints	Stress-energy conservation derived from symmetry	Closed-Conditional	Diffeomorphism Noether identity, Maxwell–Dirac exchange, scalar mass flux, and dark-sector continuity	Quantum and full kinetic extensions
CST13	Constraints	Diffeomorphism covariance or precise emergent replacement	Closed-Conditional	Off-shell covariant parent algebra and complete boundary diffeomorphism moment map	Global regulator-independent realization
CST14	Constraints	Refinement maps preserve the physical constraint surface	Closed-Conditional	Alias-free embeddings, transferred BFV charge, and adaptive canonical relation	Uniform infinite-tower stability
CST15	Constraints	Exact local 3+1 linearized Hamiltonian and momentum constraint propagation	Closed-Controlled	Task K dot C=-delta_i C_i and dot C_i=0	None at linear order
CST16	Constraints	Exact nonlinear scalar diffeomorphism-bracket defect formula	Closed-Controlled	Task L closure defect equals the discrete Leibniz defect contraction	Lift to ordered nonlinear projector-ADM brackets
CST17	Constraints	Conditional exact nonlinear closure under projector elimination	Closed-Conditional	Chapter 22 transfers a nilpotent fine BFV charge through a cyclic contraction and preserves the BV quantum master equation	Supply an anomaly-free microscopic 3+1 gravity–matter charge and establish its continuum limit
CL01	Continuum	Dirac continuum limit	Closed-Controlled	1+1 and 3+1 controlled sectors	Interacting fields
CL02	Continuum	Weak-field Poisson limit	Closed-Controlled	Variational radial action	Strong field
CL03	Continuum	FLRW reduced limit	Closed-Controlled	Background action and convergence	Perturbations
CL04	Continuum	Linear tensor-wave limit	Closed-Controlled	Fierz-Pauli and TT sector	Nonlinear waves
CL05	Continuum	Reduced nonlinear vacuum limit	Closed-Conditional	Gowdy dynamics and horizon-penetrating spherical electrovac benchmark	Axisymmetric and unrestricted 3+1 geometry
CL06	Continuum	Full Riemann tensor from projector holonomy	Partial - To Be Filled Later	Lorentzian connection curvature and reduced metric–holonomy correspondence	Uniform 3+1 contraction theorem
CL07	Continuum	Ricci tensor, scalar, and Einstein tensor	Partial - To Be Filled Later	Parent Einstein–Cartan variation and spherical Einstein diagnostics	General projector-holonomy contraction theorem

ID	Domain	Requirement	Status	Evidence	Next dependency
CL08	Continuum	Einstein-matter continuum limit	Partial - To Be Filled Later	Exact continuum parent action and multiple reduced convergence tests	Microscopic-to-continuum uniform theorem
CL09	Continuum	Uniform consistency error estimates	To Be Filled Later	Measured orders	Analytic truncation bounds
CL10	Continuum	Stability estimates	To Be Filled Later	Finite tests	Energy estimates
CL11	Continuum	Joint matter-geometry convergence	To Be Filled Later	Separate controlled sectors	Coupled local solver
CL12	Continuum	Independence from the refinement path	Partial - To Be Filled Later	Task T commuting finite-prime cylinder refinements and Task U commuting free spectral decimations	Nonlinear interacting infinite-tower theorem
CL13	Continuum	Microscopic universality across admissible atlases	To Be Filled Later	No universality class theorem	Renormalization or coarse-graining fixed point
BG01	Background	Flat FLRW redshift, horizons, and propagation	Closed-Controlled	Exact and numerical tests	None
BG02	Background	Homogeneous matter-geometry backreaction	Closed-Controlled	Packet stress-energy and action	None
BG03	Background	Homogeneous scalar cosmology	Closed-Controlled	Constraint-preserving discrete gradient	None
BG04	Background	Nonzero spatial curvature K	Closed-Controlled	Task D curved FLRW action and distance geometry	None in the homogeneous sector
BG05	Background	Cosmological constant sector	Closed-Controlled	Task D explicit Lambda action, Friedmann term, and observables	Microscopic vacuum-energy explanation
BG06	Background	Multiple interacting fluids	Closed-Controlled	Task D conservative homogeneous transfer with sum_i Q_i=0	Microphysical Q_i, positivity, and global existence
BG07	Background	Multiple scalar fields	To Be Filled Later	Single canonical field	Field-space metric and entropy modes
BG08	Background	Anisotropic Bianchi backgrounds	To Be Filled Later	No complete anisotropic solver	Directional scale factors
BG09	Background	Contracting and bouncing solutions	To Be Filled Later	Expanding tests only	Stable bounce mechanism
BG10	Background	Nonlinear stability of FLRW	To Be Filled Later	No global theorem	Perturbative and energy estimates
BG11	Background	Global existence or controlled breakdown criteria	To Be Filled Later	Finite trajectories	Well-posedness theorem
BG12	Background	Equation of state derived from microscopic packet dynamics	Partial - To Be Filled Later	Dust/radiation packet scaling	Interacting microscopic matter
MAT01	Matter	Canonical scalar matter	Closed-Controlled	Homogeneous and prescribed inhomogeneous sectors	Full local gravity coupling
MAT02	Matter	Massive and massless packet propagation	Closed-Controlled	Dirac and FLRW tests	Interactions
MAT03	Matter	External-link Abelian gauge covariance	Closed-Controlled	Exact link intertwining	None
MAT04	Matter	Dynamical 1+1 Maxwell-Dirac backreaction	Closed-Controlled	Gauss and energy balance	Constraint-preserving time discretization
MAT05	Matter	Dynamical 3+1 Maxwell-Dirac	Closed-Conditional	Unified covariant parent action, canonical constraints, and regular reduced chart	Full finite-range coupled evolution
MAT06	Matter	Transverse photons and magnetic fields	To Be Filled Later	No complete 3+1 dynamics	Maxwell plaquette action
MAT07	Matter	Dynamical SU(2) sector	To Be Filled Later	Ordered holonomy refinement only	Group-valued electric fields
MAT08	Matter	Dynamical SU(3) sector	To Be Filled Later	No color dynamics	Non-Abelian matter action
MAT09	Matter	Fermion species, masses, and mixing	To Be Filled Later	One free Dirac multiplet	Flavor structure
MAT10	Matter	Neutrino sector	To Be Filled Later	Not constructed	Mass and collision hierarchy
MAT11	Matter	Baryonic fluid limit	Partial - To Be Filled Later	Task H tight-coupling baryon inertia and acoustic loading	Full hydrodynamic and collision closure
MAT12	Matter	Photon distribution function	Partial - To Be Filled Later	Task I brightness multipoles and collisionless distribution	Thomson, polarization, and neutrino kinetic hierarchy
MAT13	Matter	Collision integrals	Partial - To Be Filled Later	Task H Thomson opacity and tight-coupling diffusion model	Full collision operator and nonequilibrium kinetics
MAT14	Matter	Particle creation and annihilation	To Be Filled Later	First-quantized dynamics	Fock space
MAT15	Matter	Quantized gauge-matter Fock space	To Be Filled Later	Classical links and first-quantized states	Canonical or path-integral quantization
MAT16	Matter	Quantum field theory on curved spacetime	To Be Filled Later	Separate Dirac and FLRW models	Common semiclassical limit
PT01	Perturbations	Scalar metric decomposition and gauge transformations	Closed-Controlled	Task B explicit convention	None
PT02	Perturbations	Gauge-invariant single-field curvature perturbation	Closed-Controlled	Task B zeta identity	None
PT03	Perturbations	Linear lapse and shift elimination in comoving gauge	Closed-Controlled	Task B ADM constraints	None
PT04	Perturbations	Quadratic curvature action	Closed-Controlled	Task B Einstein-scalar reduction	None
PT05	Perturbations	Mukhanov-Sasaki variable and mode equation	Closed-Controlled	Task B canonical action	None
PT06	Perturbations	Lattice Mukhanov-Sasaki refinement	Closed-Controlled	Second-order and fourth-order refined tests	General backgrounds

ID	Domain	Requirement	Status	Evidence	Next dependency
PT07	Perturbations	Single-field superhorizon constant mode	Closed-Controlled	Long-wavelength integral solution	Nonadiabatic sources
PT08	Perturbations	Scalar transfer function through radiation and matter eras	To Be Filled Later	No thermal history	Boltzmann background
PT09	Perturbations	Vector constraints and vorticity	Partial - To Be Filled Later	Task E gauge-invariant source-free vector mode and a^-2 decay	Matter-sourced vector constraint and vorticity hierarchy
PT10	Perturbations	Tensor modes on FLRW	Closed-Controlled	Tasks E and G: FLRW tensor action, propagation, constant-epsilon primordial spectrum, and lattice refinement	General sources and precision primordial matching
PT11	Perturbations	Matter-sourced cosmological tensor modes	Partial - To Be Filled Later	Task E sourced FLRW equation with unsolved anisotropic-stress sector	Dynamical cosmological source hierarchy
PT12	Perturbations	Scalar-vector-tensor refinement covariance	Partial - To Be Filled Later	Tasks B and E use the same cubic projector Laplacian in separate sectors	Unified sourced SVT theorem
PT13	Perturbations	Nonlinear second-order perturbations	To Be Filled Later	Linear scalar result only	Cubic action
PT14	Perturbations	Mode coupling and non-Gaussian kernels	To Be Filled Later	Not constructed	Nonlinear perturbation action
PT15	Perturbations	Multi-field entropy and isocurvature modes	To Be Filled Later	Single field only	Multiple scalar fields
PR01	Primordial	Accelerated scalar background	Closed-Controlled	Task G exact constant-epsilon power-law accelerated background	Graceful exit and microphysical potential
PR02	Primordial	Horizon problem resolution	Closed-Conditional	Task G comoving Hubble radius suppressed by exp[-(1-epsilon)N]	Initial-condition and graceful-exit completion
PR03	Primordial	Flatness problem resolution	Closed-Conditional	Task G curvature fraction suppressed by exp[-2(1-epsilon)N]	Initial curvature and exit matching
PR04	Primordial	Initial-condition attractor	To Be Filled Later	Finite examples only	Phase-space theorem
PR05	Primordial	Graceful exit	To Be Filled Later	No complete inflation model	Potential and reheating
PR06	Primordial	Reheating	To Be Filled Later	No particle-production sector	Interacting quantum matter
PR07	Primordial	Quantum origin of scalar modes	Partial - To Be Filled Later	Task G fixed-background canonical Bunch-Davies mode and Wronskian	Vacuum derived from projector axioms and quantized geometry
PR08	Primordial	Scalar power spectrum normalization	Closed-Conditional	Task G exact constant-epsilon spectrum conditional on A, epsilon, and Bunch-Davies vacuum	Absolute scale from a microphysical background and vacuum selection
PR09	Primordial	Spectral index and running	Closed-Controlled	Task G exact n_s-1=-2epsilon/(1-epsilon) and alpha_s=0 in the controlled model	General time-dependent epsilon and observational fit
PR10	Primordial	Primordial tensor spectrum and tensor-to-scalar ratio	Closed-Conditional	Task G exact two-polarization spectrum, r=16epsilon, and consistency relation	General slow-roll background and vacuum derivation
PR11	Primordial	Primordial non-Gaussianity	To Be Filled Later	No cubic perturbation action	Interaction Hamiltonian
PR12	Primordial	Primordial isocurvature constraints	To Be Filled Later	Single field only	Multi-field sector
TH01	Thermal history	Reheating temperature and particle production	To Be Filled Later	Not yet constructed	Interacting quantum matter
TH02	Thermal history	Radiation thermalization	To Be Filled Later	Not yet constructed	Collision integrals
TH03	Thermal history	Neutrino decoupling	To Be Filled Later	Not yet constructed	Neutrino kinetic sector
TH04	Thermal history	Electron-positron annihilation	To Be Filled Later	Not yet constructed	Standard particle content
TH05	Thermal history	Big-bang nucleosynthesis	To Be Filled Later	Not yet constructed	Nuclear reaction network
TH06	Thermal history	Baryon-to-photon ratio	To Be Filled Later	Not yet constructed	Baryogenesis mechanism
TH07	Thermal history	Recombination	Closed-Conditional	Task H Saha-equilibrium hydrogen recombination benchmark	Nonequilibrium multilevel hydrogen and helium kinetics
TH08	Thermal history	Photon decoupling	Closed-Conditional	Task H Saha optical depth and visibility function	Full collision hierarchy and reionization
TH09	Thermal history	Silk damping	Partial - To Be Filled Later	Task H tight-coupling diffusion scale	Full hierarchy and precision diffusion kernel
TH10	Thermal history	Reionization	To Be Filled Later	Not yet constructed	Astrophysical source model
TH11	Thermal history	Photon and neutrino Boltzmann hierarchies	Partial - To Be Filled Later	Task I collisionless photon brightness hierarchy	Thomson collisions, polarization, neutrinos, and metric sources
TH12	Thermal history	Baryon-photon acoustic oscillations	Closed-Controlled	Task H tight-coupling oscillator, baryon loading, and sound horizon	Full driven hierarchy and drag-epoch calibration
SF01	Structure formation	Linear density-growth equation	Closed-Controlled	Task F dust continuity-Euler-Poisson derivation	Relativistic radiation and pressure corrections
SF02	Structure formation	Growth factor D(a)	Closed-Controlled	Task F EdS exact modes and flat LambdaCDM Heath solution	Radiation, baryons, and neutrinos

ID	Domain	Requirement	Status	Evidence	Next dependency
SF03	Structure formation	Matter transfer function	To Be Filled Later	No thermal history	Thermal history
SF04	Structure formation	Linear matter power spectrum	To Be Filled Later	Growth factor only	Primordial spectrum and transfer function
SF05	Structure formation	Baryon acoustic oscillation scale	Closed-Conditional	Task H conditional sound horizon and geometric acoustic ruler	Precision drag epoch and full baryon transfer function
SF06	Structure formation	Weak-lensing shear	To Be Filled Later	Metric potentials and matter distribution incomplete	Metric potentials and matter distribution
SF07	Structure formation	Redshift-space distortions	Partial - To Be Filled Later	Task F growth rate f(a) available	Velocity statistics and tracer bias
SF08	Structure formation	Nonlinear mode coupling	To Be Filled Later	Not yet constructed	Second-order perturbations
SF09	Structure formation	N-body limit	To Be Filled Later	Not yet constructed	Particle-gravity solver
SF10	Structure formation	Hydrodynamic limit	To Be Filled Later	Not yet constructed	Baryonic fluid sector
SF11	Structure formation	Halo formation and profiles	To Be Filled Later	Not yet constructed	Nonlinear structure solver
SF12	Structure formation	Galaxy bias and cosmic web	To Be Filled Later	Not yet constructed	Astrophysical modeling
DM01	Dark sectors	Cold-dark-matter background scaling	Closed-Conditional	Produced Haar-detail tower and scale-covariant heavy-band limit give $\rho_D \propto a^{-3}$	Microscopic parameter derivation
DM02	Dark sectors	Dark-matter clustering	Closed-Conditional	Multi-level cold growth and scale-covariant two-fluid transfer suppression	Full collisionless Boltzmann and nonlinear structure
DM03	Dark sectors	Halo and lensing phenomenology	Partial - To Be Filled Later	Lensing source derived from the same resolved density and a finite half-mode scale	Halo solutions and shear likelihood
DM04	Dark sectors	Small-scale dark-matter constraints	Partial - To Be Filled Later	Scale-covariant Jeans and half-mode laws	Observational calibration and nonlinear phase mixing
DE01	Dark sectors	Accelerating component with w approximately -1	Closed-Conditional	Density-locked average mode yields w=-1 and transient acceleration	Microscopic selection of gamma and crossing epoch
DE02	Dark sectors	Dark-energy perturbations	Closed-Conditional	Canonical quadratic locked-mode action with vanishing linear stress source	Full coupled Einstein–Boltzmann perturbations
DE03	Dark sectors	Ghost and gradient stability	Closed-Conditional	$Q = 1$, $c_s^2 = 1$, and positive locking curvature on the declared branch	Radiative and nonlinear stability
DE04	Dark sectors	Cosmological-constant hierarchy	To Be Filled Later	The separate Ω_Λ input is removed inside the dark-sector model only	Total vacuum-energy and radiative mechanism
DE05	Dark sectors	Coincidence problem	Partial - To Be Filled Later	Dark density controls locking and predicts a release scale	First-principles parameter selection and data test
SG01	Strong field	Minkowski benchmark	Closed-Conditional	Exact Lorentzian parent vacuum and reduced finite-level stability	Global nonlinear stability
SG02	Strong field	Schwarzschild benchmark	Closed-Controlled	Horizon-penetrating Reissner–Nordstrom family includes Schwarzschild at Q=0	Nonspherical perturbations
SG03	Strong field	Interior stellar solutions	To Be Filled Later	None	Fluid and spherical constraints
SG04	Strong field	Kerr benchmark	To Be Filled Later	None	Rotation and axisymmetry
SG05	Strong field	Spherical collapse	Closed-Controlled	Smooth scalar data form a resolved marginally outer trapped sphere with refinement convergence	Regular center, critical scaling, and global theorem
SG06	Strong field	Binary inspiral and merger	Partial - To Be Filled Later	Linear waves only	3+1 numerical relativity
SG07	Strong field	Black-hole ringdown	Partial - To Be Filled Later	Linear waves only	Strong-field background
SG08	Strong field	Big-bang singularity criterion	To Be Filled Later	None	Global nonlinear theorem
SG09	Strong field	Singularity resolution or persistence	To Be Filled Later	None	Quantum or discrete strong-field closure
SG10	Strong field	Bouncing cosmology	To Be Filled Later	None	Stable high-curvature mechanism
SG11	Strong field	BKL anisotropic regime	To Be Filled Later	None	Anisotropic local solver

ID	Domain	Requirement	Status	Evidence	Next dependency
SG12	Strong field	Topology and topology change	To Be Filled Later	None	Global atlas category
SG13	Strong field	Cosmic censorship or replacement principle	To Be Filled Later	None	Collapse simulations
SG14	Strong field	Black-hole and cosmological horizon entropy	To Be Filled Later	None	Quantum geometry
SG15	Strong field	de Sitter thermodynamics	Partial - To Be Filled Later	Background only	Quantum fields and horizon states
OB01	Observables	Expansion history H(z)	Closed-Controlled	Task D general constant-w, curvature, and Lambda component library	Observational parameter calibration
OB02	Observables	Comoving distance	Closed-Controlled	Task D curved FLRW line-of-sight integral	Observational parameter calibration
OB03	Observables	Luminosity distance	Closed-Controlled	Task D transverse curvature map and D_L=(1+z)D_M	Source calibration and data likelihood
OB04	Observables	Angular-diameter distance	Closed-Controlled	Task D D_A=D_M/(1+z) and distance duality	Source calibration and data likelihood
OB05	Observables	Age of the universe	Closed-Controlled	Task D age and lookback-time functionals with exact benchmarks	Observational parameter calibration
OB06	Observables	CMB temperature spectrum	Partial - To Be Filled Later	Task I controlled low-ell Sachs-Wolfe spectrum	Thomson hierarchy, acoustic peaks, metric sources, and reionization
OB07	Observables	CMB TE and EE spectra	To Be Filled Later	No polarization hierarchy	Boltzmann solver
OB08	Observables	CMB BB spectrum	To Be Filled Later	No primordial tensor pipeline	Tensor quantization and lensing
OB09	Observables	CMB lensing	To Be Filled Later	Weak-field lensing only	Large-scale structure
OB10	Observables	Matter power spectrum	To Be Filled Later	No transfer function	Structure formation
OB11	Observables	BAO observable	Closed-Conditional	Task H conditional sound horizon and acoustic ruler	Precision drag epoch, baryon transfer, and observational calibration
OB12	Observables	Growth rate f sigma_8	Closed-Conditional	Task F f(a)D(a)sigma_8,0 functional	Transfer functions, tracer model, and fitted normalization
OB13	Observables	Weak-lensing spectra	To Be Filled Later	No source distribution	Structure and statistics
OB14	Observables	Gravitational-wave luminosity distance	Partial - To Be Filled Later	Task E unmodified FLRW tensor amplitude gives D_LˆGW=D_LˆEM after source normalization	Binary source generation and modified-friction tests
OB15	Observables	Standard-siren relation	To Be Filled Later	No source model	Binary and propagation sectors
OB16	Observables	Cosmological time-delay observables	Closed-Conditional	Task C leading finite-scale law	Source and detector systematics
PH01	Phenomenology	Local directional finite-scale dispersion	Closed-Conditional	Selected cubic walk	Universal microdynamics
PH02	Phenomenology	Cosmological redshift-integrated photon delay	Closed-Conditional	Task C proper/comoving laws	Preferred-frame transport
PH03	Phenomenology	Source-frame to observer-frame energy mapping	Closed-Controlled	E(z)=E0(1+z) within FLRW	Modified redshift if dispersion backreacts
PH04	Phenomenology	Intrinsic source-delay population model	To Be Filled Later	No source model	Astrophysical catalog
PH05	Phenomenology	Detector response and timing systematics	To Be Filled Later	No detector model	Instrument-specific likelihood
PH06	Phenomenology	Current observational bound on microscopic scale	To Be Filled Later	Synthetic inference only	Public data and systematics
PH07	Phenomenology	Universal correction forced by projector axioms	To Be Filled Later	Current law is walk-dependent	Unique microscopic dynamics
PH08	Phenomenology	Distinction from generic Lorentz-violation EFT	To Be Filled Later	Cubic angular signature identified	Multi-observable comparison
PH09	Phenomenology	Second independent observable with the same scale	To Be Filled Later	Only photon delay developed	Polarization, GW, or matter effect
PH10	Phenomenology	Joint falsifiability criterion	To Be Filled Later	No joint data test	Two or more linked observables
NUM01	Numerics	Controlled reduced solvers	Closed-Controlled	All package scripts rerun	Extend per sector
NUM02	Numerics	Constraint-preserving Gowdy map	Closed-Controlled	Task A exact global constraint	Exact local multisymplectic closure
NUM03	Numerics	Background cosmology solver	Closed-Controlled	Task D curved constant-w component library, interacting fluids, distances, and convergence	Microphysical reactions and observational calibration
NUM04	Numerics	Linear scalar perturbation solver	Partial - To Be Filled Later	Task B exact benchmark	General time-dependent backgrounds
NUM05	Numerics	Vector and tensor perturbation solver	Closed-Controlled	Task E source-free vector and vacuum FLRW tensor benchmarks	Matter sources and primordial normalization
NUM06	Numerics	Boltzmann hierarchy solver	Partial - To Be Filled Later	Task I minimal collisionless photon multipole hierarchy	Thomson collisions, polarization, neutrinos, and metric sources
NUM07	Numerics	Nonlinear local gravity solver	Partial - To Be Filled Later	Finite-range spherical electrovac and horizon-penetrating scalar-collapse solvers	Unrestricted 3+1 solver
NUM08	Numerics	Constraint-preserving adaptive refinement	Closed-Conditional	Symplectic embeddings, BFV-intertwining slab relation, and Dirac-gradient update	Level-uniform implementation
NUM09	Numerics	Comparison with numerical-relativity codes	To Be Filled Later	None	Shared benchmarks
NUM10	Numerics	N-body implementation	To Be Filled Later	None	Structure formation
NUM11	Numerics	Parameter-inference interface	To Be Filled Later	Synthetic one-parameter fit	Observable pipeline

ID	Domain	Requirement	Status	Evidence	Next dependency
NUM12	Numerics	Automatic sign, unit, gauge, and constraint checking	Partial - To Be Filled Later	Symbolic audit and correction ledger	Proof assistant and broader dimensional engine
NUM13	Numerics	Exact adic polyphase analysis, synthesis, and block-symbol solver	Closed-Controlled	Tasks S and U finite transforms, arbitrary-prime bands, channels, and Schur-Chebyshev reduction	Interacting field-theory implementation
NUM14	Numerics	Infinite-depth Haar truncation and hierarchical resolvent solver	Closed-Controlled	Task T finite-cylinder approximants, Haar-Sobolev tails, and exact resolvent-tail certificate	Adaptive nonlinear implementation
ST01	Statistics	Cosmological parameter vector	To Be Filled Later	Not yet constructed	Physical model closure
ST02	Statistics	Joint likelihood	To Be Filled Later	Not yet constructed	Observable predictions
ST03	Statistics	Priors and identifiability	To Be Filled Later	Not yet constructed	Parameter map
ST04	Statistics	Posterior sampling	To Be Filled Later	Not yet constructed	Likelihood implementation
ST05	Statistics	Nuisance parameters	To Be Filled Later	Not yet constructed	Source and detector models
ST06	Statistics	Systematic-error propagation	To Be Filled Later	Not yet constructed	Instrument and astrophysical models
ST07	Statistics	Model comparison with GR plus LambdaCDM	To Be Filled Later	Not yet constructed	Matched observable pipeline
ST08	Statistics	Out-of-sample prediction	To Be Filled Later	Not yet constructed	Frozen parameter fit
ST09	Statistics	Joint independent data classes	To Be Filled Later	Not yet constructed	CMB, BAO, SN, LSS, GW outputs
ST10	Statistics	No retuning of microscopic dynamics across datasets	To Be Filled Later	Not yet constructed	Universal parameterization
Q01	Quantum cosmology	Hilbert space of projector geometries	To Be Filled Later	Not yet constructed	Quantized geometry variables
Q02	Quantum cosmology	Quantum amplitudes for refinement histories	To Be Filled Later	Not yet constructed	Path integral or canonical quantization
Q03	Quantum cosmology	Quantum Hamiltonian constraint	To Be Filled Later	Not yet constructed	Constraint algebra
Q04	Quantum cosmology	Semiclassical FLRW states	To Be Filled Later	Not yet constructed	Geometric Hilbert space
Q05	Quantum cosmology	Quantum field theory on curved projector geometry	To Be Filled Later	Not yet constructed	Common quantization
Q06	Quantum cosmology	Vacuum-state selection	Partial - To Be Filled Later	Task G states a fixed-background Bunch-Davies choice	Vacuum selected by projector and geometric dynamics
Q07	Quantum cosmology	Quantum primordial fluctuation generation	Partial - To Be Filled Later	Task G canonical fixed-background scalar and tensor modes	Quantized projector geometry and common semiclassical limit
Q08	Quantum cosmology	Decoherence and classicalization	To Be Filled Later	Not yet constructed	Environment or coarse-graining theory
Q09	Quantum cosmology	Relational time	To Be Filled Later	Not yet constructed	Quantum constraint solution
Q10	Quantum cosmology	Arrow of time	To Be Filled Later	Not yet constructed	Entropy and initial condition
Q11	Quantum cosmology	Cosmological horizon entropy	To Be Filled Later	Not yet constructed	Quantum geometry
Q12	Quantum cosmology	Probability measure over cosmological solutions	To Be Filled Later	Not yet constructed	Quantum or statistical foundation
Q13	Quantum cosmology	Finite dyadic-odd quantum register and CRT factorization	Closed-Controlled	Task M exact finite unitary coordinate split	Continuum physical interpretation
Q14	Quantum cosmology	Model-specific anomaly-free adic quantum refinement	Closed-Conditional	Tasks N-O zero leakage and zero Feshbach self-energy	Universal interacting quantum geometry
Q15	Quantum cosmology	Cross-prime entangling quantum interaction	Closed-Conditional	Task O exact one-ebit controlled interaction	Observable field-theory realization
Q16	Quantum cosmology	Adic quantum observable tied to one physical scale	To Be Filled Later	No empirical identification	Phenomenology and independent reproduction
Q17	Quantum cosmology	Exact multi-level adic polyphase quantum transform	Closed-Controlled	Tasks S and T: recursive finite child-DFT/Haar equivalence and complete L^2(Z_p) martingale limit	Interacting field-theory implementation
Q18	Quantum cosmology	Exact polyphase low/detail Feshbach reduction	Closed-Controlled	Tasks S and U: dyadic identity and arbitrary-prime Chebyshev Schur law	Nonlinear field-theory renormalization
Q19	Quantum cosmology	Visible Dirac sector dressed by an adic radial self-energy	Closed-Conditional	Task R determinant factorization and nonlinear dispersion equation	Lorentz- and gauge-covariant microscopic coupling

ID	Domain	Requirement	Status	Evidence	Next dependency
Q20	Quantum cosmology	Infinite-depth hierarchical p-adic quantum Hamiltonian	Closed-Controlled	Task T self-adjoint level Hamiltonian, exact unitary/resolvent calculus, and compact-resolvent criterion	Microscopic derivation and relativistic coupling
Q21	Quantum cosmology	Free-sector prime-polyphase renormalization fixed measure	Closed-Conditional	Task U invariant arcsine measure, Koopman mode injection, and exact prime semigroup	Interacting quantum-field fixed point and observable identification
Q22	Quantum cosmology	Nonlinear gauge-preserving polyphase effective action	Closed-Conditional	Chapter 22 exact BFV homological transfer, BV pushforward, measure correction, and refinement semigroup	Specific microscopic quantum-gravity action, anomaly analysis, and Einstein–matter continuum theorem
VAL01	Validation	Equation-first symbolic archive	Closed-Controlled	Formula registry	Continue updates
VAL02	Validation	Explicit status and correction ledger	Closed-Controlled	Audited package	Continue updates
VAL03	Validation	Numerical residual and convergence tests	Closed-Controlled	Scientific scripts	Independent implementation
VAL04	Validation	Proof-assistant formalization	To Be Filled Later	No complete formal proof	Lean, Isabelle, or Coq
VAL05	Validation	Independent derivation of central theorems	To Be Filled Later	Internal audit only	External mathematical review
VAL06	Validation	Independent numerical reproduction	To Be Filled Later	Single codebase	Second implementation
VAL07	Validation	Complete literature-priority audit	To Be Filled Later	Partial comparisons	Specialist review
VAL08	Validation	Peer review by relativists and cosmologists	To Be Filled Later	Not yet external	Submission and open reproducibility

50.3 Critical gates

$$\mathsf{Gate}_1 : \text{one nonlinear Lorentzian gravity–matter parent action} \quad \textbf{Closed-Conditional}, \tag{50.3}$$

$$\mathsf{Gate}_2 : \{\Omega, \Omega\} = 0 \text{ under BFV–BV transfer and regular reduction} \quad \textbf{Closed-Conditional}, \tag{50.4}$$

$$\mathsf{Gate}_3 : \{\mathcal{H}, \mathcal{H}_i, \mathcal{G}\}_{\text{resolved}} \text{ closed} \quad \textbf{Closed-Conditional}, \tag{50.5}$$

$$\mathsf{Gate}_4 : \mathcal{E}_h \to G_{\mu\nu} - 8\pi G T_{\mu\nu} \text{ uniformly from microscopic packets} \quad \textbf{To Be Filled Later}, \tag{50.6}$$

$$\mathsf{Gate}_5 : \{\zeta, \text{vector}, h_{ij}^{\mathrm{TT}}\} \text{ with full kinetic sources} \quad \textbf{Partial}, \tag{50.7}$$

$$\mathsf{Gate}_6 : \{C_\ell, P(k), D_L, D_A, f\sigma_8\} \text{ from one frozen parameter set} \quad \textbf{To Be Filled Later}, \tag{50.8}$$

$$\mathsf{Gate}_7 : \delta\mathcal{O}_{\text{new}} \text{ universal and gauge invariant} \quad \textbf{Partial}, \tag{50.9}$$

$$\mathsf{Gate}_8 : \mathcal{L}(D \mid \theta), \text{ independent reproduction, and external review} \quad \textbf{To Be Filled Later}. \tag{50.10}$$

Part VI

Corrections, Reproducibility, and Verification

Chapter 51

Correction and Qualification Ledger

51.1 C1: Fierz–Pauli source sign C

Old incompatible pair:

$$S_{\mathrm{old}} = S_{\mathrm{kin}} - \frac{1}{2}\int h_{\mu\nu}T^{\mu\nu}\,\mathrm{d}^4x, \tag{51.1}$$

$$G^{(1)}_{\mu\nu} = +8\pi G T_{\mu\nu}. \tag{51.2}$$

Corrected pair for the printed kinetic convention:

$$S_{\mathrm{FP}} = S_{\mathrm{kin}} + \frac{1}{2}\int h_{\mu\nu}T^{\mu\nu}\,\mathrm{d}^4x, \tag{51.3}$$

$$\frac{\delta S_{\mathrm{FP}}}{\delta h^{\mu\nu}} = 0 \Longrightarrow G^{(1)}_{\mu\nu} = 8\pi G T_{\mu\nu}. \tag{51.4}$$

TT reduction:

$$\frac{1}{32\pi G}(\partial_t^2 - c^2\nabla^2)h_{ij}^{\mathrm{TT}} - \frac{1}{2}T_{ij}^{\mathrm{TT}} = 0 \Longrightarrow (\partial_t^2 - c^2\nabla^2)h_{ij}^{\mathrm{TT}} = 16\pi G T_{ij}^{\mathrm{TT}}. \tag{51.5}$$

51.2 C2: lattice d'Alembert uniqueness C

Correct:

$$h_j^n = F_{j-n} + G_{j+n}, \tag{51.6}$$

$$(F, G) \sim (F + \chi, G - \chi), \tag{51.7}$$

$$\chi_{r+2} = \chi_r. \tag{51.8}$$

Not generally correct:

$$(F, G) \sim (F + c, G - c) \quad \text{as the only ambiguity.} \tag{51.9}$$

Periodic characteristic packets require zero/Nyquist compatibility; otherwise:

$$F, G = \text{affine or alternating-affine on the covering lattice.} \tag{51.10}$$

51.3 C3: radial acceleration sign C

$$\Phi'(r) = +\frac{GM_{\mathrm{enc}}(r)}{r^2}, \tag{51.11}$$

$$g_+(r) := \Phi'(r) > 0, \tag{51.12}$$

$$a_r(r) := -\Phi'(r) = -\frac{GM_{\mathrm{enc}}(r)}{r^2}. \tag{51.13}$$

51.4 C4: optical deflection vector C

$$\boldsymbol{\alpha}_\perp = -\frac{2}{c^2}\int \nabla_\perp \Phi \, \mathrm{d}x, \tag{51.14}$$

$$\alpha := \|\boldsymbol{\alpha}_\perp\| > 0. \tag{51.15}$$

51.5 C5: Wilson-spacing remainder C

Universal compact expansion:

$$1 - \operatorname{Re} \mathrm{e}^{\mathrm{i}\theta} = \frac{1}{2}\theta^2 + \mathcal{O}(\theta^4). \tag{51.16}$$

Centered plaquette:

$$\theta = -q a_\mu a_\nu F_{\mu\nu} + \mathcal{O}(a^4) \Longrightarrow 1 - \operatorname{Re} W = \frac{1}{2} q^2 a_\mu^2 a_\nu^2 F_{\mu\nu}^2 + \mathcal{O}(a^6). \tag{51.17}$$

Corner nonsymmetric plaquette:

$$1 - \operatorname{Re} W = \frac{1}{2} q^2 a_\mu^2 a_\nu^2 F_{\mu\nu}^2 + \mathcal{O}(a^5) \quad \text{in general.} \tag{51.18}$$

51.6 C6: zero-over-zero diagnostic C

For pure dust:

$$\frac{\mathrm{d}}{\mathrm{d}t}(a^3\rho) = 0, \tag{51.19}$$

$$p\frac{\mathrm{d}}{\mathrm{d}t}(a^3) = 0, \tag{51.20}$$

$$\mathcal{R}_{\mathrm{abs}} := \left| \frac{\mathrm{d}}{\mathrm{d}t}(a^3\rho) + p\frac{\mathrm{d}}{\mathrm{d}t}(a^3) \right| \approx \epsilon_{\mathrm{mach}}, \tag{51.21}$$

$$\mathcal{R}_{\mathrm{rel}} := \frac{\mathcal{R}_{\mathrm{abs}}}{\left|\frac{\mathrm{d}}{\mathrm{d}t}(a^3\rho)\right| + \left|p\frac{\mathrm{d}}{\mathrm{d}t}(a^3)\right|} = \frac{0}{0} \quad \text{undefined.} \tag{51.22}$$

Reporting rule:

$$\mathcal{R}_{\mathrm{rel}} := \mathrm{NA} \quad \text{when the denominator vanishes numerically.} \tag{51.23}$$

51.7 C7: partition-incidence vector identity C

For

$$R_{A\alpha} = \mathbf{1}_{\{\alpha \in C_A\}}, \tag{51.24}$$

the correct identity is

$$R^{\mathsf{T}} \mathbf{1}_{\mathrm{c}} = \mathbf{1}_{\mathrm{f}}. \tag{51.25}$$

In general,

$$R\mathbf{1}_{\mathrm{f}} = (|C_1|, \ldots, |C_m|)^{\mathsf{T}} \neq \mathbf{1}_{\mathrm{c}}. \tag{51.26}$$

Therefore

$$\dot{E} = J\mathbf{1}_{\mathrm{f}} + s, \tag{51.27}$$

$$E^{\mathrm{c}} = RE, \qquad J^{\mathrm{c}} = RJR^{\mathsf{T}}, \tag{51.28}$$

$$\dot{E}^{\mathrm{c}} = RJ\mathbf{1}_{\mathrm{f}} + Rs \tag{51.29}$$

$$= RJR^{\mathsf{T}}\mathbf{1}_{\mathrm{c}} + Rs \tag{51.30}$$

$$= J^{\mathrm{c}}\mathbf{1}_{\mathrm{c}} + Rs. \tag{51.31}$$

51.8 C8: weighted quantum-leakage remainder C

For

$$H_\lambda = H_0 + \lambda B, \qquad QH_0P = 0, \qquad PBP = QBQ = 0, \tag{51.32}$$

Duhamel's formula gives

$$Q\mathrm{e}^{-\mathrm{i}tH_\lambda}P = -\mathrm{i}\lambda \int_0^t Q\mathrm{e}^{-\mathrm{i}(t-s)H_\lambda}B\mathrm{e}^{-\mathrm{i}sH_0}P \,\mathrm{d}s. \tag{51.33}$$

Hence

$$\boxed{\|Q\mathrm{e}^{-\mathrm{i}tH_\lambda}P\| \le |t|\,|\lambda|\,\|QBP\|.} \tag{51.34}$$

The correct generic short-time expansion is

$$Q\mathrm{e}^{-\mathrm{i}tH_\lambda}P = -\mathrm{i}t\lambda QBP + \mathcal{O}(t^2|\lambda|), \tag{51.35}$$

not $\mathcal{O}(t^2\lambda^2)$ without extra commutation or spectral-matching hypotheses. For $\lambda = p^{-w}$, leakage is still uniformly first order in p^{-w}, while the Feshbach self-energy remains second order.

51.9 C9: descendant measure versus Hamiltonian coupling C

Exact polyphase refinement fixes

$$|a_{\mathrm{leaf}}| = p^{-w/2}, \qquad P_{\mathrm{leaf}} = p^{-w}. \tag{51.36}$$

It does not force a Hamiltonian matrix element equal to p^{-w}. For equal per-leaf coupling

$$g_{\mathrm{leaf}} = gp^{-\alpha w}, \tag{51.37}$$

the bright radial matrix element is

$$\boxed{G_{\mathrm{coll}} = gp^{(1/2-\alpha)w}.} \tag{51.38}$$

A direct p^{-w} matrix element remains a valid model for one normalized aggregate detail packet, but it is not equivalent to equal coupling across all p^w descendants.

51.10 C10: resolved nonlinear closure is not strict finite-range locality Q

The exact self-dual constraint theorem is stated on the filtered Fourier algebra

$$\mathcal{T}_K \subset \mathcal{T}_{4K} \subset \mathcal{T}_{16K} \subset \cdots \tag{51.39}$$

with the sufficient headroom condition

$$16K < N. \tag{51.40}$$

It proves finite-dimensional nonlinear first-class closure before aliasing. It does not prove a finite-range microscopic lattice realization because the spectral derivative is global on the grid:

$$\text{resolved spectral closure} \neq \text{strict finite-range locality}. \tag{51.41}$$

Adaptive refinement is part of the exact statement; destructive projection to one fixed band is not.

51.11 C11: classical BV closure is not quantum anomaly cancellation Q

For the real Lorentzian parent,

$$(S_{\mathrm{BV}}, S_{\mathrm{BV}})_{\mathrm{BV}} = 0 \tag{51.42}$$

follows from off-shell gauge invariance and BRST nilpotency. The quantum equation

$$\frac{1}{2}(S_{\mathrm{BV}}, S_{\mathrm{BV}})_{\mathrm{BV}} - \mathrm{i}\hbar\Delta_{\mathrm{BV}}S_{\mathrm{BV}} = 0 \tag{51.43}$$

requires a regulator, a measure, a defined BV Laplacian, and anomaly control. Therefore

$$\text{classical master equation} \not\Longrightarrow \text{quantum master equation}. \tag{51.44}$$

No quantum anomaly-cancellation claim is made.

51.12 C12: boundary BFV closure and minimal reduction are distinct steps Q

The complete boundary charge

$$\{\Omega_\Sigma, \Omega_\Sigma\}_\Sigma = 0 \tag{51.45}$$

is established on the regular presymplectic boundary description. The later second-class chapter performs an explicit time-gauge reduction on the oriented nondegenerate sector,

$$e = \det(e_a{}^i) > 0, \tag{51.46}$$

and obtains the local canonical chart

$$\mathcal{P}_{\mathrm{red}} = \{(K_a{}^i, \widetilde{E}^a{}_i), (A_a, \mathcal{E}^a), (\Psi, \Psi^\dagger)\}. \tag{51.47}$$

The qualified implication is therefore

$$\text{presymplectic BFV closure} + \text{regular second-class reduction} \Longrightarrow \text{minimal local real chart}, \tag{51.48}$$

not a global chart through degenerate triads or every gauge orbit.

51.13 C13: additive dark-sector branch superseded by the scale-covariant model C

The first unified dark-sector construction used

$$m_h^2(\phi) = m_h^2 + 3\lambda\phi^2 \tag{51.49}$$

and assigned one produced light shell both the present abundance and the average-mode locking source. Under the one-shell, one-crossing, energy-matched, nonrelativistic assumptions, the later audit derives

$$\boxed{u_0^3 > \frac{4\sqrt{6}}{\pi^{5/2} r_D}} \tag{51.50}$$

for the present momentum-to-mass ratio. At the declared abundance this gives

$$\nu_0 > 1.1397360601, \tag{51.51}$$

contradicting the nonrelativistic premise. Chapters recording the additive mechanism are retained as conditional derivation and negative-control records. They are not the current viable branch.

The corrected model applies one scale-covariant deformation to the complete refinement operator,

$$\mathcal{L}_{\sigma,p} \longmapsto \left(1 + \gamma\frac{\phi^2}{v^2}\right)\mathcal{L}_{\sigma,p}, \tag{51.52}$$

which produces a heavy cold band and weights the locking source by refinement energy rather than by the inverse mass of one light shell. The scale-covariant coupling remains a model extension and is not claimed to follow uniquely from the primitive projector axioms.

51.14 C14: trapped-surface formation is not a global black-hole theorem Q

The spherical scalar calculation demonstrates a resolved first marginally outer trapped sphere,

$$\theta_+ = 0 \Longleftrightarrow \frac{2m}{r} = 1, \tag{51.53}$$

from smooth ingoing data in a horizon-penetrating chart. This establishes a nonlinear strong-field benchmark. It does not by itself prove global event-horizon formation, geodesic incompleteness, cosmic censorship, or nonspherical stability.

Chapter 52

Reproducibility Note

The figures and numerical residuals were produced by the Python programs included with the source. The package also contains the input tables, generated data, execution logs, and commands needed to repeat the calculations. These computations check finite-dimensional identities and convergence rates; they do not replace mathematical proofs or provide experimental validation of the physical models.

Detailed formula registries and execution records are kept with the source rather than in the main text.

Chapter 53

Symbolic and Numerical Audit Matrices

53.1 Audit totals

$$N_{\text{checks}} = 600, \tag{53.1}$$
$$N_{\text{pass}} = 586, \tag{53.2}$$
$$N_{\text{corrected}} = 10, \tag{53.3}$$
$$N_{\text{open}} = 2, \tag{53.4}$$
$$N_{\text{numerical only}} = 1, \tag{53.5}$$
$$N_{\text{conditional}} = 1, \tag{53.6}$$
$$N_{\text{scripts}} = 62, \tag{53.7}$$
$$N_{\text{script pass}} = 62, \tag{53.8}$$
$$N_{\text{formula occurrences}} = 2628, \tag{53.9}$$
$$N_{\text{canonical formulas}} = 2606. \tag{53.10}$$

53.2 Correction checks

ID	Sector	Statement	Residual	Method
W04	weak-field	physical radial acceleration a_r=-Phi'=-G M_enc/r^2	0	sign convention audit
W07	optics	vector deflection points toward decreasing impact parameter	0	sign convention audit
T06	tensor	d'Alembert converse: two-periodic ambiguity; periodic characteristic packets require zero/Nyquist compatibility	2.220×10^{-16}	affine zero-mode and parity-gauge audit
T07	tensor	Fierz-Pauli coupling sign: +1/2 int h_{mu nu}T^{mu nu} yields G^(1)=8piGT	0	reduced TT Euler-Lagrange sign audit
M01	diagnostics	relative work residual is undefined when both work terms vanish identically	0	zero/zero diagnostic audit
M03	refinement	partition incidence identity is R^T 1_c=1_f, not R 1_f=1_c	0	dimension and random partition audit
M04	weighted-adic-refinement	generic short-time leakage remainder is O(t^2 p^{-w}), while the exact Duhamel bound is linear in p^{-w}	0	Duhamel expansion and operator-norm audit
M05	projector-ADM	the shift contribution is represented either by the smeared momentum constraint or by Lie-dragged velocities, but not by both simultaneously	0	canonical ADM Legendre-form audit
M06	nonlinear-BFV-BV-bridge	with IP=I-s0 H-H s0, the transferred perturbation uses (I+H delta)^(-1) and an alternating return series	0	homological-perturbation sign audit
M07	dark-sector-model	the one-shell energy-matched additive cold-lock branch is inconsistent and is superseded by the scale-covariant refinement interaction	0	conditional abundance-coldness-locking no-go and replacement audit

53.3 All symbolic/numerical checks

ID	Sector	Statement	Method	Status	Residual	Order
P01	projector	sum_a P_a = I	random complex orthogonal atlas	PASS	8.260×10^{-16}	–
P02	projector	X = sum_ab P_a X P_b	random Hermitian X	PASS	8.890×10^{-16}	–
P03	projector	dE_a/dt = sum_b J_{a<-b}	direct derivative	PASS	9.714×10^{-17}	–
P04	projector	J_ab = -J_ba	matrix current	PASS	3.469×10^{-17}	–
P05	projector	sum_a dE_a/dt = 0	antisymmetric cancellation	PASS	1.388×10^{-16}	–
P06	projector	E_G = sum_{a in G} E_a	nonuniform refinement	PASS	5.551×10^{-17}	–
P07	projector	J_GH = sum_{a in G,b in H} J_ab	nonuniform refinement	PASS	5.551×10^{-17}	–
P08	moving-atlas	Gamma = U^* dot U is skew-adjoint	matrix exponential	PASS	1.908×10^{-16}	–
P09	moving-atlas	A_eff = U^* A U - U^* dot U is skew-adjoint	matrix exponential	PASS	2.580×10^{-16}	–
P10	nonlinear-extension	state-dependent A(psi)^*=-A(psi) preserves packet continuity	random nonlinear Hermitian mean field	PASS	9.714×10^{-17}	–
P11	balance-extension	antisymmetric local balance commutes with arbitrary grouping	finite algebra	PASS	2.887×10^{-15}	–
P12	information-extension	dot S = -1/2 sum_ab J_ab log(p_a/p_b)	random positive packet distribution	PASS	3.553×10^{-15}	–
P13	information-extension	d sum p_a^q/dt = q/2 sum_ab J_ab(p_a^{q-1}-p_b^{q-1})	random positive packet distribution	PASS	2.220×10^{-16}	–
P14	open-system-extension	dot p_a = sum_b(gamma_ab p_b-gamma_ba p_a)	gain-loss algebra	PASS	1.110×10^{-16}	–
P15	open-system-extension	sector-jump Lindblad equation reduces to gain-loss packet current	explicit jump operators	PASS	5.551×10^{-17}	–
P16	open-system-extension	dD(p||pi)/dt = -1/2 sum_ab K_ab log[(gamma_ab p_b)/(gamma_ba p_a)]	random detailed-balance jump process	PASS	0	–
P17	open-system-extension	dD(p||pi)/dt <= 0 under detailed balance	nonnegative entropy-production sum	PASS	0	–
P18	mixed-state-extension	dot p_a=sum_b 2 Im Tr(P_a H P_b rho)	random density matrix	PASS	2.776×10^{-17}	–
P19	mixed-state-extension	mixed-state packet current is antisymmetric and reduces to the pure-state current	random density matrix and rank-one state	PASS	2.602×10^{-17}	–
P20	refinement-extension	nested coarsening is associative for energies and antisymmetric currents	incidence-matrix algebra	PASS	2.887×10^{-15}	–
P21	composition-extension	tensor-product projectors form a complete atlas and reconstruct both marginals	random bipartite density matrix	PASS	6.539×10^{-16}	–
R01	dirac-1+1	omega_e = E - m^2 k^2 e^2/(6E)+O(e^4)	SymPy series matching	PASS	0	–
R02	dirac-1+1	d omega/dk = cos(me) sin(ke)/sin(omega e)	implicit symbolic differentiation	PASS	0	–
R03	dirac-1+1	tr U = 2 cos(omega e)	direct 2x2 matrix	PASS	0	–
R04	dirac-1+1	U^*U=I	direct 2x2 matrix	PASS	2.222×10^{-16}	–
C01	FLRW	dt/deta = a(eta)	symbolic differentiation	PASS	0	–
C02	FLRW	a(t(eta)) = exp(H t(eta))	symbolic substitution	PASS	0	–
C03	backreaction	d rho/da + 3(rho+p)/a = 0 packetwise	SymPy differentiation	PASS	0	–
C04	backreaction	p = -H_P'(a)/(3a^2)	SymPy differentiation	PASS	0	–
C05	backreaction	radiation: dot a = a^{-1}	symbolic solution substitution	PASS	0	–
C06	backreaction	matter: dot a = a^{-1/2}	symbolic solution substitution	PASS	0	–
C07	variational-FLRW	D_1 L_d sign and coefficients	central finite difference	PASS	3.213×10^{-10}	–
C08	variational-FLRW	D_2 L_d sign and coefficients	central finite difference	PASS	3.313×10^{-10}	–
H01	holonomy	2F_{h/2}-F_h cancels O(h)	exact polynomial algebra	PASS	0	–
H02	holonomy	(4R_{h/2}-R_h)/3 cancels O(h^2) after first stage	exact polynomial algebra	PASS	0	–
H05	refinement-extension	recursive R_m=(2^m R_{m-1}(h/2)-R_{m-1}(h))/(2^m-1) cancels h^1,...,h^m	exact polynomial algebra through m=3	PASS	0	–
H03	holonomy	h^{-2}log group-commutator = [X,Y]+O(h)	random skew matrices	PASS	0.600298	0.967197
H04	holonomy	W[X',Y']=G W[X,Y] G^*	random unitary frame	PASS	7.456×10^{-16}	–
W01	weak-field	u=r Phi => u"=r(Phi"+2Phi'/r)	SymPy differentiation	PASS	0	–
W02	weak-field	dM_enc/dr=4pi r^2 rho_G	SymPy erf differentiation	PASS	0	–
W03	weak-field	Phi_G' = G M_enc/r^2	SymPy erf differentiation	PASS	0	–

ID	Sector	Statement	Method	Status	Residual	Order
W04	weak-field	physical radial acceleration a_r=-Phi'=-G M_enc/r^2	sign convention audit	CORRECTED	0	–
W05	optics	int_{-L}^L dx/sqrt(x^2+b^2)=2 asinh(L/b)	SymPy integral	PASS	0	–
W06	optics	finite-path deflection magnitude integral	SymPy integral	PASS	0	–
W07	optics	vector deflection points toward decreasing impact parameter	sign convention audit	CORRECTED	0	–
T01	tensor	Lambda_TT^2=Lambda_TT	random symmetric tensor	PASS	6.305×10^{-16}	–
T02	tensor	k_i h_ij^TT=0	random symmetric tensor	PASS	4.454×10^{-16}	–
T03	tensor	tr h^TT=0	random symmetric tensor	PASS	1.665×10^{-16}	–
T04	tensor	plane-wave recurrence iff sin^2(omega dt/2)=lambda^2 sin^2(kh/2)	SymPy trigonometric identity	PASS	4.407×10^{-17}	–
T05	tensor	h_j^n=F_{j-n}+G_{j+n} solves critical recurrence	random periodic sequences	PASS	4.441×10^{-16}	–
T06	tensor	d'Alembert converse: two-periodic ambiguity; periodic characteristic packets require zero/Nyquist compatibility	affine zero-mode and parity-gauge audit	CORRECTED	2.220×10^{-16}	–
T07	tensor	Fierz-Pauli coupling sign: +1/2 int h_{mu nu}T^{mu nu} yields G^(1)=8piGT	reduced TT Euler-Lagrange sign audit	CORRECTED	0	–
T08	tensor	staggered discrete wave energy is exactly conserved	random periodic trajectory	PASS	1.421×10^{-14}	–
U01	countable	S_N X S_N -> X strongly for bounded X	finite truncation monotonicity witness	PASS	0	–
U02	countable	||S_N X S_N-X|| ->0 for compact X	compact diagonal monotonicity witness	PASS	0	–
D01	dirac-3+1	Clifford relations	explicit 4x4 matrices	PASS	0	–
D02	dirac-3+1	det(lambda I-U)=(lambda^2-2C lambda+1)^2	direct eigenvalue polynomial	PASS	4.006×10^{-32}	–
D03	dirac-3+1	omega=E-Delta^2 Q/(6E)+O(Delta^4)	SymPy series matching	PASS	0	–
D04	dirac-3+1	exact vector group velocity	finite-difference derivative	PASS	4.708×10^{-9}	–
D05	boost	observer shell defect has derived O(Delta^2) coefficient	direct exact shell and Lorentz boost	PASS	1.383×10^{-10}	2.00047
D06	boost	leading-subtracted defect is O(Delta^4)	direct exact shell and Lorentz boost	PASS	1.383×10^{-10}	3.99415
G01	U1	forward link L_j -> g_{j+1}L_jg_j^{-1}	random phases	PASS	6.280×10^{-16}	–
G02	U1	for forward links L=e^{+iq h A_x}, T=e^{+iq dt A_0}, right-up-left-down W=e^{-iq h dt F_01+...}	orientation audit	PASS	0	–
G03	U1	1-Re W = theta_p^2/2+O(theta_p^4)	cosine Taylor series	PASS	0	–
G04	U1	Abelian product of oriented plaquettes on cube boundary equals 1	random U(1) links	PASS	8.965×10^{-16}	–
G05	dynamic-U1	dot rho+(J_j-J_{j-1})/h=0	random semidiscrete state	PASS	5.551×10^{-17}	–
G06	dynamic-U1	dot Gauss=0	random semidiscrete state	PASS	6.286×10^{-17}	–
S01	scalar-cosmology	dot C=0	exact symbolic substitution	PASS	0	–
S02	scalar-cosmology	discrete-gradient step preserves Friedmann constraint exactly	exact symbolic factorization	PASS	0	–
S03	scalar-cosmology	quadratic-potential midpoint cubic coefficients/signs	SymPy expansion	PASS	0	–
S04	scalar-cosmology	massless stiff exact solution	symbolic ODE substitution	PASS	3.279×10^{-17}	–
S05	local-scalar	dot e_j+F_{j+1/2}-F_{j-1/2}=s_j	random finite-volume state	PASS	4.885×10^{-15}	–
S06	local-scalar	sum dot e = sum s	periodic telescoping	PASS	1.243×10^{-14}	–
P22	positive-operator	sum_a F_a=I for a positive operator partition	script21 random normalized effects	PASS	4.216×10^{-15}	–
P23	positive-operator	dot p_a=sum_b J^F_ab	script21 generalized effect current	PASS	4.163×10^{-17}	–
P24	positive-operator	J^F_ab=-J^F_ba	script21 generalized effect current	PASS	0	–
P25	positive-operator	moving effects satisfy dot p=s+J^F 1	script21 moving positive partition	PASS	6.158×10^{-17}	–
P26	positive-operator	coarse effects aggregate probabilities and currents exactly	script21 nonuniform grouping	PASS	6.939×10^{-18}	–
P27	positive-operator	positive-effect current reduces to projector current	script21 orthogonal reduction	PASS	8.674×10^{-18}	–
P28	positive-operator	every Parseval positive atlas is a compression F_a=V^* Pi_a V of an orthogonal direct-sum atlas	script21 direct-sum dilation	PASS	4.750×10^{-15}	–
P29	positive-operator	continuity is invariant under antisymmetric cycle currents K with K1=0	script21 cycle-current gauge	PASS	5.308×10^{-16}	–
G07	SU2	Delta_U transforms covariantly	script22 random SU(2) links	PASS	6.395×10^{-14}	–
G08	SU2	plaquette holonomy transforms by basepoint conjugation	script22 random SU(2) links	PASS	5.661×10^{-16}	–
G09	SU2	Wilson action is gauge invariant	script22 random SU(2) links	PASS	2.220×10^{-16}	–
G10	SU2	link matter current is gauge invariant	script22 random color doublet	PASS	3.997×10^{-15}	–
G11	SU2	color charge transforms in the adjoint representation	script22 Pauli-basis adjoint map	PASS	3.997×10^{-15}	–
G12	SU2	ordered transported fine plaquettes equal the 2x2 coarse boundary holonomy	script22 exact link-word cancellation	PASS	6.215×10^{-16}	–
G13	SU2	raw matrix-log curvature is first order	script22 group commutator	PASS	–	0.993501

ID	Sector	Statement	Method	Status	Residual	Order
G14	SU2	two-scale non-Abelian curvature refinement is second order	script22 Richardson cancellation	PASS	–	1.99234
Y01	Gowdy	d_tau lambda_theta-d_theta lambda_tau=2P_theta E_P+2e^{2P}Q_theta E_Q	SymPy jet-algebra identity	PASS	0	–
Y02	Gowdy	polarized Bessel mode constraint residual converges at second order	script23 exact mode	PASS	1.601×10^{-5}	1.97768
Y03	Gowdy	unpolarized Gowdy state converges under spatial refinement	script23 nonlinear method of lines	PASS	7.010×10^{-6}	2.12836
Y04	Gowdy	unpolarized constraint integrability converges under refinement	script23 nonlinear method of lines	PASS	2.936×10^{-5}	1.94949
Y05	Gowdy	periodic global momentum constraint is preserved	script23 parity-compatible data	PASS	1.540×10^{-17}	–
Y06	Gowdy	local constraint densities aggregate exactly under grouping	script23 fine-to-coarse sums	PASS	1.110×10^{-16}	–
D07	finite-scale-prediction	v_g=1-Delta^2 k^2 A(n)/2+O(Delta^4)	SymPy derivative of the model shell	PASS	0	–
D08	finite-scale-prediction	A(n)=1/2(1-sum_i n_i^4)	exact polynomial identity on |n|=1	PASS	0	–
D09	finite-scale-prediction	0<=A(n)<=1/3 with body-diagonal maximum	axis/face/body extremal check	PASS	1.665×10^{-16}	–
D10	finite-scale-prediction	leading-subtracted photon delay is O(Delta^4)	script24 exact lattice shell	PASS	–	4.00361
D11	finite-scale-prediction	microscopic scale is identifiable from directional energy-dependent delays	script24 synthetic least squares	PASS	7.075×10^{-6}	–
D12	finite-scale-prediction	axis direction has vanishing O(Delta^2) correction	exact directional factor	PASS	0	–
D13	finite-scale-prediction	spherical mean and second moment are <A>=1/5 and <A^2>=1/21	exact S^2 monomial moments	PASS	1.388×10^{-17}	–
D14	finite-scale-prediction	Var_{S^2}(A)=4/525	exact S^2 moment subtraction	PASS	1.908×10^{-17}	–
D15	finite-scale-prediction	A=1/5-H_4/2 with H_4 a harmonic cubic-symmetry polynomial on S^2	SymPy Laplacian and polynomial identity	PASS	0	–
Y07	Gowdy-constrained	dC_h/dtau=0 for the projected semidiscrete vector field	script25 symbolic constraint-force identity	PASS	0	–
Y08	Gowdy-constrained	the endpoint velocity projection gives C_h(z_{n+1})=C_h(z_n) exactly	script25 symbolic endpoint projection identity	PASS	0	–
Y09	Gowdy-constrained	the discrete global momentum constraint is preserved at map level	script25 constrained midpoint trajectory	PASS	6.812×10^{-18}	–
Y10	Gowdy-constrained	the constrained state converges under joint spatial-temporal refinement	script25 constrained midpoint refinement	PASS	7.864×10^{-6}	2.12804
Y11	Gowdy-constrained	the corrected local integrability diagnostic converges at second order	script25 local compatibility diagnostic	PASS	3.575×10^{-5}	1.94932
Y12	Gowdy-constrained	the constraint force vanishes in the continuum limit	script25 force refinement	PASS	2.065×10^{-7}	2.0442
Y13	Gowdy-constrained	fine global momentum contributions aggregate exactly to coarse packets	script25 exact grouping	PASS	2.602×10^{-18}	–
Y14	Gowdy-constrained	endpoint projection suppresses the control-run constraint drift by at least eight orders of magnitude	script25 projected versus unprojected control	PASS	0	–
K01	Einstein-scalar-perturbations	zeta=-Psi-H delta_phi/dot(phi) is invariant under the stated scalar time shift	script26 SymPy gauge substitution	PASS	0	–
K02	Einstein-scalar-perturbations	variation of the Mukhanov-Sasaki quadratic action gives v"-Delta v-(z"/z)v=0	script26 symbolic Euler-Lagrange variation	PASS	0	–
K03	Einstein-scalar-perturbations	the cubic-lattice momentum satisfies khat^2=k^2-h^2 sum k_i^4/12+h^4 sum k_i^6/360+O(h^6)	script26 SymPy series	PASS	0	–
K04	Einstein-scalar-perturbations	the lattice Mukhanov mode converges at second order	script26 exact de Sitter-index benchmark	PASS	6.230×10^{-5}	2.0015
K05	Einstein-scalar-perturbations	two-level Richardson refinement raises the Mukhanov mode to fourth order	script26 exact benchmark	PASS	2.014×10^{-10}	4.00117
K06	Einstein-scalar-perturbations	the lattice scalar power converges at second order	script26 exact benchmark	PASS	4.483×10^{-5}	2.0028
K07	Einstein-scalar-perturbations	the normalized canonical mode preserves the Wronskian	script26 exact mode	PASS	3.220×10^{-15}	–
L01	cosmological-finite-scale-delay	ell_phys(z)^2 E(z)^2 scales as (1+z)^(2-2 sigma)	script27 symbolic redshift exponent	PASS	0	–

ID	Sector	Statement	Method	Status	Residual	Order
L02	cosmological-finite-scale-delay	the fixed-proper-scale direct arrival delay differs from the leading redshift integral by O(ell_*^4)	script27 direct FLRW arrival integration	PASS	4.227×10^{-8}	4.00268
L03	cosmological-finite-scale-delay	the fixed-comoving-scale direct arrival delay differs from the leading redshift integral by O(ell_*^4)	script27 direct FLRW arrival integration	PASS	4.340×10^{-9}	4.00143
L04	cosmological-finite-scale-delay	a cubic-axis photon has exact directional cancellation on the principal branch	script27 exact axis shell	PASS	0	–
L05	cosmological-finite-scale-delay	the present proper microscopic scale is identifiable in a synthetic cosmological delay fit	script27 synthetic least squares	PASS	8.162×10^{-6}	–
L06	cosmological-finite-scale-delay	the present comoving microscopic scale is identifiable in a synthetic cosmological delay fit	script27 synthetic least squares	PASS	3.022×10^{-6}	–
BGD01	general-FLRW	differentiating the curved Friedmann equation gives the Raychaudhuri equation	script29 symbolic identity	PASS	0	–
BGD02	general-FLRW	sum_i Q_i=0 implies total interacting-fluid continuity	script29 symbolic identity	PASS	0	–
BGD03	general-FLRW	Etherington background identity D_L=(1+z)^2 D_A	script29 symbolic and numerical distance map	PASS	0	–
BGD04	general-FLRW	EdS, radiation, and Milne distance benchmarks	script29 adaptive quadrature	PASS	8.882×10^{-16}	–
BGD05	general-FLRW	EdS, radiation, and Milne age benchmarks	script29 adaptive quadrature	PASS	0	–
BGD06	general-FLRW	curved multicomponent background midpoint solver is second order	script29 reference integration	PASS	2.731×10^{-6}	2.02428
BGD07	general-FLRW	conservative interacting-fluid midpoint solver is second order	script29 reference integration	PASS	4.016×10^{-6}	2.05469
BGD08	general-FLRW	interacting-fluid trajectory obeys total continuity	script29 direct right-hand-side balance	PASS	4.441×10^{-16}	–
BGD09	general-FLRW	benchmark interacting densities remain positive	script29 trajectory positivity	PASS	0	–
VT01	FLRW-vector-tensor	Sigma_i=S_i-F_i' is invariant under the stated transverse gauge shift	script30 SymPy substitution	PASS	0	–
VT02	FLRW-vector-tensor	source-free vector mode satisfies (a^2 Sigma_i)'=0	script30 symbolic differentiation	PASS	0	–
VT03	FLRW-vector-tensor	variation of the FLRW tensor action gives h"+2 Hc h'+k^2 h=0	script30 symbolic Euler-Lagrange variation	PASS	0	–
VT04	FLRW-vector-tensor	mu=a h gives mu"+(k^2-a"/a)mu=0	script30 symbolic substitution	PASS	0	–
VT05	FLRW-vector-tensor	h=3 j_1(k eta)/(k eta) solves the matter-era tensor equation	script30 symbolic trigonometric form	PASS	0	–
VT06	FLRW-vector-tensor	matter-era vector decay midpoint update is second order	script30 exact a proportional eta^2 benchmark	PASS	9.997×10^{-8}	2.05295
VT07	FLRW-vector-tensor	radiation-era lattice tensor mode converges at second order	script30 exact canonical mode	PASS	3.968×10^{-5}	1.99962
VT08	FLRW-vector-tensor	radiation-era tensor Richardson refinement is fourth order	script30 exact canonical mode	PASS	2.383×10^{-10}	3.9988
VT09	FLRW-vector-tensor	matter-era lattice tensor mode converges at second order	script30 exact spherical-Bessel mode	PASS	1.335×10^{-5}	2.00269
VT10	FLRW-vector-tensor	matter-era tensor Richardson refinement is fourth order	script30 exact spherical-Bessel mode	PASS	7.801×10^{-11}	3.99638
VT11	FLRW-vector-tensor	canonical radiation tensor mode preserves the Wronskian	script30 exact mode	PASS	4.441×10^{-16}	–
VT12	FLRW-vector-tensor	radiation-era subhorizon tensor amplitude obeys a|h|=constant	script30 exact mode	PASS	4.441×10^{-16}	–
SF01	linear-structure	dust continuity, Euler, and Poisson imply delta_ddot+2H delta_dot-4piG rho delta=0	script31 symbolic elimination	PASS	0	–
SF02	linear-structure	a common lattice Laplacian cancels exactly between Euler and Poisson	script31 symbolic and finite-k tests	PASS	0	–
SF03	linear-structure	Einstein-de Sitter growing and decaying exponents are 1 and -3/2	script31 characteristic polynomial	PASS	0	–
SF04	linear-structure	Einstein-de Sitter growth midpoint solver is second order	script31 exact growth modes	PASS	6.333×10^{-5}	1.96811
SF05	linear-structure	flat LambdaCDM growth midpoint solver is second order	script31 Heath reference	PASS	2.931×10^{-5}	1.97017
SF06	linear-structure	flat LambdaCDM growth ODE agrees with the Heath integral solution	script31 high-accuracy trajectory	PASS	1.035×10^{-12}	–
SF07	linear-structure	f sigma_8 is computed conditionally from D(a), f(a), and sigma_8,0	script31 stated Omega_m0 and normalization	PASS	0	0.410237

ID	Sector	Statement	Method	Status	Residual	Order
PG01	primordial-power-law	constant-epsilon power-law background has epsilon_H=epsilon	script32 symbolic differentiation	PASS	0	–
PG02	primordial-power-law	z"/z=a"/a=(nu^2-1/4)/eta^2	script32 symbolic identity	PASS	0	–
PG03	primordial-power-law	n_s-1=-2epsilon/(1-epsilon) in exact power-law inflation	script32 exact spectrum and log-slope fit	PASS	1.180×10^{-16}	–
PG04	primordial-power-law	n_t=n_s-1 in the controlled constant-epsilon model	script32 exact tensor spectrum	PASS	2.255×10^{-16}	–
PG05	primordial-power-law	r=16epsilon	script32 exact scalar/tensor normalization ratio	PASS	2.776×10^{-17}	–
PG06	primordial-power-law	r=-8(1-epsilon)n_t	script32 symbolic consistency identity	PASS	0	–
PG07	primordial-power-law	the Bunch-Davies Hankel mode preserves the canonical Wronskian	script32 special-function derivative	PASS	3.908×10^{-14}	–
PG08	primordial-power-law	the constant-epsilon canonical-mode midpoint solver is second order	script32 exact Hankel benchmark	PASS	0.00184128	1.96461
PG09	primordial-power-law	sixty controlled e-folds exponentially suppress the curvature fraction	script32 exact background factor	PASS	0	2.546×10^{-52}
PG10	primordial-power-law	sixty controlled e-folds exponentially suppress the comoving Hubble radius	script32 exact background factor	PASS	0	1.596×10^{-26}
REC01	recombination-acoustics	the physical Saha root solves x_e^2/(1-x_e)=S	script33 SymPy algebraic root	PASS	0	–
REC02	recombination-acoustics	the conditional ionization profile satisfies the Saha relation	script33 finite redshift profile	PASS	2.208×10^{-10}	–
REC03	recombination-acoustics	the visibility function integrates to 1-exp(-tau_max)	script33 optical-depth quadrature	PASS	4.807×10^{-8}	–
REC04	recombination-acoustics	the sound horizon agrees in direct-z and log-z coordinates	script33 independent quadratures	PASS	2.093×10^{-6}	–
REC05	recombination-acoustics	the constant-loading tight-coupling oscillator has the cosine solution	script33 symbolic substitution	PASS	0	–
REC06	recombination-acoustics	the varying-loading tight-coupling acoustic solver is second order	script33 refined midpoint trajectory	PASS	3.979×10^{-5}	2.02141
REC07	recombination-acoustics	the tight-coupling diffusion length is positive	script33 positive opacity integral	PASS	0	–
REC08	recombination-acoustics	the conditional recombination, acoustic, and diffusion scales are finite	script33 stated benchmark parameters	PASS	0	340.781
CMB01	minimal-CMB-Boltzmann	the spherical-Bessel derivative obeys the collisionless hierarchy recurrence	script34 special-function identity	PASS	4.684×10^{-16}	–
CMB02	minimal-CMB-Boltzmann	the collisionless photon hierarchy reproduces j_l free streaming	script34 high-accuracy multipole ODE	PASS	6.258×10^{-13}	–
CMB03	minimal-CMB-Boltzmann	the midpoint photon hierarchy is second order	script34 exact spherical-Bessel benchmark	PASS	2.444×10^{-5}	1.99677
CMB04	minimal-CMB-Boltzmann	the truncated free-streaming multipole Parseval sum approaches unity	script34 finite ell truncation	PASS	3.365×10^{-13}	–
CMB05	minimal-CMB-Boltzmann	the scale-invariant Sachs-Wolfe plateau is l(l+1)C_l/(2pi)=A_s/25	script34 symbolic line-of-sight integral	PASS	0	–
CMB06	minimal-CMB-Boltzmann	the numerical Bessel integral agrees with 1/[2l(l+1)]	script34 quadrature plus asymptotic tail	PASS	2.499×10^{-7}	–
CMB07	minimal-CMB-Boltzmann	the conditional acoustic and diffusion multipoles are finite	script34 Task-H geometric scales	PASS	0	340.781
FD01	finite-derivation-no-go	Every derivation of the finite pointwise algebra C^N is zero (tested N=2,...,8)	exact Leibniz linear system and numerical rank	PASS	0	–
FD02	corner-differential-calculus	delta_K(XY)=delta_K(X)Y+X delta_K(Y)	random complex matrix algebra	PASS	5.283×10^{-14}	–
FD03	corner-differential-calculus	delta_K(X)^*=delta_K(X^*)	random complex matrix algebra	PASS	6.086×10^{-15}	–

ID	Sector	Statement	Method	Status	Residual	Order
FD04	corner-differential-calculus	Tr(delta_K(X)Y)=-Tr(X delta_K(Y))	trace cyclicity	PASS	2.132×10^{-14}	–
FD05	corner-differential-calculus	[delta_1,delta_2]=delta_F with F=i[K_1,K_2]	Jacobi identity in random matrix algebra	PASS	5.496×10^{-14}	–
FD06	corner-differential-calculus	The differentiated object reconstructs exactly from all projector corners	random orthogonal atlas	PASS	3.035×10^{-14}	–
FD07	corner-differential-calculus	Derivative corners aggregate exactly under nonuniform refinement	three irregular coarse blocks	PASS	5.824×10^{-15}	–
FD08	corner-differential-calculus	Each derivative corner is the sum of two-step transfer paths through intermediate corners	explicit block expansion	PASS	7.851×10^{-15}	–
ADM01	projector-ADM-linearized	The 3D linearized Einstein operator is symmetric	random periodic tensor field	PASS	8.882×10^{-16}	–
ADM02	projector-ADM-linearized	The exact discrete linearized Bianchi identity delta_j G_ij=0 holds	commuting centered derivatives	PASS	2.220×10^{-15}	–
ADM03	projector-ADM-linearized	dot C=-delta_i C_i for arbitrary lapse and shift	direct operator substitution	PASS	1.199×10^{-14}	–
ADM04	projector-ADM-linearized	dot C_i=0 for arbitrary lapse	discrete Bianchi identity	PASS	5.329×10^{-15}	–
ADM05	projector-ADM-linearized	The scalar constraint is invariant under h_ij -> h_ij+delta_i xi_j+delta_j xi_i	random spatial gauge field	PASS	1.155×10^{-14}	–
ADM06	projector-ADM-linearized	The smeared linearized Hamiltonian and momentum constraints have zero cross bracket	discrete integration by parts	PASS	1.451×10^{-14}	–
ADM07	projector-ADM-linearized	The linearized Einstein operator is self-adjoint in the lattice trace pairing	random tensor pair	PASS	2.842×10^{-14}	–
ADM08	projector-ADM-linearized	delta V/delta h=G for V=1/2<h,Gh>	centered finite variation	PASS	4.093×10^{-12}	–
ADM09	projector-ADM-linearized	The vacuum quadratic Hamiltonian is conserved at the differential level	canonical equations	PASS	0	–
ADM10	projector-ADM-linearized	Constraint energy aggregates exactly under irregular packet grouping	37 random coarse groups	PASS	1.506×10^{-16}	–
ADM11	projector-ADM-linearized	A lattice TT graviton remains on the Hamiltonian and momentum constraint surface	exact discrete transverse projector	PASS	5.107×10^{-14}	–
ADM12	projector-ADM-linearized	The exact lattice TT trajectory conserves quadratic energy	closed-form mode evolution	PASS	4.909×10^{-16}	–
ADM13	projector-ADM-linearized	The lattice graviton frequency converges to the continuum frequency	periodic centered derivative symbol	PASS	0.00480805	1.98585
NLC01	nonlinear-constraint-defect	The scalar diffeomorphism-bracket defect equals the discrete Leibniz-defect contraction	direct canonical differentiation	PASS	1.874×10^{-16}	–
NLC02	nonlinear-constraint-defect	The central-difference nonlinear closure defect converges quadratically	smooth periodic scalar constraint	PASS	3.076×10^{-4}	1.98556
NLC03	nonlinear-constraint-defect	A resolved low-band spectral window closes the scalar diffeomorphism bracket at roundoff	Fourier derivative with products below Nyquist	PASS	6.696×10^{-16}	–
NLC04	nonlinear-constraint-defect	The spectral defect formula agrees with the direct bracket	low-band Fourier window	PASS	7.282×10^{-17}	–
NLC05	nonlinear-constraint-defect	The corner-algebra inner derivative has zero Leibniz defect	random full matrix algebra	PASS	7.193×10^{-14}	–
ADQ01	adic-unit-register	Odd-prime and dyadic unit coordinates are exact bijections	complete enumeration for p=2,3,5	PASS	0	–
ADQ02	adic-unit-register	Unit multiplication becomes coordinatewise addition	finite multiplication table samples	PASS	0	–
ADQ03	dyadic-odd-CRT-quantum	The composite unit register is a unitary dyadic-odd coordinate permutation	complete unit basis modulo 144	PASS	0	–
ADQ04	dyadic-odd-CRT-quantum	Composite unit multiplication is componentwise in all adic registers	complete CRT coordinate table	PASS	0	–
ADQ05	dyadic-odd-CRT-quantum	The unit-register QFT factors into torsion and principal-phase QFTs	coordinate conjugation of tensor QFT	PASS	9.945×10^{-16}	–

ID	Sector	Statement	Method	Status	Residual	Order
ADQ06	dyadic-odd-CRT-quantum	The additive QFT obeys the exact dyadic-odd CRT factorization	N=12 input and dual-output CRT permutations	PASS	7.446×10^{-15}	–
ADQ07	dyadic-sign-channel	Omitting the dyadic sign qubit gives an exact two-to-one alias	complete unit register modulo 64	PASS	0	–
ADQ08	dyadic-sign-channel	The rational sign projectors have dyadic denominator weight one and collapse mod 2	integral involution and resultant certificate	PASS	0	–
ADQ09	adic-Haar-projectors	The p-adic averaging differences form a complete orthogonal projector family	p=2,3,5 finite phase spaces	PASS	5.621×10^{-16}	–
ADQ10	adic-Haar-projectors	The child-DFT Haar basis is unitary and reconstructs every detail projector	explicit p-ary wavelet basis	PASS	9.255×10^{-16}	–
ADQ11	adic-quantum-refinement	The refinement embedding is isometric and new detail packets annihilate the coarse image	uniform-child embedding	PASS	4.588×10^{-16}	–
ADQ12	adic-quantum-refinement	Hierarchical and translation Hamiltonians intertwine exactly across p-adic refinement	p=2 and p=3 phase towers	PASS	4.588×10^{-16}	–
ADQ13	adic-quantum-refinement	Unitary, Euclidean, and resolvent evolution commute exactly with refinement	matrix exponentials and resolvents	PASS	4.886×10^{-16}	–
ADQ14	adic-quantum-refinement	The exact refinement sector has zero leakage and zero Feshbach self-energy	orthogonal coarse/detail block decomposition	PASS	9.334×10^{-16}	–
ADQ15	adic-quantum-propagator	The hierarchical propagator is exactly isotropic on p-adic valuation shells	p=3 level-4 spectral propagator	PASS	0	–
ADQ16	adic-quantum-propagator	The p-adic Haar transform exactly diagonalizes the hierarchical Hamiltonian	explicit wavelet basis	PASS	1.098×10^{-14}	–
ADQ17	adelic-quantum-tower	A nontrivial cross-prime Hamiltonian intertwines exactly under simultaneous 2-adic and 3-adic refinement	dyadic sign, dyadic phase, 3-adic torsion, 3-adic phase	PASS	5.031×10^{-16}	–
ADQ18	adelic-quantum-tower	Unitary, Euclidean, and resolvent dynamics commute with simultaneous prime refinement	finite matrix functional calculus	PASS	4.003×10^{-16}	–
ADQ19	adelic-quantum-tower	The stable multi-prime sector has zero leakage and zero Feshbach self-energy	coarse/detail block decomposition	PASS	2.958×10^{-16}	–
ADQ20	adelic-quantum-tower	The adic coordinate Hamiltonian is unitarily equivalent to evolution on the ordinary unit basis modulo 72	CRT coordinate permutation	PASS	1.998×10^{-15}	–
ADQ21	adelic-quantum-interaction	A dyadic-sign/odd-adic detail coupling generates one ebit of cross-prime entanglement	exact controlled-phase evolution	PASS	2.220×10^{-16}	–
ADQ22	weighted-adic-refinement	A refinement-breaking coupling of adic weight w has defect proportional to p^{-w}	rank-one coarse/detail coupling for p=2,3	PASS	5.995×10^{-15}	1
ADQ23	weighted-adic-refinement	Short-time hidden-sector leakage is first order in the weighted coupling	matrix exponential at fixed small time	PASS	9.928×10^{-9}	1
ADQ24	weighted-adic-refinement	Feshbach backreaction is second order in the weighted coupling	exact finite Schur complement	PASS	5.862×10^{-14}	2
ADQ25	weighted-adic-refinement	The exact Duhamel bound controls hidden-sector leakage linearly in p^{-w}	finite-dimensional operator-norm inequality for p=2,3 and w=1,...,6	PASS	0	–
ADW-2-0.5-N	adic-normalization	collective coupling exponent is 1/2-alpha	exact rank-two reduction and depth fit	PASS	7.535×10^{-17}	7.535×10^{-17}
ADW-2-0.5-S	adic-normalization	collective Feshbach exponent is 1-2alpha	exact rank-one self-energy and depth fit	PASS	1.586×10^{-16}	1.586×10^{-16}
ADW-2-1.0-N	adic-normalization	collective coupling exponent is 1/2-alpha	exact rank-two reduction and depth fit	PASS	0	-0.5
ADW-2-1.0-S	adic-normalization	collective Feshbach exponent is 1-2alpha	exact rank-one self-energy and depth fit	PASS	2.220×10^{-16}	-1
ADW-2-1.5-N	adic-normalization	collective coupling exponent is 1/2-alpha	exact rank-two reduction and depth fit	PASS	2.220×10^{-16}	-1
ADW-2-1.5-S	adic-normalization	collective Feshbach exponent is 1-2alpha	exact rank-one self-energy and depth fit	PASS	0	-2
ADW-3-0.5-N	adic-normalization	collective coupling exponent is 1/2-alpha	exact rank-two reduction and depth fit	PASS	3.322×10^{-17}	-3.322×10^{-17}
ADW-3-0.5-S	adic-normalization	collective Feshbach exponent is 1-2alpha	exact rank-one self-energy and depth fit	PASS	1.482×10^{-17}	-1.482×10^{-17}
ADW-3-1.0-N	adic-normalization	collective coupling exponent is 1/2-alpha	exact rank-two reduction and depth fit	PASS	1.110×10^{-16}	-0.5
ADW-3-1.0-S	adic-normalization	collective Feshbach exponent is 1-2alpha	exact rank-one self-energy and depth fit	PASS	0	-1

ID	Sector	Statement	Method	Status	Residual	Order
ADW-3-1.5-N	adic-normalization	collective coupling exponent is 1/2-alpha	exact rank-two reduction and depth fit	PASS	0	-1
ADW-3-1.5-S	adic-normalization	collective Feshbach exponent is 1-2alpha	exact rank-one self-energy and depth fit	PASS	4.441×10^{-16}	-2
ADW-5-0.5-N	adic-normalization	collective coupling exponent is 1/2-alpha	exact rank-two reduction and depth fit	PASS	2.420×10^{-17}	2.420×10^{-17}
ADW-5-0.5-S	adic-normalization	collective Feshbach exponent is 1-2alpha	exact rank-one self-energy and depth fit	PASS	3.318×10^{-17}	-3.318×10^{-17}
ADW-5-1.0-N	adic-normalization	collective coupling exponent is 1/2-alpha	exact rank-two reduction and depth fit	PASS	2.220×10^{-16}	-0.5
ADW-5-1.0-S	adic-normalization	collective Feshbach exponent is 1-2alpha	exact rank-one self-energy and depth fit	PASS	2.220×10^{-16}	-1
ADW-5-1.5-N	adic-normalization	collective coupling exponent is 1/2-alpha	exact rank-two reduction and depth fit	PASS	2.220×10^{-16}	-1
ADW-5-1.5-S	adic-normalization	collective Feshbach exponent is 1-2alpha	exact rank-one self-energy and depth fit	PASS	0	-2
ADW01	adic-normalization	a normalized depth-w descendant has leaf amplitude p^{-w/2}	uniform p^w-child vector	PASS	1.110×10^{-16}	-0.5
ADW02	adic-normalization	a normalized depth-w descendant has leaf probability p^{-w}	squared normalized amplitude	PASS	1.110×10^{-16}	-1
ADW03	adic-normalization	the collective p^w-leaf coupling norm is |g|p^{(1/2-alpha)w}	rank-two singular-value formula plus direct small matrices	PASS	1.887×10^{-15}	–
ADW04	adic-normalization	the collective coupling acts only on the normalized radial descendant	radial matrix element	PASS	2.220×10^{-16}	–
ADW05	adic-normalization	resonant hidden probability is sin^2(|g|p^{(1/2-alpha)w}t)	exact two-level radial reduction	PASS	1.735×10^{-18}	–
ADW06	adic-normalization	the exact rank-one self-energy is g^2p^{(1-2alpha)w}/z	Schur-Feshbach elimination	PASS	3.496×10^{-18}	–
ADW07	adic-normalization	bounded collective refinement requires alpha>=1/2 and alpha=1/2 is scale invariant	depth monotonicity trichotomy	PASS	0	–
ADW08	adic-normalization	the previous p^{-w} single-detail model remains exact for one normalized hidden packet	two-dimensional block coupling	PASS	0	-1
ADG01	spacetime-adic-walk	all local Grover and sign/torsion coin blocks are unitary	blockwise operator check	PASS	1.310×10^{-15}	–
ADG02	spacetime-adic-walk	the flip-flop shift is an exact involutive permutation	directed-arc reversal	PASS	0	–
ADG03	spacetime-adic-walk	the locally coined walk preserves total probability	vertex-density sum over all steps	PASS	8.882×10^{-16}	–
ADG04	spacetime-adic-walk	vertex probability obeys an exact edge-flux continuity equation	post-coin directed-arc flux	PASS	3.331×10^{-16}	–
ADG05	spacetime-adic-walk	arbitrary coarse product packets obey exact boundary-flux balance	fixed physical-window and shallow-depth packet	PASS	3.331×10^{-16}	–
ADG06	spacetime-adic-walk	probability remains inside the exact product-graph causal cone	breadth-first graph distance	PASS	0	–
ADG07	spacetime-adic-walk	physical displacement and adic-depth change are each at most one per tick	active-support edge tracking	PASS	0	–
ADR01	adic-radial-self-energy	the symmetric sector of a rooted p-tree is an exact Jacobi chain	normalized level isometry	PASS	3.539×10^{-15}	–
ADR02	adic-radial-self-energy	the radial and angular Haar sectors are invariant under the symmetric Hamiltonian	coarse/detail block residual	PASS	5.308×10^{-15}	–
ADR03	adic-radial-self-energy	the root Green function equals the radial Jacobi Green function	full-tree versus reduced resolvent	PASS	1.144×10^{-16}	–
ADR04	adic-radial-self-energy	the root Green function has the exact finite continued-fraction form	Schur complement recursion	PASS	5.594×10^{-17}	–
ADR05	adic-radial-self-energy	g_r=gamma_r/sqrt(p) produces a prime-independent radial channel	p=2,3,5 root resolvents	PASS	1.144×10^{-16}	–
ADR06	adic-radial-self-energy	fixed per-edge coupling produces the expected sqrt(p) collective enhancement	direct radial matrix element	PASS	7.216×10^{-16}	0.5
ADR07	adic-radial-self-energy	the visible Dirac plus hidden adic chain obeys the exact Schur determinant identity	complex-energy determinant comparison	PASS	2.259×10^{-15}	–

ID	Sector	Statement	Method	Status	Residual	Order
ADR08	adic-radial-self-energy	visible spectral branches satisfy (E-Sigma(E))^2=k^2+m_0^2	full visible-hidden diagonalization	PASS	2.065×10^{-14}	–
ADR09	adic-radial-self-energy	the root state has zero overlap with the angular dark sector	radial projector	PASS	0	–
ADP01	adic-polyphase-quantum	standard and adic child polyphase analysis maps are unitary	exact finite matrix identity / direct numerical certificate	PASS	0	–
ADP02	adic-polyphase-quantum	adic and standard polyphase splits are conjugate by base-p digit reversal	exact finite matrix identity / direct numerical certificate	PASS	0	–
ADP03	adic-polyphase-quantum	new-digit residue selectors equal exact roots-of-unity filters	exact finite matrix identity / direct numerical certificate	PASS	2.405×10^{-16}	–
ADP04	adic-polyphase-quantum	the q=0 child-Fourier channel is exactly the coarse adjoint V*	exact finite matrix identity / direct numerical certificate	PASS	0	–
ADP05	adic-polyphase-quantum	the nonzero child-Fourier channels reconstruct the new Haar detail projector	exact finite matrix identity / direct numerical certificate	PASS	6.848×10^{-16}	–
ADP06	adic-polyphase-quantum	the recursive p-adic polyphase/Haar transform is unitary	exact finite matrix identity / direct numerical certificate	PASS	1.406×10^{-15}	–
ADP07	adic-polyphase-quantum	recursive polyphase channel selectors equal the full Haar detail projectors	exact finite matrix identity / direct numerical certificate	PASS	1.081×10^{-15}	–
ADP08	adic-polyphase-quantum	w nested refinements give descendant amplitude p^{-w/2}	exact finite matrix identity / direct numerical certificate	PASS	0	–
ADP09	adic-polyphase-quantum	w nested refinements give descendant probability p^{-w}	exact finite matrix identity / direct numerical certificate	PASS	6.939×10^{-18}	–
ADP10	adic-polyphase-quantum	every operator reconstructs exactly from its p x p polyphase blocks	exact finite matrix identity / direct numerical certificate	PASS	0	–
ADP11	adic-polyphase-quantum	self-adjointness is equivalent to H_rs^*=H_sr in polyphase blocks	exact finite matrix identity / direct numerical certificate	PASS	0	–
ADP12	adic-polyphase-quantum	unitary polyphase blocks obey exact paraunitary identities	exact finite matrix identity / direct numerical certificate	PASS	2.684×10^{-16}	–
ADP13	adic-polyphase-quantum	Hamiltonian evolution commutes exactly with polyphase readdressing	exact finite matrix identity / direct numerical certificate	PASS	1.648×10^{-16}	–
ADP14	adic-polyphase-quantum	polyphase packet probabilities obey antisymmetric Hamiltonian current balance	exact finite matrix identity / direct numerical certificate	PASS	2.776×10^{-17}	–
ADP15	adic-polyphase-quantum	coarse Fourier transformation block-diagonalizes a circulant polyphase Hamiltonian	exact finite matrix identity / direct numerical certificate	PASS	1.267×10^{-14}	–
ADP16	adic-polyphase-quantum	the union of polyphase-symbol spectra equals the full cyclic spectrum	exact finite matrix identity / direct numerical certificate	PASS	1.776×10^{-15}	–
ADP17	adic-polyphase-quantum	nearest-neighbor polyphase bands obey E_q(theta)=mu+2kappa cos((theta+2pi q)/p)	exact finite matrix identity / direct numerical certificate	PASS	9.992×10^{-16}	–
ADP18	adic-polyphase-quantum	the full p-adic Haar transform diagonalizes hierarchical Hamiltonians exactly	exact finite matrix identity / direct numerical certificate	PASS	7.656×10^{-16}	–
ADP19	adic-polyphase-quantum	CPTP completeness is invariant under polyphase quantum readdressing	exact finite matrix identity / direct numerical certificate	PASS	4.962×10^{-15}	–
ADP20	adic-polyphase-quantum	polyphase Kraus evolution is unitarily equivalent to the original quantum channel	exact finite matrix identity / direct numerical certificate	PASS	4.303×10^{-17}	–
ADP21	adic-polyphase-quantum	multi-prime child-polyphase analysis is an exact tensor-product unitary	exact finite matrix identity / direct numerical certificate	PASS	4.792×10^{-16}	–
ADP22	adic-polyphase-quantum	multi-prime interacting evolution commutes with polyphase readdressing	exact finite matrix identity / direct numerical certificate	PASS	4.928×10^{-16}	–
ADP23	adic-polyphase-quantum	the exceptional dyadic sign qubit is preserved as a separate factor under phase polyphase refinement	exact finite matrix identity / direct numerical certificate	PASS	2.872×10^{-16}	–
ADP24	adic-polyphase-quantum	the low-channel resolvent equals the exact polyphase Feshbach Schur complement	exact finite matrix identity / direct numerical certificate	PASS	1.733×10^{-16}	–
ADP25	adic-polyphase-quantum	dyadic even-odd decimation gives mu I+kappa^2(2I+T+T*)/(z-mu)	exact finite matrix identity / direct numerical certificate	PASS	1.388×10^{-17}	–
AIT01	infinite-adic-Haar	Finite cylinder conditional expectations obey E_h E_k=E_min(h,k)	p=2,3,5 cylinder towers	PASS	2.315×10^{-15}	–
AIT02	infinite-adic-Haar	The martingale differences form a complete orthogonal resolution with rank (p-1)p^(h-1)	finite restrictions of L2(Z_p)	PASS	2.315×10^{-15}	–
AIT03	infinite-adic-Haar	Uniform refinement embeddings form an exact isometric direct system	all nested depth triples in the tested towers	PASS	5.551×10^{-16}	–

ID	Sector	Statement	Method	Status	Residual	Order
AIT04	infinite-adic-Haar	The explicit child-DFT Haar basis reconstructs every martingale difference projector	p=2,3,5 wavelet bases	PASS	2.404×10^{-15}	–
AIT05	infinite-adic-Haar	The cutoff tail equals the Parseval sum of omitted detail probabilities	smooth random Haar-cylinder states	PASS	2.776×10^{-17}	–
AIT06	infinite-adic-Haar	The weighted Haar-Sobolev norm gives the bound ||(I-E_H)f|| <= p^{-s(H+1)}||f||_s	finite exact restriction of the analytic estimate	PASS	0	–
AIT07	infinite-adic-quantum	A real level-energy sequence defines exact projector spectral calculus for H, exp(-itH), and the resolvent	hierarchical Hamiltonians with increasing energies	PASS	3.392×10^{-14}	–
AIT08	infinite-adic-quantum	Cutoff Hamiltonians and unitary evolution intertwine exactly with refinement	all adjacent cutoffs in the tested towers	PASS	3.708×10^{-14}	–
AIT09	infinite-adic-quantum	The high-detail resolvent norm equals the maximum omitted spectral multiplier	hierarchical resolvent tails	PASS	6.661×10^{-16}	–
AIT10	multi-prime-direct-limit	Dyadic and odd-adic refinement paths commute exactly in the tensor direct system	2-adic depth 2->4 and 3-adic depth 1->3	PASS	0	–
AIT11	multi-prime-direct-limit	Primewise conditional expectations commute and equal the rectangular tensor expectation	finite 2-adic x 3-adic cylinder space	PASS	0	–
AIT12	dyadic-sign-register	The exceptional dyadic sign packet remains a spectator under additive Haar conditional expectations	separate sign tensor factor	PASS	0	–
PRG01	prime-polyphase-RG	Eliminating p-1 residue channels gives the exact Chebyshev Schur complement	p=2,3,5,7 finite cycles	PASS	4.595×10^{-16}	–
PRG02	prime-polyphase-RG	The normalized effective spectral variable is R_p(x)=2T_p(x/2)	Schur effective parameters	PASS	1.017×10^{-12}	–
PRG03	prime-polyphase-RG	The arbitrary-prime formula specializes to the dyadic Task-S decimation law	symbolic p=2 substitution	PASS	2.178×10^{-16}	–
PRG04	prime-polyphase-RG	The determinant factors into eliminated-detail and coarse Schur determinants	finite complex-energy cycles	PASS	2.580×10^{-14}	–
PRG05	prime-polyphase-RG	Schur poles are exactly the Dirichlet roots of U_{p-1}	open internal path spectra	PASS	6.661×10^{-16}	–
PRG06	prime-polyphase-RG	Chebyshev decimations form the exact semigroup R_m o R_n=R_mn	SymPy polynomial identities	PASS	0	–
PRG07	prime-polyphase-RG	The second-kind Chebyshev factors compose the effective hopping exactly	SymPy U_{mn-1} factorization	PASS	0	–
PRG08	prime-polyphase-RG	h repeated p-decimations equal one p^h-decimation in both center and hopping	complex-energy parameter recursion	PASS	1.112×10^{-15}	–
PRG09	polyphase-band-duality	Every folded inverse band x_q(theta) maps back to 2cos(theta) under R_p	p=2,3,5,7 dense angle grids	PASS	1.515×10^{-14}	–
PRG10	polyphase-band-duality	The fine cyclic spectrum maps exactly onto the coarse cyclic spectrum with p-fold multiplicity	N=pM Fourier spectra	PASS	1.243×10^{-14}	–
PRG11	spectral-fixed-measure	The arcsine measure dx/[pi sqrt(4-x^2)] is invariant under every R_p	Gauss-Chebyshev midpoint moments	PASS	1.184×10^{-16}	–
PRG12	spectral-fixed-measure	The Koopman map sends T_n(x/2) to T_{pn}(x/2)	Chebyshev composition on quadrature nodes	PASS	3.357×10^{-13}	–
PRG13	spectral-fixed-measure	In the orthonormal Chebyshev basis, Koopman is the exact mode injection n->pn and its adjoint is p-decimation	finite Gram matrix below truncation	PASS	3.176×10^{-15}	–
PRG14	multi-prime-RG	Dyadic and odd-prime free spectral decimations commute and depend only on the aggregate factor	R_p R_q=R_pq on [-2,2]	PASS	3.286×10^{-14}	–
QGB01	nonlinear-BFV-BV-bridge	The inner corner momentum generator produces exact conjugation on every canonical matrix pair	orthonormal symmetric-matrix coordinate bracket	PASS	5.699×10^{-15}	–
QGB02	nonlinear-BFV-BV-bridge	The inner corner momentum generators close under the matrix commutator	direct canonical bracket versus commutator generator	PASS	3.908×10^{-14}	–
QGB03	nonlinear-BFV-BV-bridge	Direct and iterated elimination of two coupled detail families have the same Schur complement	random positive block operators	PASS	1.320×10^{-15}	–
QGB04	nonlinear-BFV-BV-bridge	Reversing the two detail-family elimination orders gives the same coarse operator	random positive block operators	PASS	1.593×10^{-15}	–
QGB05	nonlinear-BFV-BV-bridge	Homological transfer cancels a nonzero naive projection defect and restores exact nilpotency	exact rational seven-dimensional contraction	PASS	0	–
QGB06	nonlinear-BFV-BV-bridge	The fine nonlinear model is exactly gauge invariant and divergence free	SymPy differentiation	PASS	0	–
QGB07	nonlinear-BFV-BV-bridge	Eliminating two detail variables induces the stated nonlinear coarse action and gauge vector	exact stationary Gaussian elimination	PASS	0	–

ID	Sector	Statement	Method	Status	Residual	Order
QGB08	nonlinear-BFV-BV-bridge	The Gaussian determinant cancels the ordinary divergence of the induced coarse gauge vector	weighted-divergence identity	PASS	0	–
QGB09	nonlinear-BFV-BV-bridge	Direct and both sequential nonlinear eliminations give the same effective action and determinant	exact SymPy elimination in both orders	PASS	0	–
GWT01	gowdy-bfv-target	The continuum momentum-momentum bracket is the Witt bracket	exact symbolic jet calculation	PASS	0	–
GWT02	gowdy-bfv-target	The continuum Hamiltonian-Hamiltonian bracket is the momentum generator	exact symbolic cancellation	PASS	0	–
GWT03	gowdy-bfv-target	The Gowdy Hamiltonian density has spatial weight two	exact symbolic Lie-derivative calculation	PASS	0	–
GWT04	gowdy-bfv-target	Centered differences recover the D-D bracket quadratically	nonlinear periodic Gowdy data	PASS	7.593×10^{-7}	1.99834
GWT05	gowdy-bfv-target	Centered differences recover the D-H bracket quadratically	nonlinear periodic Gowdy data	PASS	8.967×10^{-7}	1.99788
GWT06	gowdy-bfv-target	Centered differences recover the H-H bracket quadratically	nonlinear periodic Gowdy data	PASS	8.490×10^{-7}	1.99933
GWT07	gowdy-bfv-target	A resolved Fourier derivative reproduces all three brackets at roundoff	nonlinear analytic periodic data	PASS	1.537×10^{-15}	–
GWT08	gowdy-bfv-target	The two chiral sectors close separately and commute	spectral canonical bracket	PASS	1.033×10^{-15}	–
HCP01	analytic-residue	Residue projectors are orthogonal idempotents and sum to the identity	finite coefficient projector matrices	PASS	0	–
HCP02	factorial-lanes	The step-factorial/Pochhammer split is exact	integer and rational arithmetic	PASS	0	–
HCP03	factorial-lanes	Sparse factorial lanes equal their generalized hypergeometric forms	90-digit mpmath evaluation	PASS	2.455×10^{-91}	–
HCP04	factorial-lanes	Sparse factorial lanes equal the root-of-unity filter	90-digit mpmath evaluation	PASS	5.185×10^{-91}	–
HCP05	factorial-lanes	The residue lanes reconstruct the exponential	90-digit mpmath evaluation	PASS	4.909×10^{-91}	–
HCP06	companion-state	The factorial-lane vector satisfies E'=C_N E	matrix exponential differentiation	PASS	5.551×10^{-17}	–
HCP07	companion-state	The companion-state exponential equals the root-filter lane vector	finite matrix exponential versus root sum	PASS	1.241×10^{-16}	–
HCP08	operator-lanes	Bounded-operator factorial lanes equal the operator root filter	4 by 4 complex matrix	PASS	2.895×10^{-16}	–
HCP09	operator-lanes	Bounded-operator lanes reconstruct exp(tA)	4 by 4 complex matrix	PASS	3.908×10^{-16}	–
HCP10	prime-refinement	A parent factorial lane is the exact sum of its p child lanes	90-digit sparse series	PASS	2.455×10^{-91}	–
HCP11	p-adic-window	The normalized short factorial tail is a p-adic unit for 0<=j<p and p|N	exact binomial residues	PASS	0	–
HCP12	coherent-packets	Series, hypergeometric, and root-filter packet probabilities agree	Poisson residue probabilities	PASS	3.682×10^{-91}	–
HCP13	coherent-packets	Coherent packet weights are normalized, nonnegative, and satisfy the mixing bound	N=2,3,5,7 over a mean-occupation grid	PASS	1.110×10^{-16}	–
HCP14	gowdy-bessel-lanes	Bessel residue lanes equal their 0F_{2N-1} forms	90-digit mpmath evaluation	PASS	2.455×10^{-91}	–
HCP15	gowdy-bessel-lanes	Bessel residue lanes equal the root-of-unity filter	90-digit mpmath evaluation	PASS	3.041×10^{-91}	–
HCP16	gowdy-bessel-lanes	Bessel residue lanes reconstruct J_0(2 sqrt(y))	90-digit mpmath evaluation	PASS	1.841×10^{-91}	–
HCP17	gowdy-bessel-lanes	The Bessel lanes satisfy the cyclic Euler-operator system	termwise 90-digit series	PASS	2.455×10^{-91}	–
CTC01	companion-transfer-coordinate	The sampled transfer matrices are unimodular	exponentials of random traceless generators	PASS	3.334×10^{-16}	–
CTC02	companion-transfer-coordinate	Cayley-Hamilton reduces every unimodular two-state transfer matrix	direct 2 by 2 matrix identity	PASS	4.524×10^{-16}	–
CTC03	companion-transfer-coordinate	M^n=U_{n-1}(tr M/2)M-U_{n-2}(tr M/2)I	30 random SL(2,C) matrices and n<=15	PASS	1.502×10^{-13}	–
CTC04	companion-transfer-coordinate	tr(M^n)=2T_n(tr M/2)	30 random SL(2,C) matrices and n<=15	PASS	1.695×10^{-13}	–
CTC05	companion-transfer-coordinate	Each residue subsequence advances under the common coarse transfer M^p	direct versus polyphase state propagation	PASS	7.771×10^{-12}	–

ID	Sector	Statement	Method	Status	Residual	Order
CTC06	companion-transfer-coordinate	Iterated coarse transfer equals direct aggregate powering	(M^p)^q=M^{pq}	PASS	2.123×10^{-14}	–
CTC07	companion-transfer-coordinate	The Chebyshev trace coordinates satisfy R_q(R_p(x))=R_{pq}(x)	random complex trace coordinates	PASS	2.317×10^{-13}	–
CTC08	companion-transfer-coordinate	The free-chain normalized energy is the trace coordinate of its companion transfer	A(x)=[[x,-1],[1,0]]	PASS	0	–
UPS01	unbounded-packet-calculus	Spectral packet truncation has the stated graph-norm tail	finite harmonic spectral sum	PASS	5.551×10^{-17}	–
UPS02	unbounded-packet-calculus	Zero-padded resolvents converge in operator norm with the exact spectral-tail error	harmonic diagonal multipliers	PASS	0	–
UPS03	unbounded-packet-calculus	Zero-padded heat operators converge in operator norm with exponential tail	harmonic diagonal multipliers	PASS	0	–
UPS04	unbounded-packet-calculus	Zero-padded unitary evolution has operator-norm error one	harmonic diagonal multipliers	PASS	2.220×10^{-16}	–
UPS05	unbounded-packet-calculus	The bounded transform converges in norm after the limiting tail value is retained	tail-anchored harmonic multiplier	PASS	0	–
UPS06	unbounded-packet-calculus	The anchored bounded-transform tail decays quadratically	log-log fit over large harmonic cut-offs	PASS	0.00127043	–
UPS07	unbounded-packet-calculus	The Cayley transform converges in norm after tail anchoring	tail-anchored harmonic multiplier	PASS	5.551×10^{-17}	–
UPS08	unbounded-packet-calculus	Spectral residue packets are orthogonal projectors and reconstruct the identity	finite number-basis character projectors	PASS	0	–
UPS09	unbounded-packet-calculus	Creation and annihilation route only between adjacent residue packets	finite oscillator ladder matrices	PASS	0	–
UPS10	unbounded-packet-calculus	The resolvent commutator identity [R_z,P]=-R_z[H,P]R_z holds	random finite self-adjoint witnesses	PASS	7.880×10^{-16}	–
UPS11	unbounded-packet-calculus	The imaginary-axis resolvent commutator obeys the eta^{-2} bound	random finite self-adjoint witnesses	PASS	0	–
DM01	refinement-dark-cosmology	finite Haar multiplicities sum to p^H-1	SymPy geometric sum	PASS	0	–
DM02	refinement-dark-cosmology	the kinetic stress tensor obeys d rho/d ln a=-3(rho+p)	symbolic differentiation of the phase-space integral	PASS	0	–
DM03	refinement-dark-cosmology	generalized Laguerre moments agree with direct quadrature	160-node generalized Gauss-Laguerre versus adaptive quadrature	PASS	7.776×10^{-8}	–
DM04	refinement-dark-cosmology	the Tricomi-U formulas reproduce the kinetic moments	special-function evaluation versus adaptive quadrature	PASS	1.218×10^{-15}	–
DM05	refinement-dark-cosmology	the relativistic equation of state tends to 1/3	small-y kinetic moment	PASS	1.047×10^{-10}	–
DM06	refinement-dark-cosmology	the nonrelativistic equation of state tends to 1/(2 pi y^2)	large-y kinetic moment	PASS	6.366×10^{-5}	–
DM07	refinement-dark-cosmology	the nonrelativistic adiabatic sound speed tends to 5w/3	large-y pressure derivative	PASS	3.183×10^{-5}	–
DM08	refinement-dark-cosmology	the benchmark is normalized to the stated present dark density	direct level sum	PASS	0	–
DM09	refinement-dark-cosmology	the numerical background obeys the exact continuity law	centered derivative at 24 logarithmic scale factors	PASS	2.519×10^{-9}	–
DM10	refinement-dark-cosmology	the depth-eight background agrees with depth twelve	finite-depth comparison	PASS	0	–
DM11	refinement-dark-cosmology	the prime cumulative state count obeys N_H=m_H/Lambda-1	direct finite sums for p=2,3,5,7	PASS	0	–
DM12	refinement-dark-cosmology	the interior prime static kernels have log slope -1	finite geometric tower fit	PASS	4.441×10^{-16}	–
DM13	refinement-dark-cosmology	the cold large-scale dark growth recovers pressureless matter	multi-level growth ODE at k/H0=30	PASS	7.431×10^{-9}	–
DM14	refinement-dark-cosmology	the cold large-scale lensing source recovers pressureless matter	Einstein-Poisson lensing source at k/H0=30	PASS	7.372×10^{-9}	–

ID	Sector	Statement	Method	Status	Residual	Order
DM15	refinement-dark-cosmology	the benchmark is cold at the perturbation start	maximum level equation of state at a=10^-3	PASS	0	9.937×10^{-9}
DM16	refinement-dark-cosmology	the benchmark adiabatic sound speed is small at the perturbation start	maximum level c_a^2 at a=10^-3	PASS	0	1.656×10^{-8}
QEC01	exact-quasienergy-closure	palindromic Clifford product equals C I - i sum_r D_r Gamma_r	128 random 4x4 matrix identities	PASS	4.441×10^{-16}	–
QEC02	exact-quasienergy-closure	C^2+sum_r D_r^2=1 by telescoping normalization	128 random angle tuples	PASS	3.331×10^{-16}	–
QEC03	exact-quasienergy-closure	the characteristic factor is (lambda^2-2 C lambda+1)^2	eigenvalue polynomial residual	PASS	5.189×10^{-30}	–
QEC04	exact-quasienergy-closure	exact positive and negative quasienergy projectors are idempotent, orthogonal, and spectral	random projector identities	PASS	4.003×10^{-16}	–
QEC05	exact-quasienergy-closure	U^2-2 C U+I=0	random minimal-polynomial identities	PASS	6.753×10^{-16}	–
QEC06	exact-quasienergy-closure	omega_Delta^2=e1-e2 Delta^2/3+(6e3-e1e2)Delta^4/45+... through Delta^6	SymPy coefficient matching through Delta^8	PASS	0	–
QEC07	exact-quasienergy-closure	the corrected negative Delta^2 mass-shell term leaves an O(Delta^4) remainder	exact arccos shell refinement	PASS	2.334×10^{-8}	3.98734
QEC08	exact-quasienergy-closure	the Delta^6 mass-shell hierarchy leaves an O(Delta^8) remainder	exact arccos shell refinement	PASS	2.154×10^{-14}	7.75553
QEC09	exact-quasienergy-closure	K_Delta(omega)^2=sum_r K_Delta(a_r)^2 on the principal sector	512 random exact logarithmic identities	PASS	1.137×10^{-13}	–
QEC10	exact-quasienergy-closure	K_Delta^{-1}(K_Delta(x))=x and K_Delta(K_Delta^{-1}(P))=P	random inverse-map identities	PASS	6.232×10^{-13}	–
QEC11	exact-quasienergy-closure	cosine signs compose in Z_2 while logarithmic magnitudes add	random multi-sector phase-magnitude reconstruction	PASS	1.110×10^{-16}	–
QEC12	exact-quasienergy-closure	X oplus_Delta Y is associative because Lambda_Delta linearizes it	exact conjugation argument plus series inversion	PASS	0	–
QEC13	exact-quasienergy-closure	the exact deformed squared-energy addition law is associative and commutative	256 random exact conjugated additions	PASS	2.274×10^{-13}	–
QEC14	exact-quasienergy-closure	max_r |a_r| <= omega_Delta <= sqrt(sum_r a_r^2) on the principal branch	20000 random branch samples	PASS	0	–
QEC15	exact-quasienergy-closure	the exact three-dimensional group velocity satisfies ||v_g||<=1	20000 random exact rational velocity norms	PASS	0	–
QEC16	exact-quasienergy-closure	M_kin=tan(m Delta)/Delta from the exact small-momentum curvature	finite-difference curvature against analytic coefficient	PASS	1.250×10^{-7}	–
QEC17	exact-quasienergy-closure	the leading mass shell splits into isotropic k^4 and cubic-harmonic H_4 parts	SymPy polynomial identity	PASS	0	–
QEC18	exact-quasienergy-closure	the product of cosines is the signed-route characteristic average	exact enumeration of 16 Rademacher routes	PASS	2.220×10^{-16}	–

ID	Sector	Statement	Method	Status	Residual	Order
QEC19	exact-quasienergy-closure	Var[(sum_r epsilon_r a_r)^2]=4e_2	exact enumeration of 16 Rademacher routes	PASS	1.066×10^{-14}	–
QEC20	exact-quasienergy-closure	the axis/face-diagonal cubic route has covariance I/2 and isotropic fourth moment |k|^4/2	exact finite-route moments	PASS	2.842×10^{-14}	–
QEC21	exact-quasienergy-closure	the improved scalar cubic symbol has one zero-quasienergy corner and no pi-quasienergy corner	exact evaluation of all cubic Brillouin corners	PASS	0	–
QEC22	exact-quasienergy-closure	the conditional link shift is exactly unitary and gauge covariant	random SU(2)-valued links on a five-site periodic lattice	PASS	6.661×10^{-16}	–
QEC23	exact-quasienergy-closure	H_D^2=m^2+sum_i Pi_i^2+sum_{i<j} alpha_i alpha_j[Pi_i,Pi_j]	random noncommuting covariant-momentum matrices	PASS	3.553×10^{-15}	–
QEC24	exact-quasienergy-closure	the contraction recurrence conserves phase charge and a nonnegative quadratic energy	random 7x7 Hermitian contraction over 60 steps	PASS	1.066×10^{-14}	–
QEC25	exact-quasienergy-closure	the massless exact product shell has four zero and four pi quasienergy corners	exact enumeration of eight reduced Brillouin corners	PASS	0	–
QEC26	exact-quasienergy-closure	K_i(pi;rho)=exp(4 i rho beta) and K_i(theta)=I+i sin(2rho)theta^2 beta/2+O(theta^3)	exact corner matrix and scaled small-angle residual	PASS	1.110×10^{-16}	–
QEC27	exact-quasienergy-closure	rho=pi/16 gaps every nonorigin zero and pi corner of the Wilson-dressed walk	exact n=1,2,3 corner eigenphases	PASS	0	–
QEC28	exact-quasienergy-closure	a 21^3 momentum scan finds no additional zero or pi closing outside the origin neighborhood	coarse Wilson-dressed Brillouin scan	NUMERICAL ONLY	0	–
QEC29	exact-quasienergy-closure	A_4 matches cos(tau sqrt(m^2+k^2)) through tau^4	truncated free-symbol expansion	PASS	0	–
QEC30	exact-quasienergy-closure	the six positive local Q_j families reproduce the complete P_6 residual polynomial	symbolic leading-symbol decomposition	PASS	0	–
QEC31	exact-quasienergy-closure	the palindromic sixth-order filter cancels the P_6 free dispersion residual	truncated expansion of H_6 A_4 H_6	PASS	0	–
QEC32	exact-quasienergy-closure	the permutation-symmetric four-spinor completion satisfies sum_r d_r^2=1-C^2	512 random integral-weight completions	PASS	3.331×10^{-16}	–
QEC33	exact-quasienergy-closure	the K_Delta-conjugated Lorentz action preserves the exact auxiliary Minkowski norm	256 random finite boosts	PASS	3.155×10^{-12}	–
QEC34	exact-quasienergy-closure	the nonlinear physical boost map is an exact representation by conjugation	composition of random rapidity boosts	PASS	1.821×10^{-14}	–
RNC01	resolved-nonlinear-Ashtekar	{G[lambda],G[mu]}=G[lambda cross mu] on the resolved domain	JAX canonical differentiation on an 18^3 band-1 Fourier grid	PASS	7.676×10^{-17}	–
RNC02	resolved-nonlinear-Ashtekar	{D[xi],G[lambda]}=G[L_xi lambda] on the resolved domain	JAX canonical differentiation on an 18^3 band-1 Fourier grid	PASS	1.068×10^{-17}	–
RNC03	resolved-nonlinear-Ashtekar	{D[xi],D[eta]}=D[[xi,eta]] on the resolved domain	JAX canonical differentiation on an 18^3 band-1 Fourier grid	PASS	3.383×10^{-17}	–
RNC04	resolved-nonlinear-Ashtekar	{G[lambda],H[n]}=0 on the resolved domain	JAX canonical differentiation on an 18^3 band-1 Fourier grid	PASS	6.746×10^{-17}	–

ID	Sector	Statement	Method	Status	Residual	Order
RNC05	resolved-nonlinear-Ashtekar	{D[xi],H[n]}=H[L_xi^{(-1)}n] on the resolved domain	JAX canonical differentiation on an 18^3 band-1 Fourier grid	PASS	3.177×10^{-17}	–
RNC06	resolved-nonlinear-Ashtekar	{H[n],H[m]}=V[S(n,m)] on the resolved domain	JAX canonical differentiation on an 18^3 band-1 Fourier grid	PASS	4.987×10^{-18}	–
RNC07	resolved-nonlinear-Ashtekar	the cosmological determinant term leaves the Hamiltonian bracket unchanged	JAX canonical differentiation with nonzero Lambda	PASS	2.038×10^{-17}	–
RNC08	resolved-nonlinear-Ashtekar	the sufficient alias-free headroom condition 16K<N holds in the exact test	integer support audit	PASS	0	–
RNC09	resolved-nonlinear-Ashtekar	violating declared product headroom produces a nonzero Hamiltonian-bracket defect	band-3 aliasing negative control on the same 18^3 grid	PASS	0	–
LBP01	Lorentzian-BV-parent	Lorentz curvature transforms covariantly	resolved Fourier-form Lorentz variation	PASS	3.338×10^{-16}	–
LBP02	Lorentzian-BV-parent	the Einstein-Cartan density is Lorentz invariant	resolved tetrad-connection density variation	PASS	6.172×10^{-16}	–
LBP03	Lorentzian-BV-parent	the integrated Einstein-Cartan action is Lorentz invariant	resolved periodic trace variation	PASS	8.672×10^{-20}	–
LBP04	Lorentzian-BV-parent	the Einstein-Cartan density obeys the diffeomorphism density identity	resolved Lie derivative and exterior calculus	PASS	5.829×10^{-16}	–
LBP05	Lorentzian-BV-parent	the integrated Einstein-Cartan action is diffeomorphism invariant	resolved periodic trace variation	PASS	2.998×10^{-19}	–
LBP06	Lorentzian-BV-parent	the curvature Bianchi identity holds	resolved covariant exterior derivative	PASS	3.990×10^{-16}	–
LBP07	Lorentzian-BV-parent	the torsion Bianchi identity holds	resolved covariant exterior derivative	PASS	4.162×10^{-15}	–
LBP08	Lorentzian-BV-parent	the Lorentzian Hodge star squares to minus the identity on bivectors	exact matrix identity	PASS	0	–
LBP09	Lorentzian-BV-parent	the self-dual bivector projector is idempotent	exact matrix identity	PASS	0	–
LBP10	Lorentzian-BV-parent	the anti-self-dual bivector projector is idempotent	exact matrix identity	PASS	0	–
LBP11	Lorentzian-BV-parent	the two chiral projectors are complementary	exact matrix identity	PASS	0	–
LBP12	Lorentzian-BV-parent	real Lorentz bivectors obey conjugate chiral pairing	exact chiral reality identity	PASS	0	–
LBP13	Lorentzian-BV-parent	the gamma matrices obey the (-+++) Clifford relation	exact 4x4 matrix identity	PASS	0	–
LBP14	Lorentzian-BV-parent	the spin representation intertwines Lorentz vectors	gamma-matrix commutator identity	PASS	0	–
LBP15	Lorentzian-BV-parent	the Dirac covariant derivative transforms covariantly	resolved Lorentz-U(1) transformation	PASS	3.721×10^{-17}	–
LBP16	Lorentzian-BV-parent	the Lorentz algebra closes on the tetrad	resolved off-shell commutator	PASS	6.939×10^{-18}	–
LBP17	Lorentzian-BV-parent	the Lorentz algebra closes on the connection	resolved off-shell commutator	PASS	3.816×10^{-17}	–
LBP18	Lorentzian-BV-parent	the diffeomorphism algebra closes on the tetrad	resolved Lie-derivative commutator	PASS	2.012×10^{-16}	–
LBP19	Lorentzian-BV-parent	the diffeomorphism algebra closes on the connection	resolved Lie-derivative commutator	PASS	2.732×10^{-17}	–
LBP20	Lorentzian-BV-parent	the mixed diffeomorphism-Lorentz algebra closes on the tetrad	resolved semidirect-product commutator	PASS	3.469×10^{-17}	–
LBP21	Lorentzian-BV-parent	the mixed diffeomorphism-Lorentz algebra closes on the connection	resolved semidirect-product commutator	PASS	1.370×10^{-16}	–
LBP22	Lorentzian-BV-parent	the conjugate Dirac covariant derivative transforms covariantly	resolved Lorentz-U(1) transformation	PASS	3.956×10^{-17}	–

ID	Sector	Statement	Method	Status	Residual	Order
LBP23	Lorentzian-BV-parent	the Dirac density is Lorentz and U(1) invariant	resolved density variation	PASS	9.878×10^{-18}	–
LBP24	Lorentzian-BV-parent	the integrated Dirac action is Lorentz and U(1) invariant	resolved periodic trace variation	PASS	3.330×10^{-21}	–
LBP25	Lorentzian-BV-parent	the Abelian field strength is gauge invariant	resolved exterior derivative	PASS	4.406×10^{-16}	–
LBP26	Lorentzian-BV-parent	the first-order Maxwell density is Lorentz and U(1) invariant	resolved auxiliary-field density variation	PASS	5.388×10^{-17}	–
LBP27	Lorentzian-BV-parent	the integrated first-order Maxwell action is Lorentz and U(1) invariant	resolved periodic trace variation	PASS	8.239×10^{-21}	–
LBP28	Lorentzian-BV-parent	eliminating the Maxwell auxiliary field reproduces the on-shell Maxwell density	resolved algebraic constitutive substitution	PASS	3.469×10^{-18}	–
LBP29	Lorentzian-BV-parent	the classical BRST differential is nilpotent and the minimal BV action obeys the classical master equation	off-shell semidirect-product Jacobi identity and antifield construction	PASS	0	–
LBP30	Lorentzian-BV-parent	violating the declared multi-factor headroom produces a nonzero diffeomorphism-density defect	band-2 negative control on a 12^4 grid	PASS	0	–
LBP31	Lorentzian-BV-parent	the quantum BV master equation remains regulator- and measure-dependent	status boundary	OPEN NOT VERIFIED	–	–
BFB01	Lorentzian-3+1-BFV	the Lorentz curvature has the stated exact 3+1 split	resolved differential-form coefficient comparison	PASS	1.110×10^{-16}	–
BFB02	Lorentzian-3+1-BFV	the Einstein-Cartan action has the stated exact 3+1 split	resolved differential-form coefficient comparison	PASS	7.216×10^{-16}	–
BFB03	Lorentzian-3+1-BFV	the first-order Maxwell action has the stated exact 3+1 split	resolved differential-form coefficient comparison	PASS	5.551×10^{-17}	–
BFB04	Lorentzian-3+1-BFV	the Dirac action has the stated exact 3+1 split	resolved differential-form coefficient comparison	PASS	1.688×10^{-16}	–
BFB05	Lorentzian-3+1-BFV	the complete gravity-Maxwell-Dirac action split is exact	sum of independently checked action blocks	PASS	7.217×10^{-16}	–
BFB06	Lorentzian-3+1-BFV	spatial integration by parts gives the canonical multiplier form	periodic resolved trace identity	PASS	0	–
BFB07	Lorentzian-3+1-BFV	the spatial connection Cartan identity holds	resolved covariant differential-form identity	PASS	1.388×10^{-17}	–
BFB08	Lorentzian-3+1-BFV	the spatial tetrad Cartan identity holds	resolved covariant differential-form identity	PASS	1.110×10^{-16}	–
BFB09	Lorentzian-3+1-BFV	the shifted Lorentz-diffeomorphism identity holds on the connection	resolved moment-map parameter shift	PASS	1.702×10^{-16}	–
BFB10	Lorentzian-3+1-BFV	the shifted Lorentz-diffeomorphism identity holds on the tetrad	resolved moment-map parameter shift	PASS	1.388×10^{-16}	–
BFB11	Lorentzian-3+1-BFV	the shifted U(1)-diffeomorphism identity holds	resolved Cartan formula and Abelian gauge shift	PASS	1.423×10^{-16}	–
BFB12	Lorentzian-3+1-BFV	the exact band-1 product satisfies spectral Leibniz to roundoff	10^4 resolved Fourier grid	PASS	1.119×10^{-14}	–
BFB13	Lorentzian-3+1-BFV	direct and iterated Fourier refinement embeddings agree	coefficient-level zero-padding identity	PASS	0	–
BFB14	Lorentzian-3+1-BFV	the complete covariant boundary BFV charge is nilpotent	analytic moment-map theorem for the diffeomorphism-Lorentz-U(1) semidirect algebra	PASS	0	–
BFB15	Lorentzian-3+1-BFV	the ADM-basis BFV charge is the canonical ghost pullback of the covariant charge and is nilpotent on the regular reduction	analytic ghost-coordinate pullback	PASS	0	–
BFB16	Lorentzian-3+1-BFV	the adaptive time-slab relation intertwines the boundary BFV charges	BV-BFV master identity	CONDITIONAL THEOREM	–	–
BFB17	Lorentzian-3+1-BFV	violating product headroom produces a large spectral Leibniz defect	band-3 negative control on the 10^4 grid	PASS	0	–
BFB18	Lorentzian-3+1-BFV	explicit second-class simplicity and fermionic-primary reduction remains open	status boundary	OPEN NOT VERIFIED	–	–
SCR01	second-class-reduction	B0i dual relation	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR02	second-class-reduction	axial current spin identity	recorded deterministic symbolic/numerical reduction audit	PASS	0	–

ID	Sector	Statement	Method	Status	Residual	Order
SCR03	second-class-reduction	bosonic reduced symplectic rank	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR04	second-class-reduction	clifford axial identity	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR05	second-class-reduction	degenerate fermionic primary rank	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR06	second-class-reduction	degenerate triad determinant	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR07	second-class-reduction	dirac gradient max constraint error	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR08	second-class-reduction	dirac gradient max energy error	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR09	second-class-reduction	dirac gradient max solver residual	recorded deterministic symbolic/numerical reduction audit	PASS	1.092×10^{-16}	–
SCR10	second-class-reduction	dirac linear contorsion coefficient	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR11	second-class-reduction	fermionic primary matrix min singular	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR12	second-class-reduction	fermionic primary matrix rank	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR13	second-class-reduction	fermionic reduced symplectic rank	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR14	second-class-reduction	gravitational symplectic pullback	recorded deterministic symbolic/numerical reduction audit	PASS	1.110×10^{-16}	–
SCR15	second-class-reduction	half density symplectic identity	recorded deterministic symbolic/numerical reduction audit	PASS	3.662×10^{-15}	–
SCR16	second-class-reduction	maxwell symplectic pullback	recorded deterministic symbolic/numerical reduction audit	PASS	4.441×10^{-16}	–
SCR17	second-class-reduction	on shell four fermion coefficient	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR18	second-class-reduction	rk4 negative control max constraint error	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR19	second-class-reduction	rk4 negative control max energy error	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR20	second-class-reduction	rotation action on E	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR21	second-class-reduction	rotation action on K	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR22	second-class-reduction	rotation action on spinor	recorded deterministic symbolic/numerical reduction audit	PASS	9.155×10^{-16}	–
SCR23	second-class-reduction	spin su2 commutator	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR24	second-class-reduction	time gauge simplicity Bij	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR25	second-class-reduction	torsion equation residual	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SCR26	second-class-reduction	triad determinant	recorded deterministic symbolic/numerical reduction audit	PASS	0	–
SF01	finite-range-strong-field	Exact mass-constraint identity	recorded symbolic identity or finite-range electrovac benchmark	PASS	0	–
SF02	finite-range-strong-field	Exact PG Hamiltonian constraint	recorded symbolic identity or finite-range electrovac benchmark	PASS	0	–
SF03	finite-range-strong-field	Exact PG momentum constraint	recorded symbolic identity or finite-range electrovac benchmark	PASS	0	–
SF04	finite-range-strong-field	Periodic fourth-order derivative skew split	recorded symbolic identity or finite-range electrovac benchmark	PASS	1.395×10^{-15}	–
SF05	finite-range-strong-field	Split-form energy identity	recorded symbolic identity or finite-range electrovac benchmark	PASS	3.553×10^{-15}	–
SF06	finite-range-strong-field	Projected mass function	recorded symbolic identity or finite-range electrovac benchmark	PASS	4.441×10^{-16}	–

ID	Sector	Statement	Method	Status	Residual	Order
SF07	finite-range-strong-field	Flat-slice gauge constraint	recorded symbolic identity or finite-range electrovac benchmark	PASS	0	–
SF08	finite-range-strong-field	Algebraic mass-shell constraint	recorded symbolic identity or finite-range electrovac benchmark	PASS	2.220×10^{-15}	–
SF09	finite-range-strong-field	Reduced radial momentum constraint	recorded symbolic identity or finite-range electrovac benchmark	PASS	1.110×10^{-16}	–
SF10	finite-range-strong-field	Radial Maxwell Gauss constraint	recorded symbolic identity or finite-range electrovac benchmark	PASS	1.751×10^{-15}	–
SF11	finite-range-strong-field	Fourth-order radius convergence	recorded symbolic identity or finite-range electrovac benchmark	PASS	–	3.99856
SF12	finite-range-strong-field	Fourth-order radial metric convergence	recorded symbolic identity or finite-range electrovac benchmark	PASS	–	3.9601
SF13	finite-range-strong-field	Fourth-order mass momentum convergence	recorded symbolic identity or finite-range electrovac benchmark	PASS	–	4.02393
SF14	finite-range-strong-field	Fourth-order radial momentum convergence	recorded symbolic identity or finite-range electrovac benchmark	PASS	–	4.00788
SF15	finite-range-strong-field	Fourth-order standard ADM consistency	recorded symbolic identity or finite-range electrovac benchmark	PASS	–	3.7041
SF16	finite-range-strong-field	Outer horizon coordinate	recorded symbolic identity or finite-range electrovac benchmark	PASS	1.893×10^{-11}	–
SF17	finite-range-strong-field	Uniform refinement amplification spread	recorded symbolic identity or finite-range electrovac benchmark	PASS	0	–
SF18	finite-range-strong-field	Direct and stable split forms differ at fourth order	recorded symbolic identity or finite-range electrovac benchmark	PASS	–	4.11935
SF19	finite-range-strong-field	Unprojected negative control produces visible constraint drift	recorded symbolic identity or finite-range electrovac benchmark	PASS	0	–
SC01	spherical-scalar-collapse	Symbolic mass-flux compatibility	symbolic PG reduction or converged horizon-penetrating collapse benchmark	PASS	0	–
SC02	spherical-scalar-collapse	Canonical flat-slice substitution	symbolic PG reduction or converged horizon-penetrating collapse benchmark	PASS	0	–
SC03	spherical-scalar-collapse	All strong runs form a horizon	symbolic PG reduction or converged horizon-penetrating collapse benchmark	PASS	1	–
SC04	spherical-scalar-collapse	Weak control remains untrapped to t=5	symbolic PG reduction or converged horizon-penetrating collapse benchmark	PASS	0.146723	–
SC05	spherical-scalar-collapse	Strong event mass loss below two percent	symbolic PG reduction or converged horizon-penetrating collapse benchmark	PASS	0.00262411	–
SC06	spherical-scalar-collapse	Mass-flux relative residual	symbolic PG reduction or converged horizon-penetrating collapse benchmark	PASS	0.00126972	–
SC07	spherical-scalar-collapse	Radial mass-constraint relative residual	symbolic PG reduction or converged horizon-penetrating collapse benchmark	PASS	2.273×10^{-6}	–
SC08	spherical-scalar-collapse	Lapse-constraint relative residual	symbolic PG reduction or converged horizon-penetrating collapse benchmark	PASS	2.808×10^{-6}	–
SC09	spherical-scalar-collapse	Horizon time refinement spread	symbolic PG reduction or converged horizon-penetrating collapse benchmark	PASS	0.00347808	–
SC10	spherical-scalar-collapse	Horizon radius refinement spread	symbolic PG reduction or converged horizon-penetrating collapse benchmark	PASS	0.00932554	–
SC11	spherical-scalar-collapse	Finite lapse at first horizon	symbolic PG reduction or converged horizon-penetrating collapse benchmark	PASS	0.999829	–

ID	Sector	Statement	Method	Status	Residual	Order
SC12	spherical-scalar-collapse	Strong curvature response	symbolic PG reduction or converged horizon-penetrating collapse benchmark	PASS	7.35797	–
UDS01	superseded-additive-dark-sector	quartic Hessian is U"=3 lambda phi^2-mu^2	retained algebraic/numerical audit of superseded additive branch	PASS	0	–
UDS02	superseded-additive-dark-sector	Gaussian q^2 integral gives the analytic number density	retained algebraic/numerical audit of superseded additive branch	PASS	0	–
UDS03	superseded-additive-dark-sector	locking exit ratio is 4 m_lock^2/(3 mu^2)	retained algebraic/numerical audit of superseded additive branch	PASS	0	–
UDS04	superseded-additive-dark-sector	linear dark-energy density perturbation vanishes	retained algebraic/numerical audit of superseded additive branch	PASS	0	–
UDS05	superseded-additive-dark-sector	linear dark-energy pressure perturbation vanishes	retained algebraic/numerical audit of superseded additive branch	PASS	0	–
UDS06	superseded-additive-dark-sector	origin detail spectrum is positive	retained algebraic/numerical audit of superseded additive branch	PASS	0	–
UDS07	superseded-additive-dark-sector	quadrature matches analytic level-one number density	retained algebraic/numerical audit of superseded additive branch	PASS	0	–
UDS08	superseded-additive-dark-sector	cold exact rho/X approaches m_lock^2	retained algebraic/numerical audit of superseded additive branch	PASS	7.074×10^{-9}	–
UDS09	superseded-additive-dark-sector	present trapped phase has positive curvature	retained algebraic/numerical audit of superseded additive branch	PASS	0	–
UDS10	superseded-additive-dark-sector	present background is accelerating	retained algebraic/numerical audit of superseded additive branch	PASS	0	–
UDS11	superseded-additive-dark-sector	trap release occurs after acceleration begins	retained algebraic/numerical audit of superseded additive branch	PASS	0	–
UDS12	superseded-additive-dark-sector	stable perturbation mode remains bounded	retained algebraic/numerical audit of superseded additive branch	PASS	0	–
UDS13	superseded-additive-dark-sector	negative-control heavy tower is not trapped today	retained algebraic/numerical audit of superseded additive branch	PASS	0.290668	–
SCX01	superseded-additive-crossing	incoming crossing energy is positive	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	0	–
SCX02	superseded-additive-crossing	hypergeometric density agrees with direct quadrature	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	4.055×10^{-16}	–
SCX03	superseded-additive-crossing	hypergeometric pressure agrees with direct quadrature	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	3.540×10^{-16}	–
SCX04	superseded-additive-crossing	hypergeometric variance agrees with direct quadrature	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	1.458×10^{-16}	–
SCX05	superseded-additive-crossing	global-lock threshold exceeds local threshold	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	0	–
SCX06	superseded-additive-crossing	p=13 is below global locking	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	0	–

ID	Sector	Statement	Method	Status	Residual	Order
SCX07	superseded-additive-crossing	p=19 has positive global-lock margin	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	0	–
SCX08	superseded-additive-crossing	p=73 is below but close to direct capture	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	1.634×10^{-10}	–
SCX09	superseded-additive-crossing	p=79 exceeds direct-capture threshold	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	0	–
SCX10	superseded-additive-crossing	second refinement level is exponentially negligible	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	0	–
SCX11	superseded-additive-crossing	post-production density matches the crossing moment	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	0	–
SCX12	superseded-additive-crossing	outgoing kinetic plus produced energy equals incoming kinetic	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	0	–
SCX13	superseded-additive-crossing	continuity exchange cancels exactly along the trajectory	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	1.110×10^{-16}	–
SCX14	superseded-additive-crossing	average mode remains close to the origin before release	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	0	–
SCX15	superseded-additive-crossing	acceleration begins before lock release	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	0	–
SCX16	superseded-additive-crossing	lock release occurs before the first acceleration interval ends	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	0	–
SCX17	superseded-additive-crossing	released mode rolls away from the origin	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	0	–
SCX18	superseded-additive-crossing	release scale is stable under solver refinement	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	9.883×10^{-6}	–
SCX19	superseded-additive-crossing	first global-lock prime is 19	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	0	–
SCX20	superseded-additive-crossing	first direct-capture prime is 79	retained rapid-crossing/backreaction audit of superseded additive branch	PASS	0	–
SCW01	scale-covariant-dark-sector	additive abundance-velocity elimination identity	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	0	–
SCW02	scale-covariant-dark-sector	additive cold-lock lower bound exceeds unity at the observed ratio	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	0	–
SCW03	scale-covariant-dark-sector	continuum total-energy coefficient matches quadrature	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	1.621×10^{-12}	–
SCW04	scale-covariant-dark-sector	continuum pressure coefficient matches quadrature	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	3.493×10^{-11}	–
SCW05	scale-covariant-dark-sector	continuum production equation of state is one ninth	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	2.776×10^{-17}	–
SCW06	scale-covariant-dark-sector	continuum captured rest fraction is 525 pi / 2048	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	1.110×10^{-16}	–

ID	Sector	Statement	Method	Status	Residual	Order
SCW07	scale-covariant-dark-sector	discrete rest fraction agrees with continuum envelope	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	9.319×10^{-5}	–
SCW08	scale-covariant-dark-sector	discrete production pressure agrees with one-ninth envelope	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	4.949×10^{-5}	–
SCW09	scale-covariant-dark-sector	discrete mean mass agrees with five A over two pi	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	5.025×10^{-4}	–
SCW10	scale-covariant-dark-sector	energy projection is positive and below unity	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	0	–
SCW11	scale-covariant-dark-sector	present dark abundance reconstructs the declared ratio	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	0	–
SCW12	scale-covariant-dark-sector	weighted locking curvature is positive today	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	0	–
SCW13	scale-covariant-dark-sector	exact and asymptotic release scales agree	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	1.945×10^{-13}	–
SCW14	scale-covariant-dark-sector	exact and asymptotic acceleration scales agree	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	1.934×10^{-4}	–
SCW15	scale-covariant-dark-sector	present equation of state agrees with a_star squared over five	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	1.262×10^{-16}	–
SCW16	scale-covariant-dark-sector	exact kinetic moments recover c_a squared=a_star squared/(3a squared)	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	6.311×10^{-4}	–
SCW17	scale-covariant-dark-sector	large-scale growth is cold to one part in ten thousand	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	2.993×10^{-5}	–
SCW18	scale-covariant-dark-sector	transfer curve contains a finite half-mode	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	0	–
SCW19	scale-covariant-dark-sector	acceleration begins before locking release	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	0	–
SCW20	scale-covariant-dark-sector	discrete peak lies within one dyadic bin of the continuum peak	analytic continuum envelope or deterministic discrete dark-sector benchmark	PASS	0.0418557	–
M01	diagnostics	relative work residual is undefined when both work terms vanish identically	zero/zero diagnostic audit	CORRECTED	0	–
M02	conventions	Riemann, inner-product, gauge-link, and force-vector signs explicitly fixed	convention ledger	PASS	0	–
M03	refinement	partition incidence identity is R^T 1_c=1_f, not R 1_f=1_c	dimension and random partition audit	CORRECTED	0	–
M04	weighted-adic-refinement	generic short-time leakage remainder is O(t^2 p^{-w}), while the exact Duhamel bound is linear in p^{-w}	Duhamel expansion and operator-norm audit	CORRECTED	0	–
M05	projector-ADM	the shift contribution is represented either by the smeared momentum constraint or by Lie-dragged velocities, but not by both simultaneously	canonical ADM Legendre-form audit	CORRECTED	0	–
M06	nonlinear-BFV-BV-bridge	with IP=I-s0 H-H s0, the transferred perturbation uses (I+H delta)^(-1) and an alternating return series	homological-perturbation sign audit	CORRECTED	0	–
M07	dark-sector-model	the one-shell energy-matched additive cold-lock branch is inconsistent and is superseded by the scale-covariant refinement interaction	conditional abundance-coldness-locking no-go and replacement audit	CORRECTED	0	–

53.4 Executable rerun manifest

Script	Status	Seconds
01_projector_current_and_moving_atlas.py	PASS	1.52937
02_discrete_relativistic_packet.py	PASS	4.88856
03_expanding_flrw_projector_network.py	PASS	3.87791
04_homogeneous_projector_backreaction.py	PASS	6.26445
05_variational_projector_cosmology.py	PASS	5.23613
06_projector_holonomy_curvature.py	PASS	3.28528
07_inhomogeneous_weak_field_gravity.py	PASS	4.69579
08_weak_field_optics.py	PASS	5.6793
09_linearized_tensor_gravity.py	PASS	5.03507
10_unbounded_projector_current.py	PASS	2.94504
11_dirac_3plus1_projector_walk.py	PASS	5.39391
12_u1_gauge_projector_walk.py	PASS	2.92616
13_constraint_preserving_scalar_cosmology.py	PASS	6.46635
14_dynamical_u1_dirac.py	PASS	4.03332
15_metric_holonomy_equivalence.py	PASS	4.17954
16_local_scalar_stress_energy.py	PASS	5.43795
17_emergent_lorentz_boosts.py	PASS	3.27643
18_symbolic_formula_audit.py	PASS	2.89364
19_formula_registry.py	PASS	0.631393
20_generate_audit_tables.py	PASS	0.631938
21_povm_packet_calculus.py	PASS	1.52486
22_su2_nonabelian_refinement.py	PASS	3.1837
23_gowdy_constraint_propagation.py	PASS	4.99302
24_finite_scale_phenomenology.py	PASS	4.94539
25_constraint_preserving_gowdy_integrator.py	PASS	5.73798
26_einstein_scalar_perturbations.py	PASS	4.02896
27_cosmological_finite_scale_delay.py	PASS	4.37711
28_generate_cosmology_ledger.py	PASS	1.27173
29_general_flrw_background.py	PASS	4.68857
30_vector_tensor_flrw_perturbations.py	PASS	4.41766
31_linear_structure_growth.py	PASS	4.63593
32_primordial_power_law_inflation.py	PASS	4.38618
33_recombination_acoustics.py	PASS	6.24233
34_minimal_cmb_boltzmann.py	PASS	7.69163
35_finite_derivation_corner_calculus.py	PASS	1.32391
36_linearized_projector_adm.py	PASS	4.42791
37_nonlinear_constraint_leibniz_defect.py	PASS	2.82487
38_adic_unit_quantum_split.py	PASS	3.02382
39_adic_haar_quantum_refinement.py	PASS	2.97689
40_adelic_quantum_dynamics.py	PASS	3.32777
41_adic_normalization_weights.py	PASS	4.3417
42_spacetime_adic_product_walk.py	PASS	3.09278
43_adic_radial_self_energy.py	PASS	4.27796
44_adic_polyphase_quantum_refinement.py	PASS	4.1292
45_infinite_adic_haar_quantum_limit.py	PASS	6.48751
46_prime_polyphase_spectral_renormalization.py	PASS	4.38113
47_nonlinear_bfv_bv_polyphase_bridge.py	PASS	4.0778
48_gowdy_bfv_target.py	PASS	3.78439
49_hypergeometric_companion_packet_calculus.py	PASS	12.9694
50_companion_transfer_trace_coordinates.py	PASS	1.67337
51_unbounded_packet_surrogate_calculus.py	PASS	2.77344
52_refinement_dark_sector_cosmology.py	PASS	9.45747
53_exact_quasienergy_closure.py	PASS	15.9803
54_resolved_nonlinear_ashtekar_closure.py	PASS	0.569504
55_resolved_lorentzian_bv_parent.py	PASS	0.548735
56_resolved_lorentzian_3plus1_bfv_boundary.py	PASS	0.604448
57_resolved_second_class_reduction.py	PASS	0.600683
58_finite_range_strong_field.py	PASS	0.618156

Script	Status	Seconds
59_spherical_scalar_collapse.py	PASS	0.655308
60_unified_additive_dark_sector.py	PASS	0.629755
61_self_consistent_additive_crossing.py	PASS	0.574924
62_scale_covariant_dark_sector.py	PASS	0.576838

53.5 Formula-registry summary

Sector	Verification status	Occurrences
Einstein-scalar-perturbations	NEW_EXTENSION_PASS	31
FLRW-kinematics	FAMILY_AUDITED	29
FLRW-vector-tensor	NEW_EXTENSION_PASS	29
Gowdy-BFV-next-target	FAMILY_AUDITED	42
Gowdy-constrained-integrator	NEW_EXTENSION_PASS	27
Gowdy-reduced-gravity	NEW_EXTENSION_PASS	31
Lorentzian-3+1-BFV	BFV_BOUNDARY_PASS	55
Lorentzian-BV-parent	CLASSICAL_BV_PASS	60
Lorentzian-BV-parent	OPEN_NOT_VERIFIED	1
SU2-nonabelian	NEW_EXTENSION_PASS	17
U1-dynamical	FAMILY_AUDITED	23
U1-kinematic	FAMILY_AUDITED	32
U1-kinematic	QUALIFIED_PASS	1
adelic-quantum-dynamics	NEW_EXTENSION_PASS	30
adic-Haar-quantum-refinement	NEW_EXTENSION_PASS	40
adic-normalization-weights	NEW_EXTENSION_PASS	18
adic-polyphase-quantum	NEW_EXTENSION_PASS	51
adic-radial-self-energy	NEW_EXTENSION_PASS	20
adic-unit-quantum-split	NEW_EXTENSION_PASS	34
complete-cosmology-program	OPEN_NOT_VERIFIED	3
cosmological-finite-scale-delay	CONDITIONAL_PREDICTION_PASS	26
countable-unbounded	FAMILY_AUDITED	36
countable-unbounded-packet-calculus	NEW_EXTENSION_PASS	41
dirac-1+1	FAMILY_AUDITED	33
dirac-3+1	FAMILY_AUDITED	36
emergent-Lorentz	FAMILY_AUDITED	25
exact-quasienergy-closure	NEW_EXTENSION_PASS	179
finite-derivation-corner-calculus	NEW_EXTENSION_PASS	26
finite-range-strong-field	FINITE_RANGE_STRONG_FIELD_PASS	97
finite-scale-prediction	CONDITIONAL_PREDICTION_PASS	20
frontmatter-status	FRONTMATTER_REVIEWED	1
general-FLRW-background	NEW_EXTENSION_PASS	35
holonomy-curvature	FAMILY_AUDITED	33
homogeneous-backreaction	FAMILY_AUDITED	35
hypergeometric-companion-packets	NEW_EXTENSION_PASS	49
infinite-adic-Haar-quantum	NEW_EXTENSION_PASS	57
interpretation	OPEN_NOT_VERIFIED	12
linear-structure-growth	NEW_EXTENSION_PASS	31
linearized-tensor-gravity	CONDITIONAL_CORRECTED	1
linearized-tensor-gravity	CORRECTED_PASS	1
linearized-tensor-gravity	FAMILY_AUDITED	67
local-scalar-stress	FAMILY_AUDITED	18
metric-holonomy	FAMILY_AUDITED	24
minimal-CMB-Boltzmann	NEW_EXTENSION_PASS	29
new-symbolic-extensions	NEW_EXTENSION_PASS	23
nonlinear-BFV-BV-polyphase-bridge	CONDITIONAL_TRANSFER_THEOREM	101
nonlinear-constraint-defect	NEW_EXTENSION_PASS	26
numerical-records	RERUN_PASS	1
open-program	OPEN_NOT_VERIFIED	10
packet-dynamics	FAMILY_AUDITED	25
positive-operator-atlas	NEW_EXTENSION_PASS	19
prime-polyphase-spectral-RG	NEW_EXTENSION_PASS	62
primordial-power-law	NEW_EXTENSION_PASS	28
projector-ADM-linearized	NEW_EXTENSION_PASS	42
projector-algebra	FAMILY_AUDITED	22
proof-archive	PROOF_ARCHIVE_REVIEWED	156
recombination-acoustics	NEW_EXTENSION_PASS	35

Sector	Verification status	Occurrences
refinement-dark-cosmology	FAMILY_AUDITED	62
resolved-nonlinear-Ashtekar	OPEN_NOT_VERIFIED	1
resolved-nonlinear-Ashtekar	RESOLVED_DOMAIN_PASS	31
scalar-cosmology	FAMILY_AUDITED	37
scale-covariant-dark-sector	SCALE_COVARIANT_MODEL_PASS	85
second-class-reduction	SECOND_CLASS_REDUCTION_PASS	74
spacetime-adic-product-walk	NEW_EXTENSION_PASS	15
spherical-scalar-collapse	SCALAR_COLLAPSE_PASS	66
summary	FAMILY_AUDITED	4
superseded-additive-crossing	SUPERSEDED_MODEL_BRANCH	49
superseded-additive-dark-sector	SUPERSEDED_MODEL_BRANCH	53
symbolic-logic-ledger	CORRECTED_PASS	2
symbolic-logic-ledger	NEW_EXTENSION_PASS	11
variational-FLRW	FAMILY_AUDITED	34
weak-field-gravity	CORRECTED_PASS	4
weak-field-gravity	FAMILY_AUDITED	30
weak-field-optics	CORRECTED_PASS	1
weak-field-optics	FAMILY_AUDITED	33

53.6 Machine-readable registries

$$\mathcal{R}_{\mathrm{all}} = \texttt{data/formula_registry.csv}, \tag{53.11}$$

$$\mathcal{R}_{\mathrm{canonical}} = \texttt{data/formula_registry_canonical.csv}, \tag{53.12}$$

$$\mathcal{A} = \texttt{data/symbolic_formula_audit_checks.csv}, \tag{53.13}$$

$$\mathcal{X} = \texttt{audit/script_run_manifest.csv}. \tag{53.14}$$

Part VII

Complete Formula Corpus

Appendix A

Complete Formula-Occurrence Archive

00_frontmatter.tex

F0001 00_frontmatter.tex:26 FAMILY_AUDITED

$$\cos(\omega_\varepsilon \varepsilon) = \cos(m\varepsilon)\cos(k\varepsilon).$$

F0002 00_frontmatter.tex:34 FAMILY_AUDITED

$$H_{\mathcal{P}}(a) = \sum_{\alpha} w_\alpha \sqrt{m_\alpha^2 + k_\alpha^2/a^2}$$

F0003 00_frontmatter.tex:50 FAMILY_AUDITED

$$\cos(\omega_\Delta \Delta) = \cos(m\Delta)\cos(k_x\Delta)\cos(k_y\Delta)\cos(k_z\Delta),$$

F0004 00_frontmatter.tex:61 FAMILY_AUDITED

$$\frac{\Delta^2}{6E'}\left[Q(\boldsymbol{k}') - Q(\boldsymbol{k})\right].$$

01_foundations.tex

F0005 01_foundations.tex:17 FAMILY_AUDITED

$$\boxed{\text{project} \longrightarrow \text{transfer} \longrightarrow \text{refine} \longrightarrow \text{reconstruct.}}$$

F0006 01_foundations.tex:29 FAMILY_AUDITED

$$\mathcal{P} = \{P_\alpha\}_{\alpha \in A}$$

F0007 01_foundations.tex:33 FAMILY_AUDITED

$$P_\alpha^* = P_\alpha, \qquad P_\alpha P_\beta = \delta_{\alpha\beta} P_\alpha, \qquad \sum_{\alpha \in A} P_\alpha = I.$$

F0008 01_foundations.tex:44 FAMILY_AUDITED

$$\psi = \sum_{\alpha} \psi_\alpha, \qquad \psi_\alpha = P_\alpha \psi.$$

F0009 01_foundations.tex:50 FAMILY_AUDITED

$$\boxed{X = \sum_{\alpha,\beta} X_{\alpha\beta}, \qquad X_{\alpha\beta} = P_\alpha X P_\beta.}$$

F0010 01_foundations.tex:63 FAMILY_AUDITED

$$X = \sum_{\alpha,\beta} P_\alpha X P_\beta.$$

F0011 01_foundations.tex:70 FAMILY_AUDITED

$$X = IXI = \left(\sum_\alpha P_\alpha\right) X \left(\sum_\beta P_\beta\right) = \sum_{\alpha,\beta} P_\alpha X P_\beta.$$

F0012 01_foundations.tex:84 FAMILY_AUDITED

$$\Sigma_{\mathcal{P}}(X) = \big(X_{\alpha\beta}\big)_{\alpha,\beta}.$$

F0013 01_foundations.tex:90 FAMILY_AUDITED

$$\text{local content : } X_{\alpha\alpha}, \qquad \text{pairwise relation : } X_{\alpha\beta}, \qquad \text{whole object : } \sum_{\alpha,\beta} X_{\alpha\beta}.$$

F0014 01_foundations.tex:104 FAMILY_AUDITED

$$P^{\mathrm{c}}_\alpha = \sum_{r \in C_\alpha} P^{\mathrm{f}}_r.$$

F0015 01_foundations.tex:110 FAMILY_AUDITED

$$P^{\mathrm{c}}_\alpha \psi = \sum_{r \in C_\alpha} P^{\mathrm{f}}_r \psi.$$

F0016 01_foundations.tex:116 FAMILY_AUDITED

$$\boxed{P^{\mathrm{c}}_\alpha X P^{\mathrm{c}}_\beta = \sum_{r \in C_\alpha} \sum_{s \in C_\beta} P^{\mathrm{f}}_r X P^{\mathrm{f}}_s.}$$

F0017 01_foundations.tex:128 FAMILY_AUDITED

$$\mathcal{P}^{(0)} \preceq \mathcal{P}^{(1)} \preceq \mathcal{P}^{(2)} \preceq \cdots$$

F0018 01_foundations.tex:160 FAMILY_AUDITED

$$P'_\alpha = U P_\alpha U^*, \qquad X' = U X U^*.$$

F0019 01_foundations.tex:166 FAMILY_AUDITED

$$P'_\alpha X' P'_\beta = U(P_\alpha X P_\beta)U^*.$$

F0020 01_foundations.tex:172 FAMILY_AUDITED

$$X' = \sum_{\alpha,\beta} P'_\alpha X' P'_\beta.$$

F0021 01_foundations.tex:182 FAMILY_AUDITED

$$P_\alpha(t) = U(t) P_\alpha(0) U(t)^{-1}.$$

F0022 01_foundations.tex:186 FAMILY_AUDITED

$$K(t) = \dot{U}(t)U(t)^{-1}.$$

F0023 01_foundations.tex:190 FAMILY_AUDITED

$$\boxed{\dot{P}_\alpha = [K, P_\alpha].}$$

F0024 01_foundations.tex:196 FAMILY_AUDITED

$$X_{\alpha\beta}(t) = P_\alpha(t)X(t)P_\beta(t).$$

F0025 01_foundations.tex:200 FAMILY_AUDITED

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}X_{\alpha\beta} = P_\alpha\dot{X}P_\beta + [K, P_\alpha]XP_\beta + P_\alpha X[K, P_\beta].}$$

F0026 01_foundations.tex:213 FAMILY_AUDITED

$$\mathfrak{D}_t X = \left(\frac{\mathrm{d}}{\mathrm{d}t}P_\alpha X P_\beta\right)_{\alpha,\beta}.$$

02_conservation.tex

F0027 02_conservation.tex:13 FAMILY_AUDITED

$$\dot{\psi} = A\psi, \qquad A^* = -A,$$

F0028 02_conservation.tex:19 FAMILY_AUDITED

$$\psi_\alpha = P_\alpha\psi, \qquad E_\alpha = \|\psi_\alpha\|^2.$$

F0029 02_conservation.tex:25 FAMILY_AUDITED

$$\boxed{J_{\alpha\leftarrow\beta} = 2\,\mathrm{Re}\,\langle\psi_\alpha, A\psi_\beta\rangle\,.}$$

F0030 02_conservation.tex:34 FAMILY_AUDITED

$$\frac{\mathrm{d}E_\alpha}{\mathrm{d}t} = \sum_\beta J_{\alpha\leftarrow\beta}.$$

F0031 02_conservation.tex:40 FAMILY_AUDITED

$$J_{\alpha\leftarrow\beta} = -J_{\beta\leftarrow\alpha}.$$

F0032 02_conservation.tex:45 FAMILY_AUDITED

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_\alpha E_\alpha = 0.$$

F0033 02_conservation.tex:52 FAMILY_AUDITED

$$\dot{\psi}_\alpha = P_\alpha A\psi.$$

F0034 02_conservation.tex:56 FAMILY_AUDITED

$$\frac{\mathrm{d}E_\alpha}{\mathrm{d}t} = 2\,\mathrm{Re}\,\langle\psi_\alpha, P_\alpha A\psi\rangle\,.$$

F0035 02_conservation.tex:61 FAMILY_AUDITED

$$\frac{\mathrm{d}E_\alpha}{\mathrm{d}t} = \sum_\beta 2\,\mathrm{Re}\,\langle \psi_\alpha, A\psi_\beta\rangle = \sum_\beta J_{\alpha\leftarrow\beta}.$$

F0036 02_conservation.tex:70 FAMILY_AUDITED

$$\begin{aligned} J_{\beta\leftarrow\alpha} &= 2\,\mathrm{Re}\,\langle \psi_\beta, A\psi_\alpha\rangle \\ &= 2\,\mathrm{Re}\,\langle A^*\psi_\beta, \psi_\alpha\rangle \\ &= -2\,\mathrm{Re}\,\langle A\psi_\beta, \psi_\alpha\rangle \\ &= -J_{\alpha\leftarrow\beta}. \end{aligned}$$

F0037 02_conservation.tex:83 FAMILY_AUDITED

$$\boxed{J_{\alpha\leftarrow\beta} = 2\,\mathrm{Im}\,\langle \psi_\alpha, H\psi_\beta\rangle\,.}$$

F0038 02_conservation.tex:94 FAMILY_AUDITED

$$P^{\mathrm{c}}_\alpha = \sum_{r\in C_\alpha} P^{\mathrm{f}}_r.$$

F0039 02_conservation.tex:100 FAMILY_AUDITED

$$E^{\mathrm{c}}_\alpha = \sum_{r\in C_\alpha} E^{\mathrm{f}}_r.$$

F0040 02_conservation.tex:106 FAMILY_AUDITED

$$\begin{aligned} J^{\mathrm{c}}_{\alpha\leftarrow\beta} &= 2\,\mathrm{Re}\left\langle \sum_{r\in C_\alpha}\psi_r, A\sum_{s\in C_\beta}\psi_s\right\rangle \\ &= \sum_{r\in C_\alpha}\sum_{s\in C_\beta} J^{\mathrm{f}}_{r\leftarrow s}. \end{aligned}$$

F0041 02_conservation.tex:124 FAMILY_AUDITED

$$\boxed{J^{\mathrm{c}}_{\alpha\leftarrow\beta} = \sum_{r\in C_\alpha,\, s\in C_\beta} J^{\mathrm{f}}_{r\leftarrow s}.}$$

F0042 02_conservation.tex:140 FAMILY_AUDITED

$$P_\alpha(t) = U(t) P^{(0)}_\alpha U(t)^*.$$

F0043 02_conservation.tex:144 FAMILY_AUDITED

$$\phi = U^*\psi$$

F0044 02_conservation.tex:148 FAMILY_AUDITED

$$\Gamma = U^*\dot{U}.$$

F0045 02_conservation.tex:152 FAMILY_AUDITED

$$\Gamma^* = -\Gamma.$$

F0046 02_conservation.tex:156 FAMILY_AUDITED

$$\begin{aligned} \dot\phi &= \dot U^*\psi + U^*\dot\psi \\ &= -U^*\dot U\, U^*\psi + U^*AU\phi \\ &= (U^*AU - \Gamma)\,\phi. \end{aligned}$$

F0047 02_conservation.tex:165 FAMILY_AUDITED

$$\boxed{A_{\mathrm{eff}} = U^* A U - U^* \dot{U}.}$$

F0048 02_conservation.tex:173 FAMILY_AUDITED

$$\frac{\mathrm{d}E_\alpha}{\mathrm{d}t} = \sum_\beta 2\,\mathrm{Re}\left\langle P_\alpha^{(0)}\phi, A_{\mathrm{eff}} P_\beta^{(0)}\phi \right\rangle.$$

F0049 02_conservation.tex:191 FAMILY_AUDITED

$$\dot{P}_\alpha = [\dot{U}U^*, P_\alpha],$$

F0050 02_conservation.tex:196 FAMILY_AUDITED

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle \psi, P_\alpha \psi\rangle = \left\langle \psi, \dot{P}_\alpha \psi\right\rangle + \left\langle \dot{\psi}, P_\alpha \psi\right\rangle + \left\langle \psi, P_\alpha \dot{\psi}\right\rangle.$$

F0051 02_conservation.tex:207 FAMILY_AUDITED

$$\boxed{\dot{E}_\alpha = \left\langle \psi, \dot{P}_\alpha \psi\right\rangle + 2\,\mathrm{Re}\,\langle P_\alpha \psi, A\psi\rangle.}$$

03_discrete_relativity.tex

F0052 03_discrete_relativity.tex:7 FAMILY_AUDITED

$$\mathcal{H}_\varepsilon = \ell^2(\mathbb{Z}_N) \otimes \mathbb{C}^2.$$

F0053 03_discrete_relativity.tex:12 FAMILY_AUDITED

$$P_{j,+} = |j,+\rangle\langle j,+|, \qquad P_{j,-} = |j,-\rangle\langle j,-|.$$

F0054 03_discrete_relativity.tex:23 FAMILY_AUDITED

$$S\,|j,+\rangle = |j+1,+\rangle, \qquad S\,|j,-\rangle = |j-1,-\rangle.$$

F0055 03_discrete_relativity.tex:32 FAMILY_AUDITED

$$R_m(\varepsilon) = \exp(-\mathrm{i}m\varepsilon\sigma_x).$$

F0056 03_discrete_relativity.tex:38 FAMILY_AUDITED

$$\boxed{U_\varepsilon = R_m(\varepsilon) S.}$$

F0057 03_discrete_relativity.tex:48 FAMILY_AUDITED

$$S(k) = \exp(-\mathrm{i}k\varepsilon\sigma_z).$$

F0058 03_discrete_relativity.tex:52 FAMILY_AUDITED

$$U_\varepsilon(k) = \exp(-\mathrm{i}m\varepsilon\sigma_x)\exp(-\mathrm{i}k\varepsilon\sigma_z).$$

F0059 03_discrete_relativity.tex:58 FAMILY_AUDITED

$$\lambda_\pm(k) = \exp(\mp\mathrm{i}\omega_\varepsilon(k)\varepsilon).$$

F0060 03_discrete_relativity.tex:62 FAMILY_AUDITED

$$\mathrm{Tr}\,U_\varepsilon(k) = 2\cos(\omega_\varepsilon\varepsilon).$$

F0061 03_discrete_relativity.tex:66 FAMILY_AUDITED

$$\boxed{\cos(\omega_\varepsilon(k)\varepsilon) = \cos(m\varepsilon)\cos(k\varepsilon).}$$

F0062 03_discrete_relativity.tex:78 FAMILY_AUDITED

$$\cos(\omega_\varepsilon\varepsilon) = 1 - \frac{\omega_\varepsilon^2\varepsilon^2}{2} + \frac{\omega_\varepsilon^4\varepsilon^4}{24} + \mathcal{O}(\varepsilon^6),$$

F0063 03_discrete_relativity.tex:85 FAMILY_AUDITED

$$\cos(m\varepsilon)\cos(k\varepsilon) = 1 - \frac{m^2+k^2}{2}\varepsilon^2 + \left(\frac{m^4+k^4}{24} + \frac{m^2k^2}{4}\right)\varepsilon^4 + \mathcal{O}(\varepsilon^6).$$

F0064 03_discrete_relativity.tex:92 FAMILY_AUDITED

$$\Omega = \sqrt{m^2+k^2}$$

F0065 03_discrete_relativity.tex:96 FAMILY_AUDITED

$$\omega_\varepsilon = \Omega + c\varepsilon^2 + \mathcal{O}(\varepsilon^4).$$

F0066 03_discrete_relativity.tex:100 FAMILY_AUDITED

$$c = -\frac{m^2k^2}{6\Omega}.$$

F0067 03_discrete_relativity.tex:104 FAMILY_AUDITED

$$\boxed{\omega_\varepsilon(k) = \sqrt{m^2+k^2} - \frac{m^2k^2}{6\sqrt{m^2+k^2}}\varepsilon^2 + \mathcal{O}(\varepsilon^4).}$$

F0068 03_discrete_relativity.tex:116 FAMILY_AUDITED

$$\omega_\varepsilon^2 - k^2 - m^2 = -\frac{m^2k^2}{3}\varepsilon^2 + \mathcal{O}(\varepsilon^4).$$

F0069 03_discrete_relativity.tex:126 FAMILY_AUDITED

$$U_\varepsilon(k) = \exp\left[-\mathrm{i}\varepsilon(m\sigma_x + k\sigma_z) + \mathcal{O}(\varepsilon^2)\right].$$

F0070 03_discrete_relativity.tex:133 FAMILY_AUDITED

$$H_D(k) = k\sigma_z + m\sigma_x.$$

F0071 03_discrete_relativity.tex:137 FAMILY_AUDITED

$$\boxed{\mathrm{i}\partial_t\psi = (-\mathrm{i}\sigma_z\partial_x + m\sigma_x)\,\psi.}$$

F0072 03_discrete_relativity.tex:151 FAMILY_AUDITED

$$-\varepsilon\sin(\omega_\varepsilon\varepsilon)\frac{\partial\omega_\varepsilon}{\partial k} = -\varepsilon\cos(m\varepsilon)\sin(k\varepsilon).$$

F0073 03_discrete_relativity.tex:158 FAMILY_AUDITED

$$\boxed{v_\varepsilon(k) = \frac{\partial\omega_\varepsilon}{\partial k} = \frac{\cos(m\varepsilon)\sin(k\varepsilon)}{\sin(\omega_\varepsilon\varepsilon)}.}$$

F0074 03_discrete_relativity.tex:169 FAMILY_AUDITED

$$v_\varepsilon(k) \longrightarrow \frac{k}{\sqrt{k^2+m^2}},$$

F0075 03_discrete_relativity.tex:179 FAMILY_AUDITED

$$|j - j_0| \le n.$$

F0076 03_discrete_relativity.tex:183 FAMILY_AUDITED

$$c_{\mathrm{micro}} = \frac{\Delta x}{\Delta t} = 1.$$

F0077 03_discrete_relativity.tex:191 FAMILY_AUDITED

$$\dot{E}_{j,s} = \sum_{\ell,r} J_{(j,s)\leftarrow(\ell,r)}.$$

F0078 03_discrete_relativity.tex:200 FAMILY_AUDITED

$$Q_B = \sum_{j\in B} \sum_{s=\pm} P_{j,s}$$

F0079 03_discrete_relativity.tex:208 FAMILY_AUDITED

$$\omega(k) = \sqrt{m^2 + k^2}.$$

F0080 03_discrete_relativity.tex:230 FAMILY_AUDITED

$$\boxed{2.01119.}$$

F0081 03_discrete_relativity.tex:251 FAMILY_AUDITED

$$m = 0.7, \qquad k_0 = 0.8, \qquad x_0 = -30.$$

F0082 03_discrete_relativity.tex:277 FAMILY_AUDITED

$$\boxed{2.01260.}$$

F0083 03_discrete_relativity.tex:294 FAMILY_AUDITED

$$v = \frac{p}{\sqrt{p^2 + m^2}}$$

F0084 03_discrete_relativity.tex:302 FAMILY_AUDITED

$$\Delta\omega = -\frac{m^2 k^2}{6\sqrt{m^2 + k^2}} \varepsilon^2.$$

04_expanding_cosmology.tex

F0085 04_expanding_cosmology.tex:7 FAMILY_AUDITED

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + a(t)^2\,\mathrm{d}\chi^2$$

F0086 04_expanding_cosmology.tex:11 FAMILY_AUDITED

$$D(t) = a(t)\chi.$$

F0087 04_expanding_cosmology.tex:15 FAMILY_AUDITED

$$\mathrm{d}t = a(\eta)\,\mathrm{d}\eta.$$

F0088 04_expanding_cosmology.tex:19 FAMILY_AUDITED

$$\mathrm{d}s^2 = a(\eta)^2(-\,\mathrm{d}\eta^2 + \,\mathrm{d}\chi^2).$$

F0089 04_expanding_cosmology.tex:26 FAMILY_AUDITED

$$a(t) = \mathrm{e}^{Ht}, \qquad H = 0.01.$$

F0090 04_expanding_cosmology.tex:32 FAMILY_AUDITED

$$\boxed{a(\eta) = \frac{1}{1 - H\eta}, \qquad t(\eta) = -\frac{1}{H}\log(1 - H\eta).}$$

F0091 04_expanding_cosmology.tex:44 FAMILY_AUDITED

$$\mathrm{i}\partial_\eta \psi = (-\mathrm{i}\sigma_z \partial_\chi + m a(\eta)\sigma_x)\,\psi.$$

F0092 04_expanding_cosmology.tex:50 FAMILY_AUDITED

$$\boxed{U_n = R^{1/2}_{n+\frac{1}{2}} S R^{1/2}_{n+\frac{1}{2}},}$$

F0093 04_expanding_cosmology.tex:60 FAMILY_AUDITED

$$R^{1/2}_{n+\frac{1}{2}} = \exp\left[-\frac{\mathrm{i}}{2} m a(\eta_{n+1/2})\varepsilon\sigma_x\right].$$

F0094 04_expanding_cosmology.tex:72 FAMILY_AUDITED

$$\phi(\eta) = -k\eta.$$

F0095 04_expanding_cosmology.tex:76 FAMILY_AUDITED

$$\omega_{\mathrm{phys}} = -\frac{\mathrm{d}\phi}{\mathrm{d}t} = -\frac{\mathrm{d}\phi}{\mathrm{d}\eta}\frac{\mathrm{d}\eta}{\mathrm{d}t} = \frac{k}{a(t)}.$$

F0096 04_expanding_cosmology.tex:84 FAMILY_AUDITED

$$\boxed{a(t)\omega_{\mathrm{phys}}(t) = k}$$

F0097 04_expanding_cosmology.tex:90 FAMILY_AUDITED

$$\boxed{1 + z = \frac{\omega_{\mathrm{emit}}}{\omega_{\mathrm{obs}}} = \frac{a(t_{\mathrm{obs}})}{a(t_{\mathrm{emit}})}.}$$

F0098 04_expanding_cosmology.tex:105 FAMILY_AUDITED

$$D(t) = a(t)\chi.$$

F0099 04_expanding_cosmology.tex:109 FAMILY_AUDITED

$$\dot{D} = \dot{a}\chi = HD.$$

F0100 04_expanding_cosmology.tex:119 FAMILY_AUDITED

$$p(t) = \frac{k}{a(t)}.$$

F0101 04_expanding_cosmology.tex:123 FAMILY_AUDITED

$$E(t) = \sqrt{m^2 + \frac{k^2}{a(t)^2}}.$$

F0102 04_expanding_cosmology.tex:127 FAMILY_AUDITED

$$\boxed{\frac{\mathrm{d}\chi}{\mathrm{d}\eta} = \frac{k}{\sqrt{k^2 + m^2 a(\eta)^2}}.}$$

F0103 04_expanding_cosmology.tex:141 FAMILY_AUDITED

$$|\chi - \chi_0| \le \eta - \eta_0$$

F0104 04_expanding_cosmology.tex:146 FAMILY_AUDITED

$$D_{\mathrm{hor}}(t) = a(t) \int_{t_0}^{t} \frac{\mathrm{d}t'}{a(t')}.$$

F0105 04_expanding_cosmology.tex:155 FAMILY_AUDITED

$$m = 0.7, \qquad H = 0.01, \qquad 0 \le \eta \le 30.$$

F0106 04_expanding_cosmology.tex:163 FAMILY_AUDITED

$$\varepsilon \in \{0.4, 0.2, 0.1, 0.05\}.$$

F0107 04_expanding_cosmology.tex:186 FAMILY_AUDITED

$$\boxed{1.99763.}$$

F0108 04_expanding_cosmology.tex:200 FAMILY_AUDITED

$$\dot{D} = HD.$$

F0109 04_expanding_cosmology.tex:204 FAMILY_AUDITED

$$\boxed{1.99469.}$$

F0110 04_expanding_cosmology.tex:229 FAMILY_AUDITED

$$\boxed{2.05587.}$$

F0111 04_expanding_cosmology.tex:243 FAMILY_AUDITED

$$\boxed{0}$$

F0112 04_expanding_cosmology.tex:247 FAMILY_AUDITED

$$D_{\mathrm{hor}} = 11.111111\ldots,$$

F0113 04_expanding_cosmology.tex:262 FAMILY_AUDITED

local microscopic layer: finite unitary transfers, global macroscopic layer: expanding relativistic geometry.

05_backreaction.tex

F0114 05_backreaction.tex:13 FAMILY_AUDITED

$$p_\alpha(a) = \frac{k_\alpha}{a}.$$

F0115 05_backreaction.tex:17 FAMILY_AUDITED

$$\boxed{E_\alpha(a)=\sqrt{m_\alpha^2+\frac{k_\alpha^2}{a^2}}.}$$

F0116 05_backreaction.tex:25 FAMILY_AUDITED

$$\rho_\alpha(a)=\frac{w_\alpha E_\alpha(a)}{a^3}$$

F0117 05_backreaction.tex:31 FAMILY_AUDITED

$$\boxed{p_\alpha(a)=\frac{w_\alpha k_\alpha^2}{3a^5E_\alpha(a)}.}$$

F0118 05_backreaction.tex:40 FAMILY_AUDITED

$$\rho(a)=\sum_\alpha \rho_\alpha(a),\qquad p(a)=\sum_\alpha p_\alpha(a).$$

F0119 05_backreaction.tex:51 FAMILY_AUDITED

$$H_{\mathcal{P}}(a)=\sum_\alpha w_\alpha E_\alpha(a).$$

F0120 05_backreaction.tex:57 FAMILY_AUDITED

$$\rho(a)=\frac{H_{\mathcal{P}}(a)}{a^3}.$$

F0121 05_backreaction.tex:61 FAMILY_AUDITED

$$\frac{\mathrm{d}E_\alpha}{\mathrm{d}a}=-\frac{k_\alpha^2}{a^3E_\alpha}.$$

F0122 05_backreaction.tex:66 FAMILY_AUDITED

$$H'_{\mathcal{P}}(a)=-\sum_\alpha \frac{w_\alpha k_\alpha^2}{a^3E_\alpha}=-3a^2p(a).$$

F0123 05_backreaction.tex:72 FAMILY_AUDITED

$$\boxed{p(a)=-\frac{H'_{\mathcal{P}}(a)}{3a^2}.}$$

F0124 05_backreaction.tex:81 FAMILY_AUDITED

$$\dot\rho=\frac{H'_{\mathcal{P}}(a)\dot a}{a^3}-3\frac{H_{\mathcal{P}}(a)\dot a}{a^4}.$$

F0125 05_backreaction.tex:89 FAMILY_AUDITED

$$\boxed{\dot\rho+3H(\rho+p)=0,\qquad H=\frac{\dot a}{a}.}$$

F0126 05_backreaction.tex:97 FAMILY_AUDITED

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(a^3\rho\right)+p\frac{\mathrm{d}}{\mathrm{d}t}(a^3)=0.}$$

F0127 05_backreaction.tex:106 FAMILY_AUDITED

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(w_\alpha E_\alpha\right)=-p_\alpha\frac{\mathrm{d}}{\mathrm{d}t}(a^3).$$

F0128 05_backreaction.tex:115 FAMILY_AUDITED

$$\frac{8\pi G}{3} = 1,$$

F0129 05_backreaction.tex:119 FAMILY_AUDITED

$$\boxed{H^2 = \rho.}$$

F0130 05_backreaction.tex:123 FAMILY_AUDITED

$$\dot{a} = a\sqrt{\rho(a)}.$$

F0131 05_backreaction.tex:127 FAMILY_AUDITED

$$\boxed{\frac{y_{n+1} - y_n}{\Delta t} = \sqrt{\rho\Big(\exp\frac{y_n + y_{n+1}}{2}\Big)}.}$$

F0132 05_backreaction.tex:137 FAMILY_AUDITED

packet energies determine $\rho(a) \longrightarrow$ geometry changes $a \longrightarrow$ geometry redshifts packet energies.

F0133 05_backreaction.tex:148 FAMILY_AUDITED

$$E_\alpha(a) = \frac{|k_\alpha|}{a}.$$

F0134 05_backreaction.tex:152 FAMILY_AUDITED

$$\rho_r(a) = \frac{R}{a^4}, \qquad p_r = \frac{\rho_r}{3}.$$

F0135 05_backreaction.tex:158 FAMILY_AUDITED

$$\dot{a} = \frac{1}{a},$$

F0136 05_backreaction.tex:162 FAMILY_AUDITED

$$\boxed{a(t) = \sqrt{1 + 2t}.}$$

F0137 05_backreaction.tex:169 FAMILY_AUDITED

$$E_\alpha = m_\alpha, \qquad p_m = 0, \qquad \rho_m(a) = \frac{M}{a^3}.$$

F0138 05_backreaction.tex:177 FAMILY_AUDITED

$$\dot{a} = a^{-1/2},$$

F0139 05_backreaction.tex:181 FAMILY_AUDITED

$$\boxed{a(t) = \Big(1 + \frac{3t}{2}\Big)^{2/3}.}$$

F0140 05_backreaction.tex:189 FAMILY_AUDITED

$$\rho(a) = \frac{0.9}{a^4} + \frac{0.1}{a^3}.$$

F0141 05_backreaction.tex:193 FAMILY_AUDITED

$$w_{\text{eff}}(a) = \frac{p}{\rho} = \frac{\frac{1}{3}\,0.9a^{-4}}{0.9a^{-4} + 0.1a^{-3}} = \frac{0.3}{0.9 + 0.1a}.$$

F0142 05_backreaction.tex:205 FAMILY_AUDITED

$$q = \frac{1}{2}(1 + 3w_{\mathrm{eff}}).$$

F0143 05_backreaction.tex:233 FAMILY_AUDITED

$$\boxed{1.99361 \text{ for radiation}, \qquad 1.99606 \text{ for matter}.}$$

F0144 05_backreaction.tex:268 FAMILY_AUDITED

$$\boxed{1.96526.}$$

F0145 05_backreaction.tex:290 FAMILY_AUDITED

$$\rho_A = \sum_{\alpha \in C_A} \rho_\alpha, \qquad p_A = \sum_{\alpha \in C_A} p_\alpha.$$

F0146 05_backreaction.tex:296 FAMILY_AUDITED

$$\sum_A \rho_A = \sum_\alpha \rho_\alpha, \qquad \sum_A p_A = \sum_\alpha p_\alpha.$$

F0147 05_backreaction.tex:304 FAMILY_AUDITED

$$\boxed{1.08 \times 10^{-19}}$$

F0148 05_backreaction.tex:321 FAMILY_AUDITED

$$\boxed{\{m_\alpha, k_\alpha, w_\alpha\} \longrightarrow \rho(a), p(a) \longrightarrow \dot{a} \longrightarrow E_\alpha(a) \longrightarrow \rho(a), p(a).}$$

06_variational_closure.tex

F0149 06_variational_closure.tex:15 FAMILY_AUDITED

$$\mathrm{d}s^2 = -N(t)^2\,\mathrm{d}t^2 + a(t)^2\,\mathrm{d}\boldsymbol{x}^2,$$

F0150 06_variational_closure.tex:19 FAMILY_AUDITED

$$H = \frac{1}{N}\frac{\dot{a}}{a}.$$

F0151 06_variational_closure.tex:23 FAMILY_AUDITED

$$C_G = \frac{3}{8\pi G}.$$

F0152 06_variational_closure.tex:27 FAMILY_AUDITED

$$L_G(a, \dot{a}, N) = -C_G \frac{a\dot{a}^2}{N}.$$

F0153 06_variational_closure.tex:35 FAMILY_AUDITED

$$\boxed{H_{\mathcal{P}}(a) = \sum_\alpha w_\alpha \sqrt{m_\alpha^2 + \frac{k_\alpha^2}{a^2}}.}$$

F0154 06_variational_closure.tex:44 FAMILY_AUDITED

$$L_M(a, N) = -N H_{\mathcal{P}}(a).$$

F0155 06_variational_closure.tex:48 FAMILY_AUDITED

$$\boxed{S[a,N]=\int\left[-C_G\frac{a\dot a^2}{N}-NH_{\mathcal P}(a)\right]\,\mathrm{d}t.}$$

F0156 06_variational_closure.tex:59 FAMILY_AUDITED

$$\rho=\frac{H_{\mathcal P}}{a^3},\qquad p=-\frac{H'_{\mathcal P}}{3a^2}.$$

F0157 06_variational_closure.tex:68 FAMILY_AUDITED

$$\frac{\partial L}{\partial N}=C_G\frac{a\dot a^2}{N^2}-H_{\mathcal P}(a)=0.$$

F0158 06_variational_closure.tex:73 FAMILY_AUDITED

$$\left(\frac{\dot a}{Na}\right)^2=\frac{H_{\mathcal P}}{C_Ga^3}.$$

F0159 06_variational_closure.tex:79 FAMILY_AUDITED

$$\boxed{H^2=\frac{8\pi G}{3}\rho.}$$

F0160 06_variational_closure.tex:89 FAMILY_AUDITED

$$\frac{\partial L}{\partial\dot a}=-2C_G\frac{a\dot a}{N}$$

F0161 06_variational_closure.tex:94 FAMILY_AUDITED

$$\frac{\partial L}{\partial a}=-C_G\frac{\dot a^2}{N}-NH'_{\mathcal P}(a).$$

F0162 06_variational_closure.tex:100 FAMILY_AUDITED

$$-2C_G\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{a\dot a}{N}\right)+C_G\frac{\dot a^2}{N}+NH'_{\mathcal P}(a)=0.$$

F0163 06_variational_closure.tex:106 FAMILY_AUDITED

$$2a\ddot a+\dot a^2-2a\dot a\frac{\dot N}{N}=\frac{N^2}{C_G}H'_{\mathcal P}(a).$$

F0164 06_variational_closure.tex:113 FAMILY_AUDITED

$$\boxed{2\frac{\ddot a}{a}+\left(\frac{\dot a}{a}\right)^2-2\frac{\dot a}{a}\frac{\dot N}{N}=-8\pi GN^2p.}$$

F0165 06_variational_closure.tex:123 FAMILY_AUDITED

$$\boxed{\frac{\ddot a}{a}=-\frac{4\pi G}{3}(\rho+3p).}$$

F0166 06_variational_closure.tex:135 FAMILY_AUDITED

$$\dot\rho+3H(\rho+p)=0.$$

F0167 06_variational_closure.tex:145 FAMILY_AUDITED

$$\bar a_n=\frac{a_n+a_{n+1}}{2},\qquad \Delta a_n=a_{n+1}-a_n,$$

F0168 06_variational_closure.tex:151 FAMILY_AUDITED

$$\boxed{L_d(a_n, a_{n+1}; N_n) = -C_G \frac{\bar{a}_n (\Delta a_n)^2}{\Delta t\, N_n} - \Delta t\, N_n H_{\mathcal{P}}(\bar{a}_n).}$$

F0169 06_variational_closure.tex:161 FAMILY_AUDITED

$$S_d = \sum_{n=0}^{M-1} L_d(a_n, a_{n+1}; N_n).$$

F0170 06_variational_closure.tex:168 FAMILY_AUDITED

$$C_G \frac{\bar{a}_n (\Delta a_n)^2}{\Delta t\, N_n^2} - \Delta t H_{\mathcal{P}}(\bar{a}_n) = 0.$$

F0171 06_variational_closure.tex:174 FAMILY_AUDITED

$$\boxed{C_G \bar{a}_n \left(\frac{\Delta a_n}{\Delta t N_n} \right)^2 = H_{\mathcal{P}}(\bar{a}_n).}$$

F0172 06_variational_closure.tex:186 FAMILY_AUDITED

$$\begin{aligned} D_1 L_d(a_n, a_{n+1}; N_n) = \frac{C_G}{\Delta t N_n} \left(2\bar{a}_n \Delta a_n - \frac{(\Delta a_n)^2}{2} \right) \\ - \frac{\Delta t N_n}{2} H'_{\mathcal{P}}(\bar{a}_n), \end{aligned}$$

F0173 06_variational_closure.tex:197 FAMILY_AUDITED

$$\begin{aligned} D_2 L_d(a_n, a_{n+1}; N_n) = -\frac{C_G}{\Delta t N_n} \left(2\bar{a}_n \Delta a_n + \frac{(\Delta a_n)^2}{2} \right) \\ - \frac{\Delta t N_n}{2} H'_{\mathcal{P}}(\bar{a}_n). \end{aligned}$$

F0174 06_variational_closure.tex:208 FAMILY_AUDITED

$$\boxed{D_2 L_d(a_{n-1}, a_n; N_{n-1}) + D_1 L_d(a_n, a_{n+1}; N_n) = 0.}$$

F0175 06_variational_closure.tex:233 FAMILY_AUDITED

$$H_{\mathcal{P}}(a) = \frac{R}{a}.$$

F0176 06_variational_closure.tex:243 FAMILY_AUDITED

$$H_{\mathcal{P}}(a) = M.$$

F0177 06_variational_closure.tex:247 FAMILY_AUDITED

$$\Delta t \in \{0.2, 0.1, 0.05, 0.025, 0.0125, 0.00625\}.$$

F0178 06_variational_closure.tex:272 FAMILY_AUDITED

$$\boxed{\text{1.92428 for the scale factor,} \qquad \text{1.92486 for the constraint residual.}}$$

F0179 06_variational_closure.tex:284 FAMILY_AUDITED

$$H_{\mathcal{P}}(a) = \frac{0.9}{a} + 0.1,$$

F0180 06_variational_closure.tex:288 FAMILY_AUDITED

$$a(50) = 11.338645\ldots.$$

F0181 06_variational_closure.tex:311 FAMILY_AUDITED

1.97042 for the scale factor, 1.97256 for the constraint residual.

F0182 06_variational_closure.tex:350 FAMILY_AUDITED

packet matter Hamiltonian $H_{\mathcal{P}}(a)$,
lapse-resolved geometric action,
Friedmann constraint from δN,
acceleration equation from δa,
continuity from packet energy dependence,
discrete dynamics from $\delta S_d = 0$.

07_holonomy_curvature.tex

F0183 07_holonomy_curvature.tex:15 FAMILY_AUDITED

$$\mathcal{H} = \bigoplus_{v \in V} \mathcal{H}_v, \qquad \sum_{v \in V} P_v = I.$$

F0184 07_holonomy_curvature.tex:22 FAMILY_AUDITED

$$T_e : \mathcal{H}_v \to \mathcal{H}_w.$$

F0185 07_holonomy_curvature.tex:26 FAMILY_AUDITED

$$T_{e^{-1}} = T_e^{-1}$$

F0186 07_holonomy_curvature.tex:30 FAMILY_AUDITED

$$\widetilde{T}_e = P_w U_e P_v.$$

F0187 07_holonomy_curvature.tex:39 FAMILY_AUDITED

$$\ell = (v_0 \to v_1 \to \cdots \to v_r = v_0),$$

F0188 07_holonomy_curvature.tex:43 FAMILY_AUDITED

$$\boxed{W_\ell = T_{v_{r-1}v_r} \cdots T_{v_1 v_2} T_{v_0 v_1}.}$$

F0189 07_holonomy_curvature.tex:52 FAMILY_AUDITED

$$T_{vw} \mapsto G_w T_{vw} G_v^{-1}.$$

F0190 07_holonomy_curvature.tex:56 FAMILY_AUDITED

$$W_\ell \mapsto G_{v_0} W_\ell G_{v_0}^{-1}.$$

F0191 07_holonomy_curvature.tex:64 FAMILY_AUDITED

$$U_x(\varepsilon) = \mathrm{e}^{\varepsilon A_x}, \qquad U_y(\varepsilon) = \mathrm{e}^{\varepsilon A_y}.$$

F0192 07_holonomy_curvature.tex:70 FAMILY_AUDITED

$$\boxed{W_\varepsilon = \mathrm{e}^{\varepsilon A_x} \mathrm{e}^{\varepsilon A_y} \mathrm{e}^{-\varepsilon A_x} \mathrm{e}^{-\varepsilon A_y}.}$$

F0193 07_holonomy_curvature.tex:81 FAMILY_AUDITED

$$F_{xy} = [A_x, A_y].$$

F0194 07_holonomy_curvature.tex:85 FAMILY_AUDITED

$$\boxed{F_\varepsilon = \frac{1}{\varepsilon^2} \log W_\varepsilon.}$$

F0195 07_holonomy_curvature.tex:93 FAMILY_AUDITED

$$\log W_\varepsilon = \varepsilon^2 [A_x, A_y] + \frac{\varepsilon^3}{2} \left([A_x, [A_x, A_y]] + [A_y, [A_x, A_y]] \right) + \mathcal{O}(\varepsilon^4).$$

F0196 07_holonomy_curvature.tex:105 FAMILY_AUDITED

$$F_\varepsilon = F_{xy} + C_1 \varepsilon + C_2 \varepsilon^2 + \cdots .$$

F0197 07_holonomy_curvature.tex:114 FAMILY_AUDITED

$$F_\varepsilon = F + C_1 \varepsilon + C_2 \varepsilon^2 + C_3 \varepsilon^3 + \cdots ,$$

F0198 07_holonomy_curvature.tex:118 FAMILY_AUDITED

$$\boxed{F_\varepsilon^{[1]} = 2F_{\varepsilon/2} - F_\varepsilon}$$

F0199 07_holonomy_curvature.tex:124 FAMILY_AUDITED

$$F_\varepsilon^{[1]} = F - \frac{1}{2} C_2 \varepsilon^2 - \frac{3}{4} C_3 \varepsilon^3 + \cdots .$$

F0200 07_holonomy_curvature.tex:131 FAMILY_AUDITED

$$F_{\varepsilon/2}^{[1]} = 2F_{\varepsilon/4} - F_{\varepsilon/2},$$

F0201 07_holonomy_curvature.tex:135 FAMILY_AUDITED

$$\boxed{F_\varepsilon^{[2]} = \frac{4F_{\varepsilon/2}^{[1]} - F_\varepsilon^{[1]}}{3}.}$$

F0202 07_holonomy_curvature.tex:143 FAMILY_AUDITED

$$F_\varepsilon^{[2]} = F + \widetilde{C}_3 \varepsilon^3 + \mathcal{O}(\varepsilon^4).$$

F0203 07_holonomy_curvature.tex:152 FAMILY_AUDITED

$$F = [A_x, A_y].$$

F0204 07_holonomy_curvature.tex:174 FAMILY_AUDITED

0.99283 raw,	1.99414 after one refinement,	2.99122 after two refinements.

F0205 07_holonomy_curvature.tex:194 FAMILY_AUDITED

$$A'_x = G A_x G^*, \qquad A'_y = G A_y G^*.$$

F0206 07_holonomy_curvature.tex:200 FAMILY_AUDITED

$$W'_\varepsilon = G W_\varepsilon G^*, \qquad F'_\varepsilon = G F_\varepsilon G^*.$$

F0207 07_holonomy_curvature.tex:208 FAMILY_AUDITED

locality,	gauge covariance,	systematic refinement of curvature.

F0208 07_holonomy_curvature.tex:221 FAMILY_AUDITED

$$S_{\mathrm{quad}} = \frac{1}{2g^2} \sum_{f \in F} A_f \ \mathrm{Tr}\left(F_f^* F_f\right),$$

F0209 07_holonomy_curvature.tex:230 FAMILY_AUDITED

$$F_f = \frac{1}{A_f} \log W_f.$$

F0210 07_holonomy_curvature.tex:236 FAMILY_AUDITED

$$\boxed{S_G[P,T,B] = \frac{1}{\kappa} \sum_{f \in F} \mathrm{Re}\,\mathrm{Tr}\left(B_f \log W_f\right).}$$

F0211 07_holonomy_curvature.tex:256 FAMILY_AUDITED

$$\boxed{S_G^{\mathrm{Regge}} = \frac{1}{8\pi G} \sum_h A_h \delta_h,}$$

F0212 07_holonomy_curvature.tex:269 FAMILY_AUDITED

$$S_M = \sum_v \langle \psi_v, M_v \psi_v \rangle + \sum_{(v,w) \in E} \mathrm{Re}\,\langle \psi_w, T_{vw} \psi_v \rangle .$$

F0213 07_holonomy_curvature.tex:280 FAMILY_AUDITED

$$\boxed{\frac{\delta S_G}{\delta T_e} + \frac{\delta S_M}{\delta T_e} = 0.}$$

F0214 07_holonomy_curvature.tex:293 FAMILY_AUDITED

$$S_G \longrightarrow -\frac{3V_0}{8\pi G} \int \frac{a \dot{a}^2}{N} \, \mathrm{d}t.$$

F0215 07_holonomy_curvature.tex:301 FAMILY_AUDITED

$$S_M \longrightarrow -\int N H_{\mathcal{P}}(a) \, \mathrm{d}t.$$

08__inhomogeneous__weak__field.tex

F0216 08_inhomogeneous_weak_field.tex:21 FAMILY_AUDITED

$$\mathrm{d}s^2 = -(1+2\Phi)\,\mathrm{d}t^2 + (1-2\Phi)\,\mathrm{d}\boldsymbol{x}^2, \qquad |\Phi| \ll 1.$$

F0217 08_inhomogeneous_weak_field.tex:29 FAMILY_AUDITED

eq:weak-poisson

$$\boxed{\nabla^2 \Phi = 4\pi G \rho.} \tag{A.1}$$

F0218 08_inhomogeneous_weak_field.tex:38 FAMILY_AUDITED

$$\Phi = \Phi(r), \qquad \rho = \rho(r),$$

F0219 08_inhomogeneous_weak_field.tex:44 FAMILY_AUDITED

eq:spherical-poisson

$$\boxed{\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}\Phi}{\mathrm{d}r}\right)=4\pi G\rho(r).} \tag{A.2}$$

F0220 08_inhomogeneous_weak_field.tex:53 CORRECTED_PASS

$$g_+(r)=\frac{\mathrm{d}\Phi}{\mathrm{d}r}\geq 0, \qquad a_r(r)=-g_+(r)=-\frac{\mathrm{d}\Phi}{\mathrm{d}r}.$$

F0221 08_inhomogeneous_weak_field.tex:63 FAMILY_AUDITED

$$P_iP_j=\delta_{ij}P_i, \qquad \sum_i P_i=I.$$

F0222 08_inhomogeneous_weak_field.tex:69 FAMILY_AUDITED

$$M_i=\mathrm{Tr}(P_i\widehat{\rho}P_i)=4\pi\int_{r_i}^{r_{i+1}} r^2\rho(r)\,\mathrm{d}r.$$

F0223 08_inhomogeneous_weak_field.tex:75 FAMILY_AUDITED

eq:packet-enclosed-mass

$$\boxed{M_{\mathrm{enc}}(r_j)=\sum_{i<j}M_i.} \tag{A.3}$$

F0224 08_inhomogeneous_weak_field.tex:83 FAMILY_AUDITED

$$M_a^{\mathrm{coarse}}=\sum_{i\in C_a}M_i^{\mathrm{fine}}.$$

F0225 08_inhomogeneous_weak_field.tex:90 CORRECTED_PASS

eq:gauss-field

$$\boxed{r^2g_+(r)=GM_{\mathrm{enc}}(r), \qquad a_r(r)=-\frac{GM_{\mathrm{enc}}(r)}{r^2}.} \tag{A.4}$$

F0226 08_inhomogeneous_weak_field.tex:99 FAMILY_AUDITED

$$\boxed{\text{refine the source then compute the field} = \text{compute the field from the coarse source.}}$$

F0227 08_inhomogeneous_weak_field.tex:111 FAMILY_AUDITED

$$u(r)=r\Phi(r).$$

F0228 08_inhomogeneous_weak_field.tex:115 FAMILY_AUDITED

eq:u-poisson

$$\boxed{u''(r)=4\pi Gr\rho(r).} \tag{A.5}$$

F0229 08_inhomogeneous_weak_field.tex:124 FAMILY_AUDITED

eq:weak-field-action

$$\boxed{S[u]=\int_0^R\left[\frac{1}{2G}\big(u'(r)\big)^2+4\pi r\rho(r)u(r)\right]\mathrm{d}r.} \tag{A.6}$$

F0230 08_inhomogeneous_weak_field.tex:136 FAMILY_AUDITED

$$\delta S=\int_0^R\left[-\frac{1}{G}u''(r)+4\pi r\rho(r)\right]\delta u(r)\,\mathrm{d}r,$$

F0231 08_inhomogeneous_weak_field.tex:150 FAMILY_AUDITED

$$r_i=ih, \qquad h=R/N, \qquad u_i\approx u(r_i).$$

F0232 08_inhomogeneous_weak_field.tex:158 FAMILY_AUDITED

eq:discrete-weak-action

$$\boxed{S_h[u] = \sum_{i=0}^{N-1} \frac{(u_{i+1}-u_i)^2}{2Gh} + 4\pi h \sum_{i=1}^{N-1} r_i \rho_i u_i.} \tag{A.7}$$

F0233 08_inhomogeneous_weak_field.tex:168 FAMILY_AUDITED

eq:discrete-u-poisson

$$\boxed{\frac{u_{i+1}-2u_i+u_{i-1}}{h^2} = 4\pi G r_i \rho_i.} \tag{A.8}$$

F0234 08_inhomogeneous_weak_field.tex:178 FAMILY_AUDITED

$$\Phi_i = \frac{u_i}{r_i}, \qquad \Phi_0 = u'(0),$$

F0235 08_inhomogeneous_weak_field.tex:184 FAMILY_AUDITED

eq:discrete-acceleration

$$\boxed{g_i = \frac{r_i u_i' - u_i}{r_i^2}.} \tag{A.9}$$

F0236 08_inhomogeneous_weak_field.tex:196 FAMILY_AUDITED

eq:gaussian-density

$$\rho(r) = \frac{M}{(2\pi)^{3/2}\sigma^3} \exp\left(-\frac{r^2}{2\sigma^2}\right). \tag{A.10}$$

F0237 08_inhomogeneous_weak_field.tex:203 FAMILY_AUDITED

eq:gaussian-enclosed

$$M_{\mathrm{enc}}(r) = M\left[\operatorname{erf}\left(\frac{r}{\sqrt{2}\sigma}\right) - \sqrt{\frac{2}{\pi}}\frac{r}{\sigma}\exp\left(-\frac{r^2}{2\sigma^2}\right)\right]. \tag{A.11}$$

F0238 08_inhomogeneous_weak_field.tex:213 FAMILY_AUDITED

eq:gaussian-potential

$$\boxed{\Phi(r) = -\frac{GM}{r}\operatorname{erf}\left(\frac{r}{\sqrt{2}\sigma}\right),} \tag{A.12}$$

F0239 08_inhomogeneous_weak_field.tex:222 FAMILY_AUDITED

$$\Phi(0) = -\frac{GM}{\sigma}\sqrt{\frac{2}{\pi}}.$$

F0240 08_inhomogeneous_weak_field.tex:227 CORRECTED_PASS

eq:gaussian-acceleration

$$\boxed{g_+(r) = \frac{GM_{\mathrm{enc}}(r)}{r^2}, \qquad a_r(r) = -\frac{GM_{\mathrm{enc}}(r)}{r^2}.} \tag{A.13}$$

F0241 08_inhomogeneous_weak_field.tex:236 FAMILY_AUDITED

$$G = M = 1, \qquad \sigma = 0.6, \qquad R = 6.$$

F0242 08_inhomogeneous_weak_field.tex:265 FAMILY_AUDITED

eq:weak-orders

$$\boxed{\begin{aligned} p_\Phi &= 1.99521, \quad p_{g,\mathrm{RMS}} = 2.01134, \\ p_{\mathrm{Gauss}} &= 1.99912. \end{aligned}} \tag{A.14}$$

F0243 08_inhomogeneous_weak_field.tex:303 FAMILY_AUDITED

$$\rho_A(r) = C_A\left(1-\frac{r^2}{R_s^2}\right), \qquad \rho_B(r) = C_B\left(1-\frac{r^2}{R_s^2}\right)^4,$$

F0244 08_inhomogeneous_weak_field.tex:309 CORRECTED_PASS

$$\Phi(r) = -\frac{GM}{r}, \qquad g_+(r) = \frac{GM}{r^2}, \qquad a_r(r) = -\frac{GM}{r^2}.$$

F0245 08_inhomogeneous_weak_field.tex:333 FAMILY_AUDITED

$$\boxed{1.99485 \text{ for exterior-potential universality}, \qquad 1.99318 \text{ for exterior-acceleration universality.}}$$

F0246 08_inhomogeneous_weak_field.tex:354 FAMILY_AUDITED

$$\begin{aligned} |M_{\mathrm{fine}} - M_{\mathrm{coarse}}| &= 1.11 \times 10^{-16}, \\ \max_a |M_a^{\mathrm{coarse}} - \sum_{i \in C_a} M_i^{\mathrm{fine}}| &= 0, \\ \max_j |M_{\mathrm{enc}}^{\mathrm{coarse}}(r_j) - M_{\mathrm{enc}}^{\mathrm{fine}}(r_j)| &= 0, \\ \max_j |g_{\mathrm{coarse}}(r_j) - g_{\mathrm{fine}}(r_j)| &= 0. \end{aligned}$$

F0247 08_inhomogeneous_weak_field.tex:373 FAMILY_AUDITED

eq:cosmological-poisson

$$\nabla_\chi^2 \Phi = 4\pi G a^2 \delta\rho, \tag{A.15}$$

F0248 08_inhomogeneous_weak_field.tex:392 FAMILY_AUDITED

$$R_{0i0j} \simeq \partial_i \partial_j \Phi.$$

F0249 08_inhomogeneous_weak_field.tex:398 FAMILY_AUDITED

$$\boxed{\text{packet stress-energy} \xrightarrow{\text{variation}} \text{local geometric variables} \xrightarrow{\text{loop transport}} \text{holonomy curvature.}}$$

08_interpretation_predictions.tex

F0250 08_interpretation_predictions.tex:26 OPEN_NOT_VERIFIED

$$D(t) = a(t)\chi.$$

F0251 08_interpretation_predictions.tex:31 OPEN_NOT_VERIFIED

$$\omega = \frac{k}{a}.$$

F0252 08_interpretation_predictions.tex:64 OPEN_NOT_VERIFIED

$$|\boldsymbol{k}|\Delta \ll 1, \qquad m\Delta \ll 1.$$

F0253 08_interpretation_predictions.tex:70 OPEN_NOT_VERIFIED

$$\boxed{\omega_\Delta = E - \frac{\Delta^2}{6E}\left[m^2|\boldsymbol{k}|^2 + k_x^2 k_y^2 + k_x^2 k_z^2 + k_y^2 k_z^2\right] + \mathcal{O}(\Delta^4),}$$

F0254 08_interpretation_predictions.tex:85 OPEN_NOT_VERIFIED

$$\omega_{\mathrm{axis}} - \omega_{\mathrm{body}} = \frac{\Delta^2 |\boldsymbol{k}|^4}{18E} + \mathcal{O}(\Delta^4).$$

F0255 08_interpretation_predictions.tex:96 OPEN_NOT_VERIFIED

$$\boxed{\mathfrak{L}_\Delta(\boldsymbol{k}, \boldsymbol{v}) = \frac{\Delta^2}{6E'}\left[Q(\boldsymbol{k}') - Q(\boldsymbol{k})\right] + \mathcal{O}(\Delta^4),}$$

F0256 08_interpretation_predictions.tex:105 OPEN_NOT_VERIFIED

$$Q(\boldsymbol{k}) = m^2|\boldsymbol{k}|^2 + k_x^2 k_y^2 + k_x^2 k_z^2 + k_y^2 k_z^2.$$

F0257 08_interpretation_predictions.tex:146 OPEN_NOT_VERIFIED

$$E^2 = p^2 + m^2 + \alpha_1 \Delta p^3 + \alpha_2 \Delta^2 p^4 + \cdots ,$$

F0258 08_interpretation_predictions.tex:166 OPEN_NOT_VERIFIED

$$H_\Lambda(a) = \rho_\Lambda a^3.$$

F0259 08_interpretation_predictions.tex:170 OPEN_NOT_VERIFIED

$$p_\Lambda = -\frac{H'_\Lambda}{3a^2} = -\rho_\Lambda.$$

F0260 08_interpretation_predictions.tex:176 OPEN_NOT_VERIFIED

$$H^2 = \frac{8\pi G}{3}\left(\frac{R}{a^4} + \frac{M}{a^3} + \rho_\Lambda\right).$$

F0261 08_interpretation_predictions.tex:215 OPEN_NOT_VERIFIED

discrete:	resolved local sectors and finite transfers,
continuous:	the reconstructed large-scale geometry and field envelope.

09_research_program.tex

F0262 09_research_program.tex:35 OPEN_NOT_VERIFIED

$$\mathfrak{L}_\Delta = \frac{\Delta^2}{6E'}\left[Q(\boldsymbol{k}') - Q(\boldsymbol{k})\right] + \mathcal{O}(\Delta^4).$$

F0263 09_research_program.tex:50 OPEN_NOT_VERIFIED

$$\log W_\square = K \int_\square (\phi_{tt} - \phi_{xx})\,\mathrm{d}t\,\mathrm{d}x.$$

F0264 09_research_program.tex:59 OPEN_NOT_VERIFIED

$$\dot{e}_j + F_{j+1/2} - F_{j-1/2} = s_j$$

F0265 09_research_program.tex:67 OPEN_NOT_VERIFIED

$$\mathcal{P}_r, \quad T_e^{(r)}, \quad B_f^{(r)}, \quad N_v^{(r)}, \quad N_e^{i,(r)}, \quad \psi_v^{(r)}$$

F0266 09_research_program.tex:76 OPEN_NOT_VERIFIED

eq:updated-closure-action

$$\boxed{S_r = S_G[\mathcal{P}_r, T^{(r)}, B^{(r)}, N^{(r)}, N^{i,(r)}] + S_M[\mathcal{P}_r, T^{(r)}, B^{(r)}, N^{(r)}, N^{i,(r)}, \psi^{(r)}].} \tag{A.16}$$

F0267 09_research_program.tex:100 OPEN_NOT_VERIFIED

Hamiltonian constraint, momentum constraint, constraint propagation, energy flux, refinement order.

F0268 09_research_program.tex:143 OPEN_NOT_VERIFIED

$$\mathcal{R}_{r+1\to r} : S_{r+1} \mapsto S_r$$

F0269 09_research_program.tex:147 OPEN_NOT_VERIFIED

fine evolve then coarsen, coarse evolve, renormalize then evolve.

F0270 09_research_program.tex:178 OPEN_NOT_VERIFIED

$$\delta\omega(\boldsymbol{k}) = -\frac{\Delta^2 Q(\boldsymbol{k})}{6E(\boldsymbol{k})} + \mathcal{O}(\Delta^4),$$

$$\mathfrak{L}_\Delta(\boldsymbol{k},\boldsymbol{v}) = \frac{\Delta^2}{6E'}[Q(\boldsymbol{k}') - Q(\boldsymbol{k})] + \mathcal{O}(\Delta^4).$$

F0271 09_research_program.tex:226 OPEN_NOT_VERIFIED

atlas → corners → transfer → connection → refinement,
relativistic shell → observer defect → gauge links → dynamical fields,
local matter work → metric curvature ↔ holonomy curvature.

09_weak_field_optics.tex

F0272 09_weak_field_optics.tex:9 FAMILY_AUDITED

clock rate, gravitational redshift, Shapiro delay, light deflection.

F0273 09_weak_field_optics.tex:22 FAMILY_AUDITED

eq:weak-optical-metric

$$\boxed{\mathrm{d}s^2 = -\left(1 + \frac{2\Phi}{c^2}\right) c^2\,\mathrm{d}t^2 + \left(1 - \frac{2\Phi}{c^2}\right)\,\mathrm{d}\boldsymbol{x}^2 + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right).} \tag{A.17}$$

F0274 09_weak_field_optics.tex:32 FAMILY_AUDITED

eq:weak-lapse-optical-factor

$$N(\boldsymbol{x}) = \sqrt{1 + \frac{2\Phi(\boldsymbol{x})}{c^2}}, \qquad A(\boldsymbol{x}) = \sqrt{1 - \frac{2\Phi(\boldsymbol{x})}{c^2}}. \tag{A.18}$$

F0275 09_weak_field_optics.tex:41 FAMILY_AUDITED

$$\mathrm{d}s^2 = -N^2c^2\,\mathrm{d}t^2 + A^2\,\mathrm{d}\boldsymbol{x}^2.$$

F0276 09_weak_field_optics.tex:45 FAMILY_AUDITED

eq:local-clock-rate

$$\boxed{\mathrm{d}\tau = N(\boldsymbol{x})\,\mathrm{d}t.} \tag{A.19}$$

F0277 09_weak_field_optics.tex:54 FAMILY_AUDITED

$$N_\alpha = N(\boldsymbol{x}_\alpha).$$

F0278 09_weak_field_optics.tex:66 FAMILY_AUDITED

$$\mathcal{E} = -p_\mu \xi^\mu$$

F0279 09_weak_field_optics.tex:70 FAMILY_AUDITED

$$u^\mu = \frac{\xi^\mu}{N}.$$

F0280 09_weak_field_optics.tex:74 FAMILY_AUDITED

$$\omega = -p_\mu u^\mu = \frac{\mathcal{E}}{N}.$$

F0281 09_weak_field_optics.tex:80 FAMILY_AUDITED

eq:projector-redshift

$$\boxed{\frac{\omega_o}{\omega_e} = \frac{N_e}{N_o} = \sqrt{\frac{1 + 2\Phi_e/c^2}{1 + 2\Phi_o/c^2}}.} \tag{A.20}$$

F0282 09_weak_field_optics.tex:92 FAMILY_AUDITED

eq:redshift-first-order

$$\frac{\omega_o}{\omega_e} = 1 + \frac{\Phi_e - \Phi_o}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right). \tag{A.21}$$

F0283 09_weak_field_optics.tex:101 FAMILY_AUDITED

$$\theta_\alpha(t) = \Omega_0 \tau_\alpha(t) = \Omega_0 N_\alpha t.$$

F0284 09_weak_field_optics.tex:124 FAMILY_AUDITED

eq:weak-null-time

$$\boxed{\mathrm{d}t = \frac{1}{c}\sqrt{\frac{1-2\Phi/c^2}{1+2\Phi/c^2}}\,\mathrm{d}\ell.} \tag{A.22}$$

F0285 09_weak_field_optics.tex:137 FAMILY_AUDITED

eq:optical-index

$$\boxed{n_{\mathrm{opt}}(\boldsymbol{x}) = \sqrt{\frac{1-2\Phi(\boldsymbol{x})/c^2}{1+2\Phi(\boldsymbol{x})/c^2}} = 1 - \frac{2\Phi(\boldsymbol{x})}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right).} \tag{A.23}$$

F0286 09_weak_field_optics.tex:156 FAMILY_AUDITED

$$r(x) = \sqrt{x^2 + b^2}.$$

F0287 09_weak_field_optics.tex:160 FAMILY_AUDITED

eq:shapiro-full-weak

$$\boxed{\Delta t_{\mathrm{Shapiro}} = \frac{1}{c}\int_{-L}^{L}\left[\sqrt{\frac{1-2\Phi(r)/c^2}{1+2\Phi(r)/c^2}} - 1\right]\mathrm{d}x.} \tag{A.24}$$

F0288 09_weak_field_optics.tex:176 FAMILY_AUDITED

eq:shapiro-first-order

$$\boxed{\Delta t_{\mathrm{Shapiro}} = -\frac{2}{c^3}\int_{-L}^{L}\Phi(r(x))\,\mathrm{d}x + \mathcal{O}\left(\frac{\Phi^2}{c^5}\right).} \tag{A.25}$$

F0289 09_weak_field_optics.tex:186 FAMILY_AUDITED

$$\Phi(r) = -\frac{GM}{r},$$

F0290 09_weak_field_optics.tex:190 FAMILY_AUDITED

eq:shapiro-point-finite

$$\boxed{\Delta t^{(1)}_{\mathrm{Shapiro}} = \frac{4GM}{c^3}\,\mathrm{arsinh}\left(\frac{L}{b}\right).} \tag{A.26}$$

F0291 09_weak_field_optics.tex:199 FAMILY_AUDITED

$$\Delta t^{(1)}_{\mathrm{Shapiro}} \sim \frac{4GM}{c^3}\log\left(\frac{2L}{b}\right).$$

F0292 09_weak_field_optics.tex:216 CORRECTED_PASS

eq:weak-deflection-integral

$$\boxed{\boldsymbol{\alpha}_\perp = -\frac{2}{c^2}\int_{-L}^{L}\nabla_\perp\Phi\;\mathrm{d}x, \qquad \alpha := \|\boldsymbol{\alpha}_\perp\| = \frac{2}{c^2}\int_{-L}^{L}\frac{\partial\Phi}{\partial b}\;\mathrm{d}x.} \tag{A.27}$$

F0293 09_weak_field_optics.tex:231 FAMILY_AUDITED

$$\frac{\partial\Phi}{\partial b} = \frac{GMb}{(x^2+b^2)^{3/2}},$$

F0294 09_weak_field_optics.tex:236 FAMILY_AUDITED

eq:finite-path-deflection

$$\boxed{\alpha(L,b) = \frac{4GML}{c^2 b\sqrt{L^2+b^2}}.} \tag{A.28}$$

F0295 09_weak_field_optics.tex:244 FAMILY_AUDITED

eq:einstein-deflection

$$\boxed{\alpha_\infty = \frac{4GM}{c^2 b}.} \tag{A.29}$$

F0296 09_weak_field_optics.tex:256 FAMILY_AUDITED

$$\rho_A(r) = C_A\left(1 - \frac{r^2}{R_s^2}\right), \qquad \rho_B(r) = C_B\left(1 - \frac{r^2}{R_s^2}\right)^4,$$

F0297 09_weak_field_optics.tex:262 FAMILY_AUDITED

$$R_s = 0.625.$$

F0298 09_weak_field_optics.tex:266 FAMILY_AUDITED

$$c = 1, \qquad GM = 0.005, \qquad R_{\max} = 10.$$

F0299 09_weak_field_optics.tex:274 FAMILY_AUDITED

$$r_e = 1.2, \qquad r_o = 8,$$

F0300 09_weak_field_optics.tex:280 FAMILY_AUDITED

$$b = 1, \qquad L = 8.$$

F0301 09_weak_field_optics.tex:290 FAMILY_AUDITED

eq:optics-reference-redshift eq:optics-reference-delay eq:optics-reference-deflection

$$\frac{\omega_o}{\omega_e} = 0.996447590904\ldots, \tag{A.30}$$

$$\Delta t_{\mathrm{Shapiro}} = 0.055675087962\ldots, \tag{A.31}$$

$$\alpha = 0.019845557534\ldots. \tag{A.32}$$

F0302 09_weak_field_optics.tex:299 FAMILY_AUDITED

$$0.055529445614\ldots.$$

F0303 09_weak_field_optics.tex:324 FAMILY_AUDITED

eq:weak-optics-orders

$$\boxed{\begin{aligned} p_{\mathrm{redshift}} &= 1.998997, \\ p_{\mathrm{delay}} &= 1.998985, \\ p_{\mathrm{deflection}} &= 1.999032. \end{aligned}} \tag{A.33}$$

F0304 09_weak_field_optics.tex:365 FAMILY_AUDITED

eq:weak-optics-universality-orders

$$\boxed{1.999372, \qquad 1.999371, \qquad 1.999379} \tag{A.34}$$

F0305 09_weak_field_optics.tex:382 FAMILY_AUDITED

eq:source-to-optics-chain

$$\boxed{\text{projector source} \longrightarrow \Phi \longrightarrow (N, A) \longrightarrow \begin{cases} \text{clock rate}, \\ \text{redshift}, \\ \text{time delay}, \\ \text{deflection}. \end{cases}} \tag{A.35}$$

10_linearized_tensor_gravity.tex

F0306 10_linearized_tensor_gravity.tex:11 FAMILY_AUDITED

lapse and shift constraints, two tensor polarizations, a local discrete action,
a finite causal cone, an exact discrete energy law, source-work balance,
curvature reconstruction, geodesic deviation, refinement convergence.

F0307 10_linearized_tensor_gravity.tex:29 FAMILY_AUDITED

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad |h_{\mu\nu}| \ll 1,$$

F0308 10_linearized_tensor_gravity.tex:35 FAMILY_AUDITED

$$h = \eta^{\mu\nu} h_{\mu\nu}.$$

F0309 10_linearized_tensor_gravity.tex:39 CORRECTED_PASS

eq:fierz-pauli-action

$$\boxed{S_{\mathrm{FP}}[h] = \frac{1}{64\pi G} \int \mathrm{d}^4x \Big[- \partial_\lambda h_{\mu\nu} \partial^\lambda h^{\mu\nu} + 2\partial_\mu h^{\mu\nu} \partial^\lambda h_{\lambda\nu} - 2\partial_\mu h^{\mu\nu} \partial_\nu h + \partial_\lambda h \partial^\lambda h \Big] + \frac{1}{2} \int h_{\mu\nu} T^{\mu\nu} \, \mathrm{d}^4x.} \tag{A.36}$$

F0310 10_linearized_tensor_gravity.tex:57 FAMILY_AUDITED

eq:linearized-einstein

$$\boxed{G^{(1)}_{\mu\nu} = 8\pi G T_{\mu\nu}.} \tag{A.37}$$

F0311 10_linearized_tensor_gravity.tex:64 FAMILY_AUDITED

eq:linearized-gauge

$$h_{\mu\nu} \mapsto h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu, \tag{A.38}$$

F0312 10_linearized_tensor_gravity.tex:73 FAMILY_AUDITED

$$\partial_\mu T^{\mu\nu} = 0.$$

F0313 10_linearized_tensor_gravity.tex:83 FAMILY_AUDITED

eq:linearized-adm-decomposition

$$h_{00} = -2n, \qquad h_{0i} = n_i, \qquad h_{ij} = \gamma_{ij}. \tag{A.39}$$

F0314 10_linearized_tensor_gravity.tex:94 FAMILY_AUDITED

eq:linearized-hamiltonian-constraint

$$\boxed{\partial_i \partial_j \gamma_{ij} - \nabla^2 \gamma = 16\pi G T_{00},} \tag{A.40}$$

F0315 10_linearized_tensor_gravity.tex:103 FAMILY_AUDITED

$$\gamma = \delta^{ij} \gamma_{ij}.$$

F0316 10_linearized_tensor_gravity.tex:109 FAMILY_AUDITED

eq:linearized-momentum-constraint

$$\boxed{\partial_j \dot{\gamma}_{ij} - \partial_i \dot{\gamma} + \partial_i \partial_j n_j - \nabla^2 n_i = 16\pi G T_{0i}.} \tag{A.41}$$

F0317 10_linearized_tensor_gravity.tex:126 FAMILY_AUDITED

eq:transverse-projector

$$\Pi_{ij}(\boldsymbol{k}) = \delta_{ij} - \frac{k_i k_j}{|\boldsymbol{k}|^2}. \tag{A.42}$$

F0318 10_linearized_tensor_gravity.tex:132 FAMILY_AUDITED

eq:tt-projector

$$\boxed{\Lambda^{\mathrm{TT}}_{ij,kl} = \frac{1}{2}\left(\Pi_{ik}\Pi_{jl} + \Pi_{il}\Pi_{jk} - \Pi_{ij}\Pi_{kl}\right).} \tag{A.43}$$

F0319 10_linearized_tensor_gravity.tex:145 FAMILY_AUDITED

eq:tt-projection

$$\gamma^{\mathrm{TT}}_{ij} = \Lambda^{\mathrm{TT}}_{ij,kl}\gamma_{kl} \tag{A.44}$$

F0320 10_linearized_tensor_gravity.tex:151 FAMILY_AUDITED

eq:tt-conditions

$$k_i\gamma^{\mathrm{TT}}_{ij} = 0, \qquad \delta_{ij}\gamma^{\mathrm{TT}}_{ij} = 0. \tag{A.45}$$

F0321 10_linearized_tensor_gravity.tex:158 FAMILY_AUDITED

eq:tt-idempotent

$$\Lambda^{\mathrm{TT}}\Lambda^{\mathrm{TT}} = \Lambda^{\mathrm{TT}}. \tag{A.46}$$

F0322 10_linearized_tensor_gravity.tex:170 FAMILY_AUDITED

eq:plane-tt-metric

$$\boxed{\gamma^{\mathrm{TT}}_{ij}(t,x) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & h_+ & h_\times \\ 0 & h_\times & -h_+ \end{pmatrix}.} \tag{A.47}$$

F0323 10_linearized_tensor_gravity.tex:183 FAMILY_AUDITED

$$h_+(t,x), \qquad h_\times(t,x)$$

F0324 10_linearized_tensor_gravity.tex:191 FAMILY_AUDITED

eq:tt-reduced-action

$$\boxed{S_{\mathrm{TT}} = \frac{1}{32\pi G}\sum_{A\in\{+,\times\}}\int \mathrm{d}t\,\mathrm{d}x\left[(\partial_t h_A)^2 - c^2(\partial_x h_A)^2\right].} \tag{A.48}$$

F0325 10_linearized_tensor_gravity.tex:205 FAMILY_AUDITED

eq:tt-wave-equation

$$\boxed{\partial_t^2 h_A - c^2\partial_x^2 h_A = 0.} \tag{A.49}$$

F0326 10_linearized_tensor_gravity.tex:212 FAMILY_AUDITED

eq:tt-sourced-wave

$$\partial_t^2 h_A - c^2\partial_x^2 h_A = S_A. \tag{A.50}$$

F0327 10_linearized_tensor_gravity.tex:222 FAMILY_AUDITED

$$x_j = jh, \qquad t_n = n\Delta t, \qquad h^n_{A,j} \approx h_A(t_n, x_j).$$

F0328 10_linearized_tensor_gravity.tex:230 FAMILY_AUDITED

eq:discrete-tt-action

$$\boxed{S_d = \frac{h\Delta t}{32\pi G}\sum_{A,n,j}\left[\left(\frac{h^{n+1}_{A,j} - h^n_{A,j}}{\Delta t}\right)^2 - c^2\left(\frac{h^n_{A,j+1} - h^n_{A,j}}{h}\right)^2\right].} \tag{A.51}$$

F0329 10_linearized_tensor_gravity.tex:248 FAMILY_AUDITED

eq:discrete-tt-wave

$$\boxed{h^{n+1}_{A,j} - 2h^n_{A,j} + h^{n-1}_{A,j} = \lambda^2\left(h^n_{A,j+1} - 2h^n_{A,j} + h^n_{A,j-1}\right),} \tag{A.52}$$

F0330 10_linearized_tensor_gravity.tex:263 FAMILY_AUDITED

eq:courant-number

$$\lambda = \frac{c\Delta t}{h} \tag{A.53}$$

F0331 10_linearized_tensor_gravity.tex:274 FAMILY_AUDITED

$$h^n_{A,j} = H_A \exp\left[\mathrm{i}(kjh - \omega n \Delta t)\right]$$

F0332 10_linearized_tensor_gravity.tex:279 FAMILY_AUDITED

eq:tensor-lattice-dispersion

$$\boxed{\sin^2\left(\frac{\omega\Delta t}{2}\right) = \lambda^2 \sin^2\left(\frac{kh}{2}\right).} \tag{A.54}$$

F0333 10_linearized_tensor_gravity.tex:288 FAMILY_AUDITED

eq:tensor-dispersion-expansion

$$\omega = c|k| + \mathcal{O}(h^2). \tag{A.55}$$

F0334 10_linearized_tensor_gravity.tex:296 FAMILY_AUDITED

$$\lambda = 0.8, \qquad k = 0.7,$$

F0335 10_linearized_tensor_gravity.tex:302 FAMILY_AUDITED

$$h \in \{0.8, 0.4, 0.2, 0.1, 0.05, 0.025\}.$$

F0336 10_linearized_tensor_gravity.tex:325 FAMILY_AUDITED

eq:tensor-dispersion-orders

$$\boxed{\text{2.004464 for frequency and phase speed,} \qquad \text{2.007435 for group speed.}} \tag{A.56}$$

F0337 10_linearized_tensor_gravity.tex:346 FAMILY_AUDITED

$$\lambda = 1, \qquad c\Delta t = h.$$

F0338 10_linearized_tensor_gravity.tex:352 FAMILY_AUDITED

eq:critical-wave-equation

$$h^{n+1}_j + h^{n-1}_j = h^n_{j+1} + h^n_{j-1}. \tag{A.57}$$

F0339 10_linearized_tensor_gravity.tex:360 CONDITIONAL_CORRECTED

eq:lattice-dalembert

$$\boxed{h^n_j = F_{j-n} + G_{j+n}} \tag{A.58}$$

F0340 10_linearized_tensor_gravity.tex:370 FAMILY_AUDITED

$$F_{j-n} \quad \text{moves right by one cell per tick,}$$

F0341 10_linearized_tensor_gravity.tex:374 FAMILY_AUDITED

$$G_{j+n} \quad \text{moves left by one cell per tick.}$$

F0342 10_linearized_tensor_gravity.tex:379 FAMILY_AUDITED

$$\frac{h}{\Delta t} = c.$$

F0343 10_linearized_tensor_gravity.tex:383 FAMILY_AUDITED

$$\boxed{\begin{array}{c}\text{locally discrete geometry} + \text{ exact packet reconstruction} \\ = \text{ relativistic tensor propagation.}\end{array}}$$

F0344 10_linearized_tensor_gravity.tex:394 FAMILY_AUDITED

$$h = \Delta t = 0.025$$

F0345 10_linearized_tensor_gravity.tex:407 FAMILY_AUDITED

eq:tensor-translation-errors

$$\boxed{1.46 \times 10^{-17} \text{ for } h_+, \qquad 1.14 \times 10^{-17} \text{ for } h_\times.} \tag{A.59}$$

F0346 10_linearized_tensor_gravity.tex:422 FAMILY_AUDITED

eq:tensor-staggered-variables

$$v_{A,j}^{n+1/2} = \frac{h_{A,j}^{n+1} - h_{A,j}^{n}}{\Delta t}, \qquad d_{A,j}^{n} = \frac{h_{A,j+1}^{n} - h_{A,j}^{n}}{h}. \tag{A.60}$$

F0347 10_linearized_tensor_gravity.tex:431 FAMILY_AUDITED

eq:discrete-tensor-energy

$$\boxed{\mathcal{E}^{n+1/2} = \frac{h}{2} \sum_{A,j} \left[\left(v_{A,j}^{n+1/2} \right)^2 + c^2 d_{A,j}^{n+1} d_{A,j}^{n} \right].} \tag{A.61}$$

F0348 10_linearized_tensor_gravity.tex:447 FAMILY_AUDITED

eq:discrete-tensor-energy-conservation

$$\boxed{\mathcal{E}^{n+1/2} = \mathcal{E}^{n-1/2}.} \tag{A.62}$$

F0349 10_linearized_tensor_gravity.tex:457 FAMILY_AUDITED

eq:tensor-energy-drift

$$\boxed{1.60 \times 10^{-16}.} \tag{A.63}$$

F0350 10_linearized_tensor_gravity.tex:472 FAMILY_AUDITED

$$1.057332726873 \times 10^{-5},$$

F0351 10_linearized_tensor_gravity.tex:476 FAMILY_AUDITED

eq:tensor-energy-refinement

$$\boxed{3.39 \times 10^{-21}.} \tag{A.64}$$

F0352 10_linearized_tensor_gravity.tex:487 FAMILY_AUDITED

eq:sourced-discrete-tensor

$$\frac{h_{A,j}^{n+1} - 2h_{A,j}^{n} + h_{A,j}^{n-1}}{\Delta t^2} - c^2 \frac{h_{A,j+1}^{n} - 2h_{A,j}^{n} + h_{A,j-1}^{n}}{h^2} = S_{A,j}^{n}. \tag{A.65}$$

F0353 10_linearized_tensor_gravity.tex:494 FAMILY_AUDITED

eq:tensor-source-work

$$\boxed{\mathcal{E}^{n+1/2} - \mathcal{E}^{n-1/2} = \frac{h}{2} \sum_{A,j} S_{A,j}^{n} \left(h_{A,j}^{n+1} - h_{A,j}^{n-1} \right).} \tag{A.66}$$

F0354 10_linearized_tensor_gravity.tex:506 FAMILY_AUDITED

$$0.5 \le t \le 2.0$$

F0355 10_linearized_tensor_gravity.tex:519 FAMILY_AUDITED

$$8.185434325005 \times 10^{-9},$$

F0356 10_linearized_tensor_gravity.tex:523 FAMILY_AUDITED

$$8.185434325005 \times 10^{-9}.$$

F0357 10_linearized_tensor_gravity.tex:527 FAMILY_AUDITED

eq:tensor-source-work-residual

$$\boxed{2.98 \times 10^{-23},} \tag{A.67}$$

F0358 10_linearized_tensor_gravity.tex:534 FAMILY_AUDITED

$$3.64 \times 10^{-15}.$$

F0359 10_linearized_tensor_gravity.tex:550 FAMILY_AUDITED

$$12.25.$$

F0360 10_linearized_tensor_gravity.tex:554 FAMILY_AUDITED

eq:tensor-causal-leakage

$$\boxed{0.} \tag{A.68}$$

F0361 10_linearized_tensor_gravity.tex:565 FAMILY_AUDITED

$$\gamma = 0, \qquad \partial_i \gamma_{ij} = 0, \qquad n = 0, \qquad n_i = 0.$$

F0362 10_linearized_tensor_gravity.tex:577 FAMILY_AUDITED

eq:tensor-constraint-zero

$$\boxed{0} \tag{A.69}$$

F0363 10_linearized_tensor_gravity.tex:588 FAMILY_AUDITED

eq:tt-tidal-curvature

$$\boxed{R^{(1)}_{0i0j} = -\frac{1}{2}\partial_t^2 \gamma_{ij}^{\mathrm{TT}}.} \tag{A.70}$$

F0364 10_linearized_tensor_gravity.tex:596 FAMILY_AUDITED

eq:tt-curvature-space-time

$$R^{(1)}_{0i0j} = -\frac{c^2}{2}\partial_x^2 \gamma_{ij}^{\mathrm{TT}}. \tag{A.71}$$

F0365 10_linearized_tensor_gravity.tex:604 FAMILY_AUDITED

$$-\frac{1}{2}\delta_t^2 h_A \quad \text{with} \quad -\frac{c^2}{2}\delta_x^2 h_A.$$

F0366 10_linearized_tensor_gravity.tex:610 FAMILY_AUDITED

$$3.47 \times 10^{-16},$$

F0367 10_linearized_tensor_gravity.tex:614 FAMILY_AUDITED

eq:tensor-curvature-residual

$$\boxed{1.73 \times 10^{-16}.} \tag{A.72}$$

F0368 10_linearized_tensor_gravity.tex:626 FAMILY_AUDITED

eq:geodesic-deviation

$$\frac{\mathrm{d}^2 X^i}{\mathrm{d}t^2} = -R^i{}_{0j0} X^j. \tag{A.73}$$

F0369 10_linearized_tensor_gravity.tex:632 FAMILY_AUDITED

eq:ring-displacement

$$\boxed{\delta X^i = \frac{1}{2}\gamma^i{}_j{}^{\mathrm{TT}} X^j.} \tag{A.74}$$

F0370 10_linearized_tensor_gravity.tex:640 FAMILY_AUDITED

$$\delta Y = \frac{1}{2} h_+ Y, \qquad \delta Z = -\frac{1}{2} h_+ Z.$$

F0371 10_linearized_tensor_gravity.tex:646 FAMILY_AUDITED

$$\delta Y = \frac{1}{2} h_\times Z, \qquad \delta Z = \frac{1}{2} h_\times Y.$$

F0372 10_linearized_tensor_gravity.tex:671 FAMILY_AUDITED

eq:linear-gravity-chain

$$\boxed{\begin{array}{c}\text{source packets} \longrightarrow \text{scalar constraint geometry} \longrightarrow \text{clocks and light},\\ \text{TT source packets} \longrightarrow \text{tensor geometry} \longrightarrow \text{causal curvature waves}.\end{array}} \tag{A.75}$$

F0373 10_linearized_tensor_gravity.tex:689 FAMILY_AUDITED

$$\begin{array}{c}\text{sector selection} \longrightarrow \text{local transfer} \longrightarrow \text{constraint projection},\\ \text{variational evolution} \longrightarrow \text{refinement reconstruction}.\end{array}$$

F0374 10_linearized_tensor_gravity.tex:703 FAMILY_AUDITED

eq:tensor-summary

$$\boxed{\begin{array}{c}\text{dispersion order} \approx 2.00,\\ \text{vacuum translation error} \approx 10^{-17},\\ \text{relative energy drift} \approx 10^{-16},\\ \text{source-work residual} \approx 10^{-15},\\ \text{causal leakage} = 0,\\ \text{constraint residuals} = 0,\\ \text{curvature residual} \approx 10^{-16}.\end{array}} \tag{A.76}$$

11_unbounded_countable_calculus.tex

F0375 11_unbounded_countable_calculus.tex:9 FAMILY_AUDITED

$$H\sum_{\beta} P_\beta \psi$$

F0376 11_unbounded_countable_calculus.tex:13 FAMILY_AUDITED

$$\sum_{\beta} HP_\beta \psi$$

F0377 11_unbounded_countable_calculus.tex:26 FAMILY_AUDITED

$$\mathcal{P} = \{P_\alpha\}_{\alpha\in A}$$

F0378 11_unbounded_countable_calculus.tex:35 FAMILY_AUDITED

$$S_F\psi = \sum_{\alpha\in F} P_\alpha\psi, \qquad F \Subset A,$$

F0379 11_unbounded_countable_calculus.tex:40 FAMILY_AUDITED

$$\|\varphi\|_H^2 = \|\varphi\|^2 + \|H\varphi\|^2 .$$

F0380 11_unbounded_countable_calculus.tex:48 FAMILY_AUDITED

$$S_F\psi \longrightarrow \psi, \qquad HS_F\psi \longrightarrow H\psi$$

F0381 11_unbounded_countable_calculus.tex:58 FAMILY_AUDITED

$$\psi(t) = \mathrm{e}^{-\mathrm{i}tH}\psi_0, \qquad \psi_0 \in D(H).$$

F0382 11_unbounded_countable_calculus.tex:64 FAMILY_AUDITED

$$\psi(t) \in D(H), \qquad \dot\psi(t) = -\mathrm{i}H\psi(t),$$

F0383 11_unbounded_countable_calculus.tex:72 FAMILY_AUDITED

$$\psi_\alpha = P_\alpha\psi, \qquad E_\alpha = \|\psi_\alpha\|^2 .$$

F0384 11_unbounded_countable_calculus.tex:78 FAMILY_AUDITED

eq:unbounded-packet-current

$$\boxed{J_{\alpha\leftarrow\beta} = 2\,\mathrm{Im}\left\langle P_\alpha\psi, HP_\beta\psi\right\rangle.} \tag{A.77}$$

F0385 11_unbounded_countable_calculus.tex:90 FAMILY_AUDITED

eq:unbounded-local-continuity

$$\boxed{\frac{\mathrm{d}E_\alpha}{\mathrm{d}t} = \lim_{F\nearrow A}\sum_{\beta\in F} J_{\alpha\leftarrow\beta}.} \tag{A.78}$$

F0386 11_unbounded_countable_calculus.tex:101 FAMILY_AUDITED

eq:unbounded-current-antisymmetry

$$\boxed{J_{\alpha\leftarrow\beta} = -J_{\beta\leftarrow\alpha}.} \tag{A.79}$$

F0387 11_unbounded_countable_calculus.tex:109 FAMILY_AUDITED

eq:finite-region-boundary-flux

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\sum_{\alpha\in F} E_\alpha = \lim_{G\nearrow A}\sum_{\substack{\alpha\in F\\ \beta\in G\setminus F}} J_{\alpha\leftarrow\beta}.} \tag{A.80}$$

F0388 11_unbounded_countable_calculus.tex:122 FAMILY_AUDITED

$$\sum_\alpha \frac{\mathrm{d}E_\alpha}{\mathrm{d}t} = 0.$$

F0389 11_unbounded_countable_calculus.tex:126 FAMILY_AUDITED

$$\sum_\alpha E_\alpha(t) = \|\psi(t)\|^2 = \|\psi_0\|^2.$$

F0390 11_unbounded_countable_calculus.tex:133 FAMILY_AUDITED

$$\frac{\mathrm{d}E_\alpha}{\mathrm{d}t} = 2\,\mathrm{Re}\left\langle P_\alpha\psi, P_\alpha\dot\psi\right\rangle.$$

F0391 11_unbounded_countable_calculus.tex:139 FAMILY_AUDITED

$$\langle P_\alpha\psi, P_\alpha H\psi\rangle = \langle P_\alpha\psi, H\psi\rangle$$

F0392 11_unbounded_countable_calculus.tex:145 FAMILY_AUDITED

$$\frac{\mathrm{d}E_\alpha}{\mathrm{d}t} = 2\,\mathrm{Im}\,\langle P_\alpha\psi, H\psi\rangle.$$

F0393 11_unbounded_countable_calculus.tex:151 FAMILY_AUDITED

$$H\psi = \lim_{F\nearrow A}\sum_{\beta\in F} HP_\beta\psi.$$

F0394 11_unbounded_countable_calculus.tex:160 FAMILY_AUDITED

$$\left\langle P_\beta\psi, HP_\alpha\psi\right\rangle = \overline{\left\langle P_\alpha\psi, HP_\beta\psi\right\rangle}.$$

F0395 11_unbounded_countable_calculus.tex:170 FAMILY_AUDITED

$$\sum_\alpha \|P_\alpha\psi(t)\|^2 = \|\psi(t)\|^2.$$

F0396 11_unbounded_countable_calculus.tex:184 FAMILY_AUDITED

$$Q_A = \sum_{\alpha\in C_A} P_\alpha$$

F0397 11_unbounded_countable_calculus.tex:188 FAMILY_AUDITED

$$\psi_A = Q_A\psi, \qquad J_{A\leftarrow B} = 2\,\mathrm{Im}\,\langle Q_A\psi, HQ_B\psi\rangle\,.$$

F0398 11_unbounded_countable_calculus.tex:196 FAMILY_AUDITED

eq:unbounded-refinement-current

$$\boxed{J_{A\leftarrow B} = \sum_{\alpha\in C_A}\sum_{\beta\in C_B} J_{\alpha\leftarrow\beta}.} \tag{A.81}$$

F0399 11_unbounded_countable_calculus.tex:207 FAMILY_AUDITED

eq:unbounded-refinement-energy

$$\boxed{\|Q_A\psi\|^2 = \sum_{\alpha\in C_A}\|P_\alpha\psi\|^2\,.} \tag{A.82}$$

F0400 11_unbounded_countable_calculus.tex:221 FAMILY_AUDITED

$$S_N \longrightarrow I$$

F0401 11_unbounded_countable_calculus.tex:229 FAMILY_AUDITED

eq:strong-corner-limit

$$\boxed{S_N X S_N \longrightarrow X \quad \text{strongly.}} \tag{A.83}$$

F0402 11_unbounded_countable_calculus.tex:237 FAMILY_AUDITED

eq:compact-corner-limit

$$\boxed{\|S_N X S_N - X\| \longrightarrow 0.} \tag{A.84}$$

F0403 11_unbounded_countable_calculus.tex:248 FAMILY_AUDITED

$$S_N X S_N\psi - X\psi = S_N X(S_N - I)\psi + (S_N - I)X\psi.$$

F0404 11_unbounded_countable_calculus.tex:264 FAMILY_AUDITED

eq:fourier-unbounded-H

$$\begin{aligned}(H\psi)_n = n^2\psi_n + \frac{a}{2}(\psi_{n-1}+\psi_{n+1})\\ + \frac{b}{2\mathrm{i}}(\psi_{n-2}-\psi_{n+2}),\end{aligned} \tag{A.85}$$

F0405 11_unbounded_countable_calculus.tex:274 FAMILY_AUDITED

$$a\cos x + b\sin 2x.$$

F0406 11_unbounded_countable_calculus.tex:278 FAMILY_AUDITED

$$D(H) = \left\{\psi\in\ell^2(\mathbb{Z}) : \sum_n n^4|\psi_n|^2 < \infty\right\}.$$

F0407 11_unbounded_countable_calculus.tex:287 FAMILY_AUDITED

$$(P_r\psi)_n = \begin{cases}\psi_n, & n\equiv r \pmod 4,\\ 0, & \text{otherwise.}\end{cases}$$

F0408 11_unbounded_countable_calculus.tex:298 FAMILY_AUDITED

$$\psi_n \propto (1+n^2)^{-1.70}\exp\left[\mathrm{i}(0.173n + 0.021n^2 + 0.0007n^3)\right].$$

F0409 11_unbounded_countable_calculus.tex:346 FAMILY_AUDITED

$$H = -\Delta + V(x)$$

F0410 11_unbounded_countable_calculus.tex:350 FAMILY_AUDITED

smooth packet frames or quadratic-form currents,
boundary triples or controlled nonorthogonal localization.

12_dirac_3plus1.tex

F0411 12_dirac_3plus1.tex:9 FAMILY_AUDITED

eq:continuum-dirac-H

$$H_D(\boldsymbol{k}) = \boldsymbol{\alpha}\cdot\boldsymbol{k} + m\beta. \tag{A.86}$$

F0412 12_dirac_3plus1.tex:19 FAMILY_AUDITED

eq:dirac-matrices

$$\beta = \sigma_z \otimes I_2, \qquad \alpha_i = \sigma_x \otimes \sigma_i, \tag{A.87}$$

F0413 12_dirac_3plus1.tex:26 FAMILY_AUDITED

$$\beta^2 = \alpha_i^2 = I_4, \qquad \{\alpha_i, \alpha_j\} = 2\delta_{ij} I_4, \qquad \{\alpha_i, \beta\} = 0.$$

F0414 12_dirac_3plus1.tex:34 FAMILY_AUDITED

$$P_i^{\pm} = \frac{1}{2}(I_4 \pm \alpha_i)$$

F0415 12_dirac_3plus1.tex:38 FAMILY_AUDITED

$$P_\beta^{\pm} = \frac{1}{2}(I_4 \pm \beta)$$

F0416 12_dirac_3plus1.tex:44 FAMILY_AUDITED

eq:involution-exponential

$$\mathrm{e}^{-\mathrm{i}\theta\Gamma} = \cos\theta\, I_4 - \mathrm{i}\sin\theta\,\Gamma, \qquad \Gamma^2 = I_4. \tag{A.88}$$

F0417 12_dirac_3plus1.tex:52 FAMILY_AUDITED

$$\mathrm{e}^{-\mathrm{i}s\alpha_i p_i} = P_i^{+}\mathrm{e}^{-\mathrm{i}sp_i} + P_i^{-}\mathrm{e}^{\mathrm{i}sp_i}$$

F0418 12_dirac_3plus1.tex:61 FAMILY_AUDITED

$$M_{1/2} = \mathrm{e}^{-\mathrm{i}m\Delta\beta/2}, \quad X_{1/2} = \mathrm{e}^{-\mathrm{i}k_x\Delta\alpha_x/2}, \quad Y_{1/2} = \mathrm{e}^{-\mathrm{i}k_y\Delta\alpha_y/2}, \quad Z = \mathrm{e}^{-\mathrm{i}k_z\Delta\alpha_z}.$$

F0419 12_dirac_3plus1.tex:71 FAMILY_AUDITED

eq:3d-dirac-step

$$\boxed{U_\Delta(\boldsymbol{k}) = M_{1/2}X_{1/2}Y_{1/2}ZY_{1/2}X_{1/2}M_{1/2}.} \tag{A.89}$$

F0420 12_dirac_3plus1.tex:88 FAMILY_AUDITED

eq:3d-characteristic-polynomial

$$\boxed{\det(\lambda I_4 - U_\Delta) = \left(\lambda^2 - 2C_\Delta(\boldsymbol{k})\lambda + 1\right)^2,} \tag{A.90}$$

F0421 12_dirac_3plus1.tex:99 FAMILY_AUDITED

eq:3d-Cdelta

$$\boxed{C_\Delta(\boldsymbol{k}) = \cos(m\Delta)\cos(k_x\Delta)\cos(k_y\Delta)\cos(k_z\Delta).} \tag{A.91}$$

F0422 12_dirac_3plus1.tex:111 FAMILY_AUDITED

$$\mathrm{e}^{-\mathrm{i}\omega_\Delta\Delta}, \quad \mathrm{e}^{-\mathrm{i}\omega_\Delta\Delta}, \quad \mathrm{e}^{\mathrm{i}\omega_\Delta\Delta}, \quad \mathrm{e}^{\mathrm{i}\omega_\Delta\Delta},$$

F0423 12_dirac_3plus1.tex:121 FAMILY_AUDITED

eq:3d-exact-dispersion

$$\boxed{\cos(\omega_\Delta\Delta) = \cos(m\Delta)\prod_{i=1}^{3}\cos(k_i\Delta).} \tag{A.92}$$

F0424 12_dirac_3plus1.tex:134 FAMILY_AUDITED

$$\beta = \tau_z \otimes I_2, \qquad \alpha_i = \tau_x \otimes \sigma_i.$$

F0425 12_dirac_3plus1.tex:140 FAMILY_AUDITED

$$K = X_{1/2}Y_{1/2}ZY_{1/2}X_{1/2}$$

F0426 12_dirac_3plus1.tex:144 FAMILY_AUDITED

$$V = \mathrm{e}^{-\mathrm{i}b\sigma_x/2}\mathrm{e}^{-\mathrm{i}c\sigma_y/2}\mathrm{e}^{-\mathrm{i}d\sigma_z}\mathrm{e}^{-\mathrm{i}c\sigma_y/2}\mathrm{e}^{-\mathrm{i}b\sigma_x/2} = a_0 I_2 - \mathrm{i}\boldsymbol{a}\cdot\boldsymbol{\sigma}.$$

F0427 12_dirac_3plus1.tex:154 FAMILY_AUDITED

$$a_0 = \cos b\cos c\cos d, \qquad a_0^2 + |\boldsymbol{a}|^2 = 1.$$

F0428 12_dirac_3plus1.tex:160 FAMILY_AUDITED

$$K = a_0 I_4 - \mathrm{i}A, \qquad A^2 = |\boldsymbol{a}|^2 I_4, \qquad \{A,\beta\} = 0.$$

F0429 12_dirac_3plus1.tex:168 FAMILY_AUDITED

$$M^2 = \cos a\, I_4 - \mathrm{i}\sin a\,\beta.$$

F0430 12_dirac_3plus1.tex:172 FAMILY_AUDITED

$$U_\Delta = MKM = C_\Delta I_4 - \mathrm{i}B,$$

F0431 12_dirac_3plus1.tex:176 FAMILY_AUDITED

$$C_\Delta = a_0\cos a$$

F0432 12_dirac_3plus1.tex:180 FAMILY_AUDITED

$$B = a_0\sin a\,\beta + A, \qquad B^2 = (1-C_\Delta^2)I_4, \qquad \mathrm{Tr}\,B = 0.$$

F0433 12_dirac_3plus1.tex:195 FAMILY_AUDITED

$$E = \sqrt{m^2 + k_x^2 + k_y^2 + k_z^2}.$$

F0434 12_dirac_3plus1.tex:199 FAMILY_AUDITED

$$\omega_\Delta = E + c_2\Delta^2 + \mathcal{O}(\Delta^4).$$

F0435 12_dirac_3plus1.tex:204 FAMILY_AUDITED

eq:3d-leading-correction

$$\boxed{\omega_\Delta = E - \frac{\Delta^2}{6E}\left[m^2|\boldsymbol{k}|^2 + k_x^2k_y^2 + k_x^2k_z^2 + k_y^2k_z^2\right] + \mathcal{O}(\Delta^4).} \tag{A.93}$$

F0436 12_dirac_3plus1.tex:222 FAMILY_AUDITED

$$k_x^2k_y^2 + k_x^2k_z^2 + k_y^2k_z^2 = 0,$$

F0437 12_dirac_3plus1.tex:226 FAMILY_AUDITED

$$k_x^2 = k_y^2 = k_z^2 = \frac{k^2}{3}, \qquad k_x^2k_y^2 + k_x^2k_z^2 + k_y^2k_z^2 = \frac{k^4}{3}.$$

F0438 12_dirac_3plus1.tex:232 FAMILY_AUDITED

eq:3d-anisotropy-prediction

$$\boxed{\omega_{\mathrm{axis}} - \omega_{\mathrm{body}} = \frac{\Delta^2k^4}{18E} + \mathcal{O}(\Delta^4).} \tag{A.94}$$

F0439 12_dirac_3plus1.tex:248 FAMILY_AUDITED

eq:3d-group-velocity

$$\boxed{(v_\Delta)_i = \frac{\cos(m\Delta)\sin(k_i\Delta)\prod_{j\neq i}\cos(k_j\Delta)}{\sin(\omega_\Delta\Delta)}.} \tag{A.95}$$

F0440 12_dirac_3plus1.tex:263 FAMILY_AUDITED

eq:3d-group-limit

$$\boldsymbol{v}_\Delta = \frac{\boldsymbol{k}}{E} + \mathcal{O}(\Delta^2). \tag{A.96}$$

F0441 12_dirac_3plus1.tex:276 FAMILY_AUDITED

$$\Sigma_i = I_2 \otimes \sigma_i$$

F0442 12_dirac_3plus1.tex:280 FAMILY_AUDITED

eq:helicity-operator

$$\mathfrak{h}(\boldsymbol{k}) = \frac{\boldsymbol{\Sigma}\cdot\boldsymbol{k}}{|\boldsymbol{k}|}. \tag{A.97}$$

F0443 12_dirac_3plus1.tex:287 FAMILY_AUDITED

$$[H_D, \mathfrak{h}] = 0.$$

F0444 12_dirac_3plus1.tex:291 FAMILY_AUDITED

$$\|[U_\Delta(\boldsymbol{k}), \mathfrak{h}(\boldsymbol{k})]\| = \mathcal{O}(\Delta^3)$$

F0445 12_dirac_3plus1.tex:309 FAMILY_AUDITED

$$m = 0.7, \qquad |\boldsymbol{k}| = 1,$$

F0446 12_dirac_3plus1.tex:335 FAMILY_AUDITED

eq:3d-measured-orders

$$\boxed{\begin{aligned} p_{\text{mass}} &= 2.00239,\\ p_{\text{isotropy}} &= 1.99623,\\ p_{\text{velocity}} &= 2.00345,\\ p_{\text{corrected remainder}} &= 4.00339,\\ p_{\text{one step helicity}} &= 2.99185. \end{aligned}} \tag{A.98}$$

13_u1_gauge_matter.tex

F0447 13_u1_gauge_matter.tex:7 FAMILY_AUDITED

$$\text{site packets} \longrightarrow \text{link transfers} \longrightarrow \text{loop curvature} \longrightarrow \text{gauge-invariant reconstruction.}$$

F0448 13_u1_gauge_matter.tex:23 FAMILY_AUDITED

$$\psi_j^n \in \mathbb{C}^2.$$

F0449 13_u1_gauge_matter.tex:27 FAMILY_AUDITED

$$L_j^n \in U(1)$$

F0450 13_u1_gauge_matter.tex:31 FAMILY_AUDITED

$$T_j^n \in U(1)$$

F0451 13_u1_gauge_matter.tex:37 FAMILY_AUDITED

$$g_j^n = \mathrm{e}^{\mathrm{i}q\chi_j^n} \in U(1)$$

F0452 13_u1_gauge_matter.tex:41 FAMILY_AUDITED

eq:u1-gauge-transform

$$\boxed{\begin{aligned} &\psi_j^n \mapsto g_j^n \psi_j^n,\\ &L_j^n \mapsto g_{j+1}^n L_j^n (g_j^n)^{-1},\\ &T_j^n \mapsto g_j^{n+1} T_j^n (g_j^n)^{-1}. \end{aligned}} \tag{A.99}$$

F0453 13_u1_gauge_matter.tex:56 FAMILY_AUDITED

$$C_{1/2} = \mathrm{e}^{-\mathrm{i}m\Delta t\,\sigma_x/2}$$

F0454 13_u1_gauge_matter.tex:61 FAMILY_AUDITED

$$\phi_j^n = C_{1/2}\psi_j^n,$$

F0455 13_u1_gauge_matter.tex:65 FAMILY_AUDITED

eq:gauge-linked-shift

$$\boxed{\begin{aligned}(S_L^n\phi)_{j,+} &= L_{j-1}^n\phi_{j-1,+},\\ (S_L^n\phi)_{j,-} &= (L_j^n)^{-1}\phi_{j+1,-}.\end{aligned}} \tag{A.100}$$

F0456 13_u1_gauge_matter.tex:77 FAMILY_AUDITED

eq:gauge-linked-walk

$$\boxed{\psi_j^{n+1} = T_j^n C_{1/2}(S_L^n C_{1/2}\psi^n)_j.} \tag{A.101}$$

F0457 13_u1_gauge_matter.tex:91 FAMILY_AUDITED

eq:intertwining-gauge-walk

$$\boxed{\mathcal{U}[L'^n, T'^n]G^n = G^{n+1}\mathcal{U}[L^n, T^n],} \tag{A.102}$$

F0458 13_u1_gauge_matter.tex:104 FAMILY_AUDITED

$$L'^n_{j-1}g^n_{j-1}\phi_{j-1,+} = g_j^n L_{j-1}^n\phi_{j-1,+}.$$

F0459 13_u1_gauge_matter.tex:109 FAMILY_AUDITED

$$(L'^n_j)^{-1}g^n_{j+1}\phi_{j+1,-} = g_j^n(L_j^n)^{-1}\phi_{j+1,-}.$$

F0460 13_u1_gauge_matter.tex:114 FAMILY_AUDITED

$$T'^n_j g_j^n = g_j^{n+1}T_j^n.$$

F0461 13_u1_gauge_matter.tex:124 FAMILY_AUDITED

$$\rho_j^n = \left\|\psi_j^n\right\|^2.$$

F0462 13_u1_gauge_matter.tex:129 FAMILY_AUDITED

eq:gauge-walk-current

$$\boxed{J^n_{j+1/2} = |\phi^n_{j,+}|^2 - |\phi^n_{j+1,-}|^2.} \tag{A.103}$$

F0463 13_u1_gauge_matter.tex:142 FAMILY_AUDITED

eq:exact-gauge-continuity

$$\boxed{\rho_j^{n+1} - \rho_j^n = J^n_{j-1/2} - J^n_{j+1/2}.} \tag{A.104}$$

F0464 13_u1_gauge_matter.tex:150 FAMILY_AUDITED

$$\sum_j \rho_j^{n+1} = \sum_j \rho_j^n.$$

F0465 13_u1_gauge_matter.tex:154 FAMILY_AUDITED

eq:coarse-gauge-continuity

$$\boxed{Q_A^{n+1} - Q_A^n = J^n_{a-1/2} - J^n_{b-1/2}, \qquad Q_A^n = \sum_{j=a}^{b-1}\rho_j^n.} \tag{A.105}$$

F0466 13_u1_gauge_matter.tex:167 FAMILY_AUDITED

$$|\phi^n_{j-1,+}|^2 + |\phi^n_{j+1,-}|^2,$$

F0467 13_u1_gauge_matter.tex:171 FAMILY_AUDITED

$$\rho^n_j = |\phi^n_{j,+}|^2 + |\phi^n_{j,-}|^2.$$

F0468 13_u1_gauge_matter.tex:179 FAMILY_AUDITED

$$\partial_t \rho + \partial_x J = 0,$$

F0469 13_u1_gauge_matter.tex:187 FAMILY_AUDITED

eq:spacetime-plaquette

$$\boxed{W_{01}(j,n) = (T^n_j)^{-1}(L^{n+1}_j)^{-1}T^n_{j+1}L^n_j.} \tag{A.106}$$

F0470 13_u1_gauge_matter.tex:198 FAMILY_AUDITED

$$(j,n) \to (j+1,n) \to (j+1,n+1) \to (j,n+1) \to (j,n).$$

F0471 13_u1_gauge_matter.tex:208 FAMILY_AUDITED

$$W_{01}(j,n) = \mathrm{e}^{-\mathrm{i}q\,\Delta t\,h\,F_{01}(j,n)}$$

F0472 13_u1_gauge_matter.tex:217 FAMILY_AUDITED

eq:u1-wilson-action

$$\boxed{S_{\mathrm{EM}}[U] = \frac{1}{g^2}\sum_{x,\mu<\nu}\left[1 - \operatorname{Re} W_{\mu\nu}(x)\right].} \tag{A.107}$$

F0473 13_u1_gauge_matter.tex:227 QUALIFIED_PASS

$$1 - \operatorname{Re} W_{\mu\nu} = \frac{1}{2}\theta^2_{\mu\nu} + \mathcal{O}(\theta^4_{\mu\nu}).$$

F0474 13_u1_gauge_matter.tex:232 FAMILY_AUDITED

$$1 - \operatorname{Re} W_{\mu\nu} = \frac{1}{2}q^2 a^2_\mu a^2_\nu F^2_{\mu\nu} + \mathcal{O}(a^6).$$

F0475 13_u1_gauge_matter.tex:239 FAMILY_AUDITED

eq:lattice-noether-continuity

$$\boxed{\sum_\mu \left[J_\mu(x) - J_\mu(x-\hat{\mu})\right] = 0} \tag{A.108}$$

F0476 13_u1_gauge_matter.tex:254 FAMILY_AUDITED

eq:exact-u1-bianchi

$$\boxed{\begin{aligned} &W_{xy}(x)W_{yz}(x)W_{zx}(x) \\ &\quad\times W_{xy}(x+\hat{z})^{-1}W_{yz}(x+\hat{x})^{-1}W_{zx}(x+\hat{y})^{-1} = 1. \end{aligned}} \tag{A.109}$$

F0477 13_u1_gauge_matter.tex:270 FAMILY_AUDITED

$$\Delta_x F_{yz} + \Delta_y F_{zx} + \Delta_z F_{xy} = 0,$$

F0478 13_u1_gauge_matter.tex:276 FAMILY_AUDITED

$$\nabla \cdot \boldsymbol{B} = 0, \qquad \partial_t \boldsymbol{B} + \nabla \times \boldsymbol{E} = 0.$$

F0479 13_u1_gauge_matter.tex:306 FAMILY_AUDITED

eq:u1-bianchi-numerics

$$\boxed{\begin{aligned} \max|W_p - W'_p| &= 7.11 \times 10^{-16}, \\ \max|W_{\partial\mathrm{cube}} - 1| &= 1.00 \times 10^{-15}, \\ |S_{\mathrm{EM}}[U'] - S_{\mathrm{EM}}[U]| &= 0 \end{aligned}} \tag{A.110}$$

14_scalar_constraint_cosmology.tex

F0480 14_scalar_constraint_cosmology.tex:15 FAMILY_AUDITED

$$\mathrm{d}s^2 = -N(t)^2\,\mathrm{d}t^2 + a(t)^2\,\mathrm{d}\boldsymbol{x}^2.$$

F0481 14_scalar_constraint_cosmology.tex:19 FAMILY_AUDITED

eq:scalar-flrw-action

$$\boxed{S[a,N,\phi] = \int \mathrm{d}t\left[-\frac{3a\dot{a}^2}{N} + a^3\left(\frac{\dot{\phi}^2}{2N} - NV(\phi)\right)\right].} \tag{A.111}$$

F0482 14_scalar_constraint_cosmology.tex:34 FAMILY_AUDITED

eq:scalar-friedmann-constraint

$$\boxed{3H^2 = \rho_\phi = \frac{1}{2}\pi^2 + V(\phi),} \tag{A.112}$$

F0483 14_scalar_constraint_cosmology.tex:43 FAMILY_AUDITED

$$H = \frac{\dot{a}}{a}, \qquad \pi = \dot{\phi}.$$

F0484 14_scalar_constraint_cosmology.tex:49 FAMILY_AUDITED

eq:klein-gordon-homogeneous

$$\boxed{\dot{\pi} + 3H\pi + V'(\phi) = 0.} \tag{A.113}$$

F0485 14_scalar_constraint_cosmology.tex:56 FAMILY_AUDITED

eq:raychaudhuri-scalar

$$\boxed{\dot{H} = -\frac{1}{2}\pi^2.} \tag{A.114}$$

F0486 14_scalar_constraint_cosmology.tex:63 FAMILY_AUDITED

eq:scalar-first-order-system

$$\dot{\phi} = \pi, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\log a = H, \tag{A.115}$$

F0487 14_scalar_constraint_cosmology.tex:72 FAMILY_AUDITED

eq:scalar-pressure

$$\boxed{p_\phi = \frac{1}{2}\pi^2 - V(\phi).} \tag{A.116}$$

F0488 14_scalar_constraint_cosmology.tex:80 FAMILY_AUDITED

$$\rho_\phi + p_\phi = \pi^2.$$

F0489 14_scalar_constraint_cosmology.tex:87 FAMILY_AUDITED

eq:scalar-constraint-function

$$\mathcal{C} = 3H^2 - \frac{1}{2}\pi^2 - V(\phi). \tag{A.117}$$

F0490 14_scalar_constraint_cosmology.tex:93 FAMILY_AUDITED

$$\begin{aligned}\dot{\mathcal{C}} &= 6H\dot{H} - \pi\dot{\pi} - V'(\phi)\dot{\phi}\\ &= -3H\pi^2 - \pi(-3H\pi - V') - V'\pi\\ &= 0.\end{aligned}$$

F0491 14_scalar_constraint_cosmology.tex:102 FAMILY_AUDITED

eq:scalar-continuity

$$\boxed{\dot{\rho}_\phi + 3H(\rho_\phi + p_\phi) = 0.} \tag{A.118}$$

F0492 14_scalar_constraint_cosmology.tex:112 FAMILY_AUDITED

gauge or geometric symmetry $\Longrightarrow$ constraint propagation $\Longrightarrow$ stress-energy conservation.

F0493 14_scalar_constraint_cosmology.tex:123 FAMILY_AUDITED

$$\bar{\phi} = \frac{\phi_{n+1} + \phi_n}{2}, \quad \bar{\pi} = \frac{\pi_{n+1} + \pi_n}{2}, \quad \bar{H} = \frac{H_{n+1} + H_n}{2}.$$

F0494 14_scalar_constraint_cosmology.tex:131 FAMILY_AUDITED

eq:scalar-discrete-gradient

$$\overline{\nabla} V(\phi_n, \phi_{n+1}) = \begin{cases} \dfrac{V(\phi_{n+1}) - V(\phi_n)}{\phi_{n+1} - \phi_n}, & \phi_{n+1} \neq \phi_n, \\ V'(\phi_n), & \phi_{n+1} = \phi_n. \end{cases} \tag{A.119}$$

F0495 14_scalar_constraint_cosmology.tex:143 FAMILY_AUDITED

eq:discrete-chain-rule

$$V(\phi_{n+1}) - V(\phi_n) = \overline{\nabla} V(\phi_{n+1} - \phi_n). \tag{A.120}$$

F0496 14_scalar_constraint_cosmology.tex:152 FAMILY_AUDITED

eq:constraint-preserving-scalar-step

$$\boxed{\begin{aligned} \phi_{n+1} - \phi_n &= \Delta t\, \bar{\pi}, \\ \pi_{n+1} - \pi_n &= -\Delta t \left(3\bar{H}\bar{\pi} + \overline{\nabla} V \right), \\ H_{n+1} - H_n &= -\frac{\Delta t}{2} \bar{\pi}^2, \\ \log a_{n+1} - \log a_n &= \Delta t\, \bar{H}. \end{aligned}} \tag{A.121}$$

F0497 14_scalar_constraint_cosmology.tex:176 FAMILY_AUDITED

eq:exact-discrete-constraint

$$\boxed{\mathcal{C}_{n+1} = \mathcal{C}_n} \tag{A.122}$$

F0498 14_scalar_constraint_cosmology.tex:183 FAMILY_AUDITED

eq:exact-discrete-scalar-continuity

$$\boxed{\frac{\rho_{n+1} - \rho_n}{\Delta t} + 3\bar{H}\bar{\pi}^2 = 0.} \tag{A.123}$$

F0499 14_scalar_constraint_cosmology.tex:194 FAMILY_AUDITED

$$H_{n+1}^2 - H_n^2 = 2\bar{H}(H_{n+1} - H_n)$$

F0500 14_scalar_constraint_cosmology.tex:199 FAMILY_AUDITED

$$3(H_{n+1}^2 - H_n^2) = -3\Delta t\, \bar{H}\bar{\pi}^2.$$

F0501 14_scalar_constraint_cosmology.tex:204 FAMILY_AUDITED

$$\frac{1}{2}(\pi_{n+1}^2 - \pi_n^2) = \bar{\pi}(\pi_{n+1} - \pi_n).$$

F0502 14_scalar_constraint_cosmology.tex:209 FAMILY_AUDITED

$$\begin{aligned} &\frac{1}{2}(\pi_{n+1}^2 - \pi_n^2) + V(\phi_{n+1}) - V(\phi_n) \\ &= \bar{\pi} \left[-\Delta t (3\bar{H}\bar{\pi} + \overline{\nabla} V) \right] + \overline{\nabla} V(\Delta t \bar{\pi}) \\ &= -3\Delta t\, \bar{H}\bar{\pi}^2. \end{aligned}$$

F0503 14_scalar_constraint_cosmology.tex:225 FAMILY_AUDITED

eq:quadratic-scalar-potential

$$V(\phi) = V_0 + \frac{1}{2}\mu^2\phi^2, \tag{A.124}$$

F0504 14_scalar_constraint_cosmology.tex:230 FAMILY_AUDITED

$$\overline{\nabla}V = \frac{1}{2}\mu^2(\phi_n + \phi_{n+1}).$$

F0505 14_scalar_constraint_cosmology.tex:235 FAMILY_AUDITED

eq:scalar-midpoint-cubic

$$\boxed{-\frac{3}{4}\Delta t^2\bar{\pi}^3 + \left(2 + 3\Delta t H_n + \frac{1}{2}\mu^2\Delta t^2\right)\bar{\pi} + \mu^2\Delta t\phi_n - 2\pi_n = 0.} \tag{A.125}$$

F0506 14_scalar_constraint_cosmology.tex:251 FAMILY_AUDITED

$$\bar{H} = H_n - \frac{1}{4}\Delta t\bar{\pi}^2$$

F0507 14_scalar_constraint_cosmology.tex:260 FAMILY_AUDITED

$$H_0 = \sqrt{\frac{V_0}{3}}, \qquad \phi(t) = \phi_0, \qquad H(t) = H_0, \qquad a(t) = a_0 \mathrm{e}^{H_0 t}.$$

F0508 14_scalar_constraint_cosmology.tex:274 FAMILY_AUDITED

$$p = \rho, \qquad w = 1.$$

F0509 14_scalar_constraint_cosmology.tex:280 FAMILY_AUDITED

eq:stiff-scalar-exact

$$\boxed{\begin{aligned} H(t) &= \frac{H_0}{1 + 3H_0 t}, \\ a(t) &= a_0(1 + 3H_0 t)^{1/3}, \\ \pi(t) &= \frac{\pi_0}{1 + 3H_0 t}, \\ \phi(t) &= \phi_0 + \frac{\pi_0}{3H_0}\log(1 + 3H_0 t). \end{aligned}} \tag{A.126}$$

F0510 14_scalar_constraint_cosmology.tex:294 FAMILY_AUDITED

eq:stiff-order

$$\boxed{p_{\mathrm{stiff}} = 2.004995.} \tag{A.127}$$

F0511 14_scalar_constraint_cosmology.tex:311 FAMILY_AUDITED

$$V(\phi) = \frac{1}{2}\mu^2\phi^2, \qquad \mu = 0.25, \qquad \phi_0 = 5, \qquad \pi_0 = 0,$$

F0512 14_scalar_constraint_cosmology.tex:341 FAMILY_AUDITED

eq:massive-scalar-order

$$\boxed{p_{\mathrm{massive}} = 2.000456.} \tag{A.128}$$

F0513 14_scalar_constraint_cosmology.tex:349 FAMILY_AUDITED

$$w(t) = \frac{\frac{1}{2}\pi^2 - V}{\frac{1}{2}\pi^2 + V}$$

F0514 14_scalar_constraint_cosmology.tex:354 FAMILY_AUDITED

$$t = 21.05,$$

F0515 14_scalar_constraint_cosmology.tex:358 FAMILY_AUDITED

$$\boxed{6.755 \text{ e-folds}}$$

F0516 14_scalar_constraint_cosmology.tex:362 FAMILY_AUDITED

eq:massive-late-w

$$\boxed{\langle w \rangle = 0.00205,} \tag{A.129}$$

15_emergent_lorentz_covariance.tex

F0517 15_emergent_lorentz_covariance.tex:7 FAMILY_AUDITED
eq:boost-exact-shell

$$\cos(\omega_\Delta \Delta) = \cos(m\Delta) \prod_{i=1}^{3} \cos(k_i \Delta). \tag{A.130}$$

F0518 15_emergent_lorentz_covariance.tex:21 FAMILY_AUDITED

$$E(\boldsymbol{k}) = \sqrt{m^2 + |\boldsymbol{k}|^2} \tag{A.131}$$

F0519 15_emergent_lorentz_covariance.tex:25 FAMILY_AUDITED

eq:boost-Q

$$\boxed{Q(\boldsymbol{k}) = m^2|\boldsymbol{k}|^2 + k_x^2 k_y^2 + k_x^2 k_z^2 + k_y^2 k_z^2.} \tag{A.132}$$

F0520 15_emergent_lorentz_covariance.tex:34 FAMILY_AUDITED

eq:boost-energy-expansion

$$\boxed{\omega_\Delta(\boldsymbol{k}) = E(\boldsymbol{k}) + a(\boldsymbol{k})\Delta^2 + \mathcal{O}(\Delta^4), \qquad a(\boldsymbol{k}) = -\frac{Q(\boldsymbol{k})}{6E(\boldsymbol{k})}.} \tag{A.133}$$

F0521 15_emergent_lorentz_covariance.tex:50 FAMILY_AUDITED

$$\gamma = (1 - |\boldsymbol{v}|^2)^{-1/2}.$$

F0522 15_emergent_lorentz_covariance.tex:54 FAMILY_AUDITED

eq:boost-cont-energy eq:boost-cont-momentum

$$E' = \gamma(E - \boldsymbol{v} \cdot \boldsymbol{k}), \tag{A.134}$$

$$\boldsymbol{k}' = \boldsymbol{k} + \left[\frac{\gamma - 1}{|\boldsymbol{v}|^2} (\boldsymbol{v} \cdot \boldsymbol{k}) - \gamma E \right] \boldsymbol{v}. \tag{A.135}$$

F0523 15_emergent_lorentz_covariance.tex:66 FAMILY_AUDITED

$$p_\Delta = (\omega_\Delta(\boldsymbol{k}), \boldsymbol{k})$$

F0524 15_emergent_lorentz_covariance.tex:70 FAMILY_AUDITED

$$(\widetilde{E}_\Delta, \widetilde{\boldsymbol{k}}_\Delta) = B_{\boldsymbol{v}} p_\Delta.$$

F0525 15_emergent_lorentz_covariance.tex:75 FAMILY_AUDITED

eq:boost-defect

$$\boxed{\mathfrak{L}_\Delta(\boldsymbol{k}, \boldsymbol{v}) = \widetilde{E}_\Delta - \omega_\Delta(\widetilde{\boldsymbol{k}}_\Delta).} \tag{A.136}$$

F0526 15_emergent_lorentz_covariance.tex:84 FAMILY_AUDITED

put the state on the lattice shell, then boost,

F0527 15_emergent_lorentz_covariance.tex:87 FAMILY_AUDITED

boost the momentum, then put it on the lattice shell.

F0528 15_emergent_lorentz_covariance.tex:95 FAMILY_AUDITED

eq:boost-leading-defect

$$\boxed{\mathfrak{L}_\Delta(\boldsymbol{k}, \boldsymbol{v}) = \frac{\Delta^2}{6E'} \left[Q(\boldsymbol{k}') - Q(\boldsymbol{k}) \right] + \mathcal{O}(\Delta^4),} \tag{A.137}$$

F0529 15_emergent_lorentz_covariance.tex:112 FAMILY_AUDITED

$$p_\Delta = (E + a\Delta^2, \boldsymbol{k}) + \mathcal{O}(\Delta^4).$$

F0530 15_emergent_lorentz_covariance.tex:116 FAMILY_AUDITED

$$\widetilde{E}_\Delta = E' + \gamma a \Delta^2 + \mathcal{O}(\Delta^4),$$
$$\widetilde{\boldsymbol{k}}_\Delta = \boldsymbol{k}' - \gamma a \boldsymbol{v} \Delta^2 + \mathcal{O}(\Delta^4).$$

F0531 15_emergent_lorentz_covariance.tex:123 FAMILY_AUDITED

$$E(\widetilde{\boldsymbol{k}}_\Delta) = E' - \gamma a \frac{\boldsymbol{v}\cdot\boldsymbol{k}'}{E'} \Delta^2 + \mathcal{O}(\Delta^4).$$

F0532 15_emergent_lorentz_covariance.tex:130 FAMILY_AUDITED

$$\omega_\Delta(\widetilde{\boldsymbol{k}}_\Delta) = E' + \left[-\gamma a \frac{\boldsymbol{v}\cdot\boldsymbol{k}'}{E'} + a(\boldsymbol{k}') \right] \Delta^2 + \mathcal{O}(\Delta^4).$$

F0533 15_emergent_lorentz_covariance.tex:140 FAMILY_AUDITED

$$\mathfrak{L}_\Delta = \left[\gamma a \left(1 + \frac{\boldsymbol{v}\cdot\boldsymbol{k}'}{E'} \right) - a(\boldsymbol{k}') \right] \Delta^2 + \mathcal{O}(\Delta^4).$$

F0534 15_emergent_lorentz_covariance.tex:150 FAMILY_AUDITED

$$E = \gamma(E' + \boldsymbol{v}\cdot\boldsymbol{k}'),$$

F0535 15_emergent_lorentz_covariance.tex:154 FAMILY_AUDITED

$$a(\boldsymbol{k}) \frac{E}{E'} - a(\boldsymbol{k}').$$

F0536 15_emergent_lorentz_covariance.tex:163 FAMILY_AUDITED

eq:boost-uniform-bound eq:boost-corrected-bound

$$|\mathfrak{L}_\Delta| \le C_2 \Delta^2, \tag{A.138}$$
$$\left| \mathfrak{L}_\Delta - \frac{\Delta^2}{6E'} (Q(\boldsymbol{k}') - Q(\boldsymbol{k})) \right| \le C_4 \Delta^4 \tag{A.139}$$

F0537 15_emergent_lorentz_covariance.tex:184 FAMILY_AUDITED

$$\boxed{p_{\mathrm{boost}} = 2.002147.} \tag{A.140}$$

F0538 15_emergent_lorentz_covariance.tex:188 FAMILY_AUDITED

$$\boxed{p_{\mathrm{corrected}} = 4.002824.} \tag{A.141}$$

F0539 15_emergent_lorentz_covariance.tex:192 FAMILY_AUDITED

$$\boxed{p_{\mathrm{shell}} = 1.998823.} \tag{A.142}$$

F0540 15_emergent_lorentz_covariance.tex:196 FAMILY_AUDITED

$$8.6 \times 10^{-5},$$

F0541 15_emergent_lorentz_covariance.tex:200 FAMILY_AUDITED

$$6.20 \times 10^{-9}.$$

16_dynamical_u1_electrodynamics.tex

F0542 16_dynamical_u1_electrodynamics.tex:13 FAMILY_AUDITED

$$\psi_j(t) \in \mathbb{C}^2.$$

F0543 16_dynamical_u1_electrodynamics.tex:17 FAMILY_AUDITED

$$\alpha = \sigma_z, \qquad \beta = \sigma_x.$$

F0544 16_dynamical_u1_electrodynamics.tex:23 FAMILY_AUDITED

eq:dynamical-u-link

$$U_j = \mathrm{e}^{-\mathrm{i}hA_j}. \tag{A.143}$$

F0545 16_dynamical_u1_electrodynamics.tex:30 FAMILY_AUDITED

eq:dynamical-u-hamiltonian

$$\boxed{\begin{aligned} H_h[\psi, A, E] = hm \sum_j \psi_j^* \beta \psi_j \\ + \sum_j \mathrm{Im}\left(\psi_j^* \alpha U_j \psi_{j+1}\right) + \frac{h}{2} \sum_j E_j^2. \end{aligned}} \tag{A.144}$$

F0546 16_dynamical_u1_electrodynamics.tex:43 FAMILY_AUDITED

eq:dynamical-u-action

$$S_h = \int \left[\frac{\mathrm{i}h}{2} \sum_j \left(\psi_j^* \dot{\psi}_j - \dot{\psi}_j^* \psi_j\right) + h \sum_j E_j \dot{A}_j - H_h \right] \mathrm{d}t. \tag{A.145}$$

F0547 16_dynamical_u1_electrodynamics.tex:56 FAMILY_AUDITED

eq:dynamical-u-dirac eq:dynamical-u-A eq:dynamical-u-E

$$\dot{\psi}_j = -\mathrm{i}m\beta\psi_j - \frac{1}{2h}\alpha U_j \psi_{j+1} + \frac{1}{2h} U_{j-1}^* \alpha \psi_{j-1}, \tag{A.146}$$

$$\dot{A}_j = E_j, \tag{A.147}$$

$$\dot{E}_j = J_j, \tag{A.148}$$

F0548 16_dynamical_u1_electrodynamics.tex:68 FAMILY_AUDITED

eq:dynamical-u-current

$$\boxed{J_j = \mathrm{Re}\left(\psi_j^* \alpha U_j \psi_{j+1}\right).} \tag{A.149}$$

F0549 16_dynamical_u1_electrodynamics.tex:81 FAMILY_AUDITED

eq:dynamical-u-gauge

$$\boxed{\begin{aligned} &\psi_j \mapsto \mathrm{e}^{\mathrm{i}\chi_j}\psi_j, \\ &A_j \mapsto A_j + \frac{\chi_{j+1} - \chi_j}{h}, \\ &E_j \mapsto E_j. \end{aligned}} \tag{A.150}$$

F0550 16_dynamical_u1_electrodynamics.tex:92 FAMILY_AUDITED

$$U_j \mapsto \mathrm{e}^{\mathrm{i}\chi_j} U_j \mathrm{e}^{-\mathrm{i}\chi_{j+1}},$$

F0551 16_dynamical_u1_electrodynamics.tex:100 FAMILY_AUDITED

$$\rho_j = \psi_j^* \psi_j.$$

F0552 16_dynamical_u1_electrodynamics.tex:104 FAMILY_AUDITED

eq:dynamical-u-continuity

$$\boxed{\dot{\rho}_j + \frac{J_j - J_{j-1}}{h} = 0.} \tag{A.151}$$

F0553 16_dynamical_u1_electrodynamics.tex:116 FAMILY_AUDITED

$$-\frac{1}{h}\,\mathrm{Re}(\psi_j^* \alpha U_j \psi_{j+1}) = -\frac{J_j}{h},$$

F0554 16_dynamical_u1_electrodynamics.tex:126 FAMILY_AUDITED

$$\rho_j - \bar{\rho}, \qquad \bar{\rho} = \frac{1}{N}\sum_j \rho_j.$$

F0555 16_dynamical_u1_electrodynamics.tex:132 FAMILY_AUDITED

eq:dynamical-u-gauss

$$\boxed{G_j = \frac{E_j - E_{j-1}}{h} + \rho_j - \bar{\rho}.} \tag{A.152}$$

F0556 16_dynamical_u1_electrodynamics.tex:144 FAMILY_AUDITED

eq:dynamical-u-gauss-propagation

$$\boxed{\dot{G}_j = 0.} \tag{A.153}$$

F0557 16_dynamical_u1_electrodynamics.tex:153 FAMILY_AUDITED

$$\frac{\dot{E}_j - \dot{E}_{j-1}}{h} = \frac{J_j - J_{j-1}}{h}.$$

F0558 16_dynamical_u1_electrodynamics.tex:163 FAMILY_AUDITED

$$H_h = H_{\mathrm{m}} + H_E, \qquad H_E = \frac{h}{2}\sum_j E_j^2.$$

F0559 16_dynamical_u1_electrodynamics.tex:169 FAMILY_AUDITED

eq:dynamical-u-work

$$\boxed{\dot{H}_E = h\sum_j E_j J_j, \qquad \dot{H}_{\mathrm{m}} = -h\sum_j E_j J_j, \qquad \dot{H}_h = 0.} \tag{A.154}$$

F0560 16_dynamical_u1_electrodynamics.tex:186 FAMILY_AUDITED

eq:dynamical-u-covariant-difference

$$D_h^A \psi_j = \frac{U_j \psi_{j+1} - U_{j-1}^* \psi_{j-1}}{2h}. \tag{A.155}$$

F0561 16_dynamical_u1_electrodynamics.tex:192 FAMILY_AUDITED

eq:dynamical-u-continuum-derivative

$$\boxed{D_h^A \psi = (\partial_x - \mathrm{i}A)\psi + \mathcal{O}(h^2).} \tag{A.156}$$

F0562 16_dynamical_u1_electrodynamics.tex:204 FAMILY_AUDITED

$$\begin{aligned} \max_j |\dot{\rho}_j + (J_j - J_{j-1})/h| &= 1.39 \times 10^{-16}, \\ \max_j |\dot{G}_j| &= 1.34 \times 10^{-16}, \\ |\dot{H}_{\mathrm{m}} + \dot{H}_E| &= 2.78 \times 10^{-17}. \end{aligned}$$

F0563 16_dynamical_u1_electrodynamics.tex:213 FAMILY_AUDITED

$$\begin{aligned} \max_t \frac{|H_h(t) - H_h(0)|}{|H_h(0)|} &= 2.22 \times 10^{-12}, \\ \max_t |Q(t) - Q(0)| &= 1.13 \times 10^{-12}, \\ \max_t \|G(t) - G(0)\|_\infty &= 9.30 \times 10^{-13}, \\ \text{coupled gauge-covariance residual} &= 1.94 \times 10^{-12}. \end{aligned}$$

F0564 16_dynamical_u1_electrodynamics.tex:226 FAMILY_AUDITED

$$\boxed{p_{L^2} = 1.996761, \qquad p_{L^\infty} = 1.994685.} \tag{A.157}$$

17_metric_holonomy_equivalence.tex

F0565 17_metric_holonomy_equivalence.tex:13 FAMILY_AUDITED

eq:conformal-1plus1-metric

$$\boxed{\mathrm{d}s^2 = \mathrm{e}^{2\phi(t,x)}(-\,\mathrm{d}t^2 + \mathrm{d}x^2).} \tag{A.158}$$

F0566 17_metric_holonomy_equivalence.tex:22 FAMILY_AUDITED

$$e^0 = \mathrm{e}^{\phi}\,\mathrm{d}t, \qquad e^1 = \mathrm{e}^{\phi}\,\mathrm{d}x.$$

F0567 17_metric_holonomy_equivalence.tex:28 FAMILY_AUDITED

$$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

F0568 17_metric_holonomy_equivalence.tex:35 FAMILY_AUDITED

eq:conformal-spin-connection

$$\boxed{\omega = (\phi_x\,\mathrm{d}t + \phi_t\,\mathrm{d}x)K.} \tag{A.159}$$

F0569 17_metric_holonomy_equivalence.tex:43 FAMILY_AUDITED

eq:conformal-curvature-form

$$\boxed{\Omega = \mathrm{d}\omega = (\phi_{tt} - \phi_{xx})K\ \mathrm{d}t \wedge \mathrm{d}x.} \tag{A.160}$$

F0570 17_metric_holonomy_equivalence.tex:52 FAMILY_AUDITED

eq:conformal-scalar-curvature

$$\boxed{R = 2\mathrm{e}^{-2\phi}(\phi_{tt} - \phi_{xx}).} \tag{A.161}$$

F0571 17_metric_holonomy_equivalence.tex:64 FAMILY_AUDITED

eq:metric-holonomy-edge

$$U_e = \exp\left(\int_e \omega\right). \tag{A.162}$$

F0572 17_metric_holonomy_equivalence.tex:74 FAMILY_AUDITED

eq:metric-holonomy-exact

$$\boxed{W_\square = \exp\left[K \int_\square (\phi_{tt} - \phi_{xx})\,\mathrm{d}t\,\mathrm{d}x\right],} \tag{A.163}$$

F0573 17_metric_holonomy_equivalence.tex:86 FAMILY_AUDITED

eq:metric-holonomy-R

$$\boxed{\log W_\square = K \int_\square \frac{1}{2}\mathrm{e}^{2\phi} R\ \mathrm{d}t\,\mathrm{d}x.} \tag{A.164}$$

F0574 17_metric_holonomy_equivalence.tex:97 FAMILY_AUDITED

$$\oint_{\partial\square} \omega = K \int_\square (\phi_{tt} - \phi_{xx})\,\mathrm{d}t\,\mathrm{d}x.$$

F0575 17_metric_holonomy_equivalence.tex:109 FAMILY_AUDITED

$$G(t,x) = \mathrm{e}^{\chi K}.$$

F0576 17_metric_holonomy_equivalence.tex:113 FAMILY_AUDITED

$$U_{a\to b} \mapsto G(b) U_{a\to b} G(a)^{-1}.$$

F0577 17_metric_holonomy_equivalence.tex:117 FAMILY_AUDITED

eq:metric-holonomy-gauge

$$\boxed{W_\square \mapsto G(v_0) W_\square G(v_0)^{-1}.} \tag{A.165}$$

F0578 17_metric_holonomy_equivalence.tex:128 FAMILY_AUDITED

eq:midpoint-holonomy-angle

$$\begin{aligned}\Theta_h = {} & h\phi_x(t, x - h/2) + h\phi_t(t + h/2, x) \\ & - h\phi_x(t, x + h/2) - h\phi_t(t - h/2, x).\end{aligned} \tag{A.166}$$

F0579 17_metric_holonomy_equivalence.tex:139 FAMILY_AUDITED

$$F_h^{\mathrm{hol}} = \Theta_h / h^2$$

F0580 17_metric_holonomy_equivalence.tex:143 FAMILY_AUDITED

eq:metric-curvature-stencil

$$F_h^{\mathrm{met}} = \frac{\phi(t + h, x) - 2\phi(t, x) + \phi(t - h, x)}{h^2} - \frac{\phi(t, x + h) - 2\phi(t, x) + \phi(t, x - h)}{h^2} \tag{A.167}$$

F0581 17_metric_holonomy_equivalence.tex:152 FAMILY_AUDITED

$$R_h^{\mathrm{hol}} = 2\mathrm{e}^{-2\phi} F_h^{\mathrm{hol}}, \qquad R_h^{\mathrm{met}} = 2\mathrm{e}^{-2\phi} F_h^{\mathrm{met}}.$$

F0582 17_metric_holonomy_equivalence.tex:158 FAMILY_AUDITED

eq:metric-holonomy-difference

$$R_h^{\mathrm{hol}} - R_h^{\mathrm{met}} = \mathcal{O}(h^2). \tag{A.168}$$

F0583 17_metric_holonomy_equivalence.tex:163 FAMILY_AUDITED

eq:metric-holonomy-richardson

$$\boxed{R_h^{[4]} = \frac{4R_{h/2} - R_h}{3}} \tag{A.169}$$

F0584 17_metric_holonomy_equivalence.tex:175 FAMILY_AUDITED

holonomy curvature : 1.995438,
metric curvature : 1.987868,
metric-holonomy agreement : 1.980130,
refined holonomy curvature : 3.997011,
refined metric curvature : 3.991071.

F0585 17_metric_holonomy_equivalence.tex:183 FAMILY_AUDITED

$$3.32 \times 10^{-4} \quad \text{and} \quad 6.64 \times 10^{-4}$$

F0586 17_metric_holonomy_equivalence.tex:189 FAMILY_AUDITED

$$4.61 \times 10^{-9} \quad \text{and} \quad 2.46 \times 10^{-8}.$$

F0587 17_metric_holonomy_equivalence.tex:196 FAMILY_AUDITED

Stokes residual $\leq 2.78 \times 10^{-17}$,
matrix-log residual $\leq 2.13 \times 10^{-16}$,
local-frame covariance residual $\leq 9.04 \times 10^{-16}$,
$|\det W - 1| \leq 4.44 \times 10^{-16}$.

F0588 17_metric_holonomy_equivalence.tex:220 FAMILY_AUDITED

$$\boxed{\text{metric derivatives} \longleftrightarrow \text{frame connection} \longleftrightarrow \text{packet holonomy.}}$$

18_local_scalar_stress_energy.tex

F0589 18_local_scalar_stress_energy.tex:9 FAMILY_AUDITED
eq:local-scalar-metric

$$\boxed{\mathrm{d}s^2 = -N(t,x)^2\,\mathrm{d}t^2 + a(t,x)^2\,\mathrm{d}x^2.} \tag{A.170}$$

F0590 18_local_scalar_stress_energy.tex:22 FAMILY_AUDITED

eq:local-scalar-continuum-action

$$S_M = \frac{1}{2}\int \mathrm{d}t\,\mathrm{d}x\left[\frac{a}{N}\dot{\varphi}^2 - \frac{N}{a}\varphi_x^2 - Nam^2\varphi^2\right]. \tag{A.171}$$

F0591 18_local_scalar_stress_energy.tex:33 FAMILY_AUDITED

eq:local-scalar-stress-tensor

$$T_{\mu\nu} = \partial_\mu\varphi\partial_\nu\varphi - \frac{1}{2}g_{\mu\nu}\left(g^{\rho\sigma}\partial_\rho\varphi\partial_\sigma\varphi + m^2\varphi^2\right). \tag{A.172}$$

F0592 18_local_scalar_stress_energy.tex:45 FAMILY_AUDITED

eq:local-scalar-coefficients

$$\begin{aligned} w_j &= \frac{a_j}{N_j}, \\ c_{j+1/2} &= \frac{N_{j+1/2}}{a_{j+1/2}}, \\ v_j &= N_j a_j m^2. \end{aligned} \tag{A.173}$$

F0593 18_local_scalar_stress_energy.tex:54 FAMILY_AUDITED

eq:local-scalar-lattice-action

$$\boxed{L_h = \frac{h}{2}\sum_j w_j\dot{\varphi}_j^2 - \frac{1}{2h}\sum_j c_{j+1/2}(\varphi_{j+1}-\varphi_j)^2 - \frac{h}{2}\sum_j v_j\varphi_j^2.} \tag{A.174}$$

F0594 18_local_scalar_stress_energy.tex:64 FAMILY_AUDITED

eq:local-scalar-eom

$$\boxed{w_j\ddot{\varphi}_j + \dot{w}_j\dot{\varphi}_j = \frac{c_{j+1/2}\Delta_j - c_{j-1/2}\Delta_{j-1}}{h^2} - v_j\varphi_j,} \tag{A.175}$$

F0595 18_local_scalar_stress_energy.tex:74 FAMILY_AUDITED

$$\Delta_j = \varphi_{j+1} - \varphi_j.$$

F0596 18_local_scalar_stress_energy.tex:82 FAMILY_AUDITED

eq:local-scalar-cell-energy

$$\boxed{\begin{aligned} e_j = {} & \frac{h}{2}w_j\dot{\varphi}_j^2 + \frac{h}{2}v_j\varphi_j^2 \\ & + \frac{c_{j+1/2}}{4h}\Delta_j^2 + \frac{c_{j-1/2}}{4h}\Delta_{j-1}^2. \end{aligned}} \tag{A.176}$$

F0597 18_local_scalar_stress_energy.tex:95 FAMILY_AUDITED

eq:local-scalar-flux

$$\boxed{F_{j+1/2} = -\frac{c_{j+1/2}}{2h}\Delta_j(\dot{\varphi}_j + \dot{\varphi}_{j+1}).} \tag{A.177}$$

F0598 18_local_scalar_stress_energy.tex:105 FAMILY_AUDITED

eq:local-scalar-source

$$\boxed{\begin{aligned} s_j = {} & -\frac{h}{2}\dot{w}_j\dot{\varphi}_j^2 + \frac{h}{2}\dot{v}_j\varphi_j^2 \\ & + \frac{\dot{c}_{j+1/2}}{4h}\Delta_j^2 + \frac{\dot{c}_{j-1/2}}{4h}\Delta_{j-1}^2. \end{aligned}} \tag{A.178}$$

F0599 18_local_scalar_stress_energy.tex:121 FAMILY_AUDITED

eq:local-scalar-balance

$$\boxed{\dot{e}_j + F_{j+1/2} - F_{j-1/2} = s_j.} \tag{A.179}$$

F0600 18_local_scalar_stress_energy.tex:130 FAMILY_AUDITED

eq:local-scalar-global-work

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\sum_j e_j = \sum_j s_j = -\frac{\partial L_h}{\partial t}.} \tag{A.180}$$

F0601 18_local_scalar_stress_energy.tex:148 FAMILY_AUDITED

eq:local-scalar-coarse

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\sum_{j\in A} e_j + F_{s+1/2} - F_{r-1/2} = \sum_{j\in A} s_j.} \tag{A.181}$$

F0602 18_local_scalar_stress_energy.tex:162 FAMILY_AUDITED

eq:local-scalar-continuum

$$\frac{c_{j+1/2}\Delta_j - c_{j-1/2}\Delta_{j-1}}{h^2} = \partial_x(c\partial_x\varphi) + \mathcal{O}(h^2) \tag{A.182}$$

F0603 18_local_scalar_stress_energy.tex:172 FAMILY_AUDITED

$$\max_{j,t} |\dot{e}_j + F_{j+1/2} - F_{j-1/2} - s_j| = 1.03 \times 10^{-15},$$

$$\max_t \left| \sum_j \dot{e}_j - \sum_j s_j \right| = 8.88 \times 10^{-16},$$

$$\text{maximum nonuniform coarse-packet residual} = 9.46 \times 10^{-16}.$$

F0604 18_local_scalar_stress_energy.tex:181 FAMILY_AUDITED

$$\boxed{5.63 \times 10^{-12},} \tag{A.183}$$

F0605 18_local_scalar_stress_energy.tex:187 FAMILY_AUDITED

$$\boxed{p_{L^2} = 1.996862, \qquad p_{L^\infty} = 1.993031.} \tag{A.184}$$

F0606 18_local_scalar_stress_energy.tex:212 FAMILY_AUDITED

$$\boxed{\text{cell energy change} = \text{boundary transfer} + \text{local geometric work.}}$$

A_proofs.tex

F0607 A_proofs.tex:8 PROOF_ARCHIVE_REVIEWED

$$\|\psi\|^2 = \sum_\alpha \|P_\alpha\psi\|^2 .$$

F0608 A_proofs.tex:17 PROOF_ARCHIVE_REVIEWED

$$\|\psi\|^2 = \left\langle \psi, \sum_\alpha P_\alpha\psi \right\rangle = \sum_\alpha \langle \psi, P_\alpha\psi\rangle .$$

F0609 A_proofs.tex:23 PROOF_ARCHIVE_REVIEWED

$$\langle \psi, P_\alpha\psi\rangle = \langle P_\alpha\psi, P_\alpha\psi\rangle .$$

F0610 A_proofs.tex:32 PROOF_ARCHIVE_REVIEWED

$$\|Q_A\psi\|^2 = \sum_{\alpha\in C_A} \|P_\alpha\psi\|^2 \,.$$

F0611 A_proofs.tex:46 PROOF_ARCHIVE_REVIEWED

$$J_{\alpha\beta} = 2\,\mathrm{Im}\left\langle \psi_\alpha, H\psi_\beta \right\rangle .$$

F0612 A_proofs.tex:51 PROOF_ARCHIVE_REVIEWED

$$\left\langle \psi_\beta, H\psi_\alpha \right\rangle = \overline{\left\langle \psi_\alpha, H\psi_\beta \right\rangle}.$$

F0613 A_proofs.tex:56 PROOF_ARCHIVE_REVIEWED

$$J_{\beta\alpha} = -J_{\alpha\beta}.$$

F0614 A_proofs.tex:64 PROOF_ARCHIVE_REVIEWED

$$\dot P = \dot U P_0 U^* + U P_0 \dot U^*.$$

F0615 A_proofs.tex:69 PROOF_ARCHIVE_REVIEWED

$$\dot U^* = -U^* \dot U U^*,$$

F0616 A_proofs.tex:73 PROOF_ARCHIVE_REVIEWED

$$U P_0 \dot U^* = -U P_0 U^* \dot U U^* = -PK.$$

F0617 A_proofs.tex:79 PROOF_ARCHIVE_REVIEWED

$$\dot P = KP - PK = [K, P].$$

F0618 A_proofs.tex:87 PROOF_ARCHIVE_REVIEWED

$$\dot X_{\alpha\beta} = \dot P_\alpha X P_\beta + P_\alpha \dot X P_\beta + P_\alpha X \dot P_\beta.$$

F0619 A_proofs.tex:96 PROOF_ARCHIVE_REVIEWED

$$\sum_{\alpha,\beta} \dot X_{\alpha\beta} = \dot X,$$

F0620 A_proofs.tex:101 PROOF_ARCHIVE_REVIEWED

$$\sum_\alpha \dot P_\alpha = \frac{\mathrm{d}}{\mathrm{d}t} I = 0.$$

F0621 A_proofs.tex:110 PROOF_ARCHIVE_REVIEWED

$$\dot\phi = \dot U^* \psi + U^* A \psi.$$

F0622 A_proofs.tex:115 PROOF_ARCHIVE_REVIEWED

$$\dot U^* U + U^* \dot U = 0,$$

F0623 A_proofs.tex:119 PROOF_ARCHIVE_REVIEWED

$$\dot U^* = -U^* \dot U U^*.$$

F0624 A_proofs.tex:123 PROOF_ARCHIVE_REVIEWED

$$\dot\phi = -U^* \dot U \phi + U^* A U \phi.$$

F0625 A_proofs.tex:128 PROOF_ARCHIVE_REVIEWED

$$A_{\mathrm{eff}} = U^* A U - U^* \dot U.$$

F0626 A_proofs.tex:136 PROOF_ARCHIVE_REVIEWED

$$\mathrm{e}^{-\mathrm{i}a\sigma_x} = \cos a\, I - \mathrm{i}\sin a\, \sigma_x$$

F0627 A_proofs.tex:141 PROOF_ARCHIVE_REVIEWED

$$\mathrm{e}^{-\mathrm{i}b\sigma_z} = \cos b\, I - \mathrm{i}\sin b\, \sigma_z.$$

F0628 A_proofs.tex:146 PROOF_ARCHIVE_REVIEWED

$$\operatorname{Tr} U = 2\cos a\cos b - \sin a\sin b\operatorname{Tr}(\sigma_x\sigma_z).$$

F0629 A_proofs.tex:152 PROOF_ARCHIVE_REVIEWED

$$\operatorname{Tr} U = 2\cos a\cos b.$$

F0630 A_proofs.tex:160 PROOF_ARCHIVE_REVIEWED

$$\cos(\omega_\varepsilon \varepsilon) = \cos(m\varepsilon)\cos(k\varepsilon),$$

F0631 A_proofs.tex:165 PROOF_ARCHIVE_REVIEWED

$$1 - \frac{\Omega^2\varepsilon^2}{2} + \left(-\Omega c + \frac{\Omega^4}{24}\right)\varepsilon^4 + \mathcal{O}(\varepsilon^6).$$

F0632 A_proofs.tex:171 PROOF_ARCHIVE_REVIEWED

$$1 - \frac{m^2+k^2}{2}\varepsilon^2 + \left(\frac{m^4+k^4}{24} + \frac{m^2k^2}{4}\right)\varepsilon^4 + \mathcal{O}(\varepsilon^6).$$

F0633 A_proofs.tex:180 PROOF_ARCHIVE_REVIEWED

$$\frac{\Omega^4}{24} = \frac{m^4+k^4+2m^2k^2}{24},$$

F0634 A_proofs.tex:186 PROOF_ARCHIVE_REVIEWED

$$-\Omega c = \frac{m^2k^2}{6},$$

F0635 A_proofs.tex:191 PROOF_ARCHIVE_REVIEWED

$$c = -\frac{m^2k^2}{6\Omega}.$$

F0636 A_proofs.tex:198 PROOF_ARCHIVE_REVIEWED

$$-\sin(\omega_\varepsilon\varepsilon)\varepsilon\omega_\varepsilon' = -\cos(m\varepsilon)\sin(k\varepsilon)\varepsilon.$$

F0637 A_proofs.tex:205 PROOF_ARCHIVE_REVIEWED

$$v_\varepsilon = \frac{k}{\Omega} + \mathcal{O}(\varepsilon^2).$$

F0638 A_proofs.tex:212 PROOF_ARCHIVE_REVIEWED

$$\phi = -k\eta.$$

F0639 A_proofs.tex:216 PROOF_ARCHIVE_REVIEWED

$$\omega = -u^\mu\partial_\mu\phi = -\frac{\partial\phi}{\partial t} = k\frac{\mathrm{d}\eta}{\mathrm{d}t} = \frac{k}{a}.$$

F0640 A_proofs.tex:227 PROOF_ARCHIVE_REVIEWED

$$E_\alpha(a) = \sqrt{m_\alpha^2 + k_\alpha^2/a^2},$$

F0641 A_proofs.tex:231 PROOF_ARCHIVE_REVIEWED

$$E_\alpha'(a) = -\frac{k_\alpha^2}{a^3 E_\alpha}.$$

F0642 A_proofs.tex:236 PROOF_ARCHIVE_REVIEWED

$$H_{\mathcal{P}}(a) = \sum_\alpha w_\alpha E_\alpha(a).$$

F0643 A_proofs.tex:240 PROOF_ARCHIVE_REVIEWED

$$p(a) = -\frac{H'_{\mathcal{P}}(a)}{3a^2}.$$

F0644 A_proofs.tex:244 PROOF_ARCHIVE_REVIEWED

$$p(a) = \sum_{\alpha} \frac{w_\alpha k_\alpha^2}{3a^5 E_\alpha}.$$

F0645 A_proofs.tex:250 PROOF_ARCHIVE_REVIEWED

$$\dot{\rho} = \frac{H'_{\mathcal{P}}\dot{a}}{a^3} - 3\frac{H_{\mathcal{P}}\dot{a}}{a^4} = -3Hp - 3H\rho.$$

F0646 A_proofs.tex:260 PROOF_ARCHIVE_REVIEWED

$$L = -C_G \frac{a\dot{a}^2}{N} - N H_{\mathcal{P}}(a).$$

F0647 A_proofs.tex:264 PROOF_ARCHIVE_REVIEWED

$$C_G \frac{a\dot{a}^2}{N^2} = H_{\mathcal{P}}.$$

F0648 A_proofs.tex:268 PROOF_ARCHIVE_REVIEWED

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{a}} - \frac{\partial L}{\partial a} = 0.$$

F0649 A_proofs.tex:274 PROOF_ARCHIVE_REVIEWED

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(-2C_G\frac{a\dot{a}}{N}\right) = -2C_G\left(\frac{\dot{a}^2 + a\ddot{a}}{N} - \frac{a\dot{a}\dot{N}}{N^2}\right).$$

F0650 A_proofs.tex:285 PROOF_ARCHIVE_REVIEWED

$$\frac{\partial L}{\partial a} = -C_G\frac{\dot{a}^2}{N} - N H'_{\mathcal{P}}$$

F0651 A_proofs.tex:294 PROOF_ARCHIVE_REVIEWED

$$L_d(a,b;N) = -C_G\frac{m(b-a)^2}{hN} - hN H_{\mathcal{P}}(m), \qquad m = \frac{a+b}{2}.$$

F0652 A_proofs.tex:301 PROOF_ARCHIVE_REVIEWED

$$\frac{\partial m}{\partial a} = \frac{1}{2}, \qquad \frac{\partial d}{\partial a} = -1.$$

F0653 A_proofs.tex:307 PROOF_ARCHIVE_REVIEWED

$$\frac{\partial}{\partial a}(md^2) = \frac{1}{2}d^2 - 2md.$$

F0654 A_proofs.tex:312 PROOF_ARCHIVE_REVIEWED

$$D_1 L_d = \frac{C_G}{hN}\left(2md - \frac{1}{2}d^2\right) - \frac{hN}{2}H'_{\mathcal{P}}(m).$$

F0655 A_proofs.tex:318 PROOF_ARCHIVE_REVIEWED

$$\frac{\partial}{\partial b}(md^2) = \frac{1}{2}d^2 + 2md,$$

F0656 A_proofs.tex:327 PROOF_ARCHIVE_REVIEWED

$$W = \mathrm{e}^{X}\mathrm{e}^{Y}\mathrm{e}^{-X}\mathrm{e}^{-Y}.$$

F0657 A_proofs.tex:331 PROOF_ARCHIVE_REVIEWED

$$\log W = [X,Y] + \frac{1}{2}[X+Y,[X,Y]] + \mathcal{O}(\varepsilon^4).$$

F0658 A_proofs.tex:338 PROOF_ARCHIVE_REVIEWED

$$\log W = \varepsilon^2 [A,B] + \frac{\varepsilon^3}{2}\left([A,[A,B]] + [B,[A,B]]\right) + \mathcal{O}(\varepsilon^4).$$

F0659 A_proofs.tex:354 PROOF_ARCHIVE_REVIEWED

$$F_h = F + c_1 h + c_2 h^2 + c_3 h^3 + \mathcal{O}(h^4).$$

F0660 A_proofs.tex:358 PROOF_ARCHIVE_REVIEWED

$$2F_{h/2} - F_h = F - \frac{1}{2}c_2 h^2 - \frac{3}{4}c_3 h^3 + \mathcal{O}(h^4).$$

F0661 A_proofs.tex:363 PROOF_ARCHIVE_REVIEWED

$$G_{h/2} = F - \frac{1}{8}c_2 h^2 - \frac{3}{32}c_3 h^3 + \mathcal{O}(h^4).$$

F0662 A_proofs.tex:368 PROOF_ARCHIVE_REVIEWED

$$\frac{4G_{h/2} - G_h}{3} = F + \frac{1}{8}c_3 h^3 + \mathcal{O}(h^4).$$

F0663 A_proofs.tex:377 PROOF_ARCHIVE_REVIEWED

$$\begin{aligned} W'_\ell &= (G_{v_0} T_{v_{r-1}v_0} G_{v_{r-1}}^{-1}) \cdots (G_{v_1} T_{v_0 v_1} G_{v_0}^{-1}) \\ &= G_{v_0} W_\ell G_{v_0}^{-1}. \end{aligned}$$

F0664 A_proofs.tex:387 PROOF_ARCHIVE_REVIEWED

$$\log W'_\ell = G_{v_0} (\log W_\ell) G_{v_0}^{-1}.$$

F0665 A_proofs.tex:396 PROOF_ARCHIVE_REVIEWED

$$\Phi' = \frac{u' r - u}{r^2},$$

F0666 A_proofs.tex:400 PROOF_ARCHIVE_REVIEWED

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}(r^2 \Phi') = \frac{u''}{r}.$$

F0667 A_proofs.tex:405 PROOF_ARCHIVE_REVIEWED

$$u'' = 4\pi G r \rho.$$

F0668 A_proofs.tex:411 PROOF_ARCHIVE_REVIEWED

$$S[u] = \int_0^R \left[\frac{(u')^2}{2G} + 4\pi r \rho u\right] \mathrm{d}r,$$

F0669 A_proofs.tex:415 PROOF_ARCHIVE_REVIEWED

$$\delta S = \left[\frac{u'}{G}\delta u\right]_0^R + \int_0^R \left[-\frac{u''}{G} + 4\pi r\rho\right] \delta u \, \mathrm{d}r.$$

F0670 A_proofs.tex:425 PROOF_ARCHIVE_REVIEWED

$$S_h = \sum_{i=0}^{N-1} \frac{(u_{i+1} - u_i)^2}{2Gh} + 4\pi h \sum_{i=1}^{N-1} r_i \rho_i u_i,$$

F0671 A_proofs.tex:431 PROOF_ARCHIVE_REVIEWED

$$\frac{(u_i - u_{i-1})^2}{2Gh} + \frac{(u_{i+1} - u_i)^2}{2Gh} + 4\pi h r_i \rho_i u_i.$$

F0672 A_proofs.tex:437 PROOF_ARCHIVE_REVIEWED

$$\frac{2u_i - u_{i-1} - u_{i+1}}{Gh} + 4\pi h r_i \rho_i = 0,$$

F0673 A_proofs.tex:442 PROOF_ARCHIVE_REVIEWED

$$\frac{u_{i+1}-2u_i+u_{i-1}}{h^2}=4\pi G r_i\rho_i.$$

F0674 A_proofs.tex:451 PROOF_ARCHIVE_REVIEWED

$$M_a^{\mathrm{coarse}}=\sum_{i\in C_a}M_i^{\mathrm{fine}}.$$

F0675 A_proofs.tex:456 PROOF_ARCHIVE_REVIEWED

$$M_{\mathrm{enc}}^{\mathrm{coarse}}(r_j)=M_{\mathrm{enc}}^{\mathrm{fine}}(r_j).$$

F0676 A_proofs.tex:461 PROOF_ARCHIVE_REVIEWED

$$g_{\mathrm{coarse}}(r_j)=\frac{GM_{\mathrm{enc}}^{\mathrm{coarse}}(r_j)}{r_j^2}=\frac{GM_{\mathrm{enc}}^{\mathrm{fine}}(r_j)}{r_j^2}=g_{\mathrm{fine}}(r_j).$$

F0677 A_proofs.tex:472 PROOF_ARCHIVE_REVIEWED

$$\mathcal{E}=-p_\mu\xi^\mu$$

F0678 A_proofs.tex:476 PROOF_ARCHIVE_REVIEWED

$$p^\nu\nabla_\nu(p_\mu\xi^\mu)=p^\nu p^\mu\nabla_\nu\xi_\mu=\frac{1}{2}p^\nu p^\mu\left(\nabla_\nu\xi_\mu+\nabla_\mu\xi_\nu\right)=0.$$

F0679 A_proofs.tex:484 PROOF_ARCHIVE_REVIEWED

$$\omega=-p_\mu u^\mu=\frac{\mathcal{E}}{N}.$$

F0680 A_proofs.tex:488 PROOF_ARCHIVE_REVIEWED

$$\frac{\omega_o}{\omega_e}=\frac{N_e}{N_o}.$$

F0681 A_proofs.tex:497 PROOF_ARCHIVE_REVIEWED

$$\Pi_{ij}=\delta_{ij}-\widehat{k}_i\widehat{k}_j,\qquad \widehat{k}_i=\frac{k_i}{|\boldsymbol{k}|}.$$

F0682 A_proofs.tex:503 PROOF_ARCHIVE_REVIEWED

$$\Pi_{ik}\Pi_{kj}=\Pi_{ij}.$$

F0683 A_proofs.tex:507 PROOF_ARCHIVE_REVIEWED

$$(\Lambda^{\mathrm{TT}}s)_{ij}=\left[\frac{1}{2}(\Pi_{ik}\Pi_{jl}+\Pi_{il}\Pi_{jk})-\frac{1}{2}\Pi_{ij}\Pi_{kl}\right]s_{kl}.$$

F0684 A_proofs.tex:515 PROOF_ARCHIVE_REVIEWED

$$\delta_{ij}(\Lambda^{\mathrm{TT}}s)_{ij}=\Pi_{kl}s_{kl}-\frac{1}{2}(2)\Pi_{kl}s_{kl}=0.$$

F0685 A_proofs.tex:526 PROOF_ARCHIVE_REVIEWED

$$h_j^n=F_{j-n}+G_{j+n}$$

F0686 A_proofs.tex:532 PROOF_ARCHIVE_REVIEWED

$$G_j=h_j^0-F_j,$$

F0687 A_proofs.tex:536 PROOF_ARCHIVE_REVIEWED

$$F_{j-1}-F_{j+1}=h_j^1-h_{j+1}^0.$$

F0688 A_proofs.tex:542 PROOF_ARCHIVE_REVIEWED

$$f_{j-n}-f_{j+n}=0.$$

F0689 A_proofs.tex:546 PROOF_ARCHIVE_REVIEWED

$$f_{j+2} = f_j.$$

F0690 A_proofs.tex:550 PROOF_ARCHIVE_REVIEWED

$$F_r \mapsto F_r + \chi_r, \qquad G_r \mapsto G_r - \chi_r, \qquad \chi_{r+2} = \chi_r.$$

F0691 A_proofs.tex:560 PROOF_ARCHIVE_REVIEWED

$$\sum_j (h_j^1 - h_{j+1}^0) = 0$$

F0692 A_proofs.tex:564 PROOF_ARCHIVE_REVIEWED

$$\sum_{j \text{ even}} (h_j^1 - h_{j+1}^0) = 0, \qquad \sum_{j \text{ odd}} (h_j^1 - h_{j+1}^0) = 0$$

F0693 A_proofs.tex:574 PROOF_ARCHIVE_REVIEWED

$$\delta_t^2 h_j^n = c^2 \delta_x^2 h_j^n.$$

F0694 A_proofs.tex:579 PROOF_ARCHIVE_REVIEWED

$$\frac{h_j^{n+1} - h_j^{n-1}}{2\Delta t}$$

F0695 A_proofs.tex:583 PROOF_ARCHIVE_REVIEWED

$$\frac{1}{2\Delta t} \sum_j \left[\left(\frac{h_j^{n+1} - h_j^n}{\Delta t} \right)^2 - \left(\frac{h_j^n - h_j^{n-1}}{\Delta t} \right)^2 \right].$$

F0696 A_proofs.tex:593 PROOF_ARCHIVE_REVIEWED

$$-\frac{c^2}{2\Delta t} \sum_j \left[d_j^{n+1} d_j^n - d_j^n d_j^{n-1} \right],$$

F0697 A_proofs.tex:601 PROOF_ARCHIVE_REVIEWED

$$\mathcal{E}^{n+1/2} - \mathcal{E}^{n-1/2} = 0.$$

F0698 A_proofs.tex:609 PROOF_ARCHIVE_REVIEWED

$$\delta_t^2 h_{A,j}^n - c^2 \delta_x^2 h_{A,j}^n = S_{A,j}^n,$$

F0699 A_proofs.tex:613 PROOF_ARCHIVE_REVIEWED

$$\frac{h}{2} \sum_{A,j} S_{A,j}^n \left(h_{A,j}^{n+1} - h_{A,j}^{n-1} \right).$$

F0700 A_proofs.tex:620 PROOF_ARCHIVE_REVIEWED

$$\mathcal{E}^{n+1/2} - \mathcal{E}^{n-1/2} = \frac{h}{2} \sum_{A,j} S_{A,j}^n \left(h_{A,j}^{n+1} - h_{A,j}^{n-1} \right).$$

F0701 A_proofs.tex:631 PROOF_ARCHIVE_REVIEWED

$$S_F = \sum_{\alpha \in F} P_\alpha.$$

F0702 A_proofs.tex:635 PROOF_ARCHIVE_REVIEWED

$$\|S_F \psi - \psi\| + \|H S_F \psi - H\psi\| \longrightarrow 0.$$

F0703 A_proofs.tex:641 PROOF_ARCHIVE_REVIEWED

$$2 \operatorname{Im} \langle P_\alpha \psi, H\psi \rangle = \lim_{F \nearrow A} 2 \operatorname{Im} \langle P_\alpha \psi, H S_F \psi \rangle$$

$$= \lim_{F \nearrow A} \sum_{\beta \in F} 2 \operatorname{Im} \left\langle P_\alpha \psi, H P_\beta \psi \right\rangle .$$

F0704 A_proofs.tex:658 PROOF_ARCHIVE_REVIEWED

$$S_N X S_N - X = S_N X (S_N - I) + (S_N - I) X.$$

F0705 A_proofs.tex:663 PROOF_ARCHIVE_REVIEWED

$$\| (S_N - I) X \| \to 0.$$

F0706 A_proofs.tex:667 PROOF_ARCHIVE_REVIEWED

$$\| X (S_N - I) \| = \| (S_N - I) X^* \| \to 0.$$

F0707 A_proofs.tex:678 PROOF_ARCHIVE_REVIEWED

$$\beta = \tau_z \otimes I_2, \qquad \alpha_i = \tau_x \otimes \sigma_i.$$

F0708 A_proofs.tex:684 PROOF_ARCHIVE_REVIEWED

$$a = m\Delta, \qquad b = k_x \Delta, \qquad c = k_y \Delta, \qquad d = k_z \Delta.$$

F0709 A_proofs.tex:694 PROOF_ARCHIVE_REVIEWED

$$K = \mathrm{e}^{-\mathrm{i} b \alpha_x / 2} \mathrm{e}^{-\mathrm{i} c \alpha_y / 2} \mathrm{e}^{-\mathrm{i} d \alpha_z} \mathrm{e}^{-\mathrm{i} c \alpha_y / 2} \mathrm{e}^{-\mathrm{i} b \alpha_x / 2}.$$

F0710 A_proofs.tex:703 PROOF_ARCHIVE_REVIEWED

$$K \sim \begin{pmatrix} V & 0 \\ 0 & V^* \end{pmatrix},$$

F0711 A_proofs.tex:710 PROOF_ARCHIVE_REVIEWED

$$V = \mathrm{e}^{-\mathrm{i} b \sigma_x / 2} \mathrm{e}^{-\mathrm{i} c \sigma_y / 2} \mathrm{e}^{-\mathrm{i} d \sigma_z} \mathrm{e}^{-\mathrm{i} c \sigma_y / 2} \mathrm{e}^{-\mathrm{i} b \sigma_x / 2}.$$

F0712 A_proofs.tex:719 PROOF_ARCHIVE_REVIEWED

$$V = a_0 I_2 - \mathrm{i} \boldsymbol{a} \cdot \boldsymbol{\sigma}, \qquad a_0^2 + |\boldsymbol{a}|^2 = 1.$$

F0713 A_proofs.tex:725 PROOF_ARCHIVE_REVIEWED

$$\mathrm{Sc}(V) = \mathrm{Sc}\left(q_y(c/2) q_x(b) q_y(c/2) q_z(d) \right) = \cos b \cos c \cos d.$$

F0714 A_proofs.tex:731 PROOF_ARCHIVE_REVIEWED

$$q_y(c/2) q_x(b) q_y(c/2)$$

F0715 A_proofs.tex:735 PROOF_ARCHIVE_REVIEWED

$$a_0 = \cos b \cos c \cos d.$$

F0716 A_proofs.tex:739 PROOF_ARCHIVE_REVIEWED

$$K = a_0 I_4 - \mathrm{i} A, \qquad A = \tau_x \otimes (\boldsymbol{a} \cdot \boldsymbol{\sigma}),$$

F0717 A_proofs.tex:745 PROOF_ARCHIVE_REVIEWED

$$A^2 = |\boldsymbol{a}|^2 I_4, \qquad \{A, \beta\} = 0, \qquad \operatorname{Tr} A = 0.$$

F0718 A_proofs.tex:753 PROOF_ARCHIVE_REVIEWED

$$M = \mathrm{e}^{-\mathrm{i} a \beta / 2} = \cos(a/2) I_4 - \mathrm{i} \sin(a/2) \beta.$$

F0719 A_proofs.tex:758 PROOF_ARCHIVE_REVIEWED

$$M A M = A.$$

F0720 A_proofs.tex:762 PROOF_ARCHIVE_REVIEWED

$$M^2 = \cos a\, I_4 - \mathrm{i} \sin a\, \beta.$$

F0721 A_proofs.tex:766 PROOF_ARCHIVE_REVIEWED

$$\begin{aligned} U_\Delta = MKM &= a_0 M^2 - \mathrm{i}A \\ &= C_\Delta I_4 - \mathrm{i}B, \end{aligned}$$

F0722 A_proofs.tex:773 PROOF_ARCHIVE_REVIEWED

$$C_\Delta = a_0 \cos a = \cos(m\Delta)\cos(k_x\Delta)\cos(k_y\Delta)\cos(k_z\Delta)$$

F0723 A_proofs.tex:782 PROOF_ARCHIVE_REVIEWED

$$B = a_0 \sin a\, \beta + A.$$

F0724 A_proofs.tex:786 PROOF_ARCHIVE_REVIEWED

$$\begin{aligned} B^2 &= a_0^2 \sin^2 a\, I_4 + A^2 \\ &= \left(a_0^2 \sin^2 a + 1 - a_0^2\right) I_4 \\ &= (1 - C_\Delta^2) I_4. \end{aligned}$$

F0725 A_proofs.tex:793 PROOF_ARCHIVE_REVIEWED

$$+\sqrt{1-C_\Delta^2}, +\sqrt{1-C_\Delta^2}, -\sqrt{1-C_\Delta^2}, -\sqrt{1-C_\Delta^2}.$$

F0726 A_proofs.tex:800 PROOF_ARCHIVE_REVIEWED

$$\det(\lambda I_4 - U_\Delta) = \left(\lambda^2 - 2C_\Delta \lambda + 1\right)^2.$$

F0727 A_proofs.tex:809 PROOF_ARCHIVE_REVIEWED

$$a_0 = m, \qquad a_1 = k_x, \qquad a_2 = k_y, \qquad a_3 = k_z, \qquad E^2 = \sum_{r=0}^{3} a_r^2.$$

F0728 A_proofs.tex:821 PROOF_ARCHIVE_REVIEWED

$$\prod_{r=0}^{3} \cos(a_r \Delta) = 1 - \frac{E^2\Delta^2}{2} + \Delta^4 \left[\frac{1}{24} \sum_r a_r^4 + \frac{1}{4} \sum_{r<s} a_r^2 a_s^2 \right] + \mathcal{O}(\Delta^6).$$

F0729 A_proofs.tex:832 PROOF_ARCHIVE_REVIEWED

$$\cos(\omega_\Delta \Delta) = 1 - \frac{E^2\Delta^2}{2} + \Delta^4 \left[\frac{E^4}{24} - Ec_2 \right] + \mathcal{O}(\Delta^6).$$

F0730 A_proofs.tex:842 PROOF_ARCHIVE_REVIEWED

$$E^4 = \sum_r a_r^4 + 2 \sum_{r<s} a_r^2 a_s^2,$$

F0731 A_proofs.tex:847 PROOF_ARCHIVE_REVIEWED

$$-Ec_2 = \frac{1}{6} \sum_{r<s} a_r^2 a_s^2.$$

F0732 A_proofs.tex:856 PROOF_ARCHIVE_REVIEWED

$$S_{L'^n} G^n = G^n S_{L^n}, \qquad T'^n G^n = G^{n+1} T^n.$$

F0733 A_proofs.tex:862 PROOF_ARCHIVE_REVIEWED

$$T'^n C_{1/2} S_{L'^n} C_{1/2} G^n = G^{n+1} T^n C_{1/2} S_{L^n} C_{1/2}.$$

F0734 A_proofs.tex:869 PROOF_ARCHIVE_REVIEWED

$$\rho_j^{n+1} = |\phi_{j-1,+}|^2 + |\phi_{j+1,-}|^2.$$

F0735 A_proofs.tex:874 PROOF_ARCHIVE_REVIEWED

$$\rho_j^n = |\phi_{j,+}|^2 + |\phi_{j,-}|^2$$

F0736 A_proofs.tex:886 PROOF_ARCHIVE_REVIEWED

$$\rho_n = \frac{1}{2}\pi_n^2 + V(\phi_n).$$

F0737 A_proofs.tex:890 PROOF_ARCHIVE_REVIEWED

$$3(H_{n+1}^2 - H_n^2) = 6\bar{H}(H_{n+1} - H_n) = -3\Delta t \bar{H}\bar{\pi}^2.$$

F0738 A_proofs.tex:896 PROOF_ARCHIVE_REVIEWED

$$\begin{aligned}\rho_{n+1} - \rho_n &= \bar{\pi}(\pi_{n+1} - \pi_n) + \overline{\nabla} V(\phi_{n+1} - \phi_n) \\ &= -3\Delta t \bar{H}\bar{\pi}^2.\end{aligned}$$

F0739 A_proofs.tex:903 PROOF_ARCHIVE_REVIEWED

$$3H_{n+1}^2 - \rho_{n+1} = 3H_n^2 - \rho_n.$$

F0740 A_proofs.tex:912 PROOF_ARCHIVE_REVIEWED

$$\omega_\Delta(\boldsymbol{k}) = E + a\Delta^2 + \mathcal{O}(\Delta^4), \qquad a = -\frac{Q(\boldsymbol{k})}{6E}.$$

F0741 A_proofs.tex:918 PROOF_ARCHIVE_REVIEWED

$$(E', \boldsymbol{k}') = B_{\boldsymbol{v}}(E, \boldsymbol{k}).$$

F0742 A_proofs.tex:922 PROOF_ARCHIVE_REVIEWED

$$B_{\boldsymbol{v}}(\omega_\Delta, \boldsymbol{k}) = (E' + \gamma a \Delta^2, \boldsymbol{k}' - \gamma a \boldsymbol{v} \Delta^2) + \mathcal{O}(\Delta^4).$$

F0743 A_proofs.tex:929 PROOF_ARCHIVE_REVIEWED

$$E(\boldsymbol{k}' - \gamma a \boldsymbol{v}\Delta^2) = E' - \gamma a \frac{\boldsymbol{v}\cdot\boldsymbol{k}'}{E'}\Delta^2 + \mathcal{O}(\Delta^4).$$

F0744 A_proofs.tex:935 PROOF_ARCHIVE_REVIEWED

$$\mathfrak{L}_\Delta = \left[\gamma a\left(1 + \frac{\boldsymbol{v}\cdot\boldsymbol{k}'}{E'}\right) - a(\boldsymbol{k}')\right]\Delta^2 + \mathcal{O}(\Delta^4).$$

F0745 A_proofs.tex:943 PROOF_ARCHIVE_REVIEWED

$$\frac{Q(\boldsymbol{k}') - Q(\boldsymbol{k})}{6E'}.$$

F0746 A_proofs.tex:951 PROOF_ARCHIVE_REVIEWED

$$\dot{\psi}_j = -\mathrm{i}m\beta\psi_j - \frac{\alpha U_j \psi_{j+1}}{2h} + \frac{U_{j-1}^* \alpha \psi_{j-1}}{2h}.$$

F0747 A_proofs.tex:958 PROOF_ARCHIVE_REVIEWED

$$\dot{\rho}_j = -\frac{1}{h}\,\mathrm{Re}(\psi_j^* \alpha U_j \psi_{j+1}) + \frac{1}{h}\,\mathrm{Re}(\psi_{j-1}^* \alpha U_{j-1}\psi_j).$$

F0748 A_proofs.tex:966 PROOF_ARCHIVE_REVIEWED

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{E_j - E_{j-1}}{h} + \rho_j - \bar{\rho}\right] = 0.$$

F0749 A_proofs.tex:973 PROOF_ARCHIVE_REVIEWED

$$\dot{H}_E = h\sum_j E_j \dot{E}_j = h\sum_j E_j J_j.$$

F0750 A_proofs.tex:977 PROOF_ARCHIVE_REVIEWED

$$\dot{H}_{\mathrm{m}} = -h \sum_{j} E_j J_j .$$

F0751 A_proofs.tex:985 PROOF_ARCHIVE_REVIEWED

$$e^{0} = \mathrm{e}^{\phi}\,\mathrm{d}t, \qquad e^{1} = \mathrm{e}^{\phi}\,\mathrm{d}x,$$

F0752 A_proofs.tex:991 PROOF_ARCHIVE_REVIEWED

$$\mathrm{d}e^{a} + \omega^{a}{}_{b} \wedge e^{b} = 0$$

F0753 A_proofs.tex:995 PROOF_ARCHIVE_REVIEWED

$$\omega^{0}{}_{1} = \phi_x\,\mathrm{d}t + \phi_t\,\mathrm{d}x.$$

F0754 A_proofs.tex:999 PROOF_ARCHIVE_REVIEWED

$$\mathrm{d}\omega^{0}{}_{1} = (\phi_{tt} - \phi_{xx})\,\mathrm{d}t \wedge \mathrm{d}x.$$

F0755 A_proofs.tex:1004 PROOF_ARCHIVE_REVIEWED

$$\begin{aligned} W_{\square} &= \prod_{e \subset \partial \square} \exp\left(K \int_{e} \omega^{0}{}_{1} \right) \\ &= \exp\left(K \oint_{\partial \square} \omega^{0}{}_{1} \right) \\ &= \exp\left(K \int_{\square} (\phi_{tt} - \phi_{xx})\,\mathrm{d}t\,\mathrm{d}x \right). \end{aligned}$$

F0756 A_proofs.tex:1012 PROOF_ARCHIVE_REVIEWED

$$R = 2\mathrm{e}^{-2\phi}(\phi_{tt} - \phi_{xx}).$$

F0757 A_proofs.tex:1019 PROOF_ARCHIVE_REVIEWED

$$\Delta_j = \varphi_{j+1} - \varphi_j .$$

F0758 A_proofs.tex:1023 PROOF_ARCHIVE_REVIEWED

$$\frac{h}{2}\dot{w}_j \dot{\varphi}_j^{2} + h w_j \dot{\varphi}_j \ddot{\varphi}_j .$$

F0759 A_proofs.tex:1028 PROOF_ARCHIVE_REVIEWED

$$-\frac{h}{2}\dot{w}_j \dot{\varphi}_j^{2}.$$

F0760 A_proofs.tex:1032 PROOF_ARCHIVE_REVIEWED

$$\frac{h}{2}\dot{v}_j \varphi_j^{2}.$$

F0761 A_proofs.tex:1036 PROOF_ARCHIVE_REVIEWED

$$\frac{c_{j+1/2}}{2h}\Delta_j(\dot{\varphi}_j + \dot{\varphi}_{j+1}) + \frac{\dot{c}_{j+1/2}}{4h}\Delta_j^{2}.$$

F0762 A_proofs.tex:1042 PROOF_ARCHIVE_REVIEWED

$$\dot{e}_j + F_{j+1/2} - F_{j-1/2} = s_j .$$

B_numerical_tables.tex

F0763 B_numerical_tables.tex:281 RERUN_PASS

$$\frac{\omega_o}{\omega_e} = 0.996447590904, \qquad \Delta t = 0.055675087962, \qquad \alpha = 0.019845557534.$$

02_logic_types_units.tex

F0764 02_logic_types_units.tex:7 NEW_EXTENSION_PASS

$$\vdash \mathcal{H} : \mathsf{Hilb}, \tag{A.185}$$
$$\vdash P_\alpha : \mathcal{H} \to \mathcal{H}, \qquad P_\alpha^* = P_\alpha = P_\alpha^2, \tag{A.186}$$
$$\vdash X_{\alpha\beta} := P_\alpha X P_\beta \in P_\alpha \mathcal{B}(\mathcal{H}) P_\beta, \tag{A.187}$$
$$\vdash A \in \mathfrak{u}(\mathcal{H}) \iff A^* = -A, \tag{A.188}$$
$$\vdash H \in \mathfrak{sa}(\mathcal{H}) \iff H^* = H, \tag{A.189}$$
$$\vdash A = -\mathrm{i}H : \mathfrak{sa}(\mathcal{H}) \to \mathfrak{u}(\mathcal{H}). \tag{A.190}$$

F0765 02_logic_types_units.tex:22 NEW_EXTENSION_PASS

$$\mathfrak{A}_{\mathcal{P}} := \left\{ P_\alpha^* = P_\alpha,\ P_\alpha P_\beta = \delta_{\alpha\beta} P_\alpha,\ \sum_\alpha P_\alpha = I \right\}. \tag{A.191}$$

F0766 02_logic_types_units.tex:31 NEW_EXTENSION_PASS

$$\mathfrak{A}_{\mathcal{P}} \vdash X = \sum_{\alpha,\beta} P_\alpha X P_\beta, \tag{A.192}$$
$$\mathfrak{A}_{\mathcal{P}} \wedge A^* = -A \vdash \left\{ \dot{E} = J\mathbf{1},\ J = -J^{\mathsf{T}},\ \mathbf{1}^{\mathsf{T}} \dot{E} = 0 \right\}, \tag{A.193}$$
$$\mathfrak{A}_{\mathcal{P}} \wedge U^*U = I \vdash \left\{ \Gamma := U^* \dot{U},\ \Gamma^* = -\Gamma,\ A_{\mathrm{eff}} := U^* A U - \Gamma \right\}. \tag{A.194}$$

F0767 02_logic_types_units.tex:52 NEW_EXTENSION_PASS

$$R_{A\alpha} := \mathbf{1}_{\{\alpha \in C_A\}}, \tag{A.195}$$

F0768 02_logic_types_units.tex:55 CORRECTED_PASS

$$E^{\mathrm{c}} = RE, \tag{A.196}$$
$$J^{\mathrm{c}} = RJR^{\mathsf{T}}, \tag{A.197}$$
$$R^{\mathsf{T}} \mathbf{1}_{\mathrm{c}} = \mathbf{1}_{\mathrm{f}}, \tag{A.198}$$
$$J = -J^{\mathsf{T}} \Longrightarrow J^{\mathrm{c}} = -(J^{\mathrm{c}})^{\mathsf{T}}, \tag{A.199}$$
$$\dot{E} = J\mathbf{1}_{\mathrm{f}} + s \Longrightarrow \dot{E}^{\mathrm{c}} = J^{\mathrm{c}} \mathbf{1}_{\mathrm{c}} + Rs. \tag{A.200}$$

F0769 02_logic_types_units.tex:66 NEW_EXTENSION_PASS

$$R_{2\leftarrow 0} = R_{2\leftarrow 1} R_{1\leftarrow 0}, \tag{A.201}$$
$$\mathfrak{C}_{2\leftarrow 0} = \mathfrak{C}_{2\leftarrow 1} \circ \mathfrak{C}_{1\leftarrow 0}. \tag{A.202}$$

F0770 02_logic_types_units.tex:73 NEW_EXTENSION_PASS

$$\delta_N S[a, N, \Psi] = 0 \vdash \mathcal{H}[a, \dot{a}, \Psi] = 0, \tag{A.203}$$
$$\delta_{N^i} S[g, N, N^i, \Psi] = 0 \vdash \mathcal{H}_i[g, \dot{g}, \Psi] = 0, \tag{A.204}$$
$$\delta_\Psi S[g, \Psi] = 0 \vdash \mathcal{E}_\Psi[g, \Psi] = 0, \tag{A.205}$$
$$\delta_g S[g, \Psi] = 0 \vdash \mathcal{E}_g[g, \Psi] = 0. \tag{A.206}$$

F0771 02_logic_types_units.tex:84 NEW_EXTENSION_PASS

$$\delta_\chi S = 0 \vdash D_\mu j^\mu = 0, \tag{A.207}$$

$$\delta_\xi S = 0 \vdash \nabla_\mu T^{\mu\nu} = 0, \tag{A.208}$$

$$\nabla_\mu G^{\mu\nu} = 0 \wedge G^{\mu\nu} = 8\pi G T^{\mu\nu} \vdash \nabla_\mu T^{\mu\nu} = 0. \tag{A.209}$$

F0772 02_logic_types_units.tex:93 NEW_EXTENSION_PASS

$$\cos(\omega_\varepsilon \varepsilon) = \cos(m\varepsilon)\cos(k\varepsilon) \vdash \omega_\varepsilon = \sqrt{m^2 + k^2} + \mathcal{O}(\varepsilon^2), \tag{A.210}$$

$$\cos(\omega_\Delta \Delta) = \cos(m\Delta) \prod_i \cos(k_i \Delta) \vdash \omega_\Delta = \sqrt{m^2 + |\boldsymbol{k}|^2} + \mathcal{O}(\Delta^2), \tag{A.211}$$

$$W_\varepsilon = \mathrm{e}^{\varepsilon A_x}\mathrm{e}^{\varepsilon A_y}\mathrm{e}^{-\varepsilon A_x}\mathrm{e}^{-\varepsilon A_y} \vdash \varepsilon^{-2}\log W_\varepsilon = [A_x, A_y] + \mathcal{O}(\varepsilon), \tag{A.212}$$

$$R_0^{(m)}[F(h)] = F + \mathcal{O}(h^{m+1}). \tag{A.213}$$

F0773 02_logic_types_units.tex:112 NEW_EXTENSION_PASS

$$c = \hbar = 1, \tag{A.214}$$

$$[m] = [k_i] = [\omega] = L^{-1}, \tag{A.215}$$

$$[\varepsilon] = [\Delta] = L, \tag{A.216}$$

$$[m\Delta] = [k_i\Delta] = [\omega\Delta] = 1. \tag{A.217}$$

F0774 02_logic_types_units.tex:119 NEW_EXTENSION_PASS

$$[qA_\mu \,\mathrm{d}x^\mu] = 1, \tag{A.218}$$

$$[L_j] = [T_j] = [W_{\mu\nu}] = 1. \tag{A.219}$$

F0775 02_logic_types_units.tex:124 NEW_EXTENSION_PASS

$$[a] = 1, \qquad [H] = T^{-1}, \tag{A.220}$$

$$[p_{\mathrm{phys}}] = [\omega_{\mathrm{phys}}] = T^{-1} \quad (c = \hbar = 1), \tag{A.221}$$

$$\frac{8\pi G}{3} = 1 \quad \text{only where explicitly normalized.} \tag{A.222}$$

F0776 02_logic_types_units.tex:131 CORRECTED_PASS

$$[\Phi] = L^2 T^{-2}, \tag{A.223}$$

$$\left[\frac{\Phi}{c^2}\right] = 1, \tag{A.224}$$

$$[\boldsymbol{\alpha}_\perp] = 1, \tag{A.225}$$

$$[\Delta t_{\mathrm{Shapiro}}] = T. \tag{A.226}$$

03_new_extensions.tex

F0777 03_new_extensions.tex:8 NEW_EXTENSION_PASS

$$\dot\psi = A(\psi,t)\psi, \qquad A(\psi,t)^* = -A(\psi,t), \tag{A.227}$$

$$\psi_\alpha = P_\alpha \psi, \qquad E_\alpha = \langle \psi_\alpha, \psi_\alpha \rangle, \tag{A.228}$$

$$J_{\alpha\leftarrow\beta}(\psi,t) := 2\,\mathrm{Re}\left\langle \psi_\alpha, A(\psi,t)\psi_\beta \right\rangle, \tag{A.229}$$

$$\dot E_\alpha = 2\,\mathrm{Re}\,\langle \psi_\alpha, A(\psi,t)\psi\rangle = \sum_\beta J_{\alpha\leftarrow\beta}, \tag{A.230}$$

$$J_{\alpha\leftarrow\beta} = -J_{\beta\leftarrow\alpha}, \qquad \sum_\alpha \dot E_\alpha = 0. \tag{A.231}$$

F0778 03_new_extensions.tex:23 NEW_EXTENSION_PASS

$$\rho = \rho^* \geq 0, \qquad \mathrm{Tr}\,\rho = 1, \qquad \dot\rho = -\mathrm{i}[H,\rho], \qquad H^* = H, \tag{A.232}$$

$$p_\alpha := \mathrm{Tr}(P_\alpha \rho), \tag{A.233}$$

$$J^{(\rho)}_{\alpha\leftarrow\beta} := 2\,\mathrm{Im}\,\mathrm{Tr}(P_\alpha H P_\beta \rho), \tag{A.234}$$

$$\dot{p}_\alpha = \sum_\beta J^{(\rho)}_{\alpha\leftarrow\beta}, \tag{A.235}$$

$$J^{(\rho)}_{\alpha\leftarrow\beta} = -J^{(\rho)}_{\beta\leftarrow\alpha}, \tag{A.236}$$

$$\sum_\alpha \dot{p}_\alpha = 0. \tag{A.237}$$

F0779 03_new_extensions.tex:38 NEW_EXTENSION_PASS

$$\dot{p}_\alpha = -\mathrm{i}\,\mathrm{Tr}(P_\alpha H\rho - P_\alpha \rho H) \tag{A.238}$$

$$= -\mathrm{i}\sum_\beta \left[\mathrm{Tr}(P_\alpha H P_\beta \rho) - \mathrm{Tr}(P_\beta H P_\alpha \rho)\right]. \tag{A.239}$$

F0780 03_new_extensions.tex:48 NEW_EXTENSION_PASS

$$\mathrm{Tr}(P_\beta H P_\alpha \rho) = \overline{\mathrm{Tr}(P_\alpha H P_\beta \rho)}. \tag{A.240}$$

F0781 03_new_extensions.tex:54 NEW_EXTENSION_PASS

$$J^{(\rho)}_{\alpha\leftarrow\beta} = 2\,\mathrm{Im}\left\langle P_\alpha\psi, H P_\beta \psi\right\rangle. \tag{A.241}$$

F0782 03_new_extensions.tex:61 NEW_EXTENSION_PASS

$$Q_A := \sum_\alpha R_{A\alpha} P_\alpha, \qquad R_{A\alpha}\in\{0,1\}, \tag{A.242}$$

$$\widetilde{Q}_C := \sum_A S_{CA} Q_A, \qquad S_{CA}\in\{0,1\}. \tag{A.243}$$

F0783 03_new_extensions.tex:68 NEW_EXTENSION_PASS

$$T_{C\alpha} := (SR)_{C\alpha}, \tag{A.244}$$

$$\widetilde{Q}_C = \sum_\alpha T_{C\alpha} P_\alpha, \tag{A.245}$$

$$E^{(1)} = RE, \qquad E^{(2)} = SE^{(1)} = (SR)E, \tag{A.246}$$

$$J^{(1)} = RJR^{\mathsf{T}}, \qquad J^{(2)} = SJ^{(1)}S^{\mathsf{T}} = (SR)J(SR)^{\mathsf{T}}. \tag{A.247}$$

F0784 03_new_extensions.tex:78 NEW_EXTENSION_PASS

$$\mathfrak{C}_{P\to\widetilde{Q}} = \mathfrak{C}_{Q\to\widetilde{Q}} \circ \mathfrak{C}_{P\to Q}. \tag{A.248}$$

F0785 03_new_extensions.tex:86 NEW_EXTENSION_PASS

$$R_{\alpha\mu} := P_\alpha \otimes Q_\mu, \tag{A.249}$$

$$R_{\alpha\mu} R_{\beta\nu} = \delta_{\alpha\beta}\delta_{\mu\nu} R_{\alpha\mu}, \tag{A.250}$$

$$\sum_{\alpha,\mu} R_{\alpha\mu} = I_A \otimes I_B, \tag{A.251}$$

$$p_{\alpha\mu} := \mathrm{Tr}(R_{\alpha\mu}\rho_{AB}), \tag{A.252}$$

$$p^A_\alpha = \sum_\mu p_{\alpha\mu} = \mathrm{Tr}[(P_\alpha \otimes I_B)\rho_{AB}], \tag{A.253}$$

$$p^B_\mu = \sum_\alpha p_{\alpha\mu} = \mathrm{Tr}[(I_A \otimes Q_\mu)\rho_{AB}]. \tag{A.254}$$

F0786 03_new_extensions.tex:100 NEW_EXTENSION_PASS

$$\dot{q}_\alpha = s_\alpha + \sum_\beta F_{\alpha\beta}, \qquad F_{\alpha\beta} = -F_{\beta\alpha}, \tag{A.255}$$

$$q_C := \sum_{\alpha\in C} q_\alpha, \qquad s_C := \sum_{\alpha\in C} s_\alpha, \tag{A.256}$$

$$\dot{q}_C = s_C + \sum_{\substack{\alpha\in C\\ \beta\notin C}} F_{\alpha\beta}. \tag{A.257}$$

F0787 03_new_extensions.tex:109 NEW_EXTENSION_PASS

$$F^{\mathrm{c}}_{AB} := \sum_{\substack{\alpha\in C_A\\ \beta\in C_B}} F_{\alpha\beta}, \tag{A.258}$$

$$F^{\mathrm{c}}_{AB} = -F^{\mathrm{c}}_{BA}, \tag{A.259}$$

$$\dot{q}^{\mathrm{c}}_A = s^{\mathrm{c}}_A + \sum_B F^{\mathrm{c}}_{AB}. \tag{A.260}$$

F0788 03_new_extensions.tex:118 NEW_EXTENSION_PASS

$$p_\alpha > 0, \qquad \sum_\alpha p_\alpha = 1, \qquad \dot{p}_\alpha = \sum_\beta J_{\alpha\beta}, \qquad J_{\alpha\beta} = -J_{\beta\alpha}. \tag{A.261}$$

F0789 03_new_extensions.tex:124 NEW_EXTENSION_PASS

$$S[p] := -\sum_\alpha p_\alpha \log p_\alpha, \tag{A.262}$$

$$\dot{S} = -\sum_\alpha (1+\log p_\alpha)\dot{p}_\alpha \tag{A.263}$$

$$= -\frac{1}{2}\sum_{\alpha,\beta} J_{\alpha\beta}\log\frac{p_\alpha}{p_\beta}. \tag{A.264}$$

F0790 03_new_extensions.tex:133 NEW_EXTENSION_PASS

$$-\sum_{\alpha,\beta} J_{\alpha\beta}\log p_\alpha = -\frac{1}{2}\sum_{\alpha,\beta}\left(J_{\alpha\beta}\log p_\alpha + J_{\beta\alpha}\log p_\beta\right) \tag{A.265}$$

$$= -\frac{1}{2}\sum_{\alpha,\beta} J_{\alpha\beta}\left(\log p_\alpha - \log p_\beta\right). \tag{A.266}$$

F0791 03_new_extensions.tex:142 NEW_EXTENSION_PASS

$$M_q[p] := \sum_\alpha p_\alpha^q, \tag{A.267}$$

$$\dot{M}_q = \frac{q}{2}\sum_{\alpha,\beta} J_{\alpha\beta}\left(p_\alpha^{q-1} - p_\beta^{q-1}\right), \tag{A.268}$$

$$H_q[p] := \frac{1}{1-q}\log M_q[p], \tag{A.269}$$

$$\dot{H}_q = \frac{q}{2(1-q)M_q[p]}\sum_{\alpha,\beta} J_{\alpha\beta}\left(p_\alpha^{q-1} - p_\beta^{q-1}\right). \tag{A.270}$$

F0792 03_new_extensions.tex:156 NEW_EXTENSION_PASS

$$L_{\alpha\beta} := \sqrt{\gamma_{\alpha\beta}}\,|\alpha\rangle\langle\beta|, \qquad \gamma_{\alpha\beta}\ge 0, \tag{A.271}$$

$$\dot{\rho} = \sum_{\alpha\neq\beta}\left(L_{\alpha\beta}\rho L_{\alpha\beta}^* - \frac{1}{2}\{L_{\alpha\beta}^* L_{\alpha\beta},\rho\}\right), \tag{A.272}$$

$$p_\alpha := \rho_{\alpha\alpha}, \tag{A.273}$$

$$\dot{p}_\alpha = \sum_\beta\left(\gamma_{\alpha\beta}p_\beta - \gamma_{\beta\alpha}p_\alpha\right), \tag{A.274}$$

$$K_{\alpha\beta} := \gamma_{\alpha\beta}p_\beta - \gamma_{\beta\alpha}p_\alpha, \qquad K_{\alpha\beta} = -K_{\beta\alpha}, \tag{A.275}$$

$$\dot{p}_\alpha = \sum_\beta K_{\alpha\beta}. \tag{A.276}$$

F0793 03_new_extensions.tex:178 NEW_EXTENSION_PASS

$$L_{\alpha\beta}\rho L_{\alpha\beta}^* = \gamma_{\alpha\beta}p_\beta|\alpha\rangle\langle\alpha|, \tag{A.277}$$

$$L_{\alpha\beta}^* L_{\alpha\beta} = \gamma_{\alpha\beta}|\beta\rangle\langle\beta|. \tag{A.278}$$

F0794 03_new_extensions.tex:188 NEW_EXTENSION_PASS

$$\gamma_{\alpha\beta}\pi_\beta = \gamma_{\beta\alpha}\pi_\alpha, \qquad \pi_\alpha > 0, \qquad \sum_\alpha \pi_\alpha = 1, \tag{A.279}$$

$$D(p\|\pi) := \sum_\alpha p_\alpha \log\frac{p_\alpha}{\pi_\alpha}, \tag{A.280}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}D(p\|\pi) = -\frac{1}{2}\sum_{\alpha,\beta}\left(\gamma_{\alpha\beta}p_\beta - \gamma_{\beta\alpha}p_\alpha\right)\log\frac{\gamma_{\alpha\beta}p_\beta}{\gamma_{\beta\alpha}p_\alpha} \tag{A.281}$$

$$\le 0, \tag{A.282}$$

F0795 03_new_extensions.tex:204 NEW_EXTENSION_PASS

$$\log\frac{p_\alpha/\pi_\alpha}{p_\beta/\pi_\beta} = \log\frac{\gamma_{\beta\alpha}p_\alpha}{\gamma_{\alpha\beta}p_\beta}. \tag{A.283}$$

F0796 03_new_extensions.tex:212 NEW_EXTENSION_PASS

$$(x-y)\log\frac{x}{y} \ge 0 \qquad (x,y>0). \tag{A.284}$$

F0797 03_new_extensions.tex:218 NEW_EXTENSION_PASS

$$F(h) = F + \sum_{r\ge 1} c_r h^r, \tag{A.285}$$

$$R_j^{(0)} := F\!\left(\frac{h}{2^j}\right), \tag{A.286}$$

$$R_j^{(m)} := \frac{2^m R_{j+1}^{(m-1)} - R_j^{(m-1)}}{2^m - 1}, \qquad m \ge 1, \tag{A.287}$$

$$R_0^{(m)} = F + \mathcal{O}(h^{m+1}). \tag{A.288}$$

F0798 03_new_extensions.tex:227 NEW_EXTENSION_PASS

$$2F(h/2) - F(h) = 2\left(F + \frac{c_1 h}{2} + \frac{c_2 h^2}{4} + \cdots\right) - \left(F + c_1 h + c_2 h^2 + \cdots\right) \tag{A.289}$$

$$= F - \frac{1}{2}c_2 h^2 + \mathcal{O}(h^3), \tag{A.290}$$

F0799 03_new_extensions.tex:235 NEW_EXTENSION_PASS

$$R_0^{(1)} = 2F(h/2) - F(h), \tag{A.291}$$

$$R_0^{(2)} = \frac{8F(h/4) - 6F(h/2) + F(h)}{3}, \tag{A.292}$$

$$R_0^{(3)} = \frac{64F(h/8) - 56F(h/4) + 14F(h/2) - F(h)}{21}. \tag{A.293}$$

12_positive_operator_atlases.tex

F0800 12_positive_operator_atlases.tex:8 NEW_EXTENSION_PASS

$$F_\alpha = F_\alpha^* \geq 0, \qquad \sum_\alpha F_\alpha = I, \tag{A.294}$$

$$p_\alpha := \langle \psi, F_\alpha \psi \rangle , \qquad \dot{\psi} = A\psi, \qquad A^* = -A. \tag{A.295}$$

F0801 12_positive_operator_atlases.tex:16 NEW_EXTENSION_PASS

$$J^{(F)}_{\alpha \leftarrow \beta} := \mathrm{Re} \left\langle F_\alpha \psi, A F_\beta \psi \right\rangle - \mathrm{Re} \left\langle F_\beta \psi, A F_\alpha \psi \right\rangle . \tag{A.296}$$

F0802 12_positive_operator_atlases.tex:22 NEW_EXTENSION_PASS

$$J^{(F)}_{\alpha \leftarrow \beta} = -J^{(F)}_{\beta \leftarrow \alpha}, \tag{A.297}$$

$$\dot{p}_\alpha = 2 \,\mathrm{Re} \langle F_\alpha \psi, A\psi \rangle , \tag{A.298}$$

$$\sum_\beta J^{(F)}_{\alpha \leftarrow \beta} = \mathrm{Re} \langle F_\alpha \psi, A\psi \rangle - \mathrm{Re} \langle \psi, A F_\alpha \psi \rangle \tag{A.299}$$

$$= 2 \,\mathrm{Re} \langle F_\alpha \psi, A\psi \rangle = \dot{p}_\alpha, \tag{A.300}$$

$$\sum_\alpha \dot{p}_\alpha = 0. \tag{A.301}$$

F0803 12_positive_operator_atlases.tex:35 NEW_EXTENSION_PASS

$$\sum_\beta \mathrm{Re} \left\langle F_\alpha \psi, A F_\beta \psi \right\rangle = \mathrm{Re} \langle F_\alpha \psi, A\psi \rangle , \tag{A.302}$$

$$\sum_\beta \mathrm{Re} \left\langle F_\beta \psi, A F_\alpha \psi \right\rangle = \mathrm{Re} \langle \psi, A F_\alpha \psi \rangle . \tag{A.303}$$

F0804 12_positive_operator_atlases.tex:45 NEW_EXTENSION_PASS

$$F_\alpha = F_\alpha(t), \qquad \sum_\alpha F_\alpha(t) = I, \tag{A.304}$$

$$s_\alpha := \left\langle \psi, \dot{F}_\alpha \psi \right\rangle , \qquad \sum_\alpha s_\alpha = 0, \tag{A.305}$$

F0805 12_positive_operator_atlases.tex:52 NEW_EXTENSION_PASS

$$\dot{p}_\alpha = s_\alpha + \sum_\beta J^{(F)}_{\alpha \leftarrow \beta}. \tag{A.306}$$

F0806 12_positive_operator_atlases.tex:57 NEW_EXTENSION_PASS

$$F_\alpha(t) = U(t) F_\alpha(0) U(t)^*, \tag{A.307}$$

$$K := \dot{U} U^*, \qquad K^* = -K, \tag{A.308}$$

F0807 12_positive_operator_atlases.tex:63 NEW_EXTENSION_PASS

$$\dot{F}_\alpha = [K, F_\alpha], \qquad s_\alpha = \langle \psi, [K, F_\alpha] \psi \rangle . \tag{A.309}$$

F0808 12_positive_operator_atlases.tex:71 NEW_EXTENSION_PASS

$$G_A := \sum_\alpha R_{A\alpha} F_\alpha, \qquad R_{A\alpha} \in \{0, 1\}, \qquad R^{\mathsf{T}} \mathbf{1}_{\mathrm{c}} = \mathbf{1}_{\mathrm{f}}. \tag{A.310}$$

F0809 12_positive_operator_atlases.tex:77 NEW_EXTENSION_PASS

$$p^{\mathrm{c}} = R p, \tag{A.311}$$

$$s^{\mathrm{c}} = R s, \tag{A.312}$$

$$J^{\mathrm{c}} = R J^{(F)} R^{\mathsf{T}}, \tag{A.313}$$

$$\dot{p}^{\mathrm{c}} = s^{\mathrm{c}} + J^{\mathrm{c}} \mathbf{1}_{\mathrm{c}}. \tag{A.314}$$

F0810 12_positive_operator_atlases.tex:86 NEW_EXTENSION_PASS

$$V_\alpha : \mathcal{H} \to \mathcal{K}_\alpha, \qquad \sum_\alpha V_\alpha^* V_\alpha = I, \qquad F_\alpha := V_\alpha^* V_\alpha. \tag{A.315}$$

F0811 12_positive_operator_atlases.tex:94 NEW_EXTENSION_PASS

$$p_\alpha = \|V_\alpha \psi\|^2 , \tag{A.316}$$

$$\psi = \sum_\alpha V_\alpha^* V_\alpha \psi, \tag{A.317}$$

$$X = \sum_{\alpha,\beta} V_\alpha^* \big(V_\alpha X V_\beta^*\big) V_\beta. \tag{A.318}$$

F0812 12_positive_operator_atlases.tex:101 NEW_EXTENSION_PASS

$$X = \left(\sum_\alpha V_\alpha^* V_\alpha\right) X \left(\sum_\beta V_\beta^* V_\beta\right) \tag{A.319}$$

$$= \sum_{\alpha,\beta} V_\alpha^* V_\alpha X V_\beta^* V_\beta. \tag{A.320}$$

F0813 12_positive_operator_atlases.tex:111 NEW_EXTENSION_PASS

$$\mathcal{K} := \bigoplus_\alpha \mathcal{K}_\alpha, \tag{A.321}$$

$$V : \mathcal{H} \to \mathcal{K}, \qquad V\psi := (V_\alpha \psi)_\alpha, \tag{A.322}$$

$$\Pi_\alpha := \text{coordinate projector onto } \mathcal{K}_\alpha. \tag{A.323}$$

F0814 12_positive_operator_atlases.tex:118 NEW_EXTENSION_PASS

$$V^* V = I, \qquad F_\alpha = V^* \Pi_\alpha V. \tag{A.324}$$

F0815 12_positive_operator_atlases.tex:124 NEW_EXTENSION_PASS

$$J^{(F)}_{\alpha \leftarrow \beta} = 2 \operatorname{Re} \left\langle P_\alpha \psi, A P_\beta \psi \right\rangle . \tag{A.325}$$

F0816 12_positive_operator_atlases.tex:131 NEW_EXTENSION_PASS

$$K^{\mathsf{T}} = -K, \qquad K\mathbf{1} = 0, \tag{A.326}$$

F0817 12_positive_operator_atlases.tex:137 NEW_EXTENSION_PASS

$$J' := J^{(F)} + K \Longrightarrow J'\mathbf{1} = J^{(F)}\mathbf{1} = \dot{p}. \tag{A.327}$$

F0818 12_positive_operator_atlases.tex:145 NEW_EXTENSION_PASS

$$\left\| \textstyle\sum_\alpha F_\alpha - I \right\|_2 = 4.216049 \times 10^{-15}, \tag{A.328}$$

$$\left\| \dot{p} - J^{(F)}\mathbf{1} \right\|_\infty = 4.163336 \times 10^{-17}, \tag{A.329}$$

$$\left\| J^{(F)} + (J^{(F)})^{\mathsf{T}} \right\|_\infty = 0, \tag{A.330}$$

$$\left\| J^{\mathrm{c}} - R J^{(F)} R^{\mathsf{T}} \right\|_\infty = 6.938894 \times 10^{-18}, \tag{A.331}$$

$$\left\| \dot{p} - s - J^{(F)}\mathbf{1} \right\|_\infty = 6.158268 \times 10^{-17}, \tag{A.332}$$

$$\|V^* V - I\|_2 = 4.750450 \times 10^{-15}, \tag{A.333}$$

$$\max_\alpha \|F_\alpha - V^* \Pi_\alpha V\|_2 = 1.080132 \times 10^{-15}, \tag{A.334}$$

$$\|K\mathbf{1}\|_\infty = 4.440892 \times 10^{-16}, \tag{A.335}$$

$$\left\| \dot{p} - (J^{(F)} + K)\mathbf{1} \right\|_\infty = 5.308254 \times 10^{-16}. \tag{A.336}$$

13_su2_nonabelian_refinement.tex

F0819 13_su2_nonabelian_refinement.tex:8 NEW_EXTENSION_PASS

$$\psi_x \in \mathbb{C}^2, \qquad U_{x,\mu} \in SU(2), \qquad G_x \in SU(2), \tag{A.337}$$

$$\psi'_x = G_x \psi_x, \tag{A.338}$$

$$U'_{x,\mu} = G_x U_{x,\mu} G^{-1}_{x+\widehat{\mu}}. \tag{A.339}$$

F0820 13_su2_nonabelian_refinement.tex:16 NEW_EXTENSION_PASS

$$D^+_\mu \psi_x := \frac{U_{x,\mu}\psi_{x+\widehat{\mu}} - \psi_x}{h}, \tag{A.340}$$

$$(D^+_\mu \psi_x)' = G_x D^+_\mu \psi_x, \tag{A.341}$$

$$\Delta_U \psi_x := \frac{1}{h^2}\sum_\mu \left(U_{x,\mu}\psi_{x+\widehat{\mu}} + U^{-1}_{x-\widehat{\mu},\mu}\psi_{x-\widehat{\mu}} - 2\psi_x \right), \tag{A.342}$$

$$(\Delta_U \psi_x)' = G_x \Delta_U \psi_x. \tag{A.343}$$

F0821 13_su2_nonabelian_refinement.tex:30 NEW_EXTENSION_PASS

$$(D^+_\mu \psi_x)' = \frac{G_x U_{x,\mu} G^{-1}_{x+\widehat{\mu}} G_{x+\widehat{\mu}}\psi_{x+\widehat{\mu}} - G_x\psi_x}{h} \tag{A.344}$$

$$= G_x D^+_\mu \psi_x. \tag{A.345}$$

F0822 13_su2_nonabelian_refinement.tex:42 NEW_EXTENSION_PASS

$$W_{x,\mu\nu} := U_{x,\mu} U_{x+\widehat{\mu},\nu} U^{-1}_{x+\widehat{\nu},\mu} U^{-1}_{x,\nu}, \tag{A.346}$$

$$W'_{x,\mu\nu} = G_x W_{x,\mu\nu} G_x^{-1}, \tag{A.347}$$

$$S_W[U] := \frac{1}{g^2}\sum_{x,\mu<\nu}\left(1 - \frac{1}{2}\operatorname{Re}\operatorname{Tr} W_{x,\mu\nu}\right), \tag{A.348}$$

$$S_W[U'] = S_W[U]. \tag{A.349}$$

F0823 13_su2_nonabelian_refinement.tex:53 NEW_EXTENSION_PASS

$$W'_{x,\mu\nu} = G_x U_{x,\mu} U_{x+\widehat{\mu},\nu} U^{-1}_{x+\widehat{\nu},\mu} U^{-1}_{x,\nu} G_x^{-1}. \tag{A.350}$$

F0824 13_su2_nonabelian_refinement.tex:65 NEW_EXTENSION_PASS

$$j_{x,\mu} := 2\operatorname{Im}\left(\psi^*_x U_{x,\mu}\psi_{x+\widehat{\mu}}\right), \tag{A.351}$$

$$j'_{x,\mu} = j_{x,\mu}. \tag{A.352}$$

F0825 13_su2_nonabelian_refinement.tex:72 NEW_EXTENSION_PASS

$$(\psi'_x)^* U'_{x,\mu}\psi'_{x+\widehat{\mu}} = \psi^*_x G^*_x G_x U_{x,\mu} G^{-1}_{x+\widehat{\mu}} G_{x+\widehat{\mu}}\psi_{x+\widehat{\mu}} \tag{A.353}$$

$$= \psi^*_x U_{x,\mu}\psi_{x+\widehat{\mu}}. \tag{A.354}$$

F0826 13_su2_nonabelian_refinement.tex:81 NEW_EXTENSION_PASS

$$q^a_x := \psi^*_x \sigma^a \psi_x, \tag{A.355}$$

$$G^*_x \sigma^a G_x = R_x{}^a{}_b \sigma^b, \tag{A.356}$$

$$q'^a_x = R_x{}^a{}_b q^b_x. \tag{A.357}$$

F0827 13_su2_nonabelian_refinement.tex:89 NEW_EXTENSION_PASS

$$W_{ij} := X_{ij} Y_{i+1,j} X^{-1}_{i,j+1} Y^{-1}_{ij}, \qquad i,j \in \{0,1\}. \tag{A.358}$$

F0828 13_su2_nonabelian_refinement.tex:94 NEW_EXTENSION_PASS

$$R_0 := \left(X_{00} W_{10} X_{00}^{-1}\right) W_{00}, \tag{A.359}$$

$$R_1 := \left(X_{01} W_{11} X_{01}^{-1}\right) W_{01}. \tag{A.360}$$

F0829 13_su2_nonabelian_refinement.tex:99 NEW_EXTENSION_PASS

$$W^{(2)} := X_{00} X_{10} Y_{20} Y_{21} X_{12}^{-1} X_{02}^{-1} Y_{01}^{-1} Y_{00}^{-1} \tag{A.361}$$

$$= R_0 Y_{00} R_1 Y_{00}^{-1}. \tag{A.362}$$

F0830 13_su2_nonabelian_refinement.tex:110 NEW_EXTENSION_PASS

$$W_h := \mathrm{e}^{hA} \mathrm{e}^{hB} \mathrm{e}^{-hA} \mathrm{e}^{-hB}, \tag{A.363}$$

$$F_h := \frac{1}{h^2} \log W_h, \tag{A.364}$$

$$F_h = [A, B] + \mathcal{O}(h), \tag{A.365}$$

$$F_h^{[1]} := 2F_{h/2} - F_h = [A, B] + \mathcal{O}(h^2). \tag{A.366}$$

F0831 13_su2_nonabelian_refinement.tex:117 NEW_EXTENSION_PASS

$$\log W_h = h^2 [A, B] + h^3 C_3(A, B) + \mathcal{O}(h^4). \tag{A.367}$$

F0832 13_su2_nonabelian_refinement.tex:123 NEW_EXTENSION_PASS

$$F_h = [A, B] + hC_3 + \mathcal{O}(h^2). \tag{A.368}$$

F0833 13_su2_nonabelian_refinement.tex:127 NEW_EXTENSION_PASS

$$F_{h/2} = [A, B] + \frac{h}{2} C_3 + \mathcal{O}(h^2), \tag{A.369}$$

F0834 13_su2_nonabelian_refinement.tex:132 NEW_EXTENSION_PASS

$$F_h[GAG^{-1}, GBG^{-1}] = G F_h[A, B] G^{-1}. \tag{A.370}$$

F0835 13_su2_nonabelian_refinement.tex:137 NEW_EXTENSION_PASS

$$\left\| (\Delta_{U'} G\psi) - G\Delta_U \psi \right\|_\infty = 6.394885 \times 10^{-14}, \tag{A.371}$$

$$\left\| W' - GWG^{-1} \right\|_\infty = 5.661049 \times 10^{-16}, \tag{A.372}$$

$$\left\| W^{(2)} - R_0 Y_{00} R_1 Y_{00}^{-1} \right\|_2 = 6.214507 \times 10^{-16}, \tag{A.373}$$

$$p_{F_h} = 0.993501, \tag{A.374}$$

$$p_{F_h^{[1]}} = 1.992340. \tag{A.375}$$

14_gowdy_reduced_gravity.tex

F0836 14_gowdy_reduced_gravity.tex:8 NEW_EXTENSION_PASS

$$S_G[P, Q] := \frac{1}{2} \int \left[P_\tau^2 - \mathrm{e}^{-2\tau} P_\theta^2 + \mathrm{e}^{2P} \left(Q_\tau^2 - \mathrm{e}^{-2\tau} Q_\theta^2 \right) \right] \mathrm{d}\tau \, \mathrm{d}\theta. \tag{A.376}$$

F0837 14_gowdy_reduced_gravity.tex:17 NEW_EXTENSION_PASS

$$\delta_P S_G = \int \left[P_\tau \delta P_\tau - \mathrm{e}^{-2\tau} P_\theta \delta P_\theta + \mathrm{e}^{2P} (Q_\tau^2 - \mathrm{e}^{-2\tau} Q_\theta^2) \delta P \right] \mathrm{d}\tau \, \mathrm{d}\theta \tag{A.377}$$

$$= - \int \left[P_{\tau\tau} - \mathrm{e}^{-2\tau} P_{\theta\theta} - \mathrm{e}^{2P} (Q_\tau^2 - \mathrm{e}^{-2\tau} Q_\theta^2) \right] \delta P \, \mathrm{d}\tau \, \mathrm{d}\theta. \tag{A.378}$$

F0838 14_gowdy_reduced_gravity.tex:31 NEW_EXTENSION_PASS

$$\partial_\tau(\mathrm{e}^{2P}Q_\tau)-\partial_\theta(\mathrm{e}^{2P-2\tau}Q_\theta)=0. \tag{A.379}$$

F0839 14_gowdy_reduced_gravity.tex:36 NEW_EXTENSION_PASS

$$Q_{\tau\tau}-\mathrm{e}^{-2\tau}Q_{\theta\theta}+2(P_\tau Q_\tau-\mathrm{e}^{-2\tau}P_\theta Q_\theta)=0. \tag{A.380}$$

F0840 14_gowdy_reduced_gravity.tex:41 NEW_EXTENSION_PASS

$$\mathcal{E}_P:=P_{\tau\tau}-\mathrm{e}^{-2\tau}P_{\theta\theta}-\mathrm{e}^{2P}\left(Q_\tau^2-\mathrm{e}^{-2\tau}Q_\theta^2\right), \tag{A.381}$$

$$\mathcal{E}_Q:=Q_{\tau\tau}-\mathrm{e}^{-2\tau}Q_{\theta\theta}+2\left(P_\tau Q_\tau-\mathrm{e}^{-2\tau}P_\theta Q_\theta\right), \tag{A.382}$$

$$\delta_P S_G=0\Longleftrightarrow\mathcal{E}_P=0, \tag{A.383}$$

$$\delta_Q S_G=0\Longleftrightarrow\mathcal{E}_Q=0. \tag{A.384}$$

F0841 14_gowdy_reduced_gravity.tex:53 NEW_EXTENSION_PASS

$$\lambda_\tau:=P_\tau^2+\mathrm{e}^{-2\tau}P_\theta^2+\mathrm{e}^{2P}\left(Q_\tau^2+\mathrm{e}^{-2\tau}Q_\theta^2\right), \tag{A.385}$$

$$\lambda_\theta:=2\left(P_\tau P_\theta+\mathrm{e}^{2P}Q_\tau Q_\theta\right). \tag{A.386}$$

F0842 14_gowdy_reduced_gravity.tex:61 NEW_EXTENSION_PASS

$$\partial_\tau\lambda_\theta-\partial_\theta\lambda_\tau=2P_\theta(P_{\tau\tau}-\mathrm{e}^{-2\tau}P_{\theta\theta}) \tag{A.387}$$

$$+2\mathrm{e}^{2P}\Big[Q_\theta(Q_{\tau\tau}-\mathrm{e}^{-2\tau}Q_{\theta\theta}+2P_\tau Q_\tau) \tag{A.388}$$

$$-P_\theta Q_\tau^2-\mathrm{e}^{-2\tau}P_\theta Q_\theta^2\Big]. \tag{A.389}$$

F0843 14_gowdy_reduced_gravity.tex:72 NEW_EXTENSION_PASS

$$\boxed{\partial_\tau\lambda_\theta-\partial_\theta\lambda_\tau=2P_\theta\mathcal{E}_P+2\mathrm{e}^{2P}Q_\theta\mathcal{E}_Q.} \tag{A.390}$$

F0844 14_gowdy_reduced_gravity.tex:79 NEW_EXTENSION_PASS

$$\mathcal{E}_P=\mathcal{E}_Q=0\Longrightarrow\partial_\tau\lambda_\theta=\partial_\theta\lambda_\tau. \tag{A.391}$$

F0845 14_gowdy_reduced_gravity.tex:85 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\int_0^{2\pi}\lambda_\theta\,\mathrm{d}\theta=\int_0^{2\pi}\partial_\theta\lambda_\tau\,\mathrm{d}\theta \tag{A.392}$$

$$=\lambda_\tau(2\pi)-\lambda_\tau(0)=0. \tag{A.393}$$

F0846 14_gowdy_reduced_gravity.tex:92 NEW_EXTENSION_PASS

$$\int_0^{2\pi}\lambda_\theta(\tau,\theta)\,\mathrm{d}\theta=0. \tag{A.394}$$

F0847 14_gowdy_reduced_gravity.tex:98 NEW_EXTENSION_PASS

$$P_{\tau\tau}-\mathrm{e}^{-2\tau}P_{\theta\theta}=0. \tag{A.395}$$

F0848 14_gowdy_reduced_gravity.tex:102 NEW_EXTENSION_PASS

$$P(\tau,\theta)=AJ_0(x)\cos(n\theta). \tag{A.396}$$

F0849 14_gowdy_reduced_gravity.tex:106 NEW_EXTENSION_PASS

$$P_\tau=AxJ_1(x)\cos(n\theta). \tag{A.397}$$

F0850 14_gowdy_reduced_gravity.tex:110 NEW_EXTENSION_PASS

$$P_{\tau\tau}=A\left[x^2J_0''(x)+xJ_0'(x)\right]\cos(n\theta). \tag{A.398}$$

F0851 14_gowdy_reduced_gravity.tex:115 NEW_EXTENSION_PASS

$$x^2 J_0'' + x J_0' + x^2 J_0 = 0 \tag{A.399}$$

F0852 14_gowdy_reduced_gravity.tex:119 NEW_EXTENSION_PASS

$$P_{\tau\tau} = -x^2 P = -n^2 \mathrm{e}^{-2\tau} P, \tag{A.400}$$

$$P_{\theta\theta} = -n^2 P, \tag{A.401}$$

$$P_{\tau\tau} - \mathrm{e}^{-2\tau} P_{\theta\theta} = 0. \tag{A.402}$$

F0853 14_gowdy_reduced_gravity.tex:126 NEW_EXTENSION_PASS

$$y := \frac{x^2}{4} = \frac{n^2 \mathrm{e}^{-2\tau}}{4} \tag{A.403}$$

F0854 14_gowdy_reduced_gravity.tex:131 NEW_EXTENSION_PASS

$$\mathcal{B}(y) := J_0(2\sqrt{y}) = {}_0F_1\left(\begin{matrix}-\\ 1\end{matrix}; -y\right) = \sum_{m=0}^{\infty} \frac{(-1)^m y^m}{(m!)^2}. \tag{A.404}$$

F0855 14_gowdy_reduced_gravity.tex:140 NEW_EXTENSION_PASS

eq:gowdy-bessel-lane

$$\boxed{\mathcal{B}_{N,j}(y) := \sum_{q=0}^{\infty} \frac{(-1)^{Nq+j} y^{Nq+j}}{((Nq+j)!)^2}.} \tag{A.405}$$

F0856 14_gowdy_reduced_gravity.tex:150 NEW_EXTENSION_PASS

$$((Nq+j)!)^2 = (j!)^2 N^{2Nq} (q!)^2 \prod_{\alpha \in \mathcal{A}_{N,j}} (\alpha)_q^2. \tag{A.406}$$

F0857 14_gowdy_reduced_gravity.tex:156 NEW_EXTENSION_PASS

eq:gowdy-bessel-hypergeometric-lane

$$\boxed{\mathcal{B}_{N,j}(y) = \frac{(-1)^j y^j}{(j!)^2} {}_0F_{2N-1}\left(\begin{matrix}-\\ 1, \mathcal{A}_{N,j}, \mathcal{A}_{N,j}\end{matrix}; (-1)^N \frac{y^N}{N^{2N}}\right).} \tag{A.407}$$

F0858 14_gowdy_reduced_gravity.tex:173 NEW_EXTENSION_PASS

$$\boxed{\mathcal{B}_{N,j}(y) = \frac{1}{N} \sum_{\ell=0}^{N-1} \omega_N^{-j\ell} \mathcal{B}(\omega_N^{\ell} y),} \tag{A.408}$$

F0859 14_gowdy_reduced_gravity.tex:182 NEW_EXTENSION_PASS

$$\boxed{J_0(x) = \mathcal{B}(y) = \sum_{j=0}^{N-1} \mathcal{B}_{N,j}(y).} \tag{A.409}$$

F0860 14_gowdy_reduced_gravity.tex:189 NEW_EXTENSION_PASS

$$P(\tau, \theta) = A \cos(n\theta) \sum_{j=0}^{N-1} \mathcal{B}_{N,j}\left(\frac{n^2 \mathrm{e}^{-2\tau}}{4}\right). \tag{A.410}$$

F0861 14_gowdy_reduced_gravity.tex:200 NEW_EXTENSION_PASS

$$\vartheta^2 \mathcal{B} + y \mathcal{B} = 0, \tag{A.411}$$

F0862 14_gowdy_reduced_gravity.tex:204 NEW_EXTENSION_PASS

eq:gowdy-bessel-cyclic-lane-system

$$\boxed{\vartheta^2 \mathcal{B}_{N,j} + y \mathcal{B}_{N,j-1} = 0, \qquad j \pmod N.} \tag{A.412}$$

F0863 14_gowdy_reduced_gravity.tex:213 NEW_EXTENSION_PASS

$$\vartheta^2\left(\frac{(-1)^m y^m}{(m!)^2}\right)=\frac{(-1)^m y^m}{((m-1)!)^2}, \tag{A.413}$$

F0864 14_gowdy_reduced_gravity.tex:222 NEW_EXTENSION_PASS

$$\frac{(-1)^{m-1} y^m}{((m-1)!)^2}. \tag{A.414}$$

F0865 14_gowdy_reduced_gravity.tex:234 NEW_EXTENSION_PASS

$$p_{\mathrm{polarized}}=1.977675. \tag{A.415}$$

F0866 14_gowdy_reduced_gravity.tex:240 NEW_EXTENSION_PASS

$$p_{(P,Q,P_\tau,Q_\tau)}=2.128356, \tag{A.416}$$

$$p_{\partial_\tau\lambda_\theta-\partial_\theta\lambda_\tau}=1.949489, \tag{A.417}$$

$$\left|\frac{1}{2\pi}\int_0^{2\pi}\lambda_\theta\,\mathrm{d}\theta\right|_{\max}=1.539567\times10^{-17}, \tag{A.418}$$

$$\varepsilon_{\text{coarse constraint sum}}=1.110223\times10^{-16}. \tag{A.419}$$

15_finite_scale_prediction.tex

F0867 15_finite_scale_prediction.tex:9 CONDITIONAL_PREDICTION_PASS

$$\boldsymbol{k}=k\boldsymbol{n}, \qquad \|\boldsymbol{n}\|=1, \tag{A.420}$$

$$\cos(\omega_\Delta\Delta)=\prod_{i=1}^{3}\cos(kn_i\Delta). \tag{A.421}$$

F0868 15_finite_scale_prediction.tex:16 CONDITIONAL_PREDICTION_PASS

$$\mathcal{A}(\boldsymbol{n}):=\sum_{i<j}n_i^2n_j^2 \tag{A.422}$$

$$=\frac{1}{2}\left(1-\sum_{i=1}^{3}n_i^4\right), \tag{A.423}$$

$$0\le\mathcal{A}(\boldsymbol{n})\le\frac{1}{3}. \tag{A.424}$$

F0869 15_finite_scale_prediction.tex:23 CONDITIONAL_PREDICTION_PASS

$$1=\left(\sum_i n_i^2\right)^2 \tag{A.425}$$

$$=\sum_i n_i^4+2\sum_{i<j}n_i^2n_j^2. \tag{A.426}$$

F0870 15_finite_scale_prediction.tex:34 CONDITIONAL_PREDICTION_PASS

$$\mathcal{A}(1,0,0)=0, \tag{A.427}$$

$$\mathcal{A}\left(2^{-1/2},2^{-1/2},0\right)=\frac{1}{4}, \tag{A.428}$$

$$\mathcal{A}\left(3^{-1/2},3^{-1/2},3^{-1/2}\right)=\frac{1}{3}. \tag{A.429}$$

F0871 15_finite_scale_prediction.tex:42 CONDITIONAL_PREDICTION_PASS

$$H_4(\boldsymbol{x}) := \sum_{i=1}^{3} x_i^4 - \frac{3}{5}\left(\sum_{i=1}^{3} x_i^2\right)^2. \tag{A.430}$$

F0872 15_finite_scale_prediction.tex:48 CONDITIONAL_PREDICTION_PASS

$$\Delta \sum_i x_i^4 = 12 \sum_i x_i^2, \tag{A.431}$$

$$\Delta \left(\sum_i x_i^2\right)^2 = 20 \sum_i x_i^2, \tag{A.432}$$

F0873 15_finite_scale_prediction.tex:54 CONDITIONAL_PREDICTION_PASS

$$H_4(\boldsymbol{n}) = \sum_i n_i^4 - \frac{3}{5}, \tag{A.433}$$

$$\frac{1}{5} - \frac{1}{2} H_4(\boldsymbol{n}) = \frac{1}{2}\left(1 - \sum_i n_i^4\right) = \mathcal{A}(\boldsymbol{n}). \tag{A.434}$$

F0874 15_finite_scale_prediction.tex:64 CONDITIONAL_PREDICTION_PASS

$$\Delta_{\mathbb{R}^3} H_4 = 0, \tag{A.435}$$

$$\mathcal{A}(\boldsymbol{n}) = \frac{1}{5} - \frac{1}{2} H_4(\boldsymbol{n}), \qquad \|\boldsymbol{n}\| = 1. \tag{A.436}$$

F0875 15_finite_scale_prediction.tex:70 CONDITIONAL_PREDICTION_PASS

$$\langle \mathcal{A} \rangle_{S^2} = \frac{1}{5}, \tag{A.437}$$

$$\left\langle \mathcal{A}^2 \right\rangle_{S^2} = \frac{1}{21}, \tag{A.438}$$

$$\mathrm{Var}_{S^2}(\mathcal{A}) = \frac{4}{525}. \tag{A.439}$$

F0876 15_finite_scale_prediction.tex:78 CONDITIONAL_PREDICTION_PASS

$$\omega_\Delta(k, \boldsymbol{n}) = k - \frac{\Delta^2 k^3}{6} \mathcal{A}(\boldsymbol{n}) + \mathcal{O}(\Delta^4), \tag{A.440}$$

$$v_g(k, \boldsymbol{n}) := \frac{\partial \omega_\Delta}{\partial k} \tag{A.441}$$

$$= 1 - \frac{\Delta^2 k^2}{2} \mathcal{A}(\boldsymbol{n}) + \mathcal{O}(\Delta^4). \tag{A.442}$$

F0877 15_finite_scale_prediction.tex:86 CONDITIONAL_PREDICTION_PASS

$$\prod_i \cos(k n_i \Delta) = 1 - \frac{k^2 \Delta^2}{2} + k^4 \Delta^4 \left(\frac{1}{24} + \frac{\mathcal{A}(\boldsymbol{n})}{6}\right) + \mathcal{O}(\Delta^6). \tag{A.443}$$

F0878 15_finite_scale_prediction.tex:97 CONDITIONAL_PREDICTION_PASS

$$\cos(\omega_\Delta \Delta) = 1 - \frac{k^2 \Delta^2}{2} + \left(\frac{k^4}{24} - k a\right) \Delta^4 + \mathcal{O}(\Delta^6). \tag{A.444}$$

F0879 15_finite_scale_prediction.tex:106 CONDITIONAL_PREDICTION_PASS

$$v_g = \frac{\displaystyle\sum_i n_i \sin(k n_i \Delta) \prod_{j \neq i} \cos(k n_j \Delta)}{\sin(\omega_\Delta \Delta)}. \tag{A.445}$$

F0880 15_finite_scale_prediction.tex:116 CONDITIONAL_PREDICTION_PASS

$$\boxed{\delta t_{h\ell} = \frac{L\Delta^2}{2}\mathcal{A}(\boldsymbol{n})\left(k_h^2 - k_\ell^2\right) + \mathcal{O}(\Delta^4).} \tag{A.446}$$

F0881 15_finite_scale_prediction.tex:124 CONDITIONAL_PREDICTION_PASS

$$t(k) = \frac{L}{v_g(k)} = L\left[1 + \frac{\Delta^2 k^2}{2}\mathcal{A}(\boldsymbol{n})\right] + \mathcal{O}(\Delta^4), \tag{A.447}$$

F0882 15_finite_scale_prediction.tex:135 CONDITIONAL_PREDICTION_PASS

$$\boxed{\delta t_{h\ell} = \frac{L\,\mathcal{A}(\boldsymbol{n})\,\ell_*^2\left(E_h^2 - E_\ell^2\right)}{2\hbar^2 c^3} + \mathcal{O}(\ell_*^4).} \tag{A.448}$$

F0883 15_finite_scale_prediction.tex:144 CONDITIONAL_PREDICTION_PASS

$$k = \frac{E}{\hbar c}, \qquad t_{\mathrm{light}} = \frac{L}{c}, \qquad \Delta = \ell_*. \tag{A.449}$$

F0884 15_finite_scale_prediction.tex:153 CONDITIONAL_PREDICTION_PASS

$$\ell_* \le \left[\frac{2\hbar^2 c^3 \delta t_{\max}}{L\,\mathcal{A}(\boldsymbol{n})\,|E_h^2 - E_\ell^2|}\right]^{1/2}. \tag{A.450}$$

F0885 15_finite_scale_prediction.tex:163 CONDITIONAL_PREDICTION_PASS

$$p^{\mathrm{face}}_{\mathrm{delay\ remainder}} = 4.005520, \tag{A.451}$$
$$p^{\mathrm{body}}_{\mathrm{delay\ remainder}} = 4.003609, \tag{A.452}$$
$$\Delta_{\mathrm{true}} = 0.030000, \tag{A.453}$$
$$\Delta_{\mathrm{fit}} = 0.030007, \tag{A.454}$$
$$|\Delta_{\mathrm{fit}} - \Delta_{\mathrm{true}}| = 7 \times 10^{-6}. \tag{A.455}$$

F0886 15_finite_scale_prediction.tex:174 CONDITIONAL_PREDICTION_PASS

$$\text{projector axioms} \nvdash \delta t_{h\ell}, \tag{A.456}$$
$$\text{projector axioms} + \text{specified cubic walk} \vdash \delta t_{h\ell}. \tag{A.457}$$

16_gowdy_constraint_integrator.tex

F0887 16_gowdy_constraint_integrator.tex:9 NEW_EXTENSION_PASS

$$\theta_j = jh, \qquad h = \frac{2\pi}{N}, \qquad D_h^{\mathsf{T}} = -D_h, \tag{A.458}$$
$$(D_h f)_j := \frac{f_{j+1} - f_{j-1}}{2h}, \qquad (\Delta_h f)_j := \frac{f_{j+1} - 2f_j + f_{j-1}}{h^2}, \tag{A.459}$$
$$U_j := \partial_\tau P_j, \qquad V_j := \partial_\tau Q_j, \qquad \gamma(\tau) := \mathrm{e}^{-2\tau}. \tag{A.460}$$

F0888 16_gowdy_constraint_integrator.tex:20 NEW_EXTENSION_PASS

$$\dot{P} = U, \tag{A.461}$$
$$\dot{Q} = V, \tag{A.462}$$
$$\dot{U} = \gamma\Delta_h P + \mathrm{e}^{2P}\left[V^2 - \gamma(D_h Q)^2\right], \tag{A.463}$$
$$\dot{V} = \gamma\Delta_h Q - 2\left[UV - \gamma(D_h P)(D_h Q)\right]. \tag{A.464}$$

F0889 16_gowdy_constraint_integrator.tex:29 NEW_EXTENSION_PASS

$$\boxed{\mathcal{C}_h(P,Q,U,V) := 2h\sum_{j=0}^{N-1}\left[U_j(D_h P)_j + \mathrm{e}^{2P_j}V_j(D_h Q)_j\right].} \tag{A.465}$$

F0890 16_gowdy_constraint_integrator.tex:39 NEW_EXTENSION_PASS

$$\mathcal{C}_h = 2h\,U^{\mathsf{T}} D_h P + 2h\,(\mathrm{e}^{2P}V)^{\mathsf{T}} D_h Q. \tag{A.466}$$

F0891 16_gowdy_constraint_integrator.tex:44 NEW_EXTENSION_PASS

$$\delta_P \mathcal{C}_h = 2h\,U^{\mathsf{T}} D_h \delta P + 4h\,(\mathrm{e}^{2P}V \odot D_h Q)^{\mathsf{T}} \delta P \tag{A.467}$$

$$= \left[-2hD_hU + 4h\mathrm{e}^{2P}V(D_hQ)\right]^{\mathsf{T}} \delta P, \tag{A.468}$$

F0892 16_gowdy_constraint_integrator.tex:52 NEW_EXTENSION_PASS

$$\delta_Q \mathcal{C}_h = -2h\left[D_h(\mathrm{e}^{2P}V)\right]^{\mathsf{T}} \delta Q. \tag{A.469}$$

F0893 16_gowdy_constraint_integrator.tex:57 NEW_EXTENSION_PASS

$$\nabla_P \mathcal{C}_h = -2hD_hU + 4h\mathrm{e}^{2P}V(D_hQ), \tag{A.470}$$

$$\nabla_Q \mathcal{C}_h = -2hD_h(\mathrm{e}^{2P}V), \tag{A.471}$$

$$g_U := \nabla_U \mathcal{C}_h = 2hD_hP, \tag{A.472}$$

$$g_V := \nabla_V \mathcal{C}_h = 2h\mathrm{e}^{2P}D_hQ. \tag{A.473}$$

F0894 16_gowdy_constraint_integrator.tex:68 NEW_EXTENSION_PASS

$$G_h(z) := (0, 0, g_U, g_V). \tag{A.474}$$

F0895 16_gowdy_constraint_integrator.tex:72 NEW_EXTENSION_PASS

$$0 = \nabla\mathcal{C}_h^{\mathsf{T}}(f_h - \Lambda_h G_h) \tag{A.475}$$

$$= \nabla\mathcal{C}_h^{\mathsf{T}} f_h - \Lambda_h \nabla\mathcal{C}_h^{\mathsf{T}} G_h. \tag{A.476}$$

F0896 16_gowdy_constraint_integrator.tex:79 NEW_EXTENSION_PASS

$$\nabla\mathcal{C}_h^{\mathsf{T}} G_h = \|g_U\|_2^2 + \|g_V\|_2^2\,, \tag{A.477}$$

F0897 16_gowdy_constraint_integrator.tex:84 NEW_EXTENSION_PASS

$$\Lambda_h(\tau, z) := \frac{\nabla\mathcal{C}_h(z)^{\mathsf{T}} f_h(\tau, z)}{\|g_U\|_2^2 + \|g_V\|_2^2}, \tag{A.478}$$

$$\widehat{f}_h(\tau, z) := f_h(\tau, z) - \Lambda_h(\tau, z) G_h(z). \tag{A.479}$$

F0898 16_gowdy_constraint_integrator.tex:91 NEW_EXTENSION_PASS

$$\boxed{\nabla\mathcal{C}_h^{\mathsf{T}} \widehat{f}_h = \nabla\mathcal{C}_h^{\mathsf{T}} f_h - \Lambda_h(\|g_U\|_2^2 + \|g_V\|_2^2) = 0.} \tag{A.480}$$

F0899 16_gowdy_constraint_integrator.tex:101 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\mathcal{C}_h(z(\tau)) = 0 \tag{A.481}$$

F0900 16_gowdy_constraint_integrator.tex:108 NEW_EXTENSION_PASS

$$k_1 := \widehat{f}_h(\tau_n, z_n), \tag{A.482}$$

$$k_2 := \widehat{f}_h\left(\tau_n + \frac{\Delta\tau}{2}, z_n + \frac{\Delta\tau}{2}k_1\right), \tag{A.483}$$

$$\widetilde{z}_{n+1} := z_n + \Delta\tau k_2. \tag{A.484}$$

F0901 16_gowdy_constraint_integrator.tex:116 NEW_EXTENSION_PASS

$$\widetilde{z}_{n+1} = (\widetilde{P}, \widetilde{Q}, \widetilde{U}, \widetilde{V}) \tag{A.485}$$

F0902 16_gowdy_constraint_integrator.tex:121 NEW_EXTENSION_PASS

$$\widetilde{g}_U := 2hD_h\widetilde{P}, \tag{A.486}$$

$$\widetilde{g}_V := 2h\mathrm{e}^{2\widetilde{P}}D_h\widetilde{Q}, \tag{A.487}$$

$$\mu_n := \frac{\mathcal{C}_h(\widetilde{z}_{n+1}) - \mathcal{C}_h(z_n)}{\left\|\widetilde{g}_U\right\|_2^2 + \left\|\widetilde{g}_V\right\|_2^2}, \tag{A.488}$$

$$P_{n+1} := \widetilde{P}, \tag{A.489}$$

$$Q_{n+1} := \widetilde{Q}, \tag{A.490}$$

$$U_{n+1} := \widetilde{U} - \mu_n\widetilde{g}_U, \tag{A.491}$$

$$V_{n+1} := \widetilde{V} - \mu_n\widetilde{g}_V. \tag{A.492}$$

F0903 16_gowdy_constraint_integrator.tex:133 NEW_EXTENSION_PASS

$$\mathcal{C}_h(z_{n+1}) = \mathcal{C}_h(\widetilde{P}, \widetilde{Q}, \widetilde{U} - \mu_n\widetilde{g}_U, \widetilde{V} - \mu_n\widetilde{g}_V) \tag{A.493}$$

$$= \mathcal{C}_h(\widetilde{z}_{n+1}) - \mu_n\widetilde{g}_U^{\mathsf{T}}\widetilde{g}_U - \mu_n\widetilde{g}_V^{\mathsf{T}}\widetilde{g}_V \tag{A.494}$$

$$= \mathcal{C}_h(\widetilde{z}_{n+1}) - \mu_n(\left\|\widetilde{g}_U\right\|_2^2 + \left\|\widetilde{g}_V\right\|_2^2). \tag{A.495}$$

F0904 16_gowdy_constraint_integrator.tex:146 NEW_EXTENSION_PASS

$$\boxed{\mathcal{C}_h(z_{n+1}) = \mathcal{C}_h(z_n).} \tag{A.496}$$

F0905 16_gowdy_constraint_integrator.tex:152 NEW_EXTENSION_PASS

$$\mathcal{C}_h(z_0) = 0 \Longrightarrow \mathcal{C}_h(z_n) = 0 \qquad \forall n. \tag{A.497}$$

F0906 16_gowdy_constraint_integrator.tex:159 NEW_EXTENSION_PASS

$$\mathcal{C}_h(\widetilde{z}_{n+1}) - \mathcal{C}_h(z_n) = \mathcal{O}(\Delta\tau^3). \tag{A.498}$$

F0907 16_gowdy_constraint_integrator.tex:164 NEW_EXTENSION_PASS

$$\mu_n = \frac{\mathcal{O}(\Delta\tau^3)}{\left\|\widetilde{g}_U\right\|_2^2 + \left\|\widetilde{g}_V\right\|_2^2} = \mathcal{O}(\Delta\tau^3). \tag{A.499}$$

F0908 16_gowdy_constraint_integrator.tex:174 NEW_EXTENSION_PASS

$$\lambda_{\tau,h} := U^2 + \gamma(D_hP)^2 + \mathrm{e}^{2P}\left[V^2 + \gamma(D_hQ)^2\right], \tag{A.500}$$

$$\lambda_{\theta,h} := 2\left[U(D_hP) + \mathrm{e}^{2P}V(D_hQ)\right]. \tag{A.501}$$

F0909 16_gowdy_constraint_integrator.tex:182 NEW_EXTENSION_PASS

$$\mathcal{E}_{P,h} := \dot{U} - \dot{U}_{\mathrm{raw}}, \tag{A.502}$$

$$\mathcal{E}_{Q,h} := \dot{V} - \dot{V}_{\mathrm{raw}}. \tag{A.503}$$

F0910 16_gowdy_constraint_integrator.tex:187 NEW_EXTENSION_PASS

$$\partial_\tau\lambda_\theta - \partial_\theta\lambda_\tau = 2P_\theta\mathcal{E}_P + 2\mathrm{e}^{2P}Q_\theta\mathcal{E}_Q. \tag{A.504}$$

F0911 16_gowdy_constraint_integrator.tex:192 NEW_EXTENSION_PASS

$$\mathcal{I}_h := \partial_\tau\lambda_{\theta,h} - D_h\lambda_{\tau,h} - 2(D_hP)\mathcal{E}_{P,h} - 2\mathrm{e}^{2P}(D_hQ)\mathcal{E}_{Q,h}. \tag{A.505}$$

F0912 16_gowdy_constraint_integrator.tex:200 NEW_EXTENSION_PASS

$$p_{\mathrm{state}} = 2.128038, \tag{A.506}$$

$$p_{\mathcal{I}} = 1.949319, \tag{A.507}$$

$$p_{\text{constraint force}} = 2.044195, \tag{A.508}$$

$$\max_n |\mathcal{C}_h(z_n) - \mathcal{C}_h(z_0)| = 6.812243 \times 10^{-18}, \tag{A.509}$$

$$\max_{n} |\mathcal{C}_h(z_n^{\mathrm{unconstrained}}) - \mathcal{C}_h(z_0)| = 1.517044 \times 10^{-6} \quad (N = 256), \tag{A.510}$$

$$\varepsilon_{\text{coarse momentum sum}} = 2.602085 \times 10^{-18}. \tag{A.511}$$

F0913 16_gowdy_constraint_integrator.tex:213 NEW_EXTENSION_PASS

$$\left\| \widehat{f}_h - f_h \right\|_{\mathrm{RMS}} = \mathcal{O}(h^2), \tag{A.512}$$

17_einstein_scalar_perturbations.tex

F0914 17_einstein_scalar_perturbations.tex:9 NEW_EXTENSION_PASS

$$M_{\mathrm{Pl}}^2 := (8\pi G)^{-1} \tag{A.513}$$

F0915 17_einstein_scalar_perturbations.tex:13 NEW_EXTENSION_PASS

$$S[g,\phi] = \int \sqrt{-g} \left[\frac{M_{\mathrm{Pl}}^2}{2} R - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right] \mathrm{d}^4 x. \tag{A.514}$$

F0916 17_einstein_scalar_perturbations.tex:23 NEW_EXTENSION_PASS

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + a(t)^2\, \mathrm{d}\boldsymbol{x}^2, \qquad \phi = \bar{\phi}(t), \tag{A.515}$$

F0917 17_einstein_scalar_perturbations.tex:29 NEW_EXTENSION_PASS

$$3 M_{\mathrm{Pl}}^2 H^2 = \frac{1}{2} \dot{\bar{\phi}}^{\,2} + V(\bar{\phi}), \tag{A.516}$$

$$-2 M_{\mathrm{Pl}}^2 \dot{H} = \dot{\bar{\phi}}^{\,2}, \tag{A.517}$$

$$\ddot{\bar{\phi}} + 3H\dot{\bar{\phi}} + V_{,\phi} = 0, \tag{A.518}$$

$$\epsilon := -\frac{\dot{H}}{H^2} = \frac{\dot{\bar{\phi}}^{\,2}}{2 M_{\mathrm{Pl}}^2 H^2}. \tag{A.519}$$

F0918 17_einstein_scalar_perturbations.tex:44 NEW_EXTENSION_PASS

$$g_{ij} = a^2 \left[(1 - 2\Psi)\delta_{ij} + 2\partial_i \partial_j E \right]. \tag{A.520}$$

F0919 17_einstein_scalar_perturbations.tex:50 NEW_EXTENSION_PASS

$$\Psi \mapsto \Psi + HT, \tag{A.521}$$

$$\delta\phi \mapsto \delta\phi - \dot{\bar{\phi}} T. \tag{A.522}$$

F0920 17_einstein_scalar_perturbations.tex:55 NEW_EXTENSION_PASS

$$\boxed{\zeta := -\Psi - \frac{H}{\dot{\bar{\phi}}} \delta\phi.} \tag{A.523}$$

F0921 17_einstein_scalar_perturbations.tex:62 NEW_EXTENSION_PASS

$$\zeta \mapsto -(\Psi + HT) - \frac{H}{\dot{\bar{\phi}}} \left(\delta\phi - \dot{\bar{\phi}} T \right) \tag{A.524}$$

$$= -\Psi - HT - \frac{H}{\dot{\bar{\phi}}} \delta\phi + HT \tag{A.525}$$

$$= \zeta. \tag{A.526}$$

F0922 17_einstein_scalar_perturbations.tex:74 NEW_EXTENSION_PASS

$$\delta\phi = 0, \qquad E = 0, \qquad h_{ij} = a^2 \mathrm{e}^{2\zeta} \delta_{ij}. \tag{A.527}$$

F0923 17_einstein_scalar_perturbations.tex:84 NEW_EXTENSION_PASS

$$N = 1 + \alpha, \qquad N_i = \partial_i \beta. \tag{A.528}$$

F0924 17_einstein_scalar_perturbations.tex:90 NEW_EXTENSION_PASS

$$\alpha = \frac{\dot\zeta}{H}, \tag{A.529}$$

$$\beta = -\frac{\zeta}{H} + \chi, \tag{A.530}$$

$$\nabla^2 \chi = a^2 \epsilon \dot\zeta. \tag{A.531}$$

F0925 17_einstein_scalar_perturbations.tex:99 NEW_EXTENSION_PASS

$$\boxed{S_\zeta^{(2)} = M_{\rm Pl}^2 \int a^3 \epsilon \left[\dot\zeta^2 - \frac{(\nabla\zeta)^2}{a^2} \right] \, \mathrm{d}t \, \mathrm{d}^3x.} \tag{A.532}$$

F0926 17_einstein_scalar_perturbations.tex:110 NEW_EXTENSION_PASS

$$\delta S_\zeta^{(2)} = 2M_{\rm Pl}^2 \int \left[a^3 \epsilon \dot\zeta \, \delta\dot\zeta - a\epsilon \partial_i \zeta \, \partial_i \delta\zeta \right] \, \mathrm{d}t \, \mathrm{d}^3x \tag{A.533}$$

$$= -2M_{\rm Pl}^2 \int \left[\frac{\mathrm{d}}{\mathrm{d}t} \left(a^3 \epsilon \dot\zeta \right) - a\epsilon \nabla^2 \zeta \right] \delta\zeta \, \mathrm{d}t \, \mathrm{d}^3x, \tag{A.534}$$

F0927 17_einstein_scalar_perturbations.tex:128 NEW_EXTENSION_PASS

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t} \left(a^3 \epsilon \dot\zeta \right) - a\epsilon \nabla^2 \zeta = 0.} \tag{A.535}$$

F0928 17_einstein_scalar_perturbations.tex:137 NEW_EXTENSION_PASS

$$0 = \frac{\mathrm{d}}{\mathrm{d}t} \left(a^3 \epsilon \dot\zeta_{\boldsymbol{k}} \right) + a\epsilon k^2 \zeta_{\boldsymbol{k}} \tag{A.536}$$

$$= a^3 \epsilon \ddot\zeta_{\boldsymbol{k}} + \left(3Ha^3\epsilon + a^3 \dot\epsilon \right) \dot\zeta_{\boldsymbol{k}} + a\epsilon k^2 \zeta_{\boldsymbol{k}}. \tag{A.537}$$

F0929 17_einstein_scalar_perturbations.tex:148 NEW_EXTENSION_PASS

$$\ddot\zeta_{\boldsymbol{k}} + \left(3H + \frac{\dot\epsilon}{\epsilon} \right) \dot\zeta_{\boldsymbol{k}} + \frac{k^2}{a^2} \zeta_{\boldsymbol{k}} = 0. \tag{A.538}$$

F0930 17_einstein_scalar_perturbations.tex:155 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}}{\mathrm{d}t} \left(a^3 \epsilon \dot\zeta \right) = 0, \tag{A.539}$$

$$a^3 \epsilon \dot\zeta = C_2, \tag{A.540}$$

$$\dot\zeta = \frac{C_2}{a^3 \epsilon}, \tag{A.541}$$

$$\zeta(t) = C_1 + C_2 \int^t \frac{\mathrm{d}s}{a(s)^3 \epsilon(s)}. \tag{A.542}$$

F0931 17_einstein_scalar_perturbations.tex:166 NEW_EXTENSION_PASS

$$\mathrm{d}t = a \, \mathrm{d}\eta, \tag{A.543}$$

$$z(\eta) := a\sqrt{2\epsilon} \, M_{\rm Pl}, \tag{A.544}$$

$$v := z\zeta. \tag{A.545}$$

F0932 17_einstein_scalar_perturbations.tex:172 NEW_EXTENSION_PASS

$$S_\zeta^{(2)} = \frac{1}{2} \int z^2 \left[(\zeta')^2 - (\nabla\zeta)^2 \right] \, \mathrm{d}\eta \, \mathrm{d}^3x. \tag{A.546}$$

F0933 17_einstein_scalar_perturbations.tex:179 NEW_EXTENSION_PASS

$$z^2 (\zeta')^2 = \left(v' - \frac{z'}{z} v \right)^2 \tag{A.547}$$

$$= (v')^2 - \frac{z'}{z}(v^2)' + \left(\frac{z'}{z}\right)^2 v^2, \tag{A.548}$$

$$z^2(\nabla\zeta)^2 = (\nabla v)^2. \tag{A.549}$$

F0934 17_einstein_scalar_perturbations.tex:188 NEW_EXTENSION_PASS

$$-\int \frac{z'}{z}(v^2)'\,\mathrm{d}\eta = \int \left(\frac{z'}{z}\right)' v^2\,\mathrm{d}\eta + \text{boundary term}, \tag{A.550}$$

$$\left(\frac{z'}{z}\right)' + \left(\frac{z'}{z}\right)^2 = \frac{z''}{z}. \tag{A.551}$$

F0935 17_einstein_scalar_perturbations.tex:198 NEW_EXTENSION_PASS

$$\boxed{S_v^{(2)} = \frac{1}{2}\int \left[(v')^2 - (\nabla v)^2 + \frac{z''}{z}v^2\right]\,\mathrm{d}\eta\,\mathrm{d}^3x.} \tag{A.552}$$

F0936 17_einstein_scalar_perturbations.tex:208 NEW_EXTENSION_PASS

$$v'' - \nabla^2 v - \frac{z''}{z}v = 0, \tag{A.553}$$

$$v_{\boldsymbol{k}}'' + \left(k^2 - \frac{z''}{z}\right)v_{\boldsymbol{k}} = 0. \tag{A.554}$$

F0937 17_einstein_scalar_perturbations.tex:216 NEW_EXTENSION_PASS

$$S_{v,h}^{(2)} = \frac{1}{2}\int h^3 \sum_{\boldsymbol{j}} \left[(v_{\boldsymbol{j}}')^2 - \sum_{r=1}^{3}(\delta_r^+ v_{\boldsymbol{j}})^2 + \frac{z''}{z}v_{\boldsymbol{j}}^2\right]\,\mathrm{d}\eta. \tag{A.555}$$

F0938 17_einstein_scalar_perturbations.tex:226 NEW_EXTENSION_PASS

$$h^3 \sum_{\boldsymbol{j},r}(\delta_r^+ v_{\boldsymbol{j}})(\delta_r^+ \delta v_{\boldsymbol{j}}) = -h^3 \sum_{\boldsymbol{j}}(\Delta_h v_{\boldsymbol{j}})\delta v_{\boldsymbol{j}} \tag{A.556}$$

F0939 17_einstein_scalar_perturbations.tex:233 NEW_EXTENSION_PASS

$$v_{\boldsymbol{j}}'' - \Delta_h v_{\boldsymbol{j}} - \frac{z''}{z}v_{\boldsymbol{j}} = 0. \tag{A.557}$$

F0940 17_einstein_scalar_perturbations.tex:237 NEW_EXTENSION_PASS

$$\delta_r^+ \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}_{\boldsymbol{j}}} = \frac{\mathrm{e}^{\mathrm{i}k_r h} - 1}{h}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}_{\boldsymbol{j}}}, \tag{A.558}$$

$$\left|\frac{\mathrm{e}^{\mathrm{i}k_r h} - 1}{h}\right|^2 = \frac{(\mathrm{e}^{\mathrm{i}k_r h} - 1)(\mathrm{e}^{-\mathrm{i}k_r h} - 1)}{h^2} \tag{A.559}$$

$$= \frac{2 - 2\cos(k_r h)}{h^2} = \frac{4}{h^2}\sin^2\left(\frac{k_r h}{2}\right). \tag{A.560}$$

F0941 17_einstein_scalar_perturbations.tex:247 NEW_EXTENSION_PASS

$$\widehat{k}_h^2 := \frac{4}{h^2}\sum_{r=1}^{3}\sin^2\left(\frac{k_r h}{2}\right), \tag{A.561}$$

$$v_{\boldsymbol{k},h}'' + \left(\widehat{k}_h^2 - \frac{z''}{z}\right)v_{\boldsymbol{k},h} = 0, \tag{A.562}$$

$$\widehat{k}_h^2 = k^2 - \frac{h^2}{12}\sum_{r=1}^{3}k_r^4 + \frac{h^4}{360}\sum_{r=1}^{3}k_r^6 + \mathcal{O}(h^6). \tag{A.563}$$

F0942 17_einstein_scalar_perturbations.tex:259 NEW_EXTENSION_PASS

$$v_\kappa(\eta) = \frac{1}{\sqrt{2\kappa}}\left(1 - \frac{\mathrm{i}}{\kappa\eta}\right)\mathrm{e}^{-\mathrm{i}\kappa\eta} \tag{A.564}$$

F0943 17_einstein_scalar_perturbations.tex:266 NEW_EXTENSION_PASS

$$v_\kappa'' + \left(\kappa^2 - \frac{2}{\eta^2}\right) v_\kappa = 0. \tag{A.565}$$

F0944 17_einstein_scalar_perturbations.tex:271 NEW_EXTENSION_PASS

$$p_{v,h} = 2.001500, \tag{A.566}$$
$$p_{|v|^2,h} = 2.002803, \tag{A.567}$$
$$p_{v,h}^{\mathrm{Richardson}} = 4.001170, \tag{A.568}$$
$$\max_h \left| v_h v_h^{*\prime} - v_h^* v_h' - \mathrm{i} \right| = 3.219647 \times 10^{-15}. \tag{A.569}$$

18_cosmological_finite_scale_delay.tex

F0945 18_cosmological_finite_scale_delay.tex:9 CONDITIONAL_PREDICTION_PASS

$$\cos(\omega\ell) = \prod_{i=1}^{3} \cos(k n_i \ell), \tag{A.570}$$
$$\mathcal{A}(\boldsymbol{n}) := \sum_{i<j} n_i^2 n_j^2, \tag{A.571}$$
$$v_g(k, \boldsymbol{n}, \ell) = 1 - \frac{1}{2}\mathcal{A}(\boldsymbol{n})\ell^2 k^2 + \mathcal{O}(\ell^4). \tag{A.572}$$

F0946 18_cosmological_finite_scale_delay.tex:17 CONDITIONAL_PREDICTION_PASS

$$v_g = \frac{\left[\prod_i \cos(k n_i \ell)\right] \sum_i n_i \tan(k n_i \ell)}{\sin(\omega\ell)}. \tag{A.573}$$

F0947 18_cosmological_finite_scale_delay.tex:31 CONDITIONAL_PREDICTION_PASS

$$E(z) = E_0(1+z), \tag{A.574}$$
$$\ell_{\mathrm{phys}}(z) = \ell_{*,0}(1+z)^{-\sigma}. \tag{A.575}$$

F0948 18_cosmological_finite_scale_delay.tex:37 CONDITIONAL_PREDICTION_PASS

$$\ell_{\mathrm{phys}}(z)E(z) = \ell_{*,0}(1+z)^{-\sigma} E_0(1+z) \tag{A.576}$$
$$= \ell_{*,0} E_0 (1+z)^{1-\sigma}, \tag{A.577}$$

F0949 18_cosmological_finite_scale_delay.tex:44 CONDITIONAL_PREDICTION_PASS

$$\left[\frac{\ell_{\mathrm{phys}}(z)E(z)}{\hbar c}\right]^2 = \left[\frac{\ell_{*,0}E_0}{\hbar c}\right]^2 (1+z)^{2-2\sigma}. \tag{A.578}$$

F0950 18_cosmological_finite_scale_delay.tex:51 CONDITIONAL_PREDICTION_PASS

$$\sigma = 0 \qquad \Longleftrightarrow \text{ fixed proper microscopic length}, \tag{A.579}$$
$$\sigma = 1 \qquad \Longleftrightarrow \text{ fixed comoving microscopic length} \tag{A.580}$$

F0951 18_cosmological_finite_scale_delay.tex:63 CONDITIONAL_PREDICTION_PASS

$$\chi_s = \int_0^{z_s} \frac{v_\ell(z)}{H(z)}\, \mathrm{d}z. \tag{A.581}$$

F0952 18_cosmological_finite_scale_delay.tex:68 CONDITIONAL_PREDICTION_PASS

$$\chi_s = \int_{z_{a,h}}^{z_s} \frac{v_h(z)}{H(z)}\, \mathrm{d}z. \tag{A.582}$$

F0953 18_cosmological_finite_scale_delay.tex:73 CONDITIONAL_PREDICTION_PASS

$$\int_0^{z_s} \frac{v_\ell}{H}\,\mathrm{d}z = \int_{z_{a,h}}^{0} \frac{v_h}{H}\,\mathrm{d}z + \int_0^{z_s} \frac{v_h}{H}\,\mathrm{d}z, \tag{A.583}$$

$$\int_{z_{a,h}}^{0} \frac{v_h}{H}\,\mathrm{d}z = \int_0^{z_s} \frac{v_\ell - v_h}{H}\,\mathrm{d}z. \tag{A.584}$$

F0954 18_cosmological_finite_scale_delay.tex:81 CONDITIONAL_PREDICTION_PASS

$$\Delta t_{h\ell}^{\mathrm{direct}} = \int_{z_{a,h}}^{0} \frac{\mathrm{d}z}{(1+z)H(z)}. \tag{A.585}$$

F0955 18_cosmological_finite_scale_delay.tex:89 CONDITIONAL_PREDICTION_PASS

$$I_\sigma(z_s) := \int_0^{z_s} \frac{(1+z)^{2-2\sigma}}{H(z)}\,\mathrm{d}z. \tag{A.586}$$

F0956 18_cosmological_finite_scale_delay.tex:95 CONDITIONAL_PREDICTION_PASS

$$1 - v_g = \frac{1}{2}\mathcal{A}(\boldsymbol{n})\ell_{\mathrm{phys}}(z)^2 \frac{E(z)^2}{\hbar^2 c^2} + \mathcal{O}(\ell_{*,0}^4) \tag{A.587}$$

$$= \frac{1}{2}\mathcal{A}(\boldsymbol{n})\frac{\ell_{*,0}^2 E_0^2}{\hbar^2 c^2}(1+z)^{2-2\sigma} + \mathcal{O}(\ell_{*,0}^4). \tag{A.588}$$

F0957 18_cosmological_finite_scale_delay.tex:107 CONDITIONAL_PREDICTION_PASS

$$v_\ell - v_h = (1 - v_h) - (1 - v_\ell) \tag{A.589}$$

$$= \frac{\mathcal{A}(\boldsymbol{n})\ell_{*,0}^2}{2\hbar^2 c^2}(E_{h,0}^2 - E_{\ell,0}^2)(1+z)^{2-2\sigma} + \mathcal{O}(\ell_{*,0}^4). \tag{A.590}$$

F0958 18_cosmological_finite_scale_delay.tex:117 CONDITIONAL_PREDICTION_PASS

$$\Delta t_{h\ell} = \int_0^{z_s} \frac{v_\ell - v_h}{H(z)}\,\mathrm{d}z + \mathcal{O}(\ell_{*,0}^4). \tag{A.591}$$

F0959 18_cosmological_finite_scale_delay.tex:123 CONDITIONAL_PREDICTION_PASS

$$\boxed{\Delta t_{h\ell} = \frac{\mathcal{A}(\boldsymbol{n})\ell_{*,0}^2(E_{h,0}^2 - E_{\ell,0}^2)}{2\hbar^2 c^2} I_\sigma(z_s) + \mathcal{O}(\ell_{*,0}^4).} \tag{A.592}$$

F0960 18_cosmological_finite_scale_delay.tex:136 CONDITIONAL_PREDICTION_PASS

$$\sigma = 0: \quad \Delta t_{h\ell}^{\mathrm{proper}} = \frac{\mathcal{A}\ell_{*,0}^2 \Delta E_0^2}{2\hbar^2 c^2} \int_0^{z_s} \frac{(1+z)^2}{H(z)}\,\mathrm{d}z + \mathcal{O}(\ell_{*,0}^4), \tag{A.593}$$

$$\sigma = 1: \quad \Delta t_{h\ell}^{\mathrm{comoving}} = \frac{\mathcal{A}\ell_{*,0}^2 \Delta E_0^2}{2\hbar^2 c^2} \int_0^{z_s} \frac{\mathrm{d}z}{H(z)} + \mathcal{O}(\ell_{*,0}^4). \tag{A.594}$$

F0961 18_cosmological_finite_scale_delay.tex:157 CONDITIONAL_PREDICTION_PASS

$$H(z) = H_0 + \mathcal{O}(z), \tag{A.595}$$

$$(1+z)^{2-2\sigma} = 1 + \mathcal{O}(z), \tag{A.596}$$

$$I_\sigma(z_s) = \frac{z_s}{H_0} + \mathcal{O}(z_s^2). \tag{A.597}$$

F0962 18_cosmological_finite_scale_delay.tex:163 CONDITIONAL_PREDICTION_PASS

$$I_\sigma(z_s) = \frac{L}{c} + \mathcal{O}(z_s^2), \tag{A.598}$$

F0963 18_cosmological_finite_scale_delay.tex:171 CONDITIONAL_PREDICTION_PASS

$$\frac{\mathcal{A}(\boldsymbol{n})\ell_{*,0}^2|E_{h,0}^2-E_{\ell,0}^2|}{2\hbar^2c^2}I_\sigma(z_s)\le\Delta t_{\max}. \tag{A.599}$$

F0964 18_cosmological_finite_scale_delay.tex:180 CONDITIONAL_PREDICTION_PASS

$$\boxed{\ell_{*,0}\le\left[\frac{2\hbar^2c^2\Delta t_{\max}}{\mathcal{A}(\boldsymbol{n})|E_{h,0}^2-E_{\ell,0}^2|I_\sigma(z_s)}\right]^{1/2}.} \tag{A.600}$$

F0965 18_cosmological_finite_scale_delay.tex:195 CONDITIONAL_PREDICTION_PASS

$$\mathcal{A}(1,0,0)=0. \tag{A.601}$$

F0966 18_cosmological_finite_scale_delay.tex:199 CONDITIONAL_PREDICTION_PASS

$$\cos(\omega\ell)=\cos(k\ell)\cos(0)\cos(0) \tag{A.602}$$
$$=\cos(k\ell). \tag{A.603}$$

F0967 18_cosmological_finite_scale_delay.tex:205 CONDITIONAL_PREDICTION_PASS

$$v_g=\frac{\partial\omega}{\partial k}=1. \tag{A.604}$$

F0968 18_cosmological_finite_scale_delay.tex:212 CONDITIONAL_PREDICTION_PASS

$$H(z)=H_0\sqrt{0.3(1+z)^3+0.7},\qquad z_s=2, \tag{A.605}$$

F0969 18_cosmological_finite_scale_delay.tex:219 CONDITIONAL_PREDICTION_PASS

$$p_{\text{remainder}}^{\sigma=0}=4.002683, \tag{A.606}$$
$$p_{\text{remainder}}^{\sigma=1}=4.001430, \tag{A.607}$$
$$\varepsilon_{\text{axis}}=0, \tag{A.608}$$
$$\ell_{\text{true}}=0.030000, \tag{A.609}$$
$$\ell_{\text{fit}}^{\sigma=0}=0.030008, \tag{A.610}$$
$$\ell_{\text{fit}}^{\sigma=1}=0.030003. \tag{A.611}$$

F0970 18_cosmological_finite_scale_delay.tex:231 CONDITIONAL_PREDICTION_PASS

$$\Delta t_{h\ell}^{\text{direct}}-\Delta t_{h\ell}^{\text{leading}}=\mathcal{O}(\ell_{*,0}^4). \tag{A.612}$$

19_complete_cosmology_ledger.tex

F0971 19_complete_cosmology_ledger.tex:3 OPEN_NOT_VERIFIED

$$\boxed{S_{\text{local}}[\mathcal{P},U,q,N,N^i,\Psi]\Longrightarrow\{\mathcal{H},\mathcal{H}_i,\mathcal{G},\mathcal{E}_q,\mathcal{E}_\Psi\}=0\Longrightarrow\{H(z),C_\ell,P(k),D_L(z),\text{new observable}\}.} \tag{A.613}$$

F0972 19_complete_cosmology_ledger.tex:13 OPEN_NOT_VERIFIED

$$\textsf{incomplete}\Longrightarrow\textbf{To Be Filled Later}. \tag{A.614}$$

F0973 19_complete_cosmology_ledger.tex:259 OPEN_NOT_VERIFIED

$$\textsf{Gate}_1:\ \text{one nonlinear Lorentzian gravity–matter parent action}\quad\textbf{Closed-Conditional}, \tag{A.615}$$
$$\textsf{Gate}_2:\ \{\Omega,\Omega\}=0\ \text{under BFV–BV transfer and regular reduction}\quad\textbf{Closed-Conditional}, \tag{A.616}$$
$$\textsf{Gate}_3:\ \{\mathcal{H},\mathcal{H}_i,\mathcal{G}\}_{\text{resolved}}\ \text{closed}\quad\textbf{Closed-Conditional}, \tag{A.617}$$
$$\textsf{Gate}_4:\ \mathcal{E}_h\to G_{\mu\nu}-8\pi GT_{\mu\nu}\ \text{uniformly from microscopic packets}\quad\textbf{To Be Filled Later}, \tag{A.618}$$
$$\textsf{Gate}_5:\ \{\zeta,\text{vector},h_{ij}^{\text{TT}}\}\ \text{with full kinetic sources}\quad\textbf{Partial}, \tag{A.619}$$
$$\textsf{Gate}_6:\ \{C_\ell,P(k),D_L,D_A,f\sigma_8\}\ \text{from one frozen parameter set}\quad\textbf{To Be Filled Later}, \tag{A.620}$$
$$\textsf{Gate}_7:\ \delta\mathcal{O}_{\text{new}}\ \text{universal and gauge invariant}\quad\textbf{Partial}, \tag{A.621}$$
$$\textsf{Gate}_8:\ \mathcal{L}(D\mid\theta),\ \text{independent reproduction, and external review}\quad\textbf{To Be Filled Later}. \tag{A.622}$$

20_general_flrw_background.tex

F0974 20_general_flrw_background.tex:9 NEW_EXTENSION_PASS

$$\mathrm{d}s^2 = -N(t)^2\,\mathrm{d}t^2 + a(t)^2\gamma^{(K)}_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j, \qquad {}^{(3)}R[\gamma^{(K)}] = 6K. \tag{A.623}$$

F0975 20_general_flrw_background.tex:15 NEW_EXTENSION_PASS

$$S_{\mathrm{FLRW}} = \int \left[-\frac{3M_{\mathrm{Pl}}^2 a\dot{a}^2}{N} + 3M_{\mathrm{Pl}}^2 K a N - M_{\mathrm{Pl}}^2 \Lambda a^3 N - N H_{\mathrm{P}}(a) \right] \mathrm{d}t. \tag{A.624}$$

F0976 20_general_flrw_background.tex:25 NEW_EXTENSION_PASS

$$0 = \frac{3M_{\mathrm{Pl}}^2 a\dot{a}^2}{N^2} + 3M_{\mathrm{Pl}}^2 K a - M_{\mathrm{Pl}}^2 \Lambda a^3 - H_{\mathrm{P}}(a). \tag{A.625}$$

F0977 20_general_flrw_background.tex:36 NEW_EXTENSION_PASS

$$\rho(a) := \frac{H_{\mathrm{P}}(a)}{a^3}, \tag{A.626}$$

F0978 20_general_flrw_background.tex:40 NEW_EXTENSION_PASS

$$\boxed{H^2 = \frac{\rho}{3M_{\mathrm{Pl}}^2} + \frac{\Lambda}{3} - \frac{K}{a^2}.} \tag{A.627}$$

F0979 20_general_flrw_background.tex:49 NEW_EXTENSION_PASS

$$\dot{H} = -\frac{\rho + p}{2M_{\mathrm{Pl}}^2} + \frac{K}{a^2}, \tag{A.628}$$

$$\frac{\ddot{a}}{a} = -\frac{\rho + 3p}{6M_{\mathrm{Pl}}^2} + \frac{\Lambda}{3}, \tag{A.629}$$

$$\dot{\rho} + 3H(\rho + p) = 0. \tag{A.630}$$

F0980 20_general_flrw_background.tex:57 NEW_EXTENSION_PASS

$$2H\dot{H} = \frac{\dot{\rho}}{3M_{\mathrm{Pl}}^2} + \frac{2KH}{a^2}. \tag{A.631}$$

F0981 20_general_flrw_background.tex:64 NEW_EXTENSION_PASS

$$\frac{\ddot{a}}{a} = \dot{H} + H^2 \tag{A.632}$$

$$= -\frac{\rho + p}{2M_{\mathrm{Pl}}^2} + \frac{K}{a^2} + \frac{\rho}{3M_{\mathrm{Pl}}^2} + \frac{\Lambda}{3} - \frac{K}{a^2}, \tag{A.633}$$

F0982 20_general_flrw_background.tex:76 NEW_EXTENSION_PASS

$$\frac{a}{2}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{8\pi G}{3}\rho + \frac{\Lambda}{3} - \frac{K}{a^2}\right] = -4\pi G(\rho + p) + \frac{K}{a^2} \tag{A.634}$$

F0983 20_general_flrw_background.tex:84 NEW_EXTENSION_PASS

$$a\rho_{,a} = -3(\rho + p). \tag{A.635}$$

F0984 20_general_flrw_background.tex:90 NEW_EXTENSION_PASS

$$p_i = w_i \rho_i, \tag{A.636}$$

F0985 20_general_flrw_background.tex:93 NEW_EXTENSION_PASS

$$\dot{\rho}_i + 3H(1 + w_i)\rho_i = 0, \tag{A.637}$$

$$\rho_i(a) = \rho_{i0} a^{-3(1+w_i)}. \tag{A.638}$$

F0986 20_general_flrw_background.tex:98 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}\rho_i}{\rho_i} = -3(1+w_i)\frac{\mathrm{d}a}{a}. \tag{A.639}$$

F0987 20_general_flrw_background.tex:104 NEW_EXTENSION_PASS

$$\Omega_{i0} := \frac{\rho_{i0}}{3M_{\mathrm{Pl}}^2 H_0^2}, \tag{A.640}$$

$$\Omega_{\Lambda 0} := \frac{\Lambda}{3H_0^2}, \tag{A.641}$$

$$\Omega_{K0} := -\frac{K}{H_0^2}, \tag{A.642}$$

$$1 = \sum_i \Omega_{i0} + \Omega_{K0} + \Omega_{\Lambda 0}, \tag{A.643}$$

$$E(a)^2 := \frac{H(a)^2}{H_0^2} = \sum_i \Omega_{i0} a^{-3(1+w_i)} + \Omega_{K0} a^{-2} + \Omega_{\Lambda 0}. \tag{A.644}$$

F0988 20_general_flrw_background.tex:115 NEW_EXTENSION_PASS

$$E(z)^2 = \sum_i \Omega_{i0}(1+z)^{3(1+w_i)} + \Omega_{K0}(1+z)^2 + \Omega_{\Lambda 0}. \tag{A.645}$$

F0989 20_general_flrw_background.tex:121 NEW_EXTENSION_PASS

$$\rho_r \propto a^{-4}, \tag{A.646}$$

$$\rho_m \propto a^{-3}, \tag{A.647}$$

$$\rho_K \propto a^{-2}, \tag{A.648}$$

$$\rho_\Lambda = \text{constant}. \tag{A.649}$$

F0990 20_general_flrw_background.tex:130 NEW_EXTENSION_PASS

$$\dot\rho_i + 3H(\rho_i + p_i) = Q_i, \qquad \sum_i Q_i = 0. \tag{A.650}$$

F0991 20_general_flrw_background.tex:136 NEW_EXTENSION_PASS

$$\boxed{\dot\rho + 3H(\rho + p) = 0, \qquad \rho := \sum_i \rho_i, \quad p := \sum_i p_i.} \tag{A.651}$$

F0992 20_general_flrw_background.tex:146 NEW_EXTENSION_PASS

$$\dot\rho + 3H(\rho + p) = \sum_i Q_i = 0. \tag{A.652}$$

F0993 20_general_flrw_background.tex:153 NEW_EXTENSION_PASS

$$Q_m = -\Gamma\rho_m, \tag{A.653}$$

$$Q_r = +\Gamma\rho_m, \tag{A.654}$$

F0994 20_general_flrw_background.tex:157 NEW_EXTENSION_PASS

$$\dot\rho_m = -3H\rho_m - \Gamma\rho_m, \tag{A.655}$$

$$\dot\rho_r = -4H\rho_r + \Gamma\rho_m, \tag{A.656}$$

$$\dot\rho_m + \dot\rho_r + 3H\left(\rho_m + \rho_r + \frac{1}{3}\rho_r\right) = 0. \tag{A.657}$$

F0995 20_general_flrw_background.tex:166 NEW_EXTENSION_PASS

$$\bar\chi(z) := \int_0^z \frac{\mathrm{d}z'}{E(z')}, \tag{A.658}$$

$$D_H := \frac{c}{H_0}. \tag{A.659}$$

F0996 20_general_flrw_background.tex:171 NEW_EXTENSION_PASS

$$0 = -c^2\,\mathrm{d}t^2 + a(t)^2\,\mathrm{d}\chi_{\mathrm{com}}^2. \tag{A.660}$$

F0997 20_general_flrw_background.tex:176 NEW_EXTENSION_PASS

$$\mathrm{d}\chi_{\mathrm{com}} = \frac{c\,\mathrm{d}z}{H(z)} = D_H\,\frac{\mathrm{d}z}{E(z)}. \tag{A.661}$$

F0998 20_general_flrw_background.tex:183 NEW_EXTENSION_PASS

$$D_M(z) = D_H\,\mathcal{S}_{\Omega_K}(\bar{\chi}(z))\,, \tag{A.662}$$

F0999 20_general_flrw_background.tex:187 NEW_EXTENSION_PASS

$$\mathcal{S}_{\Omega_K}(x) := \begin{cases} \Omega_K^{-1/2}\sinh(\sqrt{\Omega_K}\,x), & \Omega_K > 0,\\ x, & \Omega_K = 0,\\ |\Omega_K|^{-1/2}\sin(\sqrt{|\Omega_K|}\,x), & \Omega_K < 0. \end{cases} \tag{A.663}$$

F1000 20_general_flrw_background.tex:197 NEW_EXTENSION_PASS

$$D_A(z) = \frac{D_M(z)}{1+z}, \tag{A.664}$$

$$D_L(z) = (1+z)D_M(z), \tag{A.665}$$

$$\boxed{D_L(z) = (1+z)^2 D_A(z)}, \tag{A.666}$$

$$H_0 t_{\mathrm{L}}(z) = \int_0^z \frac{\mathrm{d}z'}{(1+z')E(z')}, \tag{A.667}$$

$$H_0 t_0 = \int_0^1 \frac{\mathrm{d}a}{aE(a)}. \tag{A.668}$$

F1001 20_general_flrw_background.tex:207 NEW_EXTENSION_PASS

$$H_0 t_0 = \int_0^1 a^{1/2}\,\mathrm{d}a = \frac{2}{3}, \tag{A.669}$$

$$\bar{\chi}(z) = \int_0^z (1+z')^{-3/2}\,\mathrm{d}z' \tag{A.670}$$

$$= 2\left[1 - (1+z)^{-1/2}\right]. \tag{A.671}$$

F1002 20_general_flrw_background.tex:215 NEW_EXTENSION_PASS

$$H_0 t_0 = \int_0^1 a\;\mathrm{d}a = \frac{1}{2}, \tag{A.672}$$

$$\bar{\chi}(z) = \int_0^z \frac{\mathrm{d}z'}{(1+z')^2} = \frac{z}{1+z}. \tag{A.673}$$

F1003 20_general_flrw_background.tex:223 NEW_EXTENSION_PASS

$$H_0 t_0 = \int_0^1 \mathrm{d}a = 1, \tag{A.674}$$

$$\bar{\chi}(z) = \int_0^z \frac{\mathrm{d}z'}{1+z'} = \log(1+z), \tag{A.675}$$

$$\frac{D_M}{D_H} = \sinh\bar{\chi} \qquad (\Omega_K = 1). \tag{A.676}$$

F1004 20_general_flrw_background.tex:235 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}a}{\mathrm{d}\tau} = aE(a). \tag{A.677}$$

F1005 20_general_flrw_background.tex:239 NEW_EXTENSION_PASS

$$k_1 = a_n E(a_n), \tag{A.678}$$
$$k_2 = \left(a_n + \frac{\Delta\tau}{2}k_1\right) E\left(a_n + \frac{\Delta\tau}{2}k_1\right), \tag{A.679}$$
$$a_{n+1} = a_n + \Delta\tau k_2. \tag{A.680}$$

F1006 20_general_flrw_background.tex:246 NEW_EXTENSION_PASS

$$p_a = 2.024278. \tag{A.681}$$

F1007 20_general_flrw_background.tex:250 NEW_EXTENSION_PASS

$$p_{\mathrm{transfer}} = 2.054690. \tag{A.682}$$

F1008 20_general_flrw_background.tex:254 NEW_EXTENSION_PASS

$$\varepsilon_{\text{distance benchmarks}} = 8.881784 \times 10^{-16}, \tag{A.683}$$
$$\varepsilon_{\text{age benchmarks}} = 0, \tag{A.684}$$
$$\varepsilon_{D_L-(1+z)^2 D_A} = 0, \tag{A.685}$$
$$\varepsilon_{\text{total interacting continuity}} = 4.440892 \times 10^{-16}. \tag{A.686}$$

21_vector_tensor_flrw_perturbations.tex

F1009 21_vector_tensor_flrw_perturbations.tex:9 NEW_EXTENSION_PASS

$$\mathrm{d}s^2 = a(\eta)^2\left[-\mathrm{d}\eta^2 + 2S_i\,\mathrm{d}\eta\,\mathrm{d}x^i + \left(\delta_{ij} + 2\partial_{(i}F_{j)} + h_{ij}\right)\,\mathrm{d}x^i\,\mathrm{d}x^j\right], \tag{A.687}$$

F1010 21_vector_tensor_flrw_perturbations.tex:19 NEW_EXTENSION_PASS

$$\partial^i S_i = 0, \tag{A.688}$$
$$\partial^i F_i = 0, \tag{A.689}$$
$$\partial^i h_{ij} = 0, \tag{A.690}$$
$$h^i{}_i = 0. \tag{A.691}$$

F1011 21_vector_tensor_flrw_perturbations.tex:26 NEW_EXTENSION_PASS

$$S_i \mapsto S_i - L_i', \tag{A.692}$$
$$F_i \mapsto F_i - L_i. \tag{A.693}$$

F1012 21_vector_tensor_flrw_perturbations.tex:31 NEW_EXTENSION_PASS

$$\boxed{\Sigma_i := S_i - F_i'} \tag{A.694}$$

F1013 21_vector_tensor_flrw_perturbations.tex:37 NEW_EXTENSION_PASS

$$\Sigma_i \mapsto (S_i - L_i') - (F_i - L_i)' \tag{A.695}$$
$$= S_i - L_i' - F_i' + L_i' \tag{A.696}$$
$$= \Sigma_i. \tag{A.697}$$

F1014 21_vector_tensor_flrw_perturbations.tex:47 NEW_EXTENSION_PASS

$$\boxed{(a^2\Sigma_i)' = 0.} \tag{A.698}$$

F1015 21_vector_tensor_flrw_perturbations.tex:53 NEW_EXTENSION_PASS

$$(a^2\Sigma_i)' = 0, \tag{A.699}$$

$$a^2\Sigma_i = C_i, \tag{A.700}$$

$$\Sigma_i(\eta) = \frac{C_i}{a(\eta)^2}. \tag{A.701}$$

F1016 21_vector_tensor_flrw_perturbations.tex:59 NEW_EXTENSION_PASS

$$\Sigma_i \propto \eta^{-4}. \tag{A.702}$$

F1017 21_vector_tensor_flrw_perturbations.tex:63 NEW_EXTENSION_PASS

$$\Sigma' = -\frac{4}{\eta}\Sigma \tag{A.703}$$

F1018 21_vector_tensor_flrw_perturbations.tex:67 NEW_EXTENSION_PASS

$$p_{\mathrm{vector}} = 2.052951. \tag{A.704}$$

F1019 21_vector_tensor_flrw_perturbations.tex:74 NEW_EXTENSION_PASS

$$\boxed{S_T^{(2)} = \frac{M_{\mathrm{Pl}}^2}{8}\int a^2\left[(h'_{ij})^2 - (\partial_k h_{ij})^2\right]\,\mathrm{d}\eta\,\mathrm{d}^3x.} \tag{A.705}$$

F1020 21_vector_tensor_flrw_perturbations.tex:86 NEW_EXTENSION_PASS

$$\delta S_T^{(2)} = \frac{M_{\mathrm{Pl}}^2}{4}\int a^2\left[h'_{ij}\delta h'_{ij} - \partial_k h_{ij}\partial_k \delta h_{ij}\right]\,\mathrm{d}\eta\,\mathrm{d}^3x \tag{A.706}$$

$$= -\frac{M_{\mathrm{Pl}}^2}{4}\int \left[(a^2 h'_{ij})' - a^2\nabla^2 h_{ij}\right]\delta h_{ij}\,\mathrm{d}\eta\,\mathrm{d}^3x. \tag{A.707}$$

F1021 21_vector_tensor_flrw_perturbations.tex:102 NEW_EXTENSION_PASS

$$(a^2 h'_{ij})' = a^2 h''_{ij} + 2aa' h'_{ij} = a^2(h''_{ij} + 2\mathcal{H}h'_{ij}), \tag{A.708}$$

F1022 21_vector_tensor_flrw_perturbations.tex:108 NEW_EXTENSION_PASS

$$\boxed{h''_{ij} + 2\mathcal{H}h'_{ij} - \nabla^2 h_{ij} = 0, \qquad \mathcal{H} := a'/a.} \tag{A.709}$$

F1023 21_vector_tensor_flrw_perturbations.tex:116 NEW_EXTENSION_PASS

$$h''_{ij} + 2\mathcal{H}h'_{ij} - \nabla^2 h_{ij} = \frac{2a^2}{M_{\mathrm{Pl}}^2}\Pi_{ij}^{\mathrm{TT}}. \tag{A.710}$$

F1024 21_vector_tensor_flrw_perturbations.tex:121 NEW_EXTENSION_PASS

$$\mu_\lambda := \frac{aM_{\mathrm{Pl}}}{2}h_\lambda, \qquad h_\lambda = \frac{2\mu_\lambda}{aM_{\mathrm{Pl}}}. \tag{A.711}$$

F1025 21_vector_tensor_flrw_perturbations.tex:127 NEW_EXTENSION_PASS

$$h'_\lambda = \frac{2}{aM_{\mathrm{Pl}}}(\mu'_\lambda - \mathcal{H}\mu_\lambda), \tag{A.712}$$

$$h''_\lambda = \frac{2}{aM_{\mathrm{Pl}}}\left[\mu''_\lambda - 2\mathcal{H}\mu'_\lambda + (\mathcal{H}^2 - \mathcal{H}')\mu_\lambda\right]. \tag{A.713}$$

F1026 21_vector_tensor_flrw_perturbations.tex:141 NEW_EXTENSION_PASS

$$0 = \mu''_\lambda + \left[k^2 - (\mathcal{H}' + \mathcal{H}^2)\right]\mu_\lambda \tag{A.714}$$

$$= \mu''_\lambda + \left(k^2 - \frac{a''}{a}\right)\mu_\lambda. \tag{A.715}$$

F1027 21_vector_tensor_flrw_perturbations.tex:149 NEW_EXTENSION_PASS

$$\boxed{\mu''_\lambda + \left(k^2 - \frac{a''}{a}\right)\mu_\lambda = 0.} \tag{A.716}$$

F1028 21_vector_tensor_flrw_perturbations.tex:158 NEW_EXTENSION_PASS

$$\widehat{k}_h^2 = \frac{4}{h^2}\sum_{r=1}^{3}\sin^2\left(\frac{k_r h}{2}\right) \tag{A.717}$$

$$= \sum_{r=1}^{3}\left[k_r^2 - \frac{h^2 k_r^4}{12} + \frac{h^4 k_r^6}{360} + \mathcal{O}(h^6)\right] \tag{A.718}$$

$$= k^2 - \frac{h^2}{12}\sum_{r=1}^{3}k_r^4 + \frac{h^4}{360}\sum_{r=1}^{3}k_r^6 + \mathcal{O}(h^6). \tag{A.719}$$

F1029 21_vector_tensor_flrw_perturbations.tex:173 NEW_EXTENSION_PASS

$$\mu''_{\lambda,h} + \left(\widehat{k}_h^2 - \frac{a''}{a}\right)\mu_{\lambda,h} = 0. \tag{A.720}$$

F1030 21_vector_tensor_flrw_perturbations.tex:179 NEW_EXTENSION_PASS

$$a(\eta) \propto \eta, \qquad a'' = 0, \tag{A.721}$$

$$\mu_\kappa(\eta) = \frac{\mathrm{e}^{-\mathrm{i}\kappa\eta}}{\sqrt{2\kappa}}, \qquad h_\kappa = \frac{2\mu_\kappa}{aM_{\mathrm{Pl}}}. \tag{A.722}$$

F1031 21_vector_tensor_flrw_perturbations.tex:186 NEW_EXTENSION_PASS

$$a|h_\kappa| = \text{constant}. \tag{A.723}$$

F1032 21_vector_tensor_flrw_perturbations.tex:190 NEW_EXTENSION_PASS

$$a(\eta) \propto \eta^2, \tag{A.724}$$

$$h_\kappa(\eta) = 3\frac{j_1(\kappa\eta)}{\kappa\eta}, \tag{A.725}$$

$$h''_\kappa + \frac{4}{\eta}h'_\kappa + \kappa^2 h_\kappa = 0. \tag{A.726}$$

F1033 21_vector_tensor_flrw_perturbations.tex:197 NEW_EXTENSION_PASS

$$p_T^{\mathrm{radiation}} = 1.999618, \tag{A.727}$$

$$p_{T,\mathrm{R}}^{\mathrm{radiation}} = 3.998798, \tag{A.728}$$

$$p_T^{\mathrm{matter}} = 2.002687, \tag{A.729}$$

$$p_{T,\mathrm{R}}^{\mathrm{matter}} = 3.996376. \tag{A.730}$$

F1034 21_vector_tensor_flrw_perturbations.tex:204 NEW_EXTENSION_PASS

$$\varepsilon_{\mathrm{W}} = 4.440892 \times 10^{-16}, \tag{A.731}$$

$$\varepsilon_{a|h|} = 4.440892 \times 10^{-16}. \tag{A.732}$$

F1035 21_vector_tensor_flrw_perturbations.tex:211 NEW_EXTENSION_PASS

$$\mu \sim \frac{\mathrm{e}^{-\mathrm{i}k\eta}}{r}. \tag{A.733}$$

F1036 21_vector_tensor_flrw_perturbations.tex:215 NEW_EXTENSION_PASS

$$h \sim \frac{1}{ar}, \tag{A.734}$$

F1037 21_vector_tensor_flrw_perturbations.tex:220 NEW_EXTENSION_PASS

$$D_L^{\mathrm{GW}} = D_L^{\mathrm{EM}} \tag{A.735}$$

22_linear_structure_growth.tex

F1038 22_linear_structure_growth.tex:9 NEW_EXTENSION_PASS

$$\dot{\delta} = -\frac{\theta}{a}, \tag{A.736}$$

$$\dot{\theta} + H\theta = -\frac{1}{a}\nabla^2\Phi, \tag{A.737}$$

$$\nabla^2\Phi = 4\pi G a^2 \bar{\rho}_m \delta. \tag{A.738}$$

F1039 22_linear_structure_growth.tex:15 NEW_EXTENSION_PASS

$$\ddot{\delta} = -\frac{\dot{\theta}}{a} + \frac{H\theta}{a}. \tag{A.739}$$

F1040 22_linear_structure_growth.tex:21 NEW_EXTENSION_PASS

$$\dot{\theta} = -H\theta - \frac{1}{a}\nabla^2\Phi \tag{A.740}$$

F1041 22_linear_structure_growth.tex:25 NEW_EXTENSION_PASS

$$\ddot{\delta} = \frac{H\theta}{a} + \frac{1}{a^2}\nabla^2\Phi + \frac{H\theta}{a} \tag{A.741}$$

$$= -2H\dot{\delta} + \frac{1}{a^2}\nabla^2\Phi. \tag{A.742}$$

F1042 22_linear_structure_growth.tex:34 NEW_EXTENSION_PASS

$$\frac{1}{a^2}\nabla^2\Phi = 4\pi G\bar{\rho}_m\delta, \tag{A.743}$$

F1043 22_linear_structure_growth.tex:39 NEW_EXTENSION_PASS

$$\boxed{\ddot{\delta} + 2H\dot{\delta} - 4\pi G\bar{\rho}_m\delta = 0.} \tag{A.744}$$

F1044 22_linear_structure_growth.tex:45 NEW_EXTENSION_PASS

$$\dot{D} = aHD_{,a}, \tag{A.745}$$

$$\ddot{D} = a^2H^2D_{,aa} + \left(aH^2 + a^2HH_{,a}\right)D_{,a}. \tag{A.746}$$

F1045 22_linear_structure_growth.tex:52 NEW_EXTENSION_PASS

$$a^2H^2D_{,aa} + \left(3aH^2 + a^2HH_{,a}\right)D_{,a} - 4\pi G\bar{\rho}_m D = 0. \tag{A.747}$$

F1046 22_linear_structure_growth.tex:58 NEW_EXTENSION_PASS

$$4\pi G\bar{\rho}_m = \frac{3}{2}\Omega_{m0}H_0^2a^{-3} \tag{A.748}$$

F1047 22_linear_structure_growth.tex:63 NEW_EXTENSION_PASS

$$\boxed{D_{,aa} + \left(\frac{3}{a} + \frac{H_{,a}}{H}\right)D_{,a} - \frac{3\Omega_{m0}H_0^2}{2a^5H(a)^2}D = 0.} \tag{A.749}$$

F1048 22_linear_structure_growth.tex:71 NEW_EXTENSION_PASS

$$D_{,a} = \frac{1}{a}D_{,N}, \tag{A.750}$$

$$D_{,aa} = \frac{1}{a^2}\left(D_{,NN} - D_{,N}\right), \tag{A.751}$$

$$\frac{aH_{,a}}{H} = \frac{\mathrm{d}\log H}{\mathrm{d}N}, \tag{A.752}$$

$$\Omega_m(a) = \frac{\Omega_{m0}H_0^2a^{-3}}{H(a)^2}. \tag{A.753}$$

F1049 22_linear_structure_growth.tex:81 NEW_EXTENSION_PASS

$$\boxed{D_{,NN} + \left[2 + \frac{\mathrm{d}\log H}{\mathrm{d}N}\right] D_{,N} - \frac{3}{2}\Omega_m(a) D = 0.} \tag{A.754}$$

F1050 22_linear_structure_growth.tex:91 NEW_EXTENSION_PASS

$$\Omega_m(a) = 1, \qquad \frac{\mathrm{d}\log H}{\mathrm{d}N} = -\frac{3}{2}, \tag{A.755}$$

F1051 22_linear_structure_growth.tex:97 NEW_EXTENSION_PASS

$$r^2 + \frac{1}{2} r - \frac{3}{2} = 0, \tag{A.756}$$

$$(r-1)\left(r + \frac{3}{2}\right) = 0. \tag{A.757}$$

F1052 22_linear_structure_growth.tex:102 NEW_EXTENSION_PASS

$$\boxed{D_+(a) = a, \qquad D_-(a) = a^{-3/2}.} \tag{A.758}$$

F1053 22_linear_structure_growth.tex:110 NEW_EXTENSION_PASS

$$p_{\mathrm{EdS}} = 1.968111. \tag{A.759}$$

F1054 22_linear_structure_growth.tex:116 NEW_EXTENSION_PASS

$$E(a)^2 = \Omega_{m0} a^{-3} + 1 - \Omega_{m0}, \tag{A.760}$$

$$\Omega_m(a) = \frac{\Omega_{m0} a^{-3}}{E(a)^2}. \tag{A.761}$$

F1055 22_linear_structure_growth.tex:121 NEW_EXTENSION_PASS

$$I(a) := \int_0^a \frac{\mathrm{d}a'}{a'^3 E(a')^3}, \qquad I_{,a} = \frac{1}{a^3 E(a)^3}. \tag{A.762}$$

F1056 22_linear_structure_growth.tex:127 NEW_EXTENSION_PASS

$$\boxed{D_+(a) = \frac{5\Omega_{m0}}{2} E(a) \int_0^a \frac{\mathrm{d}a'}{a'^3 E(a')^3}.} \tag{A.763}$$

F1057 22_linear_structure_growth.tex:136 NEW_EXTENSION_PASS

$$\varepsilon_{\mathrm{Heath}} = 1.035117 \times 10^{-12}. \tag{A.764}$$

F1058 22_linear_structure_growth.tex:141 NEW_EXTENSION_PASS

$$p_{\Lambda\mathrm{CDM}} = 1.970169. \tag{A.765}$$

F1059 22_linear_structure_growth.tex:147 NEW_EXTENSION_PASS

$$\Delta_h \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}} = -\widehat{k}_h^2 \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}. \tag{A.766}$$

F1060 22_linear_structure_growth.tex:152 NEW_EXTENSION_PASS

$$-\widehat{k}_h^2 \Phi_h = 4\pi G a^2 \bar{\rho}_m \delta_h. \tag{A.767}$$

F1061 22_linear_structure_growth.tex:157 NEW_EXTENSION_PASS

$$\dot{\theta}_h + H\theta_h = \frac{\widehat{k}_h^2}{a} \Phi_h. \tag{A.768}$$

F1062 22_linear_structure_growth.tex:162 NEW_EXTENSION_PASS

$$\boxed{\frac{\widehat{k}_h^2}{a} \Phi_h = -4\pi G a \bar{\rho}_m \delta_h.} \tag{A.769}$$

F1063 22_linear_structure_growth.tex:169 NEW_EXTENSION_PASS

$$\ddot{\delta}_h = -\frac{\dot{\theta}_h}{a} + \frac{H\theta_h}{a} \tag{A.770}$$

$$= -2H\dot{\delta}_h - \frac{\widehat{k}_h^2}{a^2}\Phi_h. \tag{A.771}$$

F1064 22_linear_structure_growth.tex:177 NEW_EXTENSION_PASS

$$-\frac{\widehat{k}_h^2}{a^2}\Phi_h = 4\pi G\bar{\rho}_m\delta_h. \tag{A.772}$$

F1065 22_linear_structure_growth.tex:182 NEW_EXTENSION_PASS

$$\boxed{\ddot{\delta}_h + 2H\dot{\delta}_h - 4\pi G\bar{\rho}_m\delta_h = 0.} \tag{A.773}$$

F1066 22_linear_structure_growth.tex:192 NEW_EXTENSION_PASS

$$f(a) := \frac{\mathrm{d}\log D}{\mathrm{d}\log a} = \frac{(1/D)\,\mathrm{d}D}{\mathrm{d}N} = \frac{D_{,N}}{D}, \tag{A.774}$$

$$D_N(a) := \frac{D(a)}{D(1)}, \tag{A.775}$$

$$\sigma_8(a) = D_N(a)\sigma_{8,0}, \tag{A.776}$$

$$f\sigma_8(a) := f(a)\sigma_8(a) = f(a)D_N(a)\sigma_{8,0}. \tag{A.777}$$

F1067 22_linear_structure_growth.tex:203 NEW_EXTENSION_PASS

$$\Omega_{m0} = 0.3, \qquad \sigma_{8,0} = 0.8, \tag{A.778}$$

F1068 22_linear_structure_growth.tex:209 NEW_EXTENSION_PASS

$$f(1) = 0.512796, \tag{A.779}$$

$$f\sigma_8(1) = 0.410237. \tag{A.780}$$

23_primordial_power_law_inflation.tex

F1069 23_primordial_power_law_inflation.tex:9 NEW_EXTENSION_PASS

$$0 < \epsilon < 1, \qquad q := \frac{1}{1-\epsilon}, \qquad \eta < 0, \tag{A.781}$$

F1070 23_primordial_power_law_inflation.tex:17 NEW_EXTENSION_PASS

$$\boxed{a(\eta) = A(-\eta)^{-q}.} \tag{A.782}$$

F1071 23_primordial_power_law_inflation.tex:23 NEW_EXTENSION_PASS

$$a' = Aq(-\eta)^{-q-1}, \tag{A.783}$$

$$\mathcal{H} = \frac{a'}{a} = -\frac{q}{\eta}, \tag{A.784}$$

$$\mathcal{H}' = \frac{q}{\eta^2}. \tag{A.785}$$

F1072 23_primordial_power_law_inflation.tex:29 NEW_EXTENSION_PASS

$$1 - \frac{\mathcal{H}'}{\mathcal{H}^2} = 1 - \frac{1}{q} = \epsilon. \tag{A.786}$$

F1073 23_primordial_power_law_inflation.tex:36 NEW_EXTENSION_PASS

$$\mathcal{H} := \frac{a'}{a} = -\frac{q}{\eta}, \tag{A.787}$$

$$\epsilon_H := 1 - \frac{\mathcal{H}'}{\mathcal{H}^2} = \epsilon. \tag{A.788}$$

F1074 23_primordial_power_law_inflation.tex:41 NEW_EXTENSION_PASS

$$(aH)^{-1} = \mathcal{H}^{-1} = -\frac{\eta}{q}. \tag{A.789}$$

F1075 23_primordial_power_law_inflation.tex:45 NEW_EXTENSION_PASS

$$\frac{(aH)^{-1}}{(a_i H_i)^{-1}} = \mathrm{e}^{-(1-\epsilon)N}, \tag{A.790}$$

$$\frac{|\Omega_K|}{|\Omega_{K,i}|} = \mathrm{e}^{-2(1-\epsilon)N}. \tag{A.791}$$

F1076 23_primordial_power_law_inflation.tex:52 NEW_EXTENSION_PASS

$$H/H_i = \mathrm{e}^{-\epsilon N}, \qquad a/a_i = \mathrm{e}^{N}. \tag{A.792}$$

F1077 23_primordial_power_law_inflation.tex:60 NEW_EXTENSION_PASS

$$N \geq \frac{\log(S_K^{-1})}{2(1-\epsilon)} \Longrightarrow \frac{|\Omega_K|}{|\Omega_{K,i}|} \leq S_K. \tag{A.793}$$

F1078 23_primordial_power_law_inflation.tex:68 NEW_EXTENSION_PASS

$$z := a\sqrt{2\epsilon}\, M_{\mathrm{Pl}}, \qquad \nu := q + \frac{1}{2} = \frac{3-\epsilon}{2(1-\epsilon)}. \tag{A.794}$$

F1079 23_primordial_power_law_inflation.tex:74 NEW_EXTENSION_PASS

$$\boxed{\frac{z''}{z} = \frac{a''}{a} = \frac{\nu^2 - \frac{1}{4}}{\eta^2}.} \tag{A.795}$$

F1080 23_primordial_power_law_inflation.tex:82 NEW_EXTENSION_PASS

$$\frac{a''}{a} = \frac{q(q+1)}{\eta^2}, \tag{A.796}$$

$$\nu^2 - \frac{1}{4} = \left(q + \frac{1}{2}\right)^2 - \frac{1}{4} \tag{A.797}$$

$$= q(q+1). \tag{A.798}$$

F1081 23_primordial_power_law_inflation.tex:90 NEW_EXTENSION_PASS

$$S_v^{(2)} = \frac{1}{2}\int \mathrm{d}\eta\, \mathrm{d}^3k \left[|v_k'|^2 - \left(k^2 - \frac{z''}{z}\right)|v_k|^2\right]. \tag{A.799}$$

F1082 23_primordial_power_law_inflation.tex:99 NEW_EXTENSION_PASS

$$\delta S_v^{(2)} = -\frac{1}{2}\int \delta v_k^* \left[v_k'' + \left(k^2 - \frac{z''}{z}\right) v_k\right] \mathrm{d}\eta\, \mathrm{d}^3k + \mathrm{c.c.}, \tag{A.800}$$

F1083 23_primordial_power_law_inflation.tex:111 NEW_EXTENSION_PASS

$$\boxed{v_k'' + \left[k^2 - \frac{\nu^2 - \frac{1}{4}}{\eta^2}\right] v_k = 0.} \tag{A.801}$$

F1084 23_primordial_power_law_inflation.tex:118 NEW_EXTENSION_PASS

$$\boxed{v_k(\eta) = \frac{\sqrt{-\pi\eta}}{2}\mathrm{e}^{\mathrm{i}(2\nu+1)\pi/4} H_\nu^{(1)}(-k\eta).} \tag{A.802}$$

F1085 23_primordial_power_law_inflation.tex:127 NEW_EXTENSION_PASS

$$v_k v_k^{*\prime} - v_k^* v_k' = \mathrm{i}. \tag{A.803}$$

F1086 23_primordial_power_law_inflation.tex:134 NEW_EXTENSION_PASS

$$\boxed{\mathcal{P}_{\mathcal{R}}(k) = \frac{2^{2\nu-4}\Gamma(\nu)^2}{\pi^3 A^2 \epsilon M_{\rm Pl}^2} k^{3-2\nu}.} \tag{A.804}$$

F1087 23_primordial_power_law_inflation.tex:144 NEW_EXTENSION_PASS

$$H_\nu^{(1)}(x) \sim -\frac{\mathrm{i}\Gamma(\nu)}{\pi}\left(\frac{2}{x}\right)^\nu. \tag{A.805}$$

F1088 23_primordial_power_law_inflation.tex:151 NEW_EXTENSION_PASS

$$|v_k|^2 \sim \frac{2^{2\nu-2}\Gamma(\nu)^2}{\pi}(-\eta)^{1-2\nu}k^{-2\nu}. \tag{A.806}$$

F1089 23_primordial_power_law_inflation.tex:159 NEW_EXTENSION_PASS

$$\mathcal{P}_{\mathcal{R}} = \frac{k^3}{2\pi^2}|\mathcal{R}_k|^2, \tag{A.807}$$

F1090 23_primordial_power_law_inflation.tex:165 NEW_EXTENSION_PASS

$$\boxed{\mathcal{P}_T(k) = \frac{2^{2\nu}\Gamma(\nu)^2}{\pi^3 A^2 M_{\rm Pl}^2} k^{3-2\nu}.} \tag{A.808}$$

F1091 23_primordial_power_law_inflation.tex:175 NEW_EXTENSION_PASS

$$n_s - 1 = 3 - 2\nu = -\frac{2\epsilon}{1-\epsilon}, \tag{A.809}$$

$$n_t = -\frac{2\epsilon}{1-\epsilon}, \tag{A.810}$$

$$\alpha_s := \frac{\mathrm{d}n_s}{\mathrm{d}\log k} = 0, \tag{A.811}$$

$$\alpha_t := \frac{\mathrm{d}n_t}{\mathrm{d}\log k} = 0, \tag{A.812}$$

$$\boxed{r := \frac{\mathcal{P}_T}{\mathcal{P}_{\mathcal{R}}} = 16\epsilon}, \tag{A.813}$$

$$\boxed{r = -8(1-\epsilon)n_t}. \tag{A.814}$$

F1092 23_primordial_power_law_inflation.tex:184 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}\log \mathcal{P}_{\mathcal{R}}}{\mathrm{d}\log k} = 3 - 2\nu. \tag{A.815}$$

F1093 23_primordial_power_law_inflation.tex:191 NEW_EXTENSION_PASS

$$r = \frac{2^{2\nu}}{2^{2\nu-4}/\epsilon} = 16\epsilon, \tag{A.816}$$

F1094 23_primordial_power_law_inflation.tex:202 NEW_EXTENSION_PASS

$$\epsilon = 0.01, \qquad N = 60, \tag{A.817}$$

F1095 23_primordial_power_law_inflation.tex:208 NEW_EXTENSION_PASS

$$q = 1.010101, \tag{A.818}$$

$$\nu = 1.510101, \tag{A.819}$$

$$n_s = 0.979798, \tag{A.820}$$

$$n_t = -0.020202, \tag{A.821}$$

$$r = 0.16, \tag{A.822}$$

$$\frac{|\Omega_K|}{|\Omega_{K,i}|} = 2.545749 \times 10^{-52}, \tag{A.823}$$

$$\frac{(aH)^{-1}}{(a_i H_i)^{-1}} = 1.595540 \times 10^{-26}. \tag{A.824}$$

F1096 23_primordial_power_law_inflation.tex:220 NEW_EXTENSION_PASS

$$\varepsilon_{\epsilon_H} = 0, \tag{A.825}$$
$$\varepsilon_{z''/z} = 0, \tag{A.826}$$
$$\varepsilon_{n_s} = 1.179612 \times 10^{-16}, \tag{A.827}$$
$$\varepsilon_{n_t} = 2.255141 \times 10^{-16}, \tag{A.828}$$
$$\varepsilon_r = 2.775558 \times 10^{-17}, \tag{A.829}$$
$$\varepsilon_{\mathrm{W}} = 3.907985 \times 10^{-14}, \tag{A.830}$$
$$p_{v,\mathrm{mid}} = 1.964609. \tag{A.831}$$

24_recombination_acoustics.tex

F1097 24_recombination_acoustics.tex:9 NEW_EXTENSION_PASS

$$H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{A.832}$$
$$\Omega_b = 0.05, \qquad \Omega_m = 0.30, \tag{A.833}$$
$$\Omega_r = 9 \times 10^{-5}, \qquad T_0 = 2.7255\ \mathrm{K}, \tag{A.834}$$
$$Y_p = 0.24. \tag{A.835}$$

F1098 24_recombination_acoustics.tex:20 NEW_EXTENSION_PASS

$$T(z) = T_0(1+z), \tag{A.836}$$
$$n_H(z) = n_{H0}(1+z)^3, \tag{A.837}$$
$$S(z) = \frac{1}{n_H(z)} \left(\frac{m_e k_B T(z)}{2\pi\hbar^2} \right)^{3/2} \exp\left[-\frac{\chi_H}{k_B T(z)} \right]. \tag{A.838}$$

F1099 24_recombination_acoustics.tex:31 NEW_EXTENSION_PASS

$$\frac{x_e^2}{1 - x_e} = S. \tag{A.839}$$

F1100 24_recombination_acoustics.tex:35 NEW_EXTENSION_PASS

$$x_e^2 + S x_e - S = 0. \tag{A.840}$$

F1101 24_recombination_acoustics.tex:39 NEW_EXTENSION_PASS

$$x_e = \frac{-S \pm \sqrt{S^2 + 4S}}{2}. \tag{A.841}$$

F1102 24_recombination_acoustics.tex:43 NEW_EXTENSION_PASS

$$x_e(S) = \frac{-S + \sqrt{S^2 + 4S}}{2} \tag{A.842}$$
$$= \frac{2S}{S + \sqrt{S^2 + 4S}} \tag{A.843}$$
$$= \frac{2}{1 + \sqrt{1 + 4/S}}. \tag{A.844}$$

F1103 24_recombination_acoustics.tex:50 NEW_EXTENSION_PASS

$$\boxed{x_e(S) = \frac{-S + \sqrt{S^2 + 4S}}{2} = \frac{2}{1 + \sqrt{1 + 4/S}}.} \tag{A.845}$$

F1104 24_recombination_acoustics.tex:59 NEW_EXTENSION_PASS

$$n_e(z) = x_e(z) n_H(z), \tag{A.846}$$

$$\frac{\mathrm{d}\tau}{\mathrm{d}z} = \frac{c\sigma_T n_e(z)}{(1+z)H(z)}, \tag{A.847}$$

$$\tau(z) = \int_0^z \frac{c\sigma_T n_e(z')}{(1+z')H(z')}\,\mathrm{d}z'. \tag{A.848}$$

F1105 24_recombination_acoustics.tex:69 NEW_EXTENSION_PASS

$$-\frac{\mathrm{d}}{\mathrm{d}z}\mathrm{e}^{-\tau(z)} = -\left[-\mathrm{e}^{-\tau(z)}\frac{\mathrm{d}\tau}{\mathrm{d}z}\right] \tag{A.849}$$

$$= \mathrm{e}^{-\tau(z)}\frac{\mathrm{d}\tau}{\mathrm{d}z}. \tag{A.850}$$

F1106 24_recombination_acoustics.tex:75 NEW_EXTENSION_PASS

$$\boxed{g_z(z) = \mathrm{e}^{-\tau(z)}\frac{\mathrm{d}\tau}{\mathrm{d}z}.} \tag{A.851}$$

F1107 24_recombination_acoustics.tex:81 NEW_EXTENSION_PASS

$$\int_0^{z_{\max}} g_z(z)\,\mathrm{d}z = -\int_0^{z_{\max}} \frac{\mathrm{d}}{\mathrm{d}z}\mathrm{e}^{-\tau(z)}\,\mathrm{d}z \tag{A.852}$$

$$= \mathrm{e}^{-\tau(0)} - \mathrm{e}^{-\tau(z_{\max})} \tag{A.853}$$

$$= 1 - \mathrm{e}^{-\tau(z_{\max})}. \tag{A.854}$$

F1108 24_recombination_acoustics.tex:89 NEW_EXTENSION_PASS

$$z_* = 1282.15, \tag{A.855}$$

$$z_{\tau=1} = 1289.423097, \tag{A.856}$$

$$\Delta z_{\mathrm{FWHM}} = 139.2. \tag{A.857}$$

F1109 24_recombination_acoustics.tex:95 NEW_EXTENSION_PASS

$$\varepsilon_g = 4.807365 \times 10^{-8}. \tag{A.858}$$

F1110 24_recombination_acoustics.tex:102 NEW_EXTENSION_PASS

$$\rho_b(z) = \rho_{b0}(1+z)^3, \qquad \rho_\gamma(z) = \rho_{\gamma 0}(1+z)^4, \tag{A.859}$$

F1111 24_recombination_acoustics.tex:108 NEW_EXTENSION_PASS

$$R(z) := \frac{3\rho_b}{4\rho_\gamma} \tag{A.860}$$

$$= \frac{3\rho_{b0}(1+z)^3}{4\rho_{\gamma 0}(1+z)^4} \tag{A.861}$$

$$= \frac{3\Omega_b}{4\Omega_\gamma(1+z)}, \tag{A.862}$$

$$c_s(z) := \frac{c}{\sqrt{3(1+R(z))}}. \tag{A.863}$$

F1112 24_recombination_acoustics.tex:117 NEW_EXTENSION_PASS

$$r_s(t_*) = \int_0^{t_*} \frac{c_s(t)}{a(t)}\,\mathrm{d}t. \tag{A.864}$$

F1113 24_recombination_acoustics.tex:121 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}z}{\mathrm{d}t} = -(1+z)H(z), \qquad a = (1+z)^{-1}, \tag{A.865}$$

F1114 24_recombination_acoustics.tex:127 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}t}{a} = -\frac{\mathrm{d}z}{H(z)}. \tag{A.866}$$

F1115 24_recombination_acoustics.tex:131 NEW_EXTENSION_PASS

$$\boxed{r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)}{H(z)}\,\mathrm{d}z.} \tag{A.867}$$

F1116 24_recombination_acoustics.tex:138 NEW_EXTENSION_PASS

$$\chi_* := \int_0^{z_*} \frac{c}{H(z)}\,\mathrm{d}z. \tag{A.868}$$

F1117 24_recombination_acoustics.tex:142 NEW_EXTENSION_PASS

$$\boxed{\ell_A := \pi \frac{\chi_*}{r_s(z_*)}.} \tag{A.869}$$

F1118 24_recombination_acoustics.tex:148 NEW_EXTENSION_PASS

$$r_s(z_*) = 125.784218 \text{ Mpc}, \tag{A.870}$$
$$\chi_* = 13644.312614 \text{ Mpc}, \tag{A.871}$$
$$\ell_A = 340.781006. \tag{A.872}$$

F1119 24_recombination_acoustics.tex:157 NEW_EXTENSION_PASS

$$X'' + k^2 c_s^2 X = 0. \tag{A.873}$$

F1120 24_recombination_acoustics.tex:161 NEW_EXTENSION_PASS

$$X(\eta) = A\cos(\omega_s \Delta\eta) + B\sin(\omega_s \Delta\eta). \tag{A.874}$$

F1121 24_recombination_acoustics.tex:166 NEW_EXTENSION_PASS

$$X(\eta) = X_i \cos(k c_s \Delta\eta) + \frac{X_i'}{k c_s}\sin(k c_s \Delta\eta). \tag{A.875}$$

F1122 24_recombination_acoustics.tex:171 NEW_EXTENSION_PASS

$$X'' + k^2 c_s(a)^2 X = 0 \tag{A.876}$$

F1123 24_recombination_acoustics.tex:175 NEW_EXTENSION_PASS

$$p_{\mathrm{acoustic}} = 2.021414. \tag{A.877}$$

F1124 24_recombination_acoustics.tex:179 NEW_EXTENSION_PASS

$$\varepsilon_{\mathrm{acoustic}} = 3.978519 \times 10^{-5}. \tag{A.878}$$

F1125 24_recombination_acoustics.tex:186 NEW_EXTENSION_PASS

$$\kappa' := a n_e \sigma_T. \tag{A.879}$$

F1126 24_recombination_acoustics.tex:190 NEW_EXTENSION_PASS

$$\boxed{\frac{1}{k_D^2(\eta_*)} = \int_0^{\eta_*} \frac{\mathrm{d}\eta}{6\kappa'} \frac{R^2 + \frac{16}{15}(1+R)}{(1+R)^2}.} \tag{A.880}$$

F1127 24_recombination_acoustics.tex:200 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}X_k}{\mathrm{d}\eta} = -k^2 D_\gamma(\eta) X_k, \tag{A.881}$$

F1128 24_recombination_acoustics.tex:205 NEW_EXTENSION_PASS

$$\int_0^{\eta_*} D_\gamma(\eta)\,\mathrm{d}\eta = \frac{1}{k_D^2(\eta_*)}. \tag{A.882}$$

F1129 24_recombination_acoustics.tex:210 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}X_k}{X_k} = -k^2 D_\gamma\,\mathrm{d}\eta, \tag{A.883}$$

$$\log \frac{X_k(\eta_*)}{X_k(0)} = -\frac{k^2}{k_D^2}, \tag{A.884}$$

F1130 24_recombination_acoustics.tex:216 NEW_EXTENSION_PASS

$$\mathcal{D}(k) = \exp\left[-\frac{k^2}{k_D^2}\right]. \tag{A.885}$$

F1131 24_recombination_acoustics.tex:220 NEW_EXTENSION_PASS

$$r_D := k_D^{-1} = 4.554719 \text{ Mpc}, \tag{A.886}$$

$$k_D = 0.219553 \text{ Mpc}^{-1}. \tag{A.887}$$

25_minimal_cmb_boltzmann.tex

F1132 25_minimal_cmb_boltzmann.tex:9 NEW_EXTENSION_PASS

$$\Theta(\mu,\eta) = \sum_{\ell=0}^{\infty} (-\mathrm{i})^\ell (2\ell+1)\Theta_\ell(\eta) P_\ell(\mu). \tag{A.888}$$

F1133 25_minimal_cmb_boltzmann.tex:17 NEW_EXTENSION_PASS

$$\Theta' + \mathrm{i}k\mu\Theta = 0. \tag{A.889}$$

F1134 25_minimal_cmb_boltzmann.tex:21 NEW_EXTENSION_PASS

$$\Theta_\ell = \frac{\mathrm{i}^\ell}{2} \int_{-1}^{1} P_\ell(\mu)\Theta(\mu)\,\mathrm{d}\mu. \tag{A.890}$$

F1135 25_minimal_cmb_boltzmann.tex:27 NEW_EXTENSION_PASS

$$\Theta_\ell' = -\frac{\mathrm{i}^{\ell+1}k}{2} \int_{-1}^{1} \mu P_\ell(\mu)\Theta(\mu)\,\mathrm{d}\mu. \tag{A.891}$$

F1136 25_minimal_cmb_boltzmann.tex:33 NEW_EXTENSION_PASS

$$(2\ell+1)\mu P_\ell = (\ell+1)P_{\ell+1} + \ell P_{\ell-1} \tag{A.892}$$

F1137 25_minimal_cmb_boltzmann.tex:38 NEW_EXTENSION_PASS

$$\boxed{\Theta_\ell' = \frac{k}{2\ell+1}\left[\ell\Theta_{\ell-1} - (\ell+1)\Theta_{\ell+1}\right].} \tag{A.893}$$

F1138 25_minimal_cmb_boltzmann.tex:50 NEW_EXTENSION_PASS

$$\Theta(\mu,\eta) = \Theta_0(k,\eta_*)\mathrm{e}^{-\mathrm{i}k\mu(\eta-\eta_*)}. \tag{A.894}$$

F1139 25_minimal_cmb_boltzmann.tex:56 NEW_EXTENSION_PASS

$$\mathrm{e}^{-\mathrm{i}x\mu} = \sum_{\ell=0}^{\infty} (-\mathrm{i})^\ell (2\ell+1) j_\ell(x) P_\ell(\mu) \tag{A.895}$$

F1140 25_minimal_cmb_boltzmann.tex:62 NEW_EXTENSION_PASS

$$\boxed{\Theta_\ell(k,\eta) = \Theta_0(k,\eta_*) j_\ell[k(\eta-\eta_*)].} \tag{A.896}$$

F1141 25_minimal_cmb_boltzmann.tex:71 NEW_EXTENSION_PASS

$$j_{L+1}(x) = \frac{2L+1}{x} j_L(x) - j_{L-1}(x) \tag{A.897}$$

F1142 25_minimal_cmb_boltzmann.tex:76 NEW_EXTENSION_PASS

$$\Theta_{L+1} = \frac{2L+1}{x}\Theta_L - \Theta_{L-1}, \qquad x := k(\eta - \eta_*). \tag{A.898}$$

F1143 25_minimal_cmb_boltzmann.tex:84 NEW_EXTENSION_PASS

$$1 = \frac{1}{2}\int_{-1}^{1} \left|\mathrm{e}^{-\mathrm{i}x\mu}\right|^2 \,\mathrm{d}\mu \tag{A.899}$$

$$= \frac{1}{2}\int_{-1}^{1} \left|\sum_{\ell=0}^{\infty}(-\mathrm{i})^\ell(2\ell+1)j_\ell(x)P_\ell(\mu)\right|^2 \,\mathrm{d}\mu \tag{A.900}$$

$$= \sum_{\ell=0}^{\infty}(2\ell+1)j_\ell(x)^2, \tag{A.901}$$

F1144 25_minimal_cmb_boltzmann.tex:100 NEW_EXTENSION_PASS

$$L = 24, \qquad x_f = 12, \tag{A.902}$$

F1145 25_minimal_cmb_boltzmann.tex:106 NEW_EXTENSION_PASS

$$\varepsilon_{j'_\ell} = 4.683753 \times 10^{-16}, \tag{A.903}$$

$$\varepsilon_{\mathrm{hierarchy}} = 6.258188 \times 10^{-13}, \tag{A.904}$$

$$\varepsilon_{\mathrm{Parseval}} = 3.365086 \times 10^{-13}, \tag{A.905}$$

$$p_{\mathrm{hierarchy}} = 1.996773. \tag{A.906}$$

F1146 25_minimal_cmb_boltzmann.tex:116 NEW_EXTENSION_PASS

$$\Theta_0 + \Psi = \frac{1}{5}\mathcal{R}. \tag{A.907}$$

F1147 25_minimal_cmb_boltzmann.tex:120 NEW_EXTENSION_PASS

$$\boxed{\Delta_\ell^{\mathrm{SW}}(k) = \frac{1}{5}j_\ell(k\chi_*).} \tag{A.908}$$

F1148 25_minimal_cmb_boltzmann.tex:127 NEW_EXTENSION_PASS

$$C_\ell^{\mathrm{SW}} = 4\pi\int_0^\infty \frac{\mathrm{d}k}{k}\mathcal{P}_\mathcal{R}(k)\left[\Delta_\ell^{\mathrm{SW}}(k)\right]^2. \tag{A.909}$$

F1149 25_minimal_cmb_boltzmann.tex:137 NEW_EXTENSION_PASS

$$\mathcal{P}_\mathcal{R}(k) = A_s, \tag{A.910}$$

F1150 25_minimal_cmb_boltzmann.tex:141 NEW_EXTENSION_PASS

$$C_\ell^{\mathrm{SW}} = \frac{4\pi A_s}{25}\int_0^\infty \frac{\mathrm{d}k}{k}j_\ell(k\chi_*)^2 \tag{A.911}$$

$$= \frac{4\pi A_s}{25}\int_0^\infty \frac{\mathrm{d}x}{x}j_\ell(x)^2, \qquad x := k\chi_*. \tag{A.912}$$

F1151 25_minimal_cmb_boltzmann.tex:151 NEW_EXTENSION_PASS

$$\int_0^\infty \frac{\mathrm{d}x}{x}j_\ell(x)^2 = \frac{1}{2\ell(\ell+1)}, \qquad \ell \geq 1, \tag{A.913}$$

F1152 25_minimal_cmb_boltzmann.tex:158 NEW_EXTENSION_PASS

$$C_\ell^{\mathrm{SW}} = \frac{4\pi A_s}{25}\frac{1}{2\ell(\ell+1)} \tag{A.914}$$

$$= \frac{2\pi A_s}{25\ell(\ell+1)}. \tag{A.915}$$

F1153 25_minimal_cmb_boltzmann.tex:165 NEW_EXTENSION_PASS

$$\boxed{C_\ell^{\mathrm{SW}} = \frac{2\pi A_s}{25\ell(\ell+1)}.} \tag{A.916}$$

F1154 25_minimal_cmb_boltzmann.tex:173 NEW_EXTENSION_PASS

$$\boxed{\frac{\ell(\ell+1)}{2\pi} C_\ell^{\mathrm{SW}} = \frac{A_s}{25}.} \tag{A.917}$$

F1155 25_minimal_cmb_boltzmann.tex:180 NEW_EXTENSION_PASS

$$\mathrm{Var}(\widehat{C}_\ell) = \frac{2C_\ell^2}{2\ell+1}, \tag{A.918}$$

F1156 25_minimal_cmb_boltzmann.tex:185 NEW_EXTENSION_PASS

$$\sigma(C_\ell) = \sqrt{\frac{2}{2\ell+1}} C_\ell. \tag{A.919}$$

F1157 25_minimal_cmb_boltzmann.tex:193 NEW_EXTENSION_PASS

$$A_s = 2.1 \times 10^{-9}, \qquad T_0 = 2.7255 \text{ K}, \tag{A.920}$$

F1158 25_minimal_cmb_boltzmann.tex:199 NEW_EXTENSION_PASS

$$\frac{\ell(\ell+1)}{2\pi} C_\ell^{\mathrm{SW}} = 8.4 \times 10^{-11}, \tag{A.921}$$

$$\frac{\ell(\ell+1)}{2\pi} C_\ell^{\mathrm{SW}} T_0^2 = 623.981421 \ \mu\mathrm{K}^2. \tag{A.922}$$

F1159 25_minimal_cmb_boltzmann.tex:206 NEW_EXTENSION_PASS

$$\varepsilon_{j^2/x} = 2.498501 \times 10^{-7}. \tag{A.923}$$

F1160 25_minimal_cmb_boltzmann.tex:210 NEW_EXTENSION_PASS

$$\ell_A = 340.781006, \tag{A.924}$$

$$\ell_D := \frac{\chi_*}{r_D} = 2995.643251. \tag{A.925}$$

26_finite_derivation_corner_calculus.tex

F1161 26_finite_derivation_corner_calculus.tex:8 NEW_EXTENSION_PASS

$$\mathcal{A}_N := \mathbb{C}^N, \qquad (fg)_a := f_a g_a, \qquad \boldsymbol{e}_a \boldsymbol{e}_b = \delta_{ab} \boldsymbol{e}_a. \tag{A.926}$$

F1162 26_finite_derivation_corner_calculus.tex:16 NEW_EXTENSION_PASS

$$D : \mathcal{A}_N \to \mathcal{A}_N, \qquad D(fg) = D(f)g + fD(g). \tag{A.927}$$

F1163 26_finite_derivation_corner_calculus.tex:22 NEW_EXTENSION_PASS

$$D\boldsymbol{e}_a = D(\boldsymbol{e}_a^2) \tag{A.928}$$

$$= 2\boldsymbol{e}_a D\boldsymbol{e}_a. \tag{A.929}$$

F1164 26_finite_derivation_corner_calculus.tex:28 NEW_EXTENSION_PASS

$$(D\boldsymbol{e}_a)_b = 2\delta_{ab}(D\boldsymbol{e}_a)_b. \tag{A.930}$$

F1165 26_finite_derivation_corner_calculus.tex:33 NEW_EXTENSION_PASS

$$(D\boldsymbol{e}_a)_a = 2(D\boldsymbol{e}_a)_a, \tag{A.931}$$

F1166 26_finite_derivation_corner_calculus.tex:38 NEW_EXTENSION_PASS

$$(De_a)_b = 0 \qquad (1 \le a, b \le N), \tag{A.932}$$

F1167 26_finite_derivation_corner_calculus.tex:44 NEW_EXTENSION_PASS

$$\boxed{\mathrm{Der}(\mathcal{A}_N) = \{0\}.} \tag{A.933}$$

F1168 26_finite_derivation_corner_calculus.tex:50 NEW_EXTENSION_PASS

$$\operatorname{nullity} \mathcal{L}_N = 0, \tag{A.934}$$

F1169 26_finite_derivation_corner_calculus.tex:57 NEW_EXTENSION_PASS

$$\begin{cases} \text{finite pointwise algebra,} \\ \text{ordinary pointwise product,} \\ \text{exact Leibniz identity,} \\ \text{nonzero derivative} \end{cases} \tag{A.935}$$

F1170 26_finite_derivation_corner_calculus.tex:68 NEW_EXTENSION_PASS

$$\boxed{\begin{array}{c} \text{refinement limit} \quad \vee \quad \text{restricted resolved subspace} \\ \vee \quad \text{deformed product} \quad \vee \quad \text{corner/noncommutative algebra} \end{array}} \tag{A.936}$$

F1171 26_finite_derivation_corner_calculus.tex:84 NEW_EXTENSION_PASS

$$\mathcal{B}_N := M_N(\mathbb{C}), \qquad K = K^*. \tag{A.937}$$

F1172 26_finite_derivation_corner_calculus.tex:90 NEW_EXTENSION_PASS

$$\boxed{\delta_K X := \mathrm{i}[K, X].} \tag{A.938}$$

F1173 26_finite_derivation_corner_calculus.tex:96 NEW_EXTENSION_PASS

$$\delta_K(XY) = (\delta_K X)Y + X(\delta_K Y), \tag{A.939}$$
$$(\delta_K X)^* = \delta_K(X^*), \tag{A.940}$$
$$\mathrm{Tr}((\delta_K X)Y) = -\mathrm{Tr}(X\delta_K Y). \tag{A.941}$$

F1174 26_finite_derivation_corner_calculus.tex:105 NEW_EXTENSION_PASS

$$\delta_K(XY) = \mathrm{i}(KXY - XYK) \tag{A.942}$$
$$= \mathrm{i}(KX - XK)Y + \mathrm{i}X(KY - YK), \tag{A.943}$$
$$(\delta_K X)^* = -\mathrm{i}(X^*K - KX^*) = \mathrm{i}(KX^* - X^*K), \tag{A.944}$$
$$\mathrm{Tr}((\delta_K X)Y) = \mathrm{i}\,\mathrm{Tr}(KXY - XKY) \tag{A.945}$$
$$= \mathrm{i}\,\mathrm{Tr}(XYK - XKY) \tag{A.946}$$
$$= -\mathrm{Tr}(X\delta_K Y). \tag{A.947}$$

F1175 26_finite_derivation_corner_calculus.tex:119 NEW_EXTENSION_PASS

$$\delta_K \in \mathrm{Der}_*(\mathcal{B}_N), \tag{A.948}$$

F1176 26_finite_derivation_corner_calculus.tex:126 NEW_EXTENSION_PASS

$$\delta_i X := \mathrm{i}[K_i, X], \qquad F_{ij} := \mathrm{i}[K_i, K_j], \tag{A.949}$$

F1177 26_finite_derivation_corner_calculus.tex:132 NEW_EXTENSION_PASS

$$\boxed{[\delta_i, \delta_j]X = \mathrm{i}[F_{ij}, X].} \tag{A.950}$$

F1178 26_finite_derivation_corner_calculus.tex:139 NEW_EXTENSION_PASS

$$[\delta_i, \delta_j]X = -[K_i, [K_j, X]] + [K_j, [K_i, X]] \tag{A.951}$$
$$= -[[K_i, K_j], X] \tag{A.952}$$
$$= \mathrm{i}[\mathrm{i}[K_i, K_j], X] = \mathrm{i}[F_{ij}, X], \tag{A.953}$$

F1179 26_finite_derivation_corner_calculus.tex:151 NEW_EXTENSION_PASS

$$[K_i, K_j] = 0 \Longrightarrow [\delta_i, \delta_j] = 0, \tag{A.954}$$

$$[K_i, K_j] \neq 0 \Longrightarrow \text{finite corner curvature.} \tag{A.955}$$

F1180 26_finite_derivation_corner_calculus.tex:162 NEW_EXTENSION_PASS

$$P_a P_b = \delta_{ab} P_a, \qquad \sum_a P_a = I. \tag{A.956}$$

F1181 26_finite_derivation_corner_calculus.tex:168 NEW_EXTENSION_PASS

$$\delta_K X = \sum_{a,b} P_a (\delta_K X) P_b. \tag{A.957}$$

F1182 26_finite_derivation_corner_calculus.tex:173 NEW_EXTENSION_PASS

$$\boxed{P_a (\delta_K X) P_b = \mathrm{i} \sum_c \left(P_a K P_c X P_b - P_a X P_c K P_b \right).} \tag{A.958}$$

F1183 26_finite_derivation_corner_calculus.tex:183 NEW_EXTENSION_PASS

$$P_a K X P_b = \sum_c P_a K P_c X P_b, \tag{A.959}$$

$$P_a X K P_b = \sum_c P_a X P_c K P_b. \tag{A.960}$$

F1184 26_finite_derivation_corner_calculus.tex:191 NEW_EXTENSION_PASS

$$Q_A := \sum_{a \in C_A} P_a, \tag{A.961}$$

F1185 26_finite_derivation_corner_calculus.tex:195 NEW_EXTENSION_PASS

$$\boxed{Q_A (\delta_K X) Q_B = \sum_{\substack{a \in C_A \\ b \in C_B}} P_a (\delta_K X) P_b.} \tag{A.962}$$

F1186 26_finite_derivation_corner_calculus.tex:204 NEW_EXTENSION_PASS

$$\max_{2 \le N \le 8} \operatorname{nullity} \mathcal{L}_N = 0, \tag{A.963}$$

$$\| \delta_K (XY) - (\delta_K X) Y - X \delta_K Y \|_2 = 5.283117 \times 10^{-14}, \tag{A.964}$$

$$\| (\delta_K X)^* - \delta_K (X^*) \|_2 = 6.085790 \times 10^{-15}, \tag{A.965}$$

$$| \operatorname{Tr}((\delta_K X) Y) + \operatorname{Tr}(X \delta_K Y) | = 2.131628 \times 10^{-14}, \tag{A.966}$$

$$\| [\delta_1, \delta_2] X - \mathrm{i} [F_{12}, X] \|_2 = 5.495524 \times 10^{-14}, \tag{A.967}$$

$$\| \delta_K X - \sum_{a,b} P_a \delta_K X P_b \|_2 = 3.035492 \times 10^{-14}, \tag{A.968}$$

$$\max_{A,B} \left\| Q_A \delta_K X Q_B - \sum_{a \in C_A, b \in C_B} P_a \delta_K X P_b \right\|_2 = 5.824437 \times 10^{-15}. \tag{A.969}$$

27_linearized_projector_adm.tex

F1187 27_linearized_projector_adm.tex:8 NEW_EXTENSION_PASS

$$\delta_i \delta_j = \delta_j \delta_i, \tag{A.970}$$

$$\langle \delta_i X, Y \rangle_h = - \langle X, \delta_i Y \rangle_h , \tag{A.971}$$

$$\delta^2 := \sum_{i=1}^{3} \delta_i \delta_i. \tag{A.972}$$

F1188 27_linearized_projector_adm.tex:17 NEW_EXTENSION_PASS

$$h_{ij} = h_{ji}, \qquad \pi_{ij} = \pi_{ji}, \tag{A.973}$$
$$h := h_{kk}, \qquad \pi := \pi_{kk}. \tag{A.974}$$

F1189 27_linearized_projector_adm.tex:24 NEW_EXTENSION_PASS

$$\boxed{\mathcal{C} := \delta_i \delta_j h_{ij} - \delta^2 h} \tag{A.975}$$

F1190 27_linearized_projector_adm.tex:30 NEW_EXTENSION_PASS

$$\boxed{\mathcal{C}_i := -2\delta_j \pi_{ij}.} \tag{A.976}$$

F1191 27_linearized_projector_adm.tex:36 NEW_EXTENSION_PASS

$$\boxed{\begin{aligned}\mathcal{G}_{ij}(h) := \frac{1}{2}\big[&\delta_k\delta_i h_{jk} + \delta_k\delta_j h_{ik} - \delta^2 h_{ij} - \delta_i\delta_j h \\ &- \delta_{ij}(\delta_k\delta_l h_{kl} - \delta^2 h)\big].\end{aligned}} \tag{A.977}$$

F1192 27_linearized_projector_adm.tex:52 NEW_EXTENSION_PASS

$$\boxed{\delta_j \mathcal{G}_{ij}(h) = 0.} \tag{A.978}$$

F1193 27_linearized_projector_adm.tex:58 NEW_EXTENSION_PASS

$$2\delta_j \mathcal{G}_{ij} = \delta_i\delta_j\delta_k h_{jk} + \delta_j\delta_j\delta_k h_{ik} - \delta_j\delta^2 h_{ij} - \delta_i\delta_j\delta_j h \tag{A.979}$$
$$- \delta_i\delta_k\delta_l h_{kl} + \delta_i\delta^2 h. \tag{A.980}$$

F1194 27_linearized_projector_adm.tex:72 NEW_EXTENSION_PASS

$$\mathcal{T}_h(\pi) := \langle \pi_{ij}, \pi_{ij} \rangle_h - \frac{1}{2} \langle \pi, \pi \rangle_h , \tag{A.981}$$
$$\mathcal{U}_h(h) := \frac{1}{2} \langle h_{ij}, \mathcal{G}_{ij}(h) \rangle_h . \tag{A.982}$$

F1195 27_linearized_projector_adm.tex:80 NEW_EXTENSION_PASS

$$\boxed{S_h^{(2)} = \int \mathrm{d}t \left[\left\langle \pi_{ij}, \dot{h}_{ij} \right\rangle_h - \mathcal{T}_h - \mathcal{U}_h + \langle n, \mathcal{C} \rangle_h - \langle n_i, \mathcal{C}_i \rangle_h \right].} \tag{A.983}$$

F1196 27_linearized_projector_adm.tex:93 NEW_EXTENSION_PASS

$$\delta\mathcal{T}_h = 2 \langle \delta\pi_{ij}, \pi_{ij} \rangle_h - \langle \delta\pi, \pi \rangle_h \tag{A.984}$$
$$= \langle \delta\pi_{ij}, 2\pi_{ij} - \delta_{ij}\pi \rangle_h . \tag{A.985}$$

F1197 27_linearized_projector_adm.tex:103 NEW_EXTENSION_PASS

$$- \langle n_i, \mathcal{C}_i \rangle_h = 2 \langle n_i, \delta_j \pi_{ij} \rangle_h \tag{A.986}$$
$$= -2 \langle \delta_j n_i, \pi_{ij} \rangle_h \tag{A.987}$$
$$= - \langle \delta_i n_j + \delta_j n_i, \pi_{ij} \rangle_h , \tag{A.988}$$

F1198 27_linearized_projector_adm.tex:114 NEW_EXTENSION_PASS

$$\frac{\delta S_h^{(2)}}{\delta n} = 0 \Longrightarrow \mathcal{C} = 0, \tag{A.989}$$

F1199 27_linearized_projector_adm.tex:119 NEW_EXTENSION_PASS

$$\frac{\delta S_h^{(2)}}{\delta n_i} = 0 \Longrightarrow \mathcal{C}_i = 0, \tag{A.990}$$

F1200 27_linearized_projector_adm.tex:124 NEW_EXTENSION_PASS

$$\boxed{\dot{h}_{ij} = 2\pi_{ij} - \delta_{ij}\pi + \delta_i n_j + \delta_j n_i,} \tag{A.991}$$

F1201 27_linearized_projector_adm.tex:132 NEW_EXTENSION_PASS

$$\boxed{\dot{\pi}_{ij} = -\mathcal{G}_{ij}(h) + \delta_i\delta_j n - \delta_{ij}\delta^2 n.} \tag{A.992}$$

F1202 27_linearized_projector_adm.tex:141 NEW_EXTENSION_PASS

$$\left\langle n, \delta_i\delta_j\delta h_{ij} - \delta^2\delta h \right\rangle_h = \left\langle \delta_i\delta_j n - \delta_{ij}\delta^2 n, \delta h_{ij} \right\rangle_h . \tag{A.993}$$

F1203 27_linearized_projector_adm.tex:148 NEW_EXTENSION_PASS

$$\delta\mathcal{U}_h = \frac{1}{2}\left\langle \delta h, \mathcal{G}(h) \right\rangle_h + \frac{1}{2}\left\langle h, \mathcal{G}(\delta h) \right\rangle_h \tag{A.994}$$

$$= \left\langle \delta h, \mathcal{G}(h) \right\rangle_h . \tag{A.995}$$

F1204 27_linearized_projector_adm.tex:160 NEW_EXTENSION_PASS

$$\delta\mathcal{U}_h = \left\langle \delta h_{ij}, \mathcal{G}_{ij}(h) \right\rangle_h \tag{A.996}$$

F1205 27_linearized_projector_adm.tex:165 NEW_EXTENSION_PASS

$$\left\langle a_{ij}, \mathcal{G}_{ij}(b) \right\rangle_h = \left\langle \mathcal{G}_{ij}(a), b_{ij} \right\rangle_h . \tag{A.997}$$

F1206 27_linearized_projector_adm.tex:172 NEW_EXTENSION_PASS

$$\dot{\mathcal{C}} = \delta_i\delta_j\dot{h}_{ij} - \delta^2\dot{h} \tag{A.998}$$

$$= 2\delta_i\delta_j\pi_{ij} \tag{A.999}$$

$$= -\delta_i\mathcal{C}_i. \tag{A.1000}$$

F1207 27_linearized_projector_adm.tex:179 NEW_EXTENSION_PASS

$$\delta_i\delta_j(\delta_i n_j + \delta_j n_i) - \delta^2(2\delta_k n_k) = 0 \tag{A.1001}$$

F1208 27_linearized_projector_adm.tex:185 NEW_EXTENSION_PASS

$$\delta_i\delta_j(2\pi_{ij} - \delta_{ij}\pi) - \delta^2(-\pi) = 2\delta_i\delta_j\pi_{ij}. \tag{A.1002}$$

F1209 27_linearized_projector_adm.tex:191 NEW_EXTENSION_PASS

$$\dot{\mathcal{C}}_i = -2\delta_j\dot{\pi}_{ij} \tag{A.1003}$$

$$= 2\delta_j\mathcal{G}_{ij} - 2\delta_j(\delta_i\delta_j - \delta_{ij}\delta^2)n \tag{A.1004}$$

$$= 0. \tag{A.1005}$$

F1210 27_linearized_projector_adm.tex:199 NEW_EXTENSION_PASS

$$\delta_j\delta_i\delta_j n = \delta_i\delta^2 n = \delta_j\delta_{ij}\delta^2 n. \tag{A.1006}$$

F1211 27_linearized_projector_adm.tex:207 NEW_EXTENSION_PASS

$$\boxed{\dot{\mathcal{C}} = -\delta_i\mathcal{C}_i, \qquad \dot{\mathcal{C}}_i = 0.} \tag{A.1007}$$

F1212 27_linearized_projector_adm.tex:215 NEW_EXTENSION_PASS

$$\mathcal{C}(0) = 0, \qquad \mathcal{C}_i(0) = 0 \Longrightarrow \mathcal{C}(t) = 0, \qquad \mathcal{C}_i(t) = 0. \tag{A.1008}$$

F1213 27_linearized_projector_adm.tex:227 NEW_EXTENSION_PASS

$$h_{ij} \mapsto h_{ij} + \delta_i \xi_j + \delta_j \xi_i, \tag{A.1009}$$

F1214 27_linearized_projector_adm.tex:231 NEW_EXTENSION_PASS

$$\mathcal{C}[h + \delta\xi + (\delta\xi)^{\mathsf{T}}] = \mathcal{C}[h]. \tag{A.1010}$$

F1215 27_linearized_projector_adm.tex:236 NEW_EXTENSION_PASS

$$\Delta\mathcal{C} = \delta_i\delta_j(\delta_i\xi_j + \delta_j\xi_i) - \delta^2(2\delta_k\xi_k) \tag{A.1011}$$

$$= 2\delta^2\delta_k\xi_k - 2\delta^2\delta_k\xi_k = 0. \tag{A.1012}$$

F1216 27_linearized_projector_adm.tex:247 NEW_EXTENSION_PASS

$$\mathcal{C}[n] := \langle n, \mathcal{C} \rangle_h \,, \tag{A.1013}$$

$$\mathcal{C}[\boldsymbol{\xi}] := \langle \xi_i, \mathcal{C}_i \rangle_h \,. \tag{A.1014}$$

F1217 27_linearized_projector_adm.tex:252 NEW_EXTENSION_PASS

$$\boxed{\{\mathcal{C}[n], \mathcal{C}[m]\} = 0, \qquad \{\mathcal{C}[\boldsymbol{\xi}], \mathcal{C}[\boldsymbol{\eta}]\} = 0, \qquad \{\mathcal{C}[n], \mathcal{C}[\boldsymbol{\xi}]\} = 0.} \tag{A.1015}$$

F1218 27_linearized_projector_adm.tex:265 NEW_EXTENSION_PASS

$$\delta_j h_{ij} = 0, \qquad h = 0, \qquad \delta_j \pi_{ij} = 0, \qquad \pi = 0, \tag{A.1016}$$

F1219 27_linearized_projector_adm.tex:275 NEW_EXTENSION_PASS

$$\mathcal{G}_{ij}(h) = -\frac{1}{2}\delta^2 h_{ij}, \tag{A.1017}$$

$$\dot{h}_{ij} = 2\pi_{ij}, \tag{A.1018}$$

$$\dot{\pi}_{ij} = \frac{1}{2}\delta^2 h_{ij}, \tag{A.1019}$$

$$\ddot{h}_{ij} = \delta^2 h_{ij}. \tag{A.1020}$$

F1220 27_linearized_projector_adm.tex:283 NEW_EXTENSION_PASS

$$\widetilde{k}_i(h) = \frac{\sin(k_i h)}{h}, \tag{A.1021}$$

F1221 27_linearized_projector_adm.tex:287 NEW_EXTENSION_PASS

$$\frac{\sin(k_i h)}{h} = k_i - \frac{k_i^3 h^2}{6} + \mathcal{O}(h^4), \tag{A.1022}$$

F1222 27_linearized_projector_adm.tex:293 NEW_EXTENSION_PASS

$$\widetilde{k}_i(h)^2 = k_i^2 - \frac{k_i^4 h^2}{3} + \mathcal{O}(h^4). \tag{A.1023}$$

F1223 27_linearized_projector_adm.tex:300 NEW_EXTENSION_PASS

$$\boxed{\omega_h^2 = \sum_i \widetilde{k}_i(h)^2 = k^2 - \frac{h^2}{3}\sum_i k_i^4 + \mathcal{O}(h^4),} \tag{A.1024}$$

F1224 27_linearized_projector_adm.tex:308 NEW_EXTENSION_PASS

$$\boxed{\omega_h = k - \frac{h^2}{6k}\sum_i k_i^4 + \mathcal{O}(h^4).} \tag{A.1025}$$

F1225 27_linearized_projector_adm.tex:315 NEW_EXTENSION_PASS

$$p_\omega = 1.985855. \tag{A.1026}$$

F1226 27_linearized_projector_adm.tex:321 NEW_EXTENSION_PASS

$$E_A^{\mathrm{con}} := \sum_{x\in A} h^3 \left(|\mathcal{C}_x|^2 + \sum_i |\mathcal{C}_{i,x}|^2 \right). \tag{A.1027}$$

F1227 27_linearized_projector_adm.tex:329 NEW_EXTENSION_PASS

$$\boxed{\sum_A E_A^{\mathrm{con}} = \sum_x h^3 \left(|\mathcal{C}_x|^2 + \sum_i |\mathcal{C}_{i,x}|^2 \right).} \tag{A.1028}$$

F1228 27_linearized_projector_adm.tex:340 NEW_EXTENSION_PASS

$$\max_i \|\delta_j \mathcal{G}_{ij}\|_\infty = 2.220446 \times 10^{-15}, \tag{A.1029}$$

$$\|\dot{\mathcal{C}} + \delta_i \mathcal{C}_i\|_\infty = 1.199041 \times 10^{-14}, \tag{A.1030}$$

$$\max_i \|\dot{\mathcal{C}}_i\|_\infty = 5.329071 \times 10^{-15}, \tag{A.1031}$$

$$\|\mathcal{C}[h + \delta\xi + (\delta\xi)^{\mathsf{T}}] - \mathcal{C}[h]\|_\infty = 1.154632 \times 10^{-14}, \tag{A.1032}$$

$$|\{\mathcal{C}[n], \mathcal{C}[\boldsymbol{\xi}]\}| = 1.450620 \times 10^{-14}, \tag{A.1033}$$

$$|\delta\mathcal{U}_h - \langle \delta h, \mathcal{G}(h)\rangle_h| = 4.092726 \times 10^{-12}, \tag{A.1034}$$

$$\left| \frac{\mathrm{d}}{\mathrm{d}t}(\mathcal{T}_h + \mathcal{U}_h) \right| = 0, \tag{A.1035}$$

$$\frac{|E^{\mathrm{con}}_{\mathrm{coarse}} - E^{\mathrm{con}}_{\mathrm{fine}}|}{E^{\mathrm{con}}_{\mathrm{fine}}} = 1.505862 \times 10^{-16}, \tag{A.1036}$$

$$\max_h \max(\|\mathcal{C}_{\mathrm{TT}}\|_\infty, \|\mathcal{C}_{i,\mathrm{TT}}\|_\infty) = 5.107026 \times 10^{-14}, \tag{A.1037}$$

$$\max_h \frac{|E_h(t) - E_h(0)|}{|E_h(0)|} = 4.908588 \times 10^{-16}. \tag{A.1038}$$

28_nonlinear_constraint_defect.tex

F1229 28_nonlinear_constraint_defect.tex:7 NEW_EXTENSION_PASS

$$\{q_a, p_b\}_h = \frac{\delta_{ab}}{h}, \tag{A.1039}$$

F1230 28_nonlinear_constraint_defect.tex:11 NEW_EXTENSION_PASS

$$D^{\mathsf{T}} = -D. \tag{A.1040}$$

F1231 28_nonlinear_constraint_defect.tex:15 NEW_EXTENSION_PASS

$$\boxed{\mathcal{D}_h[\xi] := h \sum_a \xi_a p_a (Dq)_a.} \tag{A.1041}$$

F1232 28_nonlinear_constraint_defect.tex:22 NEW_EXTENSION_PASS

$$\frac{\partial \mathcal{D}_h[\xi]}{\partial p_a} = h\xi_a (Dq)_a, \tag{A.1042}$$

$$\frac{\partial \mathcal{D}_h[\xi]}{\partial q_b} = h \sum_a \xi_a p_a D_{ab} \tag{A.1043}$$

$$= h\left[D^{\mathsf{T}}(\xi p)\right]_b = -h[D(\xi p)]_b. \tag{A.1044}$$

F1233 28_nonlinear_constraint_defect.tex:33 NEW_EXTENSION_PASS

$$\boxed{\mathfrak{L}_D(f,g) := D(fg) - (Df)g - f(Dg).} \tag{A.1045}$$

F1234 28_nonlinear_constraint_defect.tex:40 NEW_EXTENSION_PASS

$$\boxed{\begin{aligned}\{\mathcal{D}_h[\xi], \mathcal{D}_h[\eta]\}_h &= \mathcal{D}_h[\xi D\eta - \eta D\xi] \\ &\quad + h \sum_a p_a \left[\xi_a \mathfrak{L}_D(\eta, Dq)_a - \eta_a \mathfrak{L}_D(\xi, Dq)_a\right].\end{aligned}} \tag{A.1046}$$

F1235 28_nonlinear_constraint_defect.tex:54 NEW_EXTENSION_PASS

$$\{\mathcal{D}_h[\xi],\mathcal{D}_h[\eta]\}_h = h\sum_a p_a\left[\xi_a D(\eta Dq)_a - \eta_a D(\xi Dq)_a\right]. \tag{A.1047}$$

F1236 28_nonlinear_constraint_defect.tex:64 NEW_EXTENSION_PASS

$$D(\eta Dq) = (D\eta)(Dq) + \eta D^2 q + \mathfrak{L}_D(\eta, Dq), \tag{A.1048}$$
$$D(\xi Dq) = (D\xi)(Dq) + \xi D^2 q + \mathfrak{L}_D(\xi, Dq). \tag{A.1049}$$

F1237 28_nonlinear_constraint_defect.tex:73 NEW_EXTENSION_PASS

$$h\sum_a p_a(\xi_a D\eta_a - \eta_a D\xi_a)(Dq)_a = \mathcal{D}_h[\xi D\eta - \eta D\xi], \tag{A.1050}$$

F1238 28_nonlinear_constraint_defect.tex:80 NEW_EXTENSION_PASS

$$\boxed{\text{nonlinear bracket defect} = \text{Leibniz-defect contraction.}} \tag{A.1051}$$

F1239 28_nonlinear_constraint_defect.tex:89 NEW_EXTENSION_PASS

$$\mathrm{Der}(\mathbb{C}^N) = \{0\} \tag{A.1052}$$

F1240 28_nonlinear_constraint_defect.tex:93 NEW_EXTENSION_PASS

$$D \neq 0 \Longrightarrow \exists (f,g) : \mathfrak{L}_D(f,g) \neq 0 \tag{A.1053}$$

F1241 28_nonlinear_constraint_defect.tex:99 NEW_EXTENSION_PASS

$$\boxed{\text{the undeformed nonlinear diffeomorphism algebra cannot close universally}} \tag{A.1054}$$

F1242 28_nonlinear_constraint_defect.tex:108 NEW_EXTENSION_PASS

$$\mathfrak{L}_{D_h}(f,g) = \mathcal{O}(h^2), \tag{A.1055}$$

F1243 28_nonlinear_constraint_defect.tex:112 NEW_EXTENSION_PASS

$$D_h f = f' + \frac{h^2}{6} f''' + \mathcal{O}(h^4). \tag{A.1056}$$

F1244 28_nonlinear_constraint_defect.tex:116 NEW_EXTENSION_PASS

$$\mathfrak{L}_{D_h}(f,g) = \frac{h^2}{6}\left[(fg)''' - f'''g - fg'''\right] + \mathcal{O}(h^4) \tag{A.1057}$$
$$= \frac{h^2}{2}\left(f''g' + f'g''\right) + \mathcal{O}(h^4). \tag{A.1058}$$

F1245 28_nonlinear_constraint_defect.tex:128 NEW_EXTENSION_PASS

$$\{\mathcal{D}_h[\xi],\mathcal{D}_h[\eta]\}_h - \mathcal{D}_h[\xi D_h\eta - \eta D_h\xi] = \mathcal{O}(h^2). \tag{A.1059}$$

F1246 28_nonlinear_constraint_defect.tex:134 NEW_EXTENSION_PASS

$$p_{\mathrm{closure}} = 1.985564. \tag{A.1060}$$

F1247 28_nonlinear_constraint_defect.tex:138 NEW_EXTENSION_PASS

$$\mathfrak{L}_{D_{\mathrm{spec}}}(f,g) = 0 \tag{A.1061}$$

F1248 28_nonlinear_constraint_defect.tex:145 NEW_EXTENSION_PASS

$$\delta_K X = \mathrm{i}[K, X] \tag{A.1062}$$

F1249 28_nonlinear_constraint_defect.tex:149 NEW_EXTENSION_PASS

$$\boxed{\mathfrak{L}_{\delta_K}(X,Y) = 0.} \tag{A.1063}$$

F1250 28_nonlinear_constraint_defect.tex:155 NEW_EXTENSION_PASS

$$\delta_K(XY) = \mathrm{i}(KXY - XYK) \tag{A.1064}$$
$$= \mathrm{i}(KX - XK)Y + \mathrm{i}X(KY - YK) \tag{A.1065}$$
$$= (\delta_K X)Y + X(\delta_K Y). \tag{A.1066}$$

F1251 28_nonlinear_constraint_defect.tex:166 NEW_EXTENSION_PASS

$$\boxed{S_{\mathrm{PADM},h} = \int \mathrm{d}t \left\{ \mathrm{Tr}(\Pi^{ij}\dot{Q}_{ij} + P_\Phi\dot{\Phi}) - \mathcal{H}_{\mathcal{P},h}[N] - \mathcal{D}_{\mathcal{P},h}[\boldsymbol{N}] \right\}} \tag{A.1067}$$

F1252 28_nonlinear_constraint_defect.tex:178 NEW_EXTENSION_PASS

$$\boxed{S_{\mathrm{PADM},h} = \int \mathrm{d}t \left\{ \mathrm{Tr}[\Pi^{ij}(\dot{Q}_{ij} - \mathcal{L}^{\mathcal{P}}_{\boldsymbol{N},h}Q_{ij}) + P_\Phi(\dot{\Phi} - \mathcal{L}^{\mathcal{P}}_{\boldsymbol{N},h}\Phi)] - \mathcal{H}_{\mathcal{P},h}[N] \right\}.} \tag{A.1068}$$

F1253 28_nonlinear_constraint_defect.tex:193 NEW_EXTENSION_PASS

$$\mathcal{H}_{\mathcal{P},h} := \left.\frac{\partial\Omega_{C,h}}{\partial c^\perp}\right|_{c=\varpi=0}, \tag{A.1069}$$
$$\mathcal{H}_{\mathcal{P},h,i} := \left.\frac{\partial\Omega_{C,h}}{\partial c^i}\right|_{c=\varpi=0}, \tag{A.1070}$$
$$\mathcal{L}^{\mathcal{P}}_{\boldsymbol{N},h}X := \left.\{X,\Omega_{C,h}\}_C\right|_{c^i=N^i,\ c^\perp=0,\ \varpi=0}, \tag{A.1071}$$
$$\{\cdot,\cdot\}_{\mathcal{P},\mathrm{ordered}} := \text{the canonical cyclic trace BFV bracket}, \tag{A.1072}$$
$$\text{nonlinear first-class closure} \Longleftrightarrow \{\Omega_{C,h},\Omega_{C,h}\}_C = 0. \tag{A.1073}$$

F1254 28_nonlinear_constraint_defect.tex:209 NEW_EXTENSION_PASS

$$\max_h |(\{\mathcal{D}_h[\xi],\mathcal{D}_h[\eta]\} - \mathcal{D}_h[\xi D\eta - \eta D\xi]) - \mathfrak{E}_h| = 1.873501\times10^{-16}, \tag{A.1074}$$
$$\mathfrak{E}_h := h\sum_a p_a[\xi_a\mathfrak{L}_D(\eta,Dq)_a - \eta_a\mathfrak{L}_D(\xi,Dq)_a], \tag{A.1075}$$
$$|\text{spectral low-band closure defect}| = 6.696033\times10^{-16}, \tag{A.1076}$$
$$|\text{spectral direct-defect mismatch}| = 7.281546\times10^{-17}, \tag{A.1077}$$
$$\|\mathfrak{L}_{\delta_K}(X,Y)\|_2 = 7.192762\times10^{-14}. \tag{A.1078}$$

29_adic_unit_quantum_split.tex

F1255 29_adic_unit_quantum_split.tex:8 NEW_EXTENSION_PASS

$$\mathcal{H} = \ell^2(\text{finite adic address set};\mathbb{C}). \tag{A.1079}$$

F1256 29_adic_unit_quantum_split.tex:15 NEW_EXTENSION_PASS

$$U_{p,h} := (\mathbb{Z}/p^{h+1}\mathbb{Z})^\times. \tag{A.1080}$$

F1257 29_adic_unit_quantum_split.tex:19 NEW_EXTENSION_PASS

$$\boxed{U_{p,h} \cong \mathbb{Z}/(p-1)\mathbb{Z}\times\mathbb{Z}/p^h\mathbb{Z}} \tag{A.1081}$$

F1258 29_adic_unit_quantum_split.tex:29 NEW_EXTENSION_PASS

$$(t,a) \longmapsto \zeta_{p,h}^t(1+p)^a \pmod{p^{h+1}}. \tag{A.1082}$$

F1259 29_adic_unit_quantum_split.tex:36 NEW_EXTENSION_PASS

$$|U_{p,h}| = \varphi(p^{h+1}) = p^{h+1} - p^h = (p-1)p^h. \tag{A.1083}$$

F1260 29_adic_unit_quantum_split.tex:44 NEW_EXTENSION_PASS

$$U_{2,h} := (\mathbb{Z}/2^{h+2}\mathbb{Z})^{\times}. \tag{A.1084}$$

F1261 29_adic_unit_quantum_split.tex:48 NEW_EXTENSION_PASS

$$\boxed{U_{2,h} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{h}\mathbb{Z}} \tag{A.1085}$$

F1262 29_adic_unit_quantum_split.tex:58 NEW_EXTENSION_PASS

$$(s, a) \longmapsto (-1)^{s} 5^{a} \pmod{2^{h+2}}. \tag{A.1086}$$

F1263 29_adic_unit_quantum_split.tex:65 NEW_EXTENSION_PASS

$$|U_{2,h}| = \varphi(2^{h+2}) = 2^{h+1} = 2 \cdot 2^{h}, \tag{A.1087}$$

F1264 29_adic_unit_quantum_split.tex:73 NEW_EXTENSION_PASS

$$\mathcal{H}^{\times}_{p,h} := \ell^2(U_{p,h}) \cong \mathbb{C}^{p-1}_{\mathrm{tors}} \otimes \ell^2(\mathbb{Z}/p^{h}\mathbb{Z}), \qquad p \text{ odd}, \tag{A.1088}$$

$$\mathcal{H}^{\times}_{2,h} := \ell^2(U_{2,h}) \cong \mathbb{C}^{2}_{\mathrm{sign}} \otimes \ell^2(\mathbb{Z}/2^{h}\mathbb{Z}). \tag{A.1089}$$

F1265 29_adic_unit_quantum_split.tex:92 NEW_EXTENSION_PASS

$$M = 2^{h_2+2} \prod_{p \in \mathcal{P}_{\mathrm{o}}} p^{h_p+1}. \tag{A.1090}$$

F1266 29_adic_unit_quantum_split.tex:98 NEW_EXTENSION_PASS

$$\boxed{\mathcal{U}_{\mathrm{CRT}} : \ell^2((\mathbb{Z}/M\mathbb{Z})^{\times}) \longrightarrow \mathbb{C}^{2}_{\mathrm{sign}} \otimes \ell^2(\mathbb{Z}/2^{h_2}\mathbb{Z}) \otimes \bigotimes_{p \in \mathcal{P}_{\mathrm{o}}} \left(\mathbb{C}^{p-1}_{\mathrm{tors}} \otimes \ell^2(\mathbb{Z}/p^{h_p}\mathbb{Z}) \right).} \tag{A.1091}$$

F1267 29_adic_unit_quantum_split.tex:117 NEW_EXTENSION_PASS

$$\mathcal{U}_{\mathrm{CRT}} |u\rangle = |\operatorname{coord}(u)\rangle. \tag{A.1092}$$

F1268 29_adic_unit_quantum_split.tex:123 NEW_EXTENSION_PASS

$$\langle \mathcal{U}_{\mathrm{CRT}} u | \mathcal{U}_{\mathrm{CRT}} v \rangle = \delta_{\operatorname{coord}(u), \operatorname{coord}(v)} \tag{A.1093}$$

$$= \delta_{u,v} = \langle u | v \rangle, \tag{A.1094}$$

F1269 29_adic_unit_quantum_split.tex:136 NEW_EXTENSION_PASS

$$\operatorname{coord}(uv) = \operatorname{coord}(u) + \operatorname{coord}(v) \tag{A.1095}$$

F1270 29_adic_unit_quantum_split.tex:147 NEW_EXTENSION_PASS

$$\boxed{F_{U_M} = \mathcal{U}^{*}_{\mathrm{CRT}} \left[F_2 \otimes F_{2^{h_2}} \otimes \bigotimes_{p \in \mathcal{P}_{\mathrm{o}}} (F_{p-1} \otimes F_{p^{h_p}}) \right] \mathcal{U}_{\mathrm{CRT}}.} \tag{A.1096}$$

F1271 29_adic_unit_quantum_split.tex:162 NEW_EXTENSION_PASS

$$\mathcal{P}_{\mathrm{in}} |n\rangle = |n \bmod a\rangle \otimes |n \bmod b\rangle, \tag{A.1097}$$

$$\mathcal{P}_{\mathrm{out}} |k\rangle = |k b^{-1} \bmod a\rangle \otimes |k a^{-1} \bmod b\rangle. \tag{A.1098}$$

F1272 29_adic_unit_quantum_split.tex:169 NEW_EXTENSION_PASS

$$\boxed{F_N = \mathcal{P}^{*}_{\mathrm{out}} (F_a \otimes F_b) \mathcal{P}_{\mathrm{in}}.} \tag{A.1099}$$

F1273 29_adic_unit_quantum_split.tex:180 NEW_EXTENSION_PASS

$$\frac{1}{\sqrt{a}} \mathrm{e}^{2\pi \mathrm{i} k_a n_a / a} \frac{1}{\sqrt{b}} \mathrm{e}^{2\pi \mathrm{i} k_b n_b / b} = \frac{1}{\sqrt{N}} \exp\left[2\pi \mathrm{i} \left(\frac{k_a n}{a} + \frac{k_b n}{b} \right) \right]. \tag{A.1100}$$

F1274 29_adic_unit_quantum_split.tex:192 NEW_EXTENSION_PASS

$$k_a = kb^{-1} \pmod{a}, \qquad k_b = ka^{-1} \pmod{b} \tag{A.1101}$$

F1275 29_adic_unit_quantum_split.tex:198 NEW_EXTENSION_PASS

$$\frac{k_a}{a} + \frac{k_b}{b} \equiv \frac{k}{ab} \pmod{1}, \tag{A.1102}$$

F1276 29_adic_unit_quantum_split.tex:208 NEW_EXTENSION_PASS

$$\mathcal{C}_{\mathrm{phase}} : \mathbb{C}^2_{\mathrm{sign}} \otimes \ell^2(\mathbb{Z}/2^h\mathbb{Z}) \longrightarrow \ell^2(\mathbb{Z}/2^h\mathbb{Z}) \tag{A.1103}$$

F1277 29_adic_unit_quantum_split.tex:217 NEW_EXTENSION_PASS

$$\operatorname{rank} \mathcal{C}_{\mathrm{phase}} = 2^h, \qquad \dim \mathcal{H}^{\times}_{2,h} = 2^{h+1}, \tag{A.1104}$$

F1278 29_adic_unit_quantum_split.tex:223 NEW_EXTENSION_PASS

$$(+1, a), \qquad (-1, a). \tag{A.1105}$$

F1279 29_adic_unit_quantum_split.tex:229 NEW_EXTENSION_PASS

$$\mathcal{C}_{\mathrm{phase}}|\pm, a\rangle = |a\rangle, \tag{A.1106}$$

F1280 29_adic_unit_quantum_split.tex:233 NEW_EXTENSION_PASS

$$\mathcal{C}_{\mathrm{phase}}\big(|+, a\rangle - |-, a\rangle\big) = 0. \tag{A.1107}$$

F1281 29_adic_unit_quantum_split.tex:238 NEW_EXTENSION_PASS

$$\dim \ker \mathcal{C}_{\mathrm{phase}} = 2^{h+1} - 2^h = 2^h. \tag{A.1108}$$

F1282 29_adic_unit_quantum_split.tex:244 NEW_EXTENSION_PASS

$$\boxed{\text{principal dyadic phase alone cannot reconstruct a sign-sensitive quantum observable.}} \tag{A.1109}$$

F1283 29_adic_unit_quantum_split.tex:250 NEW_EXTENSION_PASS

$$A = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \qquad A^2 = I, \tag{A.1110}$$

F1284 29_adic_unit_quantum_split.tex:258 NEW_EXTENSION_PASS

$$E_{\pm} = \frac{I \pm A}{2} \tag{A.1111}$$

F1285 29_adic_unit_quantum_split.tex:262 NEW_EXTENSION_PASS

$$E_{\pm}^2 = \frac{I \pm 2A + A^2}{4} = \frac{I \pm A}{2} = E_{\pm}, \tag{A.1112}$$

$$E_+ E_- = \frac{I - A^2}{4} = 0, \tag{A.1113}$$

$$E_+ + E_- = I. \tag{A.1114}$$

F1286 29_adic_unit_quantum_split.tex:273 NEW_EXTENSION_PASS

$$s_2(E_+) = s_2(E_-) = 1, \qquad v_2(\mathrm{Res}(x-1, x+1)) = 1. \tag{A.1115}$$

F1287 29_adic_unit_quantum_split.tex:279 NEW_EXTENSION_PASS

$$(A - I)^2 = 0, \tag{A.1116}$$

F1288 29_adic_unit_quantum_split.tex:285 NEW_EXTENSION_PASS

$$\text{local coordinate residual} = 0, \tag{A.1117}$$

$$\text{local multiplication residual} = 0, \tag{A.1118}$$

$$\|\mathcal{U}^*_{\mathrm{CRT}}\mathcal{U}_{\mathrm{CRT}} - I\|_2 = 0, \tag{A.1119}$$

$$\text{composite multiplication residual} = 0, \tag{A.1120}$$

$$\text{unit-register QFT factorization residual} = 9.945153 \times 10^{-16}, \tag{A.1121}$$
$$\text{additive CRT-QFT residual} = 7.446474 \times 10^{-15}, \tag{A.1122}$$
$$\text{dyadic sign-alias multiplicity error} = 0, \tag{A.1123}$$
$$\text{dyadic sign rank defect at } h = 4 = 16. \tag{A.1124}$$

30_adic_haar_quantum_refinement.tex

F1289 30_adic_haar_quantum_refinement.tex:8 NEW_EXTENSION_PASS

$$G_{p,h} := \mathbb{Z}/p^h\mathbb{Z}, \qquad \mathcal{H}_{p,h} := \ell^2(G_{p,h}). \tag{A.1125}$$

F1290 30_adic_haar_quantum_refinement.tex:14 NEW_EXTENSION_PASS

$$(E_r^{(h)} f)(x) := \frac{1}{p^{h-r}} \sum_{\substack{y \in G_{p,h} \\ y \equiv x \pmod{p^r}}} f(y). \tag{A.1126}$$

F1291 30_adic_haar_quantum_refinement.tex:21 NEW_EXTENSION_PASS

$$(E_r^{(h)})^* = E_r^{(h)}, \tag{A.1127}$$
$$(E_r^{(h)})^2 = E_r^{(h)}, \tag{A.1128}$$
$$E_r^{(h)} E_s^{(h)} = E_{\min(r,s)}^{(h)}, \tag{A.1129}$$
$$E_h^{(h)} = I. \tag{A.1130}$$

F1292 30_adic_haar_quantum_refinement.tex:28 NEW_EXTENSION_PASS

$$E_r^{(h)} E_s^{(h)} = E_r^{(h)}. \tag{A.1131}$$

F1293 30_adic_haar_quantum_refinement.tex:33 NEW_EXTENSION_PASS

$$D_0^{(h)} := E_0^{(h)}, \qquad D_r^{(h)} := E_r^{(h)} - E_{r-1}^{(h)} \quad (1 \le r \le h). \tag{A.1132}$$

F1294 30_adic_haar_quantum_refinement.tex:40 NEW_EXTENSION_PASS

$$\boxed{D_r^{(h)} D_s^{(h)} = \delta_{rs} D_r^{(h)}, \qquad \sum_{r=0}^{h} D_r^{(h)} = I.} \tag{A.1133}$$

F1295 30_adic_haar_quantum_refinement.tex:48 NEW_EXTENSION_PASS

$$D_r^{(h)} D_s^{(h)} = (E_r - E_{r-1})(E_s - E_{s-1}) \tag{A.1134}$$
$$= E_r - E_r - E_{r-1} + E_{r-1} = 0. \tag{A.1135}$$

F1296 30_adic_haar_quantum_refinement.tex:54 NEW_EXTENSION_PASS

$$(D_r^{(h)})^2 = E_r - 2E_{r-1} + E_{r-1} = D_r^{(h)}. \tag{A.1136}$$

F1297 30_adic_haar_quantum_refinement.tex:60 NEW_EXTENSION_PASS

$$E_0 + \sum_{r=1}^{h} (E_r - E_{r-1}) = E_h = I. \tag{A.1137}$$

F1298 30_adic_haar_quantum_refinement.tex:64 NEW_EXTENSION_PASS

$$\boxed{\operatorname{rank} D_0^{(h)} = 1, \qquad \operatorname{rank} D_r^{(h)} = (p-1)p^{r-1}.} \tag{A.1138}$$

F1299 30_adic_haar_quantum_refinement.tex:72 NEW_EXTENSION_PASS

$$\operatorname{rank} E_r^{(h)} = p^r. \tag{A.1139}$$

F1300 30_adic_haar_quantum_refinement.tex:76 NEW_EXTENSION_PASS

$$\operatorname{rank} D_r^{(h)} = p^r - p^{r-1} = (p-1)p^{r-1}. \tag{A.1140}$$

F1301 30_adic_haar_quantum_refinement.tex:85 NEW_EXTENSION_PASS

$$\psi_{r,a,q}^{(h)}(x) := \begin{cases} p^{-(h-r+1)/2}\omega_p^{qd_r(x)}, & x \equiv a \pmod{p^{r-1}}, \\ 0, & \text{otherwise}, \end{cases} \tag{A.1141}$$

F1302 30_adic_haar_quantum_refinement.tex:95 NEW_EXTENSION_PASS

$$\boxed{D_r^{(h)} = \sum_{a=0}^{p^{r-1}-1} \sum_{q=1}^{p-1} |\psi_{r,a,q}^{(h)}\rangle\langle\psi_{r,a,q}^{(h)}|.} \tag{A.1142}$$

F1303 30_adic_haar_quantum_refinement.tex:106 NEW_EXTENSION_PASS

$$\left\|\psi_{r,a,q}^{(h)}\right\|^2 = p^{h-r+1}p^{-(h-r+1)} = 1. \tag{A.1143}$$

F1304 30_adic_haar_quantum_refinement.tex:111 NEW_EXTENSION_PASS

$$\left\langle \psi_{r,a,q}, \psi_{r,a,q'} \right\rangle = \frac{1}{p}\sum_{d=0}^{p-1} \omega_p^{(q'-q)d} \tag{A.1144}$$

$$= \delta_{q,q'}. \tag{A.1145}$$

F1305 30_adic_haar_quantum_refinement.tex:121 NEW_EXTENSION_PASS

$$V_{p,h} : \mathcal{H}_{p,h} \to \mathcal{H}_{p,h+1} \tag{A.1146}$$

F1306 30_adic_haar_quantum_refinement.tex:125 NEW_EXTENSION_PASS

$$(V_{p,h}f)(a + dp^h) = \frac{1}{\sqrt{p}} f(a), \qquad 0 \le d < p. \tag{A.1147}$$

F1307 30_adic_haar_quantum_refinement.tex:133 NEW_EXTENSION_PASS

$$V_{p,h}f = f \otimes |+_p\rangle, \qquad |+_p\rangle = \frac{1}{\sqrt{p}}\sum_{d=0}^{p-1} |d\rangle. \tag{A.1148}$$

F1308 30_adic_haar_quantum_refinement.tex:142 NEW_EXTENSION_PASS

$$V_{p,h}^* V_{p,h} = I, \tag{A.1149}$$

$$D_r^{(h+1)} V_{p,h} = V_{p,h} D_r^{(h)} \quad (0 \le r \le h), \tag{A.1150}$$

$$D_{h+1}^{(h+1)} V_{p,h} = 0. \tag{A.1151}$$

F1309 30_adic_haar_quantum_refinement.tex:149 NEW_EXTENSION_PASS

$$\left\|V_{p,h}f\right\|^2 = \sum_{a=0}^{p^h-1}\sum_{d=0}^{p-1} \frac{1}{p}|f(a)|^2 \tag{A.1152}$$

$$= \sum_{a=0}^{p^h-1} |f(a)|^2 = \|f\|^2. \tag{A.1153}$$

F1310 30_adic_haar_quantum_refinement.tex:160 NEW_EXTENSION_PASS

eq:adic-intertwine

$$H_f V = V H_c, \tag{A.1154}$$

$$Q H_f P = 0, \qquad P H_f P = V H_c V^*. \tag{A.1155}$$

F1311 30_adic_haar_quantum_refinement.tex:167 NEW_EXTENSION_PASS

$$H_f P = V H_c V^*. \tag{A.1156}$$

F1312 30_adic_haar_quantum_refinement.tex:171 NEW_EXTENSION_PASS

$$\begin{aligned} H_f V &= H_f P V && \text{(A.1157)}\\ &= (P H_f P + Q H_f P) V && \text{(A.1158)}\\ &= V H_c V^* V = V H_c. && \text{(A.1159)} \end{aligned}$$

F1313 30_adic_haar_quantum_refinement.tex:179 NEW_EXTENSION_PASS

$$\boxed{H_f = V H_c V^* \oplus H_Q} \tag{A.1160}$$

F1314 30_adic_haar_quantum_refinement.tex:187 NEW_EXTENSION_PASS

$$P H_f Q = (Q H_f P)^* = 0, \tag{A.1161}$$

F1315 30_adic_haar_quantum_refinement.tex:191 NEW_EXTENSION_PASS

$$f(H_f) = V f(H_c) V^* \oplus f(H_Q). \tag{A.1162}$$

F1316 30_adic_haar_quantum_refinement.tex:195 NEW_EXTENSION_PASS

$$\boxed{f(H_f) V = V f(H_c).} \tag{A.1163}$$

F1317 30_adic_haar_quantum_refinement.tex:201 NEW_EXTENSION_PASS

$$\begin{aligned} \mathrm{e}^{-\mathrm{i}tH_f} V &= V \mathrm{e}^{-\mathrm{i}tH_c}, && \text{(A.1164)}\\ \mathrm{e}^{-\beta H_f} V &= V \mathrm{e}^{-\beta H_c}, && \text{(A.1165)}\\ (z - H_f)^{-1} V &= V (z - H_c)^{-1}. && \text{(A.1166)} \end{aligned}$$

F1318 30_adic_haar_quantum_refinement.tex:207 NEW_EXTENSION_PASS

$$Q \mathrm{e}^{-\mathrm{i}tH_f} P = 0 \tag{A.1167}$$

F1319 30_adic_haar_quantum_refinement.tex:211 NEW_EXTENSION_PASS

$$\boxed{\Sigma_P(z) = P H_f Q (z - Q H_f Q)^{-1} Q H_f P = 0.} \tag{A.1168}$$

F1320 30_adic_haar_quantum_refinement.tex:222 NEW_EXTENSION_PASS

$$\boxed{H_{p,h}^{\mathrm{Haar}} = \sum_{r=0}^{h} \epsilon_{p,r} D_r^{(h)}.} \tag{A.1169}$$

F1321 30_adic_haar_quantum_refinement.tex:231 NEW_EXTENSION_PASS

$$S_{p,h+1} V_{p,h} = V_{p,h} S_{p,h} \tag{A.1170}$$

F1322 30_adic_haar_quantum_refinement.tex:235 NEW_EXTENSION_PASS

$$L_{p,h} := 2I - S_{p,h} - S_{p,h}^* \tag{A.1171}$$

F1323 30_adic_haar_quantum_refinement.tex:239 NEW_EXTENSION_PASS

$$H_{p,h} = H_{p,h}^{\mathrm{Haar}} + \kappa_p L_{p,h} \tag{A.1172}$$

F1324 30_adic_haar_quantum_refinement.tex:246 NEW_EXTENSION_PASS

$$U_{p,h}(t) = \sum_{r=0}^{h} \mathrm{e}^{-\mathrm{i}t\epsilon_{p,r}} D_r^{(h)}. \tag{A.1173}$$

F1325 30_adic_haar_quantum_refinement.tex:253 NEW_EXTENSION_PASS

$$\left(H_{p,h}^{\mathrm{Haar}}\right)^n = \sum_{r=0}^{h} \epsilon_{p,r}^n D_r^{(h)}, \tag{A.1174}$$

F1326 30_adic_haar_quantum_refinement.tex:259 NEW_EXTENSION_PASS

$$E_r^{(h)}(x,y) = p^{-(h-r)} \mathbf{1}_{\{x \equiv y \pmod{p^r}\}}. \tag{A.1175}$$

F1327 30_adic_haar_quantum_refinement.tex:267 NEW_EXTENSION_PASS

$$\boxed{U_{p,h}(t;x,y) = \mathcal{K}_{p,h,t}(v_p(x-y)),} \tag{A.1176}$$

F1328 30_adic_haar_quantum_refinement.tex:276 NEW_EXTENSION_PASS

$$\text{detail-projector residual} = 5.620504 \times 10^{-16}, \tag{A.1177}$$
$$\text{child-DFT/Haar residual} = 9.255029 \times 10^{-16}, \tag{A.1178}$$
$$\text{Hamiltonian-intertwining residual} = 4.587620 \times 10^{-16}, \tag{A.1179}$$
$$\text{unitary-intertwining residual} = 4.885923 \times 10^{-16}, \tag{A.1180}$$
$$\text{Euclidean-intertwining residual} = 4.584318 \times 10^{-16}, \tag{A.1181}$$
$$\text{resolvent-intertwining residual} = 6.971794 \times 10^{-17}, \tag{A.1182}$$
$$\text{leakage residual} = 9.333536 \times 10^{-16}, \tag{A.1183}$$
$$\text{Feshbach self-energy residual} = 2.491706 \times 10^{-32}, \tag{A.1184}$$
$$\text{shell-isotropy residual} = 0. \tag{A.1185}$$

31_adelic_quantum_dynamics.tex

F1329 31_adelic_quantum_dynamics.tex:8 NEW_EXTENSION_PASS

$$\boxed{\mathcal{H}_{\mathcal{P},\mathbf{h}}^{\times} = \mathcal{H}_{2,h_2}^{\times} \otimes \bigotimes_{p \in \mathcal{P}_{\mathrm{o}}} \mathcal{H}_{p,h_p}^{\times}.} \tag{A.1186}$$

F1330 31_adelic_quantum_dynamics.tex:19 NEW_EXTENSION_PASS

$$D_{\mathbf{r}} := \bigotimes_{p \in \mathcal{P}} D_{p,r_p}. \tag{A.1187}$$

F1331 31_adelic_quantum_dynamics.tex:25 NEW_EXTENSION_PASS

$$\boxed{H_{\mathbf{h}} = \sum_{A,\mathbf{r}} \epsilon_{A,\mathbf{r}} T_A \otimes D_{\mathbf{r}},} \tag{A.1188}$$

F1332 31_adelic_quantum_dynamics.tex:38 NEW_EXTENSION_PASS

$$V_{\mathbf{h}}^{\mathbf{h+1}} = I_{\mathrm{tors}} \otimes \bigotimes_{p \in \mathcal{P}} V_{p,h_p}. \tag{A.1189}$$

F1333 31_adelic_quantum_dynamics.tex:46 NEW_EXTENSION_PASS

$$\boxed{H_{\mathbf{h+1}} V_{\mathbf{h}}^{\mathbf{h+1}} = V_{\mathbf{h}}^{\mathbf{h+1}} H_{\mathbf{h}}.} \tag{A.1190}$$

F1334 31_adelic_quantum_dynamics.tex:56 NEW_EXTENSION_PASS

$$D_{\mathbf{r}}^{(\mathbf{h+1})} V = \left(\bigotimes_p D_{p,r_p}^{(h_p+1)} \right) \left(I_{\mathrm{tors}} \otimes \bigotimes_p V_{p,h_p} \right) \tag{A.1191}$$
$$= \left(I_{\mathrm{tors}} \otimes \bigotimes_p V_{p,h_p} \right) \left(\bigotimes_p D_{p,r_p}^{(h_p)} \right) \tag{A.1192}$$

$$= V D_{\mathbf{r}}^{(\mathbf{h})}. \tag{A.1193}$$

F1335 31_adelic_quantum_dynamics.tex:68 NEW_EXTENSION_PASS

$$\mathrm{e}^{-\mathrm{i}tH_{\mathbf{h+1}}} V = V \mathrm{e}^{-\mathrm{i}tH_{\mathbf{h}}}, \tag{A.1194}$$

$$\mathrm{e}^{-\beta H_{\mathbf{h+1}}} V = V \mathrm{e}^{-\beta H_{\mathbf{h}}}, \tag{A.1195}$$

$$(z - H_{\mathbf{h+1}})^{-1} V = V (z - H_{\mathbf{h}})^{-1}, \tag{A.1196}$$

$$Q \mathrm{e}^{-\mathrm{i}tH_{\mathbf{h+1}}} P = 0, \tag{A.1197}$$

$$\Sigma_P(z) = 0. \tag{A.1198}$$

F1336 31_adelic_quantum_dynamics.tex:82 NEW_EXTENSION_PASS

$$B_{\mathrm{odd}} = I_{2,\mathrm{phase}} \otimes Z_{p,\mathrm{tors}} \otimes D_{p,r_p}. \tag{A.1199}$$

F1337 31_adelic_quantum_dynamics.tex:87 NEW_EXTENSION_PASS

$$\boxed{H_{\mathrm{int}} = g Z_2 \otimes B_{\mathrm{odd}}} \tag{A.1200}$$

F1338 31_adelic_quantum_dynamics.tex:93 NEW_EXTENSION_PASS

$$t_* = \frac{\pi}{2g} \tag{A.1201}$$

F1339 31_adelic_quantum_dynamics.tex:97 NEW_EXTENSION_PASS

$$\boxed{S(\rho_{\mathrm{sign}}(t_*)) = 1 \text{ bit.}} \tag{A.1202}$$

F1340 31_adelic_quantum_dynamics.tex:103 NEW_EXTENSION_PASS

$$|\Psi(0)\rangle = |+_x\rangle \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}}. \tag{A.1203}$$

F1341 31_adelic_quantum_dynamics.tex:110 NEW_EXTENSION_PASS

$$|\Psi(t)\rangle = \frac{1}{\sqrt{2}} \left(|+_x\rangle \otimes |0\rangle + \mathrm{e}^{-\mathrm{i}gtZ_2} |+_x\rangle \otimes |1\rangle \right). \tag{A.1204}$$

F1342 31_adelic_quantum_dynamics.tex:121 NEW_EXTENSION_PASS

$$\mathrm{e}^{-\mathrm{i}gt_* Z_2} |+_x\rangle = -\mathrm{i} |-_x\rangle, \tag{A.1205}$$

F1343 31_adelic_quantum_dynamics.tex:126 NEW_EXTENSION_PASS

$$\rho_{\mathrm{sign}}(t_*) = \frac{1}{2} \left(|+_x\rangle\langle +_x| + |-_x\rangle\langle -_x| \right), \tag{A.1206}$$

F1344 31_adelic_quantum_dynamics.tex:138 NEW_EXTENSION_PASS

$$PBQ \neq 0, \qquad PBP = QBQ = 0. \tag{A.1207}$$

F1345 31_adelic_quantum_dynamics.tex:144 NEW_EXTENSION_PASS

$$\lambda_{p,w} := p^{-w}, \qquad H_{p,w} := H_0 + \lambda_{p,w} B. \tag{A.1208}$$

F1346 31_adelic_quantum_dynamics.tex:150 NEW_EXTENSION_PASS

$$\boxed{\| H_{p,w} V - V H_c \| = \lambda_{p,w} \| BV \|.} \tag{A.1209}$$

F1347 31_adelic_quantum_dynamics.tex:157 NEW_EXTENSION_PASS

$$\boxed{\| Q \mathrm{e}^{-\mathrm{i}tH_{p,w}} P \| \leq |t| p^{-w} \| QBP \|.} \tag{A.1210}$$

F1348 31_adelic_quantum_dynamics.tex:165 NEW_EXTENSION_PASS

$$Q \mathrm{e}^{-\mathrm{i}tH_{p,w}} P = -\mathrm{i} p^{-w} \int_0^t Q \mathrm{e}^{-\mathrm{i}(t-s)H_{p,w}} B \mathrm{e}^{-\mathrm{i}sH_0} P \, \mathrm{d}s. \tag{A.1211}$$

F1349 31_adelic_quantum_dynamics.tex:174 NEW_EXTENSION_PASS

$$\|Q\mathrm{e}^{-\mathrm{i}tH_{p,w}}P\| \le p^{-w}\int_0^{|t|}\|QBP\|\,\mathrm{d}s \tag{A.1212}$$

$$= |t|p^{-w}\|QBP\|. \tag{A.1213}$$

F1350 31_adelic_quantum_dynamics.tex:181 NEW_EXTENSION_PASS

$$Q\mathrm{e}^{-\mathrm{i}tH_{p,w}}P = -\mathrm{i}tp^{-w}QBP + \mathcal{O}(t^2p^{-w}), \tag{A.1214}$$

F1351 31_adelic_quantum_dynamics.tex:187 NEW_EXTENSION_PASS

$$\Sigma_P(z) = PH_{p,w}Q(z - QH_{p,w}Q)^{-1}QH_{p,w}P \tag{A.1215}$$

F1352 31_adelic_quantum_dynamics.tex:192 NEW_EXTENSION_PASS

$$\boxed{\|\Sigma_P(z)\| \le \frac{p^{-2w}\|PBQ\|^2}{\mathrm{dist}(z,\sigma(QH_{p,w}Q))}.} \tag{A.1216}$$

F1353 31_adelic_quantum_dynamics.tex:201 NEW_EXTENSION_PASS

$$PH_{p,w}Q = p^{-w}PBQ, \qquad QH_{p,w}P = p^{-w}QBP. \tag{A.1217}$$

F1354 31_adelic_quantum_dynamics.tex:207 NEW_EXTENSION_PASS

$$\|\Sigma_P(z)\| \le p^{-2w}\|PBQ\|\|(z - QH_{p,w}Q)^{-1}\|\|QBP\| \tag{A.1218}$$

$$\le \frac{p^{-2w}\|PBQ\|^2}{\mathrm{dist}(z,\sigma(QH_{p,w}Q))}. \tag{A.1219}$$

F1355 31_adelic_quantum_dynamics.tex:218 NEW_EXTENSION_PASS

$$\text{refinement defect} = O(p^{-w}), \qquad \text{Feshbach backreaction} = O(p^{-2w}). \tag{A.1220}$$

F1356 31_adelic_quantum_dynamics.tex:226 NEW_EXTENSION_PASS

$$E_\pm = \frac{I \pm A}{2}, \tag{A.1221}$$

F1357 31_adelic_quantum_dynamics.tex:230 NEW_EXTENSION_PASS

$$\boxed{\text{complex quantum split} \neq \text{integral dyadic spectral lift}} \tag{A.1222}$$

F1358 31_adelic_quantum_dynamics.tex:240 NEW_EXTENSION_PASS

$$\text{multi-prime Hamiltonian-intertwining residual} = 5.030633 \times 10^{-16}, \tag{A.1223}$$

$$\text{functional-calculus residual} = 4.096089 \times 10^{-16}, \tag{A.1224}$$

$$\text{multi-prime leakage residual} = 2.909890 \times 10^{-16}, \tag{A.1225}$$

$$\text{multi-prime Feshbach residual} = 9.883696 \times 10^{-33}, \tag{A.1226}$$

$$\text{unit-basis equivalence residual} = 1.998401 \times 10^{-15}, \tag{A.1227}$$

$$S(\rho_{\mathrm{sign}}(t_*)) = 1.000000 \text{ bit}, \tag{A.1228}$$

$$\mathrm{ord}_\lambda(\text{defect}) = 1.000000, \tag{A.1229}$$

$$\mathrm{ord}_\lambda(\text{leakage}) = 1.000000, \tag{A.1230}$$

$$\mathrm{ord}_\lambda(\Sigma_P) = 2.000000, \tag{A.1231}$$

$$\text{Duhamel-bound violation} = 0. \tag{A.1232}$$

32_adic_normalization_weights.tex

F1359 32_adic_normalization_weights.tex:8 NEW_EXTENSION_PASS

$$\boxed{|R_{p,w}\rangle = \frac{1}{\sqrt{p^w}} \sum_{\mathbf{d}\in\{0,\dots,p-1\}^w} |\mathbf{d}\rangle.} \tag{A.1233}$$

F1360 32_adic_normalization_weights.tex:18 NEW_EXTENSION_PASS

$$\langle R_{p,w}|R_{p,w}\rangle = p^{-w} \sum_{\mathbf{d},\mathbf{e}} \langle \mathbf{d}|\mathbf{e}\rangle \tag{A.1234}$$

$$= p^{-w} \sum_{\mathbf{d}} 1 = 1. \tag{A.1235}$$

F1361 32_adic_normalization_weights.tex:28 NEW_EXTENSION_PASS

$$\boxed{|\langle \mathbf{d}|R_{p,w}\rangle| = p^{-w/2}, \qquad |\langle \mathbf{d}|R_{p,w}\rangle|^2 = p^{-w}.} \tag{A.1236}$$

F1362 32_adic_normalization_weights.tex:41 NEW_EXTENSION_PASS

$$g^{\mathrm{leaf}}_{p,w} := g\, p^{-\alpha w}. \tag{A.1237}$$

F1363 32_adic_normalization_weights.tex:46 NEW_EXTENSION_PASS

$$H^{(p,w)}_{\mathrm{int}} = g p^{-\alpha w} \sum_{\mathbf{d}} \Big(|c\rangle\langle \mathbf{d}| + |\mathbf{d}\rangle\langle c| \Big). \tag{A.1238}$$

F1364 32_adic_normalization_weights.tex:58 NEW_EXTENSION_PASS

$$\boxed{\langle R_{p,w}|H^{(p,w)}_{\mathrm{int}}|c\rangle = g\, p^{(1/2-\alpha)w}.} \tag{A.1239}$$

F1365 32_adic_normalization_weights.tex:66 NEW_EXTENSION_PASS

$$\langle R_{p,w}|H^{(p,w)}_{\mathrm{int}}|c\rangle = g p^{-\alpha w} p^{-w/2} \sum_{\mathbf{e},\mathbf{d}} \langle \mathbf{e}|\mathbf{d}\rangle \tag{A.1240}$$

$$= g p^{-\alpha w} p^{-w/2} p^{w} \tag{A.1241}$$

$$= g\, p^{(1/2-\alpha)w}. \tag{A.1242}$$

F1366 32_adic_normalization_weights.tex:78 NEW_EXTENSION_PASS

$$\langle A|H^{(p,w)}_{\mathrm{int}}|c\rangle = g p^{-\alpha w} \sum_{\mathbf{d}} a^*_{\mathbf{d}} = 0, \tag{A.1243}$$

F1367 32_adic_normalization_weights.tex:84 NEW_EXTENSION_PASS

$$\boxed{G^{\mathrm{coll}}_{p,w} = \sqrt{p^w}\, g^{\mathrm{leaf}}_{p,w}.} \tag{A.1244}$$

F1368 32_adic_normalization_weights.tex:95 NEW_EXTENSION_PASS

$$H_{\mathrm{rad}} = \begin{pmatrix} 0 & G^{\mathrm{coll}}_{p,w} \\ G^{\mathrm{coll}}_{p,w} & 0 \end{pmatrix}. \tag{A.1245}$$

F1369 32_adic_normalization_weights.tex:104 NEW_EXTENSION_PASS

$$\boxed{\Pr(R_{p,w};t) = \sin^2\Big(|g| p^{(1/2-\alpha)w} t\Big).} \tag{A.1246}$$

F1370 32_adic_normalization_weights.tex:114 NEW_EXTENSION_PASS

$$\mathrm{e}^{-\mathrm{i}tH_{\mathrm{rad}}} = I\cos(|G|t) - \mathrm{i}\frac{H_{\mathrm{rad}}}{|G|}\sin(|G|t). \tag{A.1247}$$

F1371 32_adic_normalization_weights.tex:122 NEW_EXTENSION_PASS

$$\boxed{\Sigma_{p,w}(z) = \frac{|g|^2 p^{(1-2\alpha)w}}{z-\epsilon}.} \tag{A.1248}$$

F1372 32_adic_normalization_weights.tex:130 NEW_EXTENSION_PASS

$$G(z-\epsilon)^{-1}G^* = \frac{|G|^2}{z-\epsilon} = \frac{|g|^2 p^{(1-2\alpha)w}}{z-\epsilon}. \tag{A.1249}$$

F1373 32_adic_normalization_weights.tex:138 NEW_EXTENSION_PASS

$$\boxed{\alpha < \frac{1}{2} \Longrightarrow \text{collective growth}, \qquad \alpha = \frac{1}{2} \Longrightarrow \text{scale-invariant collective coupling}, \qquad \alpha > \frac{1}{2} \Longrightarrow \text{collective decay}.} \tag{A.1250}$$

F1374 32_adic_normalization_weights.tex:156 NEW_EXTENSION_PASS

$$G^{\mathrm{coll}}_{p,w} = g\, p^{-w/2}, \qquad |G^{\mathrm{coll}}_{p,w}|^2 = |g|^2 p^{-w}. \tag{A.1251}$$

F1375 32_adic_normalization_weights.tex:162 NEW_EXTENSION_PASS

$$\boxed{\text{collective amplitude} = O(p^{-w/2}), \qquad \text{probability and self-energy} = O(p^{-w}).} \tag{A.1252}$$

F1376 32_adic_normalization_weights.tex:172 NEW_EXTENSION_PASS

$$\text{descendant-amplitude residual} = 1.110223 \times 10^{-16}, \tag{A.1253}$$
$$\text{descendant-probability residual} = 1.110223 \times 10^{-16}, \tag{A.1254}$$
$$\text{collective-norm residual} = 1.887379 \times 10^{-15}, \tag{A.1255}$$
$$\text{radial-reduction residual} = 2.220446 \times 10^{-16}, \tag{A.1256}$$
$$\text{two-level probability residual} = 1.734723 \times 10^{-18}, \tag{A.1257}$$
$$\text{self-energy residual} = 3.496447 \times 10^{-18}. \tag{A.1258}$$

33_spacetime_adic_product_walk.tex

F1377 33_spacetime_adic_product_walk.tex:8 NEW_EXTENSION_PASS

$$\boxed{\mathcal{G}_{p,h,N} := C_N \square T_{p,h}.} \tag{A.1259}$$

F1378 33_spacetime_adic_product_walk.tex:18 NEW_EXTENSION_PASS

$$\mathcal{H}_{\mathrm{arc}} = \ell^2(\vec{E}(\mathcal{G}_{p,h,N})) \otimes \mathbb{C}^{\tau_p}, \tag{A.1260}$$

F1379 33_spacetime_adic_product_walk.tex:26 NEW_EXTENSION_PASS

$$\tau_2 = 2 \tag{A.1261}$$

F1380 33_spacetime_adic_product_walk.tex:30 NEW_EXTENSION_PASS

$$\tau_p = p - 1, \qquad p \text{ odd}, \tag{A.1262}$$

F1381 33_spacetime_adic_product_walk.tex:38 NEW_EXTENSION_PASS

$$C = \bigoplus_v (C_v \otimes R_v). \tag{A.1263}$$

F1382 33_spacetime_adic_product_walk.tex:44 NEW_EXTENSION_PASS

$$\boxed{U_{p,h,N} = SC.} \tag{A.1264}$$

F1383 33_spacetime_adic_product_walk.tex:50 NEW_EXTENSION_PASS

$$U^*_{p,h,N} U_{p,h,N} = I. \tag{A.1265}$$

F1384 33_spacetime_adic_product_walk.tex:54 NEW_EXTENSION_PASS

$$U^*_{p,h,N}U_{p,h,N} = C^*S^*SC \tag{A.1266}$$
$$= C^*C \tag{A.1267}$$
$$= \bigoplus_v (C_v^*C_v \otimes R_v^*R_v) = I. \tag{A.1268}$$

F1385 33_spacetime_adic_product_walk.tex:65 NEW_EXTENSION_PASS

$$\boxed{\rho_v^{n+1} - \rho_v^n = \sum_{u\sim v} (F^n_{u\to v} - F^n_{v\to u})\,.} \tag{A.1269}$$

F1386 33_spacetime_adic_product_walk.tex:76 NEW_EXTENSION_PASS

$$\sum_{u\sim v} F^n_{v\to u} = \rho_v^n. \tag{A.1270}$$

F1387 33_spacetime_adic_product_walk.tex:80 NEW_EXTENSION_PASS

$$\rho_v^{n+1} = \sum_{u\sim v} F^n_{u\to v}. \tag{A.1271}$$

F1388 33_spacetime_adic_product_walk.tex:85 NEW_EXTENSION_PASS

$$\boxed{\sum_{v\in A}(\rho_v^{n+1} - \rho_v^n) = \sum_{\substack{u\notin A\\ v\in A}} F^n_{u\to v} - \sum_{\substack{v\in A\\ u\notin A}} F^n_{v\to u}.} \tag{A.1272}$$

F1389 33_spacetime_adic_product_walk.tex:101 NEW_EXTENSION_PASS

$$\boxed{d_\square(v, v_0) > n \Longrightarrow \rho_v^n = 0.} \tag{A.1273}$$

F1390 33_spacetime_adic_product_walk.tex:110 NEW_EXTENSION_PASS

$$d_{C_N}(x, x_0) \le n, \tag{A.1274}$$
$$\big|\,|\mathbf{d}| - |\mathbf{d}_0|\,\big| \le n. \tag{A.1275}$$

F1391 33_spacetime_adic_product_walk.tex:118 NEW_EXTENSION_PASS

$$\text{coin-unitarity residual} = 1.310083 \times 10^{-15}, \tag{A.1276}$$
$$\text{shift-involution residual} = 0, \tag{A.1277}$$
$$\text{probability drift} = 8.881784 \times 10^{-16}, \tag{A.1278}$$
$$\text{local continuity residual} = 3.330669 \times 10^{-16}, \tag{A.1279}$$
$$\text{coarse balance residual} = 3.330669 \times 10^{-16}, \tag{A.1280}$$
$$\text{probability outside the product cone} = 0, \tag{A.1281}$$
$$\text{physical edge excess} = 0, \tag{A.1282}$$
$$\text{adic-depth edge excess} = 0. \tag{A.1283}$$

34_adic_radial_self_energy.tex

F1392 34_adic_radial_self_energy.tex:8 NEW_EXTENSION_PASS

$$\frac{\gamma_r}{\sqrt{p}}. \tag{A.1284}$$

F1393 34_adic_radial_self_energy.tex:12 NEW_EXTENSION_PASS

$$|r\rangle := p^{-r/2} \sum_{|v|=r} |v\rangle. \tag{A.1285}$$

F1394 34_adic_radial_self_energy.tex:19 NEW_EXTENSION_PASS

$$\langle r+1|H|r\rangle = p^{-(r+1)/2}p^{-r/2}\sum_{\substack{|u|=r\\ v \text{ child of } u}}\frac{\gamma_r}{\sqrt{p}} \tag{A.1286}$$

$$= p^{-(2r+1)/2}\,p^{r+1}\,\frac{\gamma_r}{\sqrt{p}} \tag{A.1287}$$

$$= \gamma_r. \tag{A.1288}$$

F1395 34_adic_radial_self_energy.tex:33 NEW_EXTENSION_PASS

$$\boxed{J_h = \begin{pmatrix} \epsilon_0 & \gamma_0 & 0 & \cdots & 0 \\ \gamma_0 & \epsilon_1 & \gamma_1 & \ddots & \vdots \\ 0 & \gamma_1 & \epsilon_2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \gamma_{h-1} \\ 0 & \cdots & 0 & \gamma_{h-1} & \epsilon_h \end{pmatrix}.} \tag{A.1289}$$

F1396 34_adic_radial_self_energy.tex:50 NEW_EXTENSION_PASS

$$\boxed{G_{0,h}(z) = \langle 0|(z-J_h)^{-1}|0\rangle.} \tag{A.1290}$$

F1397 34_adic_radial_self_energy.tex:58 NEW_EXTENSION_PASS

$$\operatorname{rank} D_r = (p-1)p^{r-1}. \tag{A.1291}$$

F1398 34_adic_radial_self_energy.tex:65 NEW_EXTENSION_PASS

$$\Sigma_h(z) := 0, \tag{A.1292}$$

$$\Sigma_r(z) := \frac{\gamma_r^2}{z-\epsilon_{r+1}-\Sigma_{r+1}(z)}, \qquad 0 \le r < h. \tag{A.1293}$$

F1399 34_adic_radial_self_energy.tex:72 NEW_EXTENSION_PASS

$$z - J_h = \begin{pmatrix} z-\epsilon_r & -\gamma_r\langle r+1| \\ -\gamma_r|r+1\rangle & z-J_{r+1:h} \end{pmatrix}. \tag{A.1294}$$

F1400 34_adic_radial_self_energy.tex:81 NEW_EXTENSION_PASS

$$z-\epsilon_r-\gamma_r^2\langle r+1|(z-J_{r+1:h})^{-1}|r+1\rangle. \tag{A.1295}$$

F1401 34_adic_radial_self_energy.tex:88 NEW_EXTENSION_PASS

$$\boxed{G_{0,h}(z) = \frac{1}{z-\epsilon_0-\Sigma_0(z)}.} \tag{A.1296}$$

F1402 34_adic_radial_self_energy.tex:99 NEW_EXTENSION_PASS

$$H_D(k) = k\sigma_z + m_0\sigma_x \tag{A.1297}$$

F1403 34_adic_radial_self_energy.tex:103 NEW_EXTENSION_PASS

$$\boxed{\Sigma(z) = \gamma_0^2\left[(z-J_{\text{tail}})^{-1}\right]_{00}.} \tag{A.1298}$$

F1404 34_adic_radial_self_energy.tex:112 NEW_EXTENSION_PASS

$$\boxed{\det(z-H_{\text{full}}) = \det(z-H_{\text{hidden}})\det((z-\Sigma(z))I_2 - H_D(k)).} \tag{A.1299}$$

F1405 34_adic_radial_self_energy.tex:121 NEW_EXTENSION_PASS

$$z - H_{\text{full}} = \begin{pmatrix} zI_2 - H_D(k) & -\gamma_0\, I_2|0\rangle \\ -\gamma_0\, I_2\langle 0| & z-H_{\text{hidden}} \end{pmatrix}. \tag{A.1300}$$

F1406 34_adic_radial_self_energy.tex:130 NEW_EXTENSION_PASS

$$\det(z - H_{\mathrm{full}}) = \det(z - H_{\mathrm{hidden}}) \tag{A.1301}$$

$$\times \det\left[zI_2 - H_D(k) - \gamma_0^2 \langle 0|(z - H_{\mathrm{hidden}})^{-1}|0\rangle I_2\right], \tag{A.1302}$$

F1407 34_adic_radial_self_energy.tex:142 NEW_EXTENSION_PASS

$$\boxed{\left(E - \Sigma(E)\right)^2 = k^2 + m_0^2.} \tag{A.1303}$$

F1408 34_adic_radial_self_energy.tex:150 NEW_EXTENSION_PASS

$$\boxed{K(t) = \gamma_0^2 \left[\exp(-\mathrm{i}tJ_{\mathrm{tail}})\right]_{00}.} \tag{A.1304}$$

F1409 34_adic_radial_self_energy.tex:159 NEW_EXTENSION_PASS

$$\mathrm{i}\dot{\psi}_V = H_D\psi_V + \gamma_0\langle 0|\psi_H, \tag{A.1305}$$

$$\mathrm{i}\dot{\psi}_H = J_{\mathrm{tail}}\psi_H + \gamma_0|0\rangle\psi_V. \tag{A.1306}$$

F1410 34_adic_radial_self_energy.tex:164 NEW_EXTENSION_PASS

$$\psi_H(t) = -\mathrm{i}\gamma_0 \int_0^t \mathrm{e}^{-\mathrm{i}(t-s)J_{\mathrm{tail}}}|0\rangle\psi_V(s)\,\mathrm{d}s. \tag{A.1307}$$

F1411 34_adic_radial_self_energy.tex:174 NEW_EXTENSION_PASS

$$\text{radial-block residual} = 3.538833 \times 10^{-15}, \tag{A.1308}$$

$$\text{radial/dark invariance residual} = 5.306930 \times 10^{-15}, \tag{A.1309}$$

$$\text{root Green residual} = 1.144392 \times 10^{-16}, \tag{A.1310}$$

$$\text{continued-fraction residual} = 5.594315 \times 10^{-17}, \tag{A.1311}$$

$$\text{prime-universality residual} = 1.144392 \times 10^{-16}, \tag{A.1312}$$

$$\text{branching-factor residual} = 7.216450 \times 10^{-16}, \tag{A.1313}$$

$$\text{Schur determinant residual} = 2.259450 \times 10^{-15}, \tag{A.1314}$$

$$\text{visible-branch equation residual} = 2.065015 \times 10^{-14}, \tag{A.1315}$$

$$\text{root/dark overlap residual} = 0. \tag{A.1316}$$

35_adic_polyphase_quantum_refinement.tex

F1412 35_adic_polyphase_quantum_refinement.tex:8 NEW_EXTENSION_PASS

$$\boxed{\mathcal{A}^{\mathrm{std}}_{p,h}|pk + r\rangle = |r\rangle \otimes |k\rangle.} \tag{A.1317}$$

F1413 35_adic_polyphase_quantum_refinement.tex:16 NEW_EXTENSION_PASS

$$\boxed{\mathcal{A}^{\mathrm{ad}}_{p,h}|a + dp^{h-1}\rangle = |d\rangle \otimes |a\rangle.} \tag{A.1318}$$

F1414 35_adic_polyphase_quantum_refinement.tex:24 NEW_EXTENSION_PASS

$$\boxed{\mathcal{A}^{\mathrm{ad}}_{p,h} = (I_p \otimes \mathcal{R}_{p,h-1})\mathcal{A}^{\mathrm{std}}_{p,h}\mathcal{R}_{p,h}.} \tag{A.1319}$$

F1415 35_adic_polyphase_quantum_refinement.tex:34 NEW_EXTENSION_PASS

$$n = d_0 + d_1 p + \cdots + d_{h-1}p^{h-1}, \tag{A.1320}$$

F1416 35_adic_polyphase_quantum_refinement.tex:38 NEW_EXTENSION_PASS

$$d_{h-1} + d_{h-2}p + \cdots + d_0 p^{h-1}. \tag{A.1321}$$

F1417 35_adic_polyphase_quantum_refinement.tex:46 NEW_EXTENSION_PASS

$$M_{p,h}|a+dp^{h-1}\rangle=\omega_p^{d}|a+dp^{h-1}\rangle. \tag{A.1322}$$

F1418 35_adic_polyphase_quantum_refinement.tex:52 NEW_EXTENSION_PASS

$$\boxed{R_d^{(p,h)}=\frac{1}{p}\sum_{\ell=0}^{p-1}\omega_p^{-d\ell}M_{p,h}^{\ell}.} \tag{A.1323}$$

F1419 35_adic_polyphase_quantum_refinement.tex:62 NEW_EXTENSION_PASS

$$R_d^{(p,h)}|a+ep^{h-1}\rangle=\frac{1}{p}\sum_{\ell=0}^{p-1}\omega_p^{(e-d)\ell}|a+ep^{h-1}\rangle \tag{A.1324}$$

$$=\delta_{d,e}|a+ep^{h-1}\rangle, \tag{A.1325}$$

F1420 35_adic_polyphase_quantum_refinement.tex:74 NEW_EXTENSION_PASS

$$R_dR_e=\delta_{de}R_d,\qquad\sum_{d=0}^{p-1}R_d=I. \tag{A.1326}$$

F1421 35_adic_polyphase_quantum_refinement.tex:82 NEW_EXTENSION_PASS

$$\boxed{\mathcal{W}_{p,h}=(F_p\otimes I_m)\mathcal{A}_{p,h}^{\mathrm{ad}}.} \tag{A.1327}$$

F1422 35_adic_polyphase_quantum_refinement.tex:90 NEW_EXTENSION_PASS

$$\boxed{(\langle 0|\otimes I_m)\mathcal{W}_{p,h}=V_{p,h-1}^{*}.} \tag{A.1328}$$

F1423 35_adic_polyphase_quantum_refinement.tex:98 NEW_EXTENSION_PASS

$$(\langle 0|\otimes I_m)\mathcal{W}_{p,h}|a+dp^{h-1}\rangle=\langle 0|F_p|d\rangle\,|a\rangle \tag{A.1329}$$

$$=\frac{1}{\sqrt{p}}|a\rangle, \tag{A.1330}$$

F1424 35_adic_polyphase_quantum_refinement.tex:106 NEW_EXTENSION_PASS

$$V_{p,h-1}^{*}|a+dp^{h-1}\rangle=\frac{1}{\sqrt{p}}|a\rangle. \tag{A.1331}$$

F1425 35_adic_polyphase_quantum_refinement.tex:111 NEW_EXTENSION_PASS

$$\boxed{\mathcal{W}_{p,h}^{*}\Big[(I_p-|0\rangle\langle 0|)\otimes I_m\Big]\mathcal{W}_{p,h}=I-V_{p,h-1}V_{p,h-1}^{*}.} \tag{A.1332}$$

F1426 35_adic_polyphase_quantum_refinement.tex:121 NEW_EXTENSION_PASS

$$\mathcal{W}^{*}\Big[(I-|0\rangle\langle 0|)\otimes I\Big]\mathcal{W}=I-\mathcal{W}^{*}(|0\rangle\langle 0|\otimes I)\mathcal{W} \tag{A.1333}$$

$$=I-\Big[(\langle 0|\otimes I)\mathcal{W}\Big]^{*}\Big[(\langle 0|\otimes I)\mathcal{W}\Big] \tag{A.1334}$$

$$=I-VV^{*}. \tag{A.1335}$$

F1427 35_adic_polyphase_quantum_refinement.tex:136 NEW_EXTENSION_PASS

$$1,\quad (p-1),\quad (p-1)p,\quad \dots\quad ,(p-1)p^{h-1}, \tag{A.1336}$$

F1428 35_adic_polyphase_quantum_refinement.tex:146 NEW_EXTENSION_PASS

$$\boxed{\mathcal{H}_{p,h}^{*}\Pi_r\mathcal{H}_{p,h}=D_r^{(h)}.} \tag{A.1337}$$

F1429 35_adic_polyphase_quantum_refinement.tex:156 NEW_EXTENSION_PASS

$$\boxed{V_p^{(w)}|a\rangle=p^{-w/2}\sum_{\mathbf{d}\in\{0,\dots,p-1\}^w}|a,\mathbf{d}\rangle.} \tag{A.1338}$$

F1430 35_adic_polyphase_quantum_refinement.tex:166 NEW_EXTENSION_PASS

$$V_p^{(w+1)}|a\rangle = p^{-w/2}\sum_{\mathbf{d}}\frac{1}{\sqrt{p}}\sum_{e=0}^{p-1}|a,\mathbf{d},e\rangle \tag{A.1339}$$

$$= p^{-(w+1)/2}\sum_{\mathbf{d},e}|a,\mathbf{d},e\rangle. \tag{A.1340}$$

F1431 35_adic_polyphase_quantum_refinement.tex:178 NEW_EXTENSION_PASS

$$\boxed{\text{amplitude per descendant} = p^{-w/2}, \qquad \text{Born weight per descendant} = p^{-w}.} \tag{A.1341}$$

F1432 35_adic_polyphase_quantum_refinement.tex:189 NEW_EXTENSION_PASS

$$X_{rs} := J_r^* X J_s. \tag{A.1342}$$

F1433 35_adic_polyphase_quantum_refinement.tex:193 NEW_EXTENSION_PASS

$$\boxed{X = \sum_{r,s=0}^{p-1} J_r X_{rs} J_s^*.} \tag{A.1343}$$

F1434 35_adic_polyphase_quantum_refinement.tex:202 NEW_EXTENSION_PASS

$$J_r^* J_s = \delta_{rs} I_m, \qquad \sum_r J_r J_r^* = I_N. \tag{A.1344}$$

F1435 35_adic_polyphase_quantum_refinement.tex:208 NEW_EXTENSION_PASS

$$X = \left(\sum_r J_r J_r^*\right) X \left(\sum_s J_s J_s^*\right) \tag{A.1345}$$

$$= \sum_{r,s} J_r (J_r^* X J_s) J_s^*. \tag{A.1346}$$

F1436 35_adic_polyphase_quantum_refinement.tex:216 NEW_EXTENSION_PASS

$$H_{rs}^* = H_{sr}. \tag{A.1347}$$

F1437 35_adic_polyphase_quantum_refinement.tex:220 NEW_EXTENSION_PASS

$$\sum_r U_{rs}^* U_{rt} = \delta_{st} I_m, \tag{A.1348}$$

$$\sum_s U_{rs} U_{ts}^* = \delta_{rt} I_m. \tag{A.1349}$$

F1438 35_adic_polyphase_quantum_refinement.tex:225 NEW_EXTENSION_PASS

$$\sum_r U_{rs}^* U_{rt} = \sum_r J_s^* U^* J_r J_r^* U J_t \tag{A.1350}$$

$$= J_s^* U^* \left(\sum_r J_r J_r^*\right) U J_t \tag{A.1351}$$

$$= J_s^* J_t = \delta_{st} I_m. \tag{A.1352}$$

F1439 35_adic_polyphase_quantum_refinement.tex:239 NEW_EXTENSION_PASS

$$\boxed{\exp(-\mathrm{i}t\mathcal{A}H\mathcal{A}^*) = \mathcal{A}\exp(-\mathrm{i}tH)\mathcal{A}^*.} \tag{A.1353}$$

F1440 35_adic_polyphase_quantum_refinement.tex:247 NEW_EXTENSION_PASS

$$J_{r\leftarrow s} = 2\,\mathrm{Im}\langle\psi_r, H_{rs}\psi_s\rangle \tag{A.1354}$$

F1441 35_adic_polyphase_quantum_refinement.tex:254 NEW_EXTENSION_PASS

$$\dot{E}_r = \sum_s J_{r\leftarrow s}, \qquad J_{r\leftarrow s} = -J_{s\leftarrow r}. \tag{A.1355}$$

F1442 35_adic_polyphase_quantum_refinement.tex:260 NEW_EXTENSION_PASS

$$K_\mu^{\mathrm{poly}} = \mathcal{A} K_\mu \mathcal{A}^* \tag{A.1356}$$

F1443 35_adic_polyphase_quantum_refinement.tex:264 NEW_EXTENSION_PASS

$$\sum_\mu (K_\mu^{\mathrm{poly}})^* K_\mu^{\mathrm{poly}} = I \tag{A.1357}$$

F1444 35_adic_polyphase_quantum_refinement.tex:271 NEW_EXTENSION_PASS

$$\boxed{\mathcal{C}_{p,h} H \mathcal{C}_{p,h}^* = \bigoplus_{j=0}^{m-1} \mathsf{H}_p(\theta_j), \qquad \theta_j = \frac{2\pi j}{m}.} \tag{A.1358}$$

F1445 35_adic_polyphase_quantum_refinement.tex:282 NEW_EXTENSION_PASS

$$H = \mu I + \kappa (S + S^*), \tag{A.1359}$$

F1446 35_adic_polyphase_quantum_refinement.tex:286 NEW_EXTENSION_PASS

$$\boxed{E_q(\theta_j) = \mu + 2\kappa \cos\!\Big(\frac{\theta_j + 2\pi q}{p}\Big), \qquad q = 0, \dots, p-1.} \tag{A.1360}$$

F1447 35_adic_polyphase_quantum_refinement.tex:299 NEW_EXTENSION_PASS

$$k = j + qm, \qquad 0 \le j < m, \quad 0 \le q < p. \tag{A.1361}$$

F1448 35_adic_polyphase_quantum_refinement.tex:305 NEW_EXTENSION_PASS

$$\frac{2\pi k}{pm} = \frac{2\pi j/m + 2\pi q}{p} = \frac{\theta_j + 2\pi q}{p}. \tag{A.1362}$$

F1449 35_adic_polyphase_quantum_refinement.tex:318 NEW_EXTENSION_PASS

$$H_{p,h}^{\mathrm{hier}} = \sum_{r=0}^{h} \epsilon_r D_r^{(h)}, \tag{A.1363}$$

F1450 35_adic_polyphase_quantum_refinement.tex:324 NEW_EXTENSION_PASS

$$\boxed{\mathcal{H}_{p,h} H_{p,h}^{\mathrm{hier}} \mathcal{H}_{p,h}^* = \mathrm{diag}\left(\epsilon_0, \epsilon_1 I_{p-1}, \dots, \epsilon_h I_{(p-1)p^{h-1}}\right).} \tag{A.1364}$$

F1451 35_adic_polyphase_quantum_refinement.tex:340 NEW_EXTENSION_PASS

$$\boxed{\mathcal{A}_{\mathcal{P}} = I_{\mathrm{sign/torsion}} \otimes \bigotimes_{p \in \mathcal{P}} \mathcal{W}_{p,h_p}} \tag{A.1365}$$

F1452 35_adic_polyphase_quantum_refinement.tex:351 NEW_EXTENSION_PASS

$$\mathcal{A}_{\mathcal{P}} (Z_2 \otimes I) \mathcal{A}_{\mathcal{P}}^* = Z_2 \otimes I. \tag{A.1366}$$

F1453 35_adic_polyphase_quantum_refinement.tex:361 NEW_EXTENSION_PASS

$$\mathcal{W} H \mathcal{W}^* = \begin{pmatrix} H_{LL} & H_{LD} \\ H_{DL} & H_{DD} \end{pmatrix}. \tag{A.1367}$$

F1454 35_adic_polyphase_quantum_refinement.tex:370 NEW_EXTENSION_PASS

$$\boxed{P_L (z - H)^{-1} P_L = \left[z - H_{LL} - H_{LD} (z - H_{DD})^{-1} H_{DL} \right]^{-1}.} \tag{A.1368}$$

F1455 35_adic_polyphase_quantum_refinement.tex:381 NEW_EXTENSION_PASS

$$(z - H_{DD})\psi_D = H_{DL}\psi_L, \tag{A.1369}$$

F1456 35_adic_polyphase_quantum_refinement.tex:385 NEW_EXTENSION_PASS

$$\psi_D = (z - H_{DD})^{-1} H_{DL}\psi_L. \tag{A.1370}$$

F1457 35_adic_polyphase_quantum_refinement.tex:389 NEW_EXTENSION_PASS

$$\left[z - H_{LL} - H_{LD}(z - H_{DD})^{-1} H_{DL}\right] \psi_L = f_L, \tag{A.1371}$$

F1458 35_adic_polyphase_quantum_refinement.tex:396 NEW_EXTENSION_PASS

$$H = \mu I + \kappa(S + S^*), \tag{A.1372}$$

F1459 35_adic_polyphase_quantum_refinement.tex:400 NEW_EXTENSION_PASS

$$\boxed{H_{\mathrm{eff}}(z) = \mu I + \frac{\kappa^2}{z - \mu}(2I + T + T^*).} \tag{A.1373}$$

F1460 35_adic_polyphase_quantum_refinement.tex:411 NEW_EXTENSION_PASS

$$H_{LL} = H_{DD} = \mu I, \qquad H_{LD} = \kappa(I + T^*), \qquad H_{DL} = \kappa(I + T). \tag{A.1374}$$

F1461 35_adic_polyphase_quantum_refinement.tex:419 NEW_EXTENSION_PASS

$$H_{LD}(z - H_{DD})^{-1} H_{DL} = \frac{\kappa^2}{z - \mu}(I + T^*)(I + T) \tag{A.1375}$$

$$= \frac{\kappa^2}{z - \mu}(2I + T + T^*), \tag{A.1376}$$

F1462 35_adic_polyphase_quantum_refinement.tex:432 NEW_EXTENSION_PASS

$$\text{analysis-unitarity residual} = 0, \tag{A.1377}$$

$$\text{digit-reversal residual} = 0, \tag{A.1378}$$

$$\text{root-of-unity selector residual} = 2.405483 \times 10^{-16}, \tag{A.1379}$$

$$\text{low-channel residual} = 0, \tag{A.1380}$$

$$\text{detail-projector residual} = 6.848430 \times 10^{-16}, \tag{A.1381}$$

$$\text{recursive Haar residual} = 1.405796 \times 10^{-15}, \tag{A.1382}$$

$$\text{operator reconstruction residual} = 0, \tag{A.1383}$$

$$\text{paraunitary residual} = 2.684315 \times 10^{-16}, \tag{A.1384}$$

$$\text{evolution residual} = 1.648375 \times 10^{-16}, \tag{A.1385}$$

$$\text{current residual} = 2.775558 \times 10^{-17}, \tag{A.1386}$$

$$\text{band block residual} = 1.267119 \times 10^{-14}, \tag{A.1387}$$

$$\text{spectrum residual} = 1.776357 \times 10^{-15}, \tag{A.1388}$$

$$\text{analytic band residual} = 9.992007 \times 10^{-16}, \tag{A.1389}$$

$$\text{hierarchical diagonalization residual} = 7.655814 \times 10^{-16}, \tag{A.1390}$$

$$\text{CPTP completeness residual} = 4.962036 \times 10^{-15}, \tag{A.1391}$$

$$\text{multi-prime evolution residual} = 4.990483 \times 10^{-16}, \tag{A.1392}$$

$$\text{dyadic sign preservation residual} = 2.872356 \times 10^{-16}, \tag{A.1393}$$

$$\text{polyphase Feshbach residual} = 1.733301 \times 10^{-16}, \tag{A.1394}$$

$$\text{dyadic decimation residual} = 1.387779 \times 10^{-17}. \tag{A.1395}$$

36_infinite_adic_haar_quantum_limit.tex

F1463 36_infinite_adic_haar_quantum_limit.tex:8 NEW_EXTENSION_PASS

$$\mathcal{H}_p^{\infty} := L^2(\mathbb{Z}_p, \mu_p; \mathbb{C}). \tag{A.1396}$$

F1464 36_infinite_adic_haar_quantum_limit.tex:12 NEW_EXTENSION_PASS

$$B_{h,a} := a + p^h \mathbb{Z}_p, \qquad 0 \le a < p^h. \tag{A.1397}$$

F1465 36_infinite_adic_haar_quantum_limit.tex:18 NEW_EXTENSION_PASS

$$\boxed{(\mathsf{E}_h f)(x) = p^h \int_{x+p^h\mathbb{Z}_p} f(y)\, \mathrm{d}\mu_p(y).} \tag{A.1398}$$

F1466 36_infinite_adic_haar_quantum_limit.tex:26 NEW_EXTENSION_PASS

$$\mu_p(x + p^h \mathbb{Z}_p) = p^{-h}, \tag{A.1399}$$

F1467 36_infinite_adic_haar_quantum_limit.tex:30 NEW_EXTENSION_PASS

$$\mathsf{E}_h \mathbf{1}_{B_{h,a}} = \mathbf{1}_{B_{h,a}}, \tag{A.1400}$$

F1468 36_infinite_adic_haar_quantum_limit.tex:37 NEW_EXTENSION_PASS

$$\mathcal{V}_h := \operatorname{Ran} \mathsf{E}_h \tag{A.1401}$$

F1469 36_infinite_adic_haar_quantum_limit.tex:41 NEW_EXTENSION_PASS

eq:infinite-adic-martingale-product

$$\mathsf{E}_h^* = \mathsf{E}_h, \qquad \mathsf{E}_h^2 = \mathsf{E}_h, \qquad \mathsf{E}_h \mathsf{E}_k = \mathsf{E}_{\min(h,k)}, \tag{A.1402}$$

$$\mathcal{V}_0 \subset \mathcal{V}_1 \subset \cdots, \qquad \overline{\bigcup_{h \ge 0} \mathcal{V}_h} = \mathcal{H}_p^{\infty}. \tag{A.1403}$$

F1470 36_infinite_adic_haar_quantum_limit.tex:52 NEW_EXTENSION_PASS

$$\mathcal{H}_{p,h} := \ell^2(\mathbb{Z}/p^h\mathbb{Z}) \tag{A.1404}$$

F1471 36_infinite_adic_haar_quantum_limit.tex:56 NEW_EXTENSION_PASS

$$\boxed{\mathcal{U}_{p,h} e_a = p^{h/2} \mathbf{1}_{B_{h,a}}.} \tag{A.1405}$$

F1472 36_infinite_adic_haar_quantum_limit.tex:65 NEW_EXTENSION_PASS

$$\left\langle \mathcal{U}_{p,h} e_a, \mathcal{U}_{p,h} e_b \right\rangle = p^h \mu_p(B_{h,a} \cap B_{h,b}) \tag{A.1406}$$

$$= \delta_{ab}. \tag{A.1407}$$

F1473 36_infinite_adic_haar_quantum_limit.tex:78 NEW_EXTENSION_PASS

$$V_{p,h} e_a = \frac{1}{\sqrt{p}} \sum_{d=0}^{p-1} e_{a+dp^h}. \tag{A.1408}$$

F1474 36_infinite_adic_haar_quantum_limit.tex:85 NEW_EXTENSION_PASS

$$B_{h,a} = \bigsqcup_{d=0}^{p-1} B_{h+1,a+dp^h} \tag{A.1409}$$

F1475 36_infinite_adic_haar_quantum_limit.tex:91 NEW_EXTENSION_PASS

$$\boxed{\mathcal{U}_{p,h+1} V_{p,h} = \mathcal{U}_{p,h}.} \tag{A.1410}$$

F1476 36_infinite_adic_haar_quantum_limit.tex:97 NEW_EXTENSION_PASS

$$\mathcal{U}_{p,h+1}V_{p,h}e_a = \frac{1}{\sqrt{p}}\sum_{d=0}^{p-1} p^{(h+1)/2}\mathbf{1}_{B_{h+1,a+dp^h}} \tag{A.1411}$$

$$= p^{h/2}\mathbf{1}_{\bigsqcup_d B_{h+1,a+dp^h}} \tag{A.1412}$$

$$= p^{h/2}\mathbf{1}_{B_{h,a}} = \mathcal{U}_{p,h}e_a. \tag{A.1413}$$

F1477 36_infinite_adic_haar_quantum_limit.tex:112 NEW_EXTENSION_PASS

$$\boxed{\varinjlim\left(\mathcal{H}_{p,0}\xrightarrow{V_{p,0}}\mathcal{H}_{p,1}\xrightarrow{V_{p,1}}\cdots\right)\cong L^2(\mathbb{Z}_p,\mu_p).} \tag{A.1414}$$

F1478 36_infinite_adic_haar_quantum_limit.tex:130 NEW_EXTENSION_PASS

$$V_{p,h\to H} := V_{p,H-1}\cdots V_{p,h}. \tag{A.1415}$$

F1479 36_infinite_adic_haar_quantum_limit.tex:135 NEW_EXTENSION_PASS

$$V_{p,h\to H}^* V_{p,h\to H} = I, \tag{A.1416}$$

$$V_{p,k\to H}V_{p,h\to k} = V_{p,h\to H}, \tag{A.1417}$$

$$\mathsf{E}_h\big|_{\mathcal{V}_H} = \mathcal{U}_{p,H}V_{p,h\to H}V_{p,h\to H}^*\mathcal{U}_{p,H}^*. \tag{A.1418}$$

F1480 36_infinite_adic_haar_quantum_limit.tex:148 NEW_EXTENSION_PASS

$$\mathsf{D}_0 := \mathsf{E}_0, \qquad \mathsf{D}_h := \mathsf{E}_h - \mathsf{E}_{h-1} \quad (h\geq 1). \tag{A.1419}$$

F1481 36_infinite_adic_haar_quantum_limit.tex:156 NEW_EXTENSION_PASS

$$\boxed{\mathsf{D}_h\mathsf{D}_k = \delta_{hk}\mathsf{D}_h.} \tag{A.1420}$$

F1482 36_infinite_adic_haar_quantum_limit.tex:164 NEW_EXTENSION_PASS

$$\mathsf{D}_h\mathsf{D}_k = (\mathsf{E}_h - \mathsf{E}_{h-1})(\mathsf{E}_k - \mathsf{E}_{k-1}) \tag{A.1421}$$

$$= \mathsf{E}_h - \mathsf{E}_h - \mathsf{E}_{h-1} + \mathsf{E}_{h-1} \tag{A.1422}$$

$$= 0. \tag{A.1423}$$

F1483 36_infinite_adic_haar_quantum_limit.tex:172 NEW_EXTENSION_PASS

$$(\mathsf{E}_h - \mathsf{E}_{h-1})^2 = \mathsf{E}_h - \mathsf{E}_{h-1}. \tag{A.1424}$$

F1484 36_infinite_adic_haar_quantum_limit.tex:178 NEW_EXTENSION_PASS

$$\sum_{h=0}^{H}\mathsf{D}_h = \mathsf{E}_H. \tag{A.1425}$$

F1485 36_infinite_adic_haar_quantum_limit.tex:182 NEW_EXTENSION_PASS

$$\boxed{I = \sum_{h=0}^{\infty}\mathsf{D}_h \qquad \text{in the strong operator topology.}} \tag{A.1426}$$

F1486 36_infinite_adic_haar_quantum_limit.tex:191 NEW_EXTENSION_PASS

$$\boxed{\|f\|_2^2 = \sum_{h=0}^{\infty}\|\mathsf{D}_h f\|_2^2.} \tag{A.1427}$$

F1487 36_infinite_adic_haar_quantum_limit.tex:199 NEW_EXTENSION_PASS

$$\|\mathsf{E}_H f\|_2^2 = \left\|\sum_{h=0}^{H}\mathsf{D}_h f\right\|_2^2 \tag{A.1428}$$

$$= \sum_{h=0}^{H} \|\mathsf{D}_h f\|_2^2 \tag{A.1429}$$

F1488 36_infinite_adic_haar_quantum_limit.tex:210 NEW_EXTENSION_PASS

$$\dim \operatorname{Ran} \mathsf{D}_0 = 1, \qquad \boxed{\dim \operatorname{Ran} \mathsf{D}_h = (p-1)p^{h-1} \quad (h \geq 1).} \tag{A.1430}$$

F1489 36_infinite_adic_haar_quantum_limit.tex:220 NEW_EXTENSION_PASS

$$\boxed{\psi_{h,a,q}(x) = p^{(h-1)/2} \sum_{d=0}^{p-1} \omega_p^{qd} \mathbf{1}_{B_{h,a+dp^{h-1}}}(x).} \tag{A.1431}$$

F1490 36_infinite_adic_haar_quantum_limit.tex:231 NEW_EXTENSION_PASS

$$\|\psi_{h,a,q}\|_2^2 = p^{h-1} \sum_{d=0}^{p-1} p^{-h} \tag{A.1432}$$

$$= 1. \tag{A.1433}$$

F1491 36_infinite_adic_haar_quantum_limit.tex:239 NEW_EXTENSION_PASS

$$\langle \psi_{h,a,q}, \psi_{h,a,q'} \rangle = \frac{1}{p} \sum_{d=0}^{p-1} \omega_p^{(q'-q)d} \tag{A.1434}$$

$$= \delta_{q,q'}. \tag{A.1435}$$

F1492 36_infinite_adic_haar_quantum_limit.tex:248 NEW_EXTENSION_PASS

$$\boxed{\mathsf{D}_h = \sum_{a=0}^{p^{h-1}-1} \sum_{q=1}^{p-1} |\psi_{h,a,q}\rangle\langle\psi_{h,a,q}| \quad (h \geq 1).} \tag{A.1436}$$

F1493 36_infinite_adic_haar_quantum_limit.tex:262 NEW_EXTENSION_PASS

$$\mathfrak{H}_p^s := \left\{ f \in L^2(\mathbb{Z}_p) : \sum_{h=0}^{\infty} p^{2sh} \|\mathsf{D}_h f\|_2^2 < \infty \right\} \tag{A.1437}$$

F1494 36_infinite_adic_haar_quantum_limit.tex:271 NEW_EXTENSION_PASS

$$\|f\|_{\mathfrak{H}_p^s}^2 := \sum_{h=0}^{\infty} p^{2sh} \|\mathsf{D}_h f\|_2^2. \tag{A.1438}$$

F1495 36_infinite_adic_haar_quantum_limit.tex:277 NEW_EXTENSION_PASS

$$\boxed{\|(I - \mathsf{E}_H) f\|_2^2 = \sum_{h=H+1}^{\infty} \|\mathsf{D}_h f\|_2^2.} \tag{A.1439}$$

F1496 36_infinite_adic_haar_quantum_limit.tex:285 NEW_EXTENSION_PASS

$$\|(I - \mathsf{E}_H) f\|_2^2 \leq p^{-2s(H+1)} \sum_{h=H+1}^{\infty} p^{2sh} \|\mathsf{D}_h f\|_2^2, \tag{A.1440}$$

F1497 36_infinite_adic_haar_quantum_limit.tex:293 NEW_EXTENSION_PASS

$$1 \leq p^{-2s(H+1)} p^{2sh}. \tag{A.1441}$$

F1498 36_infinite_adic_haar_quantum_limit.tex:297 NEW_EXTENSION_PASS

$$\|(I - \mathsf{E}_H) f\|_2^2 \leq p^{-2s(H+1)} \|f\|_{\mathfrak{H}_p^s}^2. \tag{A.1442}$$

F1499 36_infinite_adic_haar_quantum_limit.tex:304 NEW_EXTENSION_PASS

$$\boxed{\|(I-\mathsf{E}_H)f\|_2 \le p^{-s(H+1)}\|f\|_{\mathfrak{H}_p^s}.} \tag{A.1443}$$

F1500 36_infinite_adic_haar_quantum_limit.tex:315 NEW_EXTENSION_PASS

$$\boxed{H_{p,\epsilon} := \sum_{h=0}^{\infty} \epsilon_h \mathsf{D}_h} \tag{A.1444}$$

F1501 36_infinite_adic_haar_quantum_limit.tex:323 NEW_EXTENSION_PASS

$$\boxed{\mathcal{D}(H_{p,\epsilon}) = \left\{ f \in L^2(\mathbb{Z}_p) : \sum_{h=0}^{\infty} |\epsilon_h|^2 \|\mathsf{D}_h f\|_2^2 < \infty \right\}.} \tag{A.1445}$$

F1502 36_infinite_adic_haar_quantum_limit.tex:335 NEW_EXTENSION_PASS

$$f \longmapsto (\mathsf{D}_0 f, \mathsf{D}_1 f, \ldots), \tag{A.1446}$$

F1503 36_infinite_adic_haar_quantum_limit.tex:339 NEW_EXTENSION_PASS

$$(\xi_0, \xi_1, \ldots) \longmapsto (\epsilon_0 \xi_0, \epsilon_1 \xi_1, \ldots). \tag{A.1447}$$

F1504 36_infinite_adic_haar_quantum_limit.tex:346 NEW_EXTENSION_PASS

$$L^2(\mathbb{Z}_p) \cong \bigoplus_{h=0}^{\infty} \operatorname{Ran} \mathsf{D}_h \tag{A.1448}$$

F1505 36_infinite_adic_haar_quantum_limit.tex:352 NEW_EXTENSION_PASS

$$\sup_h |\epsilon_h| < \infty. \tag{A.1449}$$

F1506 36_infinite_adic_haar_quantum_limit.tex:356 NEW_EXTENSION_PASS

$$|\epsilon_h| \longrightarrow \infty, \tag{A.1450}$$

F1507 36_infinite_adic_haar_quantum_limit.tex:360 NEW_EXTENSION_PASS

$$\boxed{(H_{p,\epsilon} - z)^{-1} \text{ is compact for } z \notin \sigma(H_{p,\epsilon}).} \tag{A.1451}$$

F1508 36_infinite_adic_haar_quantum_limit.tex:367 NEW_EXTENSION_PASS

$$R_H(z) := \sum_{h=0}^{H} (\epsilon_h - z)^{-1} \mathsf{D}_h. \tag{A.1452}$$

F1509 36_infinite_adic_haar_quantum_limit.tex:373 NEW_EXTENSION_PASS

$$\|(H_{p,\epsilon} - z)^{-1} - R_H(z)\| = \sup_{h>H} \frac{1}{|\epsilon_h - z|} \tag{A.1453}$$

$$\longrightarrow 0. \tag{A.1454}$$

F1510 36_infinite_adic_haar_quantum_limit.tex:382 NEW_EXTENSION_PASS

$$\boxed{F(H_{p,\epsilon}) = \sum_{h=0}^{\infty} F(\epsilon_h) \mathsf{D}_h} \tag{A.1455}$$

F1511 36_infinite_adic_haar_quantum_limit.tex:390 NEW_EXTENSION_PASS

eq:infinite-adic-unitary eq:infinite-adic-resolvent

$$\mathrm{e}^{-\mathrm{i}tH_{p,\epsilon}} = \sum_{h=0}^{\infty} \mathrm{e}^{-\mathrm{i}t\epsilon_h} \mathsf{D}_h, \tag{A.1456}$$

$$(z - H_{p,\epsilon})^{-1} = \sum_{h=0}^{\infty} (z - \epsilon_h)^{-1} \mathsf{D}_h. \tag{A.1457}$$

F1512 36_infinite_adic_haar_quantum_limit.tex:401 NEW_EXTENSION_PASS

$$\boxed{\left\| \mathrm{e}^{-\mathrm{i}tH_{p,\epsilon}} f - \sum_{h=0}^{H} \mathrm{e}^{-\mathrm{i}t\epsilon_h} \mathsf{D}_h f \right\|_2 = \|(I - \mathsf{E}_H) f\|_2 \le p^{-s(H+1)} \|f\|_{\mathfrak{H}_p^s}.} \tag{A.1458}$$

F1513 36_infinite_adic_haar_quantum_limit.tex:416 NEW_EXTENSION_PASS

$$\left\| \sum_{h>H} \mathrm{e}^{-\mathrm{i}t\epsilon_h} \mathsf{D}_h f \right\|_2^2 = \sum_{h>H} \left| \mathrm{e}^{-\mathrm{i}t\epsilon_h} \right|^2 \|\mathsf{D}_h f\|_2^2 \tag{A.1459}$$

$$= \sum_{h>H} \|\mathsf{D}_h f\|_2^2 \tag{A.1460}$$

$$= \|(I - \mathsf{E}_H) f\|_2^2. \tag{A.1461}$$

F1514 36_infinite_adic_haar_quantum_limit.tex:432 NEW_EXTENSION_PASS

$$\boxed{\|(I - \mathsf{E}_H)(z - H_{p,\epsilon})^{-1}\| = \sup_{h>H} \frac{1}{|z - \epsilon_h|}.} \tag{A.1462}$$

F1515 36_infinite_adic_haar_quantum_limit.tex:445 NEW_EXTENSION_PASS

$$\boxed{\mathcal{H}_{\mathcal{P}}^{\infty} = \mathcal{K}_{\mathrm{tor}} \otimes \bigotimes_{p \in \mathcal{P}} L^2(\mathbb{Z}_p, \mu_p).} \tag{A.1463}$$

F1516 36_infinite_adic_haar_quantum_limit.tex:456 NEW_EXTENSION_PASS

$$\mathsf{E}_{\mathbf{h}} = I_{\mathcal{K}_{\mathrm{tor}}} \otimes \bigotimes_{p \in \mathcal{P}} \mathsf{E}_{p,h_p}. \tag{A.1464}$$

F1517 36_infinite_adic_haar_quantum_limit.tex:464 NEW_EXTENSION_PASS

$$\boxed{\mathsf{E}_{\mathbf{h}} \mathsf{E}_{\mathbf{k}} = \mathsf{E}_{\min(\mathbf{h},\mathbf{k})}} \tag{A.1465}$$

F1518 36_infinite_adic_haar_quantum_limit.tex:472 NEW_EXTENSION_PASS

$$\boxed{(V_p \otimes I)(I \otimes V_q) = (I \otimes V_q)(V_p \otimes I) = V_p \otimes V_q.} \tag{A.1466}$$

F1519 36_infinite_adic_haar_quantum_limit.tex:484 NEW_EXTENSION_PASS

$$\text{martingale-projector residual} = 2.314960 \times 10^{-15}, \tag{A.1467}$$

$$\text{direct-system residual} = 5.551115 \times 10^{-16}, \tag{A.1468}$$

$$\text{infinite-Haar finite-restriction residual} = 2.404663 \times 10^{-15}, \tag{A.1469}$$

$$\text{Parseval tail residual} = 2.775558 \times 10^{-17}, \tag{A.1470}$$

$$\text{Haar-Sobolev bound violation} = 0, \tag{A.1471}$$

$$\text{hierarchical spectral-calculus residual} = 3.352961 \times 10^{-14}, \tag{A.1472}$$

$$\text{cutoff-intertwining residual} = 3.607583 \times 10^{-14}, \tag{A.1473}$$

$$\text{resolvent-tail residual} = 7.771561 \times 10^{-16}, \tag{A.1474}$$

$$\text{multi-prime path residual} = 0, \tag{A.1475}$$

$$\text{dyadic-sign spectator residual} = 0. \tag{A.1476}$$

37_prime_polyphase_spectral_renormalization.tex

F1520 37_prime_polyphase_spectral_renormalization.tex:8 NEW_EXTENSION_PASS

$$\boxed{H_{\mu,\kappa} = \mu I + \kappa(S + S^*)} \tag{A.1477}$$

F1521 37_prime_polyphase_spectral_renormalization.tex:16 NEW_EXTENSION_PASS

$$C := \{0, p, 2p, \dots, (M-1)p\} \tag{A.1478}$$

F1522 37_prime_polyphase_spectral_renormalization.tex:20 NEW_EXTENSION_PASS

$$\mathcal{S}_p(z) := P(z - H)P - PHQ(z - QHQ)^{-1}QHP. \tag{A.1479}$$

F1523 37_prime_polyphase_spectral_renormalization.tex:31 NEW_EXTENSION_PASS

$$x := \frac{z - \mu}{\kappa}. \tag{A.1480}$$

F1524 37_prime_polyphase_spectral_renormalization.tex:35 NEW_EXTENSION_PASS

$$T_n(\cos\theta) = \cos(n\theta), \tag{A.1481}$$

$$U_n(\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta}. \tag{A.1482}$$

F1525 37_prime_polyphase_spectral_renormalization.tex:40 NEW_EXTENSION_PASS

$$\boxed{R_p(x) := 2T_p(x/2).} \tag{A.1483}$$

F1526 37_prime_polyphase_spectral_renormalization.tex:47 NEW_EXTENSION_PASS

eq:prime-polyphase-schur

$$\boxed{\mathcal{S}_p(z) = \frac{\kappa}{U_{p-1}(x/2)}\left[R_p(x)I - (T + T^*)\right].} \tag{A.1484}$$

F1527 37_prime_polyphase_spectral_renormalization.tex:59 NEW_EXTENSION_PASS

$$\mathcal{S}_p(z) = zI - H_{\mathrm{eff}}^{(p)}(z) \coloneqq (z - \mu_p(z))I - \kappa_p(z)(T + T^*), \tag{A.1485}$$

F1528 37_prime_polyphase_spectral_renormalization.tex:66 NEW_EXTENSION_PASS

eq:prime-polyphase-hopping eq:prime-polyphase-center

$$\boxed{\kappa_p(z) = \frac{\kappa}{U_{p-1}(x/2)},} \tag{A.1486}$$

$$\boxed{\mu_p(z) = z - \kappa\frac{R_p(x)}{U_{p-1}(x/2)}.} \tag{A.1487}$$

F1529 37_prime_polyphase_spectral_renormalization.tex:82 NEW_EXTENSION_PASS

$$\boxed{\frac{z - \mu_p(z)}{\kappa_p(z)} = R_p(x).} \tag{A.1488}$$

F1530 37_prime_polyphase_spectral_renormalization.tex:90 NEW_EXTENSION_PASS

$$\psi_{n+1} = x\psi_n - \psi_{n-1} \tag{A.1489}$$

F1531 37_prime_polyphase_spectral_renormalization.tex:94 NEW_EXTENSION_PASS

$$A(x) = \begin{pmatrix} x & -1 \\ 1 & 0 \end{pmatrix}. \tag{A.1490}$$

F1532 37_prime_polyphase_spectral_renormalization.tex:103 NEW_EXTENSION_PASS

$$A(x)^p = \begin{pmatrix} U_p(x/2) & -U_{p-1}(x/2) \\ U_{p-1}(x/2) & -U_{p-2}(x/2) \end{pmatrix}. \tag{A.1491}$$

F1533 37_prime_polyphase_spectral_renormalization.tex:112 NEW_EXTENSION_PASS

$$U_{n+1}(x/2) = xU_n(x/2) - U_{n-1}(x/2), \tag{A.1492}$$

F1534 37_prime_polyphase_spectral_renormalization.tex:117 NEW_EXTENSION_PASS

$$\operatorname{Tr} A(x)^p = R_p(x), \qquad \det A(x)^p = 1. \tag{A.1493}$$

F1535 37_prime_polyphase_spectral_renormalization.tex:123 NEW_EXTENSION_PASS

$$(A^p)^2 - R_p(x)A^p + I = 0. \tag{A.1494}$$

F1536 37_prime_polyphase_spectral_renormalization.tex:127 NEW_EXTENSION_PASS

$$\chi_{j+1} - R_p(x)\chi_j + \chi_{j-1} = 0. \tag{A.1495}$$

F1537 37_prime_polyphase_spectral_renormalization.tex:132 NEW_EXTENSION_PASS

$$\frac{\kappa}{U_{p-1}(x/2)} \left[R_p(x)\psi_{jp} - \psi_{(j-1)p} - \psi_{(j+1)p}\right] = 0, \tag{A.1496}$$

F1538 37_prime_polyphase_spectral_renormalization.tex:145 NEW_EXTENSION_PASS

$$R_2(x) = x^2 - 2, \qquad U_1(x/2) = x, \tag{A.1497}$$

F1539 37_prime_polyphase_spectral_renormalization.tex:151 NEW_EXTENSION_PASS

$$\boxed{H_{\text{eff}}^{(2)}(z) = \mu I + \frac{\kappa^2}{z-\mu}(2I + T + T^*),} \tag{A.1498}$$

F1540 37_prime_polyphase_spectral_renormalization.tex:162 NEW_EXTENSION_PASS

$$\kappa_2(z) = \frac{\kappa}{x} = \frac{\kappa^2}{z-\mu}, \tag{A.1499}$$

$$\mu_2(z) = z - \kappa\frac{x^2-2}{x} \tag{A.1500}$$

$$= \mu + \frac{2\kappa^2}{z-\mu}. \tag{A.1501}$$

F1541 37_prime_polyphase_spectral_renormalization.tex:179 NEW_EXTENSION_PASS

$$M \in SL_2(\mathbb{C}), \qquad x := \operatorname{Tr} M. \tag{A.1502}$$

F1542 37_prime_polyphase_spectral_renormalization.tex:185 NEW_EXTENSION_PASS

$$M^2 - xM + I = 0. \tag{A.1503}$$

F1543 37_prime_polyphase_spectral_renormalization.tex:189 NEW_EXTENSION_PASS

$$M^n = xM^{n-1} - M^{n-2}. \tag{A.1504}$$

F1544 37_prime_polyphase_spectral_renormalization.tex:193 NEW_EXTENSION_PASS

eq:sl2-chebyshev-power

$$\boxed{M^n = U_{n-1}(x/2)M - U_{n-2}(x/2)I, \qquad n \geq 1,} \tag{A.1505}$$

F1545 37_prime_polyphase_spectral_renormalization.tex:203 NEW_EXTENSION_PASS

$$M^{n+1} = xM^n - M^{n-1} \tag{A.1506}$$

$$= \left[xU_{n-1}(x/2) - U_{n-2}(x/2)\right] M \tag{A.1507}$$

$$- \left[xU_{n-2}(x/2) - U_{n-3}(x/2)\right] I \tag{A.1508}$$

$$= U_n(x/2)M - U_{n-1}(x/2)I. \tag{A.1509}$$

F1546 37_prime_polyphase_spectral_renormalization.tex:212 NEW_EXTENSION_PASS

$$\operatorname{Tr}(M^n) = xU_{n-1}(x/2) - 2U_{n-2}(x/2) \tag{A.1510}$$

$$= 2T_n(x/2). \tag{A.1511}$$

F1547 37_prime_polyphase_spectral_renormalization.tex:218 NEW_EXTENSION_PASS

eq:trace-coordinate-renormalization

$$\boxed{\operatorname{Tr}(M^n) = R_n(\operatorname{Tr} M).} \tag{A.1512}$$

F1548 37_prime_polyphase_spectral_renormalization.tex:226 NEW_EXTENSION_PASS

eq:sl2-state-polyphase

$$\boxed{w_{pm+j} = M^j (M^p)^m w_0, \qquad 0 \le j < p.} \tag{A.1513}$$

F1549 37_prime_polyphase_spectral_renormalization.tex:234 NEW_EXTENSION_PASS

$$x_p := \operatorname{Tr}(M^p) = R_p(x). \tag{A.1514}$$

F1550 37_prime_polyphase_spectral_renormalization.tex:238 NEW_EXTENSION_PASS

$$\operatorname{Tr}((M^p)^q) = R_q(R_p(x)) = R_{pq}(x). \tag{A.1515}$$

F1551 37_prime_polyphase_spectral_renormalization.tex:249 NEW_EXTENSION_PASS

$$U_{p-1}(x/2) = 0. \tag{A.1516}$$

F1552 37_prime_polyphase_spectral_renormalization.tex:253 NEW_EXTENSION_PASS

$$\boxed{x_j = 2\cos\left(\frac{j\pi}{p}\right), \qquad j = 1, \dots, p-1.} \tag{A.1517}$$

F1553 37_prime_polyphase_spectral_renormalization.tex:261 NEW_EXTENSION_PASS

$$U_{p-1}(x/2) = \frac{\sin(p\theta)}{\sin\theta}. \tag{A.1518}$$

F1554 37_prime_polyphase_spectral_renormalization.tex:268 NEW_EXTENSION_PASS

$$\boxed{z_j = \mu + 2\kappa\cos\left(\frac{j\pi}{p}\right).} \tag{A.1519}$$

F1555 37_prime_polyphase_spectral_renormalization.tex:276 NEW_EXTENSION_PASS

$$\boxed{\det(z - H) = \det(z - QHQ)\det \mathcal{S}_p(z).} \tag{A.1520}$$

F1556 37_prime_polyphase_spectral_renormalization.tex:287 NEW_EXTENSION_PASS

$$\boxed{x_{p,q}(\theta) = 2\cos\left(\frac{\theta + 2\pi q}{p}\right), \qquad q = 0, \dots, p-1.} \tag{A.1521}$$

F1557 37_prime_polyphase_spectral_renormalization.tex:297 NEW_EXTENSION_PASS

$$\boxed{R_p\big(x_{p,q}(\theta)\big) = 2\cos\theta \quad \text{for every } q.} \tag{A.1522}$$

F1558 37_prime_polyphase_spectral_renormalization.tex:306 NEW_EXTENSION_PASS

$$R_p\big(x_{p,q}(\theta)\big) = 2T_p\left(\cos\frac{\theta + 2\pi q}{p}\right) \tag{A.1523}$$

$$= 2\cos(\theta + 2\pi q) \tag{A.1524}$$

$$= 2\cos\theta. \tag{A.1525}$$

F1559 37_prime_polyphase_spectral_renormalization.tex:319 NEW_EXTENSION_PASS

$$x_k = 2\cos\left(\frac{2\pi k}{pM}\right), \tag{A.1526}$$

F1560 37_prime_polyphase_spectral_renormalization.tex:323 NEW_EXTENSION_PASS

$$\boxed{R_p(x_k) = 2\cos\left(\frac{2\pi k}{M}\right).} \tag{A.1527}$$

F1561 37_prime_polyphase_spectral_renormalization.tex:334 NEW_EXTENSION_PASS

eq:chebyshev-semigroup

$$\boxed{R_m \circ R_n = R_{mn} = R_n \circ R_m.} \tag{A.1528}$$

F1562 37_prime_polyphase_spectral_renormalization.tex:343 NEW_EXTENSION_PASS

$$\begin{aligned} R_m(R_n(x)) &= 2T_m(T_n(\cos\theta)) && \text{(A.1529)}\\ &= 2\cos(mn\theta) && \text{(A.1530)}\\ &= R_{mn}(x). && \text{(A.1531)} \end{aligned}$$

F1563 37_prime_polyphase_spectral_renormalization.tex:353 NEW_EXTENSION_PASS

eq:chebyshev-second-kind-semigroup

$$\boxed{U_{mn-1}(t) = U_{m-1}\big(T_n(t)\big)U_{n-1}(t).} \tag{A.1532}$$

F1564 37_prime_polyphase_spectral_renormalization.tex:362 NEW_EXTENSION_PASS

$$\begin{aligned} U_{m-1}(T_n(t))U_{n-1}(t) &= \frac{\sin(mn\theta)}{\sin(n\theta)}\frac{\sin(n\theta)}{\sin\theta} && \text{(A.1533)}\\ &= \frac{\sin(mn\theta)}{\sin\theta} && \text{(A.1534)}\\ &= U_{mn-1}(t). && \text{(A.1535)} \end{aligned}$$

F1565 37_prime_polyphase_spectral_renormalization.tex:375 NEW_EXTENSION_PASS

eq:iterated-prime-R eq:iterated-prime-kappa eq:iterated-prime-mu

$$\boxed{R_p^{\circ h}(x) = R_q(x) = 2T_q(x/2),} \tag{A.1536}$$

$$\boxed{\kappa^{(h)}(z) = \frac{\kappa}{U_{q-1}(x/2)},} \tag{A.1537}$$

$$\boxed{\mu^{(h)}(z) = z - \kappa\frac{R_q(x)}{U_{q-1}(x/2)}.} \tag{A.1538}$$

F1566 37_prime_polyphase_spectral_renormalization.tex:397 NEW_EXTENSION_PASS

$$\boxed{R_pR_q = R_qR_p = R_{pq}.} \tag{A.1539}$$

F1567 37_prime_polyphase_spectral_renormalization.tex:408 NEW_EXTENSION_PASS

$$\boxed{\mathrm{d}\nu(x) = \frac{\mathrm{d}x}{\pi\sqrt{4-x^2}}.} \tag{A.1540}$$

F1568 37_prime_polyphase_spectral_renormalization.tex:416 NEW_EXTENSION_PASS

$$x = 2\cos\theta, \qquad 0 \le \theta \le \pi, \tag{A.1541}$$

F1569 37_prime_polyphase_spectral_renormalization.tex:422 NEW_EXTENSION_PASS

$$R_p(2\cos\theta) = 2\cos(p\theta). \tag{A.1542}$$

F1570 37_prime_polyphase_spectral_renormalization.tex:426 NEW_EXTENSION_PASS

$$\boxed{(R_p)_*\nu = \nu.} \tag{A.1543}$$

F1571 37_prime_polyphase_spectral_renormalization.tex:432 NEW_EXTENSION_PASS

$$\begin{aligned} \int f(R_p(x))\ \mathrm{d}\nu(x) &= \frac{1}{2\pi}\int_0^{2\pi} f(2\cos(p\theta))\ \mathrm{d}\theta && \text{(A.1544)}\\ &= \frac{1}{2\pi p}\int_0^{2\pi p} f(2\cos u)\ \mathrm{d}u && \text{(A.1545)} \end{aligned}$$

$$= \frac{1}{2\pi}\int_0^{2\pi} f(2\cos u)\ \mathrm{d}u \tag{A.1546}$$

$$= \int f(x)\ \mathrm{d}\nu(x). \tag{A.1547}$$

F1572 37_prime_polyphase_spectral_renormalization.tex:451 NEW_EXTENSION_PASS

$$\phi_0(x) = 1, \qquad \phi_n(x) = \sqrt{2}\,T_n(x/2) \quad (n \geq 1). \tag{A.1548}$$

F1573 37_prime_polyphase_spectral_renormalization.tex:458 NEW_EXTENSION_PASS

$$\mathcal{K}_p f = f \circ R_p \tag{A.1549}$$

F1574 37_prime_polyphase_spectral_renormalization.tex:462 NEW_EXTENSION_PASS

$$\boxed{\mathcal{K}_p \phi_n = \phi_{pn}.} \tag{A.1550}$$

F1575 37_prime_polyphase_spectral_renormalization.tex:468 NEW_EXTENSION_PASS

$$\boxed{\mathcal{K}_p^* \phi_m = \begin{cases} \phi_{m/p}, & p \mid m, \\ 0, & p \nmid m, \end{cases}} \tag{A.1551}$$

F1576 37_prime_polyphase_spectral_renormalization.tex:479 NEW_EXTENSION_PASS

$$\langle \phi_n, \mathcal{K}_p^* \phi_m \rangle = \langle \mathcal{K}_p \phi_n, \phi_m \rangle \tag{A.1552}$$

$$= \langle \phi_{pn}, \phi_m \rangle \tag{A.1553}$$

$$= \delta_{pn,m}. \tag{A.1554}$$

F1577 37_prime_polyphase_spectral_renormalization.tex:492 NEW_EXTENSION_PASS

$$\boxed{\int_{-2}^{2} \log |R_p'(x)|\ \mathrm{d}\nu(x) = \log p.} \tag{A.1555}$$

F1578 37_prime_polyphase_spectral_renormalization.tex:499 NEW_EXTENSION_PASS

$$R_p'(2\cos\theta) = p\frac{\sin(p\theta)}{\sin\theta}. \tag{A.1556}$$

F1579 37_prime_polyphase_spectral_renormalization.tex:504 NEW_EXTENSION_PASS

$$\int_{-2}^{2} \log |R_p'(x)|\ \mathrm{d}\nu(x) = \log p + \frac{1}{\pi}\int_0^{\pi} \log|\sin(p\theta)|\ \mathrm{d}\theta \tag{A.1557}$$

$$- \frac{1}{\pi}\int_0^{\pi} \log(\sin\theta)\ \mathrm{d}\theta. \tag{A.1558}$$

F1580 37_prime_polyphase_spectral_renormalization.tex:513 NEW_EXTENSION_PASS

$$\int_0^{\pi} \log|\sin(p\theta)|\ \mathrm{d}\theta = \int_0^{\pi} \log(\sin\theta)\ \mathrm{d}\theta, \tag{A.1559}$$

F1581 37_prime_polyphase_spectral_renormalization.tex:521 NEW_EXTENSION_PASS

$$\text{arbitrary-prime Schur residual} = 4.594889 \times 10^{-16}, \tag{A.1560}$$

$$\text{normalized Chebyshev-map residual} = 1.016846 \times 10^{-12}, \tag{A.1561}$$

$$\text{dyadic-specialization residual} = 2.177752 \times 10^{-16}, \tag{A.1562}$$

$$\text{determinant-factorization residual} = 2.580429 \times 10^{-14}, \tag{A.1563}$$

$$\text{Dirichlet-pole residual} = 6.661338 \times 10^{-16}, \tag{A.1564}$$

$$\text{semigroup symbolic residual} = 0, \tag{A.1565}$$

$$\text{second-kind composition residual} = 0, \tag{A.1566}$$

$$\text{iterated/direct flow residual} = 1.111610 \times 10^{-15}, \tag{A.1567}$$

$$\text{band-decimation residual} = 1.515454 \times 10^{-14}, \tag{A.1568}$$
$$\text{coarse-spectrum residual} = 1.243450 \times 10^{-14}, \tag{A.1569}$$
$$\text{arcsine fixed-moment residual} = 1.184238 \times 10^{-16}, \tag{A.1570}$$
$$\text{Koopman mode residual} = 3.357314 \times 10^{-13}, \tag{A.1571}$$
$$\text{multi-prime commutation residual} = 3.286260 \times 10^{-14}, \tag{A.1572}$$
$$\text{general } SL_2 \text{ power-coordinate residual} = 1.501782 \times 10^{-13}, \tag{A.1573}$$
$$\text{companion-state polyphase residual} = 7.770726 \times 10^{-12}, \tag{A.1574}$$
$$\text{trace-semigroup residual} = 2.316699 \times 10^{-13}. \tag{A.1575}$$

09_countable_unbounded.tex

F1582 09_countable_unbounded.tex:8 NEW_EXTENSION_PASS

$$S_F := \sum_{\alpha \in F} P_\alpha, \tag{A.1576}$$
$$\|\varphi\|_H^2 := \|\varphi\|^2 + \|H\varphi\|^2 . \tag{A.1577}$$

F1583 09_countable_unbounded.tex:13 NEW_EXTENSION_PASS

$$P_\alpha P_\beta = \delta_{\alpha\beta} P_\alpha, \tag{A.1578}$$
$$\sum_{\alpha \in A} P_\alpha = I \quad \text{strongly}, \tag{A.1579}$$
$$P_\alpha D(H) \subseteq D(H), \tag{A.1580}$$
$$S_F \psi \to \psi \quad \text{in } \|\cdot\|_H , \qquad \psi \in D(H). \tag{A.1581}$$

F1584 09_countable_unbounded.tex:21 NEW_EXTENSION_PASS

$$\|S_F\psi - \psi\|_H^2 = \|S_F\psi - \psi\|^2 + \|H(S_F\psi - \psi)\|^2 \longrightarrow 0, \tag{A.1582}$$
$$S_F\psi \longrightarrow \psi, \tag{A.1583}$$
$$HS_F\psi \longrightarrow H\psi. \tag{A.1584}$$

F1585 09_countable_unbounded.tex:33 NEW_EXTENSION_PASS

$$H\phi_n = \lambda_n \phi_n, \qquad \lambda_n = n + \frac{1}{2}. \tag{A.1585}$$

F1586 09_countable_unbounded.tex:39 NEW_EXTENSION_PASS

$$\boxed{f \in D(H) \Longleftrightarrow \sum_{n=0}^{\infty} \lambda_n^2 |c_n|^2 < \infty,} \tag{A.1586}$$

F1587 09_countable_unbounded.tex:47 NEW_EXTENSION_PASS

$$\boxed{f \in D(H^{1/2}) \Longleftrightarrow \sum_{n=0}^{\infty} \lambda_n |c_n|^2 < \infty.} \tag{A.1587}$$

F1588 09_countable_unbounded.tex:55 NEW_EXTENSION_PASS

$$\|f - S_M f\|_H^2 = \sum_{n=M}^{\infty} (1 + \lambda_n^2)|c_n|^2 \tag{A.1588}$$
$$\longrightarrow 0 \tag{A.1589}$$

F1589 09_countable_unbounded.tex:64 NEW_EXTENSION_PASS

$$R_{\mathrm{i}\eta} := (H - \mathrm{i}\eta)^{-1}, \tag{A.1590}$$

$$F_H := H(1 + H^2)^{-1/2}, \tag{A.1591}$$

$$U_H := (H - \mathrm{i})(H + \mathrm{i})^{-1}. \tag{A.1592}$$

F1590 09_countable_unbounded.tex:70 NEW_EXTENSION_PASS

$$\|R_{\mathrm{i}\eta} - S_M R_{\mathrm{i}\eta} S_M\| = \sup_{n \geq M} \frac{1}{\sqrt{\lambda_n^2 + \eta^2}} \longrightarrow 0, \tag{A.1593}$$

$$\|\mathrm{e}^{-tH} - S_M \mathrm{e}^{-tH} S_M\| = \mathrm{e}^{-t\lambda_M} \longrightarrow 0. \tag{A.1594}$$

F1591 09_countable_unbounded.tex:81 NEW_EXTENSION_PASS

$$(U(t) - S_M U(t) S_M)\phi_n = \mathrm{e}^{-\mathrm{i}t\lambda_n}\phi_n, \qquad n \geq M, \tag{A.1595}$$

F1592 09_countable_unbounded.tex:87 NEW_EXTENSION_PASS

$$\boxed{\|U(t) - S_M U(t) S_M\| = 1} \tag{A.1596}$$

F1593 09_countable_unbounded.tex:95 NEW_EXTENSION_PASS

$$F_H^{(M)} := S_M F_H S_M + (I - S_M), \tag{A.1597}$$

$$U_H^{(M)} := S_M U_H S_M + (I - S_M). \tag{A.1598}$$

F1594 09_countable_unbounded.tex:100 NEW_EXTENSION_PASS

$$\|F_H - F_H^{(M)}\| = \sup_{n \geq M} \left| \frac{\lambda_n}{\sqrt{1 + \lambda_n^2}} - 1 \right| \to 0, \tag{A.1599}$$

$$\|U_H - U_H^{(M)}\| = \sup_{n \geq M} \frac{2}{|\lambda_n + \mathrm{i}|} \to 0. \tag{A.1600}$$

F1595 09_countable_unbounded.tex:117 NEW_EXTENSION_PASS

$$\psi(t) := \mathrm{e}^{-\mathrm{i}tH}\psi_0, \qquad \psi_0 \in D(H), \tag{A.1601}$$

$$E_\alpha := \|P_\alpha \psi\|^2, \tag{A.1602}$$

$$J_{\alpha \leftarrow \beta} := 2\,\mathrm{Im}\left\langle P_\alpha \psi, H P_\beta \psi \right\rangle. \tag{A.1603}$$

F1596 09_countable_unbounded.tex:125 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}E_\alpha}{\mathrm{d}t} = 2\,\mathrm{Re}\left\langle P_\alpha \psi, P_\alpha \dot{\psi} \right\rangle \tag{A.1604}$$

$$= 2\,\mathrm{Im}\left\langle P_\alpha \psi, H\psi \right\rangle. \tag{A.1605}$$

F1597 09_countable_unbounded.tex:133 NEW_EXTENSION_PASS

$$\langle P_\alpha \psi, H\psi \rangle = \lim_{F \nearrow A} \langle P_\alpha \psi, H S_F \psi \rangle \tag{A.1606}$$

$$= \lim_{F \nearrow A} \sum_{\beta \in F} \left\langle P_\alpha \psi, H P_\beta \psi \right\rangle. \tag{A.1607}$$

F1598 09_countable_unbounded.tex:142 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}E_\alpha}{\mathrm{d}t} = \lim_{F \nearrow A} \sum_{\beta \in F} J_{\alpha \leftarrow \beta}. \tag{A.1608}$$

F1599 09_countable_unbounded.tex:147 NEW_EXTENSION_PASS

$$\left\langle P_\beta \psi, H P_\alpha \psi \right\rangle = \overline{\left\langle P_\alpha \psi, H P_\beta \psi \right\rangle}, \tag{A.1609}$$

$$J_{\alpha\leftarrow\beta} = -J_{\beta\leftarrow\alpha}. \tag{A.1610}$$

F1600 09_countable_unbounded.tex:154 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_{\alpha\in B} E_\alpha = \lim_{F\nearrow A}\left[\sum_{\alpha,\beta\in B} J_{\alpha\leftarrow\beta} + \sum_{\substack{\alpha\in B\\ \beta\in F\setminus B}} J_{\alpha\leftarrow\beta}\right]. \tag{A.1611}$$

F1601 09_countable_unbounded.tex:164 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_{\alpha\in B} E_\alpha = \lim_{F\nearrow A}\sum_{\substack{\alpha\in B\\ \beta\in F\setminus B}} J_{\alpha\leftarrow\beta}. \tag{A.1612}$$

F1602 09_countable_unbounded.tex:171 NEW_EXTENSION_PASS

$$\sum_\alpha E_\alpha(t) = \lim_{F\nearrow A}\sum_{\alpha\in F}\|P_\alpha\psi(t)\|^2 \tag{A.1613}$$

$$= \lim_{F\nearrow A}\|S_F\psi(t)\|^2 \tag{A.1614}$$

$$= \|\psi(t)\|^2. \tag{A.1615}$$

F1603 09_countable_unbounded.tex:179 NEW_EXTENSION_PASS

$$\sum_\alpha E_\alpha(t) = \|\psi(t)\|^2 = \|\psi_0\|^2. \tag{A.1616}$$

F1604 09_countable_unbounded.tex:184 NEW_EXTENSION_PASS

$$Q_A := \sum_{\alpha\in C_A} P_\alpha \quad \text{strongly}. \tag{A.1617}$$

F1605 09_countable_unbounded.tex:188 NEW_EXTENSION_PASS

$$\|Q_A\psi\|^2 = \lim_{F\nearrow C_A}\left\|\sum_{\alpha\in F} P_\alpha\psi\right\|^2 \tag{A.1618}$$

$$= \sum_{\alpha\in C_A}\|P_\alpha\psi\|^2. \tag{A.1619}$$

F1606 09_countable_unbounded.tex:195 NEW_EXTENSION_PASS

$$J_{A\leftarrow B} := 2\,\mathrm{Im}\,\langle Q_A\psi, HQ_B\psi\rangle \tag{A.1620}$$

$$= 2\,\mathrm{Im}\sum_{\alpha\in C_A}\sum_{\beta\in C_B}\left\langle P_\alpha\psi, HP_\beta\psi\right\rangle \tag{A.1621}$$

$$= \sum_{\alpha\in C_A}\sum_{\beta\in C_B} J_{\alpha\leftarrow\beta}. \tag{A.1622}$$

F1607 09_countable_unbounded.tex:208 NEW_EXTENSION_PASS

$$Q_r^{(M)} := \frac{1}{M}\sum_{k=0}^{M-1}\exp(-2\pi\mathrm{i}rk/M)\exp(2\pi\mathrm{i}k\widehat{N}/M). \tag{A.1623}$$

F1608 09_countable_unbounded.tex:216 NEW_EXTENSION_PASS

$$\boxed{Q_r^{(M)} = \sum_{n\equiv r\ (\mathrm{mod}\ M)} P_n.} \tag{A.1624}$$

F1609 09_countable_unbounded.tex:224 NEW_EXTENSION_PASS

$$a\phi_n = \sqrt{n}\,\phi_{n-1}, \qquad a^\dagger \phi_n = \sqrt{n+1}\,\phi_{n+1}, \tag{A.1625}$$

F1610 09_countable_unbounded.tex:230 NEW_EXTENSION_PASS

$$\boxed{Q_r^{(M)} a Q_s^{(M)} = 0 \quad \text{unless} \quad r \equiv s-1 \pmod{M},} \tag{A.1626}$$

F1611 09_countable_unbounded.tex:237 NEW_EXTENSION_PASS

$$\boxed{Q_r^{(M)} a^\dagger Q_s^{(M)} = 0 \quad \text{unless} \quad r \equiv s+1 \pmod{M}.} \tag{A.1627}$$

F1612 09_countable_unbounded.tex:247 NEW_EXTENSION_PASS

$$[R_z, P] = -(H-z)^{-1}[H-z, P](H-z)^{-1} \tag{A.1628}$$
$$= -R_z[H,P]R_z. \tag{A.1629}$$

F1613 09_countable_unbounded.tex:253 NEW_EXTENSION_PASS

$$\boxed{\|[R_z,P]\| \le \|R_z\|^2 \|[H,P]\|.} \tag{A.1630}$$

F1614 09_countable_unbounded.tex:260 NEW_EXTENSION_PASS

$$\boxed{\|[R_{i\eta},P]\| \le \frac{\|[H,P]\|}{\eta^2}.} \tag{A.1631}$$

F1615 09_countable_unbounded.tex:270 NEW_EXTENSION_PASS

$$X \in \mathcal{B}(\mathcal{H}) \Longrightarrow S_N X S_N \to X \quad \text{strongly}, \tag{A.1632}$$
$$X \in \mathcal{K}(\mathcal{H}) \Longrightarrow \|S_N X S_N - X\| \to 0, \tag{A.1633}$$
$$X \in \mathcal{S}_2(\mathcal{H}) \Longrightarrow \|S_N X S_N - X\|_{\mathcal{S}_2} \to 0. \tag{A.1634}$$

F1616 09_countable_unbounded.tex:282 NEW_EXTENSION_PASS

$$S_N X S_N - X = S_N X (S_N - I) + (S_N - I) X. \tag{A.1635}$$

F1617 09_countable_unbounded.tex:287 NEW_EXTENSION_PASS

$$\|(S_N X S_N - X)\psi\| \le \|X\|\,\|(S_N - I)\psi\| + \|(S_N - I)X\psi\| \tag{A.1636}$$
$$\longrightarrow 0, \tag{A.1637}$$

F1618 09_countable_unbounded.tex:298 NEW_EXTENSION_PASS

$$\|S_N X S_N - X\| \le 2\,\|X - K\| + \|S_N K S_N - K\| \tag{A.1638}$$
$$\longrightarrow 2\varepsilon. \tag{A.1639}$$

F1619 09_countable_unbounded.tex:306 NEW_EXTENSION_PASS

$$\|(S_N - I)X\|_{\mathcal{S}_2} \longrightarrow 0, \qquad \|X(S_N - I)\|_{\mathcal{S}_2} \longrightarrow 0. \tag{A.1640}$$

F1620 09_countable_unbounded.tex:314 NEW_EXTENSION_PASS

$$S_N H S_N \psi - H\psi = S_N(HS_N\psi - H\psi) + (S_N - I)H\psi \tag{A.1641}$$
$$\longrightarrow 0, \qquad \psi \in D(H), \tag{A.1642}$$

F1621 09_countable_unbounded.tex:325 NEW_EXTENSION_PASS

$$7.879615 \times 10^{-16}. \tag{A.1643}$$

F1622 09_countable_unbounded.tex:329 NEW_EXTENSION_PASS

$$1.998730. \tag{A.1644}$$

41_hypergeometric_companion_packet_calculus.tex

F1623 41_hypergeometric_companion_packet_calculus.tex:8 NEW_EXTENSION_PASS

$$\omega_N := \exp(2\pi \mathrm{i}/N). \tag{A.1645}$$

F1624 41_hypergeometric_companion_packet_calculus.tex:12 NEW_EXTENSION_PASS

eq:analytic-residue-projector

$$\boxed{(\mathfrak{R}_{N,j}F)(z) := \frac{1}{N}\sum_{\ell=0}^{N-1} \omega_N^{-j\ell} F(\omega_N^{\ell} z), \qquad 0 \le j < N.} \tag{A.1646}$$

F1625 41_hypergeometric_companion_packet_calculus.tex:23 NEW_EXTENSION_PASS

$$F(z) = \sum_{n=0}^{\infty} a_n z^n. \tag{A.1647}$$

F1626 41_hypergeometric_companion_packet_calculus.tex:27 NEW_EXTENSION_PASS

$$(\mathfrak{R}_{N,j}F)(z) = \frac{1}{N}\sum_{\ell=0}^{N-1} \omega_N^{-j\ell} \sum_{n=0}^{\infty} a_n \omega_N^{\ell n} z^n \tag{A.1648}$$

$$= \sum_{n=0}^{\infty} a_n z^n \left(\frac{1}{N}\sum_{\ell=0}^{N-1} \omega_N^{\ell(n-j)} \right). \tag{A.1649}$$

F1627 41_hypergeometric_companion_packet_calculus.tex:41 NEW_EXTENSION_PASS

$$\frac{1}{N}\sum_{\ell=0}^{N-1} \omega_N^{\ell(n-j)} = \begin{cases} 1, & n \equiv j \pmod N, \\ 0, & n \not\equiv j \pmod N. \end{cases} \tag{A.1650}$$

F1628 41_hypergeometric_companion_packet_calculus.tex:51 NEW_EXTENSION_PASS

eq:analytic-residue-lane

$$\boxed{(\mathfrak{R}_{N,j}F)(z) = \sum_{q=0}^{\infty} a_{Nq+j} z^{Nq+j}.} \tag{A.1651}$$

F1629 41_hypergeometric_companion_packet_calculus.tex:60 NEW_EXTENSION_PASS

$$\mathfrak{R}_{N,j}\mathfrak{R}_{N,k} = \delta_{jk}\mathfrak{R}_{N,j}, \qquad \sum_{j=0}^{N-1} \mathfrak{R}_{N,j} = I. \tag{A.1652}$$

F1630 41_hypergeometric_companion_packet_calculus.tex:67 NEW_EXTENSION_PASS

$$\boxed{F(z) = \sum_{j=0}^{N-1} (\mathfrak{R}_{N,j}F)(z).} \tag{A.1653}$$

F1631 41_hypergeometric_companion_packet_calculus.tex:75 NEW_EXTENSION_PASS

eq:factorial-residue-lane

$$\boxed{E_{N,j}(z) := \mathfrak{R}_{N,j}(\mathrm{e}^z) = \sum_{q=0}^{\infty} \frac{z^{Nq+j}}{(Nq+j)!}.} \tag{A.1654}$$

F1632 41_hypergeometric_companion_packet_calculus.tex:87 NEW_EXTENSION_PASS

$$\frac{(Nq+j)!}{j!} = \prod_{m=1}^{Nq} (j+m) \tag{A.1655}$$

$$= \prod_{r=1}^{N} \prod_{k=0}^{q-1} (j + r + Nk) \tag{A.1656}$$

$$= N^{Nq} \prod_{r=1}^{N} \left(\frac{j+r}{N} \right)_q . \tag{A.1657}$$

F1633 41_hypergeometric_companion_packet_calculus.tex:100 NEW_EXTENSION_PASS

$$(1)_q = q!, \tag{A.1658}$$

F1634 41_hypergeometric_companion_packet_calculus.tex:104 NEW_EXTENSION_PASS

eq:factorial-lane-parameters

$$\boxed{\mathcal{A}_{N,j} := \left\{ \frac{j+r}{N} : 1 \le r \le N \right\} \setminus \{1\}.} \tag{A.1659}$$

F1635 41_hypergeometric_companion_packet_calculus.tex:116 NEW_EXTENSION_PASS

eq:factorial-lane-split

$$\boxed{(Nq + j)! = j!\, N^{Nq} q! \prod_{\alpha \in \mathcal{A}_{N,j}} (\alpha)_q .} \tag{A.1660}$$

F1636 41_hypergeometric_companion_packet_calculus.tex:126 NEW_EXTENSION_PASS

$$E_{N,j}(z) = \frac{z^j}{j!} \sum_{q=0}^{\infty} \frac{(z^N/N^N)^q}{q! \prod_{\alpha \in \mathcal{A}_{N,j}} (\alpha)_q}. \tag{A.1661}$$

F1637 41_hypergeometric_companion_packet_calculus.tex:135 NEW_EXTENSION_PASS

eq:factorial-lane-hypergeometric

$$\boxed{E_{N,j}(z) = \frac{z^j}{j!}\, {}_0F_{N-1}\left(\begin{matrix} - \\ \mathcal{A}_{N,j} \end{matrix} ; \frac{z^N}{N^N} \right).} \tag{A.1662}$$

F1638 41_hypergeometric_companion_packet_calculus.tex:150 NEW_EXTENSION_PASS

$$\mathcal{A}_{3,0} = \left\{ \frac{1}{3}, \frac{2}{3} \right\}, \tag{A.1663}$$

$$\mathcal{A}_{3,1} = \left\{ \frac{2}{3}, \frac{4}{3} \right\}, \tag{A.1664}$$

$$\mathcal{A}_{3,2} = \left\{ \frac{4}{3}, \frac{5}{3} \right\}. \tag{A.1665}$$

F1639 41_hypergeometric_companion_packet_calculus.tex:156 NEW_EXTENSION_PASS

$$E_{3,1}(z) = z\, {}_0F_2\left(\begin{matrix} - \\ \frac{2}{3}, \frac{4}{3} \end{matrix} ; \frac{z^3}{27} \right). \tag{A.1666}$$

F1640 41_hypergeometric_companion_packet_calculus.tex:167 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}}{\mathrm{d}z} E_{N,j}(z) = \sum_{q=0}^{\infty} \frac{z^{Nq+j-1}}{(Nq + j - 1)!} \tag{A.1667}$$

$$= E_{N,j-1}(z). \tag{A.1668}$$

F1641 41_hypergeometric_companion_packet_calculus.tex:175 NEW_EXTENSION_PASS

$$\frac{\mathrm{d}}{\mathrm{d}z} E_{N,0}(z) = \sum_{q=1}^{\infty} \frac{z^{Nq-1}}{(Nq-1)!} \tag{A.1669}$$

$$= \sum_{r=0}^{\infty} \frac{z^{Nr+N-1}}{(Nr+N-1)!} \tag{A.1670}$$

$$= E_{N,N-1}(z). \tag{A.1671}$$

F1642 41_hypergeometric_companion_packet_calculus.tex:186 NEW_EXTENSION_PASS

$$\mathbf{E}_N(z) := \begin{pmatrix} E_{N,0}(z) & E_{N,1}(z) & \cdots & E_{N,N-1}(z) \end{pmatrix}^{\mathsf{T}} \tag{A.1672}$$

F1643 41_hypergeometric_companion_packet_calculus.tex:194 NEW_EXTENSION_PASS

$$(C_N v)_0 = v_{N-1}, \qquad (C_N v)_j = v_{j-1} \quad (1 \le j < N). \tag{A.1673}$$

F1644 41_hypergeometric_companion_packet_calculus.tex:200 NEW_EXTENSION_PASS

eq:factorial-lane-companion-system

$$\boxed{\mathbf{E}_N'(z) = C_N \mathbf{E}_N(z), \qquad \mathbf{E}_N(0) = e_0, \qquad C_N^N = I.} \tag{A.1674}$$

F1645 41_hypergeometric_companion_packet_calculus.tex:211 NEW_EXTENSION_PASS

$$\boxed{\mathbf{E}_N(z) = \mathrm{e}^{zC_N} e_0.} \tag{A.1675}$$

F1646 41_hypergeometric_companion_packet_calculus.tex:217 NEW_EXTENSION_PASS

eq:factorial-lane-root-filter

$$\boxed{E_{N,j}(z) = \frac{1}{N} \sum_{\ell=0}^{N-1} \omega_N^{-j\ell} \mathrm{e}^{\omega_N^{\ell} z}.} \tag{A.1676}$$

F1647 41_hypergeometric_companion_packet_calculus.tex:229 NEW_EXTENSION_PASS

$$v_{n+1} = M v_n, \tag{A.1677}$$

F1648 41_hypergeometric_companion_packet_calculus.tex:233 NEW_EXTENSION_PASS

eq:companion-state-polyphase

$$\boxed{v_{Nq+j} = M^j (M^N)^q v_0.} \tag{A.1678}$$

F1649 41_hypergeometric_companion_packet_calculus.tex:243 NEW_EXTENSION_PASS

$$E_{N,j}(tA) := \sum_{q=0}^{\infty} \frac{(tA)^{Nq+j}}{(Nq+j)!}. \tag{A.1679}$$

F1650 41_hypergeometric_companion_packet_calculus.tex:250 NEW_EXTENSION_PASS

eq:operator-root-filter

$$\boxed{E_{N,j}(tA) = \frac{1}{N} \sum_{\ell=0}^{N-1} \omega_N^{-j\ell} \mathrm{e}^{\omega_N^{\ell} tA}.} \tag{A.1680}$$

F1651 41_hypergeometric_companion_packet_calculus.tex:260 NEW_EXTENSION_PASS

eq:operator-lane-reconstruction

$$\boxed{\mathrm{e}^{tA} = \sum_{j=0}^{N-1} E_{N,j}(tA).} \tag{A.1681}$$

F1652 41_hypergeometric_companion_packet_calculus.tex:267 NEW_EXTENSION_PASS

eq:operator-hypergeometric-lane

$$\boxed{E_{N,j}(tA) = \frac{(tA)^j}{j!} {}_0F_{N-1}\left(\begin{matrix} - \\ \mathcal{A}_{N,j} \end{matrix} ; \frac{(tA)^N}{N^N} \right).} \tag{A.1682}$$

F1653 41_hypergeometric_companion_packet_calculus.tex:282 NEW_EXTENSION_PASS

$$I, A, \ldots, A^{N-1}, \tag{A.1683}$$

F1654 41_hypergeometric_companion_packet_calculus.tex:291 NEW_EXTENSION_PASS

$$0 \le j < p. \tag{A.1684}$$

F1655 41_hypergeometric_companion_packet_calculus.tex:295 NEW_EXTENSION_PASS

$$g_q := \frac{j!(Nq)!}{(Nq+j)!} = \binom{Nq+j}{j}^{-1}. \tag{A.1685}$$

F1656 41_hypergeometric_companion_packet_calculus.tex:303 NEW_EXTENSION_PASS

$$\binom{Nq+j}{j} = \prod_{r=1}^{j} \frac{Nq+r}{r}. \tag{A.1686}$$

F1657 41_hypergeometric_companion_packet_calculus.tex:309 NEW_EXTENSION_PASS

eq:standard-window-unit

$$\boxed{g_q \in \mathbb{Z}_p^{\times} \qquad (q \ge 0).} \tag{A.1687}$$

F1658 41_hypergeometric_companion_packet_calculus.tex:319 NEW_EXTENSION_PASS

$$E_{N,j}(z) = \sum_{a=0}^{p-1} \sum_{r=0}^{\infty} \frac{z^{N(pr+a)+j}}{(N(pr+a)+j)!} \tag{A.1688}$$

$$= \sum_{a=0}^{p-1} \sum_{r=0}^{\infty} \frac{z^{pNr+(j+aN)}}{(pNr+j+aN)!}. \tag{A.1689}$$

F1659 41_hypergeometric_companion_packet_calculus.tex:331 NEW_EXTENSION_PASS

eq:factorial-lane-prime-refinement

$$\boxed{E_{N,j}(z) = \sum_{a=0}^{p-1} E_{pN,j+aN}(z).} \tag{A.1690}$$

F1660 41_hypergeometric_companion_packet_calculus.tex:343 NEW_EXTENSION_PASS

eq:number-residue-projector

$$\boxed{Q_j^{(N)} := \frac{1}{N} \sum_{\ell=0}^{N-1} \omega_N^{-j\ell} \mathrm{e}^{2\pi \mathrm{i} \ell \widehat{N}/N} = \sum_{n \equiv j \pmod N} |n\rangle \langle n|.} \tag{A.1691}$$

F1661 41_hypergeometric_companion_packet_calculus.tex:356 NEW_EXTENSION_PASS

$$|\alpha\rangle = \mathrm{e}^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \qquad \mu := |\alpha|^2, \tag{A.1692}$$

F1662 41_hypergeometric_companion_packet_calculus.tex:366 NEW_EXTENSION_PASS

$$|\langle n \mid \alpha\rangle|^2 = \mathrm{e}^{-\mu} \frac{\mu^n}{n!}. \tag{A.1693}$$

F1663 41_hypergeometric_companion_packet_calculus.tex:372 NEW_EXTENSION_PASS

$$p_{N,j}(\mu) := \langle \alpha | Q_j^{(N)} | \alpha \rangle \tag{A.1694}$$

$$= \mathrm{e}^{-\mu} \sum_{q=0}^{\infty} \frac{\mu^{Nq+j}}{(Nq+j)!} \tag{A.1695}$$

$$= \mathrm{e}^{-\mu} E_{N,j}(\mu). \tag{A.1696}$$

F1664 41_hypergeometric_companion_packet_calculus.tex:384 NEW_EXTENSION_PASS

eq:coherent-packet-hypergeometric

$$\boxed{p_{N,j}(\mu) = \frac{\mathrm{e}^{-\mu}\mu^{j}}{j!}\,{}_0F_{N-1}\left(\begin{matrix}-\\ \mathcal{A}_{N,j}\end{matrix}; \frac{\mu^{N}}{N^{N}}\right)} \tag{A.1697}$$

F1665 41_hypergeometric_companion_packet_calculus.tex:397 NEW_EXTENSION_PASS

eq:coherent-packet-root-sum

$$\boxed{p_{N,j}(\mu) = \frac{1}{N}\sum_{\ell=0}^{N-1} \omega_N^{-j\ell}\,\exp\!\big(\mu(\omega_N^{\ell}-1)\big).} \tag{A.1698}$$

F1666 41_hypergeometric_companion_packet_calculus.tex:408 NEW_EXTENSION_PASS

$$\sum_{j=0}^{N-1} p_{N,j}(\mu) = 1. \tag{A.1699}$$

F1667 41_hypergeometric_companion_packet_calculus.tex:412 NEW_EXTENSION_PASS

eq:coherent-packet-mixing-bound

$$\left|p_{N,j}(\mu) - \frac{1}{N}\right| \le \frac{1}{N}\sum_{\ell=1}^{N-1} \exp\!\big[-\mu\big(1-\cos(2\pi\ell/N)\big)\big] \tag{A.1700}$$

$$\le \boxed{\frac{N-1}{N}\exp\!\big[-\mu\big(1-\cos(2\pi/N)\big)\big].} \tag{A.1701}$$

F1668 41_hypergeometric_companion_packet_calculus.tex:427 NEW_EXTENSION_PASS

$$\mathrm{e}^{2\pi\mathrm{i}\widehat{N}/N} a \mathrm{e}^{-2\pi\mathrm{i}\widehat{N}/N} = \omega_N^{-1} a, \tag{A.1702}$$

F1669 41_hypergeometric_companion_packet_calculus.tex:434 NEW_EXTENSION_PASS

$$\boxed{Q_r^{(N)} a Q_s^{(N)} = 0 \quad \text{unless} \quad r \equiv s-1 \pmod N,} \tag{A.1703}$$

F1670 41_hypergeometric_companion_packet_calculus.tex:442 NEW_EXTENSION_PASS

$$\boxed{Q_r^{(N)} a^{\dagger} Q_s^{(N)} = 0 \quad \text{unless} \quad r \equiv s+1 \pmod N.} \tag{A.1704}$$

F1671 41_hypergeometric_companion_packet_calculus.tex:460 NEW_EXTENSION_PASS

$$3.908382 \times 10^{-16}. \tag{A.1705}$$

39_nonlinear_bfv_bv_polyphase_bridge.tex

F1672 39_nonlinear_bfv_bv_polyphase_bridge.tex:8 CONDITIONAL_TRANSFER_THEOREM

$$(Z^A, P_A) \in \mathcal{A}_h \times \mathcal{A}_h, \qquad \mathcal{A}_h = M_{N_h}(\mathbb{C}), \tag{A.1706}$$

F1673 39_nonlinear_bfv_bv_polyphase_bridge.tex:14 CONDITIONAL_TRANSFER_THEOREM

$$(Q_{ij}, \Pi^{ij}), \qquad (\Phi, P_\Phi), \qquad \text{connection, frame, or clock pairs as required.} \tag{A.1707}$$

F1674 39_nonlinear_bfv_bv_polyphase_bridge.tex:22 CONDITIONAL_TRANSFER_THEOREM

$$\Omega_h^{\mathrm{can}} := \sum_A \mathrm{Tr}(\,\mathrm{d}P_A \wedge \mathrm{d}Z^A), \tag{A.1708}$$

F1675 39_nonlinear_bfv_bv_polyphase_bridge.tex:27 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\{F,G\}_h = \sum_A \mathrm{Tr}\left(\nabla_{Z^A} F\, \nabla_{P_A} G - \nabla_{P_A} F\, \nabla_{Z^A} G\right).} \tag{A.1709}$$

F1676 39_nonlinear_bfv_bv_polyphase_bridge.tex:40 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\mathcal{D}_h[\Lambda] := \mathrm{i}\sum_A \mathrm{Tr}\Big(P_A[\Lambda, Z^A]\Big).} \tag{A.1710}$$

F1677 39_nonlinear_bfv_bv_polyphase_bridge.tex:47 CONDITIONAL_TRANSFER_THEOREM

$$\delta\mathcal{D}_h[\Lambda] = \mathrm{i}\,\mathrm{Tr}(\delta P[\Lambda, Z]) + \mathrm{i}\,\mathrm{Tr}(P[\Lambda, \delta Z]) \tag{A.1711}$$
$$= \mathrm{Tr}(\mathrm{i}[\Lambda, Z]\,\delta P) + \mathrm{Tr}(\mathrm{i}[P, \Lambda]\,\delta Z)\,. \tag{A.1712}$$

F1678 39_nonlinear_bfv_bv_polyphase_bridge.tex:57 CONDITIONAL_TRANSFER_THEOREM

$$\nabla_P \mathcal{D}_h = \mathrm{i}[\Lambda, Z], \qquad \nabla_Z \mathcal{D}_h = \mathrm{i}[P, \Lambda]. \tag{A.1713}$$

F1679 39_nonlinear_bfv_bv_polyphase_bridge.tex:63 CONDITIONAL_TRANSFER_THEOREM

$$\{Z, \mathcal{D}_h\} = \nabla_P \mathcal{D}_h, \qquad \{P, \mathcal{D}_h\} = -\nabla_Z \mathcal{D}_h. \tag{A.1714}$$

F1680 39_nonlinear_bfv_bv_polyphase_bridge.tex:69 CONDITIONAL_TRANSFER_THEOREM

$$\{Z^A, \mathcal{D}_h[\Lambda]\}_h = \mathrm{i}[\Lambda, Z^A], \tag{A.1715}$$
$$\{P_A, \mathcal{D}_h[\Lambda]\}_h = \mathrm{i}[\Lambda, P_A]. \tag{A.1716}$$

F1681 39_nonlinear_bfv_bv_polyphase_bridge.tex:79 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\{\mathcal{D}_h[\Lambda], \mathcal{D}_h[\Gamma]\}_h = \mathcal{D}_h[\mathrm{i}[\Lambda, \Gamma]].} \tag{A.1717}$$

F1682 39_nonlinear_bfv_bv_polyphase_bridge.tex:86 CONDITIONAL_TRANSFER_THEOREM

$$\{\mathcal{E}, \mathcal{D}_h[\Lambda]\}_h = \mathrm{i}[\Lambda, \mathcal{E}], \tag{A.1718}$$

F1683 39_nonlinear_bfv_bv_polyphase_bridge.tex:91 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\{\mathcal{D}_h[\Lambda], H_h[N]\}_h = H_h[\mathrm{i}[\Lambda, N]].} \tag{A.1719}$$

F1684 39_nonlinear_bfv_bv_polyphase_bridge.tex:101 CONDITIONAL_TRANSFER_THEOREM

$$\{\{Z^A, \mathcal{D}_h[\Gamma]\}, \mathcal{D}_h[\Lambda]\} - \{\{Z^A, \mathcal{D}_h[\Lambda]\}, \mathcal{D}_h[\Gamma]\} = -\Big[\Lambda, [\Gamma, Z^A]\Big] + \Big[\Gamma, [\Lambda, Z^A]\Big] \tag{A.1720}$$
$$= \mathrm{i}\Big[\mathrm{i}[\Lambda, \Gamma], Z^A\Big]. \tag{A.1721}$$

F1685 39_nonlinear_bfv_bv_polyphase_bridge.tex:112 CONDITIONAL_TRANSFER_THEOREM

$$\{\mathcal{D}_h[\Lambda], \mathcal{D}_h[\Gamma]\} - \mathcal{D}_h[\mathrm{i}[\Lambda, \Gamma]] \tag{A.1722}$$

F1686 39_nonlinear_bfv_bv_polyphase_bridge.tex:119 CONDITIONAL_TRANSFER_THEOREM

$$\{\mathcal{D}_h[\Lambda], H_h[N]\} = -\{H_h[N], \mathcal{D}_h[\Lambda]\} \tag{A.1723}$$
$$= -\,\mathrm{Tr}(N\,\mathrm{i}[\Lambda, \mathcal{E}]) \tag{A.1724}$$
$$= \mathrm{Tr}(\mathrm{i}[\Lambda, N]\mathcal{E}) \tag{A.1725}$$
$$= H_h[\mathrm{i}[\Lambda, N]], \tag{A.1726}$$

F1687 39_nonlinear_bfv_bv_polyphase_bridge.tex:136 CONDITIONAL_TRANSFER_THEOREM

$$s_F^2 = 0. \tag{A.1727}$$

F1688 39_nonlinear_bfv_bv_polyphase_bridge.tex:140 CONDITIONAL_TRANSFER_THEOREM

$$P : \mathcal{C}_F \to \mathcal{C}_C, \qquad I : \mathcal{C}_C \to \mathcal{C}_F, \qquad PI = I_C, \tag{A.1728}$$

F1689 39_nonlinear_bfv_bv_polyphase_bridge.tex:148 CONDITIONAL_TRANSFER_THEOREM

$$Q_F := I_F - IP. \tag{A.1729}$$

F1690 39_nonlinear_bfv_bv_polyphase_bridge.tex:152 CONDITIONAL_TRANSFER_THEOREM

$$\bar{s}_C := P s_F I. \tag{A.1730}$$

F1691 39_nonlinear_bfv_bv_polyphase_bridge.tex:159 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\bar{s}_C^2 = -P s_F Q_F s_F I.} \tag{A.1731}$$

F1692 39_nonlinear_bfv_bv_polyphase_bridge.tex:169 CONDITIONAL_TRANSFER_THEOREM

$$\begin{aligned} \bar{s}_C^2 &= P s_F (IP) s_F I && \text{(A.1732)}\\ &= P s_F (I_F - Q_F) s_F I && \text{(A.1733)}\\ &= -P s_F Q_F s_F I. && \text{(A.1734)} \end{aligned}$$

F1693 39_nonlinear_bfv_bv_polyphase_bridge.tex:181 CONDITIONAL_TRANSFER_THEOREM

$$\Omega_F = c^A \mathcal{C}_{F,A} - \frac{1}{2} c^A c^B U_{F,AB}{}^C \varpi_C + \Omega_F^{(\geq 3)}, \tag{A.1735}$$

F1694 39_nonlinear_bfv_bv_polyphase_bridge.tex:188 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\{\Omega_F, \Omega_F\}_F = 0.} \tag{A.1736}$$

F1695 39_nonlinear_bfv_bv_polyphase_bridge.tex:194 CONDITIONAL_TRANSFER_THEOREM

$$s_F := \{\Omega_F, \cdot\}_F. \tag{A.1737}$$

F1696 39_nonlinear_bfv_bv_polyphase_bridge.tex:200 CONDITIONAL_TRANSFER_THEOREM

$$s_F = s_0 + \delta, \tag{A.1738}$$

F1697 39_nonlinear_bfv_bv_polyphase_bridge.tex:204 CONDITIONAL_TRANSFER_THEOREM

$$\begin{aligned} PI &= I_C, && \text{(A.1739)}\\ IP &= I_F - s_0 H - H s_0, && \text{(A.1740)}\\ PH &= 0, \qquad HI = 0, \qquad H^2 = 0. && \text{(A.1741)} \end{aligned}$$

F1698 39_nonlinear_bfv_bv_polyphase_bridge.tex:218 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{s_C := s_{0,C} + P\delta (I_F + H\delta)^{-1} I, \qquad s_{0,C} := P s_0 I.} \tag{A.1742}$$

F1699 39_nonlinear_bfv_bv_polyphase_bridge.tex:227 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{s_C^2 = 0.} \tag{A.1743}$$

F1700 39_nonlinear_bfv_bv_polyphase_bridge.tex:233 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\begin{aligned} s_C = s_{0,C} &+ P\delta I - P\delta H \delta I \\ &+ P\delta H \delta H \delta I - \cdots . \end{aligned}} \tag{A.1744}$$

F1701 39_nonlinear_bfv_bv_polyphase_bridge.tex:246 CONDITIONAL_TRANSFER_THEOREM

$$(I_F + H\delta)^{-1} = I_F - H\delta + (H\delta)^2 - \cdots . \tag{A.1745}$$

F1702 39_nonlinear_bfv_bv_polyphase_bridge.tex:253 CONDITIONAL_TRANSFER_THEOREM

$$s_C = \{\Omega_C, \cdot\}_C, \qquad \boxed{\{\Omega_C, \Omega_C\}_C = 0.} \tag{A.1746}$$

F1703 39_nonlinear_bfv_bv_polyphase_bridge.tex:264 CONDITIONAL_TRANSFER_THEOREM

$$I_\delta := (I_F + H\delta)^{-1} I. \tag{A.1747}$$

F1704 39_nonlinear_bfv_bv_polyphase_bridge.tex:268 CONDITIONAL_TRANSFER_THEOREM

$$I_\delta = I - H\delta I_\delta. \tag{A.1748}$$

F1705 39_nonlinear_bfv_bv_polyphase_bridge.tex:272 CONDITIONAL_TRANSFER_THEOREM

$$PI_\delta = PI - PH\delta I_\delta = I_C. \tag{A.1749}$$

F1706 39_nonlinear_bfv_bv_polyphase_bridge.tex:280 CONDITIONAL_TRANSFER_THEOREM

$$K := (s_0 + \delta)I_\delta - I_\delta s_C. \tag{A.1750}$$

F1707 39_nonlinear_bfv_bv_polyphase_bridge.tex:286 CONDITIONAL_TRANSFER_THEOREM

$$K = H\left(s_0\delta I_\delta + \delta I_\delta s_{0,C} + \delta I_\delta P\delta I_\delta\right). \tag{A.1751}$$

F1708 39_nonlinear_bfv_bv_polyphase_bridge.tex:296 CONDITIONAL_TRANSFER_THEOREM

$$s_0\delta + \delta s_0 + \delta^2 = 0. \tag{A.1752}$$

F1709 39_nonlinear_bfv_bv_polyphase_bridge.tex:300 CONDITIONAL_TRANSFER_THEOREM

$$K = -H\delta K, \qquad (I_F + H\delta)K = 0. \tag{A.1753}$$

F1710 39_nonlinear_bfv_bv_polyphase_bridge.tex:306 CONDITIONAL_TRANSFER_THEOREM

$$(s_0 + \delta)I_\delta = I_\delta s_C. \tag{A.1754}$$

F1711 39_nonlinear_bfv_bv_polyphase_bridge.tex:310 CONDITIONAL_TRANSFER_THEOREM

$$0 = (s_0 + \delta)^2 I_\delta = I_\delta s_C^2. \tag{A.1755}$$

F1712 39_nonlinear_bfv_bv_polyphase_bridge.tex:320 CONDITIONAL_TRANSFER_THEOREM

$$-P\delta H\delta I. \tag{A.1756}$$

F1713 39_nonlinear_bfv_bv_polyphase_bridge.tex:326 CONDITIONAL_TRANSFER_THEOREM

$$\bar{s}_C = \begin{pmatrix} -1/25 & -1/25 & -26/625 \\ 1 & 0 & 1/25 \\ 0 & 1/25 & 0 \end{pmatrix}, \qquad \bar{s}_C^2 \neq 0. \tag{A.1757}$$

F1714 39_nonlinear_bfv_bv_polyphase_bridge.tex:337 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{s_C = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1/25 \\ 0 & 0 & 0 \end{pmatrix}, \qquad s_C^2 = 0.} \tag{A.1758}$$

F1715 39_nonlinear_bfv_bv_polyphase_bridge.tex:353 CONDITIONAL_TRANSFER_THEOREM

$$c^A = (c^\perp, c^i, c^{\mathrm{int}}). \tag{A.1759}$$

F1716 39_nonlinear_bfv_bv_polyphase_bridge.tex:357 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\mathcal{H}_{\mathcal{P},h} := \left.\frac{\partial \Omega_{C,h}}{\partial c^\perp}\right|_{c=\varpi=0},} \tag{A.1760}$$

F1717 39_nonlinear_bfv_bv_polyphase_bridge.tex:365 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\mathcal{H}_{\mathcal{P},h,i} := \left.\frac{\partial \Omega_{C,h}}{\partial c^i}\right|_{c=\varpi=0}.} \tag{A.1761}$$

F1718 39_nonlinear_bfv_bv_polyphase_bridge.tex:374 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\mathcal{L}^{\mathcal{P}}_{\boldsymbol{N},h} X := \{X, \Omega_{C,h}\}_C\Big|_{c^i = N^i,\ c^\perp = 0,\ \varpi = 0}.} \tag{A.1762}$$

F1719 39_nonlinear_bfv_bv_polyphase_bridge.tex:385 CONDITIONAL_TRANSFER_THEOREM

$$\begin{aligned}\Omega_{C,h} = \mathcal{H}_h[c^{\perp}] + \mathcal{D}_h[\boldsymbol{c}] \\ - \frac{1}{2}\left\langle \varpi_A, \mathbb{B}^A_h(c,c)\right\rangle + \Omega^{(\geq 3)}_{C,h}.\end{aligned} \tag{A.1763}$$

F1720 39_nonlinear_bfv_bv_polyphase_bridge.tex:395 CONDITIONAL_TRANSFER_THEOREM

$$\{\mathcal{D}_h[\boldsymbol{\xi}], \mathcal{D}_h[\boldsymbol{\eta}]\}_C = \mathcal{D}_h[\mathbb{B}^h_{DD}(\boldsymbol{\xi},\boldsymbol{\eta})], \tag{A.1764}$$

$$\{\mathcal{D}_h[\boldsymbol{\xi}], \mathcal{H}_h[N]\}_C = \mathcal{H}_h[\mathbb{B}^h_{DH}(\boldsymbol{\xi},N)], \tag{A.1765}$$

$$\{\mathcal{H}_h[N], \mathcal{H}_h[M]\}_C = \mathcal{D}_h[\mathbb{B}^h_{HH}(N,M)]. \tag{A.1766}$$

F1721 39_nonlinear_bfv_bv_polyphase_bridge.tex:404 CONDITIONAL_TRANSFER_THEOREM

$$\{\mathcal{C}_A, \mathcal{C}_B\}_C - \mathbb{B}^C_{AB}\mathcal{C}_C. \tag{A.1767}$$

F1722 39_nonlinear_bfv_bv_polyphase_bridge.tex:411 CONDITIONAL_TRANSFER_THEOREM

$$\mathbb{B}^h_{DD}(\boldsymbol{\xi},\boldsymbol{\eta}) \longrightarrow [\boldsymbol{\xi},\boldsymbol{\eta}], \tag{A.1768}$$

$$\mathbb{B}^h_{DH}(\boldsymbol{\xi},N) \longrightarrow \mathcal{L}_{\boldsymbol{\xi}}N, \tag{A.1769}$$

$$\mathbb{B}^{h,i}_{HH}(N,M) \longrightarrow q^{ij}(N\partial_j M - M\partial_j N). \tag{A.1770}$$

F1723 39_nonlinear_bfv_bv_polyphase_bridge.tex:423 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\frac{1}{2}(S_F,S_F)_F - \mathrm{i}\hbar\Delta_F S_F = 0.} \tag{A.1771}$$

F1724 39_nonlinear_bfv_bv_polyphase_bridge.tex:431 CONDITIONAL_TRANSFER_THEOREM

$$\Delta_F \exp\left(\frac{\mathrm{i}}{\hbar}S_F\right) = 0. \tag{A.1772}$$

F1725 39_nonlinear_bfv_bv_polyphase_bridge.tex:435 CONDITIONAL_TRANSFER_THEOREM

$$\Delta_F \mathrm{e}^{\mathrm{i}S_F/\hbar} = \left[\frac{\mathrm{i}}{\hbar}\Delta_F S_F - \frac{1}{2\hbar^2}(S_F,S_F)_F\right]\mathrm{e}^{\mathrm{i}S_F/\hbar}. \tag{A.1773}$$

F1726 39_nonlinear_bfv_bv_polyphase_bridge.tex:446 CONDITIONAL_TRANSFER_THEOREM

$$\mathcal{F}_F = \mathcal{F}_C \oplus \mathcal{F}_D, \qquad \Delta_F = \Delta_C + \Delta_D, \tag{A.1774}$$

F1727 39_nonlinear_bfv_bv_polyphase_bridge.tex:452 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\exp\left(\frac{\mathrm{i}}{\hbar}S_C\right) := \mathcal{N}^{-1}\int_{\mathcal{L}_D}\exp\left(\frac{\mathrm{i}}{\hbar}S_F\right)\mathcal{D}\chi_D.} \tag{A.1775}$$

F1728 39_nonlinear_bfv_bv_polyphase_bridge.tex:464 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\frac{1}{2}(S_C,S_C)_C - \mathrm{i}\hbar\Delta_C S_C = 0.} \tag{A.1776}$$

F1729 39_nonlinear_bfv_bv_polyphase_bridge.tex:475 CONDITIONAL_TRANSFER_THEOREM

$$\Delta_C \exp\left(\frac{\mathrm{i}}{\hbar}S_C\right) = \mathcal{N}^{-1}\int_{\mathcal{L}_D}\Delta_C \exp\left(\frac{\mathrm{i}}{\hbar}S_F\right)\mathcal{D}\chi_D \tag{A.1777}$$

$$= -\mathcal{N}^{-1}\int_{\mathcal{L}_D}\Delta_D \exp\left(\frac{\mathrm{i}}{\hbar}S_F\right)\mathcal{D}\chi_D \tag{A.1778}$$

$$= 0. \tag{A.1779}$$

F1730 39_nonlinear_bfv_bv_polyphase_bridge.tex:493 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\exp\left(\frac{\mathrm{i}}{\hbar}S^{\mathrm{BV}}_h[\chi_h]\right) := \mathcal{N}_h^{-1}\int_{\mathcal{L}_{>h}}\exp\left(\frac{\mathrm{i}}{\hbar}S^{\mathrm{BV}}_F[\chi_h,\chi_{>h}]\right)\mathcal{D}\chi_{>h}.} \tag{A.1780}$$

F1731 39_nonlinear_bfv_bv_polyphase_bridge.tex:503 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\frac{1}{2}(S_h^{\mathrm{BV}},S_h^{\mathrm{BV}})_h-\mathrm{i}\hbar\Delta_h S_h^{\mathrm{BV}}=0} \tag{A.1781}$$

F1732 39_nonlinear_bfv_bv_polyphase_bridge.tex:512 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\mathfrak{R}_{h_0\leftarrow h_2}=\mathfrak{R}_{h_0\leftarrow h_1}\circ\mathfrak{R}_{h_1\leftarrow h_2}.} \tag{A.1782}$$

F1733 39_nonlinear_bfv_bv_polyphase_bridge.tex:521 CONDITIONAL_TRANSFER_THEOREM

$$\int \mathcal{D}\chi_{h_2:h_1}\mathcal{D}\chi_{h_1:h_0}\,\mathrm{e}^{\mathrm{i}S_{h_2}/\hbar}. \tag{A.1783}$$

F1734 39_nonlinear_bfv_bv_polyphase_bridge.tex:527 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\mathfrak{R}_{D_1}\circ\mathfrak{R}_{D_2}=\mathfrak{R}_{D_2}\circ\mathfrak{R}_{D_1}=\mathfrak{R}_{D_1\oplus D_2},} \tag{A.1784}$$

F1735 39_nonlinear_bfv_bv_polyphase_bridge.tex:538 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{L_{\mathrm{eff}}=L_{CC}-L_{CD}L_{DD}^{-1}L_{DC}.} \tag{A.1785}$$

F1736 39_nonlinear_bfv_bv_polyphase_bridge.tex:545 CONDITIONAL_TRANSFER_THEOREM

$$\frac{1}{2}\begin{pmatrix}x_C\\x_D\end{pmatrix}^{*}\begin{pmatrix}L_{CC}&L_{CD}\\L_{DC}&L_{DD}\end{pmatrix}\begin{pmatrix}x_C\\x_D\end{pmatrix}, \tag{A.1786}$$

F1737 39_nonlinear_bfv_bv_polyphase_bridge.tex:555 CONDITIONAL_TRANSFER_THEOREM

$$L_{DC}x_C+L_{DD}x_D=0,\qquad x_D=-L_{DD}^{-1}L_{DC}x_C. \tag{A.1787}$$

F1738 39_nonlinear_bfv_bv_polyphase_bridge.tex:561 CONDITIONAL_TRANSFER_THEOREM

$$\frac{1}{2}x_C^{*}\left(L_{CC}-L_{CD}L_{DD}^{-1}L_{DC}\right)x_C. \tag{A.1788}$$

F1739 39_nonlinear_bfv_bv_polyphase_bridge.tex:566 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\mathrm{Schur}_{D_1\oplus D_2}\,L=\mathrm{Schur}_{D_1}\left(\mathrm{Schur}_{D_2}\,L\right)=\mathrm{Schur}_{D_2}\left(\mathrm{Schur}_{D_1}\,L\right).} \tag{A.1789}$$

F1740 39_nonlinear_bfv_bv_polyphase_bridge.tex:581 CONDITIONAL_TRANSFER_THEOREM

$$S_F=\frac{1}{2}(x_2-x_1y)^2+\frac{m}{2}(y-z)^2+\frac{n}{2}z^2. \tag{A.1790}$$

F1741 39_nonlinear_bfv_bv_polyphase_bridge.tex:588 CONDITIONAL_TRANSFER_THEOREM

$$R_F=\partial_{x_1}+y\partial_{x_2}. \tag{A.1791}$$

F1742 39_nonlinear_bfv_bv_polyphase_bridge.tex:593 CONDITIONAL_TRANSFER_THEOREM

$$R_FS_F=0,\qquad \operatorname{div}R_F=0. \tag{A.1792}$$

F1743 39_nonlinear_bfv_bv_polyphase_bridge.tex:599 CONDITIONAL_TRANSFER_THEOREM

$$R_F(x_2-x_1y)=y-y=0, \tag{A.1793}$$

F1744 39_nonlinear_bfv_bv_polyphase_bridge.tex:604 CONDITIONAL_TRANSFER_THEOREM

$$M_D=\begin{pmatrix}x_1^2+m&-m\\-m&m+n\end{pmatrix},\qquad D=(m+n)x_1^2+mn. \tag{A.1794}$$

F1745 39_nonlinear_bfv_bv_polyphase_bridge.tex:614 CONDITIONAL_TRANSFER_THEOREM

$$y_*=\frac{(m+n)x_1x_2}{D}, \tag{A.1795}$$

$$z_*=\frac{mx_1x_2}{D}. \tag{A.1796}$$

F1746 39_nonlinear_bfv_bv_polyphase_bridge.tex:619 CONDITIONAL_TRANSFER_THEOREM

$$(x_1^2+m)y - mz = x_1x_2, \tag{A.1797}$$
$$-my + (m+n)z = 0. \tag{A.1798}$$

F1747 39_nonlinear_bfv_bv_polyphase_bridge.tex:624 CONDITIONAL_TRANSFER_THEOREM

$$\left(x_1^2 + \frac{mn}{m+n}\right) y = x_1x_2, \tag{A.1799}$$

F1748 39_nonlinear_bfv_bv_polyphase_bridge.tex:629 CONDITIONAL_TRANSFER_THEOREM

$$\mu := \frac{mn}{m+n}, \tag{A.1800}$$

F1749 39_nonlinear_bfv_bv_polyphase_bridge.tex:633 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{S_C = \frac{\mu x_2^2}{2(x_1^2+\mu)}.} \tag{A.1801}$$

F1750 39_nonlinear_bfv_bv_polyphase_bridge.tex:640 CONDITIONAL_TRANSFER_THEOREM

$$\frac{m}{2}(y-z)^2 + \frac{n}{2}z^2\Big|_{z=my/(m+n)} = \frac{\mu}{2}y^2. \tag{A.1802}$$

F1751 39_nonlinear_bfv_bv_polyphase_bridge.tex:647 CONDITIONAL_TRANSFER_THEOREM

$$\frac{1}{2}(x_2 - x_1y)^2 + \frac{\mu}{2}y^2. \tag{A.1803}$$

F1752 39_nonlinear_bfv_bv_polyphase_bridge.tex:651 CONDITIONAL_TRANSFER_THEOREM

$$y_* = \frac{x_1x_2}{x_1^2+\mu}. \tag{A.1804}$$

F1753 39_nonlinear_bfv_bv_polyphase_bridge.tex:656 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{R_C = \partial_{x_1} + \frac{x_1x_2}{x_1^2+\mu}\partial_{x_2},} \tag{A.1805}$$

F1754 39_nonlinear_bfv_bv_polyphase_bridge.tex:664 CONDITIONAL_TRANSFER_THEOREM

$$R_C S_C = 0. \tag{A.1806}$$

F1755 39_nonlinear_bfv_bv_polyphase_bridge.tex:668 CONDITIONAL_TRANSFER_THEOREM

$$R_C x_1 = 1, \qquad R_C x_2 = y_* = \frac{x_1x_2}{x_1^2+\mu}. \tag{A.1807}$$

F1756 39_nonlinear_bfv_bv_polyphase_bridge.tex:674 CONDITIONAL_TRANSFER_THEOREM

$$\partial_{x_1} S_C = -\frac{\mu x_1 x_2^2}{d^2}, \tag{A.1808}$$
$$\partial_{x_2} S_C = \frac{\mu x_2}{d}. \tag{A.1809}$$

F1757 39_nonlinear_bfv_bv_polyphase_bridge.tex:679 CONDITIONAL_TRANSFER_THEOREM

$$R_C S_C = -\frac{\mu x_1 x_2^2}{d^2} + \frac{x_1x_2}{d}\frac{\mu x_2}{d} = 0. \tag{A.1810}$$

F1758 39_nonlinear_bfv_bv_polyphase_bridge.tex:687 CONDITIONAL_TRANSFER_THEOREM

$$\operatorname{div} R_C = \frac{x_1}{x_1^2+\mu}. \tag{A.1811}$$

F1759 39_nonlinear_bfv_bv_polyphase_bridge.tex:692 CONDITIONAL_TRANSFER_THEOREM

$$\rho_C \propto D^{-1/2} \propto (x_1^2+\mu)^{-1/2}. \tag{A.1812}$$

F1760 39_nonlinear_bfv_bv_polyphase_bridge.tex:697 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\operatorname{div} R_C + R_C(\log \rho_C) = 0.} \tag{A.1813}$$

F1761 39_nonlinear_bfv_bv_polyphase_bridge.tex:703 CONDITIONAL_TRANSFER_THEOREM

$$\operatorname{div} R_C = \partial_{x_1}(1) + \partial_{x_2}\left(\frac{x_1 x_2}{d}\right) = \frac{x_1}{d}, \tag{A.1814}$$

$$R_C(\log \rho_C) = -\frac{1}{2} R_C \log\Big((m+n)d\Big) \tag{A.1815}$$

$$= -\frac{1}{2}\frac{2x_1}{d} = -\frac{x_1}{d}. \tag{A.1816}$$

F1762 39_nonlinear_bfv_bv_polyphase_bridge.tex:718 CONDITIONAL_TRANSFER_THEOREM

$$S_{C,\hbar} = S_C + \frac{\hbar}{2}\log D \tag{A.1817}$$

F1763 39_nonlinear_bfv_bv_polyphase_bridge.tex:725 CONDITIONAL_TRANSFER_THEOREM

$$\mu = \frac{mn}{m+n} \tag{A.1818}$$

F1764 39_nonlinear_bfv_bv_polyphase_bridge.tex:740 CONDITIONAL_TRANSFER_THEOREM

$$\begin{aligned} s_C - (s_{0,C} + P\delta I) &= P\delta\Big[(I_F + H\delta)^{-1} - I_F\Big] I \\ &= -P\delta(I_F + H\delta)^{-1} H\delta I. \end{aligned} \tag{A.1819}$$

F1765 39_nonlinear_bfv_bv_polyphase_bridge.tex:748 CONDITIONAL_TRANSFER_THEOREM

$$\|H\delta\| \le \rho < 1, \qquad \|Q_F \delta I\| \le C h^r. \tag{A.1820}$$

F1766 39_nonlinear_bfv_bv_polyphase_bridge.tex:754 CONDITIONAL_TRANSFER_THEOREM

$$\boxed{\|s_C - (s_{0,C} + P\delta I)\| \le \frac{\|P\delta\|\,\|H\|}{1-\rho} C h^r.} \tag{A.1821}$$

F1767 39_nonlinear_bfv_bv_polyphase_bridge.tex:763 CONDITIONAL_TRANSFER_THEOREM

$$\|(I_F + H\delta)^{-1}\| \le \sum_{n=0}^{\infty} \|H\delta\|^n \le \frac{1}{1-\rho}. \tag{A.1822}$$

F1768 39_nonlinear_bfv_bv_polyphase_bridge.tex:769 CONDITIONAL_TRANSFER_THEOREM

$$HIP = 0, \qquad H = H(I_F - IP) = HQ_F. \tag{A.1823}$$

F1769 39_nonlinear_bfv_bv_polyphase_bridge.tex:775 CONDITIONAL_TRANSFER_THEOREM

$$\|H\delta I\| = \|HQ_F \delta I\| \tag{A.1824}$$

$$\le \|H\|\, C h^r. \tag{A.1825}$$

F1770 39_nonlinear_bfv_bv_polyphase_bridge.tex:784 CONDITIONAL_TRANSFER_THEOREM

$$\mathsf{P}_h \longrightarrow I, \tag{A.1826}$$

$$\Omega_{C,h} \longrightarrow \Omega_{\mathrm{ADM}}, \tag{A.1827}$$

$$S_h^{\mathrm{BV}} \longrightarrow S_{\mathrm{GR}}^{\mathrm{BV}}, \tag{A.1828}$$

$$\Delta_h S_h^{\mathrm{BV}} \longrightarrow 0 \quad \text{after the required renormalization.} \tag{A.1829}$$

F1771 39_nonlinear_bfv_bv_polyphase_bridge.tex:795 CONDITIONAL_TRANSFER_THEOREM

$$\text{exact finite spatial closure,} \qquad \{\mathcal{D}_h[\Lambda], \mathcal{D}_h[\Gamma]\} = \mathcal{D}_h[\mathrm{i}[\Lambda, \Gamma]], \tag{A.1830}$$

$$\text{an exact nonlinear coarse differential,} \qquad s_C^2 = 0, \tag{A.1831}$$

$$\text{an exact coarse BFV charge,} \qquad \{\Omega_{C,h}, \Omega_{C,h}\}_C = 0, \tag{A.1832}$$

$$\text{an exact quantum effective action,} \qquad \frac{1}{2}(S_h, S_h)_h - \mathrm{i}\hbar\Delta_h S_h = 0, \tag{A.1833}$$

$$\text{and direct–iterated refinement equivalence.} \tag{A.1834}$$

F1772 39_nonlinear_bfv_bv_polyphase_bridge.tex:819 CONDITIONAL_TRANSFER_THEOREM

$$\max \varepsilon_{\text{inner action}} = 5.699160 \times 10^{-15}, \tag{A.1835}$$

$$\max \varepsilon_{\text{inner closure}} = 3.907985 \times 10^{-14}, \tag{A.1836}$$

$$\max \varepsilon_{\text{Schur direct/iterated}} = 1.320281 \times 10^{-15}, \tag{A.1837}$$

$$\max \varepsilon_{\text{detail order}} = 1.593034 \times 10^{-15}. \tag{A.1838}$$

40_gowdy_bfv_target.tex

F1773 40_gowdy_bfv_target.tex:11 FAMILY_AUDITED

$$t := \mathrm{e}^{-\tau} > 0. \tag{A.1839}$$

F1774 40_gowdy_bfv_target.tex:15 FAMILY_AUDITED

$$\frac{\partial}{\partial\tau} = -t\frac{\partial}{\partial t}, \qquad \mathrm{d}\tau = -\frac{\mathrm{d}t}{t}, \qquad \mathrm{e}^{-2\tau} = t^2. \tag{A.1840}$$

F1775 40_gowdy_bfv_target.tex:24 FAMILY_AUDITED

$$S_G = \frac{1}{2}\int \left[P_\tau^2 - \mathrm{e}^{-2\tau}P_\theta^2 + \mathrm{e}^{2P}\left(Q_\tau^2 - \mathrm{e}^{-2\tau}Q_\theta^2\right)\right] \mathrm{d}\tau\,\mathrm{d}\theta, \tag{A.1841}$$

F1776 40_gowdy_bfv_target.tex:33 FAMILY_AUDITED

$$\boxed{S_G[P,Q] = \frac{1}{2}\int t\left[P_t^2 - P_\theta^2 + \mathrm{e}^{2P}\left(Q_t^2 - Q_\theta^2\right)\right] \mathrm{d}t\,\mathrm{d}\theta.} \tag{A.1842}$$

F1777 40_gowdy_bfv_target.tex:43 FAMILY_AUDITED

$$\Pi_P := tP_t, \tag{A.1843}$$

$$\Pi_Q := t\mathrm{e}^{2P}Q_t. \tag{A.1844}$$

F1778 40_gowdy_bfv_target.tex:48 FAMILY_AUDITED

$$P_t = \frac{\Pi_P}{t}, \qquad Q_t = \frac{\mathrm{e}^{-2P}\Pi_Q}{t}. \tag{A.1845}$$

F1779 40_gowdy_bfv_target.tex:54 FAMILY_AUDITED

$$\mathcal{H}_t = \Pi_P P_t + \Pi_Q Q_t - \mathcal{L}_t \tag{A.1846}$$

$$= \frac{1}{2t}\left(\Pi_P^2 + \mathrm{e}^{-2P}\Pi_Q^2\right) + \frac{t}{2}\left(P_\theta^2 + \mathrm{e}^{2P}Q_\theta^2\right). \tag{A.1847}$$

F1780 40_gowdy_bfv_target.tex:62 FAMILY_AUDITED

$$\mathcal{H}_t := \frac{1}{2t}\left(\Pi_P^2 + \mathrm{e}^{-2P}\Pi_Q^2\right) + \frac{t}{2}\left(P_\theta^2 + \mathrm{e}^{2P}Q_\theta^2\right), \tag{A.1848}$$

$$\mathcal{D} := \Pi_P P_\theta + \Pi_Q Q_\theta. \tag{A.1849}$$

F1781 40_gowdy_bfv_target.tex:70 FAMILY_AUDITED

$$H_t[N] := \int_0^{2\pi} N\mathcal{H}_t\,\mathrm{d}\theta, \tag{A.1850}$$

$$D[\xi] := \int_0^{2\pi} \xi\mathcal{D}\,\mathrm{d}\theta. \tag{A.1851}$$

F1782 40_gowdy_bfv_target.tex:79 FAMILY_AUDITED

$$\{P(\theta),\Pi_P(\vartheta)\}=\{Q(\theta),\Pi_Q(\vartheta)\}=\delta(\theta-\vartheta), \tag{A.1852}$$

F1783 40_gowdy_bfv_target.tex:85 FAMILY_AUDITED

$$\mathbb{B}(f,g):=fg_\theta-gf_\theta. \tag{A.1853}$$

F1784 40_gowdy_bfv_target.tex:91 FAMILY_AUDITED

$$\boxed{\{D[\xi],D[\eta]\}=D[\mathbb{B}(\xi,\eta)],} \tag{A.1854}$$

$$\boxed{\{D[\xi],H_t[N]\}=H_t[\mathbb{B}(\xi,N)],} \tag{A.1855}$$

$$\boxed{\{H_t[N],H_t[M]\}=D[\mathbb{B}(N,M)].} \tag{A.1856}$$

F1785 40_gowdy_bfv_target.tex:109 FAMILY_AUDITED

$$\delta\int\frac{t}{2}NP_\theta^2\,\mathrm{d}\theta=\int tNP_\theta\,\partial_\theta(\delta P)\,\mathrm{d}\theta \tag{A.1857}$$

$$=-\int\partial_\theta(tNP_\theta)\,\delta P\,\mathrm{d}\theta. \tag{A.1858}$$

F1786 40_gowdy_bfv_target.tex:117 FAMILY_AUDITED

$$\frac{\delta H_t[N]}{\delta\Pi_P}=\frac{N}{t}\Pi_P,\qquad \frac{\delta H_t[N]}{\delta\Pi_Q}=\frac{N}{t}\mathrm{e}^{-2P}\Pi_Q, \tag{A.1859}$$

$$\frac{\delta H_t[N]}{\delta P}=-\partial_\theta(tNP_\theta)+tN\mathrm{e}^{2P}Q_\theta^2-\frac{N}{t}\mathrm{e}^{-2P}\Pi_Q^2, \tag{A.1860}$$

$$\frac{\delta H_t[N]}{\delta Q}=-\partial_\theta(tN\mathrm{e}^{2P}Q_\theta), \tag{A.1861}$$

$$\frac{\delta D[\xi]}{\delta\Pi_P}=\xi P_\theta,\qquad \frac{\delta D[\xi]}{\delta\Pi_Q}=\xi Q_\theta, \tag{A.1862}$$

$$\frac{\delta D[\xi]}{\delta P}=-\partial_\theta(\xi\Pi_P),\qquad \frac{\delta D[\xi]}{\delta Q}=-\partial_\theta(\xi\Pi_Q). \tag{A.1863}$$

F1787 40_gowdy_bfv_target.tex:142 FAMILY_AUDITED

$$\{D[\xi],D[\eta]\}=\int\Big[-\partial_\theta(\xi\Pi_P)\,\eta P_\theta+\xi P_\theta\,\partial_\theta(\eta\Pi_P) \tag{A.1864}$$

$$-\partial_\theta(\xi\Pi_Q)\,\eta Q_\theta+\xi Q_\theta\,\partial_\theta(\eta\Pi_Q)\Big]\,\mathrm{d}\theta. \tag{A.1865}$$

F1788 40_gowdy_bfv_target.tex:155 FAMILY_AUDITED

$$\{D[\xi],D[\eta]\}=\int(\xi\eta_\theta-\eta\xi_\theta)(\Pi_PP_\theta+\Pi_QQ_\theta)\,\mathrm{d}\theta \tag{A.1866}$$

$$=D[\mathbb{B}(\xi,\eta)]. \tag{A.1867}$$

F1789 40_gowdy_bfv_target.tex:164 FAMILY_AUDITED

$$\{P,D[\xi]\}=\xi P_\theta,\qquad \{Q,D[\xi]\}=\xi Q_\theta, \tag{A.1868}$$

$$\{\Pi_P,D[\xi]\}=\partial_\theta(\xi\Pi_P),\qquad \{\Pi_Q,D[\xi]\}=\partial_\theta(\xi\Pi_Q). \tag{A.1869}$$

F1790 40_gowdy_bfv_target.tex:173 FAMILY_AUDITED

$$\{\mathcal{H}_t,D[\xi]\}=\xi\partial_\theta\mathcal{H}_t+2\xi_\theta\mathcal{H}_t. \tag{A.1870}$$

F1791 40_gowdy_bfv_target.tex:179 FAMILY_AUDITED

$$\{D[\xi],H_t[N]\}=-\int N\left(\xi\mathcal{H}_{t,\theta}+2\xi_\theta\mathcal{H}_t\right)\,\mathrm{d}\theta \tag{A.1871}$$

$$= \int (\xi N_\theta - N\xi_\theta)\mathcal{H}_t \, \mathrm{d}\theta \tag{A.1872}$$

$$= H_t[\mathbb{B}(\xi, N)], \tag{A.1873}$$

F1792 40_gowdy_bfv_target.tex:191 FAMILY_AUDITED

$$-\frac{M\Pi_P}{t}\partial_\theta(tNP_\theta) + \frac{N\Pi_P}{t}\partial_\theta(tMP_\theta) \tag{A.1874}$$

$$= (NM_\theta - MN_\theta)\Pi_P P_\theta. \tag{A.1875}$$

F1793 40_gowdy_bfv_target.tex:198 FAMILY_AUDITED

$$-\frac{M\mathrm{e}^{-2P}\Pi_Q}{t}\partial_\theta(tN\mathrm{e}^{2P}Q_\theta) + \frac{N\mathrm{e}^{-2P}\Pi_Q}{t}\partial_\theta(tM\mathrm{e}^{2P}Q_\theta) \tag{A.1876}$$

$$= (NM_\theta - MN_\theta)\Pi_Q Q_\theta. \tag{A.1877}$$

F1794 40_gowdy_bfv_target.tex:208 FAMILY_AUDITED

$$\{H_t[N], H_t[M]\} = \int_0^{2\pi} \mathbb{B}(N, M)\left(\Pi_P P_\theta + \Pi_Q Q_\theta\right) \mathrm{d}\theta \tag{A.1878}$$

$$= D[\mathbb{B}(N, M)]. \tag{A.1879}$$

F1795 40_gowdy_bfv_target.tex:222 FAMILY_AUDITED

$$C_\pm[f] := \frac{1}{2}\left(H_t[f] \pm D[f]\right). \tag{A.1880}$$

F1796 40_gowdy_bfv_target.tex:227 FAMILY_AUDITED

$$\boxed{\{C_+[f], C_+[g]\} = C_+[\mathbb{B}(f, g)],} \tag{A.1881}$$

$$\boxed{\{C_-[f], C_-[g]\} = -C_-[\mathbb{B}(f, g)],} \tag{A.1882}$$

$$\boxed{\{C_+[f], C_-[g]\} = 0.} \tag{A.1883}$$

F1797 40_gowdy_bfv_target.tex:241 FAMILY_AUDITED

$$\{C_+[f], C_+[g]\} = \frac{1}{4}\big(\{H[f], H[g]\} + \{H[f], D[g]\} \tag{A.1884}$$

$$+ \{D[f], H[g]\} + \{D[f], D[g]\}\big) \tag{A.1885}$$

$$= \frac{1}{2}\left(H[\mathbb{B}(f, g)] + D[\mathbb{B}(f, g)]\right) \tag{A.1886}$$

$$= C_+[\mathbb{B}(f, g)]. \tag{A.1887}$$

F1798 40_gowdy_bfv_target.tex:264 FAMILY_AUDITED

$$(T, X; \Pi_T, \Pi_X) \tag{A.1888}$$

F1799 40_gowdy_bfv_target.tex:268 FAMILY_AUDITED

$$(P, Q; \Pi_P, \Pi_Q). \tag{A.1889}$$

F1800 40_gowdy_bfv_target.tex:272 FAMILY_AUDITED

$$\mathcal{C}_\perp = \mathcal{E}_\perp[T, X, \Pi_T, \Pi_X] + \mathcal{H}^G_\perp[T, X; P, Q, \Pi_P, \Pi_Q], \tag{A.1890}$$

$$\mathcal{C}_\parallel = \Pi_T T_\theta + \Pi_X X_\theta + \Pi_P P_\theta + \Pi_Q Q_\theta, \tag{A.1891}$$

F1801 40_gowdy_bfv_target.tex:281 FAMILY_AUDITED

$$T = t, \qquad X = \theta \tag{A.1892}$$

F1802 40_gowdy_bfv_target.tex:293 FAMILY_AUDITED

$$\boxed{\mathbb{B}_G(u, v) = \left(\mathbb{B}(\xi, M) + \mathbb{B}(N, \eta), \mathbb{B}(\xi, \eta) + \mathbb{B}(N, M)\right).} \tag{A.1893}$$

F1803 40_gowdy_bfv_target.tex:304 FAMILY_AUDITED

$$\mathcal{C}[(N,\xi)] := H_t[N] + D[\xi], \tag{A.1894}$$

F1804 40_gowdy_bfv_target.tex:308 FAMILY_AUDITED

$$\{\mathcal{C}[u], \mathcal{C}[v]\} = \mathcal{C}[\mathbb{B}_G(u,v)]. \tag{A.1895}$$

F1805 40_gowdy_bfv_target.tex:313 FAMILY_AUDITED

$$\Omega_{G,F} = \mathcal{C}_{F,\perp}[c^{\perp}] + \mathcal{C}_{F,\parallel}[c^{\parallel}] - \frac{1}{2}\left\langle \varpi, \mathbb{B}_{G,F}(c,c)\right\rangle + \Omega^{(\geq 3)}_{G,F}. \tag{A.1896}$$

F1806 40_gowdy_bfv_target.tex:322 FAMILY_AUDITED

$$\boxed{\{\Omega_{G,F}, \Omega_{G,F}\}_F = 0.} \tag{A.1897}$$

F1807 40_gowdy_bfv_target.tex:329 FAMILY_AUDITED

$$s_{G,h} = s_{0,h} + P_h \delta_G (I_F + H_h \delta_G)^{-1} I_h. \tag{A.1898}$$

F1808 40_gowdy_bfv_target.tex:335 FAMILY_AUDITED

$$\boxed{\{\Omega_{G,h}, \Omega_{G,h}\}_h = 0} \tag{A.1899}$$

F1809 40_gowdy_bfv_target.tex:360 FAMILY_AUDITED

$$\boxed{\begin{aligned}&\|\mathcal{C}_{\perp,h} - \mathcal{C}_{\perp}\|_{s-1} + \|\mathcal{C}_{\parallel,h} - \mathcal{C}_{\parallel}\|_{s-1} \\ &+ \|\mathbb{B}_{G,h} - \mathbb{B}_G\|_{s-1} \leq C_s h^r\end{aligned}} \tag{A.1900}$$

F1810 40_gowdy_bfv_target.tex:374 FAMILY_AUDITED

$$\boxed{\mathfrak{R}_{h_0 \leftarrow h_2}(\Omega_{G,h_2}) = \mathfrak{R}_{h_0 \leftarrow h_1}\left(\mathfrak{R}_{h_1 \leftarrow h_2}(\Omega_{G,h_2})\right).} \tag{A.1901}$$

F1811 40_gowdy_bfv_target.tex:389 FAMILY_AUDITED

$$p_{DD} = 1.998341, \tag{A.1902}$$
$$p_{DH} = 1.997880, \tag{A.1903}$$
$$p_{HH} = 1.999334. \tag{A.1904}$$

F1812 40_gowdy_bfv_target.tex:395 FAMILY_AUDITED

$$\max(\varepsilon^{\mathrm{spec}}_{DD}, \varepsilon^{\mathrm{spec}}_{DH}, \varepsilon^{\mathrm{spec}}_{HH}) = 1.536965 \times 10^{-15}, \tag{A.1905}$$

F1813 40_gowdy_bfv_target.tex:402 FAMILY_AUDITED

$$\varepsilon_{\mathrm{chiral}} = 1.032576 \times 10^{-15}. \tag{A.1906}$$

F1814 40_gowdy_bfv_target.tex:415 FAMILY_AUDITED

$$\boxed{\begin{aligned}&\text{ungauge-fixed nonlinear Gowdy action} \\ &+ \text{explicit fine BFV charge} \\ &+ \text{cyclic projector/polyphase contraction} \\ &\Downarrow \\ &\text{exact finite master equation at every scale} \\ &+ \text{refinement-path independence} \\ &+ \text{controlled Gowdy continuum limit.}\end{aligned}} \tag{A.1907}$$

42_refinement_dark_sector_cosmology.tex

F1815 42_refinement_dark_sector_cosmology.tex:9 FAMILY_AUDITED

$$d_h := \dim \mathrm{Ran}\, \mathsf{D}_h = (p-1)p^{h-1}. \tag{A.1908}$$

F1816 42_refinement_dark_sector_cosmology.tex:13 FAMILY_AUDITED

$$\mathcal{L}_{\sigma,p} := \Lambda^2 \sum_{h\geq 1} p^{2\sigma h} \mathsf{D}_h. \tag{A.1909}$$

F1817 42_refinement_dark_sector_cosmology.tex:18 FAMILY_AUDITED

$$\ell_h = \ell_0 p^{-h}. \tag{A.1910}$$

F1818 42_refinement_dark_sector_cosmology.tex:22 FAMILY_AUDITED

$$\ell_h^{-2\sigma} = \ell_0^{-2\sigma} p^{2\sigma h}. \tag{A.1911}$$

F1819 42_refinement_dark_sector_cosmology.tex:26 FAMILY_AUDITED

$$\lambda_{h+1} = p^{2\sigma}\lambda_h \tag{A.1912}$$

F1820 42_refinement_dark_sector_cosmology.tex:30 FAMILY_AUDITED

$$\lambda_h = \Lambda^2 p^{2\sigma h}. \tag{A.1913}$$

F1821 42_refinement_dark_sector_cosmology.tex:35 FAMILY_AUDITED

$$S_\Phi = -\frac{1}{2}\int_{M^{3,1}} \sqrt{-g}\, \mathrm{d}^4x \int_{\mathbb{Z}_p} \mathrm{d}\mu_p(u) \Big[g^{\mu\nu}\partial_\mu\Phi\, \partial_\nu\Phi + m_0^2\Phi^2 + \Phi\mathcal{L}_{\sigma,p}\Phi \Big]. \tag{A.1914}$$

F1822 42_refinement_dark_sector_cosmology.tex:46 FAMILY_AUDITED

$$\Phi(x,u) = \phi_0(x) + \sum_{h\geq 1}\sum_{a=1}^{d_h} \chi_{h,a}(x)\psi_{h,a}(u). \tag{A.1915}$$

F1823 42_refinement_dark_sector_cosmology.tex:53 FAMILY_AUDITED

$$S_\Phi = -\frac{1}{2}\int \sqrt{-g}\, \mathrm{d}^4x \Big[(\partial\phi_0)^2 + m_0^2\phi_0^2 \tag{A.1916}$$

$$+ \sum_{h\geq 1}\sum_{a=1}^{d_h} \big((\partial\chi_{h,a})^2 + m_h^2\chi_{h,a}^2 \big) \Big], \tag{A.1917}$$

F1824 42_refinement_dark_sector_cosmology.tex:65 FAMILY_AUDITED

$$\boxed{m_h^2 = m_0^2 + \Lambda^2 p^{2\sigma h}, \qquad d_h = (p-1)p^{h-1}.} \tag{A.1918}$$

F1825 42_refinement_dark_sector_cosmology.tex:75 FAMILY_AUDITED

$$\int_{\mathbb{Z}_p} \Phi(x,u)\, \mathrm{d}\mu_p(u) = \phi_0(x), \tag{A.1919}$$

F1826 42_refinement_dark_sector_cosmology.tex:81 FAMILY_AUDITED

$$\mathcal{A}_{\mathrm{fib}}[g,\bar\phi] = m_0^2 + \xi R + U''(\bar\phi) + \mathcal{L}_{\sigma,p}. \tag{A.1920}$$

F1827 42_refinement_dark_sector_cosmology.tex:89 FAMILY_AUDITED

$$\omega_{h,k}^2(t) = m_h^2 + k_{\mathrm{phys}}^2 + \nu^2(t-t_\star)^2, \tag{A.1921}$$

F1828 42_refinement_dark_sector_cosmology.tex:94 FAMILY_AUDITED

$$t_\pm - t_\star = \pm\frac{\mathrm{i}\sqrt{m_h^2 + k_{\mathrm{phys}}^2}}{\nu}. \tag{A.1922}$$

F1829 42_refinement_dark_sector_cosmology.tex:99 FAMILY_AUDITED

$$2\,\mathrm{Im}\int_{t_-}^{t_+}\omega_{h,k}(t)\,\mathrm{d}t \;=\frac{2}{\nu}\int_{-\sqrt{m_h^2+k_{\mathrm{phys}}^2}}^{\sqrt{m_h^2+k_{\mathrm{phys}}^2}}\sqrt{m_h^2+k_{\mathrm{phys}}^2-y^2}\;\mathrm{d}y \tag{A.1923}$$

$$=\frac{\pi(m_h^2+k_{\mathrm{phys}}^2)}{\nu}. \tag{A.1924}$$

F1830 42_refinement_dark_sector_cosmology.tex:108 FAMILY_AUDITED

$$\boxed{f_h(q)=\mathcal{N}_\star\exp\left[-\frac{\pi m_h^2}{\nu}-\frac{\pi q^2}{a_\star^2\nu}\right].} \tag{A.1925}$$

F1831 42_refinement_dark_sector_cosmology.tex:120 FAMILY_AUDITED

$$\epsilon_h(q,a)=\sqrt{q^2+a^2m_h^2}, \tag{A.1926}$$

F1832 42_refinement_dark_sector_cosmology.tex:124 FAMILY_AUDITED

$$\rho_h(a)=\frac{d_h}{2\pi^2a^4}\int_0^\infty q^2\epsilon_h(q,a)f_h(q)\;\mathrm{d}q, \tag{A.1927}$$

$$p_h(a)=\frac{d_h}{6\pi^2a^4}\int_0^\infty\frac{q^4}{\epsilon_h(q,a)}f_h(q)\;\mathrm{d}q. \tag{A.1928}$$

F1833 42_refinement_dark_sector_cosmology.tex:134 FAMILY_AUDITED

$$x:=\frac{q}{a_\star\sqrt{\nu}},\qquad y_h(a):=\frac{am_h}{a_\star\sqrt{\nu}}. \tag{A.1929}$$

F1834 42_refinement_dark_sector_cosmology.tex:140 FAMILY_AUDITED

$$I_\rho(y):=\int_0^\infty x^2\sqrt{x^2+y^2}\,\mathrm{e}^{-\pi x^2}\,\mathrm{d}x, \tag{A.1930}$$

$$I_p(y):=\int_0^\infty\frac{x^4}{\sqrt{x^2+y^2}}\,\mathrm{e}^{-\pi x^2}\,\mathrm{d}x, \tag{A.1931}$$

$$J_p(y):=\int_0^\infty\frac{x^4y^2}{(x^2+y^2)^{3/2}}\,\mathrm{e}^{-\pi x^2}\,\mathrm{d}x. \tag{A.1932}$$

F1835 42_refinement_dark_sector_cosmology.tex:152 FAMILY_AUDITED

$$\rho_h(a)=\frac{\mathcal{N}_\star d_h a_\star^4\nu^2}{2\pi^2a^4}\mathrm{e}^{-\pi m_h^2/\nu}I_\rho(y_h), \tag{A.1933}$$

$$p_h(a)=\frac{\mathcal{N}_\star d_h a_\star^4\nu^2}{6\pi^2a^4}\mathrm{e}^{-\pi m_h^2/\nu}I_p(y_h). \tag{A.1934}$$

F1836 42_refinement_dark_sector_cosmology.tex:162 FAMILY_AUDITED

$$I_\rho(y)=\frac{y^4}{2}\Gamma\left(\frac{3}{2}\right)U\left(\frac{3}{2},3,\pi y^2\right), \tag{A.1935}$$

$$I_p(y)=\frac{y^4}{2}\Gamma\left(\frac{5}{2}\right)U\left(\frac{5}{2},3,\pi y^2\right). \tag{A.1936}$$

F1837 42_refinement_dark_sector_cosmology.tex:172 FAMILY_AUDITED

$$\rho_h(a)=\frac{\mathcal{N}_\star d_h m_h^4}{4\pi^2}\mathrm{e}^{-\pi m_h^2/\nu}\Gamma\left(\frac{3}{2}\right)U\left(\frac{3}{2},3,\pi y_h^2\right), \tag{A.1937}$$

$$p_h(a)=\frac{\mathcal{N}_\star d_h m_h^4}{12\pi^2}\mathrm{e}^{-\pi m_h^2/\nu}\Gamma\left(\frac{5}{2}\right)U\left(\frac{5}{2},3,\pi y_h^2\right). \tag{A.1938}$$

F1838 42_refinement_dark_sector_cosmology.tex:187 FAMILY_AUDITED

$$\rho_D := \sum_{h\geq 1} \rho_h, \qquad p_D := \sum_{h\geq 1} p_h, \qquad w_D := \frac{p_D}{\rho_D}. \tag{A.1939}$$

F1839 42_refinement_dark_sector_cosmology.tex:195 FAMILY_AUDITED

$$\frac{\mathrm{d}\epsilon_h}{\mathrm{d}N} = \frac{a^2 m_h^2}{\epsilon_h} = \epsilon_h - \frac{q^2}{\epsilon_h}, \tag{A.1940}$$

F1840 42_refinement_dark_sector_cosmology.tex:201 FAMILY_AUDITED

$$\frac{\mathrm{d}\rho_h}{\mathrm{d}N} = -4\rho_h + \frac{d_h}{2\pi^2 a^4}\int q^2\left(\epsilon_h - \frac{q^2}{\epsilon_h}\right) f_h \,\mathrm{d}q \tag{A.1941}$$

$$= -4\rho_h + \rho_h - 3p_h \tag{A.1942}$$

$$= -3(\rho_h + p_h). \tag{A.1943}$$

F1841 42_refinement_dark_sector_cosmology.tex:214 FAMILY_AUDITED

$$\boxed{\frac{\mathrm{d}\rho_D}{\mathrm{d}N} + 3(\rho_D + p_D) = 0.} \tag{A.1944}$$

F1842 42_refinement_dark_sector_cosmology.tex:223 FAMILY_AUDITED

$$w_h(a) = \frac{I_p(y_h)}{3I_\rho(y_h)}. \tag{A.1945}$$

F1843 42_refinement_dark_sector_cosmology.tex:227 FAMILY_AUDITED

$$\frac{\mathrm{d}p_h}{\mathrm{d}N} = -\frac{\mathcal{C}_h(a)}{3}\left[4I_p(y_h) + J_p(y_h)\right], \tag{A.1946}$$

F1844 42_refinement_dark_sector_cosmology.tex:233 FAMILY_AUDITED

$$\mathcal{C}_h(a) := \frac{\mathcal{N}_\star d_h a_\star^4 \nu^2}{2\pi^2 a^4}\mathrm{e}^{-\pi m_h^2/\nu}. \tag{A.1947}$$

F1845 42_refinement_dark_sector_cosmology.tex:239 FAMILY_AUDITED

$$\frac{\mathrm{d}\rho_h}{\mathrm{d}N} = -\mathcal{C}_h(a)\left[3I_\rho(y_h) + I_p(y_h)\right], \tag{A.1948}$$

F1846 42_refinement_dark_sector_cosmology.tex:244 FAMILY_AUDITED

$$\boxed{c_{a,h}^2 := \frac{\mathrm{d}p_h/\,\mathrm{d}N}{\mathrm{d}\rho_h/\,\mathrm{d}N} = \frac{4I_p(y_h) + J_p(y_h)}{3\left[3I_\rho(y_h) + I_p(y_h)\right]}.} \tag{A.1949}$$

F1847 42_refinement_dark_sector_cosmology.tex:254 FAMILY_AUDITED

$$w_h \longrightarrow \frac{1}{3}, \qquad c_{a,h}^2 \longrightarrow \frac{1}{3}, \qquad \rho_h \propto a^{-4}. \tag{A.1950}$$

F1848 42_refinement_dark_sector_cosmology.tex:262 FAMILY_AUDITED

$$w_h(a) = \frac{a_\star^2 \nu}{2\pi a^2 m_h^2} + \mathcal{O}(y_h^{-4}), \tag{A.1951}$$

$$c_{a,h}^2 = \frac{5}{3} w_h + \mathcal{O}(y_h^{-4}), \tag{A.1952}$$

$$\rho_h(a) = m_h n_h(a)\left[1 + \mathcal{O}(y_h^{-2})\right], \tag{A.1953}$$

F1849 42_refinement_dark_sector_cosmology.tex:274 FAMILY_AUDITED

$$\boxed{n_h(a) = \frac{\mathcal{N}_\star d_h a_\star^3 \nu^{3/2}}{8\pi^3 a^3}\mathrm{e}^{-\pi m_h^2/\nu}.} \tag{A.1954}$$

F1850 42_refinement_dark_sector_cosmology.tex:286 FAMILY_AUDITED

$$\rho_D(1) = 3M_{\mathrm{Pl}}^2 H_0^2 \Omega_{D0}. \tag{A.1955}$$

F1851 42_refinement_dark_sector_cosmology.tex:290 FAMILY_AUDITED

$$\mathcal{N}_\star = \frac{3M_{\rm Pl}^2 H_0^2 \Omega_{D0}}{\widetilde{\rho}_D(1)}. \tag{A.1956}$$

F1852 42_refinement_dark_sector_cosmology.tex:296 FAMILY_AUDITED

$$\boxed{E(a)^2 = \Omega_{r0} a^{-4} + \Omega_{b0} a^{-3} + \Omega_{D0} \frac{\rho_D(a)}{\rho_D(1)} + \Omega_{\Lambda 0}.} \tag{A.1957}$$

F1853 42_refinement_dark_sector_cosmology.tex:306 FAMILY_AUDITED

$$\frac{\mathrm{d}\log H}{\mathrm{d}\log a} = -\frac{1}{2}\left[4\Omega_r(a) + 3\Omega_b(a) + 3(1 + w_D(a))\Omega_D(a)\right]. \tag{A.1958}$$

F1854 42_refinement_dark_sector_cosmology.tex:317 FAMILY_AUDITED

$$p = 2, \qquad \sigma = 1, \qquad H_{\max} = 8, \qquad \Lambda = \frac{1}{2}, \tag{A.1959}$$

$$\nu = 2\pi, \qquad a_\star = 10^{-7}, \qquad \Omega_{r0} = 9 \times 10^{-5}, \qquad \Omega_{b0} = 0.049, \tag{A.1960}$$

$$\Omega_{D0} = 0.261, \qquad \Omega_{\Lambda 0} = 1 - \Omega_{r0} - \Omega_{b0} - \Omega_{D0}. \tag{A.1961}$$

F1855 42_refinement_dark_sector_cosmology.tex:344 FAMILY_AUDITED

$$w_D(a_\star) = 0.1961573, \qquad w_D(10^{-6}) = 0.0062439, \tag{A.1962}$$

$$w_D(10^{-5}) = 6.4338 \times 10^{-5}, \qquad w_D(10^{-3}) = 6.4358 \times 10^{-9}. \tag{A.1963}$$

F1856 42_refinement_dark_sector_cosmology.tex:367 FAMILY_AUDITED

$$N_H = \sum_{h=1}^{H} (p-1) p^{h-1} = p^H - 1. \tag{A.1964}$$

F1857 42_refinement_dark_sector_cosmology.tex:373 FAMILY_AUDITED

$$m_H = \Lambda p^H, \tag{A.1965}$$

F1858 42_refinement_dark_sector_cosmology.tex:377 FAMILY_AUDITED

$$\boxed{N_H = \frac{m_H}{\Lambda} - 1.} \tag{A.1966}$$

F1859 42_refinement_dark_sector_cosmology.tex:383 FAMILY_AUDITED

$$\boxed{\rho_{\rm DOS}(m) = \frac{\mathrm{d}N}{\mathrm{d}m} = \frac{1}{\Lambda}.} \tag{A.1967}$$

F1860 42_refinement_dark_sector_cosmology.tex:391 FAMILY_AUDITED

$$T_h \propto d_h m_h \mathrm{e}^{-\pi m_h^2/\nu}. \tag{A.1968}$$

F1861 42_refinement_dark_sector_cosmology.tex:395 FAMILY_AUDITED

$$\frac{T_{h+1}}{T_h} = p^2 \exp\left[-\frac{\pi \Lambda^2}{\nu}(p^2 - 1) p^{2h}\right]. \tag{A.1969}$$

F1862 42_refinement_dark_sector_cosmology.tex:402 FAMILY_AUDITED

$$H = 1: \quad 4.740634 \times 10^{-1}, \tag{A.1970}$$

$$H = 2: \quad 4.654197 \times 10^{-3}, \tag{A.1971}$$

$$H = 3: \quad 7.030165 \times 10^{-13}, \tag{A.1972}$$

$$H \geq 4: \quad 2.3 \times 10^{-16} \text{ or smaller.} \tag{A.1973}$$

F1863 42_refinement_dark_sector_cosmology.tex:419 FAMILY_AUDITED

$$K_{p,H}(k) := \sum_{h=1}^{H} \frac{d_h}{k^2 + \Lambda^2 p^{2h}}. \tag{A.1974}$$

F1864 42_refinement_dark_sector_cosmology.tex:425 FAMILY_AUDITED

$$K_p(pk) = \sum_h \frac{(p-1)p^{h-1}}{p^2k^2 + \Lambda^2 p^{2h}} \tag{A.1975}$$

$$= p^{-1} K_p(k). \tag{A.1976}$$

F1865 42_refinement_dark_sector_cosmology.tex:434 FAMILY_AUDITED

$$K_p(k) = \frac{1}{k} F_p\!\left(\frac{\log k}{\log p}\right), \tag{A.1977}$$

F1866 42_refinement_dark_sector_cosmology.tex:449 FAMILY_AUDITED

$$a_i = 10^{-3}, \tag{A.1978}$$

F1867 42_refinement_dark_sector_cosmology.tex:453 FAMILY_AUDITED

$$\max_h w_h(a_i) = 9.937370 \times 10^{-9}, \qquad \max_h c_{a,h}^2(a_i) = 1.656228 \times 10^{-8}. \tag{A.1979}$$

F1868 42_refinement_dark_sector_cosmology.tex:459 FAMILY_AUDITED

$$\delta_b'' + \left(2 + \frac{\mathrm{d}\log H}{\mathrm{d}N}\right)\delta_b' = \frac{3}{2}\mathcal{S}, \tag{A.1980}$$

$$\delta_h'' + \left(2 + \frac{\mathrm{d}\log H}{\mathrm{d}N}\right)\delta_h' + \frac{c_{a,h}^2 k^2}{a^2 H^2}\delta_h = \frac{3}{2}\mathcal{S}, \tag{A.1981}$$

F1869 42_refinement_dark_sector_cosmology.tex:470 FAMILY_AUDITED

$$\mathcal{S}(a,k) := \Omega_b(a)\delta_b(a,k) + \sum_h \Omega_h(a)\delta_h(a,k). \tag{A.1982}$$

F1870 42_refinement_dark_sector_cosmology.tex:476 FAMILY_AUDITED

$$\boxed{\delta_D(a,k) = \frac{\sum_h \rho_h(a)\delta_h(a,k)}{\rho_D(a)}.} \tag{A.1983}$$

F1871 42_refinement_dark_sector_cosmology.tex:485 FAMILY_AUDITED

$$\delta_b(a_i) = \delta_h(a_i) = a_i, \qquad \delta_b'(a_i) = \delta_h'(a_i) = a_i. \tag{A.1984}$$

F1872 42_refinement_dark_sector_cosmology.tex:513 FAMILY_AUDITED

$$\mathrm{d}s^2 = -(1+2\Psi)\,\mathrm{d}t^2 + a^2(1-2\Phi)\,\mathrm{d}\boldsymbol{x}^2. \tag{A.1985}$$

F1873 42_refinement_dark_sector_cosmology.tex:519 FAMILY_AUDITED

$$\Phi - \Psi = \mathcal{O}(w_D \Phi). \tag{A.1986}$$

F1874 42_refinement_dark_sector_cosmology.tex:523 FAMILY_AUDITED

$$\boxed{k^2(\Phi+\Psi) = -8\pi G a^2 \left[\rho_b\delta_b + \sum_h \rho_h\delta_h\right] + \mathcal{O}(w_D).} \tag{A.1987}$$

F1875 42_refinement_dark_sector_cosmology.tex:535 FAMILY_AUDITED

$$\Phi + \Psi = -\frac{3a^2}{k^2}\left[\Omega_{b0}a^{-3}\delta_b + \sum_h \rho_h\delta_h\right] + \mathcal{O}(w_D). \tag{A.1988}$$

F1876 42_refinement_dark_sector_cosmology.tex:554 FAMILY_AUDITED

$$m_h^2 = m_0^2 + \Lambda^2 p^{2h}, \qquad d_h = (p-1)p^{h-1}, \tag{A.1989}$$

$$\rho_D(a) = \sum_h \rho_h(a), \qquad p_D(a) = \sum_h p_h(a), \tag{A.1990}$$

$$w_D(a) = p_D/\rho_D, \qquad c_{a,h}^2 = \frac{\mathrm{d}p_h}{\mathrm{d}\rho_h}, \tag{A.1991}$$

$$\delta_D(a,k) = \frac{\sum_h \rho_h \delta_h}{\rho_D}, \qquad k^2(\Phi+\Psi) = -8\pi G a^2 \delta\rho_{\mathrm{resolved}} + \mathcal{O}(w_D). \tag{A.1992}$$

43_exact_quasienergy_closure.tex

F1877 43_exact_quasienergy_closure.tex:5 NEW_EXTENSION_PASS

$$\text{exact discrete dynamics} \not\Longrightarrow \text{new fundamental law without independent physical identification.} \tag{A.1993}$$

F1878 43_exact_quasienergy_closure.tex:13 NEW_EXTENSION_PASS

$$a_0 := m, \qquad a_1 := k_x, \qquad a_2 := k_y, \qquad a_3 := k_z, \tag{A.1994}$$

$$E^2 := \sum_{r=0}^{3} a_r^2, \qquad C_\Delta := \prod_{r=0}^{3} \cos(a_r \Delta), \tag{A.1995}$$

$$\cos(\omega_\Delta \Delta) = C_\Delta. \tag{A.1996}$$

F1879 43_exact_quasienergy_closure.tex:23 NEW_EXTENSION_PASS

$$\omega_\Delta = E + c_2 \Delta^2 + \mathcal{O}(\Delta^4), \tag{A.1997}$$

F1880 43_exact_quasienergy_closure.tex:27 NEW_EXTENSION_PASS

$$C_\Delta = 1 - \frac{E^2\Delta^2}{2} + \Delta^4 \left[\frac{1}{24} \sum_r a_r^4 + \frac{1}{4} \sum_{r<s} a_r^2 a_s^2 \right] + \mathcal{O}(\Delta^6), \tag{A.1998}$$

$$\cos(\omega_\Delta \Delta) = 1 - \frac{E^2\Delta^2}{2} + \Delta^4 \left[\frac{E^4}{24} - E c_2 \right] + \mathcal{O}(\Delta^6), \tag{A.1999}$$

$$E^4 = \sum_r a_r^4 + 2 \sum_{r<s} a_r^2 a_s^2. \tag{A.2000}$$

F1881 43_exact_quasienergy_closure.tex:45 NEW_EXTENSION_PASS

$$\boxed{E c_2 = -\frac{1}{6} \sum_{r<s} a_r^2 a_s^2} \tag{A.2001}$$

F1882 43_exact_quasienergy_closure.tex:51 NEW_EXTENSION_PASS

$$\boxed{\omega_\Delta = E - \frac{\Delta^2}{6E} \sum_{r<s} a_r^2 a_s^2 + \mathcal{O}(\Delta^4).} \tag{A.2002}$$

F1883 43_exact_quasienergy_closure.tex:60 NEW_EXTENSION_PASS

$$\omega_\Delta = E - c_2 \Delta^2 + \mathcal{O}(\Delta^4), \qquad E c_2 = \frac{1}{6} \sum_{r<s} a_r^2 a_s^2. \tag{A.2003}$$

F1884 43_exact_quasienergy_closure.tex:67 NEW_EXTENSION_PASS

$$x_r := a_r^2, \tag{A.2004}$$

$$e_1 := \sum_r x_r, \tag{A.2005}$$

$$e_2 := \sum_{r<s} x_r x_s, \tag{A.2006}$$

$$e_3 := \sum_{r<s<t} x_r x_s x_t, \tag{A.2007}$$

$$e_4 := x_0 x_1 x_2 x_3. \tag{A.2008}$$

F1885 43_exact_quasienergy_closure.tex:75 NEW_EXTENSION_PASS

$$\boxed{\begin{aligned}\omega_\Delta^2 = e_1 - \frac{e_2}{3}\Delta^2 + \frac{6e_3 - e_1 e_2}{45}\Delta^4 \\ - \frac{2e_1^2 e_2 - 9e_1 e_3 - 8e_2^2 + 51 e_4}{945}\Delta^6 + \mathcal{O}(\Delta^8).\end{aligned}} \tag{A.2009}$$

F1886 43_exact_quasienergy_closure.tex:89 NEW_EXTENSION_PASS

$$\boxed{\omega_\Delta = E - \frac{e_2}{6E}\Delta^2 + \left[\frac{6e_3 - E^2 e_2}{90E} - \frac{e_2^2}{72E^3}\right]\Delta^4 + \mathcal{O}(\Delta^6).} \tag{A.2010}$$

F1887 43_exact_quasienergy_closure.tex:101 NEW_EXTENSION_PASS

$$\cos\left(\Delta\sqrt{\sum_{j\geq 0} b_{2j}\Delta^{2j}}\right) = \prod_{r=0}^{3}\cos(a_r\Delta), \qquad b_0 = e_1, \tag{A.2011}$$

F1888 43_exact_quasienergy_closure.tex:114 NEW_EXTENSION_PASS

$$\sigma_\Delta(x) := \operatorname{sgn}[\cos(\Delta x)], \tag{A.2012}$$

$$\kappa_\Delta(x) := \frac{1}{\Delta}\sqrt{-2\log|\cos(\Delta x)|}. \tag{A.2013}$$

F1889 43_exact_quasienergy_closure.tex:121 NEW_EXTENSION_PASS

$$\cos(\Delta x) = \sigma_\Delta(x)\exp\left[-\frac{\Delta^2}{2}\kappa_\Delta(x)^2\right]. \tag{A.2014}$$

F1890 43_exact_quasienergy_closure.tex:127 NEW_EXTENSION_PASS

$$\sigma_\Delta(\omega_\Delta) = \prod_{r=0}^{3}\sigma_\Delta(a_r), \tag{A.2015}$$

$$\kappa_\Delta(\omega_\Delta)^2 = \sum_{r=0}^{3}\kappa_\Delta(a_r)^2. \tag{A.2016}$$

F1891 43_exact_quasienergy_closure.tex:136 NEW_EXTENSION_PASS

$$\mathcal{I}_\Delta := \left(-\frac{\pi}{2\Delta}, \frac{\pi}{2\Delta}\right), \tag{A.2017}$$

F1892 43_exact_quasienergy_closure.tex:141 NEW_EXTENSION_PASS

$$K_\Delta(x) := \operatorname{sgn}(x)\frac{\sqrt{-2\log\cos(\Delta x)}}{\Delta}. \tag{A.2018}$$

F1893 43_exact_quasienergy_closure.tex:147 NEW_EXTENSION_PASS

$$K_\Delta^{-1}(P) = \operatorname{sgn}(P)\frac{1}{\Delta}\arccos\left(\mathrm{e}^{-\Delta^2 P^2/2}\right). \tag{A.2019}$$

F1894 43_exact_quasienergy_closure.tex:154 NEW_EXTENSION_PASS

$$K_\Delta(x) = x + \frac{x^3\Delta^2}{12} + \frac{3x^5\Delta^4}{160} + \frac{209x^7\Delta^6}{40320} + \mathcal{O}(\Delta^8 x^9). \tag{A.2020}$$

F1895 43_exact_quasienergy_closure.tex:162 NEW_EXTENSION_PASS

$$\boxed{K_\Delta(\omega_\Delta)^2 - \sum_{i=1}^{3} K_\Delta(k_i)^2 = K_\Delta(m)^2.} \tag{A.2021}$$

F1896 43_exact_quasienergy_closure.tex:171 NEW_EXTENSION_PASS

$$\Lambda_\Delta(X) := -\frac{2}{\Delta^2}\log\cos(\Delta\sqrt{X}), \qquad 0 \le X < \frac{\pi^2}{4\Delta^2}. \tag{A.2022}$$

F1897 43_exact_quasienergy_closure.tex:179 NEW_EXTENSION_PASS

$$\Lambda_\Delta(\omega_\Delta^2) = \sum_{r=0}^{3} \Lambda_\Delta(a_r^2). \tag{A.2023}$$

F1898 43_exact_quasienergy_closure.tex:184 NEW_EXTENSION_PASS

$$X \oplus_\Delta Y := \Lambda_\Delta^{-1}\left[\Lambda_\Delta(X) + \Lambda_\Delta(Y)\right] \tag{A.2024}$$

F1899 43_exact_quasienergy_closure.tex:190 NEW_EXTENSION_PASS

$$X \oplus_\Delta Y = Y \oplus_\Delta X, \tag{A.2025}$$

$$(X \oplus_\Delta Y) \oplus_\Delta Z = X \oplus_\Delta (Y \oplus_\Delta Z). \tag{A.2026}$$

F1900 43_exact_quasienergy_closure.tex:196 NEW_EXTENSION_PASS

$$\boxed{\begin{aligned} X \oplus_\Delta Y = X + Y - \frac{\Delta^2}{3}XY - \frac{\Delta^4}{45}XY(X+Y) \\ - \frac{2\Delta^6}{945}XY(X-Y)^2 + \mathcal{O}(\Delta^8). \end{aligned}} \tag{A.2027}$$

F1901 43_exact_quasienergy_closure.tex:208 NEW_EXTENSION_PASS

$$\omega_\Delta^2 = m^2 \oplus_\Delta k_x^2 \oplus_\Delta k_y^2 \oplus_\Delta k_z^2. \tag{A.2028}$$

F1902 43_exact_quasienergy_closure.tex:217 NEW_EXTENSION_PASS

$$g(X) := -\log\cos\sqrt{X}, \qquad 0 \le X < \frac{\pi^2}{4}. \tag{A.2029}$$

F1903 43_exact_quasienergy_closure.tex:223 NEW_EXTENSION_PASS

$$g'(X) = \frac{\tan\sqrt{X}}{2\sqrt{X}} \tag{A.2030}$$

F1904 43_exact_quasienergy_closure.tex:227 NEW_EXTENSION_PASS

$$g(X+Y) \ge g(X) + g(Y). \tag{A.2031}$$

F1905 43_exact_quasienergy_closure.tex:231 NEW_EXTENSION_PASS

$$\cos(E\Delta) \le \prod_r \cos(a_r\Delta), \tag{A.2032}$$

$$\prod_r \cos(a_r\Delta) \le \min_r \cos(|a_r|\Delta), \tag{A.2033}$$

F1906 43_exact_quasienergy_closure.tex:238 NEW_EXTENSION_PASS

$$\boxed{\max_r |a_r| \le \omega_\Delta \le \sqrt{\sum_r a_r^2}.} \tag{A.2034}$$

F1907 43_exact_quasienergy_closure.tex:248 NEW_EXTENSION_PASS

$$\mathcal{E} := K_\Delta(\omega), \tag{A.2035}$$

$$\mathcal{P}_i := K_\Delta(k_i), \tag{A.2036}$$
$$\mathcal{M} := K_\Delta(m). \tag{A.2037}$$

F1908 43_exact_quasienergy_closure.tex:254 NEW_EXTENSION_PASS

$$\mathcal{B}_\Lambda(p)_\mu := K_\Delta^{-1}\left[\Lambda_\mu{}^\nu K_\Delta(p_\nu)\right]. \tag{A.2038}$$

F1909 43_exact_quasienergy_closure.tex:262 NEW_EXTENSION_PASS

$$\mathcal{B}_{\Lambda_1}\circ\mathcal{B}_{\Lambda_2} = \mathcal{B}_{\Lambda_1\Lambda_2}, \tag{A.2039}$$
$$K_\Delta(\omega')^2 - \sum_i K_\Delta(k_i')^2 = \mathcal{M}^2. \tag{A.2040}$$

F1910 43_exact_quasienergy_closure.tex:270 NEW_EXTENSION_PASS

$$(p\boxplus_\Delta q)_\mu := K_\Delta^{-1}\left[K_\Delta(p_\mu)+K_\Delta(q_\mu)\right]. \tag{A.2041}$$

F1911 43_exact_quasienergy_closure.tex:276 NEW_EXTENSION_PASS

$$(p\boxplus_\Delta q)\boxplus_\Delta r = p\boxplus_\Delta (q\boxplus_\Delta r), \tag{A.2042}$$
$$\mathcal{B}_\Lambda(p\boxplus_\Delta q) = \mathcal{B}_\Lambda(p)\boxplus_\Delta \mathcal{B}_\Lambda(q). \tag{A.2043}$$

F1912 43_exact_quasienergy_closure.tex:283 NEW_EXTENSION_PASS

$$\mathrm{d}\mu(p) = \prod_{\mu=0}^{3} K_\Delta'(p_\mu)\,\mathrm{d}p_\mu, \tag{A.2044}$$

F1913 43_exact_quasienergy_closure.tex:288 NEW_EXTENSION_PASS

$$K_\Delta'(x) = \frac{|\tan(\Delta x)|}{\sqrt{-2\log\cos(\Delta x)}}, \qquad K_\Delta'(0)=1. \tag{A.2045}$$

F1914 43_exact_quasienergy_closure.tex:296 NEW_EXTENSION_PASS

$$\mathrm{d}s^2 = K_\Delta'(E)^2\,\mathrm{d}E^2 - \sum_i K_\Delta'(p_i)^2\,\mathrm{d}p_i^2. \tag{A.2046}$$

F1915 43_exact_quasienergy_closure.tex:305 NEW_EXTENSION_PASS

$$\Gamma_r^* = \Gamma_r, \qquad \Gamma_r^2 = I, \qquad \{\Gamma_r,\Gamma_s\} = 0 \quad (r\neq s). \tag{A.2047}$$

F1916 43_exact_quasienergy_closure.tex:314 NEW_EXTENSION_PASS

$$W_r := \mathrm{e}^{-\mathrm{i}\theta_r\Gamma_r/2} \tag{A.2048}$$

F1917 43_exact_quasienergy_closure.tex:318 NEW_EXTENSION_PASS

$$\mathcal{U}_n := W_0W_1\cdots W_{n-1}\mathrm{e}^{-\mathrm{i}\theta_n\Gamma_n}W_{n-1}\cdots W_1W_0. \tag{A.2049}$$

F1918 43_exact_quasienergy_closure.tex:325 NEW_EXTENSION_PASS

$$\mathrm{e}^{-\mathrm{i}\theta_r\Gamma_r/2}\Gamma_s\mathrm{e}^{-\mathrm{i}\theta_r\Gamma_r/2} = \Gamma_s, \tag{A.2050}$$

F1919 43_exact_quasienergy_closure.tex:332 NEW_EXTENSION_PASS

$$\mathrm{e}^{-\mathrm{i}\theta_r\Gamma_r/2}I\mathrm{e}^{-\mathrm{i}\theta_r\Gamma_r/2} = \cos\theta_r\,I - \mathrm{i}\sin\theta_r\,\Gamma_r. \tag{A.2051}$$

F1920 43_exact_quasienergy_closure.tex:340 NEW_EXTENSION_PASS

$$\boxed{\mathcal{U}_n = C_nI - \mathrm{i}\sum_{r=0}^{n} D_r\Gamma_r,} \tag{A.2052}$$

F1921 43_exact_quasienergy_closure.tex:346 NEW_EXTENSION_PASS

$$C_n := \prod_{r=0}^{n} \cos\theta_r, \tag{A.2053}$$

$$D_r := \sin\theta_r \prod_{s=r+1}^{n} \cos\theta_s. \tag{A.2054}$$

F1922 43_exact_quasienergy_closure.tex:352 NEW_EXTENSION_PASS

$$C_n^2 + \sum_{r=0}^{n} D_r^2 = 1. \tag{A.2055}$$

F1923 43_exact_quasienergy_closure.tex:358 NEW_EXTENSION_PASS

$$\Gamma_0 := \alpha_x, \qquad \Gamma_1 := \alpha_y, \qquad \Gamma_2 := \alpha_z, \qquad \Gamma_3 := \beta, \tag{A.2056}$$

$$\theta_0 := k_x\Delta, \qquad \theta_1 := k_y\Delta, \qquad \theta_2 := k_z\Delta, \qquad \theta_3 := m\Delta. \tag{A.2057}$$

F1924 43_exact_quasienergy_closure.tex:369 NEW_EXTENSION_PASS

$$\boxed{\begin{aligned} U_\Delta^{(xyz)} &= \mathrm{e}^{-\mathrm{i}k_x\Delta\alpha_x/2}\mathrm{e}^{-\mathrm{i}k_y\Delta\alpha_y/2}\mathrm{e}^{-\mathrm{i}k_z\Delta\alpha_z/2}\mathrm{e}^{-\mathrm{i}m\Delta\beta} \\ &\quad\times \mathrm{e}^{-\mathrm{i}k_z\Delta\alpha_z/2}\mathrm{e}^{-\mathrm{i}k_y\Delta\alpha_y/2}\mathrm{e}^{-\mathrm{i}k_x\Delta\alpha_x/2}. \end{aligned}} \tag{A.2058}$$

F1925 43_exact_quasienergy_closure.tex:385 NEW_EXTENSION_PASS

$$U_\Delta^{(xyz)} = CI - \mathrm{i}\left(D_x\alpha_x + D_y\alpha_y + D_z\alpha_z + D_m\beta\right), \tag{A.2059}$$

F1926 43_exact_quasienergy_closure.tex:392 NEW_EXTENSION_PASS

$$C = \cos(m\Delta)\cos(k_x\Delta)\cos(k_y\Delta)\cos(k_z\Delta), \tag{A.2060}$$

$$D_x = \sin(k_x\Delta)\cos(k_y\Delta)\cos(k_z\Delta)\cos(m\Delta), \tag{A.2061}$$

$$D_y = \sin(k_y\Delta)\cos(k_z\Delta)\cos(m\Delta), \tag{A.2062}$$

$$D_z = \sin(k_z\Delta)\cos(m\Delta), \tag{A.2063}$$

$$D_m = \sin(m\Delta). \tag{A.2064}$$

F1927 43_exact_quasienergy_closure.tex:410 NEW_EXTENSION_PASS

$$R := \sqrt{1 - C^2}, \qquad \widehat{\mathcal{D}} := \frac{D_x\alpha_x + D_y\alpha_y + D_z\alpha_z + D_m\beta}{R}. \tag{A.2065}$$

F1928 43_exact_quasienergy_closure.tex:417 NEW_EXTENSION_PASS

$$\widehat{\mathcal{D}}^{\,2} = I, \tag{A.2066}$$

$$\Pi_\pm := \frac{1}{2}(I \pm \widehat{\mathcal{D}}), \tag{A.2067}$$

$$U_\Delta \Pi_\pm = \mathrm{e}^{\mp \mathrm{i}\omega_\Delta \Delta}\Pi_\pm, \tag{A.2068}$$

$$\mathrm{spec}(U_\Delta) = \left\{\mathrm{e}^{-\mathrm{i}\omega_\Delta\Delta}, \mathrm{e}^{-\mathrm{i}\omega_\Delta\Delta}, \mathrm{e}^{+\mathrm{i}\omega_\Delta\Delta}, \mathrm{e}^{+\mathrm{i}\omega_\Delta\Delta}\right\}, \tag{A.2069}$$

$$\cos(\omega_\Delta\Delta) = C. \tag{A.2070}$$

F1929 43_exact_quasienergy_closure.tex:432 NEW_EXTENSION_PASS

$$\boxed{H_{\mathrm{eff}}(\boldsymbol{k}) = \frac{\omega_\Delta}{\sqrt{1-C^2}}\left(D_x\alpha_x + D_y\alpha_y + D_z\alpha_z + D_m\beta\right).} \tag{A.2071}$$

F1930 43_exact_quasienergy_closure.tex:442 NEW_EXTENSION_PASS

$$U_\Delta^2 - 2CU_\Delta + I = 0 \tag{A.2072}$$

F1931 43_exact_quasienergy_closure.tex:446 NEW_EXTENSION_PASS

$$\Psi_{n+1} + \Psi_{n-1} = 2C\Psi_n. \tag{A.2073}$$

F1932 43_exact_quasienergy_closure.tex:451 NEW_EXTENSION_PASS

$$D_i = k_i\Delta + \mathcal{O}(\Delta^3), \tag{A.2074}$$

$$D_m = m\Delta + \mathcal{O}(\Delta^3), \tag{A.2075}$$

$$R = E\Delta + \mathcal{O}(\Delta^3), \tag{A.2076}$$

$$\Pi_\pm \longrightarrow \frac{1}{2}\left[I \pm \frac{\boldsymbol{\alpha}\cdot\boldsymbol{k} + \beta m}{E}\right], \tag{A.2077}$$

$$H_{\text{eff}} = \boldsymbol{\alpha}\cdot\boldsymbol{k} + \beta m + \mathcal{O}(\Delta^2). \tag{A.2078}$$

F1933 43_exact_quasienergy_closure.tex:465 NEW_EXTENSION_PASS

$$H_{\text{eff}} = \sum_r a_r\Gamma_r + \Delta^2 Q_{\text{ord}} + \mathcal{O}(\Delta^4), \tag{A.2079}$$

F1934 43_exact_quasienergy_closure.tex:472 NEW_EXTENSION_PASS

$$\boxed{Q_{\text{ord}} = \sum_r a_r\left[\frac{1}{6}\sum_{s<r} a_s^2 - \frac{1}{3}\sum_{s>r} a_s^2\right]\Gamma_r.} \tag{A.2080}$$

F1935 43_exact_quasienergy_closure.tex:483 NEW_EXTENSION_PASS

$$\frac{1}{E}\sum_r a_r^2\left[\frac{1}{6}\sum_{s<r} a_s^2 - \frac{1}{3}\sum_{s>r} a_s^2\right] = -\frac{e_2}{6E}. \tag{A.2081}$$

F1936 43_exact_quasienergy_closure.tex:492 NEW_EXTENSION_PASS

$$\boxed{\overline{Q} = \frac{m|\boldsymbol{k}|^2}{6}\beta - \sum_i k_i\left[\frac{m^2}{3} + \frac{|\boldsymbol{k}|^2 - k_i^2}{12}\right]\alpha_i.} \tag{A.2082}$$

F1937 43_exact_quasienergy_closure.tex:508 NEW_EXTENSION_PASS

$$q_r := \sin^2\theta_r, \tag{A.2083}$$

$$C := \prod_r \cos\theta_r, \tag{A.2084}$$

$$\rho_r := \int_0^1 \prod_{s\neq r}(1 - tq_s)\,\mathrm{d}t, \tag{A.2085}$$

$$d_r := \sin\theta_r\sqrt{\rho_r}. \tag{A.2086}$$

F1938 43_exact_quasienergy_closure.tex:515 NEW_EXTENSION_PASS

$$\rho_r = 1 - \frac{1}{2}\sum_{s\neq r} q_s + \frac{1}{3}\sum_{\substack{s<t\\ s,t\neq r}} q_s q_t - \frac{1}{4}\prod_{s\neq r} q_s. \tag{A.2087}$$

F1939 43_exact_quasienergy_closure.tex:522 NEW_EXTENSION_PASS

$$-\frac{\mathrm{d}}{\mathrm{d}t}\prod_s(1 - tq_s) = \sum_r q_r\prod_{s\neq r}(1 - tq_s), \tag{A.2088}$$

F1940 43_exact_quasienergy_closure.tex:527 NEW_EXTENSION_PASS

$$\sum_r d_r^2 = 1 - C^2. \tag{A.2089}$$

F1941 43_exact_quasienergy_closure.tex:531 NEW_EXTENSION_PASS

$$U_{\text{sym}} = CI - \mathrm{i}\sum_r d_r\Gamma_r \tag{A.2090}$$

F1942 43_exact_quasienergy_closure.tex:538 NEW_EXTENSION_PASS

$$\mathbb{U}_{\mathrm{loc}} := \bigoplus_{\pi \in S_3} U_\pi. \tag{A.2091}$$

F1943 43_exact_quasienergy_closure.tex:559 NEW_EXTENSION_PASS

$$\boxed{\frac{\partial \omega_\Delta}{\partial k_i} = \frac{\sin(k_i\Delta)\cos(m\Delta)\prod_{j\neq i}\cos(k_j\Delta)}{\sin(\omega_\Delta\Delta)}.} \tag{A.2092}$$

F1944 43_exact_quasienergy_closure.tex:569 NEW_EXTENSION_PASS

$$q_0 := \sin^2(m\Delta), \qquad q_i := \sin^2(k_i\Delta). \tag{A.2093}$$

F1945 43_exact_quasienergy_closure.tex:575 NEW_EXTENSION_PASS

$$\|\nabla_{\boldsymbol{k}}\omega_\Delta\|^2 = \frac{\displaystyle\sum_{i=1}^{3} q_i \prod_{r\neq i}(1-q_r)}{\displaystyle 1-\prod_{r=0}^{3}(1-q_r)}. \tag{A.2094}$$

F1946 43_exact_quasienergy_closure.tex:585 NEW_EXTENSION_PASS

$$\boxed{\|\nabla_{\boldsymbol{k}}\omega_\Delta\| \le 1.} \tag{A.2095}$$

F1947 43_exact_quasienergy_closure.tex:593 NEW_EXTENSION_PASS

$$\omega_\Delta(\mathbf{0}) = m \tag{A.2096}$$

F1948 43_exact_quasienergy_closure.tex:597 NEW_EXTENSION_PASS

$$\omega_\Delta(\boldsymbol{k}) = m + \frac{\Delta}{2}\cot(m\Delta)|\boldsymbol{k}|^2 + \mathcal{O}(|\boldsymbol{k}|^4). \tag{A.2097}$$

F1949 43_exact_quasienergy_closure.tex:603 NEW_EXTENSION_PASS

$$\boxed{M_{\mathrm{kin}} = \frac{\tan(m\Delta)}{\Delta} = m + \frac{m^3\Delta^2}{3} + \frac{2m^5\Delta^4}{15} + \mathcal{O}(\Delta^6).} \tag{A.2098}$$

F1950 43_exact_quasienergy_closure.tex:613 NEW_EXTENSION_PASS

$$0 < m\Delta < \frac{\pi}{2}. \tag{A.2099}$$

F1951 43_exact_quasienergy_closure.tex:618 NEW_EXTENSION_PASS

$$H_4(\boldsymbol{k}) := \sum_i k_i^4 - \frac{3}{5}|\boldsymbol{k}|^4. \tag{A.2100}$$

F1952 43_exact_quasienergy_closure.tex:623 NEW_EXTENSION_PASS

$$\boxed{\omega_\Delta^2 = m^2 + |\boldsymbol{k}|^2 - \frac{\Delta^2}{3}m^2|\boldsymbol{k}|^2 - \frac{\Delta^2}{15}|\boldsymbol{k}|^4 + \frac{\Delta^2}{6}H_4(\boldsymbol{k}) + \mathcal{O}(\Delta^4).} \tag{A.2101}$$

F1953 43_exact_quasienergy_closure.tex:634 NEW_EXTENSION_PASS

$$\boldsymbol{k} = (k,0,0) \quad\Longrightarrow\quad \omega_\Delta = |k| \tag{A.2102}$$

F1954 43_exact_quasienergy_closure.tex:640 NEW_EXTENSION_PASS

$$\boldsymbol{k} = \frac{k}{\sqrt{3}}(1,1,1) \tag{A.2103}$$

F1955 43_exact_quasienergy_closure.tex:644 NEW_EXTENSION_PASS

$$\omega_\Delta^2 = k^2 - \frac{k^4\Delta^2}{9} - \frac{k^6\Delta^4}{405} + \frac{k^8\Delta^6}{1701} + \mathcal{O}(\Delta^8). \tag{A.2104}$$

F1956 43_exact_quasienergy_closure.tex:654 NEW_EXTENSION_PASS

$$S := \sum_{r=0}^{3} \epsilon_r a_r. \tag{A.2105}$$

F1957 43_exact_quasienergy_closure.tex:658 NEW_EXTENSION_PASS

$$\boxed{\prod_{r=0}^{3} \cos(a_r \Delta) = \mathbb{E}_{\boldsymbol{\epsilon}} \left[\cos(\Delta S)\right].} \tag{A.2106}$$

F1958 43_exact_quasienergy_closure.tex:666 NEW_EXTENSION_PASS

$$\mathbb{E}[S^2] = e_1, \tag{A.2107}$$
$$\mathbb{E}[S^4] = e_1^2 + 4e_2, \tag{A.2108}$$
$$\mathrm{Var}(S^2) = 4e_2. \tag{A.2109}$$

F1959 43_exact_quasienergy_closure.tex:672 NEW_EXTENSION_PASS

$$\boxed{\omega_\Delta^2 = E^2 - \frac{\Delta^2}{12} \mathrm{Var}_{\boldsymbol{\epsilon}}(S^2) + \mathcal{O}(\Delta^4).} \tag{A.2110}$$

F1960 43_exact_quasienergy_closure.tex:684 NEW_EXTENSION_PASS

$$\mathbb{E}[V] = 0, \qquad \mathbb{E}[V_i V_j] = a\delta_{ij}. \tag{A.2111}$$

F1961 43_exact_quasienergy_closure.tex:690 NEW_EXTENSION_PASS

$$S_h(\boldsymbol{k}) := \mathbb{E}\left[\cos(h\boldsymbol{k}\cdot V)\right], \tag{A.2112}$$
$$\tau := h\sqrt{a}, \tag{A.2113}$$
$$\cos(\tau\omega) := \cos(\tau m) S_h(\boldsymbol{k}). \tag{A.2114}$$

F1962 43_exact_quasienergy_closure.tex:696 NEW_EXTENSION_PASS

$$\nabla_{\boldsymbol{k}}\omega = \frac{h\cos(\tau m)\mathbb{E}[V\sin(h\boldsymbol{k}\cdot V)]}{\tau\sqrt{1-\cos^2(\tau m)S_h(\boldsymbol{k})^2}}. \tag{A.2115}$$

F1963 43_exact_quasienergy_closure.tex:706 NEW_EXTENSION_PASS

$$|n\cdot\mathbb{E}[V\sin(h\boldsymbol{k}\cdot V)]|^2 \le \mathbb{E}[(n\cdot V)^2]\mathbb{E}[\sin^2(h\boldsymbol{k}\cdot V)] \tag{A.2116}$$
$$\le a[1 - S_h(\boldsymbol{k})^2]. \tag{A.2117}$$

F1964 43_exact_quasienergy_closure.tex:716 NEW_EXTENSION_PASS

$$\boxed{\|\nabla_{\boldsymbol{k}}\omega\|^2 \le \frac{\cos^2(\tau m)[1-S_h^2]}{1-\cos^2(\tau m)S_h^2} \le 1.} \tag{A.2118}$$

F1965 43_exact_quasienergy_closure.tex:730 NEW_EXTENSION_PASS

$$\beta(\widehat{\boldsymbol{k}}) := \frac{\mathbb{E}[(\widehat{\boldsymbol{k}}\cdot V)^4]}{a^2}. \tag{A.2119}$$

F1966 43_exact_quasienergy_closure.tex:737 NEW_EXTENSION_PASS

$$S_h(\boldsymbol{k}) = 1 - \frac{\tau^2|\boldsymbol{k}|^2}{2} + \frac{\tau^4}{24}\beta(\widehat{\boldsymbol{k}})|\boldsymbol{k}|^4 + \mathcal{O}(\tau^6), \tag{A.2120}$$

F1967 43_exact_quasienergy_closure.tex:745 NEW_EXTENSION_PASS

$$\boxed{\omega^2 = m^2 + |\boldsymbol{k}|^2 - \tau^2\left[\frac{1}{3}m^2|\boldsymbol{k}|^2 + \frac{\beta(\widehat{\boldsymbol{k}})-1}{12}|\boldsymbol{k}|^4\right] + \mathcal{O}(\tau^4).} \tag{A.2121}$$

F1968 43_exact_quasienergy_closure.tex:757 NEW_EXTENSION_PASS

$$\beta(\widehat{\boldsymbol{k}}) \geq 1. \tag{A.2122}$$

F1969 43_exact_quasienergy_closure.tex:763 NEW_EXTENSION_PASS

$$\mathbb{E}[V_i V_j V_k V_l] = b(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}). \tag{A.2123}$$

F1970 43_exact_quasienergy_closure.tex:771 NEW_EXTENSION_PASS

$$\beta = \frac{3b}{a^2}, \tag{A.2124}$$

$$\mathbb{E}|V|^2 = da, \tag{A.2125}$$

$$\mathbb{E}|V|^4 = d(d+2)b. \tag{A.2126}$$

F1971 43_exact_quasienergy_closure.tex:777 NEW_EXTENSION_PASS

$$\mathbb{E}|V|^4 \geq (\mathbb{E}|V|^2)^2, \tag{A.2127}$$

F1972 43_exact_quasienergy_closure.tex:781 NEW_EXTENSION_PASS

$$\boxed{\beta \geq \frac{3d}{d+2}.} \tag{A.2128}$$

F1973 43_exact_quasienergy_closure.tex:787 NEW_EXTENSION_PASS

$$\beta \geq \frac{9}{5}, \tag{A.2129}$$

F1974 43_exact_quasienergy_closure.tex:791 NEW_EXTENSION_PASS

$$\omega^2 \leq |\boldsymbol{k}|^2 - \frac{\tau^2}{15}|\boldsymbol{k}|^4 + \mathcal{O}(\tau^4) \tag{A.2130}$$

F1975 43_exact_quasienergy_closure.tex:800 NEW_EXTENSION_PASS

$$c_i := \cos(hk_i). \tag{A.2131}$$

F1976 43_exact_quasienergy_closure.tex:804 NEW_EXTENSION_PASS

$$\boxed{S_{\mathrm{cub}}(\boldsymbol{k}) = \frac{1}{6}\left[c_x + c_y + c_z + c_x c_y + c_x c_z + c_y c_z\right].} \tag{A.2132}$$

F1977 43_exact_quasienergy_closure.tex:814 NEW_EXTENSION_PASS

$$\mathbb{E}[V_i V_j] = \frac{1}{2}\delta_{ij}, \tag{A.2133}$$

$$\mathbb{E}[(\boldsymbol{k}\cdot V)^4] = \frac{1}{2}|\boldsymbol{k}|^4. \tag{A.2134}$$

F1978 43_exact_quasienergy_closure.tex:820 NEW_EXTENSION_PASS

$$\tau = \frac{h}{\sqrt{2}}, \tag{A.2135}$$

$$\beta = 2, \tag{A.2136}$$

$$\omega^2 = m^2 + |\boldsymbol{k}|^2 - \tau^2\left[\frac{1}{3}m^2|\boldsymbol{k}|^2 + \frac{1}{12}|\boldsymbol{k}|^4\right] + \mathcal{O}(\tau^4). \tag{A.2137}$$

F1979 43_exact_quasienergy_closure.tex:834 NEW_EXTENSION_PASS

$$S_{\mathrm{cub}} = 1 \tag{A.2138}$$

F1980 43_exact_quasienergy_closure.tex:841 NEW_EXTENSION_PASS

$$a := \frac{\Delta}{2}. \tag{A.2139}$$

F1981 43_exact_quasienergy_closure.tex:845 NEW_EXTENSION_PASS

$$L_i(x) \mapsto g(x + a\widehat{i})L_i(x)g(x)^{-1}. \tag{A.2140}$$

F1982 43_exact_quasienergy_closure.tex:850 NEW_EXTENSION_PASS

$$P_i^{\pm} := \frac{1}{2}(I \pm \alpha_i). \tag{A.2141}$$

F1983 43_exact_quasienergy_closure.tex:854 NEW_EXTENSION_PASS

$$\boxed{\begin{aligned}(\mathsf{S}_i\Psi)(x) = P_i^{+} L_i(x - a\widehat{i})\Psi(x - a\widehat{i}) \\ + P_i^{-} L_i(x)^{-1}\Psi(x + a\widehat{i}).\end{aligned}} \tag{A.2142}$$

F1984 43_exact_quasienergy_closure.tex:864 NEW_EXTENSION_PASS

$$\mathsf{S}_i^{*}\mathsf{S}_i = I, \tag{A.2143}$$

$$\mathsf{S}_i[L^{g}] = \mathcal{G}\mathsf{S}_i[L]\mathcal{G}^{-1}. \tag{A.2144}$$

F1985 43_exact_quasienergy_closure.tex:869 NEW_EXTENSION_PASS

$$\mathsf{S}_i(\boldsymbol{k}) = \mathrm{e}^{-\mathrm{i}ak_i\alpha_i} = \mathrm{e}^{-\mathrm{i}k_i\Delta\alpha_i/2}. \tag{A.2145}$$

F1986 43_exact_quasienergy_closure.tex:875 NEW_EXTENSION_PASS

$$\mathsf{M} := \mathrm{e}^{-\mathrm{i}m\Delta\beta}, \tag{A.2146}$$

F1987 43_exact_quasienergy_closure.tex:879 NEW_EXTENSION_PASS

$$\boxed{\mathsf{U}_\Delta = \mathsf{S}_x\mathsf{S}_y\mathsf{S}_z\mathsf{M}\mathsf{S}_z\mathsf{S}_y\mathsf{S}_x} \tag{A.2147}$$

F1988 43_exact_quasienergy_closure.tex:890 NEW_EXTENSION_PASS

$$\mathsf{S}_i = \exp[-\mathrm{i}a\alpha_i\Pi_i] + \mathcal{O}(a^2\partial L). \tag{A.2148}$$

F1989 43_exact_quasienergy_closure.tex:896 NEW_EXTENSION_PASS

$$\mathsf{U}_\Delta(-\Delta) = \mathsf{U}_\Delta(\Delta)^{*}, \tag{A.2149}$$

F1990 43_exact_quasienergy_closure.tex:901 NEW_EXTENSION_PASS

$$\mathsf{U}_\Delta = \exp\left[-\mathrm{i}\Delta\left(H_D + \Delta^2 Q + \mathcal{O}(\Delta^4)\right)\right], \tag{A.2150}$$

F1991 43_exact_quasienergy_closure.tex:909 NEW_EXTENSION_PASS

$$H_D := \boldsymbol{\alpha}\cdot\boldsymbol{\Pi} + \beta m. \tag{A.2151}$$

F1992 43_exact_quasienergy_closure.tex:913 NEW_EXTENSION_PASS

$$\boxed{H_D^2 = m^2 + \sum_i \Pi_i^2 + \sum_{i<j}\alpha_i\alpha_j[\Pi_i,\Pi_j].} \tag{A.2152}$$

F1993 43_exact_quasienergy_closure.tex:921 NEW_EXTENSION_PASS

$$\Pi_i = -\mathrm{i}\partial_i - qA_i, \qquad [\Pi_i,\Pi_j] = \mathrm{i}q\epsilon_{ijk}B_k, \tag{A.2153}$$

F1994 43_exact_quasienergy_closure.tex:927 NEW_EXTENSION_PASS

$$H_D^2 = m^2 + \boldsymbol{\Pi}^2 - q\boldsymbol{\Sigma}\cdot\boldsymbol{B}. \tag{A.2154}$$

F1995 43_exact_quasienergy_closure.tex:934 NEW_EXTENSION_PASS

$$\mathcal{A}^{U}_{\mathrm{prod}} := \frac{\cos(m\Delta)}{48}\sum_{\pi\in S_3}\sum_{\boldsymbol{\sigma}\in\{\pm1\}^3} P_{\pi,\boldsymbol{\sigma}}. \tag{A.2155}$$

F1996 43_exact_quasienergy_closure.tex:942 NEW_EXTENSION_PASS

$$(\mathcal{A}^{U}_{\mathrm{prod}})^{*} = \mathcal{A}^{U}_{\mathrm{prod}}, \tag{A.2156}$$

$$\left\|\mathcal{A}^{U}_{\mathrm{prod}}\right\| \le 1, \tag{A.2157}$$

$$\mathcal{A}^{U,g}_{\mathrm{prod}} = \mathcal{G}\mathcal{A}^{U}_{\mathrm{prod}}\mathcal{G}^{-1}. \tag{A.2158}$$

F1997 43_exact_quasienergy_closure.tex:949 NEW_EXTENSION_PASS

$$\mathcal{A}^{U}_{\mathrm{prod}}(\boldsymbol{k}) = \cos(m\Delta)\prod_i \cos(k_i\Delta). \tag{A.2159}$$

F1998 43_exact_quasienergy_closure.tex:955 NEW_EXTENSION_PASS

$$S^U = \sum_{\ell=1}^{L} w_\ell \frac{P_\ell + P_\ell^*}{2}, \qquad w_\ell > 0, \qquad \sum_\ell w_\ell = 1. \tag{A.2160}$$

F1999 43_exact_quasienergy_closure.tex:965 NEW_EXTENSION_PASS

$$J_\ell := \begin{pmatrix} 0 & P_\ell \\ P_\ell^* & 0 \end{pmatrix}. \tag{A.2161}$$

F2000 43_exact_quasienergy_closure.tex:973 NEW_EXTENSION_PASS

$$|\chi\rangle := \sum_\ell \sqrt{w_\ell}\,|\ell\rangle\,|+\rangle\,, \tag{A.2162}$$

$$\mathcal{J} := \sum_\ell |\ell\rangle\langle\ell| \otimes J_\ell. \tag{A.2163}$$

F2001 43_exact_quasienergy_closure.tex:980 NEW_EXTENSION_PASS

$$\langle\chi|\,\mathcal{J}\,|\chi\rangle = S^U. \tag{A.2164}$$

F2002 43_exact_quasienergy_closure.tex:984 NEW_EXTENSION_PASS

$$M_m := \begin{pmatrix} \cos(m\tau) & \sin(m\tau) \\ \sin(m\tau) & -\cos(m\tau) \end{pmatrix}, \qquad M_m^2 = I. \tag{A.2165}$$

F2003 43_exact_quasienergy_closure.tex:994 NEW_EXTENSION_PASS

$$\Pi\mathcal{U}\Pi = \cos(m\tau) S^U =: A^U. \tag{A.2166}$$

F2004 43_exact_quasienergy_closure.tex:999 NEW_EXTENSION_PASS

$$R := 2\Pi - I, \tag{A.2167}$$

$$\mathcal{W} := R\mathcal{U}. \tag{A.2168}$$

F2005 43_exact_quasienergy_closure.tex:1004 NEW_EXTENSION_PASS

$$A^U\phi = \lambda\phi, \tag{A.2169}$$

F2006 43_exact_quasienergy_closure.tex:1008 NEW_EXTENSION_PASS

$$\mathrm{e}^{\pm \mathrm{i}\arccos\lambda}. \tag{A.2170}$$

F2007 43_exact_quasienergy_closure.tex:1015 NEW_EXTENSION_PASS

$$\psi_{n+1} + \psi_{n-1} = 2A\psi_n. \tag{A.2171}$$

F2008 43_exact_quasienergy_closure.tex:1019 NEW_EXTENSION_PASS

$$\mathcal{S} = \sum_n \left[\langle\psi_n, A\psi_n\rangle - \mathrm{Re}\,\langle\psi_{n+1}, \psi_n\rangle\right]. \tag{A.2172}$$

F2009 43_exact_quasienergy_closure.tex:1027 NEW_EXTENSION_PASS

$$Q_n := 2\,\mathrm{Im}\,\langle\psi_n, \psi_{n+1}\rangle\,. \tag{A.2173}$$

F2010 43_exact_quasienergy_closure.tex:1032 NEW_EXTENSION_PASS

$$\mathcal{E}_n := \|\psi_{n+1}\|^2 + \|\psi_n\|^2 - 2\,\mathrm{Re}\,\langle\psi_{n+1}, A\psi_n\rangle\,. \tag{A.2174}$$

F2011 43_exact_quasienergy_closure.tex:1038 NEW_EXTENSION_PASS

$$\boxed{\mathcal{E}_n = \|\psi_{n+1} - A\psi_n\|^2 + \left\langle \psi_n, (I - A^2)\psi_n \right\rangle \geq 0.} \tag{A.2175}$$

F2012 43_exact_quasienergy_closure.tex:1047 NEW_EXTENSION_PASS

$$\psi_n = T_n(A)\psi_0 + U_{n-1}(A)(\psi_1 - A\psi_0), \tag{A.2176}$$

F2013 43_exact_quasienergy_closure.tex:1055 NEW_EXTENSION_PASS

$$\begin{pmatrix} \psi_{n+1} \\ \psi_n \end{pmatrix} = L_A \begin{pmatrix} \psi_n \\ \psi_{n-1} \end{pmatrix}, \qquad L_A := \begin{pmatrix} 2A & -I \\ I & 0 \end{pmatrix}. \tag{A.2177}$$

F2014 43_exact_quasienergy_closure.tex:1073 NEW_EXTENSION_PASS

$$M_A := \begin{pmatrix} I & -A \\ -A & I \end{pmatrix}, \tag{A.2178}$$

F2015 43_exact_quasienergy_closure.tex:1080 NEW_EXTENSION_PASS

$$L_A^* M_A L_A = M_A. \tag{A.2179}$$

F2016 43_exact_quasienergy_closure.tex:1087 NEW_EXTENSION_PASS

$$k_i = \frac{\pi}{\Delta}\epsilon_i + p_i, \qquad \epsilon_i \in \{0, 1\}. \tag{A.2180}$$

F2017 43_exact_quasienergy_closure.tex:1093 NEW_EXTENSION_PASS

$$\cos(k_i\Delta) = (-1)^{\epsilon_i}\cos(p_i\Delta). \tag{A.2181}$$

F2018 43_exact_quasienergy_closure.tex:1098 NEW_EXTENSION_PASS

$$C = (-1)^{\epsilon_x+\epsilon_y+\epsilon_z}. \tag{A.2182}$$

F2019 43_exact_quasienergy_closure.tex:1102 NEW_EXTENSION_PASS

$$U_\Delta = (-1)^{\epsilon_x+\epsilon_y+\epsilon_z} I. \tag{A.2183}$$

F2020 43_exact_quasienergy_closure.tex:1107 NEW_EXTENSION_PASS

$$\boxed{N_{\omega=0} = 4, \qquad N_{\omega=\pi/\Delta} = 4.} \tag{A.2184}$$

F2021 43_exact_quasienergy_closure.tex:1115 NEW_EXTENSION_PASS

$$U + U^* = 2CI \tag{A.2185}$$

F2022 43_exact_quasienergy_closure.tex:1122 NEW_EXTENSION_PASS

$$A := \alpha_i, \qquad B := \beta, \qquad \{A, B\} = 0. \tag{A.2186}$$

F2023 43_exact_quasienergy_closure.tex:1130 NEW_EXTENSION_PASS

$$Q_i(\theta) := \mathrm{e}^{-\mathrm{i}\theta A/2}, \tag{A.2187}$$

$$V_i(\theta;\rho) := \mathrm{e}^{+\mathrm{i}\rho B} Q_i(\theta) \mathrm{e}^{-\mathrm{i}\rho B} Q_i(\theta)^*, \tag{A.2188}$$

$$K_i(\theta;\rho) := V_i(\theta;\rho) V_i(-\theta;\rho). \tag{A.2189}$$

F2024 43_exact_quasienergy_closure.tex:1141 NEW_EXTENSION_PASS

$$\boxed{K_i(\theta;\rho) = I + \frac{\mathrm{i}}{2}\sin(2\rho)\theta^2\beta + \mathcal{O}(\theta^3).} \tag{A.2190}$$

F2025 43_exact_quasienergy_closure.tex:1149 NEW_EXTENSION_PASS

$$r := \frac{1}{2}\sin(2\rho). \tag{A.2191}$$

F2026 43_exact_quasienergy_closure.tex:1153 NEW_EXTENSION_PASS

$$U_{\Delta,W} := (K_x K_y K_z)^* \, U_\Delta, \tag{A.2192}$$

F2027 43_exact_quasienergy_closure.tex:1158 NEW_EXTENSION_PASS

$$H_{\mathrm{eff},W} = \boldsymbol{\alpha}\cdot\boldsymbol{k} + \beta\left(m + r\Delta|\boldsymbol{k}|^2\right) + \mathcal{O}(\Delta^2). \tag{A.2193}$$

F2028 43_exact_quasienergy_closure.tex:1165 NEW_EXTENSION_PASS

$$\boxed{K_i(\pi;\rho) = \mathrm{e}^{4\mathrm{i}\rho\beta}.} \tag{A.2194}$$

F2029 43_exact_quasienergy_closure.tex:1171 NEW_EXTENSION_PASS

$$U_{\Delta,W} = (-1)^n \mathrm{e}^{-4\mathrm{i}n\rho\beta}. \tag{A.2195}$$

F2030 43_exact_quasienergy_closure.tex:1176 NEW_EXTENSION_PASS

$$\rho = \frac{\pi}{16}, \tag{A.2196}$$

F2031 43_exact_quasienergy_closure.tex:1180 NEW_EXTENSION_PASS

n	$\lvert\omega\Delta\rvert$
1	$3\pi/4$
2	$\pi/2$
3	$\pi/4$

(A.2197)

F2032 43_exact_quasienergy_closure.tex:1192 NEW_EXTENSION_PASS

$$\delta_{\mathrm{grid}} = 2.582434 \times 10^{-1} \tag{A.2198}$$

F2033 43_exact_quasienergy_closure.tex:1199 NEW_EXTENSION_PASS

$$C_i := \frac{T_i + T_i^*}{2}, \tag{A.2199}$$

$$S^U_{\mathrm{cub}} := \frac{1}{6}\sum_i C_i + \frac{1}{12}\sum_{i<j}\{C_i, C_j\}. \tag{A.2200}$$

F2034 43_exact_quasienergy_closure.tex:1206 NEW_EXTENSION_PASS

$$T_i = \mathrm{e}^{hD_i}, \qquad h = \sqrt{2}\tau, \tag{A.2201}$$

F2035 43_exact_quasienergy_closure.tex:1212 NEW_EXTENSION_PASS

$$S^U_{\mathrm{cub}} = I + \frac{\tau^2}{2}D^2 + \frac{\tau^4}{12}(D^2)^2 + \mathcal{O}(\tau^6), \qquad D^2 := \sum_i D_i^2. \tag{A.2202}$$

F2036 43_exact_quasienergy_closure.tex:1221 NEW_EXTENSION_PASS

$$W_h := 3I - (C_x + C_y + C_z), \tag{A.2203}$$

$$G_m := I - \frac{m^2\tau^2}{12}W_h, \tag{A.2204}$$

$$G_4 := I - \frac{1}{48}W_h^2. \tag{A.2205}$$

F2037 43_exact_quasienergy_closure.tex:1227 NEW_EXTENSION_PASS

$$0 \le W_h \le 6I, \tag{A.2206}$$

F2038 43_exact_quasienergy_closure.tex:1231 NEW_EXTENSION_PASS

$$\|G_m\| \le 1, \tag{A.2207}$$

$$\|G_4\| \le 1. \tag{A.2208}$$

F2039 43_exact_quasienergy_closure.tex:1236 NEW_EXTENSION_PASS

$$\boxed{A_4^U := \cos(m\tau) G_4 G_m S^U_{\mathrm{cub}} G_m G_4.} \tag{A.2209}$$

F2040 43_exact_quasienergy_closure.tex:1244 NEW_EXTENSION_PASS

$$K := m^2 - D^2, \tag{A.2210}$$

F2041 43_exact_quasienergy_closure.tex:1248 NEW_EXTENSION_PASS

$$\boxed{A_4^U = I - \frac{\tau^2}{2}K + \frac{\tau^4}{24}K^2 + \mathcal{O}(\tau^6).} \tag{A.2211}$$

F2042 43_exact_quasienergy_closure.tex:1257 NEW_EXTENSION_PASS

$$A_4^U = \cos(\tau\sqrt{K}) + \mathcal{O}(\tau^6) \tag{A.2212}$$

F2043 43_exact_quasienergy_closure.tex:1262 NEW_EXTENSION_PASS

$$\Omega_4^2 = m^2 - D^2 + \mathcal{O}(\tau^4). \tag{A.2213}$$

F2044 43_exact_quasienergy_closure.tex:1270 NEW_EXTENSION_PASS

$$A_4 = \cos(\tau E) + \frac{\tau^6}{360}\mathcal{P}_6(m, \boldsymbol{k}) + \mathcal{O}(\tau^8), \tag{A.2214}$$

F2045 43_exact_quasienergy_closure.tex:1277 NEW_EXTENSION_PASS

$$\boxed{\begin{aligned}\mathcal{P}_6 = 11\sum_i k_i^6 + 19\sum_{i\neq j} k_i^4 k_j^2 + 48 k_x^2 k_y^2 k_z^2 \\ + 34m^2\sum_i k_i^4 + 48m^2\sum_{i<j} k_i^2 k_j^2 + 24m^4|\boldsymbol{k}|^2.\end{aligned}} \tag{A.2215}$$

F2046 43_exact_quasienergy_closure.tex:1291 NEW_EXTENSION_PASS

$$\omega_4^2 = m^2 + |\boldsymbol{k}|^2 - \frac{\tau^4}{180}\mathcal{P}_6 + \mathcal{O}(\tau^6). \tag{A.2216}$$

F2047 43_exact_quasienergy_closure.tex:1299 NEW_EXTENSION_PASS

$$w_i := I - C_i, \tag{A.2217}$$

$$Y_i := \frac{T_i - T_i^*}{2\mathrm{i}}, \tag{A.2218}$$

$$\mu := m\tau. \tag{A.2219}$$

F2048 43_exact_quasienergy_closure.tex:1305 NEW_EXTENSION_PASS

$$Q_1 := 11\sum_i w_i^3, \tag{A.2220}$$

$$Q_2 := 19\sum_{i\neq j} w_i w_j w_i, \tag{A.2221}$$

$$Q_3 := \sum_{\pi\in S_3} (Y_{\pi_1}Y_{\pi_2}Y_{\pi_3})^*(Y_{\pi_1}Y_{\pi_2}Y_{\pi_3}), \tag{A.2222}$$

$$Q_4 := 34\mu^2\sum_i w_i^2, \tag{A.2223}$$

$$Q_5 := 6\mu^2\sum_{i<j}\left[(Y_iY_j)^*(Y_iY_j) + (Y_jY_i)^*(Y_jY_i)\right], \tag{A.2224}$$

$$Q_6 := 24\mu^4\sum_i w_i. \tag{A.2225}$$

F2049 43_exact_quasienergy_closure.tex:1320 NEW_EXTENSION_PASS

$$\sum_{j=1}^{6} Q_j = \tau^6\mathcal{P}_6(m, \boldsymbol{k}) + \mathcal{O}(\tau^8). \tag{A.2226}$$

F2050 43_exact_quasienergy_closure.tex:1326 NEW_EXTENSION_PASS

$$H_j := I - \frac{Q_j}{720}, \tag{A.2227}$$
$$H := H_1 H_2 H_3 H_4 H_5 H_6, \tag{A.2228}$$
$$A_6^U := H^* A_4^U H. \tag{A.2229}$$

F2051 43_exact_quasienergy_closure.tex:1332 NEW_EXTENSION_PASS

$$\|Q_1\| \le 264, \tag{A.2230}$$
$$\|Q_2\| \le 912, \tag{A.2231}$$
$$\|Q_3\| \le 6, \tag{A.2232}$$
$$\|Q_4\| \le 102\pi^2, \tag{A.2233}$$
$$\|Q_5\| \le 9\pi^2, \tag{A.2234}$$
$$\|Q_6\| \le 9\pi^4, \tag{A.2235}$$

F2052 43_exact_quasienergy_closure.tex:1341 NEW_EXTENSION_PASS

$$\|H_j\| \le 1 \tag{A.2236}$$

F2053 43_exact_quasienergy_closure.tex:1345 NEW_EXTENSION_PASS

$$\boxed{A_6 = \cos\!\left(\tau\sqrt{m^2 + |\boldsymbol{k}|^2}\right) + \mathcal{O}(\tau^8),} \tag{A.2237}$$

F2054 43_exact_quasienergy_closure.tex:1353 NEW_EXTENSION_PASS

$$\omega_6^2 = m^2 + |\boldsymbol{k}|^2 + \mathcal{O}(\tau^6). \tag{A.2238}$$

F2055 43_exact_quasienergy_closure.tex:1388 NEW_EXTENSION_PASS

$$\mathcal{Q}_{\mathrm{checks}} = \texttt{data/exact_quasienergy_closure_checks.csv}, \tag{A.2239}$$
$$\mathcal{Q}_{\mathrm{summary}} = \texttt{data/exact_quasienergy_closure_summary.csv}, \tag{A.2240}$$
$$\mathcal{Q}_{\mathrm{script}} = \texttt{code/53_exact_quasienergy_closure.py}. \tag{A.2241}$$

44_resolved_nonlinear_ashtekar_closure.tex

F2056 44_resolved_nonlinear_ashtekar_closure.tex:5 OPEN_NOT_VERIFIED

$$\boxed{\begin{array}{c}\text{nonlinear vacuum Hamiltonian constraint: closed on an alias-free resolved domain,}\\ \text{three momentum constraints: closed on the same domain,}\\ \text{Gauss–momentum–Hamiltonian algebra: exact at finite dimension,}\\ \text{strict finite-range locality, Lorentzian reality preservation, and unified matter: open.}\end{array}} \tag{A.2242}$$

F2057 44_resolved_nonlinear_ashtekar_closure.tex:20 RESOLVED_DOMAIN_PASS

$$\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3, \qquad \mathcal{T}_K = \left\{ f(x) = \sum_{\|k\|_\infty \le K} \widehat{f}_k e^{\mathrm{i}k\cdot x} \right\}. \tag{A.2243}$$

F2058 44_resolved_nonlinear_ashtekar_closure.tex:30 RESOLVED_DOMAIN_PASS

$$\mathcal{T}_K \mathcal{T}_L \subseteq \mathcal{T}_{K+L}, \qquad \partial_a \mathcal{T}_K \subseteq \mathcal{T}_K, \tag{A.2244}$$

F2059 44_resolved_nonlinear_ashtekar_closure.tex:36 RESOLVED_DOMAIN_PASS

$$\mathcal{T}_{\mathrm{fin}} := \bigcup_{K \ge 0} \mathcal{T}_K \tag{A.2245}$$

F2060 44_resolved_nonlinear_ashtekar_closure.tex:42 RESOLVED_DOMAIN_PASS

$$\Gamma_N = \left\{ \frac{2\pi}{N}(r_1, r_2, r_3) : 0 \le r_a < N \right\}, \tag{A.2246}$$

F2061 44_resolved_nonlinear_ashtekar_closure.tex:50 RESOLVED_DOMAIN_PASS

$$\delta_a^{(N)} S_N f := S_N(\partial_a f), \tag{A.2247}$$

F2062 44_resolved_nonlinear_ashtekar_closure.tex:54 RESOLVED_DOMAIN_PASS

$$\tau_N(u) := \frac{1}{N^3} \sum_{x \in \Gamma_N} u(x). \tag{A.2248}$$

F2063 44_resolved_nonlinear_ashtekar_closure.tex:62 RESOLVED_DOMAIN_PASS

$$\sum_{r=1}^{q} K_r < \frac{N}{2}, \tag{A.2249}$$

F2064 44_resolved_nonlinear_ashtekar_closure.tex:66 RESOLVED_DOMAIN_PASS

$$\delta_a^{(N)} \prod_{r=1}^{q} S_N f_r = S_N \partial_a \left(\prod_{r=1}^{q} f_r \right), \tag{A.2250}$$

$$= \sum_{s=1}^{q} \left(\prod_{r<s} S_N f_r \right) S_N(\partial_a f_s) \left(\prod_{r>s} S_N f_r \right), \tag{A.2251}$$

$$\tau_N \left(\prod_{r=1}^{q} S_N f_r \right) = \frac{1}{(2\pi)^3} \int_{\mathbb{T}^3} \prod_{r=1}^{q} f_r \, \mathrm{d}^3 x. \tag{A.2252}$$

F2065 44_resolved_nonlinear_ashtekar_closure.tex:87 RESOLVED_DOMAIN_PASS

$$A_a^i, E_i^a, \lambda^i, \xi^a, n \in \mathcal{T}_K, \tag{A.2253}$$

F2066 44_resolved_nonlinear_ashtekar_closure.tex:93 RESOLVED_DOMAIN_PASS

$$F_{ab}^i := \delta_a A_b^i - \delta_b A_a^i + \epsilon^i{}_{jk} A_a^j A_b^k, \tag{A.2254}$$

$$\mathcal{G}_i := \delta_a E_i^a + \epsilon_{ij}{}^k A_a^j E_k^a, \tag{A.2255}$$

$$\mathcal{V}_a := F_{ab}^i E_i^b, \tag{A.2256}$$

$$\mathcal{D}_a := \mathcal{V}_a - A_a^i \mathcal{G}_i, \tag{A.2257}$$

$$\det E := \frac{1}{6} \epsilon_{abc} \epsilon^{ijk} E_i^a E_j^b E_k^c, \tag{A.2258}$$

$$\mathcal{H} := \frac{1}{2} \epsilon^{ij}{}_k E_i^a E_j^b F_{ab}^k + \Lambda \det E. \tag{A.2259}$$

F2067 44_resolved_nonlinear_ashtekar_closure.tex:117 RESOLVED_DOMAIN_PASS

$$G_N[\lambda] := \tau_N(\lambda^i \mathcal{G}_i), \tag{A.2260}$$

$$V_N[\xi] := \tau_N(\xi^a \mathcal{V}_a), \tag{A.2261}$$

$$D_N[\xi] := \tau_N(\xi^a \mathcal{D}_a), \tag{A.2262}$$

$$H_N[n] := \tau_N(n\mathcal{H}). \tag{A.2263}$$

F2068 44_resolved_nonlinear_ashtekar_closure.tex:125 RESOLVED_DOMAIN_PASS

$$\boxed{S_{N,K}^{\mathrm{SD}} = \int \mathrm{d}t \left[\tau_N(E_i^a \dot{A}_a^i) - G_N[\lambda] - D_N[\xi] - H_N[n] \right].} \tag{A.2264}$$

F2069 44_resolved_nonlinear_ashtekar_closure.tex:139 RESOLVED_DOMAIN_PASS

$$\boxed{\{F, G\}_N = N^3 \sum_{x,a,i} \left(\frac{\partial F}{\partial A_a^i(x)} \frac{\partial G}{\partial E_i^a(x)} - \frac{\partial F}{\partial E_i^a(x)} \frac{\partial G}{\partial A_a^i(x)} \right).} \tag{A.2265}$$

F2070 44_resolved_nonlinear_ashtekar_closure.tex:156 RESOLVED_DOMAIN_PASS

$$(\lambda \times \mu)^i := \epsilon^i{}_{jk} \lambda^j \mu^k. \tag{A.2266}$$

F2071 44_resolved_nonlinear_ashtekar_closure.tex:162 RESOLVED_DOMAIN_PASS

$$[\xi, \eta]^a := \xi^b \delta_b \eta^a - \eta^b \delta_b \xi^a, \tag{A.2267}$$
$$\mathcal{L}_\xi \lambda^i := \xi^a \delta_a \lambda^i, \tag{A.2268}$$
$$\mathcal{L}_\xi^{(-1)} n := \xi^a \delta_a n - n \delta_a \xi^a. \tag{A.2269}$$

F2072 44_resolved_nonlinear_ashtekar_closure.tex:174 RESOLVED_DOMAIN_PASS

$$S^a(n, m) := E^a_i E^{bi} \left(m \delta_b n - n \delta_b m \right). \tag{A.2270}$$

F2073 44_resolved_nonlinear_ashtekar_closure.tex:183 RESOLVED_DOMAIN_PASS

$$\boxed{16K < N.} \tag{A.2271}$$

F2074 44_resolved_nonlinear_ashtekar_closure.tex:187 RESOLVED_DOMAIN_PASS

$$\{G_N[\lambda], G_N[\mu]\}_N = G_N[\lambda \times \mu], \tag{A.2272}$$
$$\{D_N[\xi], G_N[\lambda]\}_N = G_N[\mathcal{L}_\xi \lambda], \tag{A.2273}$$
$$\{D_N[\xi], D_N[\eta]\}_N = D_N[[\xi, \eta]], \tag{A.2274}$$
$$\{G_N[\lambda], H_N[n]\}_N = 0, \tag{A.2275}$$
$$\{D_N[\xi], H_N[n]\}_N = H_N[\mathcal{L}_\xi^{(-1)} n], \tag{A.2276}$$
$$\{H_N[n], H_N[m]\}_N = V_N[S(n, m)] \tag{A.2277}$$
$$= D_N[S(n, m)] + G_N[S^a(n, m) A_a]. \tag{A.2278}$$

F2075 44_resolved_nonlinear_ashtekar_closure.tex:208 RESOLVED_DOMAIN_PASS

$$V_N[S] = D_N[S] + G_N[S^a A_a], \tag{A.2279}$$

F2076 44_resolved_nonlinear_ashtekar_closure.tex:216 RESOLVED_DOMAIN_PASS

$$H_{\mathrm{tot}} := G_N[\lambda] + D_N[\xi] + H_N[n]. \tag{A.2280}$$

F2077 44_resolved_nonlinear_ashtekar_closure.tex:222 RESOLVED_DOMAIN_PASS

$$\frac{\mathrm{d}}{\mathrm{d}t} G_N[\alpha] = G_N[\alpha \times \lambda - \mathcal{L}_\xi \alpha], \tag{A.2281}$$
$$\frac{\mathrm{d}}{\mathrm{d}t} D_N[\eta] = G_N[\mathcal{L}_\eta \lambda] + D_N[[\eta, \xi]] + H_N[\mathcal{L}_\eta^{(-1)} n], \tag{A.2282}$$
$$\frac{\mathrm{d}}{\mathrm{d}t} H_N[m] = -H_N[\mathcal{L}_\xi^{(-1)} m] + V_N[S(m, n)]. \tag{A.2283}$$

F2078 44_resolved_nonlinear_ashtekar_closure.tex:237 RESOLVED_DOMAIN_PASS

$$G_N = D_N = H_N = 0 \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t} (G_N, D_N, H_N) = 0 \tag{A.2284}$$

F2079 44_resolved_nonlinear_ashtekar_closure.tex:247 RESOLVED_DOMAIN_PASS

$$I_{K \to L} : \mathcal{T}_K \hookrightarrow \mathcal{T}_L \tag{A.2285}$$

F2080 44_resolved_nonlinear_ashtekar_closure.tex:251 RESOLVED_DOMAIN_PASS

$$I_{L \to M} I_{K \to L} = I_{K \to M}. \tag{A.2286}$$

F2081 44_resolved_nonlinear_ashtekar_closure.tex:257 RESOLVED_DOMAIN_PASS

$$\mathcal{T}_K \longrightarrow \mathcal{T}_{4K} \longrightarrow \mathcal{T}_{16K} \longrightarrow \cdots, \tag{A.2287}$$

F2082 44_resolved_nonlinear_ashtekar_closure.tex:270 RESOLVED_DOMAIN_PASS

$$B_{GG}(\lambda,\mu)=\lambda\times\mu, \tag{A.2288}$$
$$B_{DG}(\xi,\lambda)=\mathcal{L}_\xi\lambda, \tag{A.2289}$$
$$B_{DD}(\xi,\eta)=[\xi,\eta], \tag{A.2290}$$
$$B_{DH}(\xi,n)=\mathcal{L}_\xi^{(-1)}n, \tag{A.2291}$$
$$B_{HH}(n,m)=S(n,m). \tag{A.2292}$$

F2083 44_resolved_nonlinear_ashtekar_closure.tex:278 RESOLVED_DOMAIN_PASS

$$s_C=s_{0,C}+P\delta(I+H\delta)^{-1}I \tag{A.2293}$$

F2084 44_resolved_nonlinear_ashtekar_closure.tex:303 RESOLVED_DOMAIN_PASS

$$3.288599\times10^{-8},\qquad \varepsilon_{\mathrm{rel}}=1.742589\times10^{-5}. \tag{A.2294}$$

F2085 44_resolved_nonlinear_ashtekar_closure.tex:313 RESOLVED_DOMAIN_PASS

$$E^a_i=\overline{E^a_i},\qquad A^i_a+\overline{A^i_a}=2\Gamma^i_a(E). \tag{A.2295}$$

F2086 44_resolved_nonlinear_ashtekar_closure.tex:321 RESOLVED_DOMAIN_PASS

$$\begin{gathered}\text{strictly local nonlinear realization},\\ \text{Lorentzian reality-constraint propagation},\\ \text{unified Maxwell–Dirac and scalar matter},\\ \text{uniform Einstein–matter continuum estimates},\\ \text{constraint-preserving discrete-time integration}.\end{gathered} \tag{A.2296}$$

F2087 44_resolved_nonlinear_ashtekar_closure.tex:332 RESOLVED_DOMAIN_PASS

$$\mathcal{R}_{\mathrm{RNC}}=\texttt{data/resolved_nonlinear_ashtekar_checks.csv}, \tag{A.2297}$$
$$\mathcal{S}_{\mathrm{RNC}}=\texttt{data/resolved_nonlinear_ashtekar_summary.csv}, \tag{A.2298}$$
$$\mathcal{X}_{\mathrm{RNC}}=\texttt{code/54_resolved_nonlinear_ashtekar_closure.py}. \tag{A.2299}$$

45_resolved_lorentzian_bv_parent.tex

F2088 45_resolved_lorentzian_bv_parent.tex:5 OPEN_NOT_VERIFIED

$$\boxed{\begin{gathered}\text{real Lorentzian tetrad–connection action: constructed},\\ \text{Dirac and first-order Maxwell matter in the same polynomial action: constructed},\\ \text{off-shell diffeomorphism–Lorentz–U(1) BRST differential: nilpotent},\\ \text{classical BV master equation: exact on an alias-free resolved domain},\\ \text{strict finite-range locality, canonical matter closure, and quantum anomaly control: open}.\end{gathered}} \tag{A.2300}$$

F2089 45_resolved_lorentzian_bv_parent.tex:21 CLASSICAL_BV_PASS

$$\mathbb{T}^4=(\mathbb{R}/2\pi\mathbb{Z})^4,\qquad \mathcal{T}^{(4)}_K=\left\{f(x)=\sum_{\|k\|_\infty\le K}\widehat{f}_k\mathrm{e}^{\mathrm{i}k\cdot x}\right\}. \tag{A.2301}$$

F2090 45_resolved_lorentzian_bv_parent.tex:31 CLASSICAL_BV_PASS

$$\mathcal{T}^{(4)}_K\mathcal{T}^{(4)}_L\subseteq\mathcal{T}^{(4)}_{K+L},\qquad \partial_\mu\mathcal{T}^{(4)}_K\subseteq\mathcal{T}^{(4)}_K. \tag{A.2302}$$

F2091 45_resolved_lorentzian_bv_parent.tex:41 CLASSICAL_BV_PASS

$$\Gamma^{(4)}_N=\left\{\frac{2\pi}{N}(r_0,r_1,r_2,r_3):0\le r_\mu<N\right\}, \tag{A.2303}$$

F2092 45_resolved_lorentzian_bv_parent.tex:49 CLASSICAL_BV_PASS

$$\delta_\mu^{(N)} S_N f = S_N(\partial_\mu f), \qquad \tau_N^{(4)}(u) = \frac{1}{N^4} \sum_{x \in \Gamma_N^{(4)}} u(x). \tag{A.2304}$$

F2093 45_resolved_lorentzian_bv_parent.tex:55 CLASSICAL_BV_PASS

$$\sum_r K_r < \frac{N}{2}, \tag{A.2305}$$

F2094 45_resolved_lorentzian_bv_parent.tex:65 CLASSICAL_BV_PASS

$$\eta_{IJ} = \mathrm{diag}(-1,1,1,1), \qquad \epsilon_{0123} = 1. \tag{A.2306}$$

F2095 45_resolved_lorentzian_bv_parent.tex:71 CLASSICAL_BV_PASS

$$\begin{aligned} &e^I \in \Omega^1(\mathbb{T}^4), && \omega^{IJ} = -\omega^{JI} \in \Omega^1(\mathbb{T}^4), \\ &a \in \Omega^1(\mathbb{T}^4), && \varphi_{IJ} = -\varphi_{JI} \in \Omega^0(\mathbb{T}^4), \\ &\psi, \bar\psi \in \Omega^0(\mathbb{T}^4) \otimes \mathbb{C}^4. \end{aligned} \tag{A.2307}$$

F2096 45_resolved_lorentzian_bv_parent.tex:81 CLASSICAL_BV_PASS

$$R^{IJ} := \mathrm{d}\omega^{IJ} + \omega^I{}_K \wedge \omega^{KJ}, \tag{A.2308}$$

$$T^I := \mathrm{d}e^I + \omega^I{}_J \wedge e^J, \tag{A.2309}$$

$$F := \mathrm{d}a, \tag{A.2310}$$

$$\mathrm{vol}_e := \frac{1}{4!}\epsilon_{IJKL} e^I \wedge e^J \wedge e^K \wedge e^L, \tag{A.2311}$$

$$\Sigma_I := \frac{1}{3!}\epsilon_{IJKL} e^J \wedge e^K \wedge e^L. \tag{A.2312}$$

F2097 45_resolved_lorentzian_bv_parent.tex:94 CLASSICAL_BV_PASS

$$\{\gamma^I, \gamma^J\} = 2\eta^{IJ} I, \qquad \gamma_{IJ} := \frac{1}{2}[\gamma_I, \gamma_J]. \tag{A.2313}$$

F2098 45_resolved_lorentzian_bv_parent.tex:100 CLASSICAL_BV_PASS

$$\mathcal{D}\psi := \mathrm{d}\psi + \frac{1}{4}\omega^{IJ}\gamma_{IJ}\psi - \mathrm{i}q a \psi, \tag{A.2314}$$

$$\mathcal{D}\bar\psi := \mathrm{d}\bar\psi - \frac{1}{4}\bar\psi\omega^{IJ}\gamma_{IJ} + \mathrm{i}q a \bar\psi. \tag{A.2315}$$

F2099 45_resolved_lorentzian_bv_parent.tex:110 CLASSICAL_BV_PASS

$$S_{\mathrm{G}} = \frac{1}{4\kappa}\int \epsilon_{IJKL} e^I \wedge e^J \wedge R^{KL} - \frac{\Lambda}{24\kappa}\int \epsilon_{IJKL} e^I \wedge e^J \wedge e^K \wedge e^L. \tag{A.2316}$$

F2100 45_resolved_lorentzian_bv_parent.tex:122 CLASSICAL_BV_PASS

$$S_{\mathrm{M}}^{(1)} = \int \left[\frac{1}{2}\varphi_{IJ} F \wedge e^I \wedge e^J - \frac{1}{4}\varphi_{IJ}\varphi^{IJ}\,\mathrm{vol}_e\right]. \tag{A.2317}$$

F2101 45_resolved_lorentzian_bv_parent.tex:133 CLASSICAL_BV_PASS

$$S_{\mathrm{D}} = \frac{\mathrm{i}}{2}\int \Sigma_I \wedge \left[\bar\psi\gamma^I\mathcal{D}\psi - (\mathcal{D}\bar\psi)\gamma^I\psi\right] - m\int \mathrm{vol}_e\ \bar\psi\psi. \tag{A.2318}$$

F2102 45_resolved_lorentzian_bv_parent.tex:148 CLASSICAL_BV_PASS

$$\boxed{S_0[e,\omega,a,\varphi,\psi,\bar\psi] := S_{\mathrm{G}} + S_{\mathrm{M}}^{(1)} + S_{\mathrm{D}}.} \tag{A.2319}$$

F2103 45_resolved_lorentzian_bv_parent.tex:158 CLASSICAL_BV_PASS

$$F \wedge e^I \wedge e^J = \varphi^{IJ}\,\mathrm{vol}_e\,. \tag{A.2320}$$

F2104 45_resolved_lorentzian_bv_parent.tex:164 CLASSICAL_BV_PASS

$$\varphi^{IJ} = (\star_e F)^{IJ}. \tag{A.2321}$$

F2105 45_resolved_lorentzian_bv_parent.tex:168 CLASSICAL_BV_PASS

$$S_{\mathrm{M}}^{(1)}\big|_{\varphi=\star_e F} = -\frac{1}{4}\int F_{IJ}F^{IJ}\,\mathrm{vol}_e = -\frac{1}{2}\int F\wedge\star_e F. \tag{A.2322}$$

F2106 45_resolved_lorentzian_bv_parent.tex:180 CLASSICAL_BV_PASS

$$\delta e^I = \mathcal{L}_\xi e^I + \lambda^I{}_J e^J, \tag{A.2323}$$
$$\delta\omega^{IJ} = \mathcal{L}_\xi\omega^{IJ} - D_\omega\lambda^{IJ}, \tag{A.2324}$$
$$\delta a = \mathcal{L}_\xi a + \mathrm{d}u, \tag{A.2325}$$
$$\delta\varphi_{IJ} = \mathcal{L}_\xi\varphi_{IJ} - \lambda^K{}_I\varphi_{KJ} - \lambda^K{}_J\varphi_{IK}, \tag{A.2326}$$
$$\delta\psi = \mathcal{L}_\xi\psi + \frac{1}{4}\lambda^{IJ}\gamma_{IJ}\psi + \mathrm{i}qu\psi, \tag{A.2327}$$
$$\delta\bar\psi = \mathcal{L}_\xi\bar\psi - \frac{1}{4}\bar\psi\lambda^{IJ}\gamma_{IJ} - \mathrm{i}qu\bar\psi. \tag{A.2328}$$

F2107 45_resolved_lorentzian_bv_parent.tex:201 CLASSICAL_BV_PASS

$$\delta R^{IJ} = \mathcal{L}_\xi R^{IJ} + \lambda^I{}_K R^{KJ} + \lambda^J{}_K R^{IK}, \tag{A.2329}$$

F2108 45_resolved_lorentzian_bv_parent.tex:207 CLASSICAL_BV_PASS

$$\delta(\mathcal{D}\psi) = \mathcal{L}_\xi(\mathcal{D}\psi) + \frac{1}{4}\lambda^{IJ}\gamma_{IJ}\mathcal{D}\psi + \mathrm{i}qu\mathcal{D}\psi, \tag{A.2330}$$

F2109 45_resolved_lorentzian_bv_parent.tex:214 CLASSICAL_BV_PASS

$$\delta F = \mathcal{L}_\xi F. \tag{A.2331}$$

F2110 45_resolved_lorentzian_bv_parent.tex:218 CLASSICAL_BV_PASS

$$\boxed{\delta S_0 = 0} \tag{A.2332}$$

F2111 45_resolved_lorentzian_bv_parent.tex:225 CLASSICAL_BV_PASS

$$[\![U,V]\!] := \big([\xi,\eta], \mathcal{L}_\xi\mu - \mathcal{L}_\eta\lambda + [\lambda,\mu], \mathcal{L}_\xi\upsilon - \mathcal{L}_\eta u\big). \tag{A.2333}$$

F2112 45_resolved_lorentzian_bv_parent.tex:235 CLASSICAL_BV_PASS

$$[\delta_U,\delta_V]_{\mathrm{fld}} = -\delta_{[\![U,V]\!]}. \tag{A.2334}$$

F2113 45_resolved_lorentzian_bv_parent.tex:244 CLASSICAL_BV_PASS

$$c^\mu, \qquad \chi^{IJ} = -\chi^{JI}, \qquad \upsilon. \tag{A.2335}$$

F2114 45_resolved_lorentzian_bv_parent.tex:252 CLASSICAL_BV_PASS

$$\mathfrak{c} := (c,\chi,\upsilon), \qquad s\Phi := \delta_{\mathfrak{c}}\Phi, \qquad s\mathfrak{c} := \frac{1}{2}[\![\mathfrak{c},\mathfrak{c}]\!]. \tag{A.2336}$$

F2115 45_resolved_lorentzian_bv_parent.tex:260 CLASSICAL_BV_PASS

$$sc^\mu = c^\nu\partial_\nu c^\mu, \tag{A.2337}$$
$$s\chi = \mathcal{L}_c\chi + \chi^2, \tag{A.2338}$$
$$s\upsilon = \mathcal{L}_c\upsilon, \tag{A.2339}$$

F2116 45_resolved_lorentzian_bv_parent.tex:269 CLASSICAL_BV_PASS

$$se^I = \mathcal{L}_c e^I + \chi^I{}_J e^J, \tag{A.2340}$$
$$s\omega^{IJ} = \mathcal{L}_c\omega^{IJ} - D_\omega\chi^{IJ}, \tag{A.2341}$$
$$sa = \mathcal{L}_c a + \mathrm{d}\upsilon, \tag{A.2342}$$
$$s\varphi_{IJ} = \mathcal{L}_c\varphi_{IJ} - \chi^K{}_I\varphi_{KJ} - \chi^K{}_J\varphi_{IK}, \tag{A.2343}$$

$$s\psi = \mathcal{L}_c\psi + \frac{1}{4}\chi^{IJ}\gamma_{IJ}\psi + \mathrm{i}q\upsilon\psi, \tag{A.2344}$$

$$s\bar{\psi} = \mathcal{L}_c\bar{\psi} - \frac{1}{4}\bar{\psi}\chi^{IJ}\gamma_{IJ} - \mathrm{i}q\upsilon\bar{\psi}. \tag{A.2345}$$

F2117 45_resolved_lorentzian_bv_parent.tex:289 CLASSICAL_BV_PASS

$$\boxed{s^2 = 0.} \tag{A.2346}$$

F2118 45_resolved_lorentzian_bv_parent.tex:295 CLASSICAL_BV_PASS

$$s^2\Phi = \frac{1}{2}\left([\delta_{\mathfrak{c}}, \delta_{\mathfrak{c}}]_{\mathrm{gr}} + \delta_{[\![\mathfrak{c},\mathfrak{c}]\!]}\right)\Phi = 0. \tag{A.2347}$$

F2119 45_resolved_lorentzian_bv_parent.tex:311 CLASSICAL_BV_PASS

$$\{e, \omega, a, \varphi, \psi, \bar{\psi}\}. \tag{A.2348}$$

F2120 45_resolved_lorentzian_bv_parent.tex:315 CLASSICAL_BV_PASS

$$\Phi^\ddagger, \qquad c_\mu^\ddagger, \qquad \chi_{IJ}^\ddagger, \qquad \upsilon^\ddagger \tag{A.2349}$$

F2121 45_resolved_lorentzian_bv_parent.tex:325 CLASSICAL_BV_PASS

$$\boxed{S_{\mathrm{BV}} = S_0 + \int\left[\sum_\Phi \langle\Phi^\ddagger, s\Phi\rangle + c_\mu^\ddagger sc^\mu + \frac{1}{2}\chi_{IJ}^\ddagger s\chi^{IJ} + \upsilon^\ddagger s\upsilon\right].} \tag{A.2350}$$

F2122 45_resolved_lorentzian_bv_parent.tex:340 CLASSICAL_BV_PASS

$$(F,G)_{\mathrm{BV}} := \sum_\Xi \int\left(\frac{\delta_r F}{\delta\Xi}\frac{\delta_l G}{\delta\Xi^\ddagger} - \frac{\delta_r F}{\delta\Xi^\ddagger}\frac{\delta_l G}{\delta\Xi}\right), \tag{A.2351}$$

F2123 45_resolved_lorentzian_bv_parent.tex:357 CLASSICAL_BV_PASS

$$\boxed{(S_{\mathrm{BV}}, S_{\mathrm{BV}})_{\mathrm{BV}} = 0.} \tag{A.2352}$$

F2124 45_resolved_lorentzian_bv_parent.tex:369 CLASSICAL_BV_PASS

$$B_{IJ} := \frac{1}{4\kappa}\epsilon_{KLIJ}e^K \wedge e^L, \qquad \Pi := \frac{1}{2}\varphi_{IJ}e^I \wedge e^J. \tag{A.2353}$$

F2125 45_resolved_lorentzian_bv_parent.tex:379 CLASSICAL_BV_PASS

$$\boxed{\Theta_\Sigma = \int_\Sigma\left[B_{IJ}\wedge\delta\omega^{IJ} + \Pi\wedge\delta a + \frac{\mathrm{i}}{2}\Sigma_I\left(\bar{\psi}\gamma^I\delta\psi - \delta\bar{\psi}\gamma^I\psi\right)\right].} \tag{A.2354}$$

F2126 45_resolved_lorentzian_bv_parent.tex:395 CLASSICAL_BV_PASS

$$\delta_\omega S_{\mathrm{D}} = \int \delta\omega^{IJ}\wedge\mathcal{S}_{IJ}, \qquad \delta_a S_{\mathrm{D}} = \int \delta a\wedge\mathcal{J}. \tag{A.2355}$$

F2127 45_resolved_lorentzian_bv_parent.tex:405 CLASSICAL_BV_PASS

$$\mathcal{G}_{IJ} := D_\omega B_{IJ} + \mathcal{S}_{IJ}, \qquad \mathcal{G}_{\mathrm{U}(1)} := \mathrm{d}\Pi + \mathcal{J}. \tag{A.2356}$$

F2128 45_resolved_lorentzian_bv_parent.tex:415 CLASSICAL_BV_PASS

$$\{G_{\mathrm{L}}[\lambda], G_{\mathrm{L}}[\mu]\}_\Sigma = G_{\mathrm{L}}[[\lambda,\mu]], \tag{A.2357}$$

$$\{G_{\mathrm{L}}[\lambda], G_{\mathrm{U}}[u]\}_\Sigma = 0, \tag{A.2358}$$

$$\{G_{\mathrm{U}}[u], G_{\mathrm{U}}[v]\}_\Sigma = 0. \tag{A.2359}$$

F2129 45_resolved_lorentzian_bv_parent.tex:426 CLASSICAL_BV_PASS

$$\boxed{\Omega_{\Sigma,\mathrm{int}} = G_{\mathrm{L}}[\chi] + G_{\mathrm{U}}[\upsilon] - \frac{1}{2}\int_\Sigma \pi_{\chi,IJ}[\chi,\chi]^{IJ}.} \tag{A.2360}$$

F2130 45_resolved_lorentzian_bv_parent.tex:438 CLASSICAL_BV_PASS

$$\boxed{\{\Omega_{\Sigma,\mathrm{int}},\Omega_{\Sigma,\mathrm{int}}\}_\Sigma = 0.} \tag{A.2361}$$

F2131 45_resolved_lorentzian_bv_parent.tex:447 CLASSICAL_BV_PASS

$$\mathcal{T}_K^{(4)} \otimes \mathfrak{G}, \tag{A.2362}$$

F2132 45_resolved_lorentzian_bv_parent.tex:454 CLASSICAL_BV_PASS

$$\boxed{16K < N} \tag{A.2363}$$

F2133 45_resolved_lorentzian_bv_parent.tex:458 CLASSICAL_BV_PASS

$$\boxed{\begin{aligned} s_N^2 &= 0, \\ s_N S_{0,N} &= 0, \\ (S_{\mathrm{BV},N}, S_{\mathrm{BV},N})_{\mathrm{BV},N} &= 0. \end{aligned}} \tag{A.2364}$$

F2134 45_resolved_lorentzian_bv_parent.tex:471 CLASSICAL_BV_PASS

$$I_{K\to L} : \mathcal{T}_K^{(4)} \hookrightarrow \mathcal{T}_L^{(4)}, \tag{A.2365}$$

F2135 45_resolved_lorentzian_bv_parent.tex:475 CLASSICAL_BV_PASS

$$I_{L\to M} I_{K\to L} = I_{K\to M}. \tag{A.2366}$$

F2136 45_resolved_lorentzian_bv_parent.tex:483 CLASSICAL_BV_PASS

$$\boxed{D_\omega R^{IJ} = 0,} \tag{A.2367}$$

F2137 45_resolved_lorentzian_bv_parent.tex:486 CLASSICAL_BV_PASS

$$\boxed{D_\omega T^I = R^I{}_J \wedge e^J.} \tag{A.2368}$$

F2138 45_resolved_lorentzian_bv_parent.tex:492 CLASSICAL_BV_PASS

$$\sum_\Phi \mathcal{E}_\Phi \, s\Phi + \mathrm{d}\Theta_s = 0. \tag{A.2369}$$

F2139 45_resolved_lorentzian_bv_parent.tex:500 CLASSICAL_BV_PASS

$$(\star X)^{IJ} := \frac{1}{2}\epsilon^{IJ}{}_{KL} X^{KL}. \tag{A.2370}$$

F2140 45_resolved_lorentzian_bv_parent.tex:506 CLASSICAL_BV_PASS

$$\star^2 = -I. \tag{A.2371}$$

F2141 45_resolved_lorentzian_bv_parent.tex:510 CLASSICAL_BV_PASS

$$P_\pm := \frac{1}{2}(I \mp \mathrm{i}\star), \tag{A.2372}$$

F2142 45_resolved_lorentzian_bv_parent.tex:516 CLASSICAL_BV_PASS

$$P_\pm^2 = P_\pm, \qquad P_+ P_- = 0, \qquad P_+ + P_- = I. \tag{A.2373}$$

F2143 45_resolved_lorentzian_bv_parent.tex:524 CLASSICAL_BV_PASS

$$\boxed{\overline{P_+ X} = P_- X.} \tag{A.2374}$$

F2144 45_resolved_lorentzian_bv_parent.tex:529 CLASSICAL_BV_PASS

$$\mathfrak{C} S_{\mathrm{BV}} = S_{\mathrm{BV}}, \qquad \mathfrak{C} s = s \mathfrak{C}. \tag{A.2375}$$

F2145 45_resolved_lorentzian_bv_parent.tex:535 CLASSICAL_BV_PASS

$$X(0) \in \mathrm{Fix}(\mathfrak{C}) \quad \Longrightarrow \quad X(t) \in \mathrm{Fix}(\mathfrak{C}) \tag{A.2376}$$

F2146 45_resolved_lorentzian_bv_parent.tex:572 CLASSICAL_BV_PASS

$$4.425261 \times 10^{-3}. \tag{A.2377}$$

F2147 45_resolved_lorentzian_bv_parent.tex:603 CLASSICAL_BV_PASS

$$\boxed{\begin{array}{c} S_{\mathrm{BV},N}^{\mathrm{Lorentzian}} \xrightarrow{\text{3+1 split}} \{G_{IJ}, G_{\mathrm{U}(1)}, D_a, H\}_{\mathrm{matter}}, \\ \text{derive the complete real canonical bracket algebra,} \\ \text{construct one adaptive-refinement time map preserving all constraints,} \\ \text{prove convergence to Einstein–Maxwell–Dirac evolution.} \end{array}} \tag{A.2378}$$

F2148 45_resolved_lorentzian_bv_parent.tex:616 CLASSICAL_BV_PASS

$$\mathcal{R}_{\mathrm{LBP}} = \texttt{data/resolved_lorentzian_bv_parent_checks.csv}, \tag{A.2379}$$
$$\mathcal{S}_{\mathrm{LBP}} = \texttt{data/resolved_lorentzian_bv_parent_summary.csv}, \tag{A.2380}$$
$$\mathcal{X}_{\mathrm{LBP}} = \texttt{code/55_resolved_lorentzian_bv_parent.py}. \tag{A.2381}$$

46_resolved_lorentzian_3plus1_bfv_boundary.tex

F2149 46_resolved_lorentzian_3plus1_bfv_boundary.tex:14 BFV_BOUNDARY_PASS

$$M = I \times \Sigma, \qquad \mathrm{or}(M) = \mathrm{d}t \wedge \mathrm{or}(\Sigma). \tag{A.2382}$$

F2150 46_resolved_lorentzian_3plus1_bfv_boundary.tex:20 BFV_BOUNDARY_PASS

$$e^I = e^I_\perp \, \mathrm{d}t + E^I, \tag{A.2383}$$
$$\omega^{IJ} = \omega^{IJ}_\perp \, \mathrm{d}t + \Omega^{IJ}, \tag{A.2384}$$
$$a = a_\perp \, \mathrm{d}t + A. \tag{A.2385}$$

F2151 46_resolved_lorentzian_3plus1_bfv_boundary.tex:26 BFV_BOUNDARY_PASS

$$\mathcal{R}^{IJ} := \mathrm{d}_\Sigma \Omega^{IJ} + \Omega^I{}_K \wedge \Omega^{KJ}, \tag{A.2386}$$
$$\mathcal{K}^{IJ} := \dot{\Omega}^{IJ} - D_\Omega \omega^{IJ}_\perp, \tag{A.2387}$$
$$\mathcal{F} := \mathrm{d}_\Sigma A, \tag{A.2388}$$
$$\mathcal{E} := \dot{A} - \mathrm{d}_\Sigma a_\perp. \tag{A.2389}$$

F2152 46_resolved_lorentzian_3plus1_bfv_boundary.tex:37 BFV_BOUNDARY_PASS

$$\boxed{R^{IJ} = \mathrm{d}t \wedge \mathcal{K}^{IJ} + \mathcal{R}^{IJ}, \qquad F = \mathrm{d}t \wedge \mathcal{E} + \mathcal{F}.} \tag{A.2390}$$

F2153 46_resolved_lorentzian_3plus1_bfv_boundary.tex:44 BFV_BOUNDARY_PASS

$$\mathcal{D}_\Sigma \psi := \mathrm{d}_\Sigma \psi + \frac{1}{4} \Omega^{IJ} \gamma_{IJ} \psi - \mathrm{i} q A \psi, \tag{A.2391}$$
$$\mathcal{D}_\Sigma \bar{\psi} := \mathrm{d}_\Sigma \bar{\psi} - \frac{1}{4} \bar{\psi} \Omega^{IJ} \gamma_{IJ} + \mathrm{i} q A \bar{\psi}, \tag{A.2392}$$
$$\mathcal{D}_\perp \psi := \dot{\psi} + \frac{1}{4} \omega^{IJ}_\perp \gamma_{IJ} \psi - \mathrm{i} q a_\perp \psi, \tag{A.2393}$$
$$\mathcal{D}_\perp \bar{\psi} := \dot{\bar{\psi}} - \frac{1}{4} \bar{\psi} \omega^{IJ}_\perp \gamma_{IJ} + \mathrm{i} q a_\perp \bar{\psi}. \tag{A.2394}$$

F2154 46_resolved_lorentzian_3plus1_bfv_boundary.tex:55 BFV_BOUNDARY_PASS

$$\mathcal{D}^{(4)} \psi = \mathrm{d}t \, \mathcal{D}_\perp \psi + \mathcal{D}_\Sigma \psi, \qquad \mathcal{D}^{(4)} \bar{\psi} = \mathrm{d}t \, \mathcal{D}_\perp \bar{\psi} + \mathcal{D}_\Sigma \bar{\psi}. \tag{A.2395}$$

F2155 46_resolved_lorentzian_3plus1_bfv_boundary.tex:62 BFV_BOUNDARY_PASS

$$B_{IJ} := \frac{1}{4\kappa} \epsilon_{KLIJ} E^K \wedge E^L, \tag{A.2396}$$

$$\Pi := \frac{1}{2}\varphi_{IJ}E^I \wedge E^J, \tag{A.2397}$$

$$\sigma_I := \frac{1}{3!}\epsilon_{IJKL}E^J \wedge E^K \wedge E^L, \tag{A.2398}$$

$$\rho_I(e_\perp) := \frac{1}{2}\epsilon_{IJKL}e_\perp^J E^K \wedge E^L. \tag{A.2399}$$

F2156 46_resolved_lorentzian_3plus1_bfv_boundary.tex:73 BFV_BOUNDARY_PASS

$$\mathrm{vol}_e = \mathrm{d}t \wedge e_\perp^I \sigma_I, \qquad \Sigma_I = \sigma_I + \mathrm{d}t \wedge \rho_I(e_\perp). \tag{A.2400}$$

F2157 46_resolved_lorentzian_3plus1_bfv_boundary.tex:86 BFV_BOUNDARY_PASS

$$\boxed{S_{\mathrm{G}} = \int_I \mathrm{d}t \int_\Sigma \left[B_{IJ} \wedge \mathcal{K}^{IJ} + e_\perp^I \mathcal{C}_I^{\mathrm{G}} \right],} \tag{A.2401}$$

F2158 46_resolved_lorentzian_3plus1_bfv_boundary.tex:97 BFV_BOUNDARY_PASS

$$\boxed{\mathcal{C}_I^{\mathrm{G}} = \frac{1}{2\kappa}\epsilon_{IJKL}E^J \wedge \mathcal{R}^{KL} - \frac{\Lambda}{6\kappa}\epsilon_{IJKL}E^J \wedge E^K \wedge E^L.} \tag{A.2402}$$

F2159 46_resolved_lorentzian_3plus1_bfv_boundary.tex:109 BFV_BOUNDARY_PASS

$$\boxed{S_{\mathrm{M}}^{(1)} = \int_I \mathrm{d}t \int_\Sigma \left[\Pi \wedge \mathcal{E} + e_\perp^I \mathcal{C}_I^{\mathrm{M}} \right],} \tag{A.2403}$$

F2160 46_resolved_lorentzian_3plus1_bfv_boundary.tex:120 BFV_BOUNDARY_PASS

$$\boxed{\mathcal{C}_I^{\mathrm{M}} = \varphi_{IJ}\mathcal{F} \wedge E^J - \frac{1}{4}\varphi_{KL}\varphi^{KL}\sigma_I.} \tag{A.2404}$$

F2161 46_resolved_lorentzian_3plus1_bfv_boundary.tex:133 BFV_BOUNDARY_PASS

$$\boxed{\begin{aligned} S_{\mathrm{D}} = \int_I \mathrm{d}t \int_\Sigma \Bigg[& -\frac{\mathrm{i}}{2}\sigma_I \left(\bar{\psi}\gamma^I \mathcal{D}_\perp \psi - (\mathcal{D}_\perp \bar{\psi})\gamma^I \psi \right) \\ & + e_\perp^I \mathcal{C}_I^{\mathrm{D}} \Bigg]. \end{aligned}} \tag{A.2405}$$

F2162 46_resolved_lorentzian_3plus1_bfv_boundary.tex:150 BFV_BOUNDARY_PASS

$$\boxed{\begin{aligned} \mathcal{C}_I^{\mathrm{D}} = {} & \frac{\mathrm{i}}{4}\epsilon_{JIKL}E^K \wedge E^L \wedge \left[\bar{\psi}\gamma^J \mathcal{D}_\Sigma \psi - (\mathcal{D}_\Sigma \bar{\psi})\gamma^J \psi \right] \\ & - m\sigma_I \bar{\psi}\psi. \end{aligned}} \tag{A.2406}$$

F2163 46_resolved_lorentzian_3plus1_bfv_boundary.tex:168 BFV_BOUNDARY_PASS

$$\boxed{S_0 = \int_I \mathrm{d}t \left[\Theta_\Sigma(\dot{\Phi}) + G_{\mathrm{L}}[\omega_\perp] + G_{\mathrm{U}}[a_\perp] + C[e_\perp] \right],} \tag{A.2407}$$

F2164 46_resolved_lorentzian_3plus1_bfv_boundary.tex:184 BFV_BOUNDARY_PASS

$$\Theta_\Sigma = \int_\Sigma \Big[B_{IJ} \wedge \delta\Omega^{IJ} + \Pi \wedge \delta A \tag{A.2408}$$

$$- \frac{\mathrm{i}}{2}\sigma_I \left(\bar{\psi}\gamma^I \delta\psi - \delta\bar{\psi}\gamma^I \psi \right) \Big], \tag{A.2409}$$

F2165 46_resolved_lorentzian_3plus1_bfv_boundary.tex:202 BFV_BOUNDARY_PASS

$$C[e_\perp] := \int_\Sigma e_\perp^I \mathcal{C}_I, \qquad \mathcal{C}_I := \mathcal{C}_I^{\mathrm{G}} + \mathcal{C}_I^{\mathrm{M}} + \mathcal{C}_I^{\mathrm{D}}, \tag{A.2410}$$

F2166 46_resolved_lorentzian_3plus1_bfv_boundary.tex:209 BFV_BOUNDARY_PASS

$$G_{\mathrm{L}}[\lambda] := \int_\Sigma \lambda^{IJ}\mathcal{G}_{IJ}, \qquad G_{\mathrm{U}}[u] := \int_\Sigma u\mathcal{G}_{\mathrm{U}}, \tag{A.2411}$$

F2167 46_resolved_lorentzian_3plus1_bfv_boundary.tex:217 BFV_BOUNDARY_PASS

$$\mathcal{G}_{IJ} := D_\Omega B_{IJ} - \frac{\mathrm{i}}{8}\sigma_K \bar{\psi}\{\gamma^K, \gamma_{IJ}\}\psi, \tag{A.2412}$$

$$\mathcal{G}_{\mathrm{U}} := \mathrm{d}_\Sigma \Pi - q\sigma_I \bar{\psi}\gamma^I \psi. \tag{A.2413}$$

F2168 46_resolved_lorentzian_3plus1_bfv_boundary.tex:226 BFV_BOUNDARY_PASS

$$\boxed{\mathcal{G}_{IJ} = 0, \qquad \mathcal{G}_{\mathrm{U}} = 0, \qquad \mathcal{C}_I = 0.} \tag{A.2414}$$

F2169 46_resolved_lorentzian_3plus1_bfv_boundary.tex:238 BFV_BOUNDARY_PASS

$$\zeta^I(c) := \iota_c E^I, \tag{A.2415}$$

$$\widehat{\chi}^{IJ}(c,\chi) := \chi^{IJ} - \iota_c \Omega^{IJ}, \tag{A.2416}$$

$$\widehat{u}(c,u) := u + \iota_c A. \tag{A.2417}$$

F2170 46_resolved_lorentzian_3plus1_bfv_boundary.tex:244 BFV_BOUNDARY_PASS

$$\mathcal{L}_c \Omega^{IJ} = \iota_c \mathcal{R}^{IJ} + D_\Omega(\iota_c \Omega^{IJ}), \tag{A.2418}$$

$$\mathcal{L}_c A = \iota_c \mathcal{F} + \mathrm{d}_\Sigma(\iota_c A), \tag{A.2419}$$

$$\mathcal{L}_c E^I = \iota_c \mathcal{T}^I + D_\Omega(\iota_c E^I) - (\iota_c \Omega^I{}_J) E^J, \tag{A.2420}$$

F2171 46_resolved_lorentzian_3plus1_bfv_boundary.tex:254 BFV_BOUNDARY_PASS

$$\mathcal{T}^I := \mathrm{d}_\Sigma E^I + \Omega^I{}_J \wedge E^J. \tag{A.2421}$$

F2172 46_resolved_lorentzian_3plus1_bfv_boundary.tex:258 BFV_BOUNDARY_PASS

$$\boxed{J_\Sigma[c,\chi,u] = C[\zeta(c)] + G_{\mathrm{L}}[\widehat{\chi}(c,\chi)] + G_{\mathrm{U}}[\widehat{u}(c,u)].} \tag{A.2422}$$

F2173 46_resolved_lorentzian_3plus1_bfv_boundary.tex:273 BFV_BOUNDARY_PASS

$$(c^\mu, \chi^{IJ}, u) \tag{A.2423}$$

F2174 46_resolved_lorentzian_3plus1_bfv_boundary.tex:277 BFV_BOUNDARY_PASS

$$(b_\mu, b_{\chi,IJ}, b_u) \tag{A.2424}$$

F2175 46_resolved_lorentzian_3plus1_bfv_boundary.tex:281 BFV_BOUNDARY_PASS

$$sc^\mu = c^\nu \partial_\nu c^\mu, \tag{A.2425}$$

$$s\chi = \mathcal{L}_c \chi + \chi^2, \tag{A.2426}$$

$$su = \mathcal{L}_c u, \tag{A.2427}$$

F2176 46_resolved_lorentzian_3plus1_bfv_boundary.tex:287 BFV_BOUNDARY_PASS

$$\boxed{\begin{aligned}\Omega_\Sigma &= J_\Sigma[c,\chi,u] \\ &\quad + \int_\Sigma \left[b_\mu s c^\mu + \frac{1}{2} b_{\chi,IJ} s\chi^{IJ} + b_u s u \right].\end{aligned}} \tag{A.2428}$$

F2177 46_resolved_lorentzian_3plus1_bfv_boundary.tex:303 BFV_BOUNDARY_PASS

$$\boxed{\{\Omega_\Sigma, \Omega_\Sigma\}_\Sigma = 0.} \tag{A.2429}$$

F2178 46_resolved_lorentzian_3plus1_bfv_boundary.tex:310 BFV_BOUNDARY_PASS

$$\mathrm{Vect}(M) \ltimes [\mathfrak{so}(1,3) \oplus \mathfrak{u}(1)] \tag{A.2430}$$

F2179 46_resolved_lorentzian_3plus1_bfv_boundary.tex:323 BFV_BOUNDARY_PASS

$$q_{ab} := \eta_{IJ} E^I_a E^J_b, \tag{A.2431}$$

$$\widetilde{n}_I := \frac{1}{3!}\epsilon_{IJKL}\epsilon^{abc}E^J_a E^K_b E^L_c, \tag{A.2432}$$

$$\widetilde{q}^{ab} := \frac{1}{2}\epsilon^{acd}\epsilon^{bef}q_{ce}q_{df}. \tag{A.2433}$$

F2180 46_resolved_lorentzian_3plus1_bfv_boundary.tex:331 BFV_BOUNDARY_PASS

$$\widetilde{n}_I E^I_a = 0, \qquad -\eta^{IJ}\widetilde{n}_I\widetilde{n}_J = \det q, \qquad \widetilde{q}^{ab} = (\det q)q^{ab}. \tag{A.2434}$$

F2181 46_resolved_lorentzian_3plus1_bfv_boundary.tex:339 BFV_BOUNDARY_PASS

$$\zeta^I = \nu\widetilde{n}^I + \eta^a E^I_a, \tag{A.2435}$$

F2182 46_resolved_lorentzian_3plus1_bfv_boundary.tex:344 BFV_BOUNDARY_PASS

$$H[\nu] := C[\nu\widetilde{n}], \tag{A.2436}$$

$$V[\eta] := C[\eta^a E_a], \tag{A.2437}$$

$$D[\eta] := V[\eta] - G_{\mathrm{L}}[\eta^a\Omega_a] + G_{\mathrm{U}}[\eta^a A_a]. \tag{A.2438}$$

F2183 46_resolved_lorentzian_3plus1_bfv_boundary.tex:354 BFV_BOUNDARY_PASS

$$S^a(\nu,\mu) := \widetilde{q}^{ab}\left(\nu\partial_b\mu - \mu\partial_b\nu\right). \tag{A.2439}$$

F2184 46_resolved_lorentzian_3plus1_bfv_boundary.tex:362 BFV_BOUNDARY_PASS

$$\{D[\eta], D[\xi]\} = D[[\eta,\xi]], \tag{A.2440}$$

$$\{D[\eta], H[\nu]\} = H[\mathcal{L}^{(-1)}_\eta\nu], \tag{A.2441}$$

$$\{H[\nu], H[\mu]\} = V[S(\nu,\mu)] \tag{A.2442}$$

$$= D[S(\nu,\mu)] + G_{\mathrm{L}}[S^a\Omega_a] - G_{\mathrm{U}}[S^a A_a], \tag{A.2443}$$

$$\{G_{\mathrm{L}}[\lambda], H[\nu]\} = 0, \tag{A.2444}$$

$$\{G_{\mathrm{U}}[u], H[\nu]\} = 0. \tag{A.2445}$$

F2185 46_resolved_lorentzian_3plus1_bfv_boundary.tex:378 BFV_BOUNDARY_PASS

$$\mathcal{L}^{(-1)}_\eta\nu = \eta^a\partial_a\nu - \nu\partial_a\eta^a. \tag{A.2446}$$

F2186 46_resolved_lorentzian_3plus1_bfv_boundary.tex:385 BFV_BOUNDARY_PASS

$$\varepsilon = (\lambda, u, \eta, \nu). \tag{A.2447}$$

F2187 46_resolved_lorentzian_3plus1_bfv_boundary.tex:389 BFV_BOUNDARY_PASS

$$B_{\mathrm{L}}(\varepsilon_1,\varepsilon_2) = [\lambda,\mu] + \mathcal{L}_\eta\mu - \mathcal{L}_\xi\lambda + S^a(\nu,\rho)\Omega_a, \tag{A.2448}$$

$$B_{\mathrm{U}}(\varepsilon_1,\varepsilon_2) = \mathcal{L}_\eta v - \mathcal{L}_\xi u - S^a(\nu,\rho)A_a, \tag{A.2449}$$

$$B_D(\varepsilon_1,\varepsilon_2) = [\eta,\xi] + S(\nu,\rho), \tag{A.2450}$$

$$B_H(\varepsilon_1,\varepsilon_2) = \mathcal{L}^{(-1)}_\eta\rho - \mathcal{L}^{(-1)}_\xi\nu. \tag{A.2451}$$

F2188 46_resolved_lorentzian_3plus1_bfv_boundary.tex:404 BFV_BOUNDARY_PASS

$$\boxed{\Omega^{3+1}_\Sigma = G_{\mathrm{L}}[\gamma] + G_{\mathrm{U}}[\alpha] + D[\eta] + H[\nu] - \frac{1}{2}\langle b, B_{\mathrm{gr}}(c,c)\rangle.} \tag{A.2452}$$

F2189 46_resolved_lorentzian_3plus1_bfv_boundary.tex:416 BFV_BOUNDARY_PASS

$$\boxed{\{\Omega^{3+1}_\Sigma, \Omega^{3+1}_\Sigma\}_\Sigma = 0.} \tag{A.2453}$$

F2190 46_resolved_lorentzian_3plus1_bfv_boundary.tex:424 BFV_BOUNDARY_PASS

$$\boxed{16K < N.} \tag{A.2454}$$

F2191 46_resolved_lorentzian_3plus1_bfv_boundary.tex:429 BFV_BOUNDARY_PASS

$$\boxed{32K < N.} \tag{A.2455}$$

F2192 46_resolved_lorentzian_3plus1_bfv_boundary.tex:433 BFV_BOUNDARY_PASS

$$\boxed{\{\Omega^{3+1}_{\Sigma,N},\Omega^{3+1}_{\Sigma,N}\}_N=0.} \tag{A.2456}$$

F2193 46_resolved_lorentzian_3plus1_bfv_boundary.tex:442 BFV_BOUNDARY_PASS

$$I_{K\to L}:\mathcal{P}_{K,n}\hookrightarrow\mathcal{P}_{L,n} \tag{A.2457}$$

F2194 46_resolved_lorentzian_3plus1_bfv_boundary.tex:446 BFV_BOUNDARY_PASS

$$I^*_{K\to L}\omega_{L,n}=\omega_{K,n},\qquad I^*_{K\to L}\Omega_{L,n}=\Omega_{K,n} \tag{A.2458}$$

F2195 46_resolved_lorentzian_3plus1_bfv_boundary.tex:454 BFV_BOUNDARY_PASS

$$\mathcal{U}_{L,n}\subset\overline{\mathcal{P}}_{L,n}\times\mathcal{P}_{L,n+1}. \tag{A.2459}$$

F2196 46_resolved_lorentzian_3plus1_bfv_boundary.tex:460 BFV_BOUNDARY_PASS

$$\boxed{\Omega_{L,n+1}\big|_{\mathcal{U}_{L,n}}=\Omega_{L,n}\big|_{\mathcal{U}_{L,n}}.} \tag{A.2460}$$

F2197 46_resolved_lorentzian_3plus1_bfv_boundary.tex:467 BFV_BOUNDARY_PASS

$$\boxed{U^*_{L,n}\Omega_{L,n+1}=\Omega_{L,n}.} \tag{A.2461}$$

F2198 46_resolved_lorentzian_3plus1_bfv_boundary.tex:474 BFV_BOUNDARY_PASS

$$\boxed{U_{K\to L,n}:=U_{L,n}\circ I_{K\to L}.} \tag{A.2462}$$

F2199 46_resolved_lorentzian_3plus1_bfv_boundary.tex:480 BFV_BOUNDARY_PASS

$$U_{L\to M,n+1}\circ U_{K\to L,n}=U_{K\to M,[n,n+2]} \tag{A.2463}$$

F2200 46_resolved_lorentzian_3plus1_bfv_boundary.tex:514 BFV_BOUNDARY_PASS

$$1.118955\times10^{-14}. \tag{A.2464}$$

F2201 46_resolved_lorentzian_3plus1_bfv_boundary.tex:518 BFV_BOUNDARY_PASS

$$\boxed{\|D(fg)-(Df)g-fDg\|_\infty=6.258742\times10^{2}.} \tag{A.2465}$$

F2202 46_resolved_lorentzian_3plus1_bfv_boundary.tex:550 BFV_BOUNDARY_PASS

$$\boxed{\begin{array}{c}\text{solve the algebraic simplicity and fermionic primary sector,}\\ \text{obtain one minimal real canonical phase space,}\\ \text{construct a computable multisymplectic or gauge-fixed time step,}\\ \text{prove stability and uniform convergence across the refinement tower.}\end{array}} \tag{A.2466}$$

F2203 46_resolved_lorentzian_3plus1_bfv_boundary.tex:561 BFV_BOUNDARY_PASS

$$\mathcal{R}_{\mathrm{BFB}}=\texttt{data/resolved_lorentzian_3plus1_bfv_checks.csv}, \tag{A.2467}$$
$$\mathcal{S}_{\mathrm{BFB}}=\texttt{data/resolved_lorentzian_3plus1_bfv_summary.csv}, \tag{A.2468}$$
$$\mathcal{X}_{\mathrm{BFB}}=\texttt{code/56_resolved_lorentzian_3plus1_bfv_boundary.py}. \tag{A.2469}$$

47_resolved_second_class_reduction_integrator.tex

F2204 47_resolved_second_class_reduction_integrator.tex:12 SECOND_CLASS_REDUCTION_PASS

$$\eta_{IJ}=\mathrm{diag}(-1,1,1,1),\qquad \epsilon_{0123}=1, \tag{A.2470}$$

F2205 47_resolved_second_class_reduction_integrator.tex:18 SECOND_CLASS_REDUCTION_PASS

$$E^0 = 0, \qquad E^i = e_a{}^i \, \mathrm{d}x^a, \qquad e := \det(e_a{}^i) > 0. \tag{A.2471}$$

F2206 47_resolved_second_class_reduction_integrator.tex:26 SECOND_CLASS_REDUCTION_PASS

$$\widetilde{E}^a{}_i := \frac{1}{2}\epsilon^{abc}\epsilon_{ijk}e_b{}^j e_c{}^k, \qquad \widetilde{E}^a{}_i e_a{}^j = e\,\delta_i{}^j, \tag{A.2472}$$

F2207 47_resolved_second_class_reduction_integrator.tex:32 SECOND_CLASS_REDUCTION_PASS

$$q_{ab} := e_a{}^i e_{bi}, \qquad q := \det q_{ab} = e^2, \tag{A.2473}$$

F2208 47_resolved_second_class_reduction_integrator.tex:38 SECOND_CLASS_REDUCTION_PASS

$$K^i := \Omega^{0i}, \qquad \Gamma^i := -\frac{1}{2}\epsilon^i{}_{jk}\Omega^{jk}. \tag{A.2474}$$

F2209 47_resolved_second_class_reduction_integrator.tex:48 SECOND_CLASS_REDUCTION_PASS

$$B_{IJ} = \frac{1}{4\kappa}\epsilon_{KLIJ}E^K \wedge E^L. \tag{A.2475}$$

F2210 47_resolved_second_class_reduction_integrator.tex:52 SECOND_CLASS_REDUCTION_PASS

$$\boxed{B_{ij} = 0, \qquad B_{0i} = \frac{1}{4\kappa}\epsilon_{ijk}E^j \wedge E^k.} \tag{A.2476}$$

F2211 47_resolved_second_class_reduction_integrator.tex:59 SECOND_CLASS_REDUCTION_PASS

$$b^a{}_i := \frac{1}{2}\epsilon^{abc}(B_{0i})_{bc}, \tag{A.2477}$$

F2212 47_resolved_second_class_reduction_integrator.tex:63 SECOND_CLASS_REDUCTION_PASS

$$\boxed{b^a{}_i = \frac{1}{2\kappa}\widetilde{E}^a{}_i.} \tag{A.2478}$$

F2213 47_resolved_second_class_reduction_integrator.tex:68 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\int_\Sigma B_{IJ} \wedge \delta\Omega^{IJ} = \frac{1}{\kappa}\int_\Sigma \widetilde{E}^a{}_i \, \delta K_a{}^i \, \mathrm{d}^3x.} \tag{A.2479}$$

F2214 47_resolved_second_class_reduction_integrator.tex:80 SECOND_CLASS_REDUCTION_PASS

$$\mathcal{E}^a := \frac{1}{2}\epsilon^{abc}\Pi_{bc}, \qquad \mathcal{B}^a := \frac{1}{2}\epsilon^{abc}\mathcal{F}_{bc}. \tag{A.2480}$$

F2215 47_resolved_second_class_reduction_integrator.tex:86 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\int_\Sigma \Pi \wedge \delta A = \int_\Sigma \mathcal{E}^a \delta A_a \, \mathrm{d}^3x.} \tag{A.2481}$$

F2216 47_resolved_second_class_reduction_integrator.tex:93 SECOND_CLASS_REDUCTION_PASS

$$\varphi^{IJ} = (\star_e F)^{IJ}. \tag{A.2482}$$

F2217 47_resolved_second_class_reduction_integrator.tex:97 SECOND_CLASS_REDUCTION_PASS

$$\mathcal{H}_{\mathrm{M}} = \frac{1}{2e}q_{ab}\left(\mathcal{E}^a\mathcal{E}^b + \mathcal{B}^a\mathcal{B}^b\right), \tag{A.2483}$$

F2218 47_resolved_second_class_reduction_integrator.tex:106 SECOND_CLASS_REDUCTION_PASS

$$\mathcal{V}^{\mathrm{M}}_a = \mathcal{F}_{ab}\mathcal{E}^b. \tag{A.2484}$$

F2219 47_resolved_second_class_reduction_integrator.tex:113 SECOND_CLASS_REDUCTION_PASS

$$\bar{\psi} = \psi^\dagger\gamma^0, \qquad \psi^\dagger = -\bar{\psi}\gamma^0, \tag{A.2485}$$

F2220 47_resolved_second_class_reduction_integrator.tex:119 SECOND_CLASS_REDUCTION_PASS

$$\sigma_0 = e \, \mathrm{d}^3x, \qquad \sigma_i = 0. \tag{A.2486}$$

F2221 47_resolved_second_class_reduction_integrator.tex:125 SECOND_CLASS_REDUCTION_PASS

$$\Theta_{\mathrm{D}} = \frac{\mathrm{i}}{2}\int_{\Sigma} e\left(\psi^{\dagger}\delta\psi - \delta\psi^{\dagger}\psi\right)\,\mathrm{d}^{3}x. \tag{A.2487}$$

F2222 47_resolved_second_class_reduction_integrator.tex:135 SECOND_CLASS_REDUCTION_PASS

$$\chi_A := \pi_A - \frac{\mathrm{i}}{2}e\psi_A^{\dagger}, \qquad \bar{\chi}_A := \bar{\pi}_A + \frac{\mathrm{i}}{2}e\psi_A, \tag{A.2488}$$

F2223 47_resolved_second_class_reduction_integrator.tex:143 SECOND_CLASS_REDUCTION_PASS

$$\{\chi_A(x), \bar{\chi}_B(y)\}_{\mathrm{P}} = -\mathrm{i}e(x)\delta_{AB}\delta^{3}(x-y). \tag{A.2489}$$

F2224 47_resolved_second_class_reduction_integrator.tex:150 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\Psi := e^{1/2}\psi, \qquad \Psi^{\dagger} := e^{1/2}\psi^{\dagger}.} \tag{A.2490}$$

F2225 47_resolved_second_class_reduction_integrator.tex:157 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\Theta_{\mathrm{D}} = \frac{\mathrm{i}}{2}\int_{\Sigma}\left(\Psi^{\dagger}\delta\Psi - \delta\Psi^{\dagger}\Psi\right)\,\mathrm{d}^{3}x.} \tag{A.2491}$$

F2226 47_resolved_second_class_reduction_integrator.tex:168 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\{\Psi_A(x), \Psi_B^{\dagger}(y)\}_{\mathrm{D}} = -\mathrm{i}\delta_{AB}\delta^{3}(x-y).} \tag{A.2492}$$

F2227 47_resolved_second_class_reduction_integrator.tex:178 SECOND_CLASS_REDUCTION_PASS

$$\gamma_5 := \mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3, \qquad J_5^I := \bar{\psi}\gamma^I\gamma_5\psi. \tag{A.2493}$$

F2228 47_resolved_second_class_reduction_integrator.tex:184 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\{\gamma^I, \gamma_{KL}\} = -2\mathrm{i}\epsilon^I{}_{KLM}\gamma^M\gamma_5.} \tag{A.2494}$$

F2229 47_resolved_second_class_reduction_integrator.tex:190 SECOND_CLASS_REDUCTION_PASS

$$-\frac{1}{2\kappa}\epsilon_{IJKL}T^I\wedge e^J + \frac{1}{4}\Sigma_M\epsilon^M{}_{KLN}J_5^N = 0. \tag{A.2495}$$

F2230 47_resolved_second_class_reduction_integrator.tex:197 SECOND_CLASS_REDUCTION_PASS

$$\boxed{T^I = \frac{\kappa}{4}\epsilon^I{}_{JKL}J_5^J e^K\wedge e^L,} \tag{A.2496}$$

F2231 47_resolved_second_class_reduction_integrator.tex:202 SECOND_CLASS_REDUCTION_PASS

$$\boxed{C^{IJ} = \frac{\kappa}{4}\epsilon^{IJ}{}_{KL}J_5^K e^L, \qquad \omega^{IJ} = \mathring{\omega}^{IJ}(e) + C^{IJ}.} \tag{A.2497}$$

F2232 47_resolved_second_class_reduction_integrator.tex:210 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\Gamma_a{}^i = \mathring{\Gamma}_a{}^i(E) - \frac{\kappa}{4}J_5^0 e_a{}^i,} \tag{A.2498}$$

F2233 47_resolved_second_class_reduction_integrator.tex:218 SECOND_CLASS_REDUCTION_PASS

$$K_a{}^i = \mathring{K}_a{}^i - \frac{\kappa}{4}\epsilon^i{}_{jk}J_5^j e_a{}^k. \tag{A.2499}$$

F2234 47_resolved_second_class_reduction_integrator.tex:227 SECOND_CLASS_REDUCTION_PASS

$$-\frac{3\kappa}{8}J_{5I}J_5^I\,\mathrm{vol}_e\,. \tag{A.2500}$$

F2235 47_resolved_second_class_reduction_integrator.tex:231 SECOND_CLASS_REDUCTION_PASS

$$\boxed{S_{\mathrm{EC+D}}\big|_{\omega} = S_{\mathrm{EH}} + S_{\mathrm{D}}[\mathring{\omega}] - \frac{3\kappa}{16}\int J_{5I}J_5^I\,\mathrm{vol}_e\,.} \tag{A.2501}$$

F2236 47_resolved_second_class_reduction_integrator.tex:241 SECOND_CLASS_REDUCTION_PASS

$$\mathcal{H}_{4\psi} = \frac{3\kappa}{16} e\, J_{5I} J_5^I. \tag{A.2502}$$

F2237 47_resolved_second_class_reduction_integrator.tex:251 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\mathcal{P}_{\mathrm{red}} = \left\{ (K_a{}^i, \widetilde{E}^a{}_i), (A_a, \mathcal{E}^a), (\Psi, \Psi^\dagger) \right\}.} \tag{A.2503}$$

F2238 47_resolved_second_class_reduction_integrator.tex:262 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\begin{aligned}\Theta_{\mathrm{red}} = \int_\Sigma \mathrm{d}^3x \Big[& \frac{1}{\kappa} \widetilde{E}^a{}_i \delta K_a{}^i + \mathcal{E}^a \delta A_a \\ & + \frac{\mathrm{i}}{2} \left(\Psi^\dagger \delta \Psi - \delta \Psi^\dagger \Psi \right) \Big].\end{aligned}} \tag{A.2504}$$

F2239 47_resolved_second_class_reduction_integrator.tex:279 SECOND_CLASS_REDUCTION_PASS

$$\{K_a{}^i(x), \widetilde{E}^b{}_j(y)\}_{\mathrm{D}} = \kappa \delta_a{}^b \delta^i{}_j \delta^3(x-y), \tag{A.2505}$$

$$\{A_a(x), \mathcal{E}^b(y)\}_{\mathrm{D}} = \delta_a{}^b \delta^3(x-y), \tag{A.2506}$$

$$\{\Psi_A(x), \Psi_B^\dagger(y)\}_{\mathrm{D}} = -\mathrm{i}\delta_{AB} \delta^3(x-y). \tag{A.2507}$$

F2240 47_resolved_second_class_reduction_integrator.tex:292 SECOND_CLASS_REDUCTION_PASS

$$\mathsf{S}_i := -\frac{\mathrm{i}}{4} \epsilon_{ijk} \gamma^{jk}. \tag{A.2508}$$

F2241 47_resolved_second_class_reduction_integrator.tex:297 SECOND_CLASS_REDUCTION_PASS

$$[\mathsf{S}_i, \mathsf{S}_j] = \mathrm{i}\epsilon_{ij}{}^k \mathsf{S}_k, \qquad \gamma^0 \gamma^i \gamma_5 = -2\mathsf{S}_i. \tag{A.2509}$$

F2242 47_resolved_second_class_reduction_integrator.tex:304 SECOND_CLASS_REDUCTION_PASS

$$\boxed{G_{\mathrm{rot}}[\theta] = \int_\Sigma \mathrm{d}^3x\, \theta^i \left[\frac{1}{\kappa} \epsilon_{ij}{}^k K_a{}^j \widetilde{E}^a{}_k + \Psi^\dagger \mathsf{S}_i \Psi \right].} \tag{A.2510}$$

F2243 47_resolved_second_class_reduction_integrator.tex:315 SECOND_CLASS_REDUCTION_PASS

$$\delta_\theta K_a{}^i = \epsilon^i{}_{jk} \theta^j K_a{}^k, \tag{A.2511}$$

$$\delta_\theta \widetilde{E}^a{}_i = \epsilon_{ij}{}^k \theta^j \widetilde{E}^a{}_k, \tag{A.2512}$$

$$\delta_\theta \Psi = -\mathrm{i}\theta^i \mathsf{S}_i \Psi. \tag{A.2513}$$

F2244 47_resolved_second_class_reduction_integrator.tex:324 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\mathcal{G}_{\mathrm{U}}^{\mathrm{red}} = \partial_a \mathcal{E}^a + q \Psi^\dagger \Psi.} \tag{A.2514}$$

F2245 47_resolved_second_class_reduction_integrator.tex:330 SECOND_CLASS_REDUCTION_PASS

$$Q[u] := -\int_\Sigma u\, \mathcal{G}_{\mathrm{U}}^{\mathrm{red}}\, \mathrm{d}^3x, \tag{A.2515}$$

F2246 47_resolved_second_class_reduction_integrator.tex:334 SECOND_CLASS_REDUCTION_PASS

$$\delta_u A_a = \partial_a u, \qquad \delta_u \Psi = \mathrm{i}q u \Psi. \tag{A.2516}$$

F2247 47_resolved_second_class_reduction_integrator.tex:341 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\begin{aligned}D[N] = \int_\Sigma \mathrm{d}^3x \Big[& \frac{1}{\kappa} \widetilde{E}^a{}_i \mathcal{L}_N K_a{}^i + \mathcal{E}^a \mathcal{L}_N A_a \\ & + \frac{\mathrm{i}}{2} \left(\Psi^\dagger \mathcal{L}_N^{(1/2)} \Psi - (\mathcal{L}_N^{(1/2)} \Psi^\dagger) \Psi \right) \Big],\end{aligned}} \tag{A.2517}$$

F2248 47_resolved_second_class_reduction_integrator.tex:357 SECOND_CLASS_REDUCTION_PASS

$$\mathcal{L}_N^{(1/2)}\Psi = N^a\partial_a\Psi + \frac{1}{2}(\partial_a N^a)\Psi. \tag{A.2518}$$

F2249 47_resolved_second_class_reduction_integrator.tex:363 SECOND_CLASS_REDUCTION_PASS

$$R^i(\Gamma) := \mathrm{d}_\Sigma\Gamma^i + \frac{1}{2}\epsilon^i{}_{jk}\Gamma^j\wedge\Gamma^k. \tag{A.2519}$$

F2250 47_resolved_second_class_reduction_integrator.tex:369 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\mathcal{H}_{\mathrm{G}}\,\mathrm{d}^3x = \frac{1}{\kappa}E^i\wedge R_i(\Gamma) - \frac{1}{2\kappa}\epsilon_{ijk}E^i\wedge K^j\wedge K^k + \frac{\Lambda}{\kappa}\,\mathrm{vol}_\Sigma,} \tag{A.2520}$$

F2251 47_resolved_second_class_reduction_integrator.tex:378 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\begin{aligned}\mathcal{H}_{\mathrm{D}}\,\mathrm{d}^3x = {}& \frac{\mathrm{i}}{4}\epsilon_{jkl}E^k\wedge E^l\wedge\left[\bar\psi\gamma^j\mathcal{D}\psi - (\mathcal{D}\bar\psi)\gamma^j\psi\right]\\ &+ m\,\mathrm{vol}_\Sigma\,\bar\psi\psi,\end{aligned}} \tag{A.2521}$$

F2252 47_resolved_second_class_reduction_integrator.tex:392 SECOND_CLASS_REDUCTION_PASS

$$\boxed{H[N] = \int_\Sigma N\left(\mathcal{H}_{\mathrm{G}} + \mathcal{H}_{\mathrm{M}} + \mathcal{H}_{\mathrm{D}}\right)\,\mathrm{d}^3x.} \tag{A.2522}$$

F2253 47_resolved_second_class_reduction_integrator.tex:410 SECOND_CLASS_REDUCTION_PASS

$$\Omega_{\mathrm{red}} := \iota^*\Omega_\Sigma \tag{A.2523}$$

F2254 47_resolved_second_class_reduction_integrator.tex:414 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\{\Omega_{\mathrm{red}},\Omega_{\mathrm{red}}\}_{\mathrm{D}} = 0.} \tag{A.2524}$$

F2255 47_resolved_second_class_reduction_integrator.tex:419 SECOND_CLASS_REDUCTION_PASS

$$\iota^*\{\Omega_\Sigma,\Omega_\Sigma\}_\Sigma = 0. \tag{A.2525}$$

F2256 47_resolved_second_class_reduction_integrator.tex:427 SECOND_CLASS_REDUCTION_PASS

$$z\in\mathbb{R}^{2n}, \qquad J^T = -J, \tag{A.2526}$$

F2257 47_resolved_second_class_reduction_integrator.tex:433 SECOND_CLASS_REDUCTION_PASS

$$\Phi(z) = (C_1,\dots,C_m,\chi_1,\dots,\chi_m)^T \tag{A.2527}$$

F2258 47_resolved_second_class_reduction_integrator.tex:439 SECOND_CLASS_REDUCTION_PASS

$$\Delta(z) := \nabla\Phi(z)J\nabla\Phi(z)^T \tag{A.2528}$$

F2259 47_resolved_second_class_reduction_integrator.tex:446 SECOND_CLASS_REDUCTION_PASS

$$\overline{\nabla}F(z_0,z_1) := \int_0^1 \nabla F((1-s)z_0 + sz_1)\,\mathrm{d}s. \tag{A.2529}$$

F2260 47_resolved_second_class_reduction_integrator.tex:452 SECOND_CLASS_REDUCTION_PASS

$$F(z_1) - F(z_0) = \overline{\nabla}F(z_0,z_1)^T(z_1 - z_0). \tag{A.2530}$$

F2261 47_resolved_second_class_reduction_integrator.tex:457 SECOND_CLASS_REDUCTION_PASS

$$G_\Delta := \begin{pmatrix}\overline{\nabla}\Phi_1^T\\ \vdots\\ \overline{\nabla}\Phi_{2m}^T\end{pmatrix}, \qquad \Delta_\Delta := G_\Delta J G_\Delta^T. \tag{A.2531}$$

F2262 47_resolved_second_class_reduction_integrator.tex:469 SECOND_CLASS_REDUCTION_PASS

$$\boxed{J_{\mathrm{D}}^\Delta := J - JG_\Delta^T\Delta_\Delta^{-1}G_\Delta J.} \tag{A.2532}$$

F2263 47_resolved_second_class_reduction_integrator.tex:475 SECOND_CLASS_REDUCTION_PASS

$$(J_{\mathrm{D}}^{\Delta})^{T} = -J_{\mathrm{D}}^{\Delta}, \qquad G_{\Delta} J_{\mathrm{D}}^{\Delta} = 0. \tag{A.2533}$$

F2264 47_resolved_second_class_reduction_integrator.tex:481 SECOND_CLASS_REDUCTION_PASS

$$\boxed{z_1 - z_0 = \Delta t\, J_{\mathrm{D}}^{\Delta}(z_0, z_1)\overline{\nabla} H(z_0, z_1).} \tag{A.2534}$$

F2265 47_resolved_second_class_reduction_integrator.tex:491 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\Phi(z_1) = \Phi(z_0), \qquad H(z_1) = H(z_0).} \tag{A.2535}$$

F2266 47_resolved_second_class_reduction_integrator.tex:498 SECOND_CLASS_REDUCTION_PASS

$$\Phi(z_1) - \Phi(z_0) = G_{\Delta}(z_1 - z_0) = 0, \tag{A.2536}$$

$$H(z_1) - H(z_0) = \Delta t\, \overline{\nabla} H^{T} J_{\mathrm{D}}^{\Delta} \overline{\nabla} H = 0. \tag{A.2537}$$

F2267 47_resolved_second_class_reduction_integrator.tex:512 SECOND_CLASS_REDUCTION_PASS

$$\sigma_{\min}(\Delta(z)) \geq \delta > 0 \tag{A.2538}$$

F2268 47_resolved_second_class_reduction_integrator.tex:518 SECOND_CLASS_REDUCTION_PASS

$$H(z) \geq a\|z\|^{p} - b, \qquad a > 0, \tag{A.2539}$$

F2269 47_resolved_second_class_reduction_integrator.tex:524 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\|z_n\| \leq \left(\frac{H(z_0) + b}{a}\right)^{1/p}} \tag{A.2540}$$

F2270 47_resolved_second_class_reduction_integrator.tex:535 SECOND_CLASS_REDUCTION_PASS

$$I_{K \to L} : \mathcal{P}_K \hookrightarrow \mathcal{P}_L \tag{A.2541}$$

F2271 47_resolved_second_class_reduction_integrator.tex:539 SECOND_CLASS_REDUCTION_PASS

$$U_{K \to L}^{\Delta t} := U_{L}^{\Delta t} \circ I_{K \to L}. \tag{A.2542}$$

F2272 47_resolved_second_class_reduction_integrator.tex:568 SECOND_CLASS_REDUCTION_PASS

$$\frac{\mathcal{L}_{\mathrm{D},C}}{\kappa J_5^2} = -\frac{3}{8}, \qquad \frac{\mathcal{L}_{\text{on-shell},C}}{\kappa J_5^2} = -\frac{3}{16}. \tag{A.2543}$$

F2273 47_resolved_second_class_reduction_integrator.tex:575 SECOND_CLASS_REDUCTION_PASS

$$q \cdot q = 1, \qquad q \cdot p = 0 \tag{A.2544}$$

F2274 47_resolved_second_class_reduction_integrator.tex:581 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\epsilon_H = 7.771561 \times 10^{-16}, \qquad \epsilon_{\Phi} = 3.885781 \times 10^{-15}, \qquad \epsilon_{\mathrm{solve}} = 1.092187 \times 10^{-16}.} \tag{A.2545}$$

F2275 47_resolved_second_class_reduction_integrator.tex:590 SECOND_CLASS_REDUCTION_PASS

$$\epsilon_H^{\mathrm{RK4}} = 1.396105 \times 10^{-6}, \qquad \epsilon_{\Phi}^{\mathrm{RK4}} = 2.594613 \times 10^{-6}. \tag{A.2546}$$

F2276 47_resolved_second_class_reduction_integrator.tex:624 SECOND_CLASS_REDUCTION_PASS

$$\boxed{\begin{array}{c}\text{implement the reduced Einstein–Maxwell–Dirac Hamiltonian on a finite-range complex,}\\ \text{choose a globally controlled gauge and solve the implicit Dirac-gradient slab step,}\\ \text{derive level-independent coercivity and constraint-regularity estimates,}\\ \text{prove convergence and run the first nonlinear strong-field benchmark.}\end{array}} \tag{A.2547}$$

F2277 47_resolved_second_class_reduction_integrator.tex:636 SECOND_CLASS_REDUCTION_PASS

$$R_{\mathrm{SC}} := \texttt{data/resolved_second_class_reduction_checks.csv}, \tag{A.2548}$$

$$S_{\mathrm{SC}} := \texttt{data/resolved_second_class_reduction_summary.csv}, \tag{A.2549}$$

$$X_{\mathrm{SC}} := \texttt{code/57_resolved_second_class_reduction.py}. \tag{A.2550}$$

48_finite_range_spherical_electrovac_strong_field.tex

F2278 48_finite_range_spherical_electrovac_strong_field.tex:10 FINITE_RANGE_STRONG_FIELD_PASS

$$G = c = 4\pi\epsilon_0 = 1. \tag{A.2551}$$

F2279 48_finite_range_spherical_electrovac_strong_field.tex:18 FINITE_RANGE_STRONG_FIELD_PASS

$$\mathrm{d}s^2 = -N^2\,\mathrm{d}t^2 + \Lambda^2(\,\mathrm{d}r + N^r\,\mathrm{d}t)^2 + R^2\,\mathrm{d}\Omega^2, \qquad \mathcal{A} = A_r\,\mathrm{d}r + \Phi\,\mathrm{d}t. \tag{A.2552}$$

F2280 48_finite_range_spherical_electrovac_strong_field.tex:27 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{S_{\mathrm{sph}} = \int \mathrm{d}t \int \mathrm{d}r \left[P_\Lambda \dot{\Lambda} + P_R \dot{R} + Q\dot{A}_r - N\mathcal{H} - N^r\mathcal{D} - \Phi\mathcal{G}\right] + B_\partial.} \tag{A.2553}$$

F2281 48_finite_range_spherical_electrovac_strong_field.tex:38 FINITE_RANGE_STRONG_FIELD_PASS

$$\mathcal{D} := P_R R' - \Lambda P'_\Lambda, \tag{A.2554}$$

F2282 48_finite_range_spherical_electrovac_strong_field.tex:43 FINITE_RANGE_STRONG_FIELD_PASS

$$\mathcal{G} := Q'. \tag{A.2555}$$

F2283 48_finite_range_spherical_electrovac_strong_field.tex:47 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{\mathcal{H} = -\frac{P_R P_\Lambda}{R} + \frac{\Lambda P_\Lambda^2}{2R^2} + \frac{RR''}{\Lambda} - \frac{RR'\Lambda'}{\Lambda^2} + \frac{R'^2}{2\Lambda} - \frac{\Lambda}{2} + \frac{\Lambda Q^2}{2R^2}.} \tag{A.2556}$$

F2284 48_finite_range_spherical_electrovac_strong_field.tex:62 FINITE_RANGE_STRONG_FIELD_PASS

$$P_\Lambda = -\frac{R}{N}(\dot{R} - N^r R'), \tag{A.2557}$$

F2285 48_finite_range_spherical_electrovac_strong_field.tex:66 FINITE_RANGE_STRONG_FIELD_PASS

$$P_R = -\frac{1}{N}\left[\Lambda(\dot{R} - N^r R') + R\big(\dot{\Lambda} - (N^r\Lambda)'\big)\right]. \tag{A.2558}$$

F2286 48_finite_range_spherical_electrovac_strong_field.tex:78 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{\mathcal{M} := \frac{R}{2}\left[1 - \left(\frac{R'}{\Lambda}\right)^2 + \left(\frac{P_\Lambda}{R}\right)^2\right] + \frac{Q^2}{2R}.} \tag{A.2559}$$

F2287 48_finite_range_spherical_electrovac_strong_field.tex:93 FINITE_RANGE_STRONG_FIELD_PASS

eq:spherical-mass-identity

$$\boxed{\mathcal{M}' = -\frac{R'}{\Lambda}\mathcal{H} - \frac{P_\Lambda}{R\Lambda}\mathcal{D} + \frac{Q}{R}\mathcal{G}.} \tag{A.2560}$$

F2288 48_finite_range_spherical_electrovac_strong_field.tex:103 FINITE_RANGE_STRONG_FIELD_PASS

$$\mathcal{M}' = \frac{R'}{2} - \frac{R'^3}{2\Lambda^2} - \frac{RR'R''}{\Lambda^2} + \frac{RR'^2\Lambda'}{\Lambda^3} + \frac{P_\Lambda P'_\Lambda}{R} - \frac{P_\Lambda^2 R'}{2R^2} + \frac{QQ'}{R} - \frac{Q^2R'}{2R^2}. \tag{A.2561}$$

F2289 48_finite_range_spherical_electrovac_strong_field.tex:117 FINITE_RANGE_STRONG_FIELD_PASS

$$\frac{R'}{\Lambda} \neq 0, \tag{A.2562}$$

F2290 48_finite_range_spherical_electrovac_strong_field.tex:121 FINITE_RANGE_STRONG_FIELD_PASS

$$\begin{pmatrix} \mathcal{M}' \\ \mathcal{D} \\ \mathcal{G} \end{pmatrix} = \begin{pmatrix} -R'/\Lambda & -P_\Lambda/(R\Lambda) & Q/R \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mathcal{H} \\ \mathcal{D} \\ \mathcal{G} \end{pmatrix}. \tag{A.2563}$$

F2291 48_finite_range_spherical_electrovac_strong_field.tex:140 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{\mathcal{H} = \mathcal{D} = \mathcal{G} = 0 \quad\Longleftrightarrow\quad \mathcal{M}' = \mathcal{D} = \mathcal{G} = 0} \tag{A.2564}$$

F2292 48_finite_range_spherical_electrovac_strong_field.tex:154 FINITE_RANGE_STRONG_FIELD_PASS

$$v(R) := \sigma\sqrt{\frac{2M}{R} - \frac{Q^2}{R^2}}, \qquad \sigma \in \{+1, -1\}. \tag{A.2565}$$

F2293 48_finite_range_spherical_electrovac_strong_field.tex:160 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{\mathrm{d}s^2 = -\,\mathrm{d}t_{\mathrm{P}}^2 + \Big(\mathrm{d}R + v(R)\,\mathrm{d}t_{\mathrm{P}}\Big)^2 + R^2\,\mathrm{d}\Omega^2.} \tag{A.2566}$$

F2294 48_finite_range_spherical_electrovac_strong_field.tex:169 FINITE_RANGE_STRONG_FIELD_PASS

$$R = F(t, r), \qquad F_r > 0, \qquad t_{\mathrm{P}} = t. \tag{A.2567}$$

F2295 48_finite_range_spherical_electrovac_strong_field.tex:177 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{N = 1, \qquad \Lambda = F_r, \qquad N^r = \frac{F_t + v(F)}{F_r}.} \tag{A.2568}$$

F2296 48_finite_range_spherical_electrovac_strong_field.tex:187 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{P_\Lambda = F v(F) = \sigma\sqrt{2MF - Q^2},} \tag{A.2569}$$

F2297 48_finite_range_spherical_electrovac_strong_field.tex:194 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{P_R = F_r\left[v(F) + F v'(F)\right] = \sigma\frac{MF_r}{\sqrt{2MF - Q^2}}.} \tag{A.2570}$$

F2298 48_finite_range_spherical_electrovac_strong_field.tex:202 FINITE_RANGE_STRONG_FIELD_PASS

$$A_r = 0, \qquad \Phi = -\frac{Q}{F}, \qquad Q' = 0. \tag{A.2571}$$

F2299 48_finite_range_spherical_electrovac_strong_field.tex:210 FINITE_RANGE_STRONG_FIELD_PASS

$$\Phi' = \frac{QF_r}{F^2}, \tag{A.2572}$$

F2300 48_finite_range_spherical_electrovac_strong_field.tex:214 FINITE_RANGE_STRONG_FIELD_PASS

$$\dot{A}_r = \frac{N\Lambda Q}{R^2} - \Phi' \tag{A.2573}$$

F2301 48_finite_range_spherical_electrovac_strong_field.tex:222 FINITE_RANGE_STRONG_FIELD_PASS

$$\frac{R'}{\Lambda} = 1, \qquad \frac{P_\Lambda^2}{R^2} = \frac{2M}{R} - \frac{Q^2}{R^2}. \tag{A.2574}$$

F2302 48_finite_range_spherical_electrovac_strong_field.tex:229 FINITE_RANGE_STRONG_FIELD_PASS

$$\mathcal{M} = M. \tag{A.2575}$$

F2303 48_finite_range_spherical_electrovac_strong_field.tex:233 FINITE_RANGE_STRONG_FIELD_PASS

$$P'_\Lambda = \sigma\frac{MF_r}{\sqrt{2MF - Q^2}} = P_R, \tag{A.2576}$$

F2304 48_finite_range_spherical_electrovac_strong_field.tex:239 FINITE_RANGE_STRONG_FIELD_PASS

$$\mathcal{D} = P_R R' - \Lambda P'_\Lambda = 0. \tag{A.2577}$$

F2305 48_finite_range_spherical_electrovac_strong_field.tex:243 FINITE_RANGE_STRONG_FIELD_PASS

$$\mathcal{H} = 0. \tag{A.2578}$$

F2306 48_finite_range_spherical_electrovac_strong_field.tex:250 FINITE_RANGE_STRONG_FIELD_PASS

$$\dot{R} = -\frac{P_\Lambda}{R} + N^r R', \tag{A.2579}$$

F2307 48_finite_range_spherical_electrovac_strong_field.tex:253 FINITE_RANGE_STRONG_FIELD_PASS

$$\dot{\Lambda} = \frac{\Lambda P_\Lambda}{R^2} - \frac{P_R}{R} + (N^r \Lambda)', \tag{A.2580}$$

F2308 48_finite_range_spherical_electrovac_strong_field.tex:259 FINITE_RANGE_STRONG_FIELD_PASS

$$\dot{P}_\Lambda = N^r P_\Lambda' + \frac{1}{2}\left[1 - \frac{P_\Lambda^2 + Q^2}{R^2} - \left(\frac{R'}{\Lambda}\right)^2\right], \tag{A.2581}$$

F2309 48_finite_range_spherical_electrovac_strong_field.tex:268 FINITE_RANGE_STRONG_FIELD_PASS

$$\dot{P}_R = \frac{\Lambda(P_\Lambda^2 + Q^2)}{R^3} - \frac{P_\Lambda P_R}{R^2} + (N^r P_R)' - \frac{R''}{\Lambda} + \frac{\Lambda' R'}{\Lambda^2}. \tag{A.2582}$$

F2310 48_finite_range_spherical_electrovac_strong_field.tex:279 FINITE_RANGE_STRONG_FIELD_PASS

$$\dot{R} = F_t, \qquad \dot{\Lambda} = F_{tr}, \qquad \dot{P}_\Lambda = \frac{M F_t}{P_\Lambda}, \tag{A.2583}$$

F2311 48_finite_range_spherical_electrovac_strong_field.tex:286 FINITE_RANGE_STRONG_FIELD_PASS

$$\dot{P}_R = M\left[\frac{F_{tr}}{P_\Lambda} - \frac{M F_r F_t}{P_\Lambda^3}\right], \tag{A.2584}$$

F2312 48_finite_range_spherical_electrovac_strong_field.tex:300 FINITE_RANGE_STRONG_FIELD_PASS

$$\Delta_M := M^2 - Q^2. \tag{A.2585}$$

F2313 48_finite_range_spherical_electrovac_strong_field.tex:304 FINITE_RANGE_STRONG_FIELD_PASS

$$r_\pm = M \pm \sqrt{\Delta_M}. \tag{A.2586}$$

F2314 48_finite_range_spherical_electrovac_strong_field.tex:308 FINITE_RANGE_STRONG_FIELD_PASS

$$1 - \frac{2M}{R} + \frac{Q^2}{R^2} = 0. \tag{A.2587}$$

F2315 48_finite_range_spherical_electrovac_strong_field.tex:312 FINITE_RANGE_STRONG_FIELD_PASS

$$P_\Lambda^2 = R^2. \tag{A.2588}$$

F2316 48_finite_range_spherical_electrovac_strong_field.tex:316 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{R_{\mathrm{turn}} = \frac{Q^2}{2M},} \tag{A.2589}$$

F2317 48_finite_range_spherical_electrovac_strong_field.tex:324 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{\mathcal{K} = \frac{48M^2}{R^6} - \frac{96MQ^2}{R^7} + \frac{56Q^4}{R^8}.} \tag{A.2590}$$

F2318 48_finite_range_spherical_electrovac_strong_field.tex:333 FINITE_RANGE_STRONG_FIELD_PASS

$$M = 1, \qquad Q = 0.8, \qquad R \in [0.6, 6], \tag{A.2591}$$

F2319 48_finite_range_spherical_electrovac_strong_field.tex:341 FINITE_RANGE_STRONG_FIELD_PASS

$$r_+ = 1.6, \qquad r_- = 0.4, \qquad R_{\mathrm{turn}} = 0.32, \tag{A.2592}$$

F2320 48_finite_range_spherical_electrovac_strong_field.tex:348 FINITE_RANGE_STRONG_FIELD_PASS

$$\mathcal{K}(0.6) = 199.664685261393, \qquad \mathcal{K}(r_+) = 1.10626220703125. \tag{A.2593}$$

F2321 48_finite_range_spherical_electrovac_strong_field.tex:358 FINITE_RANGE_STRONG_FIELD_PASS

$$r_j = r_0 + jh. \tag{A.2594}$$

F2322 48_finite_range_spherical_electrovac_strong_field.tex:362 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{(D_4 f)_j = \frac{-f_{j+2} + 8f_{j+1} - 8f_{j-1} + f_{j-2}}{12h},} \tag{A.2595}$$

F2323 48_finite_range_spherical_electrovac_strong_field.tex:369 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{(D_4^{(2)} f)_j = \frac{-f_{j+2} + 16f_{j+1} - 30f_j + 16f_{j-1} - f_{j-2}}{12h^2}.} \tag{A.2596}$$

F2324 48_finite_range_spherical_electrovac_strong_field.tex:378 FINITE_RANGE_STRONG_FIELD_PASS

$$\mathcal{M}_j^{(h)} := \frac{R_j}{2}\left[1 - \left(\frac{D_4 R_j}{\Lambda_j}\right)^2 + \left(\frac{P_{\Lambda,j}}{R_j}\right)^2\right] + \frac{Q_j^2}{2R_j}. \tag{A.2597}$$

F2325 48_finite_range_spherical_electrovac_strong_field.tex:391 FINITE_RANGE_STRONG_FIELD_PASS

$$\mathcal{C}_{\Lambda,j} := \Lambda_j - D_4 R_j, \tag{A.2598}$$

F2326 48_finite_range_spherical_electrovac_strong_field.tex:395 FINITE_RANGE_STRONG_FIELD_PASS

$$\mathcal{C}_{M,j} := P_{\Lambda,j}^2 - 2MR_j + Q^2, \tag{A.2599}$$

F2327 48_finite_range_spherical_electrovac_strong_field.tex:399 FINITE_RANGE_STRONG_FIELD_PASS

$$\mathcal{C}_{P,j} := P_{R,j} P_{\Lambda,j} - M\Lambda_j, \tag{A.2600}$$

F2328 48_finite_range_spherical_electrovac_strong_field.tex:403 FINITE_RANGE_STRONG_FIELD_PASS

$$\mathcal{C}_{Q,j} := Q_j - Q. \tag{A.2601}$$

F2329 48_finite_range_spherical_electrovac_strong_field.tex:408 FINITE_RANGE_STRONG_FIELD_PASS

$$D_4 R_j > 0, \qquad 2MR_j - Q^2 > 0, \tag{A.2602}$$

F2330 48_finite_range_spherical_electrovac_strong_field.tex:414 FINITE_RANGE_STRONG_FIELD_PASS

eq:finite-strong-field-projection

$$\boxed{\Pi_h[R]_j = \left(R_j, D_4 R_j, \sigma\sqrt{2MR_j - Q^2}, \sigma\frac{MD_4 R_j}{\sqrt{2MR_j - Q^2}}, Q\right).} \tag{A.2603}$$

F2331 48_finite_range_spherical_electrovac_strong_field.tex:428 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{\mathcal{C}_{\Lambda,j} = \mathcal{C}_{M,j} = \mathcal{C}_{P,j} = \mathcal{C}_{Q,j} = 0} \tag{A.2604}$$

F2332 48_finite_range_spherical_electrovac_strong_field.tex:438 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{\mathcal{M}_j^{(h)} = M} \tag{A.2605}$$

F2333 48_finite_range_spherical_electrovac_strong_field.tex:450 FINITE_RANGE_STRONG_FIELD_PASS

eq:radius-evolution-direct

$$\boxed{\dot{R} = -\frac{\sqrt{2MR - Q^2}}{R} + N^r D_4 R.} \tag{A.2606}$$

F2334 48_finite_range_spherical_electrovac_strong_field.tex:461 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{\mathcal{A}_h[N^r]R := \frac{1}{2}\left[N^r D_4 R + D_4(N^r R) - (D_4 N^r)R\right].} \tag{A.2607}$$

F2335 48_finite_range_spherical_electrovac_strong_field.tex:471 FINITE_RANGE_STRONG_FIELD_PASS

eq:radius-evolution-split

$$\dot{R} = -\frac{\sqrt{2MR - Q^2}}{R} + \mathcal{A}_h[N^r]R. \tag{A.2608}$$

F2336 48_finite_range_spherical_electrovac_strong_field.tex:478 FINITE_RANGE_STRONG_FIELD_PASS

$$p_{\mathrm{direct-split}} = 4.119353. \tag{A.2609}$$

F2337 48_finite_range_spherical_electrovac_strong_field.tex:487 FINITE_RANGE_STRONG_FIELD_PASS

$$R_j \geq R_* > 0, \qquad 2MR_j - Q^2 \geq p_*^2 > 0, \tag{A.2610}$$

F2338 48_finite_range_spherical_electrovac_strong_field.tex:492 FINITE_RANGE_STRONG_FIELD_PASS

$$0 < d_* \leq D_4 R_j \leq d^*, \qquad \|D_4 N^r\|_\infty \leq S_1. \tag{A.2611}$$

F2339 48_finite_range_spherical_electrovac_strong_field.tex:501 FINITE_RANGE_STRONG_FIELD_PASS

$$\|\delta\Lambda\|_\infty \leq \|D_4 \delta R\|_\infty, \tag{A.2612}$$

F2340 48_finite_range_spherical_electrovac_strong_field.tex:505 FINITE_RANGE_STRONG_FIELD_PASS

$$\|\delta P_\Lambda\|_\infty \leq \frac{M}{p_*} \|\delta R\|_\infty, \tag{A.2613}$$

F2341 48_finite_range_spherical_electrovac_strong_field.tex:509 FINITE_RANGE_STRONG_FIELD_PASS

$$\|\delta P_R\|_\infty \leq \frac{M}{p_*} \|D_4 \delta R\|_\infty + \frac{M^2 d^*}{p_*^3} \|\delta R\|_\infty. \tag{A.2614}$$

F2342 48_finite_range_spherical_electrovac_strong_field.tex:515 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{\|\Pi_h[R] - \Pi_h[\widetilde{R}]\| \leq C_\Pi \left(\|\delta R\|_\infty + \|D_4 \delta R\|_\infty \right),} \tag{A.2615}$$

F2343 48_finite_range_spherical_electrovac_strong_field.tex:530 FINITE_RANGE_STRONG_FIELD_PASS

$$D_4^{\mathsf{T}} = -D_4. \tag{A.2616}$$

F2344 48_finite_range_spherical_electrovac_strong_field.tex:534 FINITE_RANGE_STRONG_FIELD_PASS

$$\left\langle R, \frac{1}{2} \left[N^r D_4 R + D_4 (N^r R) \right] \right\rangle_h = 0, \tag{A.2617}$$

F2345 48_finite_range_spherical_electrovac_strong_field.tex:540 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{\langle R, \mathcal{A}_h[N^r] R \rangle_h = -\frac{1}{2} \langle R, (D_4 N^r) R \rangle_h .} \tag{A.2618}$$

F2346 48_finite_range_spherical_electrovac_strong_field.tex:548 FINITE_RANGE_STRONG_FIELD_PASS

$$g(R) := -\frac{\sqrt{2MR - Q^2}}{R}. \tag{A.2619}$$

F2347 48_finite_range_spherical_electrovac_strong_field.tex:552 FINITE_RANGE_STRONG_FIELD_PASS

$$g'(R) = \frac{MR - Q^2}{R^2 \sqrt{2MR - Q^2}}, \tag{A.2620}$$

F2348 48_finite_range_spherical_electrovac_strong_field.tex:557 FINITE_RANGE_STRONG_FIELD_PASS

$$\frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \|R - \widetilde{R}\|_h^2 \leq \left(L_g + \frac{1}{2} S_1 \right) \|R - \widetilde{R}\|_h^2. \tag{A.2621}$$

F2349 48_finite_range_spherical_electrovac_strong_field.tex:564 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{\|R(t) - \widetilde{R}(t)\|_h \leq \exp\left[\left(L_g + \frac{1}{2} S_1 \right) t \right] \|R(0) - \widetilde{R}(0)\|_h .} \tag{A.2622}$$

F2350 48_finite_range_spherical_electrovac_strong_field.tex:577 FINITE_RANGE_STRONG_FIELD_PASS

$$\widehat{D}_4(\theta) = \frac{\mathrm{i}}{h} \frac{8 \sin\theta - \sin 2\theta}{6}, \tag{A.2623}$$

F2351 48_finite_range_spherical_electrovac_strong_field.tex:583 FINITE_RANGE_STRONG_FIELD_PASS

$$\max_{\theta}\left|\frac{8\sin\theta-\sin 2\theta}{6}\right| = 1.37222197980322. \tag{A.2624}$$

F2352 48_finite_range_spherical_electrovac_strong_field.tex:591 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{\frac{|N^r|\Delta t}{h} \le 2.06120231739167.} \tag{A.2625}$$

F2353 48_finite_range_spherical_electrovac_strong_field.tex:598 FINITE_RANGE_STRONG_FIELD_PASS

$$\frac{\Delta t}{h} = 0.1. \tag{A.2626}$$

F2354 48_finite_range_spherical_electrovac_strong_field.tex:607 FINITE_RANGE_STRONG_FIELD_PASS

$$F(R) := 1 - \frac{2M}{R} + \frac{Q^2}{R^2}. \tag{A.2627}$$

F2355 48_finite_range_spherical_electrovac_strong_field.tex:611 FINITE_RANGE_STRONG_FIELD_PASS

$$r_+ = M + d, \qquad d := \sqrt{M^2 - Q^2} > 0, \tag{A.2628}$$

F2356 48_finite_range_spherical_electrovac_strong_field.tex:617 FINITE_RANGE_STRONG_FIELD_PASS

$$F'(r_+) = \frac{2d}{r_+^2}. \tag{A.2629}$$

F2357 48_finite_range_spherical_electrovac_strong_field.tex:621 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{|\delta r_+| \le \frac{r_+^2}{2d}\epsilon + \mathcal{O}(\epsilon^2).} \tag{A.2630}$$

F2358 48_finite_range_spherical_electrovac_strong_field.tex:629 FINITE_RANGE_STRONG_FIELD_PASS

$$\frac{r_+^2}{2d} = 2.13333333333333. \tag{A.2631}$$

F2359 48_finite_range_spherical_electrovac_strong_field.tex:637 FINITE_RANGE_STRONG_FIELD_PASS

$$F(t,r) = r + 0.08\sin(0.7t)\sin^2\left(\pi\frac{r-0.6}{5.4}\right). \tag{A.2632}$$

F2360 48_finite_range_spherical_electrovac_strong_field.tex:645 FINITE_RANGE_STRONG_FIELD_PASS

$$\min_{t,r} F_r > 0.987138, \qquad \min_{t,r}(2MF - Q^2) > 0.559999. \tag{A.2633}$$

F2361 48_finite_range_spherical_electrovac_strong_field.tex:667 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{\begin{aligned} p_R^{\mathrm{state}} &= 3.998556,\\ p_\Lambda^{\mathrm{metric}} &= 3.960101,\\ p_{P_\Lambda} &= 4.023926,\\ p_{P_R} &= 4.007880. \end{aligned}} \tag{A.2634}$$

F2362 48_finite_range_spherical_electrovac_strong_field.tex:677 FINITE_RANGE_STRONG_FIELD_PASS

$$p_{\mathcal{H}} = 3.704099. \tag{A.2635}$$

F2363 48_finite_range_spherical_electrovac_strong_field.tex:681 FINITE_RANGE_STRONG_FIELD_PASS

$$\max_j |\mathcal{M}_j^{(h)} - M| = 4.440892 \times 10^{-16}, \tag{A.2636}$$

F2364 48_finite_range_spherical_electrovac_strong_field.tex:685 FINITE_RANGE_STRONG_FIELD_PASS

$$\max_j |\mathcal{C}_{\Lambda,j}| = 0, \qquad \max_j |\mathcal{C}_{M,j}| = 2.220446 \times 10^{-15}, \tag{A.2637}$$

F2365 48_finite_range_spherical_electrovac_strong_field.tex:691 FINITE_RANGE_STRONG_FIELD_PASS

$$\max_j |\mathcal{C}_{P,j}| = 1.110223 \times 10^{-16}, \qquad \max_j |D_4 Q_j| = 1.750999 \times 10^{-15}. \tag{A.2638}$$

F2366 48_finite_range_spherical_electrovac_strong_field.tex:699 FINITE_RANGE_STRONG_FIELD_PASS

$$|r_{+,h} - r_+^{\mathrm{exact}}| = 1.892619 \times 10^{-11}. \tag{A.2639}$$

F2367 48_finite_range_spherical_electrovac_strong_field.tex:705 FINITE_RANGE_STRONG_FIELD_PASS

$$1.03345462, \quad 1.03347604, \quad 1.03347993, \quad 1.03348106 \tag{A.2640}$$

F2368 48_finite_range_spherical_electrovac_strong_field.tex:715 FINITE_RANGE_STRONG_FIELD_PASS

$$2.64438 \times 10^{-5}. \tag{A.2641}$$

F2369 48_finite_range_spherical_electrovac_strong_field.tex:721 FINITE_RANGE_STRONG_FIELD_PASS

$$\max|\mathcal{M} - M| = 2.251251 \times 10^{-4}, \tag{A.2642}$$

F2370 48_finite_range_spherical_electrovac_strong_field.tex:725 FINITE_RANGE_STRONG_FIELD_PASS

$$\max|\mathcal{D}| = 1.276876 \times 10^{-3}, \qquad \max|\mathcal{H}| = 1.257350 \times 10^{-2}. \tag{A.2643}$$

F2371 48_finite_range_spherical_electrovac_strong_field.tex:783 FINITE_RANGE_STRONG_FIELD_PASS

$$\boxed{\begin{array}{c}\text{spherical scalar or spinor collapse with dynamical mass transfer,}\\ \text{a trapped-surface formation criterion preserved by the finite update,}\\ \text{or a nonspherical finite-range gravitational wave benchmark.}\end{array}} \tag{A.2644}$$

F2372 48_finite_range_spherical_electrovac_strong_field.tex:795 FINITE_RANGE_STRONG_FIELD_PASS

$$R_{\mathrm{SF}} = \texttt{data/finite_range_strong_field_results.json}, \tag{A.2645}$$

F2373 48_finite_range_spherical_electrovac_strong_field.tex:799 FINITE_RANGE_STRONG_FIELD_PASS

$$C_{\mathrm{SF}} = \texttt{data/finite_range_strong_field_checks.csv}, \tag{A.2646}$$

F2374 48_finite_range_spherical_electrovac_strong_field.tex:803 FINITE_RANGE_STRONG_FIELD_PASS

$$X_{\mathrm{SF}} = \texttt{verification/Finite_Range_Spherical_Electrovac_Verification.py}. \tag{A.2647}$$

49_horizon_penetrating_spherical_scalar_collapse.tex

F2375 49_horizon_penetrating_spherical_scalar_collapse.tex:12 SCALAR_COLLAPSE_PASS

$$G = c = 1, \qquad \varphi := \sqrt{4\pi}\,\phi_{\mathrm{phys}}. \tag{A.2648}$$

F2376 49_horizon_penetrating_spherical_scalar_collapse.tex:22 SCALAR_COLLAPSE_PASS

$$\mathrm{d}s^2 = -N^2\,\mathrm{d}t^2 + \Lambda^2(\,\mathrm{d}r + N^r\,\mathrm{d}t)^2 + R^2\,\mathrm{d}\Omega^2. \tag{A.2649}$$

F2377 49_horizon_penetrating_spherical_scalar_collapse.tex:29 SCALAR_COLLAPSE_PASS

$$(\Lambda, P_\Lambda), \qquad (R, P_R), \qquad (\varphi, P_\varphi), \tag{A.2650}$$

F2378 49_horizon_penetrating_spherical_scalar_collapse.tex:37 SCALAR_COLLAPSE_PASS

$$\boxed{S_{\mathrm{ES}} = \int \mathrm{d}t \int \mathrm{d}r \left[P_\Lambda \dot{\Lambda} + P_R \dot{R} + P_\varphi \dot{\varphi} - N\mathcal{H} - N^r \mathcal{D}\right] + B_\partial.} \tag{A.2651}$$

F2379 49_horizon_penetrating_spherical_scalar_collapse.tex:47 SCALAR_COLLAPSE_PASS

$$\begin{aligned}\mathcal{H} = &-\frac{P_R P_\Lambda}{R} + \frac{\Lambda P_\Lambda^2}{2R^2} + \frac{R R''}{\Lambda} - \frac{R R' \Lambda'}{\Lambda^2} + \frac{R'^2}{2\Lambda} - \frac{\Lambda}{2}\\ &+ \frac{P_\varphi^2}{2\Lambda R^2} + \frac{R^2 \varphi'^2}{2\Lambda},\end{aligned} \tag{A.2652}$$

F2380 49_horizon_penetrating_spherical_scalar_collapse.tex:64 SCALAR_COLLAPSE_PASS

$$\boxed{\mathcal{D} = P_R R' - \Lambda P'_\Lambda + P_\varphi \varphi'.} \tag{A.2653}$$

F2381 49_horizon_penetrating_spherical_scalar_collapse.tex:73 SCALAR_COLLAPSE_PASS

$$\Lambda = 1, \qquad R = r. \tag{A.2654}$$

F2382 49_horizon_penetrating_spherical_scalar_collapse.tex:79 SCALAR_COLLAPSE_PASS

$$N = \sigma, \qquad N^r = \sigma v, \qquad P_\Lambda = rv, \qquad P_\varphi = r^2 \Pi, \qquad \Phi := \varphi'. \tag{A.2655}$$

F2383 49_horizon_penetrating_spherical_scalar_collapse.tex:91 SCALAR_COLLAPSE_PASS

$$\boxed{m := \frac{r}{2} v^2, \qquad v = \sqrt{\frac{2m}{r}}.} \tag{A.2656}$$

F2384 49_horizon_penetrating_spherical_scalar_collapse.tex:98 SCALAR_COLLAPSE_PASS

$$\boxed{\mathrm{d}s^2 = -\sigma^2 \,\mathrm{d}t^2 + (\,\mathrm{d}r + \sigma v \,\mathrm{d}t)^2 + r^2 \,\mathrm{d}\Omega^2.} \tag{A.2657}$$

F2385 49_horizon_penetrating_spherical_scalar_collapse.tex:110 SCALAR_COLLAPSE_PASS

$$\boxed{P_R = P'_\Lambda - P_\varphi \Phi.} \tag{A.2658}$$

F2386 49_horizon_penetrating_spherical_scalar_collapse.tex:119 SCALAR_COLLAPSE_PASS

eq:scalar-mass-constraint

$$\boxed{m' = r^2 \left[\frac{1}{2}(\Pi^2 + \Phi^2) + v \Pi \Phi \right].} \tag{A.2659}$$

F2387 49_horizon_penetrating_spherical_scalar_collapse.tex:129 SCALAR_COLLAPSE_PASS

$$-\frac{P_\Lambda P'_\Lambda}{r} + \frac{P_\Lambda^2}{2r^2} = -m', \tag{A.2660}$$

F2388 49_horizon_penetrating_spherical_scalar_collapse.tex:135 SCALAR_COLLAPSE_PASS

$$\frac{P_\Lambda P_\varphi \Phi}{r} = r^2 v \Pi \Phi. \tag{A.2661}$$

F2389 49_horizon_penetrating_spherical_scalar_collapse.tex:144 SCALAR_COLLAPSE_PASS

$$0 = \dot{\Lambda} = \sigma \left(\frac{v}{r} - \frac{P_R}{r} \right) + (\sigma v)'. \tag{A.2662}$$

F2390 49_horizon_penetrating_spherical_scalar_collapse.tex:149 SCALAR_COLLAPSE_PASS

eq:scalar-lapse-constraint

$$\boxed{\frac{\sigma'}{\sigma} = -\frac{r \Pi \Phi}{v}.} \tag{A.2663}$$

F2391 49_horizon_penetrating_spherical_scalar_collapse.tex:160 SCALAR_COLLAPSE_PASS

$$\boxed{\dot{\varphi} = \sigma(\Pi + v\Phi),} \tag{A.2664}$$

F2392 49_horizon_penetrating_spherical_scalar_collapse.tex:165 SCALAR_COLLAPSE_PASS

eq:phi-evolution

$$\boxed{\dot{\Phi} = \partial_r \left[\sigma(\Pi + v\Phi) \right],} \tag{A.2665}$$

F2393 49_horizon_penetrating_spherical_scalar_collapse.tex:171 SCALAR_COLLAPSE_PASS

eq:pi-evolution

$$\boxed{\dot{\Pi} = \frac{1}{r^2} \partial_r \left[r^2 \sigma(\Phi + v\Pi) \right].} \tag{A.2666}$$

F2394 49_horizon_penetrating_spherical_scalar_collapse.tex:185 SCALAR_COLLAPSE_PASS

eq:scalar-mass-flux

$$\boxed{\dot{m} = r^2\sigma(\Pi + v\Phi)(\Phi + v\Pi).} \tag{A.2667}$$

F2395 49_horizon_penetrating_spherical_scalar_collapse.tex:197 SCALAR_COLLAPSE_PASS

$$\mathcal{E} := \frac{1}{2}(\Pi^2 + \Phi^2) + v\Pi\Phi, \tag{A.2668}$$

F2396 49_horizon_penetrating_spherical_scalar_collapse.tex:201 SCALAR_COLLAPSE_PASS

$$\mathcal{J} := \sigma(\Pi + v\Phi)(\Phi + v\Pi). \tag{A.2669}$$

F2397 49_horizon_penetrating_spherical_scalar_collapse.tex:206 SCALAR_COLLAPSE_PASS

eq:mass-compatibility

$$\boxed{\partial_t(r^2\mathcal{E}) = \partial_r(r^2\mathcal{J})} \tag{A.2670}$$

F2398 49_horizon_penetrating_spherical_scalar_collapse.tex:213 SCALAR_COLLAPSE_PASS

$$\frac{v^2}{2} + rvv' = r^2\mathcal{E}, \qquad \frac{\sigma'}{\sigma} = -\frac{r\Pi\Phi}{v}, \qquad rv\dot{v} = r^2\mathcal{J}. \tag{A.2671}$$

F2399 49_horizon_penetrating_spherical_scalar_collapse.tex:221 SCALAR_COLLAPSE_PASS

$$\partial_t m' = \partial_r \dot{m}. \tag{A.2672}$$

F2400 49_horizon_penetrating_spherical_scalar_collapse.tex:227 SCALAR_COLLAPSE_PASS

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left[m(t,r_o) - m(t,r_i)\right] = r_o^2\mathcal{J}(t,r_o) - r_i^2\mathcal{J}(t,r_i).} \tag{A.2673}$$

F2401 49_horizon_penetrating_spherical_scalar_collapse.tex:238 SCALAR_COLLAPSE_PASS

$$w_+ := \Pi + \Phi, \qquad w_- := \Pi - \Phi. \tag{A.2674}$$

F2402 49_horizon_penetrating_spherical_scalar_collapse.tex:244 SCALAR_COLLAPSE_PASS

eq:wplus

$$\begin{aligned}\dot{w}_+ = \sigma(v+1)w_+' + \left[\sigma v' + \sigma'(v+1)\right] w_+ \\ + \frac{\sigma}{r}\left[(v+1)w_+ + (v-1)w_-\right],\end{aligned} \tag{A.2675}$$

F2403 49_horizon_penetrating_spherical_scalar_collapse.tex:255 SCALAR_COLLAPSE_PASS

eq:wminus

$$\begin{aligned}\dot{w}_- = \sigma(v-1)w_-' + \left[\sigma v' + \sigma'(v-1)\right] w_- \\ + \frac{\sigma}{r}\left[(v-1)w_- + (v+1)w_+\right].\end{aligned} \tag{A.2676}$$

F2404 49_horizon_penetrating_spherical_scalar_collapse.tex:267 SCALAR_COLLAPSE_PASS

$$\boxed{c_+ = -\sigma(v+1), \qquad c_- = \sigma(1-v).} \tag{A.2677}$$

F2405 49_horizon_penetrating_spherical_scalar_collapse.tex:278 SCALAR_COLLAPSE_PASS

$$\ell_\pm = \sigma^{-1}\partial_t + (-v \pm 1)\partial_r. \tag{A.2678}$$

F2406 49_horizon_penetrating_spherical_scalar_collapse.tex:283 SCALAR_COLLAPSE_PASS

$$\theta_+ = \frac{2}{r}(1-v), \qquad \theta_- = -\frac{2}{r}(1+v). \tag{A.2679}$$

F2407 49_horizon_penetrating_spherical_scalar_collapse.tex:291 SCALAR_COLLAPSE_PASS

eq:apparent-horizon

$$\boxed{\theta_+ = 0 \quad\Longleftrightarrow\quad v = 1 \quad\Longleftrightarrow\quad \frac{2m}{r} = 1.} \tag{A.2680}$$

F2408 49_horizon_penetrating_spherical_scalar_collapse.tex:303 SCALAR_COLLAPSE_PASS

$$\boxed{\mathcal{R} = 2(\Phi^2 - \Pi^2).} \tag{A.2681}$$

F2409 49_horizon_penetrating_spherical_scalar_collapse.tex:315 SCALAR_COLLAPSE_PASS

$$m(r_{\mathrm{exc}}) = 0. \tag{A.2682}$$

F2410 49_horizon_penetrating_spherical_scalar_collapse.tex:319 SCALAR_COLLAPSE_PASS

$$\sigma(r_{\max}) = 1. \tag{A.2683}$$

F2411 49_horizon_penetrating_spherical_scalar_collapse.tex:327 SCALAR_COLLAPSE_PASS

$$(D_+^{(2)} f)_j = \frac{-3f_j + 4f_{j+1} - f_{j+2}}{2h}, \tag{A.2684}$$

F2412 49_horizon_penetrating_spherical_scalar_collapse.tex:331 SCALAR_COLLAPSE_PASS

$$(D_-^{(2)} f)_j = \frac{3f_j - 4f_{j-1} + f_{j-2}}{2h}. \tag{A.2685}$$

F2413 49_horizon_penetrating_spherical_scalar_collapse.tex:342 SCALAR_COLLAPSE_PASS

$$r \in [r_{\mathrm{exc}}, r_{\max}] = [0.05, 30]. \tag{A.2686}$$

F2414 49_horizon_penetrating_spherical_scalar_collapse.tex:347 SCALAR_COLLAPSE_PASS

$$w_-(r_{\mathrm{exc}}) = 0, \qquad w_+(r_{\max}) = 0. \tag{A.2687}$$

F2415 49_horizon_penetrating_spherical_scalar_collapse.tex:357 SCALAR_COLLAPSE_PASS

$$\varphi(0, r) = A \exp\left[- \left(\frac{r - r_0}{w} \right)^2 \right], \qquad r_0 = 10, \qquad w = 1. \tag{A.2688}$$

F2416 49_horizon_penetrating_spherical_scalar_collapse.tex:366 SCALAR_COLLAPSE_PASS

$$\boxed{\Pi(0, r) = \Phi(0, r) + \frac{\varphi(0, r)}{r}.} \tag{A.2689}$$

F2417 49_horizon_penetrating_spherical_scalar_collapse.tex:374 SCALAR_COLLAPSE_PASS

$$A_{\mathrm{strong}} = 0.10, \qquad A_{\mathrm{weak}} = 0.05. \tag{A.2690}$$

F2418 49_horizon_penetrating_spherical_scalar_collapse.tex:384 SCALAR_COLLAPSE_PASS

$$N_r = 2400, \qquad h = 0.0124843684869, \tag{A.2691}$$

F2419 49_horizon_penetrating_spherical_scalar_collapse.tex:390 SCALAR_COLLAPSE_PASS

$$\boxed{t_{\mathrm{H}} = 4.69182094580,} \tag{A.2692}$$

F2420 49_horizon_penetrating_spherical_scalar_collapse.tex:394 SCALAR_COLLAPSE_PASS

$$\boxed{r_{\mathrm{H}} = 3.11341154797,} \tag{A.2693}$$

F2421 49_horizon_penetrating_spherical_scalar_collapse.tex:398 SCALAR_COLLAPSE_PASS

$$\boxed{m_{\mathrm{H}} = \frac{r_{\mathrm{H}}}{2} = 1.55670577399.} \tag{A.2694}$$

F2422 49_horizon_penetrating_spherical_scalar_collapse.tex:404 SCALAR_COLLAPSE_PASS

$$M_{\mathrm{ADM}}(0) = 1.72992821318, \tag{A.2695}$$

F2423 49_horizon_penetrating_spherical_scalar_collapse.tex:407 SCALAR_COLLAPSE_PASS

$$M_{\mathrm{ADM}}(t_{\mathrm{H}}) = 1.72538868709. \tag{A.2696}$$

F2424 49_horizon_penetrating_spherical_scalar_collapse.tex:411 SCALAR_COLLAPSE_PASS

$$\boxed{\frac{M(t_{\mathrm{H}})-M(0)}{M(0)} = -2.6241124\times10^{-3}.} \tag{A.2697}$$

F2425 49_horizon_penetrating_spherical_scalar_collapse.tex:419 SCALAR_COLLAPSE_PASS

$$\min_r \sigma = 0.999829382946, \qquad \sigma(r_{\mathrm{H}}) = 1.02582462096. \tag{A.2698}$$

F2426 49_horizon_penetrating_spherical_scalar_collapse.tex:425 SCALAR_COLLAPSE_PASS

$$\max_r |\mathcal{R}| = 7.35797274672. \tag{A.2699}$$

F2427 49_horizon_penetrating_spherical_scalar_collapse.tex:462 SCALAR_COLLAPSE_PASS

$$\boxed{t_{\mathrm{H}}(h) = 4.68935223701 + 33.8857733838\,h^{2.17385328224},} \tag{A.2700}$$

F2428 49_horizon_penetrating_spherical_scalar_collapse.tex:468 SCALAR_COLLAPSE_PASS

$$\boxed{r_{\mathrm{H}}(h) = 3.12049657441 - 65.5931060303\,h^{2.08439496457}.} \tag{A.2701}$$

F2429 49_horizon_penetrating_spherical_scalar_collapse.tex:485 SCALAR_COLLAPSE_PASS

$$A = 0.05, \qquad N_r = 1200, \qquad t = 5, \tag{A.2702}$$

F2430 49_horizon_penetrating_spherical_scalar_collapse.tex:493 SCALAR_COLLAPSE_PASS

$$\boxed{\max_r \frac{2m}{r} = 0.146722872367.} \tag{A.2703}$$

F2431 49_horizon_penetrating_spherical_scalar_collapse.tex:498 SCALAR_COLLAPSE_PASS

$$-4.5106042\times10^{-3}. \tag{A.2704}$$

F2432 49_horizon_penetrating_spherical_scalar_collapse.tex:511 SCALAR_COLLAPSE_PASS

$$\frac{\|m' - r^2[(\Pi^2+\Phi^2)/2 + v\Pi\Phi]\|_\infty}{\|r^2[(\Pi^2+\Phi^2)/2 + v\Pi\Phi]\|_\infty} = 2.27319745\times10^{-6}, \tag{A.2705}$$

F2433 49_horizon_penetrating_spherical_scalar_collapse.tex:516 SCALAR_COLLAPSE_PASS

$$\frac{\|(\log\sigma)' + r\Pi\Phi/v\|_\infty}{\|r\Pi\Phi/v\|_\infty} = 2.80774843\times10^{-6}, \tag{A.2706}$$

F2434 49_horizon_penetrating_spherical_scalar_collapse.tex:521 SCALAR_COLLAPSE_PASS

$$\frac{\|\Delta_t m - r^2\sigma(\Pi+v\Phi)(\Phi+v\Pi)\|_\infty}{\|r^2\sigma(\Pi+v\Phi)(\Phi+v\Pi)\|_\infty} = 1.26971871\times10^{-3}. \tag{A.2707}$$

F2435 49_horizon_penetrating_spherical_scalar_collapse.tex:533 SCALAR_COLLAPSE_PASS

$$\boxed{\begin{array}{c}\text{an exact nonlinear Einstein–scalar PG reduction,}\\ \text{triangular mass and lapse constraints,}\\ \text{an exact local Misner–Sharp flux theorem,}\\ \text{exact continuum constraint propagation,}\\ \text{a horizon-penetrating characteristic system,}\\ \text{a radius-two scalar evolution stencil,}\\ \text{numerical formation of a marginally trapped sphere,}\\ \text{second-order refinement of the event coordinates.}\end{array}} \tag{A.2708}$$

F2436 49_horizon_penetrating_spherical_scalar_collapse.tex:550 SCALAR_COLLAPSE_PASS

$$\begin{array}{c}\text{regular-center evolution through reflection or dispersion,}\\ \text{a critical amplitude or universal critical exponent,}\\ \text{continuation to singularity formation,}\\ \text{global event-horizon identification,}\\ \text{nonspherical stability,}\\ \text{scalar self-interaction or Dirac collapse,}\\ \text{a complete finite-range } 3+1 \text{ strong-field solver.}\end{array} \tag{A.2709}$$

F2437 49_horizon_penetrating_spherical_scalar_collapse.tex:566 SCALAR_COLLAPSE_PASS

$$\boxed{\begin{array}{c}\text{replace the excision core by a regular-center formulation,}\\ \text{continue supercritical data beyond first horizon formation,}\\ \text{bracket the critical amplitude and test scaling,}\\ \text{then add nonspherical perturbations and ringdown.}\end{array}} \tag{A.2710}$$

F2438 49_horizon_penetrating_spherical_scalar_collapse.tex:578 SCALAR_COLLAPSE_PASS

$$R_{\mathrm{collapse}} = \texttt{data/Finite_Range_Spherical_Scalar_Collapse_Verification.json}, \tag{A.2711}$$

F2439 49_horizon_penetrating_spherical_scalar_collapse.tex:582 SCALAR_COLLAPSE_PASS

$$T_{\mathrm{collapse}} = \texttt{data/scalar_collapse_convergence.csv}, \tag{A.2712}$$

F2440 49_horizon_penetrating_spherical_scalar_collapse.tex:586 SCALAR_COLLAPSE_PASS

$$X_{\mathrm{collapse}} = \texttt{verification/Finite_Range_Spherical_Scalar_Collapse_Verification.py}. \tag{A.2713}$$

50_unified_refinement_dark_sector.tex

F2441 50_unified_refinement_dark_sector.tex:15 SUPERSEDED_MODEL_BRANCH

$$\mathsf{P}_0\Phi(x,u) := \int_{\mathbb{Z}_p} \Phi(x,v)\ \mathrm{d}\mu_p(v) =: \phi(x), \qquad \chi := (I-\mathsf{P}_0)\Phi, \tag{A.2714}$$

F2442 50_unified_refinement_dark_sector.tex:23 SUPERSEDED_MODEL_BRANCH

$$\int_{\mathbb{Z}_p} \chi(x,u)\ \mathrm{d}\mu_p(u) = 0. \tag{A.2715}$$

F2443 50_unified_refinement_dark_sector.tex:27 SUPERSEDED_MODEL_BRANCH

$$\mathcal{L}_{\sigma,p} = \Lambda^2 \sum_{h\geq 1} p^{2\sigma h}\mathsf{D}_h, \qquad d_h = (p-1)p^{h-1}. \tag{A.2716}$$

F2444 50_unified_refinement_dark_sector.tex:34 SUPERSEDED_MODEL_BRANCH

$$U(\phi) = \frac{\lambda}{4}(\phi^2 - v^2)^2, \qquad \mu^2 := \lambda v^2, \qquad \rho_V := U(0) = \frac{\mu^4}{4\lambda}. \tag{A.2717}$$

F2445 50_unified_refinement_dark_sector.tex:43 SUPERSEDED_MODEL_BRANCH

$$\boxed{U''(\phi) = 3\lambda\phi^2 - \mu^2.} \tag{A.2718}$$

F2446 50_unified_refinement_dark_sector.tex:49 SUPERSEDED_MODEL_BRANCH

$$S_{\mathrm{UDS}} = -\int \sqrt{-g}\ \mathrm{d}^4x \Bigg[\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + U(\phi) \tag{A.2719}$$

$$+ \frac{1}{2}\int_{\mathbb{Z}_p} \mathrm{d}\mu_p(u)\left(g^{\mu\nu}\partial_\mu\chi\partial_\nu\chi + \chi[\mathcal{L}_{\sigma,p} + U''(\phi)]\chi\right)\Bigg]. \tag{A.2720}$$

F2447 50_unified_refinement_dark_sector.tex:65 SUPERSEDED_MODEL_BRANCH

$$\chi(x,u) = \sum_{h\geq 1}\sum_{a=1}^{d_h} \chi_{h,a}(x)\psi_{h,a}(u), \tag{A.2721}$$

F2448 50_unified_refinement_dark_sector.tex:70 SUPERSEDED_MODEL_BRANCH

$$\boxed{m_h^2(\phi) = \Lambda^2 p^{2\sigma h} - \mu^2 + 3\lambda\phi^2.} \tag{A.2722}$$

F2449 50_unified_refinement_dark_sector.tex:77 SUPERSEDED_MODEL_BRANCH

$$m_h^2 = \Lambda^2 p^{2\sigma h} - \mu^2. \tag{A.2723}$$

F2450 50_unified_refinement_dark_sector.tex:82 SUPERSEDED_MODEL_BRANCH

$$\boxed{\Lambda^2 p^{2\sigma} > \mu^2.} \tag{A.2724}$$

F2451 50_unified_refinement_dark_sector.tex:88 SUPERSEDED_MODEL_BRANCH

$$U(\pm v) = 0, \qquad m_h^2(\pm v) = \Lambda^2 p^{2\sigma h} + 2\mu^2 > 0. \tag{A.2725}$$

F2452 50_unified_refinement_dark_sector.tex:97 SUPERSEDED_MODEL_BRANCH

$$\phi(t) = \dot{\phi}_\star (t - t_\star) + \mathcal{O}((t - t_\star)^2). \tag{A.2726}$$

F2453 50_unified_refinement_dark_sector.tex:101 SUPERSEDED_MODEL_BRANCH

$$\omega_{h,k}^2(t) = k_{\mathrm{phys}}^2 + m_h^2 + 3\lambda \dot{\phi}_\star^2 (t - t_\star)^2 + \mathcal{O}((t - t_\star)^3). \tag{A.2727}$$

F2454 50_unified_refinement_dark_sector.tex:108 SUPERSEDED_MODEL_BRANCH

$$\boxed{\nu := \sqrt{3\lambda}\, |\dot{\phi}_\star|.} \tag{A.2728}$$

F2455 50_unified_refinement_dark_sector.tex:114 SUPERSEDED_MODEL_BRANCH

$$\boxed{f_h(q) = \mathcal{N}_\star \exp\left[-\frac{\pi m_h^2}{\nu} - \frac{\pi q^2}{a_\star^2 \nu} \right].} \tag{A.2729}$$

F2456 50_unified_refinement_dark_sector.tex:125 SUPERSEDED_MODEL_BRANCH

$$\mathcal{N}_\star = \frac{\Delta\rho_\phi}{\widetilde{\rho}_D(a_\star)}, \tag{A.2730}$$

F2457 50_unified_refinement_dark_sector.tex:130 SUPERSEDED_MODEL_BRANCH

$$\Delta\rho_\phi = \frac{1}{2}\dot{\phi}_{\star,-}^2 \qquad (\dot{\phi}_{\star,+} = 0). \tag{A.2731}$$

F2458 50_unified_refinement_dark_sector.tex:139 SUPERSEDED_MODEL_BRANCH

$$\epsilon_h(q, a) = \sqrt{q^2 + a^2 m_h^2}, \tag{A.2732}$$

F2459 50_unified_refinement_dark_sector.tex:143 SUPERSEDED_MODEL_BRANCH

$$\rho_h(a) = \frac{d_h}{2\pi^2 a^4} \int_0^\infty q^2 \epsilon_h(q, a) f_h(q)\, \mathrm{d}q, \tag{A.2733}$$

$$p_h(a) = \frac{d_h}{6\pi^2 a^4} \int_0^\infty \frac{q^4}{\epsilon_h(q, a)} f_h(q)\, \mathrm{d}q. \tag{A.2734}$$

F2460 50_unified_refinement_dark_sector.tex:153 SUPERSEDED_MODEL_BRANCH

$$\boxed{X_h(a) := \langle \chi_h^2 \rangle = \frac{d_h}{2\pi^2 a^2} \int_0^\infty \frac{q^2}{\epsilon_h(q, a)} f_h(q)\, \mathrm{d}q.} \tag{A.2735}$$

F2461 50_unified_refinement_dark_sector.tex:162 SUPERSEDED_MODEL_BRANCH

$$\rho_D := \sum_h \rho_h, \qquad p_D := \sum_h p_h, \qquad X_D := \sum_h X_h. \tag{A.2736}$$

F2462 50_unified_refinement_dark_sector.tex:170 SUPERSEDED_MODEL_BRANCH

$$\boxed{\frac{\mathrm{d}\rho_D}{\mathrm{d}\log a} + 3(\rho_D + p_D) = 0.} \tag{A.2737}$$

F2463 50_unified_refinement_dark_sector.tex:177 SUPERSEDED_MODEL_BRANCH

$$n_h(a) = \frac{\mathcal{N}_\star d_h a_\star^3 \nu^{3/2}}{8\pi^3 a^3} \exp\!\left(-\frac{\pi m_h^2}{\nu}\right), \tag{A.2738}$$

F2464 50_unified_refinement_dark_sector.tex:184 SUPERSEDED_MODEL_BRANCH

$$\rho_D = \sum_h m_h n_h, \qquad X_D = \sum_h \frac{n_h}{m_h}. \tag{A.2739}$$

F2465 50_unified_refinement_dark_sector.tex:190 SUPERSEDED_MODEL_BRANCH

$$\boxed{m_{\rm lock}^2 := \frac{\displaystyle\sum_{h\geq 1} d_h m_h \exp(-\pi m_h^2/\nu)}{\displaystyle\sum_{h\geq 1} d_h m_h^{-1} \exp(-\pi m_h^2/\nu)}.} \tag{A.2740}$$

F2466 50_unified_refinement_dark_sector.tex:201 SUPERSEDED_MODEL_BRANCH

$$\boxed{X_D = \frac{\rho_D}{m_{\rm lock}^2}} \tag{A.2741}$$

F2467 50_unified_refinement_dark_sector.tex:210 SUPERSEDED_MODEL_BRANCH

$$\ddot{\phi} + 3H\dot{\phi} + \left[-\mu^2 + 3\lambda X_D(a)\right]\phi + \lambda\phi^3 = 0. \tag{A.2742}$$

F2468 50_unified_refinement_dark_sector.tex:216 SUPERSEDED_MODEL_BRANCH

$$\boxed{M_\phi^2(a) := 3\lambda X_D(a) - \mu^2.} \tag{A.2743}$$

F2469 50_unified_refinement_dark_sector.tex:222 SUPERSEDED_MODEL_BRANCH

$$M_\phi^2(a) > 0, \tag{A.2744}$$

F2470 50_unified_refinement_dark_sector.tex:226 SUPERSEDED_MODEL_BRANCH

$$\boxed{\rho_{\rm DE} = \rho_V = \frac{\mu^4}{4\lambda}, \qquad p_{\rm DE} = -\rho_V, \qquad w_{\rm DE} = -1.} \tag{A.2745}$$

F2471 50_unified_refinement_dark_sector.tex:236 SUPERSEDED_MODEL_BRANCH

$$\boxed{H^2 = \frac{1}{3M_{\rm Pl}^2}\left(\rho_r + \rho_b + \rho_D + \rho_V\right).} \tag{A.2746}$$

F2472 50_unified_refinement_dark_sector.tex:247 SUPERSEDED_MODEL_BRANCH

$$S_{\delta\phi}^{(2)} = \frac{1}{2}\int \mathrm{d}t\ \mathrm{d}^3x\, a^3 \left[\delta\dot{\phi}^2 - \frac{(\nabla\delta\phi)^2}{a^2} - M_\phi^2(a)\delta\phi^2\right]. \tag{A.2747}$$

F2473 50_unified_refinement_dark_sector.tex:257 SUPERSEDED_MODEL_BRANCH

$$\boxed{Q_{\delta\phi} = 1 > 0, \qquad c_{s,\rm DE}^2 = 1, \qquad M_\phi^2 > 0.} \tag{A.2748}$$

F2474 50_unified_refinement_dark_sector.tex:269 SUPERSEDED_MODEL_BRANCH

$$\delta\rho_\phi = \dot{\phi}\,\delta\dot{\phi} - \dot{\phi}^2\Psi + U_{,\phi}\delta\phi, \tag{A.2749}$$

$$\delta p_\phi = \dot{\phi}\,\delta\dot{\phi} - \dot{\phi}^2\Psi - U_{,\phi}\delta\phi. \tag{A.2750}$$

F2475 50_unified_refinement_dark_sector.tex:281 SUPERSEDED_MODEL_BRANCH

$$\dot{\phi} = 0, \qquad U_{,\phi} = 0, \tag{A.2751}$$

F2476 50_unified_refinement_dark_sector.tex:287 SUPERSEDED_MODEL_BRANCH

$$\boxed{\delta\rho_{\rm DE}^{(1)} = \delta p_{\rm DE}^{(1)} = \delta q_{\rm DE}^{(1)} = 0.} \tag{A.2752}$$

F2477 50_unified_refinement_dark_sector.tex:300 SUPERSEDED_MODEL_BRANCH

$$M_\phi^2 = 3\lambda \frac{\rho_D}{m_{\rm lock}^2} - \mu^2. \tag{A.2753}$$

F2478 50_unified_refinement_dark_sector.tex:305 SUPERSEDED_MODEL_BRANCH

$$\boxed{\rho_{D,\rm exit} = \frac{\mu^2 m_{\rm lock}^2}{3\lambda}.} \tag{A.2754}$$

F2479 50_unified_refinement_dark_sector.tex:312 SUPERSEDED_MODEL_BRANCH

$$\boxed{\frac{\rho_{D,\rm exit}}{\rho_V} = \frac{4m_{\rm lock}^2}{3\mu^2}.} \tag{A.2755}$$

F2480 50_unified_refinement_dark_sector.tex:321 SUPERSEDED_MODEL_BRANCH

$$r_D(a) := \frac{\rho_D(a)}{\rho_V}, \qquad r_b(a) := \frac{\rho_b(a)}{\rho_V}, \qquad \xi := \frac{m_{\rm lock}^2}{\mu^2}. \tag{A.2756}$$

F2481 50_unified_refinement_dark_sector.tex:329 SUPERSEDED_MODEL_BRANCH

$$\text{locked phase:} \qquad r_D > \frac{4\xi}{3}, \tag{A.2757}$$

$$\text{accelerated expansion:} \qquad r_D + r_b < 2. \tag{A.2758}$$

F2482 50_unified_refinement_dark_sector.tex:337 SUPERSEDED_MODEL_BRANCH

$$r_D(a) = r_{D0} a^{-3}, \qquad r_b(a) = r_{b0} a^{-3}, \tag{A.2759}$$

F2483 50_unified_refinement_dark_sector.tex:343 SUPERSEDED_MODEL_BRANCH

$$\boxed{a_{\rm acc} = \left(\frac{r_{D0} + r_{b0}}{2}\right)^{1/3},} \tag{A.2760}$$

$$\boxed{a_{\rm exit} = \left(\frac{3r_{D0}}{4\xi}\right)^{1/3}.} \tag{A.2761}$$

F2484 50_unified_refinement_dark_sector.tex:355 SUPERSEDED_MODEL_BRANCH

$$\boxed{\frac{m_{\rm lock}}{\mu} < \sqrt{\frac{3r_{D0}}{4}}.} \tag{A.2762}$$

F2485 50_unified_refinement_dark_sector.tex:365 SUPERSEDED_MODEL_BRANCH

$$\Omega_{r0} = 9 \times 10^{-5}, \qquad \Omega_{b0} = 0.049, \qquad \Omega_{D0} = 0.261, \qquad \Omega_{V0} = 0.68991. \tag{A.2763}$$

F2486 50_unified_refinement_dark_sector.tex:375 SUPERSEDED_MODEL_BRANCH

$$r_{D0} = 0.3783102144, \qquad r_{b0} = 0.0710237567. \tag{A.2764}$$

F2487 50_unified_refinement_dark_sector.tex:381 SUPERSEDED_MODEL_BRANCH

$$p = 2, \qquad \sigma = 1, \qquad \frac{m_1}{\mu} = 0.45, \qquad \frac{\nu}{\mu^2} = 0.3, \qquad H_{\max} = 8. \tag{A.2765}$$

F2488 50_unified_refinement_dark_sector.tex:393 SUPERSEDED_MODEL_BRANCH

$$\frac{\Lambda}{\mu} = 0.5482928050. \tag{A.2766}$$

F2489 50_unified_refinement_dark_sector.tex:397 SUPERSEDED_MODEL_BRANCH

$$\boxed{\frac{m_{\rm lock}}{\mu} = 0.45, \qquad \xi = 0.2025.} \tag{A.2767}$$

F2490 50_unified_refinement_dark_sector.tex:405 SUPERSEDED_MODEL_BRANCH

$$\boxed{\frac{M_\phi^2(1)}{\mu^2} = 0.4011489421 > 0.} \tag{A.2768}$$

F2491 50_unified_refinement_dark_sector.tex:412 SUPERSEDED_MODEL_BRANCH

$$\boxed{a_{\rm acc} = 0.6079199831, \qquad a_{\rm exit} = 1.1189948843.} \tag{A.2769}$$

F2492 50_unified_refinement_dark_sector.tex:422 SUPERSEDED_MODEL_BRANCH

$$\frac{M_\phi^2(1)}{\mu^2} = -0.2906683480, \tag{A.2770}$$

F2493 50_unified_refinement_dark_sector.tex:468 SUPERSEDED_MODEL_BRANCH

$$\boxed{\begin{array}{c}\text{one refinement-field action for clustered dark matter and vacuum-like dark energy,}\\ \text{a fixed false-vacuum density } \rho_V = \mu^4/(4\lambda),\\ \text{exact } w_{\rm DE} = -1 \text{ while the average mode is locked,}\\ \text{ghost and gradient stability with } c_s^2 = 1,\\ \text{vanishing linear dark-energy stress perturbations,}\\ \text{a mass-moment criterion for current trapping,}\\ \text{and a derived future release density.}\end{array}} \tag{A.2771}$$

51_self_consistent_refinement_crossing_backreaction.tex

F2494 51_self_consistent_refinement_crossing_backreaction.tex:15 SUPERSEDED_MODEL_BRANCH

$$U(\phi) = \frac{\lambda}{4}(\phi^2 - v^2)^2, \qquad \mu^2 = \lambda v^2, \qquad \rho_V = U(0) = \frac{\mu^4}{4\lambda}. \tag{A.2772}$$

F2495 51_self_consistent_refinement_crossing_backreaction.tex:23 SUPERSEDED_MODEL_BRANCH

$$m_h^2 = \Lambda^2 p^{2\sigma h} - \mu^2, \qquad d_h = (p-1)p^{h-1}. \tag{A.2773}$$

F2496 51_self_consistent_refinement_crossing_backreaction.tex:29 SUPERSEDED_MODEL_BRANCH

$$\frac{H_*}{\sqrt{\nu}} \ll 1, \qquad \nu := \sqrt{3\lambda}\,|\dot\phi_*|. \tag{A.2774}$$

F2497 51_self_consistent_refinement_crossing_backreaction.tex:35 SUPERSEDED_MODEL_BRANCH

$$\phi(t) = \dot\phi_*(t - t_*) + \mathcal{O}((t-t_*)^2), \tag{A.2775}$$

F2498 51_self_consistent_refinement_crossing_backreaction.tex:39 SUPERSEDED_MODEL_BRANCH

$$\ddot u_{h,k} + \left[k^2 + m_h^2 + \nu^2(t-t_*)^2\right] u_{h,k} = 0. \tag{A.2776}$$

F2499 51_self_consistent_refinement_crossing_backreaction.tex:44 SUPERSEDED_MODEL_BRANCH

$$\boxed{f_h(k) = \exp\left[-\frac{\pi(k^2 + m_h^2)}{\nu}\right].} \tag{A.2777}$$

F2500 51_self_consistent_refinement_crossing_backreaction.tex:53 SUPERSEDED_MODEL_BRANCH

$$y_h := \frac{m_h}{\sqrt{\nu}}. \tag{A.2778}$$

F2501 51_self_consistent_refinement_crossing_backreaction.tex:57 SUPERSEDED_MODEL_BRANCH

$$I_\rho(y) := \int_0^\infty x^2\sqrt{x^2+y^2}\,e^{-\pi x^2}\,\mathrm{d}x, \tag{A.2779}$$

$$I_p(y) := \int_0^\infty \frac{x^4}{\sqrt{x^2+y^2}} e^{-\pi x^2}\,\mathrm{d}x, \tag{A.2780}$$

$$I_X(y) := \int_0^\infty \frac{x^2}{\sqrt{x^2+y^2}} e^{-\pi x^2}\ \mathrm{d}x. \tag{A.2781}$$

F2502 51_self_consistent_refinement_crossing_backreaction.tex:68 SUPERSEDED_MODEL_BRANCH

$$I_\rho(y) = \frac{y^4}{2}\Gamma\left(\frac{3}{2}\right) U\left(\frac{3}{2}, 3, \pi y^2\right), \tag{A.2782}$$

$$I_p(y) = \frac{y^4}{2}\Gamma\left(\frac{5}{2}\right) U\left(\frac{5}{2}, 3, \pi y^2\right), \tag{A.2783}$$

$$I_X(y) = \frac{y^2}{2}\Gamma\left(\frac{3}{2}\right) U\left(\frac{3}{2}, 2, \pi y^2\right). \tag{A.2784}$$

F2503 51_self_consistent_refinement_crossing_backreaction.tex:82 SUPERSEDED_MODEL_BRANCH

$$n_{h,*} = \frac{d_h \nu^{3/2}}{8\pi^3} e^{-\pi m_h^2/\nu}, \tag{A.2785}$$

$$\rho_{h,*} = \frac{d_h \nu^2}{2\pi^2} e^{-\pi m_h^2/\nu} I_\rho(y_h), \tag{A.2786}$$

$$p_{h,*} = \frac{d_h \nu^2}{6\pi^2} e^{-\pi m_h^2/\nu} I_p(y_h), \tag{A.2787}$$

$$X_{h,*} = \frac{d_h \nu}{2\pi^2} e^{-\pi m_h^2/\nu} I_X(y_h). \tag{A.2788}$$

F2504 51_self_consistent_refinement_crossing_backreaction.tex:103 SUPERSEDED_MODEL_BRANCH

$$K_* := \frac{1}{2}\dot\phi_*^2 \tag{A.2789}$$

F2505 51_self_consistent_refinement_crossing_backreaction.tex:107 SUPERSEDED_MODEL_BRANCH

$$\rho_* := \sum_h \rho_{h,*}. \tag{A.2790}$$

F2506 51_self_consistent_refinement_crossing_backreaction.tex:111 SUPERSEDED_MODEL_BRANCH

$$\boxed{\dot\phi_{*,+}^2 = \max\left(\dot\phi_{*,-}^2 - 2\rho_*, 0\right).} \tag{A.2791}$$

F2507 51_self_consistent_refinement_crossing_backreaction.tex:118 SUPERSEDED_MODEL_BRANCH

$$\eta_{\mathrm{cap}} := \frac{\rho_*}{K_*}. \tag{A.2792}$$

F2508 51_self_consistent_refinement_crossing_backreaction.tex:122 SUPERSEDED_MODEL_BRANCH

$$\eta_{\mathrm{cap}} < 1 \Longrightarrow \dot\phi_{*,+} \neq 0, \tag{A.2793}$$

F2509 51_self_consistent_refinement_crossing_backreaction.tex:128 SUPERSEDED_MODEL_BRANCH

$$\eta_{\mathrm{cap}} \geq 1 \tag{A.2794}$$

F2510 51_self_consistent_refinement_crossing_backreaction.tex:134 SUPERSEDED_MODEL_BRANCH

$$\rho_D(a,\phi) = \sum_h \frac{d_h}{2\pi^2 a^4} \int_0^\infty q^2 \epsilon_h(q,a,\phi) f_h(q)\ \mathrm{d}q, \tag{A.2795}$$

F2511 51_self_consistent_refinement_crossing_backreaction.tex:140 SUPERSEDED_MODEL_BRANCH

$$\epsilon_h(q,a,\phi) = \sqrt{q^2 + a^2[m_h^2 + 3\lambda\phi^2]}. \tag{A.2796}$$

F2512 51_self_consistent_refinement_crossing_backreaction.tex:148 SUPERSEDED_MODEL_BRANCH

$$\widehat{X}(\phi) := \frac{1}{2\pi^2} \int_0^\infty \frac{k^2 f(k)}{\sqrt{k^2 + m_1^2 + 3\lambda\phi^2}}\ \mathrm{d}k \tag{A.2797}$$

F2513 51_self_consistent_refinement_crossing_backreaction.tex:155 SUPERSEDED_MODEL_BRANCH

$$\widehat{\rho}(\phi) := \frac{1}{2\pi^2} \int_0^\infty k^2 f(k) \sqrt{k^2 + m_1^2 + 3\lambda\phi^2}\ \mathrm{d}k. \tag{A.2798}$$

F2514 51_self_consistent_refinement_crossing_backreaction.tex:162 SUPERSEDED_MODEL_BRANCH

$$\boxed{V_{\rm eff}(\phi) = U(\phi) + g\widehat{\rho}(\phi).} \tag{A.2799}$$

F2515 51_self_consistent_refinement_crossing_backreaction.tex:169 SUPERSEDED_MODEL_BRANCH

$$V'_{\rm eff}(\phi) = \lambda\phi\left[\phi^2 - v^2 + 3g\widehat{X}(\phi)\right]. \tag{A.2800}$$

F2516 51_self_consistent_refinement_crossing_backreaction.tex:175 SUPERSEDED_MODEL_BRANCH

$$\boxed{g_{\rm local} = \frac{\mu^2}{3\lambda\widehat{X}(0)}.} \tag{A.2801}$$

F2517 51_self_consistent_refinement_crossing_backreaction.tex:182 SUPERSEDED_MODEL_BRANCH

$$\boxed{g_{\rm global} = \sup_{0\le\phi\le v}\frac{v^2-\phi^2}{3\widehat{X}(\phi)}.} \tag{A.2802}$$

F2518 51_self_consistent_refinement_crossing_backreaction.tex:192 SUPERSEDED_MODEL_BRANCH

$$\boxed{g_{\rm capture} = \frac{K_*}{\widehat{\rho}(0)}.} \tag{A.2803}$$

F2519 51_self_consistent_refinement_crossing_backreaction.tex:198 SUPERSEDED_MODEL_BRANCH

$$g < g_{\rm local}: \quad \text{origin unstable,} \tag{A.2804}$$
$$g_{\rm local} < g < g_{\rm global}: \quad \text{local metastable well,} \tag{A.2805}$$
$$g_{\rm global} < g < g_{\rm capture}: \quad \text{global lock with finite outgoing oscillation,} \tag{A.2806}$$
$$g \ge g_{\rm capture}: \quad \text{direct-capture regime.} \tag{A.2807}$$

F2520 51_self_consistent_refinement_crossing_backreaction.tex:215 SUPERSEDED_MODEL_BRANCH

$$I_\rho(0) = \frac{1}{2\pi^2}, \qquad I_X(0) = \frac{1}{2\pi}. \tag{A.2808}$$

F2521 51_self_consistent_refinement_crossing_backreaction.tex:221 SUPERSEDED_MODEL_BRANCH

$$\widehat{\rho}(0) \sim \frac{\nu^2}{4\pi^4}, \qquad \widehat{X}(0) \sim \frac{\nu}{4\pi^3}. \tag{A.2809}$$

F2522 51_self_consistent_refinement_crossing_backreaction.tex:229 SUPERSEDED_MODEL_BRANCH

$$\boxed{g_{\rm capture} \longrightarrow \frac{2\pi^4}{3\lambda}.} \tag{A.2810}$$

F2523 51_self_consistent_refinement_crossing_backreaction.tex:240 SUPERSEDED_MODEL_BRANCH

$$\rho_D(a,\phi) = \sum_h \frac{d_h}{2\pi^2 a^4}\int q^2\epsilon_h f_h\ \mathrm{d}q, \tag{A.2811}$$
$$p_D(a,\phi) = \sum_h \frac{d_h}{6\pi^2 a^4}\int \frac{q^4}{\epsilon_h} f_h\ \mathrm{d}q, \tag{A.2812}$$
$$X_D(a,\phi) = \sum_h \frac{d_h}{2\pi^2 a^2}\int \frac{q^2}{\epsilon_h} f_h\ \mathrm{d}q. \tag{A.2813}$$

F2524 51_self_consistent_refinement_crossing_backreaction.tex:254 SUPERSEDED_MODEL_BRANCH

$$H^2 = \frac{1}{3M_{\rm Pl}^2}\left[\frac{1}{2}\dot\phi^2 + U(\phi) + \rho_D\right], \tag{A.2814}$$
$$\ddot\phi + 3H\dot\phi + U_{,\phi} + 3\lambda\phi X_D = 0. \tag{A.2815}$$

F2525 51_self_consistent_refinement_crossing_backreaction.tex:263 SUPERSEDED_MODEL_BRANCH

$$\dot{\rho}_D = -3H(\rho_D + p_D) + 3\lambda\phi\dot{\phi}X_D, \tag{A.2816}$$

F2526 51_self_consistent_refinement_crossing_backreaction.tex:269 SUPERSEDED_MODEL_BRANCH

$$\dot{\rho}_\phi = -3H\dot{\phi}^2 - 3\lambda\phi\dot{\phi}X_D. \tag{A.2817}$$

F2527 51_self_consistent_refinement_crossing_backreaction.tex:275 SUPERSEDED_MODEL_BRANCH

$$\boxed{\dot{\rho}_{\rm tot} + 3H(\rho_{\rm tot} + p_{\rm tot}) = 0.} \tag{A.2818}$$

F2528 51_self_consistent_refinement_crossing_backreaction.tex:285 SUPERSEDED_MODEL_BRANCH

$$\mu = \lambda = 1, \qquad m_1 = 0.45, \qquad \phi_i = -2, \qquad \dot{\phi}_i = 0. \tag{A.2819}$$

F2529 51_self_consistent_refinement_crossing_backreaction.tex:295 SUPERSEDED_MODEL_BRANCH

$$K_* = 2, \qquad |\dot{\phi}_*| = 2, \qquad \nu = \sqrt{12} = 3.4641016151. \tag{A.2820}$$

F2530 51_self_consistent_refinement_crossing_backreaction.tex:303 SUPERSEDED_MODEL_BRANCH

$$g_{\rm local} = 17.2875822920, \tag{A.2821}$$
$$g_{\rm global} = 17.9043887055, \tag{A.2822}$$
$$g_{\rm capture} = 72.0989329581. \tag{A.2823}$$

F2531 51_self_consistent_refinement_crossing_backreaction.tex:311 SUPERSEDED_MODEL_BRANCH

$$\phi_{\rm global} = 0.3252548920. \tag{A.2824}$$

F2532 51_self_consistent_refinement_crossing_backreaction.tex:315 SUPERSEDED_MODEL_BRANCH

$$\boxed{p_{\rm global} = 19,} \tag{A.2825}$$

F2533 51_self_consistent_refinement_crossing_backreaction.tex:319 SUPERSEDED_MODEL_BRANCH

$$\boxed{p_{\rm capture} = 79.} \tag{A.2826}$$

F2534 51_self_consistent_refinement_crossing_backreaction.tex:324 SUPERSEDED_MODEL_BRANCH

$$n_* = 1.5574778055, \tag{A.2827}$$
$$\rho_* = 1.9972556332, \tag{A.2828}$$
$$X_* = 1.3882797256, \tag{A.2829}$$
$$\eta_{\rm cap} = 0.9986278166, \tag{A.2830}$$
$$\dot{\phi}_{*,+} = 0.0740859876. \tag{A.2831}$$

F2535 51_self_consistent_refinement_crossing_backreaction.tex:351 SUPERSEDED_MODEL_BRANCH

$$\boxed{a_{\rm acc} = 1.6596778467, \qquad t_{\rm acc} = 8.7878307745.} \tag{A.2832}$$

F2536 51_self_consistent_refinement_crossing_backreaction.tex:359 SUPERSEDED_MODEL_BRANCH

$$\boxed{a_{\rm release} = 1.8424690117, \qquad t_{\rm release} = 11.3673608512.} \tag{A.2833}$$

F2537 51_self_consistent_refinement_crossing_backreaction.tex:367 SUPERSEDED_MODEL_BRANCH

$$\max|\phi| = 0.0394198188. \tag{A.2834}$$

F2538 51_self_consistent_refinement_crossing_backreaction.tex:371 SUPERSEDED_MODEL_BRANCH

$$w_D = 0.2189375169, \qquad w_{\rm tot} = -0.4568362431. \tag{A.2835}$$

F2539 51_self_consistent_refinement_crossing_backreaction.tex:377 SUPERSEDED_MODEL_BRANCH

$$\boxed{a_{\rm end} = 2.3209537346, \qquad t_{\rm end} = 17.8431038770.} \tag{A.2836}$$

F2540 51_self_consistent_refinement_crossing_backreaction.tex:385 SUPERSEDED_MODEL_BRANCH

$$\Delta N_{\rm acc} = \log\left(\frac{a_{\rm end}}{a_{\rm acc}}\right) = 0.3353546784. \tag{A.2837}$$

F2541 51_self_consistent_refinement_crossing_backreaction.tex:391 SUPERSEDED_MODEL_BRANCH

$$\boxed{\begin{array}{c}\text{detail production} \Longrightarrow \text{global average-mode lock} \\ \Longrightarrow \text{accelerated expansion} \Longrightarrow \text{density-driven release}\end{array}} \tag{A.2838}$$

F2542 51_self_consistent_refinement_crossing_backreaction.tex:469 SUPERSEDED_MODEL_BRANCH

$$\boxed{\begin{array}{c}\text{dynamically derive the crossing epoch and active light-shell content,} \\ \text{propagate the resulting distribution through the full collisionless Boltzmann hierarchy,} \\ \text{compute CMB, matter-power, and weak-lensing observables with one parameter set,} \\ \text{and perform the first likelihood-level falsification test.}\end{array}} \tag{A.2839}$$

52_scale_covariant_refinement_dark_sector.tex

F2543 52_scale_covariant_refinement_dark_sector.tex:7 SCALE_COVARIANT_MODEL_PASS

eq:scw-additive-mass

$$m_h^2(\phi) = m_h^2 + 3\lambda\phi^2, \tag{A.2840}$$

F2544 52_scale_covariant_refinement_dark_sector.tex:14 SCALE_COVARIANT_MODEL_PASS

eq:scw-scale-covariant-rule

$$\mathcal{L}_{\sigma,p} \longmapsto \left(1 + \gamma\frac{\phi^2}{v^2}\right)\mathcal{L}_{\sigma,p}. \tag{A.2841}$$

F2545 52_scale_covariant_refinement_dark_sector.tex:31 SCALE_COVARIANT_MODEL_PASS

eq:scw-additive-definitions

$$\rho_V = \frac{\mu^4}{4\lambda}, \qquad m = \alpha\mu, \qquad \beta = \frac{\nu}{\mu^2}, \qquad r_D = \frac{\rho_{D0}}{\rho_V}. \tag{A.2842}$$

F2546 52_scale_covariant_refinement_dark_sector.tex:42 SCALE_COVARIANT_MODEL_PASS

eq:scw-additive-distribution

$$f(q) = \eta\exp\left[-\frac{\pi m^2}{\nu} - \frac{\pi q^2}{a_\star^2\nu}\right], \qquad 0 < \eta \le 1. \tag{A.2843}$$

F2547 52_scale_covariant_refinement_dark_sector.tex:54 SCALE_COVARIANT_MODEL_PASS

eq:scw-additive-abundance

$$r_D = \frac{\eta g\lambda\alpha\beta^{3/2}}{2\pi^3}e^{-\pi\alpha^2/\beta}a_\star^3, \tag{A.2844}$$

F2548 52_scale_covariant_refinement_dark_sector.tex:63 SCALE_COVARIANT_MODEL_PASS

eq:scw-additive-speed

$$u_0^2 := \frac{\langle p^2\rangle_0}{m^2} = \frac{3\beta a_\star^2}{2\pi\alpha^2}. \tag{A.2845}$$

F2549 52_scale_covariant_refinement_dark_sector.tex:70 SCALE_COVARIANT_MODEL_PASS

eq:scw-additive-elimination

$$u_0^3 = \frac{(3\pi)^{3/2}}{\sqrt{2}}\frac{r_D}{\eta g\lambda\alpha^4}e^{\pi\alpha^2/\beta}. \tag{A.2846}$$

F2550 52_scale_covariant_refinement_dark_sector.tex:80 SCALE_COVARIANT_MODEL_PASS

eq:scw-additive-unit-energy

$$\widetilde{\rho}_\star = \frac{g\nu^2}{4\pi^4}, \tag{A.2847}$$

F2551 52_scale_covariant_refinement_dark_sector.tex:86 SCALE_COVARIANT_MODEL_PASS

eq:scw-additive-kinetic

$$K_\star = \frac{\nu^2}{6\lambda}. \tag{A.2848}$$

F2552 52_scale_covariant_refinement_dark_sector.tex:91 SCALE_COVARIANT_MODEL_PASS

eq:scw-additive-energy-bound

$$\eta g \lambda \le \frac{2\pi^4}{3}. \tag{A.2849}$$

F2553 52_scale_covariant_refinement_dark_sector.tex:96 SCALE_COVARIANT_MODEL_PASS

eq:scw-additive-lock-bound

$$-\mu^2 + 3\lambda \frac{\rho_{D0}}{m^2} > 0 \quad \Longrightarrow \quad \alpha^2 < \frac{3 r_D}{4}. \tag{A.2850}$$

F2554 52_scale_covariant_refinement_dark_sector.tex:103 SCALE_COVARIANT_MODEL_PASS

eq:scw-additive-no-go

$$\boxed{u_0^3 > \frac{4\sqrt{6}}{\pi^{5/2} r_D}.} \tag{A.2851}$$

F2555 52_scale_covariant_refinement_dark_sector.tex:111 SCALE_COVARIANT_MODEL_PASS

eq:scw-rd-benchmark

$$r_D = 0.3783102144, \tag{A.2852}$$

F2556 52_scale_covariant_refinement_dark_sector.tex:116 SCALE_COVARIANT_MODEL_PASS

eq:scw-additive-numeric-bound

$$\boxed{u_0 > 1.1397360601.} \tag{A.2853}$$

F2557 52_scale_covariant_refinement_dark_sector.tex:132 SCALE_COVARIANT_MODEL_PASS

eq:scw-average-potential

$$U(\phi) = \frac{\lambda}{4}(\phi^2 - v^2)^2, \qquad \mu^2 = \lambda v^2, \tag{A.2854}$$

F2558 52_scale_covariant_refinement_dark_sector.tex:139 SCALE_COVARIANT_MODEL_PASS

eq:scw-fiber-operator

$$\mathcal{L}_{\sigma,p} = \Lambda^2 \sum_{h \ge 1} p^{2\sigma h} \mathsf{D}_h. \tag{A.2855}$$

F2559 52_scale_covariant_refinement_dark_sector.tex:145 SCALE_COVARIANT_MODEL_PASS

eq:scw-action

$$\boxed{\begin{aligned} S_{\mathrm{SC}} = -\int \sqrt{-g}\, d^4x \Bigg[& \frac{1}{2}(\partial\phi)^2 + U(\phi) + \frac{1}{2}\langle \partial\chi, \partial\chi \rangle_p \\ & + \frac{1}{2}\left\langle \chi, \left(m_0^2 - \mu^2 + \left(1 + \gamma \frac{\phi^2}{v^2}\right) \mathcal{L}_{\sigma,p} \right) \chi \right\rangle_p \Bigg]. \end{aligned}} \tag{A.2856}$$

F2560 52_scale_covariant_refinement_dark_sector.tex:165 SCALE_COVARIANT_MODEL_PASS

eq:scw-level-masses

$$m_h^2 = m_0^2 - \mu^2 + \Lambda^2 p^{2\sigma h}, \tag{A.2857}$$

F2561 52_scale_covariant_refinement_dark_sector.tex:171 SCALE_COVARIANT_MODEL_PASS

eq:scw-level-degeneracy

$$d_h = (p-1)p^{h-1}. \tag{A.2858}$$

F2562 52_scale_covariant_refinement_dark_sector.tex:176 SCALE_COVARIANT_MODEL_PASS

eq:scw-average-eom

$$\ddot{\phi} + 3H\dot{\phi} + \lambda\phi(\phi^2 - v^2) + \frac{\gamma\phi}{v^2} Y_D = 0, \tag{A.2859}$$

F2563 52_scale_covariant_refinement_dark_sector.tex:183 SCALE_COVARIANT_MODEL_PASS

eq:scw-weighted-lock-source

$$Y_D := \sum_h \Lambda^2 p^{2\sigma h} X_h, \qquad X_h := \langle \chi_h^2 \rangle. \tag{A.2860}$$

F2564 52_scale_covariant_refinement_dark_sector.tex:191 SCALE_COVARIANT_MODEL_PASS

eq:scw-lock-curvature

$$\boxed{M_\phi^2 = -\mu^2 + \frac{\gamma}{v^2} Y_D.} \tag{A.2861}$$

F2565 52_scale_covariant_refinement_dark_sector.tex:198 SCALE_COVARIANT_MODEL_PASS

eq:scw-weighted-energy-limit

$$\Lambda^2 p^{2\sigma h} X_h = \rho_h \left[1 + O(w_h) + O\left(\frac{\mu^2 + m_0^2}{m_h^2} \right) \right]. \tag{A.2862}$$

F2566 52_scale_covariant_refinement_dark_sector.tex:208 SCALE_COVARIANT_MODEL_PASS

eq:scw-crossing-linearization

$$\phi(t) = \dot{\phi}_\star (t - t_\star) + O((t - t_\star)^2), \tag{A.2863}$$

F2567 52_scale_covariant_refinement_dark_sector.tex:213 SCALE_COVARIANT_MODEL_PASS

eq:scw-crossing-frequency

$$\omega_{h,k}^2 = k^2 + m_h^2 + \frac{\gamma \Lambda^2 p^{2\sigma h}}{v^2} \dot{\phi}_\star^2 (t - t_\star)^2. \tag{A.2864}$$

F2568 52_scale_covariant_refinement_dark_sector.tex:221 SCALE_COVARIANT_MODEL_PASS

eq:scw-weighted-rate

$$A_\star := \frac{\sqrt{\gamma} |\dot{\phi}_\star|}{v}, \qquad \nu_h := A_\star \Lambda p^{\sigma h}. \tag{A.2865}$$

F2569 52_scale_covariant_refinement_dark_sector.tex:228 SCALE_COVARIANT_MODEL_PASS

eq:scw-weighted-occupation

$$\boxed{f_h(k) = \eta \exp\left[-\frac{\pi(k^2 + m_h^2)}{\nu_h} \right],} \tag{A.2866}$$

F2570 52_scale_covariant_refinement_dark_sector.tex:236 SCALE_COVARIANT_MODEL_PASS

eq:scw-energy-projection

$$\eta = \min\left(1, \frac{K_\star}{\widetilde{\rho}_\star} \right). \tag{A.2867}$$

F2571 52_scale_covariant_refinement_dark_sector.tex:244 SCALE_COVARIANT_MODEL_PASS

eq:scw-high-level-law

$$m_h \simeq \Lambda p^{\sigma h}, \qquad \nu_h \simeq A_\star m_h. \tag{A.2868}$$

F2572 52_scale_covariant_refinement_dark_sector.tex:251 SCALE_COVARIANT_MODEL_PASS

eq:scw-linear-mass-suppression

$$\frac{\pi m_h^2}{\nu_h} \simeq \frac{\pi m_h}{A_\star}. \tag{A.2869}$$

F2573 52_scale_covariant_refinement_dark_sector.tex:257 SCALE_COVARIANT_MODEL_PASS

eq:scw-heavy-colder

$$\frac{p_{\mathrm{rms},\star}}{m_h} \simeq \sqrt{\frac{3A_\star}{2\pi m_h}}, \tag{A.2870}$$

F2574 52_scale_covariant_refinement_dark_sector.tex:268 SCALE_COVARIANT_MODEL_PASS

eq:scw-state-count

$$N(m) = \frac{m}{\Lambda} - 1 \tag{A.2871}$$

F2575 52_scale_covariant_refinement_dark_sector.tex:273 SCALE_COVARIANT_MODEL_PASS

eq:scw-density-of-levels

$$\frac{dN}{dm} = \frac{1}{\Lambda}. \tag{A.2872}$$

F2576 52_scale_covariant_refinement_dark_sector.tex:278 SCALE_COVARIANT_MODEL_PASS

eq:scw-number-spectrum

$$\frac{dn}{dm} = \eta \frac{A^{3/2}}{8\pi^3 \Lambda} m^{3/2} e^{-\pi m/A}, \tag{A.2873}$$

F2577 52_scale_covariant_refinement_dark_sector.tex:284 SCALE_COVARIANT_MODEL_PASS

eq:scw-rest-spectrum

$$\boxed{\frac{d\rho_R}{dm} = \eta \frac{A^{3/2}}{8\pi^3 \Lambda} m^{5/2} e^{-\pi m/A}.} \tag{A.2874}$$

F2578 52_scale_covariant_refinement_dark_sector.tex:292 SCALE_COVARIANT_MODEL_PASS

eq:scw-mass-peak

$$\boxed{m_{\mathrm{peak}} = \frac{5A}{2\pi}.} \tag{A.2875}$$

F2579 52_scale_covariant_refinement_dark_sector.tex:298 SCALE_COVARIANT_MODEL_PASS

eq:scw-number-constant

$$n_\star = C_N \eta \frac{A^4}{\Lambda}, \qquad C_N = \frac{3}{32\pi^5}, \tag{A.2876}$$

F2580 52_scale_covariant_refinement_dark_sector.tex:304 SCALE_COVARIANT_MODEL_PASS

eq:scw-rest-constant

$$\rho_{R,\star} = C_R \eta \frac{A^5}{\Lambda}, \qquad C_R = \frac{15}{64\pi^6}, \tag{A.2877}$$

F2581 52_scale_covariant_refinement_dark_sector.tex:310 SCALE_COVARIANT_MODEL_PASS

eq:scw-energy-constant

$$\rho_\star = C_E \eta \frac{A^5}{\Lambda}, \qquad C_E = \frac{32}{35\pi^7}, \tag{A.2878}$$

F2582 52_scale_covariant_refinement_dark_sector.tex:316 SCALE_COVARIANT_MODEL_PASS

eq:scw-pressure-constant

$$p_\star = C_P \eta \frac{A^5}{\Lambda}, \qquad C_P = \frac{32}{315\pi^7}. \tag{A.2879}$$

F2583 52_scale_covariant_refinement_dark_sector.tex:323 SCALE_COVARIANT_MODEL_PASS

eq:scw-energy-beta-proof

$$\begin{aligned} C_E &= \frac{1}{2\pi^2} \int_0^\infty y^2 e^{-\pi y} I_\rho(\sqrt{y})\, dy \\ &= \frac{6}{\pi^7} B\left(\frac{3}{2}, 3\right) = \frac{32}{35\pi^7}. \end{aligned} \tag{A.2880}$$

F2584 52_scale_covariant_refinement_dark_sector.tex:334 SCALE_COVARIANT_MODEL_PASS

eq:scw-production-w

$$\boxed{w_\star = \frac{p_\star}{\rho_\star} = \frac{1}{9},} \tag{A.2881}$$

F2585 52_scale_covariant_refinement_dark_sector.tex:339 SCALE_COVARIANT_MODEL_PASS

eq:scw-rest-fraction

$$\boxed{f_R := \frac{\rho_{R,\star}}{\rho_\star} = \frac{525\pi}{2048} = 0.8053399136\ldots,} \tag{A.2882}$$

F2586 52_scale_covariant_refinement_dark_sector.tex:347 SCALE_COVARIANT_MODEL_PASS

eq:scw-mean-mass

$$\frac{\rho_{R,\star}}{n_\star} = \frac{5A}{2\pi} = m_{\mathrm{peak}}. \tag{A.2883}$$

F2587 52_scale_covariant_refinement_dark_sector.tex:355 SCALE_COVARIANT_MODEL_PASS

eq:scw-discrete-ratios

$$f_R^{(2)} = 0.8054331018, \qquad w_\star^{(2)} = 0.1110616168, \tag{A.2884}$$

F2588 52_scale_covariant_refinement_dark_sector.tex:362 SCALE_COVARIANT_MODEL_PASS

eq:scw-discrete-peak

$$\frac{m_{\mathrm{peak}}^{(2)}}{A} = 0.8192, \tag{A.2885}$$

F2589 52_scale_covariant_refinement_dark_sector.tex:378 SCALE_COVARIANT_MODEL_PASS

eq:scw-captured-rest

$$\rho_{R,\star} = f_R K_\star. \tag{A.2886}$$

F2590 52_scale_covariant_refinement_dark_sector.tex:383 SCALE_COVARIANT_MODEL_PASS

eq:scw-present-abundance

$$\rho_{D0} = f_R K_\star a_\star^3. \tag{A.2887}$$

F2591 52_scale_covariant_refinement_dark_sector.tex:388 SCALE_COVARIANT_MODEL_PASS

eq:scw-F-definition

$$F := \frac{K_\star}{\rho_V}. \tag{A.2888}$$

F2592 52_scale_covariant_refinement_dark_sector.tex:393 SCALE_COVARIANT_MODEL_PASS

eq:scw-F-calibration

$$\boxed{F = \frac{r_D}{f_R a_\star^3}.} \tag{A.2889}$$

F2593 52_scale_covariant_refinement_dark_sector.tex:399 SCALE_COVARIANT_MODEL_PASS

eq:scw-A-over-mu

$$\frac{A_\star}{\mu} = \sqrt{\frac{\gamma F}{2}}, \tag{A.2890}$$

F2594 52_scale_covariant_refinement_dark_sector.tex:404 SCALE_COVARIANT_MODEL_PASS

eq:scw-peak-hierarchy

$$\boxed{\frac{m_{\mathrm{peak}}}{\mu} = \frac{5}{2\pi}\sqrt{\frac{\gamma r_D}{2 f_R a_\star^3}}.} \tag{A.2891}$$

F2595 52_scale_covariant_refinement_dark_sector.tex:412 SCALE_COVARIANT_MODEL_PASS

eq:scw-initial-energy

$$F = \left(\frac{\phi_i}{v}\right)^4 - 2\left(\frac{\phi_i}{v}\right)^2, \tag{A.2892}$$

F2596 52_scale_covariant_refinement_dark_sector.tex:419 SCALE_COVARIANT_MODEL_PASS

eq:scw-initial-amplitude

$$\boxed{\frac{|\phi_i|}{v} = \sqrt{1+\sqrt{1+F}}.} \tag{A.2893}$$

F2597 52_scale_covariant_refinement_dark_sector.tex:428 SCALE_COVARIANT_MODEL_PASS

eq:scw-benchmark-inputs

$$a_\star = 10^{-6}, \qquad \gamma = 16, \qquad p = 2, \qquad \frac{\Lambda}{A_\star} = 10^{-4}, \tag{A.2894}$$

F2598 52_scale_covariant_refinement_dark_sector.tex:439 SCALE_COVARIANT_MODEL_PASS

eq:scw-benchmark-F

$$\frac{K_\star}{\rho_V} = 4.6969787254 \times 10^{17}, \tag{A.2895}$$

F2599 52_scale_covariant_refinement_dark_sector.tex:444 SCALE_COVARIANT_MODEL_PASS

eq:scw-benchmark-hierarchy

$$\frac{A_\star}{\mu} = 1.9384486014 \times 10^{9}, \qquad \frac{|\phi_i|}{v} = 2.6179096175 \times 10^{4}, \tag{A.2896}$$

F2600 52_scale_covariant_refinement_dark_sector.tex:451 SCALE_COVARIANT_MODEL_PASS

eq:scw-benchmark-peak

$$\boxed{\frac{m^{(2)}_{\mathrm{peak}}}{\mu} = 1.5879770943 \times 10^{9}.} \tag{A.2897}$$

F2601 52_scale_covariant_refinement_dark_sector.tex:458 SCALE_COVARIANT_MODEL_PASS

eq:scw-benchmark-eta

$$\eta = 3.8062069178 \times 10^{-21}. \tag{A.2898}$$

F2602 52_scale_covariant_refinement_dark_sector.tex:467 SCALE_COVARIANT_MODEL_PASS

eq:scw-late-w

$$\boxed{w_D(a) = \frac{a_\star^2}{5a^2},} \tag{A.2899}$$

F2603 52_scale_covariant_refinement_dark_sector.tex:472 SCALE_COVARIANT_MODEL_PASS

eq:scw-late-ca

$$\boxed{c^2_{a,D}(a) = \frac{a_\star^2}{3a^2},} \tag{A.2900}$$

F2604 52_scale_covariant_refinement_dark_sector.tex:478 SCALE_COVARIANT_MODEL_PASS

eq:scw-late-speed

$$\boxed{v_{\mathrm{rms}}(a) = \sqrt{\frac{3}{5}}\frac{a_\star}{a}.} \tag{A.2901}$$

F2605 52_scale_covariant_refinement_dark_sector.tex:487 SCALE_COVARIANT_MODEL_PASS

eq:scw-benchmark-coldness

$$w_{D0} = 1.9987378335 \times 10^{-13}, \qquad v_{\mathrm{rms},0} = 7.7459666924 \times 10^{-7}. \tag{A.2902}$$

F2606 52_scale_covariant_refinement_dark_sector.tex:494 SCALE_COVARIANT_MODEL_PASS

eq:scw-jeans-definition

$$\frac{k_J(a)}{aH(a)} = \sqrt{\frac{3\Omega_m(a)}{2c^2_{a,D}(a)}}, \tag{A.2903}$$

F2607 52_scale_covariant_refinement_dark_sector.tex:500 SCALE_COVARIANT_MODEL_PASS

eq:scw-jeans-law

$$\boxed{\frac{k_J(a)}{H_0} = \frac{3}{\sqrt{2}}\frac{\sqrt{\Omega_{m0}a}}{a_\star}.} \tag{A.2904}$$

F2608 52_scale_covariant_refinement_dark_sector.tex:508 SCALE_COVARIANT_MODEL_PASS

eq:scw-benchmark-jeans

$$\frac{k_{J0}}{H_0} = 1.1811011811 \times 10^6. \tag{A.2905}$$

F2609 52_scale_covariant_refinement_dark_sector.tex:516 SCALE_COVARIANT_MODEL_PASS

eq:scw-lock-density-law

$$\frac{M_\phi^2}{\mu^2} = -1 + \frac{\gamma}{4} r_D(a) + O(w_D) + O\left(\frac{\mu^2}{m_{\rm eff}^2}\right). \tag{A.2906}$$

F2610 52_scale_covariant_refinement_dark_sector.tex:523 SCALE_COVARIANT_MODEL_PASS

eq:scw-gamma-lock

$$\boxed{\gamma > \frac{4}{r_{D0}}.} \tag{A.2907}$$

F2611 52_scale_covariant_refinement_dark_sector.tex:529 SCALE_COVARIANT_MODEL_PASS

eq:scw-present-lock-curvature

$$\boxed{\frac{M_\phi^2(1)}{\mu^2} = 0.5132408576.} \tag{A.2908}$$

F2612 52_scale_covariant_refinement_dark_sector.tex:535 SCALE_COVARIANT_MODEL_PASS

eq:scw-weighted-source-numeric

$$\frac{Y_D(1)}{\rho_V} = 0.3783102144 \tag{A.2909}$$

F2613 52_scale_covariant_refinement_dark_sector.tex:542 SCALE_COVARIANT_MODEL_PASS

eq:scw-release-law

$$a_{\rm release} = \left(\frac{\gamma r_{D0}}{4}\right)^{1/3}. \tag{A.2910}$$

F2614 52_scale_covariant_refinement_dark_sector.tex:548 SCALE_COVARIANT_MODEL_PASS

eq:scw-release-numeric

$$\boxed{a_{\rm release} = 1.1480726020.} \tag{A.2911}$$

F2615 52_scale_covariant_refinement_dark_sector.tex:554 SCALE_COVARIANT_MODEL_PASS

eq:scw-rb-benchmark

$$r_{b0} = 0.0710237567, \tag{A.2912}$$

F2616 52_scale_covariant_refinement_dark_sector.tex:559 SCALE_COVARIANT_MODEL_PASS

eq:scw-acceleration-numeric

$$\boxed{a_{\rm acc} = 0.6081134084,} \tag{A.2913}$$

F2617 52_scale_covariant_refinement_dark_sector.tex:567 SCALE_COVARIANT_MODEL_PASS

eq:scw-perturbation-action

$$S_{\delta\phi}^{(2)} = \frac{1}{2}\int a^3 dt\, d^3x \left[\delta\dot{\phi}^2 - \frac{(\nabla\delta\phi)^2}{a^2} - M_\phi^2 \delta\phi^2\right]. \tag{A.2914}$$

F2618 52_scale_covariant_refinement_dark_sector.tex:578 SCALE_COVARIANT_MODEL_PASS

eq:scw-perturbation-stability

$$Q_{\delta\phi} = 1, \qquad c_{s,\phi}^2 = 1, \qquad M_\phi^2 > 0 \tag{A.2915}$$

F2619 52_scale_covariant_refinement_dark_sector.tex:598 SCALE_COVARIANT_MODEL_PASS

eq:scw-baryon-growth

$$\delta_b'' + \left(2 + \frac{d\log H}{dN}\right)\delta_b' = \frac{3}{2}S, \tag{A.2916}$$

F2620 52_scale_covariant_refinement_dark_sector.tex:604 SCALE_COVARIANT_MODEL_PASS

eq:scw-dark-growth

$$\delta_D'' + \left(2 + \frac{d\log H}{dN}\right)\delta_D' + \frac{c_{a,D}^2 k^2}{a^2 H^2}\delta_D = \frac{3}{2}S, \tag{A.2917}$$

F2621 52_scale_covariant_refinement_dark_sector.tex:612 SCALE_COVARIANT_MODEL_PASS

eq:scw-growth-source

$$S = \Omega_b(a)\delta_b + \Omega_D(a)\delta_D. \tag{A.2918}$$

F2622 52_scale_covariant_refinement_dark_sector.tex:634 SCALE_COVARIANT_MODEL_PASS

eq:scw-half-mode

$$\boxed{\frac{k_{1/2}}{H_0} = 4.4076353413 \times 10^4.} \tag{A.2919}$$

F2623 52_scale_covariant_refinement_dark_sector.tex:640 SCALE_COVARIANT_MODEL_PASS

eq:scw-half-mode-mpc

$$k_{1/2} = 10.2916022620\,\mathrm{Mpc}^{-1}. \tag{A.2920}$$

F2624 52_scale_covariant_refinement_dark_sector.tex:675 SCALE_COVARIANT_MODEL_PASS

eq:scw-json-record

$$R_{\mathrm{SC}} = \texttt{data/Scale_Covariant_Dark_Sector_Verification.json}, \tag{A.2921}$$

F2625 52_scale_covariant_refinement_dark_sector.tex:679 SCALE_COVARIANT_MODEL_PASS

eq:scw-check-record

$$C_{\mathrm{SC}} = \texttt{data/scale_covariant_dark_checks.csv}, \tag{A.2922}$$

F2626 52_scale_covariant_refinement_dark_sector.tex:683 SCALE_COVARIANT_MODEL_PASS

eq:scw-script-record

$$X_{\mathrm{SC}} = \texttt{verification/Scale_Covariant_Dark_Sector_Verification.py}. \tag{A.2923}$$

F2627 52_scale_covariant_refinement_dark_sector.tex:712 SCALE_COVARIANT_MODEL_PASS

eq:scw-result-chain

$$\boxed{\begin{array}{c}\text{average-mode dilation of the refinement spectrum}\\ \Longrightarrow\ \text{heavy cold detail band}\\ \Longrightarrow\ \text{density-weighted vacuum locking}\end{array}} \tag{A.2924}$$

00b_revision_and_claim_status.tex

F2628 00b_revision_and_claim_status.tex:7 FRONTMATTER_REVIEWED

P : proved in the declared mathematical setting,
N : numerically reproduced with the declared discretization,
C : conditional on a stated model or regularity hypothesis,
Q : qualified or corrected statement,
O : open target or unverified physical interpretation.